



Data Communication Network Basis



Foreword

- Communication has always been with us ever since the origin of human society. Communication has been playing an increasingly important role especially since human society entered the information era in the 1970s or 1980s.
- The communication mentioned in this course refers to the communication implemented through a data communication network. This course describes the concepts related to communication and a data communication network, information transfer process, network devices and their functions, network types, and typical networking. In addition, this course briefly introduces the concepts related to network engineering and network engineers.



Objectives

- On completion of this course, you will be able to:
 - Understand the concepts related to communication and a data communication network.
 - Be able to describe the information transfer process.
 - Differentiate network devices of different types and understand their basic functions.
 - Understand different network types and topology types.
 - Understand the concepts related to network engineering and network engineers.



Huawei Device Icons



General router



General switch



Core switch



Aggregation switch



Access switch



Stacked switches



Firewall



General NMS



AP



Base station



General server



Cluster



FTP server



Authentication server



PN user



Enterprise network user



Enterprise



Business trip



AC



Wi-Fi signals



Internet



Network cloud 1



Network cloud 2



IP phone



PC



Pad



Mobile phone



Laptop



Contents

- 1. Communication and Networks**
2. Network Types and Topology Types
3. Network Engineering and Network Engineers



Concept of Network Communication

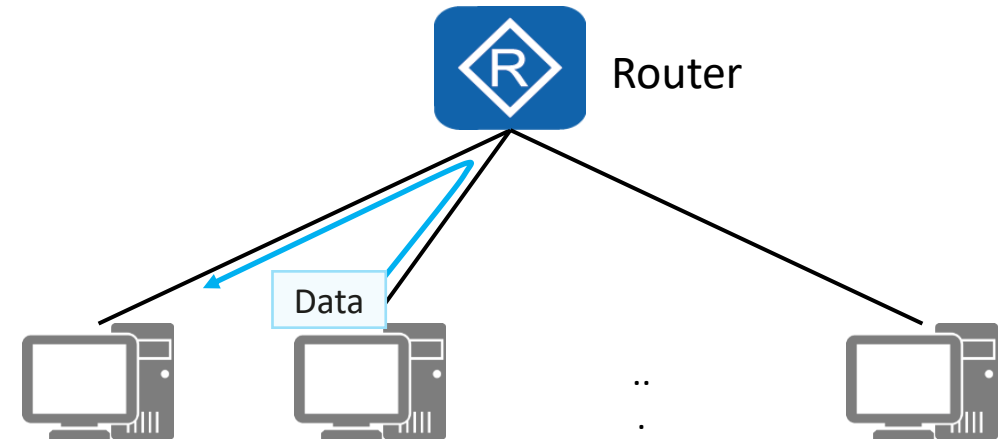
- Communication refers to the information transfer and exchange between people, between people and things, and between things through a certain medium and behavior.
- Network communication refers to communication between terminal devices through a computer network.
- Examples of network communication:



A. Files are transferred between two computers (terminals) through a network cable.



C. A computer (terminal) downloads files through the Internet.

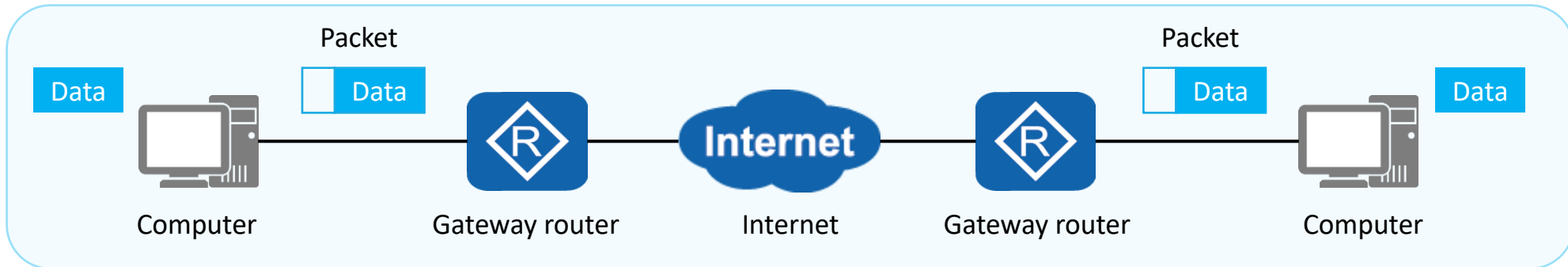
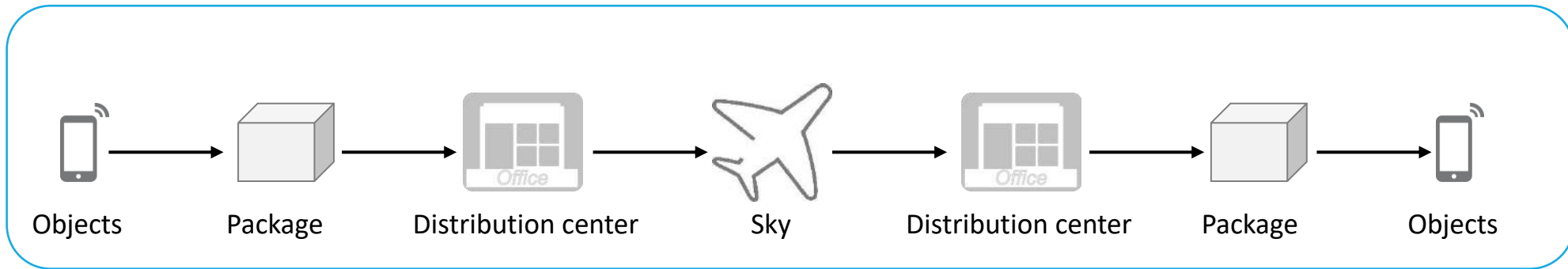


B. Files are transferred among multiple computers (terminals) through a router.



Information Transfer Process

- Virtual information transfer is similar to real object transfer.





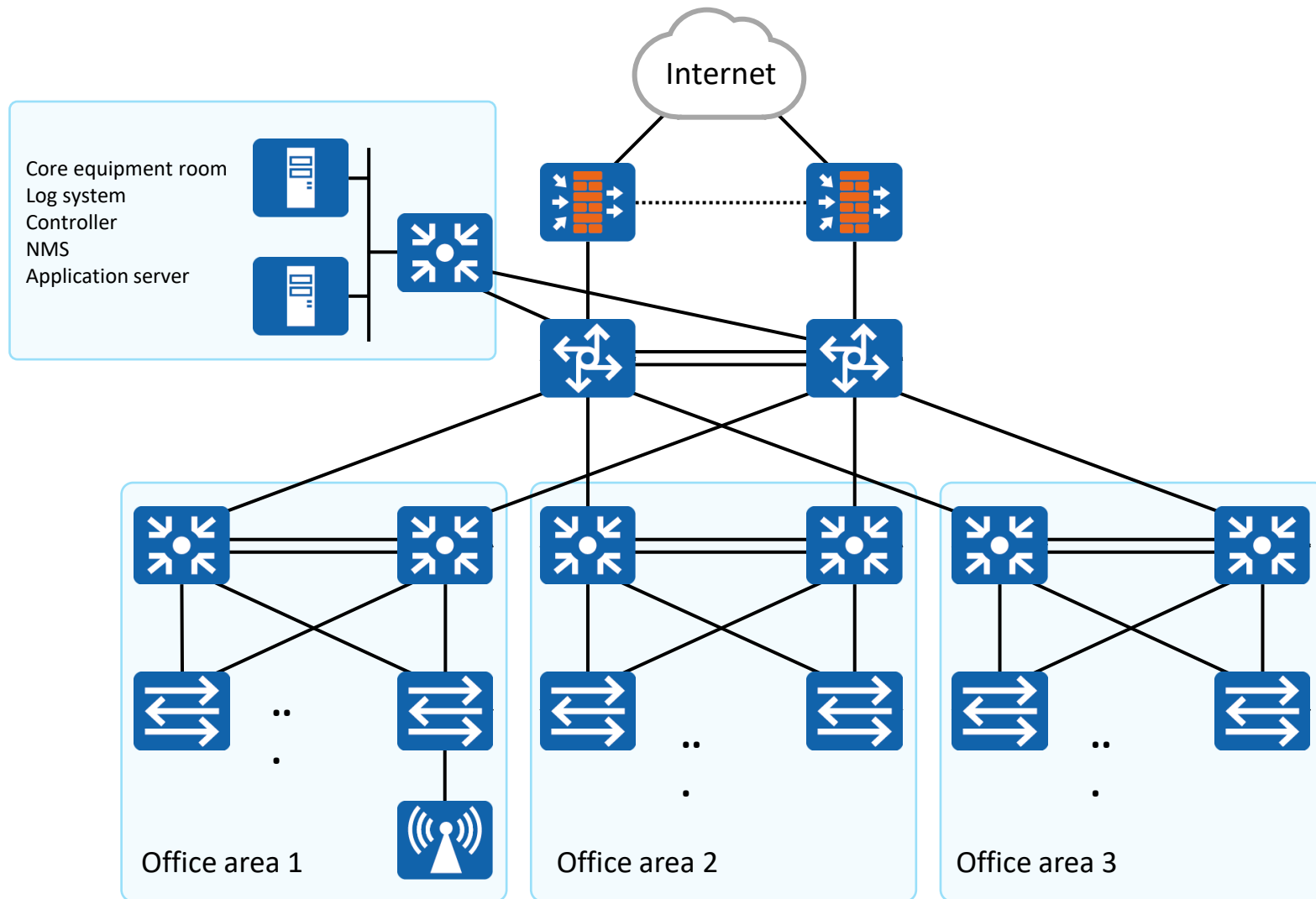
Common Terms

Term	Description
Data payload	Information conveyed
Packet	Data unit switched and transmitted on the network
Header	Information segment added before the data payload
Tail	Information segment added after the data payload
Encapsulation	Process of adding a header and a tail to a data payload to form a new packet
Decapsulation	Process of removing the header and tail from a packet to obtain the data payload
Gateway	Network device that provides functions such as protocol conversion, route selection, and data exchange
Router	Network device that selects a forwarding path for packets
Terminal device	End device of a data communication system, used as a sender or receiver of data



Concept of the Data Communication Network

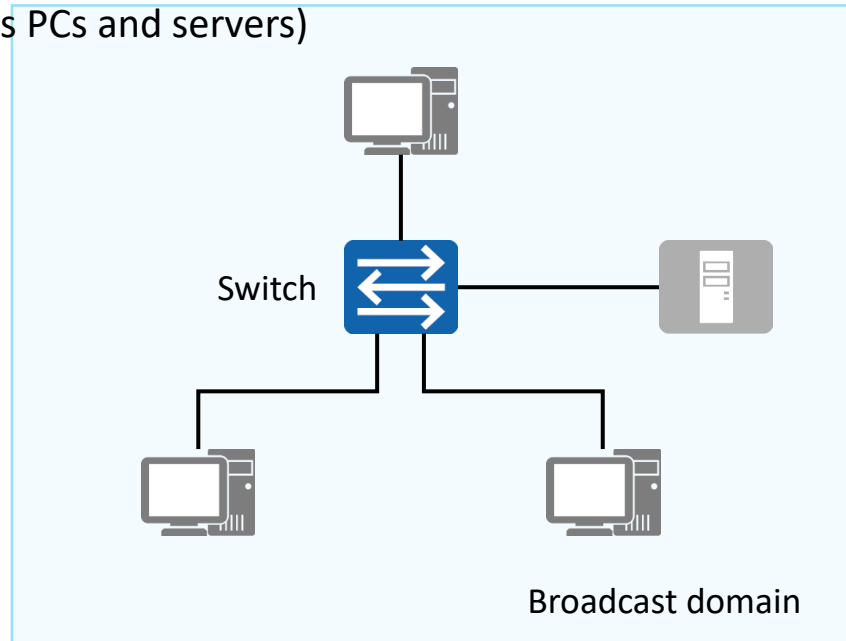
- Data communication network:
Communication network that consists of routers, switches, firewalls, access controllers (ACs), access points (APs), PCs, network printers, and servers
- Function:
To implement data communication





Switches

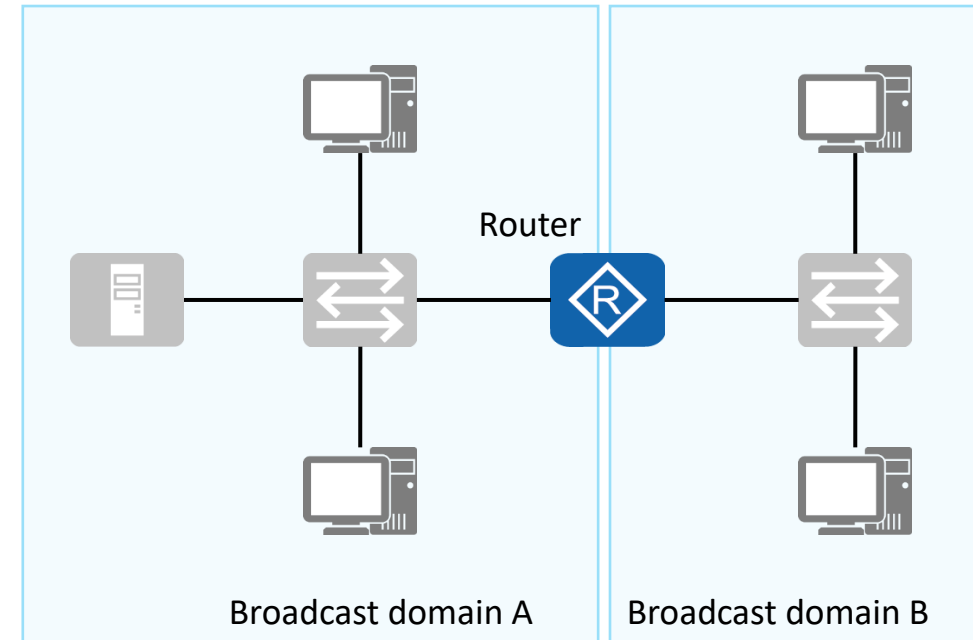
- A switch is a network device closest to end users, used to access the network and switch data frames. It operates at Layer 2 (Data Link Layer) with a single broadcast domain.
- Switch: a device closest to end users, used to access the network and switch data frames
 - Network access of terminals (such as PCs and servers)
 - Layer 2 switching





Routers

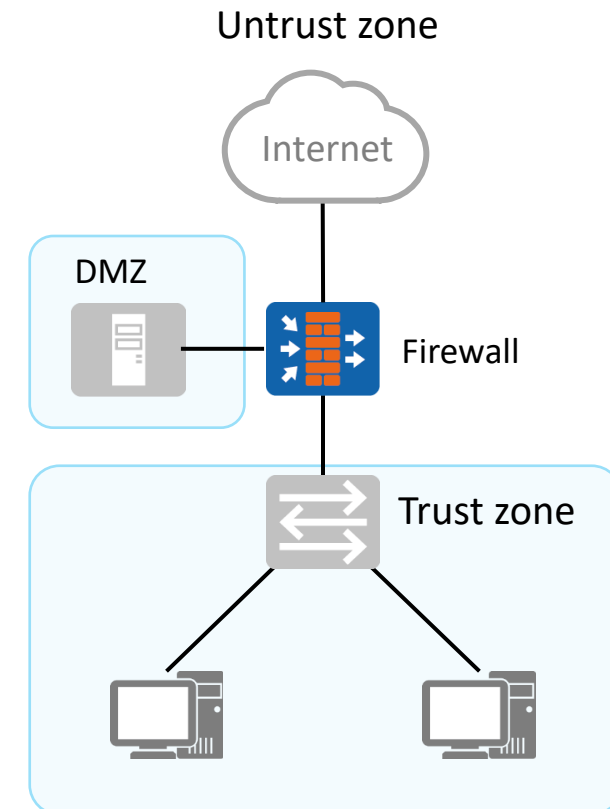
- Router: A router is a network-layer (Layer 3) device that forwards data packets on the Internet. It selects a path based on the destination address in a received packet and sends it to the next router or destination host.
 - Implementing communication between networks of the same type or different types
 - Isolating broadcast domains
 - Maintaining the routing table and running routing protocols
 - Selecting routes and forwarding IP packets
 - Implementing WAN access and network address translation
 - Connecting Layer 2 networks established through switches





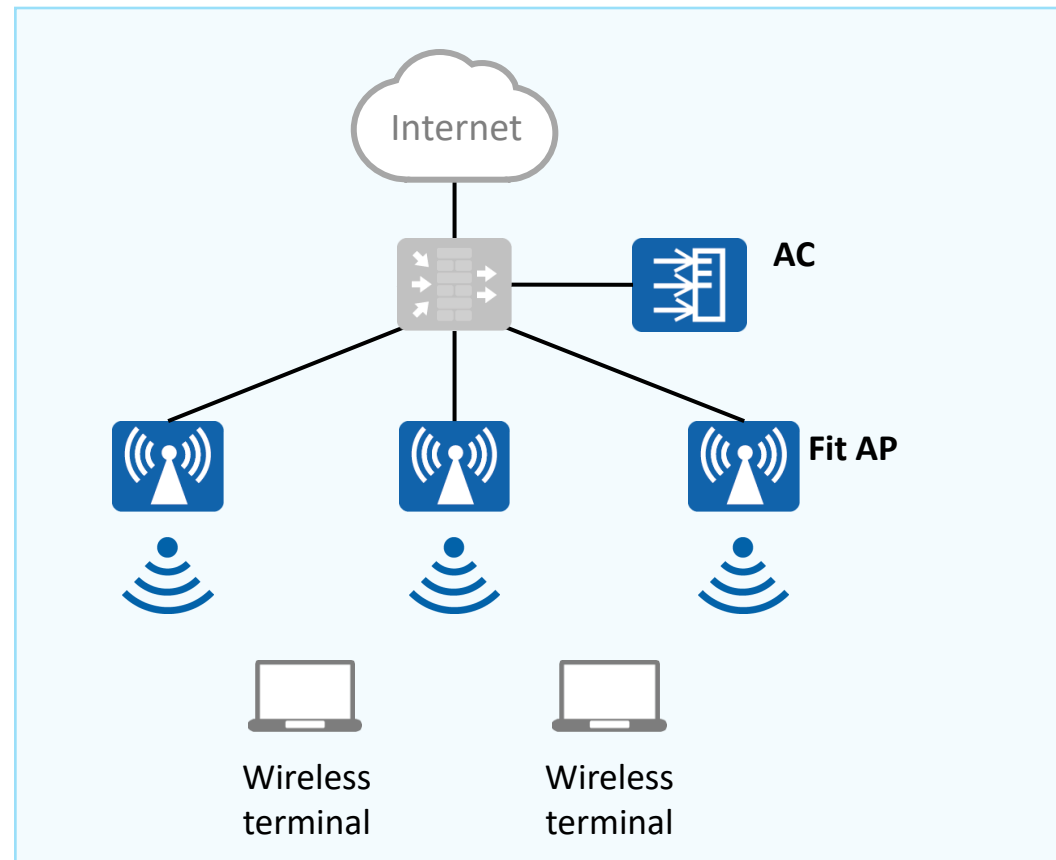
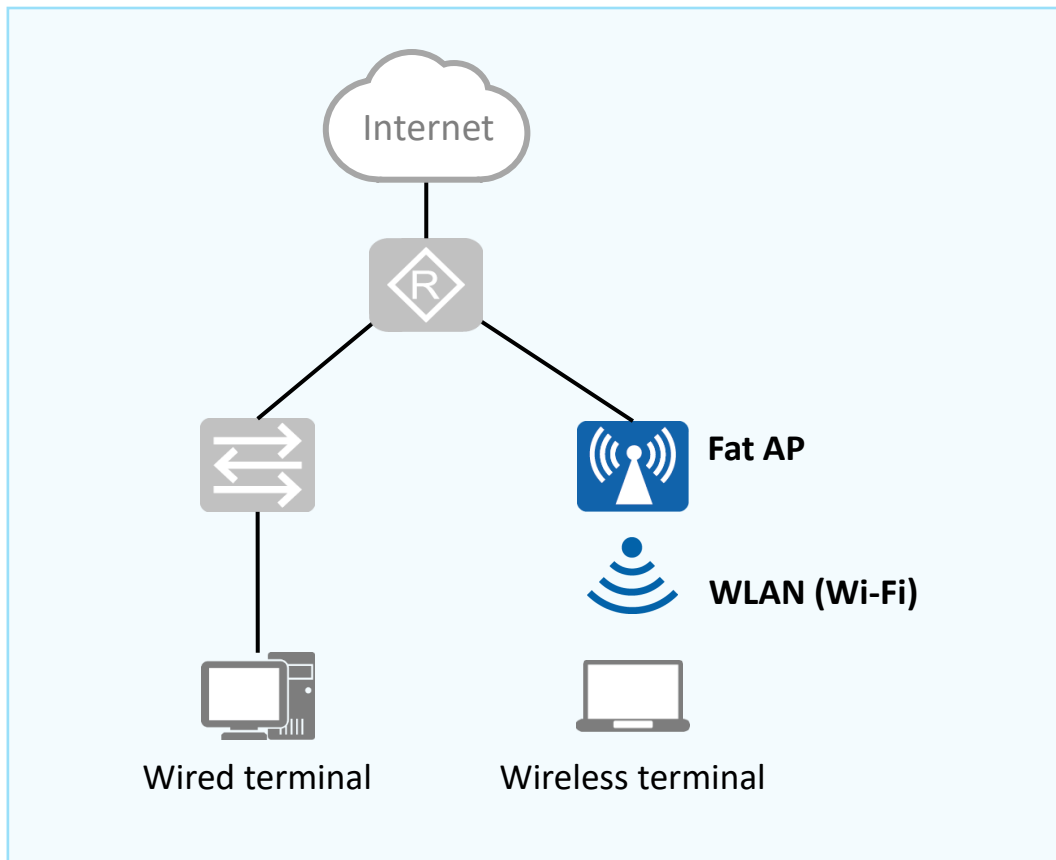
Firewalls

- Firewall: a network security device used to ensure secure communication between two networks. It monitors, restricts, and modifies data flows passing through it to shield internal network information from the public network.
 - Isolating networks of different security levels
 - Implementing access control (using security policies) between networks of different security levels
 - Implementing user identity authentication
 - Implementing remote access
 - Supporting data encryption and VPN services
 - Implementing network address translation
 - Implementing other security functions





Wireless devices





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- 2. Network Types and Topology Types**
3. Network Engineering and Network Engineers

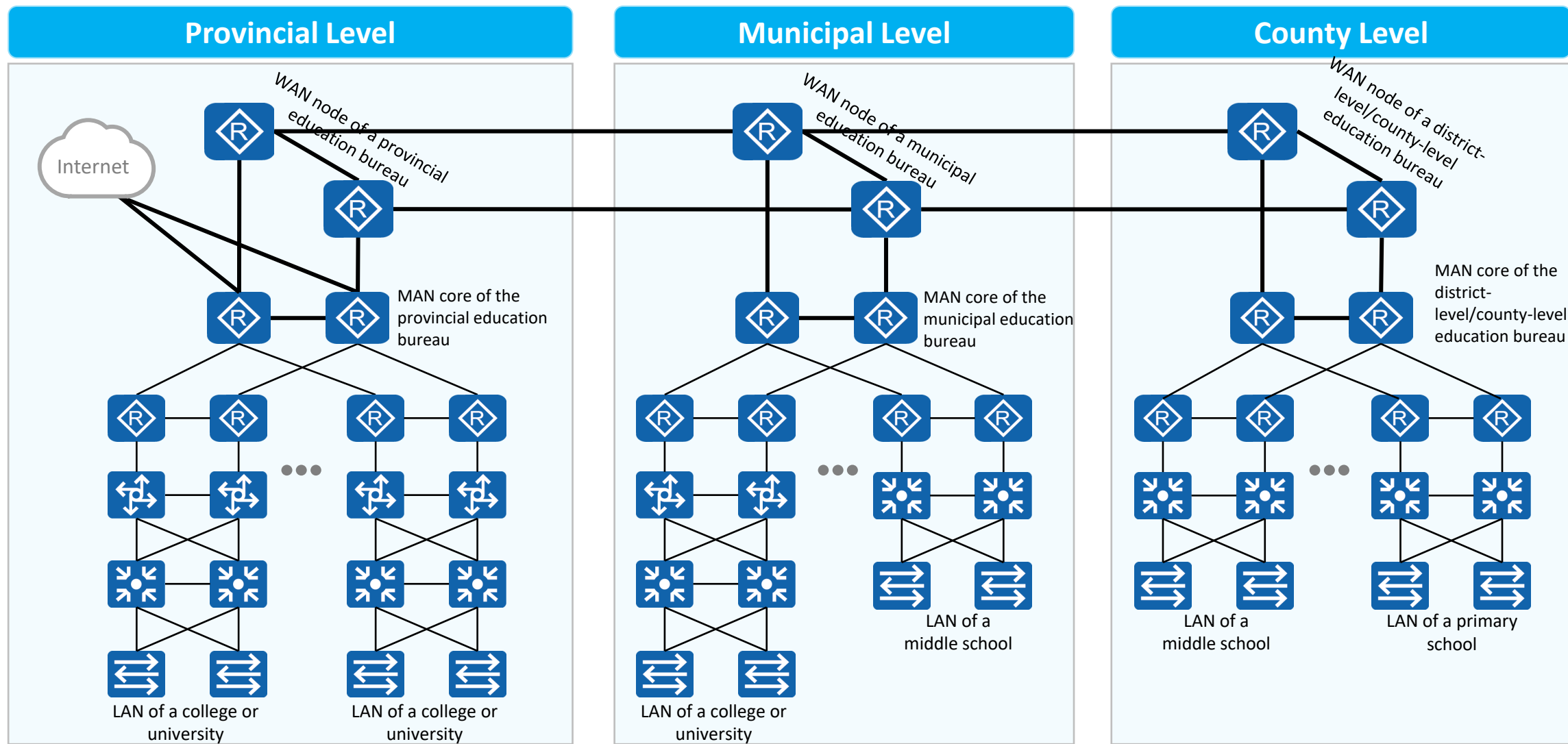


LAN, MAN, and WAN

- Based on the geographical coverage, networks can be classified into local area networks (LANs), metropolitan area networks (MANs), and wide area networks (WANs).
 - LAN
 - A LAN is a network that consists of computers, servers, and network devices in a geographic area. The coverage of a LAN is generally within several thousand square meters.
 - Typical LANs include a company's office network, a cyber bar network, a home network.
 - MAN
 - A MAN is a computer communication network established within a city.
 - Typical MANs include broadband MANs, education MANs, and municipal or provincial e-government private lines.
 - WAN
 - A WAN generally covers a large geographical area ranging from tens of square kilometers to thousands of square kilometers. It can connect networks of multiple cities or even networks of countries (as an international large-scale network) and provide long-distance communication.
 - The Internet is a typical WAN.



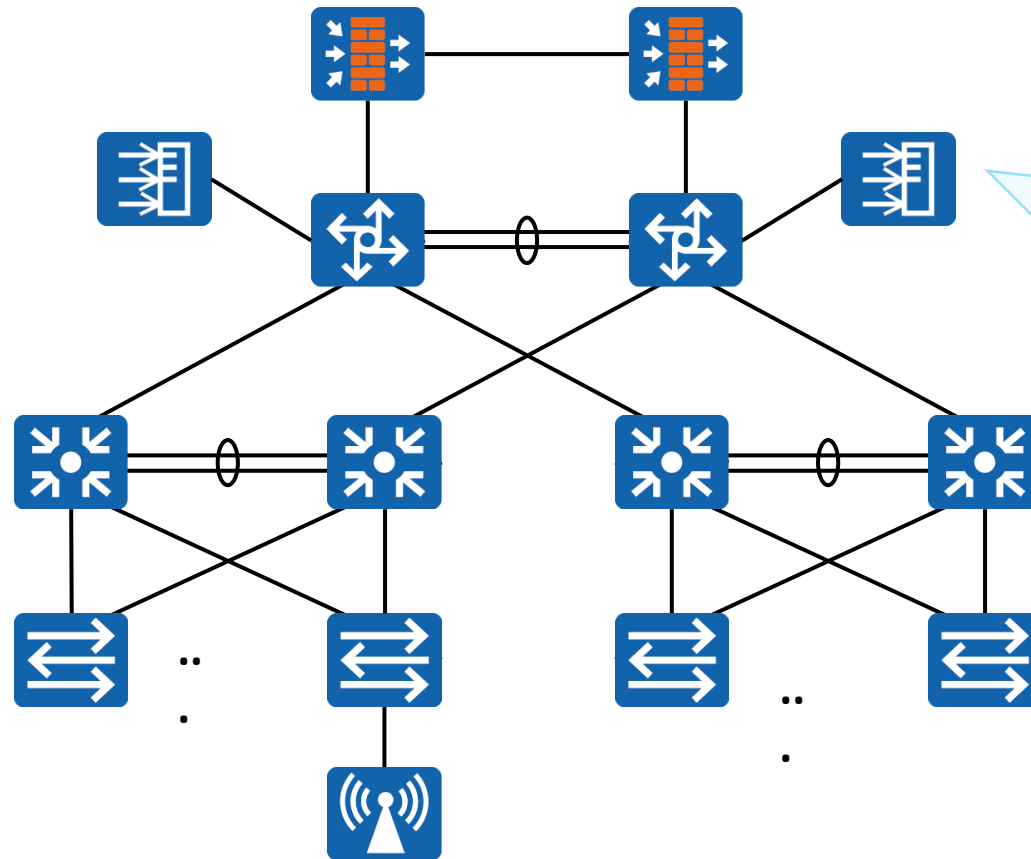
LAN, MAN, and WAN in the Education Industry





Network Topologies

- A network topology is a structured layout presented using transmission media (such as twisted pairs and optical fibers) to interconnect various devices (such as computer terminals, routers, and switches).

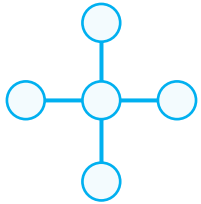


The network topology is used to describe the physical or logical structure of a network in the network engineering field, and is a very important network concept.

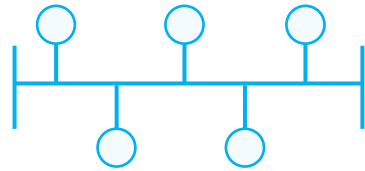


Network Topology Types

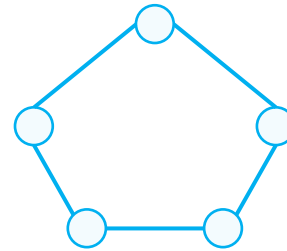
- Network topologies are classified into star, bus, ring, tree, full-mesh, and partial-mesh network topologies.



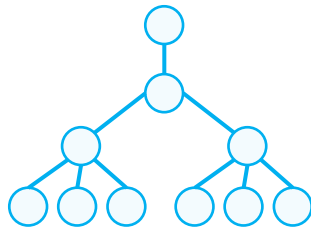
Star network topology



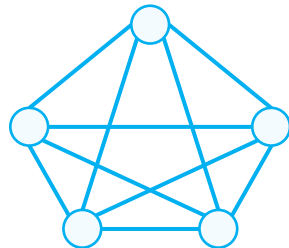
Bus network topology



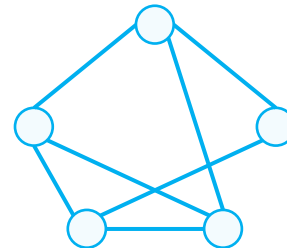
Ring network topology



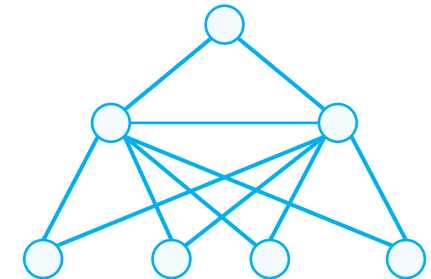
Tree network topology



Full-mesh network topology



Partial-mesh network topology



Combined network topology



Bus Topology:

Advantages: Simple, easy to install, cost-effective for small networks. Disadvantages: Limited cable length and number of stations, difficult to troubleshoot, and performance degrades with heavy traffic.

Star Topology:

Advantages: Easy to manage and troubleshoot, failure of one node doesn't affect the rest. Disadvantages: Requires more cable, central hub represents a single point of failure.

Ring Topology:

Advantages: Data packets travel at great speed, no collisions. Disadvantages: A failure in any cable or device breaks the loop and can take down the entire network.

Mesh Topology:

Advantages: High fault tolerance, data can be transmitted from different devices simultaneously. Disadvantages: Expensive due to high cabling and complex setup.

Tree Topology:

Advantages: Hierarchical, easy to expand and manage. Disadvantages: Difficult to configure, single point of failure if the root node fails.



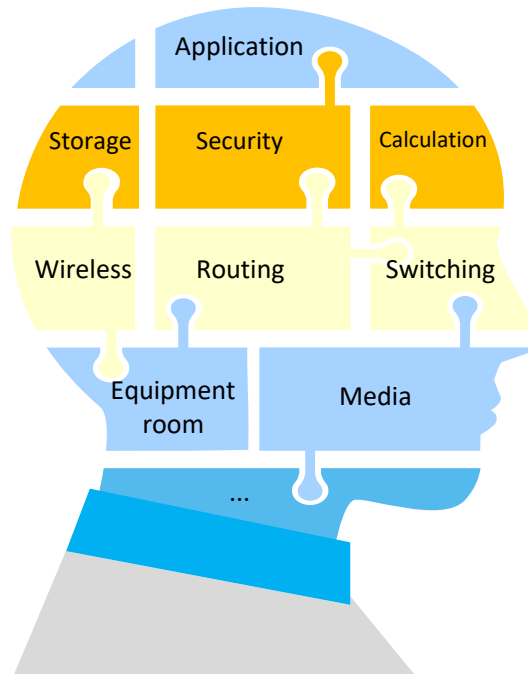
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Network Engineering

- Network engineering
 - Network engineering refers to planning and designing feasible solutions based on network application requirements and computer network system standards, specifications, and technologies under the guidance of information system engineering methods and complete organizations, as well as integrating computer network hardware devices, software, and technologies to form a cost-effective network system that meets user requirements.
- Technical modules covered by network engineering:





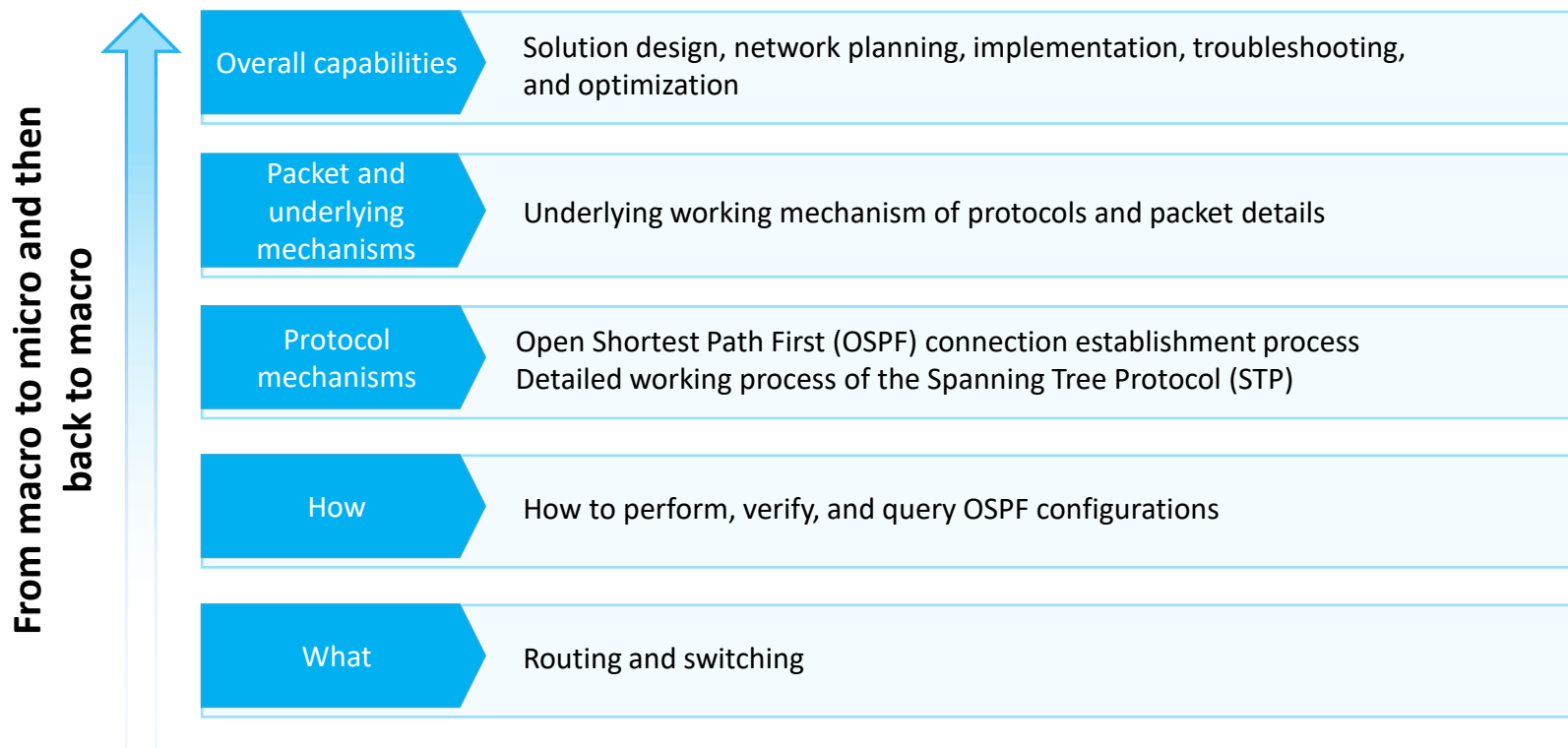
Network Engineer

- Network engineer
 - Network engineers are technology professionals who master professional network technologies, have professional skills, professionalism, and project implementation experience in the network engineering field, and are able to fully communicate with customers or other project stakeholders onsite. In addition, they can develop implementation solutions and project plans (recognized by project stakeholders) based on customer requirements and environment factors, fully mobilize resources of all parties to ensure timely and high-quality project implementation, and provide training for stakeholders and deliver engineering documents after the project is implemented.
- Comprehensive capability models for network engineers:

Process specification	Business etiquette	Team collaboration
Industry knowledge	Values	Business management
Engineering knowledge	Service awareness	Presentation capability
Product knowledge	Information collection	Problem solving
Technical knowledge	Learning competency	Communication capability
Professional knowledge	Basic qualification	Professional skills



Network Engineers' Technology Development Path





Huawei Certification Injects Vitality into Talent Development for Enterprises



Certification exam

- Cultivate experts who understand both business and technologies.
- Cultivate platform construction and service application experts based on HUAWEI CLOUD.
- Focuses on ICT infrastructure and cultivates architecture talent in all ICT fields.



Providing talent with career development paths

- Meets enterprise talent's career evolution from an engineer to a senior engineer, and then to an expert.
- Provides a hierarchical certification system that provides customized talent growth paths in accordance with job-based capability requirements and supports in-depth professional development, integration, and expansion, reducing the talent cultivation cost for enterprises.



Facilitating enterprise innovation and transformation

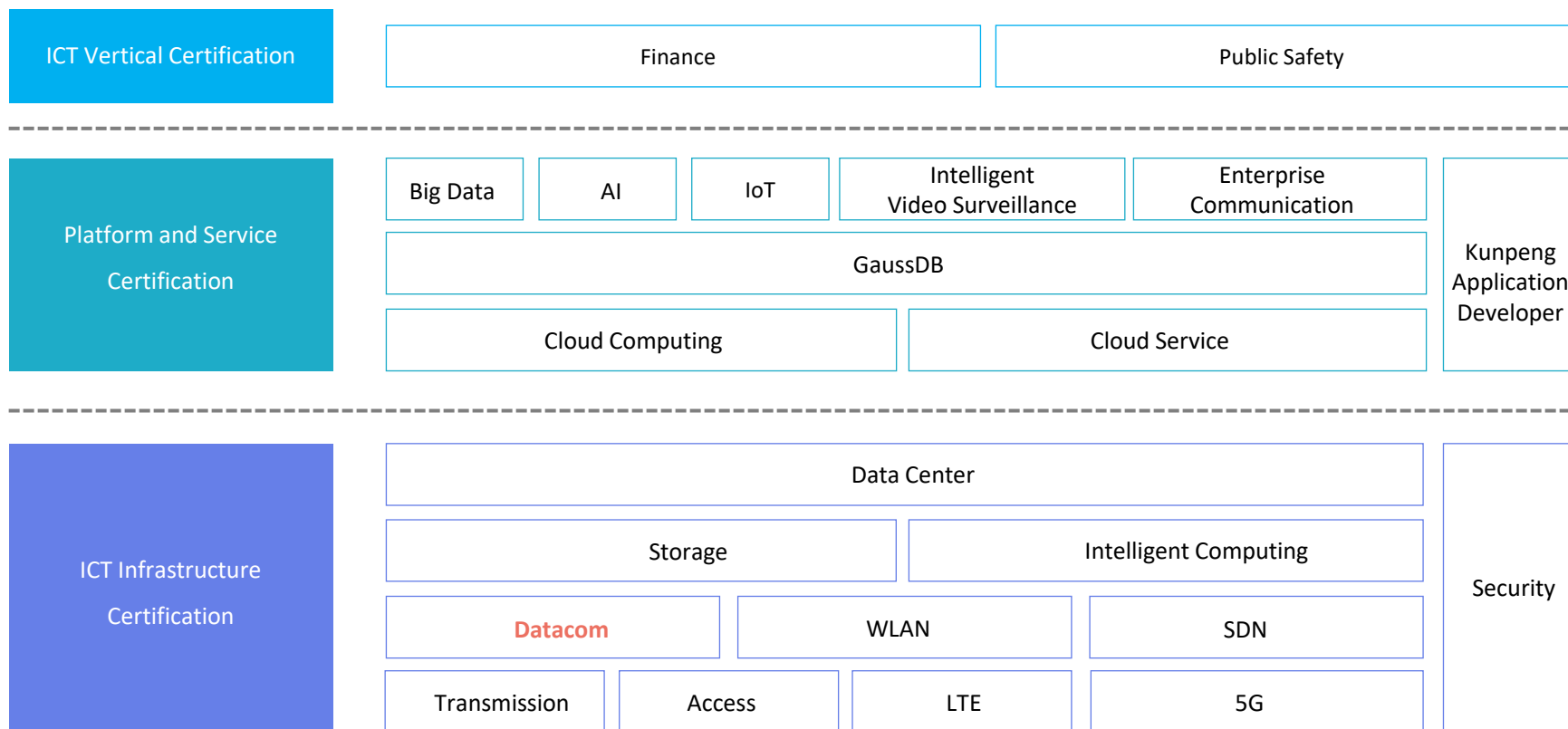
- Provides authoritative certification for ICT talent. The ICT talent with authoritative certification helps ensure project delivery quality and improve customer satisfaction.
- Enhances the overall performance and productivity of enterprises.
- Accelerates business innovation and transformation, and improve the overall operational efficiency.



Huawei Certification Portfolio

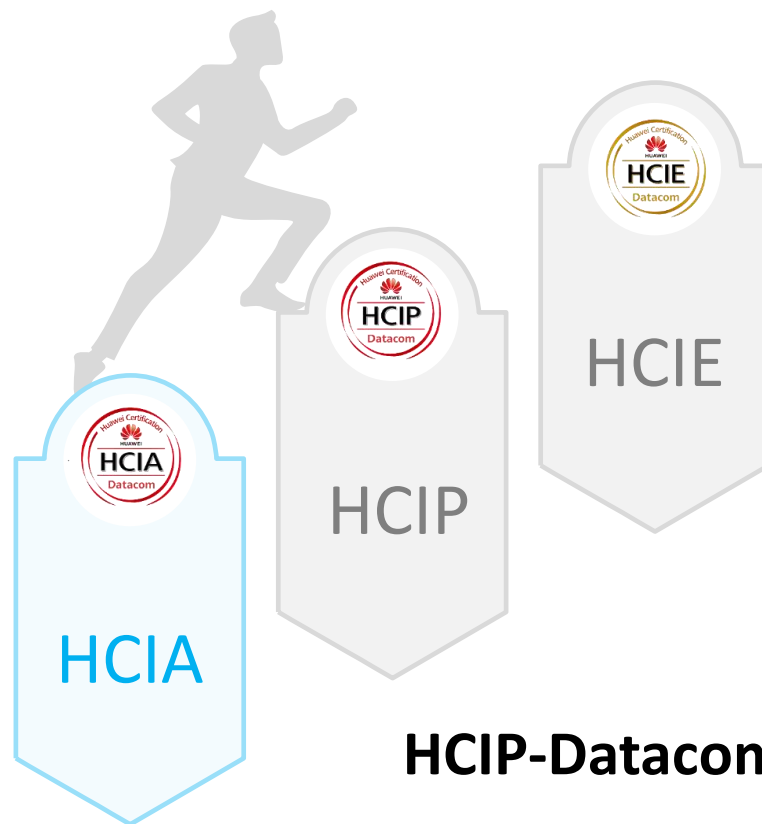
- Huawei certification covers all ICT fields and is committed to providing a leading talent cultivation architecture and certification standards, cultivating ICT professionals in the digital era, and building a healthy ICT talent ecosystem.

Huawei Certification





Huawei Datacom Certification Portfolio



HCIA-Datacom

Aiming to cultivate network engineers with basic datacom theories and skills

HCIE-Datacom

Aiming to cultivate network experts with solid theoretical foundation and deployment capabilities for cross-field solutions

HCIP-Datacom

Aiming to cultivate senior network engineers for cross-field solution planning and design or single-field planning and deployment



Quiz

1. (Single) Which of the following type of network topology has the highest reliability? ()

- A. Star network topology
- B. Ring network topology
- C. Full-mesh network topology
- D. Tree network topology

- **Full-mesh network topology** (highest reliability)
- **Star network topology**
- **Tree network topology**
- **Ring network topology** (lowest reliability)



Summary

- This section describes the concepts of network communication and data communication network. The basic function of a data communication network is to implement network communication.
- This section also introduces various network devices, the differences between LAN, MAN and WAN, and various network topologies. In actual networking, multiple topologies are combined according to the requirements of multiple parties.
- This section also describes network engineering and network engineers and introduces Huawei datacom certification system.

The background of the slide features a blue-tinted image of several business professionals in a modern office environment. They are standing on a highly reflective floor, and their silhouettes are clearly visible against the lighter background. The overall aesthetic is professional and corporate.

Thank You
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Network Reference Model



Foreword

- In the digital era, various information is presented as data in our life. What is data? How is data transmitted?
- In this course, we will use the network reference model to understand the "life" of data.



Objectives

- On completion of this course, you will be able to:
 - Understand the data definition and transmission process.
 - Understand the concepts and advantages of the network reference model.
 - Understand common standard protocols.
 - Understand the data encapsulation and decapsulation processes.



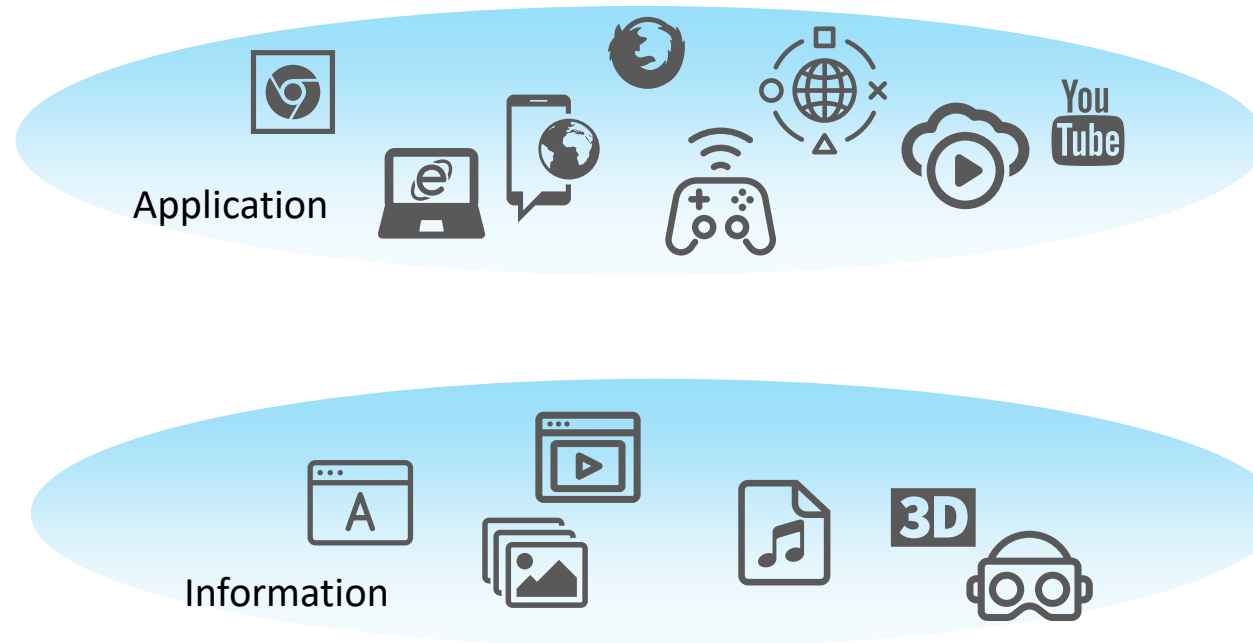
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Origin of the Story - Applications

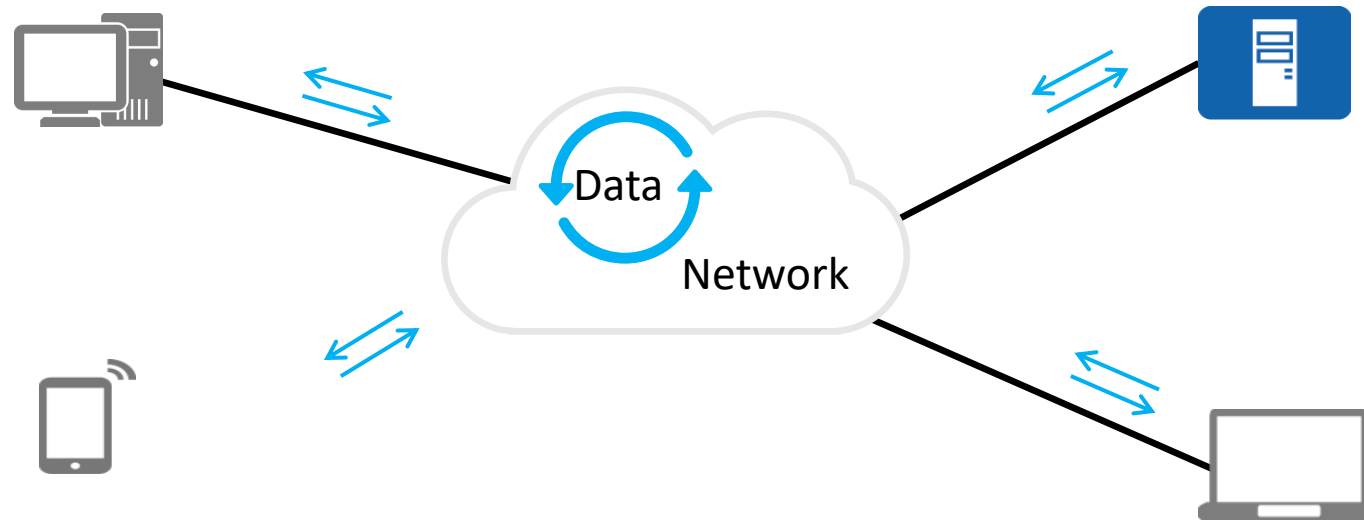
- Applications are used to meet various requirements of people, such as web page access, online gaming, and online video playback.
- Information is generated along with applications. Texts, pictures, and videos are all information presentation modes.





Application Implementation - Data

- Data generation
 - In the computer field, data is the carrier of all kinds of information.
- Data transmission
 - Data generated by most applications needs to be transmitted between devices.



Does an application need to complete the entire process from data generation to data transmission?



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OSI Reference Model

The OSI Model is a framework that explains how data travels from one device to another over a network.

It divides the communication process into 7 layers, where each layer has a specific role in sending and receiving data.

It helps us understand, design, and troubleshoot networks more easily.

7. Application Layer	Provides interfaces for applications.
6. Presentation Layer	Translates data formats to ensure that the application-layer data of one system can be identified by the application layer of another system.
5. Session Layer	Establishes, manages, and terminates sessions between communicating parties.
4. Transport Layer	Establishes, maintains, and cancels an end-to-end data transmission process; controls transmission speeds and adjusts data sequences.
3. Network Layer	Defines logical addresses and transfers data from sources to destinations.
2. Data Link Layer	Encapsulates packets into frames, transmits frames in P2P or P2MP mode, and implements error checking.
1. Physical Layer	Transmits bitstreams over transmission media and defines electrical and physical specifications.



TCP/IP Reference Model

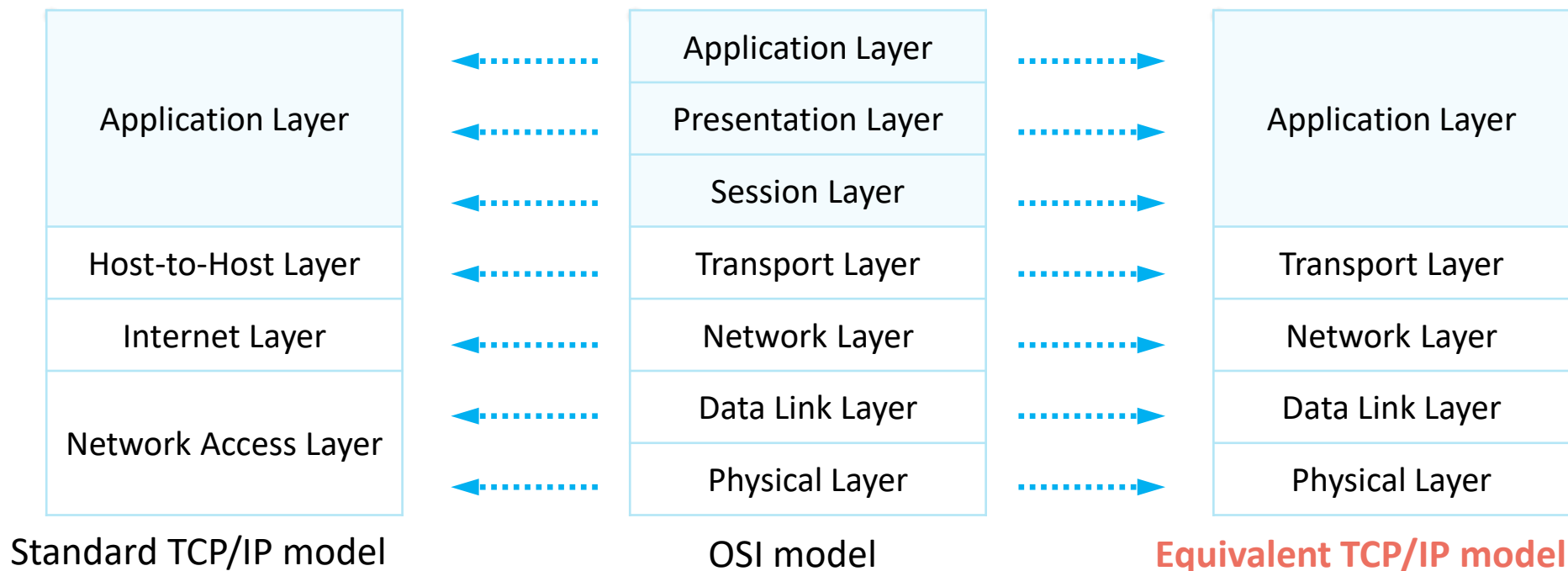
- The OSI protocol stack is complex, and the TCP and IP protocols are widely used in the industry. Therefore, the TCP/IP reference model becomes the mainstream reference model of the Internet.

The TCP/IP model is the main communication model used on the Internet.

Although the OSI model is useful for understanding networking concepts, it is more complex and mostly theoretical.

TCP/IP became the mainstream model because:

- It was practical and already implemented
- It powered the early Internet (ARPANET)
- It is simpler than OSI
- It is widely adopted in real-world networking





Common TCP/IP Protocols

- The TCP/IP protocol stack defines a series of standard protocols.

Application Layer	Telnet	FTP	TFTP	SNMP
	HTTP	SMTP	DNS	DHCP
Transport Layer	TCP		UDP	
Network Layer	ICMP		IGMP	
	IP			
Data Link Layer	PPPoE			
	Ethernet		PPP	
Physical Layer	...			



Common Protocol Standardization Organizations

- Internet Engineering Task Force (IETF)
 - IETF is a voluntary organization responsible for developing and promoting Internet protocols (especially protocols that constitute the TCP/IP protocol suite), and releasing new or replacing old protocol standards through RFCs.
- Institute of Electrical and Electronics Engineers (IEEE)
 - IEEE has formulated about 30% of standards in the electronics, electrical, and computer science fields worldwide. Those standards include well-known IEEE802.3 (Ethernet) and IEEE802.11 (Wi-Fi).
- International Organization for Standardization (ISO)
 - ISO is an international organization that plays an important role in the formulation of computer network standards, such as the OSI model defined in ISO/IEC 7498-1.



Application Layer

Application
Layer

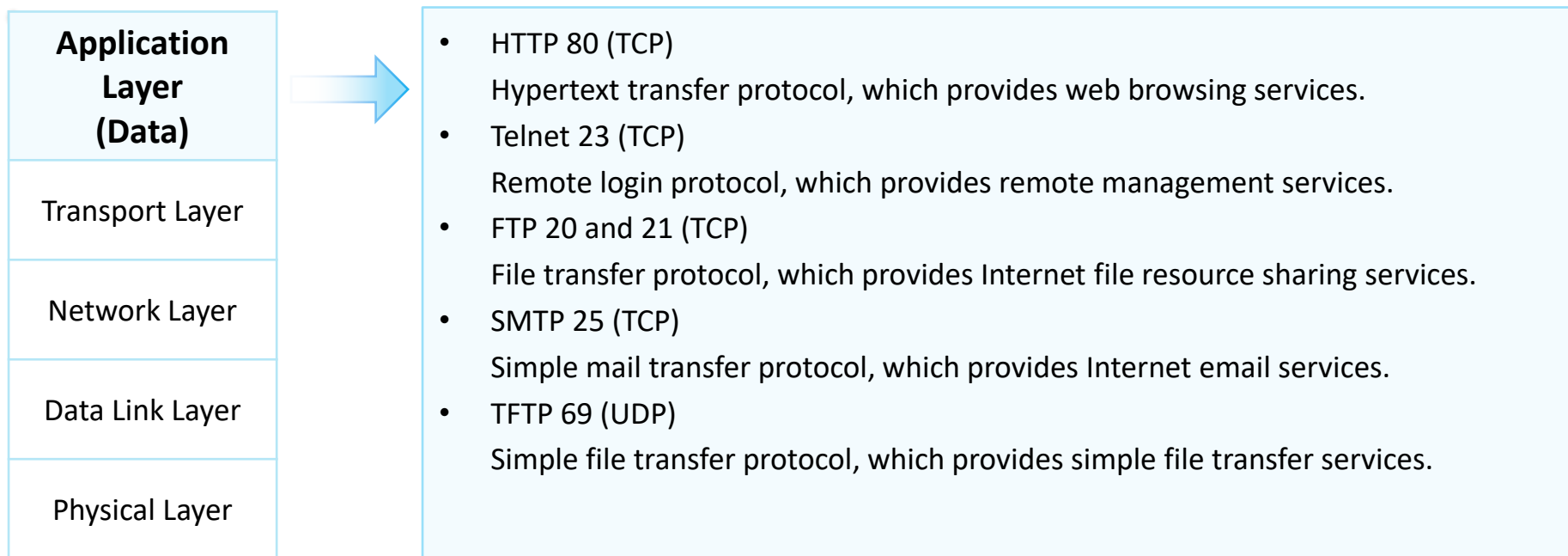
Transport
Layer

Network
Layer

Data Link
Layer

Physical
Layer

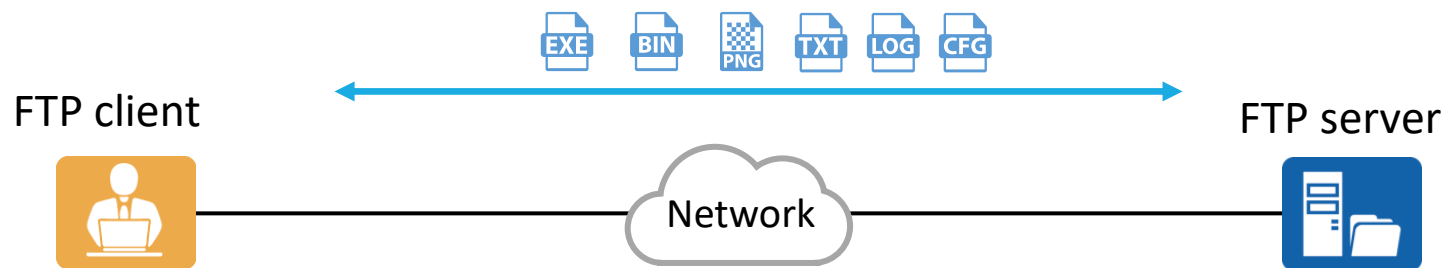
- The application layer provides interfaces for application software so that applications can use network services. The application layer protocol designates transport layer protocols and ports.
- PDUs transmitted at the network layer are called data.





Common Application Layer Protocols - FTP

- The File Transfer Protocol (FTP) transfers files from one host to another to implement file download and upload. This protocol adopts the client/server (C/S) structure.

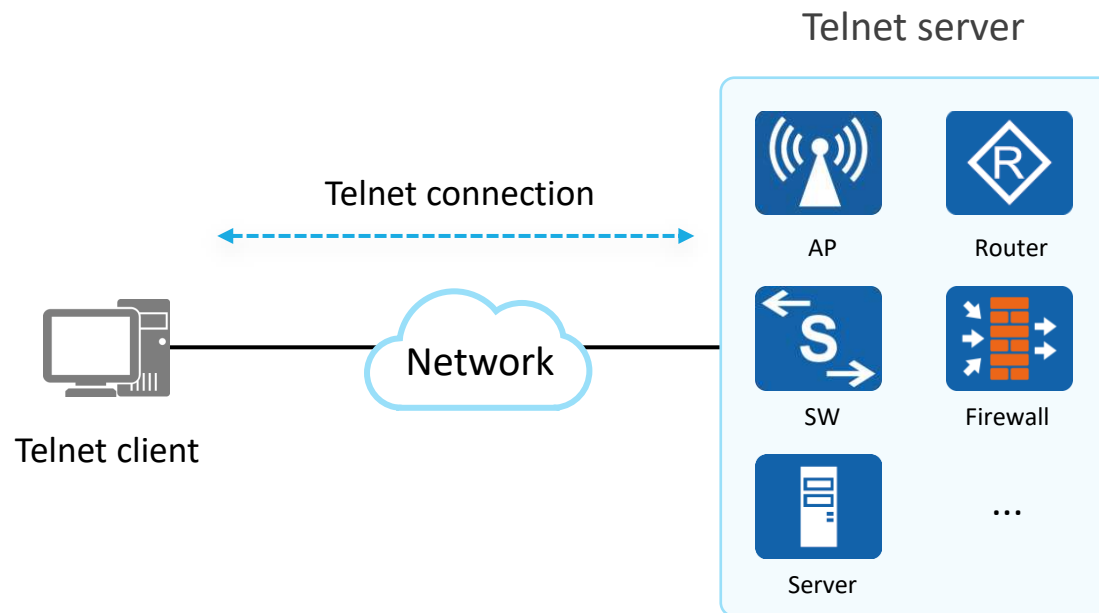


FTP client: provides commands for local users to operate files on a remote server. A user can install an FTP client program on a PC and set up a connection with an FTP server to operate files on the server.

FTP server: a device that runs the FTP service. It provides the access and operation functions for remote clients, allowing users to access the FTP server through the FTP client program and access files on the server.

Common Application Layer Protocols - Telnet

- Telnet is a standard protocol that provides remote login services on a network. It provides users with the ability to operate remote devices through local PCs.

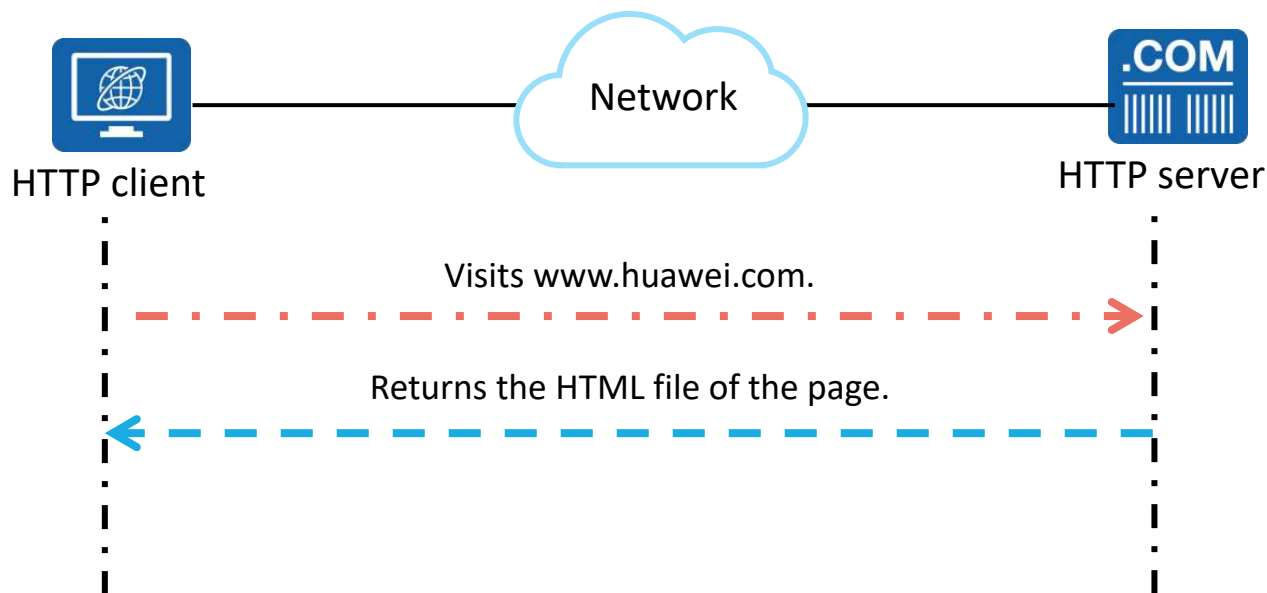


A user connects to a Telnet server through the Telnet client program. The commands entered on the Telnet client are executed on the server, as if the commands were entered on the console of the server.



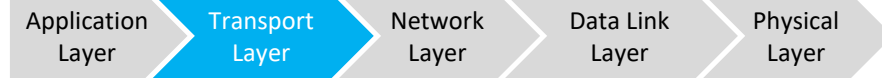
Common Application Layer Protocols - HTTP

- Hypertext Transfer Protocol (HTTP): is one of the most widely used network protocols on the Internet. HTTP was originally designed to provide a method for publishing and receiving HTML pages.

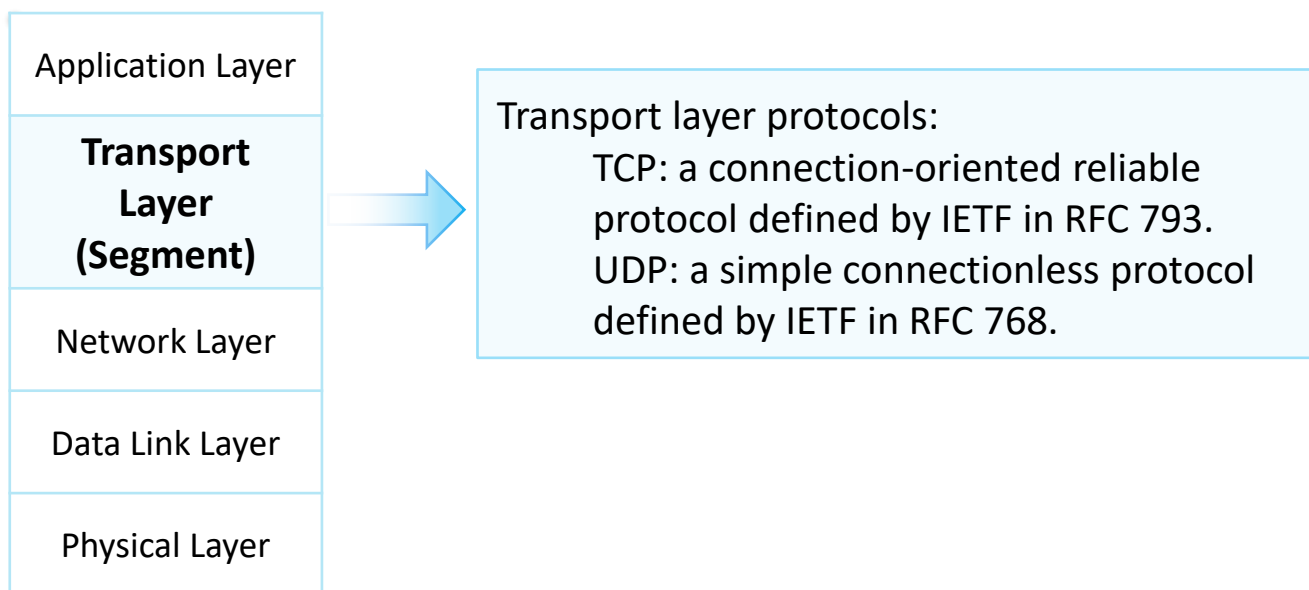




Transport Layer



- A transport layer protocol receives data from an application layer protocol, encapsulates the data with the corresponding transport layer protocol header, and helps establish an end-to-end (port-to-port) connection.
- PDUs transmitted at the transport layer are called segments.





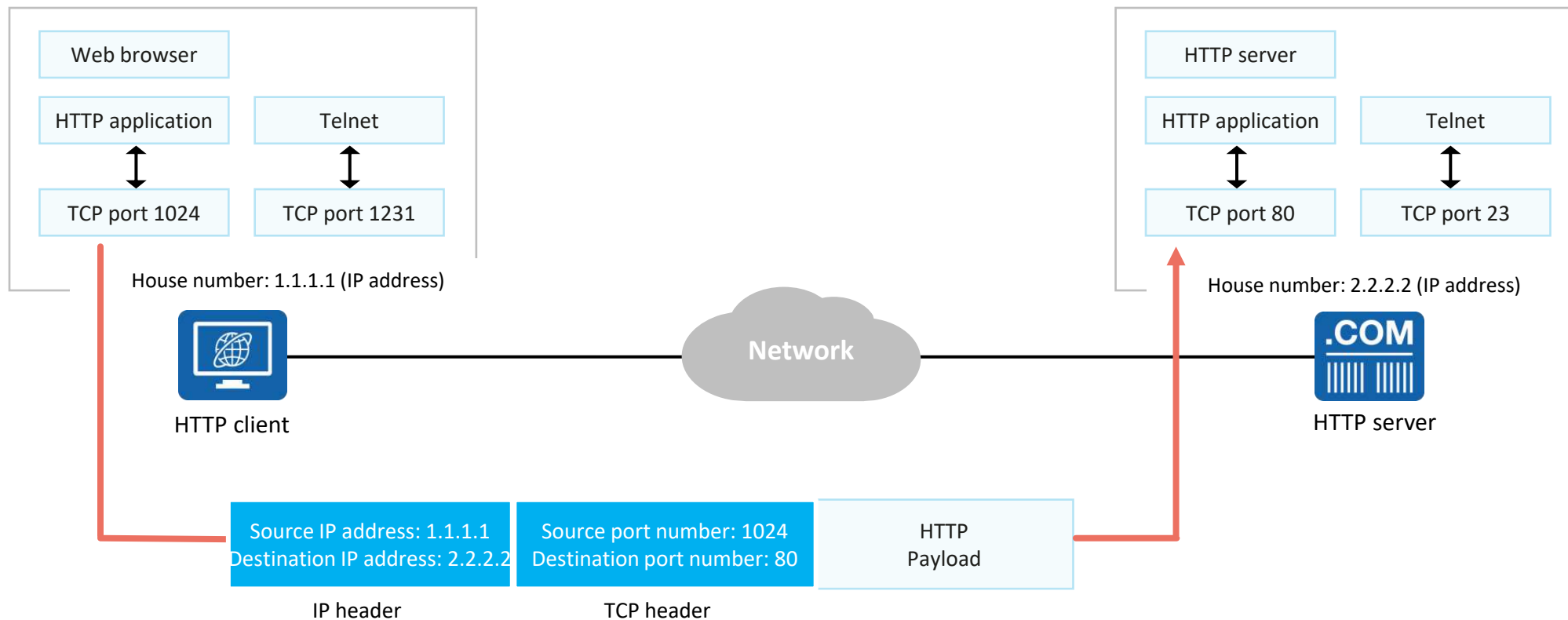
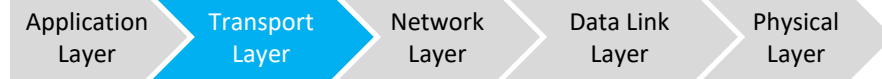
TCP and UDP - Header Formats

Source Port (16)			Destination Port (16)		TCP header 20 bytes
Sequence Number (32)					
Acknowledgement Number (32)					
Header Length (4)	Reserved (6)	Control Bits (6)	Window (16)		
Checksum (16)			Urgent (16)		
Options					
Data (varies)					

Source Port (16)	Destination Port (16)	} UDP header 8 bytes
Length (16)	Checksum (16)	
Data (if any)		



TCP and UDP - Port Numbers



- Generally, the source port used by a client is randomly allocated, and the destination port is specified by the application of a server.
- The system generally selects a source port number that is greater than 1023 and is not being used.
- The destination port number is the listening port of the application (service) enabled on the server. For example, the default port number for HTTP is 80.



TCP (TRANSMISSION CONTROL PROTOCOL)

TCP is a connection-oriented and reliable transport protocol. It ensures that data is delivered accurately and in the correct order.

- Uses a 3-way handshake to establish connection
- Requires acknowledgments (ACKs)
- Performs error checking and retransmission
- Slower but highly reliable

Examples:

Web browsing (HTTP/HTTPS), Email, File transfer (FTP)

UDP (USER DATAGRAM PROTOCOL)

UDP is a connectionless and faster transport protocol. It sends data without guaranteeing delivery or order.

- No handshake required
- No acknowledgments
- No retransmission of lost data
- Faster but less reliable

Examples:

Online gaming, Live streaming, VoIP calls



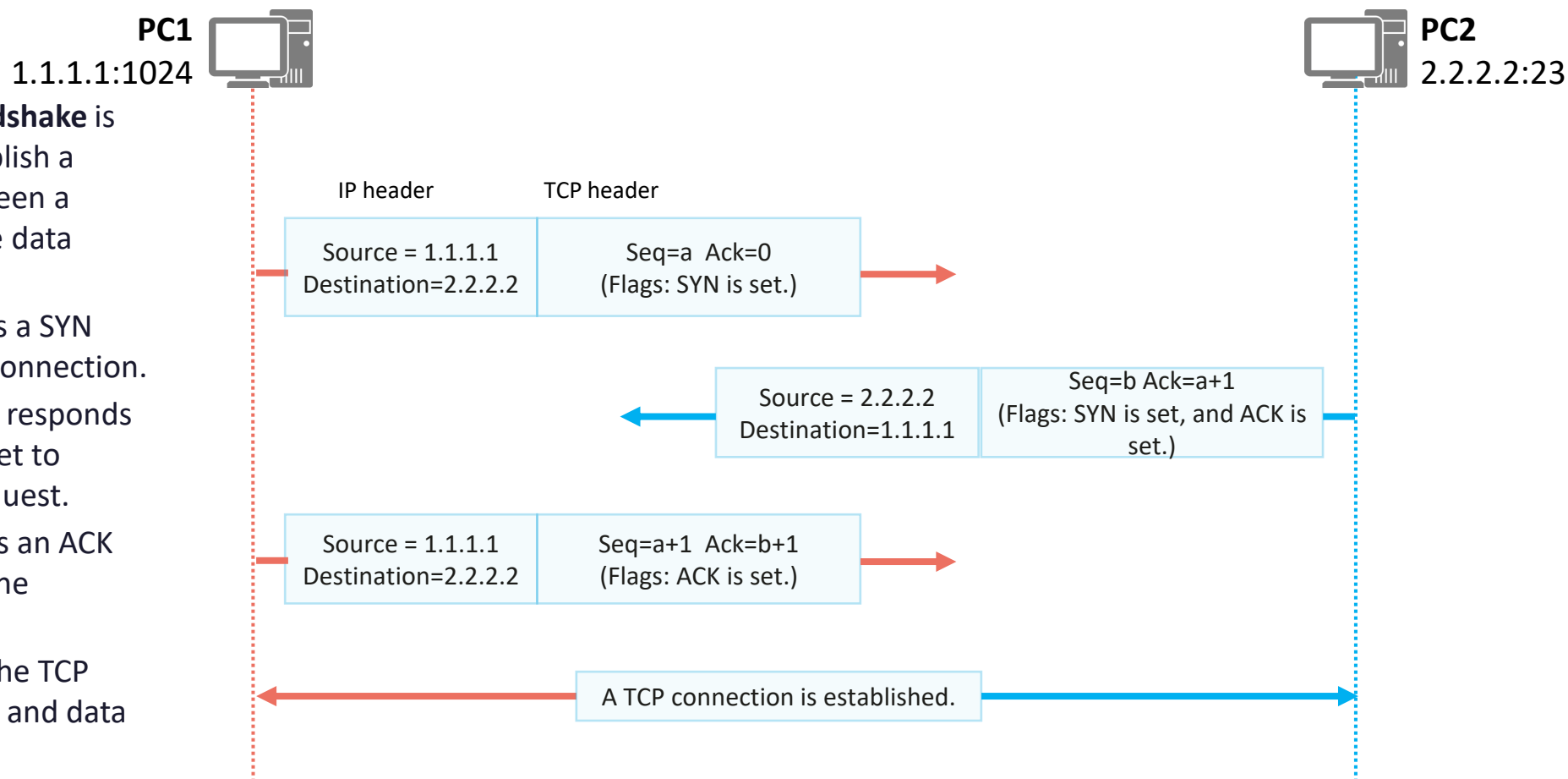
TCP Connection Setup - Three-Way Handshake

- Before sending data, a TCP-based application needs to establish a connection through three-way handshake.

The **TCP Three-Way Handshake** is the process used to establish a reliable connection between a client and a server before data transmission.

- 1. SYN:** The client sends a SYN packet to request a connection.
- 2. SYN-ACK:** The server responds with a SYN-ACK packet to acknowledge the request.
- 3. ACK:** The client sends an ACK packet, completing the connection.

After these three steps, the TCP connection is established and data transfer can begin.





TCP Sequence Number and Acknowledgment Number

Application Layer

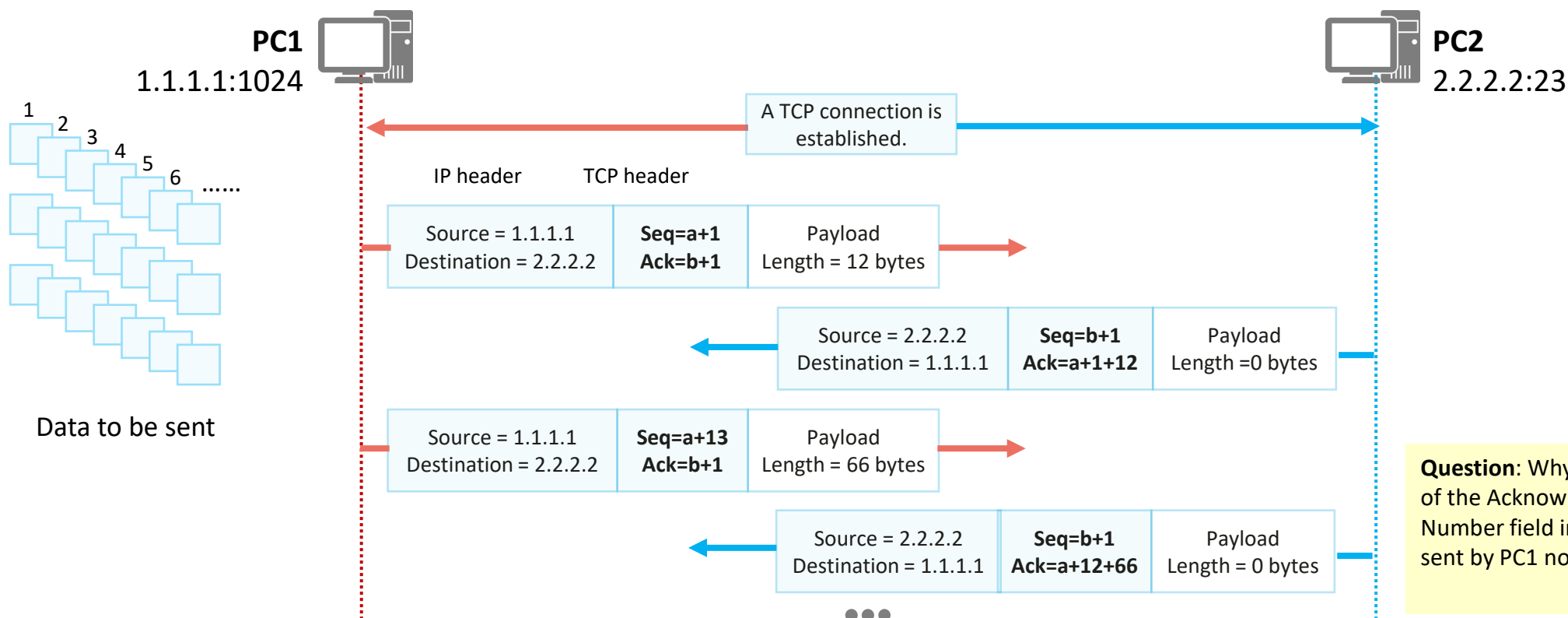
Transport Layer

Network Layer

Data Link Layer

Physical Layer

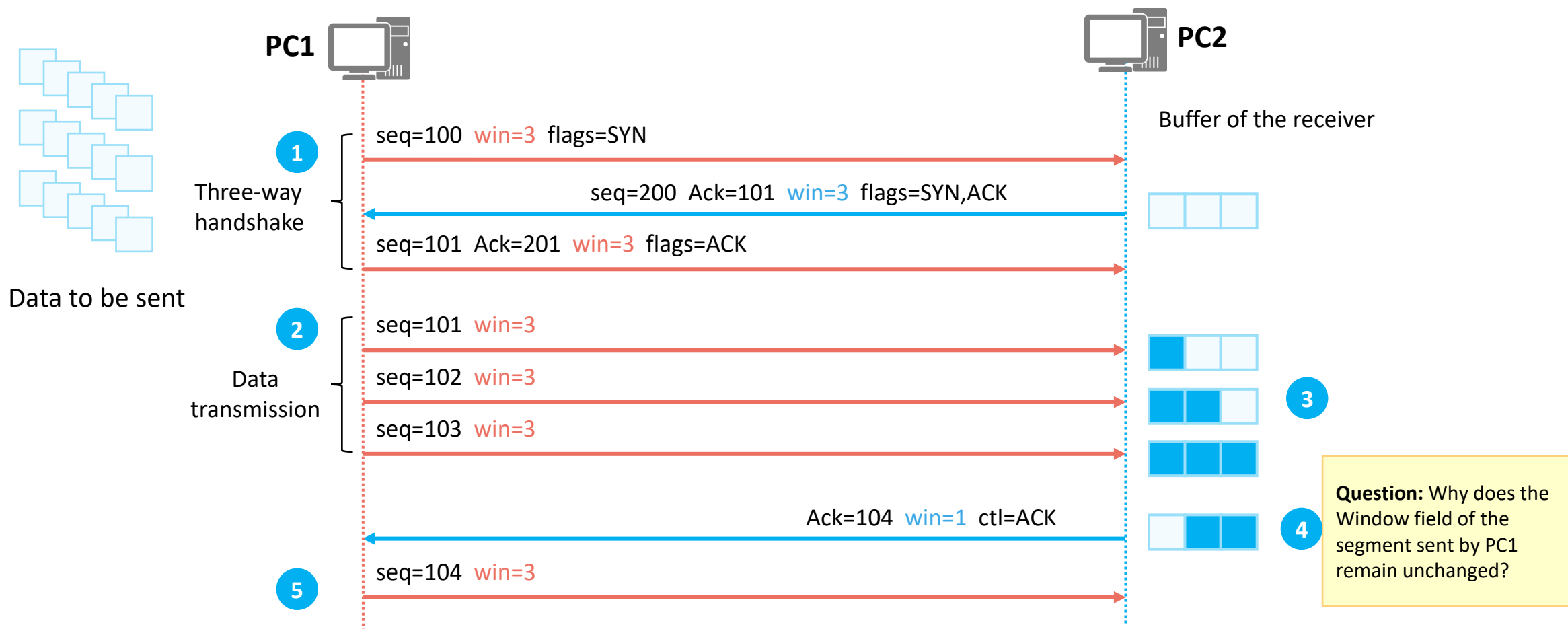
- TCP uses the Sequence Number and Acknowledgment Number fields to implement reliable and ordered data transmission.





TCP Window Sliding Mechanism

- TCP uses the sliding window mechanism to control the data transmission rate.





TCP Shutdown - Four-Way Handshake

Application Layer

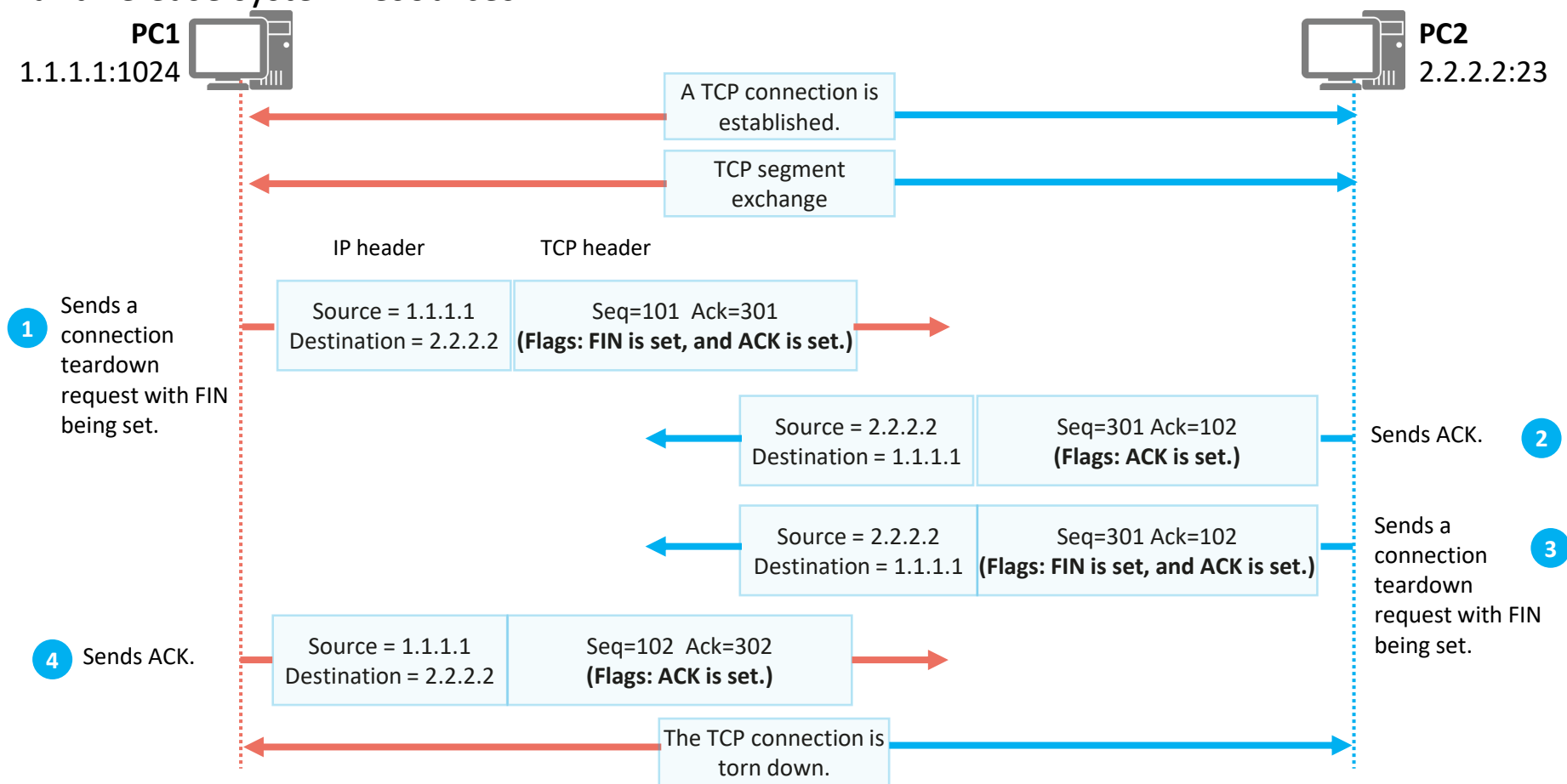
Transport Layer

Network Layer

Data Link Layer

Physical Layer

- After data transmission is complete, TCP needs to use the four-way handshake mechanism to disconnect the TCP connection and release system resources.

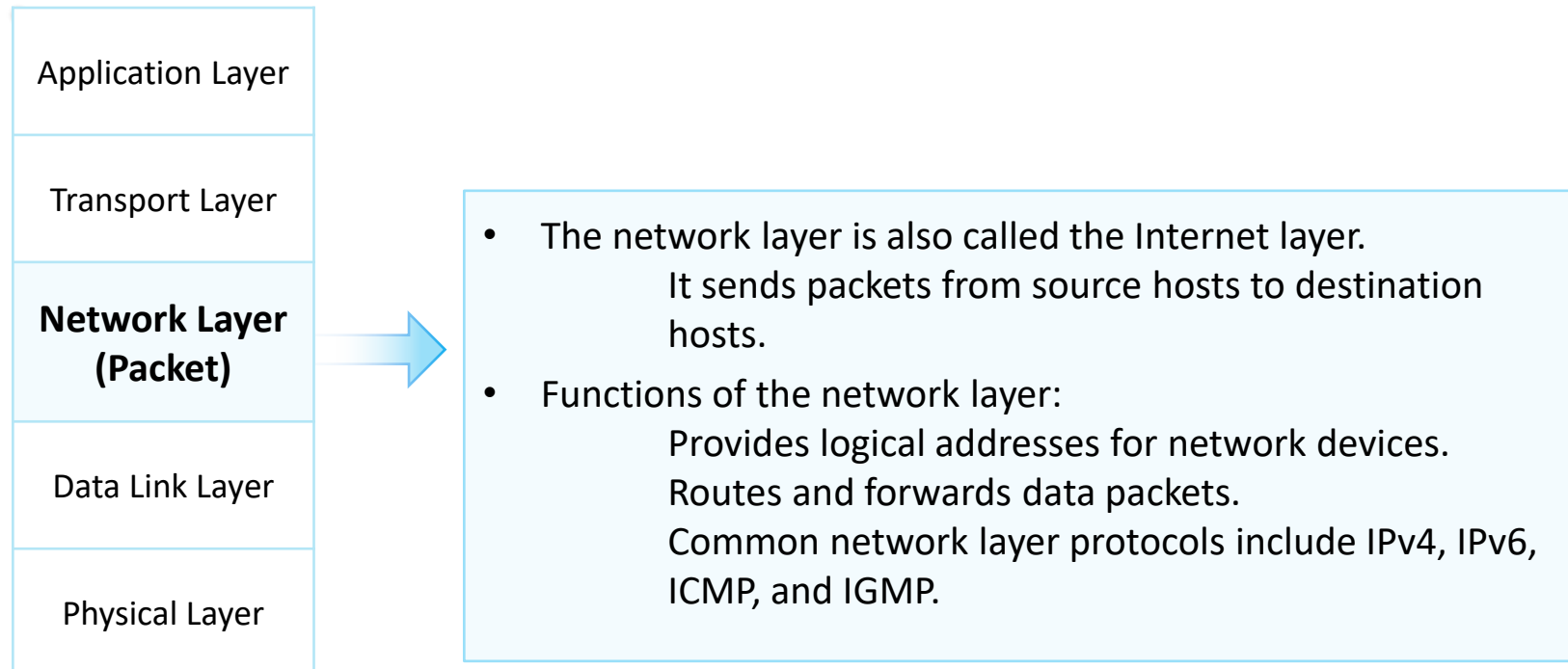




Network Layer



- The transport layer is responsible for establishing connections between processes on hosts, and the network layer is responsible for transmitting data from one host to another.
- PDUs transmitted at the network layer are called packets.





Working Process of a Network Layer Protocol

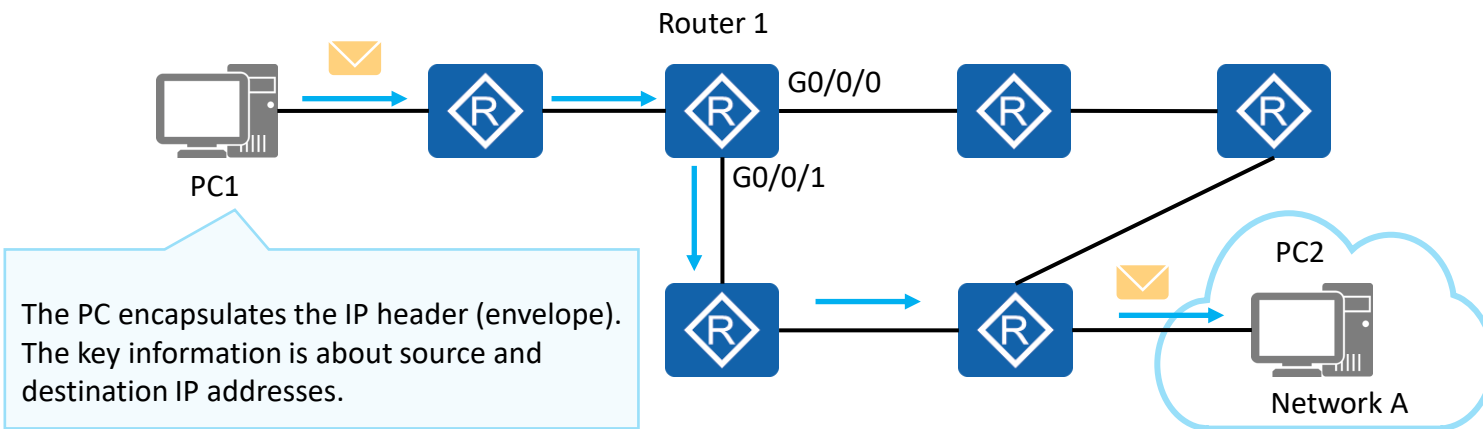
Packet Encapsulation

Letter: data sent by an upper layer (for example, the transport layer)



Envelope: IP packet header
Sender: source IP address
Receiver: destination IP address

Packet Forwarding Based on Network Layer Addresses



Routing table of Router 1

Network	Outbound Interface
Network A	G0/0/1
...	...
...	...

- The network layer header of a packet sent by a source device carries the network layer addresses of the source and destination devices.
- Each network device (such as a router) that has the routing function maintains a **routing table** (like a map of the network device).
- After receiving a packet, the network device reads the network layer **destination address** of the packet, searches the routing table for the matching entry of the destination address, and forwards the packet according to the instruction of the matching entry.



Data Link Layer

Application
Layer

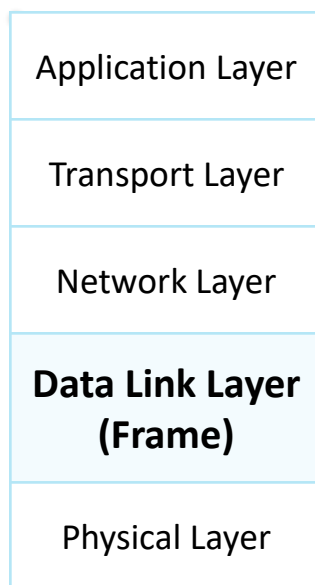
Transport
Layer

Network
Layer

Data Link
Layer

Physical
Layer

- The data link layer is located between the network layer and the physical layer and provides services for protocols such as IP and IPv6 at the network layer. PDUs transmitted at the data link layer are called frames.
- Ethernet is the most common data link layer protocol.



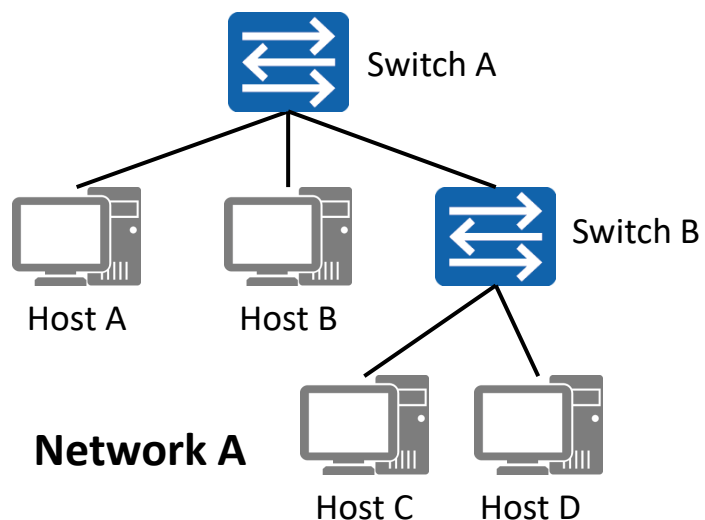
The data link layer is located between the network layer and the physical layer.

- The data link layer provides intra-segment communication for the network layer.
- The functions of the data link layer include framing, physical addressing, and error control.
- Common data link layer protocols include Ethernet, PPPoE, and PPP.



Ethernet and Source MAC Addresses

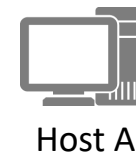
Ethernet Definition



- Ethernet is a broadcast multiple access protocol that works at the data link layer protocol.
- The network interfaces of PCs comply with the Ethernet standard.
- Generally, a broadcast domain corresponds to an IP network segment.

Ethernet Source MAC Addresses

I have a MAC address when I leave the factory.



Host A

Name: Host A

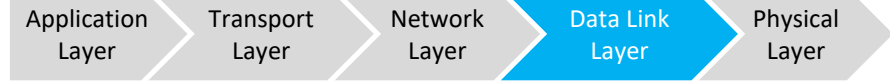


MAC address/Ethernet address/physical address:

- A media access control (MAC) address uniquely identifies a NIC on a network. Each NIC requires and has a unique MAC address.
- MAC addresses are used to locate specific physical devices in an IP network segment.
- A device that works at the data link layer, such as an Ethernet switch, maintains a MAC address table to guide data frame forwarding.



ARP



- Address Resolution Protocol (ARP):
 - ARP is used to discover the MAC address associated with a known IP address. Process:
 - Before sending a datagram, a device searches its local ARP table for the destination MAC address of the datagram.
 - If found, the device encapsulates the MAC address in the frame and sends out the frame.
 - If not found: sends ARP broadcast request with (dest MAC = FF-FF-FF-FF-FF-FF).
 - Target host replies with unicast ARP Reply containing its MAC address.
 - Both hosts update their ARP tables.



ARP Implementation Principles (1)

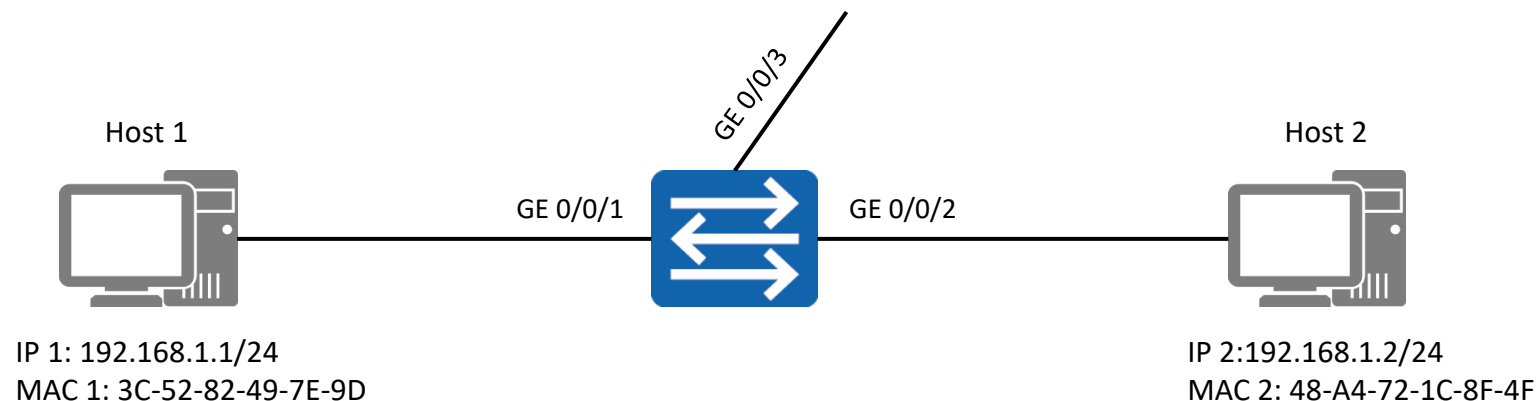
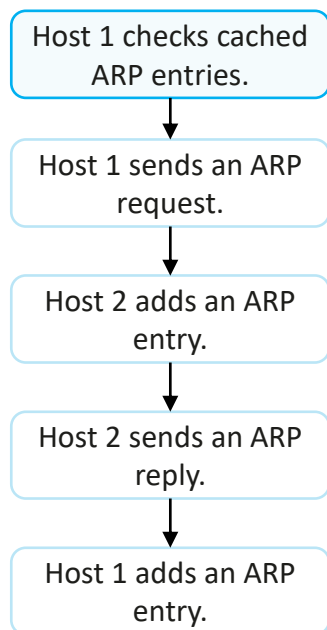
Application Layer

Transport Layer

Network Layer

Data Link Layer

Physical Layer



1

```
Host 1>arp -a
Internet Address  Physical Address  Type
```

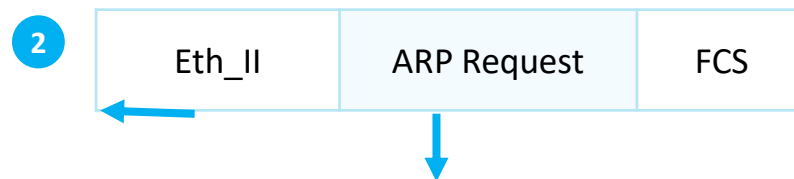
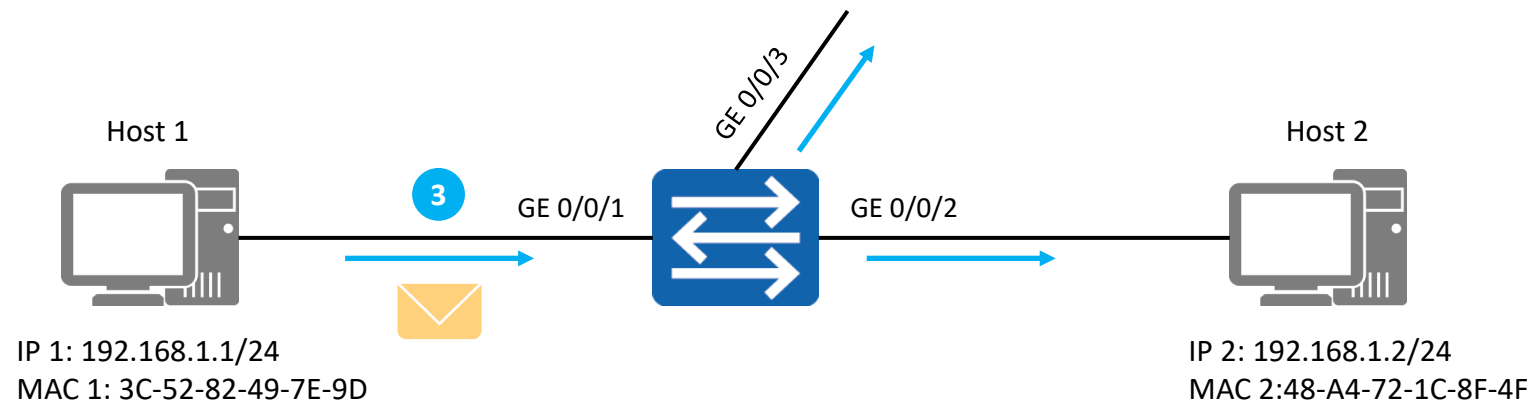
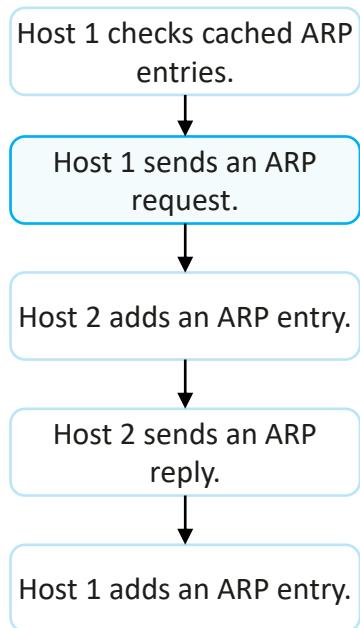
The ARP cache table is empty.

Step 1:

- Before sending a datagram, a device searches its ARP table for the destination MAC address of the datagram.
- If the destination MAC address exists in the ARP table, the device encapsulates the MAC address in the frame and sends out the frame. If the destination MAC address does not exist in the ARP table, the device sends an ARP request to discover the MAC address.



ARP Implementation Principles (2)



Destination MAC address:
FF-FF-FF-FF-FF-FF
Source MAC address:
MAC 1

Operation type: ARP request
MAC address of the sender: MAC 1
IP address of the sender: IP 1
Destination MAC address: 00-00-00-00-00-00
Destination IP address: IP 2

Step 2:

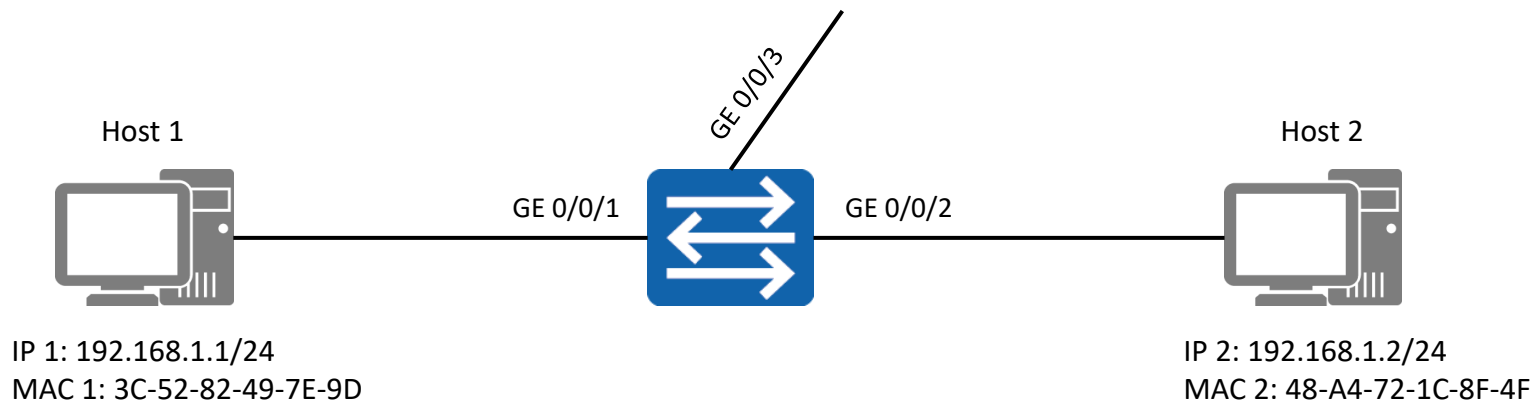
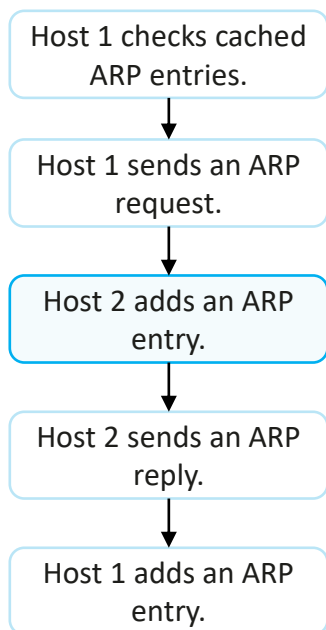
- Host 1 sends an ARP request to discover the MAC address of Host 2.
- The destination MAC address in the ARP request is 0 because the destination MAC address is unknown.

Step 3:

- The ARP request message is a broadcast data frame. After receiving the ARP request message, the switch floods it.



ARP Implementation Principles (3)



Step 4:

- After receiving the ARP request message, each host checks whether it is the destination of the message based on the carried destination IP address.
- Host 2 finds that it is the destination of the message and then records the mapping between the sender's MAC and IP addresses in its ARP table.

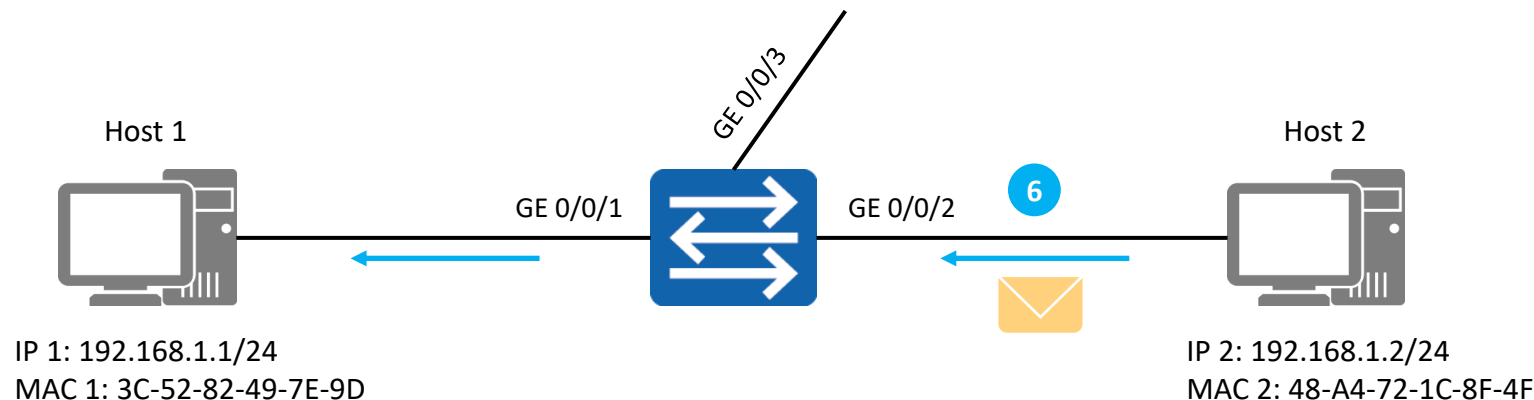
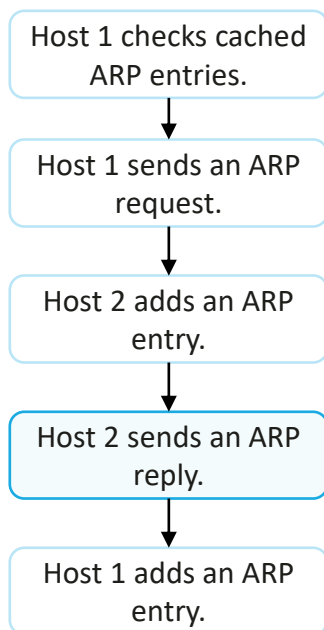
4

```
Host 2>arp -a
```

Internet Address	Physical Address	Type
192.168.1.1	3C-52-82-49-7E-9D	Dynamic



ARP Implementation Principles (4)



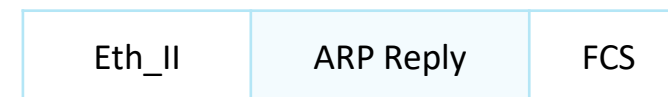
Step 5:

- Host 2 sends an ARP reply to Host 1.
- In this step, Host 2 has discovered the MAC address of Host 1, so the ARP reply is a unicast data frame.

Step 6:

- After receiving the unicast data frame, the switch forwards the frame.

5



Destination MAC address: MAC-1
Source MAC address: MAC 2

Operation type: **ARP reply**
Sender's MAC address: **MAC 2**
Sender's IP address: **IP 2**
Receiver's MAC address: MAC 1
Receiver's IP address: IP 1



ARP Implementation Principles (5)

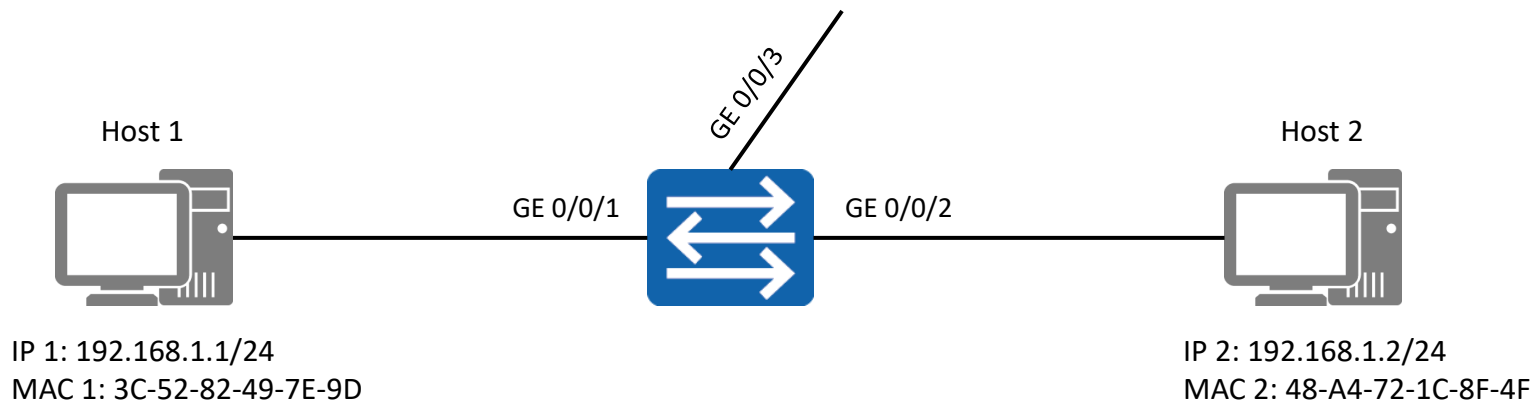
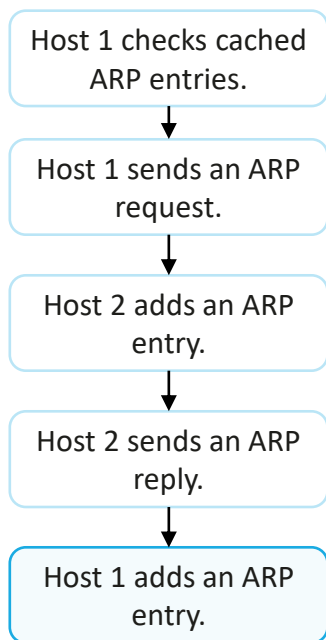
Application Layer

Transport Layer

Network Layer

Data Link Layer

Physical Layer



7

Internet Address	Physical Address	Type
192.168.1.2	48-A4-72-1C-8F-4F	Dynamic

Step 7:

- After receiving the ARP reply message, Host 1 checks whether it is the destination of the message based on the carried destination IP address.
- If so, Host 1 records the carried sender's MAC and IP addresses in its ARP table.



Physical Layer

Application
Layer

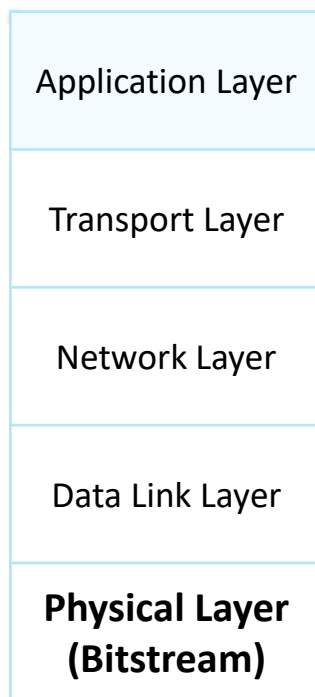
Transport
Layer

Network
Layer

Data Link
Layer

Physical
Layer

- After data arrives at the physical layer, the physical layer converts a digital signal into an optical signal, an electrical signal, or an electromagnetic wave signal based on the physical media.
- PDUs transmitted at the physical layer are called bitstreams.



The physical layer is at the bottom of the model.

- This layer transmits bitstreams on media.
- It standardizes physical features such as cables, pins, voltages, and interfaces.
- Common transmission media include twisted pairs, optical fibers, and electromagnetic waves.



Common Transmission Media

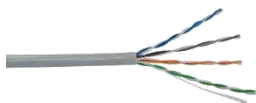
Application
Layer

Transport
Layer

Network
Layer

Data Link
Layer

Physical
Layer



Twisted pair



RJ45 connector

Data transmission through twisted pairs

1



Fiber



Optical module



Data transmission through optical fibers

2



Synchronous/asynchronous
serial cable: V.24 on the left and
V.35 on the right

Data transmission through serial cables

3



PAD



Mobile phone



Laptop



Wireless router

Data transmission between terminal and wireless
routers through wireless signals

4

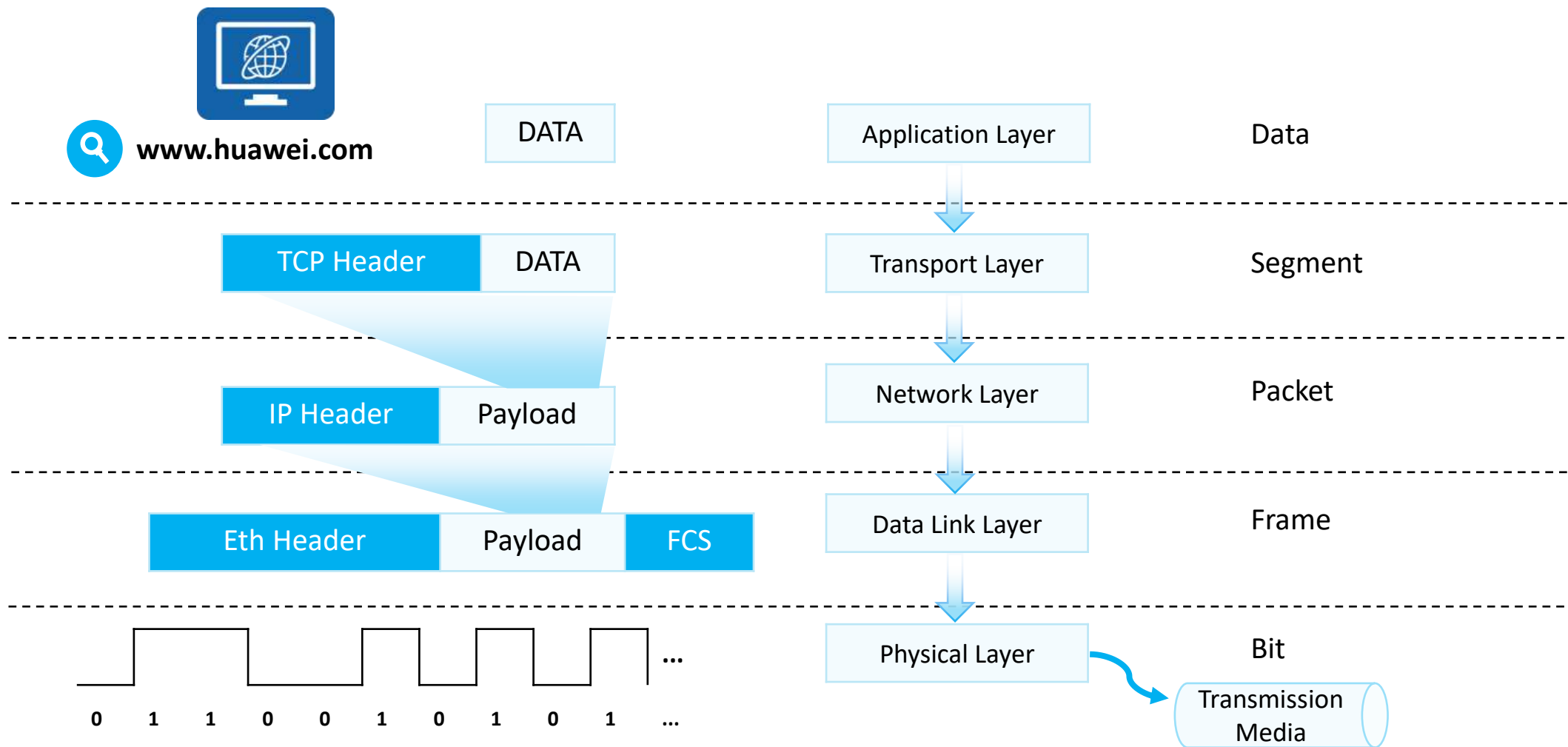


Contents

1. Applications and Data
2. Network Reference Model and Standard Protocols
- 3. Data Communication Process**



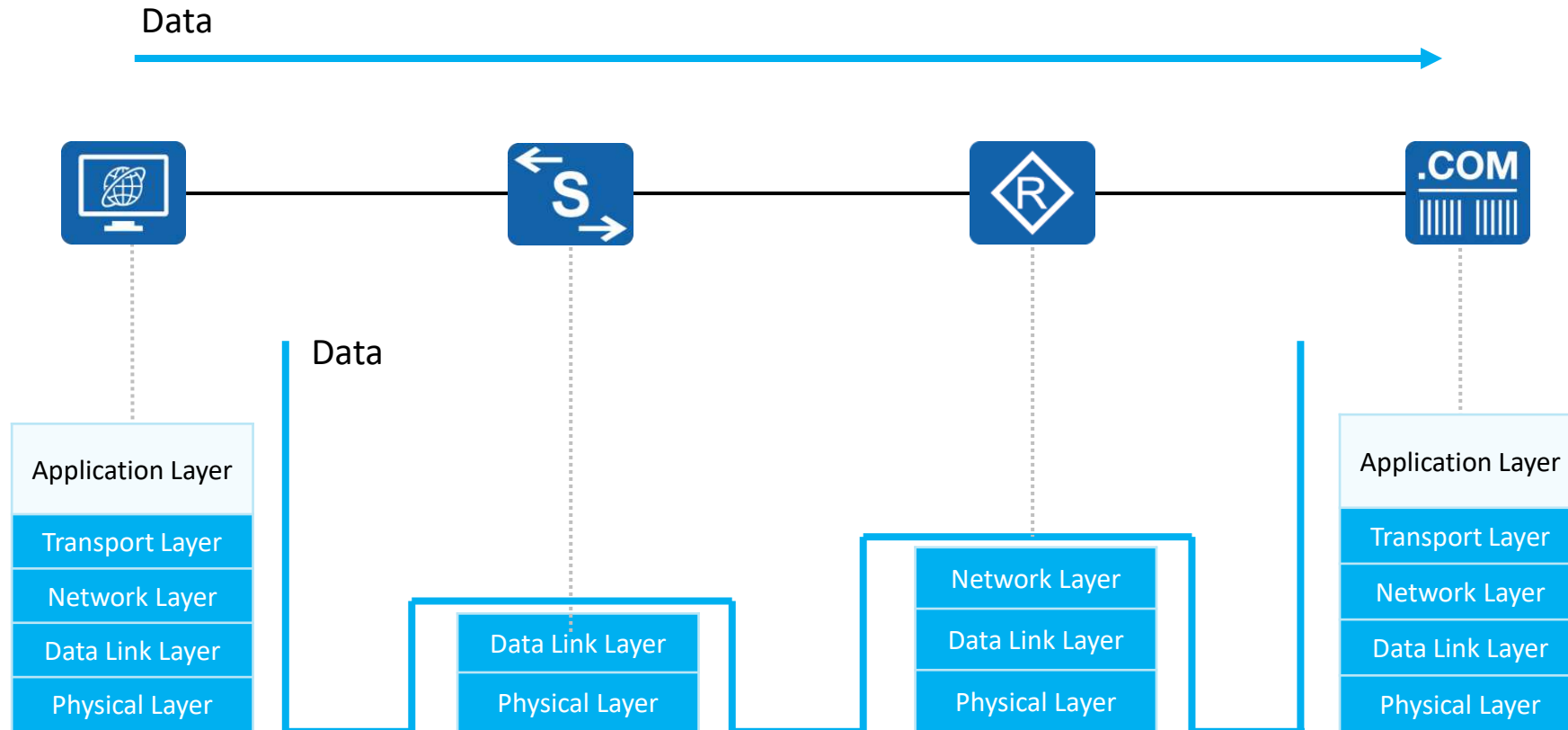
Data Encapsulation on the Sender





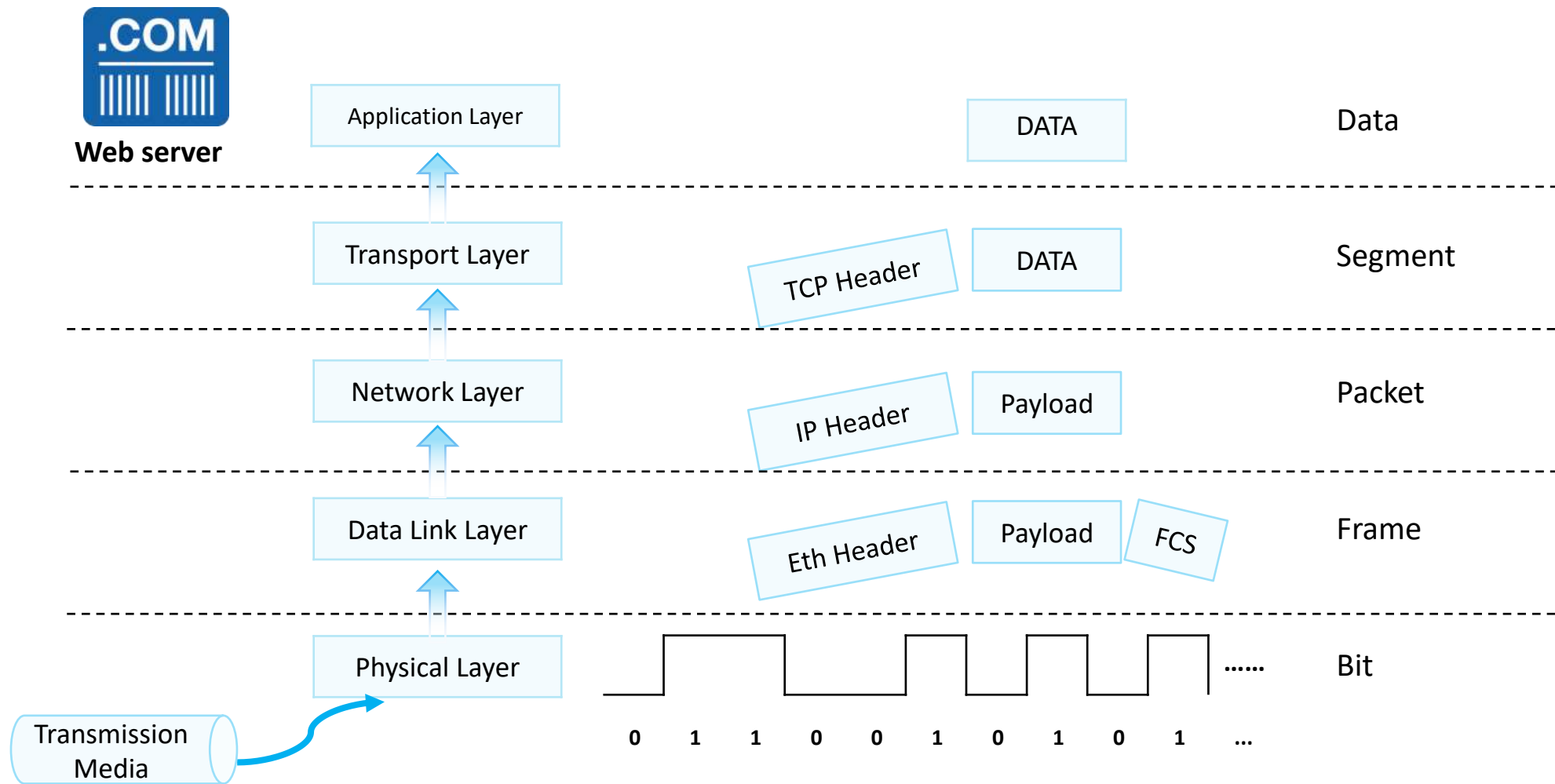
Data Transmission on the Intermediate Network

- Encapsulated data is transmitted on the network.





Data Decapsulation on the Receiver





Summary

- Both the OSI reference model and the TCP/IP reference model adopt the layered design concept.
 - Clear division of functions and boundaries between layers facilitates the development, design, and troubleshooting of each component.
 - The functions of each layer can be defined to impel industry standardization.
 - Interfaces can be provided to enable communication between hardware and software on various networks, improving compatibility.
- Data generation and transmission require collaboration between modules. Meanwhile, each module must fulfill its own responsibilities.

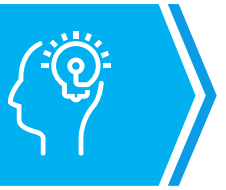


Quiz

1. What are the benefits of the layered model?
2. What are the common protocols at the application layer, transport layer, network layer, and data link layer?

Benefits of Layered Model (Short)

1. Simplifies networking → breaks complex tasks into layers
2. Easy troubleshooting → isolate problems by layer
3. Interoperability → different hardware vendors can work together
4. Flexibility → changes in one layer don't affect others
5. Standardization → clear structure for protocols



Quiz

Common Protocols by Layer (Short)

1. Application Layer

- A. HTTP/S
- B. Telnet
- C. FTP
- D. SMTP
- E. DNS

2. Transport Layer

- A. Transmission Control Protocol
- B. User Datagram Protocol

3. Network Layer

- A. Internet Protocol
- B. Internet Control Message Protocol
- C. IPsec

4. Data Link Layer

- A. Ethernet
- B. Point-to-Point Protocol
- C. Address Resolution Protocol

The background of the slide features a blue-tinted image of several business professionals in a modern office environment. They are standing on a highly reflective floor, and their silhouettes are clearly visible against the lighter background. The overall aesthetic is professional and corporate.

Thank You
www.huawei.com



Huawei VRP Basics



Foreword

- The Versatile Routing Platform (VRP) is a universal operating system (OS) platform for Huawei datacom products. It is based on IP and adopts a component-based architecture. It provides rich features and functions, including application-based tailorable and extensible functions, greatly improving the running efficiency of the devices that use this OS. To efficiently manage such devices, you must be familiar with VRP and VRP-based configuration.
- This course describes the basic concepts, common commands, and command line interface (CLI) of VRP.



Objectives

- On completion of this course, you will be able to:
 - Understand VRP basics.
 - Learn how to use CLI.
 - Master basic CLI commands.

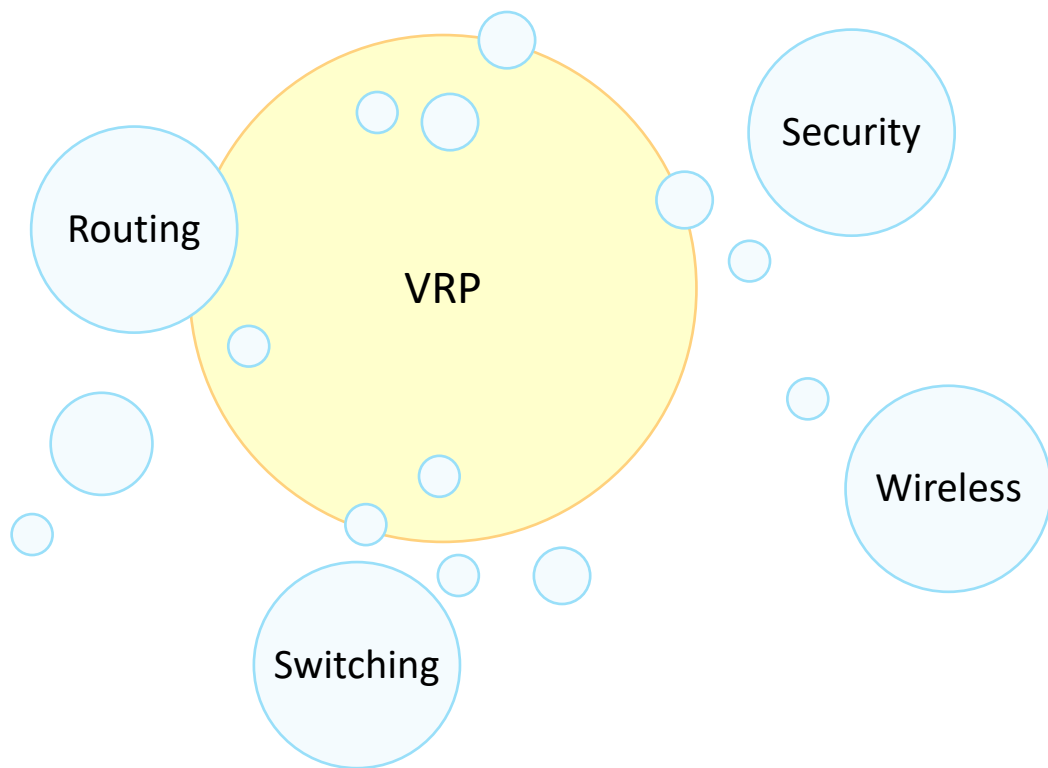


Contents

- 1. VRP Overview**
2. Command Line Basics



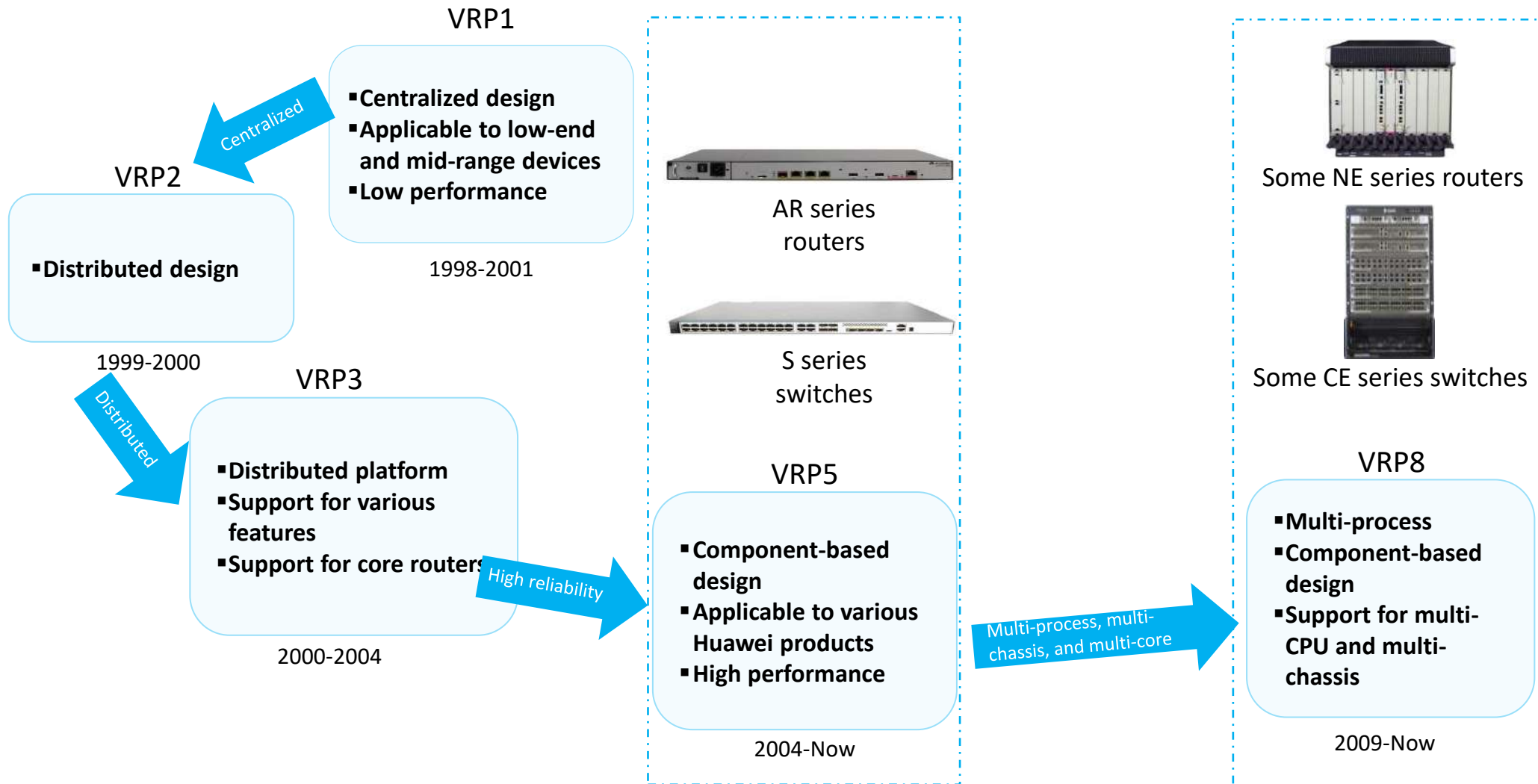
What Is VRP?



- VRP is a universal OS platform for Huawei datacom products. It serves as the software core engine of Huawei's full series of routers from low-end to core ones, Ethernet switches, service gateways, and so on.
- VRP provides the following functions:
 - Provides a unified user interface and a unified management interface.
 - Implements the functions of the control plane and defines the interface specifications of the forwarding plane.
 - Implements communication between the device forwarding plane and VRP control plane.



Development of the VRP





File System

- The file system manages files and directories in storage media, allowing users to view, create, rename, and delete directories and copy, move, rename, and delete files.
- Mastering the basic operations of the file system is crucial for network engineers to efficiently manage the configuration files and VRP system files of devices.

The system software is a must for device startup and operation, providing support, management, and services for a device. The common file name extension is .cc.

System
Software

Configuration
File

A configuration file stores configuration commands, enabling a device to start with the configurations in the file. The common file name extensions are .cfg, zip, and .dat.

A patch is a kind of software compatible with the system software. It is used to fix bugs in system software. The common file name extension is .pat.

Patch File

PAF
File

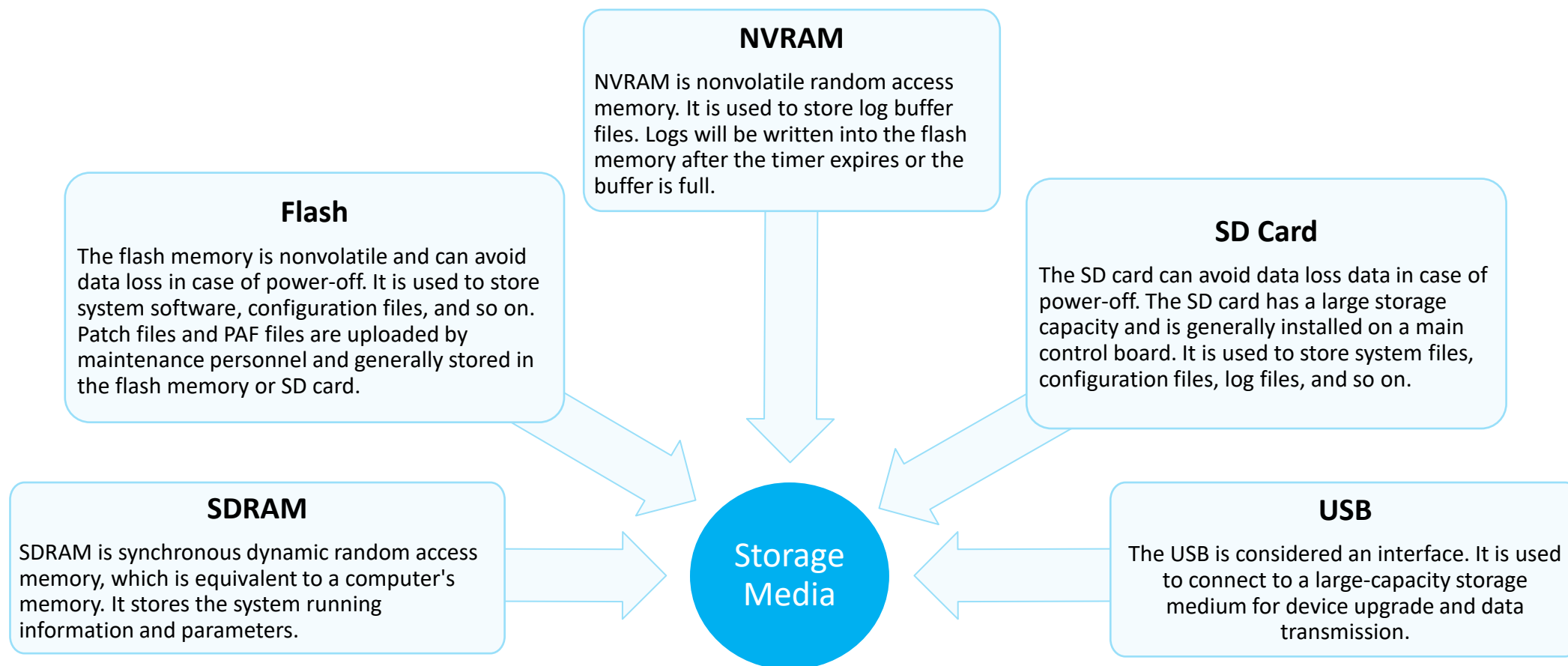
A PAF file effectively controls product features and resources. The common file name extension is .bin.

Common File Types



Storage Media

- Storage media include SDRAM, flash memory, NVRAM, SD card, and USB.





Device Initialization Process

- After a device is powered on, it runs the BootROM software to initialize the hardware and display hardware parameters. Then, it runs the system software and reads the configuration file from the default storage path to perform initialization.

BIOS Creation Date : Jan 5 2013, 18:00:24

DDR DRAM init : OK

Start Memory Test ? ('t' or 'T' is test):skip

Copying Data : Done

Uncompressing : Done

.....

Press Ctrl+B to break auto startup ... 1

Now boot from flash:/AR2220E-V200R007C00SPC600.cc,

.....



Device Management

- There are two commonly used device management modes: CLI and web system.
- To use a device management mode, you must first log in to a device through a login mode supported by this device management mode.

Web System

- The web system provides a graphical user interface (GUI) for easy device management and maintenance. This method, however, can be used to manage and maintain only some, not all, device functions.
- The web system supports the HTTP and HTTPS login modes.

CLI

- The CLI requires users to use commands provided by a device to manage and maintain the device. This mode implements refined device management but requires users to be familiar with the commands.
- The CLI supports the console port, Telnet, and SSH login modes.



VRP User Interfaces

- When a user logs in to a device through a CLI-supported mode, the system allocates a user interface to manage and monitor the current session between the user terminal and device.
- Such a user interface can be a console user interface or virtual type terminal (VTY) user interface.

Console User Interface

- A console user interface is used to manage and monitor users who log in to a device through the console port.
- The serial port of a user terminal can be directly connected to the console port of a device for local access.

VTY User Interface

- The VTY user interface is used to manage and monitor users who log in to a device by means of VTY.
- After a Telnet or STelnet connection is established between a user terminal and a device, a VTY channel is established to implement remote access to the device.



VRP User Levels

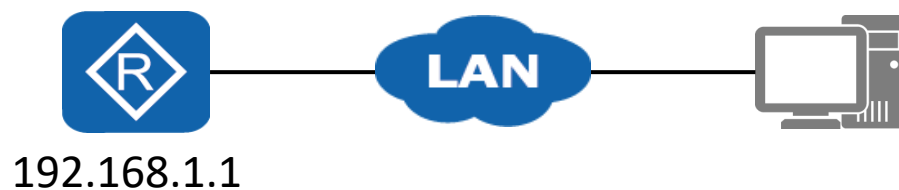
- VRP provides basic permission control functions. It defines the levels of commands that each level of users can execute to restrict the operations of users at different levels.

User Level	Command Level	Name	Available Command
0	0	Visit level	Network diagnosis commands (such as ping and tracert), commands for accessing external devices from the local device (such as Telnet client commands), and some display commands
1	0 and 1	Monitoring level	System maintenance commands, including display commands
2	0, 1, and 2	Configuration level	Service configuration commands, including routing commands and IP configuration commands, to directly provide users with network services
3-15	0, 1, 2, and 3	Management level	Commands for controlling basic system operations and providing support for services, including the file system, FTP, TFTP download, user management, and command level commands, as well as debugging commands for fault diagnosis



Login to the Web System

Take the web system for a Huawei AR router as an example. Start a browser on a PC, enter **https://192.168.1.1** in the address bar, and press **Enter**. Then, the web system login page is displayed.

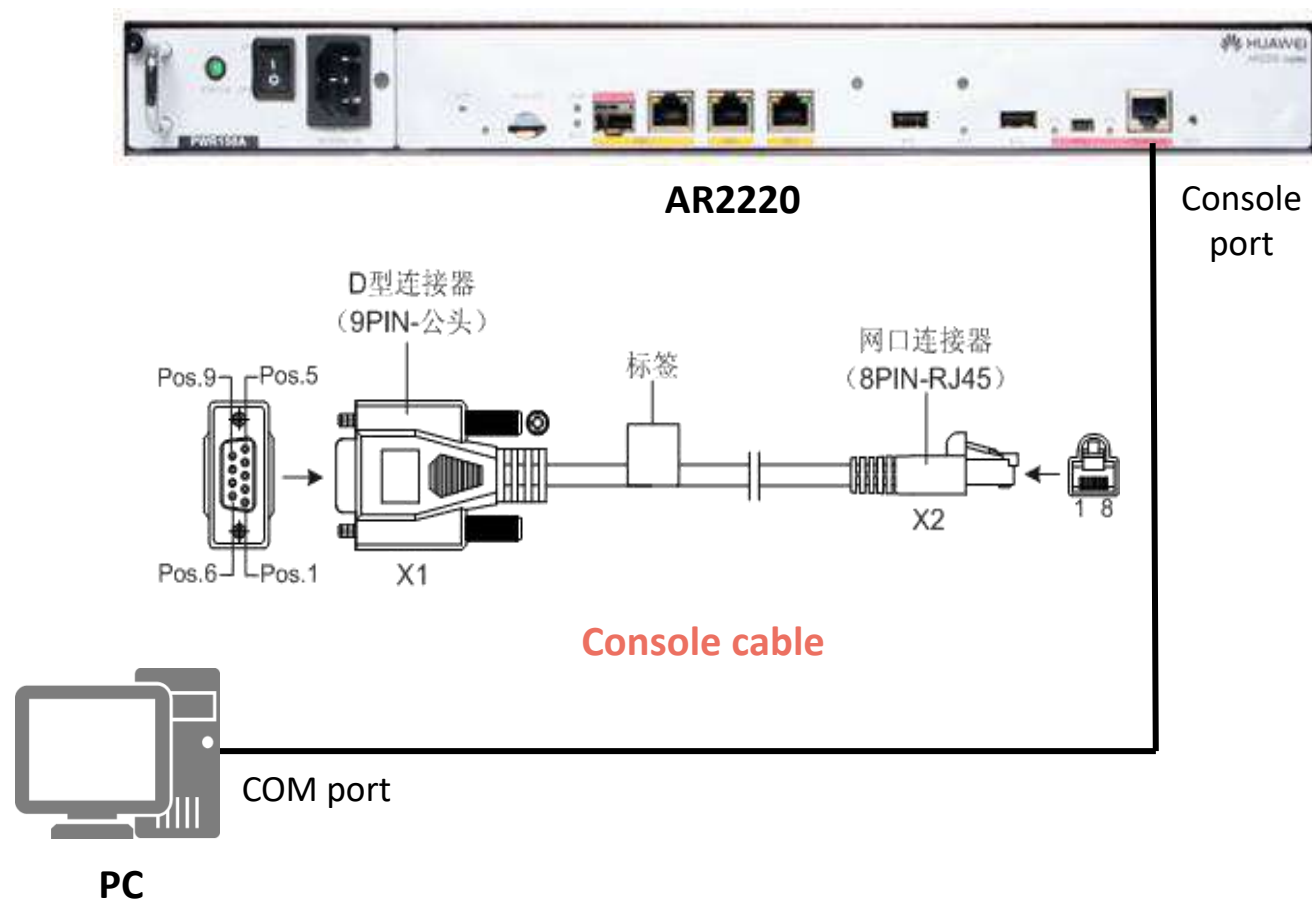




CLI - Local Login (1)

You can log in to a device in local or remote mode. Local login mode:

- Use this mode when you need to configure a device that is powered on for the first time. You can use the console port of the device for a local login.
- The console port is a serial port provided by the main control board of a device.
- To implement the login, directly connect your terminal's serial port to the device's console port, and use PuTTY to log in to the device. You can then configure the device after the login succeeds.

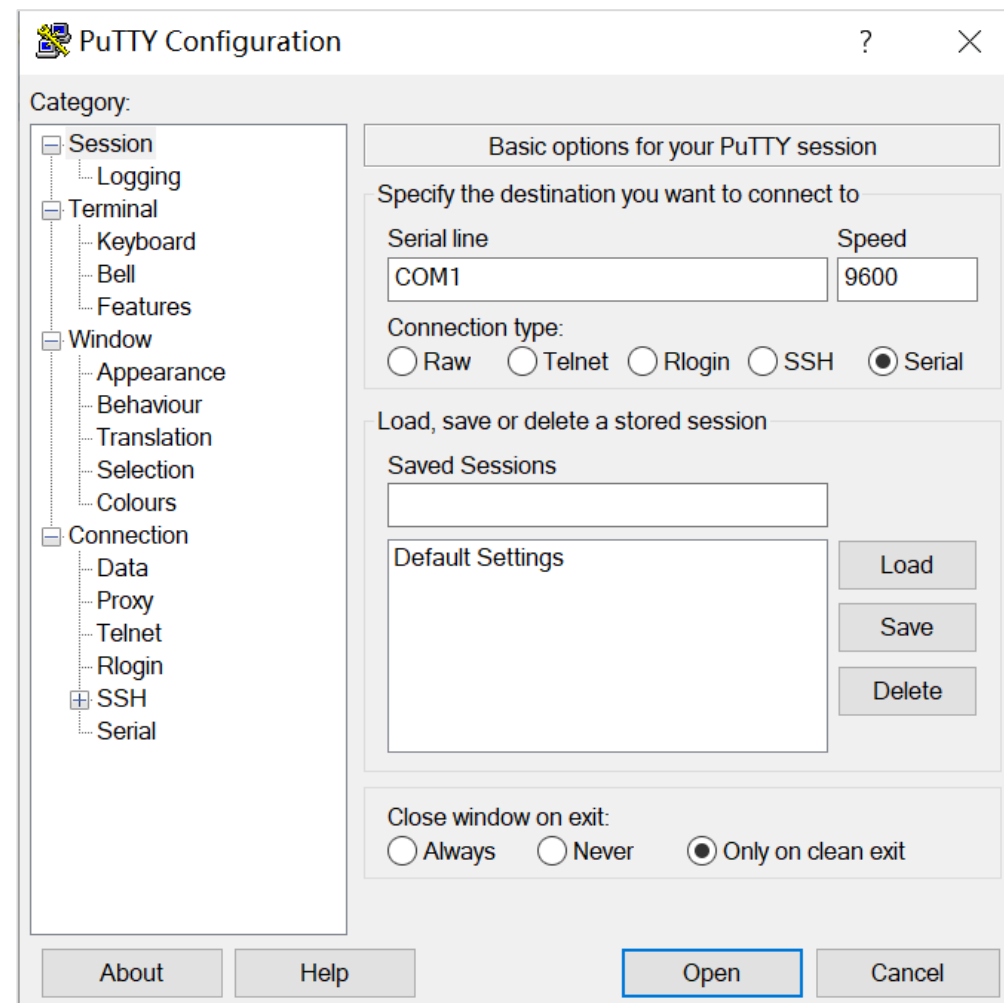




CLI - Local Login (2)

PuTTY is a connection software for login through Telnet, SSH, serial interfaces, and so on.

In local login, the terminal is connected to the console port of the Huawei device through a serial port. Therefore, set **Connection type** to **Serial**. Set **Serial line** based on the actually used port on the terminal. Set **Speed** to **9600**.

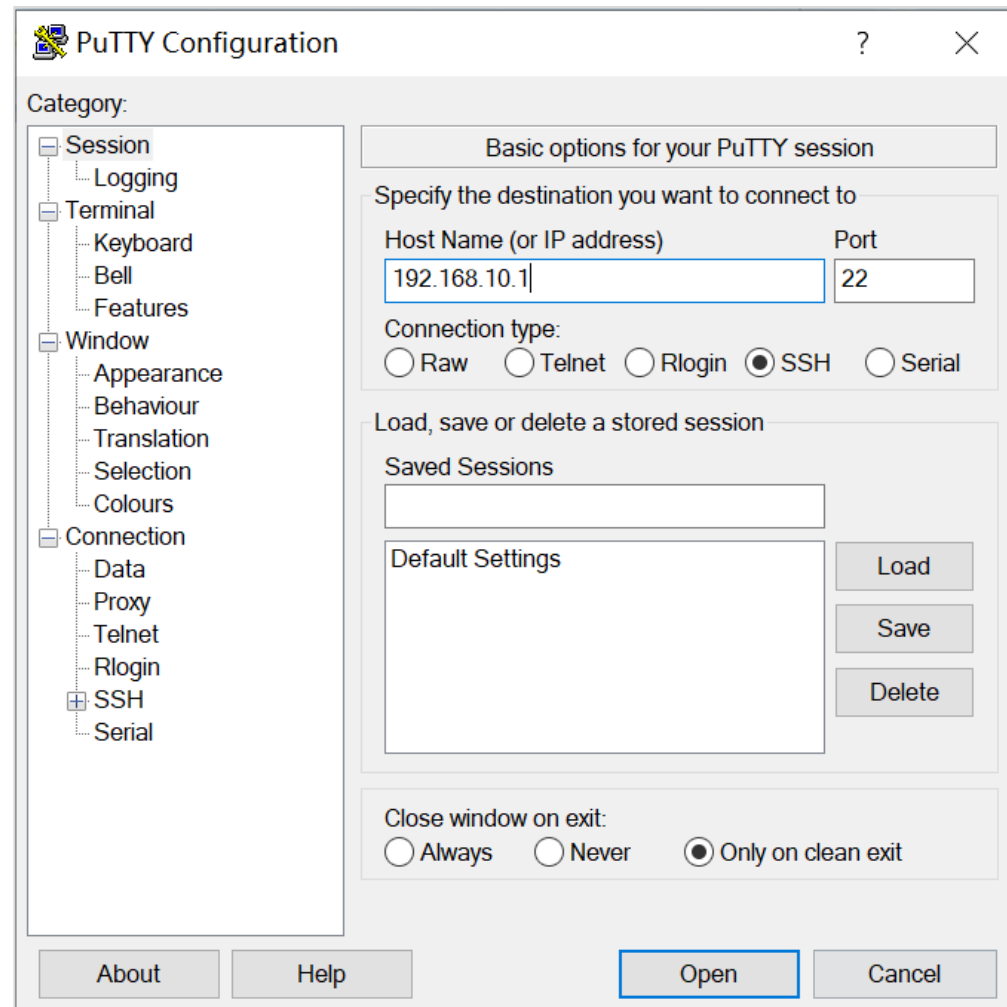
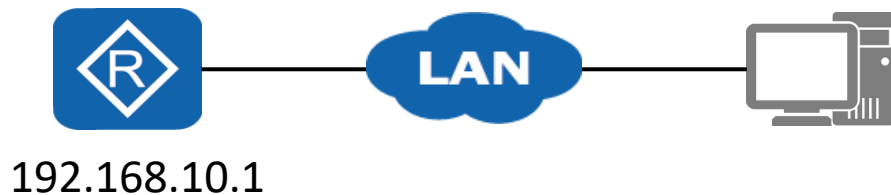




CLI - Remote Login

Remote login means that you log in to a device that can function as a remote login server, allowing you to centrally manage and maintain network devices. Remote login methods include Telnet and SSH.

- If you use the SSH login mode, set **Connection type** to **SSH**, enter the IP address of the remote login server, and use the default port number 22.
- If you use the Telnet login mode, set **Connection type** to **Telnet**, enter the IP address of the remote login server, and use the default port number 23.





CLI

- After a login succeeds, the command line interface (CLI) is displayed.
- The CLI is a common tool for engineers to interact with network devices.

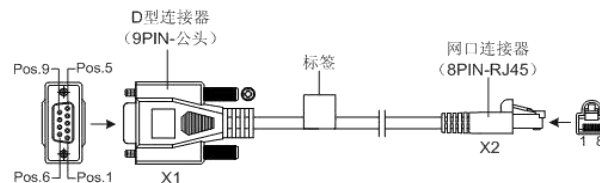
127.0.0.1 - PuTTY

```
<Huawei>sys
sys
Enter system view, return user view with Ctrl+Z.
[Huawei]
```



AR2220

Console port



Console cable



COM port

PC



Contents

1. VRP Basics

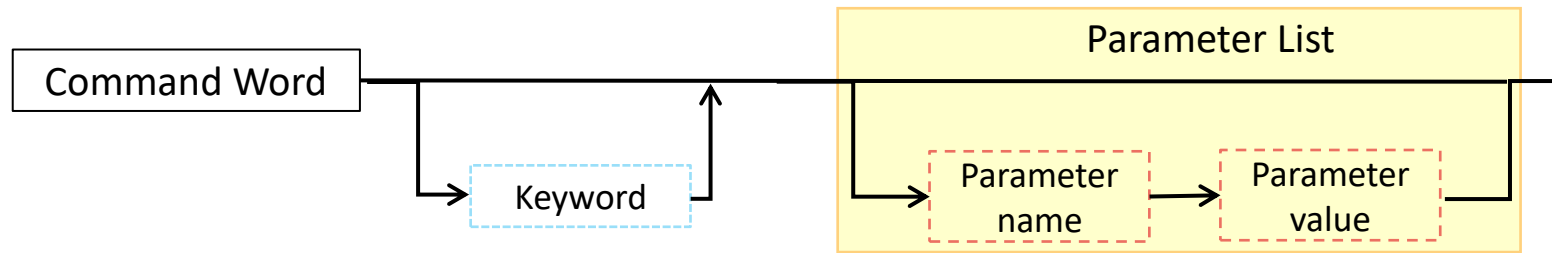
2. Command Line Basics

- **Command Line Overview**
 - Basic Configuration Commands
 - Case Analysis



Basic Command Structure

- CLI commands follow a unified structure. After a command is entered on the CLI, the CLI parses the command and executes it to implement the function of the command, such as query, configuration, or management.



- Command word:** specifies the operation to be executed in a command, such as display (device status query) or reboot (device restart).
- Keyword:** a special character string that is used to further restrict a command. It is an extension of a command and can also be used to express the command composition logic.
- Parameter list:** is composed of parameter names and values to further restrict the command function. It can contain one or more pairs of parameter names and values.

Example 1:

display ip interface GE0/0/0: displays interface information.

Command word: display

Keyword: ip

Parameter name: interface

Parameter value: GE0/0/0

Example 2:

reboot: restarts a device.

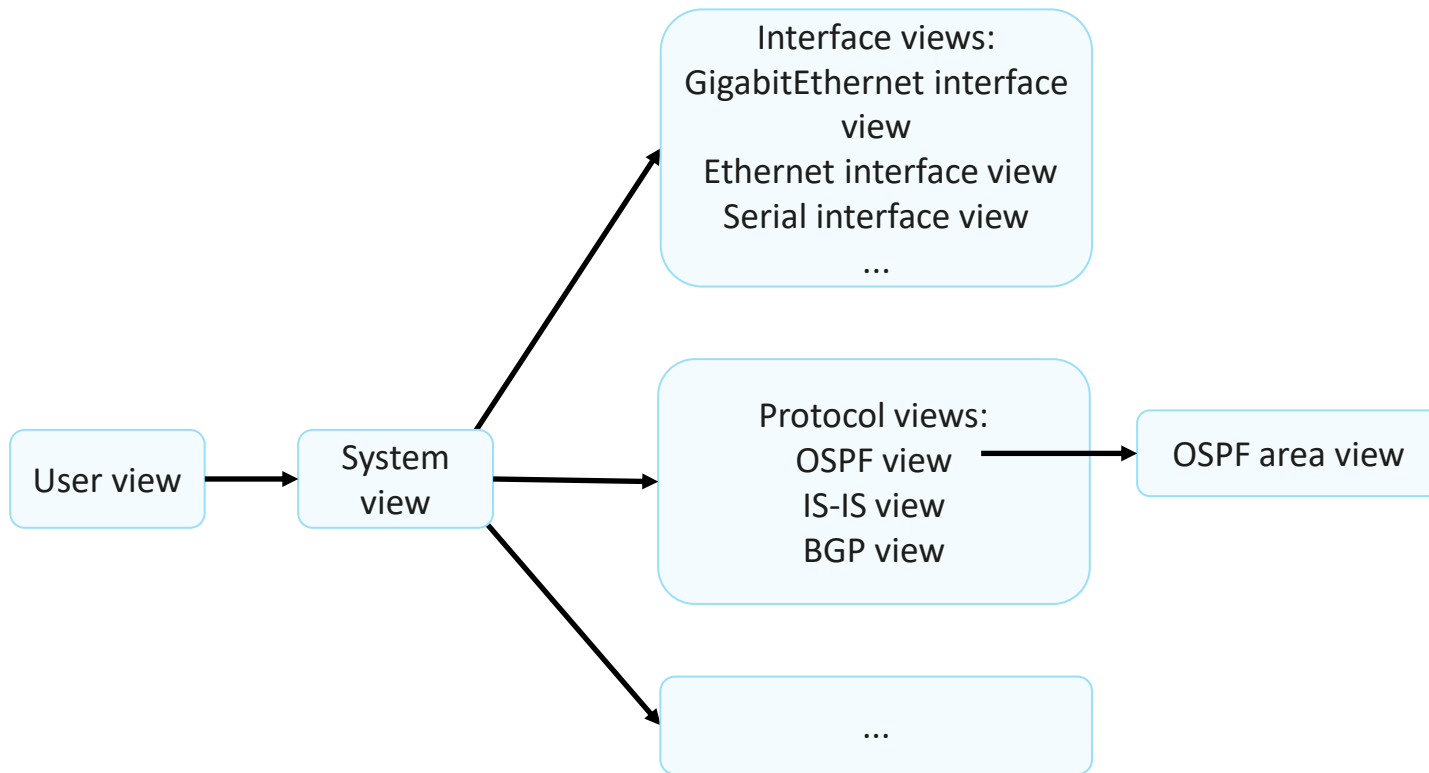
Command word: reboot

Each operation command must start with a command word, and the command word is selected from the standard command word list.



Command Views (1)

- A device provides various configuration and query commands. To facilitate the use of these commands, VRP registers the commands in different views according to their functions.



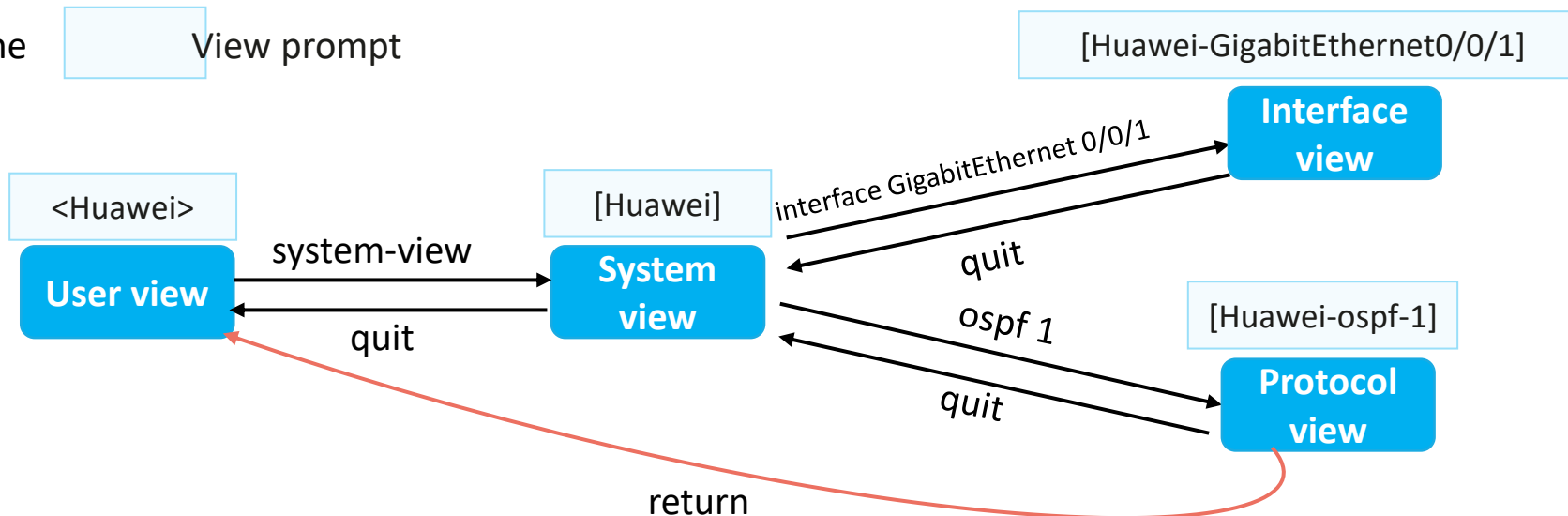
- User view: In this view, you can check the running status and statistics of a device.
- System view: In this view, you can set system parameters and enter the configuration views of other commands.
- Other views: In other views, such as the interface view and protocol view, you can set interface parameters and protocol parameters.



Command Views (2)

View name

View prompt



Command examples:

`<Huawei>system-view` #This command is used to enter the system view from the user view. The user view is the first view that is displayed after you log in to a device.

`[Huawei]interface GigabitEthernet 0/0/1` #This command is used to enter the interface view from the system view.

`[Huawei-GigabitEthernet0/0/1]ip address 192.168.1.1 24` #This command is used to set an IP address.

`[Huawei-GigabitEthernet0/0/1]quit` #This command is used to return to the previous view.

`[Huawei]ospf 1` #This command is used to enter the protocol view from the system view.

`[Huawei-ospf-1]area 0` #This command is used to enter the OSPF area view from the OSPF view.

`[Huawei-ospf-1-area-0.0.0.0]return` #This command is used to return to the user view.



Editing a Command (1)

- The CLI of a device provides basic command editing functions. Common editing functions are as follows:

1. Command editing through function keys

- Backspace: deletes the character before the cursor and moves the cursor to the left. When the cursor reaches the beginning of the command, an alarm is generated.
- Left cursor key ← or **Ctrl+B**: moves the cursor one character to the left. When the cursor reaches the beginning of the command, an alarm is generated.
- Right cursor key → or **Ctrl+F**: moves the cursor one character to the right. When the cursor reaches the end of the command, an alarm is generated.

2. Incomplete keyword input

- A device allows the input of incomplete keywords. Specifically, if an entered character string can match a unique keyword, you do not need to enter the remaining characters of the keyword.

```
<Huawei>d cu
<Huawei>di cu
<Huawei>dis cu
<Huawei>d c
    ^
Error:Ambiguous command found at '^' position.
<Huawei>dis c
    ^
Error:Ambiguous command found at '^' position.
```

For example, the **display current-configuration** command is identified when you enter **d cu**, **di cu**, or **dis cu**. However, the command cannot be identified if you enter **d c** or **dis c** because the character string **d c** or **dis c** matches more than one command.



Editing a Command (2)

3. Command editing through the Tab key

- If an entered character string matches a unique keyword, the system automatically supplements the keyword after you press **Tab**. If the keyword is complete, it remains unchanged even if you press **Tab** repeatedly.

```
[Huawei] info-          #Press Tab.  
[Huawei] info-center
```

- If an entered character string matches more than one keyword, you can press **Tab** repeatedly. The system will then circularly display the keywords beginning with the entered character string to help you find the desired keyword.

```
[Huawei] info-center log      #Press Tab.  
[Huawei] info-center logbuffer #Press Tab repeatedly to circularly display all matched keywords.  
[Huawei] info-center logfile  
[Huawei] info-center loghost
```

- If an entered character string cannot identify any keyword, the entered string remains unchanged after you press **Tab**.

```
[Huawei] info-center loglog    #Enter an incorrect keyword and press Tab.  
[Huawei] info-center loglog
```



Using Command Line Online Help

- You can use command line online help to obtain real-time help without memorizing a large number of complex commands.
- The online help can be classified into full help and partial help. To obtain the online help, enter a question mark (?) when using a command.

Full Help

- To obtain full help, press ? after a view displayed. The system will then display all commands in the view and their descriptions.

<Huawei> ?

User view commands:

arp-ping	ARP-ping
autosave	<Group> autosave command group
backup	Backup information
cd	Change current directory
clear	Clear
clock	Specify the system clock

...

Partial Help

- To obtain partial help, press ? after you enter the start character or character string of a command. The system will then display all the commands that start with this character or character string.

<Huawei> d?

debugging	<Group> debugging command group
delete	Delete a file
dialer	Dialer
dir	List files on a filesystem
display	Display information



Interpreting Command Line Error Messages

- If a command passes the syntax check, the system executes it. Otherwise, the system reports an error message.

[Huawei] sysname

^

Error: Incomplete command found at '^' position. #A supplement needs to be made at the position pointed by the arrow.

[Huawei] router if 1.1.1.1

^

Error: Unrecognized command found at '^' position. #An identification failure occurs at the position pointed by the arrow.
Check whether the command is correct.

[Huawei] a

^

Error: Ambiguous command found at '^' position. #More than one command matches the keyword at the position pointed by the arrow. In this example, it indicates that there are multiple keywords starting with a.

[Huawei-GigabitEthernet0/0/0] ospf cost 800000

^

#The parameter value at the position pointed by the arrow is invalid.

Error: Wrong parameter found at '^' position.



Using Undo Command Lines

- If a command begins with the keyword **undo**, it is an undo command. An undo command is generally used to restore a default configuration, disable a function, or delete a configuration. For example:

- Run an undo command to restore a default configuration.

```
<Huawei> system-view  
[Huawei] sysname Server  
[Server] undo sysname  
[Huawei]
```

- Run an undo command disable a function.

```
<Huawei> system-view  
[Huawei] ftp server enable  
[Huawei] undo ftp server
```

- Run an undo command to delete a configuration.

```
[Huawei] interface g0/0/1  
[Huawei-GigabitEthernet0/0/1] ip address 192.168.1.1 24  
[Huawei-GigabitEthernet0/0/1] undo ip address
```




Using Command Line Shortcut Keys

- A device provides command shortcut keys to speed up and simplify command input.
- Command shortcut keys are classified into user-defined shortcut keys and system shortcut keys.

User-defined Shortcut Keys

- There are four user-defined shortcut keys: **Ctrl+G**, **Ctrl+L**, **Ctrl+O**, and **Ctrl+U**.
- You can associate a user-defined shortcut key with any command. After you press a shortcut key, the system will automatically run the command associated with the shortcut key.

```
<Huawei> system-view  
[Huawei] hotkey ctrl_l "display tcp status"
```

System Shortcut Keys

- **CTRL_A**: moves the cursor to the beginning of the current line.
- **CTRL_B**: moves the cursor one character to the left.
- **CTRL_C**: stops the running of the current command.
- **CTRL_E**: moves the cursor to the end of the current line.
- **CTRL_X**: deletes all characters on the left of the cursor.
- **CTRL_Y**: deletes the character at the cursor and all characters on the right of the cursor.
- **CTRL_Z**: returns to the user view.
- **CTRL+]**: terminates the current connection or switches to another connection.



Contents

1. VRP Basics

2. Command Line Basics

- Command Views and Use of Command Views
- **Basic Configuration Commands**
- Case Analysis



Common File System Operation Commands (1)

1. Check the current directory.

```
<Huawei>pwd
```

2. Display information about files in the current directory.

```
<Huawei>dir
```

3. Display the content of a text file.

```
<Huawei>more
```

4. Change the current working directory.

```
<Huawei>acd
```

5. Create a directory.

```
<Huawei>mkdir
```



Common File System Operation Commands (2)

6. Delete a directory.

```
<Huawei>rmdir
```

7. Copy a file.

```
<Huawei>copy
```

8. Move a file.

```
<Huawei>move
```

9. Rename a file.

```
<Huawei>rename
```

10. Delete a file.

```
<Huawei>delete
```



Common File System Operation Commands (3)

11. Restore a deleted file.

```
<Huawei>undelete
```

12. Permanently delete a file in the recycle bin.

```
<Huawei>reset recycle-bin
```



Basic Configuration Commands (1)

1. Configure a system name.

```
[Huawei] sysname name
```

2. Configure a system clock.

```
<Huawei> clock timezone time-zone-name { add | minus } offset
```

This command configures a local time zone.

```
<Huawei> clock datetime [ utc ] HH:MM:SS YYYY-MM-DD
```

This command configures the current or UTC date and time.

```
<Huawei> clock daylight-saving-time
```

This command configures the daylight saving time.



Basic Configuration Commands (2)

3. Configure a command level.

```
[Huawei] command-privilege level level view view-name command-key
```

This command configures a level for commands in a specified view. Command levels are classified into visit, monitoring, configuration, and management, which are identified by the numbers 0, 1, 2, and 3, respectively.

4. Configure the password-based login mode.

```
[Huawei] user-interface vty 0 4  
[Huawei-ui-vty0-4] set authentication password cipher information
```

This **user-interface vty** command displays the virtual type terminal (VTY) user interface view, and the **set authentication password** command configures the password authentication mode. The system supports the console user interface and VTY user interface. The console user interface is used for local login, and the VTY user interface is used for remote login. By default, a device supports a maximum of 15 concurrent VTY-based user accesses.

5. Configure user interface parameters.

```
[Huawei] idle-timeout minutes [ seconds ]
```

This command sets a timeout period to disconnect from the user interface. If no command is entered within the specified period, the system tears down the current connection. The default timeout period is 10 minutes.



Basic Configuration Commands (3)

6. Configure an IP address for an interface.

```
[Huawei]interface interface-number  
[Huawei-interface-number]ip address ip address
```

This command configures an IP address for a physical or logical interface on a device.

7. Display currently effective configurations.

```
<Huawei>display current-configuration
```

8. Save a configuration file.

```
<Huawei>save
```

9. Check saved configurations.

```
<Huawei>display saved-configuration
```




Basic Configuration Commands (4)

10. Clear saved configurations.

```
<Huawei>reset saved-configuration
```

11. Check system startup configuration parameters.

```
<Huawei> display startup
```

This command displays the system software for the current and next startup, backup system software, configuration file, license file, and patch file, as well as voice file.

12. Configure the configuration file for next startup.

```
<Huawei>startup saved-configuration configuration-file
```

During a device upgrade, you can run this command to configure the device to load the specified configuration file for the next startup.

13. Restart a device.

```
<Huawei>reboot
```



Contents

1. VRP Basics

2. Command Line Basics

- Command Views and Use of Command Views
- Basic Configuration Commands
- **Case Analysis**



Case 1: File Query Commands and Directory Operations

Requirement description:

- Check information about files and directories in the current directory of a router named RTA.
- Create a directory named **test**, and then delete the directory.



RTA

```
<Huawei>pwd
```

```
flash:
```

```
<Huawei>dir
```

```
Directory of flash:/
```

Idx	Attr	Size(Byte)	Date	Time(LMT)	FileName
0	drw-	-	Dec 27 2019	02:54:09	dhcp
1	-rw-	121,802	May 26 2014	09:20:58	portalpage.zip
2	-rw-	2,263	Dec 27 2019	02:53:59	statemach.efs
3	-rw-	828,482	May 26 2014	09:20:58	sslvpn.zip

```
1,090,732 KB total (784,464 KB free)
```

```
<Huawei>mkdir test
```

```
<Huawei>dir
```

```
Directory of flash:/
```

Idx	Attr	Size(Byte)	Date	Time(LMT)	FileName
0	drw-	-	Dec 27 2019	02:54:39	test
1	drw-	-	Dec 27 2019	02:54:09	dhcp
2	-rw-	121,802	May 26 2014	09:20:58	portalpage.zip
3	-rw-	2,263	Dec 27 2019	02:53:59	statemach.efs
4	-rw-	828,482	May 26 2014	09:20:58	sslvpn.zip

```
1,090,732 KB total (784,460 KB free)
```

```
<Huawei>rmdir test
```



Case 2: File Operations (1)

Requirement description:

- Rename the **huawei.txt** file **save.zip**.
- Make a copy for the **save.zip** file and name the copy **file.txt**.
- Move the **file.txt** file to the **dhcp** directory.
- Delete the **file.txt** file.
- Restore the deleted file **file.txt**.



RTA

```
<Huawei>rename huawei.txt save.zip
```

```
<Huawei>dir
```

Directory of flash:/

Idx	Attr	Size(Byte)	Date	Time(LMT)	FileName
0	drw-	-	Mar 04 2020	04:39:52	dhcp
1	-rw-	121,802	May 26 2014	09:20:58	portalpage.zip
2	-rw-	828,482	Mar 04 2020	04:51:45	save.zip
3	-rw-	2,263	Mar 04 2020	04:39:45	statemach.efs
4	-rw-	828,482	May 26 2014	09:20:58	sslvpn.zip

1,090,732 KB total (784,464 KB free)

```
<Huawei>copy save.zip file.txt
```

```
<Huawei>dir
```

Directory of flash:/

Idx	Attr	Size(Byte)	Date	Time(LMT)	FileName
0	drw-	-	Mar 04 2020	04:39:52	dhcp
1	-rw-	121,802	May 26 2014	09:20:58	portalpage.zip
2	-rw-	828,482	Mar 04 2020	04:51:45	save.zip
3	-rw-	2,263	Mar 04 2020	04:39:45	statemach.efs
4	-rw-	828,482	May 26 2014	09:20:58	sslvpn.zip
5	-rw-	828,482	Mar 04 2020	04:56:05	file.txt

1,090,732 KB total (784,340 KB free)



Case 2: File Operations (2)

Requirement description:

- Rename the **huawei.txt** file **save.zip**.
- Copy the **save.zip** file to the **file.txt** file.
- Move the **file.txt** file to the **dhcp** directory.
- Delete the **file.txt** file.
- Restore the deleted file **file.txt**.



RTA

```
<Huawei>move file.txt flash:/dhcp/
```

```
<Huawei>cd dhcp
```

```
<Huawei>dir
```

Directory of flash:/dhcp/

Idx	Attr	Size(Byte)	Date	Time(LMT)	FileName
0	-rw-	98	Dec 27 2019	02:54:09	dhcp-duid.txt
1	-rw-	121,802	Dec 27 2019	03:13:50	file.txt

1,090,732 KB total (784,344 KB free)

```
<Huawei>delete file.txt
```

```
<Huawei>dir
```

Directory of flash:/dhcp/

Idx	Attr	Size(Byte)	Date	Time(LMT)	FileName
0	-rw-	98	Dec 27 2019	02:54:09	dhcp-duid.txt

1,090,732 KB total (784,340 KB free)

```
<Huawei>undelete file.txt
```

```
<Huawei>dir
```

Directory of flash:/dhcp/

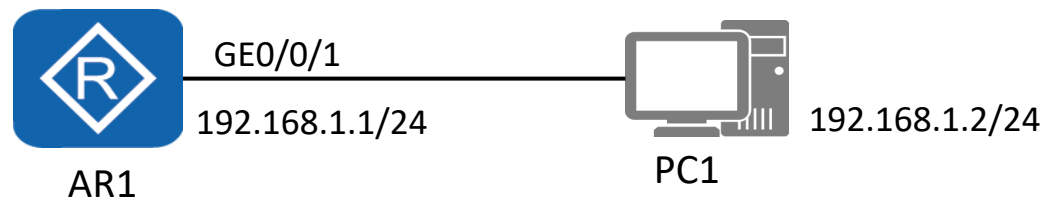
Idx	Attr	Size(Byte)	Date	Time(LMT)	FileName
0	-rw-	98	Dec 27 2019	02:54:09	dhcp-duid.txt
1	-rw-	121,802	Dec 27 2019	03:13:50	file.txt

1,090,732 KB total (784,340 KB free)



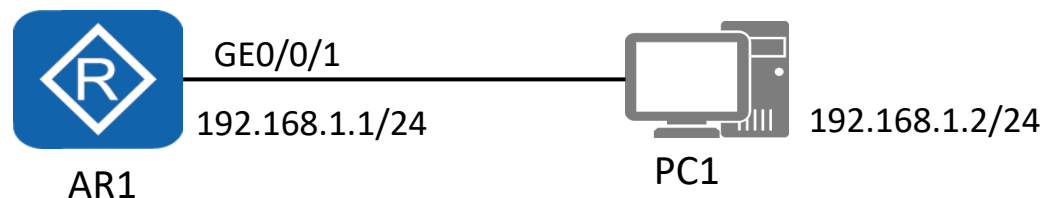
Case 3: VRP Basic Configuration Commands

- As shown in the figure, an engineer needs to configure a router. The requirements are as follows:
 - Connect the router and PC. Assign the IP addresses shown in the figure to the router and PC.
 - Allow other employees of the company to use the password **huawei123** to remotely log in to the router through the PC. Allow them to view configurations but disable them from modifying configurations.
 - Save current configurations and name the configuration file **huawei.zip**. Configure this file as the configuration file for the next startup.





Configuration Procedure (1)



Configure an interface IP address.

```
<Huawei>system-view
[Huawei]sysname AR1
[AR1]interface GigabitEthernet 0/0/1
[AR1-GigabitEthernet0/0/1]ip address 192.168.1.1 24
[AR1-GigabitEthernet0/0/1]quit
```

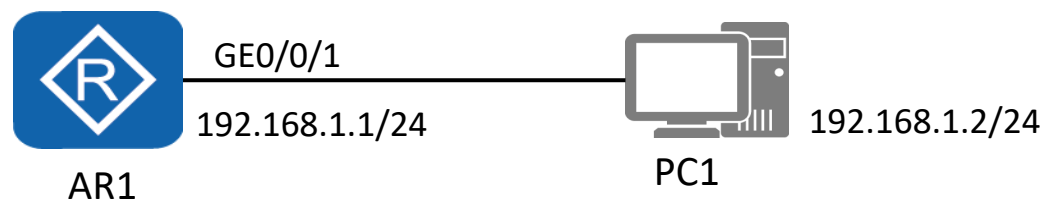
Configuring a user level and a user authentication mode.

```
[AR1]user-interface vty 0 4
[Huawei-ui-vty0-4]authentication-mode password
Please configure the login password (maximum length 16):huawei123
[AR1-ui-vty0-4]user privilege level 1
[AR1-ui-vty0-4]quit
```

The password configuration command may vary according to devices. For details, see the product documentation.



Configuration Procedure (2)



Specify the configuration file for next startup.

```
<HUAWEI>save huawei.zip
```

```
Are you sure to save the configuration to huawei.zip? (y/n)[n]:y
```

```
It will take several minutes to save configuration file, please wait.....
```

```
Configuration file had been saved successfully
```

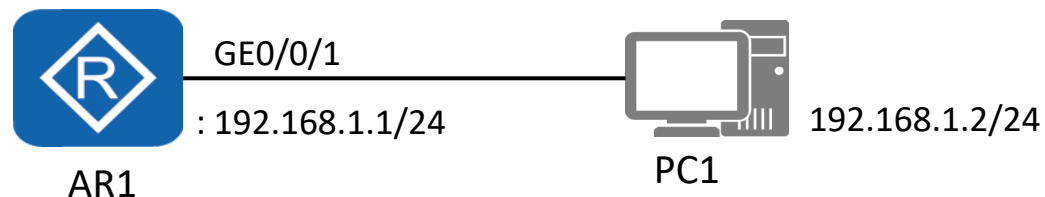
```
Note: The configuration file will take effect after being activated
```

```
<HUAWEI>startup saved-configuration huawei.zip
```

By default, configurations are saved in the **vrpcfg.cfg** file. You can also create a file for saving the configurations. VRPv5 and VRPv8 have the same command that is used to specify the configuration file for the next startup, but different directories for saving the file.



Checking Configurations



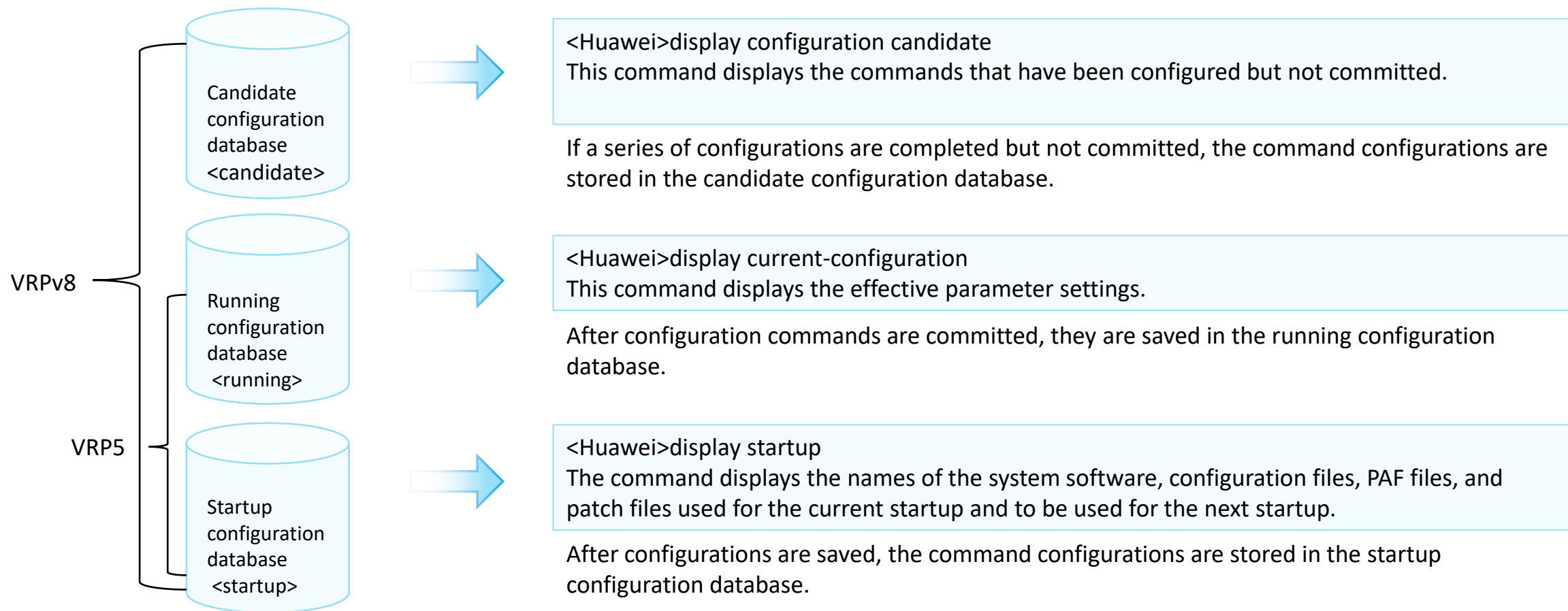
<AR1>**display startup**

MainBoard:

Startup system software:	flash:/vrpcfg.zip	null
Next startup system software:	flash:/vrpcfg.zip	null
Backup system software for next startup:	flash:/vrpcfg.zip	null
Startup saved-configuration file:	flash:/vrpcfg.zip	
Next startup saved-configuration file:	flash:/huawei.zip	
Startup license file:	flash:/vrpcfg.zip	
Next startup license file:	flash:/vrpcfg.zip	null
Startup patch package:	flash:/vrpcfg.zip	null
Next startup patch package:	flash:/vrpcfg.zip	null
Startup voice-files:	flash:/vrpcfg.zip	null
Next startup voice-files:	flash:/vrpcfg.zip	null



More Information



VRPv5 has the running and startup configuration databases but does not have the candidate configuration database. Therefore, a command configuration takes effect immediately after the command is executed, without being committed. However, in VRPv8, the configuration command takes effect only after the command committed.



Quiz

1. What is the VRP version currently used by Huawei datacom devices?
2. What is the maximum number of users that are allowed to log in to a Huawei device through the console port concurrently?
3. How do I specify the configuration file for next startup if a device has multiple configuration files?
 1. Currently, most Huawei datacom products use VRPv5, and a few products such as NE series routers use VRPv8.
 2. A Huawei device allows only one user to log in through the console interface at a time. Therefore, the console user ID is fixed at 0.
 3. To specify a configuration file for next startup, run the **startup saved-configuration** [*configuration-file*] command. The value of *configuration-file* should contain both the file name and extension.



Summary

- VRP is a Huawei proprietary network OS that can run on various hardware platforms. VRP has unified network, user, and management interfaces. To efficiently manage Huawei devices, you need to be familiar with VRP commands and configurations.
- You also need to understand some common commands and shortcut keys and learn how to use them.
- After learning this course, you need to know basic VRP concepts, functions of common commands, and CLI.

The background of the slide features a blue-tinted image of several business professionals in a modern office environment. They are standing on a highly reflective floor, and their silhouettes are clearly visible against the lighter background. The overall aesthetic is professional and corporate.

Thank You
www.huawei.com



Network Layer Protocols and IP Addressing



Foreword

- Internet Protocol Version 4 (IPv4) is the core protocol suite in the TCP/IP protocol suite. It works at the network layer in the TCP/IP protocol stack and this layer corresponds to the network layer in the Open System Interconnection Reference Model (OSI RM).
- The network layer provides connectionless data transmission services. A network does not need to establish a connection before sending data packets. Each IP data packet is sent separately.
- This presentation describes the basic concepts of IPv4 addresses, subnetting, network IP address planning, and basic IP address configuration.



Objectives

- On completion of this course, you will be able:
 - Describe main protocols at the network layer.
 - Describe the concepts and classification of IPv4 addresses and special IPv4 addresses.
 - Calculate IP networks and subnets.
 - Use the IP network address planning method.



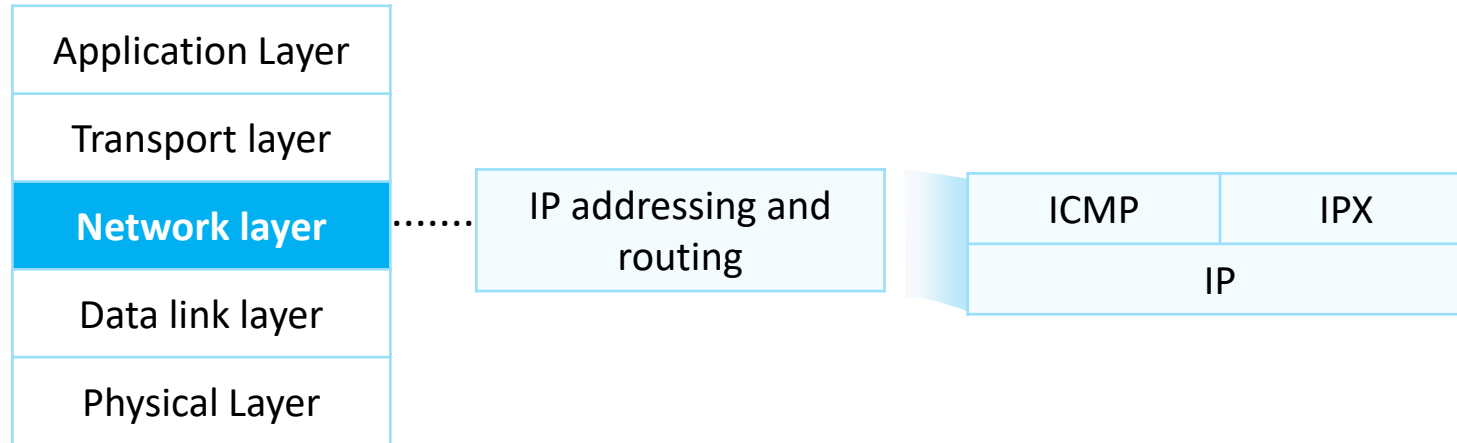
Contents

- 1. Network Layer Protocols**
2. Introduction to IPv4 Addresses
3. Subnetting
4. ICMP
5. IPv4 Address Configuration and Basic Application



Network Layer Protocols

- The network layer is often called the IP layer. Network layer protocols include Internet Control Message Protocol (ICMP) and Internet Packet Exchange (IPX), in addition to IP.



Equivalent TCP/IP model



Internet Protocol

- IP is short for the Internet Protocol. IP is the name of a protocol file with small content. It is used to identify a node on a network. It defines and describes the format of IP packets.
- The frequently mentioned IP refers to any content related directly or indirectly to the Internet Protocol, instead of the Internet Protocol itself.

Function

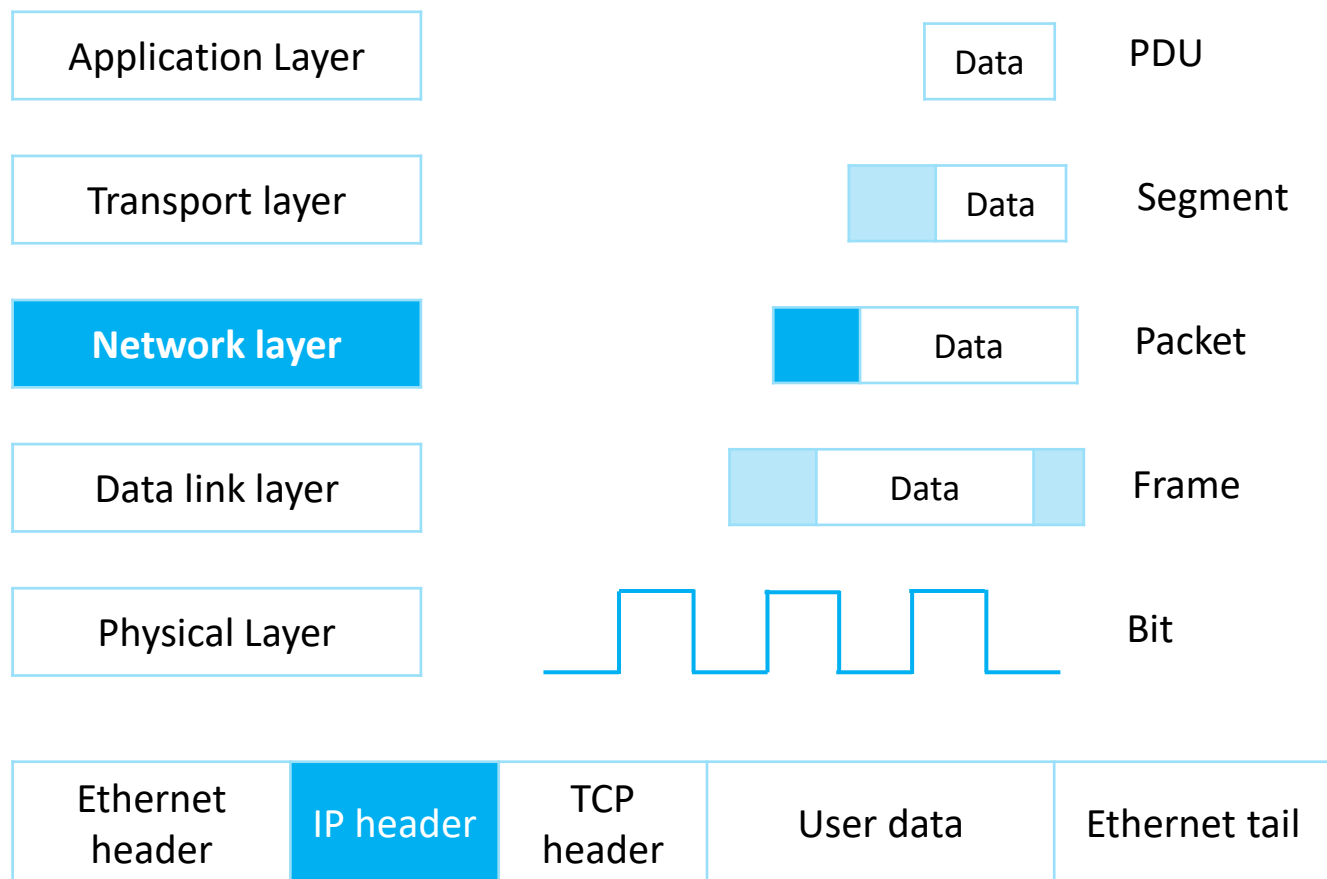
- Provides logical addresses for devices at the network layer.
- Is responsible for addressing and forwarding data packets.

Version

- IP Version 4 (IPv4)
- IP Version 6 (IPv6)

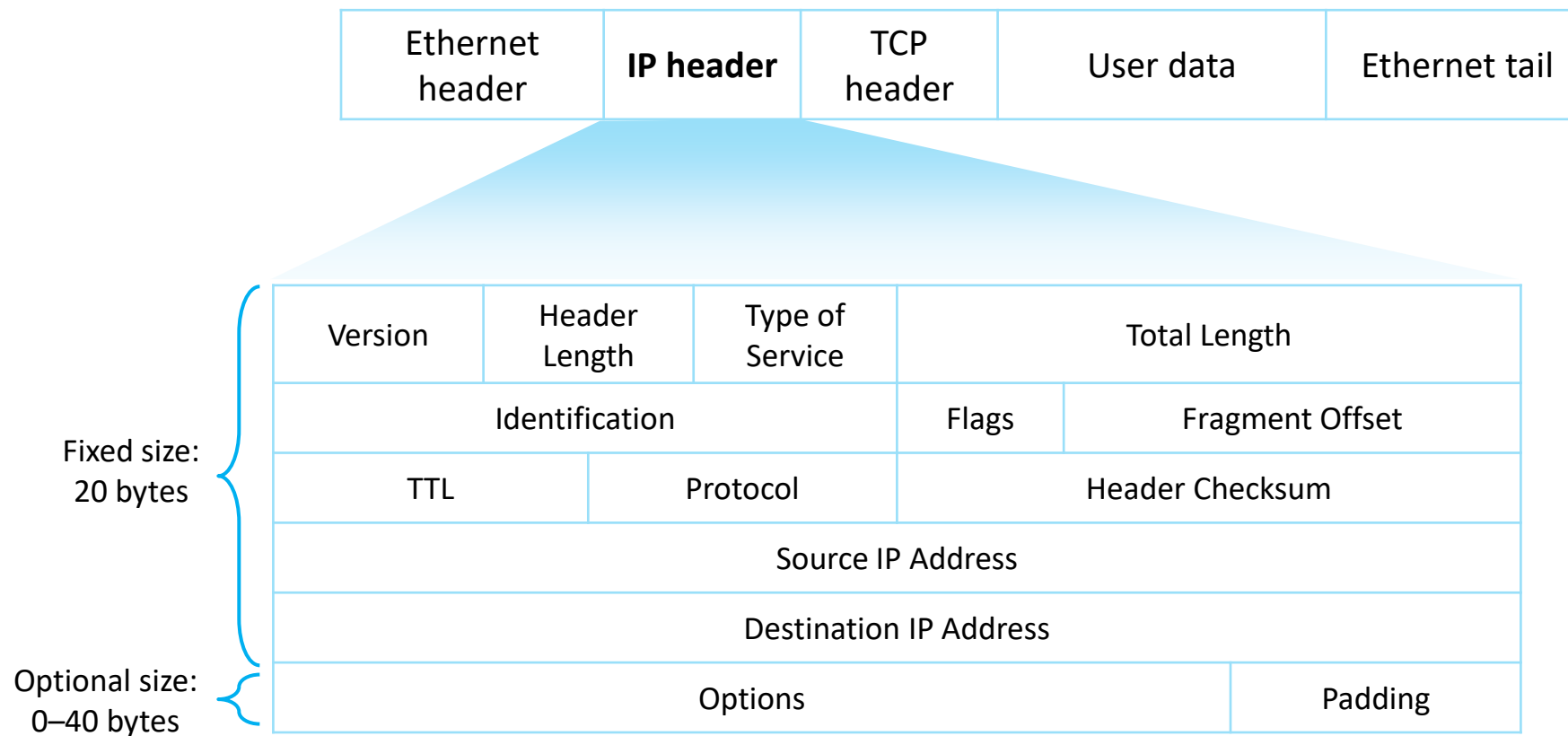


Data Encapsulation





IPv4 Packet Format

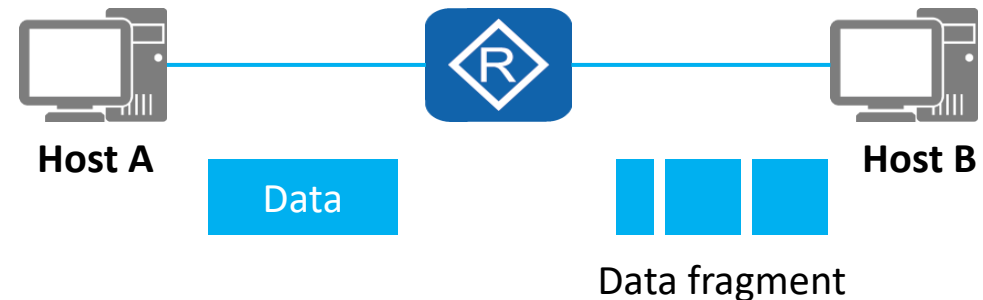




Data Packet Fragmentation

- The process of dividing a packet into multiple fragments is called fragmentation.
- The sizes of IP packets forwarded on a network may be different. If the size of an IP packet exceeds the maximum size supported by a data link, the packet needs to be divided into several smaller fragments before being transmitted on the link.

Version	Header Length	Type of Service	Total Length	
Identification			Flags	Fragment Offset
TTL	Protocol		Header Checksum	
Source IP Address				
Destination IP Address				
Options				Padding

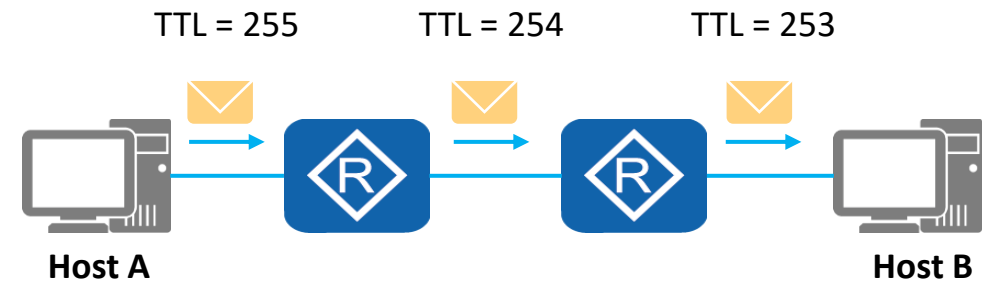




Time to Live

- The TTL field specifies the number of routers that a packet can pass through.
- Once a packet passes through a router, the TTL is reduced by 1. If the TTL value is reduced to 0, a data packet is discarded.

Version	Header Length	Type of Service	Total Length	
Identification			Flags	Fragment Offset
TTL	Protocol		Header Checksum	
Source IP Address				
Destination IP Address				
Options				Padding

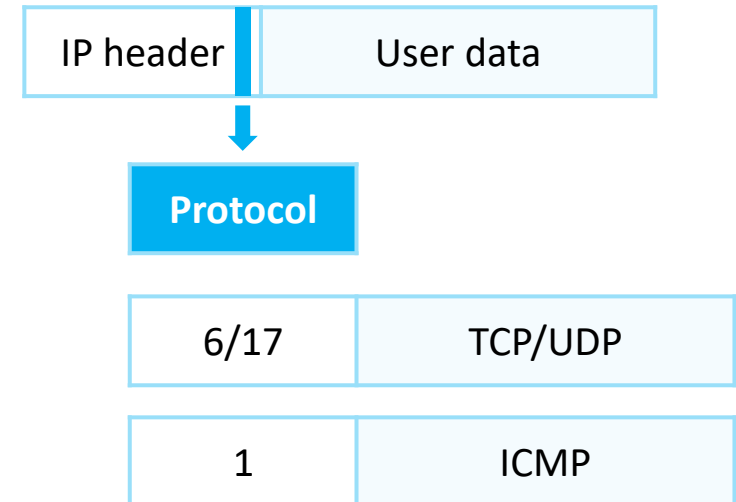




Protocol

- The Protocol field in the IP packet header identifies a protocol that will continue to process the packet.
- This field identifies the protocol used by the data carried in the data packet so that the IP layer of the destination host is also mapped to that Protocol field.

Version	Header Length	Type of Service	Total Length	
Identification			Flags	Fragment Offset
TTL	Protocol		Header Checksum	
Source IP Address				
Destination IP Address				
Options				Padding





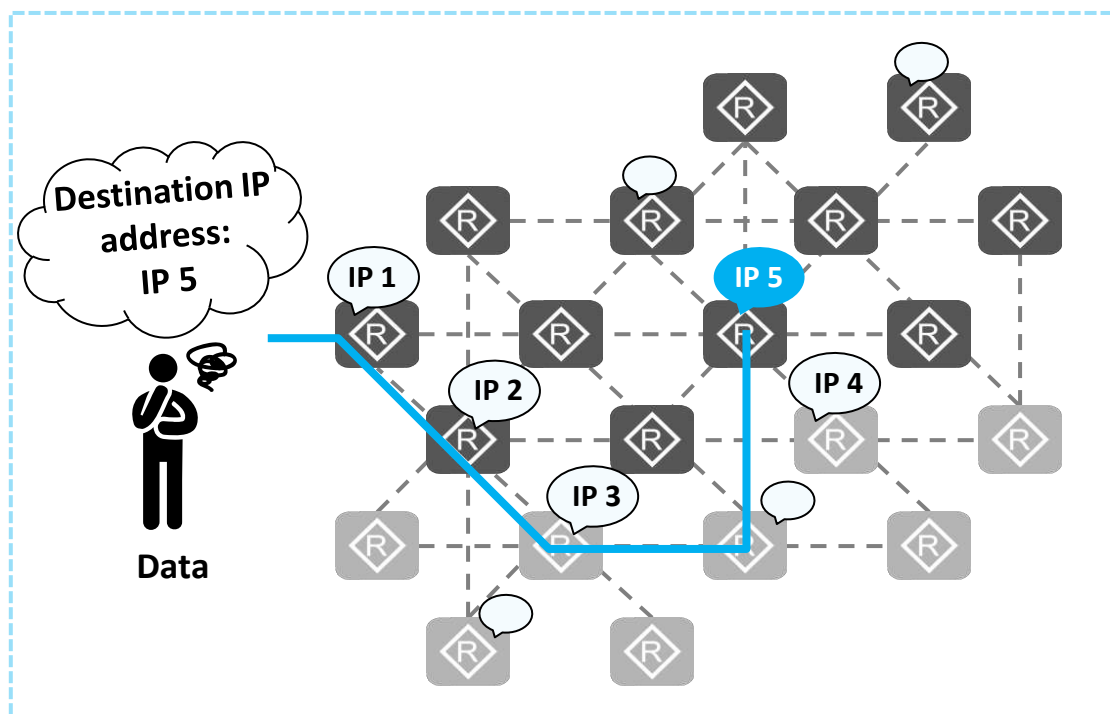
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1. Network Layer Protocols
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What Is an IP Address?

- An IP address identifies a node (or an interface on a network device) on a network.
- IP addresses are used to forward IP packets on the network.



IP Address

An IP address identifies a node on a network and is used to find the destination for data.



IP address Notation

- An IPv4 address is 32 bits long.
- It is in dotted decimal notation.

Dotted decimal notation	Decimal	192.				168.				10.				1				4 bytes			
	Binary	11000000				10101000				00001010				00000001				32 bits			
Conversion between decimal and binary systems																					
	Power	2 ⁷		2 ⁶		2 ⁵		2 ⁴		2 ³		2 ²		2 ¹		2 ⁰					
		128		64		32		16		8		4		2		1					
	Bit	1		1		0		0		0		0		0		0					
		= 128 + 64 = 192																			

- IPv4 address range is 0.0.0.0–255.255.255.255.

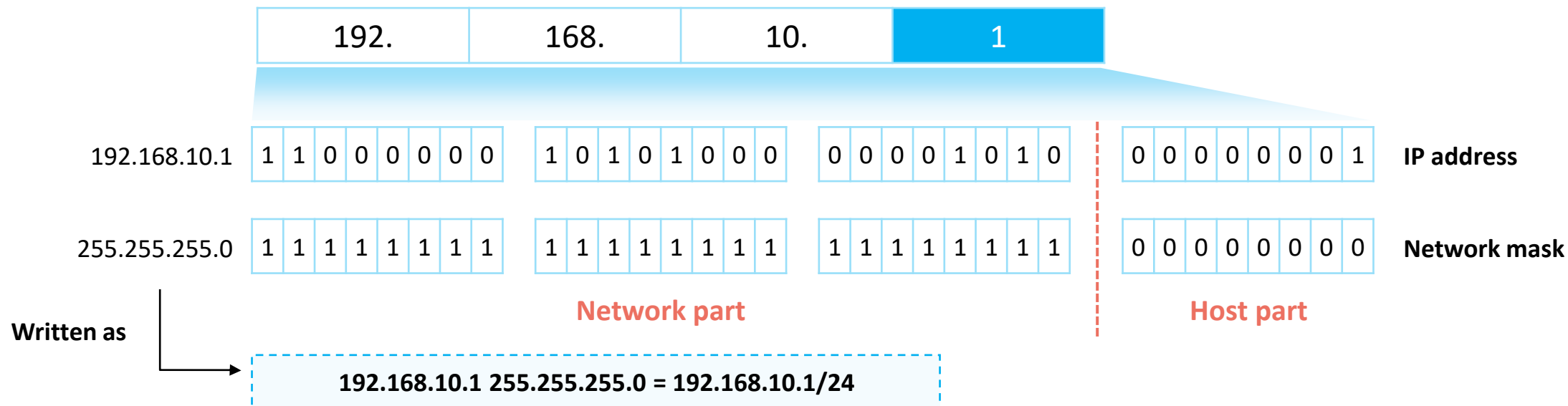


IP Address Structure

- **Network part:** identifies a network.
- **Host part:** identifies a host and is used to differentiate hosts on a network.



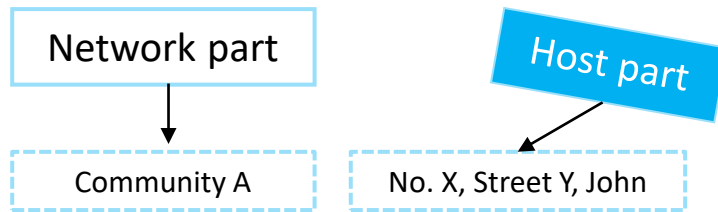
- **Network mask:** is used to distinguish the network part from the host part in an IP address.





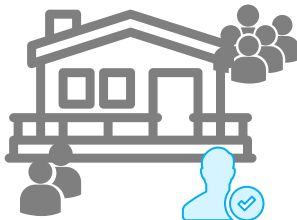
IP Addressing

- **Network part (network ID):** identifies a network.
- **Host part:** identifies a host and is used to differentiate hosts on a network.



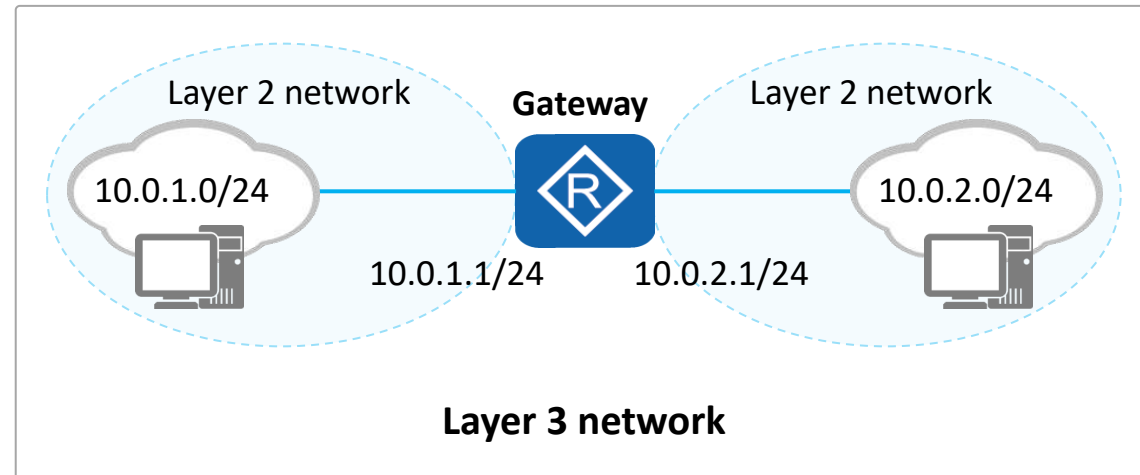
Layer 2 network addressing

Community A (network bits)



No. X, Street Y, John
(host bits)

Layer 3 network addressing



IP Address Classification (Classful Addressing)

- To facilitate IP address management and networking, IP addresses are classified into the following classes:

Class A	0NNNNNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	0.0.0.0–127.255.255.255	Assigned to hosts
Class B	10NNNNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	128.0.0.0–191.255.255.255	
Class C	110NNNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	192.0.0.0–223.255.255.255	
Class D	1110NNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	224.0.0.0–239.255.255.255	Used for multicast
Class E	1111NNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	240.0.0.0–255.255.255.255	Used for research

- Default subnet masks of classes A, B, and C
 - Class A: 8 bits, 0.0.0.0–127.255.255.255/8
 - Class B: 16 bits, 128.0.0.0–191.255.255.255/16
 - Class C: 24 bits, 192.0.0.0–223.255.255.255/24

Network part

Host part



IP Address Types

- A network range defined by a network ID is called a network segment.

- **Network address:** identifies a network.

Example: 192.168.10.0/24

192.	168.	10.	00000000
------	------	-----	----------

- **Broadcast address:** a special address used to send data to all hosts on a network.

Example: 192.168.10.255/24

192.	168.	10.	11111111
------	------	-----	----------

- **Available addresses:** IP addresses that can be allocated to device interfaces on a network.

Example: 192.168.10.1/24

192.	168.	10.	00000001
------	------	-----	----------

Note

- Network and broadcast addresses cannot be directly used by devices or their interfaces.
- Number of available addresses on a network segment is $2^n - 2$ (n is the number of bits in the host part).



IP Address Calculation

- Example: What are the network address, broadcast address, and number of available addresses of class B address 172.16.10.1/16?

	172.	16.	00001010.	00000001
IP address	1 0 1 0 1 1 0 0	0 0 0 1 0 0 0 0	0 0 0 0 1 0 1 0	0 0 0 0 0 0 0 1
Network mask	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0
Network address	1 0 1 0 1 1 0 0	0 0 0 1 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0
Broadcast address	1 0 1 0 1 1 0 0	0 0 0 1 0 0 0 0	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1

The network address is obtained, with all host bits set to 0s.
172.16.0.0/16

The broadcast address is obtained, with all host bits set to 1s.
172.16.255.255/16

Number of IP addresses $2^{16} = 65536$

Number of available addresses $2^{16} - 2 = 65534$

Range of available addresses 172.16.0.1/16–172.16.255.254/16

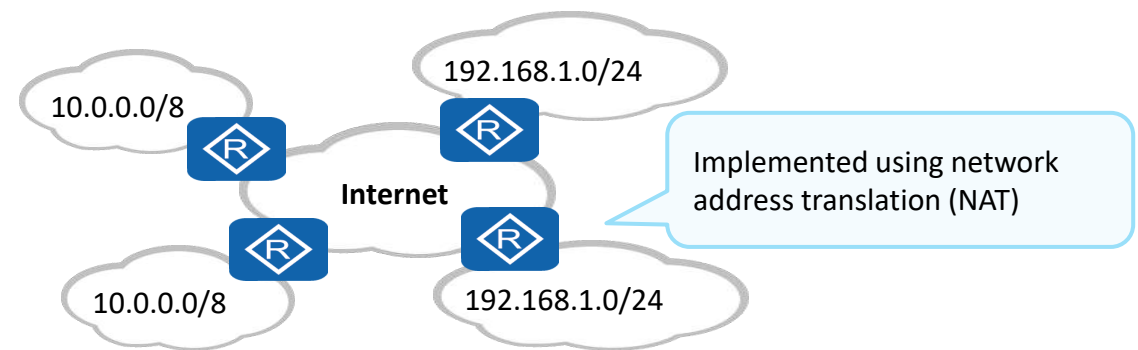
Quiz

Example: What are the network address, broadcast address, and number of available addresses of class A address 10.128.20.10/8?



Private IP Addresses

- **Public IP address:** An IP address is assigned by the Internet Assigned Numbers Authority (IANA), and this address allocation mode ensures that each IP address is unique on the Internet. Such an IP address is a public IP address.
- **Private IP address:** In practice, some networks do not need to connect to the Internet. For example, on a network of a lab in a college, IP addresses of devices need to avoid conflicting with each other only within the same network. In the IP address space, some IP addresses of class A, B, and C addresses are reserved for the preceding situations. These IP addresses are called private IP addresses.
 - Class A: 10.0.0.0–10.255.255.255
 - Class B: 172.16.0.0–172.31.255.255
 - Class C: 192.168.0.0–192.168.255.255



Connecting a private network to the Internet



Special IP Addresses

- Some IP addresses in the IP address space are of special meanings and functions.
- For example:

Special IP Address	Address Scope	Function
Limited broadcast address	255.255.255.255	It can be used as a destination address and traffic destined for it is sent to all hosts on the network segment to which the address belongs. (Its usage is restricted by a gateway).
Any IP address	0.0.0.0	It is an address of any network. Addresses in this block refer to source hosts on "this" network.
Loopback address	127.0.0.0/8	It is used to test the software system of a test device.
Link-local address	169.254.0.0/24	If a host fails to automatically obtain an IP address, the host can use an IP address in this address block for temporary communication.



IPv4 vs. IPv6

- IPv4 addresses managed by the IANA were exhausted in 2011. As the last public IPv4 address was allocated and more and more users and devices access the public network, IPv4 addresses were exhausted. This is the biggest driving force for IPv6 to replace IPv4.

IPv4

- Address length: 32 bits
- Address types: unicast address, broadcast address, and multicast address
- Characteristics:
 - IPv4 address depletion
 - Inappropriate packet header design
 - ARP dependency-induced flooding
 - ...

IPv6

- Address length: 128 bits
- Address types: unicast address, multicast address, and anycast address
- Characteristics:
 - Unlimited number of addresses
 - Simplified packet header
 - Automatic IPv6 address allocation
 - ...

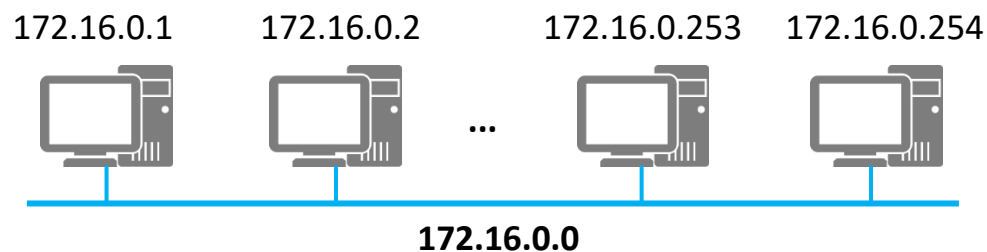


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5. IPv4 Address Configuration and Basic Application

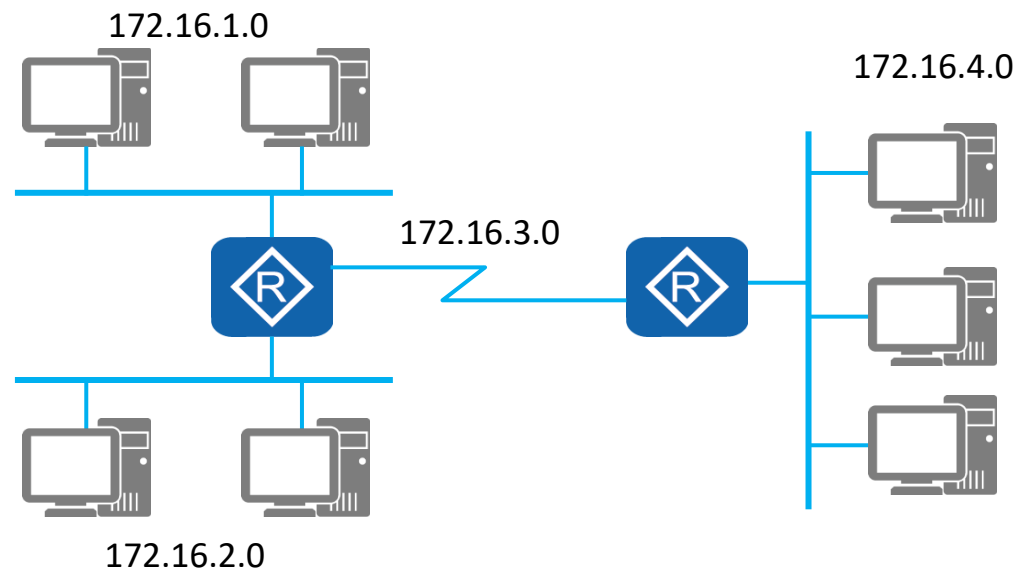


Why Subnetting?



$2^{16} = 65536$ IP addresses

- A class B address is used for a broadcast domain, wasting addresses.
- The broadcast domain is too large. Once broadcast occurs, an internal network is overloaded.



- A network number is divided into multiple subnets, and each subnet is allocated to a separate broadcast domain.
- In this way, the broadcast domain is smaller, and the network planning is more reasonable.
- IP addresses are properly used.



Why Subnetting ?

Subnetting is done to divide a large network into smaller, manageable networks (subnets). It's a practical concept used within the **TCP/IP Model**. Here are the main reasons:

1. Efficient Use of IP Addresses

Without subnetting, a network might waste a lot of IP addresses.

Subnetting lets you **divide one large network into smaller ones**, so addresses are used more efficiently allocating only the required number of IPs to each subnet

2. Reduces Network Traffic (Less Congestion)

Each subnet is a separate broadcast domain which limits broadcast messages thus increasing performance.

3. Improves Security

Each subnet can act like a **separate mini-network**, so:

- You can **restrict access** between subnets
- Sensitive departments (e.g., finance, HR) can be isolated and controlled via routers.

4. Better Network Management

It becomes easier to troubleshoot problems, monitor network activity and organize devices based on logical grouping (e.g., HR, IT, Sales)

5. Supports Scalability

Subnetting allows a network to **grow in an organized way** without becoming messy or inefficient. It also enables efficient routing (used with longest prefix match)



Subnetting - Analyzing the Original Network Segment

- Example: 192.168.10.0/24

192.168.10.1											
IP address	192.	168.	10.	0	0	0	0	0	0	0	1
Default subnet mask	255.	255.	255.	0	0	0	0	0	0	0	0
192.168.10.255				...							
IP address	192.	168.	10.	1	1	1	1	1	1	1	1
Default subnet mask	255.	255.	255.	0	0	0	0	0	0	0	0
Network part				Host part							

One class C network:

192.168.10.0/24

Default subnet mask:

255.255.255.0

Network address: 192.168.10.0/24

Broadcast address: 192.168.10.255

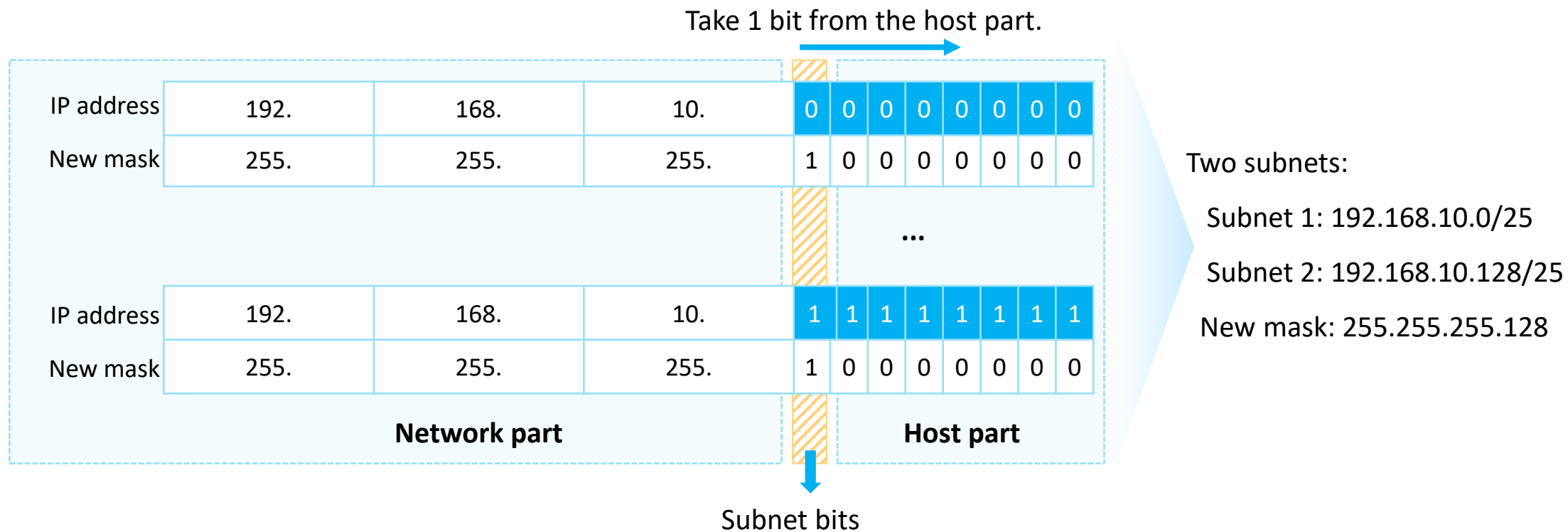
Total IP addresses: $2^8 = 256$

Available IP addresses: $2^8 - 2 = 254$



Subnetting - Taking Bits from the Host Part

- Bits can be taken from the host part to create subnets.

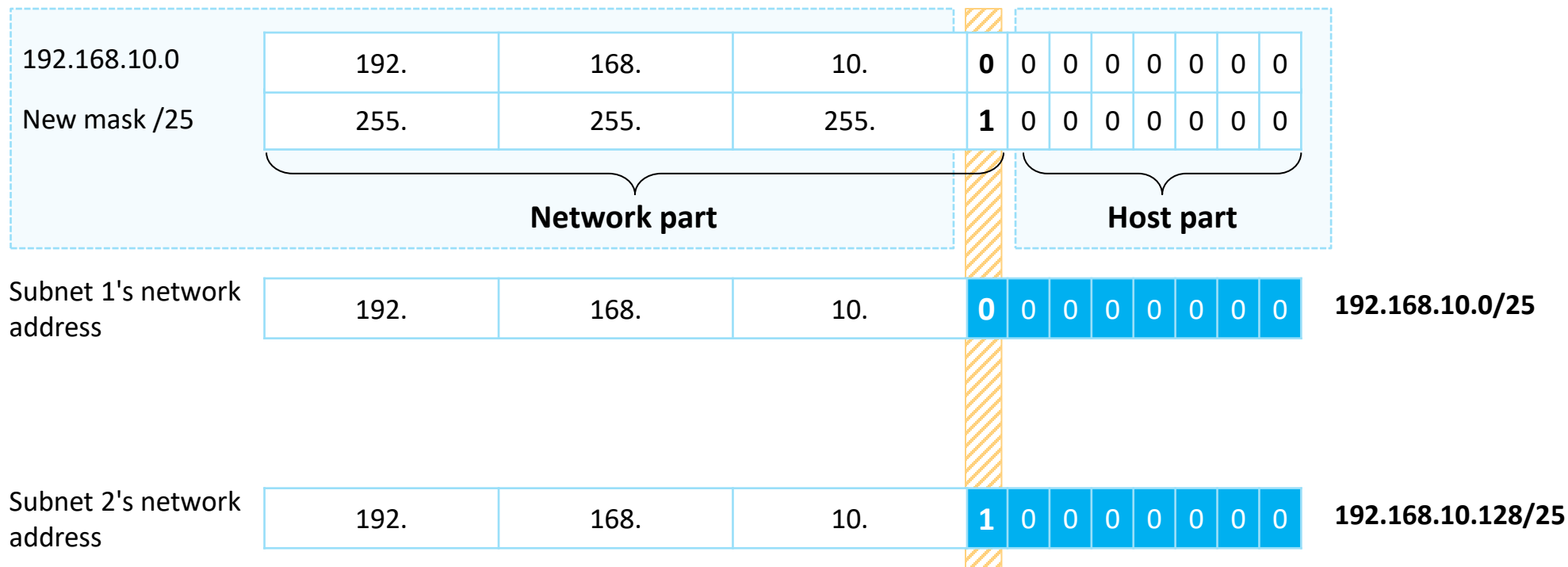


- Variable length subnet mask (VLSM)



Subnetting - Calculating the Subnet Network Address

- The network address is obtained, with all host bits set to 0s.





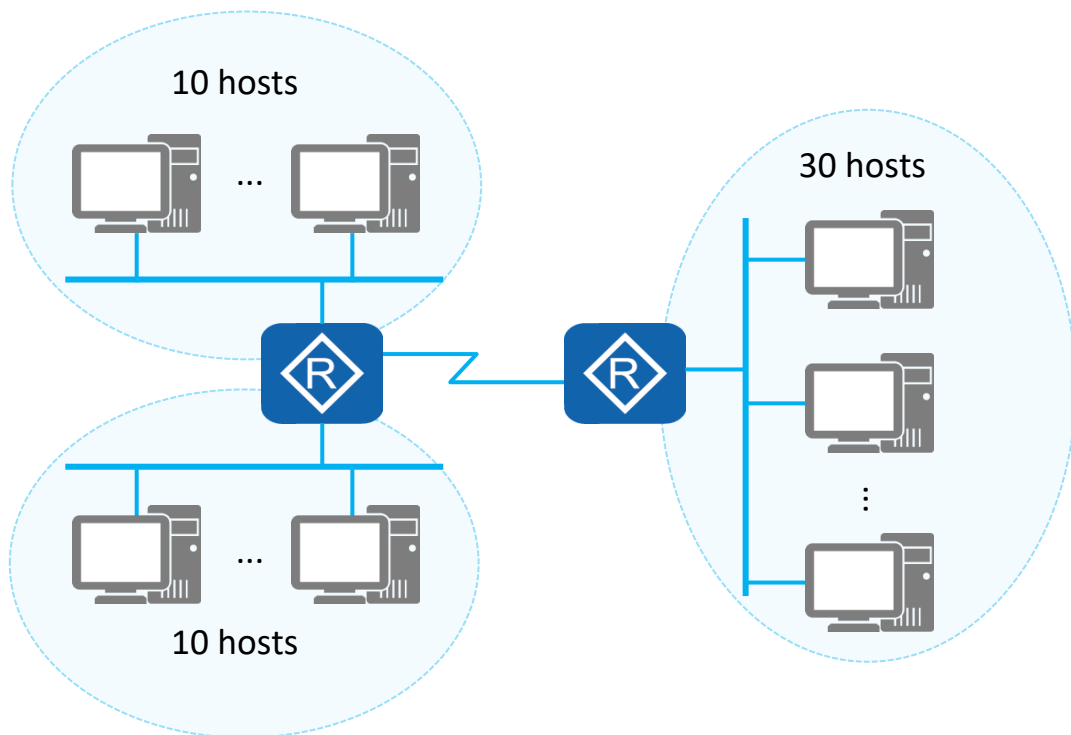
Subnetting - Calculating the Subnet Broadcast Address

- The broadcast address is obtained, with all host bits set to 1s.

192.168.10.0	192.	168.	10.	0	0	0	0	0	0	0	0
New mask /25	255.	255.	255.	1	0	0	0	0	0	0	0
	Network part				Host part						
Subnet 1's network address	192.	168.	10.	0	0	0	0	0	0	0	0
Subnet 1's broadcast address	192.	168.	10.	0	1	1	1	1	1	1	1
Subnet 2's network address	192.	168.	10.	1	0	0	0	0	0	0	0
Subnet 2's broadcast address	192.	168.	10.	1	1	1	1	1	1	1	1



Practice: Computing Subnets (1)



- Question:** An existing class C network segment is 192.168.1.0/24. Use the VLSM to allocate IP addresses to three subnets.

- Answer:** (Use a network with 10 hosts as an example.)

Step 1: Calculate the number of host bits to be taken.

$$2^n - 2 \geq 10$$

$$n \geq 4, \text{ host bits}$$

Step 2: Take bits from the host part.

Take 4 bits from the host part.

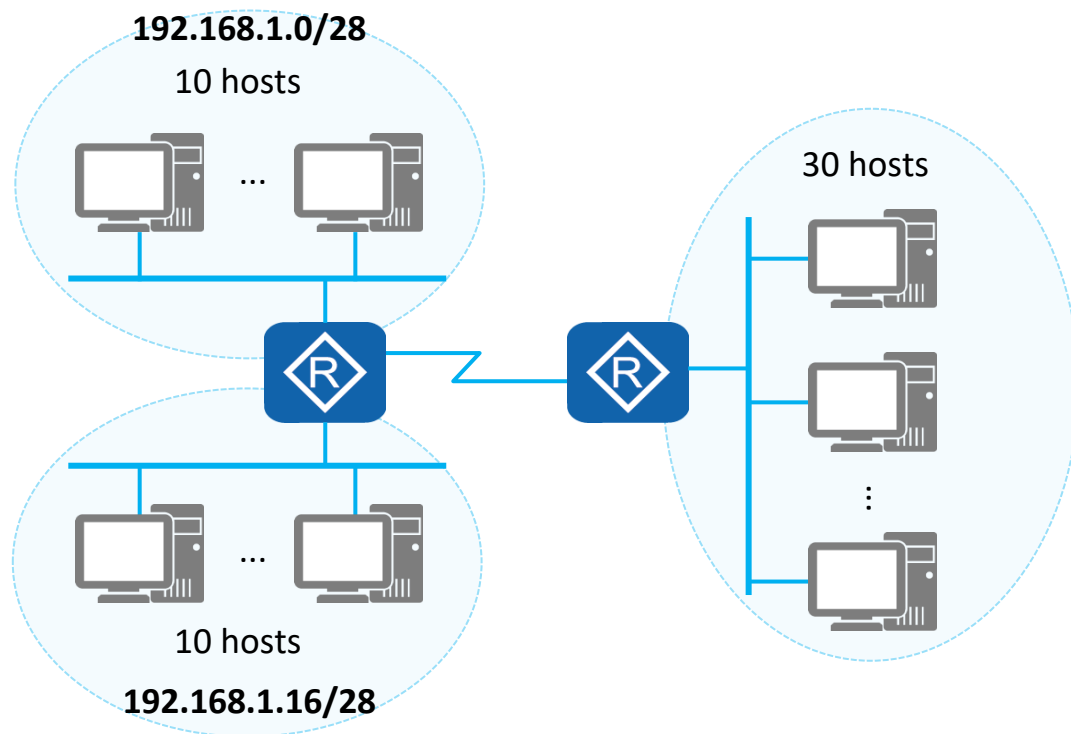
IP address	192.	168.	1.	0	0	0	0	0	0	0	0
Subnet mask	255.	255.	255.	1	1	1	1	0	0	0	0

Subnet bits

Number of subnets:
 $2^4 = 16$ subnets



Practice: Computing Subnets (2)



- Question:** An existing class C network segment is 192.168.1.0/24. Use the VLSM to allocate IP addresses to three subnets.

- Answer:** (Use a network with 10 hosts as an example.)

Step 3: Calculate subnet network addresses.

IP address	192.	168.	1.	0	0	0	0	0	0	0	0	0	
New mask	255.	255.	255.	1	1	1	1	0	0	0	0	0	
													Network address
Subnet 1	192.	168.	1.	0	0	0	0	0	0	0	0	0	192.168.1.0/28
Subnet 2	192	168.	1.	0	0	0	1	0	0	0	0	0	192.168.1.16/28
Subnet 3	192.	168.	1.	0	0	1	0	0	0	0	0	0	192.168.1.32/28
Subnet 16	192.	168.	1.	1	1	1	1	0	0	0	0	0	192.168.1.240/28



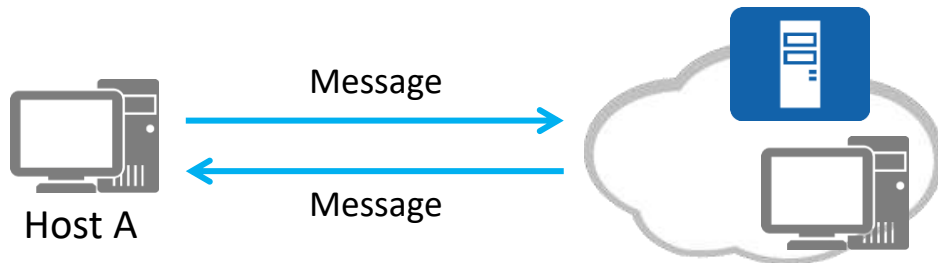
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5. IPv4 Address Configuration and Basic Application



ICMP

- The Internet Control Message Protocol (ICMP) is an auxiliary protocol of the IP protocol.
- ICMP is used to transmit error and control information between network devices. It plays an important role in collecting network information, diagnosing and rectifying network faults.



Ethernet header	IP header	ICMP message	Ethernet tail
-----------------	-----------	--------------	---------------

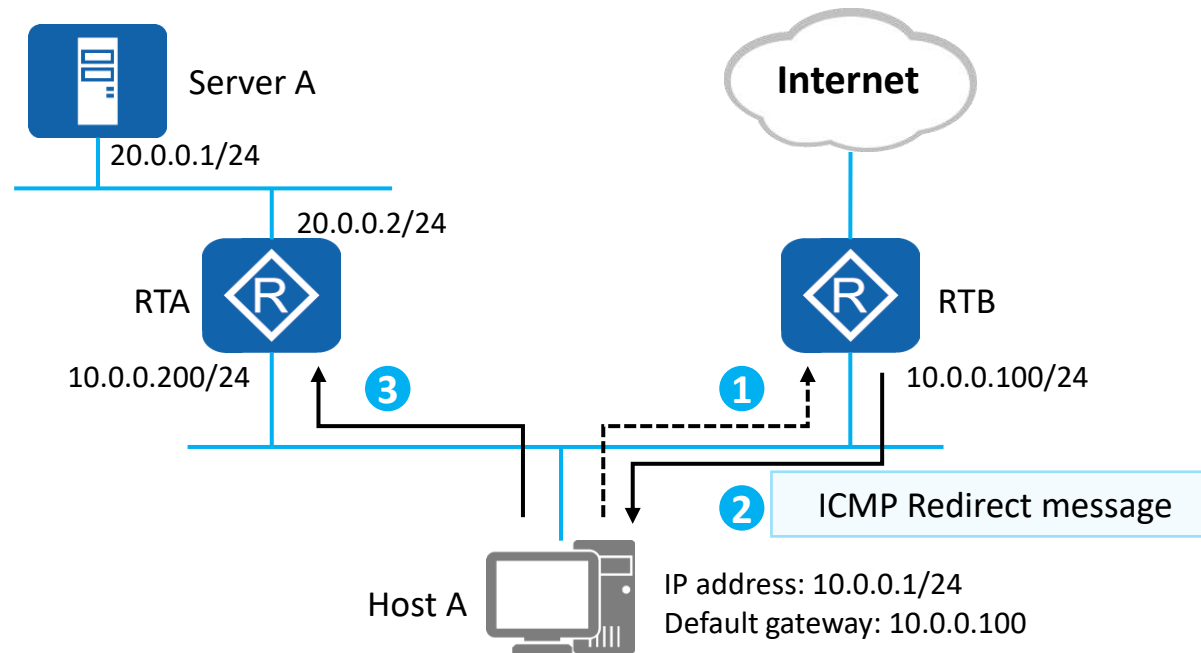
Type	Code	Checksum
ICMP message content		

Type	Code	Description
0	0	Echo Reply
3	0	Network Unreachable
3	1	Host Unreachable
3	2	Protocol Unreachable
3	3	Port Unreachable
5	0	Redirect
8	0	Echo Request



ICMP Redirection

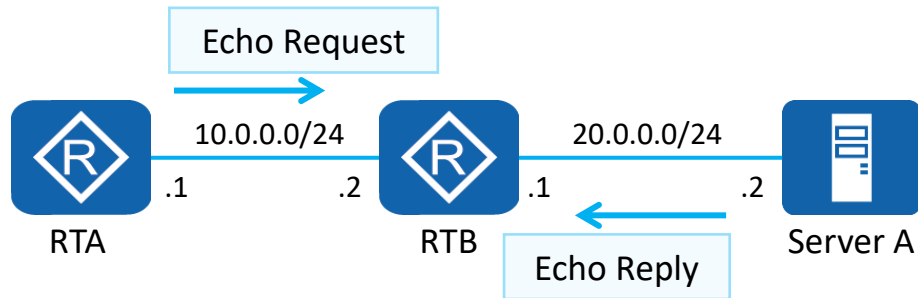
- ICMP Redirect messages are a type of ICMP control message. When a router detects that a host uses a non-optimal route in a specific scenario, the router sends an ICMP Redirect message requesting the host to change the route.





ICMP Error Detection

- ICMP Echo messages are used to check network connectivity between the source and destination and provide other information, such as the round-trip time.



Function: Ping

Ping is a command used on network devices, Windows OS, Unix OS, and Linux OS. Ping is a small and useful application based on the ICMP protocol.

A ping tests the reachability of a destination node.

```
[RTA]ping 20.0.0.2
```

```
PING 20.0.0.2: 56 data bytes, press CTRL_C to break
```

```
Reply from 20.0.0.2: bytes=56 Sequence=1 ttl=254 time=70 ms
```

```
Reply from 20.0.0.2: bytes=56 Sequence=2 ttl=254 time=30 ms
```

```
Reply from 20.0.0.2: bytes=56 Sequence=3 ttl=254 time=30 ms
```

```
Reply from 20.0.0.2: bytes=56 Sequence=4 ttl=254 time=40 ms
```

```
Reply from 20.0.0.2: bytes=56 Sequence=5 ttl=254 time=30 ms
```

```
--- 20.0.0.2 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

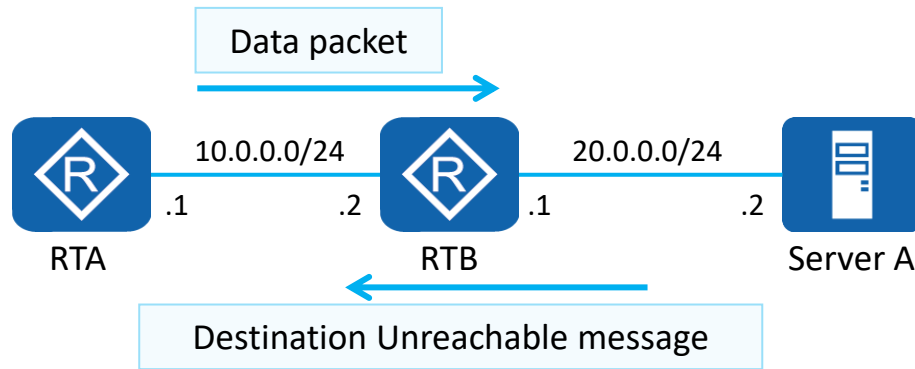
```
0.00% packet loss
```

```
round-trip min/avg/max = 30/40/70 ms
```




ICMP Error Report

- ICMP defines various error messages for diagnosing network connectivity problems. The source can determine the cause for a data transmission failure based on the received error messages. For example, after a network device receives a packet, it cannot access the network where the destination device resides, the network device automatically sends an ICMP Destination Unreachable message to the source.



Function: Tracert

Tracert checks the reachability of each hop on a forwarding path based on the TTL value carried in the packet header.

Tracert is an effective method to detect packet loss and delay on a network and helps administrators discover routing loops on the network.

```
[RTA]tracert 20.0.0.2
```

```
traceroute to 20.0.0.2(20.0.0.2), max hops: 30 ,packet length:
40,press CTRL_C
to break
```

1	10.0.0.2	80 ms	10 ms	10 ms
2	20.0.0.2	30 ms	30 ms	20 ms



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4. ICMP
5. **IPv4 Address Configuration and Basic Application**



Basic IP Address Configuration Commands

1. Enter the interface view.

```
[Huawei] interface interface-type interface-number
```

You can run this command to enter the view of a specified interface and configure attributes for the interface.

- *interface-type interface-number*: specifies the type and number of an interface. The interface type and number can be closely next to each other or separated by a space character.

2. Configure an IP address for the interface.

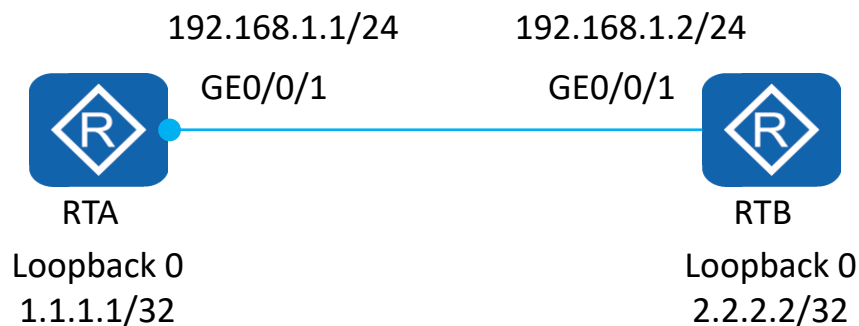
```
[Huawei-GigabitEthernet0/0/1] ip address ip-address { mask | mask-length }
```

You can run this command in the interface view to assign an IP address to the interface on the network devices to implement network interconnection.

- *ip-address*: specifies the IP address of an interface. The value is in dotted decimal notation.
- *mask*: specifies a subnet mask. The value is in dotted decimal notation.
- *mask-length*: specifies a mask length. The value is an integer ranging from 0 to 32.



Case: Configuring an IP address for an Interface



On the preceding network where the two routers are interconnected, configure IP addresses for the interconnected physical interfaces and logical IP addresses.

Configure an IP address for a physical interface.

```
[RTA] interface gigabitethernet 0/0/1
[RTA-GigabitEthernet0/0/1] ip address 192.168.1.1 255.255.255.0

Or,

[RTA-GigabitEthernet0/0/1] ip address 192.168.1.1 24
```

Configure an IP address for a logical interface.

```
[RTA] interface LoopBack 0
[RTA-LoopBack0] ip address 1.1.1.1 255.255.255.255

Or,

[RTA-LoopBack0] ip address 1.1.1.1 32
```



Network IP Address Planning

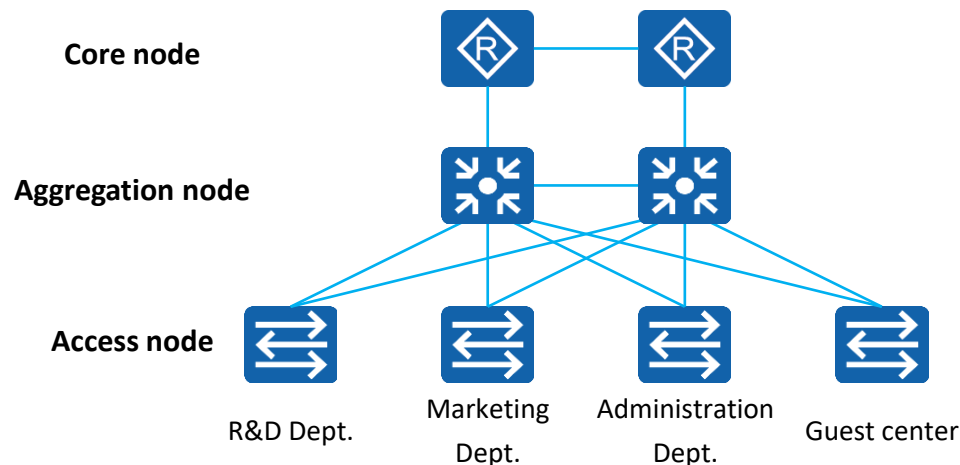
- IP address planning must be considered together with the network structure, routing protocols, traffic planning, and service rules. In addition, IP address planning should be corresponding to the network hierarchy and performed in a top-bottom way.
- In conclusion, IP address planning objectives are to achieve easy management, easy scalability, and high utilization.

- IP Address Planning Example**

Background	Address Type	Address Scope
Example: A company is assigned 192.168.0.0/16 as a network segment address.	Network segment of the R&D department	192.168.1.0/24
	Network segment of the marketing department	192.168.2.0/24
	Network segment of the administrative department	192.168.3.0/24
	Network segment of the guest center	192.168.4.0/24
	Others	...

Reference Planning Rules

Uniqueness, continuity, and scalability
Structured and service-related





Quiz

1. (Multiple) Which class does 201.222.5.64 belong? ()
 - A. Class A
 - B. Class B
 - C. Class C
 - D. Class D

2. (Multiple) A company is assigned a class C network segment 192.168.20.0/24. One of its departments has 40 hosts. Which of the following subnets can be allocated? ()
 - A. 192.168.20.64/26
 - B. 192.168.20.64/27
 - C. 192.168.20.128/26
 - D. 192.168.20.190/26 → not a valid network address for /26 as inside a subnet



Summary

- To connect a PC to the Internet, apply an IP address from the Internet Service Provider (ISP).
- This presentation provides an overview of the IP protocol and describes concepts related to IPv4 addresses and subnetting.
- This presentation also describes the planning and basic configuration of IP addresses.

The image features a blue-tinted background showing silhouettes of several groups of business professionals in a modern office environment. They are standing on a reflective floor, and their reflections are visible below them. The background includes architectural elements like glass walls and windows, suggesting a high-rise building. Overlaid on this scene is the text "Thank You" and the website address "www.huawei.com".

Thank You

www.huawei.com



IP Routing Basics



Foreword

- There are typically multiple IP subnets on a typical data communication network. Layer 3 devices are required to exchange data between these IP subnets. These devices have the routing capability and can forward data across subnets.
- Routing is the basic element of data communication networks. It is the process of selecting paths on a network along which packets are sent from a source to a destination.
- This course introduces the basic concepts of routing.



Objectives

- On completion of this course, you will be able to:
 - Understand the basic principles of routers.
 - Know how routers select optimal routes.
 - Understand the contents of routing tables.
 - Master advanced routing features.



Contents

1. Overview of IP Routing

▪ Basic Concepts of Routing

- Generation of Routing Entries
- Optimal Route Selection
- Route-based Forwarding

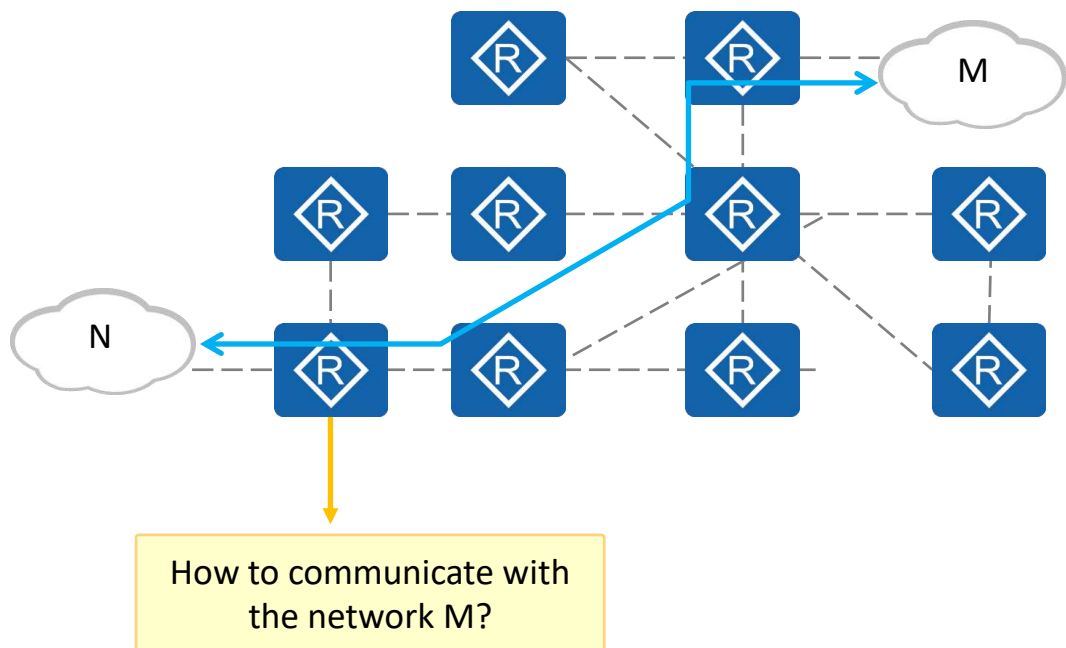
2. Static Routing

3. Dynamic Routing

4. Advanced Routing Features



Background: Inter-Subnet Communication

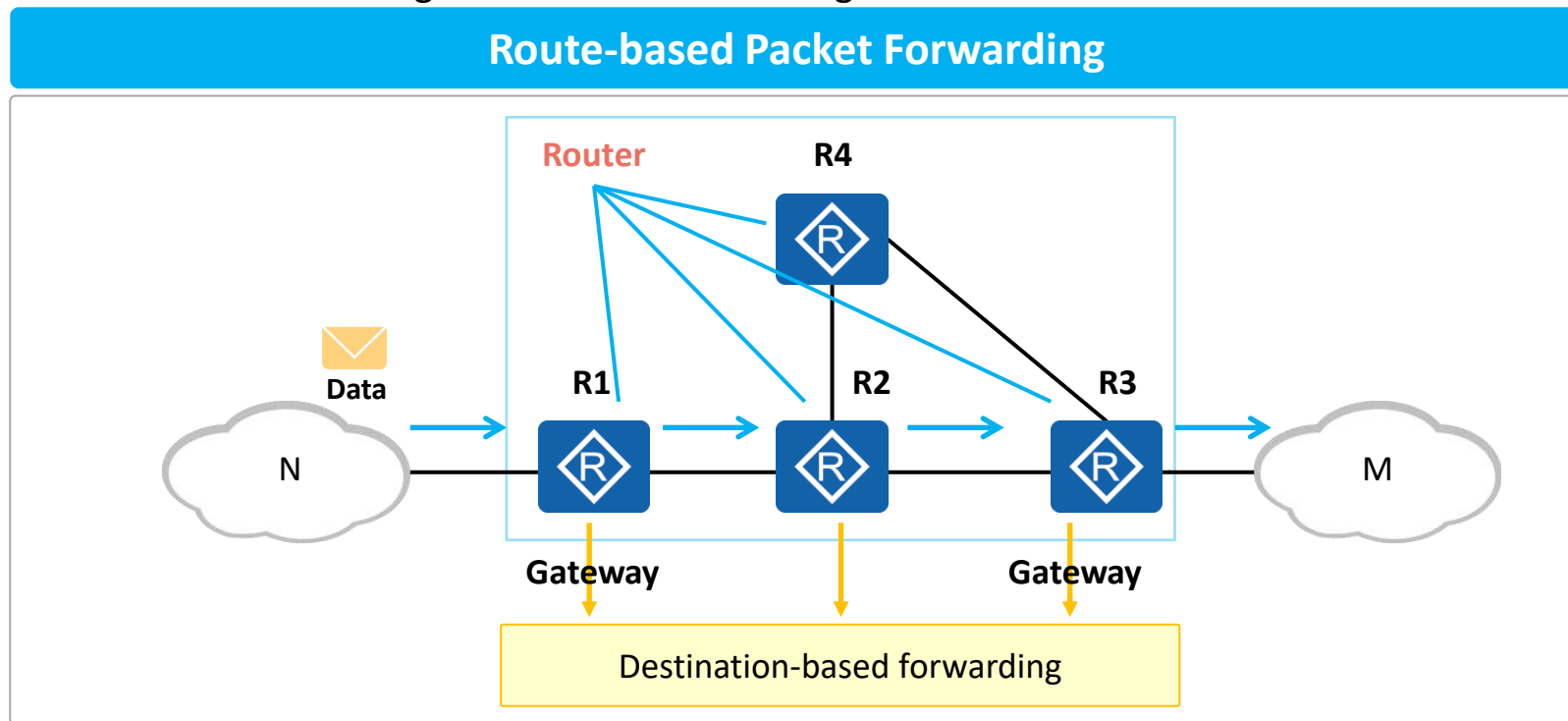


- An IP address uniquely identifies a node on a network. Each IP address belongs to a unique subnet, and each subnet may belong to a different area of the network.
- To implement IP addressing, subnets in different areas need to communicate with each other.



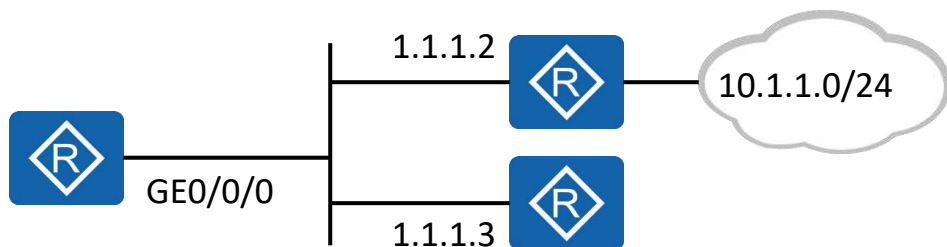
Routes

- Routes are the path information used to guide packet forwarding.
- A routing device is a network device that forwards packets to a destination subnet based on routes. The most common routing device is a router.
- A routing device maintains an IP routing table that stores routing information.





Routing Information



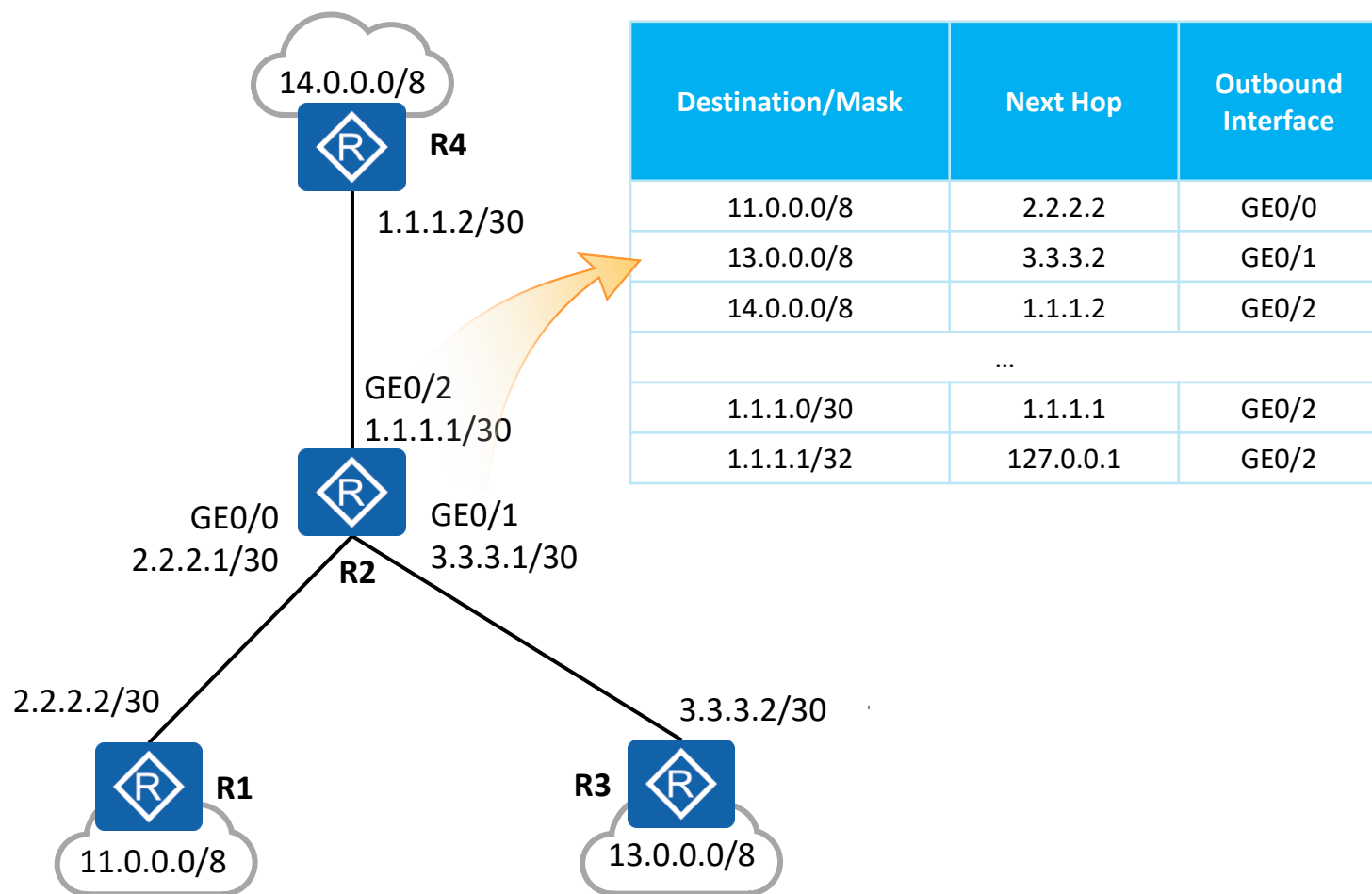
IP routing table

Destination/Mask	Outbound Interface	Next Hop
10.1.1.0/24	GEO/0/0	1.1.1.2

- A route contains the following information:
 - Destination: identifies a destination subnet.
 - Mask: identifies a subnet together with a destination IP address.
 - Outbound interface: indicates the interface through which a data packet is sent out of the local router.
 - Next hop: indicates the next-hop address used by the router to forward the data packet to the destination subnet.
- The information identifies the destination subnet and specifies the path for forwarding data packets.



IP Routing Table



- Routers discover routes using multiple methods.
- A router selects the optimal route and installs it in its IP routing table.
- The router forwards IP packets based on routes in the IP routing table.
- Routers manage path information by managing their IP routing tables.



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1. Overview of IP Routing

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- **Generation of Routing Entries**
- Optimal Route Selection
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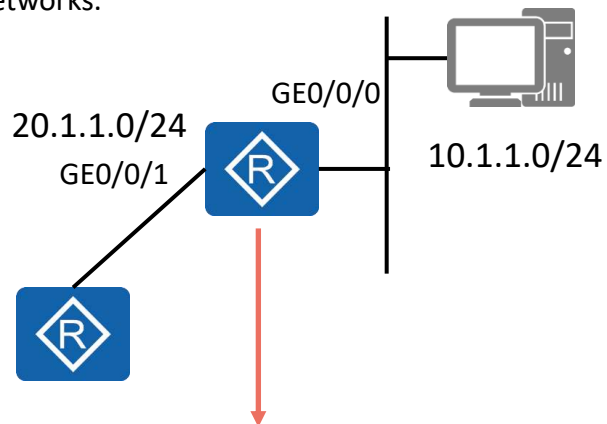


How to Obtain Routing Information

- A router forwards packets based on its IP routing table. To implement route-based packet forwarding, the router needs to obtain routes. The following describes the common methods of obtaining routes.

Direct Routes

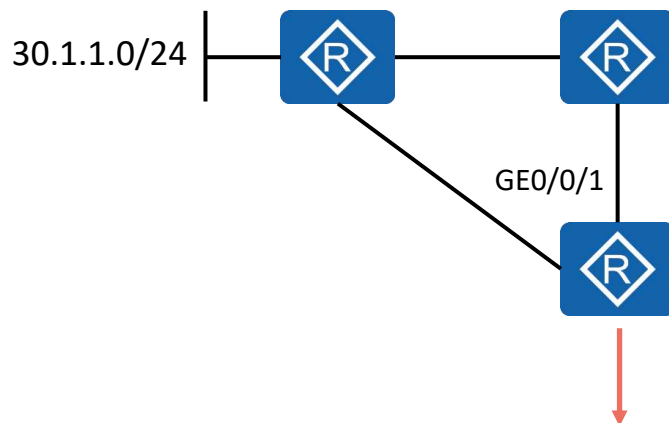
- Direct routes are automatically generated by devices and point to local directly connected networks.



Protocol	Destination/Mask	Outbound Interface
Direct	10.1.1.0/24	GE0/0/0
Direct	20.1.1.0/24	GE0/0/1

Static Routes

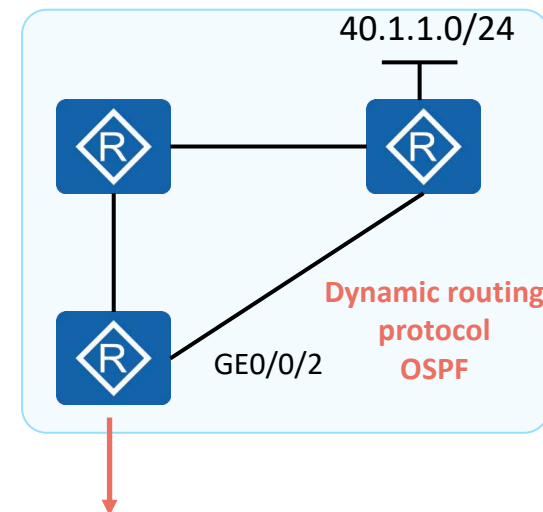
- Static routes are manually configured by network administrators.



Protocol	Destination/Mask	Outbound Interface
Static	30.1.1.0/24	GE0/0/1

Dynamic Routes

- Dynamic routes are learned by dynamic routing protocols running on routers.

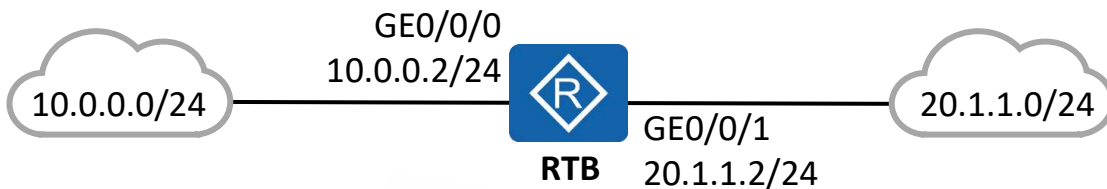


Protocol	Destination/Mask	Outbound Interface
OSPF	40.1.1.0/24	GE0/0/2



Direct Routes (1)

Direct Routes



Direct routes in the IP routing table of RTB

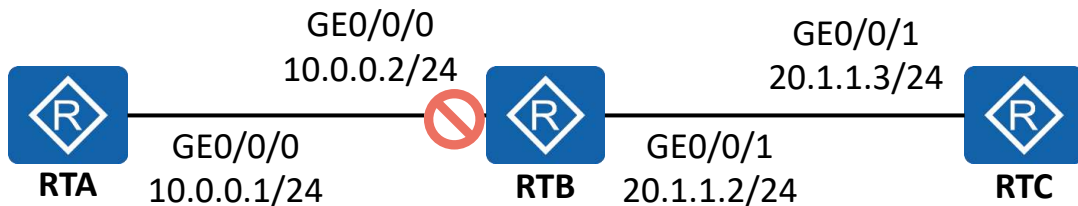
Destination/Mask	Protocol	Next Hop	Outbound Interface
10.0.0.0/24	Direct	10.0.0.2	GE0/0/0
20.1.1.0/24	Direct	20.1.1.2	GE0/0/1

- A direct route is automatically generated by a device and points to a local directly-connected network.
- When a router is the last hop router, IP packets to be forwarded will match a direct route and the router will directly forward the IP packet to the destination host.
- When a direct route is used for packet forwarding, the destination IP address of a packet to be forwarded and the IP address of the router's outbound interface are in the same subnet.



Direct Routes (2)

Direct routes



Direct routes in the IP routing table of RTB

Destination/Mask	Protocol	Next Hop	Outbound Interface
20.1.1.0/24	Direct	20.1.1.2	GE0/0/1

- When GE0/0/0 goes down, the direct route for this interface is not installed in the IP routing table.

- Not all the direct routes generated for interfaces are installed in the IP routing table. Only the direct routes of which the physical status and protocol status of interfaces are up are installed in the IP routing table.



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Examining the IP Routing Table

<Quidway> display ip routing-table

Route Flags: R - relay, D - download to fib

-----Routing Tables: Public

Destinations : 6

Routes : 6

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
1.1.1.1/32	Static	60	0	D	0.0.0.0	NULL0
2.2.2.2/32	Static	60	0	D	100.0.0.2	Vlanif100
100.0.0.0/24	Direct	0	0	D	100.0.0.1	Vlanif100
100.0.0.1/32	Direct	0	0	D	127.0.0.1	Vlanif100
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Destination/Mask

Protocol

Route
preference

Cost
(Metric)

Flag

Next-hop
address

Outbound
interface



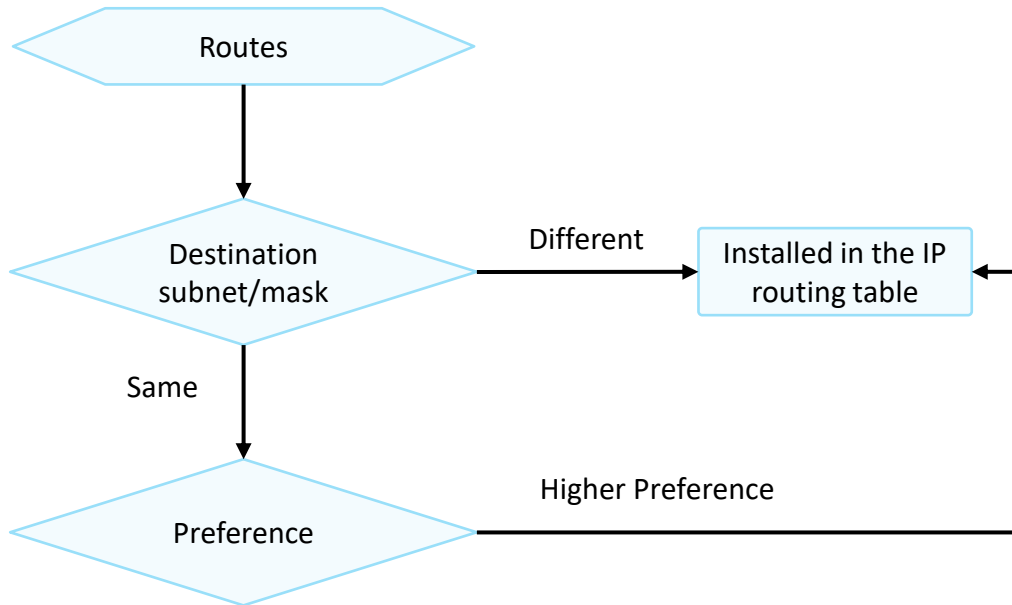
Fields in the IP Routing Table

- **Destination/Mask:** indicates the destination network address and mask of a specific route. The subnet address of a destination host or router is obtained through the AND operation on the destination address and mask. For example, if the destination address is 1.1.1.1 and the mask is 255.255.255.0, the IP address of the subnet to which the host or router belongs is 1.1.1.0.
- **Proto (Protocol):** indicates the protocol type of the route, that is, the protocol through which a router learns the route.
- **Pre (Preference):** indicates the routing protocol preference of the route. There may be multiple routes to the same destination, which have different next hops and outbound interfaces. These routes may be discovered by different routing protocols or be manually configured. A router selects the route with the highest preference (with the lowest preference value) as the optimal route.
- **Cost:** indicates the cost of the route. When multiple routes to the same destination have the same preference, the route with the lowest cost is selected as the optimal route.
- **NextHop:** indicates the local router's next-hop address of the route to the destination network. This field specifies the next-hop device to which packets are forwarded.
- **Interface:** indicates the outbound interface of the route. This field specifies the local interface through which the local router forwards packets.



Route Preference - Basic Concepts

Comparing Route Preferences

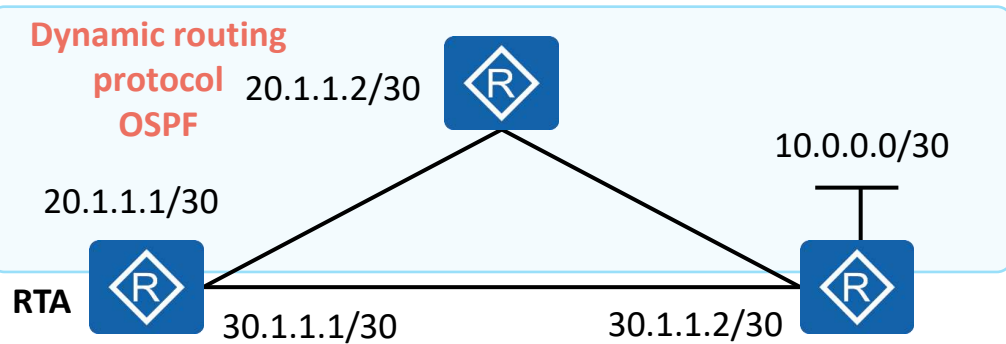


- When a router learns routes to the same destination subnet from different routing protocols, it compares their preference values (administrative distance).
- A lower preference value indicates a higher preference.
- The route with the highest preference is installed in the IP routing table.



Route Preference - Comparison Process

Comparing Route Preferences



- RTA discovers two routes to 10.0.0.0/30, one is an OSPF route and the other a static route. In this case, RTA compares the preferences of the two routes and selects the route with the lowest preference value.
- Each routing protocol has a unique preference.
- OSPF has a higher preference. Therefore, the OSPF route is installed in the IP routing table.

IP routing table of RTA

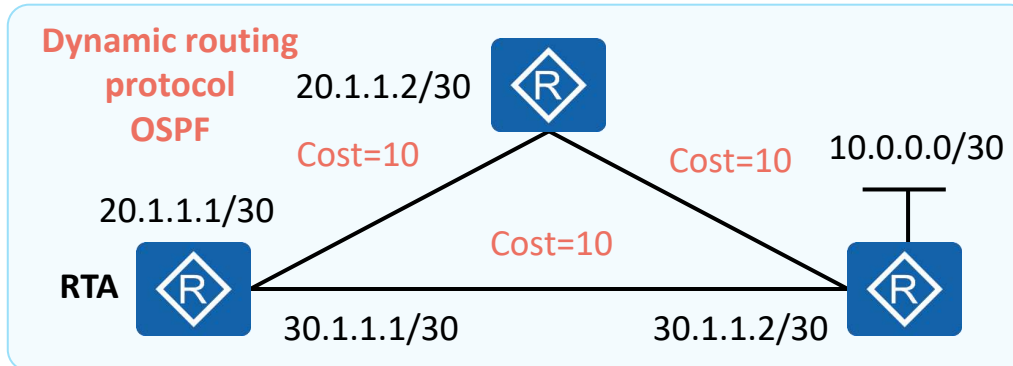
Destination/Mask	Protocol	Preference	Next Hop
10.0.0.0/30	Static	60	30.1.1.2
10.0.0.0/30	OSPF	10	20.1.1.2

Installed in the IP routing table



Metric - Comparison Process

Metric comparison



IP routing table of RTA

Destination/Mask	Protocol	Cost	Next Hop
10.0.0.0/30	OSPF	20	20.1.1.2
10.0.0.0/30	OSPF	10	30.1.1.2

Installed in
the IP
routing
table

- RTA learns two routes with the same destination address (10.0.0.0/30) and preference through OSPF. In this case, RTA needs to compare the metrics of the two routes.
- The two routes have different metrics. The OSPF route with the next hop being 30.1.1.2 has a lower metric (with the cost 10), so it is installed in the IP routing table.



Route Preference - Common Default Values

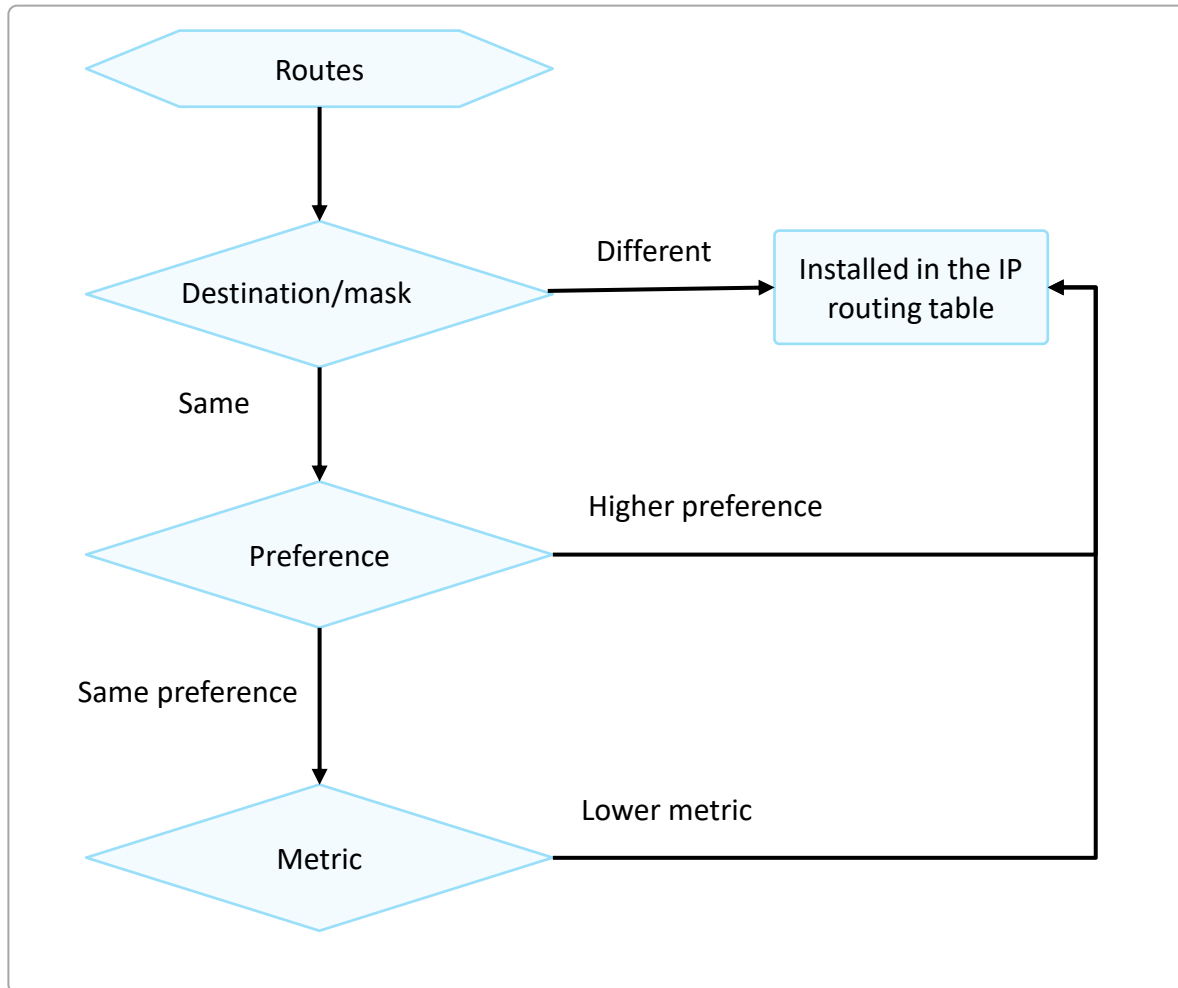
- The following table lists the default preference values of common route types:

Protocol	Route Type	Default Preference
Direct	Direct route	0
Static	Static route	60
Dynamic routing protocol	OSPF internal route	10
	OSPF external route	150



Metric - Basic Concepts

Comparing Metrics



- When a router learns multiple routes to the same destination through the same routing protocol, the preference values are usually the same. The router then compares the metric(costs) of the routes.
- The metric indicates the cost of reaching the destination address of the route.
- Common metrics include the hop count, bandwidth, delay, cost, load, and reliability.
- The route with the lowest metric is installed in the IP routing table.



Contents

1. Overview of IP Routing

- Basic Concepts of Routing
- Generation of Routing Entries
- Optimal Route Selection
- **Route-based Forwarding**

2. Static Routing

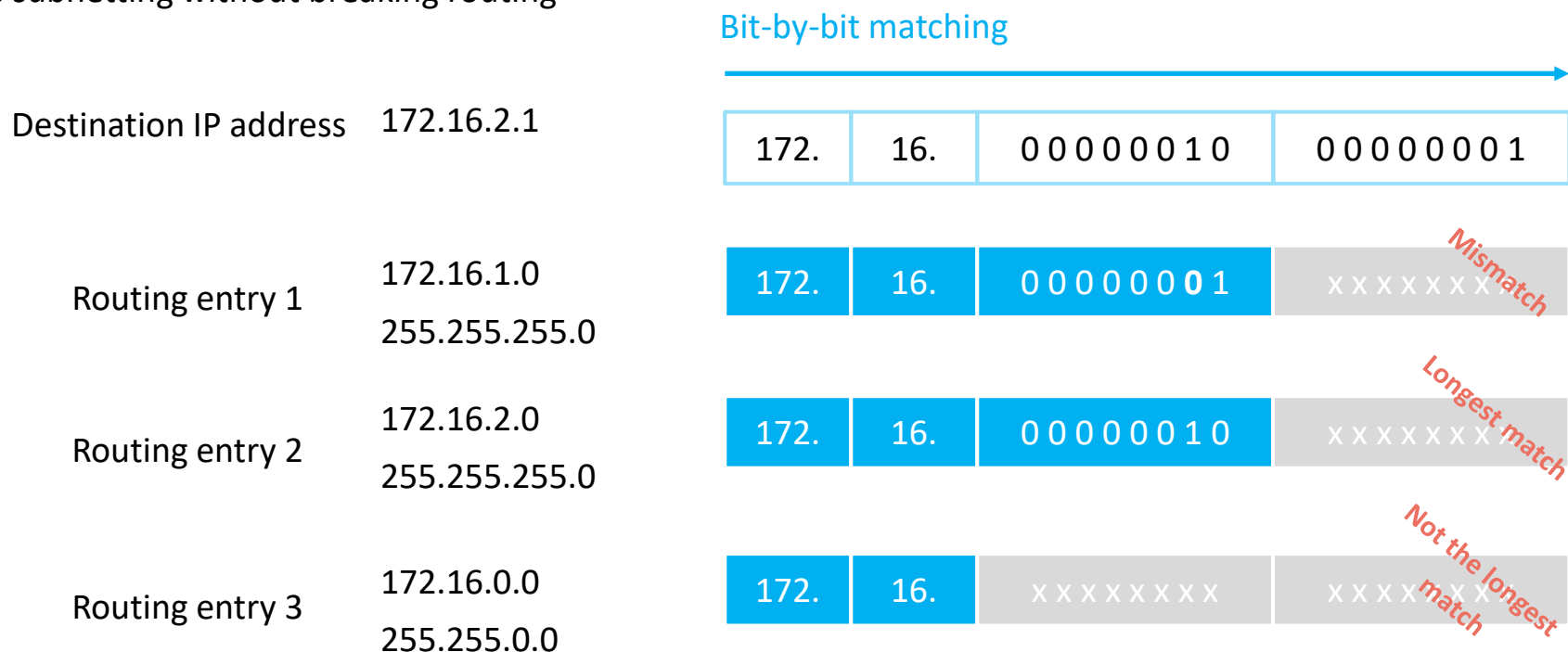
3. Dynamic Routing

4. Advanced Routing Features



Longest Matching

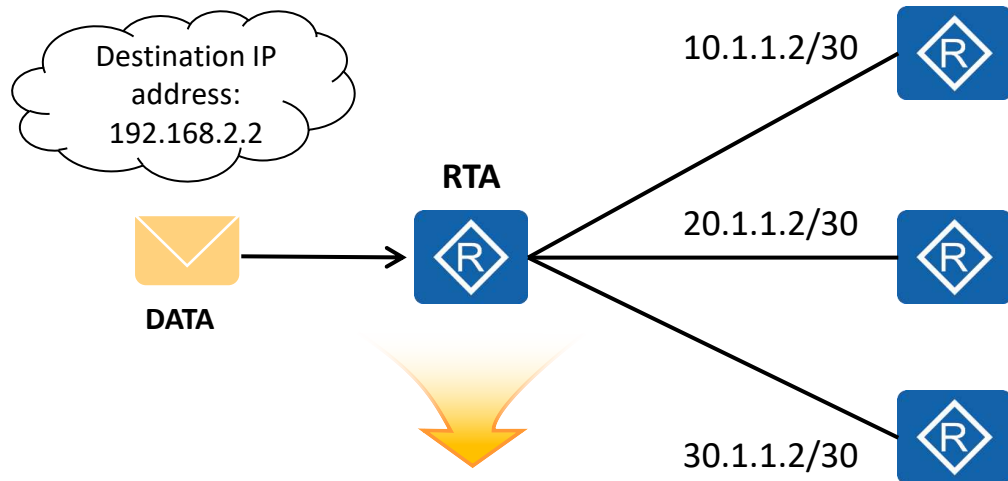
- When a router receives an IP packet, it compares the destination IP address of the packet with all routing entries in the local routing table bit by bit until the longest matching entry is found. This is the longest matching mechanism. **Importance:**
 - Ensures packets take the most specific and correct path “Specific” means **how detailed a network address is**.
 - Allows subnetting without breaking routing





Example of Longest Matching (1)

Example of Longest Matching



IP routing table of RTA

Destination/Mask	Next Hop
192.168.0.0/16	10.1.1.2
192.168.2.0/24	20.1.1.2
192.168.3.0/24	30.1.1.2

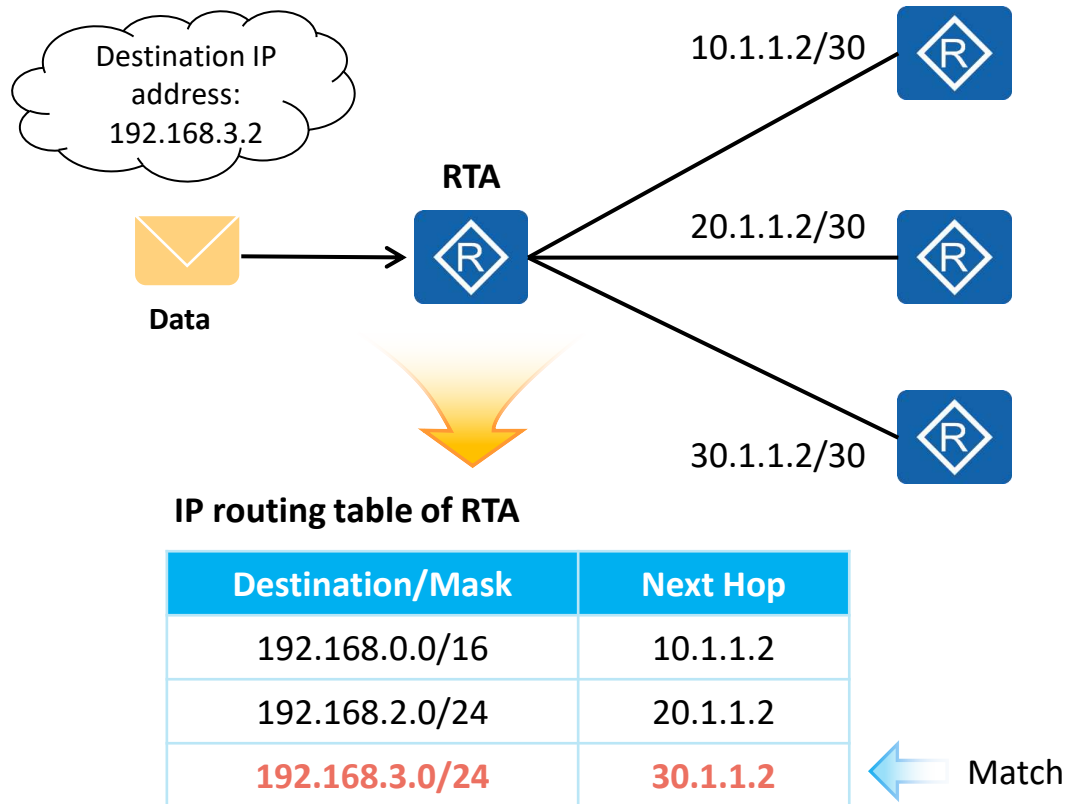


- There are two routes to 192.168.2.2 in the IP routing table of RTA, one has the 16-bit mask and the other has the 24-bit mask. According to the longest matching rule, the route with the 24-bit mask is preferred to guide the forwarding of packets destined for 192.168.2.2.



Example of Longest Matching (2)

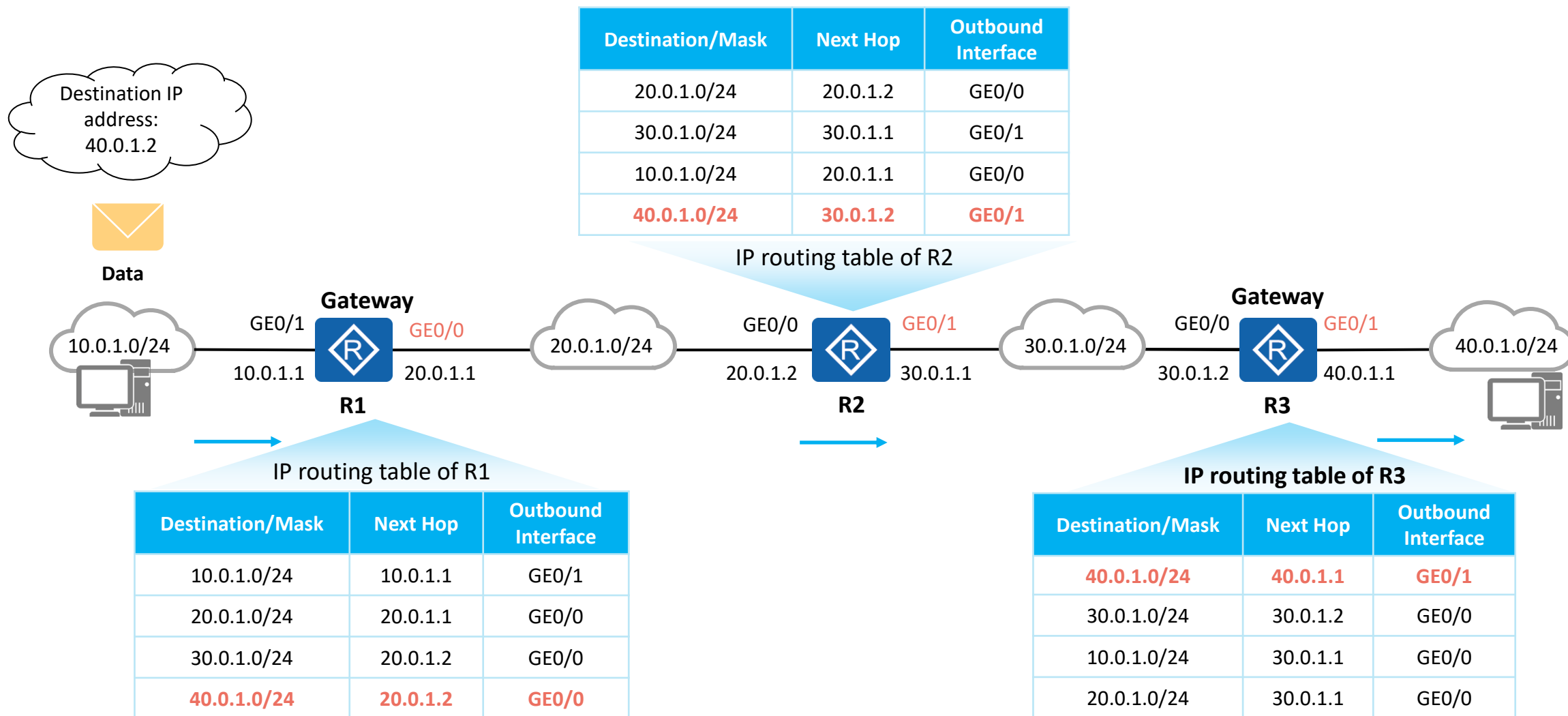
Example of Longest Matching



- According to the longest matching rule, only the route to 192.168.3.0/24 in the IP routing table matches the destination IP address 192.168.3.2. Therefore, this route is used to forward packets destined for 192.168.3.2.



Route-based Forwarding Process





Summary of the IP Routing Table

- When a router obtains routes to the same destination subnet with the same mask from different routing protocols, the router prefers the route with the lowest preference value of these routing protocols. If these routes are learned from the same routing protocol, the router prefers the route with the lowest cost. In summary, only the optimal route is installed in the IP routing table.
- When a router receives a packet, it searches its IP routing table for the outbound interface and next hop based on the destination IP address of the packet. If it finds a matching routing entry, it forwards the packet according to the outbound interface and next hop specified by this entry. Otherwise, it discards the packet.
- Packets are forwarded hop by hop. Therefore, all the routers along the path from the source to the destination must have routes destined for the destination. Otherwise, packet loss occurs.
- Data communication is bidirectional. Therefore, both forward and backward routes must be available.



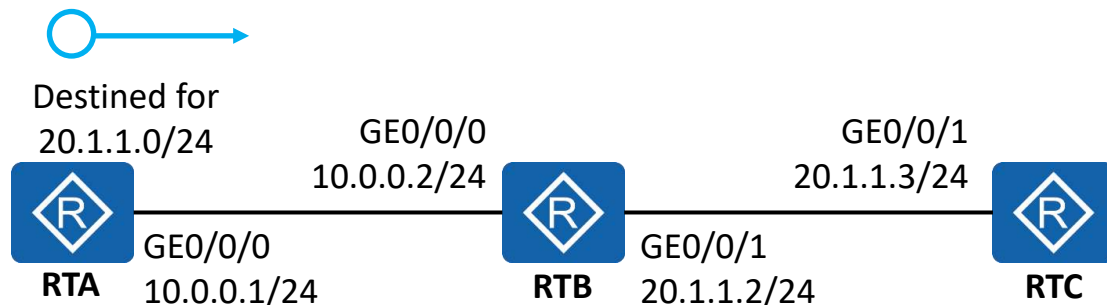
Contents

1. Overview of IP Routing
- 2. Static Routing**
3. Dynamic Routing
4. Advanced Routing Features



Application Scenarios of Static Routes

Static Routes



Destination	Protocol	Next Hop
20.1.1.0	Static	10.0.0.2
10.0.0.0	Direct	10.0.0.1

- Static routes are manually configured by network administrators, have low system requirements, and apply to simple, stable, and small networks.
- The disadvantage of static routes is that they cannot automatically adapt to network topology changes and so require manual intervention.
- RTA needs to forward the packets with the destination address 20.1.1.0/24. However, the IP routing table of RTA has only one direct route, which does not match 20.1.1.0/24. In this case, a static route needs to be manually configured so that the packets sent from RTA to 20.1.1.0/24 can be forwarded to the next hop 10.0.0.2.



Static Route Configuration

- Specify a next-hop IP address for a static route.

```
[Huawei] ip route-static ip-address { mask | mask-length } nexthop-address
```

- Specify an outbound interface for a static route.

```
[Huawei] ip route-static ip-address { mask | mask-length } interface-type interface-number
```

- Specify both the outbound interface and next hop for a static route.

```
[Huawei] ip route-static ip-address { mask | mask-length } interface-type interface-number [ nexthop-address ]
```

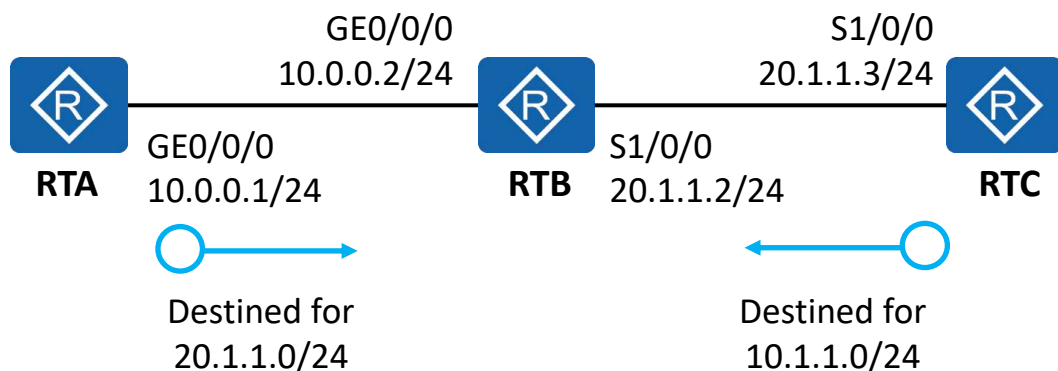
When creating a static route, you can specify both the outbound interface and next hop. Alternatively, you can specify either the outbound interface or next hop, depending on the interface type:

For a point-to-point interface (such as a serial interface), you must specify the outbound interface.

For a broadcast interface (for example, an Ethernet interface) or a virtual template (VT) interface, you must specify the next hop.



Configuration Example



- Configure static routes on RTA and RTC for communication between 10.0.0.0/24 and 20.1.1.0/24.
- Packets are forwarded hop by hop. Therefore, all the routers along the path from the source to the destination must have routes destined for the destination.
- Data communication is bidirectional. Therefore, both forward and backward routes must be available.

Configure RTA.

```
[RTA] ip route-static 20.1.1.0 255.255.255.0 10.0.0.2
```

Configure RTC.

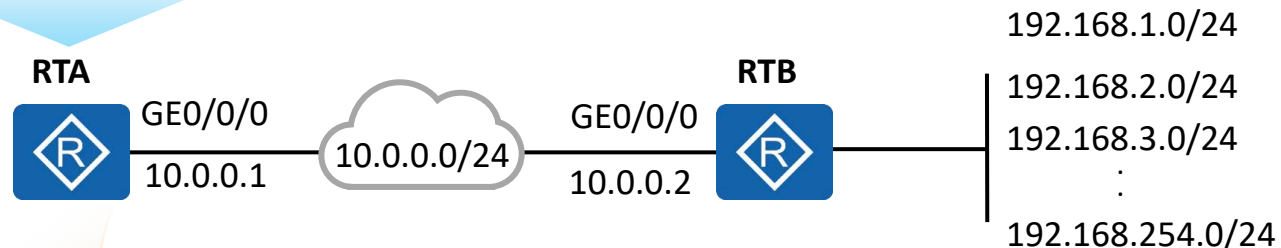
```
[RTC] ip route-static 10.0.0.0 255.255.255.0 S1/0/0
```



Default Routes

- Default routes are used only when packets to be forwarded do not match any routing entry in an IP routing table.
- In an IP routing table, a default route is the route to network 0.0.0.0 (with the mask 0.0.0.0), namely, 0.0.0.0/0.

RTA needs to forward packets to a subnet that is not directly connected to it and forwards the packets to 10.0.0.2.

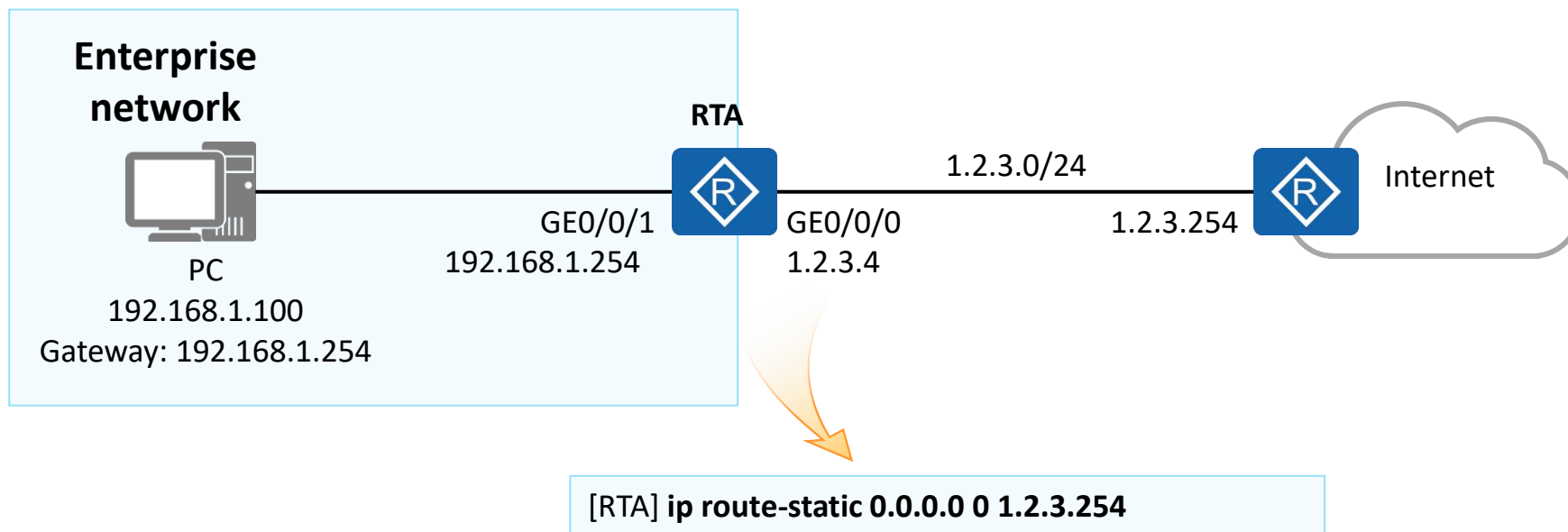


[RTA] ip route-static 0.0.0.0 0 10.0.0.2



Application Scenarios of Default Routes

- Default routes are typically used at the egress of an enterprise network. For example, you can configure a default route on an egress device to enable the device to forward IP packets destined for any address on the Internet.





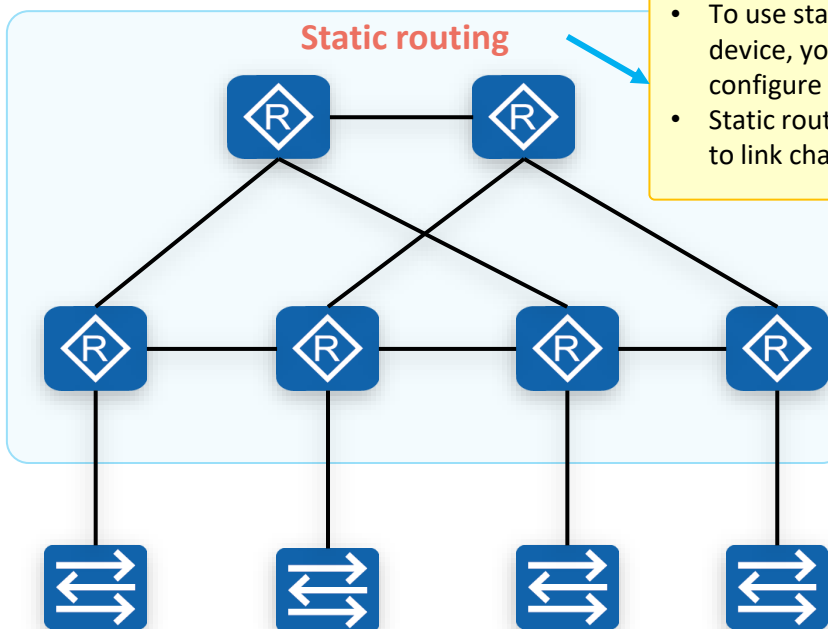
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1. Overview of IP Routing
2. Static Routing
- 3. Dynamic Routing**
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Overview of Dynamic Routing

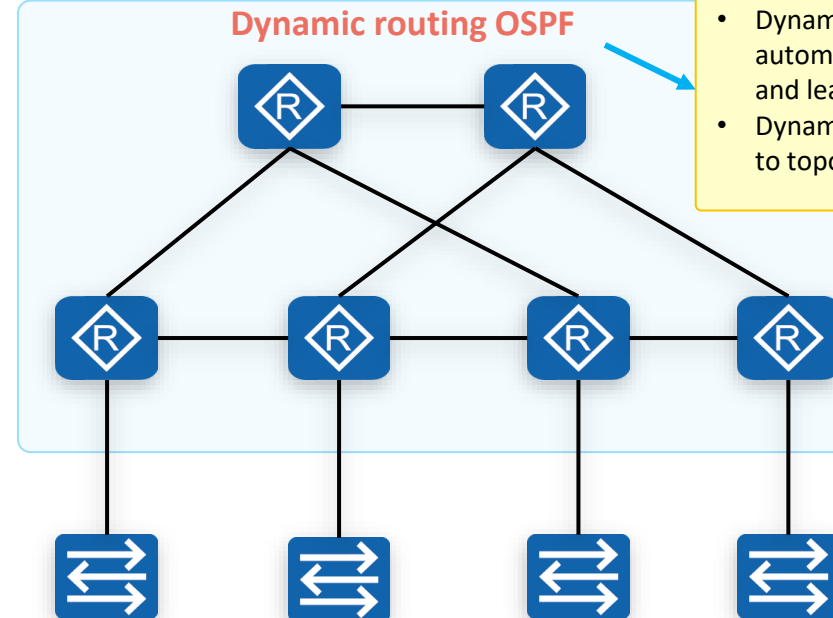
Static Routing



- To use static routes on any device, you must manually configure them.
- Static routes cannot adapt to link changes.

- When the network scale expands, it becomes increasingly complex to manually configure static routes. In addition, when the network topology changes, static routes cannot adapt to these changes in a timely and flexible manner.

Dynamic Routing



- Dynamic routes can be automatically discovered and learned.
- Dynamic routes can adapt to topology changes.

- Dynamic routing protocols automatically discover and generate routes, and update routes when the topology changes. These protocols effectively reduce the workload of network administrators and are widely used on large networks.



Classification of Dynamic Routing Protocols

Classification by the application scope

Interior Gateway Protocol (IGP)

RIP

OSPF

IS-IS

Exterior Gateway Protocol (EGP)

BGP

Classification by working mechanism and routing algorithm

Distance-vector routing protocol

RIP

Link-state routing protocol

OSPF

IS-IS



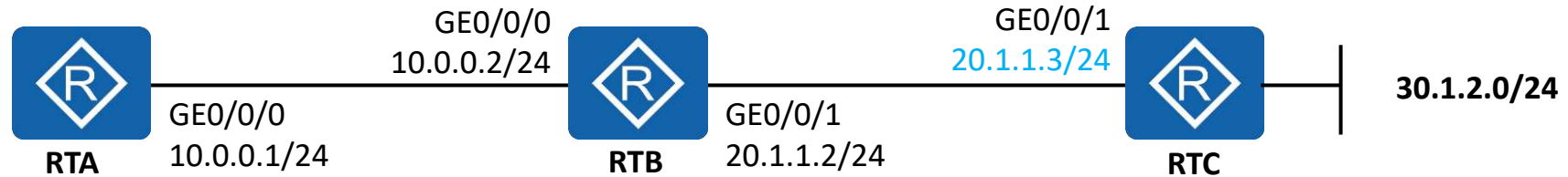
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- 4. Advanced Routing Features**



Route Recursion (1)

Route recursion is a recursive search process where a router performs a recursive lookup in its IP routing table to find the outgoing interface for a route's next-hop IP address. It occurs when the next-hop IP address is not on a directly connected network. The router searches its routing table again and again until a directly connected route is found.



[RTA] ip route-static 30.1.2.0 24 20.1.1.3

The next hop of the route to 30.1.2.0/24 is 20.1.1.3, which is not on a directly connected network of RTA. If the IP routing table does not have a route to 20.1.1.3, this static route does not take effect and cannot be installed in the IP routing table.



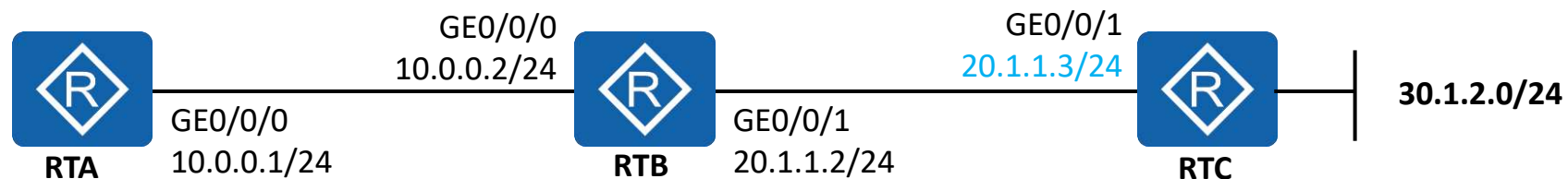
Route Recursion (2)

Route Recursion

Equal-Cost Route

Floating Route

Route Summarization



[RTA] ip route-static 30.1.2.0 24 20.1.1.3
[RTA] ip route-static 20.1.1.0 24 10.0.0.2

Recursion

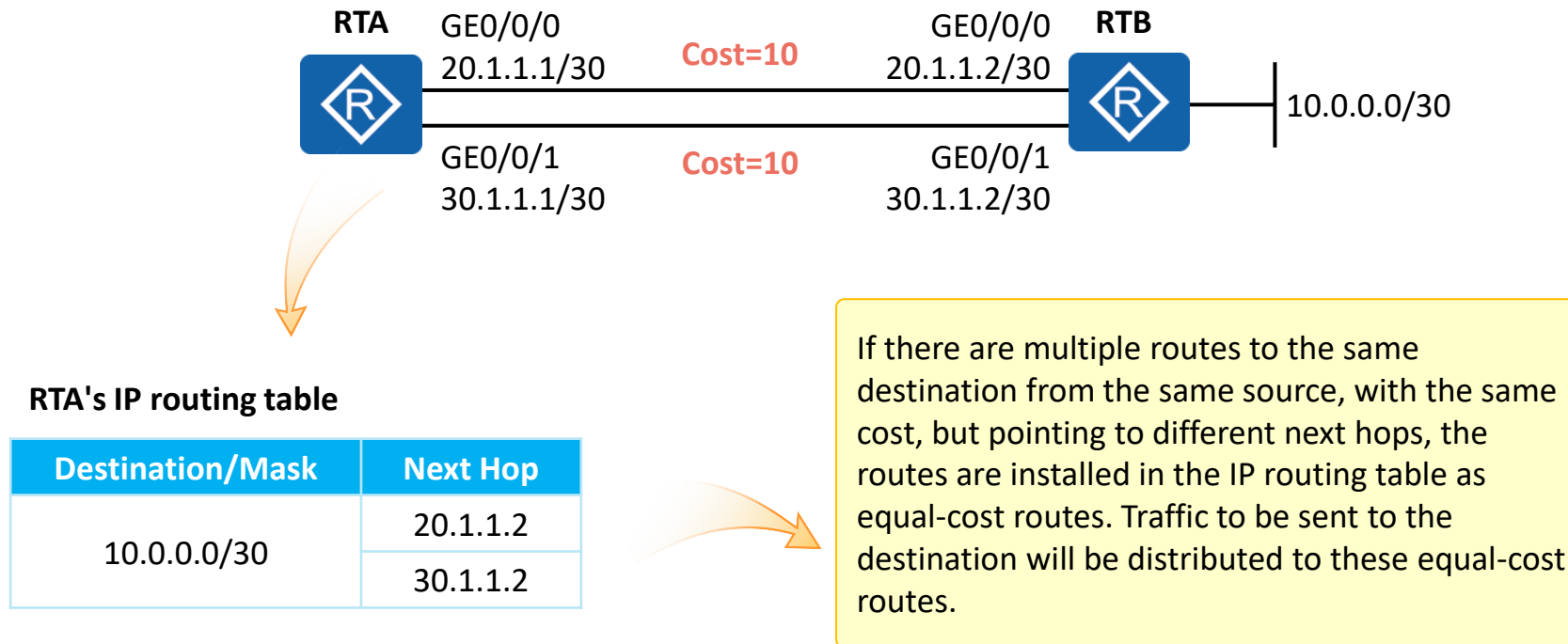
Destination/Mask	Next Hop	Outbound Interface
30.1.2.0/24	20.1.1.3	GE0/0/0
20.1.1.0/24	10.0.0.2	GE0/0/0

Configure a route to 20.1.1.3, with the next hop pointing to 10.0.0.2 on the directly connected network.
In this way, RTA can recurse the route with the destination 30.1.2.0/24 to the route with the destination 10.0.0.2.



Equal-Cost Route

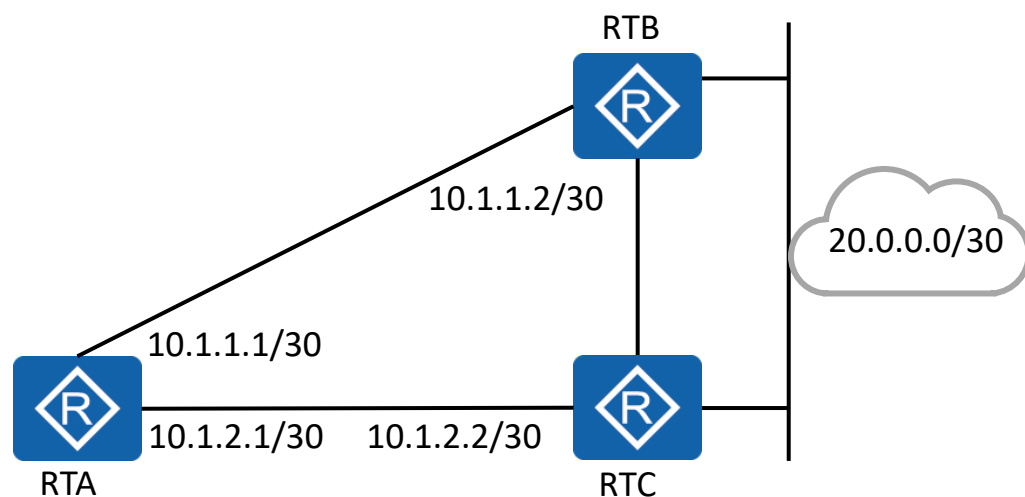
- Equal-cost routes are multiple routes to the same destination that have the same preference and the same metric (cost).
- When there are equal-cost routes in the IP routing table, a router forwards IP packets sent to the destination subnet through all valid outbound interfaces and next hops, achieving load balancing.





Floating Route - Basic Concepts

Floating Route



Configure a floating route on RTA.

```
[RTA] ip route-static 20.0.0.0 30 10.1.1.2
[RTA] ip route-static 20.0.0.0 30 10.1.2.2 preference 70
```

- A floating route is a backup route that is used only when the primary route becomes unavailable.
- It is configured with a higher preference value (administrative distance) than the primary route.
- Different preferences can be manually configured for static routes. Therefore, you can configure two static routes with the same destination address/mask but different preferences and next hops to implement backup of forwarding paths.
- That is, a floating route is installed in the IP routing table only when the next hop of the primary route is unreachable.



Floating Route - Example

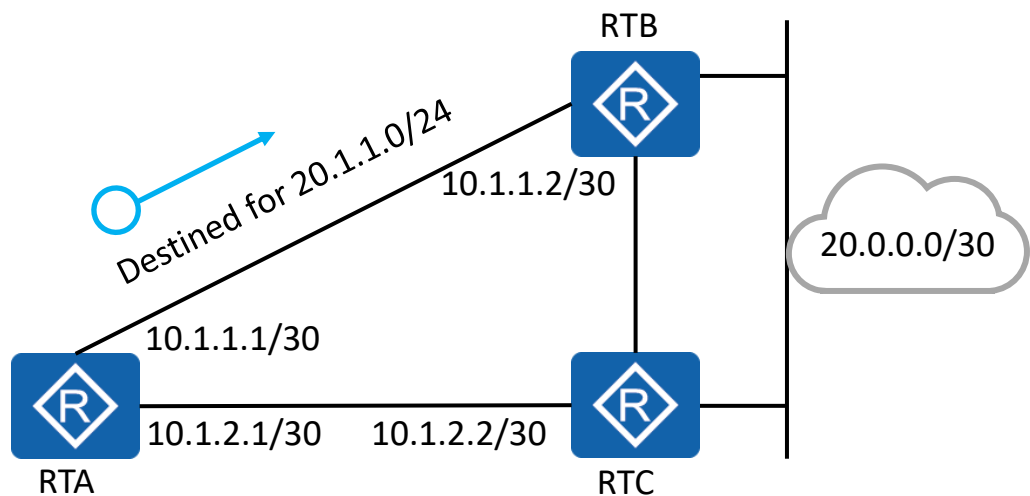
Route Recursion

Equal-Cost Route

Floating Route

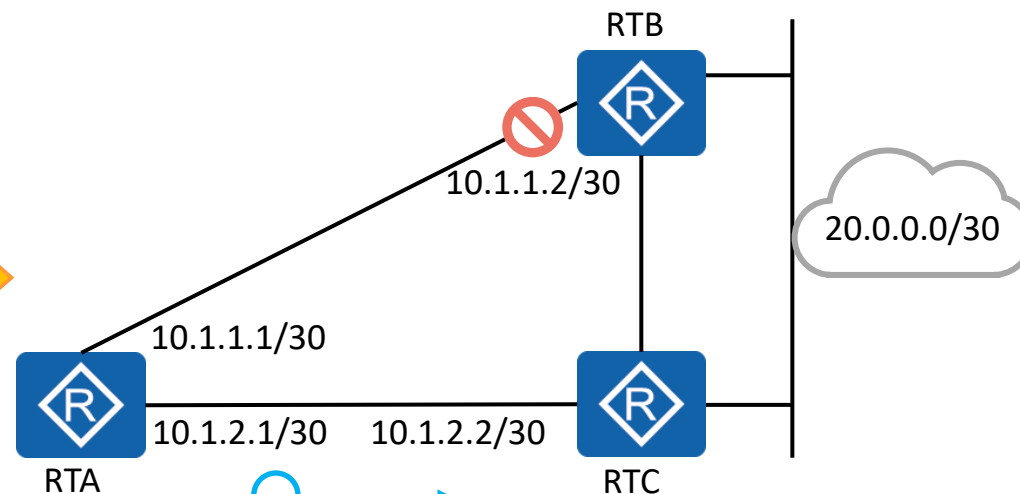
Route Summarization

Floating Route Switching



RTA's IP routing table when the primary link is available

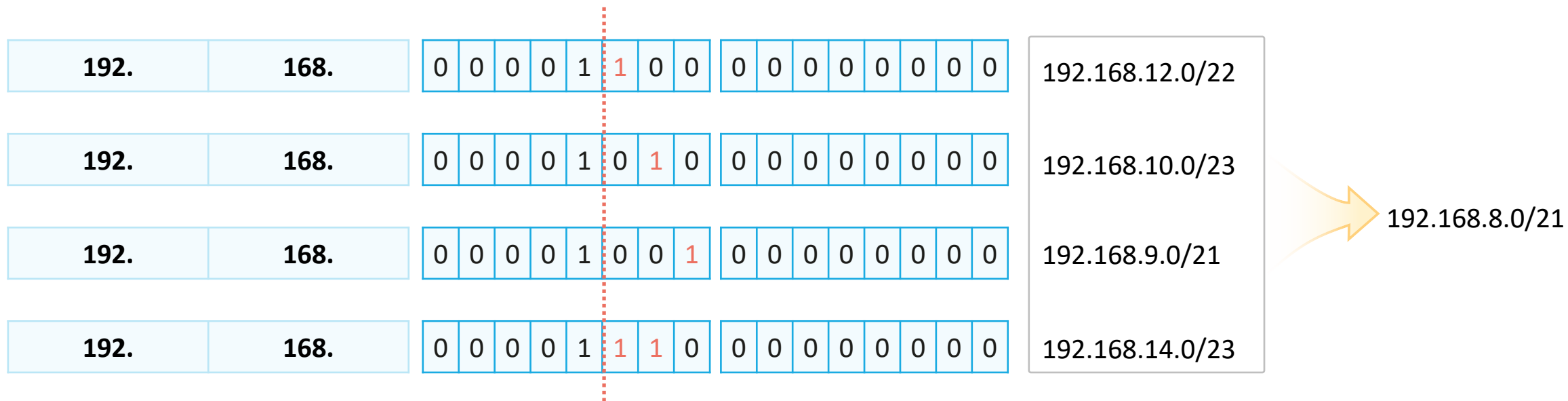
Destination	Next Hop	Preference
20.0.0.0	10.1.1.2	60



RTA's IP routing table when the primary link fails

Destination	Next Hop	Preference
20.0.0.0	10.1.2.2	70

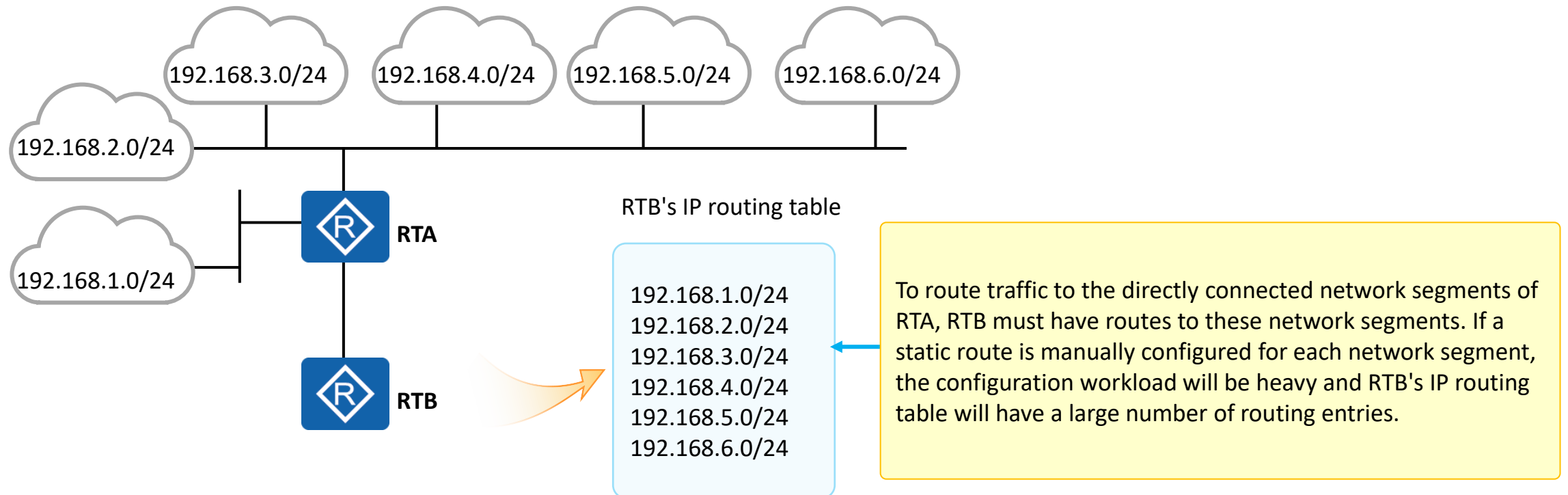
- Classless Inter-Domain Routing (CIDR) uses IP addresses and masks to identify networks and subnets replacing the previous addressing architecture of classful network design (such as classes A, B, and C addresses).
- CIDR is based on variable length subnet mask (VLSM). CIDR uses prefixes of any lengths to divide the address space with continuous reduced IP addresses. Multiple address segments with continuous prefixes can be summarized into a network, effectively reducing the number of routing entries.





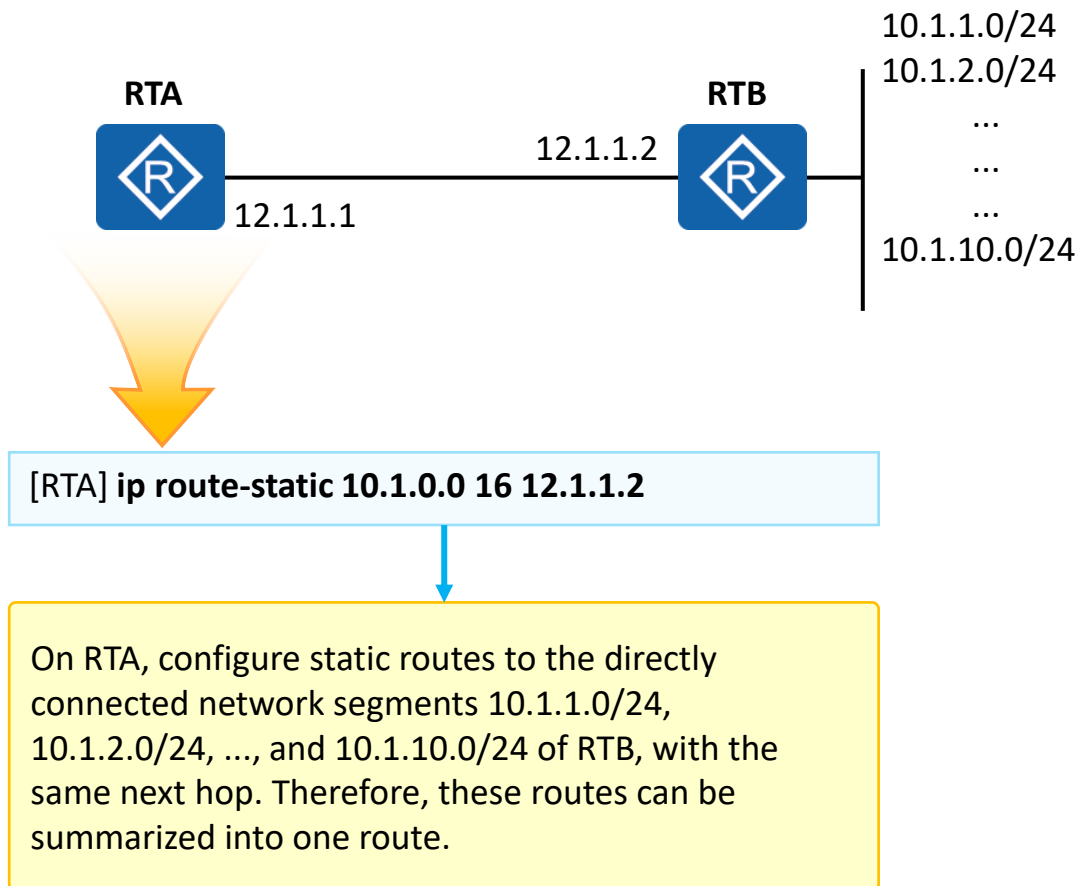
Background of Route Summarization

- Subnet division and VLSM resolve the problem of address space waste, but also bring a new challenge: increasing routing entries in the IP routing table.
- Route summarization can minimize routing entries.





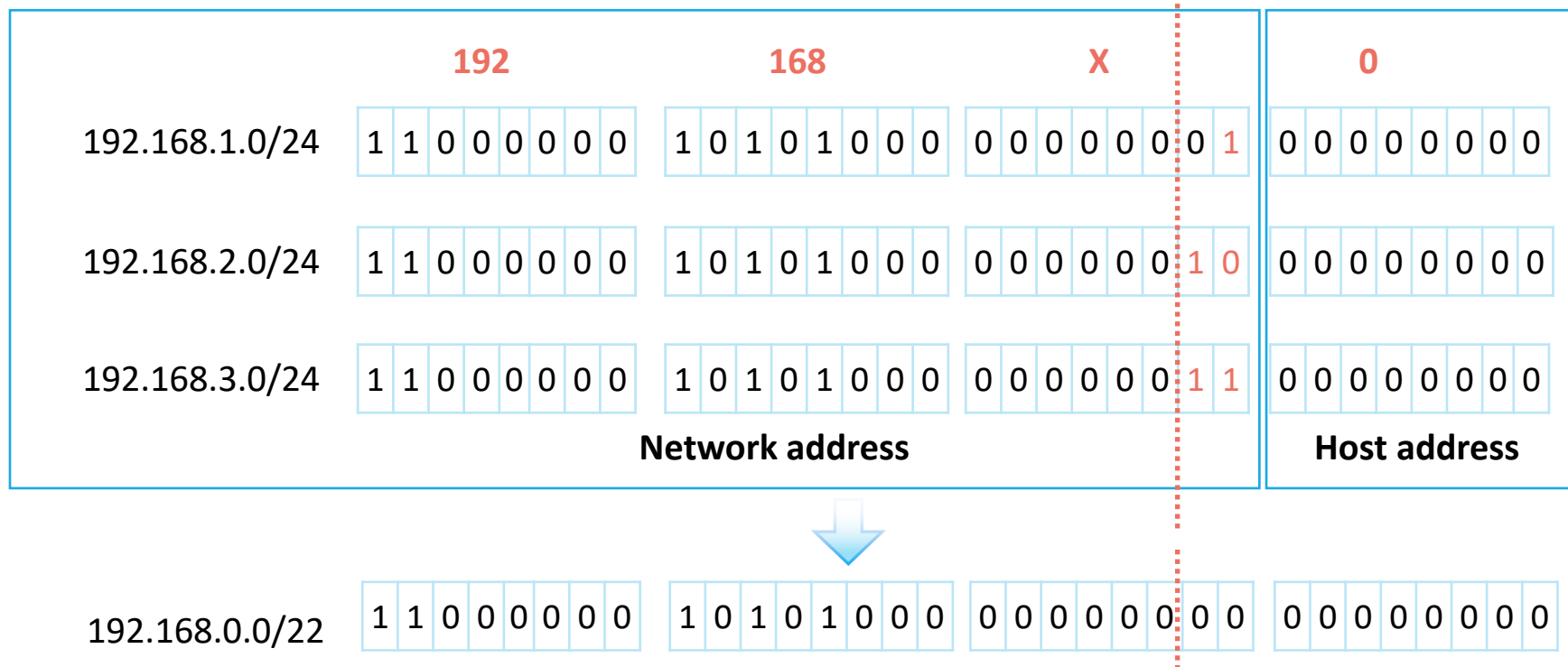
Overview of Route Summarization



- Route summarization is an approach of summarizing routes with the same prefix into one summary route to minimize the IP routing table size and improve device resource usage.
- Route summarization uses CIDR to summarize network segments with the same prefix into a single one.
- The routes before being summarized are known as specific routes, and the routes created after summarization are known as summarized routes or summary routes.



Summarization and Calculation



- To summarize routes to multiple continuous network segments into one summary route that just includes these network segments, ensure that the mask length of the summary route is as long as possible.
- The key to achieve this is to convert the destination addresses of specific routes into binary numbers and then find out the identical bits in these binary numbers.



Problems Caused by Route Summarization (1)

Routing Loop

```
[RTB] ip route-static 0.0.0.0 0 12.1.1.2
```

```
[RTA] ip route-static 10.1.0.0 16 12.1.1.1
```

10.1.1.0/24
10.1.2.0/24
...
...
...
10.1.10.0/24

RTB



1

12.1.1.1

12.1.1.2

RTA



2

Internet

RTB receives traffic destined for 10.1.20.0/24 and forwards the traffic to RTA according to the default route.

Routes are summarized on RTA. Therefore, RTA forwards the traffic back to RTB according to the summary route to 10.1.0.0/16.

Loop

3



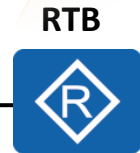
Problems Caused by Route Summarization (2)

Solution for Preventing Routing Loops

```
[RTB] ip route-static 0.0.0.0 0 12.1.1.2
```

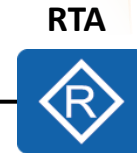
```
[RTA] ip route-static 10.1.0.0 16 12.1.1.1
```

10.1.1.0/24
10.1.2.0/24
...
...
...
10.1.10.0/24



12.1.1.1

12.1.1.2



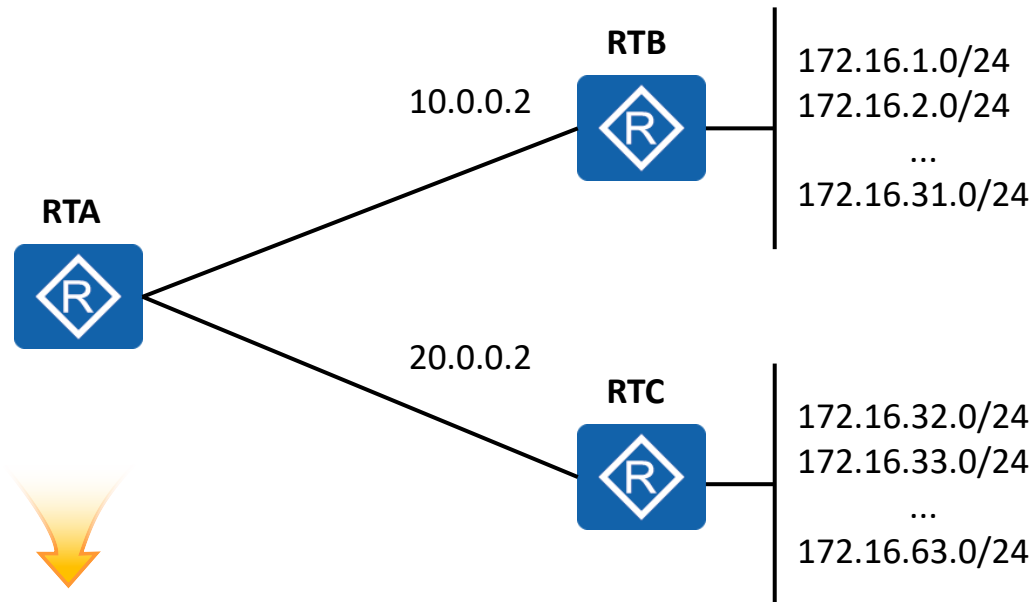
Internet

```
[RTB] ip route-static 10.1.0.0 16 0 NULL0
```

- Configure a route pointing to Null0 (blackhole to discard) on RTB to prevent routing loops when summarizing routes.



Accurate Route Summarization (1)



```
[RTA] ip route-static 172.16.0 16 10.0.0.2
```

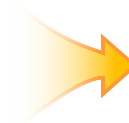
- To simplify the configuration, an administrator may configure a static summary route on RTA to allow RTA to reach network segments 172.16.1.0/24 to 172.16.31.0/24 of RTB. However, this summary route also includes the network segments of RTC. As a result, RTA forwards the traffic destined for network segments of RTC to RTB, causing data packet loss. This problem is caused by inaccurate route summarization. To resolve this problem, the summary route must be as accurate as possible; that is, it just covers all specific routes that are to be summarized, with no extra route included.



Accurate Route Summarization (2)

	10	1	0	0
10.1.1.0/24	0 0 0 0 1 0 1 0	0 0 0 0 0 0 0 1	0 0 0 0 0 0 0 1	0 0 0 0 0 0 0 0
10.1.2.0/24	0 0 0 0 1 0 1 0	0 0 0 0 0 0 0 1	0 0 0 0 0 0 1 0	0 0 0 0 0 0 0 0
10.1.3.0/24	0 0 0 0 1 0 1 0	0 0 0 0 0 0 0 1	0 0 0 0 0 0 1 1	0 0 0 0 0 0 0 0

```
ip route-static 10.1.1.0 24 12.1.1.2  
ip route-static 10.1.2.0 24 12.1.1.2  
ip route-static 10.1.3.0 24 12.1.1.2
```

**/22**

```
ip route-static 10.1.1.0 22 12.1.1.2
```

Accurately calculate the summarized network address and mask to ensure accurate route summarization.



Quiz

1. How does a router select the optimal route?
 2. How do I configure a floating route?
 3. What is the summary route for routes to 10.1.1.0/24, 10.1.3.0/24, and 10.1.9.0/24?
-
1. The router first compares preferences of routes. The route with the lowest preference value is selected as the optimal route. If the routes have the same preferences, the router compares their metrics. If the routes have the same metric, they are installed in the IP routing table as equal-cost routes.
 2. To configure a floating route, configure a static route with the same destination network segment and mask as the primary route but a different next hop and a larger preference value.
 3. The summary route is 10.1.0.0/20.



Summary

- This section presents the basic concepts of routes, how routes instruct routers to forward IP packets, common route attributes, and default routes (special static routes).
- In addition, this section describes advanced routing features including route recursion, floating routes, and equal-cost routes, which are widely used on live networks.

The image features a blue-tinted background showing silhouettes of several groups of business professionals in a modern office environment. They are standing on a reflective floor, and their reflections are visible below them. The background also shows a blurred view of city buildings through large windows. Overlaid on this scene is the text "Thank You" and the website address "www.huawei.com".

Thank You

www.huawei.com



OSPF Basics



Foreword

- Static routes are manually configured. If a network topology changes, static routes have to be manually adjusted, which restricts the large-scale application of static routes on the live network.
- Dynamic routing protocols are widely used on live networks because of their high flexibility, high reliability, and easy scalability. The Open Shortest Path First (OSPF) protocol is a widely used dynamic routing protocol.
- This course describes basic concepts, working mechanism, and basic configurations of OSPF.



Objectives

- On completion of this course, you will be able to:
 - Describe the advantages and classification of dynamic routing protocols.
 - Describe basic OSPF concepts and usage scenarios.
 - Describe the working mechanism of OSPF.
 - Implement basic OSPF configurations.



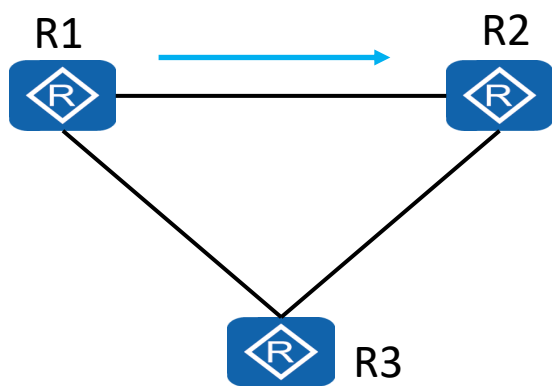
Contents

- 1. OSPF Overview**
2. OSPF Working Mechanism
3. Typical OSPF Configuration



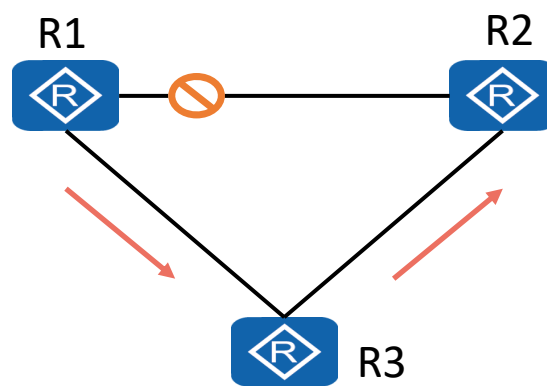
Why Are Dynamic Routing Protocol Used?

- Static routes are manually configured and maintained, and the command lines are simple and clear. They apply to small-scale or stable networks. Static routes have the following disadvantages:
 - Unable to adapt to large-scale networks: As the number of devices increases, the configuration workload increases sharply.
 - Unable to dynamically respond to network changes: If the network topology changes, the network cannot automatically converge, and static routes must be manually modified.



R1-to-R2 static route

Link fault
→



Manually configured static route R1-R3-R2



Classification of Dynamic Routing Protocols

By ASs (Autonomous System)

Interior Gateway Protocols (IGPs)

RIP

OSPF

IS-IS

Exterior Gateway Protocols (EGPs)

BGP

By working mechanisms and algorithms

Distance Vector Routing Protocols

RIP

Link-State Routing Protocols

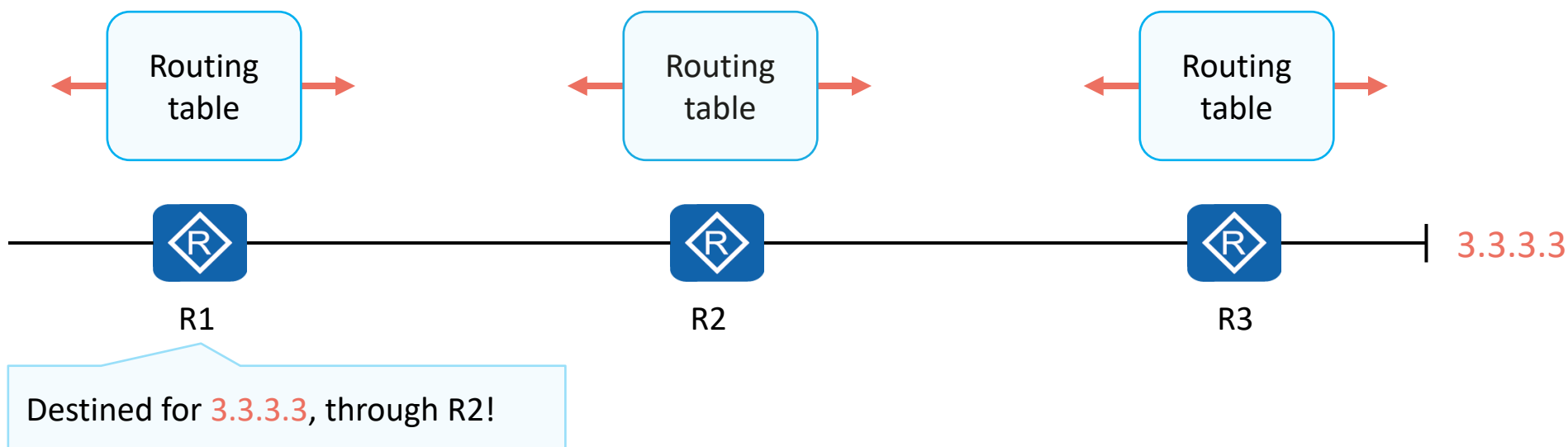
OSPF

IS-IS



Distance-Vector Routing Protocol

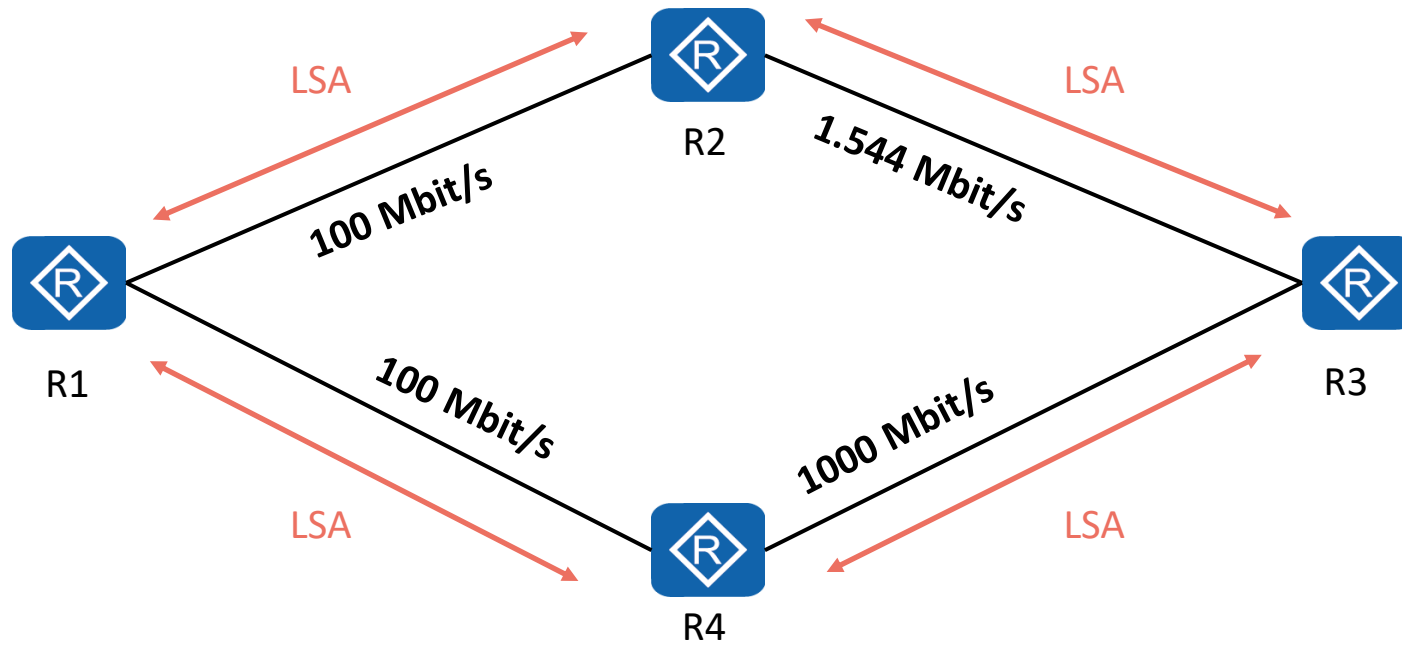
- A router running a distance-vector routing protocol periodically floods routes. Through route exchange, each router learns routes from neighboring routers and installs the routes into its routing table.
- Each router on a network is clear only about where the destination is and how far the destination is, but unclear about the whole network topology. This is the essence of the distance-vector algorithm.





Link-State Routing Protocol - LSA Flooding

- Different from a distance-vector routing protocol, a link-state routing protocol advertises link status information rather than routes in the routing table. Routers that run a link-state routing protocol establish a neighbor relationship and then exchange Link State Advertisements (LSAs).

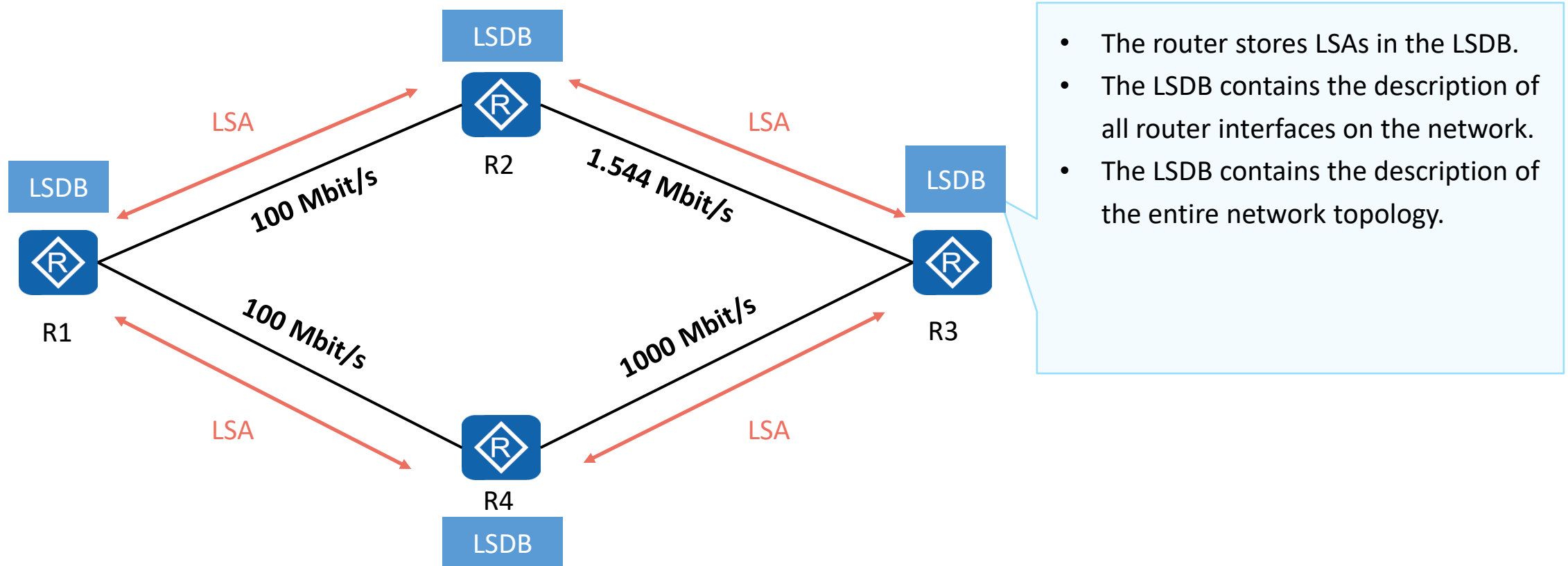


- LSAs, instead of routes, are advertised.
- An LSA describes a router interface's status information, such as the cost of the interface and a connected interface name.



Link-State Routing Protocol - LSDB Creation

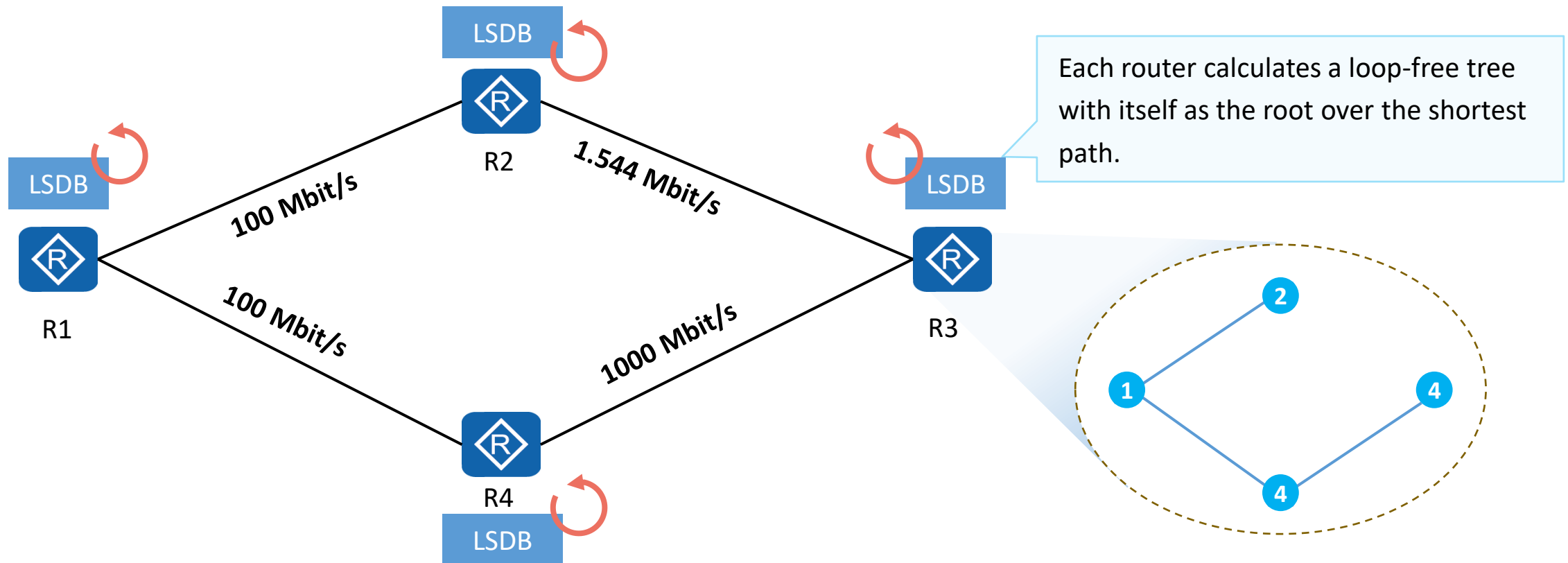
- Each router generates LSAs and adds the received LSAs to its own link state database (LSDB). Routers learn the whole network topology through the LSDB.





Link-State Routing Protocol - SPF Calculation

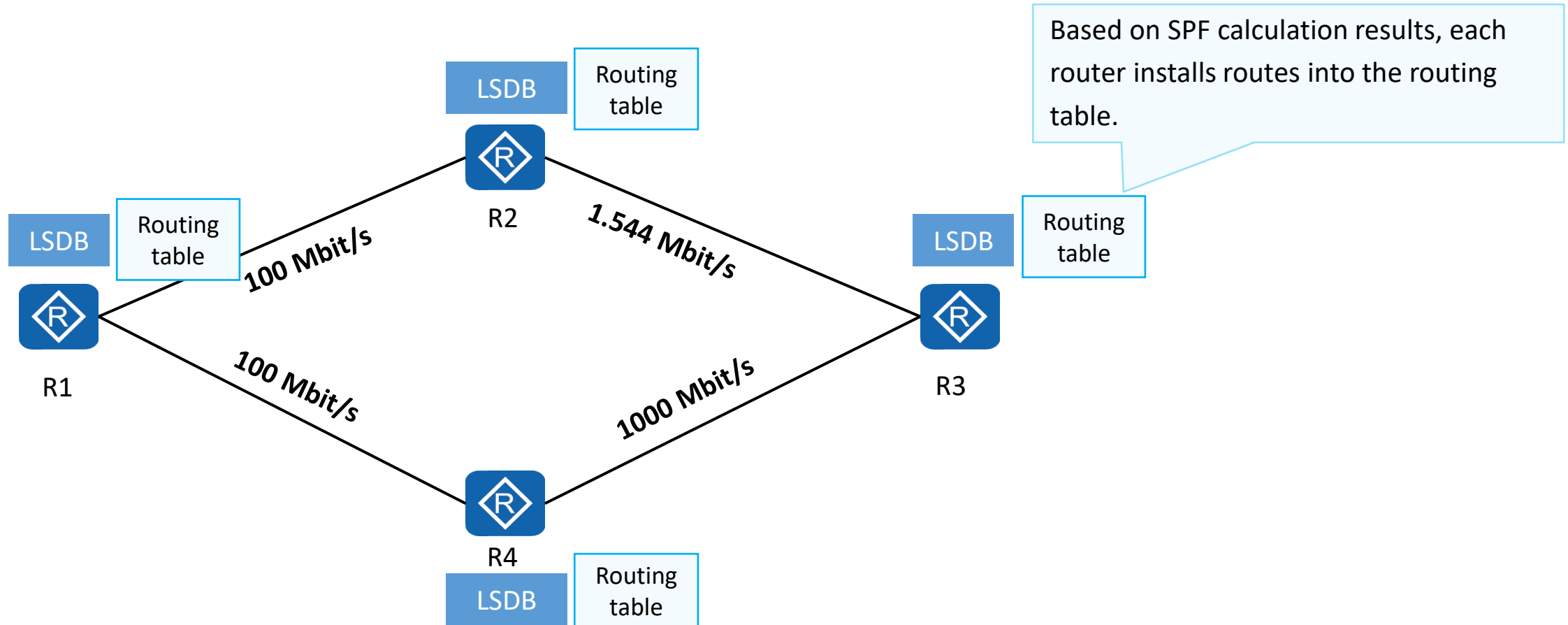
- Each router uses the Shortest Path First (SPF) algorithm and LSDB information to calculate routes. Each router calculates a loop-free tree with itself as the root and the shortest path. With this tree, a router determines the optimal path to each corner of a network.





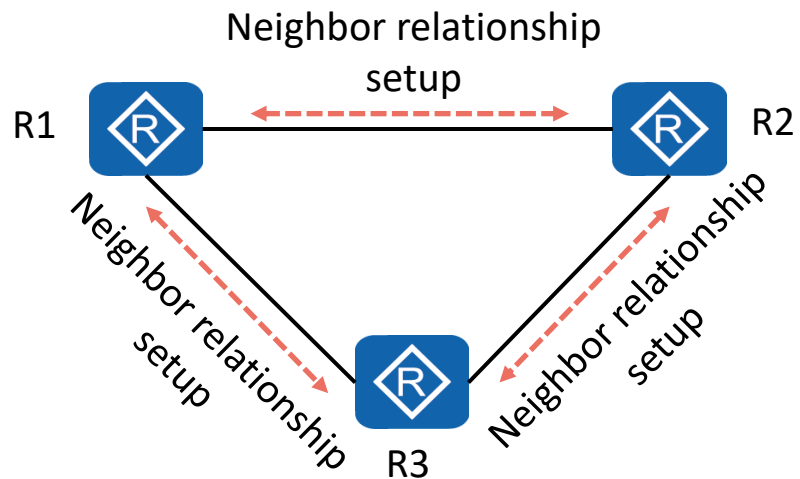
Link-State Routing Protocol - Routing Table Generation

- Ultimately, the router installs routes for the calculated preferred paths into its routing table.



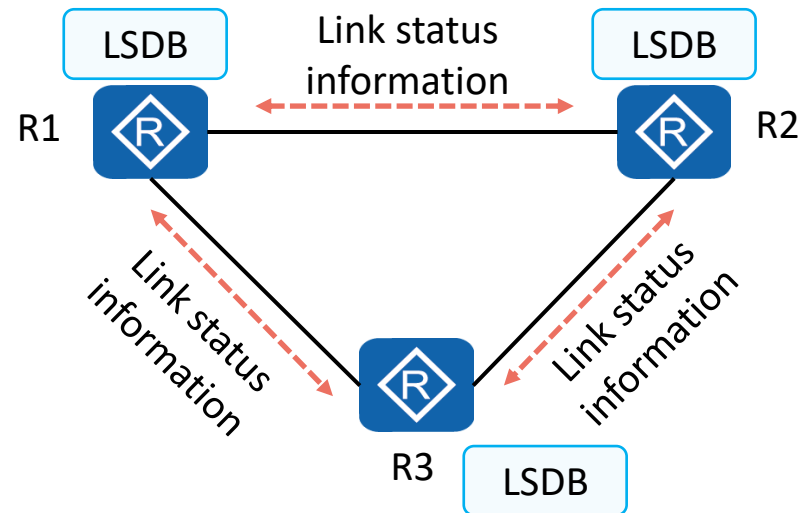


Summary of Link-State Routing Protocols



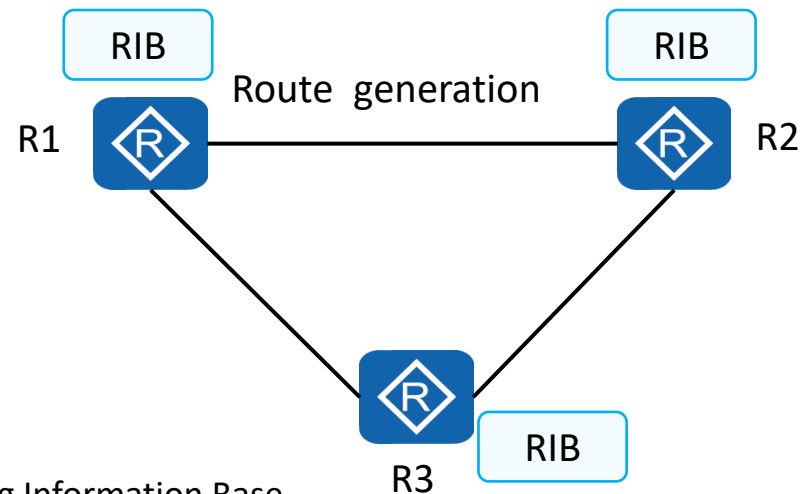
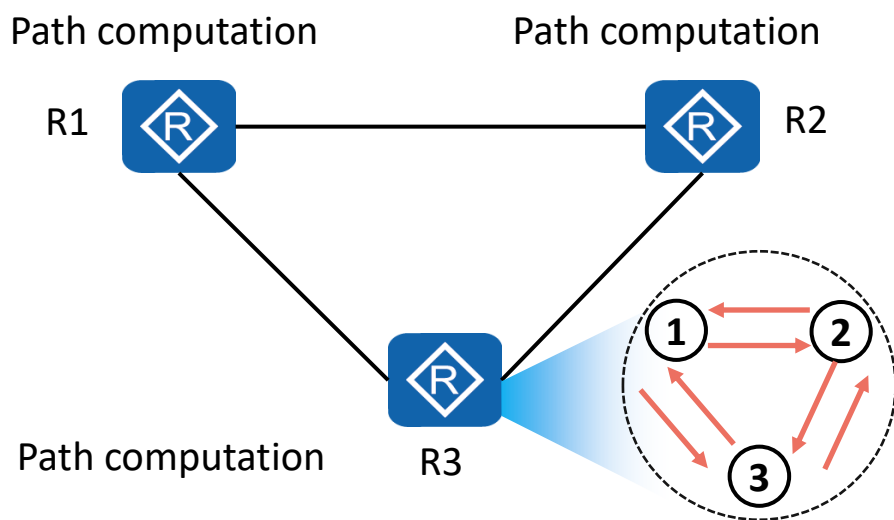
1

2



3

4



RIB: Routing Information Base

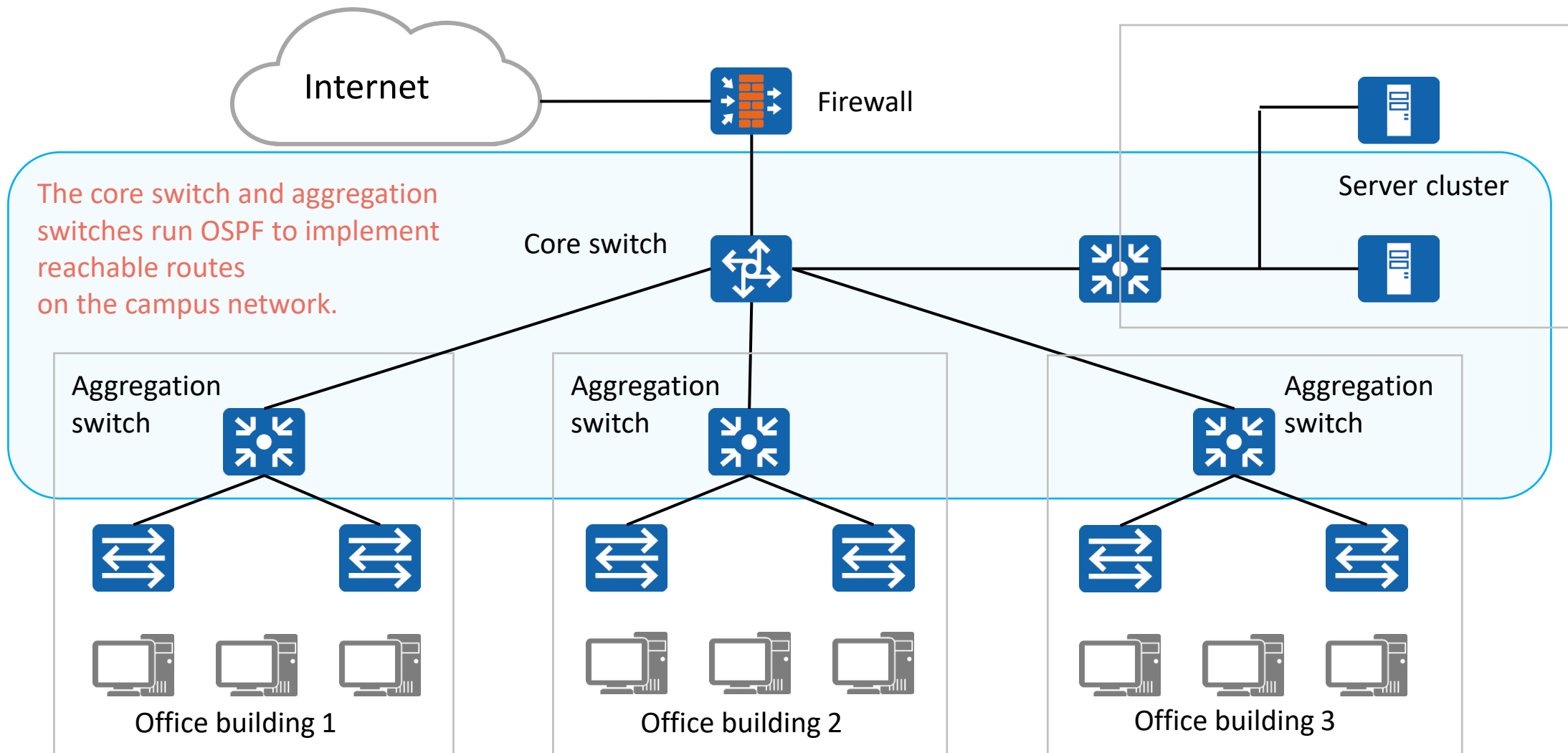


Introduction to OSPF

- OSPF is a typical link-state routing protocol and one of the widely used IGPs in the industry.
- OSPFv2, as defined in RFC 2328, is designed for IPv4. OSPFv3, as defined in RFC 2740, is designed for IPv6. Unless otherwise specified, OSPF in this presentation refers to OSPFv2.
- OSPF routers exchange link status information, but not routes. Link status information is key information for OSPF to perform topology and route calculation.
- An OSPF router collects link status information on a network and stores the information in the LSDB. Routers are aware of the intra-area network topology and be able to calculate loop-free paths.
- Each OSPF router uses the SPF algorithm to calculate the shortest path to a specific destination. Routers generate routes based on these paths and install the routes to the routing table.
- OSPF supports the variable length subnet mask (VLSM) mechanism and manual route summarization.
- The multi-area design enables OSPF to support a larger network.



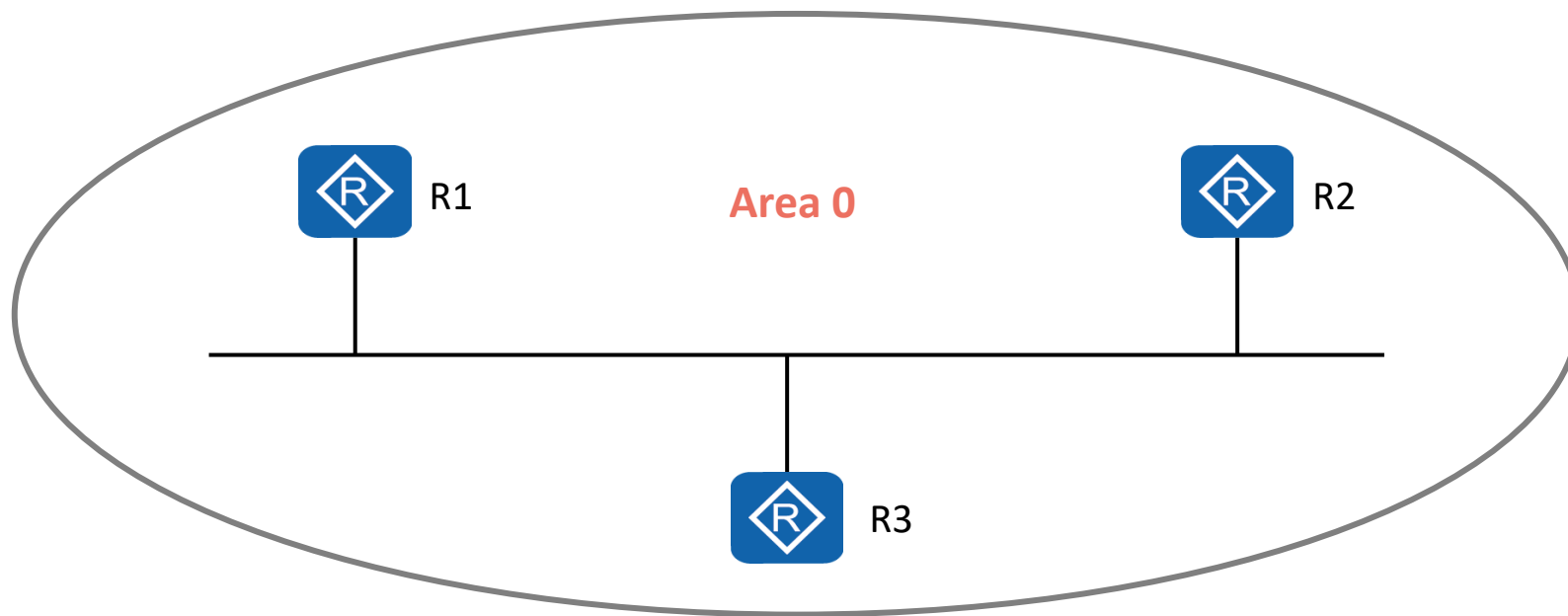
OSPF Application on a Campus Network





Basic OSPF Concepts: Area

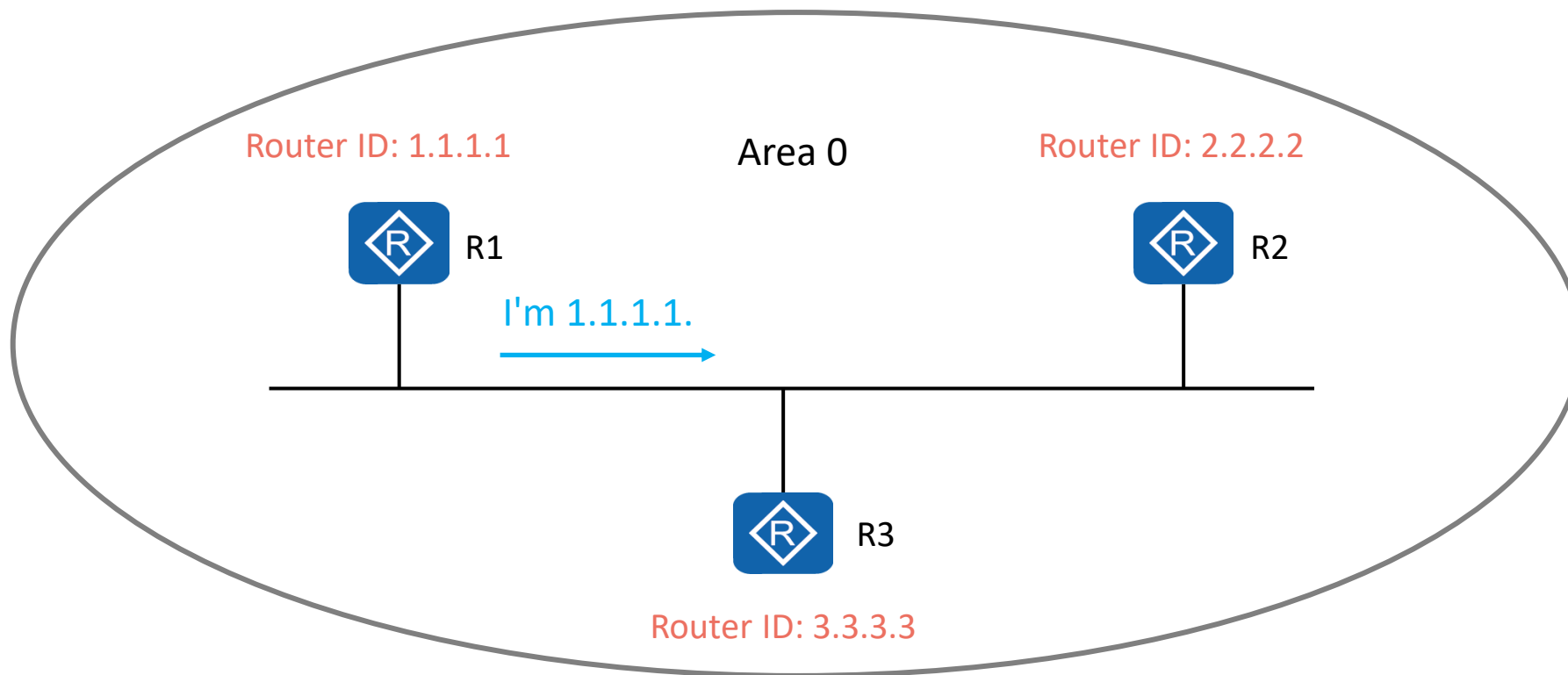
- The OSPF area keyword identifies an OSPF area.
- The area is considered as a logical group, and each group is identified by an area ID.





Basic OSPF Concepts: Router ID

- A router ID uniquely identifies a router in an OSPF area.
- The router ID can be manually specified or automatically assigned by the system.

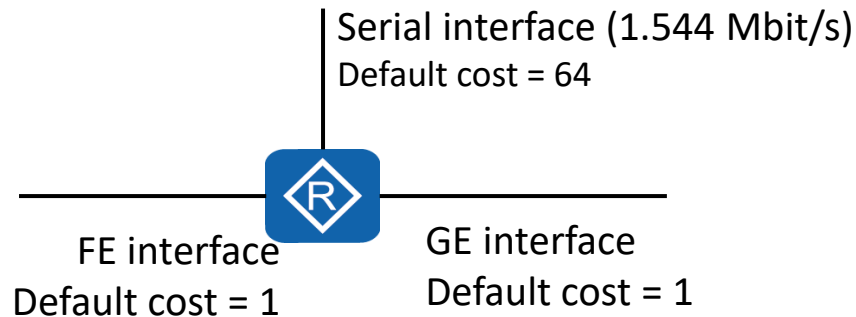




Basic OSPF Concepts: Cost Value

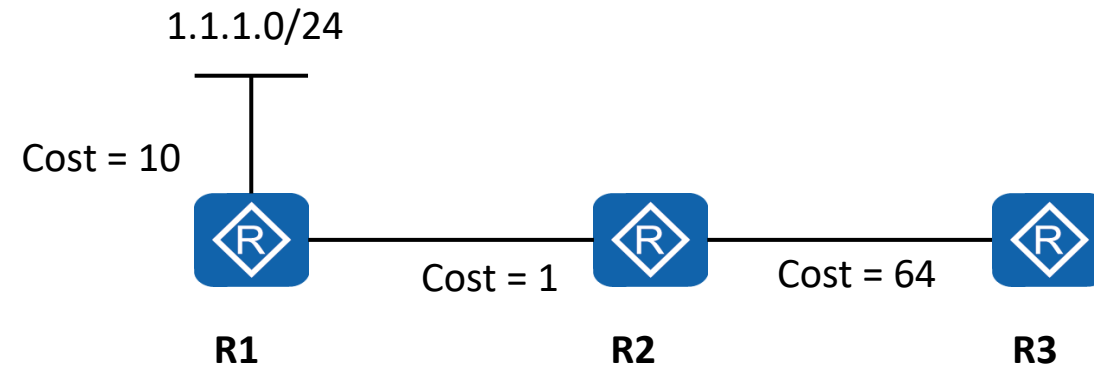
- OSPF uses costs as route metric values. Each OSPF-enabled interface maintains a cost value. Default cost value = $\frac{100 \text{ Mbit/s}}{\text{Interface bandwidth}}$, where, 100 Mbit/s is the default reference value specified by OSPF and is configurable.
- Generally, the cost of an OSPF route is the sum of costs of all inbound interfaces along a path from the destination network segment to the local router.

Cost Value of an OSPF Interface



- Each OSPF interface has a specific cost because of the particular bandwidth value.

Accumulated Costs on an OSPF Path



- In the routing table of R3, the cost of the OSPF route to 1.1.1.0/24 is 75 (10 + 1 + 64).



OSPF Packet Types

- There are five types of OSPF protocol packets and implement different functions in interaction between OSPF routers.

Packet Name	Function
Hello	Is periodically sent to discover and maintain OSPF neighbor relationships.
Database Description	Describes the summary of the local LSDB, which is used to synchronize the LSDBs of two devices.
Link State Request	Requests a needed LSA from a neighbor. LSRs are sent only after DD packets have been successfully exchanged.
Link State Update	Is sent to advertise a requested LSA to a neighbor.
Link State ACK	Is used to acknowledge the receipt of an LSA.



Three Types of OSPF Entries - Entries in the Neighbor Table

- OSPF provides entries in three important tables: OSPF neighbor table, LSDB table, and OSPF routing table. For the OSPF neighbor table, you need to know:
 - Before OSPF transmits link status information, OSPF neighbor relationships must be established.
 - OSPF neighbor relationships are established by exchanging Hello packets.
 - The OSPF neighbor table describes the status of the neighbor relationship between OSPF routers. You can run the **display ospf peer** command to view status information.

[R1]display ospf peer

Router ID: 1.1.1.1



R1

GE 1/0/0

10.1.1.1/30

Router ID: 2.2.2.2



R2

GE 1/0/0

10.1.1.2/30

<R1> display ospf peer

OSPF Process 1 with Router ID 1.1.1.1

Neighbors

Area 0.0.0.0 interface 10.1.1.1(GigabitEthernet1/0/0)'s neighbors

Router ID: 2.2.2.2 Address: 10.1.1.2 GR State: Normal

State: Full Mode:Nbr is Master Priority: 1

DR: 10.1.1.1 BDR: 10.1.1.2 MTU: 0

Dead timer due in 35 sec

Retrans timer interval: 5

Neighbor is up for 00:00:05

Authentication Sequence: [0]



Three Types OSPF Entries - Entries in the LSDB Table

- For the OSPF LSDB table, you need to know:
 - An LSDB stores LSAs generated by a router itself and received from neighbors. In this example, the LSDB of R1 contains three LSAs.
 - The **Type** field indicates an LSA type, and the **AdvRouter** field indicates the router that sends the LSA.
 - Run the **display ospf lsdb** command to query the LSDB.

[R1]display ospf lsdb



Router ID: 1.1.1.1

Router ID: 2.2.2.2



R1

GE 1/0/0

10.1.1.1/30



R2

GE 1/0/0

10.1.1.2/30

<R1> display ospf lsdb

OSPF Process 1 with Router ID 1.1.1.1

Link State Database

Router ID: 0.0.0.0

Type	LinkState ID	AdvRouter	Age	Len	Sequence	Metric
Router	2.2.2.2	2.2.2.2	98	36	8000000B	1
Router	1.1.1.1	1.1.1.1	92	36	80000005	1
Network	10.1.1.2	2.2.2.2	98	32	80000004	0



Three Types of OSPF Entries - Entries in the OSPF Routing Table

Neighbor
Table

LSDB Table

OSPF Routing Table

- For the OSPF routing table, you need to know:
 - The OSPF routing table and the router routing table are different. In this example, the OSPF routing table contains three routes.
 - An OSPF routing table contains information, such as the destination IP address, cost, and next-hop IP address, which guides packet forwarding.
 - Run the **display ospf routing** command to query the OSPF routing table.

[R1]display ospf routing

Router ID: 1.1.1.1



R1

GE 1/0/0

10.1.1.1/30

Router ID: 2.2.2.2



R2

GE 1/0/0

10.1.1.2/30

<R1> display ospf routing

OSPF Process 1 with Router ID 1.1.1.1

Routing tables

Routing for Network

Destination	Cost	Type	NextHop	AdvRouter	Area
1.1.1.1/32	0	stub	1.1.1.1	1.1.1.1	0.0.0.0
10.1.1.0/20	1	Transit	10.1.1.1	1.1.1.1	0.0.0.0
2.2.2.2/32	1	stub	10.1.1.2	2.2.2.2	0.0.0.0

Total Nets: 3

Intra Area: 3 Inter Area: 0 ASE: 0 NSSA: 0



Contents

1. OSPF Overview
- 2. OSPF Working Mechanism**
3. Typical OSPF Configuration



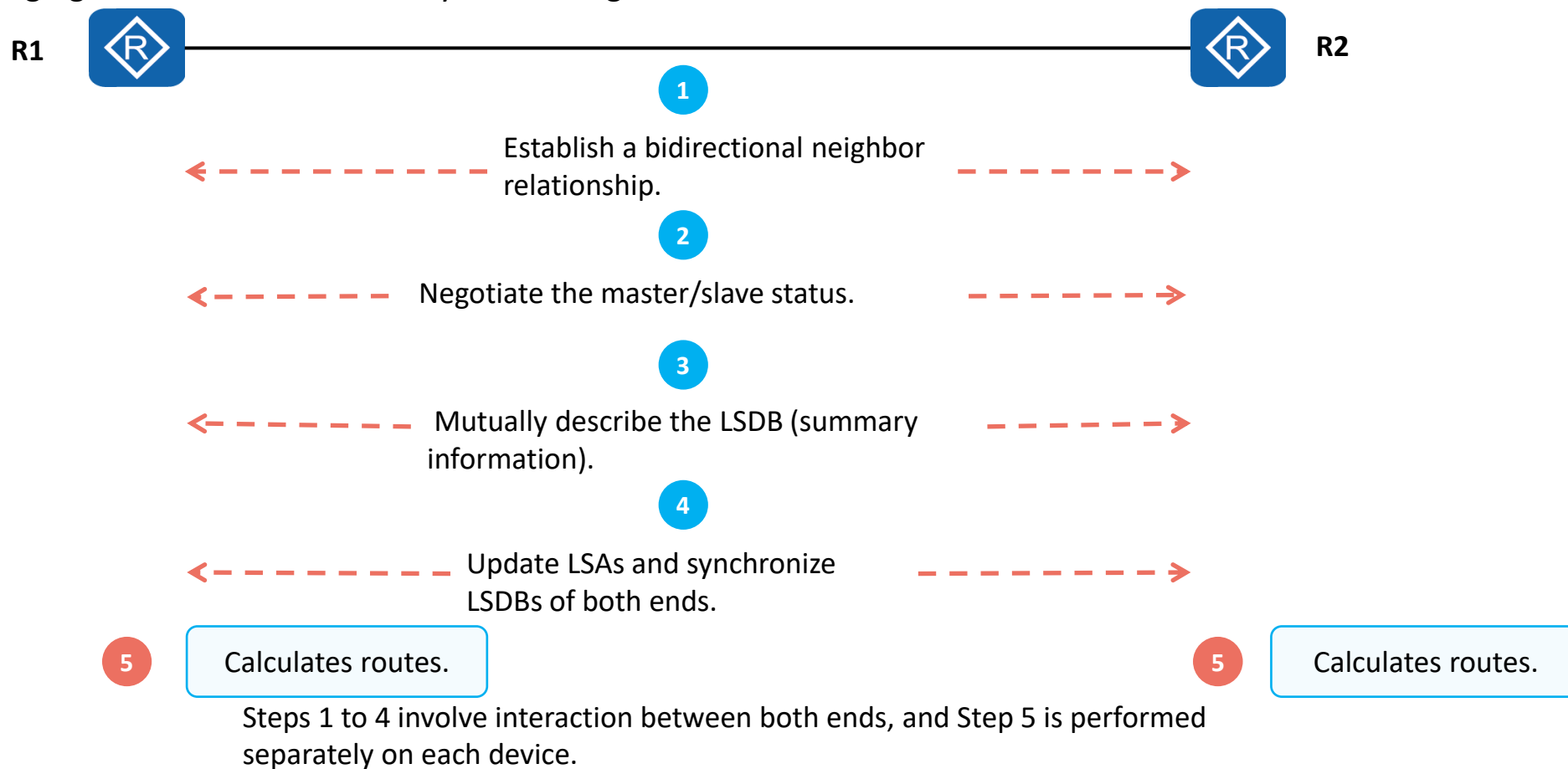
Relationships Between OSPF Routers

- There are two important concepts about the relationship between OSPF routers: neighbor relationship and adjacency.
- On a simple network, two routers are directly connected. OSPF is enabled on interconnected interfaces. The routers start to send and listen to Hello packets. After the two routers discover each other through Hello packets, they establish a neighbor relationship.
- The establishment of a neighbor relationship is just the beginning. A series of packets, such as DD, LSR, LSU, and LSAck packets, will be exchanged later. When LSDB synchronization between two routers is complete and the two routers start to calculate routes independently, the two routers establish an adjacency.



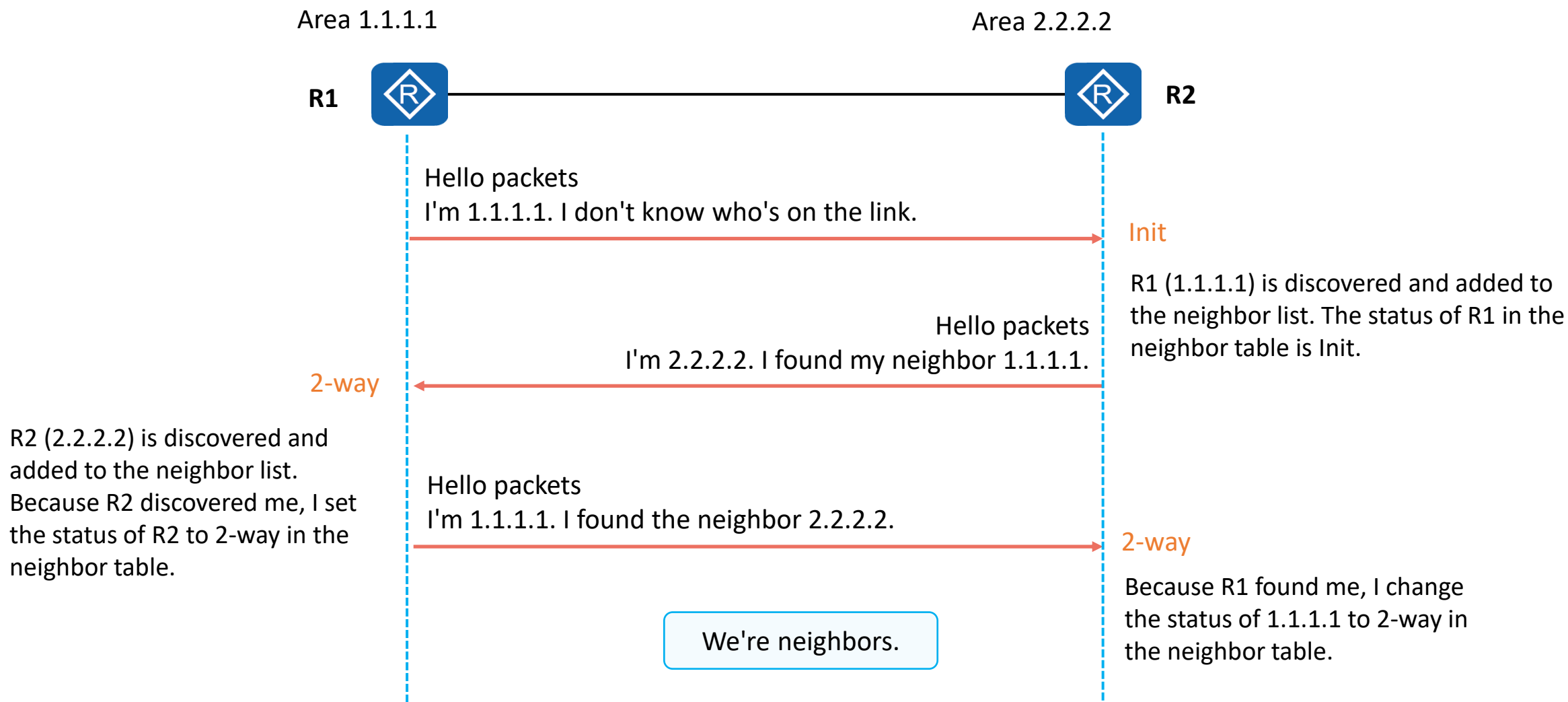
Process of Establishing an OSPF Adjacency Relationship

- OSPF adjacency relationship establishment involves four steps: establishing a neighbor relationship, negotiating the master/slave status, exchanging LSDB information, and synchronizing LSDBs.



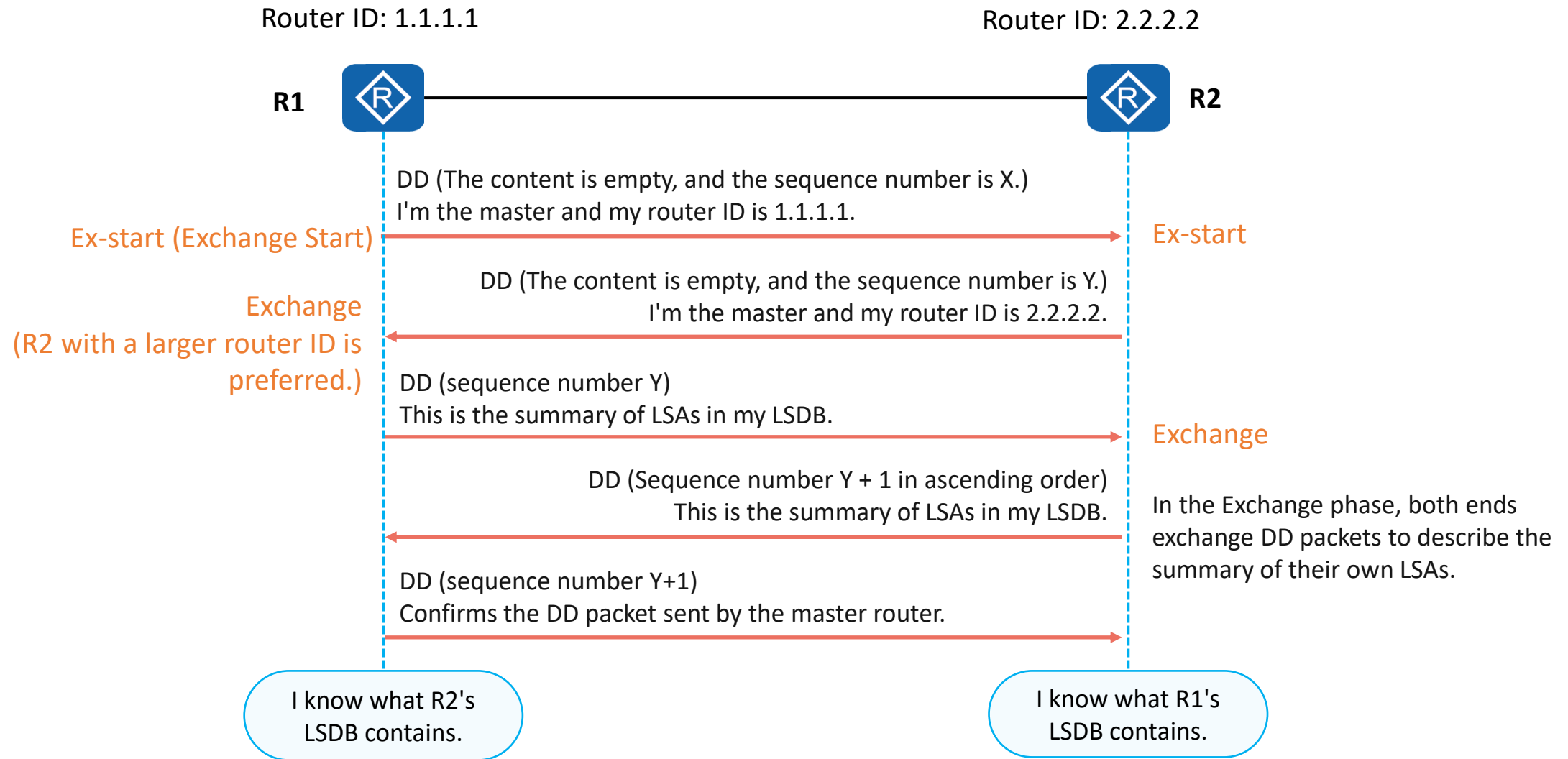


Process of Establishing an OSPF Adjacency - Step 1



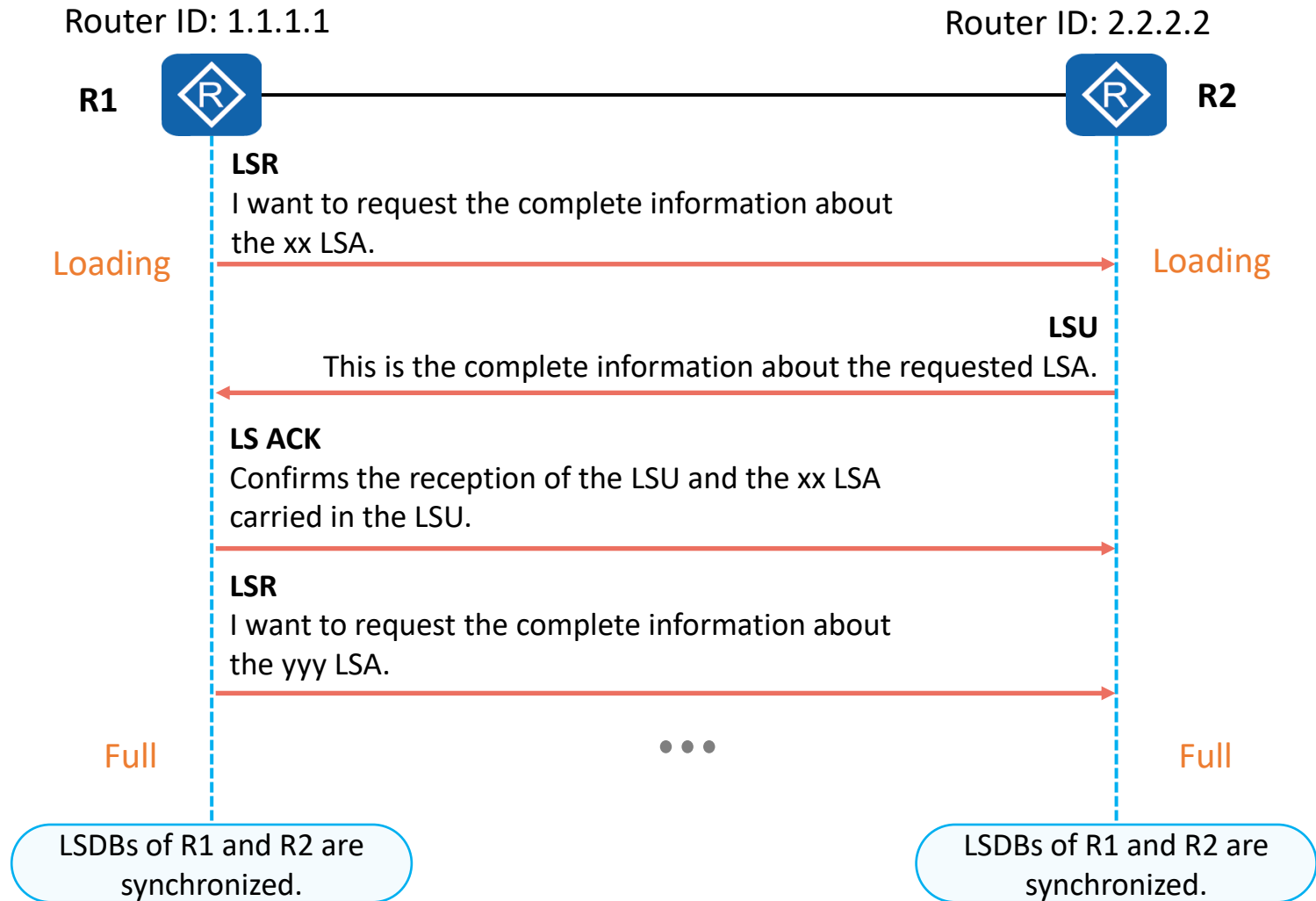


Process of Establishing an OSPF Adjacency - Steps 2 and 3





Process of Establishing an OSPF Adjacency - Step 4

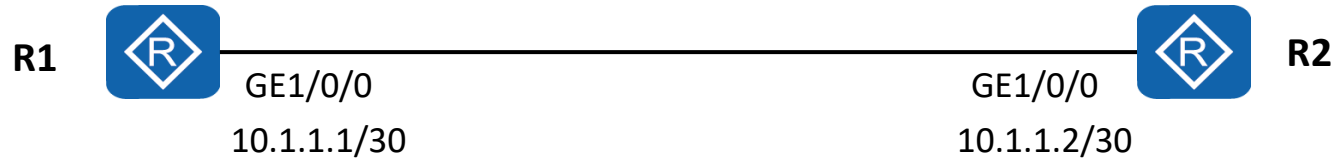




Review of the OSPF Neighbor Table

Router ID: 1.1.1.1

Router ID: 2.2.2.2



```
<R1> display ospf peer
OSPF Process 1 with Router ID 1.1.1.1
Neighbors
Area 0.0.0.0 interface 10.1.1.1(GigabitEthernet1/0/0)'s neighbors
Area 2.2.2.2 Address: 10.1.1.2 GR State: Normal
State: Full Mode:Nbr is Master Priority: 1
DR: 10.1.1.1 BDR: 10.1.1.2 MTU: 0
Dead timer due in 35 sec
Retrans timer interval: 5
Neighbor is up for 00:00:05
Authentication Sequence: [ 0 ]
```

Router ID of the
neighbor: 2.2.2.2

The neighbor status
is Full.

R1 discovers neighbors
in area 0 on GE 1/0/0.

The neighbor at 2.2.2.2
is the master.

Quiz: What is the DR/BDR in
the neighbor table?

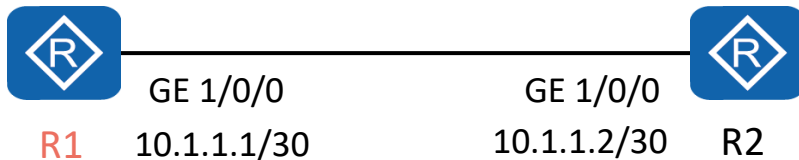


OSPF Network Types

- Before learning concepts of the DR and BDR, understand OSPF network types.
- The OSPF network type is a very important interface variable. This variable affects OSPF operations on interfaces. For example, it determines how to send OSPF packets and whether to elect a DR or BDR.
- The default OSPF network type of an interface depends on the data link layer encapsulation used by the interface.
- As shown in the figure, OSPF has four network types: broadcast, NBMA, P2MP, and P2P.

Router ID: 1.1.1.1

Router ID: 2.2.2.2



[R1-GigabitEthernet1/0/0] ospf network-type ?

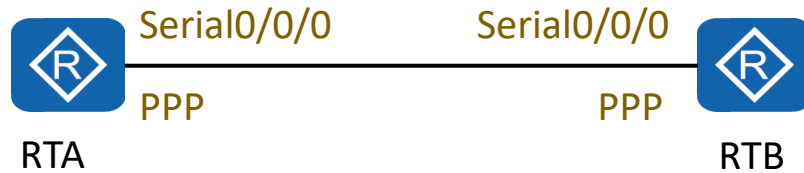
- | | |
|------------------|--|
| broadcast | Specify OSPF broadcast network |
| nbma | Specify OSPF NBMA network |
| p2mp | Specify OSPF point-to-multipoint network |
| p2p | Specify OSPF point-to-point network |



OSPF Network Types (1)

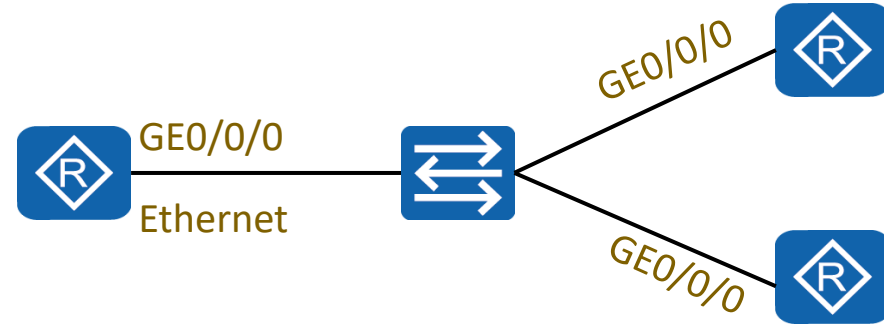
- Generally, the network types of OSPF interfaces at both ends of a link must be the same. Otherwise, the two interfaces cannot establish a neighbor relationship.
- An OSPF network type can be manually changed on an interface to adapt to different network scenarios. For example, you can change the BMA network type to P2P.

Point-to-Point (P2P)



- P2P indicates that only two network devices can be connected on a link.
- A typical example is a PPP link. When an interface uses PPP encapsulation, the default network type of the OSPF interface is P2P.

Broadcast Multiple Access (BMA)

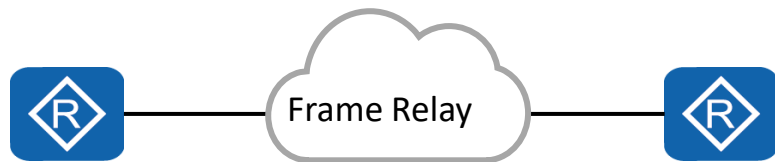


- BMA is also called broadcast. It refers to an environment that allows multiple devices to access and supports broadcast.
- A typical example is an Ethernet network. When an interface uses Ethernet encapsulation, the default network type of the OSPF interface is BMA.



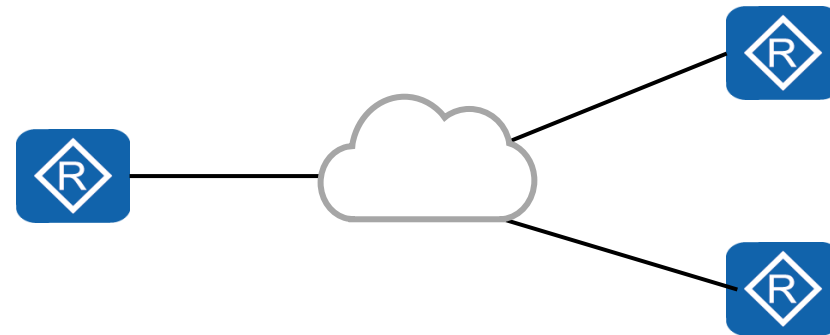
OSPF Network Types (2)

Non-Broadcast Multiple Access (NBMA)



- NBMA refers to an environment that allows multiple network devices to access but does not support broadcast.
- A typical example is a Frame Relay (FR) network.

Point-to-Multipoint (P2MP)

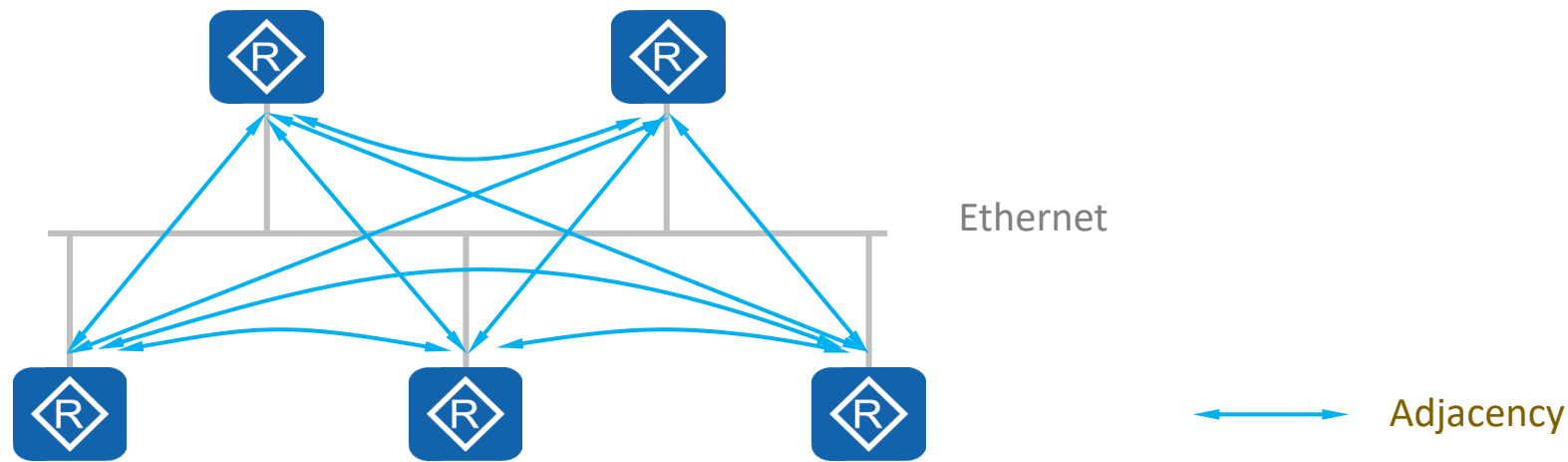


- A P2MP network is formed by bundling endpoints of multiple P2P links.
- No link layer protocol is considered as a P2MP network by default. This type must be manually changed from another network type.
- For example, a non-full-mesh NBMA network can be changed to a P2MP network.



Background of DR and BDR

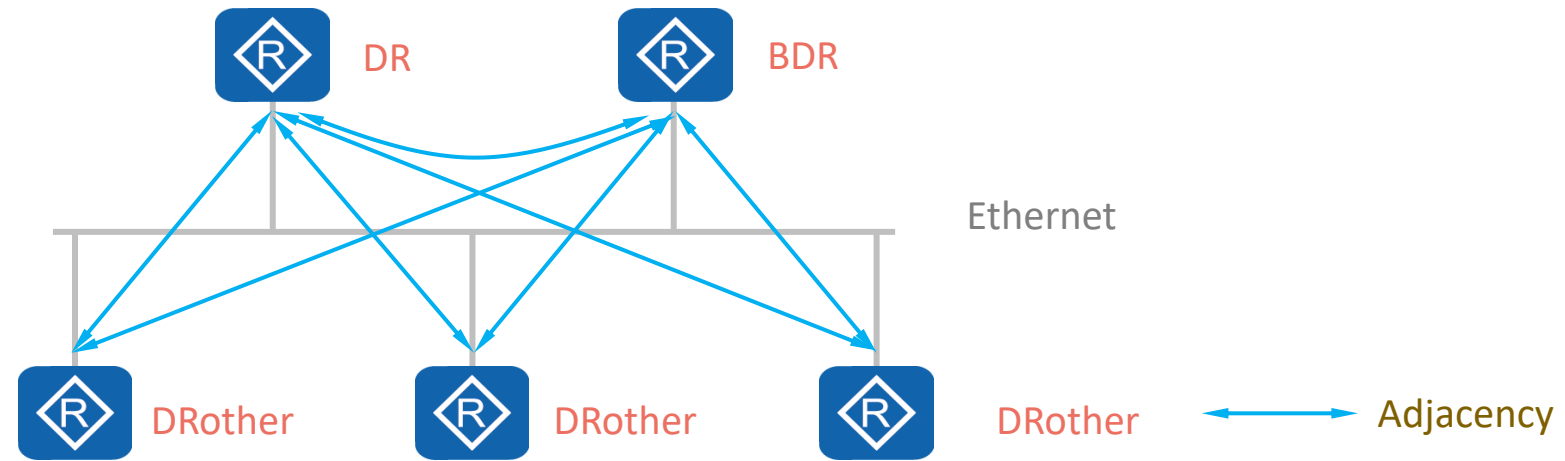
- Multi-access (MA) networks are classified into BMA and NBMA networks. Ethernet is a typical broadcast multi-access network.
- On an MA network, if each OSPF router establishes OSPF adjacencies with all the other routers, excessive OSPF adjacencies exist on the network, which increases the load on the devices and the number of OSPF packets flooded on the network.
- Once the network topology changes, LSA flooding on the network may waste bandwidth and device resources.





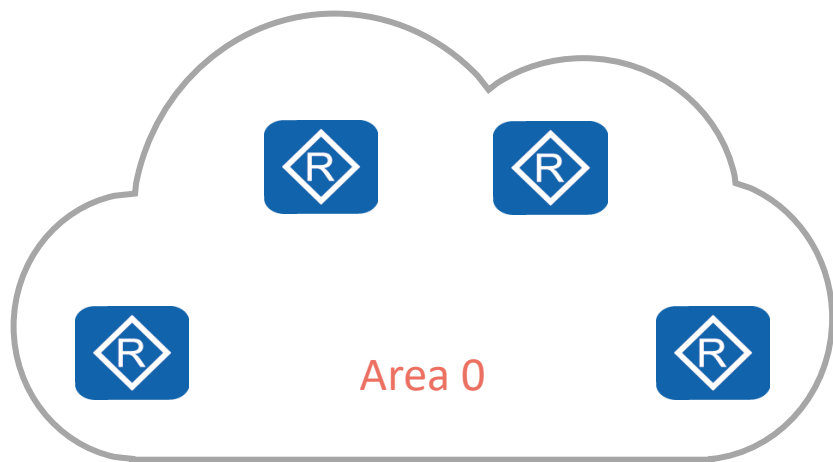
DR and BDR

- To optimize OSPF neighbor relationships on an MA network, the OSPF protocol specifies three types of OSPF routers: DR, BDR, and DRoother.
- Only the DR and BDR can establish adjacencies with other OSPF routers. DRoother does not establish OSPF adjacencies with one another, and their relationship is in the 2-way state.
- The BDR monitors the status of the DR and takes over the role of the DR if the existing DR fails.





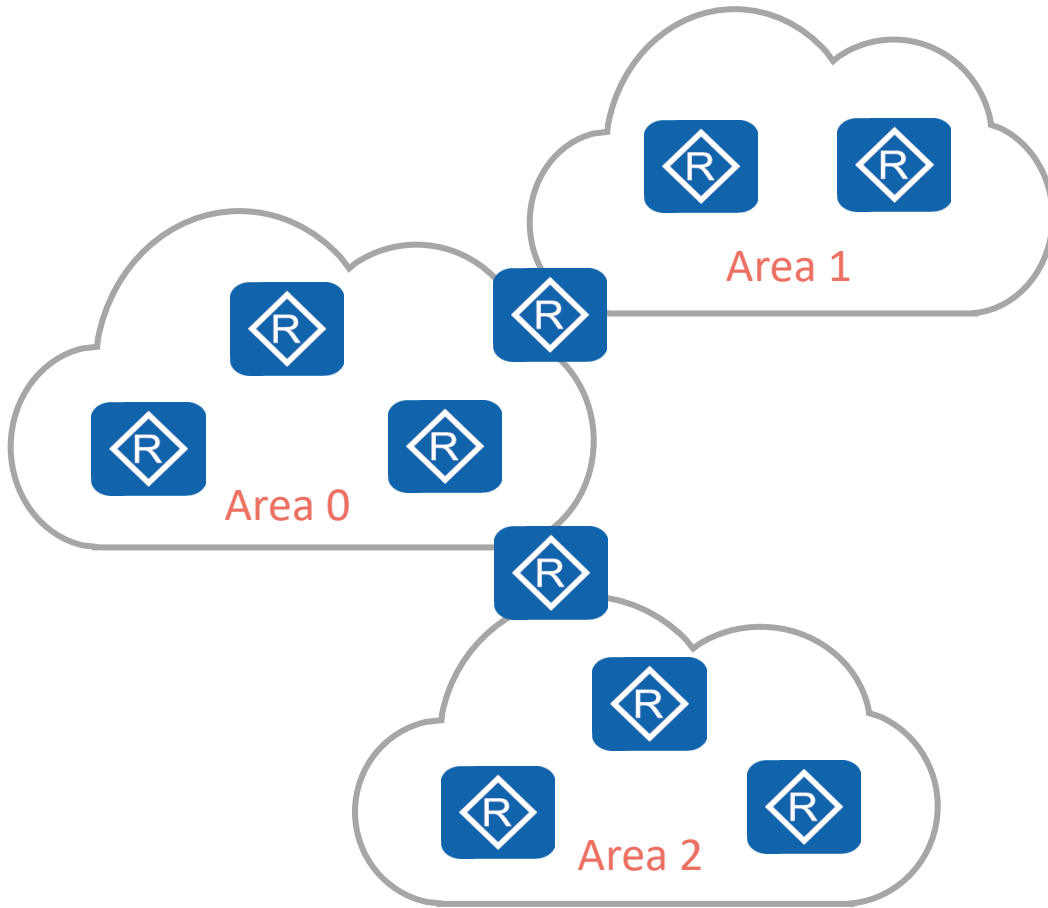
OSPF Domain and Single Area



- An OSPF domain is a network that consists of a series of contiguous OSPF network devices that use the same policy.
- An OSPF router floods LSAs in the same area. To ensure that all routers have the same understanding of the network topology, LSDBs need to be synchronized within an area.
- If there is only one OSPF area, the number of OSPF routers increases with the network scale. This causes the following problems:
 - The LSDB becomes larger and larger, and the size of the OSPF routing table increases. A large number of router resources are consumed, device performance deteriorates, and data forwarding is affected.
 - It is difficult to calculate routes based on a large LSDB.
 - When the network topology changes, LSA flooding and SPF recalculation on the entire network bring heavy loads.



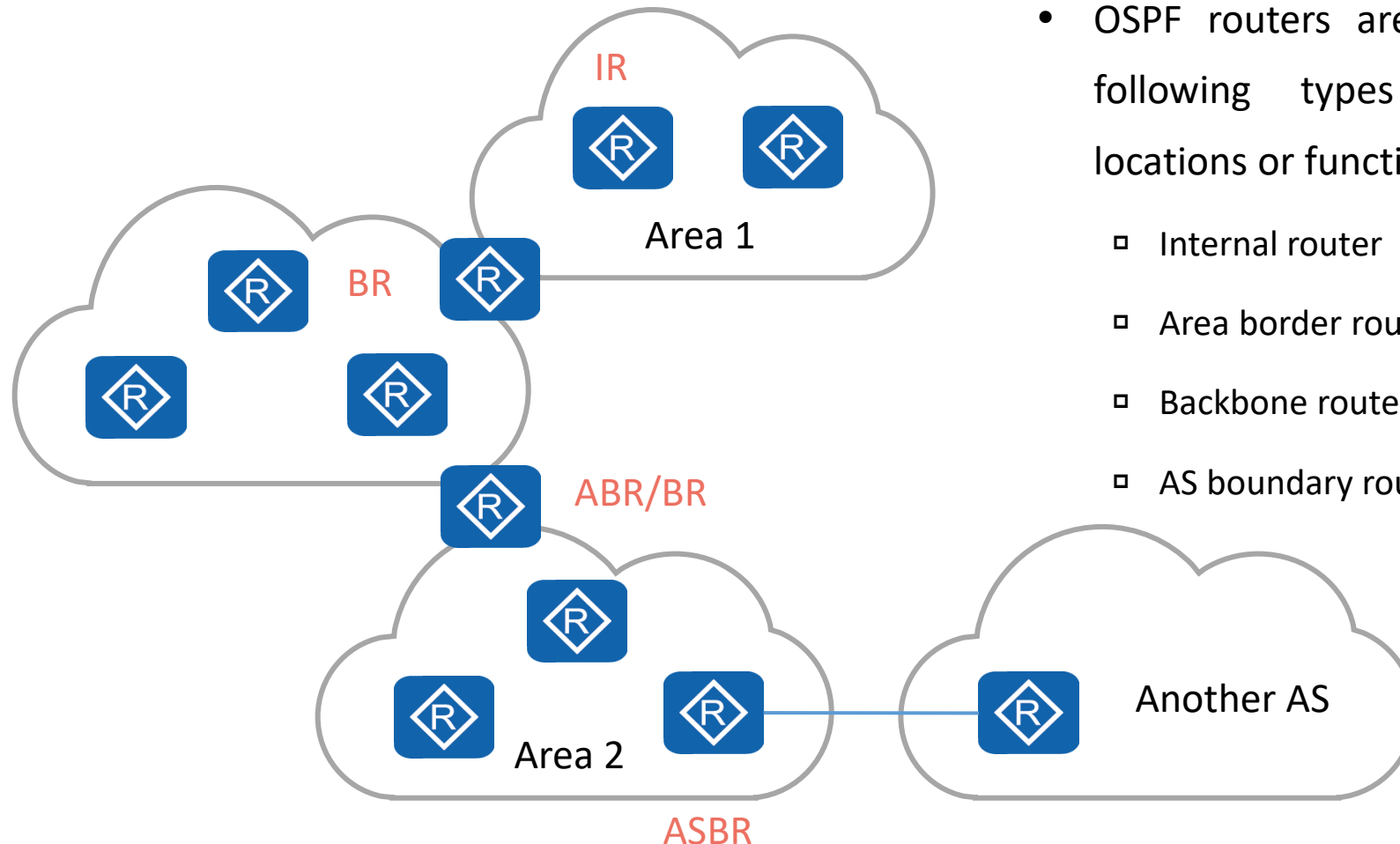
Multi-Area OSPF



- OSPF introduces the concept of area. An OSPF domain is divided into multiple areas to support larger-scale networking.
- The OSPF multi-area design reduces the flooding scope of LSAs and effectively controls the impact of topology changes within an area, optimizing the network.
- Routes can be summarized at the area border to reduce the size of the routing table.
- Multi-area improves network scalability and facilitates large-scale network construction.



Types of OSPF Routers

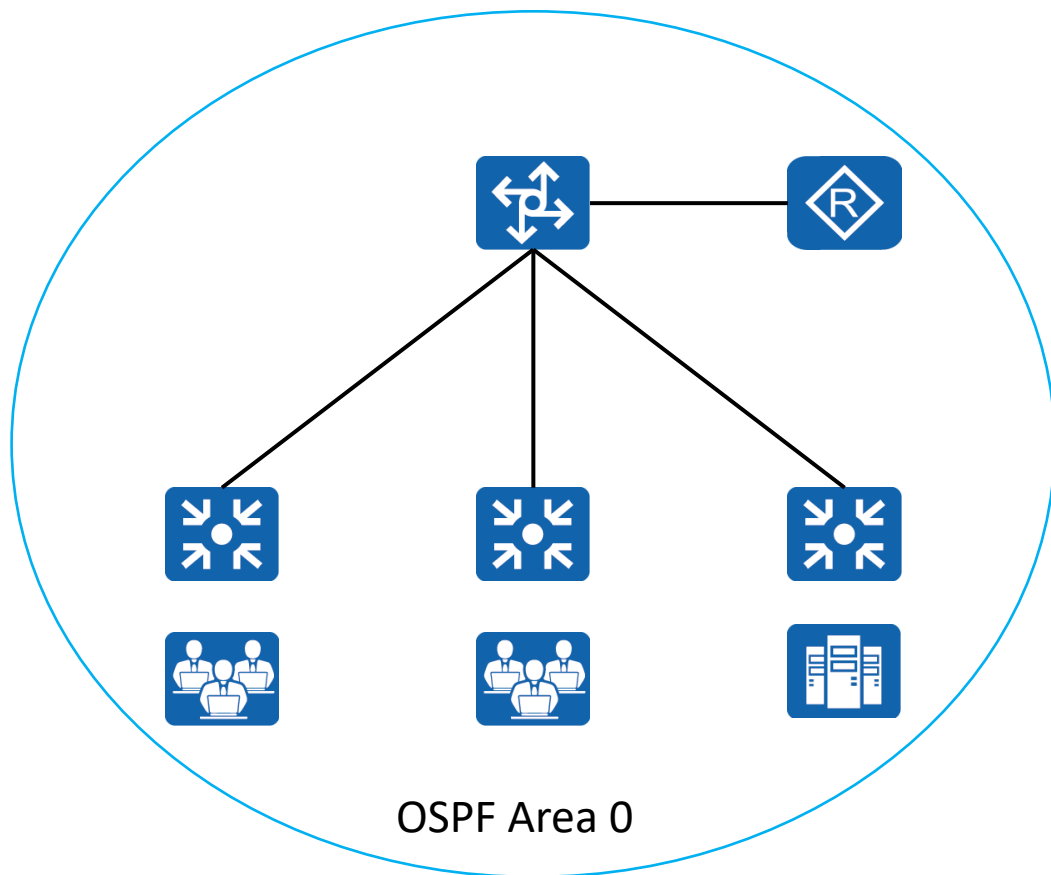


- OSPF routers are classified into the following types based on their locations or functions:

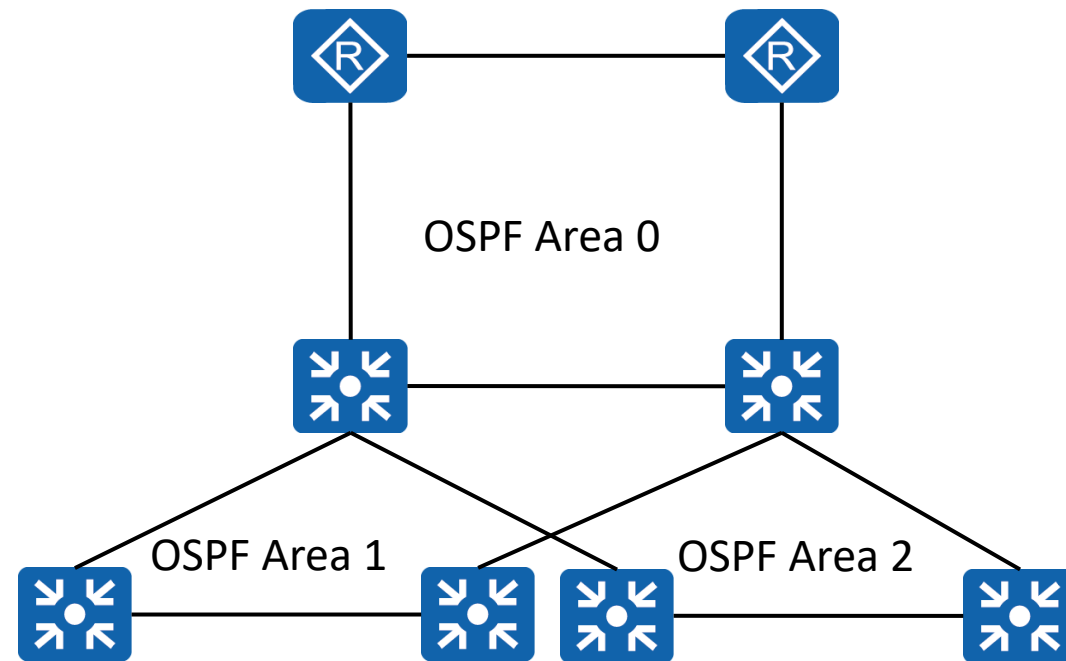
- Internal router
- Area border router (ABR)
- Backbone router
- AS boundary router (ASBR)



Typical OSPF Single-Area and Multi-Area Networking



Small- and medium-sized enterprise network (single area)



Large enterprise network (multiple areas)



Contents

1. OSPF Overview
2. OSPF Working Mechanism
- 3. Typical OSPF Configuration**



Basic OSPF Configuration Commands (1)

1. (System view) Create and run an OSPF process.

```
[Huawei] ospf [ process-id | router-id router-id ]
```

The *process-id* parameter specifies an OSPF process. The default process ID is 1. OSPF supports multiple processes. Multiple OSPF processes can separately run on the same device. The **router-id** command is used to manually specify the ID of a device. If no ID is specified, the system automatically selects the IP address of an interface as the device ID.

2. (OSPF view) Create an OSPF area and enter the OSPF area view.

```
[Huawei] area area-id
```

The **area** command creates an OSPF area and displays the OSPF area view.

The *area-id* value can be a decimal integer or in dotted decimal notation. If the value is an integer, it ranges from 0 to 4294967295.

3. (OSPF area view) Specify the interface that runs OSPF.

```
[Huawei-ospf-1-area-0.0.0.0] network network-address wildcard-mask
```

The **network** command specifies the interface that runs OSPF and the area to which the interface belongs. The *network-address* parameter specifies the network segment address of the interface. The *wildcard-mask* parameter is the wildcard of an IP address, which is equivalent to the reverse mask of the IP address (0 is converted to 1, and 1 to 0). For example, 0.0.0.255 indicates that the mask length is 24 bits.



Basic OSPF Configuration Commands (2)

4. (Interface view) Set an OSPF interface cost.

```
[Huawei-GE1/0/1] ospf cost cost
```

The **ospf cost** command sets a cost for an OSPF interface. By default, OSPF automatically calculates the cost of an interface based on the interface bandwidth. The cost value is an integer ranging from 1 to 65535.

5. (OSPF view) Set an OSPF bandwidth reference value.

```
[Huawei-ospf-1] bandwidth-reference value
```

The **bandwidth-reference** command sets a bandwidth reference value that is used to calculate interface costs. The value ranges from 1 to 2147483648, in Mbit/s. The default value is 100 Mbit/s.

6. (Interface view) Set the priority of an interface for DR election.

```
[Huawei-GigabitEthernet0/0/0/0] ospf dr-priority priority
```

The **ospf dr-priority** command sets a priority for an interface that participates in DR election. A larger value indicates a higher priority. The value ranges from 0 to 255.

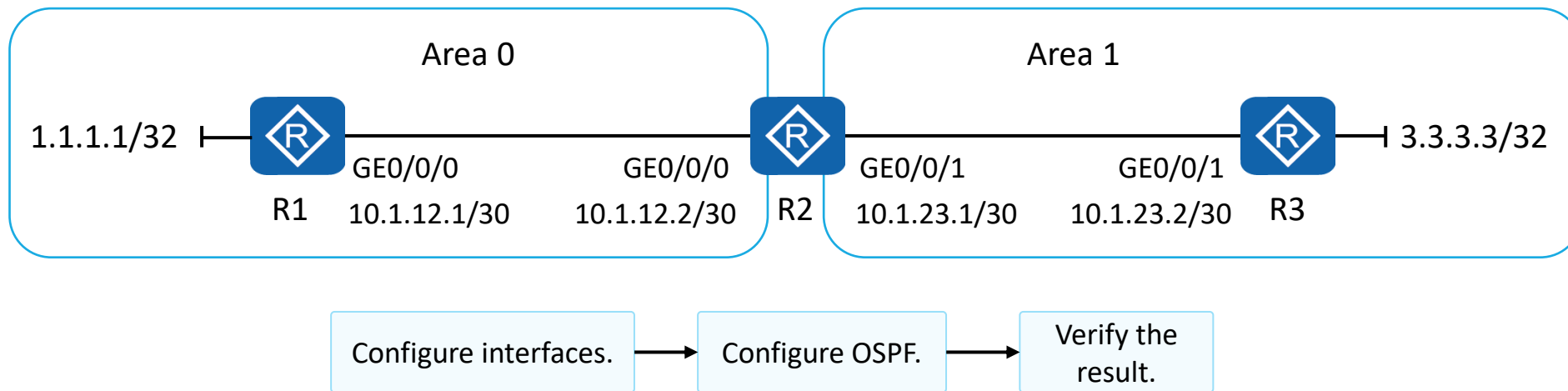


OSPF Configuration Example

Description:

- There are three routers, R1, R2, and R3. R1 and R3 are connected to networks 1.1.1.1/32 and 3.3.3.3/32 (simulated by Loopback 0), respectively. OSPF needs to be used to implement interworking between the two networks.

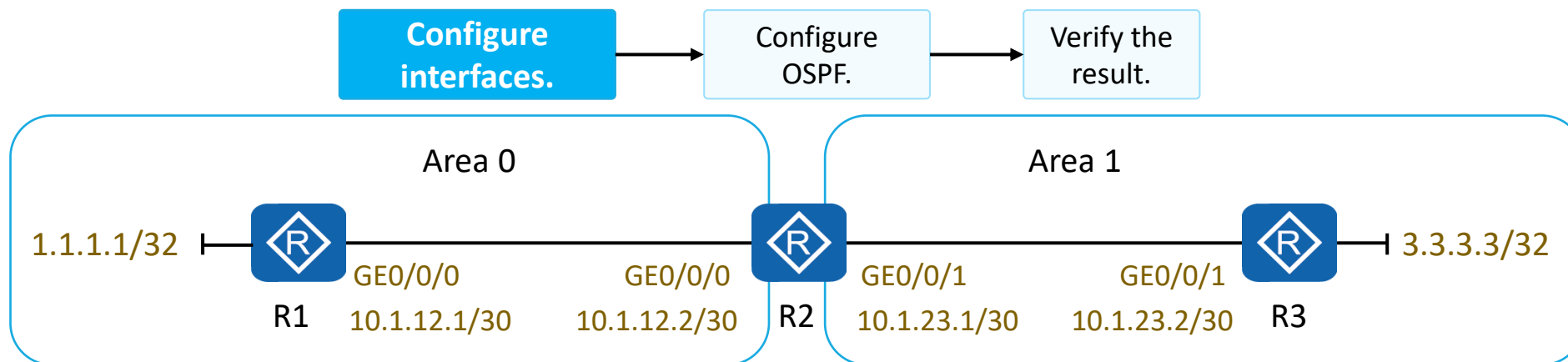
Detailed topology was as follows:



- The configuration process consists of three steps: configuring device interfaces, configuring OSPF, and verifying the result.



OSPF Configuration Example - Configuring Interfaces



- Set IP addresses for R1's, R2's, and R3's interfaces according to the plan.

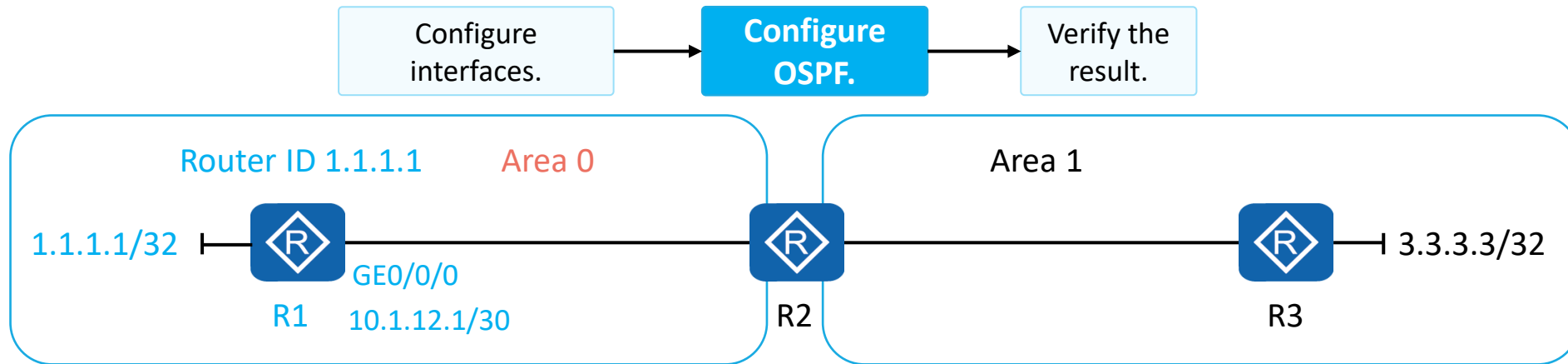
```
# Configure interfaces of R1.  
[R1] interface LoopBack 0  
[R1-LoopBack0] ip address 1.1.1.1 32  
[R1-LoopBack0] interface GigabitEthernet 0/0/0  
[R1-GigabitEthernet0/0/0] ip address 10.1.12.1 30
```

```
# Configure interfaces of R3.  
[R3] interface LoopBack 0  
[R3-LoopBack0] ip address 3.3.3.3 32  
[R3-LoopBack0] interface GigabitEthernet 0/0/1  
[R3-GigabitEthernet0/0/1] ip address 10.1.23.2 30
```

Assign IP addresses for GE0/0/0 and GE0/0/1 on R2. For details, see comment in this slide.



OSPF Configuration Example - Configuring OSPF (1)



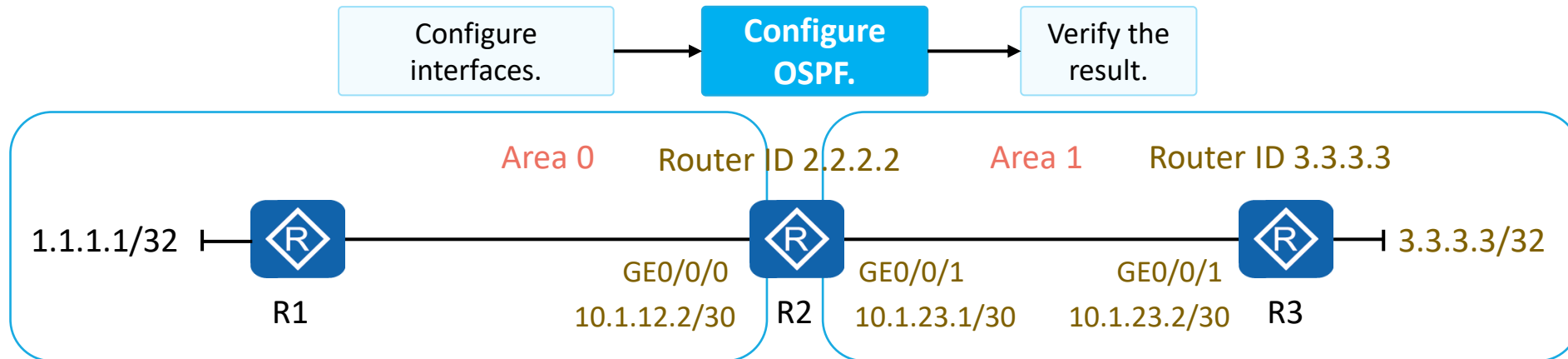
- Planned OSPF parameters: The OSPF process ID is 1. Router IDs of R1, R2, and R3 are 1.1.1.1, 2.2.2.2, and 3.3.3.3 respectively.
- Procedure:
 - Create and run an OSPF process.
 - Create an OSPF area and enter the OSPF area view.
 - Specify the interface that runs OSPF.

```
# Configure OSPF on R1.  
[R1] ospf 1 router-id 1.1.1.1  
[R1-ospf-1] area 0  
[R1-ospf-1-area-0.0.0.0] network 1.1.1.1 0.0.0.0  
[R1-ospf-1-area-0.0.0.0] network 10.1.12.0 0.0.0.3
```

← An inverse mask is specified here.



OSPF Configuration Example - Configuring OSPF (2)



- When configuring OSPF multi-area, be sure to advertise the route destined for a network segment that responds to a specified area.

Configure OSPF on R2.

```
[R2] ospf 1 router-id 2.2.2.2
```

```
[R2-ospf-1] area 0
```

```
[R2-ospf-1-area-0.0.0.0] network 10.1.12.0 0.0.0.3
```

```
[R2-ospf-1-area-0.0.0.0] area 1
```

```
[R2-ospf-1-area-0.0.0.1] network 10.1.23.0 0.0.0.3
```

Configure OSPF on R3.

```
[R3] ospf 1 router-id 3.3.3.3
```

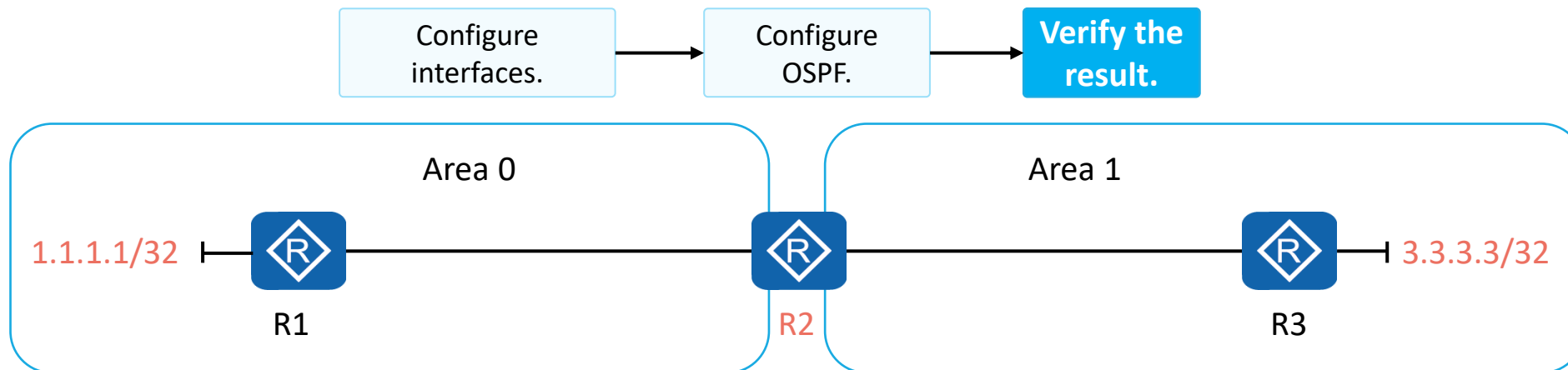
```
[R3-ospf-1] area 1
```

```
[R3-ospf-1-area-0.0.0.1] network 3.3.3.3 0.0.0.0
```

```
[R3-ospf-1-area-0.0.0.1] network 10.1.23.0 0.0.0.3
```




OSPF Configuration Example - Verification (1)



- Check the OSPF neighbor table on R2.

<R2> display ospf peer brief

OSPF Process 1 with Router ID 2.2.2.2

Peer Statistic Information

Area Id	Interface	Neighbor id	State
0.0.0.0	GigabitEthernet0/0/0	1.1.1.1	Full
0.0.0.1	GigabitEthernet0/0/1	3.3.3.3	Full

Area ID of a neighbor

Neighbor status
Verify that the neighbor status is **Full**, indicating that the adjacency has been established successfully.



OSPF Configuration Example - Verification (2)

- Check the routing table on R1 and ping 3.3.3.3 from 1.1.1.1.

Route to
3.3.3.3/32
learned using
OSPF

Set the source
IP address to
1.1.1.1 and
ping 3.3.3.3.

```
<R1>display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing tables: Public

Destinations : 10 Routes : 10

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
1.1.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
3.3.3.3/32	OSPF	10	2	D	10.1.12.2	GigabitEthernet 0/0/0
10.1.12.0/30	Direct	0	0	D	10.1.12.1	GigabitEthernet 0/0/0

...

```
<R1>ping -a 1.1.1.1 3.3.3.3
```

PING 3.3.3.3: 56 data bytes, press CTRL_C to break

Reply from 3.3.3.3: bytes = 56 Sequence = 1 ttl = 254 time = 50 ms

...



Quiz

1. (Multiple) In the process of establishing OSPF neighbor relationships and adjacencies, which of the following states are stable? ()
 - A. Exstart
 - B. Two-way
 - C. Exchange
 - D. Full

2. (Multiple) In which of the following situation will the establishment of adjacencies between routers be triggered? ()
 - A. Two routers on a point-to-point link
 - B. DR and BDR on a broadcast network
 - C. DRouter and DRouter on an NBMA network
 - D. BDR and DRouter on a broadcast network



Summary

- OSPF is a widely used routing protocol on the live network. This presentation describes basic concepts, application scenarios, and basic configurations of OSPF.
- The router ID, area, OSPF neighbor table, LSDB table, and OSPF routing table are basic OSPF concepts. Describe the establishment of OSPF neighbor relationships and adjacencies, which helps you better understand the link-state routing protocol.
- OSPF has more interesting details, such as LSA types, the SPF calculation process, and the OSPF special area. For more OSPF information, please continue your Huawei HCIP-Datacom certification courses.

The image features a blue-tinted background showing silhouettes of several groups of business professionals in a modern office environment. They are standing on a reflective floor, and their reflections are visible. The background includes architectural elements like glass walls and windows, suggesting a high-rise building. Overlaid on this scene is the text "Thank You" and the website address "www.huawei.com".

Thank You

www.huawei.com

OSPF

Open Shortest Path First (OSPF) Characteristics

- Open standard
- Establishes **adjacencies** with other routers
- Sends **Link State Advertisements (LSAs)** to other routers in an area
- Constructs a **Link State Database** from received LSAs
- Runs the **Dijkstra Shortest Path First (SPF) Algorithm** to determine the shortest path to a network
- Attempts to inject the best path for each network into a router's IP routing table

OSPF Terminology

- **Hello:** A protocol used to discover OSPF neighbors and confirm reachability to those neighbors (also used in the election of a Designated Router)
- **Link State Advertisement (LSA):** Information a router sends and receives about network reachability (used to construct a router's Link State Database)
- **Link State Update (LSU):** A packet that carries LSAs
- **Link State Request (LSR):** Used by a router to request specific LSA information from a neighbor
- **Link State Acknowledgement (LSAck):** Used by a router to confirm it received an LSU

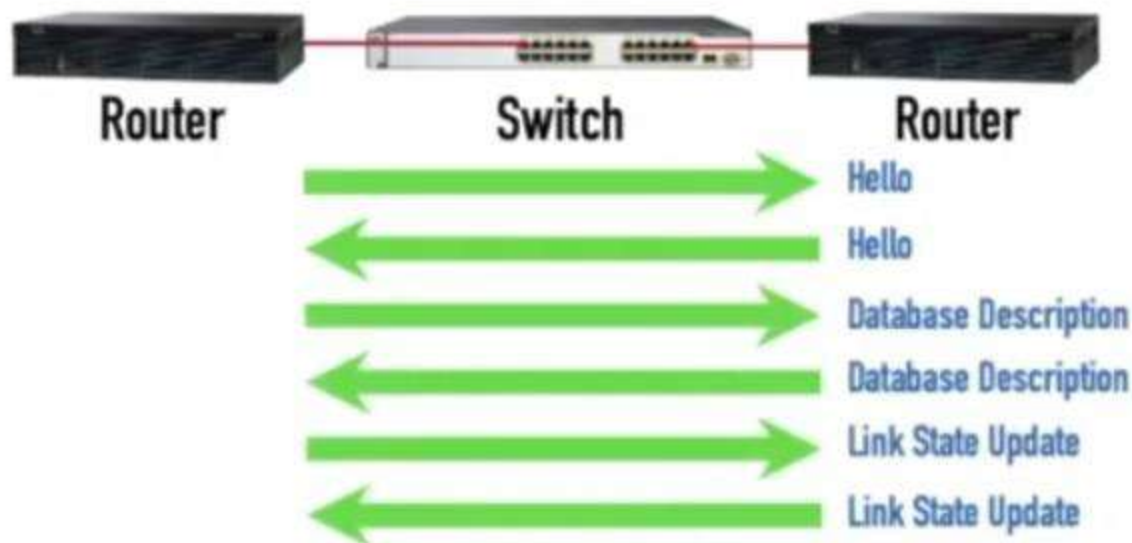
Neighborhood vs. Adjacencies

Neighbors are routers that:

- Reside on the same network link
- Exchange Hello messages

Adjacencies are routers that:

- Are neighbors
- Have exchanged Link State Updates (LSUs) and Database Description (DD) packets



Neighborhood Requirements

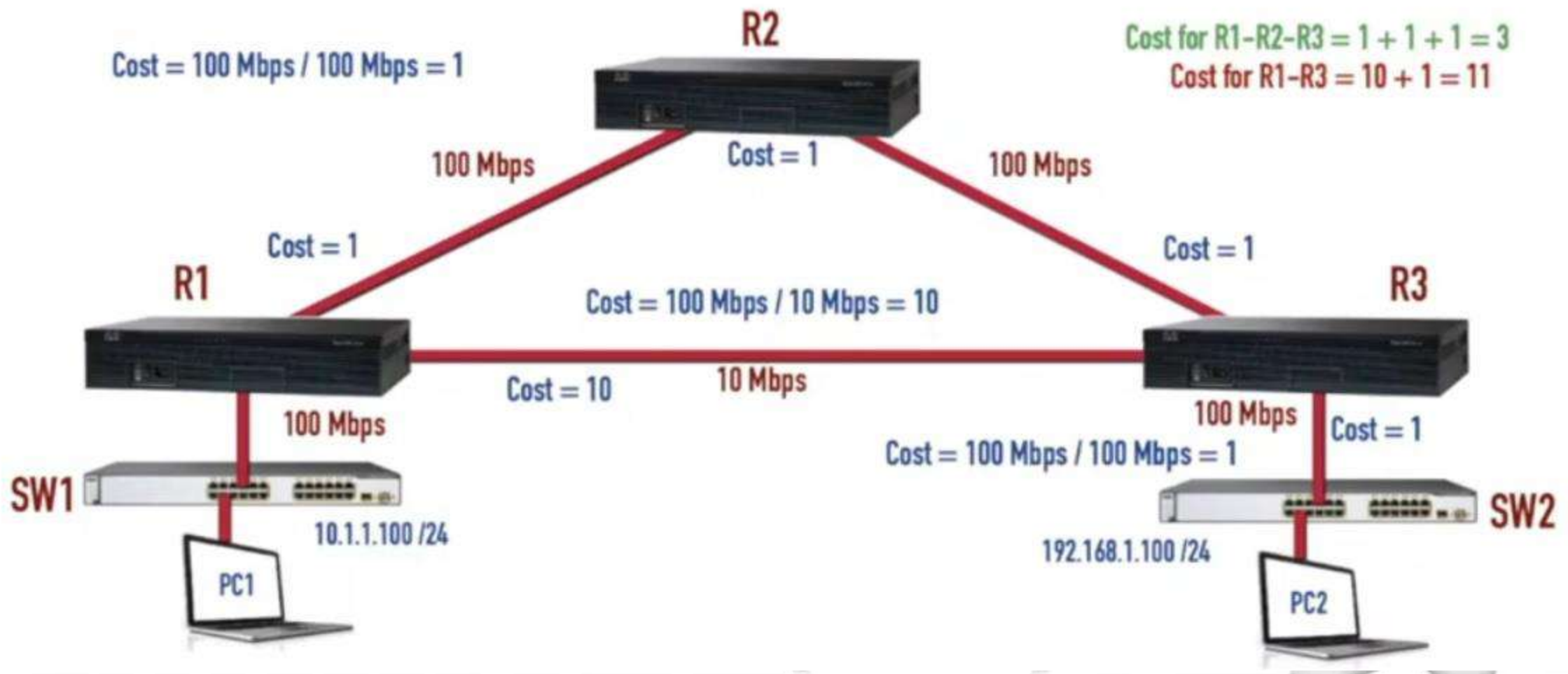
- Matching Area
- Matching Authentication
- Matching Subnet
- Matching Timers
- Matching Stub Flags
- Matching MTU (EXSTART/EXCHANGE State)



OSPF Cost

The default reference bandwidth is 100,000,000 bits per second (100 Mbps).

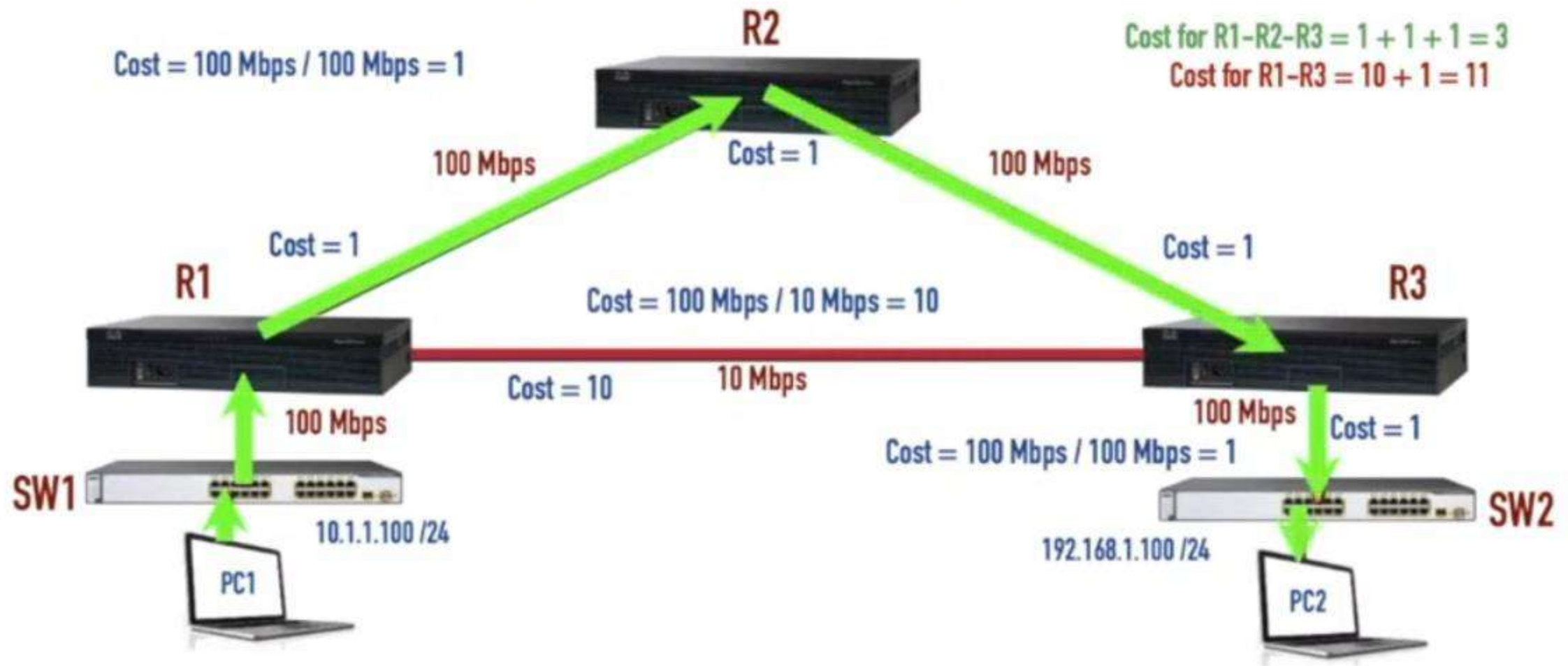
$$\text{Cost} = \text{Reference BW} / \text{Interface BW}$$



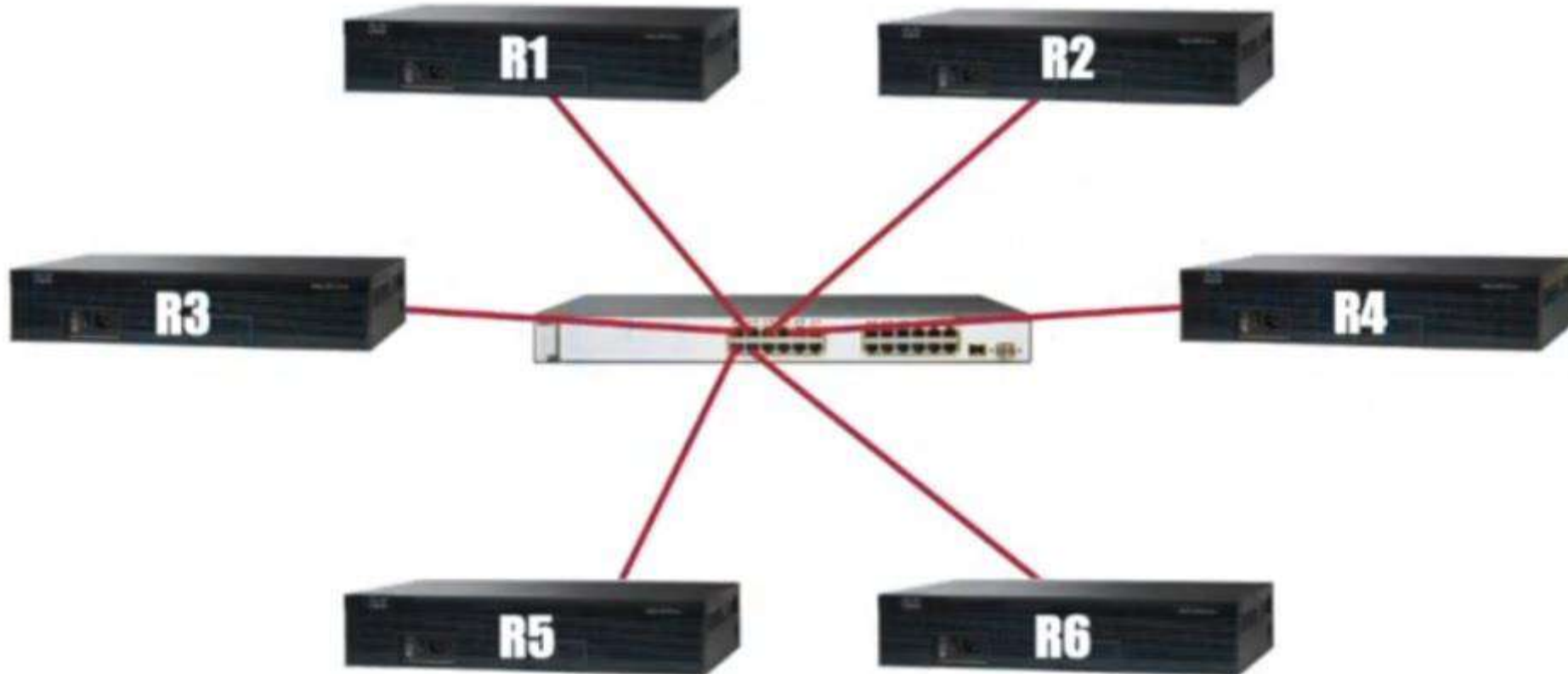
OSPF Cost

The default reference bandwidth is 100,000,000 bits per second (100 Mbps).

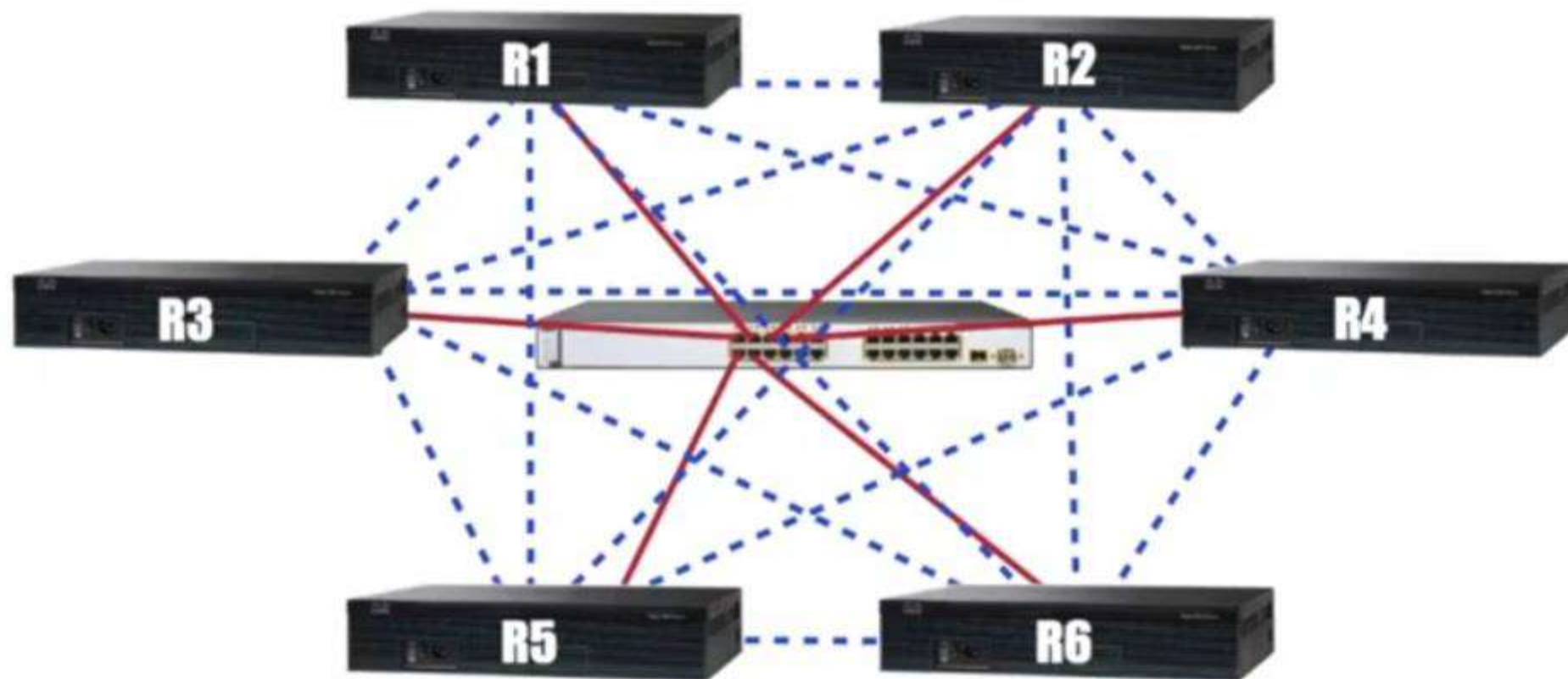
$$\text{Cost} = \text{Reference BW} / \text{Interface BW}$$



The Need for Designated Routers



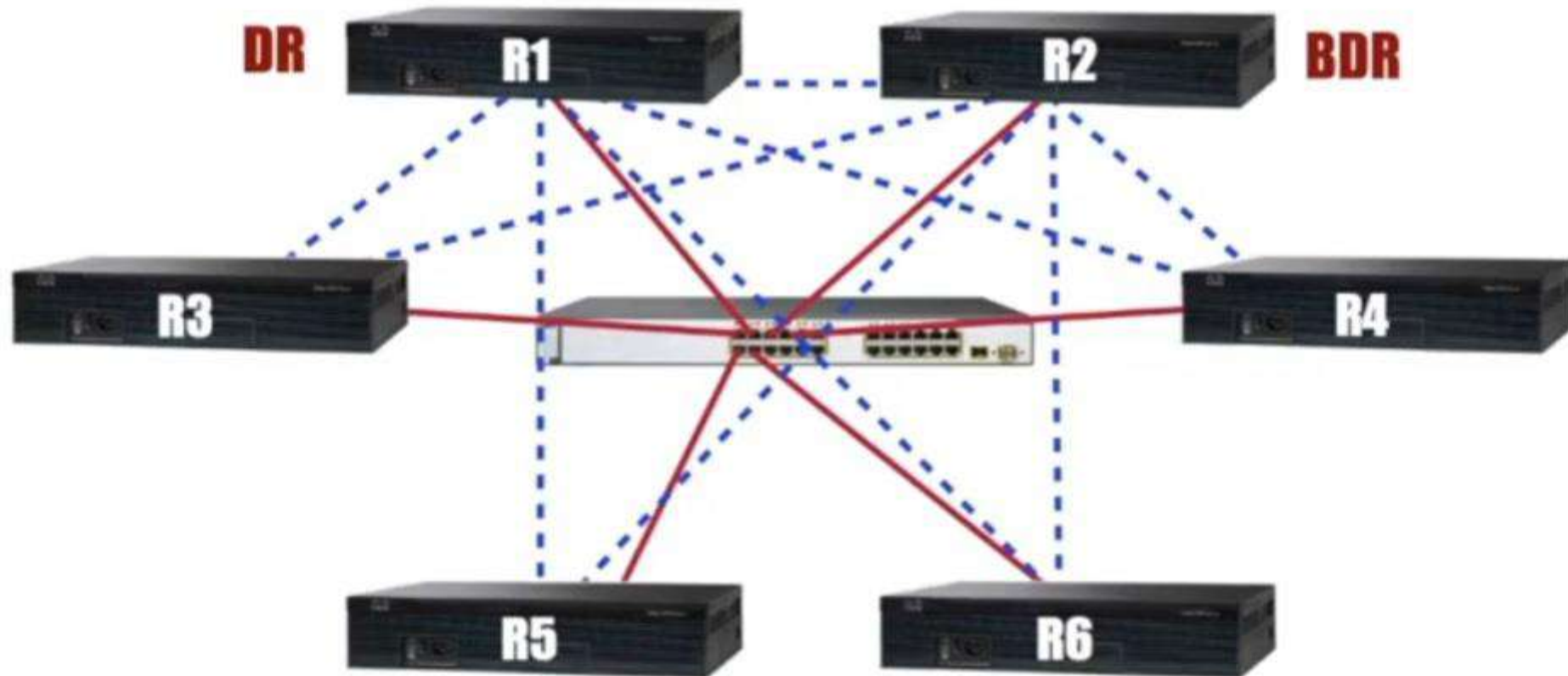
The Need for Designated Routers



of Adjacencies = $[n * (n - 1)] / 2$, where n is the number of routers.

The Need for Designated Routers

Adjacencies only need to be formed with the DR and BDR.



- 224.0.0.5 or FF02::5 – All OSPF routers
- 224.0.0.6 or FF02::6 – All designated routers

DR and BDR Election



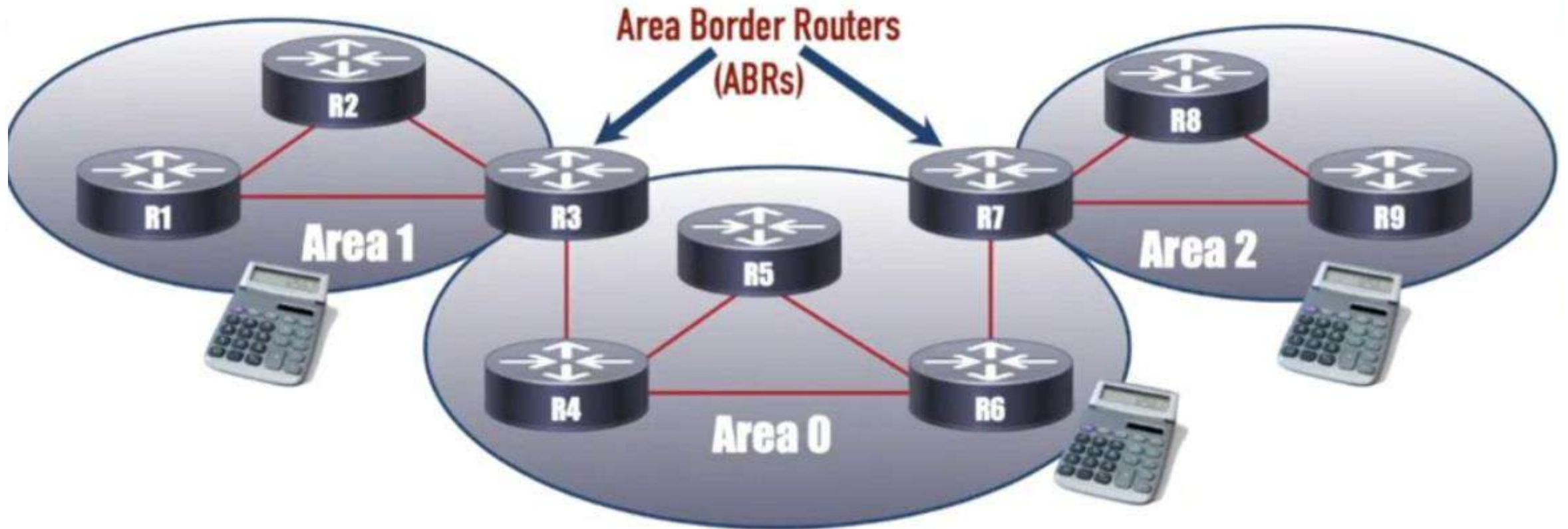
Highest Router Priority Wins

- Carried in Hello packet
- Configured in interface configuration mode:
 - `Router(config-if)# ip ospf priority number`
 - A priority of 0 prevents a router from participating in the election.

TIE BREAKER: Highest Router ID Wins

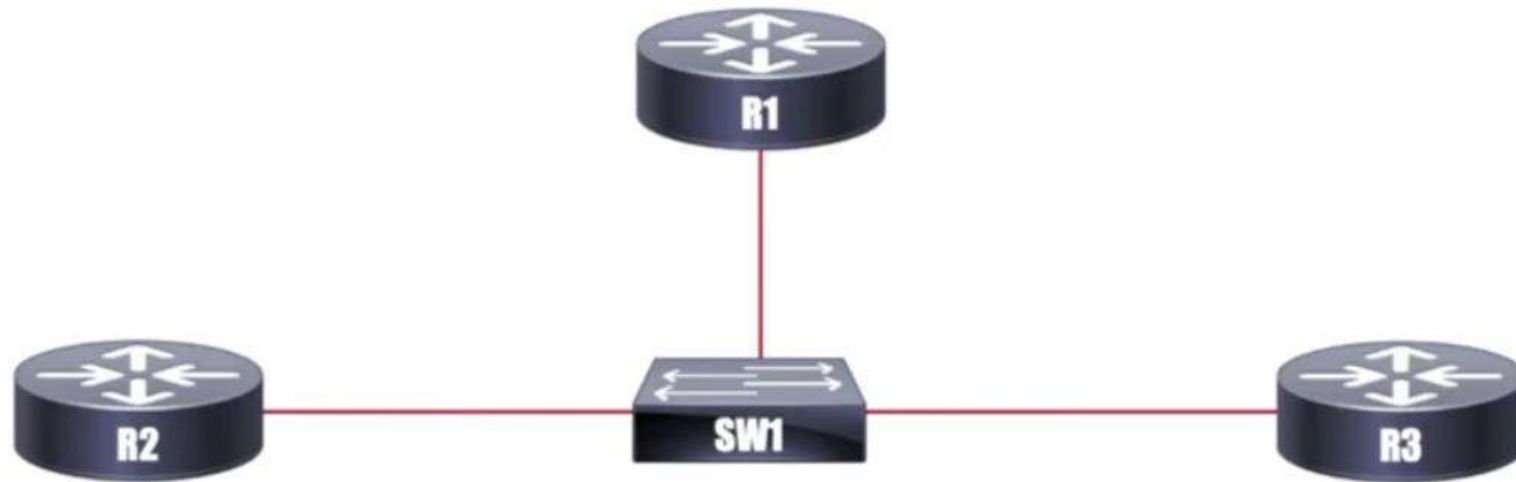
- Configured in router configuration mode:
 - `Router(config-router)# router-id id`
- If there's no configured Router ID, the highest IP address on a Loopback interface wins.
- If there's no Loopback interface, the highest IP address on an interface that's up wins.

OSPF Areas



Multi-Area OSPF networks must have a **Backbone Area** numbered **0** or **0.0.0.0**

Broadcast Network Type



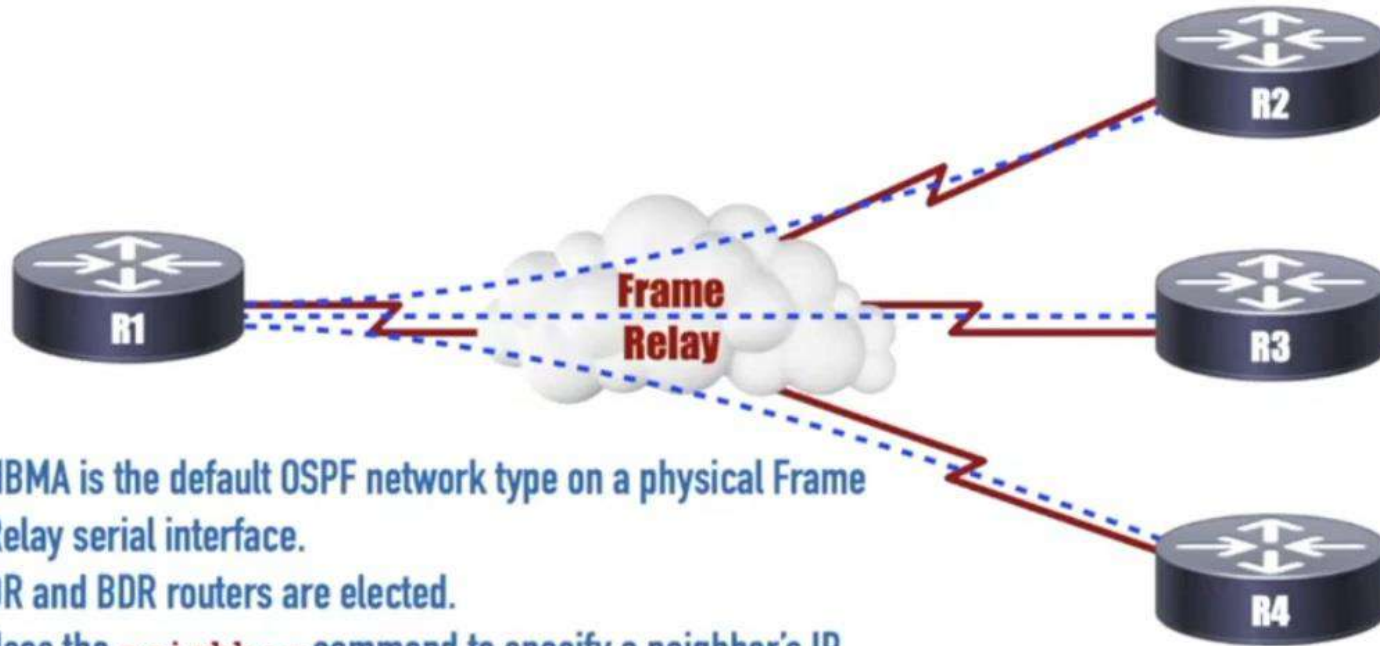
- Broadcast is the default OSPF network type for any Ethernet interface.
- DR and BDR routers are elected.
- Default HELLO interval: 10 seconds.

Point-to-Point Network Type



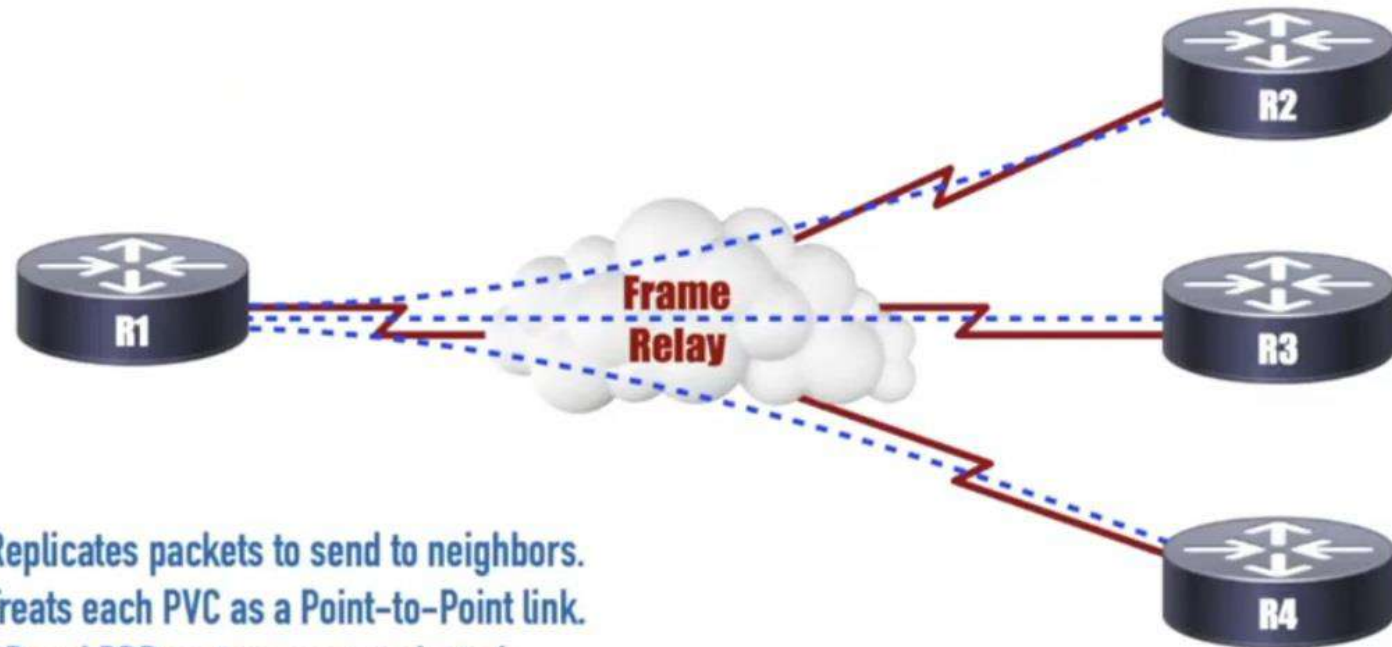
- Point-to-Point is the default OSPF network type on a non-Frame Relay serial interface.
- DR and BDR routers are not elected.
- Default HELLO interval: 10 seconds.

Non-Broadcast (NBMA) Network Type



- NBMA is the default OSPF network type on a physical Frame Relay serial interface.
- DR and BDR routers are elected.
- Uses the **neighbor** command to specify a neighbor's IP address.
- Default HELLO interval: 30 seconds.

Point-to-Multipoint Network Type



- Replicates packets to send to neighbors.
- Treats each PVC as a Point-to-Point link.
- DR and BDR routers are not elected.
- Default HELLO interval: 30 seconds.

OSPF Timers

- **Hello Timer:** The interval (in seconds) at which a router sends Hello messages out of an OSPF-enabled interface
- **Dead Timers:** The time (in seconds) that an OSPF-enabled interface will wait to receive a Hello message from a neighbor, before considering that neighbor to be down
- $\text{Dead Timer} = \text{Hello Timer} * 4$
- Timers must match between neighbors



OSPF Neighbor States



- **Down:** A router interface hasn't received any Hellos (within the Dead Interval), but it can send Hellos
- **Attempt:** This is only seen in an NBMA network, where an interface is sending Hellos to a configured neighbor, but has not received Hellos from that neighbor.
- **Init:** An interface received a Hello packet from a neighbor, but it didn't contain the neighbor's Router ID. That means the neighbor hasn't received a Hello from us.
- **2-Way:** There is bidirectional communication between the routers, and a neighborhood has been formed. Then, a decision is made whether or not to form an adjacency.
- **Exstart:** Routers on Broadcast or NBMA networks are elected.
- **Exchange:** Adjacent routers exchange Database Descriptor (DBD) packets, but not the contents of the Link State Database.
- **Loading:** Based on what was learned by the Database Descriptor packets, routers send LSRs to request any needed information.
- **Full:** Routers are fully adjacent with each other.

Route Summarization

192.168.0.0 /24

192.168.1.0 /24

192.168.2.0 /24

192.168.3.0 /24

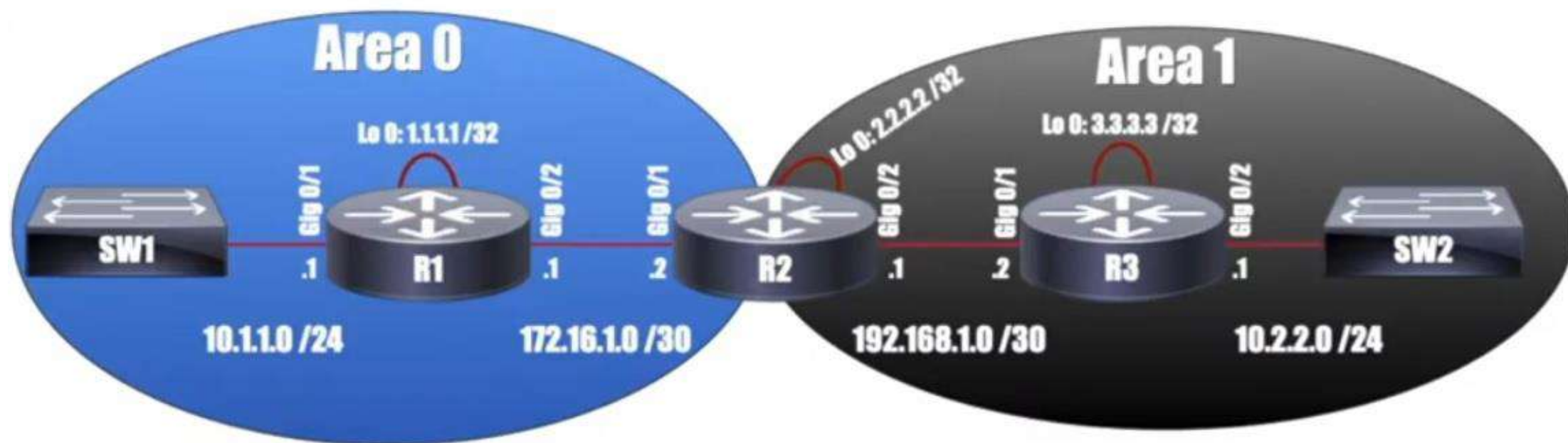


Route Summarization

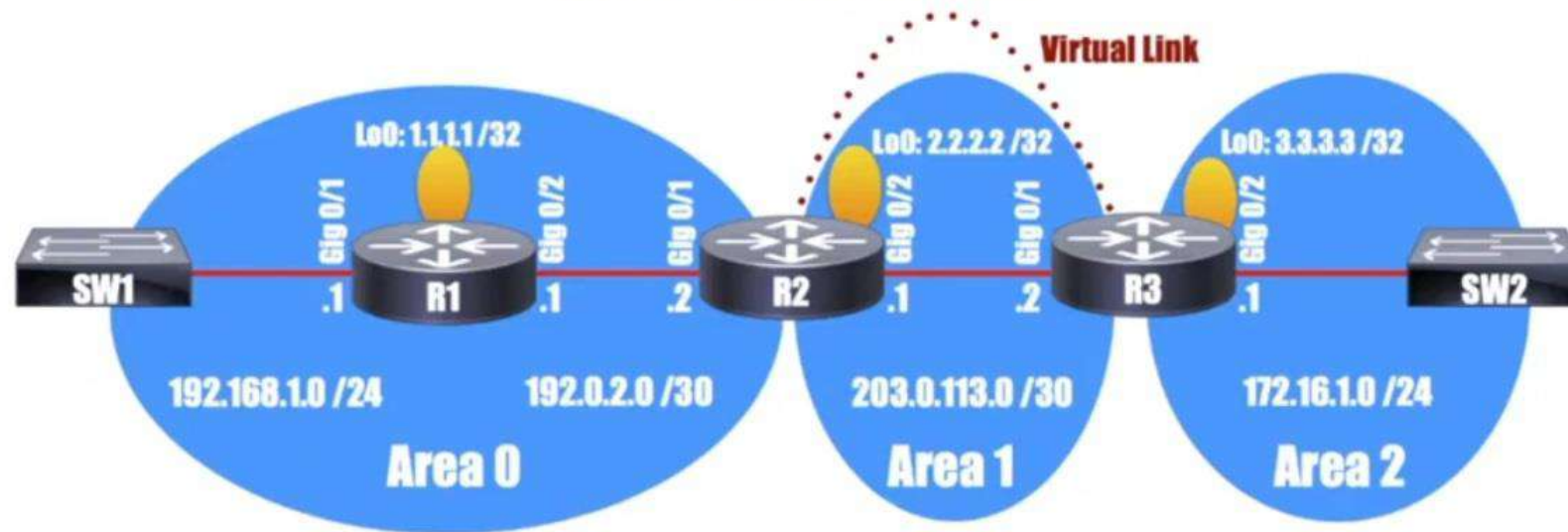
Network Address	Octet 1	Octet 2	Octet 3	Octet 4
192.168.0.0 /24	11000000	10101000	00000000	00000000
192.168.1.0 /24	11000000	10101000	00000001	00000000
192.168.2.0 /24	11000000	10101000	00000010	00000000
192.168.3.0 /24	11000000	10101000	00000011	00000000
All Networks Have Their First 22 Bits In Common				
Subnet Mask (Binary)	11111111	11111111	11111100	00000000
Subnet Mask (Decimal)	255	255	252	0
Network Address (Binary)	11000000	10101000	00000000	00000000
Network Address (Decimal)	192	168	0	0

Route Summarization





OSPF Virtual Links



- All OSPF areas must be adjacent to the backbone area (Area 0 or Area 0.0.0.0) physically or logically.
- An OSPF Virtual Link logically spans a transit area (Area 1 in this topology).
- Routing updates are tunneled, while data is sent natively, resulting in no tunnel overhead for data.



Ethernet Switching Basics



Foreword

- Data transmission on networks must comply with certain standards. Ethernet protocols define how data frames are transmitted over an Ethernet network. Understanding Ethernet protocols is the basis for fully understanding communication at the data link layer. An Ethernet switch is the main device for implementing data link layer communication. It is essential to understand how an Ethernet switch works.
- This course describes the concepts related to Ethernet protocols, MAC address types, and working process and mechanism of Layer 2 switches.



Objectives

- On completion of this course, you will be able to:
 - Describe the basic concepts of an Ethernet network.
 - Distinguish MAC address types.
 - Get familiar with the working process of a Layer 2 switch.
 - Get familiar with the structure and generation process of a MAC address table.



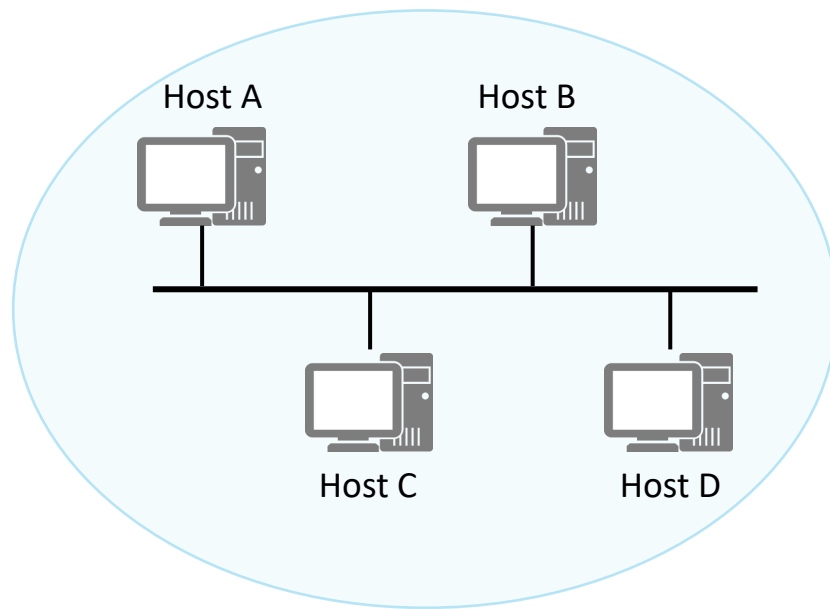
Contents

- 1. Overview of Ethernet Protocols**
2. Overview of Ethernet Frames
3. Overview of Ethernet Switches
4. Process of Data Communication Within a Network Segment

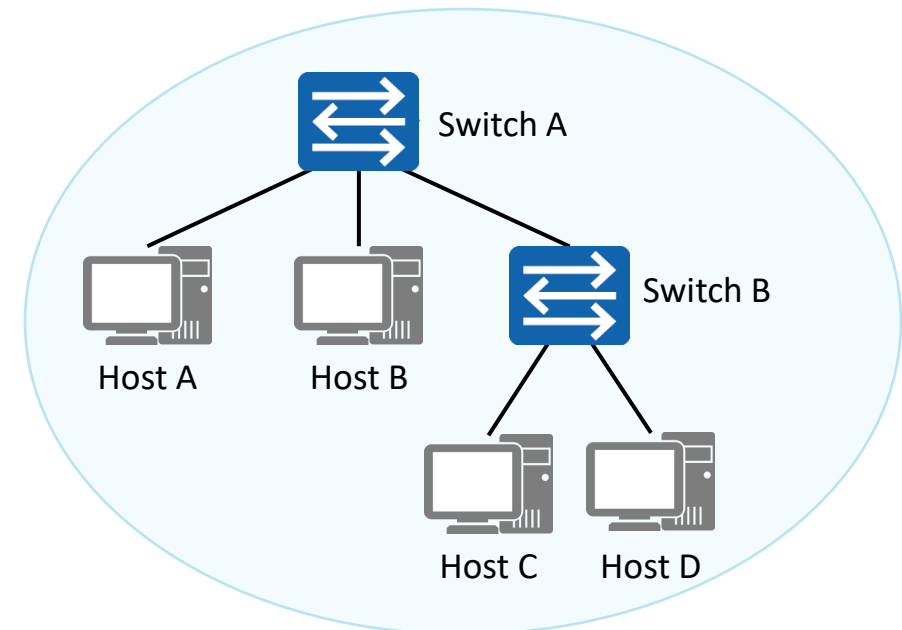


Ethernet Protocols

- Ethernet is the most common communication protocol standard used by existing local area networks (LANs). It defines the cable types and signal processing methods that are used on a LAN.
- An Ethernet network is a broadcast network built based on the carrier sense multiple access/collision detection (CSMA/CD) mechanism.



Early Ethernet

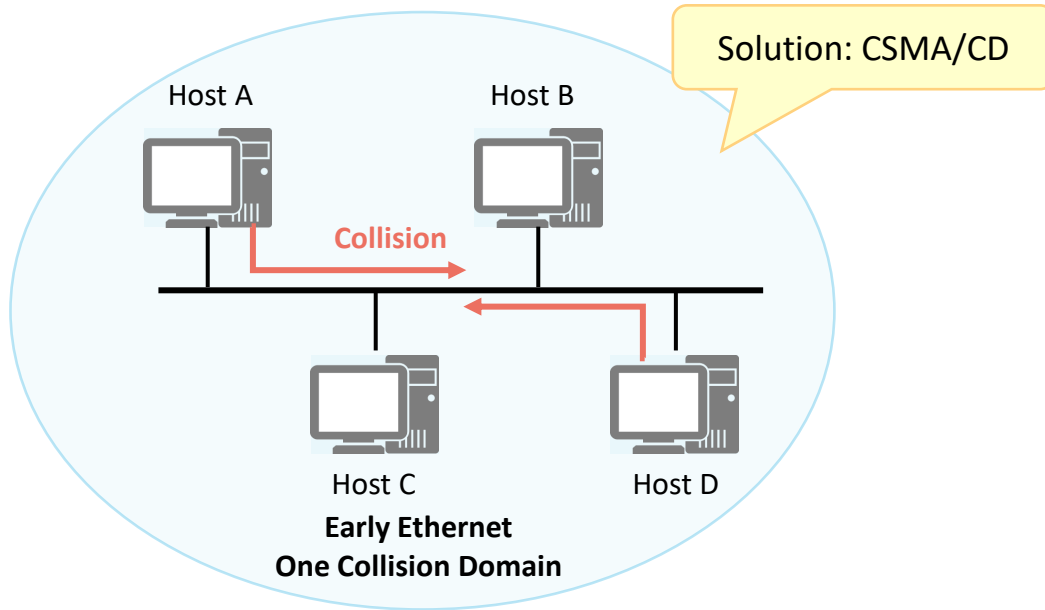


Switch Networking

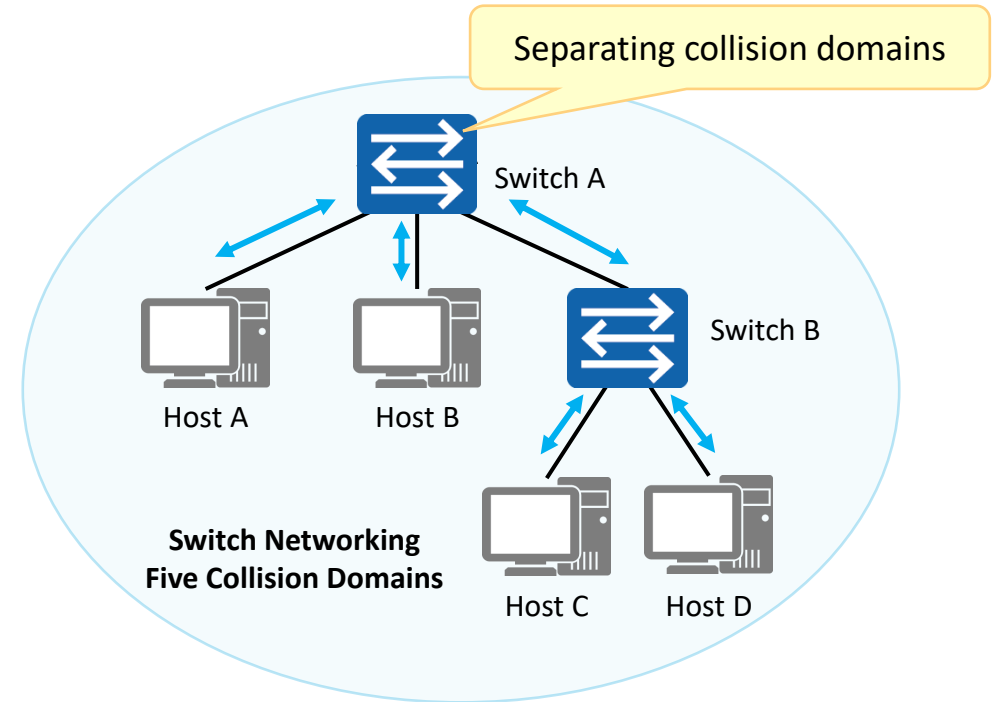


Collision Domain

- A collision domain is a set of nodes connected to the same shared medium. All nodes in a collision domain compete for the same bandwidth. Packets (unicast, multicast, or broadcast) sent by a node can be received by other nodes.



- On a traditional Ethernet network, multiple nodes on the same medium share the link bandwidth and compete for the right to use the link. As a result, collision occurs.
- The probability that collision occurs increases when more nodes are deployed on a shared medium.

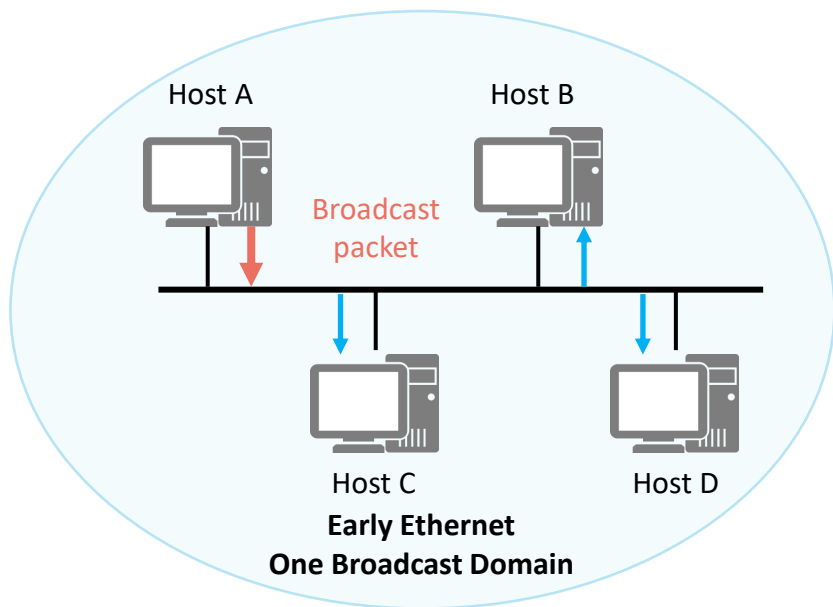


The switch interfaces used to send and receive data are independent of each other and belong to different collision domains. Therefore, collisions do not occur between hosts (or networks) connected through switch interfaces.

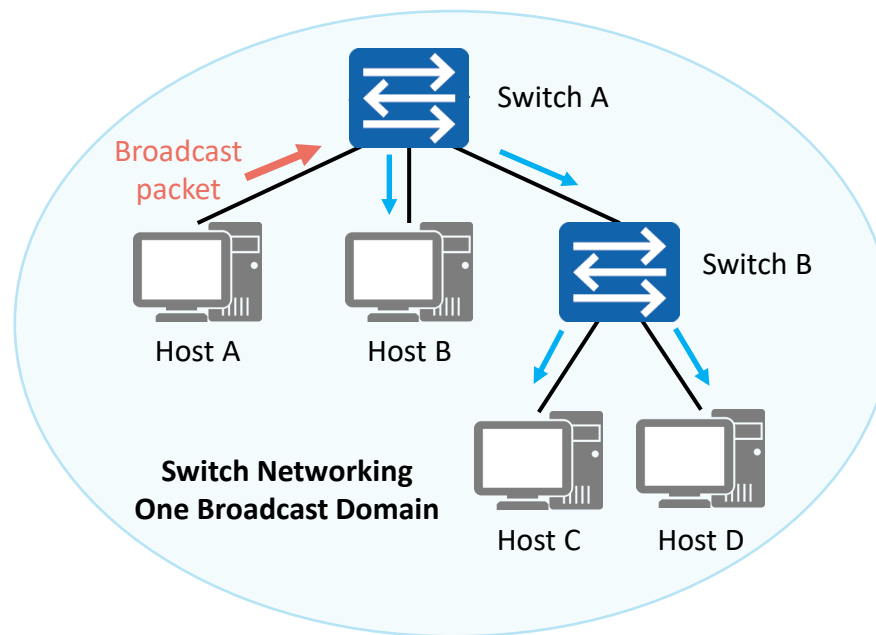


Broadcast Domain

- The entire access scope of broadcast packets is called a Layer 2 broadcast domain, which is also called a broadcast domain. All hosts in the same broadcast domain can receive broadcast packets.



On a traditional Ethernet network, multiple nodes on the same medium share a link. The broadcast packets sent by a device can be received by all the other devices.

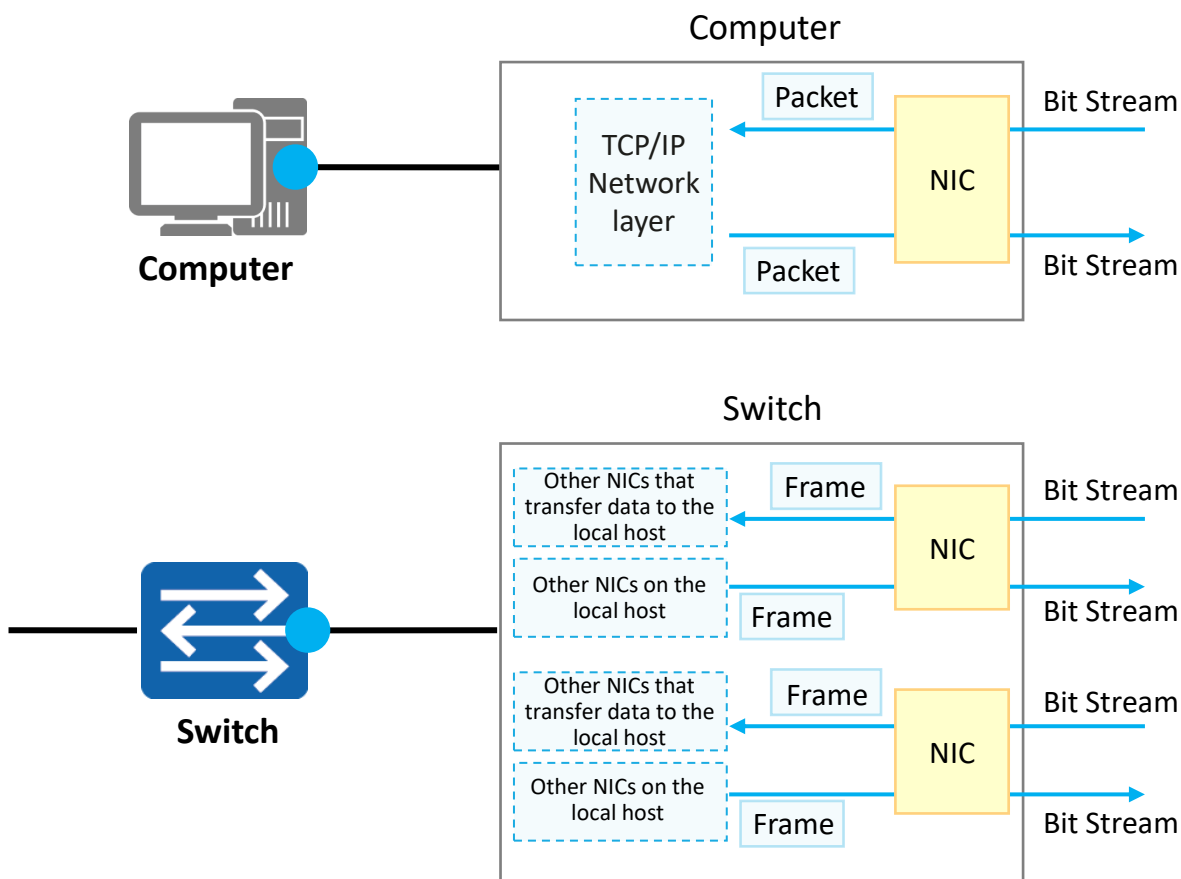


A switch forwards broadcast packets to all interfaces. Therefore, the nodes connected to all interfaces of the switch belong to the same broadcast domain.



Ethernet NIC

- A network interface card (NIC) is a key component that connects a network device (such as a computer, a switch, or a router) to an external network.



- **Network Port**
 - A network port is also called a network interface, interface, or port.
- **NIC**
 - Each network port corresponds to a NIC.
 - A computer or switch forwards data through a NIC.



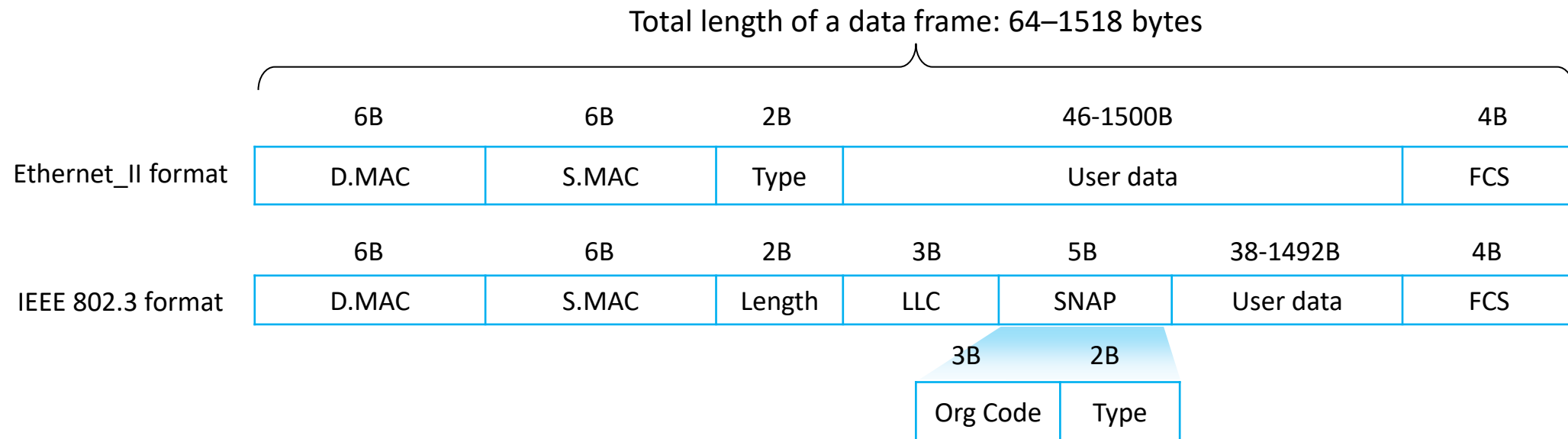
Contents

1. Overview of Ethernet Protocols
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Ethernet Frame Format

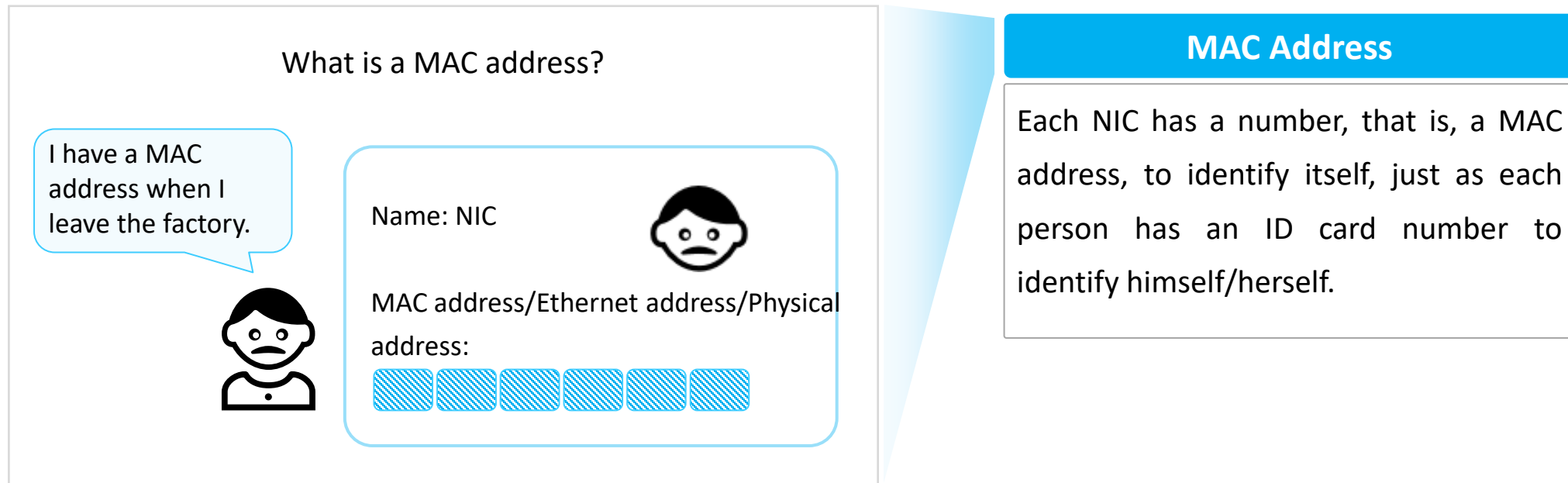
- The frames used by Ethernet technology are referred to as Ethernet frames.
- Ethernet frames are in two formats: Ethernet_II and IEEE 802.3.





What Is a MAC Address?

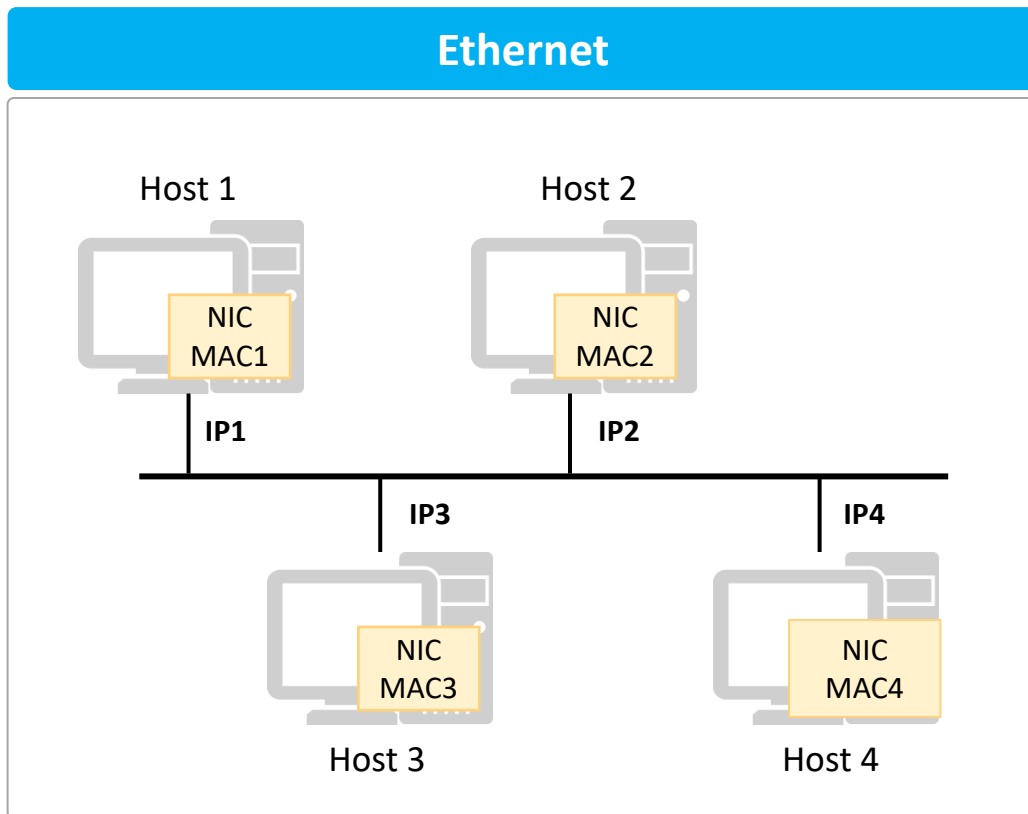
- A media access control (MAC) address uniquely identifies a NIC on a network. Each NIC must have a globally unique MAC address.





IP Address Vs. MAC Address

- Each Ethernet device has a unique MAC address before delivery. When the device accesses the network, it assigns an IP address to each host. Why?



Characteristics of IP addresses:

- IP addresses are unique.
- IP addresses are **changeable**.
- IP addresses are assigned based on **network topology**.

Characteristics of MAC addresses:

- MAC addresses are unique.
- MAC addresses **cannot be changed**.
- MAC addresses are assigned based on the **manufacturer**.

Can a network device have either a MAC address or an IP address?





MAC Address Presentation

- A MAC address is 48 bits (6 bytes) in length.
- As typically represented, MAC addresses are recognizable as six groups of two hexadecimal digits, separated by hyphens, colons, or without a separator.

For example, 00-1E-10-DD-DD-02 or 001E-10DD-DD02

Hexadecimal

00	1E	10	DD	DD	02
0000 0000	0001 1110	0001 0000	1101 1101	1101 1101	0000 0010

6-byte
48-bit

Binary

Conversion between
hexadecimal and
binary digits

Power

2^3	2^2	2^1	2^0	2^3	2^2	2^1	2^0
8	4	2	1	8	4	2	1
0	0	0	1	1	1	1	0

Bit

= 1

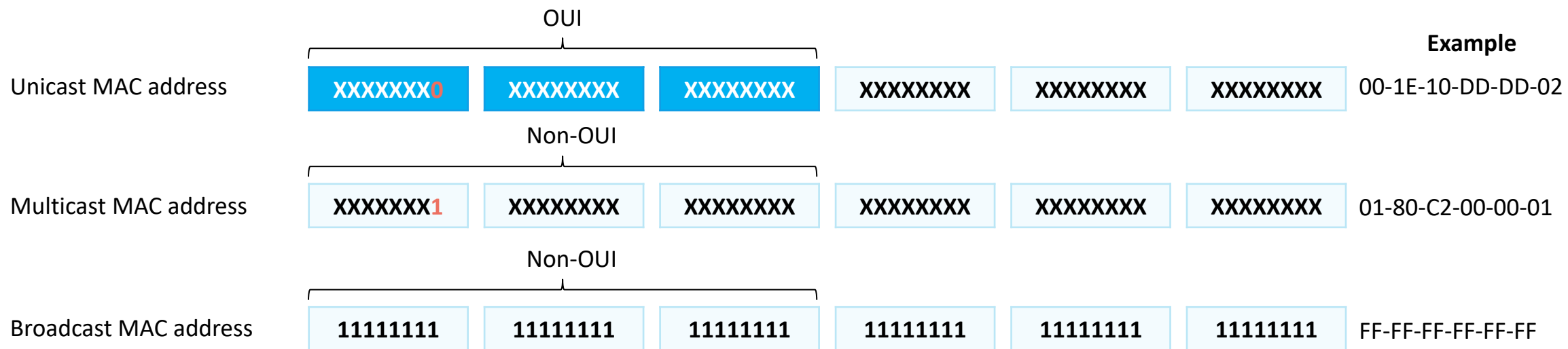
= 8+4+2=14=E

MAC Address Composition and Classification

- Organizational unique identifier (OUI): a 24-bit (3-byte) number. It is a globally unique identifier assigned by the IEEE.
- Company ID (CID): a 24-bit (3-byte) number. It is assigned by a manufacturer.



- MAC address classification:





Unicast Ethernet Frame

- A unicast Ethernet frame is also called a unicast frame.
- The destination MAC address of a unicast frame is a unicast MAC address.

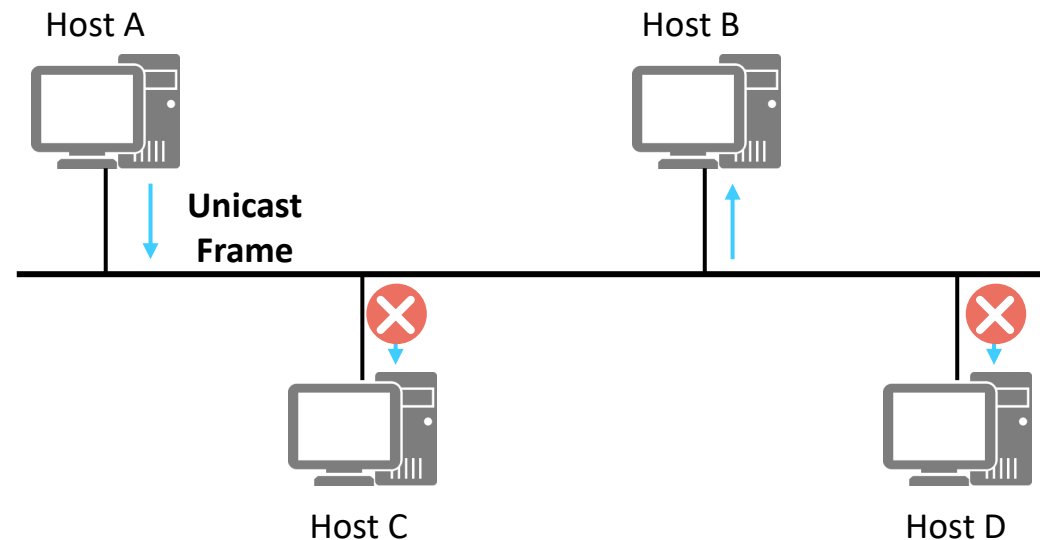


D.MAC:
00-1E-10-DD-DD-02

S.MAC

DATA

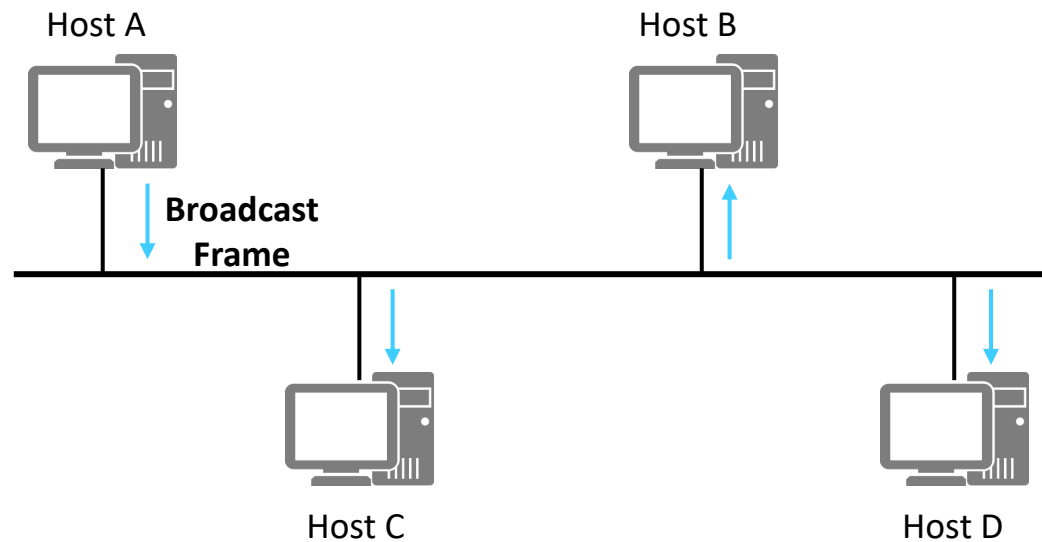
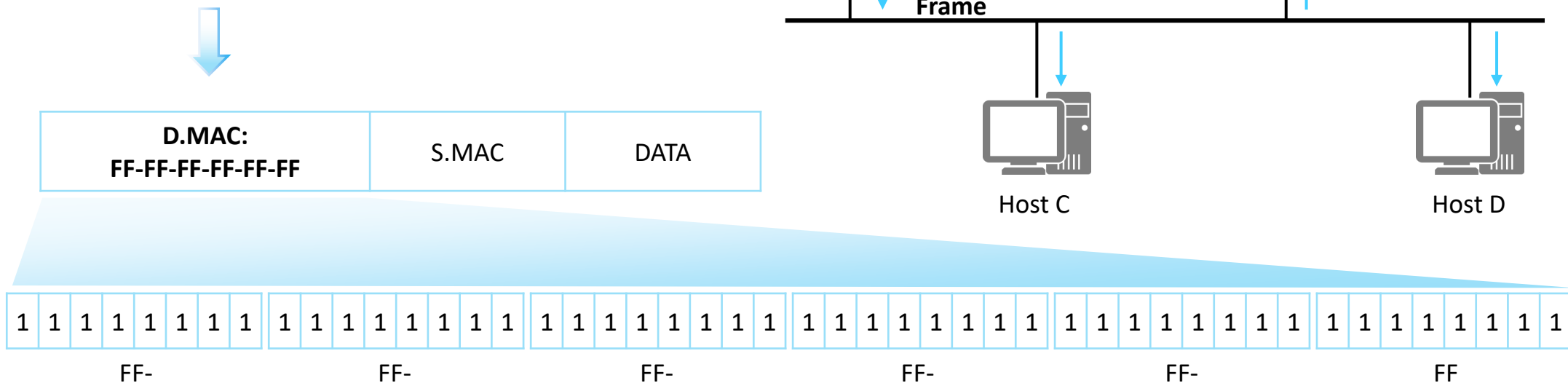
0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	1	0	0	0	0	1	1	0	1	1	1	0	1	1	1	0	1	0	0	0	0	0	1	0							
00-								1E-								10-								DD-								DD-								02							





Broadcast Ethernet Frame

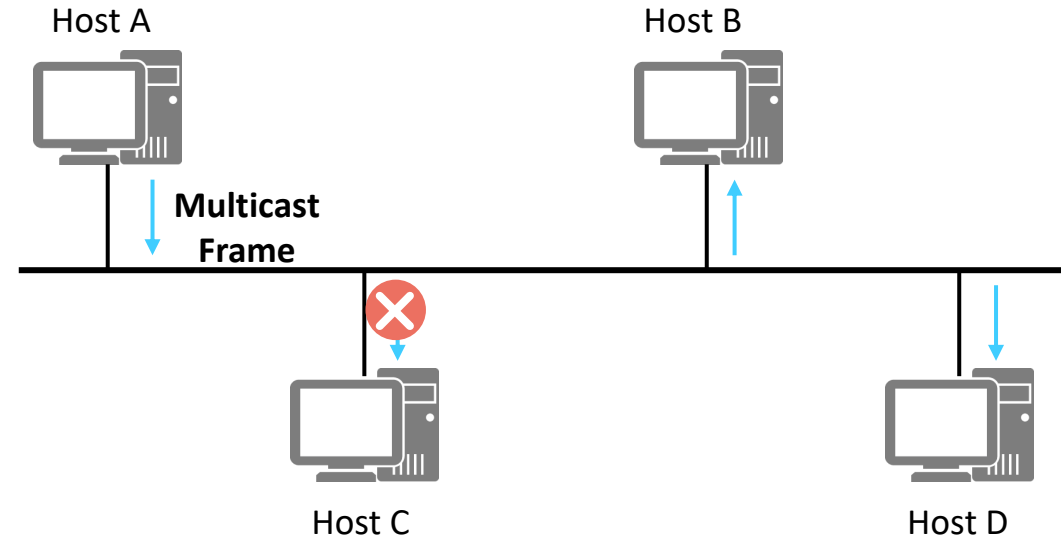
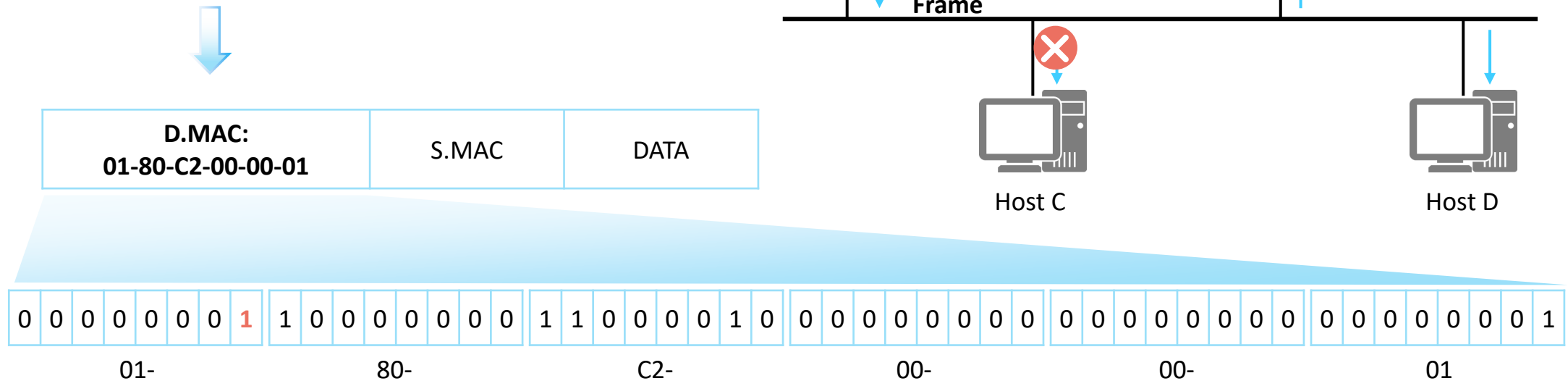
- A broadcast Ethernet frame is also called a broadcast frame.
- The destination MAC address of a broadcast frame is a broadcast MAC address.





Multicast Ethernet Frame

- A multicast Ethernet frame is also called a multicast frame.
- The destination MAC address of a multicast frame is a unicast MAC address.



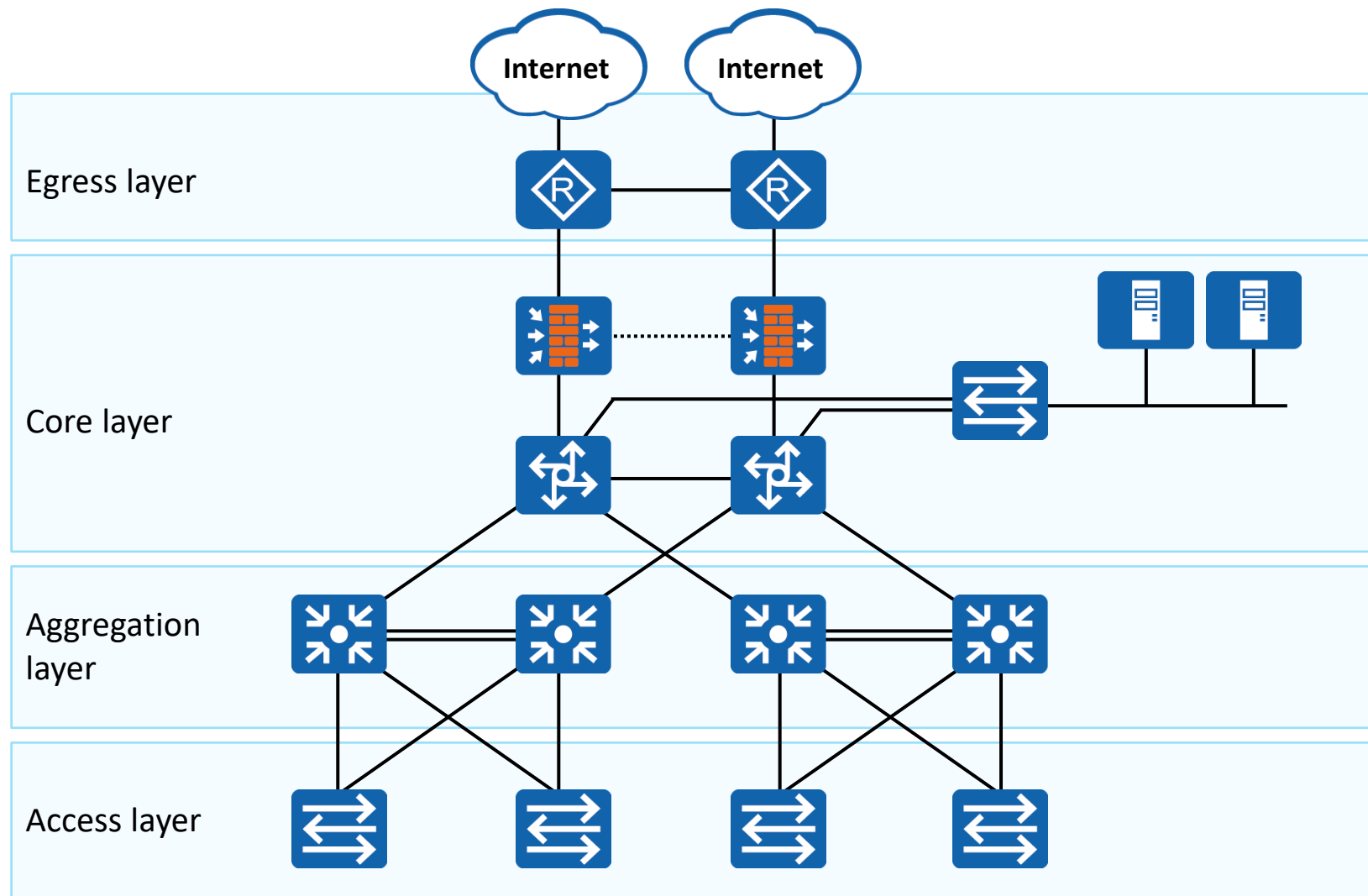


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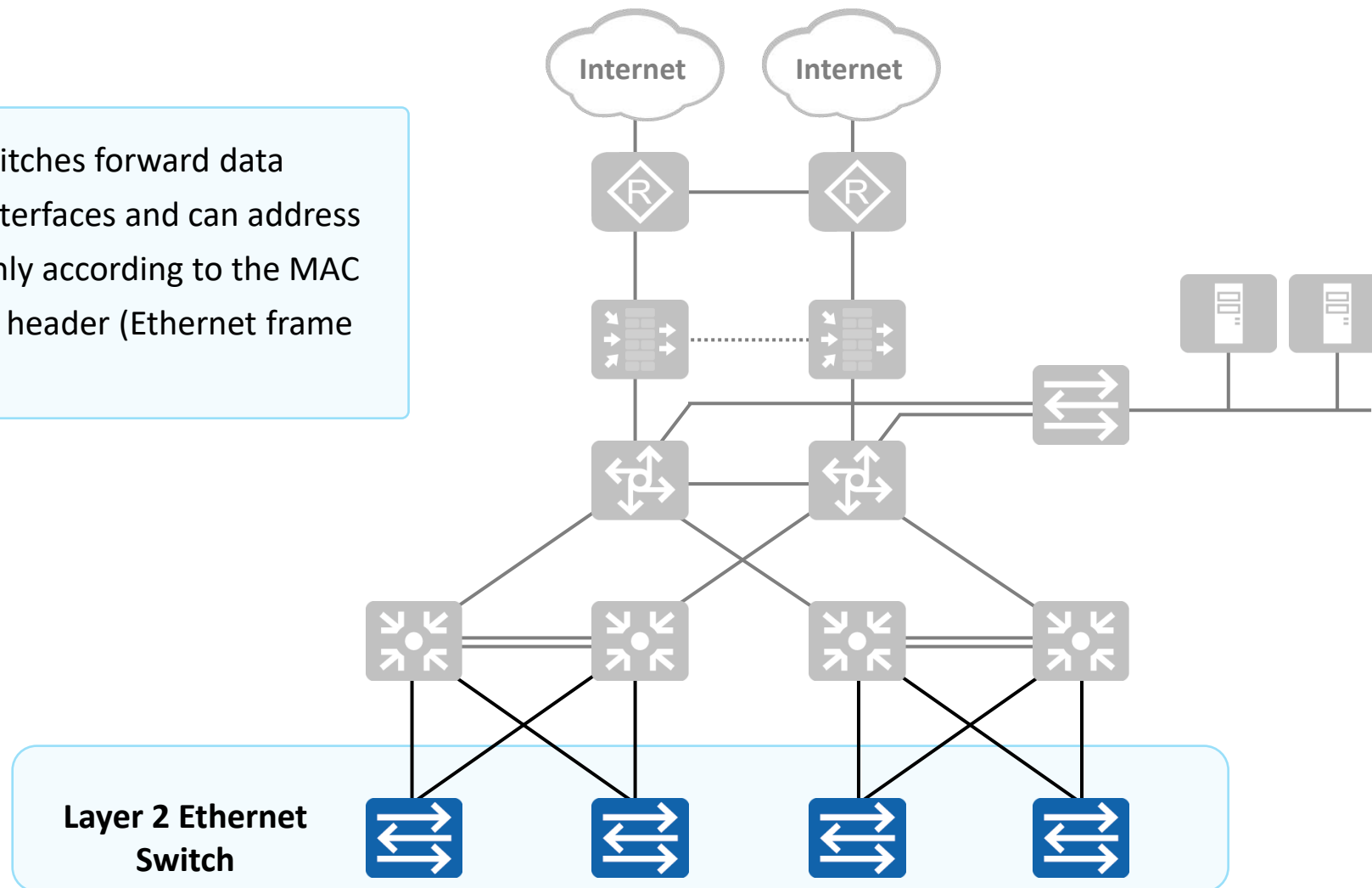
Typical Architecture of a Campus Network





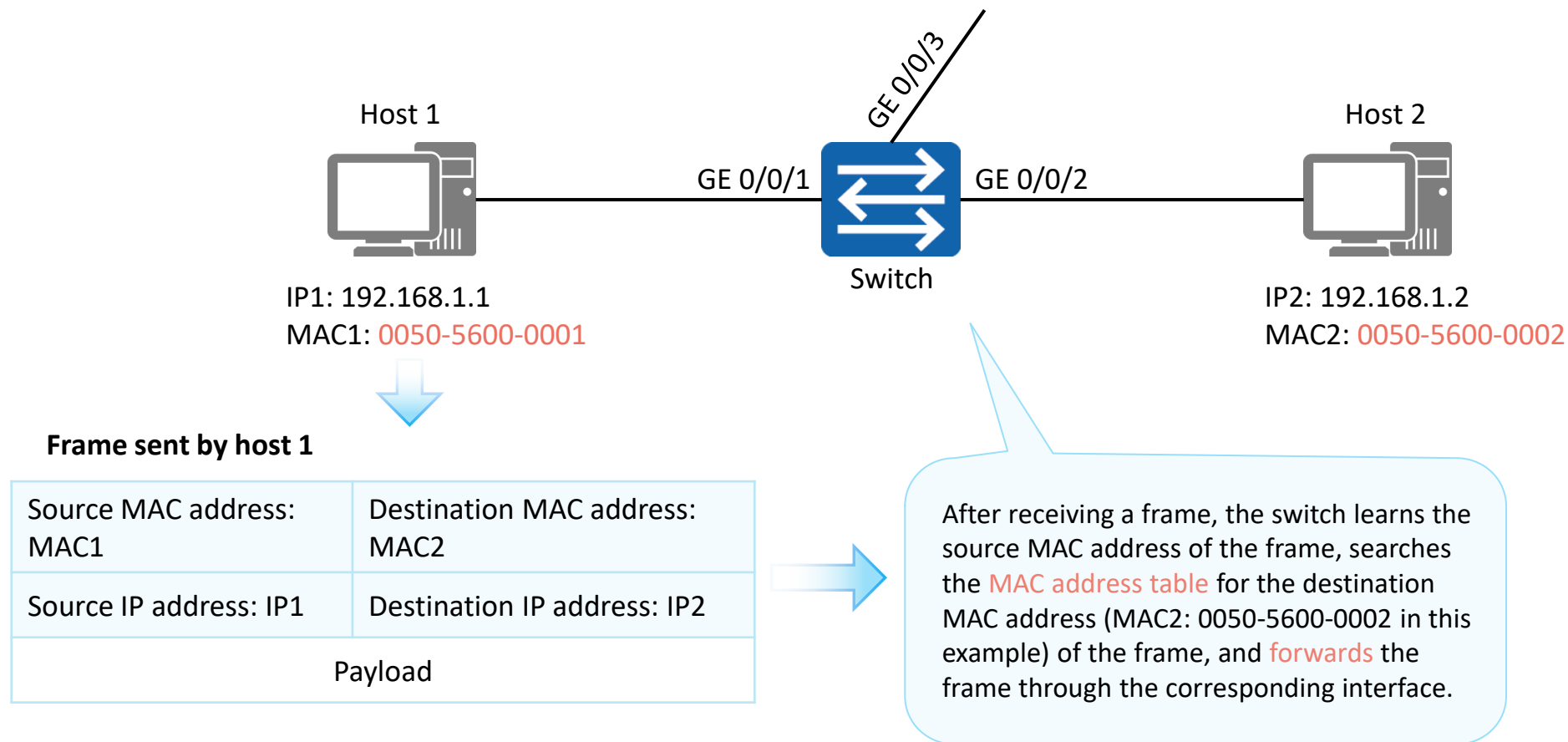
Layer 2 Ethernet switch

Layer 2 Ethernet switches forward data through Ethernet interfaces and can address and forward data only according to the MAC address in a Layer 2 header (Ethernet frame header).





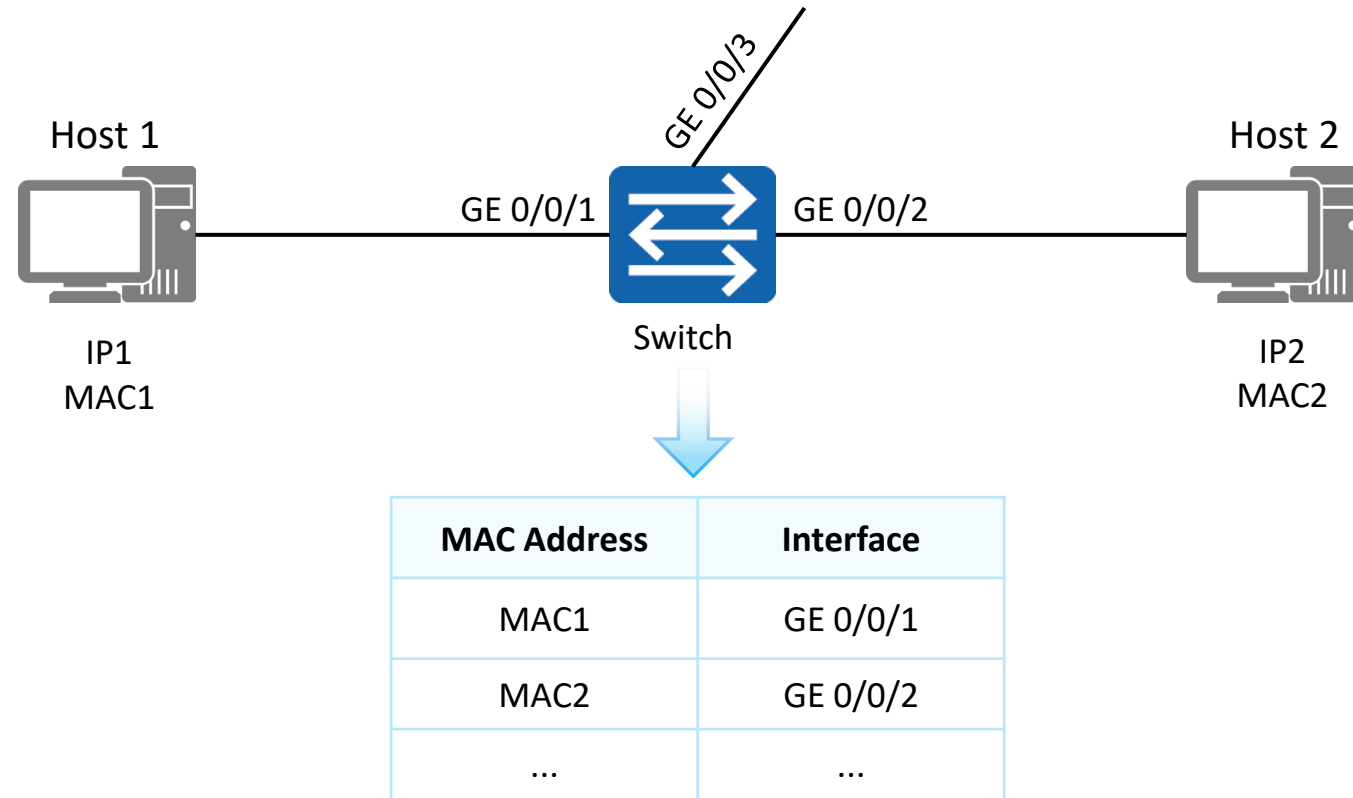
Working Principles of Switches





MAC Address Table

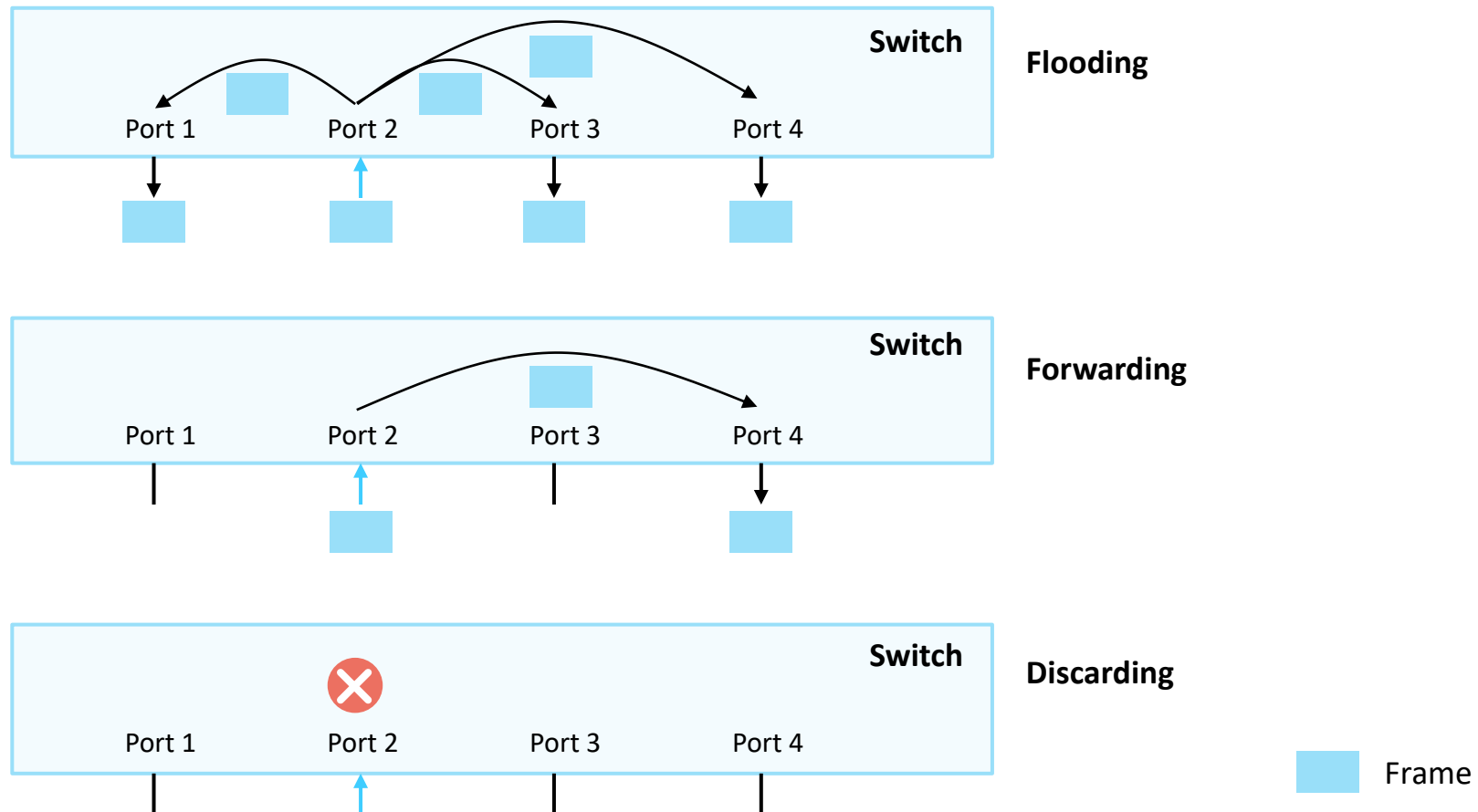
- Each switch has a MAC address table that stores the mapping between MAC addresses and switch interfaces.





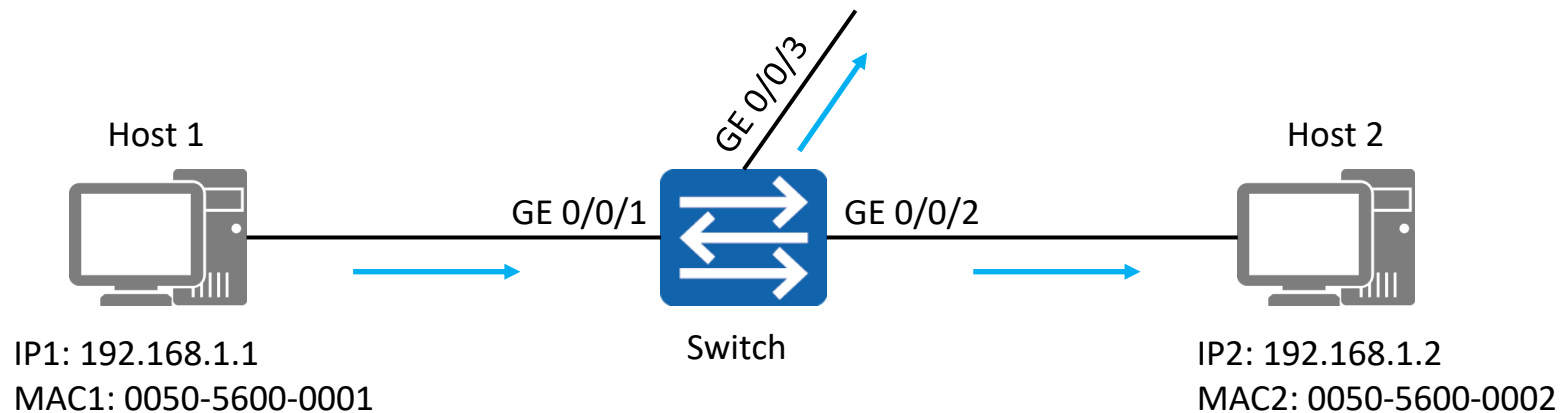
Three Frame Processing Behaviors of a Switch

- A switch processes the frames entering an interface over a transmission medium in three ways:





Flooding



1 Frame sent by host 1

Source MAC: MAC1
Destination MAC: **MAC2**

or

Source MAC: MAC1
Destination MAC: **FF-FF-FF-FF-FF-FF**

2 MAC address table searched by the switch

MAC Address	Interface
MAC1	GE 0/0/1

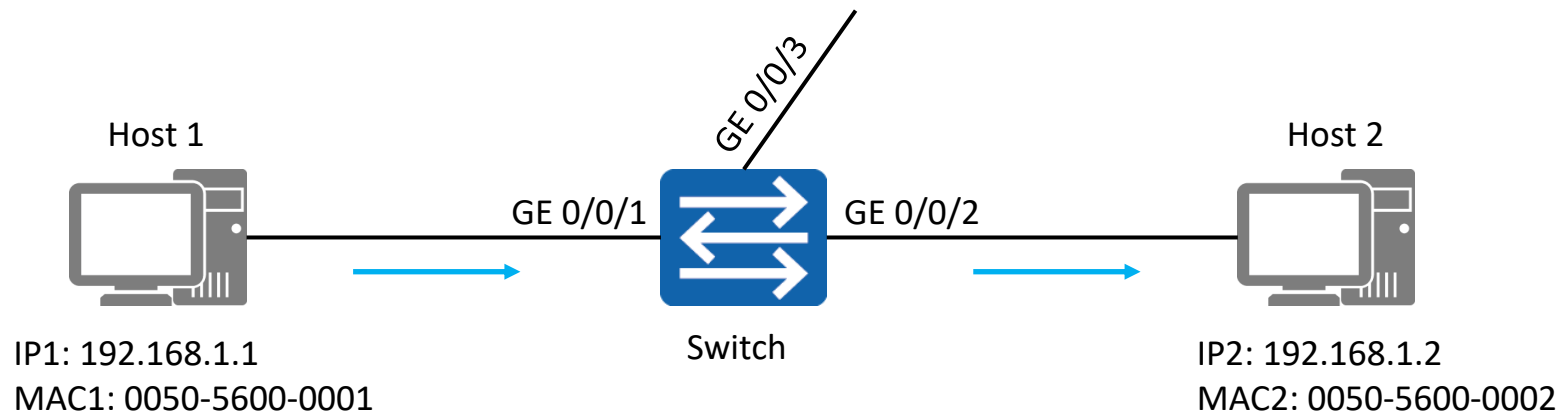
3 Frame processing behavior of the switch

- **If a unicast frame is received:**
If the switch cannot find the destination MAC address of the frame in the MAC address table, the switch floods the unicast frame.
- **If a broadcast frame is received:**
The switch directly floods the broadcast frame without searching the MAC address table.

Unknown unicast frame



Forwarding



1 Frame sent by host 1

Source MAC: MAC1
Destination MAC: MAC2

2 MAC address table searched by the switch

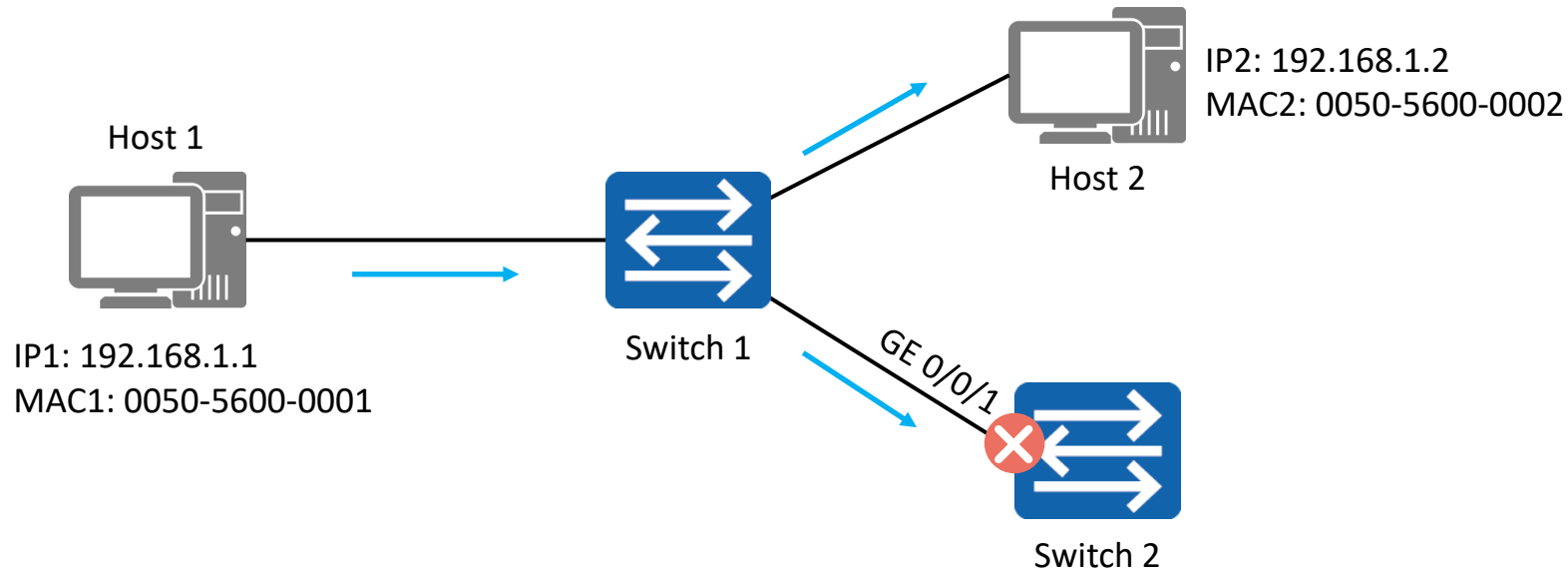
MAC Address	Interface
MAC1	GE 0/0/1
MAC2	GE 0/0/2

3 Frame processing behavior of the switch

- If a unicast frame is received:**
If the switch finds the destination MAC address of the frame in the MAC address table and the interface number in the table is not the number of the interface through which the frame enters over the transmission medium, the switch forwards the unicast frame.



Discarding



1 Frame sent by host 1

Source MAC: MAC1
Destination MAC: MAC2

2 MAC address table queried by switch 2

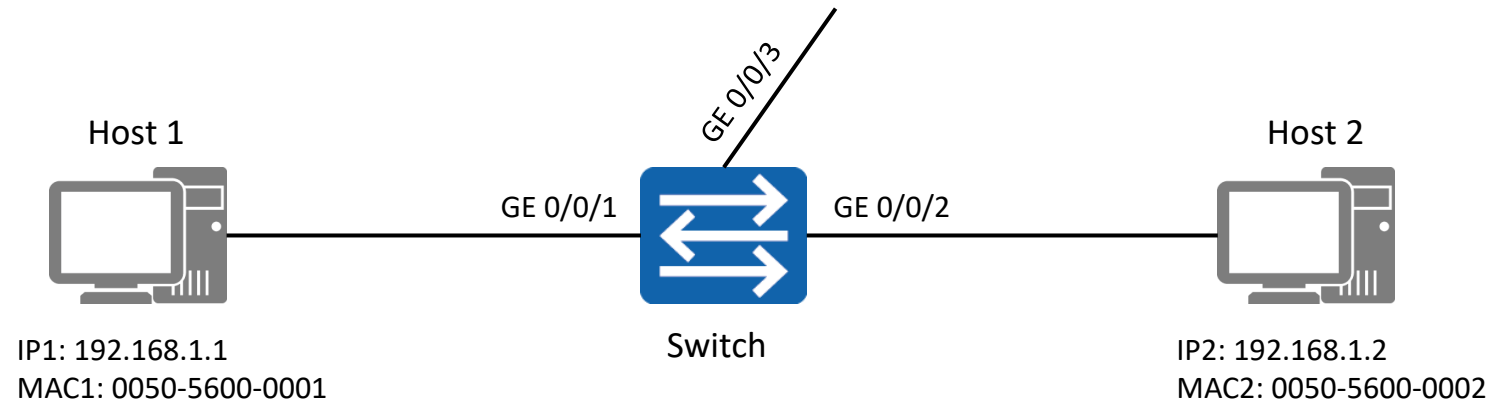
MAC Address	Interface
MAC2	GE 0/0/1

3 Frame processing behavior of the switch

- If a unicast frame is received:
- The switch finds the destination MAC address of the frame in the MAC address table, but the interface number in the table is the number of the interface through which the frame enters the switch over the transmission medium. In this case, the switch discards the unicast frame.



MAC Address Learning on a Switch (1)



MAC address table of the switch

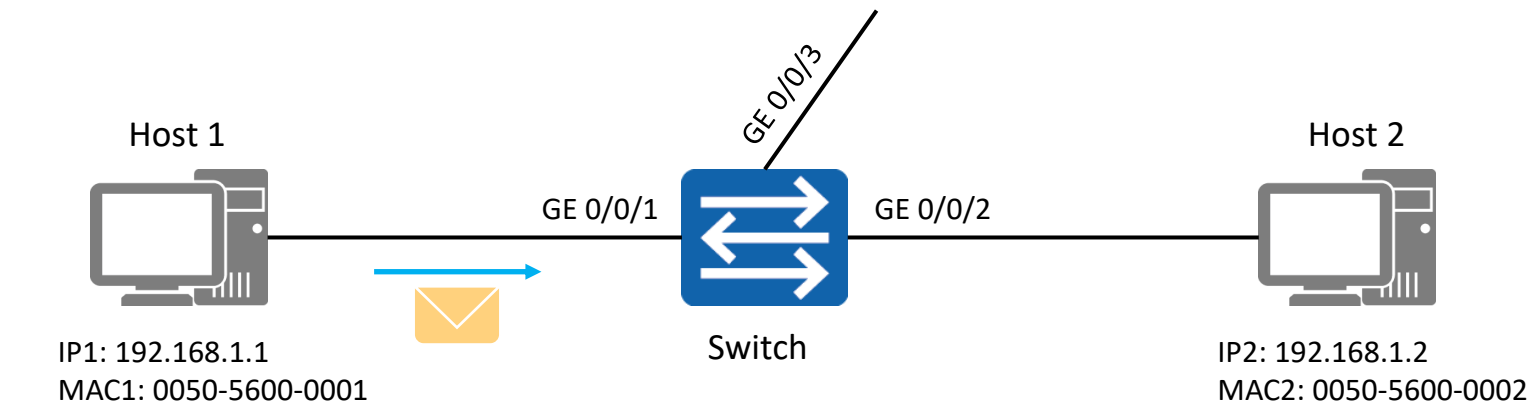
MAC Address	Interface

1

Initially, the MAC address table of the switch is empty.



MAC Address Learning on a Switch (2)



Frame sent by host 1

Source MAC: MAC1
Destination MAC: **MAC2**

(Assume that host 1 has obtained the MAC address of host 2.)

MAC address table searched by the switch

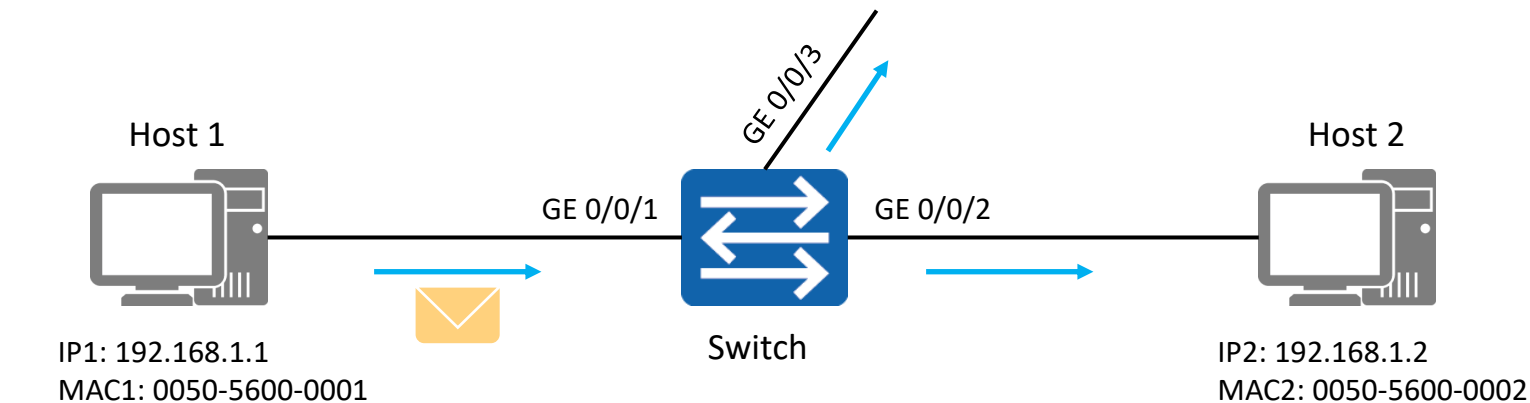
MAC Address	Interface

2

- Host 1 sends a frame to host 2.
- After the frame is received on the switch's GE 0/0/1, the switch searches the MAC address table for the destination MAC address of the frame. If no matching entry is found, the switch considers the frame an unknown unicast frame.



MAC Address Learning on a Switch (3)



Frame sent by host 1

Source MAC: **MAC1**
Destination MAC: MAC2

MAC address table searched by the switch

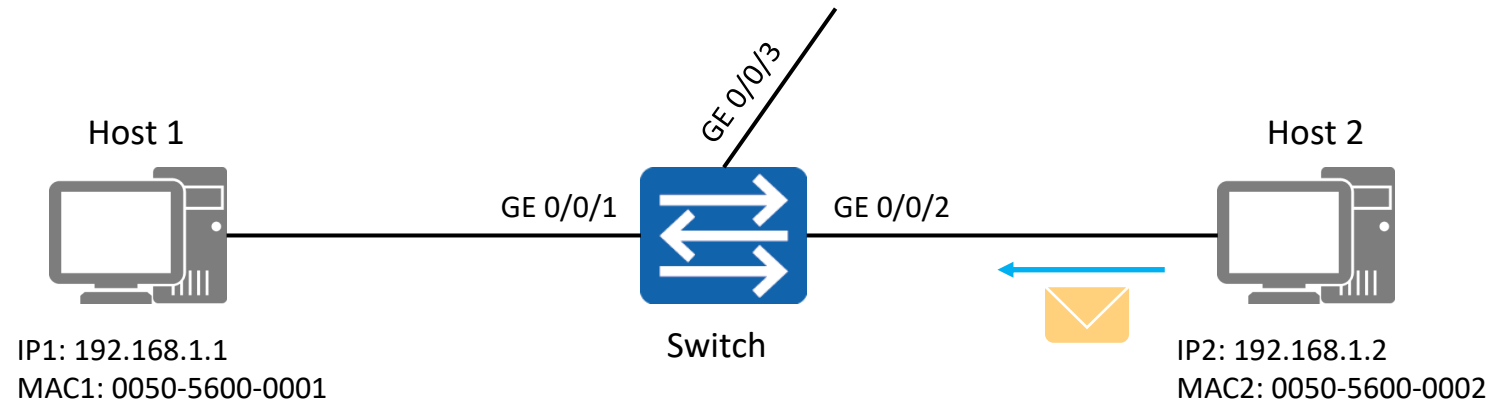
MAC Address	Interface
MAC1	GE 0/0/1

3

- If the corresponding entry is not found in the MAC address table, the switch **floods** the unicast frame.
- At the same time, the switch learns the source **MAC** address of the frame, creates the corresponding MAC address entry, and associates the MAC address entry with **GE 0/0/1**.



MAC Address Learning on a Switch (4)



4

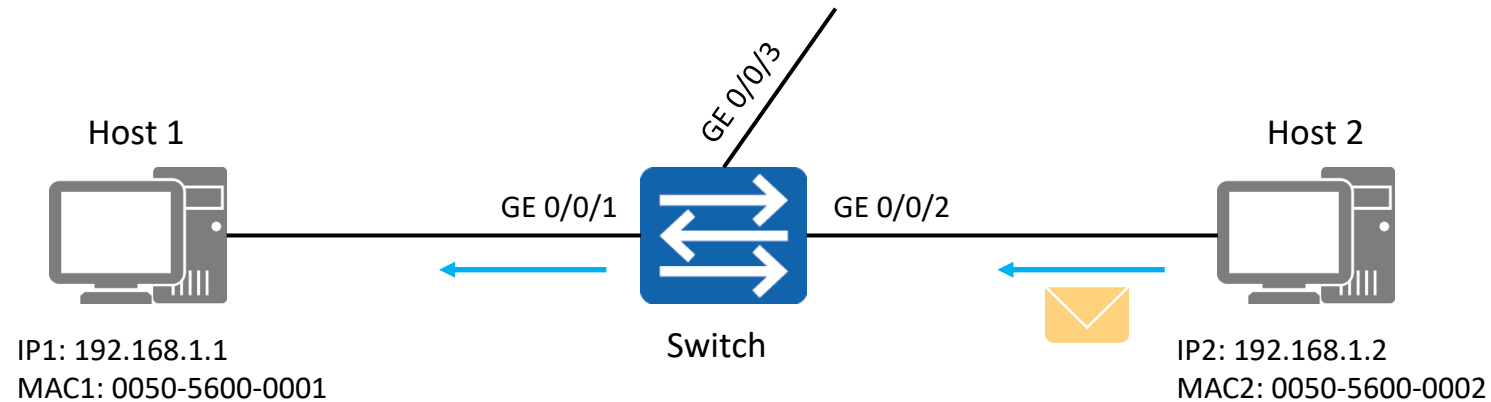
- The frame is also received by the hosts connected to other interfaces on the switch. These hosts, however, discard the frame.
- Host 2 receives and processes the frame, responds to host 1, and sends the frame to the switch.

Frame sent by host 2

Source MAC: MAC2
Destination MAC: MAC1



MAC Address Learning on a Switch (5)



MAC address table searched by the switch

MAC Address	Interface
MAC1	GE 0/0/1
MAC2	GE 0/0/2

Frame sent by host 2

Source MAC: MAC2
Destination MAC: MAC1

5

- If the switch finds the corresponding entry in the MAC address table, the switch forwards the unicast frame through GE 0/0/1.
- At the same time, the switch learns the source MAC address of the frame, creates the corresponding MAC address entry, and associates the MAC address entry with GE 0/0/2.



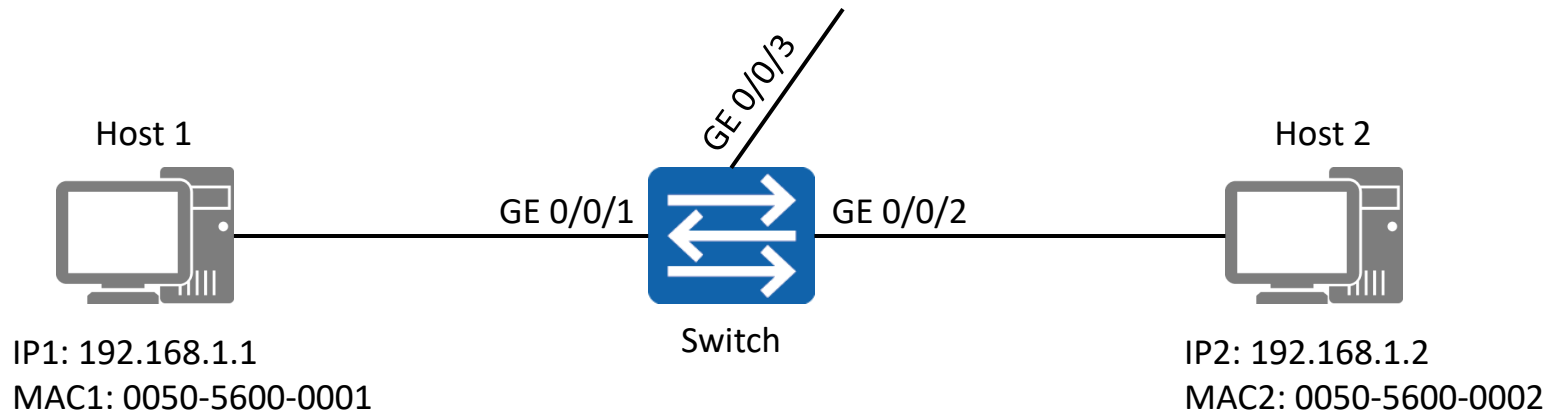
Contents

1. Overview of Ethernet Protocols
2. Overview of Ethernet Frames
3. Overview of Ethernet Switches
- 4. Process of Data communication Within a Network Segment**



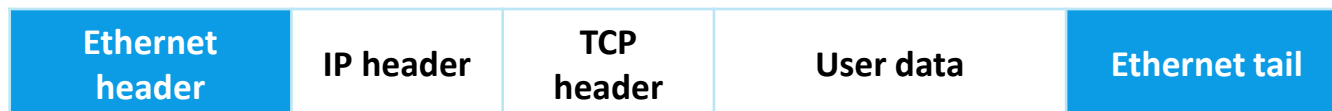
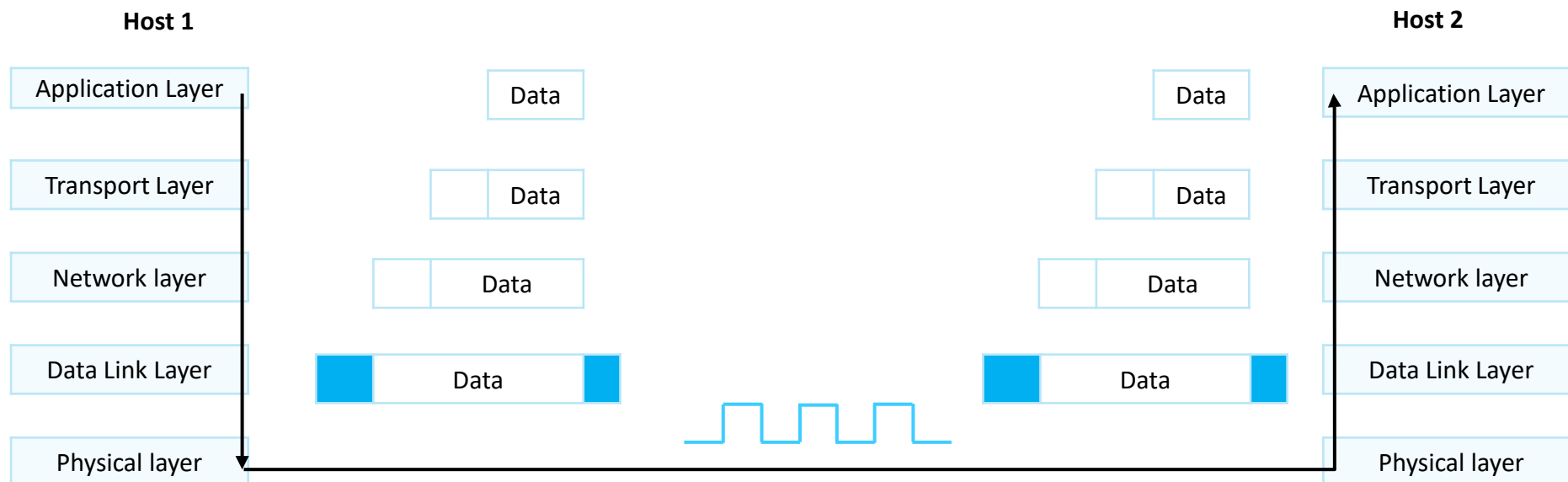
Process of Data Communication Within a Network Segment

- Scenario description:
 - Task: Host 1 wants to access host 2.
 - Host: The host is in the initialized state and only knows its own IP address and MAC address (assume that the IP address of the peer host has been obtained).
 - Switch: The switch is just powered on and in the initialized state.





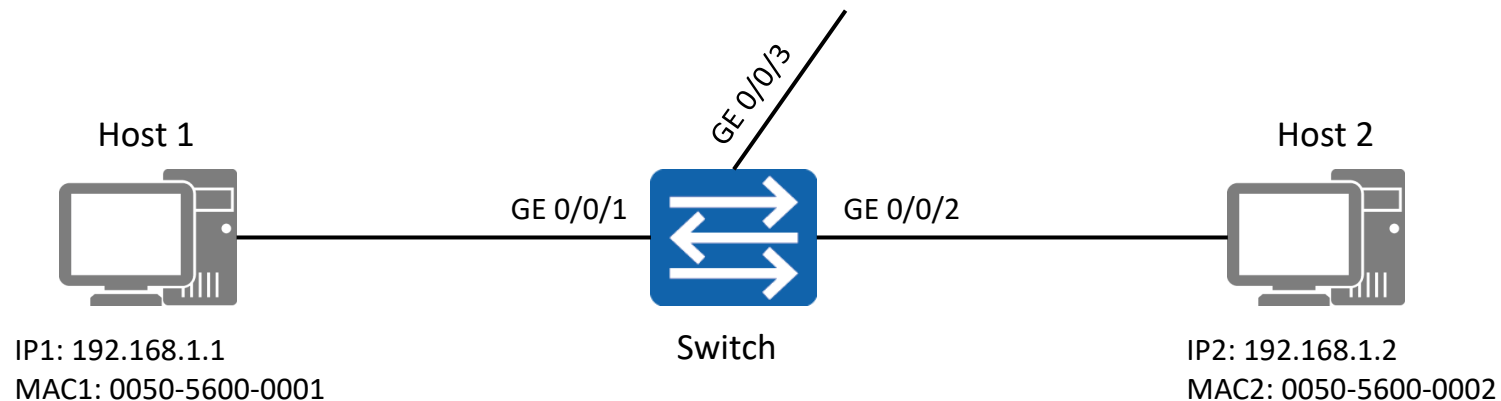
Data Encapsulation Process



- **Information that needs to be encapsulated:**
- Source MAC address
- Destination MAC address



Initialization



ARP cache table of host 1

```
Host 1>arp -a
```

Internet Address	Physical Address	Type

MAC address table of the switch

```
[Switch]display mac-address verbose
```

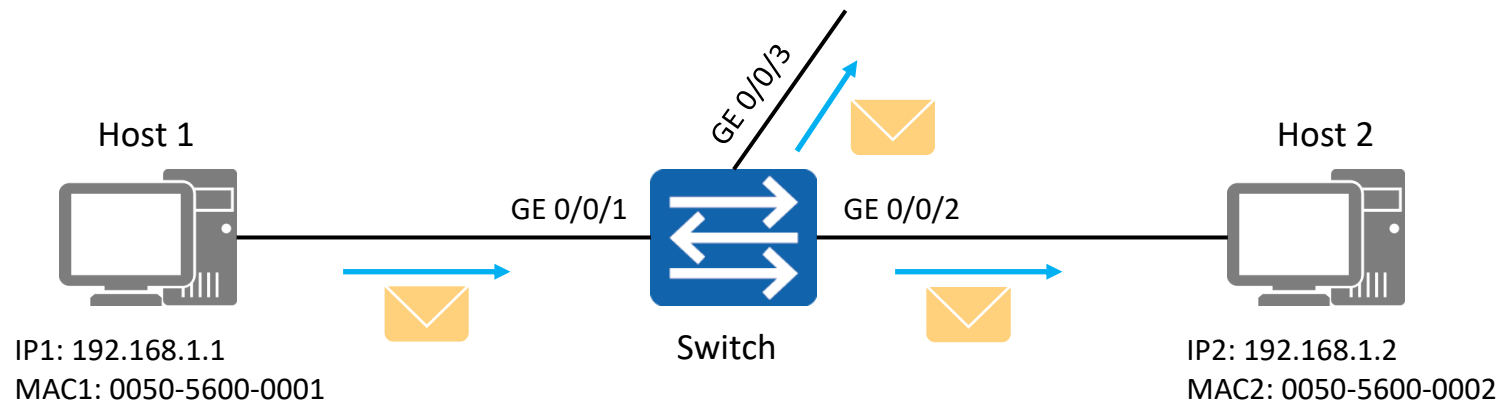
MAC address table of slot 0:

-----MAC

Address	Port	Type



Flooding Frames



ARP Request packet sent by host 1

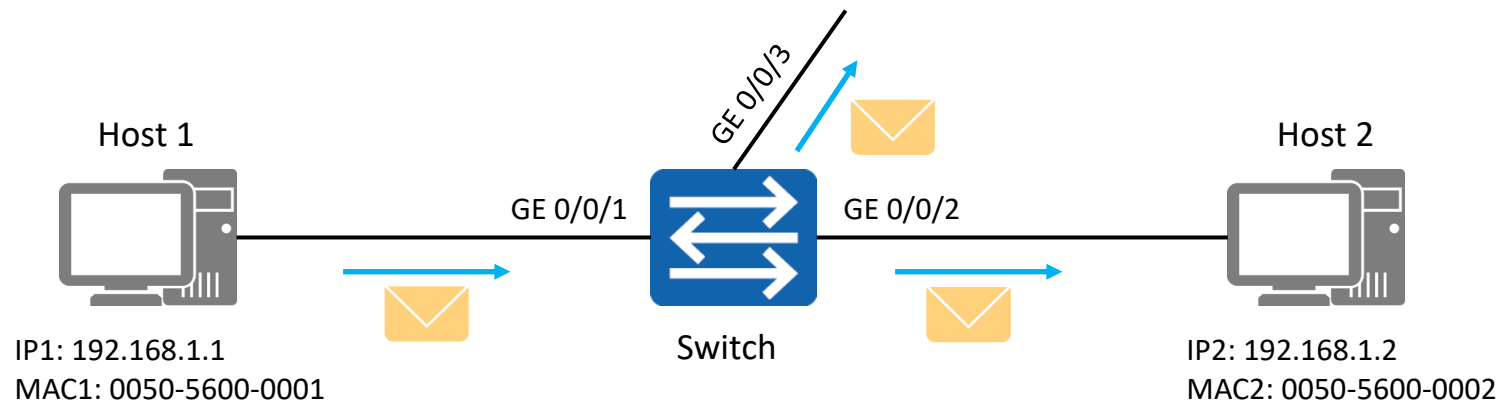
Source MAC address: MAC1	Destination MAC address: FF- FF-FF-FF-FF
Source IP address: IP1	Destination IP address: IP2
Operation type: ARP Request Sender's MAC address: MAC1 Sender's IP address: IP1 Destination MAC address: 00-00-00-00-00-00 Destination IP address: IP2	

MAC address table of the switch

```
[Switch]display mac-address verbose
MAC address table of slot 0:
-----MAC
Address Port      Type
-----
-----
```




MAC Address Learning



ARP Request packet sent by host 1

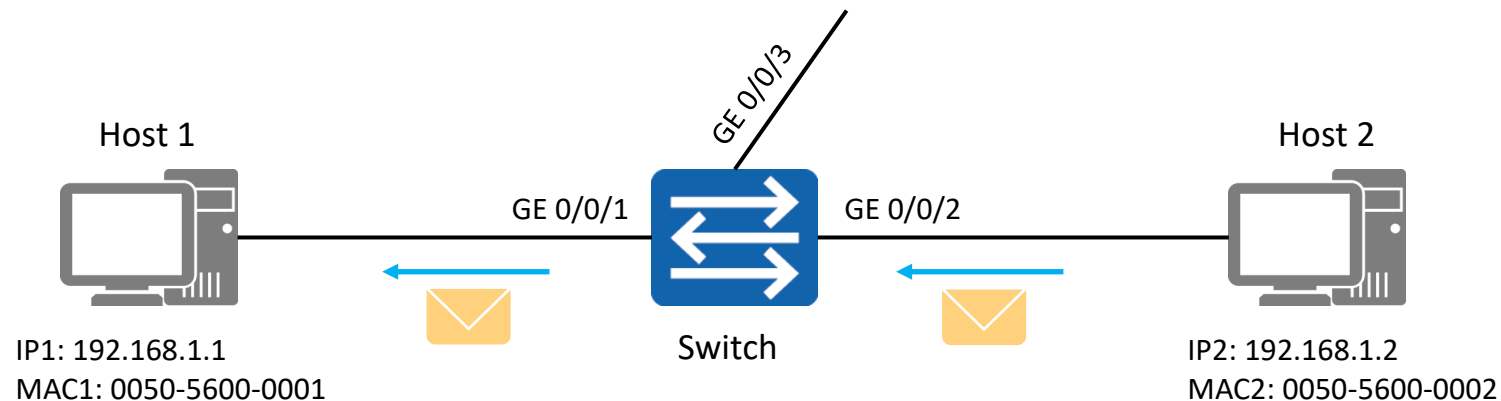
Source MAC address: MAC1	Destination MAC address: FF- FF-FF-FF-FF
Source IP address: IP1	Destination IP address: IP2
Operation type: ARP Request Sender's MAC address: MAC1 Sender's IP address: IP1 Destination MAC address: 00-00-00-00-00-00 Destination IP address: IP2	

MAC address table of the switch

```
[Switch]display mac-address verbose
MAC address table of slot 0:
-----MAC
Address Port      Type
-----0050-5600-
0001    GE0/0/1    dynamic
-----
```



Reply of the Target Host



MAC address table of the switch

```
[Switch]display mac-address verbose
MAC address table of slot 0:
-----MAC
Address Port      Type
-----0050-5600-
0001   GE0/0/1    dynamic
0050-5600-0002   GE0/0/2    dynamic
-----
```

ARP Reply packet sent by host 2

Source MAC address: MAC2	Destination MAC address: MAC1
Source IP address: IP2	Destination IP address: IP1
Operation type: ARP Reply Sender's MAC address: MAC2 Sender's IP address: IP2 Destination MAC address: MAC1 Destination IP address: IP1	



Quiz

1. (Single) A Layer 2 Ethernet switch generates a MAC address table entry based on the () of the packet received by an interface.
 - A. Source MAC address
 - B. Destination MAC address
 - C. Source IP address
 - D. Destination IP address.
2. (Single) A switch has eight interfaces. A unicast frame enters the switch through one of the eight interfaces, but the switch cannot find the destination MAC address entry of the frame in the MAC address table. In this case, which of the following operations is performed by the switch? ()
 - A. Discarding
 - B. Flooding
 - C. Point-to-point forwarding



Summary

- This section describes the basic information about the Ethernet protocol, Ethernet frame formats, MAC address, and working principles of Layer 2 switches. Specifically, after receiving a frame, a switch learns the source MAC address of the frame and searches the destination MAC address of the frame in the MAC address table. If the destination MAC address exists in the table, the switch forwards the frame through the corresponding interface.
- This course also describes the whole process of data communication within the same network segment based on the working principles of switches.

The image features a blue-tinted background showing silhouettes of several groups of business professionals in a modern office environment. They are standing on a reflective floor, and their reflections are visible. The background includes architectural elements like glass walls and windows, suggesting a high-rise building. Overlaid on this scene is the text "Thank You" and the website address "www.huawei.com".

Thank You

www.huawei.com



VLAN Principles and Configuration



Foreword

- Ethernet technology implements data communication over shared media based on carrier sense multiple access with collision detection (CSMA/CD). If there are a large number of PCs on the Ethernet, security risks and broadcast storms may occur, deteriorating network performance and even causing network breakdowns.
- The virtual local area network (VLAN) technology is therefore introduced to solve the preceding problem.
- This course describes basic VLAN principles, working principles of different Layer 2 interfaces, VLAN applications, data forwarding principles, and basic VLAN configuration methods.



Objectives

- On completion of this course, you will be able to:
 - Understand the background of the VLAN technology.
 - Identify the VLAN to which data belongs.
 - Master different VLAN assignment modes.
 - Describe how data communication is implemented through VLANs.
 - Master basic VLAN configuration methods.

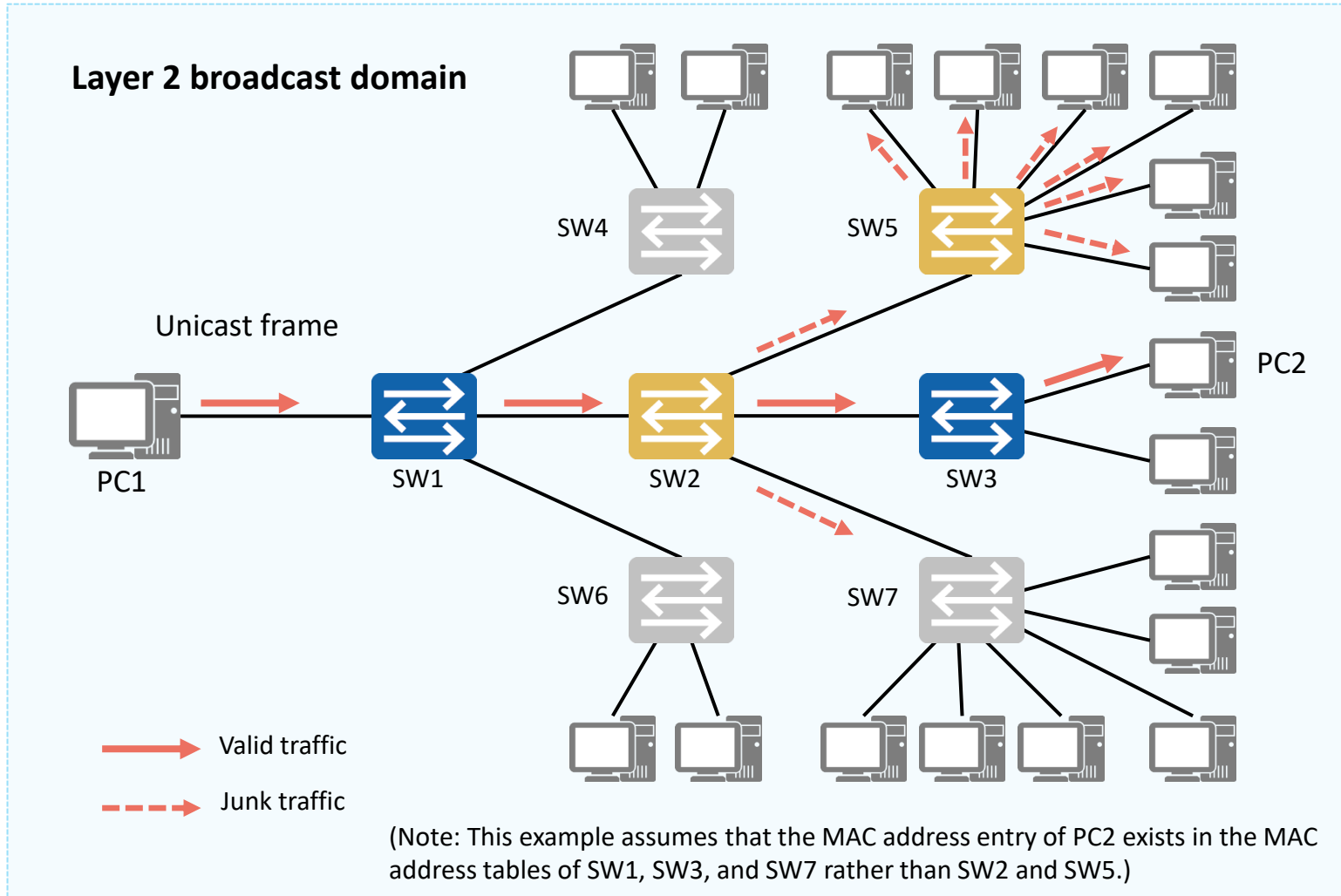


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- 1. What Is VLAN**
2. VLAN Principles
3. VLAN Applications
4. VLAN Configuration Examples



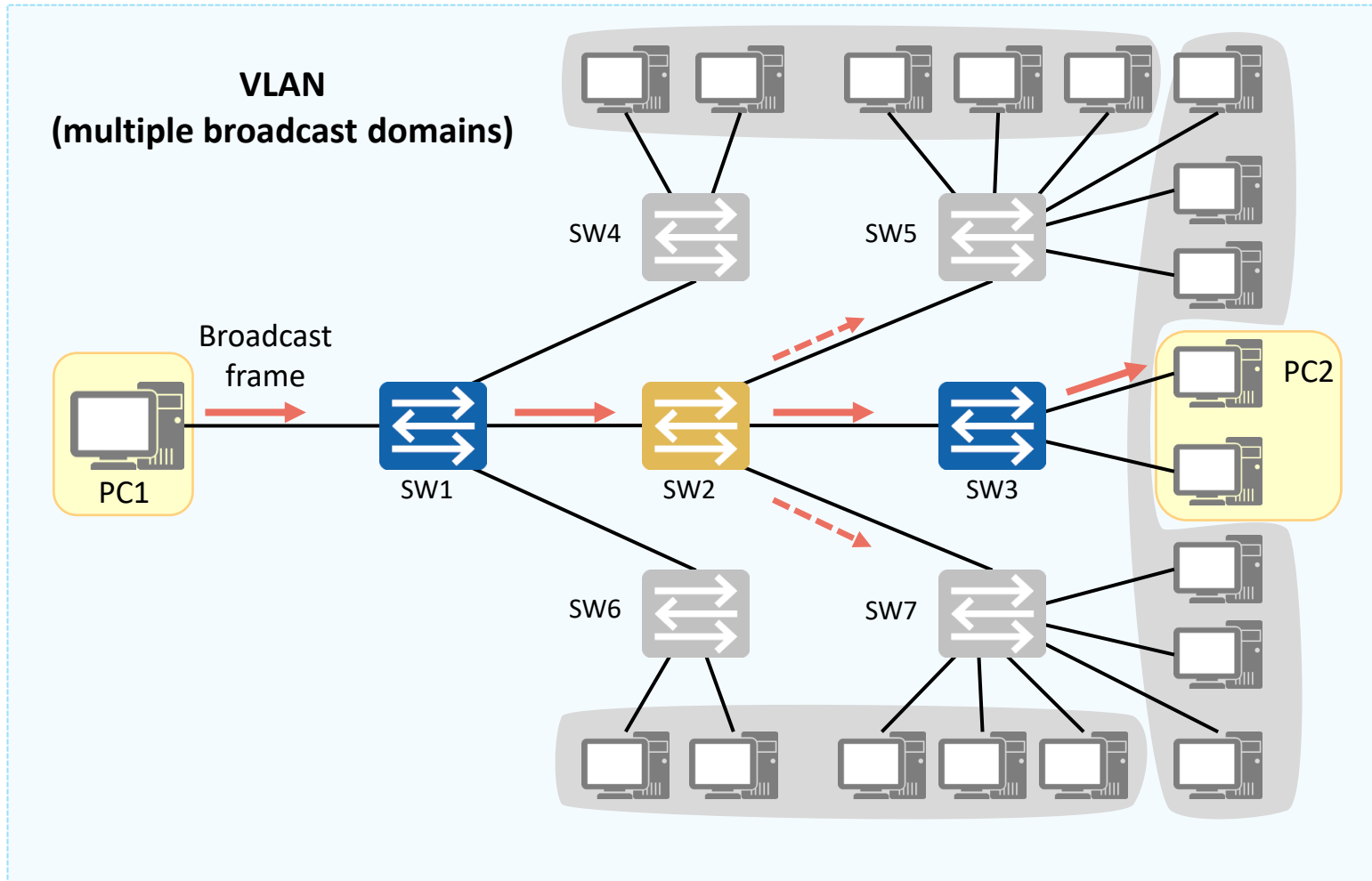
Issues Facing a Traditional Ethernet



- On a typical switching network, broadcast frames or unknown unicast frames sent by a PC are flooded in the entire broadcast domain.
- The larger the broadcast domain is, the more serious network security and junk traffic problems are.



VLAN



- A **Virtual Local Area Network (VLAN)** is a logical grouping of devices within a network that allows them to communicate as if they were on the same physical LAN, regardless of their physical location. VLANs improve network management, security, and performance by separating broadcast domains.
- The VLAN technology isolates broadcast domains.
- Characteristics:
 - Geographically independent.
 - Only devices in the same VLAN can directly communicate at Layer 2.

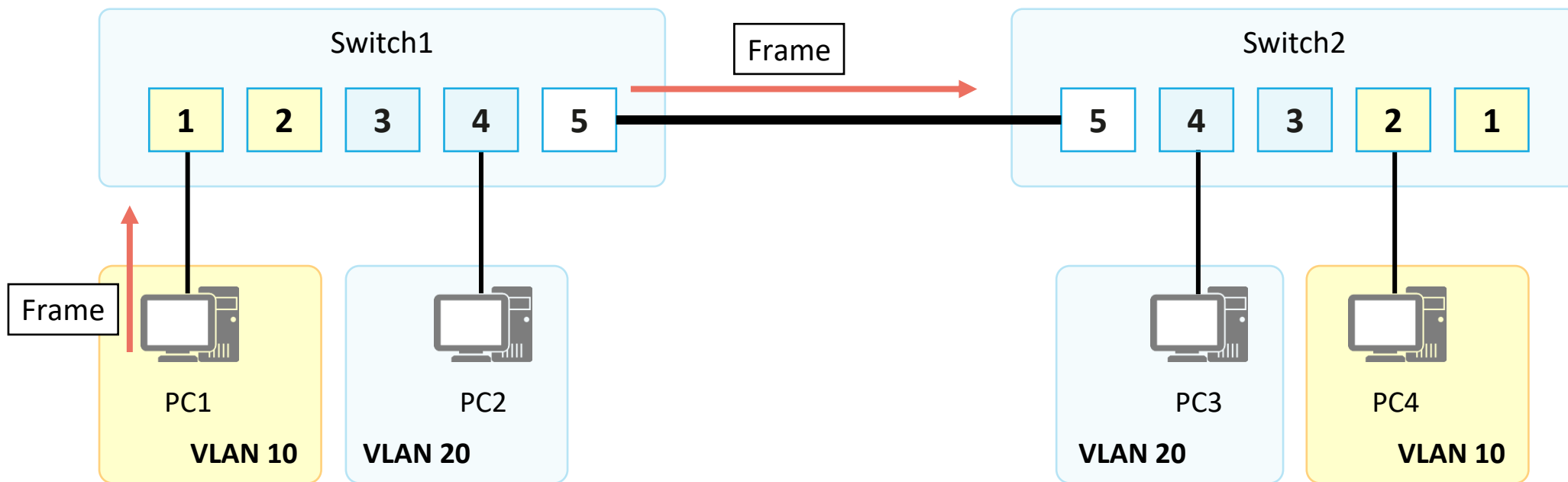


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- 2. VLAN Principles**
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VLAN Implementation

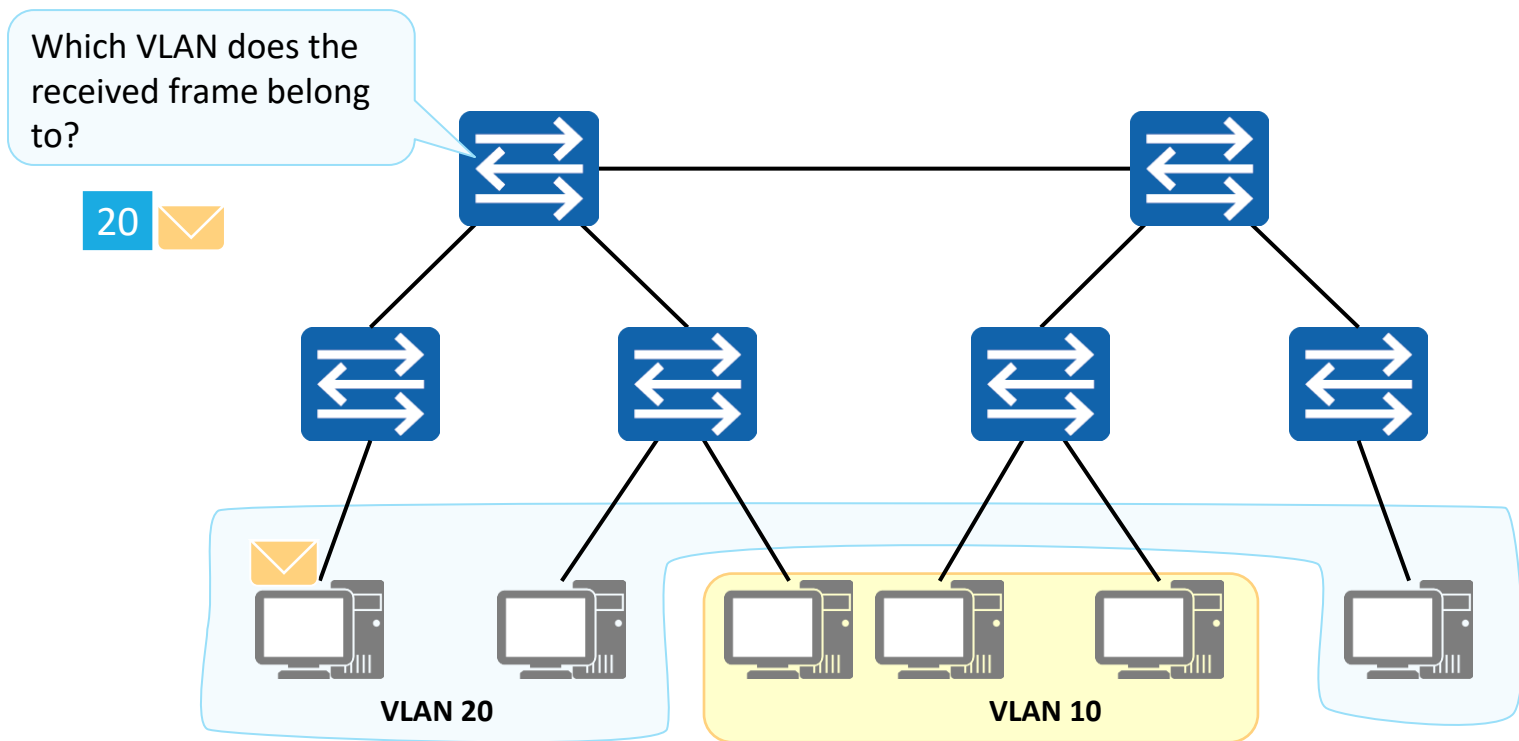


- Switch1 and Switch2 belong to the network of the same enterprise. VLANs are planned for the network, with VLAN 10 for department A and VLAN 20 for department B. Employees in departments A and B are connected to both Switch1 and Switch2.
- Assume that a frame sent from PC1 reaches Switch2 through the link between Switch1 and Switch2. **If no processing is implemented, Switch2 can neither identify the VLAN to which the frame belongs nor determine the local VLAN to which the frame should be sent.**



VLAN Tag

- How does a switch identify the VLAN to which a received frame belongs?



VLAN Tag

IEEE 802.1Q defines a 4-byte VLAN tag for Ethernet frames, enabling switches to identify the VLANs to which received frames belong.



VLAN Frame

Original Ethernet frame
(untagged frame)

Destination MAC address	Source MAC address	Length/Type	Data	FCS
-------------------------	--------------------	-------------	------	-----

802.1Q tag inserted between
the two fields

TPID (0x8100)	PRI	CFI	VLAN ID
16 bits	3 bits	1 bit	12 bits

802.1Q tag

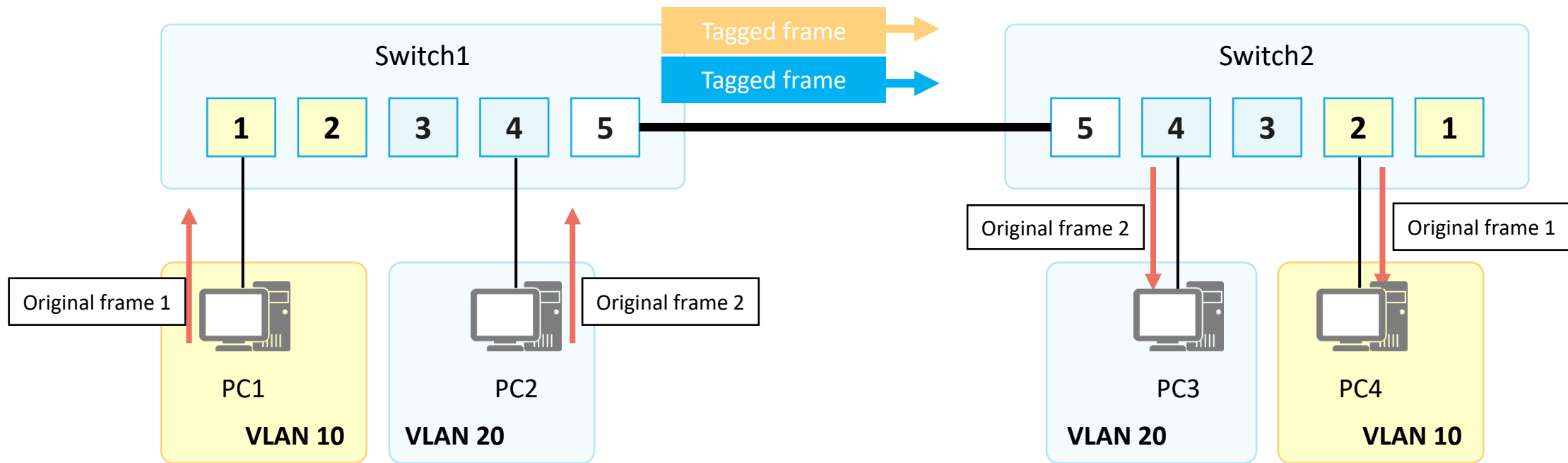
- Tag protocol identifier (TPID): identifies the type of a frame. The value 0x8100 indicates an IEEE 802.1Q frame.
- PRI: identifies the priority of a frame, which is mainly used for QoS.
- Canonical format indicator (CFI): indicates whether a MAC address is in the canonical format. For Ethernet frames, the value of this field is 0.
- VLAN ID: identifies the VLAN to which a frame belongs.

802.1Q frame
(tagged frame)

Destination MAC address	Source MAC address	Tag	Length/Type	Data	FCS
-------------------------	--------------------	-----	-------------	------	-----



VLAN Implementation

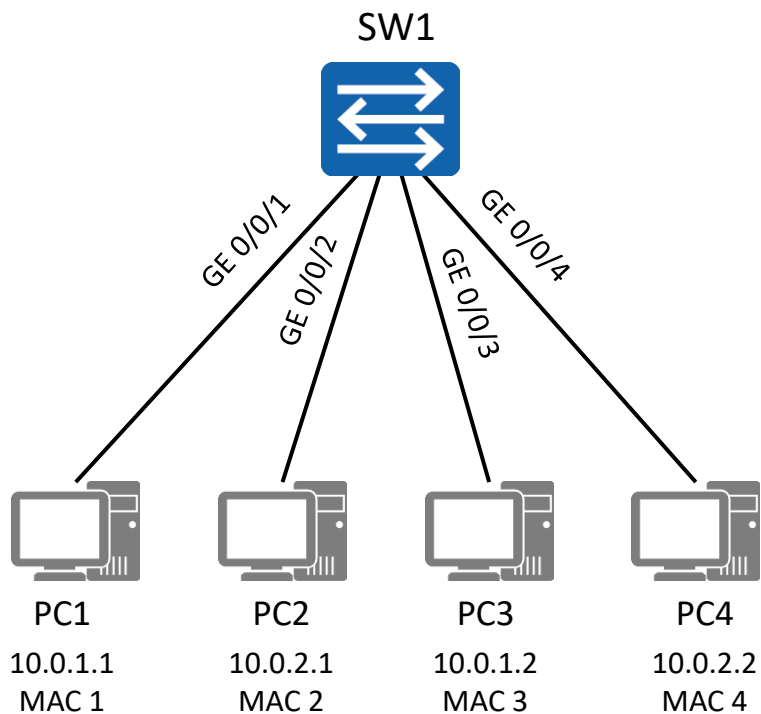


- The link between Switch1 and Switch2 carries data of multiple VLANs. In this situation, a VLAN-based data tagging method is required to distinguish the frames of different VLANs.
- IEEE 802.1Q, often referred to as Dot1q, defines a system of VLAN tagging for Ethernet frames by inserting an 802.1Q tag into the frame header to carry VLAN information.



VLAN Assignment Methods

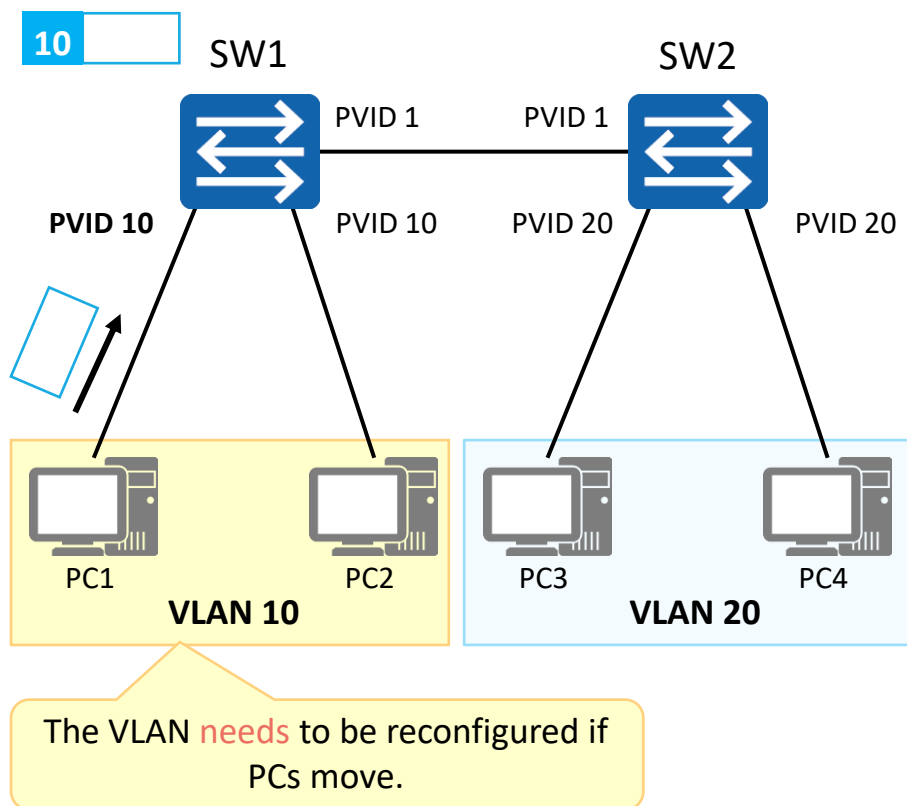
- How are VLANs assigned on a network?



VLAN Assignment Method	VLAN 10	VLAN 20
Interface-based assignment	GE 0/0/1 and GE 0/0/3	GE 0/0/2 and GE 0/0/4
MAC address-based assignment	MAC 1 and MAC 3	MAC 2 and MAC 4
IP subnet-based assignment	10.0.1.*	10.0.2.*
Protocol-based assignment	IP	IPv6
Policy-based assignment	10.0.1.* + GE 0/0/1 + MAC 1	10.0.2.* + GE 0/0/2 + MAC 2



Interface-based VLAN Assignment



Interface-based VLAN Assignment

- **Principles**

- VLANs are assigned based on interfaces.
- A network administrator preconfigures a PVID for each switch interface and assigns each interface to a VLAN corresponding to the PVID.
- After an interface receives an untagged frame, the switch adds a tag carrying the PVID of the interface to the frame. The frame is then transmitted in the specified VLAN.

- **Port Default VLAN ID: PVID**

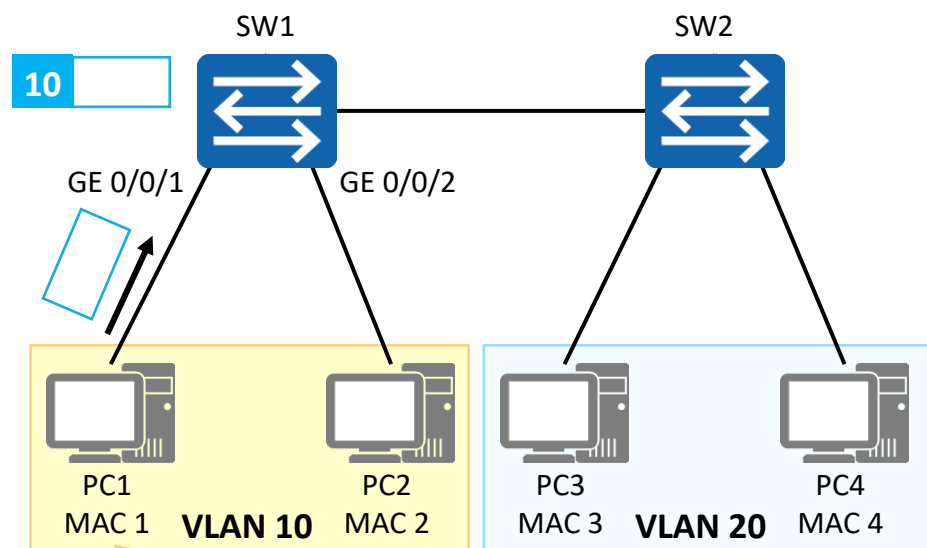
- Default VLAN ID for an interface
- Value range: 1–4094



MAC Address-based VLAN Assignment

Mapping Between MAC Addresses and
VLAN IDs on SW1

MAC Address	VLAN ID
MAC 1	10
MAC 2	10
...	...



The VLAN **does not need** to be reconfigured
even if PCs move.

MAC Address-based VLAN Assignment

• Principles

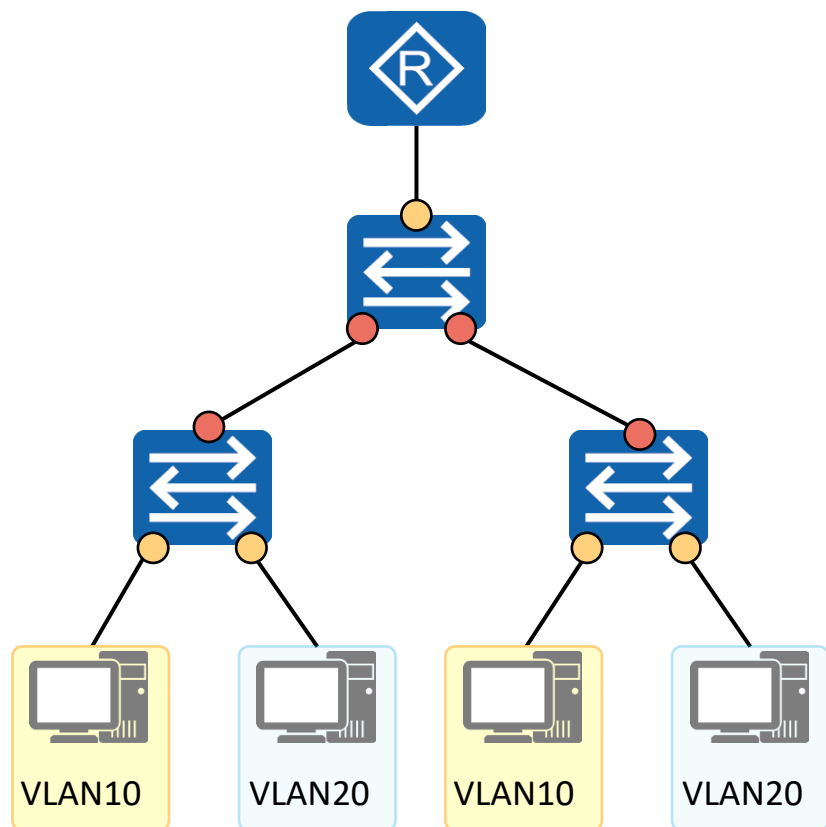
- VLANs are assigned based on the source MAC addresses of frames.
- A network administrator preconfigures the mapping between MAC addresses and VLAN IDs.
- After receiving an untagged frame, a switch adds the VLAN tag mapping the source MAC address of the frame to the frame. The frame is then transmitted in the specified VLAN.

• Mapping table

- Records the mapping between MAC addresses and VLAN IDs.



Layer 2 Ethernet Interface Types



● Access interface ● Trunk interface

Interface Types

- **Access interface**

An access interface is used to connect a switch to a terminal, such as a PC or server and is used to receive or send only untagged frames. An access interface can be added to only one VLAN.

- **Trunk interface**

A trunk interface allows frames that belong to multiple VLANs to pass through and differentiates the frames using the 802.1Q tag. This type of interface is used to connect a switch to another switch or a sub-interface on a device, such as a router or firewall.

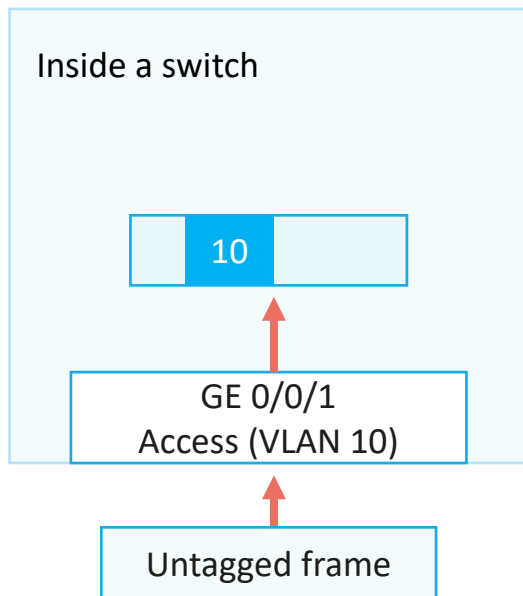
- **Hybrid interface**

Similar to a trunk interface, a hybrid interface also allows frames that belong to multiple VLANs to pass through and differentiates the frames using the 802.1Q tag. You can determine whether to allow a hybrid interface to carry VLAN tags when sending the frames of one or more VLANs.



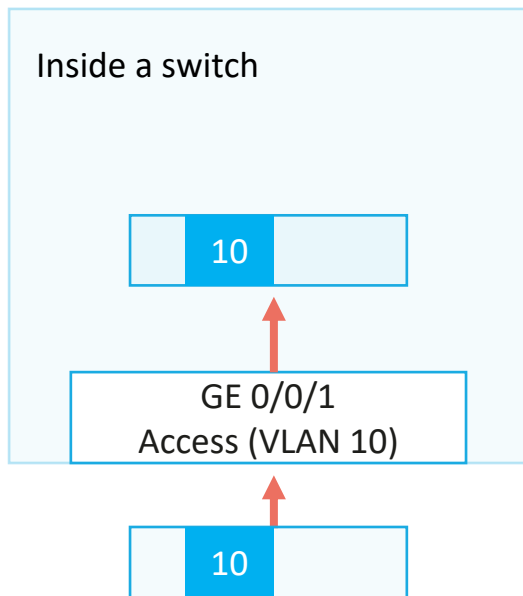
Access Interface

Frame receiving



After receiving an untagged frame:

The interface permits the frame and adds a VLAN tag carrying the PVID of the interface.

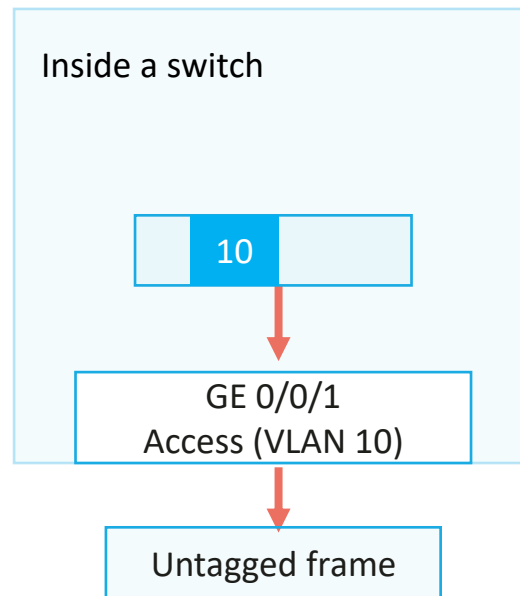


After receiving a tagged frame:

If the VLAN ID of the frame is the same as the PVID of the interface, the interface permits the frame.

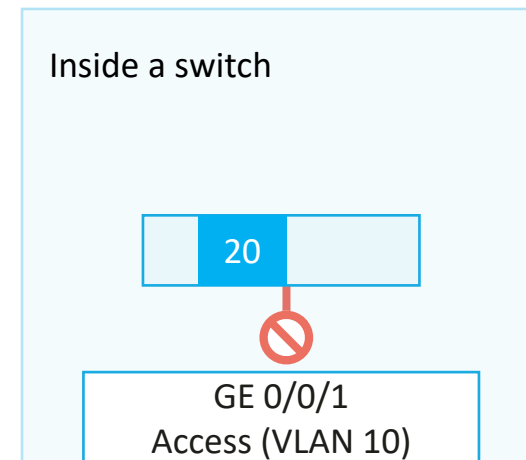
If the VLAN ID of the frame is different from the PVID of the interface, the interface discards the frame.

Frame sending



If the VLAN ID of the frame is the same as the PVID of the interface:

The interface removes the VLAN tag from the frame and then sends the frame.



If the VLAN ID of the frame is different from the PVID of the interface:

The interface discards the frame.

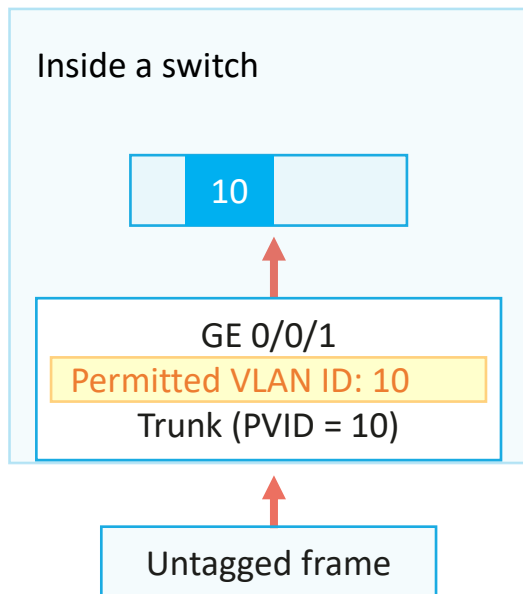
Untagged frame

10 Tagged frame



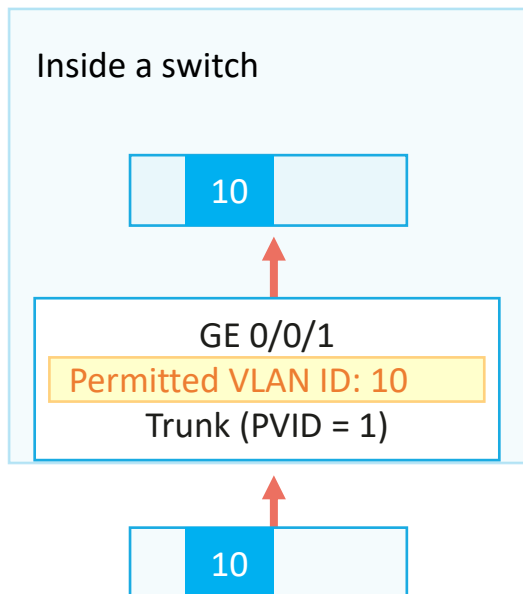
Trunk interface

Frame receiving



After receiving an untagged frame:

The interface adds a VLAN tag with the VID being the PVID of the interface to the frame and permits the frame only when the **VID is in the list of VLAN IDs permitted by the interface**. If the VID is not in the list, the interface discards the frame.



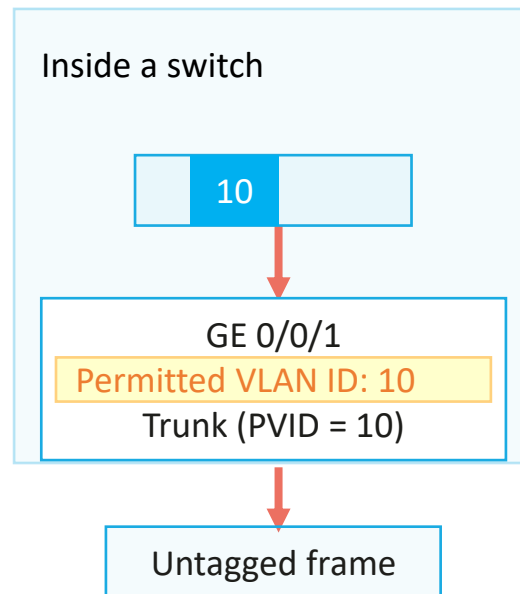
After receiving a tagged frame:

If the VLAN ID of the frame is in the list of VLAN IDs permitted by the interface, the interface permits the frame. Otherwise, the interface discards the frame.

Untagged frame

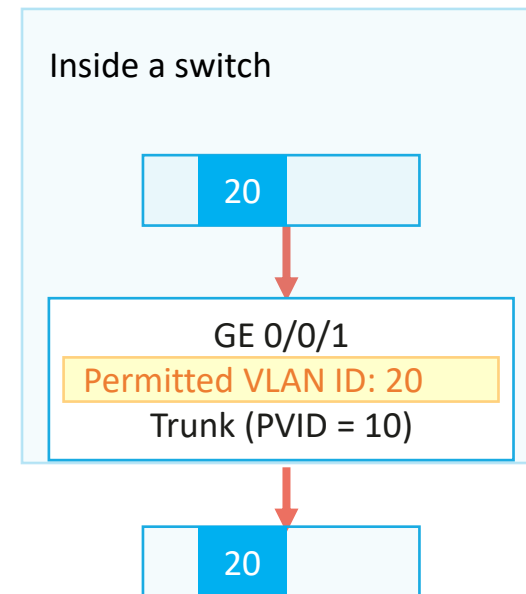
10 Tagged frame

Frame sending



If the VLAN ID of the frame is the same as the PVID of the interface:

If the VLAN ID of the frame is in the list of VLAN IDs permitted by the interface, the interface removes the tag from the frame and sends the frame out. Otherwise, the interface discards the frame.



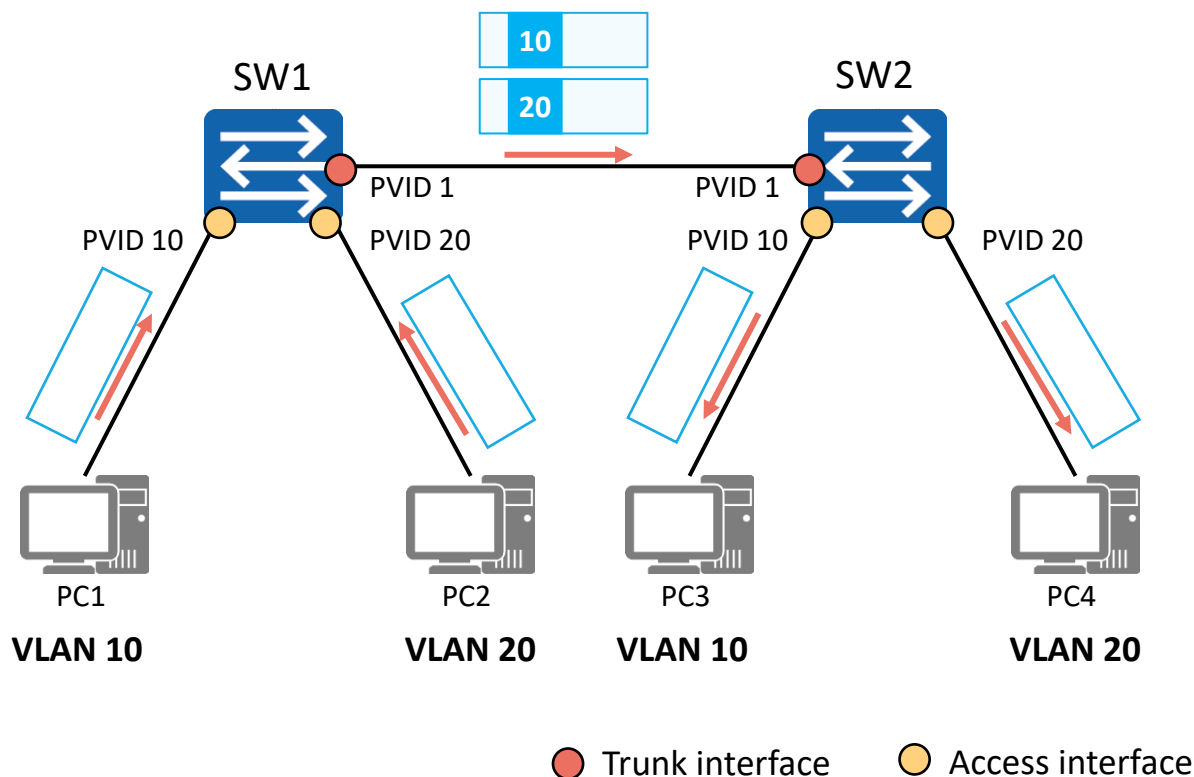
If the VLAN ID of the frame is different from the PVID of the interface:

If the VLAN ID of the frame is in the list of VLAN IDs permitted by the interface, the interface sends the frame out without removing the tag of the frame. Otherwise, the interface discards the frame.



Example for Frame Processing on Access and Trunk Interfaces

- Describe how inter-PC access is implemented in this example.



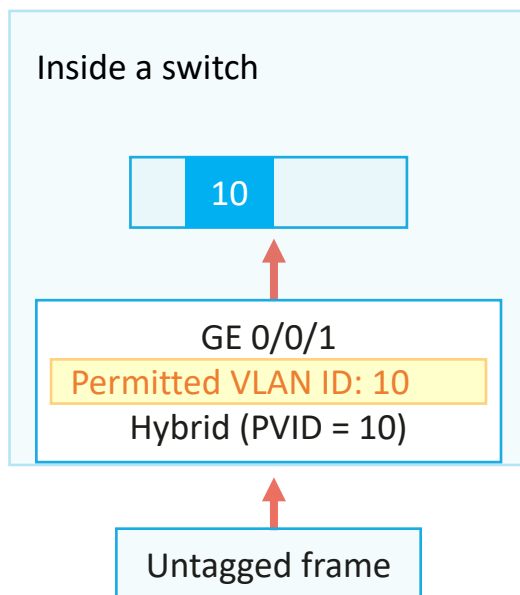
Trunk Interfaces on SW1 and SW2

List of Permitted VLAN IDs	
VLAN ID	1
	10
	20



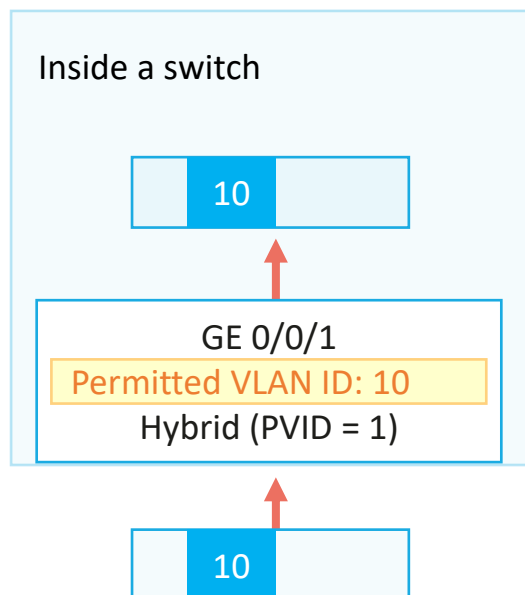
Hybrid Interface

Frame receiving



After receiving an untagged frame:

The interface adds a VLAN tag with the VID being the PVID of the interface to the frame and permits the frame only when the VID is in the list of VLAN IDs permitted by the interface. If the VID is not in the list, the interface discards the frame.

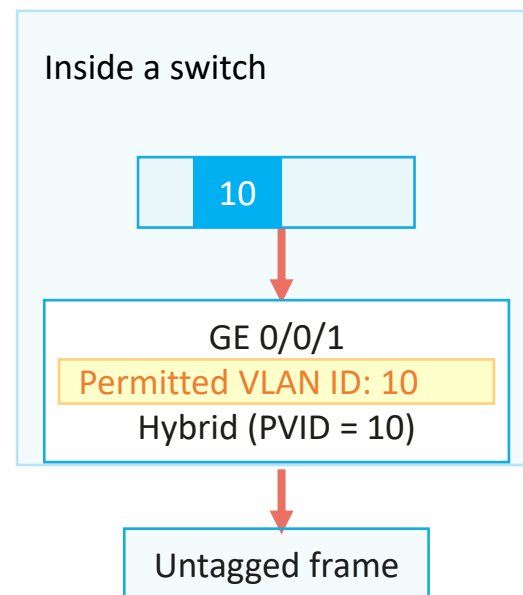


After receiving a tagged frame:

If the VLAN ID of the frame is in the list of VLAN IDs permitted by the interface, the interface permits the frame. Otherwise, the interface discards the frame.

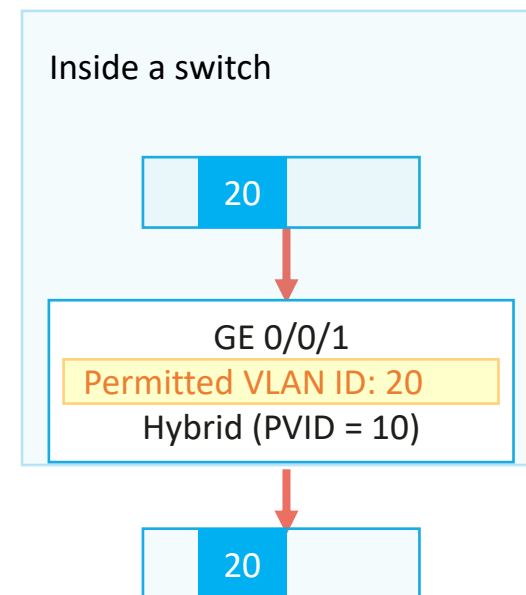


Frame sending



If the VLAN ID of the frame is in the list of VLAN IDs permitted by the interface:

If the interface has been configured not to carry VLAN tags when sending frames, it removes the tag from the frame and then sends the frame out.



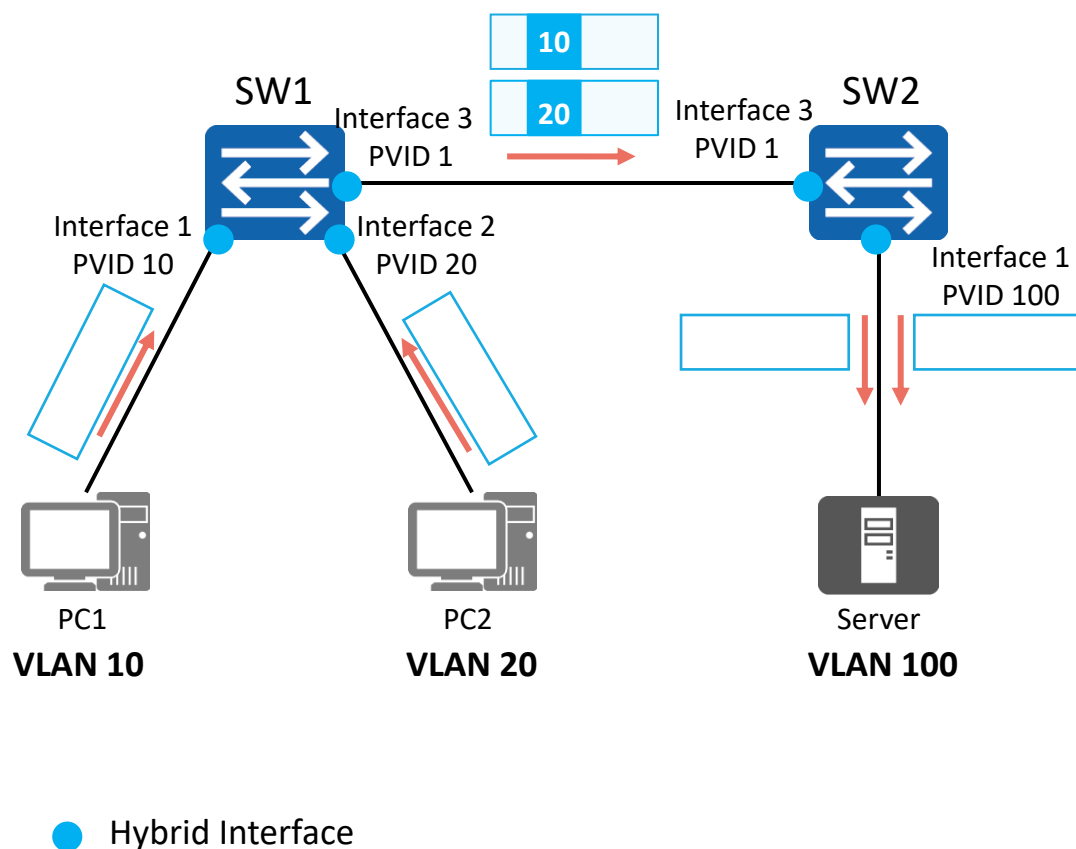
If the VLAN ID of the frame is in the list of VLAN IDs permitted by the interface:

If the interface has been configured to carry VLAN tags when sending frames, it sends the frame out without removing the tag of the frame.



Example for Frame Processing on Hybrid Interfaces

- Describe how PCs access the server in this example.



List of VLAN IDs Permitted by Interfaces on SW1

Interface 1		Interface 2		Interface 3	
Untagged		Untagged		Tagged	
VLAN ID	1	VLAN ID	1	VLAN ID	10
	10		20		10
	100		100		100

List of VLAN IDs Permitted by Interfaces on SW2

Interface 1		Interface 3	
Untagged		Tagged	
VLAN ID	1	VLAN ID	10
	10		20
	20		100
	100		



Summary

Access Interface

Frame receiving

- Untagged frame: adds a tag with the VID being the PVID of the interface and permits the frame.
- Tagged frame: checks whether the VID in the tag of the frame is the same as the PVID of the interface. If they are the same, permits the frame; otherwise, discards the frame.

Frame sending

- Checks whether the VID in the tag of the frame is the same as the PVID of the interface. If they are the same, removes the tag and sends the frame out; otherwise, discards the frame.

Trunk Interface

Frame receiving

- Untagged frame: adds a tag with the VID being the PVID of the interface and checks whether the VID is in the list of permitted VLAN IDs. If yes, permits the frame. If not, discards it.
- Tagged frame: checks whether the VID is in the list of permitted VLAN IDs. If yes, permits the frame. If not, discards it.

Frame sending

- If the VID is in the list of permitted VLAN IDs and the same as the PVID of the interface, removes the tag and sends the frame out.
- If the VID is in the list of permitted VLAN IDs but different from the PVID of the interface, sends the frame out without removing the tag.
- If the VID is not in the list of permitted VLAN IDs, discards the frame.

Hybrid Interface

Frame receiving

- Untagged frame: adds a tag with the VID being the PVID of the interface and checks whether the VID is in the list of permitted VLAN IDs. If yes, permits the frame. If not, discards it.
- Tagged frame: checks whether the VID is in the list of permitted VLAN IDs. If yes, permits the frame. If not, discards it.

Frame sending

- If the VID is not in the list of permitted VLAN IDs, discards the frame.
- If the VID is in the untagged VLAN ID list, removes the tag and sends the frame out.
- If the VID is in the tagged VLAN ID list, sends the frame out without removing the tag.



Contents

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VLAN Planning

- **VLAN assignment rules**

- By service: voice, video, and data VLANs
- By department: e.g. VLANs for engineering, marketing, and financing departments
- By application: e.g. VLANs for servers, offices, and classrooms

- **Example for VLAN planning**

- Assume that there are three buildings: administrative building with offices, classrooms, and financing sections, teaching building with offices and classrooms, and office building with offices and financing sections. Each building has one access switch, and the core switch is deployed in the administrative building.
- The following table describes the VLAN plan.

VLAN ID	IP Address Segment	Description
1	X.16.10.0/24	VLAN to which office users belong
2	X.16.20.0/24	VLAN to which the users of the financing department belong
3	X.16.30.0/24	VLAN to which classroom users belong
100	Y.16.100.0/24	VLAN to which the device management function belongs

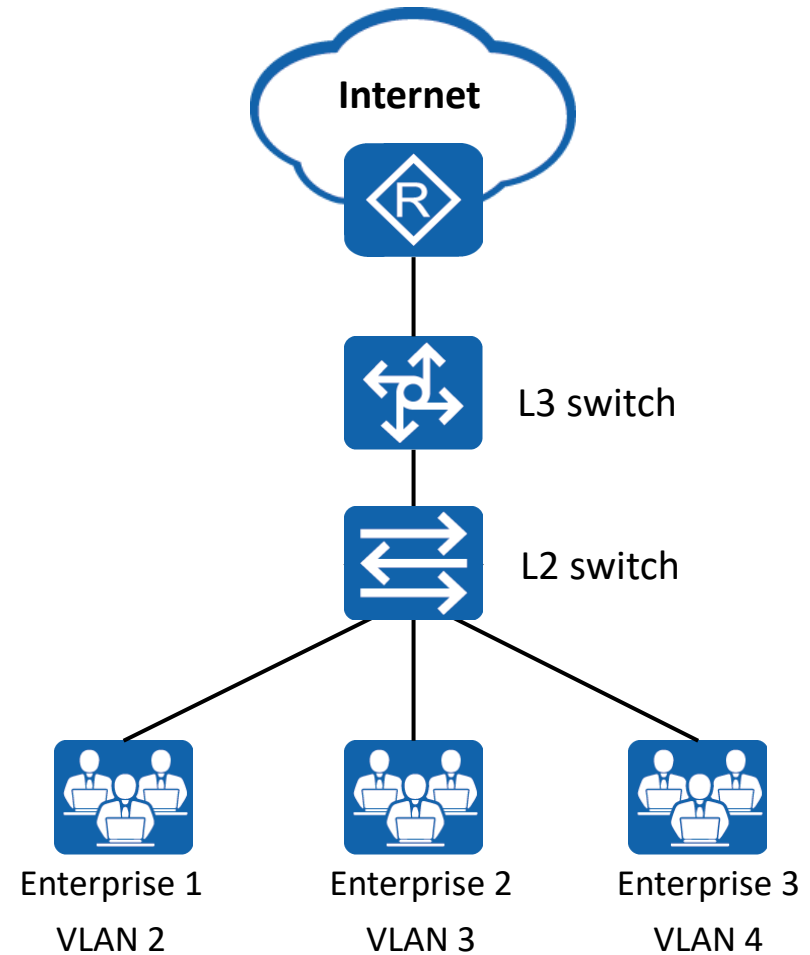
- **Tips for VLAN assignment**

VLAN IDs can be randomly assigned within the supported range. To improve VLAN ID continuity, you can associate VLAN IDs with subnets during VLAN assignment.



Interface-based VLAN Assignment

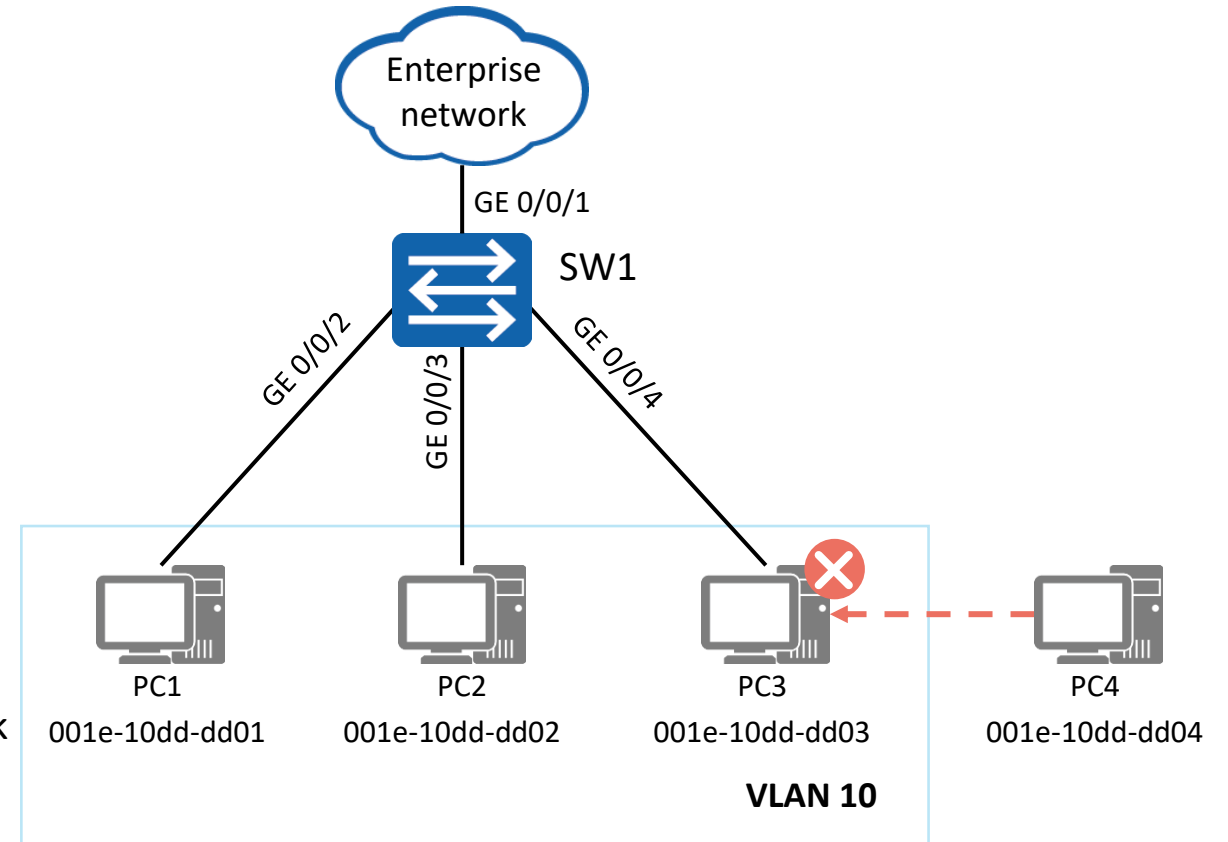
- Applicable scenario:
 - There are multiple enterprises in a building. These enterprises share network resources to reduce costs. Networks of the enterprises connect to different interfaces of the same Layer 2 switch and access the Internet through the same egress device.
- VLAN assignment:
 - To isolate the services of different enterprises and ensure service security, assign interfaces connected to the enterprises' networks to different VLANs. In this way, each enterprise has an independent network, and each VLAN works as a virtual work group.





MAC Address-based VLAN Assignment

- Applicable scenario:
 - The network administrator of an enterprise assigns PCs in the same department to the same VLAN. To improve information security, the enterprise requires that only employees in the specified department be allowed to access specific network resources.
- VLAN assignment:
 - To meet the preceding requirement, configure MAC address-based VLAN assignment on SW1, preventing new PCs connected to the network from accessing the network resources.





Contents

1. What Is VLAN
2. VLAN Principles
3. VLAN Applications
- 4. VLAN Configuration Examples**



Basic VLAN Configuration Commands

1. Create one or more VLANs.

```
[Huawei] vlan vlan-id
```

This command creates a VLAN and displays the VLAN view. If the VLAN to be created already exists, this command directly displays the VLAN view.

- The value of *vlan-id* is an integer ranging from 1 to 4094.

```
[Huawei] vlan batch { vlan-id1 [ to vlan-id2 ] }
```

This command creates VLANs in a batch. In this command:

- **batch**: creates VLANs in a batch.
- *vlan-id1*: specifies a start VLAN ID.
- *vlan-id2*: specifies an end VLAN ID.



Basic Access Interface Configuration Commands

1. Set the link type of an interface.

```
[Huawei-GigabitEthernet0/0/1] port link-type access
```

In the interface view, set the link type of the interface to access.

2. Configure a default VLAN for the access interface.

```
[Huawei-GigabitEthernet0/0/1] port default vlan vlan-id
```

In the interface view, configure a default VLAN for the interface and add the interface to the VLAN.

- *vlan-id*: specifies an ID for the default VLAN. The value is an integer ranging from 1 to 4094.



Basic Trunk Interface Configuration Commands

1. Set the link type of an interface.

```
[Huawei-GigabitEthernet0/0/1] port link-type trunk
```

In the interface view, set the link type of the interface to trunk.

2. Add the trunk interface to specified VLANs.

```
[Huawei-GigabitEthernet0/0/1] port trunk allow-pass vlan { { vlan-id1 [ to vlan-id2 ] } | all }
```

In the interface view, add the trunk interface to specified VLANs.

3. (Optional) Configure a default VLAN for the trunk interface.

```
[Huawei-GigabitEthernet0/0/1] port trunk pvid vlan vlan-id
```

In the interface view, configure a default VLAN for the trunk interface.



Basic Hybrid Interface Configuration Commands

1. Set the link type of an interface.

```
[Huawei-GigabitEthernet0/0/1] port link-type hybrid
```

In the interface view, set the link type of the interface to hybrid.

2. Add the hybrid interface to specified VLANs.

```
[Huawei-GigabitEthernet0/0/1] port hybrid untagged vlan { { vlan-id1 [ to vlan-id2 ] } | all }
```

In the interface view, add the hybrid interface to specified VLANs in untagged mode.

```
[Huawei-GigabitEthernet0/0/1] port hybrid tagged vlan { { vlan-id1 [ to vlan-id2 ] } | all }
```

In the interface view, add the hybrid interface to specified VLANs in tagged mode.

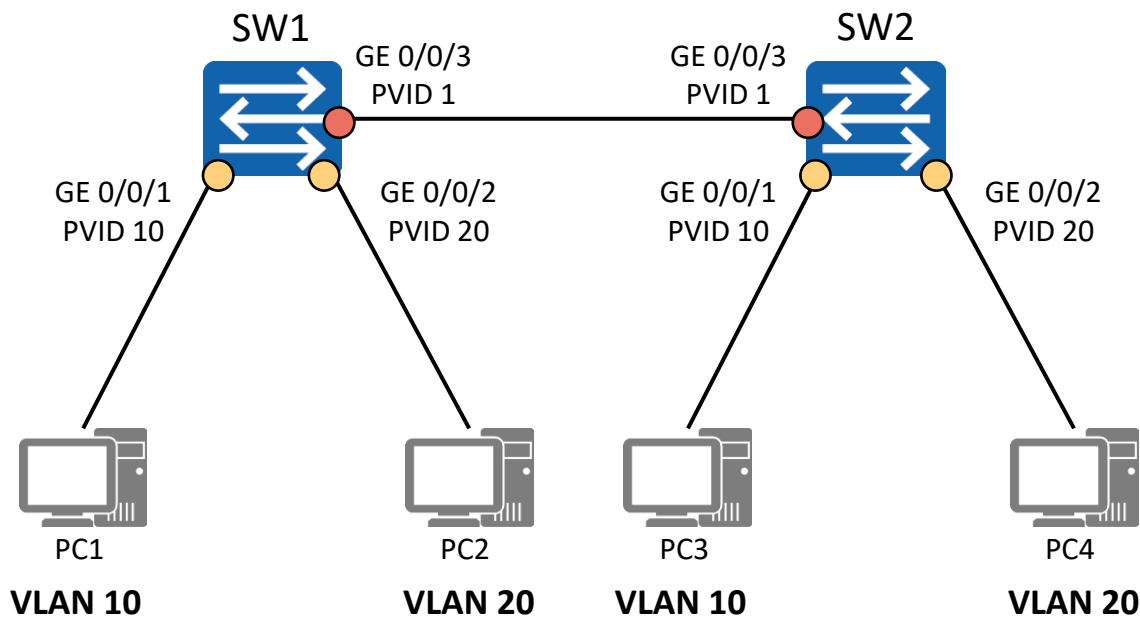
3. (Optional) Configure a default VLAN for the hybrid interface.

```
[Huawei-GigabitEthernet0/0/1] port hybrid pvid vlan vlan-id
```

In the interface view, configure a default VLAN for the hybrid interface.



Case1:Configuring Interface-based VLAN Assignment



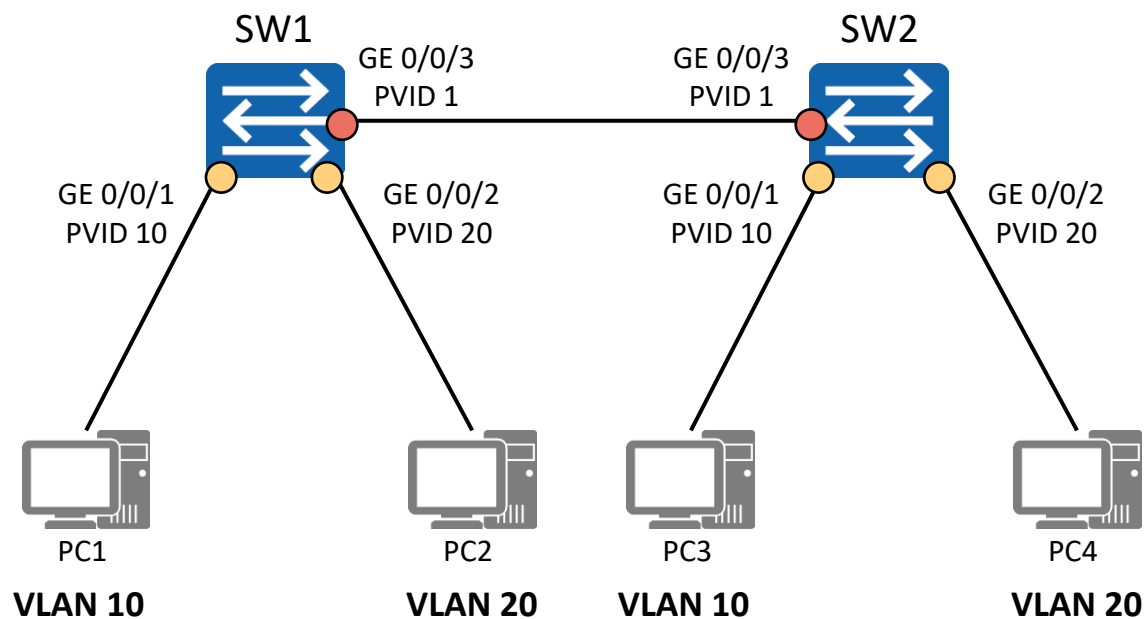
● Access interface

● Trunk interface

- Networking requirements:
 - On the network shown in the left figure, the switches (SW1 and SW2) of an enterprise are connected to multiple PCs, and PCs with the same services access the network using different devices. To ensure communication security, the enterprise requires that only PCs with the same service can directly communicate.
 - To meet this requirement, configure interface-based VLAN assignment on the switches and add interfaces connected to PCs with the same service to the same VLAN. In this way, PCs in different VLANs cannot directly communicate at Layer 2, but PCs in the same VLAN can directly communicate.



Creating VLANs



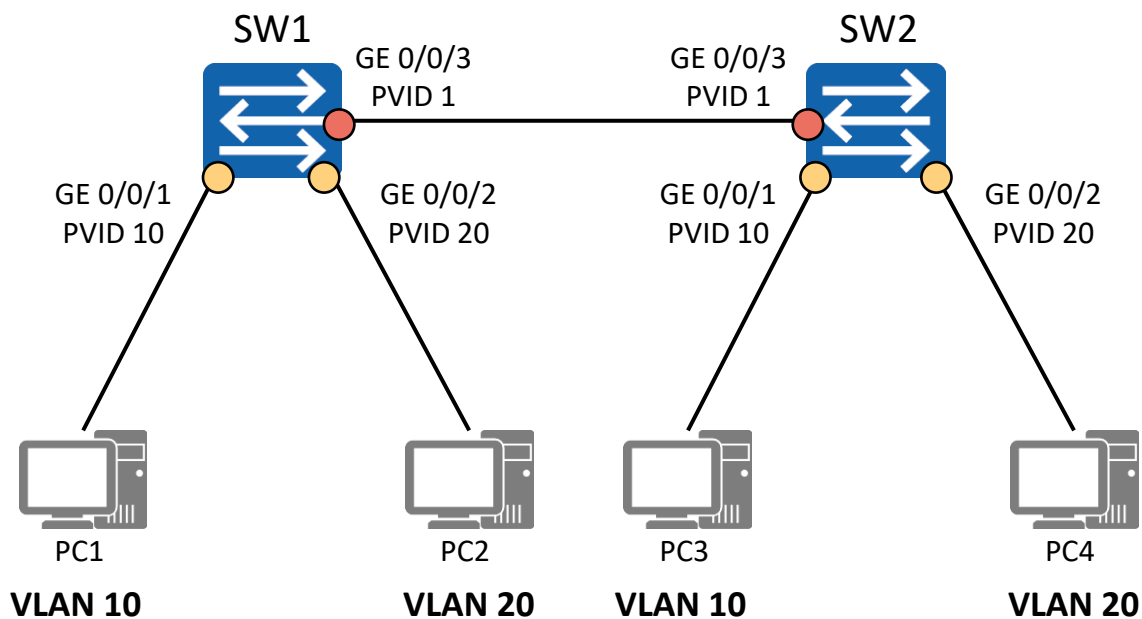
Create VLANs.

```
[SW1] vlan 10  
[SW1-vlan10] quit  
[SW1] vlan 20  
[SW1-vlan20] quit
```

```
[SW2] vlan batch 10 20
```



Configuring Access and Trunk Interfaces



Configure access interfaces and add the interfaces to corresponding VLANs.

```
[SW1] interface GigabitEthernet 0/0/1
[SW1-GigabitEthernet0/0/1] port link-type access
[SW1-GigabitEthernet0/0/1] port default vlan 10
```

```
[SW1] interface GigabitEthernet 0/0/2
[SW1-GigabitEthernet0/0/2] port link-type access
[SW1] vlan 20
[SW1-vlan20] port GigabitEthernet0/0/2
[SW1-vlan20] quit
```

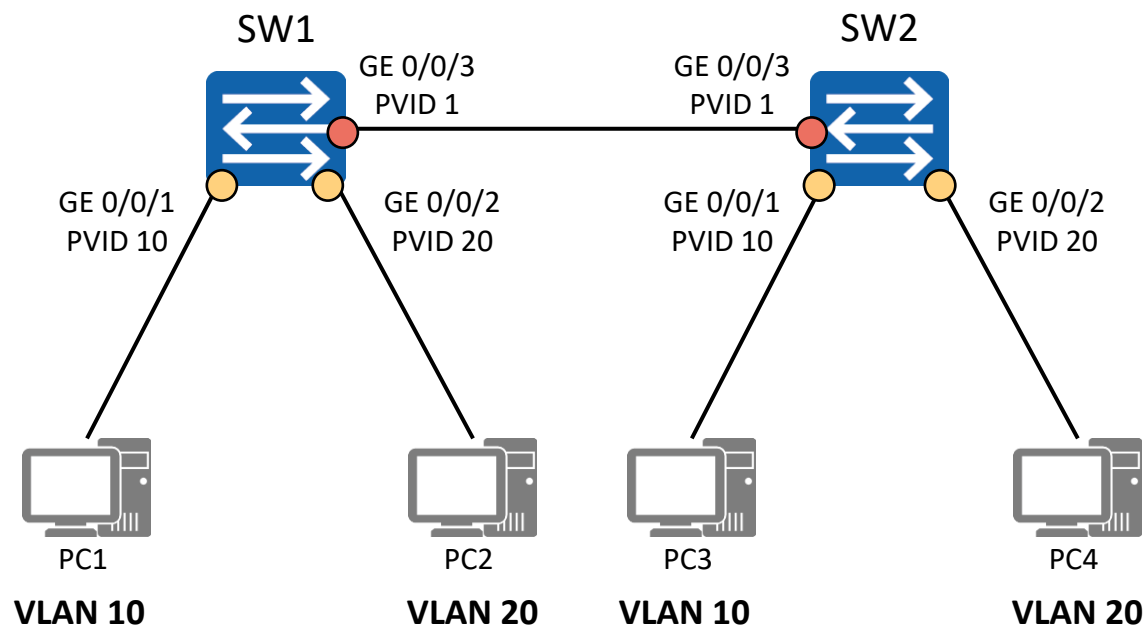
Configure a trunk interface and specify a list of VLAN IDs permitted by the interface.

```
[SW1] interface GigabitEthernet 0/0/3
[SW1-GigabitEthernet0/0/3] port link-type trunk
[SW1-GigabitEthernet0/0/3] port trunk pvid vlan 1
[SW1-GigabitEthernet0/0/3] port trunk allow-pass vlan 10 20
```

Note: The configuration on SW2 is similar to that on SW1.



Verifying the Configuration



[SW1]display vlan

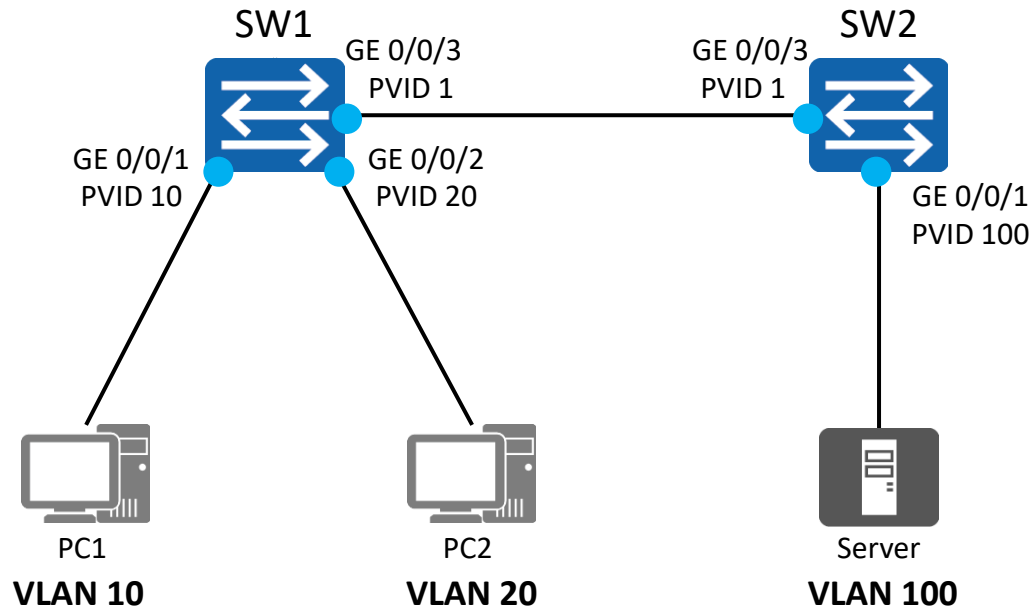
The total number of vlans is : 3

U: Up; D: Down; TG: Tagged; UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;

VID	Type	Ports
1	common	UT:GE0/0/3(U)
10	common	UT:GE0/0/1(U) TG:GE0/0/3(U)
20	common	UT:GE0/0/2(U) TG:GE0/0/3(U)
.....		



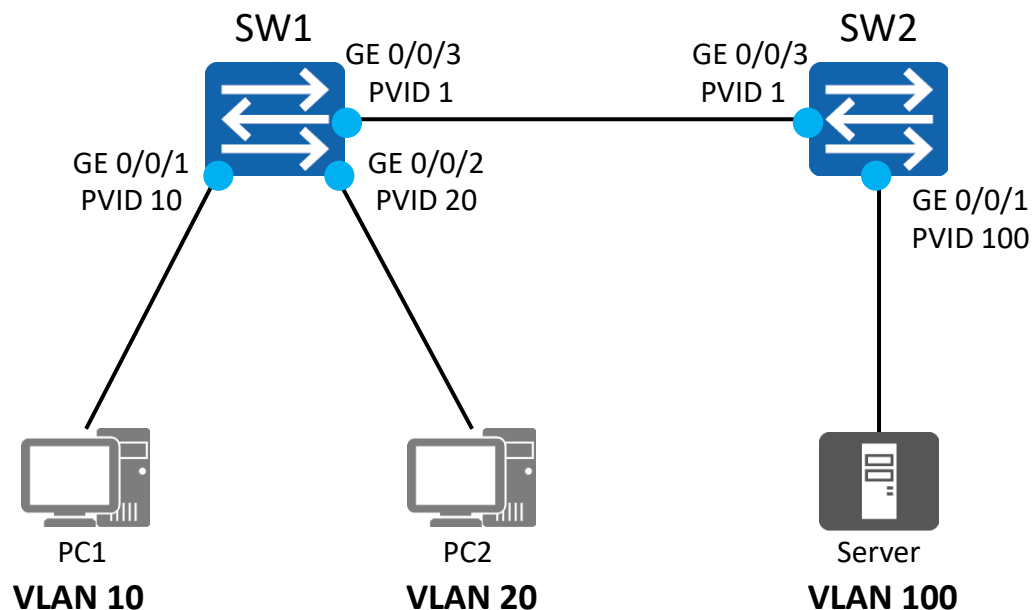
Case2:Configuring Interface-based VLAN Assignment



- Networking requirements:
 - On the network shown in the left figure, the switches (SW1 and SW2) of an enterprise are connected to multiple PCs, and PCs in different departments need to access the server of the enterprise. To ensure communication security, the enterprise requires that PCs in different departments cannot directly communicate.
 - To meet this requirement, configure interface-based VLAN assignment and hybrid interfaces on the switches to enable PCs in different departments to access the server but disable them from directly communicating at Layer 2.



Configuring Hybrid Interfaces (1)

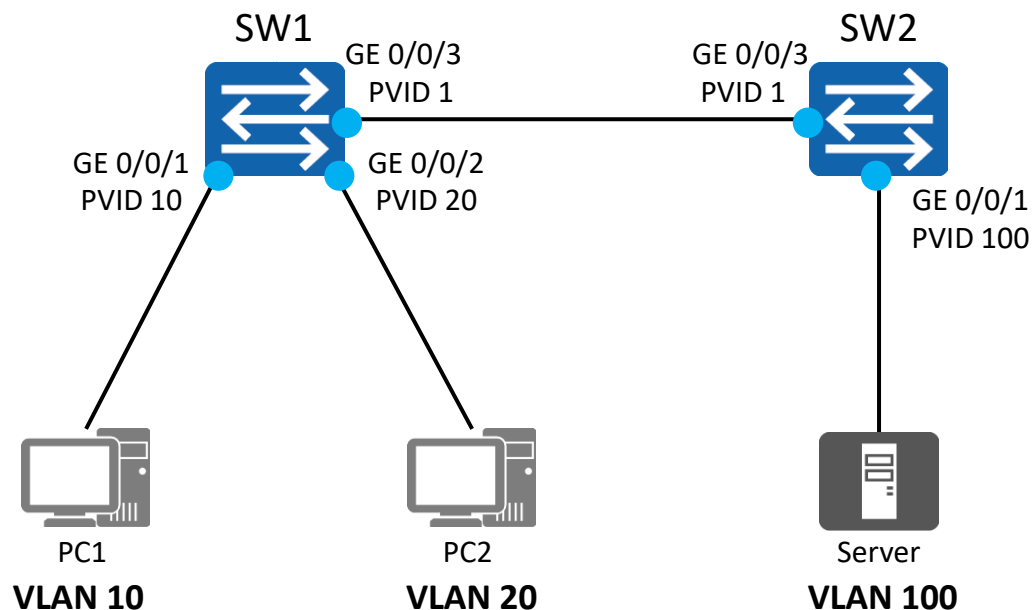


SW1 configuration:

```
[SW1] vlan batch 10 20 100
[SW1] interface GigabitEthernet 0/0/1
[SW1-GigabitEthernet0/0/1] port link-type hybrid
[SW1-GigabitEthernet0/0/1] port hybrid pvid vlan 10
[SW1-GigabitEthernet0/0/1] port hybrid untagged vlan 10 100
[SW1-GigabitEthernet0/0/1] interface GigabitEthernet 0/0/2
[SW1-GigabitEthernet0/0/2] port link-type hybrid
[SW1-GigabitEthernet0/0/2] port hybrid pvid vlan 20
[SW1-GigabitEthernet0/0/2] port hybrid untagged vlan 20 100
[SW1-GigabitEthernet0/0/2] interface GigabitEthernet 0/0/3
[SW1-GigabitEthernet0/0/3] port link-type hybrid
[SW1-GigabitEthernet0/0/3] port hybrid tagged vlan 10 20 100
```



Configuring Hybrid Interfaces (2)

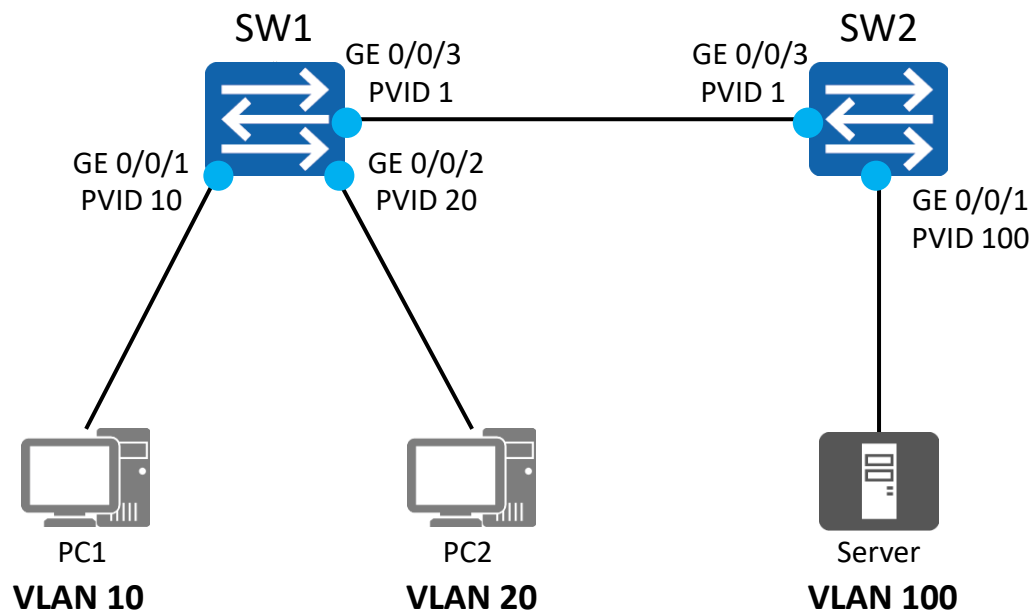


SW2 configuration:

```
[SW2] vlan batch 10 20 100
[SW2] interface GigabitEthernet 0/0/1
[SW2-GigabitEthernet0/0/1] port link-type hybrid
[SW2-GigabitEthernet0/0/1] port hybrid pvid vlan 100
[SW2-GigabitEthernet0/0/1] port hybrid untagged vlan 10 20 100
[SW2-GigabitEthernet0/0/1] interface GigabitEthernet 0/0/3
[SW2-GigabitEthernet0/0/3] port link-type hybrid
[SW2-GigabitEthernet0/0/3] port hybrid tagged vlan 10 20 100
```



Verifying the Configuration



```
[SW1]display vlan
```

The total number of vlans is : 4

U: Up; D: Down; TG: Tagged; UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;

VID	Type	Ports
1	common	UT:GE0/0/1(U) GE0/0/2(U) GE0/0/3(U)
10	common	UT:GE0/0/1(U) TG:GE0/0/3(U)
20	common	UT:GE0/0/2(U) TG:GE0/0/3(U)
100	common	UT:GE0/0/1(U) GE0/0/2(U) TG:GE0/0/3(U)
.....		



Basic VLAN Configuration Commands

1. Associate a MAC address with a VLAN.

```
[Huawei-vlan10] mac-vlan mac-address mac-address [ mac-address-mask | mac-address-mask-length ]
```

This command associates a MAC address with a VLAN.

- *mac-address*: specifies the MAC address to be associated with a VLAN. The value is a hexadecimal number in the format of H-H-H. Each H contains one to four digits, such as 00e0 or fc01. If an H contains less than four digits, the left-most digits are padded with zeros. For example, e0 is displayed as 00e0. The MAC address cannot be 0000-0000-0000, FFFF-FFFF-FFFF, or any multicast address.
- *mac-address-mask*: specifies the mask of a MAC address. The value is a hexadecimal number in the format of H-H-H. Each H contains one to four digits.
- *mac-address-mask-length*: specifies the mask length of a MAC address. The value is an integer ranging from 1 to 48.

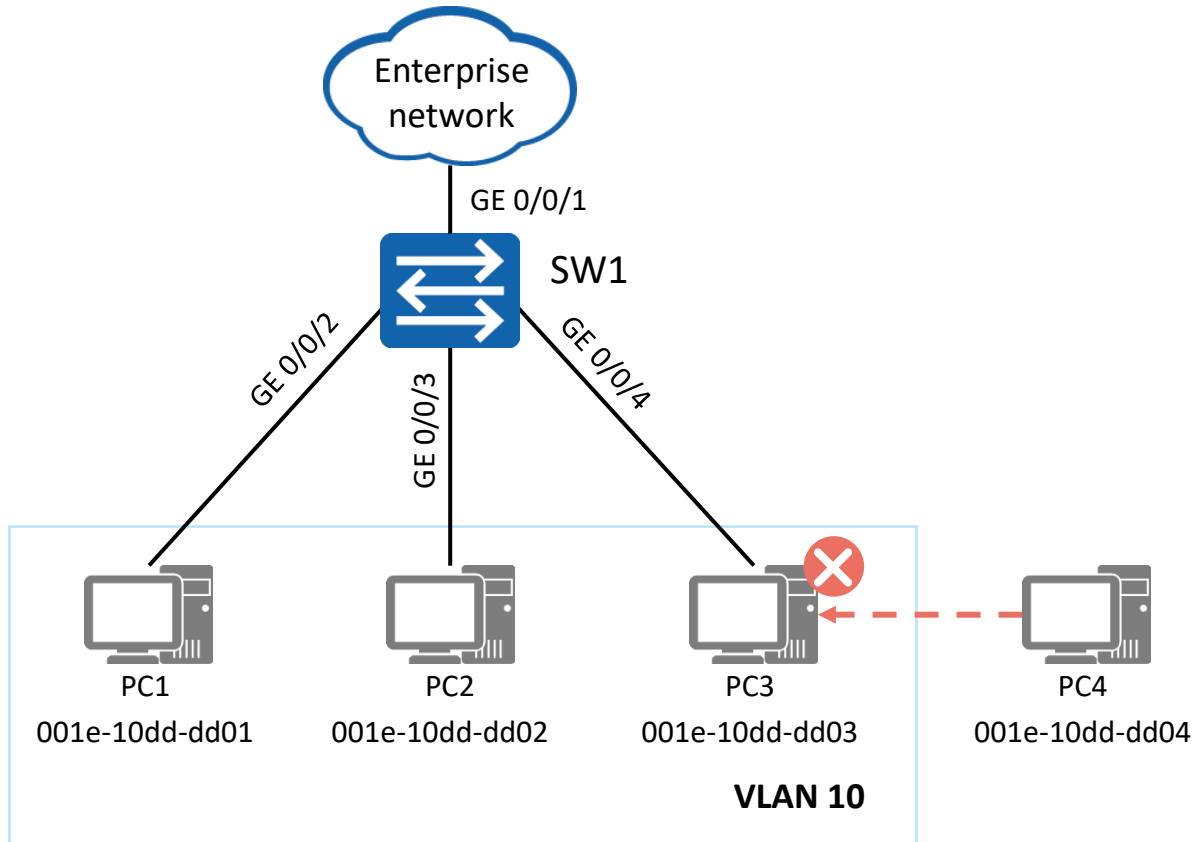
2. Enable MAC address-based VLAN assignment on an interface.

```
[Huawei-GigabitEthernet0/0/1] mac-vlan enable
```

This command enables MAC address-based VLAN assignment on an interface.



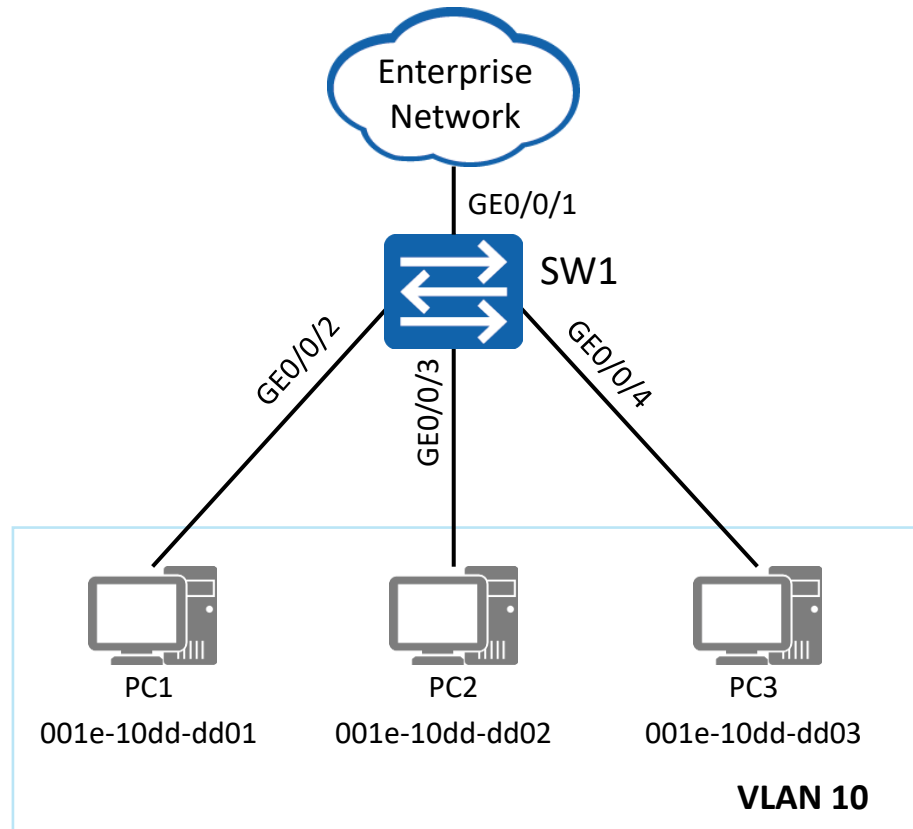
Example for Configuring MAC Address-based VLAN Assignment



- Networking requirements:
 - The network administrator of an enterprise assigns PCs in the same department to the same VLAN. To improve information security, the enterprise requires that only employees in the department be allowed to access the network resources of the enterprise.
 - PCs 1 through 3 belong to the same department. According to the enterprise' requirement, only the three PCs can access the enterprise network through SW1.
 - To meet this requirement, configure MAC address-based VLAN assignment and associate the MAC addresses of the three PCs with the specified VLAN.



Creating a VLAN and Associating MAC Addresses with the VLAN



Create a VLAN.

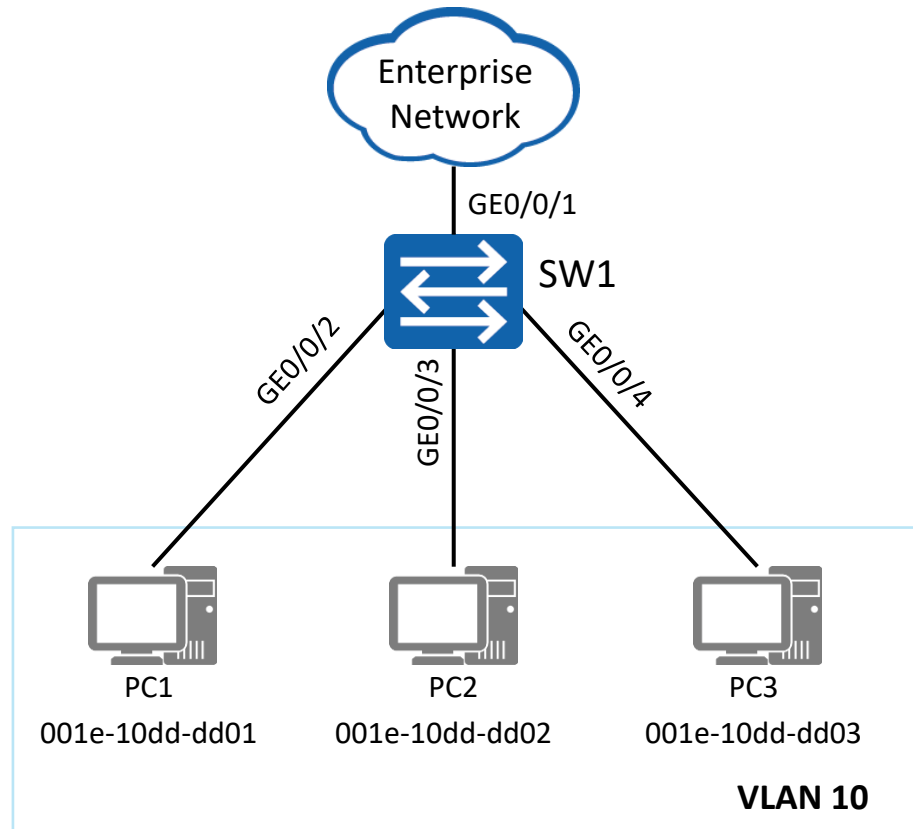
```
[SW1] vlan 10  
[SW1-vlan10] quit
```

Associate MAC addresses with the VLAN.

```
[SW1] vlan 10  
[SW1-vlan10] mac-vlan mac-address 001e-10dd-dd01 [SW1-vlan10]  
mac-vlan mac-address 001e-10dd-dd02 [SW1-vlan10] mac-vlan  
mac-address 001e-10dd-dd03 [SW1-vlan10] quit
```



Adding Interfaces to the VLAN and Enabling MAC Address-based VLAN Assignment



Add interfaces to the VLAN.

```
[SW1] interface gigabitethernet 0/0/1
[SW1-GigabitEthernet0/0/1] port link-type hybrid
[SW1-GigabitEthernet0/0/1] port hybrid tagged vlan 10
```

```
[SW1] interface gigabitethernet 0/0/2
[SW1-GigabitEthernet0/0/2] port link-type hybrid
[SW1-GigabitEthernet0/0/2] port hybrid untagged vlan 10
```

Enable MAC address-based VLAN assignment on the specified interface.

```
[SW1] interface gigabitethernet 0/0/2
[SW1-GigabitEthernet0/0/2] mac-vlan enable
[SW1-GigabitEthernet0/0/2] quit
```

Note: The configuration of GE 0/0/3 and GE 0/0/4 is similar to that of GE 0/0/2.



Verifying the Configuration

```
[SW1]display vlan
```

```
The total number of vlans is : 2
```

```
-----  
U: Up;          D: Down;          TG: Tagged;   UT: Untagged;
```

```
MP: Vlan-mapping;          ST: Vlan-stacking;
```

```
#: ProtocolTransparent-vlan; *: Management-vlan;
```

```
-----  
VID      Type      Ports  
-----  
1         common    UT:GE0/0/1(U)      GE0/0/2(U)  GE0/0/3(U) .....  
10        common    UT:GE0/0/2(U)      GE0/0/3(U)  GE0/0/4(U)  
          TG:GE0/0/1(U)  
.....
```

```
[SW1]display mac-vlan mac-address all
```

```
-----  
MAC Address  MASK      VLAN      Priority  
-----  
001e-10dd-dd01      ffff-ffff-ffff      10          0  
001e-10dd-dd02      ffff-ffff-ffff      10          0  
001e-10dd-dd03      ffff-ffff-ffff      10          0
```

```
Total MAC VLAN address count: 3
```




Quiz

1. (Multiple) Which of the following statements about the VLAN technology are incorrect? ()
 - A. The VLAN technology can isolate a large collision domain into several small collision domains. → Collision domains are determined by switch ports. VLANs isolate **broadcast domains**, not collisions.
 - B. The VLAN technology can isolate a large Layer 2 broadcast domain into several small Layer 2 broadcast domains.
 - C. PCs in different VLANs cannot communicate.
 - D. PCs in the same VLAN can communicate at Layer 2.
2. If the PVID of a trunk interface is 5 and the port trunk allow-pass vlan 2 3 command is run on the interface, which VLANs' frames can be transmitted through the trunk interface?
 - VLANs **2, 3 (tagged)** can pass
 - VLAN **5 (untagged/native)** can also pass



Summary

- This course describes the VLAN technology, including the functions, identification, assignment, data exchange, planning, application, and basic configuration of VLANs.
- The VLAN technology can divide a physical LAN into multiple broadcast domains so that network devices in the same VLAN can directly communicate at Layer 2, while devices in different VLANs cannot.

The background of the slide features a blue-tinted image of several business professionals in a modern office environment. They are standing on a highly reflective floor, and their silhouettes are clearly visible against the lighter background. The overall aesthetic is professional and corporate.

Thank You
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STP Principles and Configuration



Foreword

- On an Ethernet switching network, redundant links are used to implement link backup and enhance network reliability. However, the use of redundant links may produce loops, leading to broadcast storms and an unstable MAC address table. As a result, communication on the network may deteriorate or even be interrupted. To prevent loops, IEEE introduced the Spanning Tree Protocol (STP).
- Devices running STP exchange STP Bridge Protocol Data Units (BPDUs) to discover loops on the network and block appropriate ports. This enables a ring topology to be trimmed into a loop-free tree topology, preventing infinite looping of packets and ensuring packet processing capabilities of devices.
- IEEE introduced the Rapid Spanning Tree Protocol (RSTP) to improve the network convergence speed.



Objectives

- Upon completion of this course, you will be able to:
 - Describe the causes and problems of Layer 2 loops on a campus switching network.
 - Describe basic concepts and working mechanism of STP.
 - Distinguish STP from RSTP and describe the improvement of RSTP on STP.
 - Complete basic STP configurations.
 - Understand other methods to eliminate Layer 2 loops on the switching network except STP.



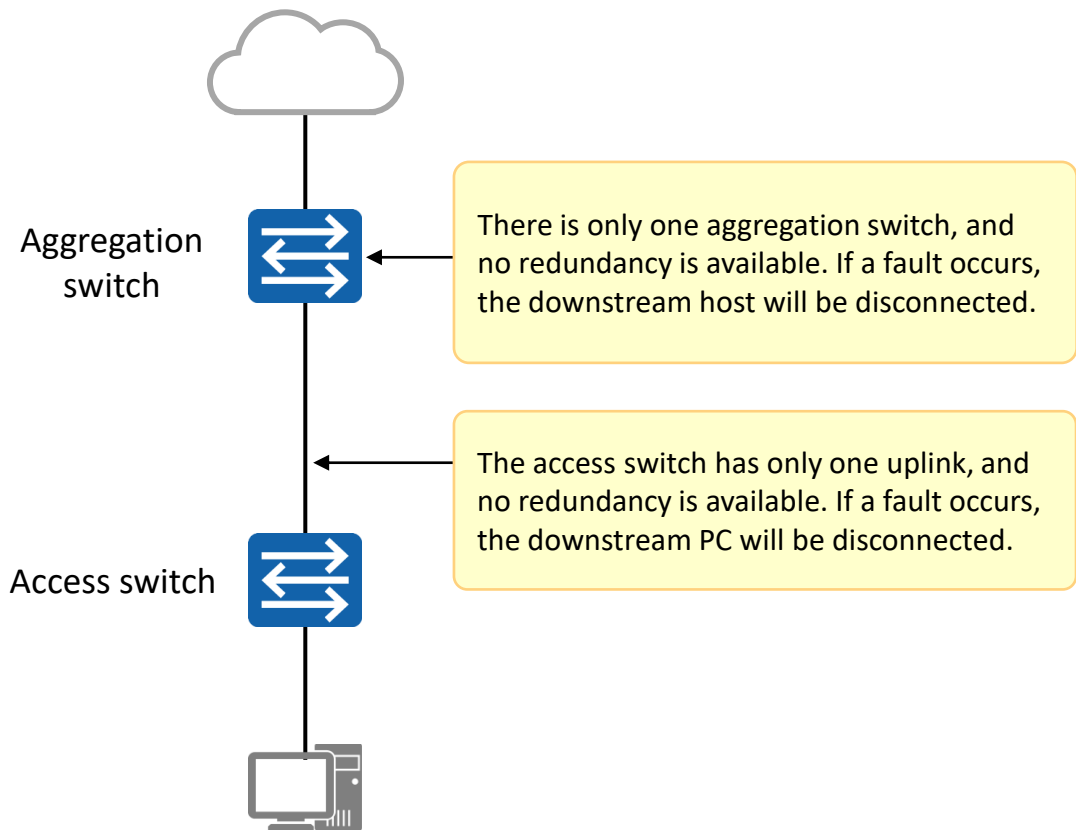
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- 1. STP Overview**
2. Basic Concepts and Working Mechanism of STP
3. Basic STP Configurations
4. Improvements Made in RSTP
5. STP Advancement

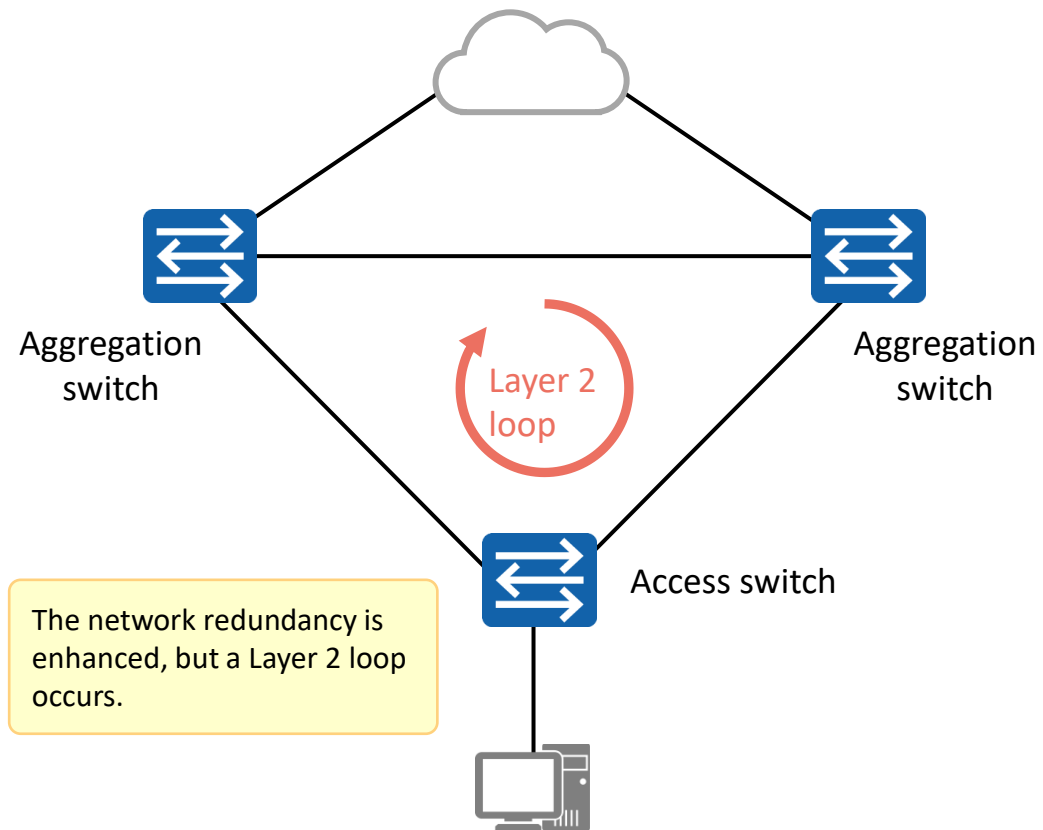


Technical Background: Redundancy and Loops on a Layer 2 Switching Network

A network without redundancy design



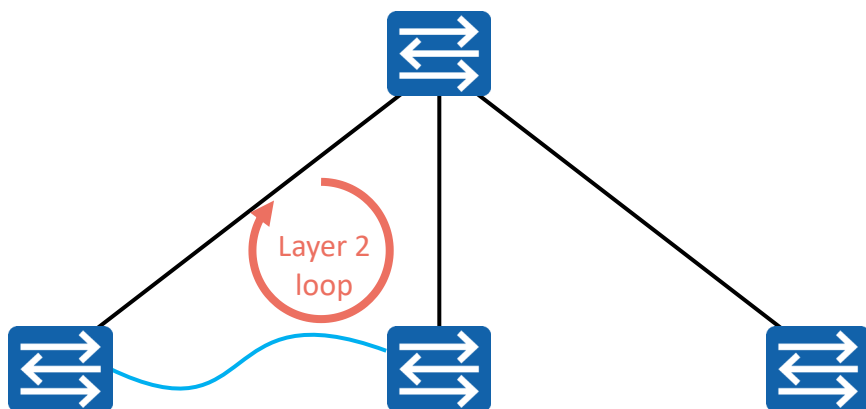
Layer 2 loops introduced along with redundancy





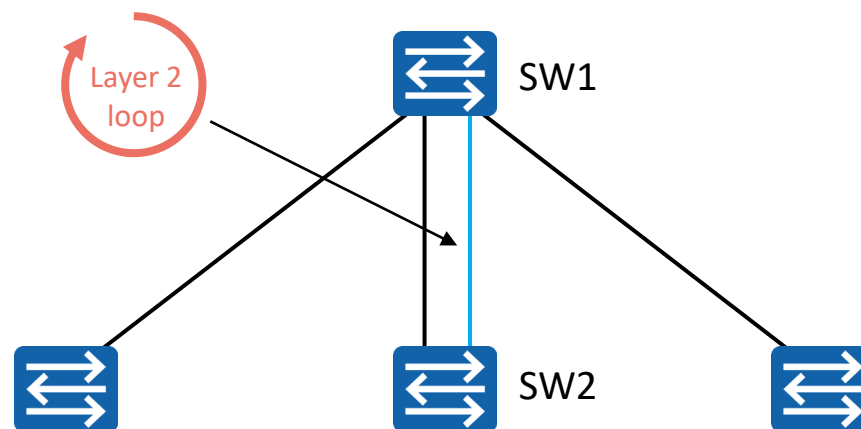
Technical Background: Layer 2 Loops Caused by Human Errors

Case 1



Incorrect operations: For example, connections of cables between devices are incorrect.

Case 2

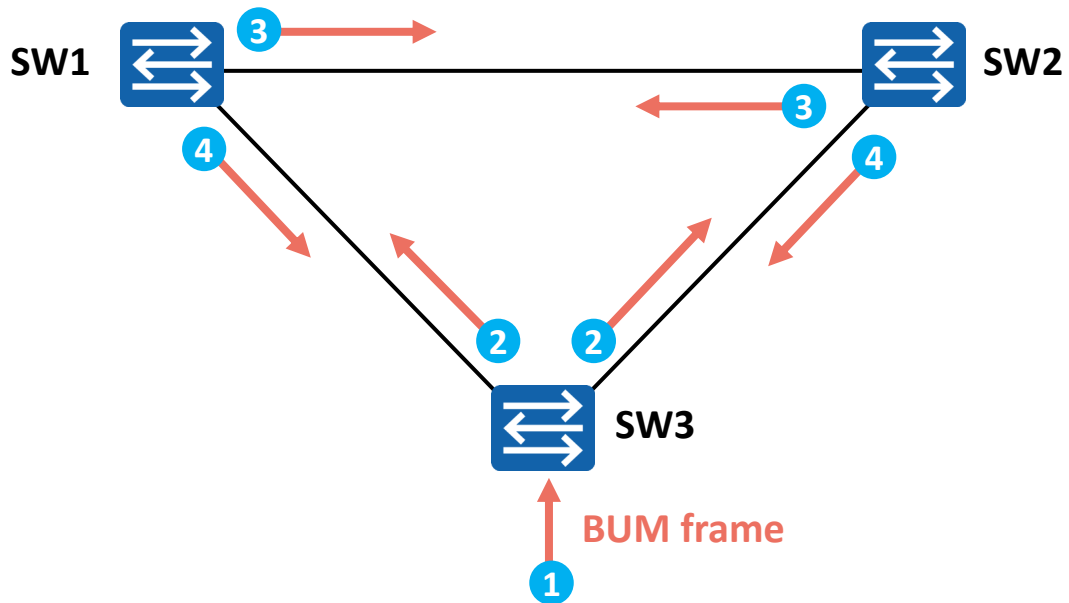


Incorrect manual configurations: For example, the network administrator does not bind the link between SW1 and SW2 to a logical link (aggregation link), causing Layer 2 loops.



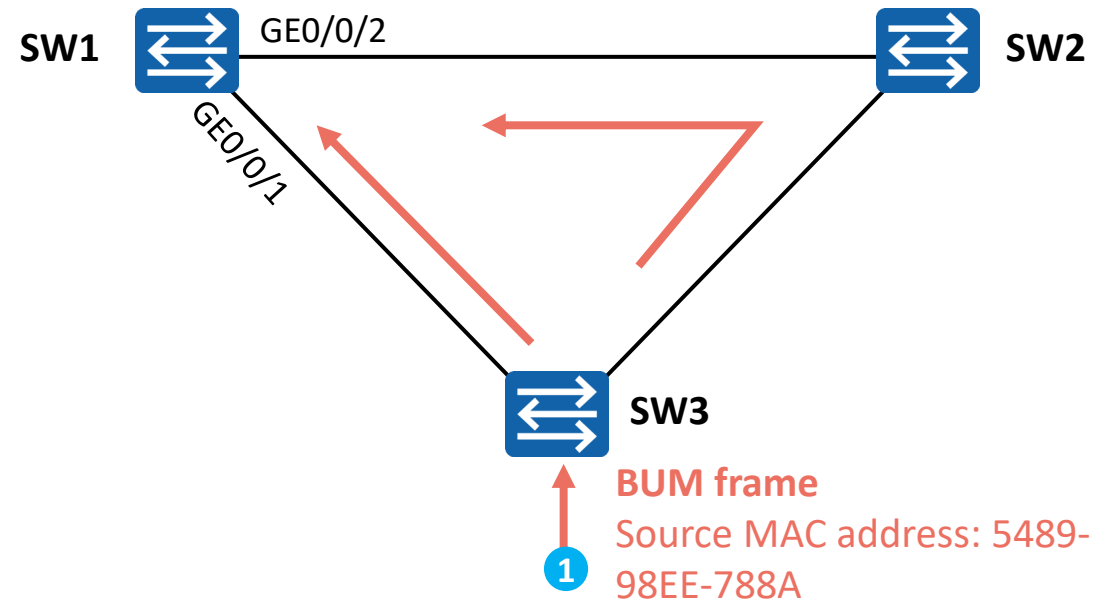
Issues Caused by Layer 2 Loops

Typical Issue 1: Broadcast Storm



When SW3 receives the BUM frames, it floods the frames. After SW1 and SW2 receive the BUM frames, they flood the frames again. As a result, network resources are exhausted and the network is unavailable.

Typical Issue 2: MAC Address Flapping

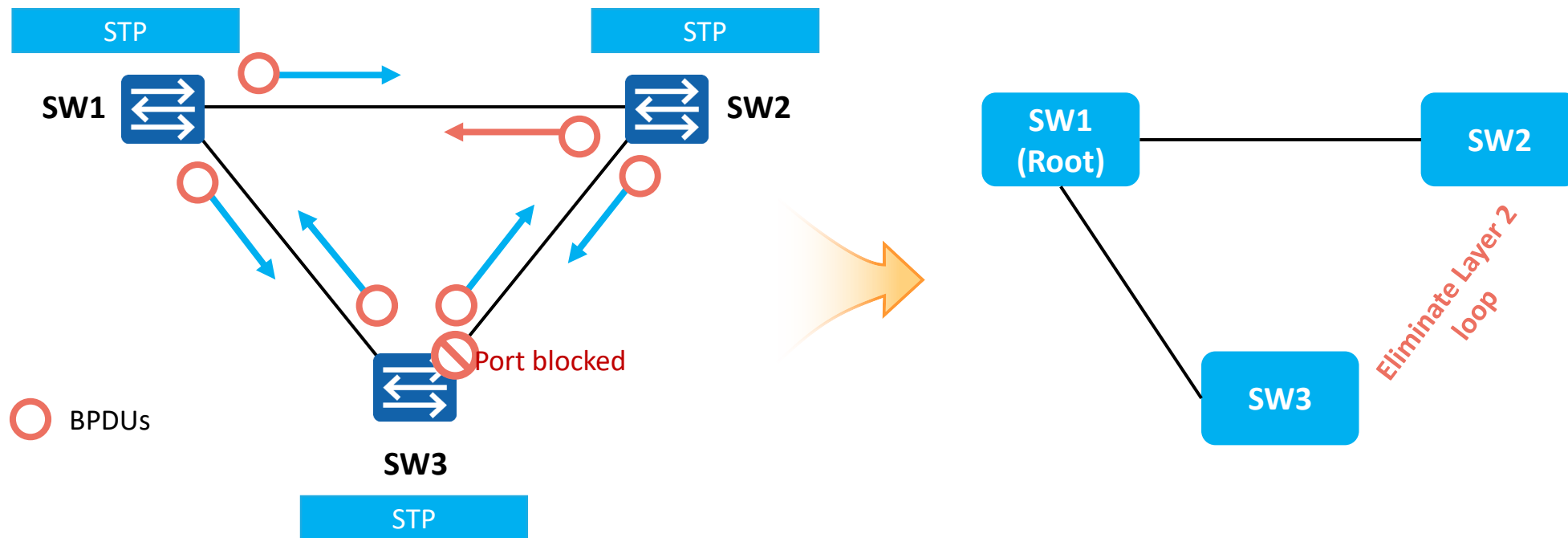


SW1 is used as an example. The MAC address of 5489-98EE-788A is frequently switched between GE0/0/1 and GE0/0/2, causing MAC address flapping.

BUM frames: broadcast, unknown unicast, and multicast frames



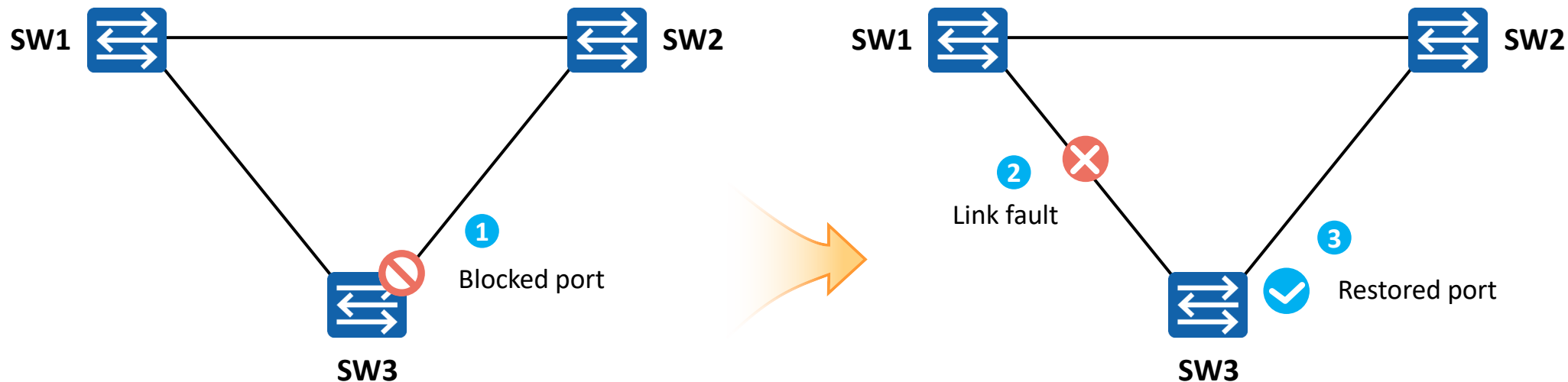
Introduction to STP



When STP is deployed on a network, switches exchange STP BPDUs and calculate a loop-free topology. Finally, one or more ports on the network are blocked to eliminate loops.



STP Can Dynamically Respond to Network Topology Changes and Adjust Blocked Ports



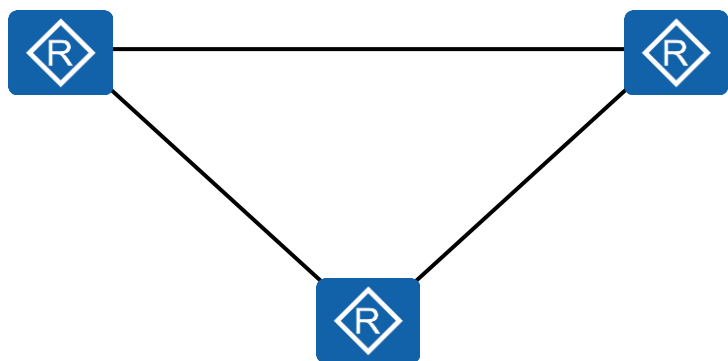
STP running on a switch continuously monitors the network topology. When the network topology changes, STP can detect the changes and automatically adjust the network topology.

Therefore, STP can solve the Layer 2 loop problem and provide a solution for network redundancy.



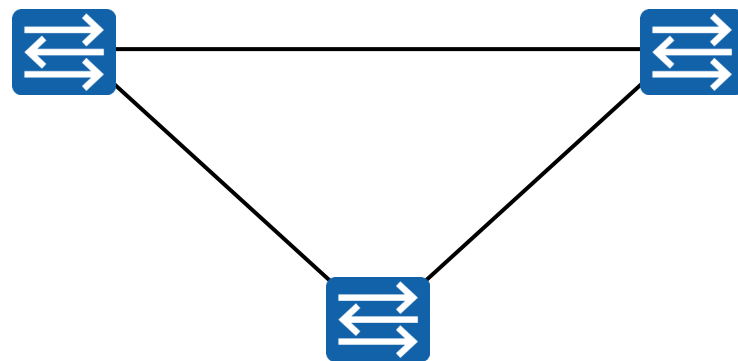
Q&A: Layer 2 and Layer 3 loops

Layer 3 loop



- Common root cause: routing loop
- Dynamic routing protocols have certain loop prevention capabilities.
- The TTL field in the IP packet header can be used to prevent infinite packet forwarding.

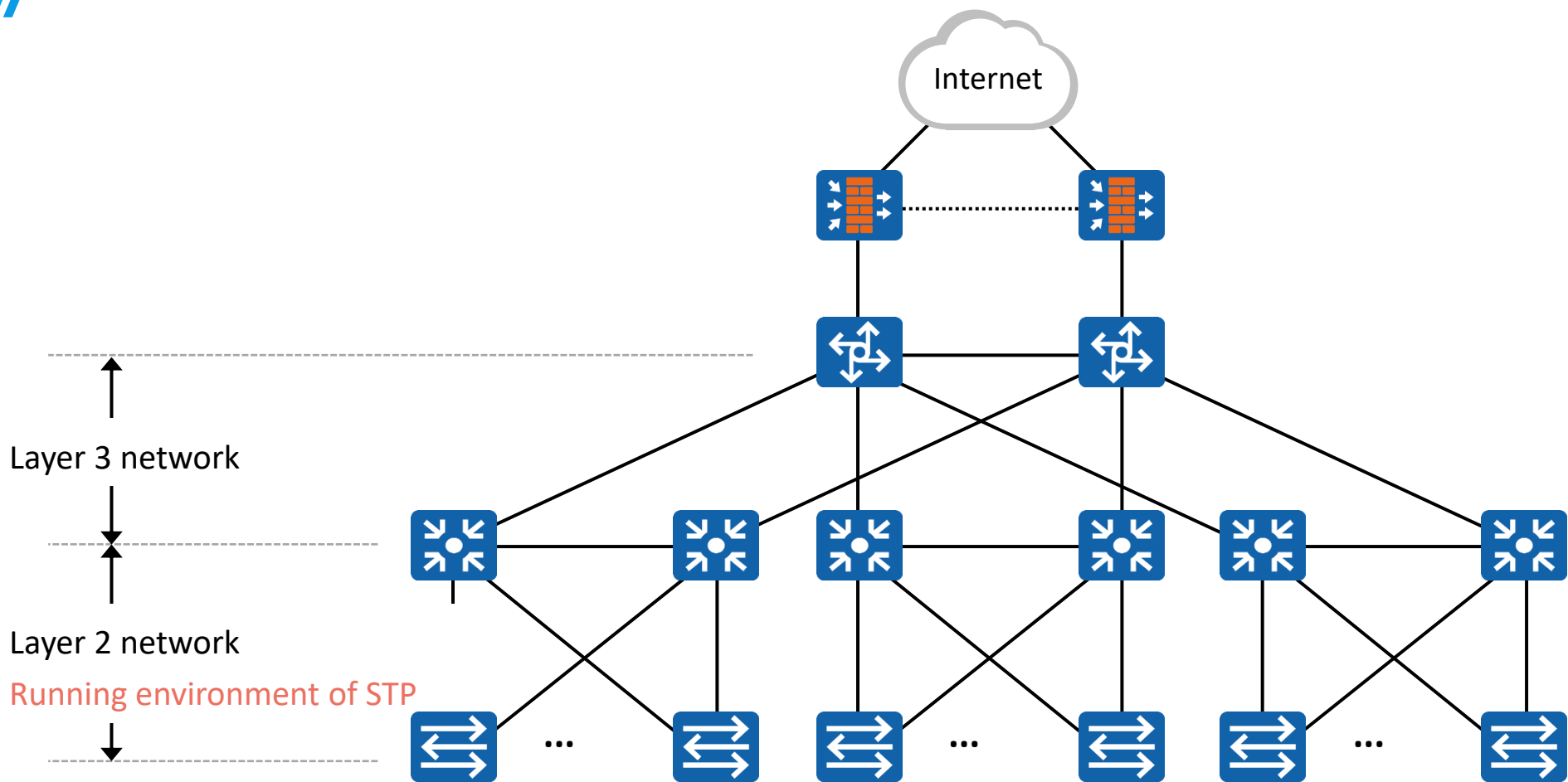
Layer 2 loop



- Common root cause: Layer 2 redundancy is deployed on the network, or cables are incorrectly connected.
- Specific protocols or mechanisms are required to implement Layer 2 loop prevention.
- The Layer 2 frame header does not contain any information to prevent data frames from being forwarded infinitely.



Application of STP on a Campus Network





STP Overview

- STP is used on a LAN to prevent loops.
- Devices running STP exchange information with one another to discover loops on the network, and block certain ports to eliminate loops.
- After running on a network, STP continuously monitors the network status. When the network topology changes, STP can detect the change and automatically respond to the change. In this way, the network status can adapt to the new topology, ensuring network reliability.
- With the growth in scale of LANs, STP has become an important protocol for a LAN.



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- 2. Basic Concepts and Working Mechanism of STP**
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5. STP Advancement



STP Basic Concepts: BID

BID

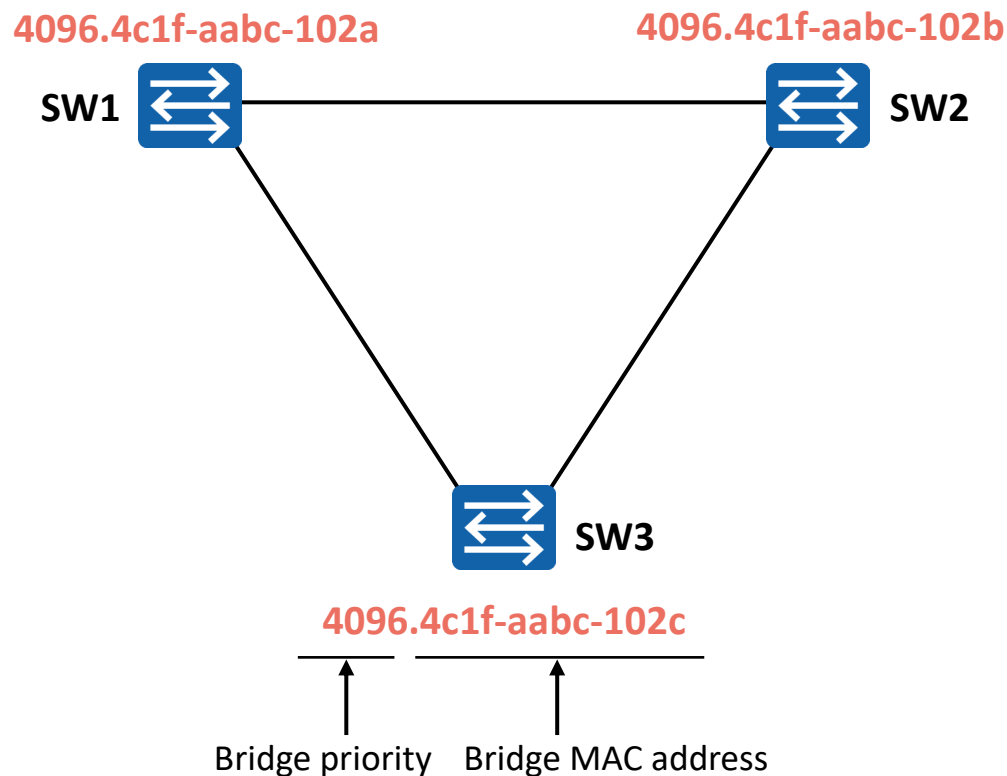
Root
Bridge

Cost

RPC

PID

BPDU



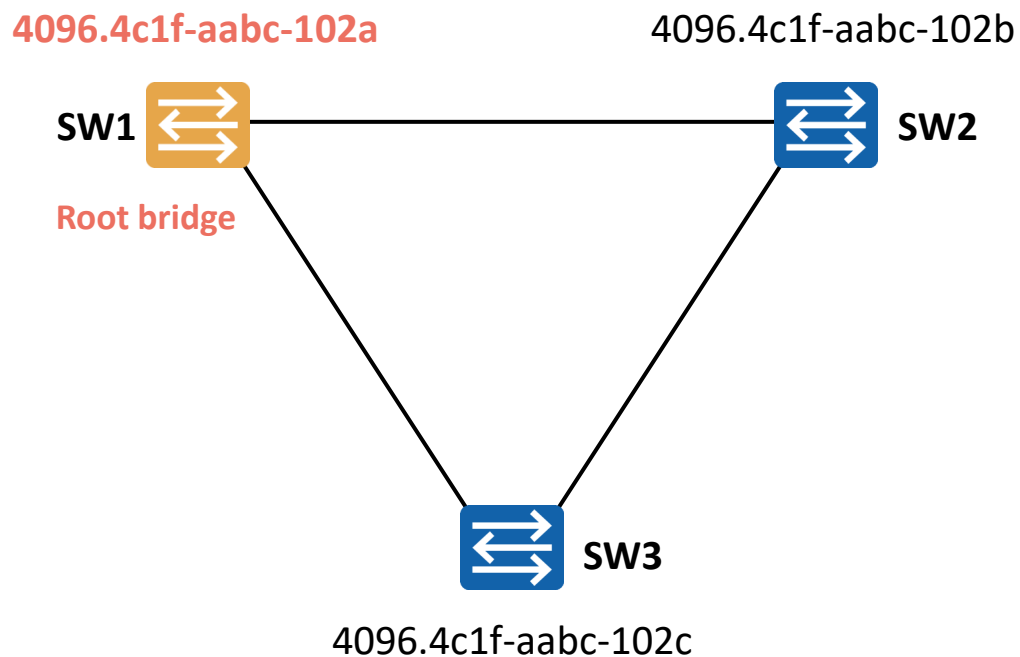
Bridge ID (BID)

- As defined in IEEE 802.1D, a BID consists of a 16-bit bridge priority and a bridge MAC address.
- Each switch running STP has a unique BID.
- The bridge priority occupies the leftmost 16 bits and the MAC address occupies the rightmost 48 bits.
- On an STP network, the device with the smallest BID acts as the root bridge.

Note: A bridge is a switch.



STP Basic Concepts: Root Bridge



Root Bridge

- One of the main functions of STP is to calculate a loop-free STP tree on the entire switching network.
- The root bridge is the root of an STP network.
- After STP starts to work, it elects a root bridge on the switching network. The root bridge is the key for topology calculation of the spanning tree and is the root of the loop-free topology calculated by STP.
- On an STP network, the device with the smallest BID acts as the root bridge.

During BID comparison, devices first compare bridge priorities. A smaller priority value indicates a higher priority of a device. The switch with the smallest priority value becomes the root bridge. If priority values are the same, the switch with the smallest MAC address becomes the root bridge.



STP Basic Concepts: Cost

BID

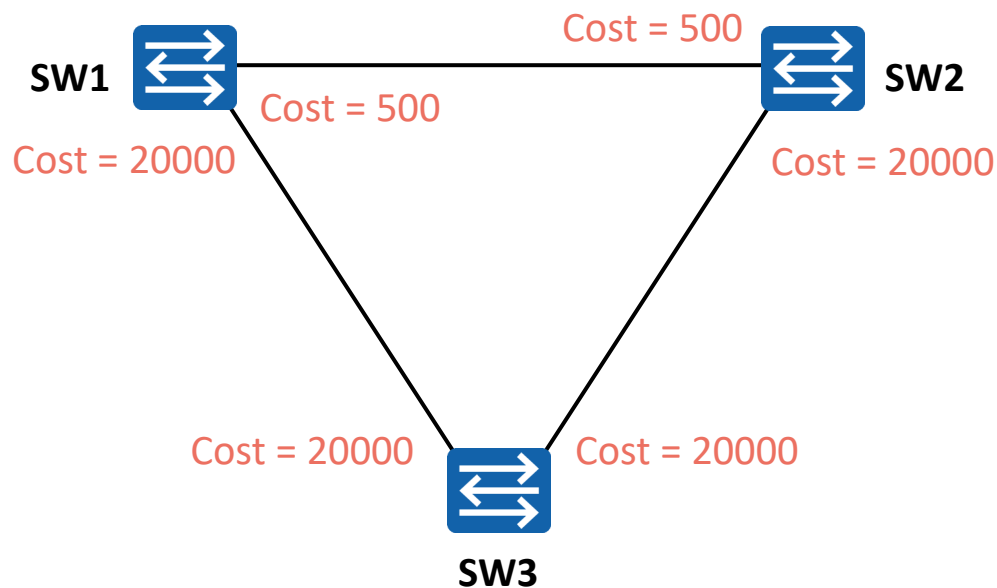
Root
Bridge

Cost

RPC

PID

BPDU



Cost

- Each STP-enabled port maintains a cost. The cost of a port is used to calculate the root path cost (RPC), that is, the cost of the path to the root.
- The default cost of a port is related to the rate, working mode, and STP cost calculation method used by a switch.
- A higher port bandwidth indicates a smaller cost.
- You can also run commands to adjust the cost of a port as required.



STP Basic Concepts: Cost Calculation Methods

Port Rate	Port Mode	Recommended STP Cost		
		IEEE 802.1d-1998	IEEE 802.1t	Huawei Legacy Standard
100 Mbit/s	Half-duplex	19	200,000	200
	Full-duplex	18	199,999	199
	Aggregated link: two ports	15	100,000	180
1000 Mbit/s	Full-duplex	4	20,000	20
	Aggregated link: two ports	3	10,000	18
10 Gbit/s	Full-duplex	2	2000	2
	Aggregated link: two ports	1	1000	1
40 Gbit/s	Full-duplex	1	500	1
	Aggregated link: two ports	1	250	1
100 Gbit/s	Full-duplex	1	200	1
	Aggregated link: two ports	1	100	1
...				

The cost has a default value and is associated with the port rate. When the device uses different algorithms, the same port rate corresponds to different cost values.



STP Basic Concepts: RPC

BID

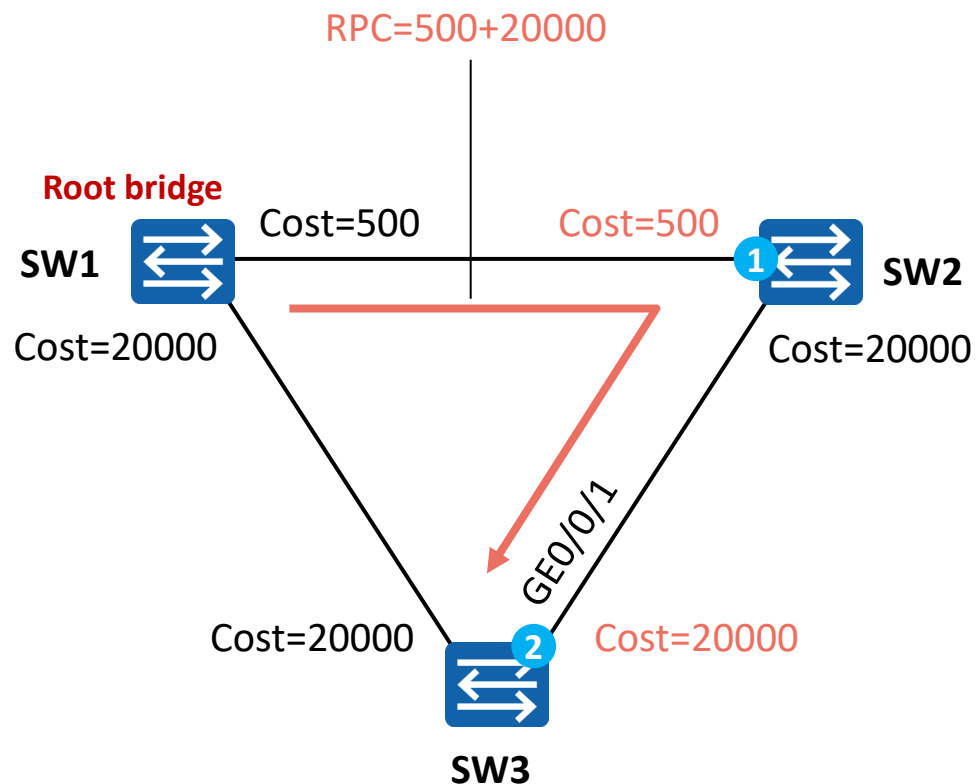
Root
Bridge

Cost

RPC

PID

BPDU



RPC

- The cost from a switch port to the root bridge, that is, RPC, is important during STP topology calculation.
- The RPC from a port to the root bridge is the sum of costs of all inbound ports along the path from the root bridge to the device.
- In this example, the RPC for SW3 to reach the root bridge through GE0/0/1 is equal to the cost of port 1 plus the cost of port 2.



STP Basic Concepts: PID

BID

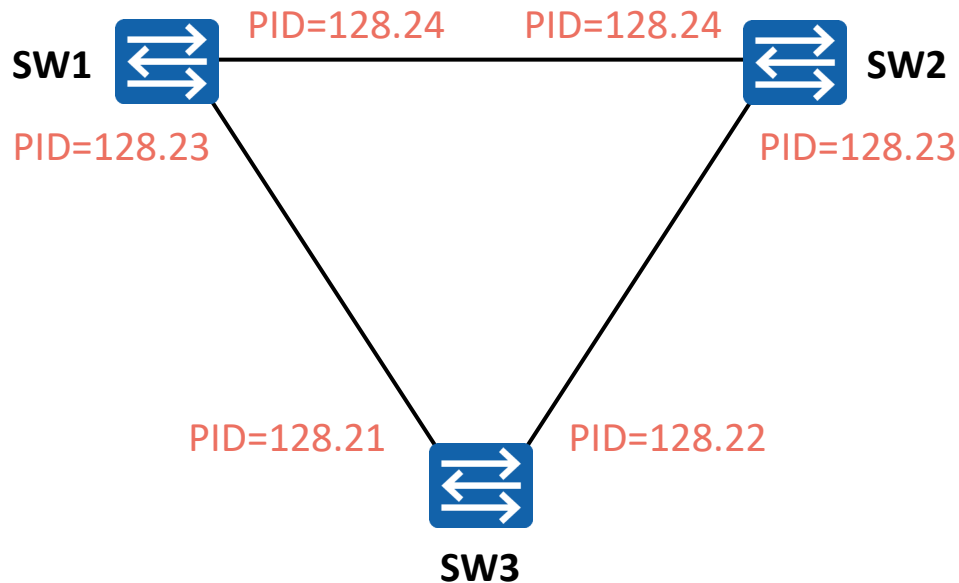
Root
Bridge

Cost

RPC

PID

BPDU



Port ID (PID)

- An STP-enabled switch uses PIDs to identify ports. A PID is used to elect a designated port in a specific scenario.
- A PID consists of the leftmost four bits (port priority) and the rightmost 12 bits (port number).
- An STP-enabled port maintains a default port priority, which is 128 on Huawei switches. You can run a command to change the priority as required.



STP Basic Concepts: BPDU

BID

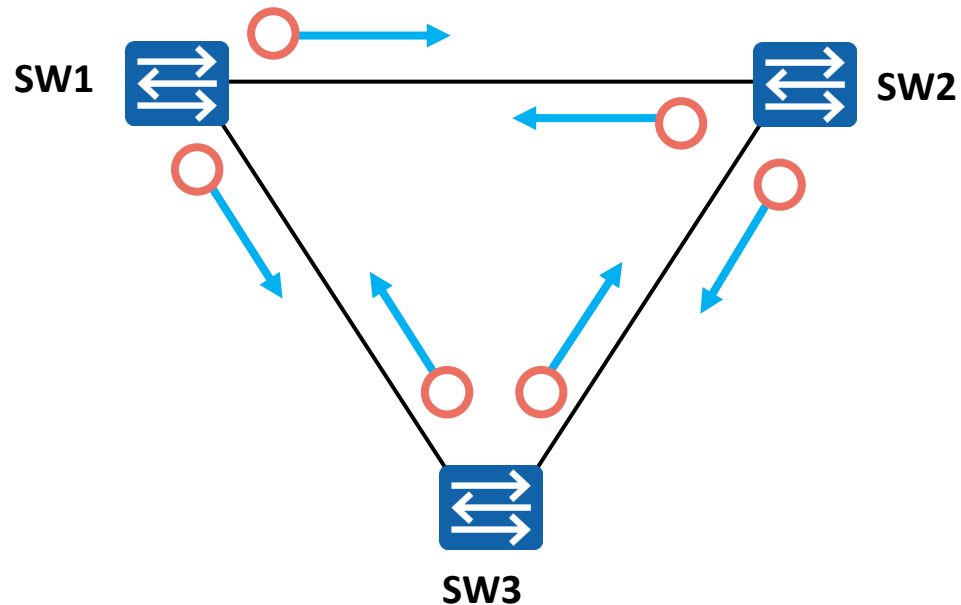
Root
Bridge

Cost

RPC

PID

BPDU



○ Configuration BPDU

Bridge Protocol Data Unit (BPDU)

- BPDU is the basis for STP to work normally.
- STP-enabled switches exchange BPDUs that carry important information.
- There are two types of BPDUs:
 - Configuration BPDU
 - Topology Change Notification (TCN) BPDU
- Configuration BPDUs are the key to STP topology calculation. TCN BPDUs are triggered only when the network topology changes.



Format of Configuration BPDUs

PID	PVI	BPDU Type	Flags	Root ID	RPC	Bridge ID	Port ID	Message Age	Max Age	Hello Time	Forward Delay
-----	-----	-----------	-------	---------	-----	-----------	---------	-------------	---------	------------	---------------

Byte	Field	Description
2	PID	For STP, the value of this field is always 0.
1	PVI	For STP, the value of this field is always 0.
1	BPDU Type	Type of BPDUs. The value 0x00 indicates a configuration BPDU and the value 0x80 indicates a TCN BPDU.
1	Flags	STP uses only the leftmost two bits and the rightmost two bits: Topology Change Acknowledgment (TCA) and Topology Change (TC).
8	Root D	BID of the root bridge.
4	RPC	STP cost of the path from the current port to the root bridge.
8	Bridge ID	BID of the sender.
2	Port ID	ID of the port that sends this BPDU, which consists of the port priority and port number.
2	Message Age	Number of seconds after a BPDU is sent from the root bridge. The value increases by 1 each time the BPDU passes through a network bridge. It refers to the number of hops to the root bridge.
2	Max Age	If the bridge does not receive any BPDU for a period of time and the lifetime of the network bridge reaches the maximum, the network bridge considers that the link connected to the port is faulty. The default value is 20s.
2	Hello Time	Interval at which the root bridge sends configuration BPDUs. The default value is 2s.
2	Forward Delay	Time that is spent in Listening or Learning state. The default value is 15s.



BPDU Comparison Rules

BID

Root
Bridge

Cost

RPC

PID

BPDU

Field

Protocol Identifier

Protocol Version Identifier

BPDU Type

Flags

Root Identifier

Root Path Cost

Bridge Identifier

Port Identifier

Message Age

Max Age

Hello Time

Forward Delay

The core of STP is to calculate a loop-free topology on a switching network. During topology calculation, the comparison of configuration BPDUs is important. The Root Identifier, Root Path Cost, Bridge Identifier, and Port Identifier fields are the main fields of a configuration BPDUs. STP-enabled switches compare the four fields.

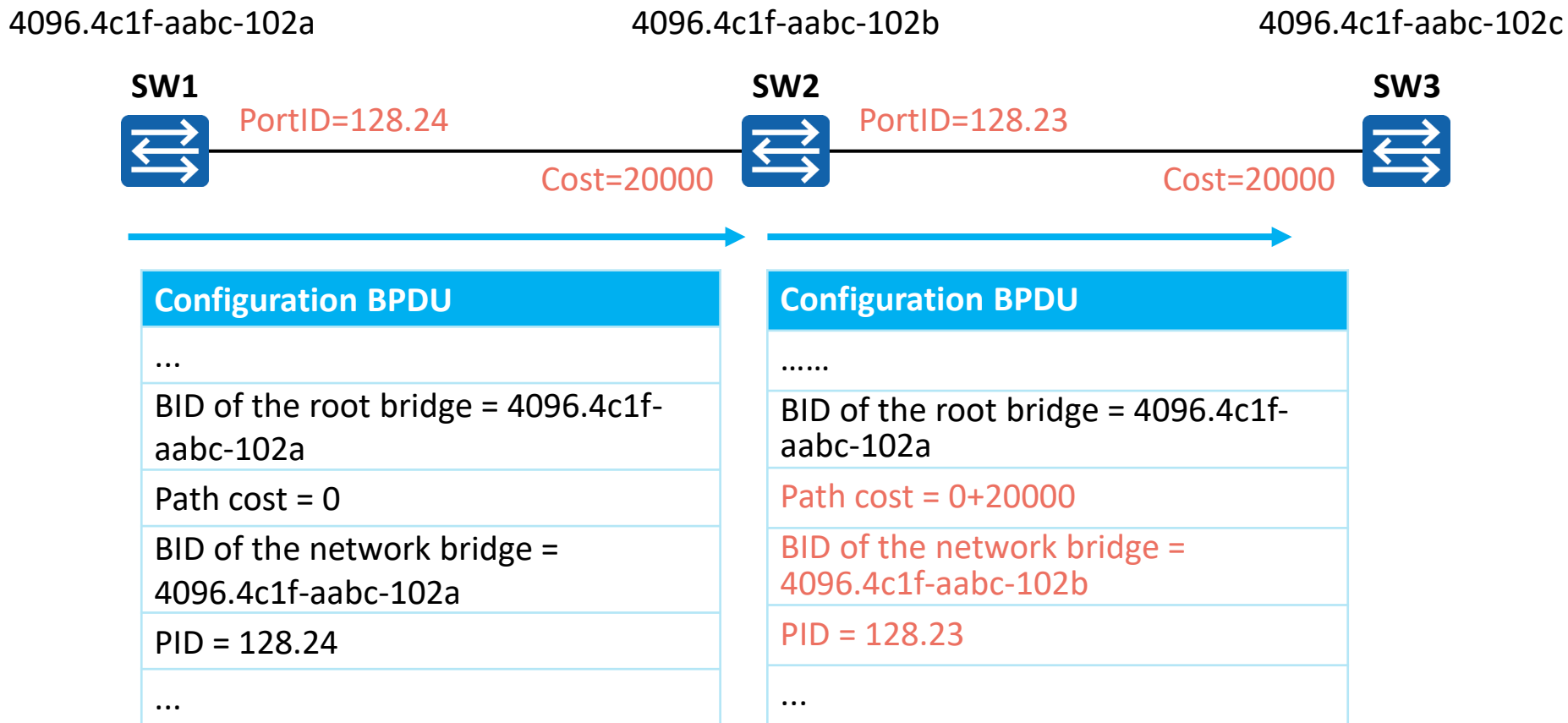
STP selects the optimal configuration BPDUs in the following sequence:

1. Smallest BID of the root bridge
2. Smallest RPC
3. Smallest BID of the network bridge
4. Smallest PID

Among the four rules (each rule corresponds to a field in a configuration BPDUs), the first rule is used to elect the root bridge on the network, and the following rules are used to elect the root port and designated port.



Configuration BPDU Forwarding Process





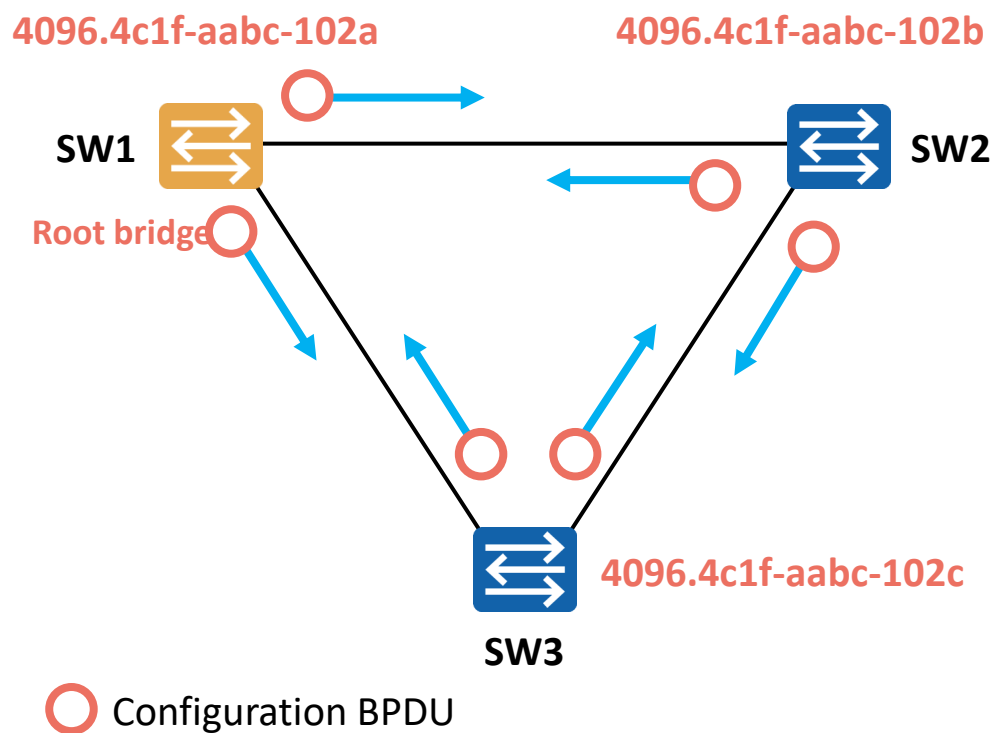
STP Calculation (1)

Select Root
Bridge

Select Root Port

Select Designated
Port

Block Non-designated
Port



Select a Root Bridge on the Switching Network

- After STP starts to work on a switching network, each switch sends configuration BPDUs to the network. The configuration BPDU contains the BID of a switch.
- The switch with the smallest bridge ID becomes the root bridge.
- Only one root bridge exists on a contiguous STP switching network.
- The role of the root bridge can be preempted.
- To ensure the stability of the switching network, you are advised to plan the STP network in advance and set the bridge priority of the switch that is planned as the root bridge to the minimum value 0.



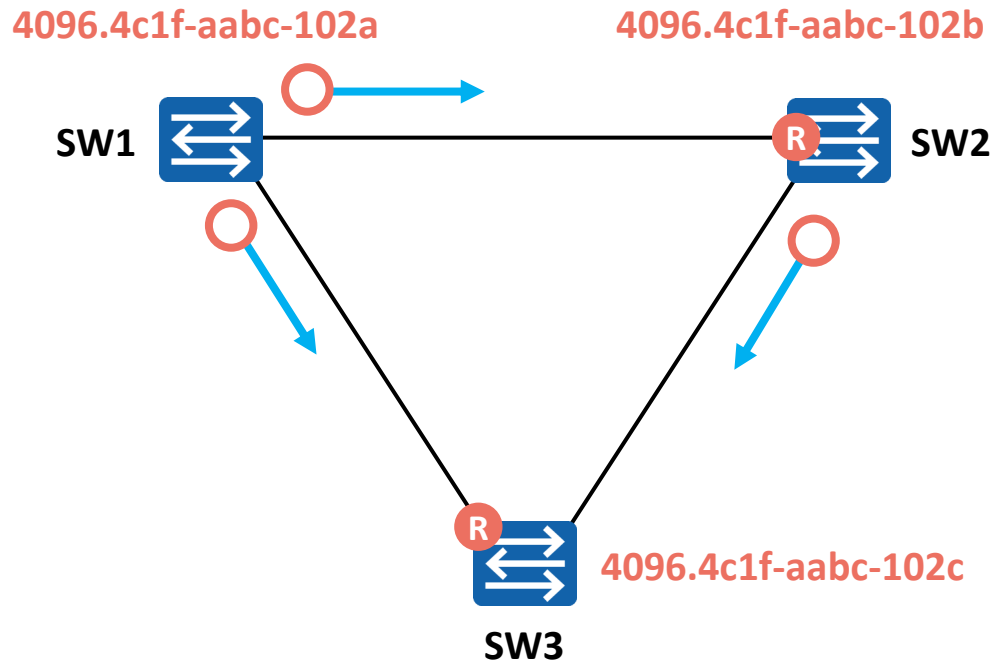
STP Calculation (2)

Select Root Bridge

Select Root Port

Select Designated Port

Block Non-designated Port



○ Configuration BPDU ● Root port

Select a Root Port on Each Non-root Bridge

- Each non-root bridge selects a root port from its ports.
- A non-root bridge has only one root port.
- When a non-root-bridge switch has multiple ports connected to the network, the root port receives the optimal configuration BPDU.
- The root port is located on each non-root bridge and has the shortest distance away from the root bridge.



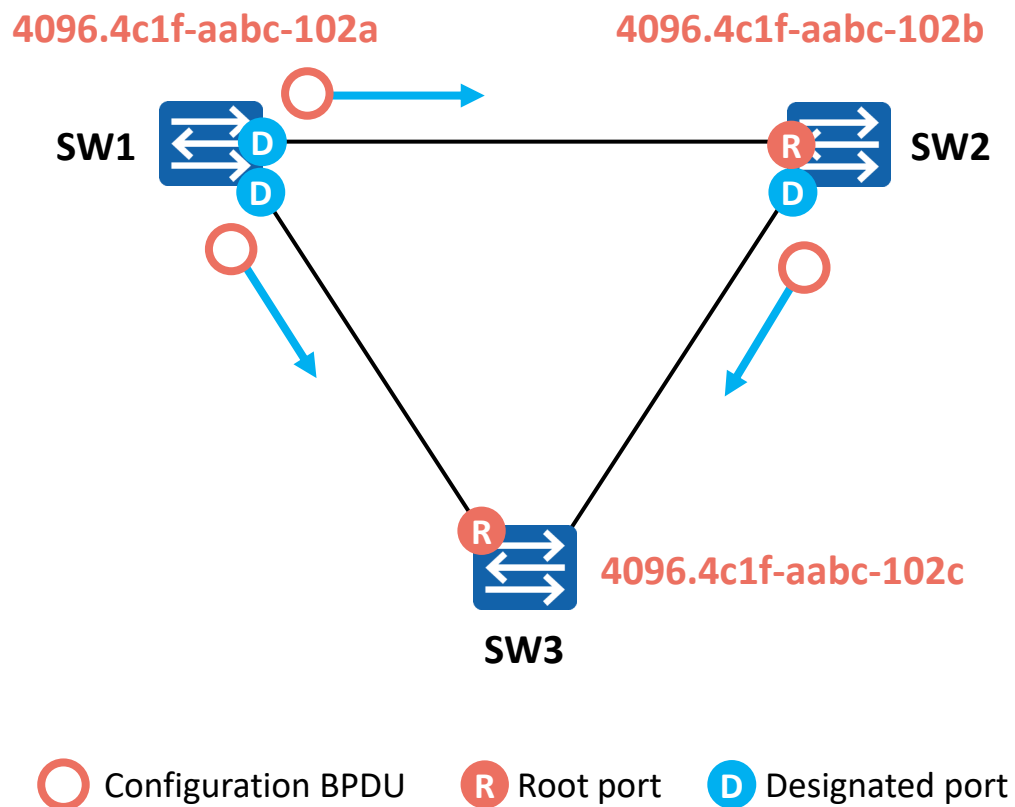
STP Calculation (3)

Select Root Bridge

Select Root Port

Select Designated Port

Block Non-designated Port



A designated port is elected on each link.

- After the root port is elected, the non-root bridge uses the optimal BPDU received on the port to calculate the configuration BPDU and compares the calculated configuration BPDU with the configuration BPDUs received by all ports except the root port.
 - If the former is better, the port is a designated port.
 - If the latter is better, the port is not a designated port.
- In most cases, all ports on the root bridge are designated ports.



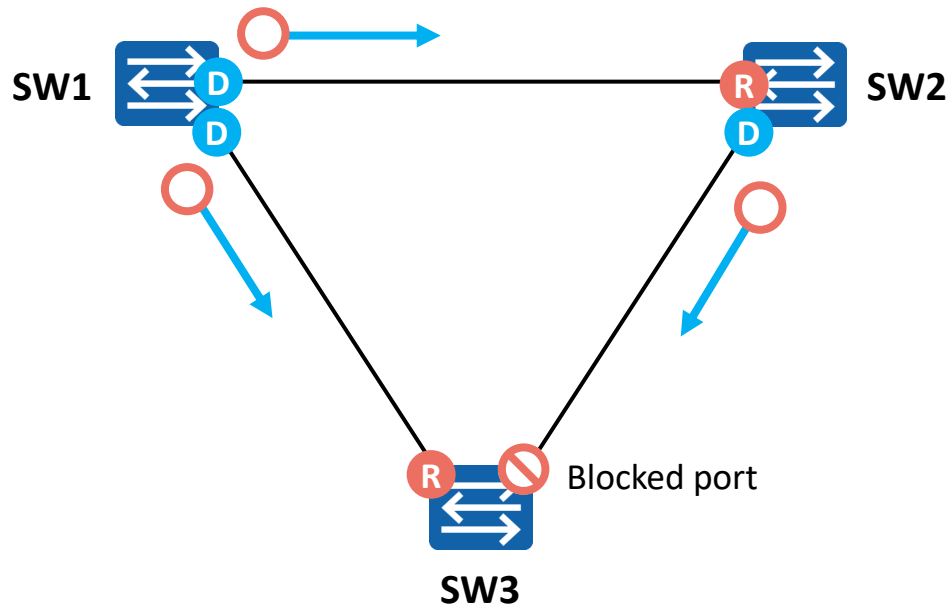
STP Calculation (4)

Select Root Bridge

Select Root Port

Select Designated Port

Block Non-designated Port



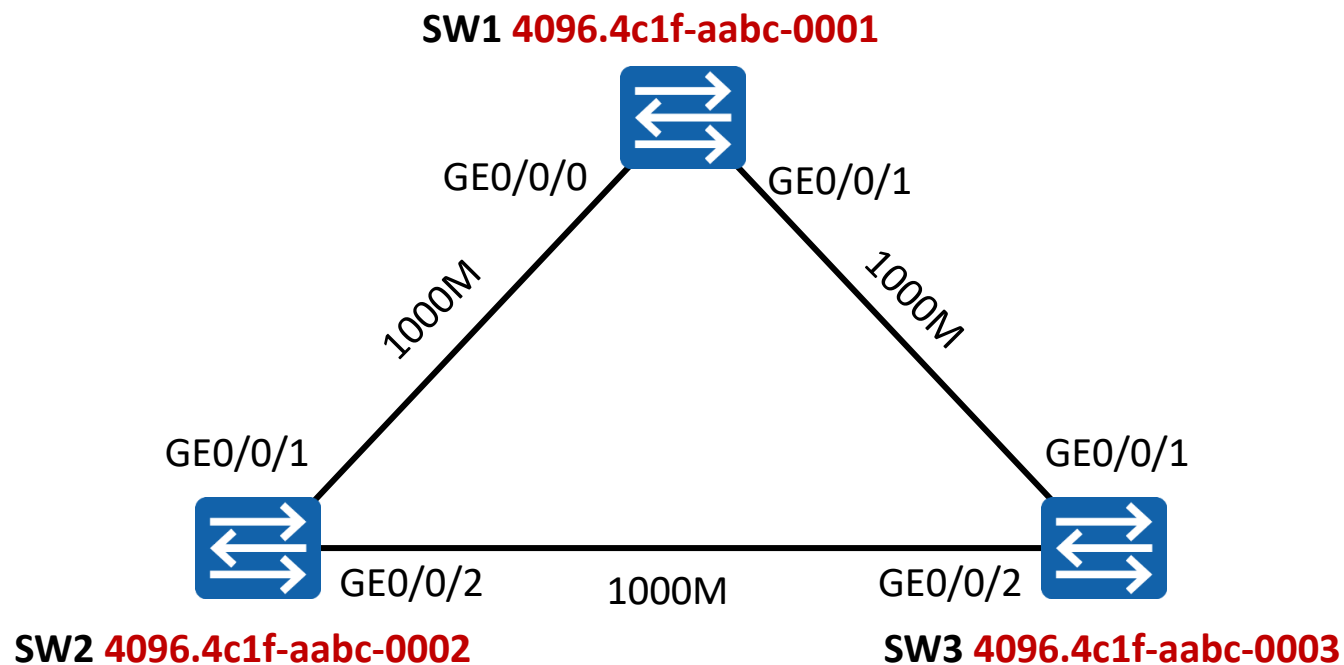
○ Configuration BPDU ● Root port ● Designated port

Block Non-designated Port

- On a switch, a port that is neither a root port nor a designated port is called a non-designated port.
- The last step of STP operations is to block the non-designated port on the network. After this step is complete, the Layer 2 loop on the network is eliminated.

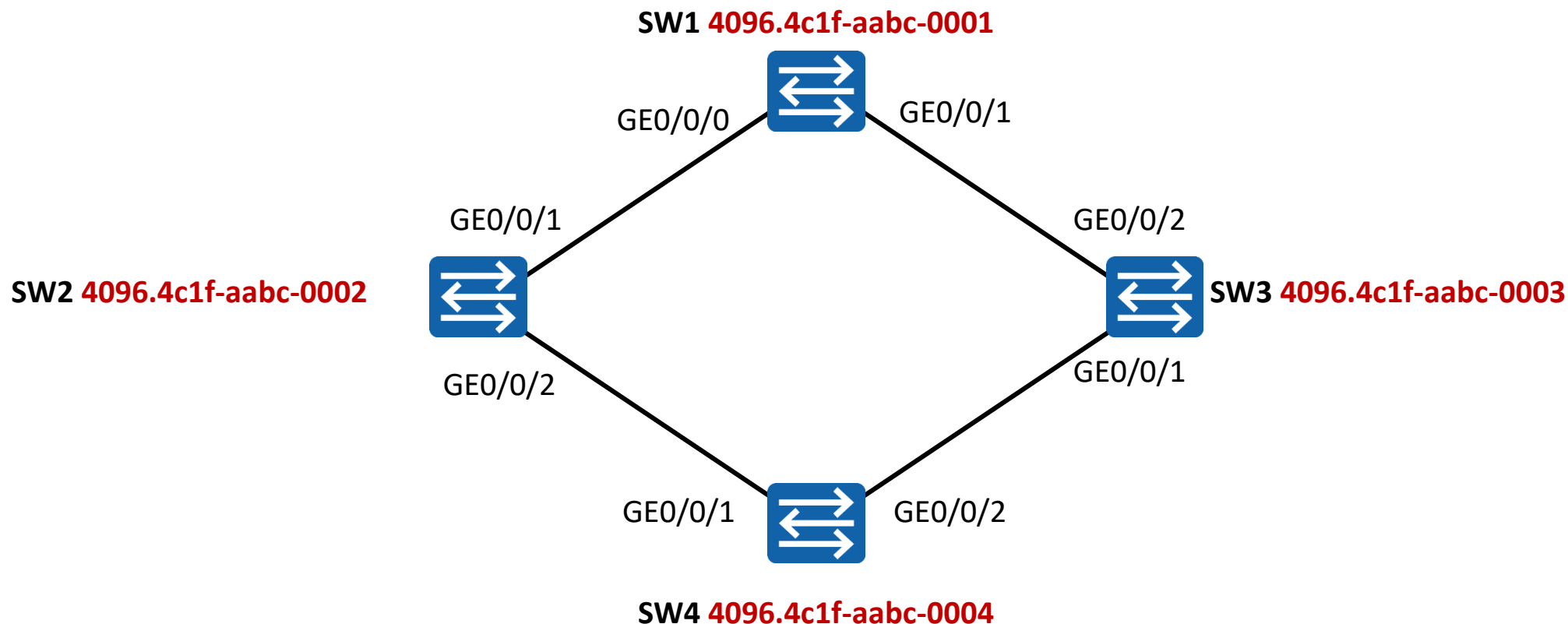


Quiz 1: Identify the Root Bridge and Port Roles



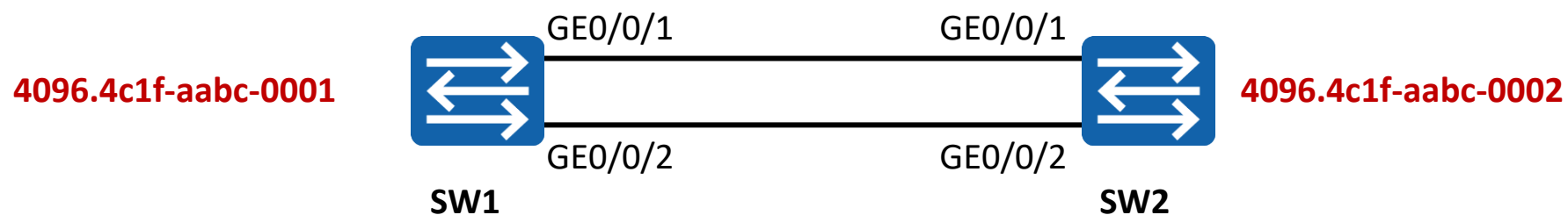


Quiz 2: Identify the Root Bridge and Port Roles in the Following Topology





Quiz 3: Identify the Root Bridge and Port Roles in the Following Topology



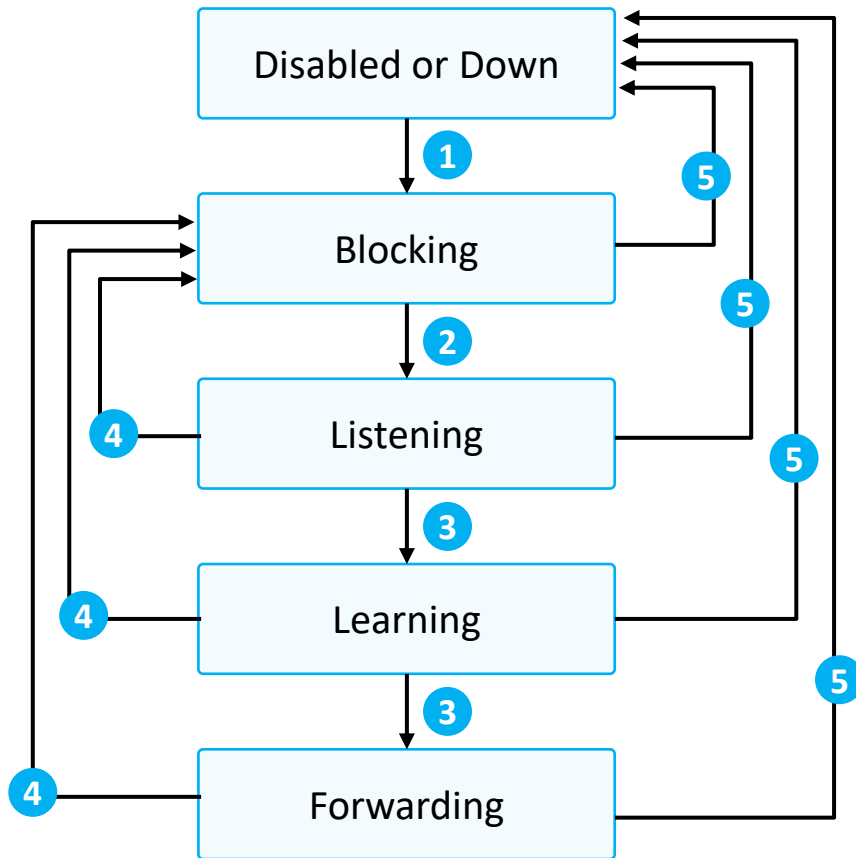


STP Port States

Port State	Description
Disabled	The port cannot send or receive BPDUs or service data frames. That is, the port is Down.
Blocking	The port is blocked by STP. A blocked port cannot send BPDUs but listens to BPDUs. In addition, the blocked port cannot send or receive service data frames or learn MAC addresses.
Listening	STP considers the port in Listening state as the root port or designated port, but the port is still in the STP calculation process. In this case, the port can send and receive BPDUs but cannot send or receive service data frames or learn MAC addresses.
Learning	A port in Learning state listens to service data frames but cannot forward them. After receiving service data frames, the port learns MAC addresses.
Forwarding	A port in Forwarding state can send and receive service data frames and process BPDUs. Only the root port or designated port can enter the Forwarding state.



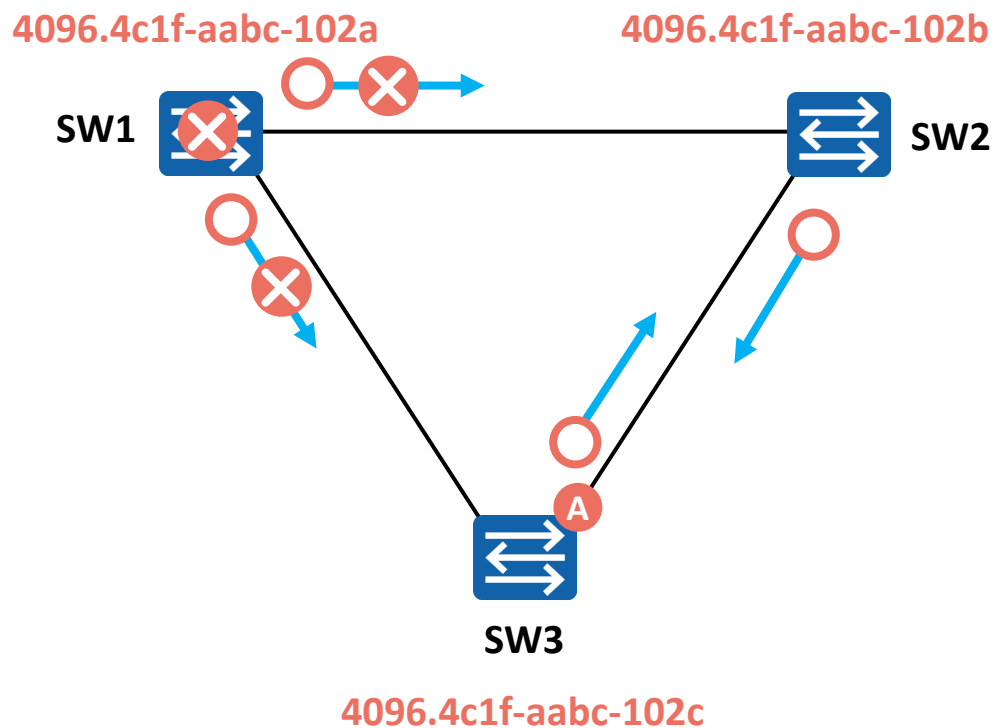
STP Port State Transition



- 1 When a port is initialized or activated, it automatically enters the blocking state.
- 2 The port is elected as the root port or designated port and automatically enters the Listening state.
- 3 The Forward Delay timer expires and the port is still the root port or designated port.
- 4 The port is no longer the root port or designated port.
- 5 The port is disabled or the link fails.



Topology Change: Root Bridge Fault

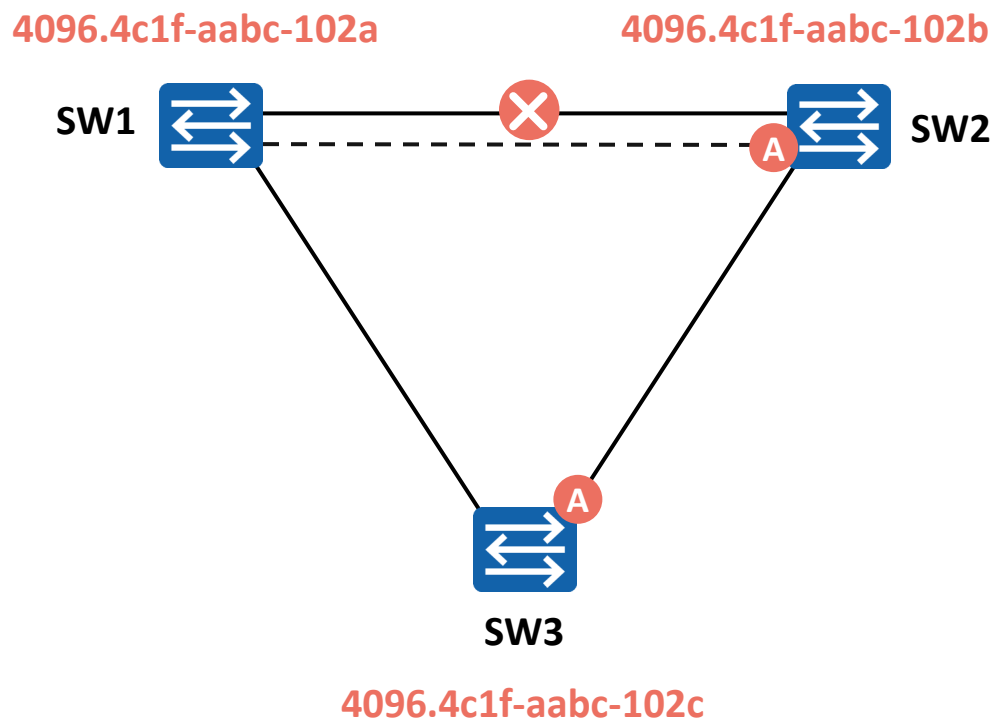


Root Bridge Fault Rectification Process

1. SW1 (root bridge) is faulty and stops sending BPDUs.
2. SW2 waits for the Max Age timer (20s) to expire. In this case, the record about the received BPDUs becomes invalid, and SW2 cannot receive new BPDUs from the root bridge. SW2 learns that the upstream device is faulty.
3. Non-root bridges send configuration BPDUs to each other to elect a new root bridge.
4. After re-election, port A of SW3 transitions to the Forwarding state after two intervals of the Forward Delay timer (the default interval is 15s).
 - A non-root bridge starts root bridge re-election after BPDUs age.
 - Due to the root bridge failure, it takes about 50s to recover from a root bridge failure.



Topology Change: Direct Link Fault



Direct Link Fault Rectification Process

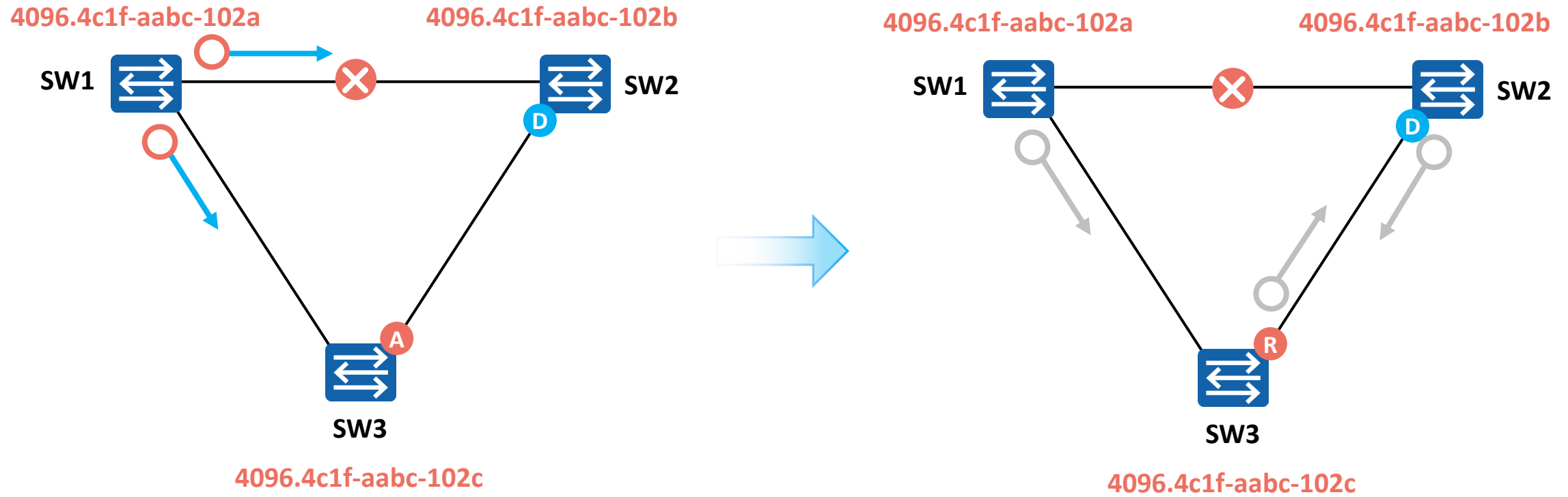
On a stable network, when SW2 detects that the link of the root port is faulty, the alternate port of SW2 enters the Forwarding state after twice the value of the Forward Delay timer (the default value is 15s).

- After SW2 detects a fault on the direct link, it switches the alternate port to the root port.
- If a direct link fails, the alternate port restores to the Forwarding state after 30s.



Topology Change: Indirect Link Fault

- When the indirect link fails, the alternate port on SW3 restores to the Forwarding state. It takes about 50s to recover from an indirect link failure.



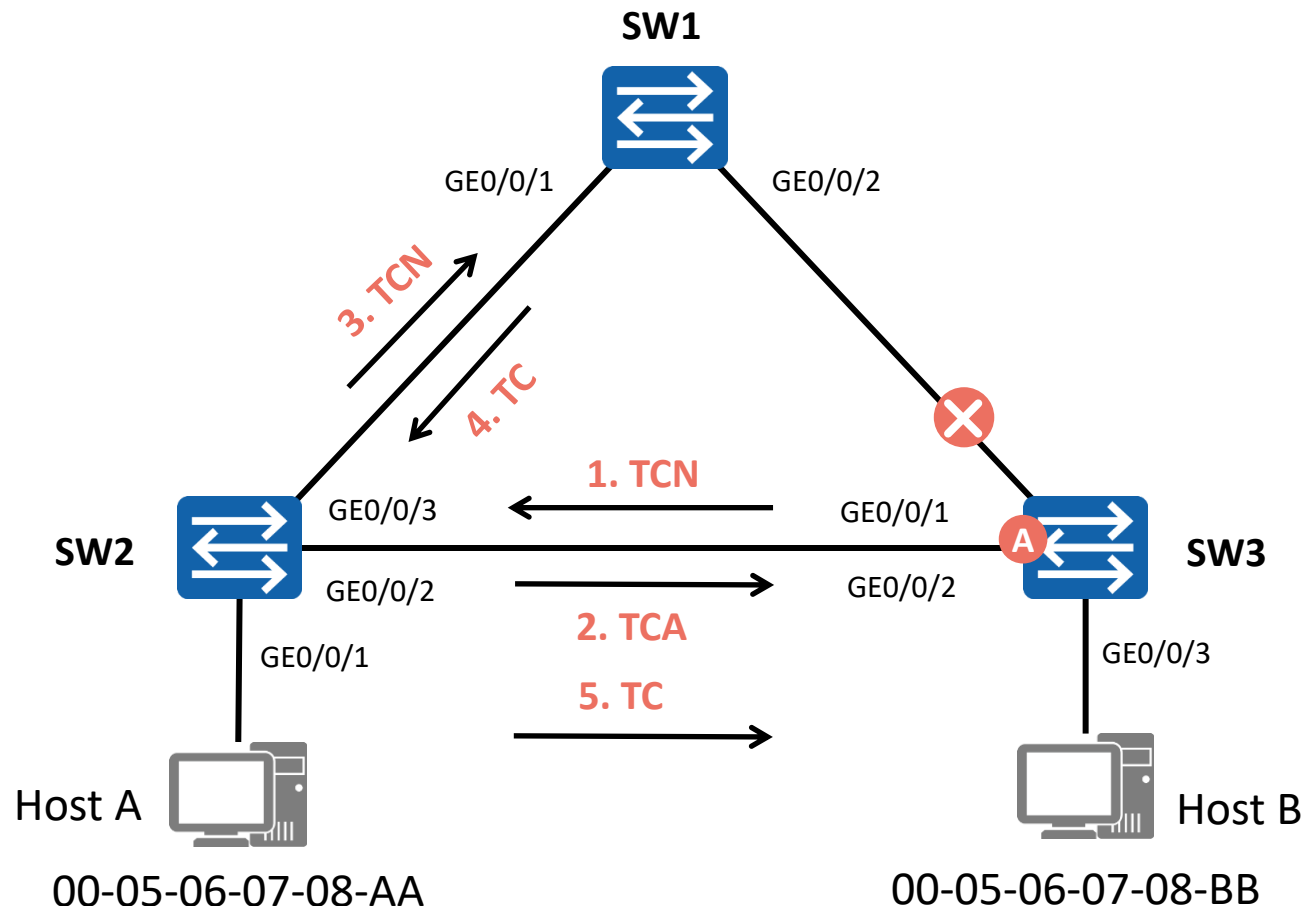


The MAC Address Table Is Incorrect Because the Topology Changes

MAC address table

MAC	Port
00-05-06-07-08-AA	GE0/0/3
00-05-06-07-08-BB	GE0/0/1
00-05-06-07-08-BB	GE0/0/2

- TCN BPDUs are generated when the network topology changes.
- Packet format: protocol identifier, version number, and type
- Topology change: The TCA and TC bits in the Flags field of configuration BPDUs are used.





Contents

1. STP Overview
2. Basic Concepts and Working Mechanism of STP
- 3. Basic STP Configurations**
4. Improvements Made in RSTP
5. STP Advancement



Basic STP Configuration Commands (1)

1. Configure a working mode.

```
[Huawei] stp mode { stp | rstp | mstp }
```

The switch supports three working modes: STP, RSTP, and Multiple Spanning Tree Protocol (MSTP). By default, a switch works in MSTP mode. On a ring network running only STP, the working mode of a switch is configured as STP; on a ring network running RSTP, the working mode of a switch is configured as RSTP.

2. (Optional) Configure the root bridge.

```
[Huawei] stp root primary
```

Configure the switch as the root bridge. By default, a switch does not function as the root bridge of any spanning tree. After you run this command, the priority value of the switch is set to 0 and cannot be changed.

3. (Optional) Configure the switch as the secondary root bridge.

```
[Huawei] stp root secondary
```

Configure the switch as the secondary root bridge. By default, a switch does not function as the secondary root bridge of any spanning tree. After you run this command, the priority value of the switch is set to 4096 and cannot be changed.



Basic STP Configuration Commands (2)

1. (Optional) Configure the STP priority of a switch.

```
[Huawei] stp priority priority
```

By default, the priority value of a switch is 32768.

2. (Optional) Configure a path cost for a port.

```
[Huawei] stp pathcost-standard { dot1d-1998 | dot1t | legacy }
```

Configure a path cost calculation method. By default, the IEEE 802.1t standard (**dot1t**) is used to calculate path costs. All switches on a network must use the same path cost calculation method.

```
[Huawei-GigabitEthernet0/0/1] stp cost cost
```

Set the path cost of the port.



Basic STP Configuration Commands (3)

1. (Optional) Configure a priority for a port.

```
[Huawei-GigabitEthernet0/0/1] stp priority priority
```

Configure a priority for a port. By default, the priority of a switch port is 128.

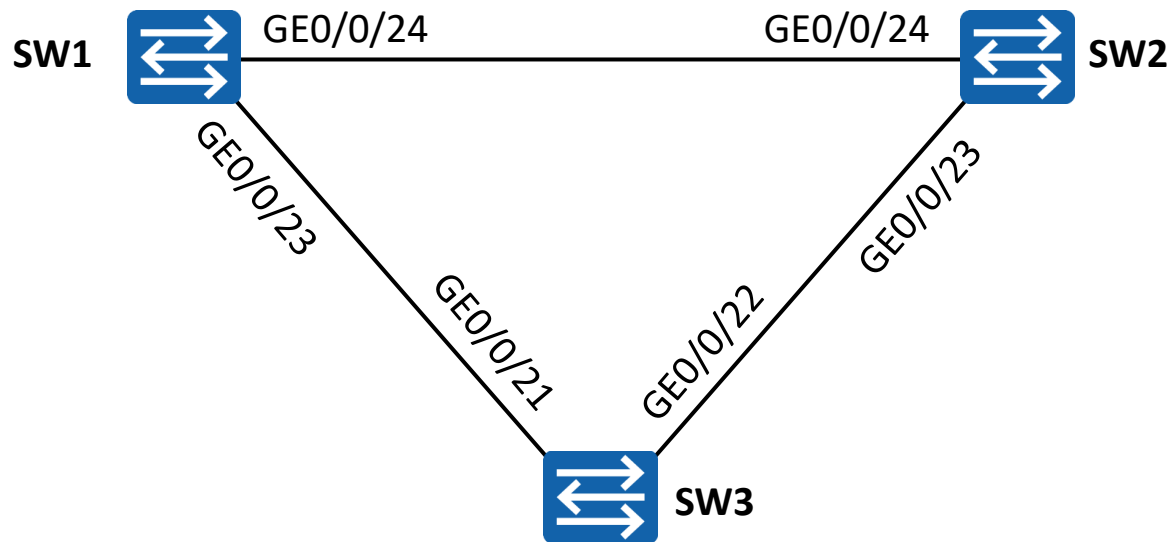
2. Enable STP, RSTP, or MSTP.

```
[Huawei] stp enable
```

Enable STP, RSTP, or MSTP on a switch. By default, STP, RSTP, or MSTP is enabled on a switch.



Case 1: Basic STP Configurations



- Deploy STP on the three switches to eliminate Layer 2 loops on the network.
- Configure SW1 as the root bridge and block GE0/0/22 on SW3.

SW1 configuration:

```
[SW1] stp mode stp
[SW1] stp enable
[SW1] stp priority 0
```

SW2 configuration:

```
[SW2] stp mode stp
[SW2] stp enable
[SW2] stp priority 4096
```

SW3 configuration:

```
[SW3] stp mode stp
[SW3] stp enable
```



Case 1: Basic STP Configurations

Check brief information about STP states of ports on SW3.

```
<SW3> display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/21	ROOT	FORWARDING	NONE
0	GigabitEthernet0/0/22	ALTE	DISCARDING	NONE



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Disadvantages of STP

- STP ensures a loop-free network but is slow to converge, leading to service quality deterioration. If the network topology changes frequently, connections on the STP network are frequently torn down, causing frequent service interruption.
- STP does not differentiate between port roles according to their states, making it difficult for less experienced administrators to learn about and deploy this protocol.
 - Ports in Listening, Learning, and Blocking states are the same for users because none of these ports forwards service traffic.
 - In terms of port use and configuration, the essential differences between ports lie in the port roles but not port states.
 - Both root and designated ports can be in Listening state or Forwarding state, so the port roles cannot be differentiated according to their states.
- The STP algorithm does not determine topology changes until the timer expires, delaying network convergence.
- The STP algorithm requires the root bridge to send configuration BPDUs after the network topology becomes stable, and other devices process and spread the configuration BPDUs through the entire network. This also delays convergence.



RSTP Overview

- RSTP defined in IEEE 802.1w is an enhancement to STP. RSTP optimizes STP in many aspects, provides faster convergence, and is compatible with STP.
- RSTP introduces new port roles. When the root port fails, the switch can enable the alternate port to obtain an alternate path from the designated bridge to the root bridge. RSTP defines three states for a port based on whether the port forwards user traffic and learns MAC addresses. In addition, RSTP introduces the edge port. The port connecting a switch to a terminal is configured as an edge port that enters the Forwarding state immediately after initialization, thus improving the working efficiency.



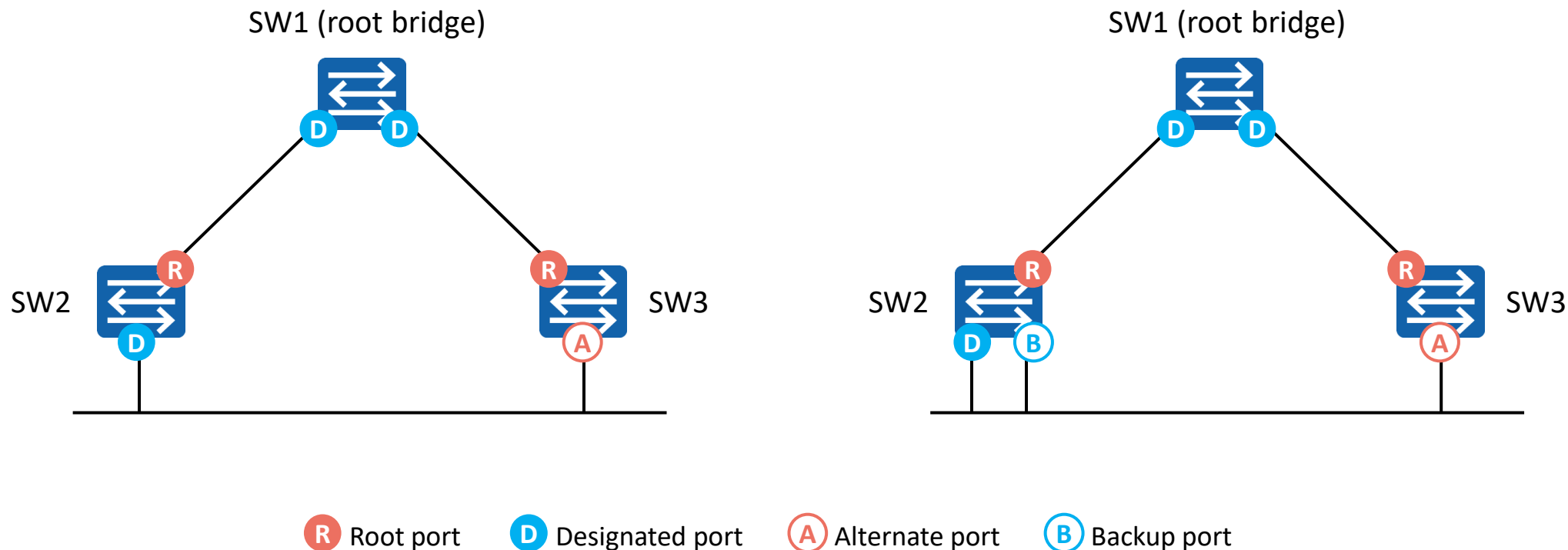
Improvements Made in RSTP

- RSTP processes configuration BPDUs differently from STP.
 - When the topology becomes stable, the mode of sending configuration BPDUs is optimized.
 - RSTP uses a shorter timeout interval of BPDUs.
 - RSTP optimizes the method of processing inferior BPDUs.
- RSTP changes the configuration BPDU format and uses the Flags field to describe port roles.
- RSTP topology change processing: Compared with STP, RSTP is optimized to accelerate the response to topology changes.



Port Roles in RSTP

- RSTP adds port roles to help understand RSTP and simplify RSTP deployment.

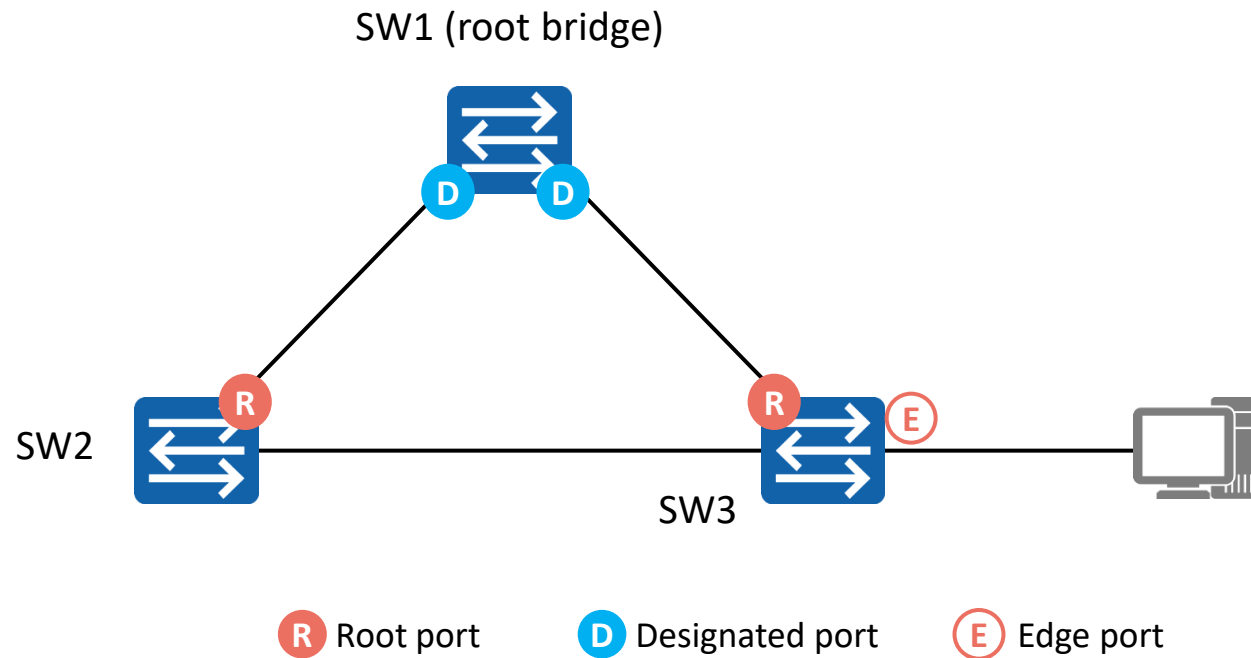


RSTP defines four port roles: root port, designated port, alternate port, and backup port.



Edge Port

- An edge port is located at the edge of a region and does not connect to any switching device.



Generally, an edge port is directly connected to a user terminal. The edge port can transition from the Disabled state to the Forwarding state.



Port States in RSTP

- RSTP deletes two port states defined in STP, reducing the number of port states to three.
 - If the port does not forward user traffic or learn MAC addresses, it is in Discarding state.
 - If the port does not forward user traffic but learns MAC addresses, it is in Learning state.
 - If the port forwards user traffic and learns MAC addresses, it is in Forwarding state.

STP Port State	RSTP Port State	Port Role
Forwarding	Forwarding	Root port or designated port
Learning	Learning	Root port or designated port
Listening	Discarding	Root port or designated port
Blocking	Discarding	Alternate port or backup port
Disabled	Discarding	Disabled port



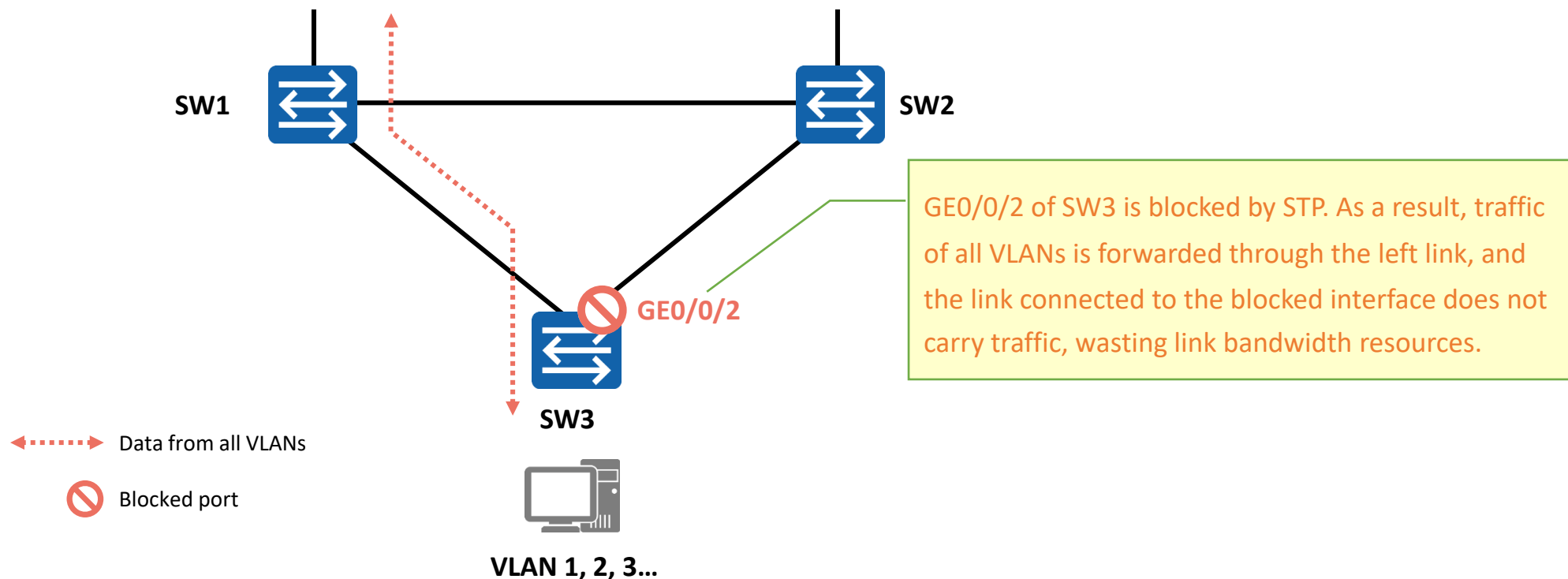
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Defects of STP/RSTP: All VLANs Share One Spanning Tree

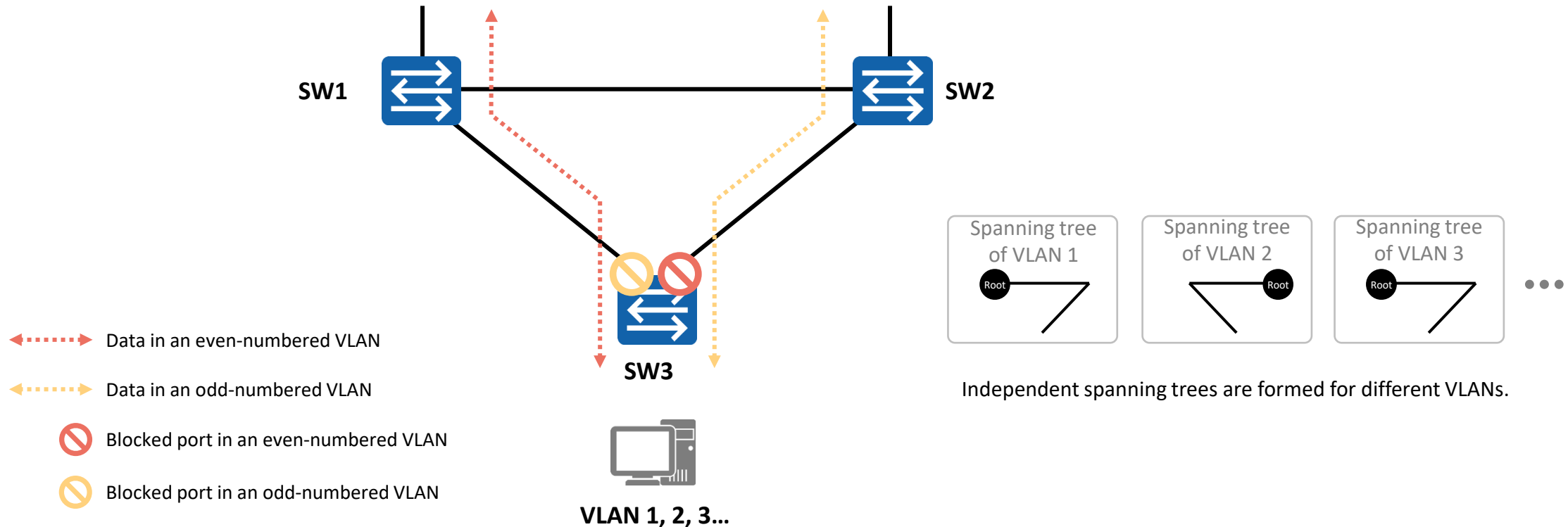
- RSTP, an enhancement to STP, allows for fast network topology convergence.
- STP and RSTP both have a defect: All VLANs on a LAN share one spanning tree. As a result, inter-VLAN load balancing cannot be performed, and blocked links cannot transmit any traffic, which may lead to VLAN packet transmission failures.





VBST

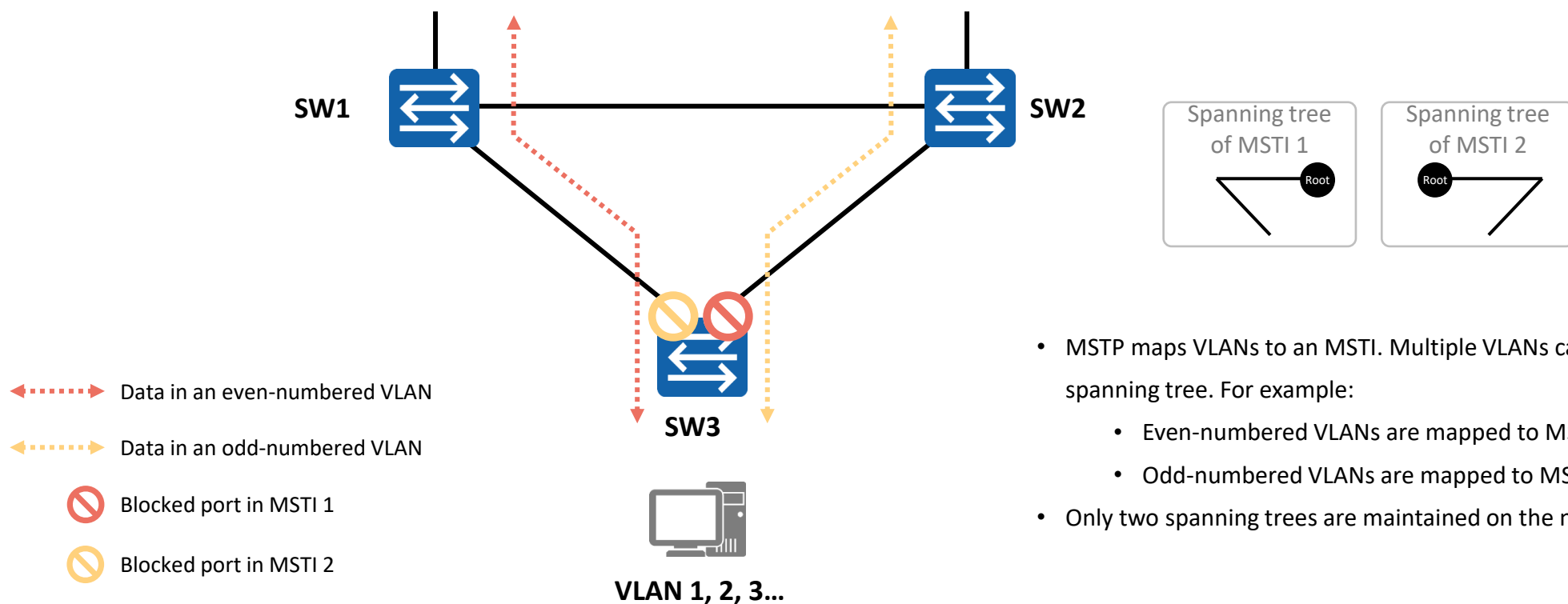
- Huawei provides the VLAN-based Spanning Tree (VBST). VBST constructs a spanning tree in each VLAN so that traffic from different VLANs is load balanced along different spanning trees.





MSTP

- To fix the defects, the IEEE released the 802.1s standard that defines the Multiple Spanning Tree Protocol (MSTP) in 2002.
- MSTP is compatible with STP and RSTP, and can rapidly converge traffic and provides multiple paths to load balance VLAN traffic.



- MSTP maps VLANs to an MSTI. Multiple VLANs can share one spanning tree. For example:
 - Even-numbered VLANs are mapped to MSTI 1.
 - Odd-numbered VLANs are mapped to MSTI 2.
- Only two spanning trees are maintained on the network.



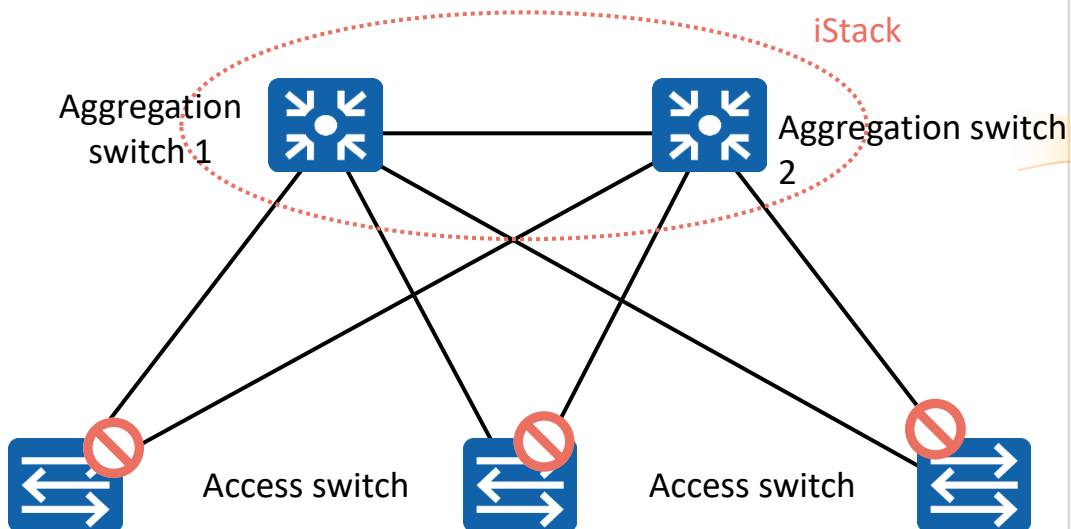
MSTP Overview

- MSTP divides a switching network into multiple regions, each of which has multiple spanning trees that are independent of each other.
- Each spanning tree is called a multiple spanning tree instance (MSTI).
- An MSTI is the spanning tree corresponding to a set of VLANs.
- Binding multiple VLANs to a single MSTI reduces communication costs and resource usage.
- The topology of each MSTI is calculated independently, and traffic can be balanced among MSTIs.
- Multiple VLANs with the same topology can be mapped to a single MSTI. The forwarding state of the VLANs for an interface is determined by the interface state in the MSTI.



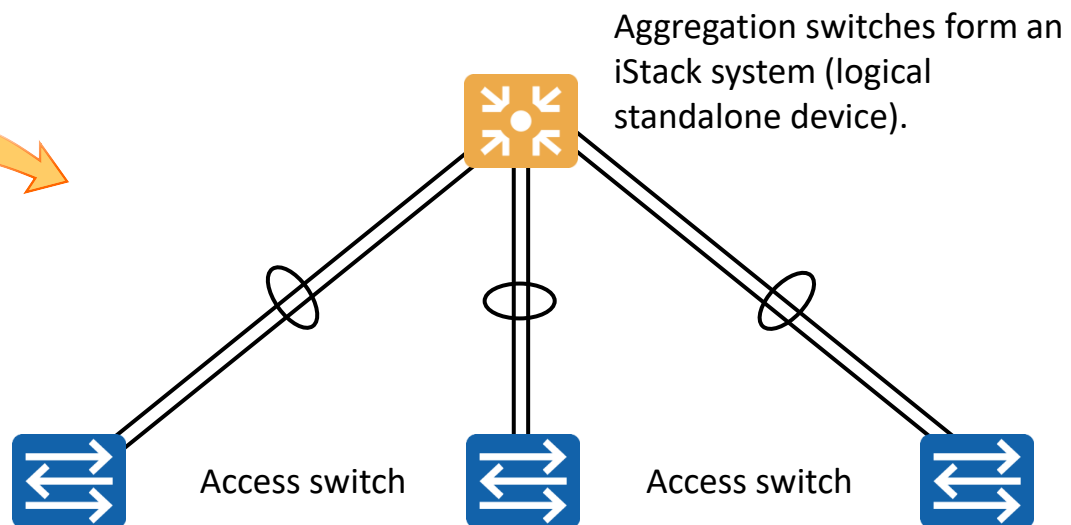
Stack and Tree Networking of Campus Networks

Traditional STP Networking



Two aggregation switches form a triangle Layer 2 loop with access switches, so STP must be deployed on the network. However, STP blocks ports on the network, causing a failure to fully utilize link bandwidth.

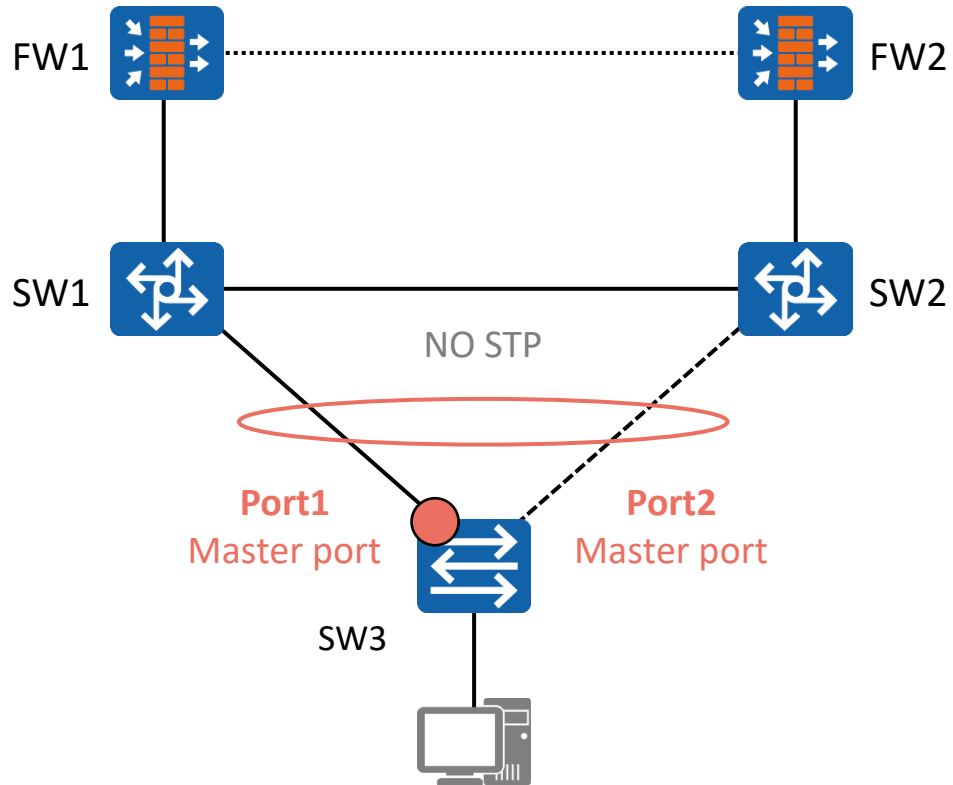
iStack Networking



Aggregation switches are stacked to form a single logical device, simplifying the network topology. In addition, link aggregation is deployed between aggregation switches and access switches to simplify the network topology to a tree topology, eliminating Layer 2 loops and improving link bandwidth utilization.



Smart Link



 Smart Link group

 Active status

Smart Link is tailored for dual-uplink networking.

- Smart Link is deployed on two switches where a host is dual-homed. When the network is normal, one of the two uplinks is active, and the other is in standby state (does not carry service traffic). In this way, a Layer 2 loop is eliminated.
- When the active link is faulty, traffic is switched to the standby link in milliseconds. This ensures proper data forwarding.
- Smart Link is easy to configure.
- Smart Link does not involve protocol packet exchange, therefore greatly improving speed and reliability.



Quiz

1. (Single Choice) Which statement about the STP port state is false? ()
 - A. The blocked port does not listen to or send BPDUs. (Correct)
 - B. A port in Learning state learns MAC addresses but does not forward data.
 - C. A port in Listening state keeps listening to BPDUs.
 - D. If a blocked port does not receive BPDUs within a specified period, the port automatically switches to the Listening state.

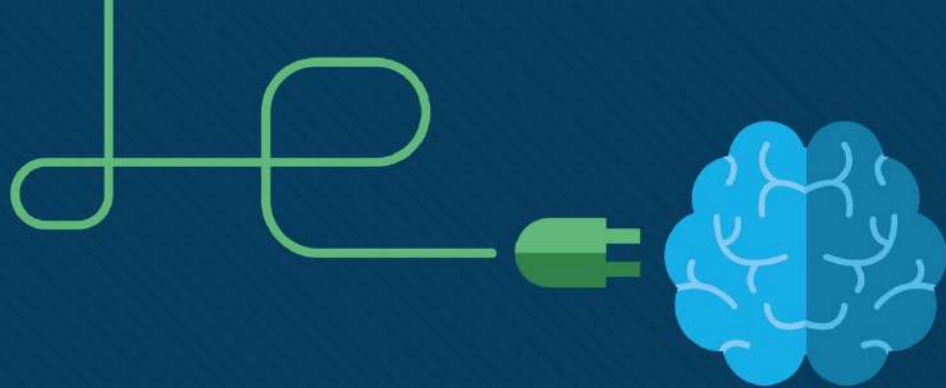


Summary

- STP prevents loops on a LAN. Devices running STP exchange information with one another to discover loops on the network, and block certain ports to eliminate loops. With the growth in scale of LANs, STP has become an important protocol for a LAN.
- After STP is configured on an Ethernet switching network, the protocol calculates the network topology to implement the following functions:
 - Loop prevention: The spanning tree protocol blocks redundant links to prevent potential loops on the network.
 - Link redundancy: If an active link fails and a redundant link exists, the spanning tree protocol activates the redundant link to ensure network connectivity.
- STP cannot meet requirements of modern campus networks. However, understanding the working mechanism of STP helps you better understand the working mechanism and deployment of RSTP and MSTP.

The background of the slide features a blue-tinted image of several business professionals in a modern office. They are standing on a highly reflective floor, and their silhouettes are clearly visible. The individuals are engaged in various interactions, some holding documents or tablets. The overall aesthetic is professional and corporate.

Thank You
www.huawei.com



Module 5: STP Concepts

Switching, Routing and Wireless
Essentials v7.0 (SRWE)



Module Objectives

Module Title: STP Concepts

Module Objective: Explain how STP enables redundancy in a Layer 2 network.

Topic Title	Topic Objective
Purpose of STP	Explain common problems in a redundant, L2 switched network.
STP Operations	Explain how STP operates in a simple switched network.
Evolution of STP	Explain how Rapid PVST+ operates.

5.1 Purpose of STP

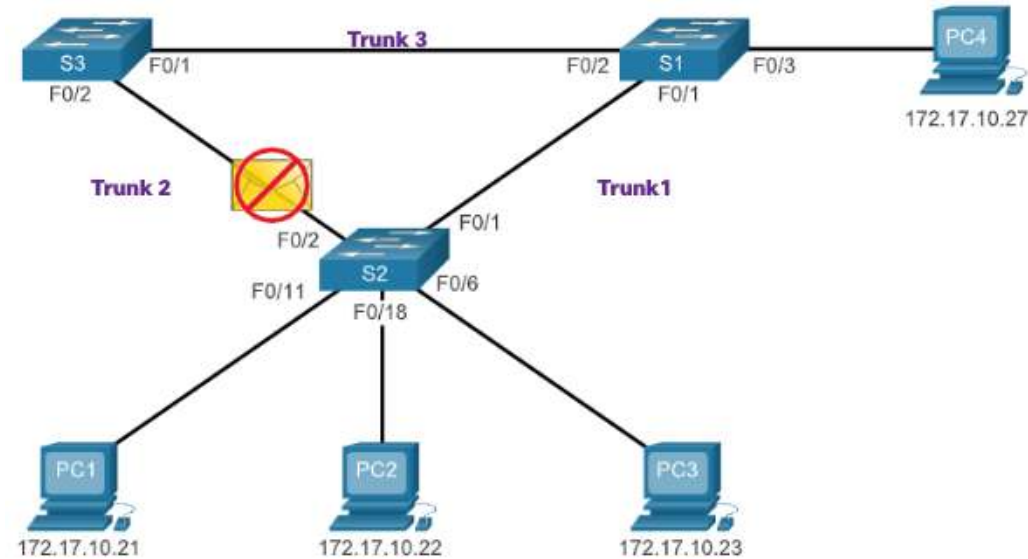
Redundancy in Layer 2 Switched Networks

- This topic covers the causes of loops in a Layer 2 network and briefly explains how spanning tree protocol works. Redundancy is an important part of the hierarchical design for eliminating single points of failure and preventing disruption of network services to users. Redundant networks require the addition of physical paths, but logical redundancy must also be part of the design. Having alternate physical paths for data to traverse the network makes it possible for users to access network resources, despite path disruption. However, redundant paths in a switched Ethernet network may cause both physical and logical Layer 2 loops.
- Ethernet LANs require a loop-free topology with a single path between any two devices. A loop in an Ethernet LAN can cause continued propagation of Ethernet frames until a link is disrupted and breaks the loop.

Purpose of STP

Spanning Tree Protocol

- Spanning Tree Protocol (STP) is a loop-prevention network protocol that allows for redundancy while creating a loop-free Layer 2 topology.
- STP logically blocks physical loops in a Layer 2 network, preventing frames from circling the network forever.

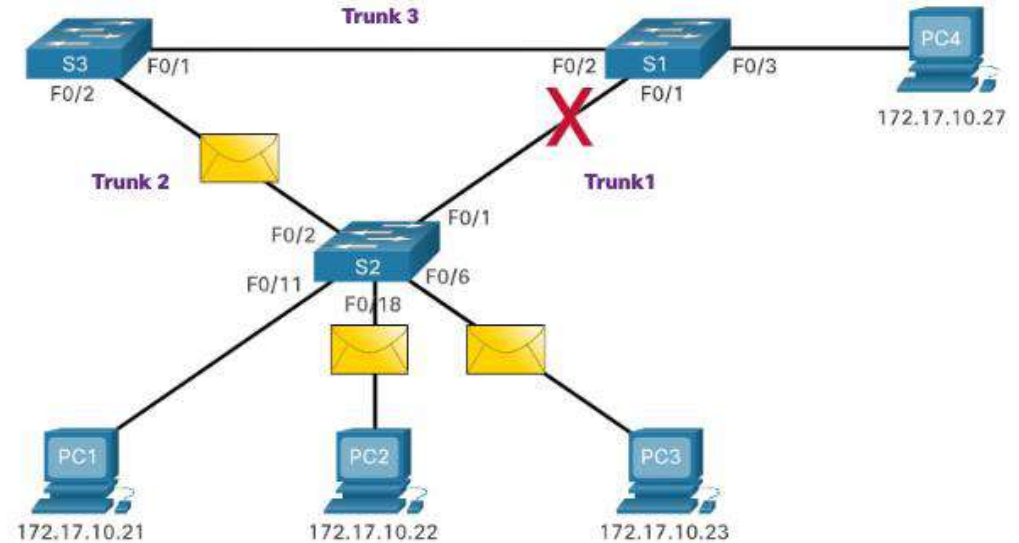


S2 drops the frame because it received it on a blocked port.

Purpose of STP

STP Recalculation

STP compensates for a failure in the network by recalculating and opening up previously blocked ports.



Purpose of STP

Issues with Redundant Switch Links

- Path redundancy provides multiple network services by eliminating the possibility of a single point of failure. When multiple paths exist between two devices on an Ethernet network, and there is no spanning tree implementation on the switches, a Layer 2 loop occurs. A Layer 2 loop can result in MAC address table instability, link saturation, and high CPU utilization on switches and end-devices, resulting in the network becoming unusable.
- Layer 2 Ethernet does not include a mechanism to recognize and eliminate endlessly looping frames. Both IPv4 and IPv6 include a mechanism that limits the number of times a Layer 3 networking device can retransmit a packet. A router will decrement the TTL (Time to Live) in every IPv4 packet, and the Hop Limit field in every IPv6 packet. When these fields are decremented to 0, a router will drop the packet. Ethernet and Ethernet switches have no comparable mechanism for limiting the number of times a switch retransmits a Layer 2 frame. STP was developed specifically as a loop prevention mechanism for Layer 2 Ethernet.

Purpose of STP

Layer 2 Loops

- Without STP enabled, Layer 2 loops can form, causing broadcast, multicast and unknown unicast frames to loop endlessly. This can bring down a network quickly.
- When a loop occurs, the MAC address table on a switch will constantly change with the updates from the broadcast frames, which results in MAC database instability. This can cause high CPU utilization, which makes the switch unable to forward frames.
- An unknown unicast frame is when the switch does not have the destination MAC address in its MAC address table and must forward the frame out all ports, except the ingress port.

Purpose of STP

Broadcast Storm

- A broadcast storm is an abnormally high number of broadcasts overwhelming the network during a specific amount of time. Broadcast storms can disable a network within seconds by overwhelming switches and end devices. Broadcast storms can be caused by a hardware problem such as a faulty NIC or from a Layer 2 loop in the network.
- Layer 2 broadcasts in a network, such as ARP Requests are very common. Layer 2 multicasts are typically forwarded the same way as a broadcast by the switch. IPv6 packets are never forwarded as a Layer 2 broadcast, ICMPv6 Neighbor Discovery uses Layer 2 multicasts.
- A host caught in a Layer 2 loop is not accessible to other hosts on the network. Additionally, due to the constant changes in its MAC address table, the switch does not know out of which port to forward unicast frames.
- To prevent these issues from occurring in a redundant network, some type of spanning tree must be enabled on the switches. Spanning tree is enabled, by default, on Cisco switches to prevent Layer 2 loops from occurring.

Purpose of STP

The Spanning Tree Algorithm

- STP is based on an algorithm invented by Radia Perlman while working for Digital Equipment Corporation, and published in the 1985 paper "An Algorithm for Distributed Computation of a Spanning Tree in an Extended LAN." Her spanning tree algorithm (STA) creates a loop-free topology by selecting a single root bridge where all other switches determine a single least-cost path.
- STP prevents loops from occurring by configuring a loop-free path through the network using strategically placed "blocking-state" ports. The switches running STP are able to compensate for failures by dynamically unblocking the previously blocked ports and permitting traffic to traverse the alternate paths.

The Spanning Tree Algorithm (Cont.)

How does the STA create a loop-free topology?

- **Selecting a Root Bridge:** This bridge (switch) is the reference point for the entire network to build a spanning tree around.
- **Block Redundant Paths:** STP ensures that there is only one logical path between all destinations on the network by intentionally blocking redundant paths that could cause a loop. When a port is blocked, user data is prevented from entering or leaving that port.
- **Create a Loop-Free Topology:** A blocked port has the effect of making that link a non-forwarding link between the two switches. This creates a topology where each switch has only a single path to the root bridge, similar to branches on a tree that connect to the root of the tree.
- **Recalculate in case of Link Failure:** The physical paths still exist to provide redundancy, but these paths are disabled to prevent the loops from occurring. If the path is ever needed to compensate for a network cable or switch failure, STP recalculates the paths and unblocks the necessary ports to allow the redundant path to become active. STP recalculations can also occur any time a new switch or new inter-switch link is added to the network.

5.2 STP Operations

Steps to a Loop-Free Topology

Using the STA, STP builds a loop-free topology in a four-step process:

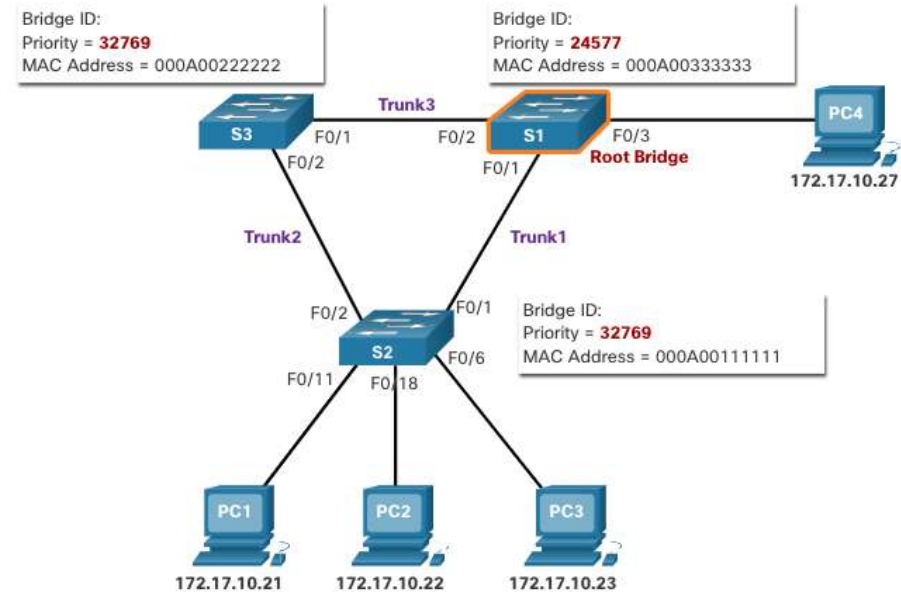
1. Elect the root bridge.
 2. Elect the root ports.
 3. Elect designated ports.
 4. Elect alternate (blocked) ports.
- During STA and STP functions, switches use Bridge Protocol Data Units (BPDUs) to share information about themselves and their connections. BPDUs are used to elect the root bridge, root ports, designated ports, and alternate ports.
 - Each BPDU contains a bridge ID (BID) that identifies which switch sent the BPDU. The BID is involved in making many of the STA decisions including root bridge and port roles.
 - The BID contains a priority value, the MAC address of the switch, and an extended system ID. The lowest BID value is determined by the combination of these three fields.

Steps to a Loop-Free Topology (Cont.)

- **Bridge Priority:** The default priority value for all Cisco switches is the decimal value 32768. The range is 0 to 61440 in increments of 4096. A lower bridge priority is preferable. A bridge priority of 0 takes precedence over all other bridge priorities.
- **Extended System ID:** The extended system ID value is a decimal value added to the bridge priority value in the BID to identify the VLAN for this BPDU.
- **MAC address:** When two switches are configured with the same priority and have the same extended system ID, the switch having the MAC address with the lowest value, expressed in hexadecimal, will have the lower BID.

1. Elect the Root Bridge

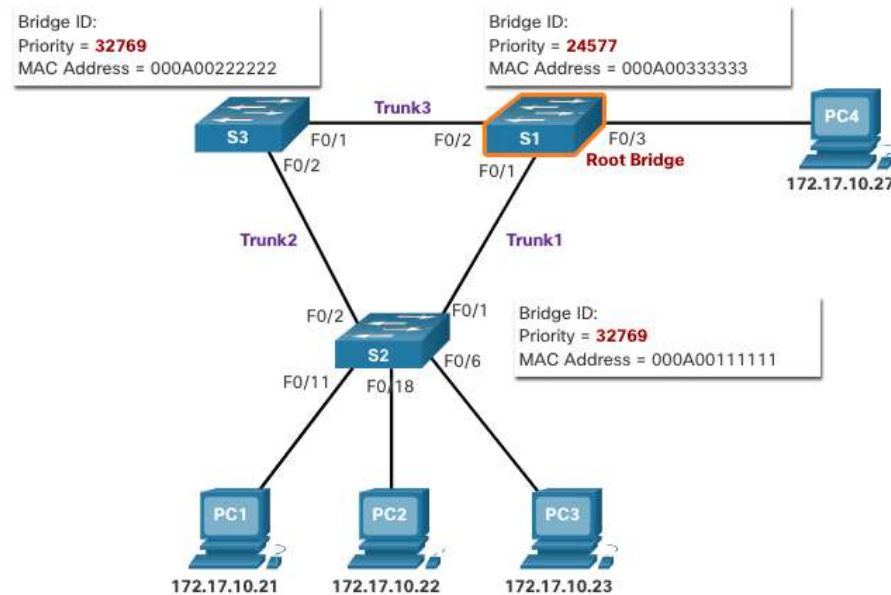
- The STA designates a single switch as the root bridge and uses it as the reference point for all path calculations. Switches exchange BPDUs to build the loop-free topology beginning with selecting the root bridge.
- All switches in the broadcast domain participate in the election process. After a switch boots, it begins to send out BPDU frames every two seconds. These BPDU frames contain the BID of the sending switch and the BID of the root bridge, known as the Root ID.
- The switch with the lowest BID will become the root bridge. At first, all switches declare themselves as the root bridge with their own BID set as the Root ID. Eventually, the switches learn through the exchange of BPDUs which switch has the lowest BID and will agree on one root bridge.



STP Operations

Impact of Default BIDs

- Because the default BID is 32768, it is possible for two or more switches to have the same priority. In this scenario, where the priorities are the same, the switch with the lowest MAC address will become the root bridge. The administrator should configure the desired root bridge switch with a lower priority.
- In the figure, all switches are configured with the same priority of 32769. Here the MAC address becomes the deciding factor as to which switch becomes the root bridge. The switch with the lowest hexadecimal MAC address value is the preferred root bridge. In this example, S2 has the lowest value for its MAC address and is elected as the root bridge for that spanning tree instance.
- Note:** The priority of all the switches is 32769. The value is based on the 32768 default bridge priority and the extended system ID (VLAN 1 assignment) associated with each switch (32768+1).



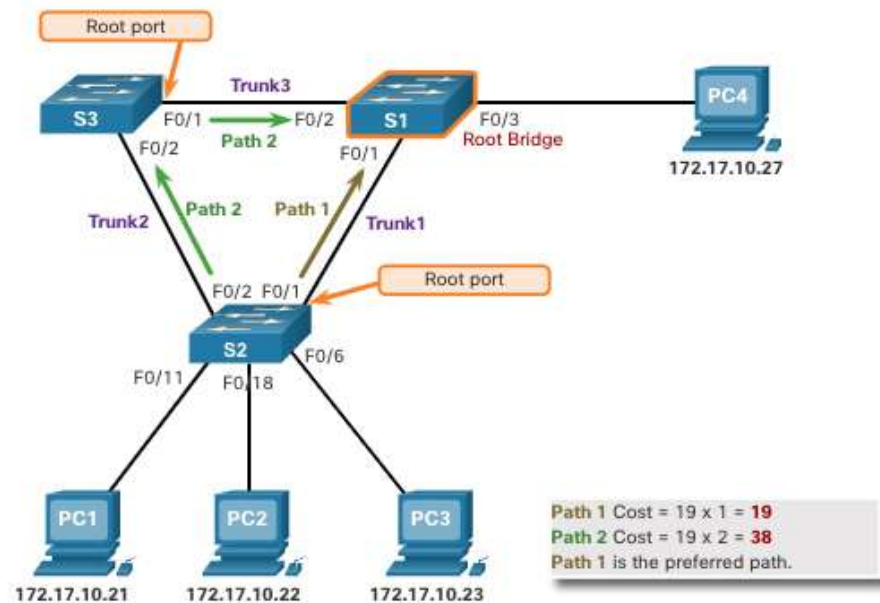
Determine the Root Path Cost

- When the root bridge has been elected for a given spanning tree instance, the STA starts determining the best paths to the root bridge from all destinations in the broadcast domain. The path information, known as the internal root path cost, is determined by the sum of all the individual port costs along the path from the switch to the root bridge.
- When a switch receives the BPDU, it adds the ingress port cost of the segment to determine its internal root path cost.
- The default port costs are defined by the speed at which the port operates. The table shows the default port costs suggested by IEEE. Cisco switches by default use the values as defined by the IEEE 802.1D standard, also known as the short path cost, for both STP and RSTP.
- Although switch ports have a default port cost associated with them, the port cost is configurable. The ability to configure individual port costs gives the administrator the flexibility to manually control the spanning tree paths to the root bridge.

Link Speed	STP Cost: IEEE 802.1D-1998	RSTP Cost: IEEE 802.1w-2004
10 Gbps	2	2,000
1 Gbps	4	20,000
100 Mbps	19	200,000
10 Mbps	100	2,000,000

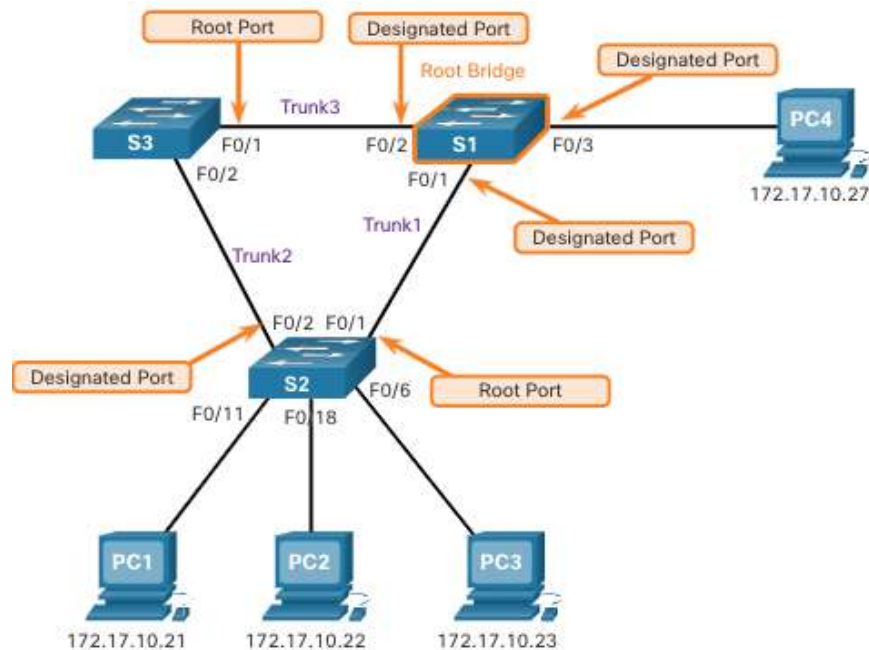
2. Elect the Root Ports

- After the root bridge has been determined, the STA algorithm is used to select the root port. Every non-root switch will select one root port. The root port is the port closest to the root bridge in terms of overall cost to the root bridge. This overall cost is known as the internal root path cost.
- The internal root path cost is equal to the sum of all the port costs along the path to the root bridge, as shown in the figure. Paths with the lowest cost become preferred, and all other redundant paths are blocked. In the example, the internal root path cost from S2 to the root bridge S1 over path 1 is 19 while the internal root path cost over path 2 is 38. Because path 1 has a lower overall path cost to the root bridge, it is the preferred path and F0/1 becomes the root port on S2.



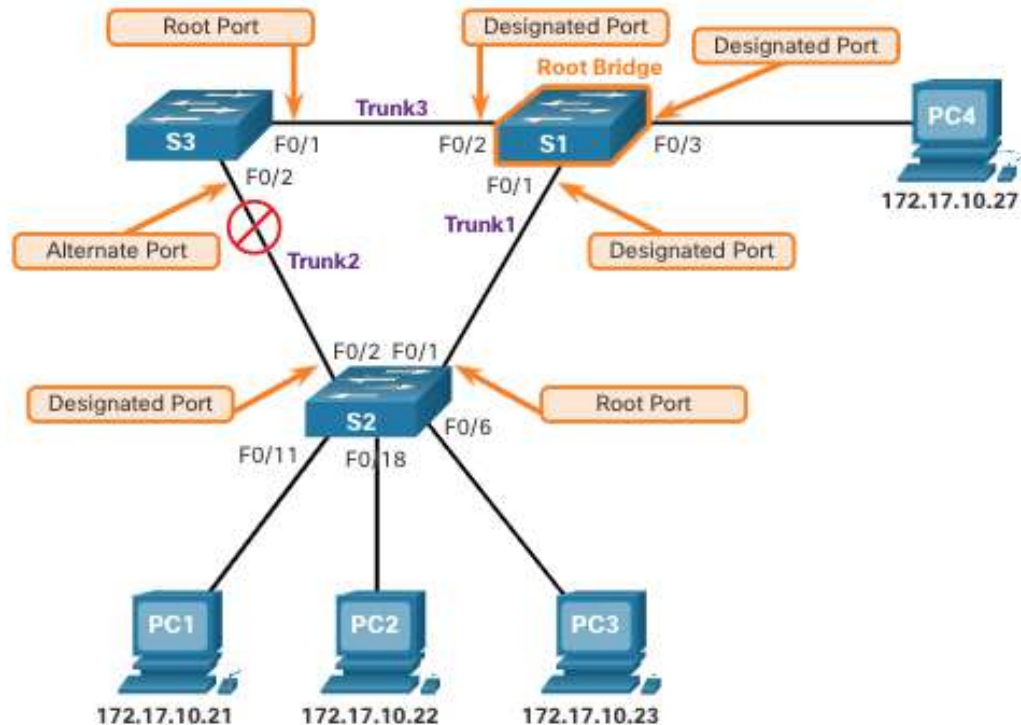
3. Elect Designated Ports

- Every segment between two switches will have one designated port. The designated port is a port on the segment that has the internal root path cost to the root bridge. In other words, the designated port has the best path to receive traffic leading to the root bridge.
- What is not a root port or a designated port becomes an alternate or blocked port.
- All ports on the root bridge are designated ports.
- If one end of a segment is a root port, the other end is a designated port.
- All ports attached to end devices are designated ports.
- On segments between two switches where neither of the switches is the root bridge, the port on the switch with the least-cost path to the root bridge is a designated port.



4. Elect Alternate (Blocked) Ports

If a port is not a root port or a designated port, then it becomes an alternate (or backup) port. Alternate ports are in discarding or blocking state to prevent loops. In the figure, the STA has configured port F0/2 on S3 in the alternate role. Port F0/2 on S3 is in the blocking state and will not forward Ethernet frames. All other inter-switch ports are in forwarding state. This is the loop-prevention part of STP.



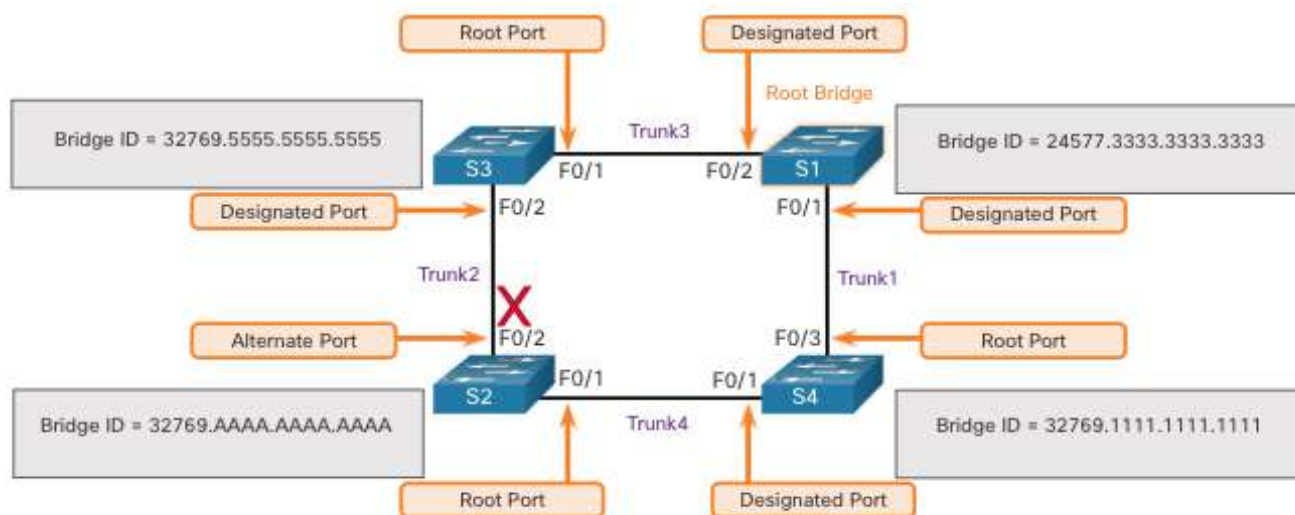
Elect a Root Port from Multiple Equal-Cost Paths

When a switch has multiple equal-cost paths to the root bridge, the switch will determine a port using the following criteria:

- Lowest sender BID
- Lowest sender port priority
- Lowest sender port ID

Elect a Root Port from Multiple Equal-Cost Paths (Cont.)

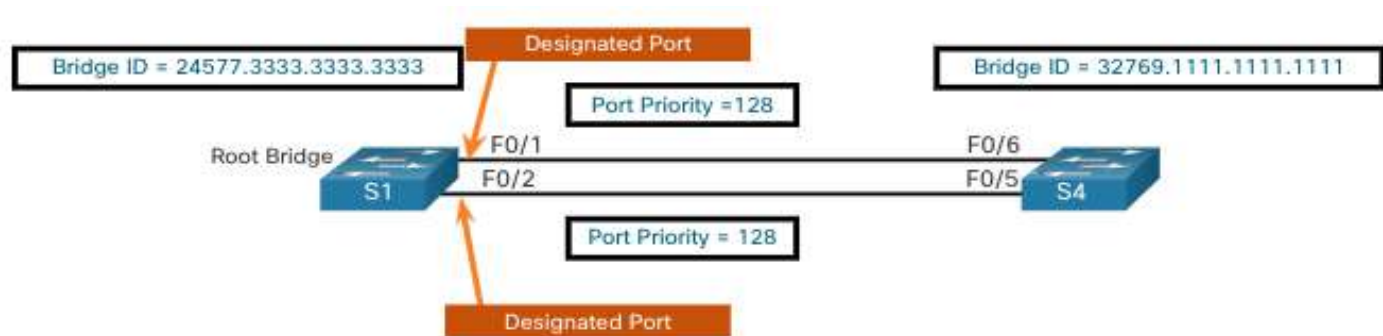
Lowest Sender BID: This topology has four switches with switch S1 as the root bridge. Port F0/1 on switch S3 and port F0/3 on switch S4 have been selected as root ports because they have the root path cost for their respective switches. S2 has two ports, F0/1 and F0/2 with equal cost paths to the root bridge. The bridge IDs of S3 and S4, will be used to break the tie. This is known as the sender's BID. S3 has a BID of 32769.5555.5555.5555 and S4 has a BID of 32769.1111.1111.1111. Because S4 has a lower BID, the F0/1 port of S2, which is the port connected to S4, will be the root port.



Elect a Root Port from Multiple Equal-Cost Paths (Cont.)

Lowest Sender Port Priority: This topology has two switches which are connected with two equal-cost paths between them. S1 is the root bridge, so both of its ports are designated ports.

- S4 has two ports with equal-cost paths to the root bridge. Because both ports are connected to the same switch, the sender's BID (S1) is equal. So the first step is a tie.
- Next, is the sender's (S1) port priority. The default port priority is 128, so both ports on S1 have the same port priority. This is also a tie. However, if either port on S1 was configured with a lower port priority, S4 would put its adjacent port in forwarding state. The other port on S4 would be a blocking state.



Elect a Root Port from Multiple Equal-Cost Paths (Cont.)

- **Lowest Sender Port ID:** The last tie-breaker is the lowest sender's port ID. Switch S4 has received BPDUs from port F0/1 and port F0/2 on S1. The decision is based on the sender's port ID, not the receiver's port ID. Because the port ID of F0/1 on S1 is lower than port F0/2, the port F0/6 on switch S4 will be the root port. This is the port on S4 that is connected to the F0/1 port on S1.
- Port F0/5 on S4 will become an alternate port and placed in the blocking state.



STP Operations

STP Timers and Port States

STP convergence requires three timers, as follows:

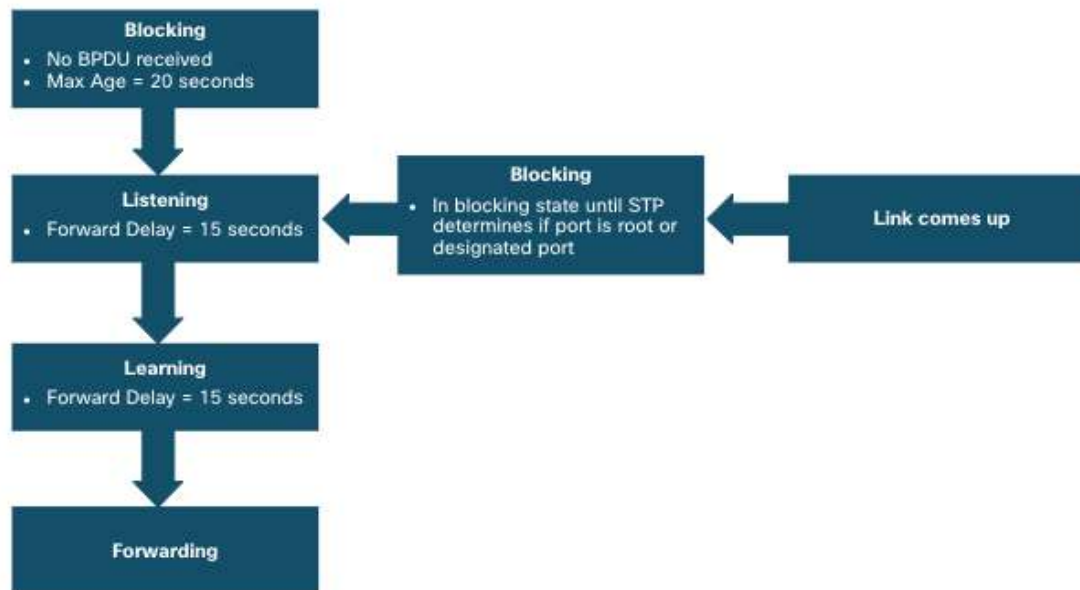
- **Hello Timer** -The hello time is the interval between BPDUs. The default is 2 seconds but can be modified to between 1 and 10 seconds.
- **Forward Delay Timer** -The forward delay is the time that is spent in the listening and learning state. The default is 15 seconds but can be modified to between 4 and 30 seconds.
- **Max Age Timer** -The max age is the maximum length of time that a switch waits before attempting to change the STP topology. The default is 20 seconds but can be modified to between 6 and 40 seconds.

Note: The default times can be changed on the root bridge, which dictates the value of these timers for the STP domain.

STP Operations

STP Timers and Port States (Cont.)

STP facilitates the logical loop-free path throughout the broadcast domain. The spanning tree is determined through the information learned by the exchange of the BPDU frames between the interconnected switches. If a switch port transitions directly from the blocking state to the forwarding state without information about the full topology during the transition, the port can temporarily create a data loop. For this reason, STP has five port states, four of which are operational port states as shown in the figure. The disabled state is considered non-operational.



STP Operations

Operational Details of Each Port State

The table summarizes the operational details of each port state

Port State	BPDU	MAC Address Table	Forwarding Data Frames
Blocking	Receive only	No update	No
Listening	Receive and send	No update	No
Learning	Receive and send	Updating table	No
Forwarding	Receive and send	Updating table	Yes
Disabled	None sent or received	No update	No

STP Operations

Per-VLAN Spanning Tree

STP can be configured to operate in an environment with multiple VLANs. In Per-VLAN Spanning Tree (PVST) versions of STP, there is a root bridge elected for each spanning tree instance. This makes it possible to have different root bridges for different sets of VLANs. STP operates a separate instance of STP for each individual VLAN. If all ports on all switches are members of VLAN 1, then there is only one spanning tree instance.

5.3 Evolution of STP

Evolution of STP

Different Versions of STP

- Many professionals generically use spanning tree and STP to refer to the various implementations of spanning tree, such as Rapid Spanning Tree Protocol (RSTP) and Multiple Spanning Tree Protocol (MSTP). In order to communicate spanning tree concepts correctly, it is important to refer to the implementation or standard of spanning tree in context.
- The latest IEEE documentation on spanning tree (IEEE-802-1D-2004) says, "STP has now been superseded by the Rapid Spanning Tree Protocol (RSTP)." The IEEE uses "STP" to refer to the original implementation of spanning tree and "RSTP" to describe the version of spanning tree specified in IEEE-802.1D-2004.
- Because the two protocols share much of the same terminology and methods for the loop-free path, the primary focus will be on the current standard and the Cisco proprietary implementations of STP and RSTP.
- Cisco switches running IOS 15.0 or later, run PVST+ by default. This version incorporates many of the specifications of IEEE 802.1D-2004, such as alternate ports in place of the former non-designated ports. Switches must be explicitly configured for rapid spanning tree mode in order to run the rapid spanning tree protocol.

Evolution of STP

Different Versions of STP (Cont.)

STP Variety	Description
STP	This is the original IEEE 802.1D version (802.1D-1998 and earlier) that provides a loop-free topology in a network with redundant links. Also called Common Spanning Tree (CST), it assumes one spanning tree instance for the entire bridged network, regardless of the number of VLANs.
PVST+	Per-VLAN Spanning Tree (PVST+) is a Cisco enhancement of STP that provides a separate 802.1D spanning tree instance for each VLAN configured in the network. PVST+ supports PortFast, UplinkFast, BackboneFast, BPDU guard, BPDU filter, root guard, and loop guard.
802.1D-2004	This is an updated version of the STP standard, incorporating IEEE 802.1w.
RSTP	Rapid Spanning Tree Protocol (RSTP) or IEEE 802.1w is an evolution of STP that provides faster convergence than STP.
Rapid PVST+	This is a Cisco enhancement of RSTP that uses PVST+ and provides a separate instance of 802.1w per VLAN. Each separate instance supports PortFast, BPDU guard, BPDU filter, root guard, and loop guard.
MSTP	Multiple Spanning Tree Protocol (MSTP) is an IEEE standard inspired by the earlier Cisco proprietary Multiple Instance STP (MISTP) implementation. MSTP maps multiple VLANs into the same spanning tree instance.
MST	Multiple Spanning Tree (MST) is the Cisco implementation of MSTP, which provides up to 16 instances of RSTP and combines many VLANs with the same physical and logical topology into a common RSTP instance. Each instance supports PortFast, BPDU guard, BPDU filter, root guard, and loop guard.

Evolution of STP

RSTP Concepts

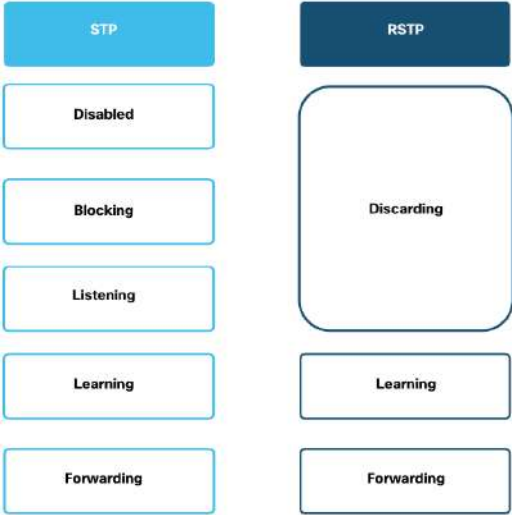
- RSTP (IEEE 802.1w) supersedes the original 802.1D while retaining backward compatibility. The 802.1w STP terminology remains primarily the same as the original IEEE 802.1D STP terminology. Most parameters have been left unchanged. Users that are familiar with the original STP standard can easily configure RSTP. The same spanning tree algorithm is used for both STP and RSTP to determine port roles and topology.
- RSTP increases the speed of the recalculation of the spanning tree when the Layer 2 network topology changes. RSTP can achieve much faster convergence in a properly configured network, sometimes in as little as a few hundred milliseconds. If a port is configured to be an alternate port it can immediately change to a forwarding state without waiting for the network to converge.

Note: Rapid PVST+ is the Cisco implementation of RSTP on a per-VLAN basis. With Rapid PVST+ an independent instance of RSTP runs for each VLAN.

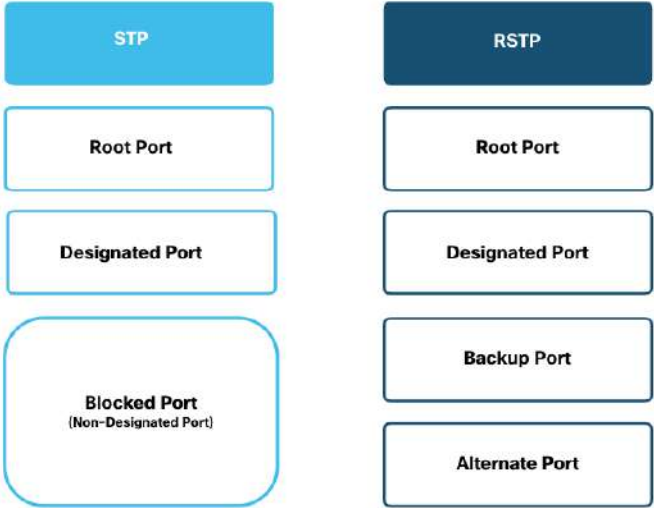
Evolution of STP

RSTP Port States and Port Roles

There are only three port states in RSTP that correspond to the three possible operational states in STP. The 802.1D disabled, blocking, and listening states are merged into a unique 802.1w discarding state.

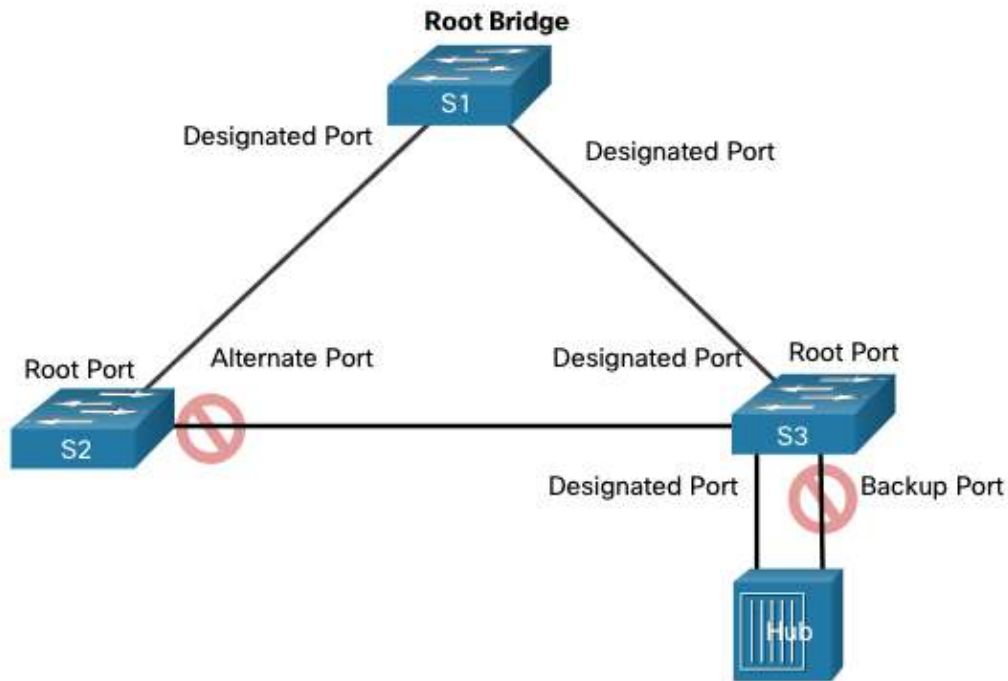


Root ports and designated ports are the same for both STP and RSTP. However, there are two RSTP port roles that correspond to the blocking state of STP. In STP, a blocked port is defined as not being the designated or root port. RSTP has two port roles for this purpose.



RSTP Port States and Port Roles (Cont.)

The alternate port has an alternate path to the root bridge. The backup port is a backup to a shared medium, such as a hub. A backup port is less common because hubs are now considered legacy devices.



Evolution of STP

PortFast and BPDU Guard

- When a device is connected to a switch port or when a switch powers up, the switch port goes through both the listening and learning states, each time waiting for the Forward Delay timer to expire. This delay is 15 seconds for each state for a total of 30 seconds. This can present a problem for DHCP clients trying to discover a DHCP server because the DHCP process may timeout. The result is that an IPv4 client will not receive a valid IPv4 address.
- When a switch port is configured with PortFast, that port transitions from blocking to forwarding state immediately, avoiding the 30 second delay. You can use PortFast on access ports to allow devices connected to these ports to access the network immediately. PortFast should only be used on access ports. If you enable PortFast on a port connecting to another switch, you risk creating a spanning tree loop.
- A PortFast-enabled switch port should never receive BPDUs because that would indicate that switch is connected to the port, potentially causing a spanning tree loop. Cisco switches support a feature called BPDU guard. When enabled, it immediately puts the switch port in an errdisabled (error-disabled) state upon receipt of any BPDU. This protects against potential loops by effectively shutting down the port. The administrator must manually put the interface back into service.

Evolution of STP

Alternatives to STP

- Over the years, organizations required greater resiliency and availability in the LAN. Ethernet LANs went from a few interconnected switches connected to a single router, to a sophisticated hierarchical network design including access, distribution and core layer switches.
- Depending on the implementation, Layer 2 may include not only the access layer, but also the distribution or even the core layers. These designs may include hundreds of switches, with hundreds or even thousands of VLANs. STP has adapted to the added redundancy and complexity with enhancements, as part of RSTP and MSTP.
- An important aspect to network design is fast and predictable convergence when there is a failure or change in the topology. Spanning tree does not offer the same efficiencies and predictabilities provided by routing protocols at Layer 3.
- Layer 3 routing allows for redundant paths and loops in the topology, without blocking ports. For this reason, some environments are transitioning to Layer 3 everywhere except where devices connect to the access layer switch. In other words, the connections between access layer switches and distribution switches would be Layer 3 instead of Layer 2.



Revision Record

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Course Code	Product	Product Version	Course Version
		V5R2	V1R1

Author/ID	Date	Reviewer/ID	New/ Update
Shi Miaomiao/swx791350	2019.10.23		





Inter-VLAN Communication



Foreword

- By default, a Layer 2 switching network is a broadcast domain, which brings many problems. Virtual local area network (VLAN) technology isolates such broadcast domains, preventing users in different VLANs from communicating with each other. However, such users sometimes need to communicate.
- This course describes how to implement inter-VLAN communication.



Objectives

- On completion of this course, you will be able to understand:
 - Methods of implementing inter-VLAN communication.
 - How to use routers (physical interfaces or sub-interfaces) to implement inter-VLAN communication.
 - How to use Layer 3 switches to implement inter-VLAN communication.
 - How Layer 3 packets are forwarded.



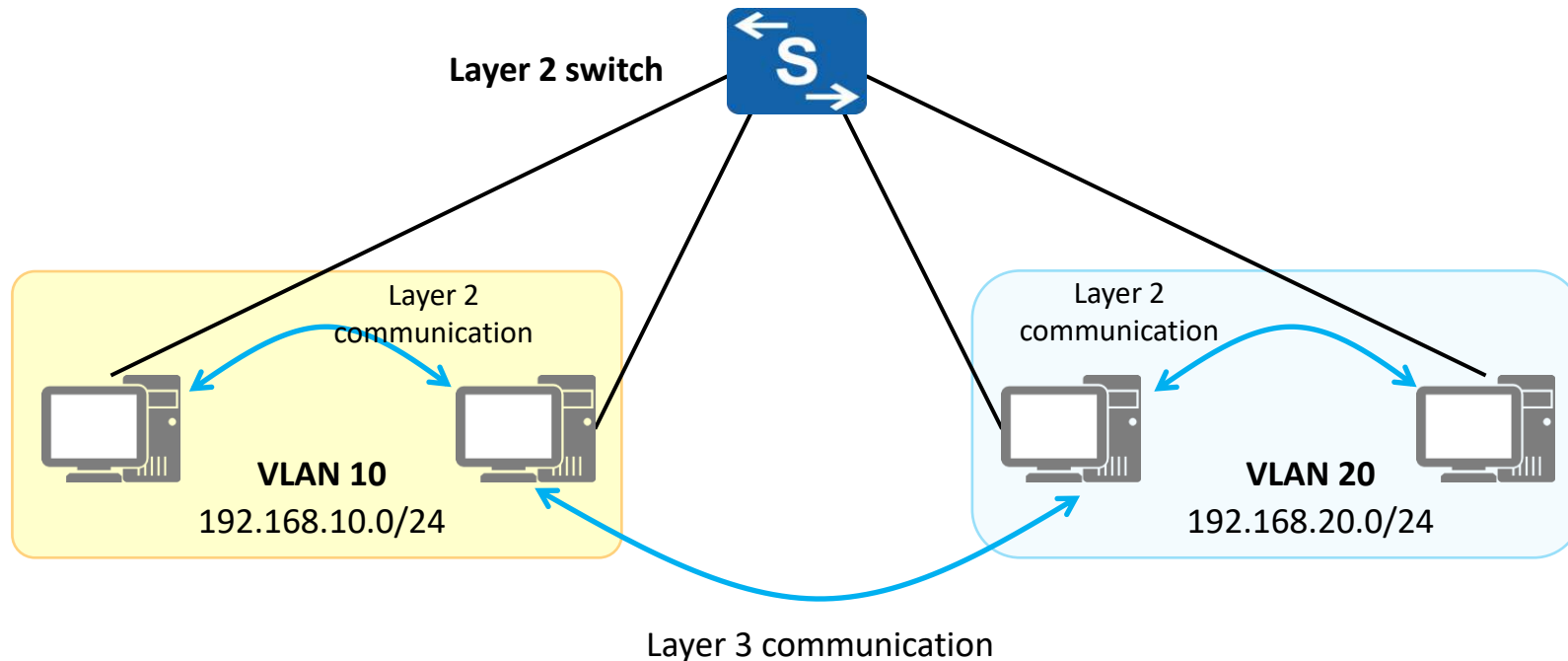
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- 1. Background**
2. Using Routers' Physical Interfaces or Sub-interfaces to Implement Inter-VLAN Communication
3. Using VLANIF Interfaces to Implement Inter-VLAN Communication
4. Layer 3 Communication Process



Inter-VLAN Communication (1)

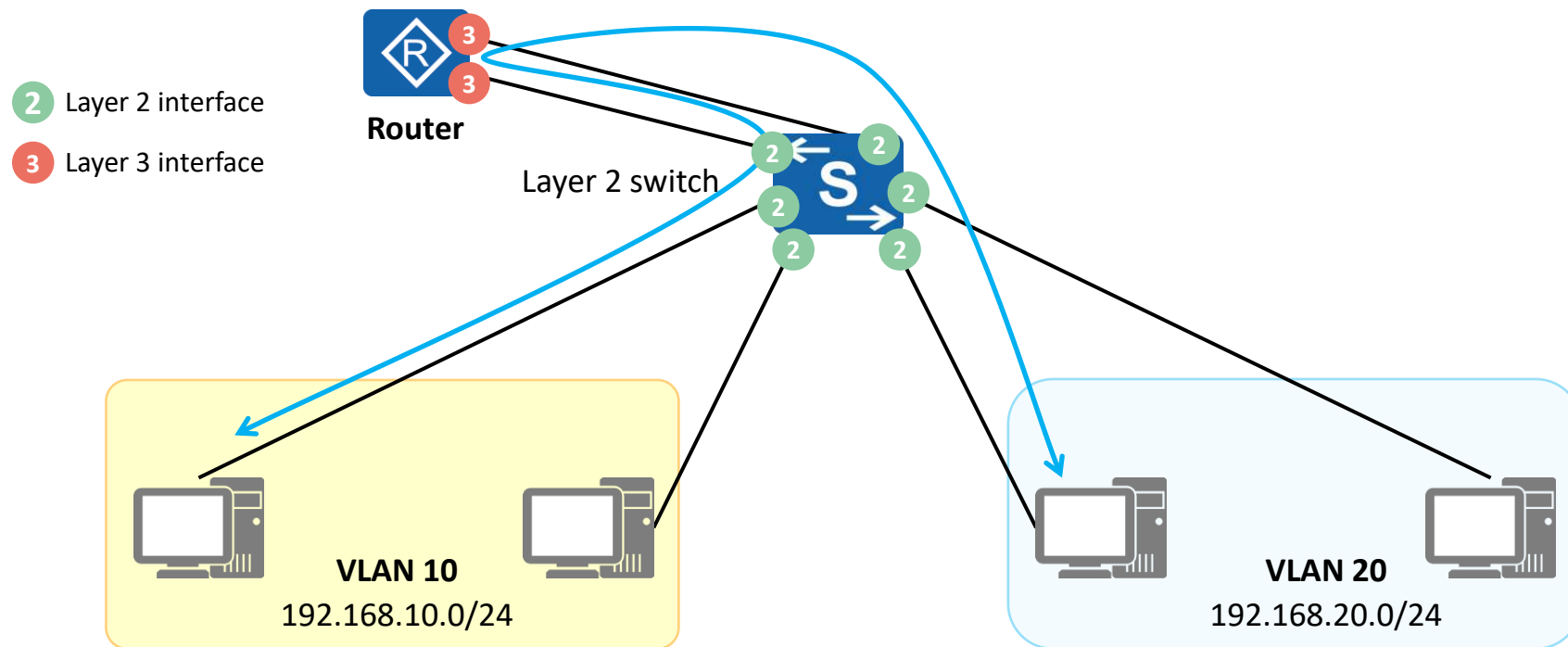
- In real-world network deployments, different IP address segments are assigned to different VLANs.
- PCs on the same network segment in the same VLAN can directly communicate with each other without the need for Layer 3 forwarding devices. This communication mode is called Layer 2 communication.
- Inter-VLAN communication belongs to Layer 3 communication, which requires Layer 3 devices.





Inter-VLAN Communication (2)

- Common Layer 3 devices: routers, Layer 3 switches, firewalls, etc.
- Inter-VLAN communication is implemented by connecting a Layer 2 switch to a Layer 3 interface of a Layer 3 device. The communication packets are routed by the Layer 3 device.





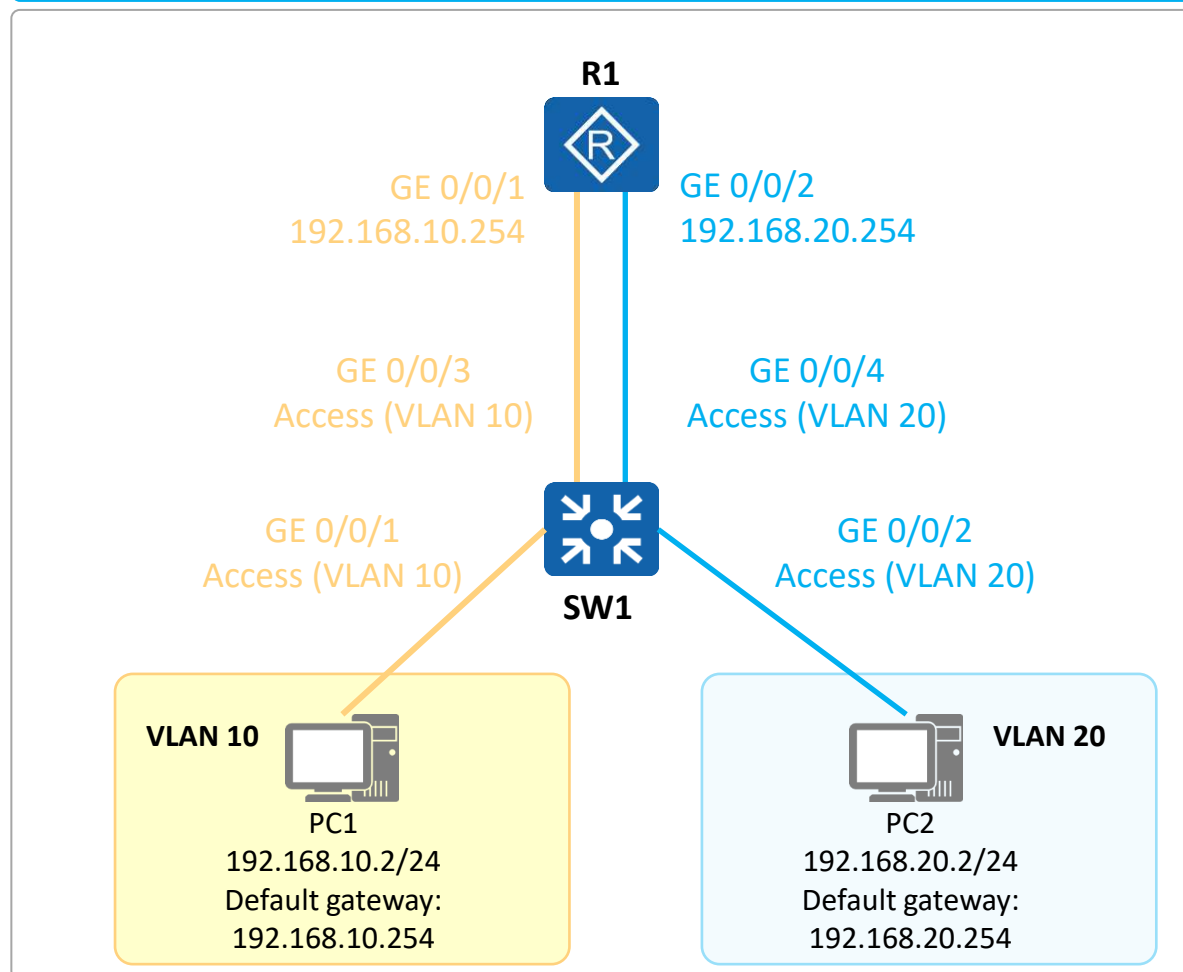
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Using a Router's Physical Interfaces

Physical Connection



- In this method, each VLAN is connected to a separate **physical interface** on the router. The router's Layer 3 interfaces act as **gateways** and forward traffic between different network segments.
- Since router physical interfaces cannot process VLAN-tagged frames, the switch ports connected to the router must be configured as Access ports.
- Each router interface can serve as the gateway for **only one VLAN**. Therefore, the number of router interfaces required increases with the number of VLANs.

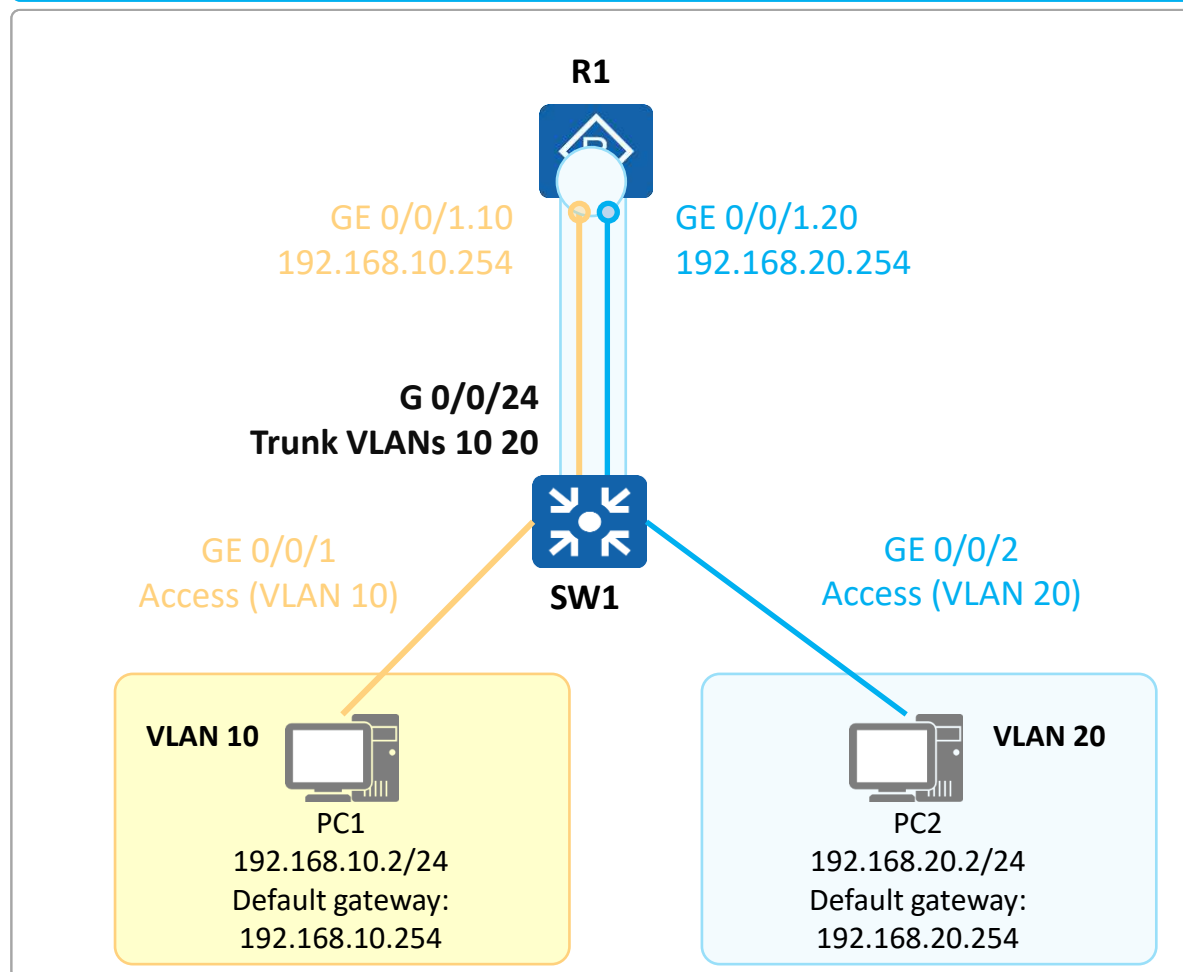
Disadvantages

- Requires one physical router interface per VLAN.
- Routers usually have a limited number of physical interfaces.
- Poor scalability for networks with many VLANs.
- Higher hardware requirements and costs.



Using a Router's Sub-interfaces

Physical Connection

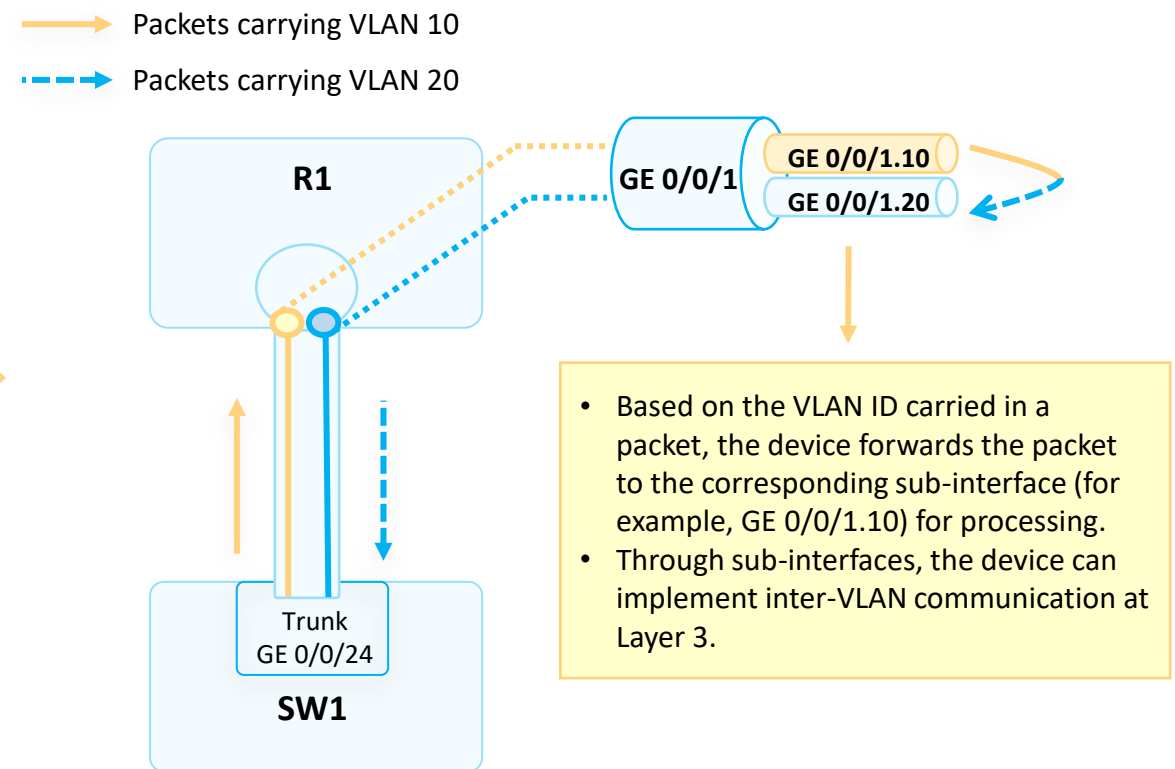
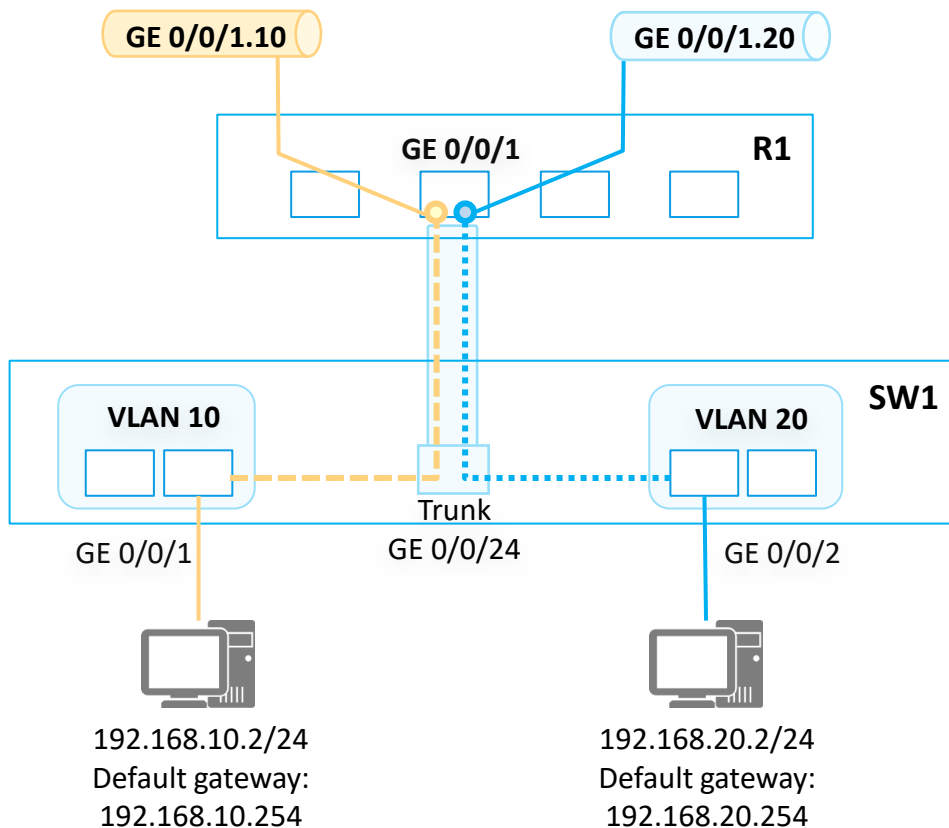


- A sub-interface is a logical interface created on a router's physical Ethernet interface. It is identified by a **physical interface number and a sub-interface number** (for example, G0/0/1.10).
- Like a physical interface, a sub-interface can perform Layer 3 forwarding. Unlike a physical interface, it can also process VLAN-tagged frames.
- Multiple sub-interfaces can be created on a single physical interface, with each sub-interface serving as the gateway for a different VLAN. The router's physical interface is connected to a **Trunk port** on the switch, allowing traffic from multiple VLANs to pass through a single link. The router then uses sub-interfaces to route traffic between VLANs.
- It has better scalability than using separate physical interfaces and is more cost effective.



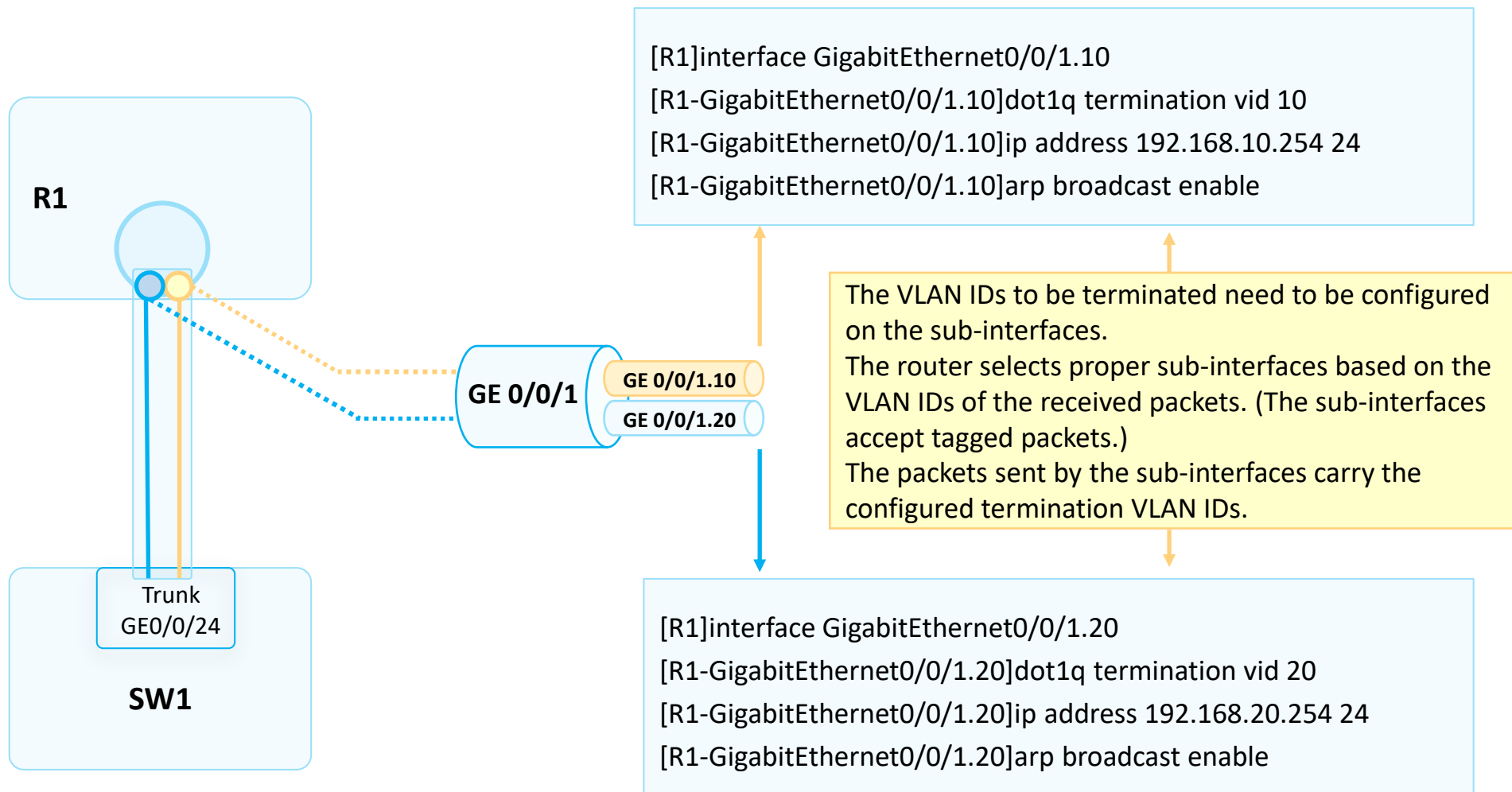
Sub-Interface Processing

- The interface connecting the switch to the router is set to a trunk interface. The router forwards the received packets to the corresponding sub-interfaces according to the VLAN tags in the packets.





Example for Configuring Sub-interfaces



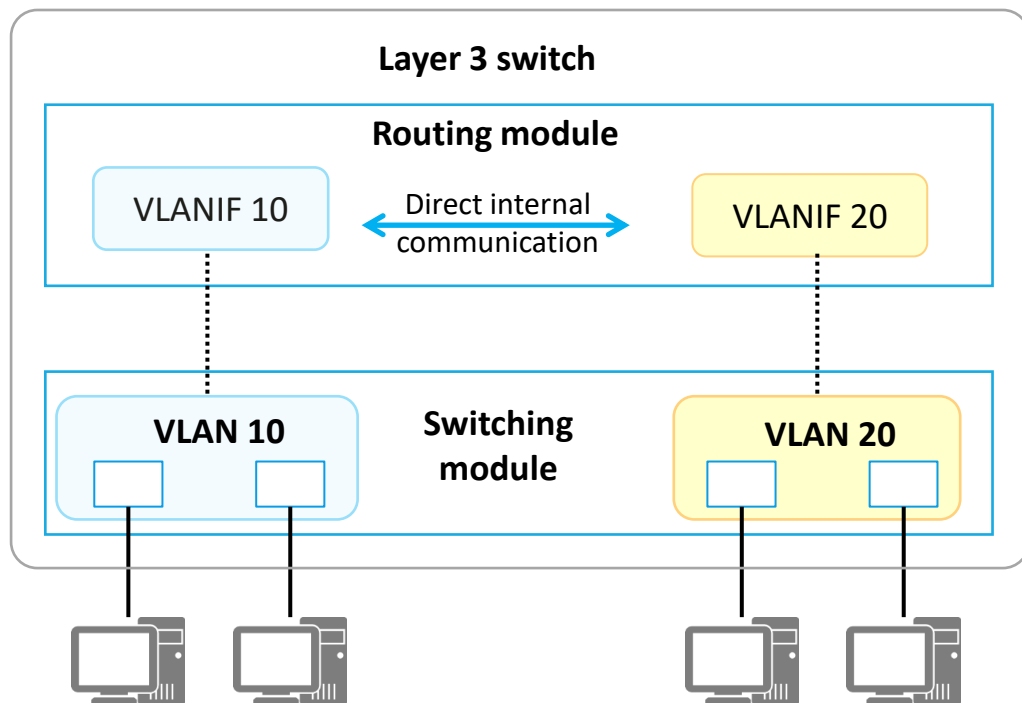


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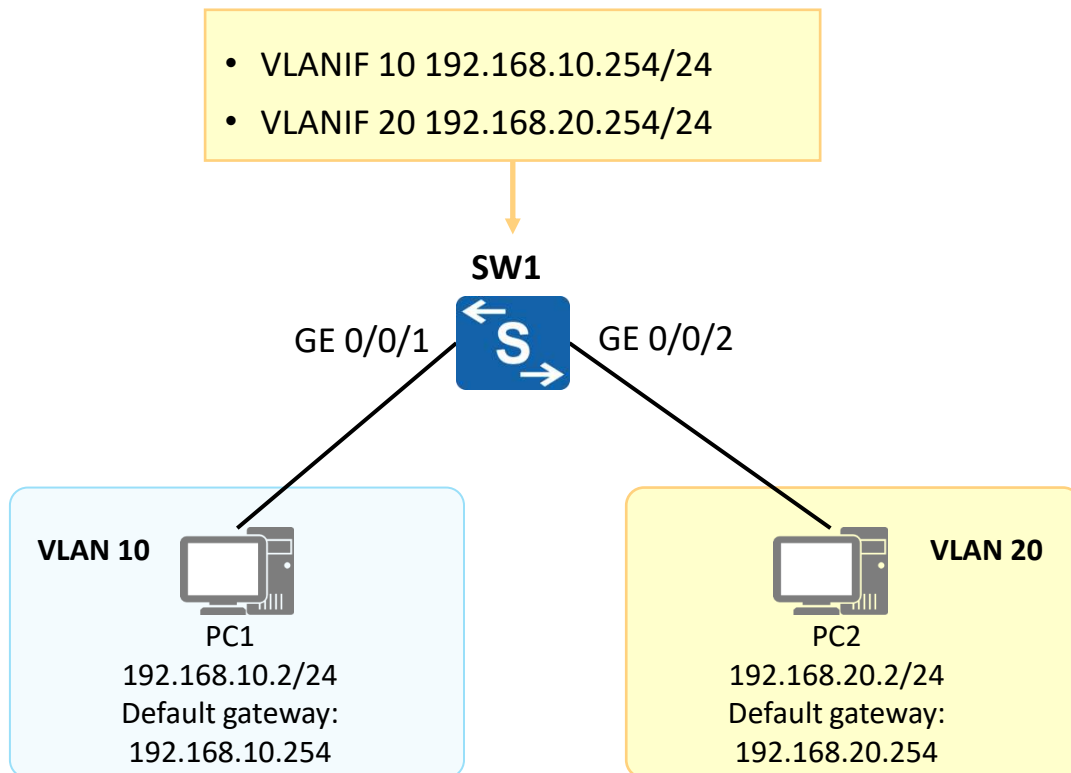
Layer 3 Switch and VLANIF Interfaces



- A **Layer 3 switch** combines the functions of both a **Layer 2 switch** and a **router**. In addition to Layer 2 switching, it can perform routing using VLANIF interfaces.
- A **VLANIF (VLAN Interface)** is a logical Layer 3 interface associated with a VLAN. It acts as the **gateway** for devices within that VLAN and can process VLAN-tagged traffic, enabling communication between different VLANs.
- Each VLANIF interface corresponds to a specific VLAN, and its interface number is the same as the VLAN ID. For example, VLANIF 10 corresponds to VLAN 10.
- When devices in different VLANs need to communicate, traffic is sent to their respective VLANIF interfaces, and the Layer 3 switch routes the traffic between VLANs.



Example for Configuring VLANIF Interfaces



- **Configuration Requirements**
Configure VLANs 10 and 20 for the interfaces connecting to PC1 and PC2, respectively. Configure the Layer 3 switch to allow the two PCs to communicate with each other.

Basic configurations:

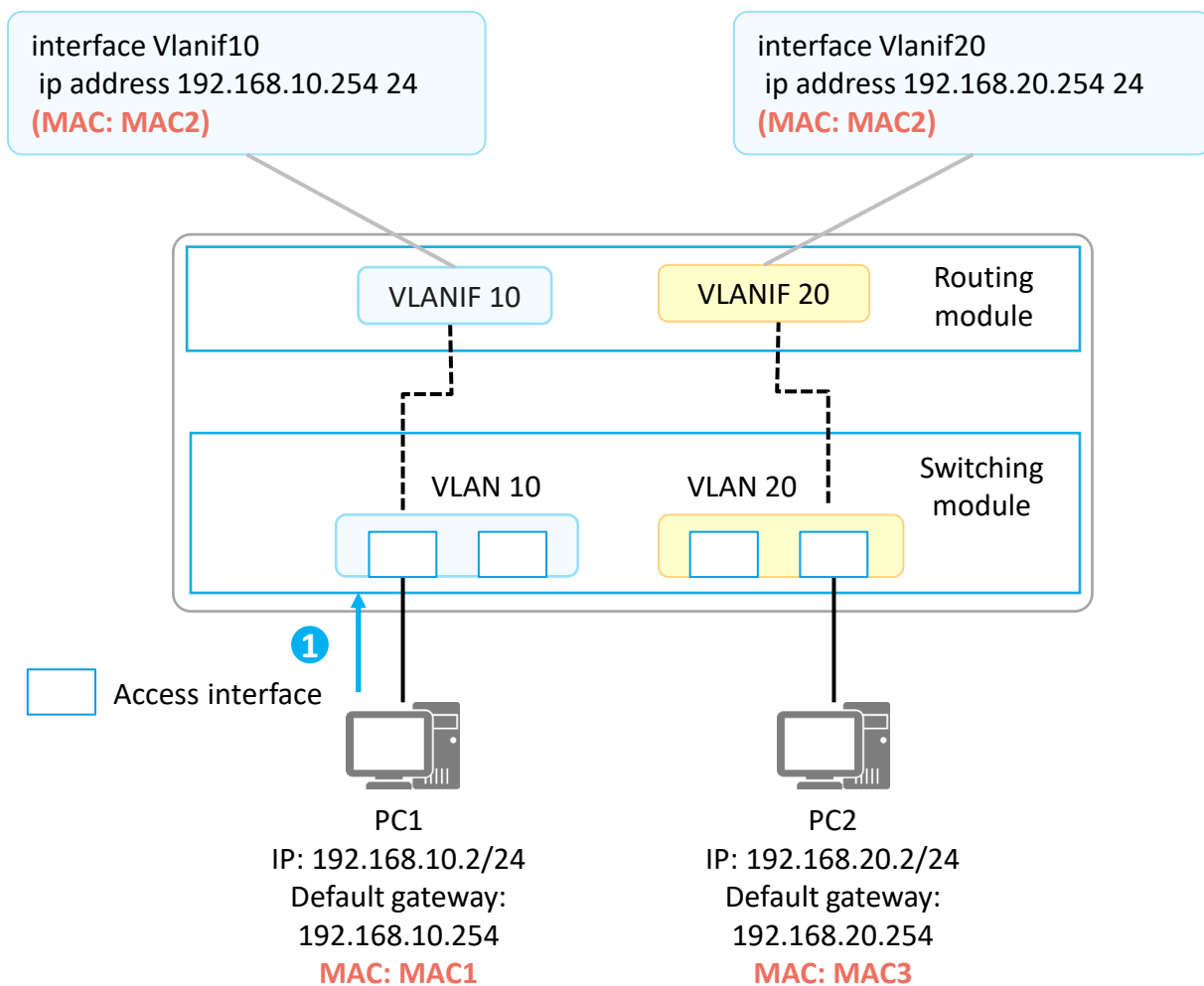
```
[SW1]vlan batch 10 20
[SW1] interface GigabitEthernet 0/0/1
[SW1-GigabitEthernet0/0/1] port link-type access
[SW1-GigabitEthernet0/0/1] port default vlan 10
[SW1] interface GigabitEthernet 0/0/2
[SW1-GigabitEthernet0/0/2] port link-type access
[SW1-GigabitEthernet0/0/2] port default vlan 20
```

Configure VLANIF interfaces:

```
[SW1]interface Vlanif 10
[SW1-Vlanif10]ip address 192.168.10.254 24
[SW1]interface Vlanif 20
[SW1-Vlanif20]ip address 192.168.20.254 24
```



VLANIF Forwarding Process (1)



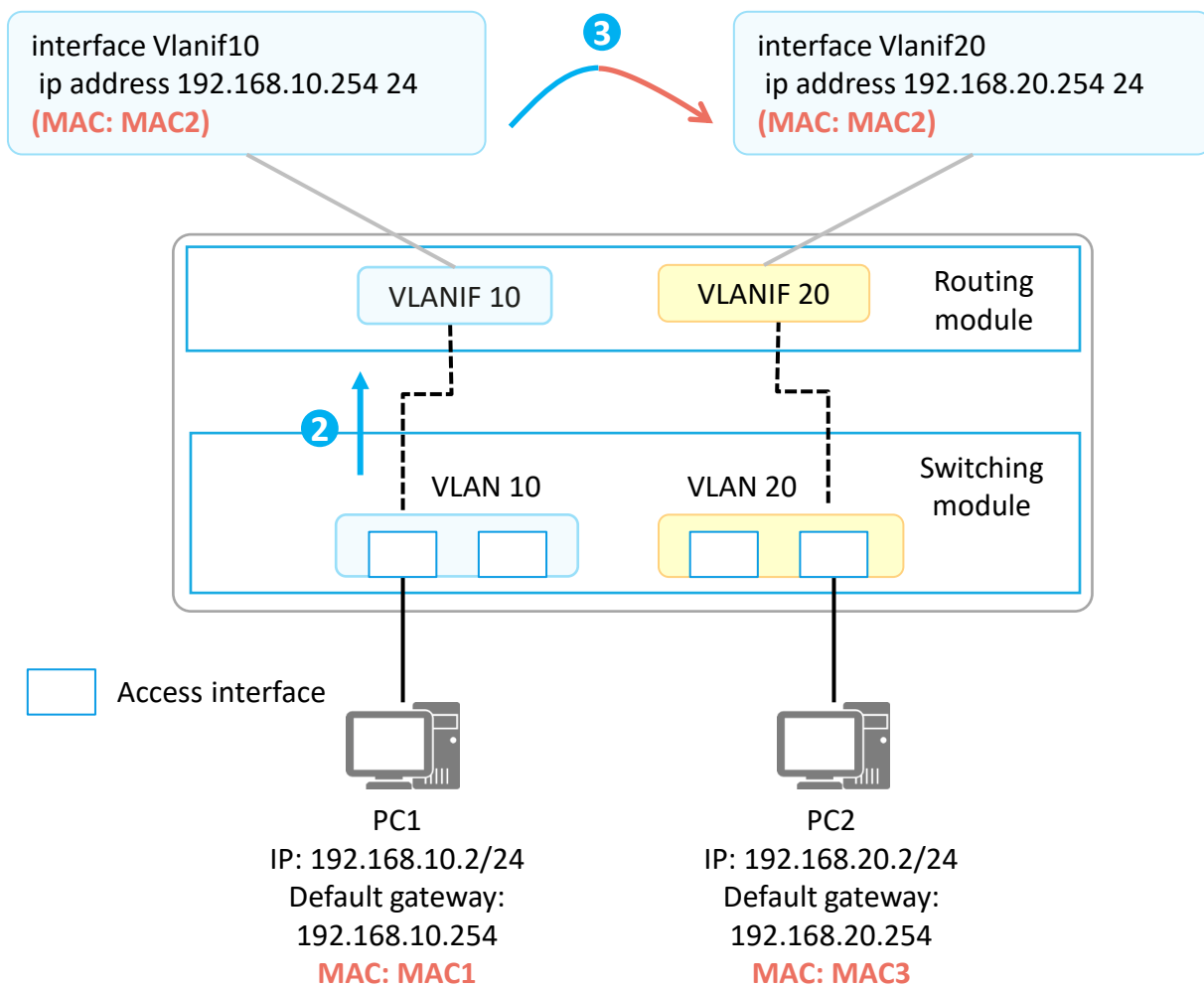
This example assumes that the required ARP or MAC address entries already exist on the PCs and the Layer 3 switch.

The communication process between PC1 and PC2 is as follows:

1. PC1 performs calculation based on its local IP address, local subnet mask, and destination IP address, and finds that the destination device PC2 is not on its network segment. PC1 then determines that Layer 3 communication is required and sends the traffic destined for PC2 to its gateway. Data frame sent by PC1: source MAC = MAC1, destination MAC = MAC2



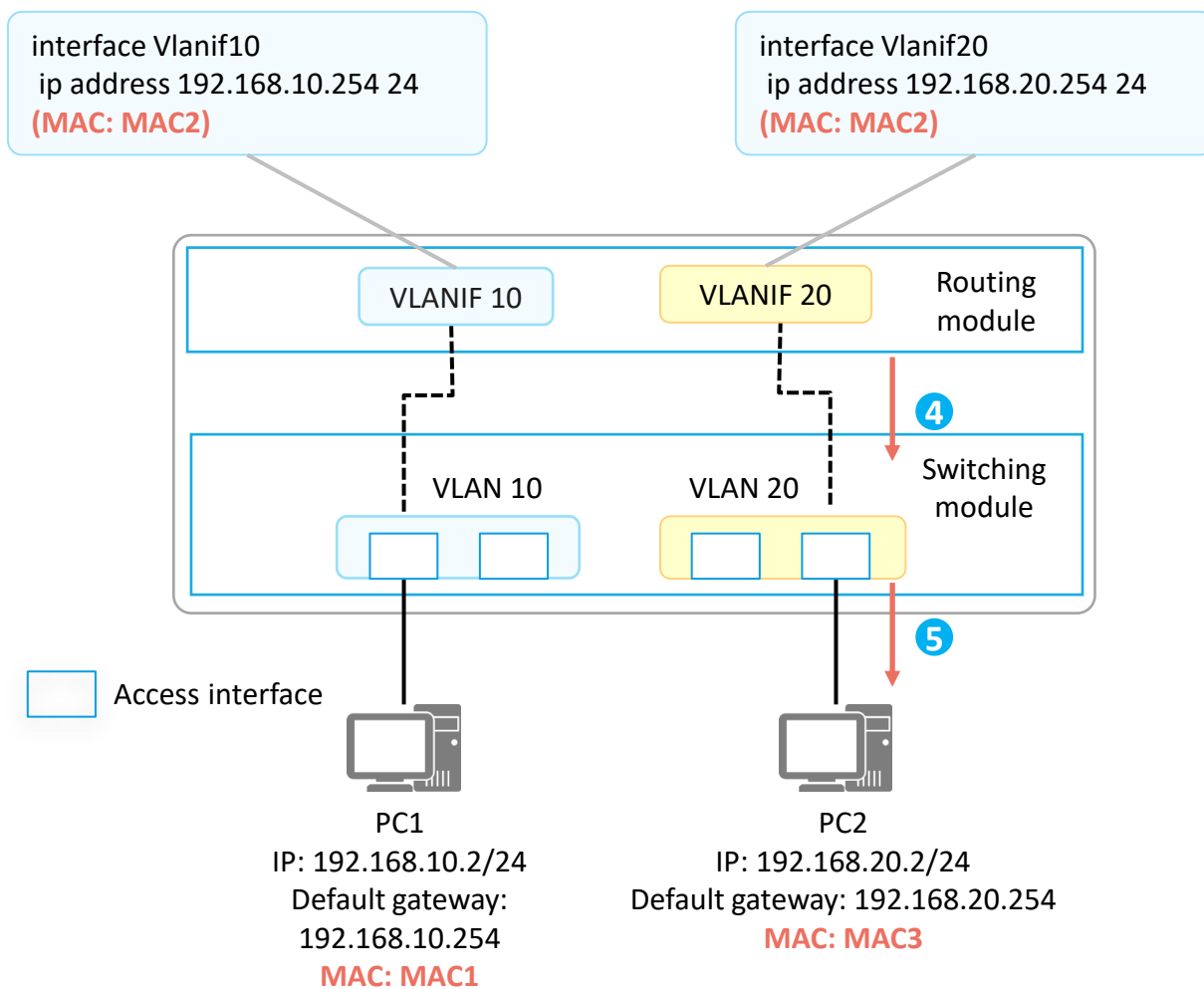
VLANIF Forwarding Process (2)



2. After receiving the packet sent from PC1 to PC2, the switch decapsulates the packet and finds that the destination MAC address is the MAC address of VLANIF 10. The switch then sends the packet to the routing module for further processing.
3. The routing module finds that the destination IP address is 192.168.20.2, which is not the IP address of its local interface, and determines that this packet needs to be forwarded at Layer 3. By searching the routing table, the routing module finds a matching route – the direct route generated by VLANIF 20 – for this packet.



VLANIF Forwarding Process (3)



4. Because the matching route is a direct route, the switch determines that the packet has reached the last hop. It searches its ARP table for 192.168.20.2, obtains the corresponding MAC address, and sends the packet to the switching module for re-encapsulation.
5. The switching module searches its MAC address table to determine the outbound interface of the frame and whether the frame needs to carry a VLAN tag. Data frame sent by the switching module: source MAC = MAC2, destination MAC = MAC3, VLAN tag = None

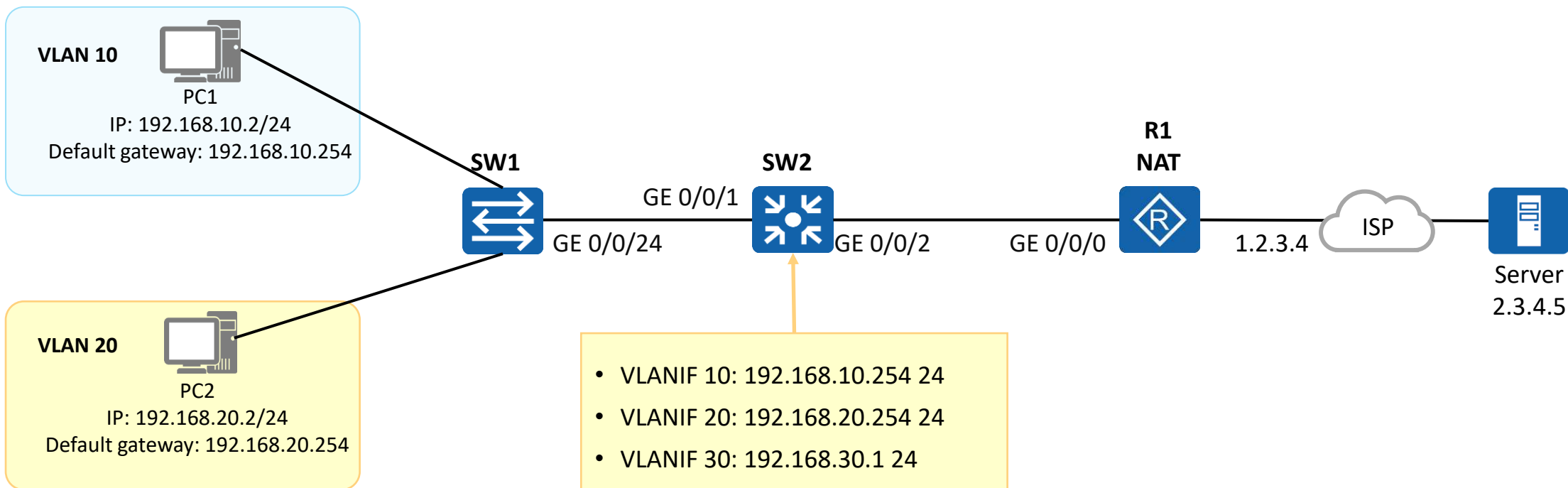


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1. Background
2. Using Routers' Physical Interfaces or Sub-interfaces to Implement Inter-VLAN Communication
3. Using VLANIF Interfaces to Implement Inter-VLAN Communication
- 4. Layer 3 Communication Process**



Network Topology

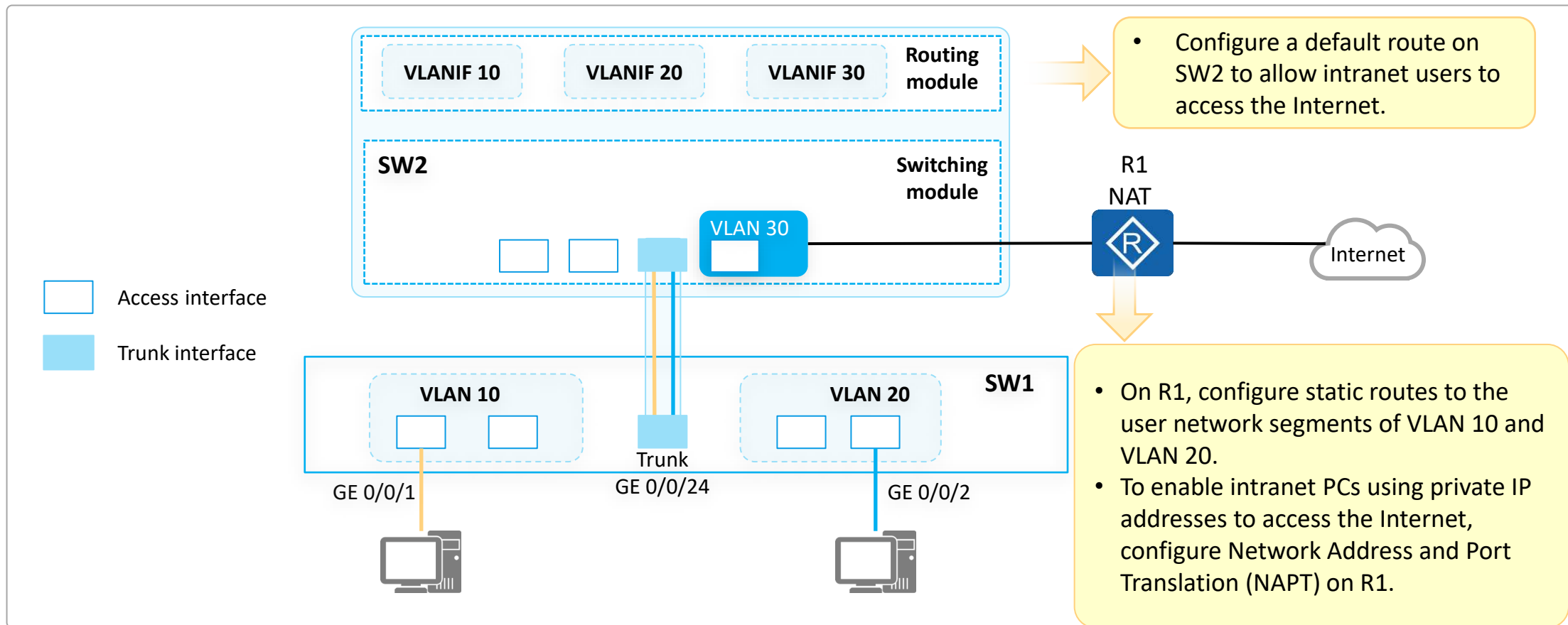


This topology is used as an example to describe the communication process from PC1 in VLAN 10 to the server (2.3.4.5) on the Internet.



Logical Connection

Logical Connection



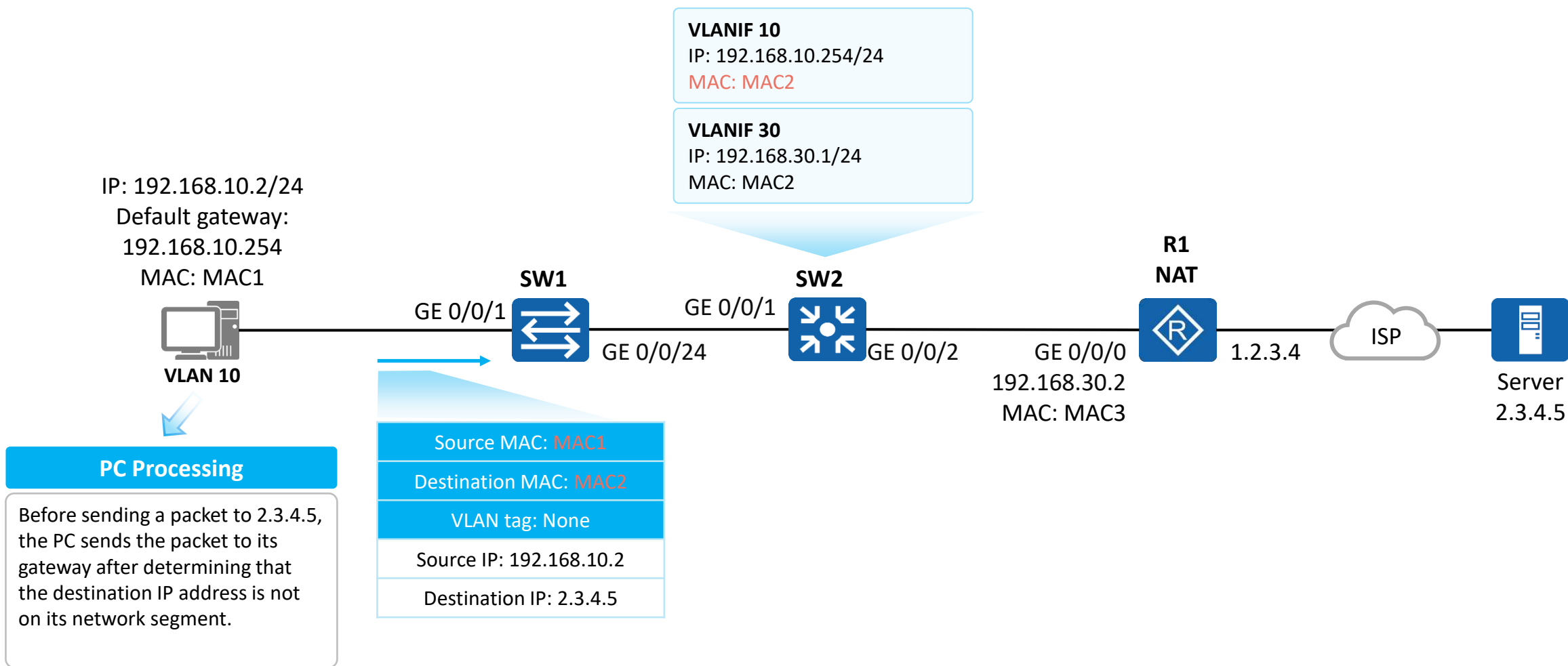


Communication Process (1)

Network
Topology

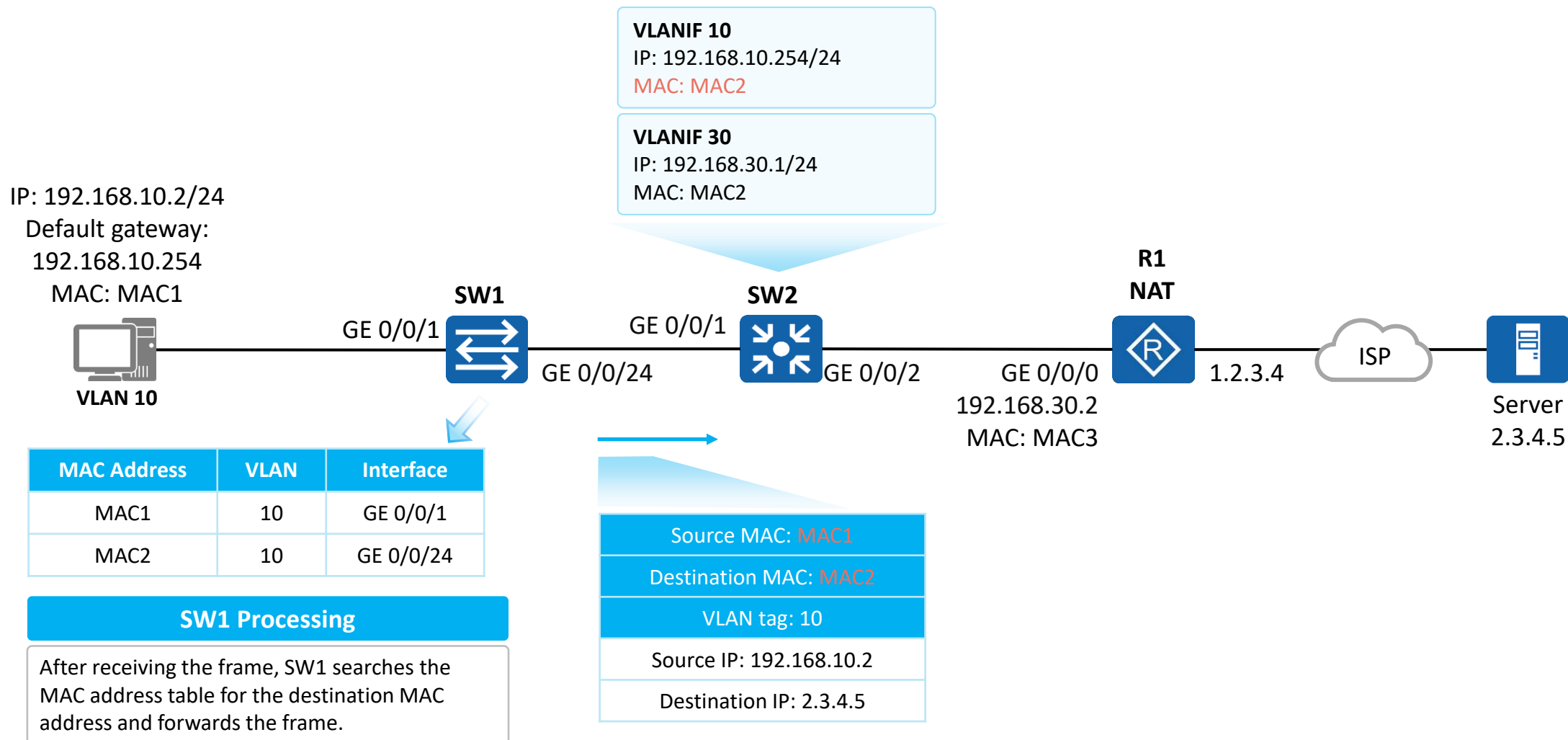
Logical
Connection

Communication
Process



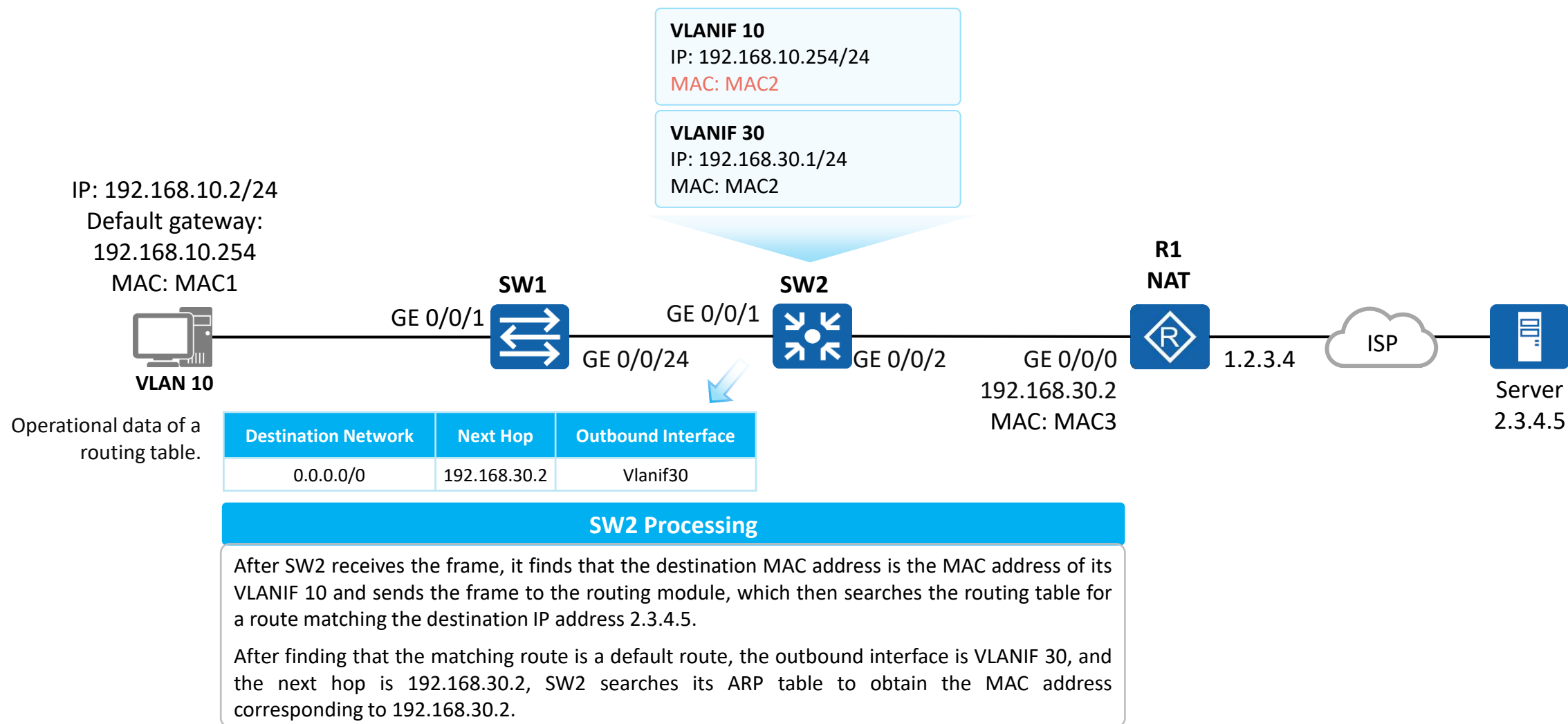


Communication Process (2)



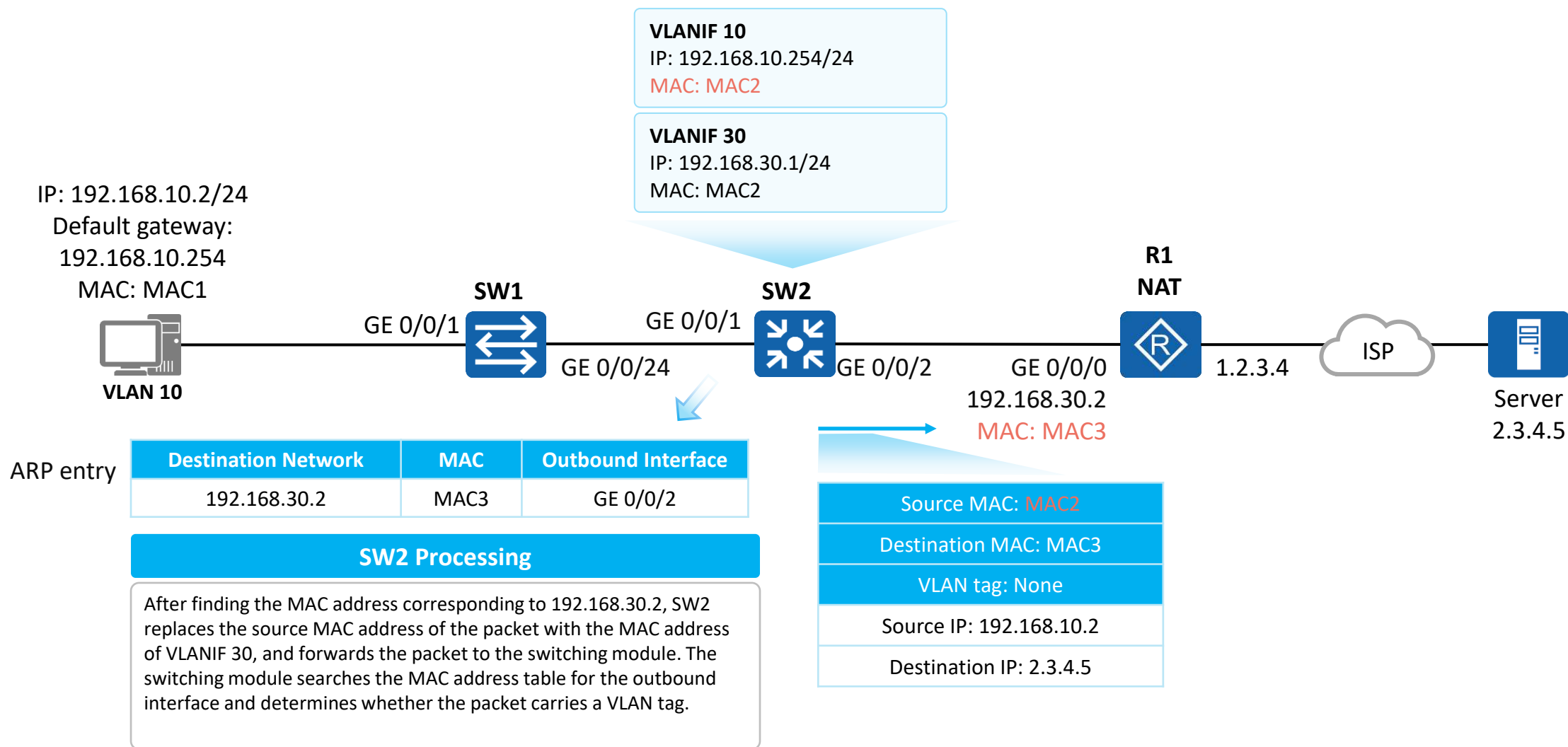


Communication Process (3)



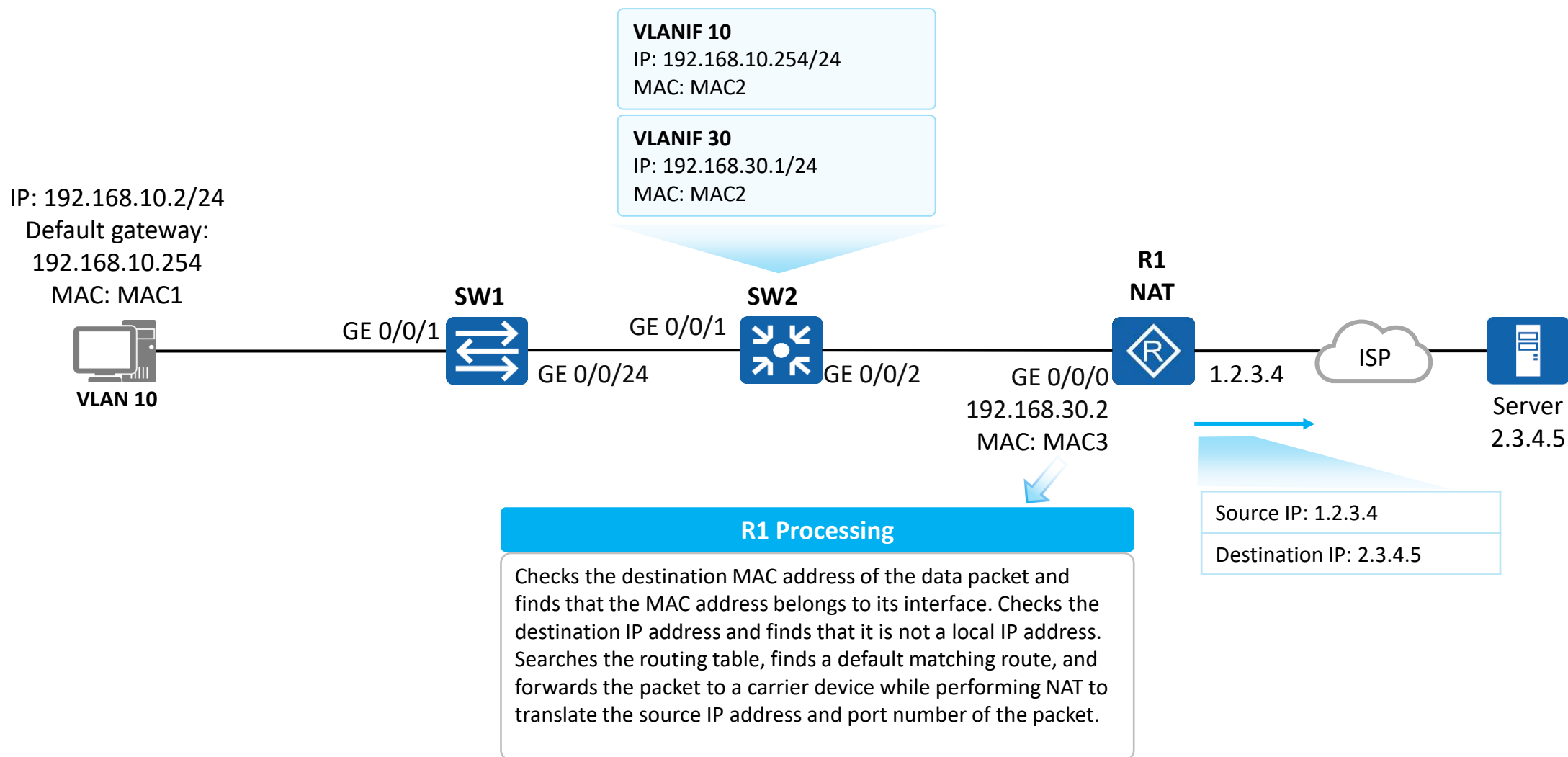


Communication Process (4)





Communication Process (5)





Quiz

1. When a sub-interface is used to implement inter-VLAN communication, how does the switch interface connected to the router need to be configured?

2. How are packets changed when being forwarded at Layer 3?

Step	Action
Incoming packet	Tagged with source VLAN
Router sub-interface	VLAN tag removed, IP routing applied
Outgoing packet	Tagged with destination VLAN
Switch	Delivers frame to destination VLAN hosts



Quiz

3. How packets change at Layer 3

When a packet is forwarded **between VLANs**:

1. At the switch:

1. Packet arrives tagged with its **VLAN ID** (or gets tagged if coming from access port)

2. At the router (Layer 3):

1. Router **removes the VLAN tag**
2. Reads the **IP header** and determines the **destination VLAN/subnet**
3. Router **forwards the packet out the appropriate sub-interface**

3. Back at the switch:

1. Frame is **re-tagged** with the VLAN ID of the destination VLAN
2. Delivered to the correct VLAN hosts



Summary

- This course describes three methods of implementing inter-VLAN communication: through physical interfaces, sub-interfaces, and VLANIF interfaces.
- It also elaborates the Layer 3 communication process, and device processing mechanism and packet header changes during the communication.



More Information

- Comparison between Layer 2 and Layer 3 interfaces

Layer 2 Interface	Layer 3 Interface
An IP address cannot be configured for a Layer 2 interface.	An IP address can be configured for a Layer 3 interface
A Layer 2 interface does not have a MAC address.	A Layer 3 interface has a MAC address.
After a Layer 2 interface receives a data frame, it searches its MAC address table for the destination MAC address of the frame. If a matching MAC address entry is found, it forwards the frame according to the entry. If no matching MAC address entry is found, it floods the frame.	After a Layer 3 interface receives a data frame, if the destination MAC address of the data frame is the same as the local MAC address, it decapsulates the data frame and looks up the destination IP address of the data packet in the routing table. If a matching route is found, it forwards the data frame according to the instruction of the route. If no matching route is found, it discards the packet.
A physical interface on a Layer 2 switch (has only Layer 2 switching capabilities) is a typical Layer 2 interface. By default, the physical interfaces of most Layer 3 switches (have both Layer 2 and Layer 3 switching capabilities) work at Layer 2.	A Layer 3 interface on a router is a typical Layer 3 interface. Physical interfaces on some Layer 3 switches can be switched to Layer 3 mode. In addition to Layer 3 physical interfaces, there are Layer 3 logical interfaces, such as VLANIF interfaces on switches or logical sub-interfaces on other network devices, such as GE 0/0/1.10.
Layer 2 interfaces do not isolate broadcast domains. They flood received broadcast frames.	Layer 3 interfaces isolate broadcast domains. They directly terminate received broadcast frames instead of flooding them.

The background of the slide features a blue-tinted image of several business professionals in a modern office environment. They are standing on a highly reflective floor, and their silhouettes are clearly visible against the bright background. The overall aesthetic is professional and corporate.

Thank You
www.huawei.com



Eth-Trunk, iStack, and CSS



Foreword

- As services develop and the campus network scale expands, users have increasingly demanding requirements on network bandwidth and reliability. Traditional solutions improve network bandwidth by upgrading devices and implement high reliability by deploying redundant links and using the Spanning Tree Protocol (STP), leading to low flexibility, time-consuming troubleshooting, and complex configuration.
- This chapter describes how to use Eth-Trunk, intelligent stack (iStack), and cluster switch system (CSS) technologies to improve network bandwidth and reliability.



Objectives

- On completion of this course, you will be able to:
 - Understand the functions of link aggregation.
 - Understand the link aggregation types.
 - Understand the link aggregation negotiation process in Link Aggregation Control Protocol (LACP) mode.
 - Understand the advantages and principles of iStack and CSS.
 - Understand the common applications and networking of link aggregation and stacking technologies.



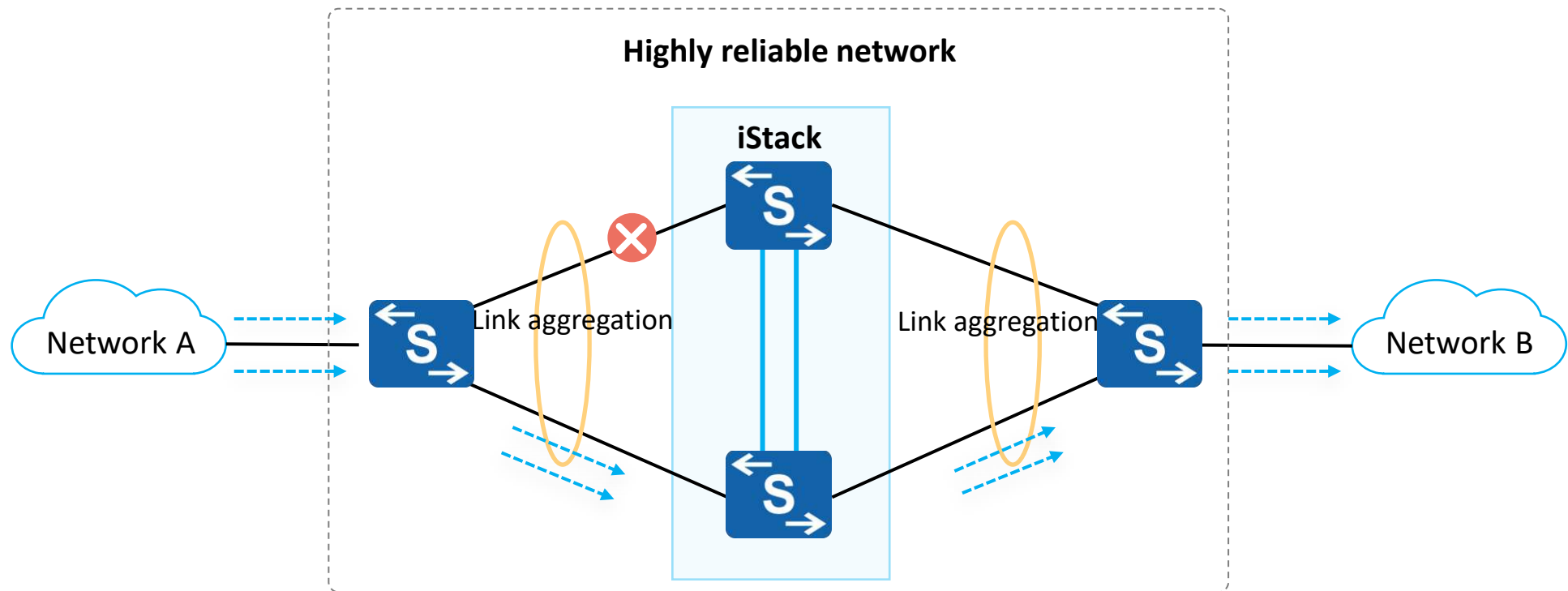
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- 1. Network Reliability Requirements**
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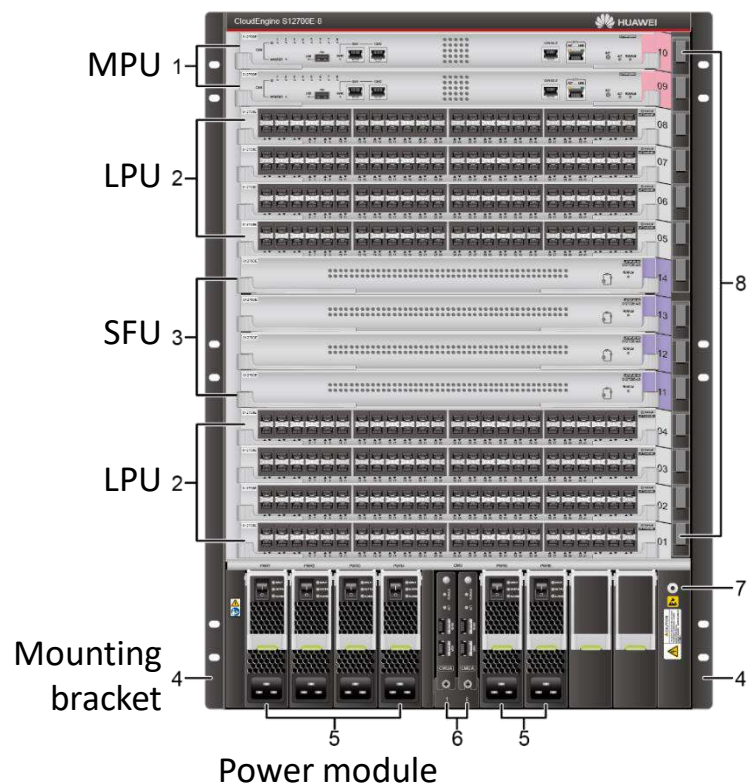
Network Reliability

- Network reliability refers to the capability of ensuring nonstop network services when a single point or multiple points of failure occur on a device or link.
- Network reliability can be implemented at the card, device, and link levels.





Card Reliability (1)

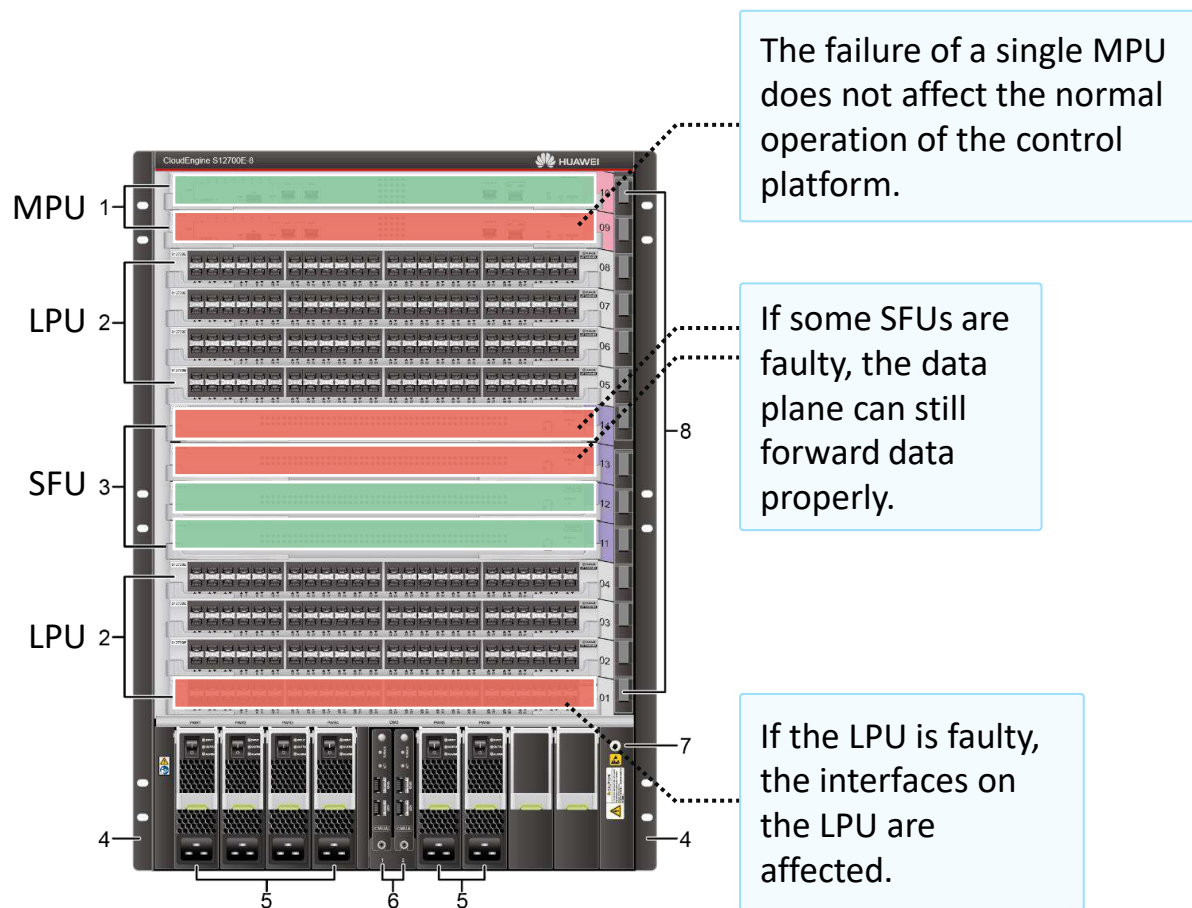


**Front view of the
S12700E-8 chassis**

- A modular switch consists of a chassis, power modules, fan modules, main processing units (MPUs), switch fabric units (SFUs), and line processing units (LPUs).
- Chassis: provides slots for various cards and modules to implement inter-card communication.
- Power module: power supply system of the device
- Fan module: heat dissipation system
- MPU: responsible for the control plane and management plane of the entire system.
- SFU: responsible for the data plane of the entire system. The data plane provides high-speed non-blocking data channels for data switching between service modules.
- LPU: provides data forwarding functions on a physical device and provides optical and electrical interfaces of different rates.



Card Reliability (2)



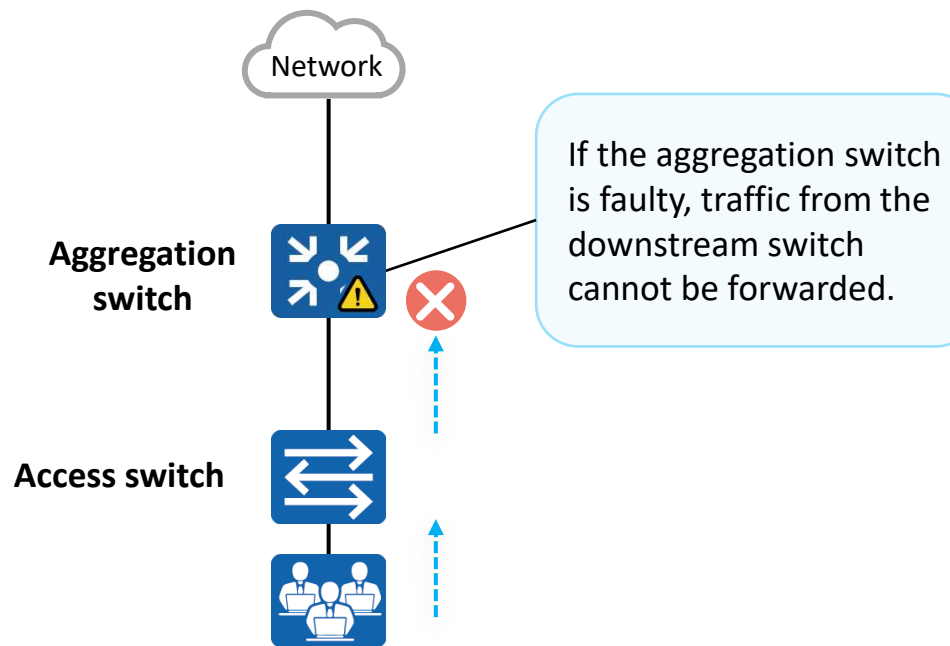
Front view of the
S12700E-8 chassis

- For example, the S12700E-8 provides eight LPU slots, four SFU slots, two MPU slots, six power module slots, and four fan module slots.
- A modular switch can be configured with multiple MPUs and SFUs to ensure device reliability. If an SFU or MPU in a single slot is faulty, the switch can still run properly.
- After an LPU of a modular switch is damaged, interfaces on the LPU cannot forward data.



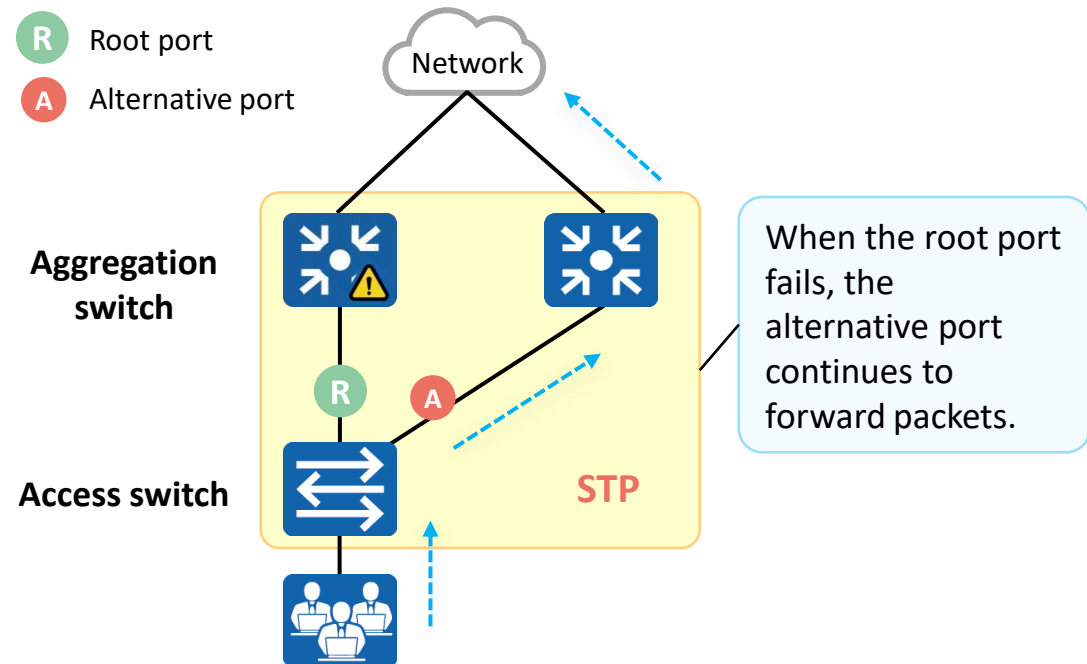
Device Reliability

No backup



On a network without the device redundancy design, a downstream switch uses a single uplink. If the upstream switch or its interfaces fail, all downstream networks are interrupted.

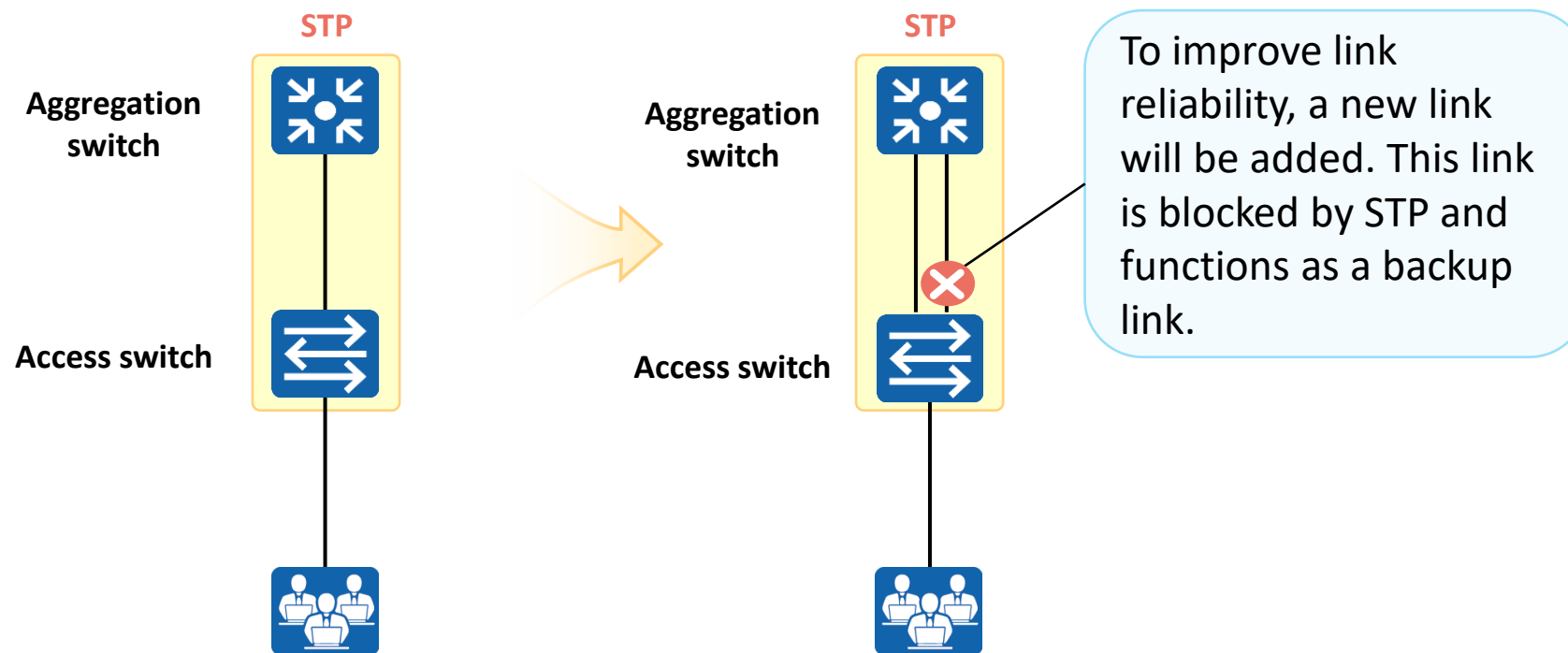
Master/Backup mode



On a network with the device redundancy design, a downstream switch is dual-homed to two upstream switches. The links work in active/backup mode. If the active link or upstream switch fails, traffic is switched to the backup link and forwarded through the backup device.



Link Reliability



- To ensure link reliability, deploy multiple physical links between devices. To prevent loops, configure STP to ensure that traffic is forwarded on only one link, and other links function as backup links.



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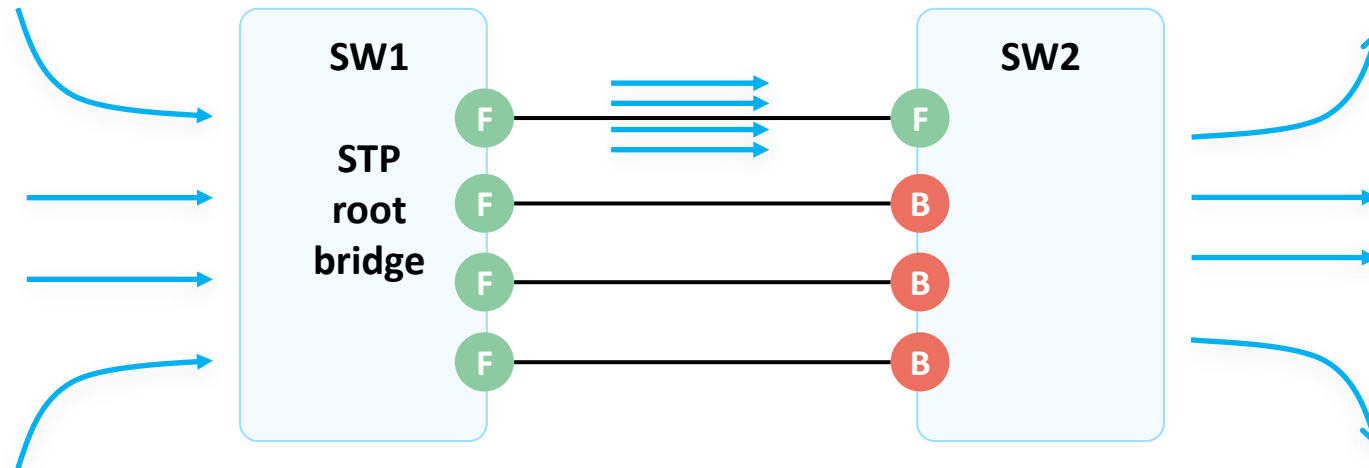
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3. Overview of iStack and CSS



Increasing Link Bandwidth

- When multiple links exist between devices, traffic is forwarded on only one link due to STP. In this case, the inter-device link bandwidth remains unchanged.

- F** Interface that forwards traffic
- B** Interface blocked due to STP

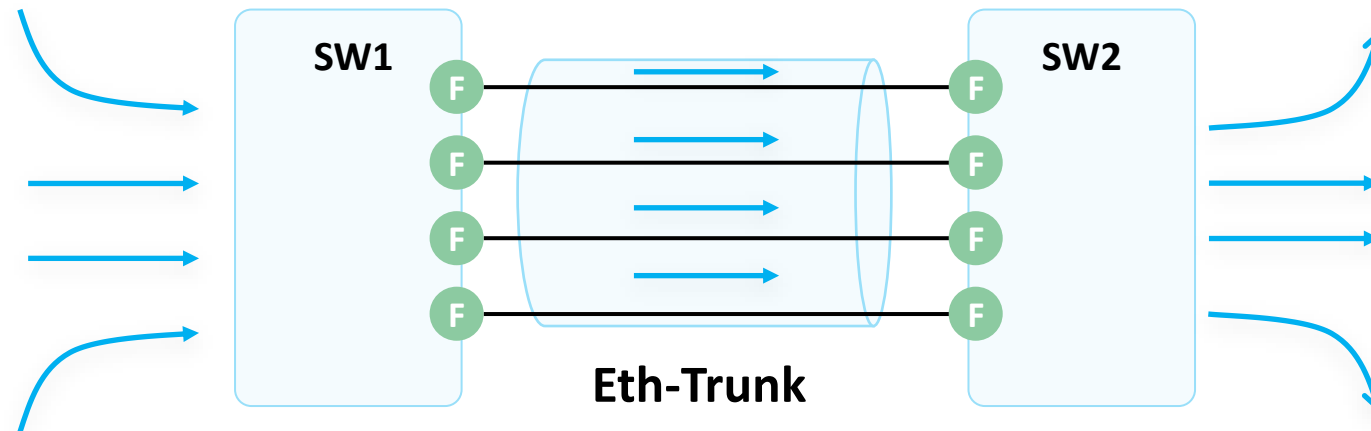




Eth-Trunk

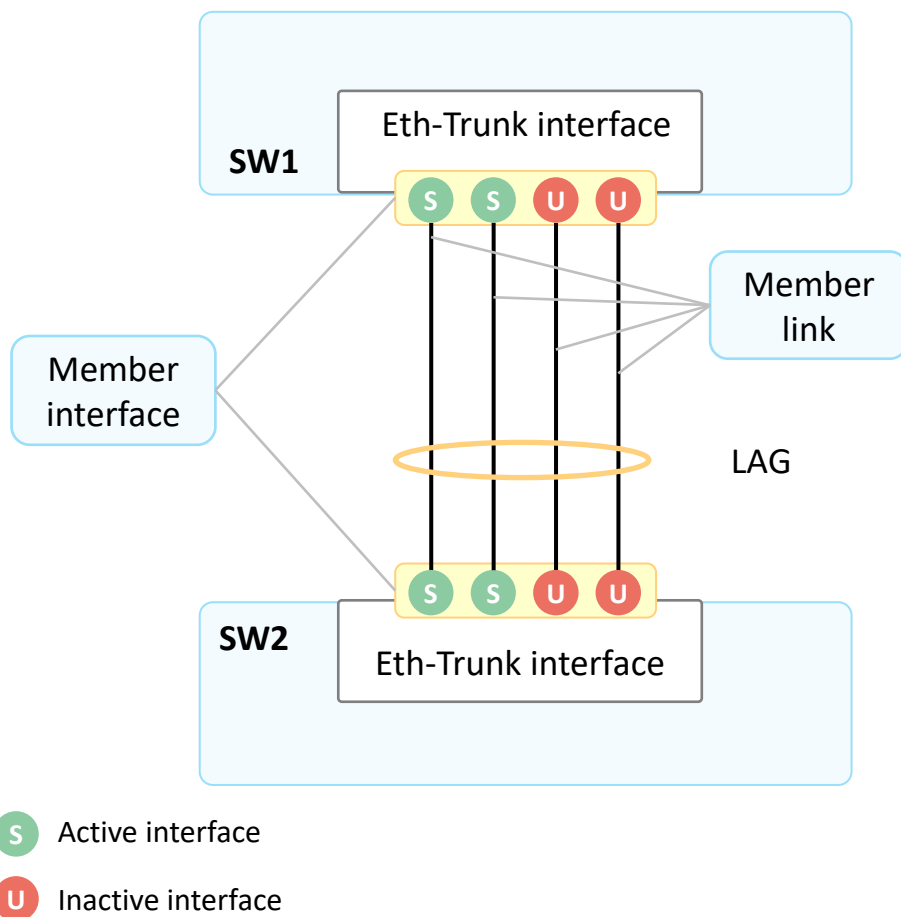
- Ethernet link aggregation, also called Eth-Trunk, bundles multiple physical links into a logical link to increase link bandwidth, without having to upgrade hardware.

F Traffic forwarding interface





Basic Concepts of Eth-Trunk



- A link aggregation group (LAG) is a logical link formed by bundling several links. Each LAG has one logical interface, known as an LAG interface or Eth-Trunk interface.
- Member interface and member link: Physical interfaces that constitute an Eth-Trunk interface are called member interfaces, and the link corresponding to a member interface is known as a member link.
- Active interface and active link: An active interface is also called a selected interface and is a member interface that participates in data forwarding. The link corresponding to an active interface is called an active link.
- Inactive interface and inactive link: An inactive interface is also called an unselected interface and is a member interface that does not participate in data forwarding. A link corresponding to an inactive interface is referred to as an inactive link.
- Link aggregation mode: Based on whether the Link Aggregation Control Protocol (LACP) is enabled, link aggregation can be classified into manual mode and LACP mode.
- Other concepts: upper and lower thresholds for the number of active interfaces

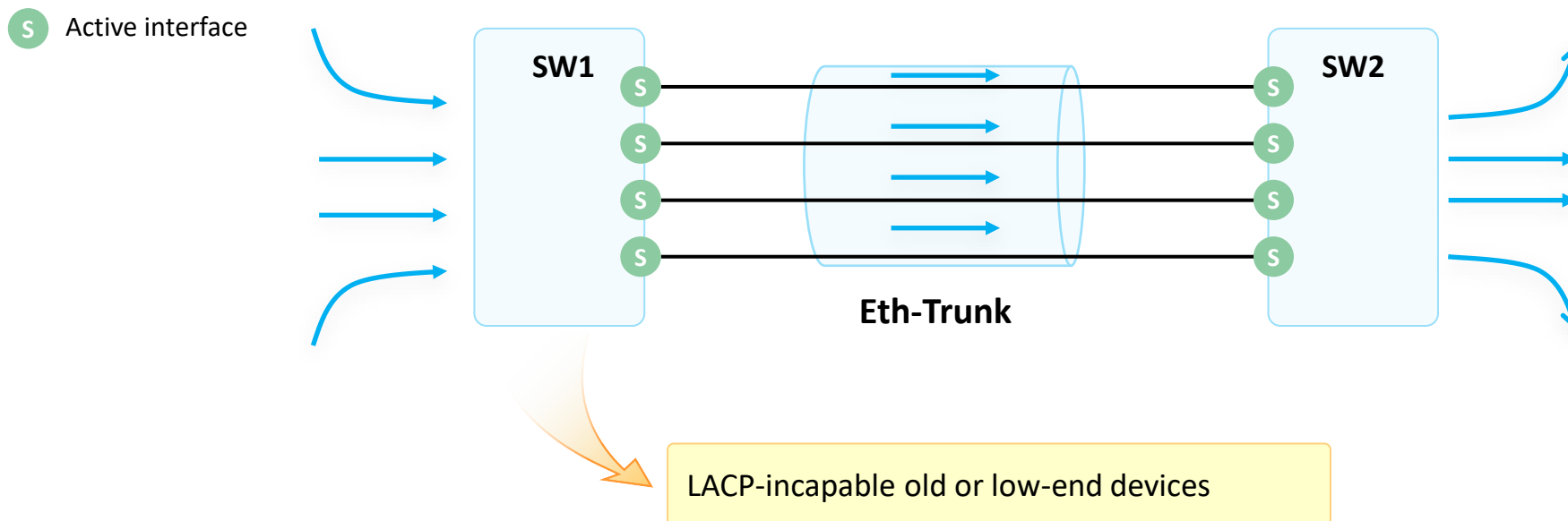


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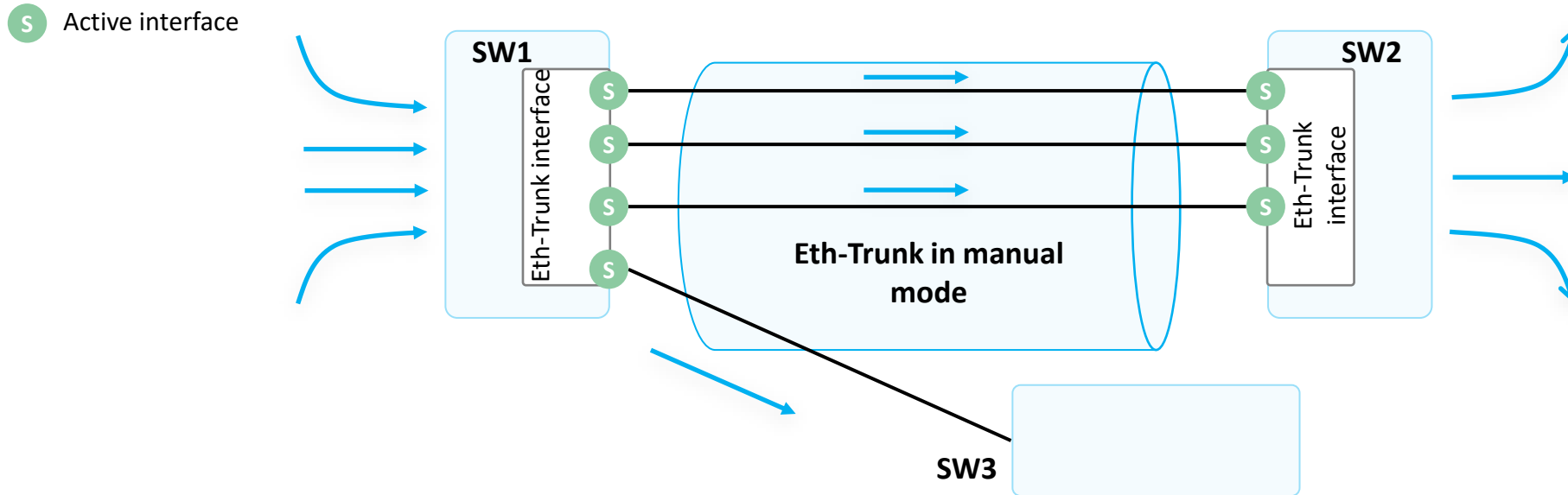
Manual Mode



- Manual mode: An Eth-Trunk is manually created, and its member interfaces are manually configured. LACP is not used for negotiation between the two systems.
- In most cases, all links are active links. In this mode, all active links forward data and evenly share traffic. If an active link is faulty, the LAG automatically evenly shares traffic among the remaining active links.
- If one of the devices at both ends of an LAG does not support LACP, you can use the manual mode.



Defects of the Manual Mode (1)



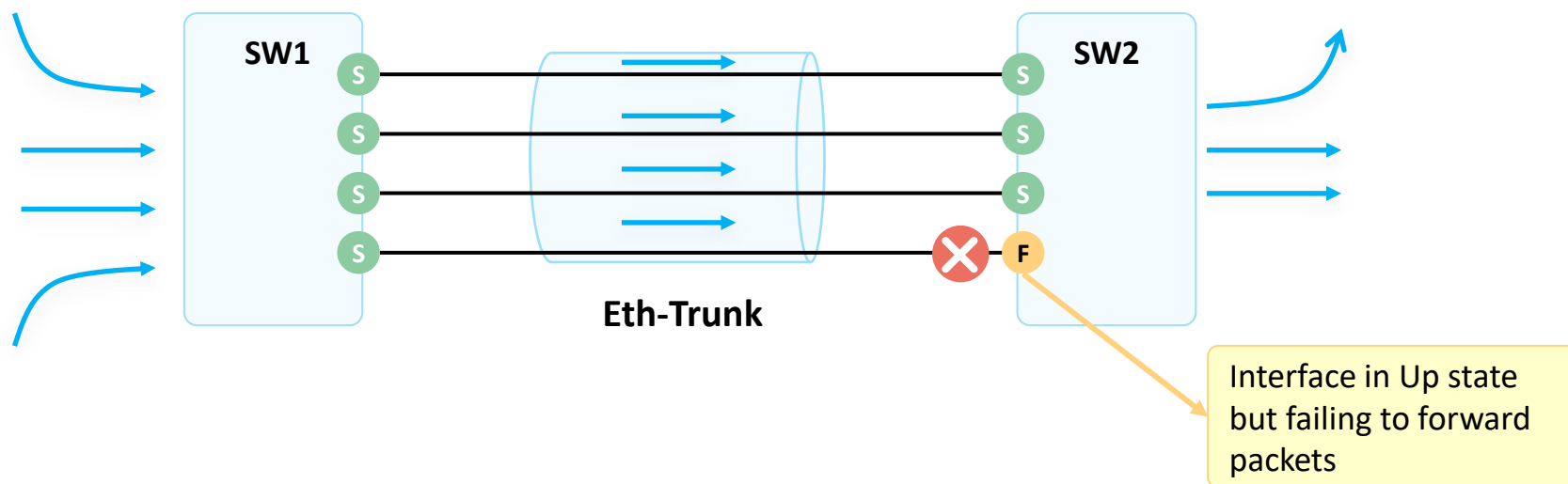
- To ensure that the Eth-Trunk works properly, ensure that the peer interfaces of all member interfaces in the Eth-Trunk meet the following requirements:
 - The peer interfaces reside on the same device.
 - The peer interfaces are added to the same Eth-Trunk.
- In manual mode, devices do not exchange packets. Therefore, the configuration needs to be manually confirmed.



Defects of the Manual Mode (2)

S Active interface

F Faulty interface



- In manual mode, the device can determine whether the peer interface is working properly based only on the physical layer status.

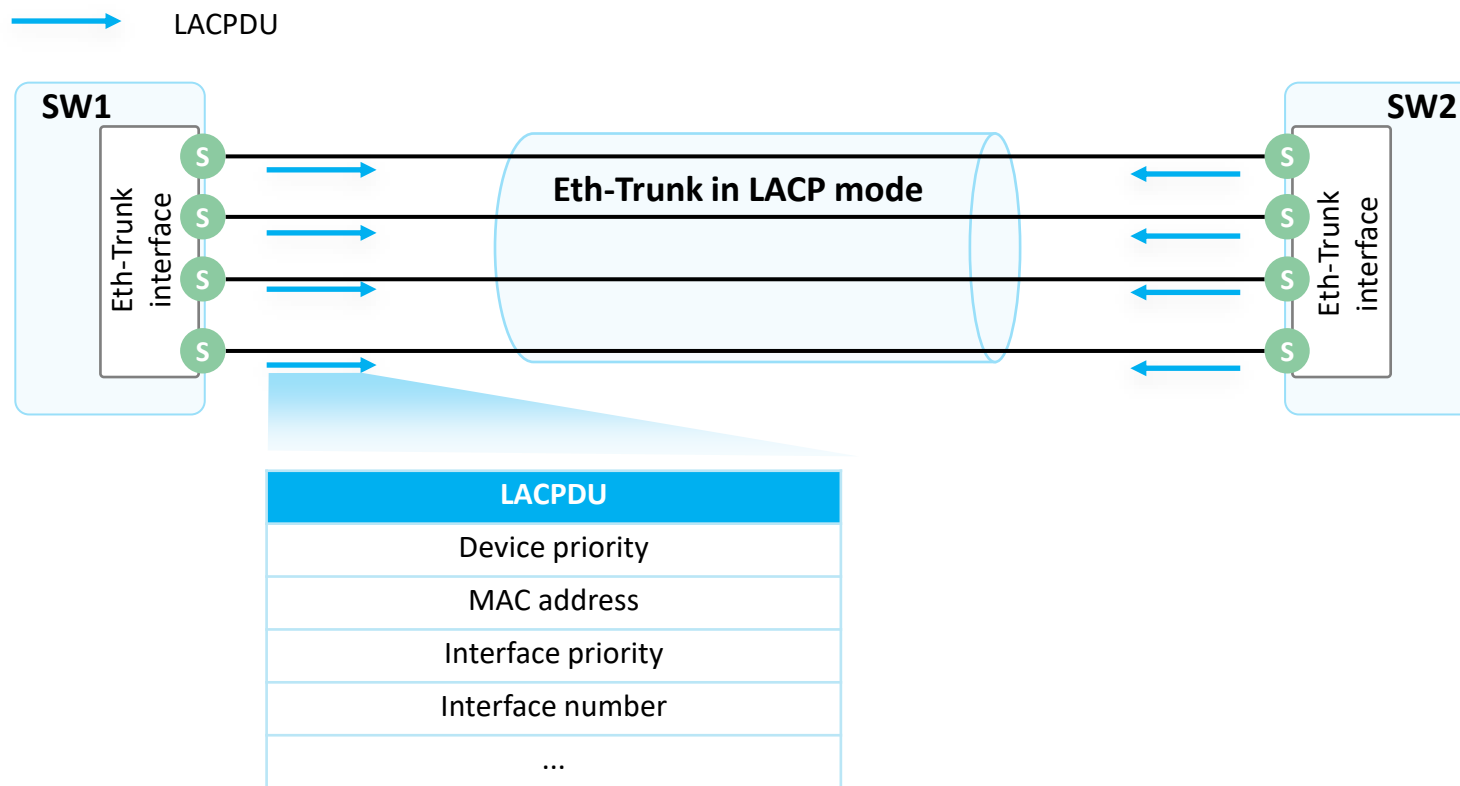


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LACPDU

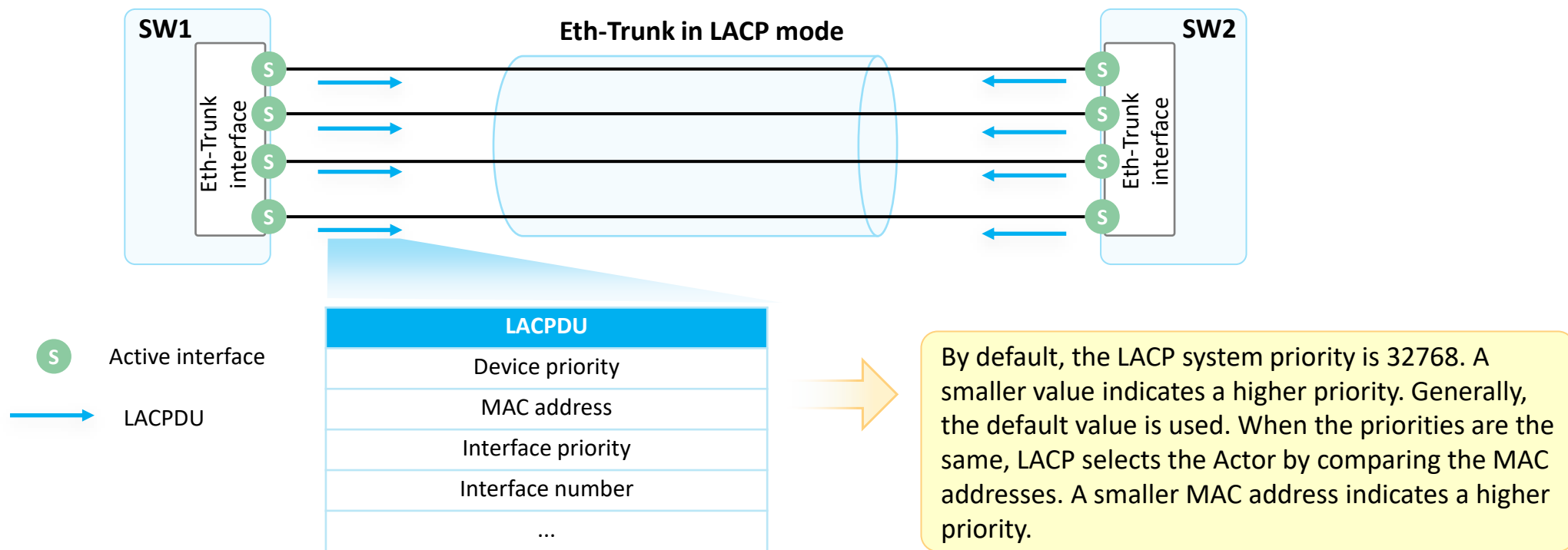


- LACP mode: A link aggregation mode that uses the LACP protocol. Devices exchange Link Aggregation Control Protocol Data Units (LACPDU) to ensure that the peer interfaces are member interfaces that belong to the same Eth-Trunk and are on the same device.
- An LACPDU contains the device priority, MAC address, interface priority, and interface number.



System Priority

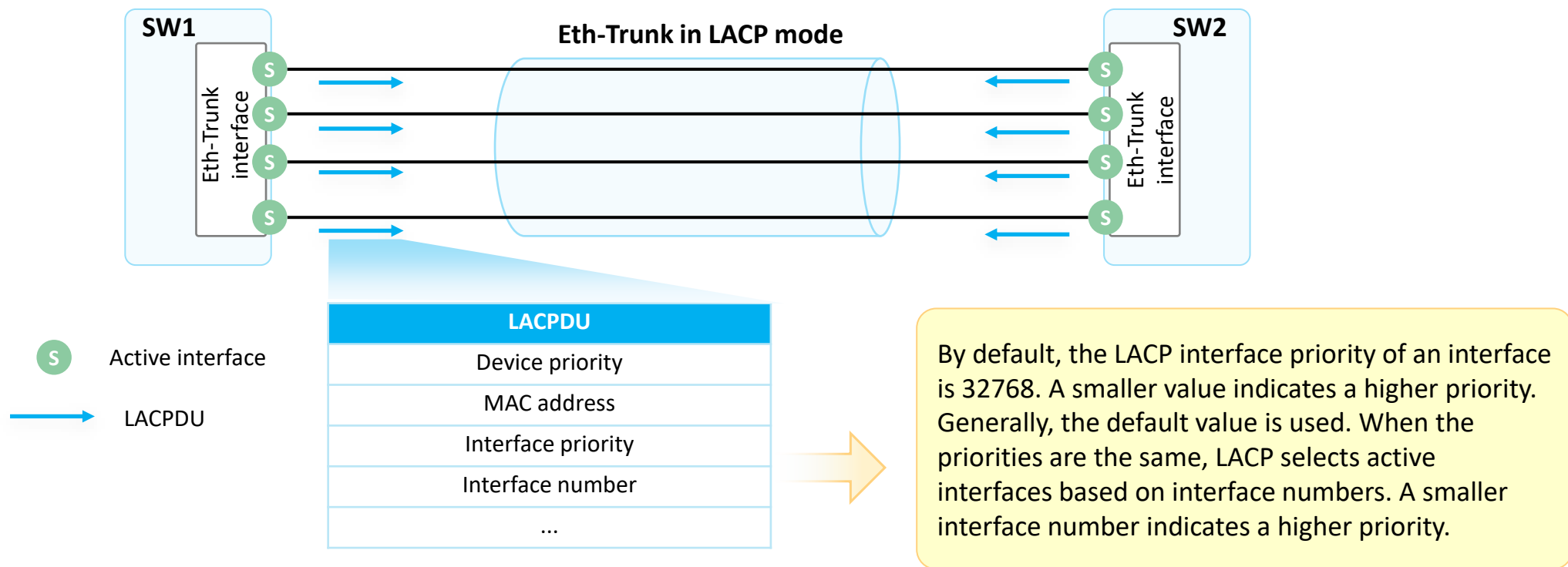
- In LACP mode, the number of active interfaces selected by devices at both ends must be consistent; otherwise, the Eth-Trunk cannot be set up. In this case, configure one end as the Actor. Then the other end selects active interfaces according to the Actor.
- The Actor is determined based on the LACP system priority. A smaller value indicates a higher priority.





Interface Priority

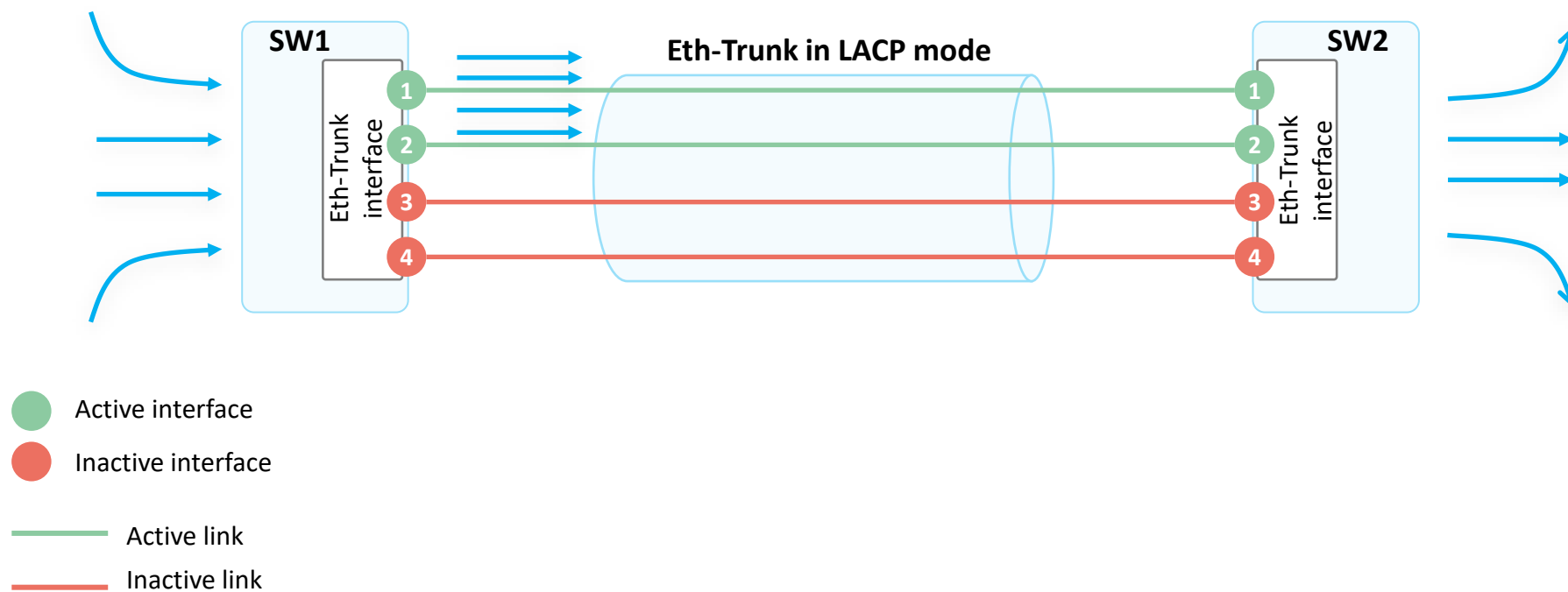
- After the Actor is selected, both devices select active interfaces based on the interface priorities of the Actor. A smaller LACP interface priority value indicates a higher priority.





Maximum Number of Active Interfaces (1)

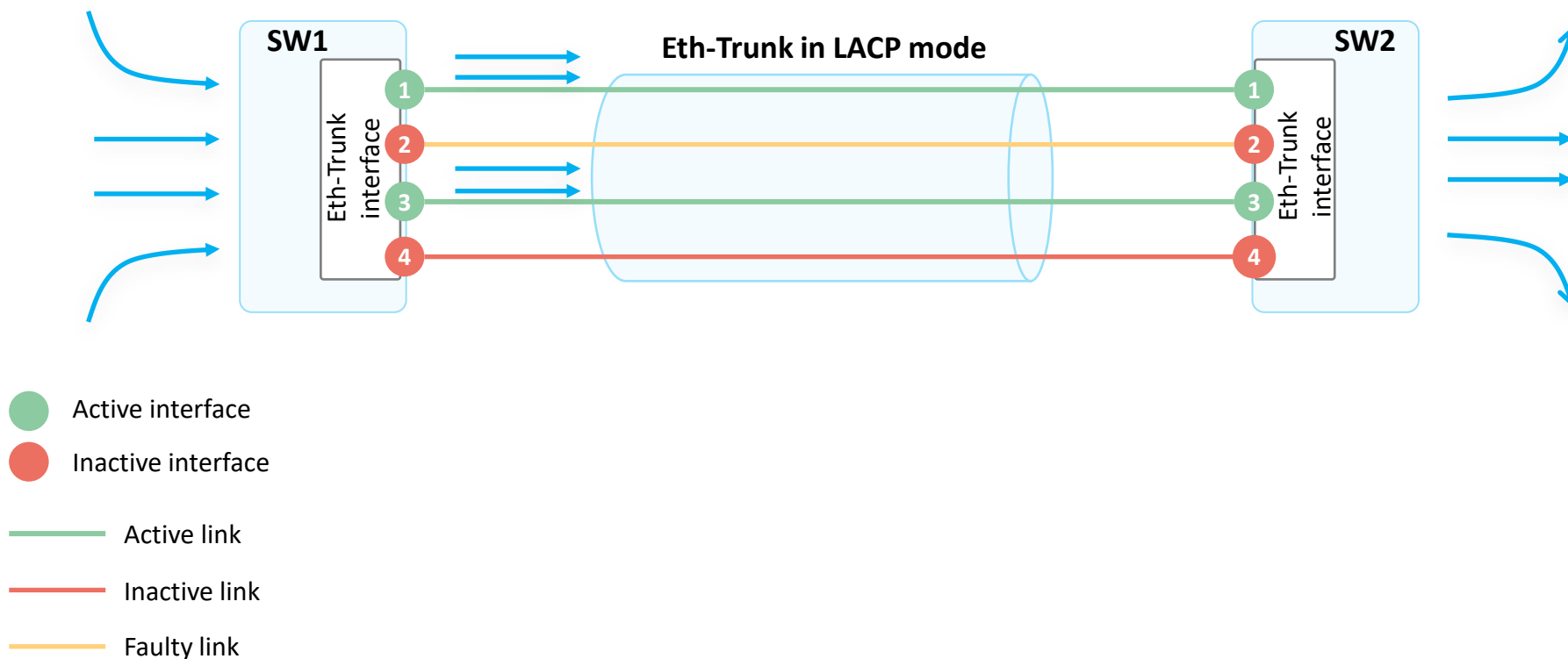
- In LACP mode, the maximum number of active interfaces can be configured. When the number of member interfaces exceeds the maximum number of active interfaces, the interfaces with higher priorities and smaller interface numbers are selected as active interfaces, and the other interfaces function as backup interfaces (inactive interfaces). In addition, the links corresponding to active interfaces become active links, and the links corresponding to inactive interfaces become inactive links. The switch sends and receives packets only through active interfaces.





Maximum Number of Active Interfaces (2)

- If an active link fails, an inactive link with the highest priority (based on the interface priority and interface number) is selected to replace the faulty link. This ensures that the overall bandwidth does not change and services are not interrupted.

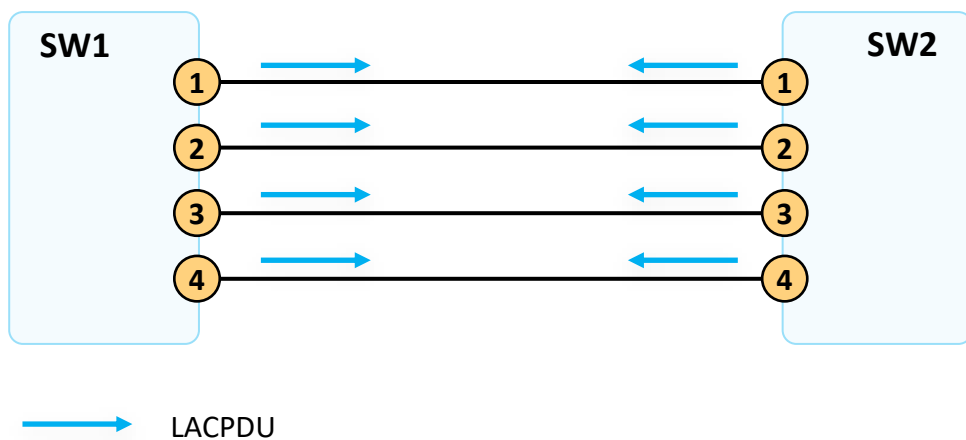




Active Link Election (1)

Bridge MAC:
4c1f-cc58-6d64

Bridge MAC:
4c1f-cc58-6d65



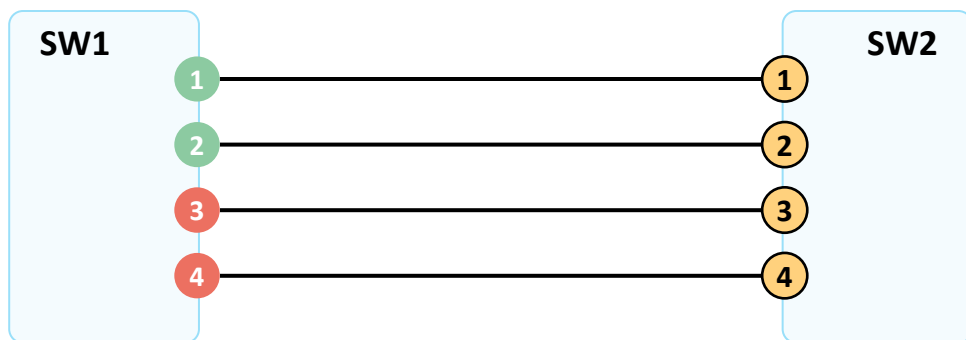
- An Eth-Trunk in LACP mode is set up between SW1 and SW2. The maximum number of active interfaces is set to 2 on SW1 and SW2.
- SW1 with a higher priority is elected as the Actor through LACPDUs.



Active Link Election (2)

Bridge MAC:
4c1f-cc58-6d64

Bridge MAC:
4c1f-cc58-6d65



- Active interface
- Inactive interface

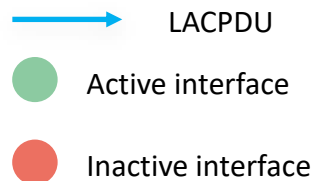
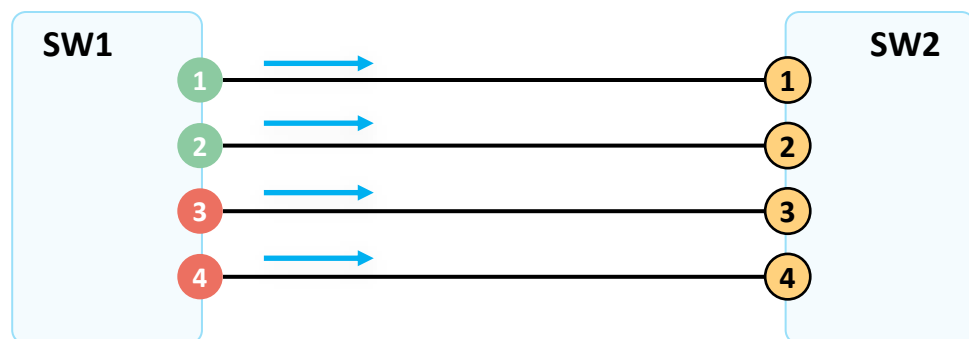
- SW1 compares the interface priorities and interface numbers to select active interfaces. Under the same interface priority, interfaces 1 and 2 have smaller interface numbers and are elected as active interfaces.



Active Link Election (3)

Bridge MAC:
4c1f-cc58-6d64

Bridge MAC:
4c1f-cc58-6d65



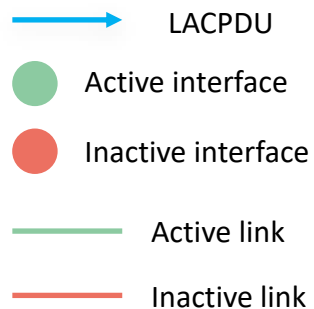
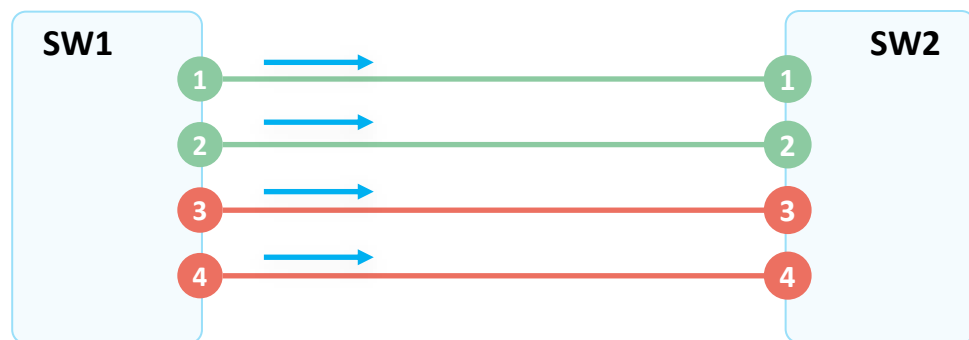
- SW1 notifies the peer end of the elected active interfaces through LACPDU.



Active Link Election (4)

Bridge MAC:
4c1f-cc58-6d64

Bridge MAC:
4c1f-cc58-6d65



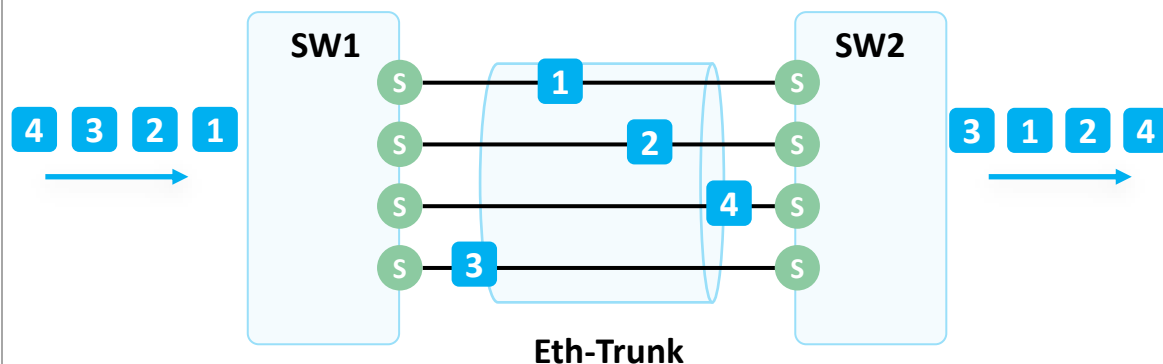
- SW2 determines the local active interfaces based on the election result of SW1 and the corresponding links become active links.
- In this way, the election of active links is complete.



Load Balancing

Per-packet load balancing

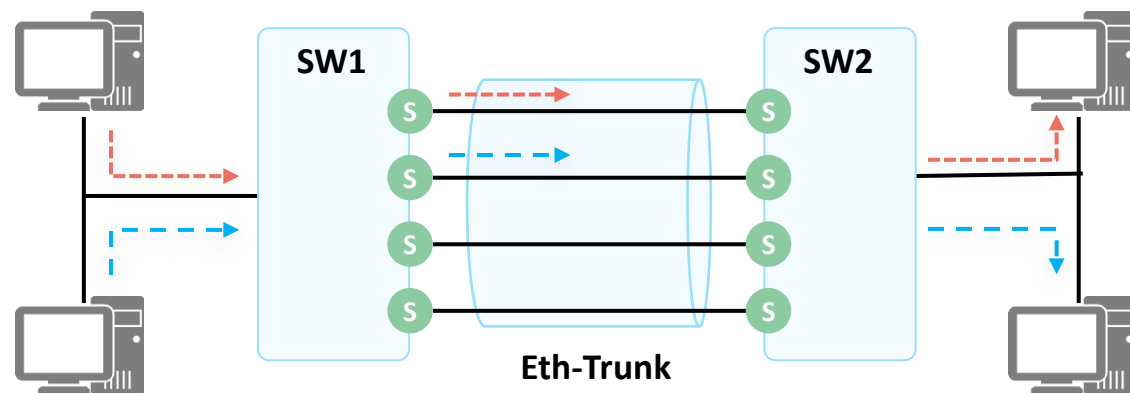
S Active interface



When an Eth-Trunk is used to forward data, there are multiple physical links between devices at both ends of the Eth-Trunk. If data frames are forwarded on different links, data frames may arrive at the peer end in a different order in which they were transmitted, resulting in out-of-order packets.

Per-flow load balancing

S Active interface



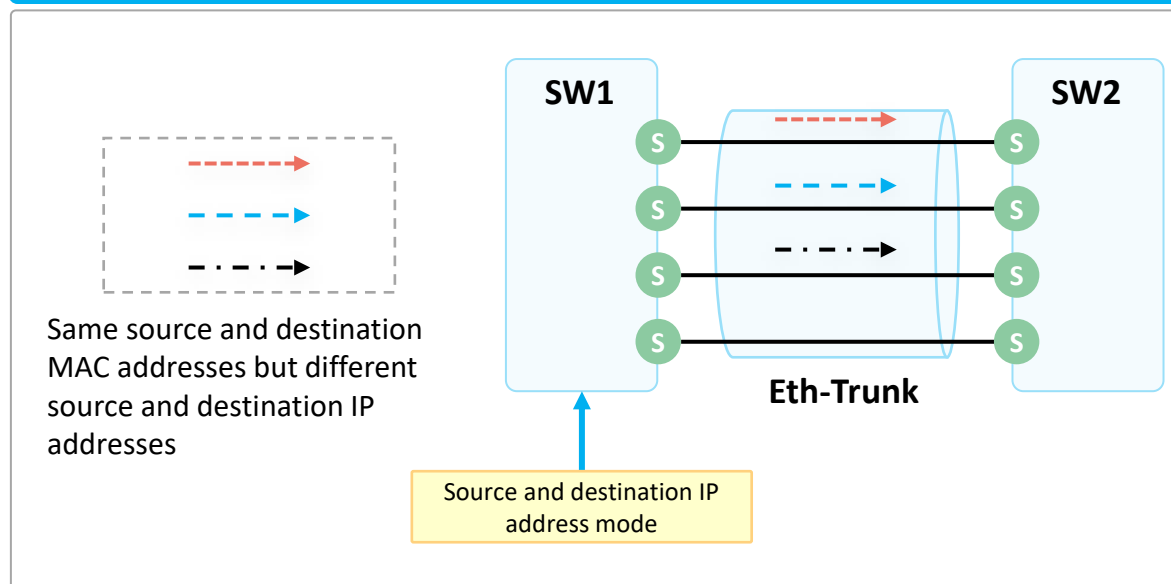
Load balancing based on flows is recommended for an Eth-Trunk. In this mode, a flow is load balanced to the same link. This ensures that frames of the same flow are transmitted over the same physical link and implements load balancing among physical links in an Eth-Trunk.



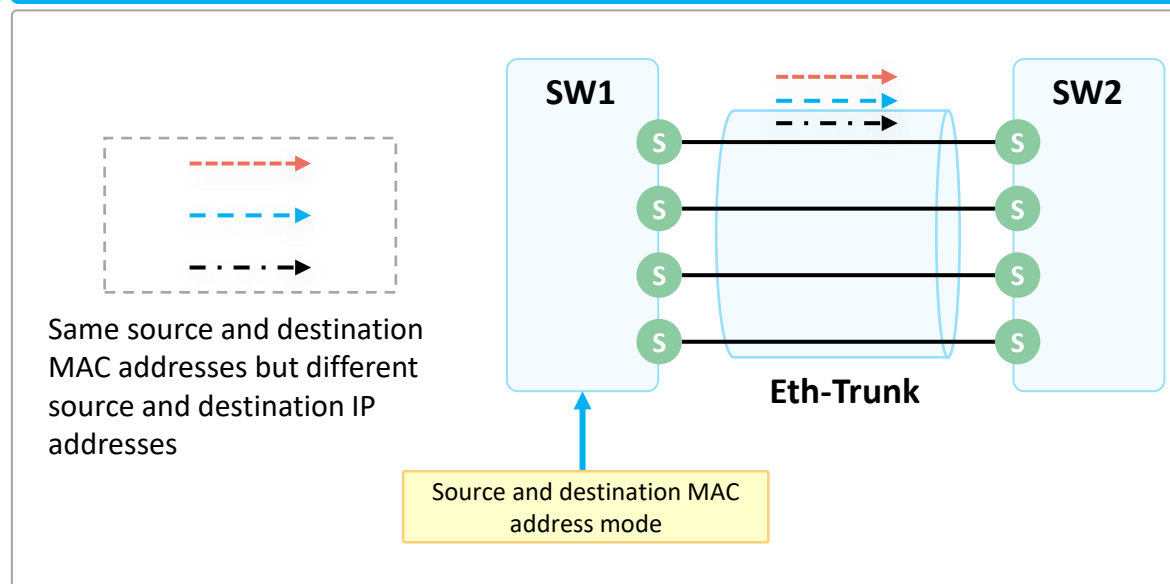
Load Balancing Mode

- An Eth-Trunk can load balance traffic based on IP addresses or MAC addresses of packets. You can configure different load balancing modes (valid locally only for outgoing packets) to distribute data flows to different member interfaces.
- Traffic can be load balanced based on: source IP address, source MAC address, destination IP address, destination MAC address, source and destination IP addresses, and source and destination MAC addresses.
- For actual services, you need to configure a proper load balancing mode based on traffic characteristics. If a service traffic parameter changes frequently, it is easier to load balance traffic if you use the load balancing mode based on this frequently-changing parameter.

Proper load balancing algorithm



Improper load balancing algorithm





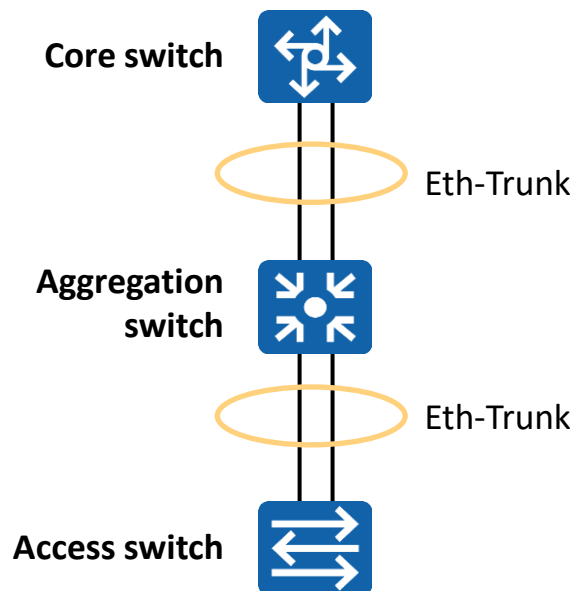
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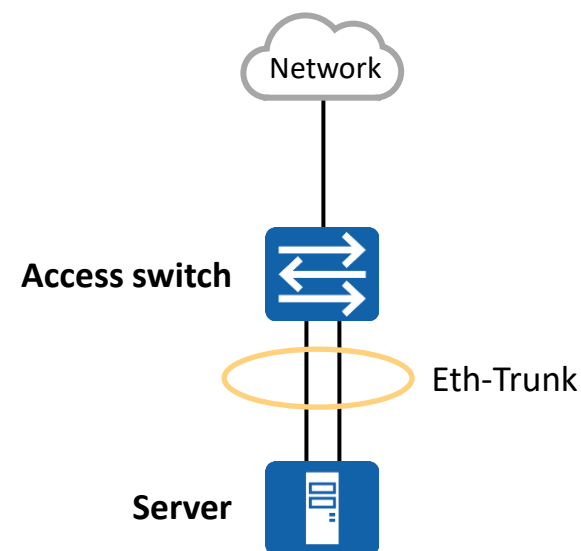
Typical Application Scenario (1)

Between switches



To ensure the bandwidth and reliability of links between switches, deploy multiple physical links between switches and add them to an Eth-Trunk.

Between the switch and server

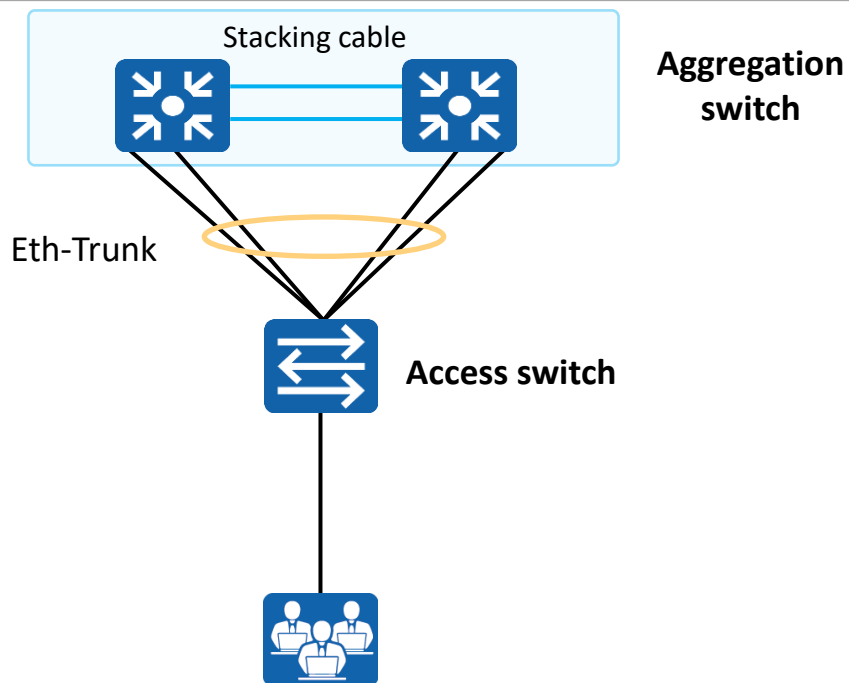


To improve the access bandwidth and reliability of the server, bind two or more physical NICs into a NIC group and establish an Eth-Trunk with the switch.



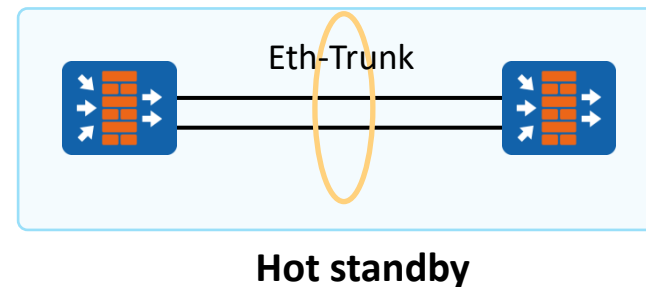
Typical Application Scenario (2)

Between a switch and stack



An iStack is a logical device consisting of two switches. A switch can be connected to the iStack through an Eth-Trunk to form a highly reliable loop-free network.

Heartbeat link of firewalls in hot standby mode



If two firewalls are deployed in hot standby mode, the heartbeat link is used to detect the status of the peer device. To prevent status detection errors caused by single-interface or single-link faults, you can create an Eth-Trunk and use it as the heartbeat link for status detection.



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Configuration Commands (1)

1. Create an Eth-Trunk.

```
[Huawei] interface eth-trunk trunk-id
```

An Eth-Trunk interface is created, and the Eth-Trunk interface view is displayed.

2. Configure a link aggregation mode.

```
[Huawei-Eth-Trunk1] mode {lacp | manual load-balance }
```

To enable the LACP mode, run **mode lacp**. To enable the manual mode, run **mode manual load-balance**.

Note: The link aggregation modes at both ends must be the same.

3. Add an interface to the Eth-Trunk (Ethernet interface view).

```
[Huawei-GigabitEthernet0/0/1] eth-trunk trunk-id
```

In the interface view, the interface is added to the Eth-Trunk.



Configuration Commands (2)

4. Add an interface to the Eth-Trunk (Eth-Trunk view).

```
[Huawei-Eth-Trunk1] trunkport interface-type { interface-number}
```

In the Eth-Trunk view, the interface is added to the Eth-Trunk. You can use either of the preceding commands to add an interface to an Eth-Trunk.

5. Enable interfaces at different rates to join the same Eth-Trunk interface.

```
[Huawei-Eth-Trunk1] mixed-rate link enable
```

By default, interfaces at different rates are not allowed to join the same Eth-Trunk, and only interfaces at the same rate can be added to the same Eth-Trunk.

6. Configure the LACP system priority.

```
[Huawei] lacp priority priority
```

A smaller priority value indicates a higher LACP system priority. By default, the LACP priority is 32768.



Configuration Commands (3)

7. Configure the LACP interface priority.

```
[Huawei-GigabitEthernet0/0/1] lACP priority priority
```

The LACP interface priority is set in the interface view. By default, the LACP interface priority is 32768. A smaller priority value indicates a higher LACP interface priority.

You can run this command only after an interface is added to the Eth-Trunk.

8. Configure the maximum number of active interfaces.

```
[Huawei-Eth-Trunk1] max active-linknumber {number}
```

Ensure that the maximum number of active interfaces on the local end is the same as that on the peer end. The maximum number of active interfaces can be configured only in LACP mode.

9. Configure the minimum number of active interfaces.

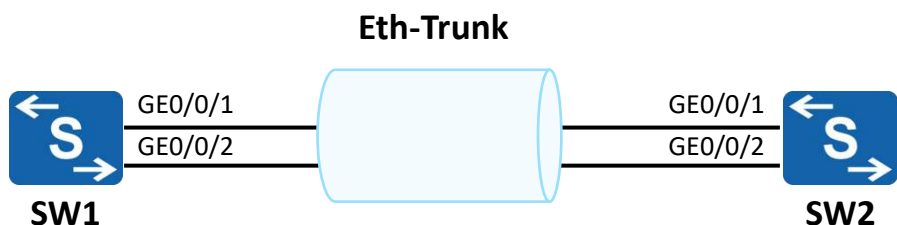
```
[Huawei-Eth-Trunk1] least active-linknumber {number}
```

The minimum number of active interfaces can be different on the local end and peer end and can be configured in both manual and LACP modes.

The minimum number of active interfaces is configured to ensure the minimum bandwidth. When the number of active links is smaller than the lower threshold, the Eth-Trunk interface goes down.



Example for Configuring an Eth-Trunk in Manual Mode



- Requirement description:
 - SW1 and SW2 are connected to the networks of VLAN 10 and VLAN 20.
 - SW1 and SW2 are connected through two Ethernet links. To provide link redundancy and enhance transmission reliability, configure an Eth-Trunk in manual mode between SW1 and SW2.

SW1 configuration:

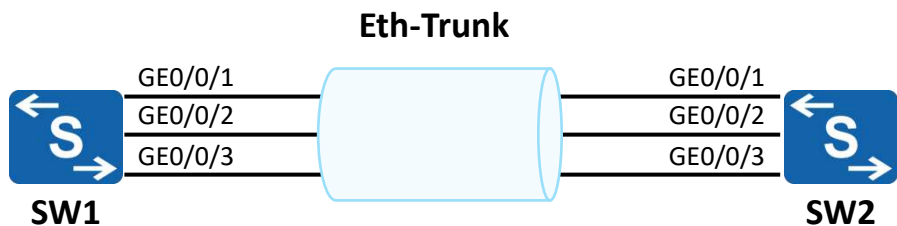
```
[SW1] interface eth-trunk 1
[SW1-Eth-Trunk1] trunkport gigabitethernet 0/0/1 to 0/0/2
[SW1-Eth-Trunk1] port link-type trunk
[SW1-Eth-Trunk1] port trunk allow-pass vlan 10 20
```

SW2 configuration:

```
[SW2] interface eth-trunk 1
[SW2-Eth-Trunk1] trunkport gigabitethernet 0/0/1 to 0/0/2
[SW2-Eth-Trunk1] port link-type trunk
[SW2-Eth-Trunk1] port trunk allow-pass vlan 10 20
```




Example for Configuring an Eth-Trunk in LACP Mode (1)



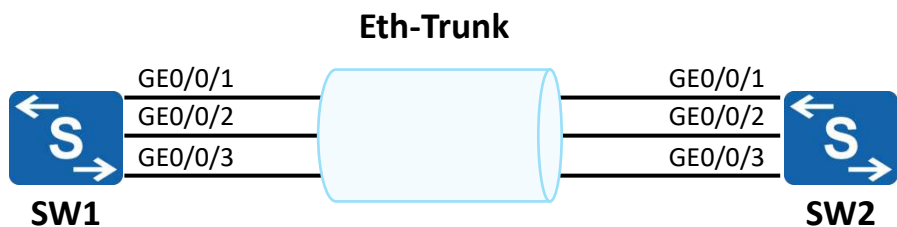
- Requirement description:
 - SW1 and SW2 are connected to the networks of VLAN 10 and VLAN 20.
 - SW1 and SW2 are connected through three Ethernet links. To provide link redundancy and enhance transmission reliability, configure an Eth-Trunk in LACP mode between SW1 and SW2, manually adjust the priority to configure SW1 as the Actor, and set the maximum number of active interfaces to 2. The other link functions as the backup link.

SW1 configuration:

```
[SW1] interface eth-trunk 1
[SW1-Eth-Trunk1] mode lacp
[SW1-Eth-Trunk1] max active-linknumber 2
[SW1-Eth-Trunk1] trunkport gigabitethernet 0/0/1 to 0/0/3
[SW1-Eth-Trunk1] port link-type trunk
[SW1-Eth-Trunk1] port trunk allow-pass vlan 10 20
[SW1-Eth-Trunk1] quit
[SW1] lacp priority 30000
```



Example for Configuring an Eth-Trunk in LACP Mode (2)



- Requirement description:
 - SW1 and SW2 are connected to the networks of VLAN 10 and VLAN 20.
 - SW1 and SW2 are connected through three Ethernet links. To provide link redundancy and enhance transmission reliability, configure an Eth-Trunk in LACP mode between SW1 and SW2, manually adjust the priority to configure SW1 as the Actor, and set the maximum number of active interfaces to 2. The other link functions as the backup link.

SW1 configuration:

```
[SW2] interface eth-trunk 1
[SW2-Eth-Trunk1] mode lacp
[SW2-Eth-Trunk1] max active-linknumber 2
[SW2-Eth-Trunk1] trunkport gigabitethernet 0/0/1 to 0/0/3
[SW2-Eth-Trunk1] port link-type trunk
[SW2-Eth-Trunk1] port trunk allow-pass vlan 10 20
[SW2-Eth-Trunk1] quit
```

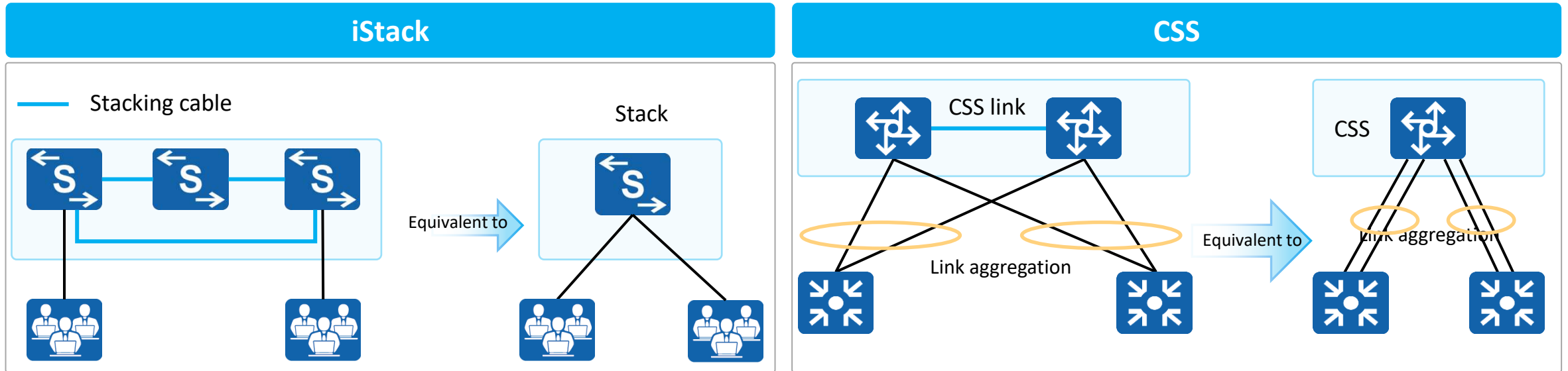


Contents

1. Network Reliability Requirements
2. Principle and Configuration of Link Aggregation
- 3. Overview of iStack and CSS**



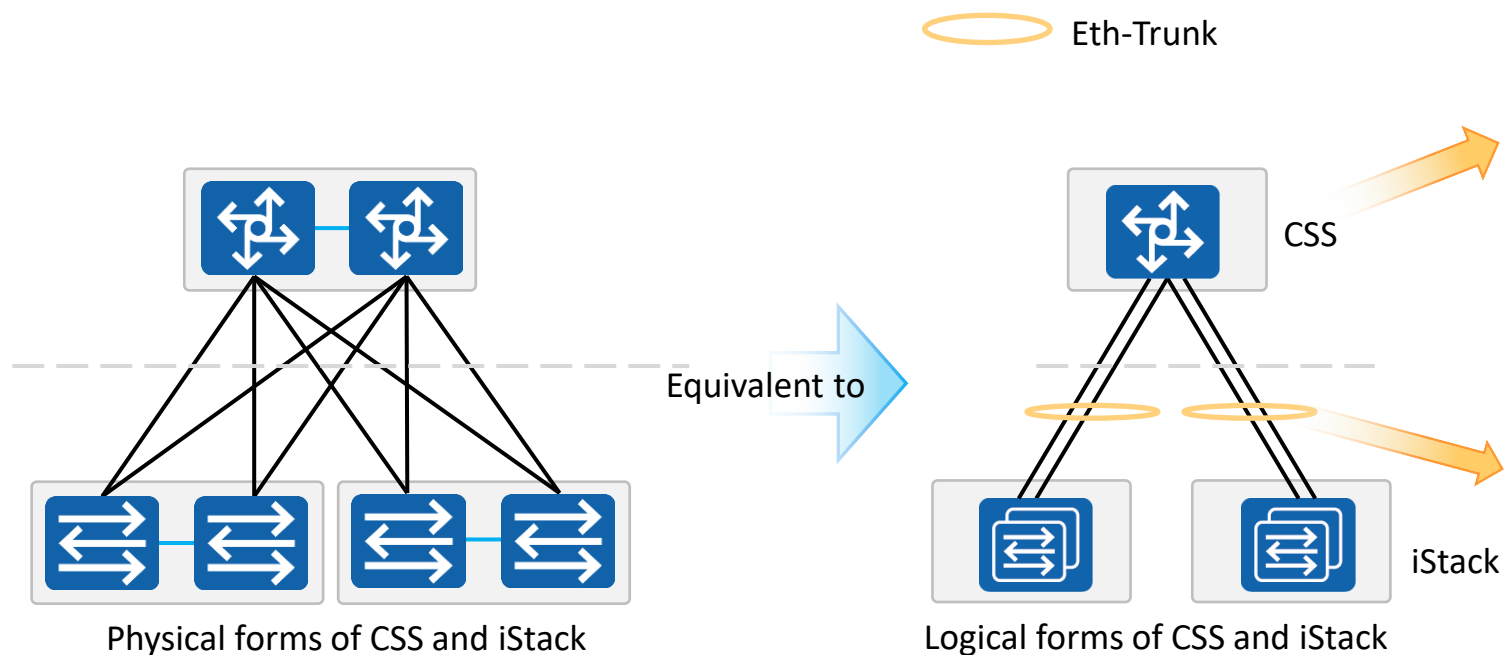
Introduction to iStack and CSS



- iStack: Multiple iStack-capable switches are connected using stacking cables to form a logical switch that participates in data forwarding.
- Cluster switch system (CSS): Two CSS-capable switches are bundled into one logical switch.
- A CSS consists of only two switches. Generally, modular switches support CSS, and fixed switches support iStack.



Advantages of iStack and CSS



- One logical device simplifies O&M and facilitates management.
- If a physical device fails, the other device can take over the forwarding and control functions, preventing single points of failure.

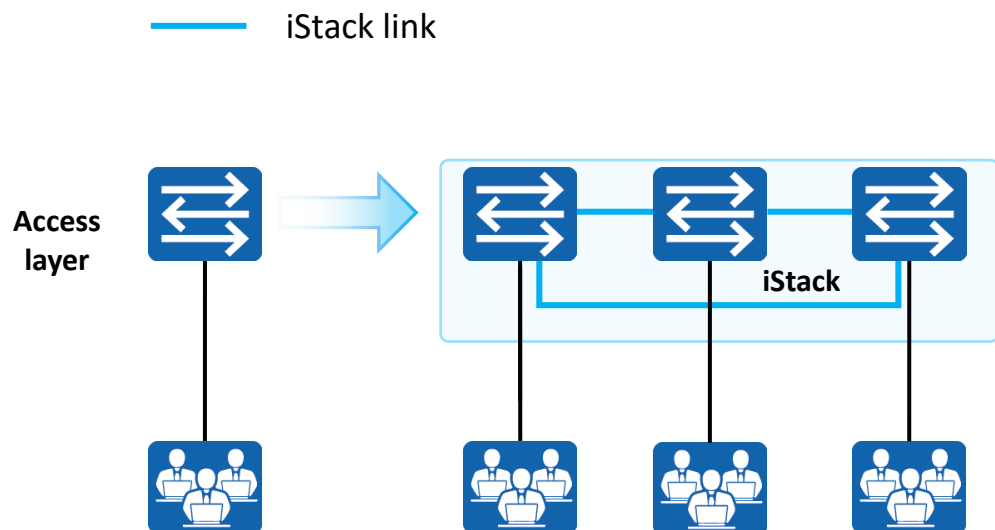
- Inter-device link aggregation is implemented on a loop-free physical network, so STP does not need to be deployed.
- All links in the Eth-Trunk are used, and the link usage is 100%.

- Many-to-one virtualization: Switches can be virtualized into one logical switch (CSS) that has a unified control plane for unified management.
- Unified forwarding plane: Physical switches in a CSS use a unified forwarding plane, and share and synchronize forwarding information in real time.
- Inter-device link aggregation: Links between physical switches are aggregated into a single Eth-Trunk interface to interconnect with downstream devices.



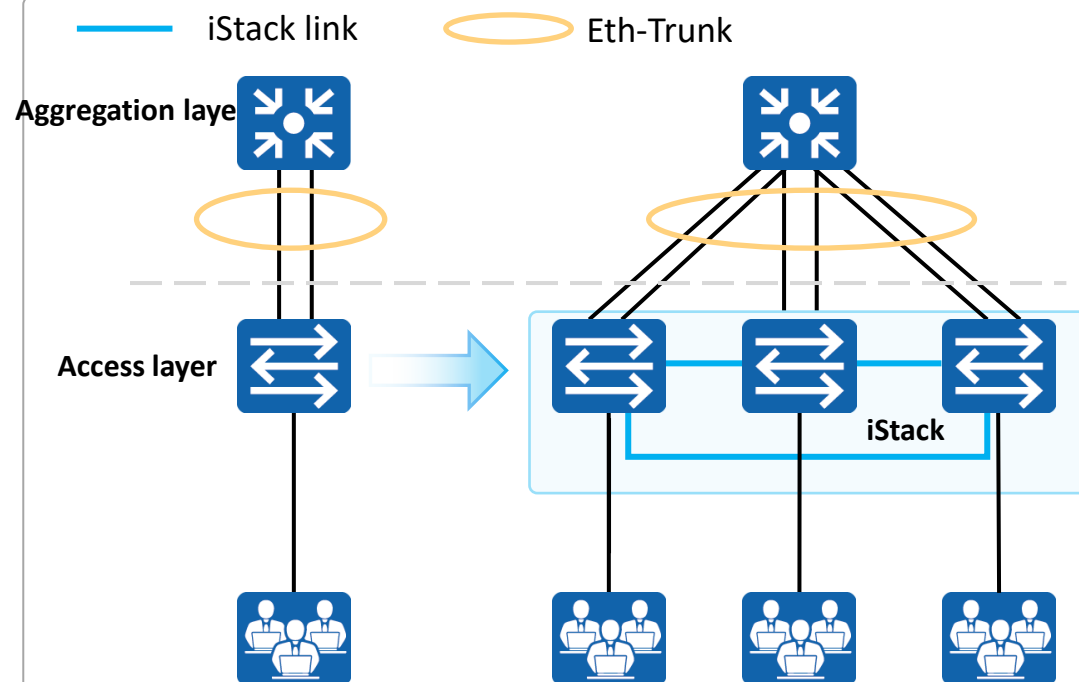
Application (1)

Extending the port quantity



- When the port density of a switch cannot meet the access requirements, add new switches to set up an iStack to increase the number of ports.

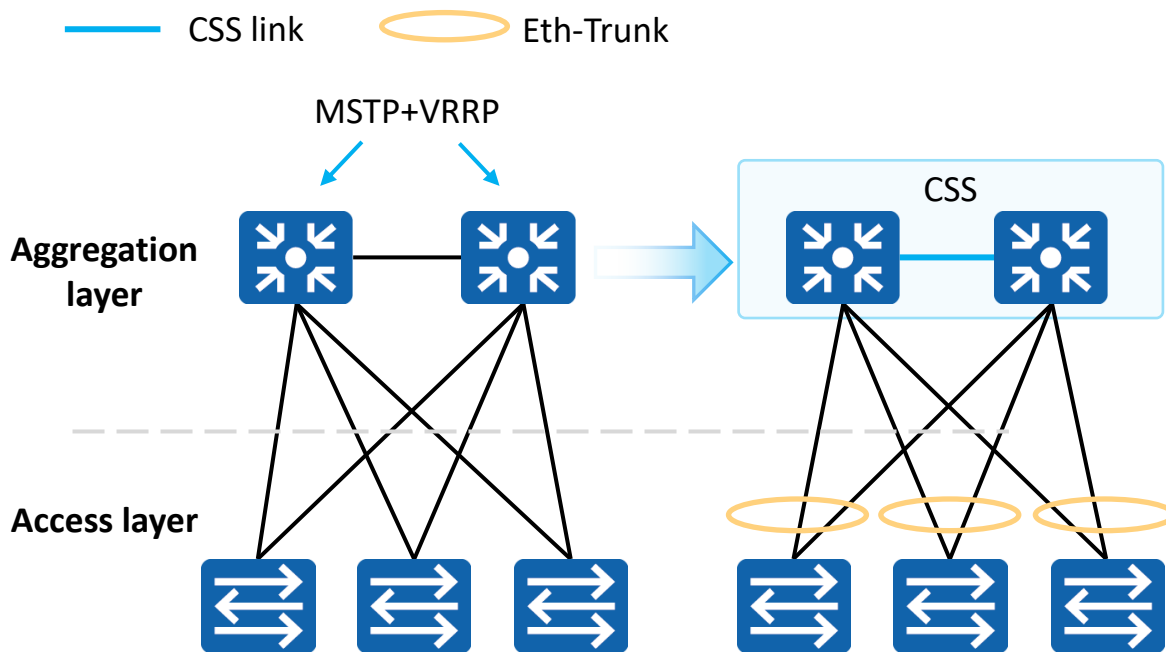
Extending the bandwidth and implementing redundancy backup



- To increase the uplink bandwidth, add new switches to set up an iStack and add multiple physical links of the member switches to an Eth-Trunk. This increases the uplink bandwidth, implements inter-device backup and inter-device link redundancy, and improves reliability.



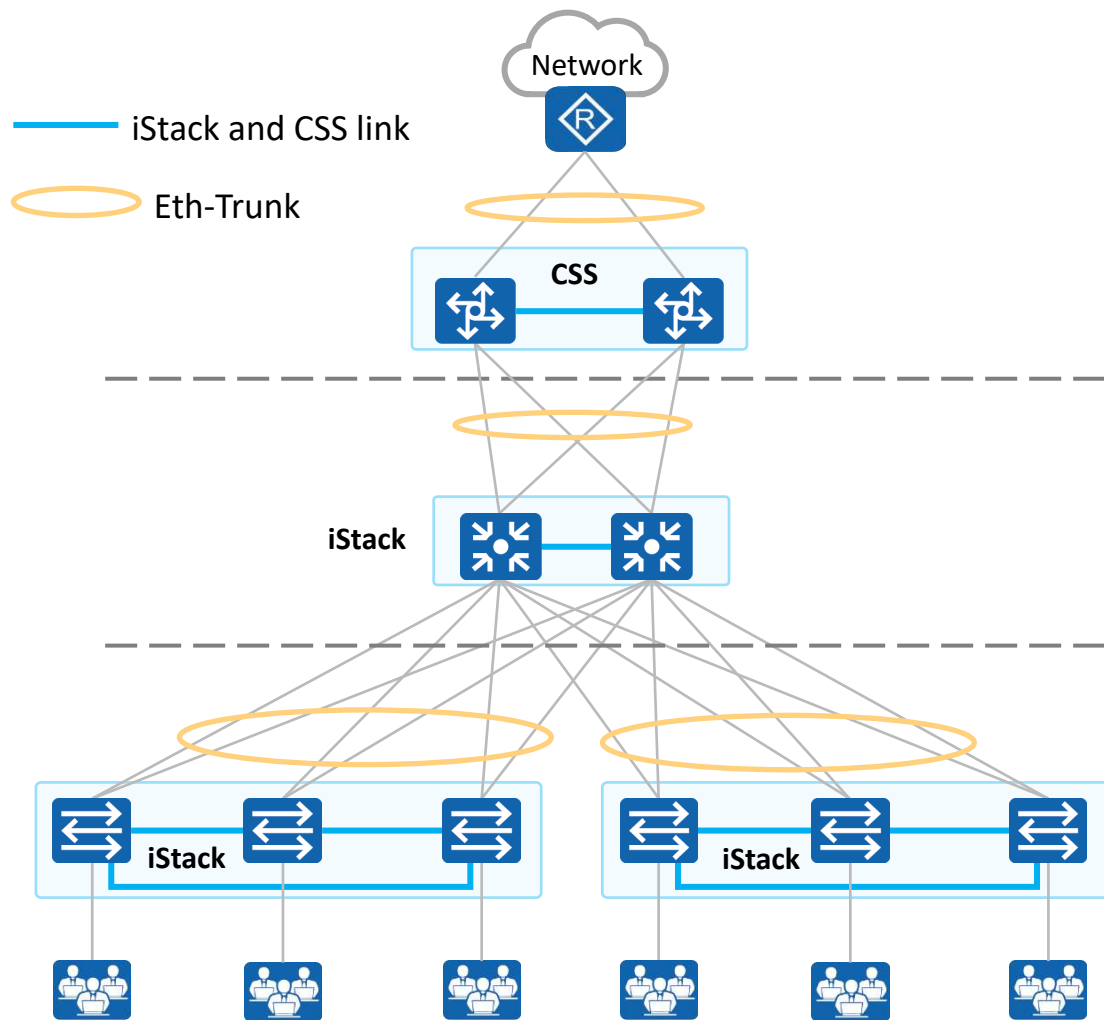
Application (2)



- Two devices form a CSS and are virtualized into a single logical device. This simplified network does not require Multiple Spanning Tree Protocol (MSTP) or Virtual Router Redundancy Protocol (VRRP), so network configuration is much simpler. Additionally, inter-device link aggregation speeds up network convergence and improves network reliability.



Recommended Architecture



Core layer

- Core switches set up a CSS and use Eth-Trunks to connect to uplink and downlink devices, building a highly reliable and loop-free network.

Aggregation layer

- Aggregation switches set up an iStack and use Eth-Trunks to connect to uplink and downlink devices, building a highly reliable loop-free network.

Access layer

- Access devices that are geographically close to each other (such as access switches in a building) are virtualized into one logical device using iStack. This adds interfaces and simplifies management.
- An Eth-Trunk is used to connect to the aggregation layer. The logical network architecture is simple, and STP and VRRP are not required. This networking offers high reliability, high uplink bandwidth, and fast convergence.



Quiz

1. What are the differences between per-packet load balancing and per-flow load balancing?
2. How does an Actor be elected in LACP mode?
3. What are the advantages of CSS and iStack?
 1. Packet disorder may occur if packets are load balanced to different links based on packets. If packets are load balanced to the same link based on flows, packet disorder will not occur. However, a single flow cannot make full use of the bandwidth of the entire Eth-Trunk.
 2. Switches compare system priorities. A smaller value indicates a higher priority. If the system priorities are the same, the bridge MAC addresses are compared. A smaller bridge MAC address indicates a higher priority. The device with a higher priority becomes the Actor.
 3. CSS and iStack simplify network management, improve network reliability, make full use of network link bandwidth, and use inter-device Eth-Trunk to construct a loop-free physical network.



Summary

- Link aggregation can be used to improve link reliability, utilization, and bandwidth. Link aggregation can be classified into static and LACP aggregation based on the aggregation mode.
- LACP uses packet negotiation to implement backup for active links. When a link fails, the backup link is elected as the active link to forward packets.
- To ensure the sequence in which packets arrive, link aggregation uses per-flow load balancing.
- iStack and CSS simplify network management and network structure, and improve network reliability.

The background of the slide features a blue-tinted image of several business professionals in a modern office environment. They are standing on a highly reflective floor, and their silhouettes are clearly visible against the lighter background. The overall aesthetic is professional and corporate.

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ACL Principles and Configuration



Foreword

- Rapid network development brings challenges to network security and quality of service (QoS). Access control lists (ACLs) are closely related to network security and QoS.
- By accurately identifying packet flows on a network and working with other technologies, ACLs can control network access behaviors, prevent network attacks, and improve network bandwidth utilization, thereby ensuring network environment security and QoS reliability.
- This course describes the basic principles and functions of ACLs, types and characteristics of ACLs, basic composition of ACLs, ACL rule ID matching order, usage of wildcards, and ACL configurations.



Objectives

- On completion of this course, you will be able to:
 - Describe the basic principles and functions of ACLs.
 - Understand the types and characteristics of ACLs.
 - Describe the basic composition of ACLs and ACL rule ID matching order.
 - Understand how to use wildcards in ACLs.
 - Complete the basic configurations of ACLs.

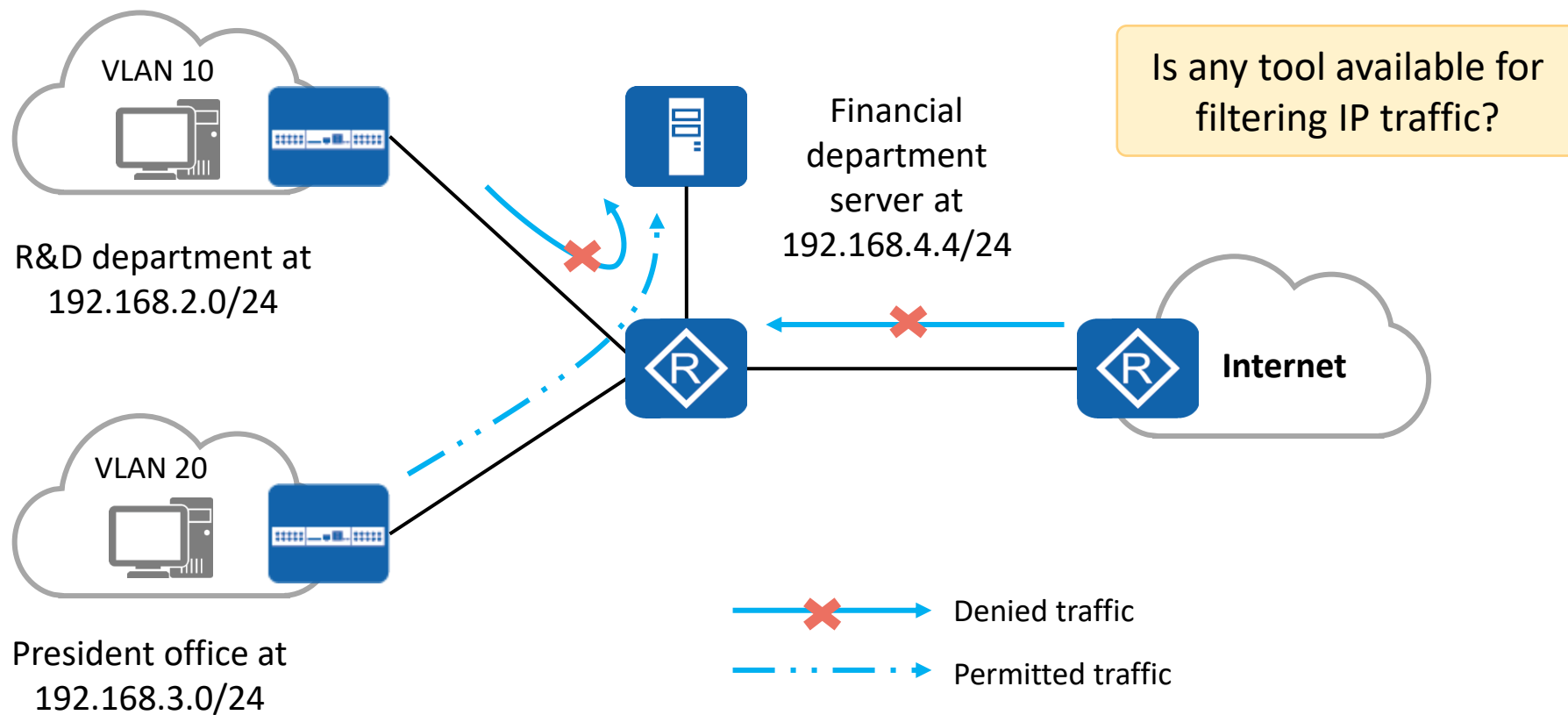


Contents

- 1. ACL Overview**
2. Basic Concepts and Working Mechanism of ACLs
3. Basic Configurations and Applications of ACLs



Background: A Tool Is Required to Filter Traffic

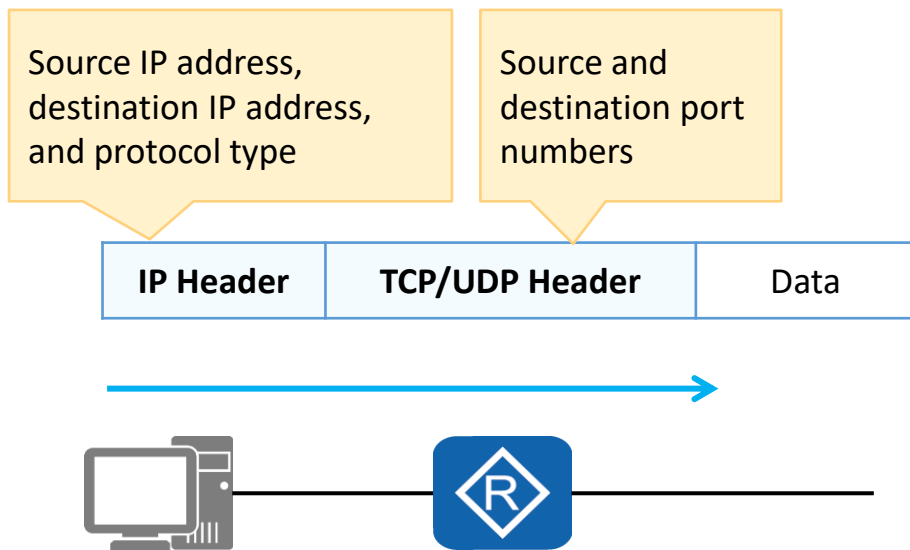


- To ensure financial data security, an enterprise prohibits the R&D department's access to the financial department server but allows the president office's access to the financial department server.



ACL Overview

- An ACL is a set of sequential rules composed of permit or deny statements.
- ACLs accurately identify and control packets on a network to manage network access behaviors, prevent network attacks, and improve bandwidth utilization.



ACL Application

- Matching IP traffic
- Invoked in a traffic filter
- Invoked in network address translation (NAT)
- Invoked in a routing policy
- Invoked in a firewall policy
- Invoked in QoS
- Others



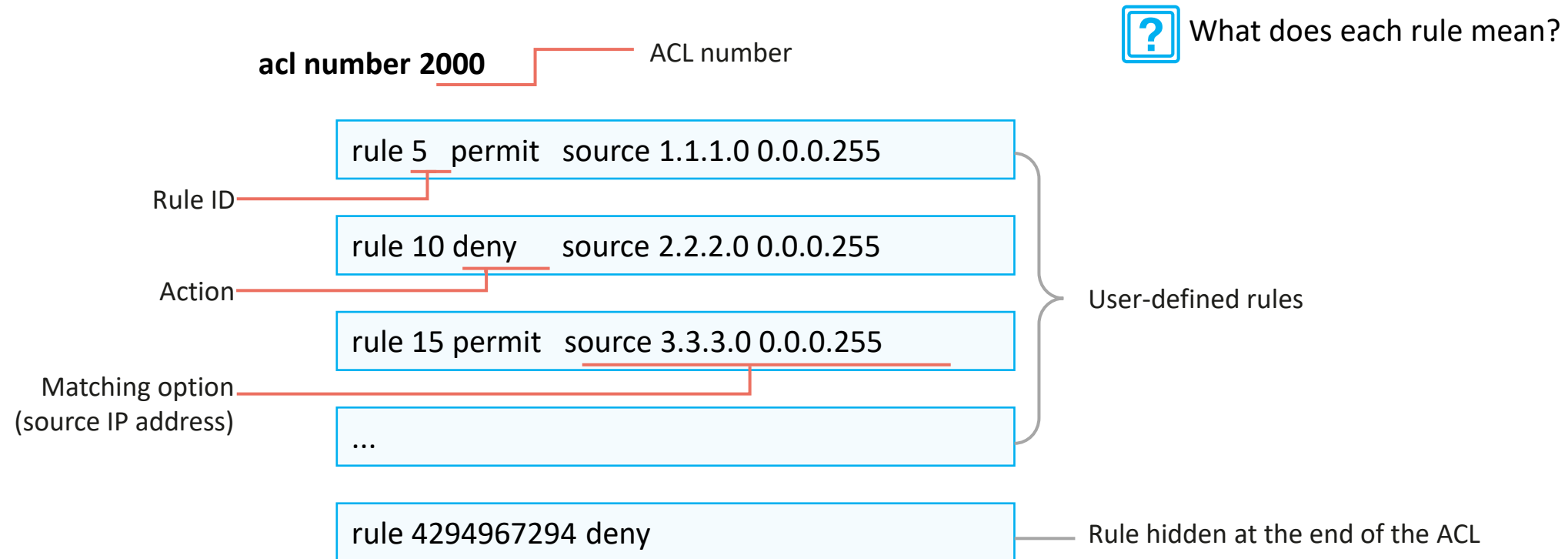
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ACL Composition

- An ACL consists of several permit or deny statements. Each statement is a rule of the ACL, and permit or deny in each statement is the action corresponding to the rule.





Rule ID

acl number 2000

	Rule ID		
rule	5	deny	source 10.1.1.1 0
rule	10	deny	source 10.1.1.2 0
rule	15	permit	source 10.1.1.0 0.0.0.255

Step = 5



How do I add a rule?

rule 11 deny source 10.1.1.3 0

acl number 2000

rule	5	deny	source 10.1.1.1 0
rule	10	deny	source 10.1.1.2 0
rule	11	deny	source 10.1.1.3 0
rule	15	permit	source 10.1.1.0 0.0.0.255

Rule ID and Step

- **Rule ID**
Each rule in an ACL has an ID.
- **Step**
A step is an increment between neighboring rule IDs automatically allocated by the system. The default step is 5. Setting a step facilitates rule insertion between existing rules of an ACL.
- **Rule ID allocation**
If a rule is added to an empty ACL but no ID is manually specified for the rule, the system allocates a step value (5 for example) as the ID of the rule. If an ACL contains rules with manually specified IDs and a rule with no manually specified ID is added, the system allocates to this rule an ID that is greater than the largest rule ID in the ACL and is the smallest integer multiple of the step value.



Wildcard (1)

acl number 2000

rule	5	deny	source	10.1.1.1	0
rule	10	deny	source	10.1.1.2	0
rule	15	permit	source	10.1.1.0	0.0.255

Wildcard

Wildcard

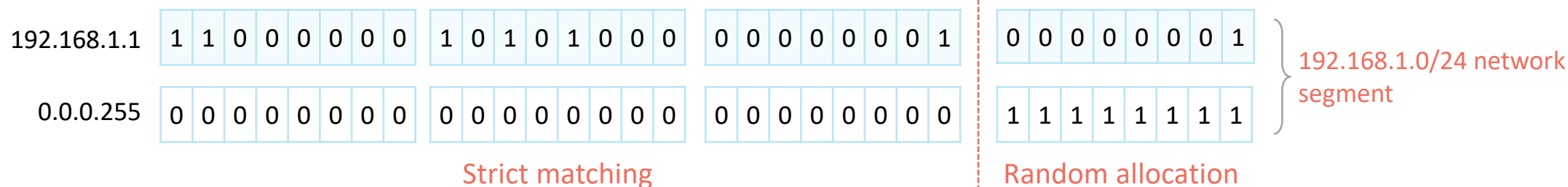
- A wildcard is a 32-bit number that indicates which bits in an IP address need to be strictly matched and which bits do not need to be matched.
- A wildcard is usually expressed in dotted decimal notation, as a network mask is expressed. However, their meanings are different.

- Matching rule

0: matching; 1: random allocation



How do I match the network segment address corresponding to 192.168.1.1/24?





Wildcard (2)

- A wildcard can be used to match odd IP addresses in the network segment 192.168.1.0/24, such as 192.168.1.1, 192.168.1.3, and 192.168.1.5.

Strict matching	Random allocation								Strict matching
192.168.1	1								
192.168.1	0	0	0	0	0	0	0	1	
192.168.1	3								
192.168.1	0	0	0	0	0	0	1	1	
192.168.1	5								
192.168.1	0	0	0	0	0	1	0	1	
...									
Wildcard									
0.0.0.	1	1	1	1	1	1	1	0	

192.168.1.1 0.0.0.254

The value 1 or 0 in the wildcard can be inconsecutive.

Special Wildcard

- Exactly match the IP address 192.168.1.1.
192.168.1.1 0.0.0.0 = 192.168.1.1 0
- Match All IP addresses.
0.0.0.0 255.255.255 = any



ACL Classification and Identification

- ACL classification based on ACL rule definition methods

Category	Number Range	Description
Basic ACL	2000 to 2999	Defines rules based on source IPv4 addresses, fragmentation information, and effective time ranges.
Advanced ACL	3000 to 3999	Defines rules based on source and destination IPv4 addresses, IPv4 protocol types, ICMP types, TCP source/destination port numbers, UDP source/destination port numbers, and effective time ranges.
Layer 2 ACL	4000 to 4999	Defines rules based on information in Ethernet frame headers of packets, such as source and destination MAC addresses and Layer 2 protocol types.
User-defined ACL	5000 to 5999	Defines rules based on packet headers, offsets, character string masks, and user-defined character strings.
User ACL	6000 to 6999	Defines rules based on source IPv4 addresses or user control list (UCL) groups, destination IPv4 addresses or destination UCL groups, IPv4 protocol types, ICMP types, TCP source/destination port numbers, and UDP source/destination port numbers.

- ACL classification based on ACL identification methods

Category	Description
Numbered ACL	Traditional ACL identification method. A numbered ACL is identified by a number.
Named ACL	A named ACL is identified by a name.



Basic and Advanced ACLs

- Basic ACL

Number range:
2000 to 2999

Source IP address

IP Header		TCP/UDP Header		Data
acl number 2000				
rule	5	deny	source 10.1.1.1 0	
rule	10	deny	source 10.1.1.2 0	
rule	15	permit	source 10.1.1.0 0.0.0.255	

- Advanced ACL

Number range:
3000-3999

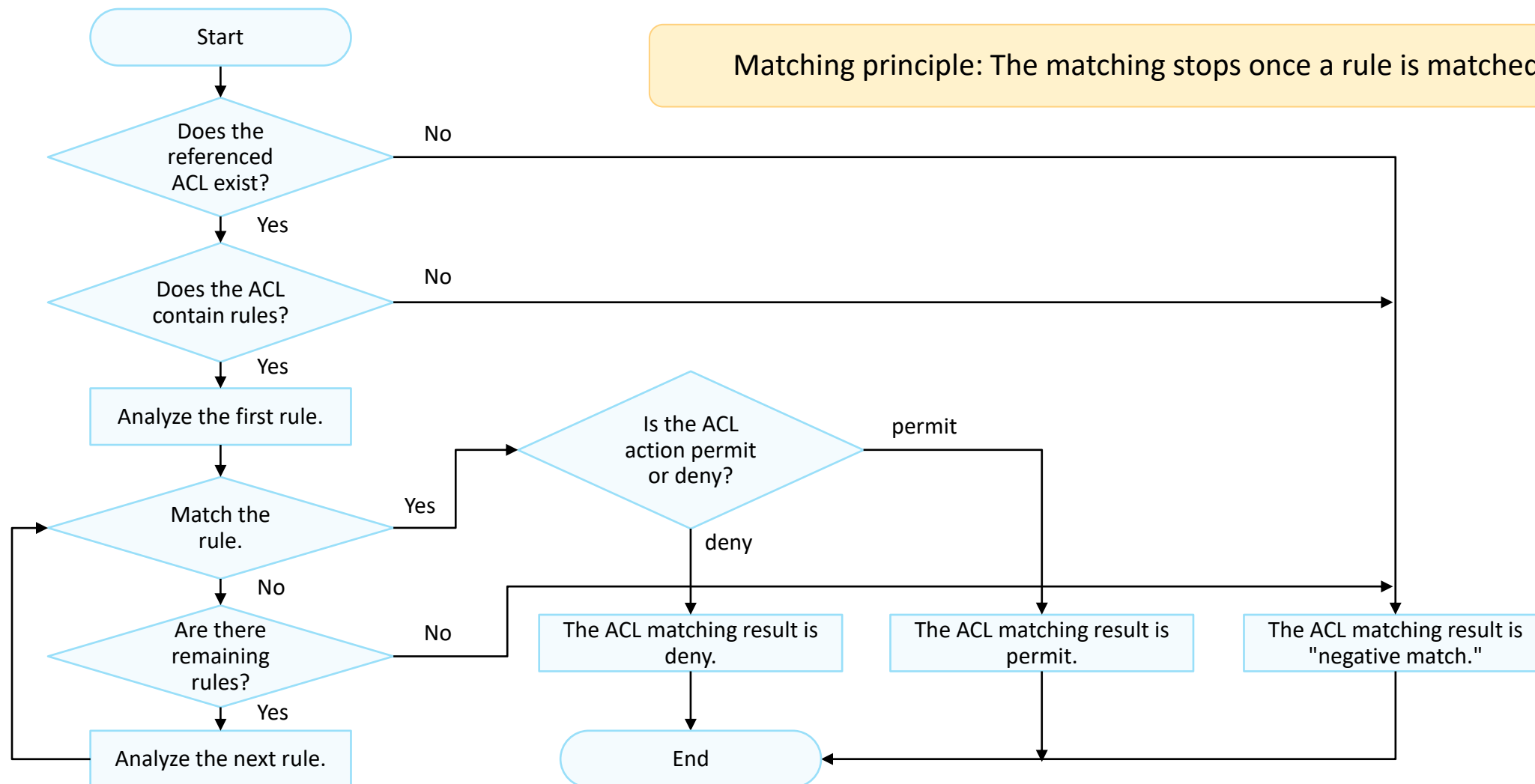
Source IP address,
destination IP address,
and protocol type

Source and
destination port
numbers

IP Header	TCP/UDP Header	Data
acl number 3000		
rule 5 permit ip source 10.1.1.0 0.0.0.255 destination 10.1.3.0 0.0.0.255		
rule 10 permit tcp source 10.1.2.0 0.0.0.255 destination 10.1.3.0 0.0.0.255 destination-port eq 21		



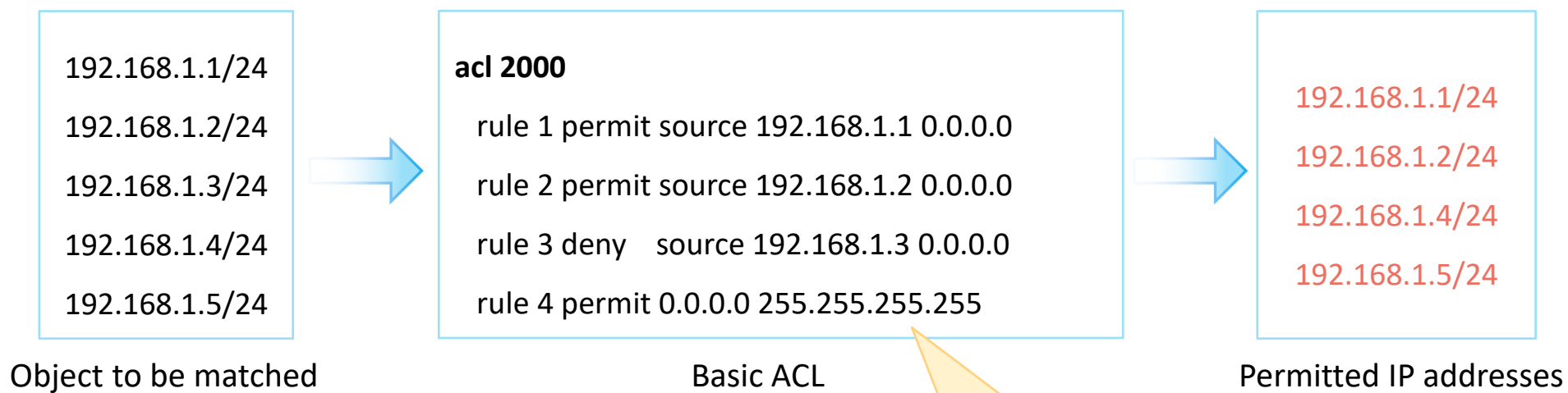
ACL Matching Mechanism





ACL Matching Order and Result

- Configuration order (config mode)
 - The system matches packets against ACL rules in ascending order of rule ID. That is, the rule with the smallest ID is processed first.

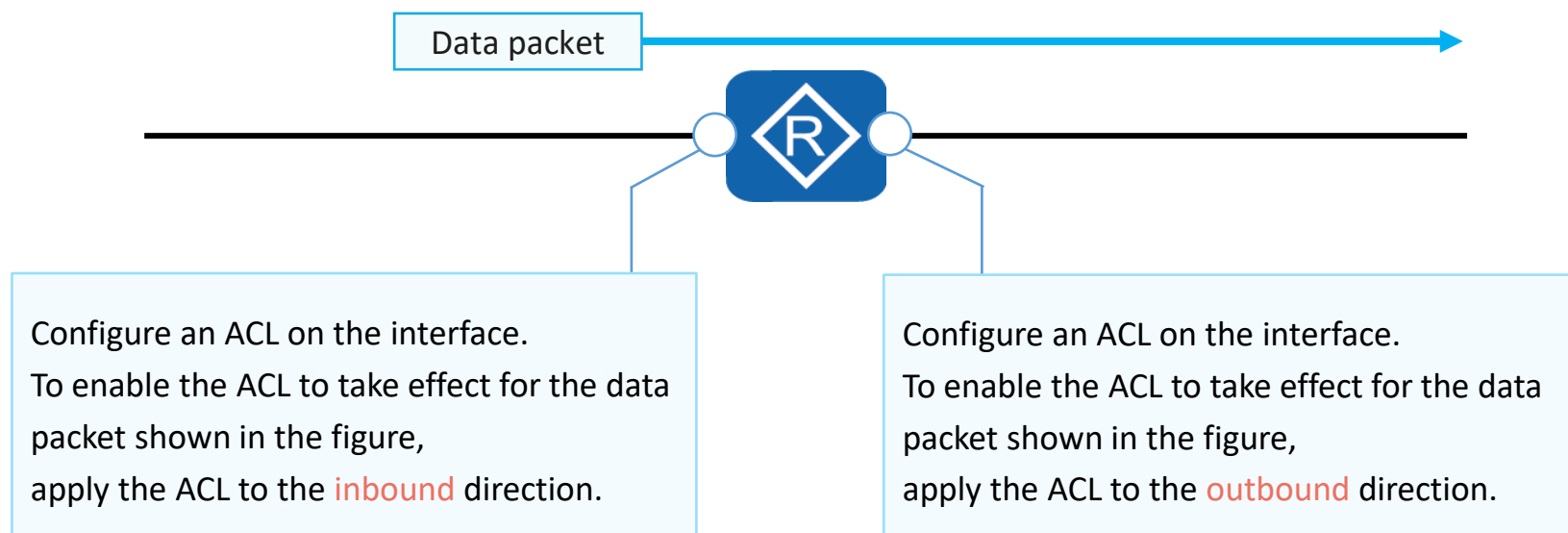


Does "permit" mean that traffic is allowed to pass?

rule 1: permits packets with the source IP address 192.168.1.1.
rule 2: permits packets with the source IP address 192.168.1.2.
rule 3: denies packets with the source IP address 192.168.1.3.
rule 4: permits packets from all other IP addresses.



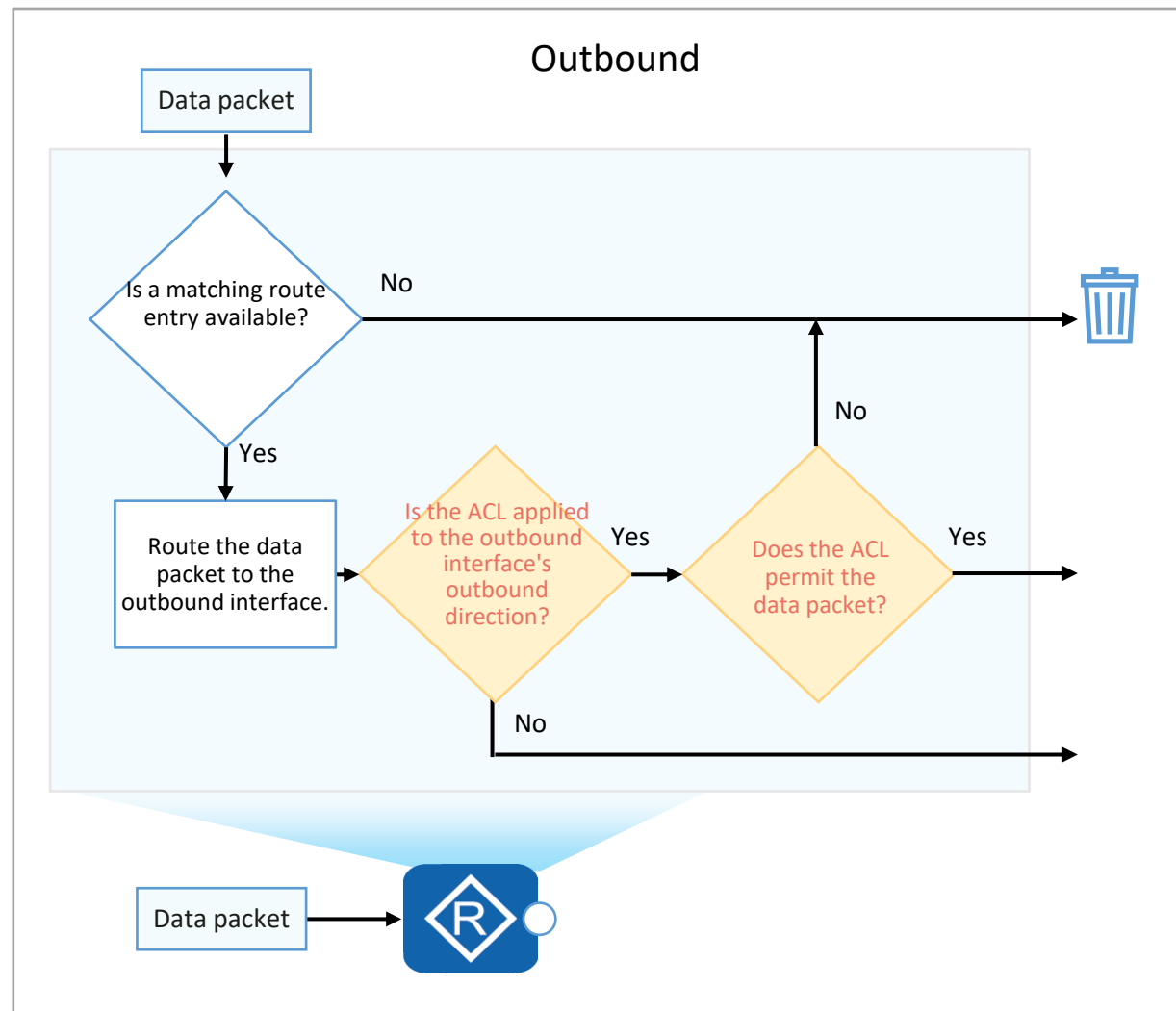
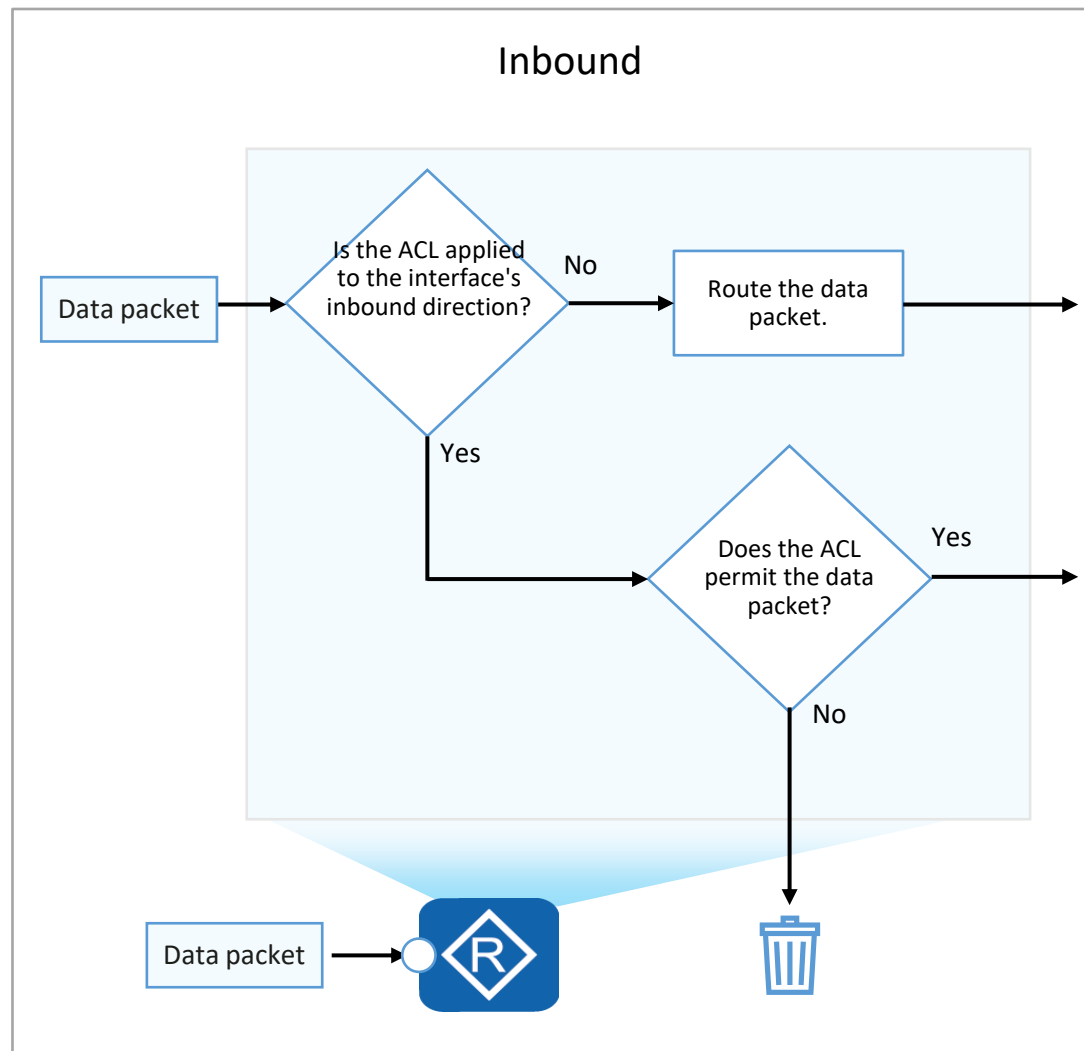
ACL Matching Position



Inbound = stop traffic coming in
Outbound = control traffic going out



Inbound and Outbound Directions





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1. ACL Overview
2. Basic Concepts and Working Mechanism of ACLs
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Basic Configuration Commands of Basic ACLs

1. Create a basic ACL.

```
[Huawei] acl [ number ] acl-number [ match-order config ]
```

Create a numbered basic ACL and enter its view.

```
[Huawei] acl name acl-name { basic | acl-number } [ match-order config ]
```

Create a named basic ACL and enter its view.

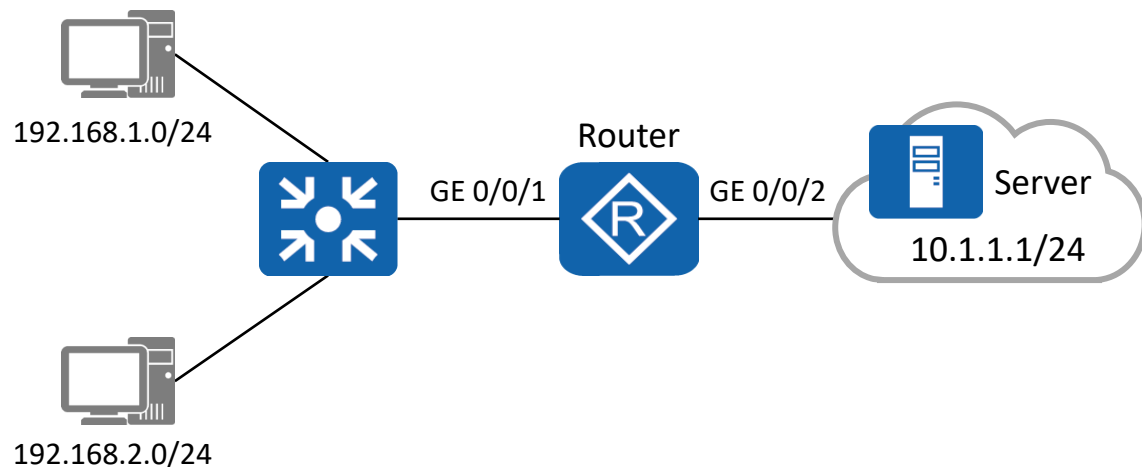
2. Configure a rule for the basic ACL.

```
[Huawei-acl-basic-2000] rule [ rule-id ] { deny | permit } [ source { source-address source-wildcard | any } | time-range time-name ]
```

In the basic ACL view, you can run this command to configure a rule for the basic ACL.



Case: Use a Basic ACL to Filter Data Traffic



- Requirements:

To prevent the user host on the network segment 192.168.1.0/24 from accessing the network where the server resides, configure a basic ACL on the router. After the configuration is complete, the ACL filters out the data packets whose source IP addresses are on the network segment 192.168.1.0/24 and permits other data packets.

1. Configure IP addresses and routes on the router.

2. Create a basic ACL on the router to prevent the network segment 192.168.1.0/24 from accessing the network where the server resides.

```
[Router] acl 2000
[Router-acl-basic-2000] rule deny source 192.168.1.0 0.0.0.255
[Router-acl-basic-2000] rule permit source any
```

3. Configure traffic filtering in the inbound direction of GE 0/0/1.

```
[Router] interface GigabitEthernet 0/0/1
[Router-GigabitEthernet0/0/1] traffic-filter inbound acl 2000
[Router-GigabitEthernet0/0/1] quit
```



Basic Configuration Commands of Advanced ACLs (1)

1. Create an advanced ACL.

```
[Huawei] acl [ number ] acl-number [ match-order config ]
```

Create a numbered advanced ACL and enter its view.

```
[Huawei] acl name acl-name { advance | acl-number } [ match-order config ]
```

Create a named advanced ACL and enter its view.



Basic Configuration Commands of Advanced ACLs (2)

2. Configure a rule for the advanced ACL.

You can configure advanced ACL rules according to the protocol types of IP packets. The parameters vary according to the protocol types.

- When the protocol type is **IP**, the command format is:

```
rule [ rule-id ] { deny | permit } ip [ destination { destination-address destination-wildcard | any } | source { source-address source-wildcard | any } | time-range time-name | [ dscp dscp | [ tos tos | precedence precedence ] ] ]
```

In the advanced ACL view, you can run this command to configure a rule for the advanced ACL.

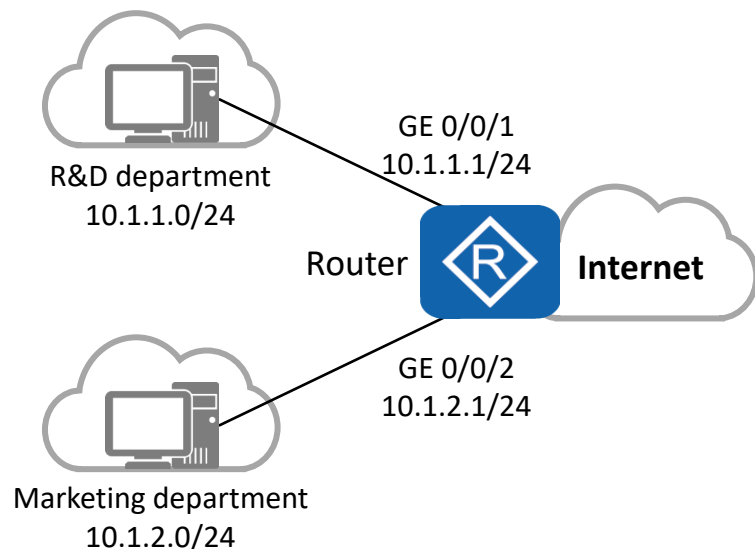
- When the protocol type is **TCP**, the command format is:

```
rule [ rule-id ] { deny | permit } { protocol-number | tcp } [ destination { destination-address destination-wildcard | any } | destination-port { eq port | gt port | lt port | range port-start port-end } | source { source-address source-wildcard | any } | source-port { eq port | gt port | lt port | range port-start port-end } | tcp-flag { ack | fin | syn } * | time-range time-name ] *
```

In the advanced ACL view, you can run this command to configure a rule for the advanced ACL.



Case: Use Advanced ACLs to Prevent User Hosts on Different Network Segments from Communicating (1)



Requirements:

- The departments of a company are connected through the router. To facilitate network management, the administrator allocates IP addresses of different network segments to the R&D and marketing departments.
- The company requires that the router prevent the user hosts on different network segments from communicating to ensure information security.

1. Configure IP addresses and routes on the router.
2. Create ACL 3001 and configure rules for the ACL to deny packets from the R&D department to the marketing department.

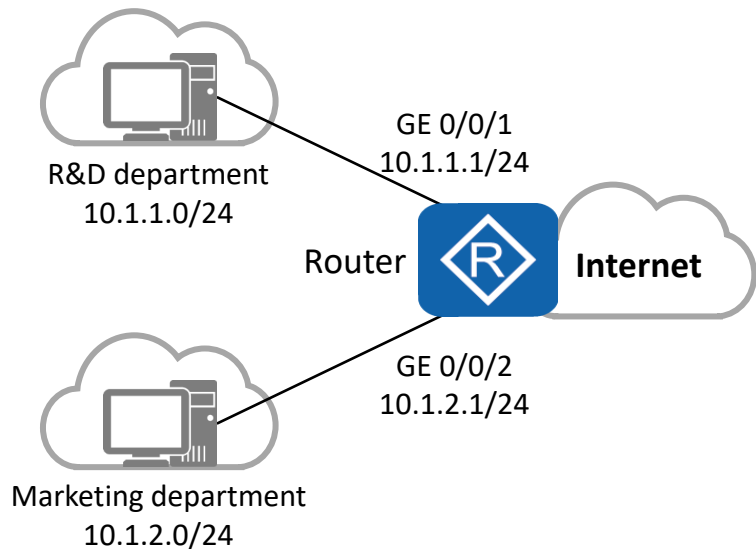
```
[Router] acl 3001
[Router-acl-adv-3001] rule deny ip source 10.1.1.0 0.0.0.255 destination
10.1.2.0 0.0.0.255
[Router-acl-adv-3001] quit
```

3. Create ACL 3002 and configure rules for the ACL to deny packets from the marketing department to the R&D department.

```
[Router] acl 3002
[Router-acl-adv-3002] rule deny ip source 10.1.2.0 0.0.0.255 destination
10.1.1.0 0.0.0.255
[Router-acl-adv-3002] quit
```



Case: Use Advanced ACLs to Prevent User Hosts on Different Network Segments from Communicating (2)



Requirements:

- The departments of a company are connected through the router. To facilitate network management, the administrator allocates IP addresses of different network segments to the R&D and marketing departments.
- The company requires that the router prevent the user hosts on different network segments from communicating to ensure information security.

4. Configure traffic filtering in the inbound direction of GE 0/0/1 and GE 0/0/2.

```
[Router] interface GigabitEthernet 0/0/1
[Router-GigabitEthernet0/0/1] traffic-filter inbound acl 3001
[Router-GigabitEthernet0/0/1] quit

[Router] interface GigabitEthernet 0/0/2
[Router-GigabitEthernet0/0/2] traffic-filter inbound acl 3002
[Router-GigabitEthernet0/0/2] quit
```



Quiz

1. (Single) Which one of the following rules is a valid basic ACL rule? ()
 - A. rule permit ip
 - B. rule deny ip
 - C. rule permit source any (Correct) Basic ACLs mainly allow/deny based on **IP addresses**, not specific protocols like TCP/UDP.
 - D. rule deny tcp source any
2. Which parameters can you use to define advanced ACL rules?

parameters such as the source/destination IP address, source/destination port number, protocol type, and TCP flag (SYN, ACK, or FIN).



Summary

- ACL is a widely used network technology. Its principle is as follows: packets are matched against configured ACL rules and actions are taken on the packets as configured in the ACL rules. The matching rules and actions are configured based on network requirements. Due to the variety of matching rules and actions, ACLs can implement a lot of functions.
- ACLs are often used with other technologies, such as firewall, routing policy, QoS, and traffic filtering.

The background of the slide features a blue-tinted image of several business professionals in a modern office environment. They are standing on a highly reflective floor, and their silhouettes are clearly visible against the lighter background. The overall aesthetic is professional and corporate.

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AAA Principles and Configuration



Foreword

- User management is one of the most basic security management requirements for any network.
- Authentication, authorization, and accounting (AAA) is a management framework that provides a security mechanism for authorizing some users to access specified resources and recording the operations of these users. AAA is widely used because of its good scalability and easy implementation of centralized management of user information. AAA can be implemented through multiple protocols. In actual applications, the Remote Authentication Dial-In User Service (RADIUS) protocol is the most commonly used to implement AAA.
- This course describes the basic concepts, implementation, basic configurations, and typical application scenarios of AAA.



Objectives

- Upon completion of this course, you will be able to:
 - Understand the fundamentals of AAA.
 - Describe the application scenarios of AAA.
 - Understand the fundamentals of RADIUS.
 - Get familiar with the basic configurations of AAA.



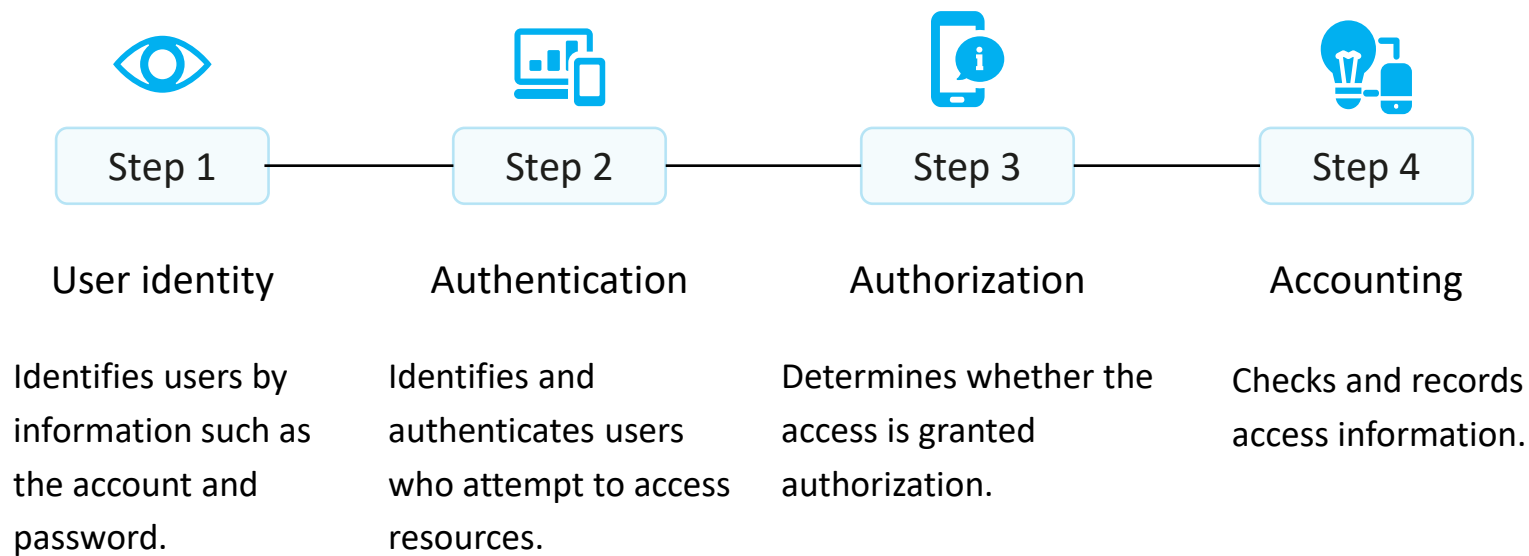
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- 1. AAA Overview**
2. AAA Configuration



Basic Concepts of AAA

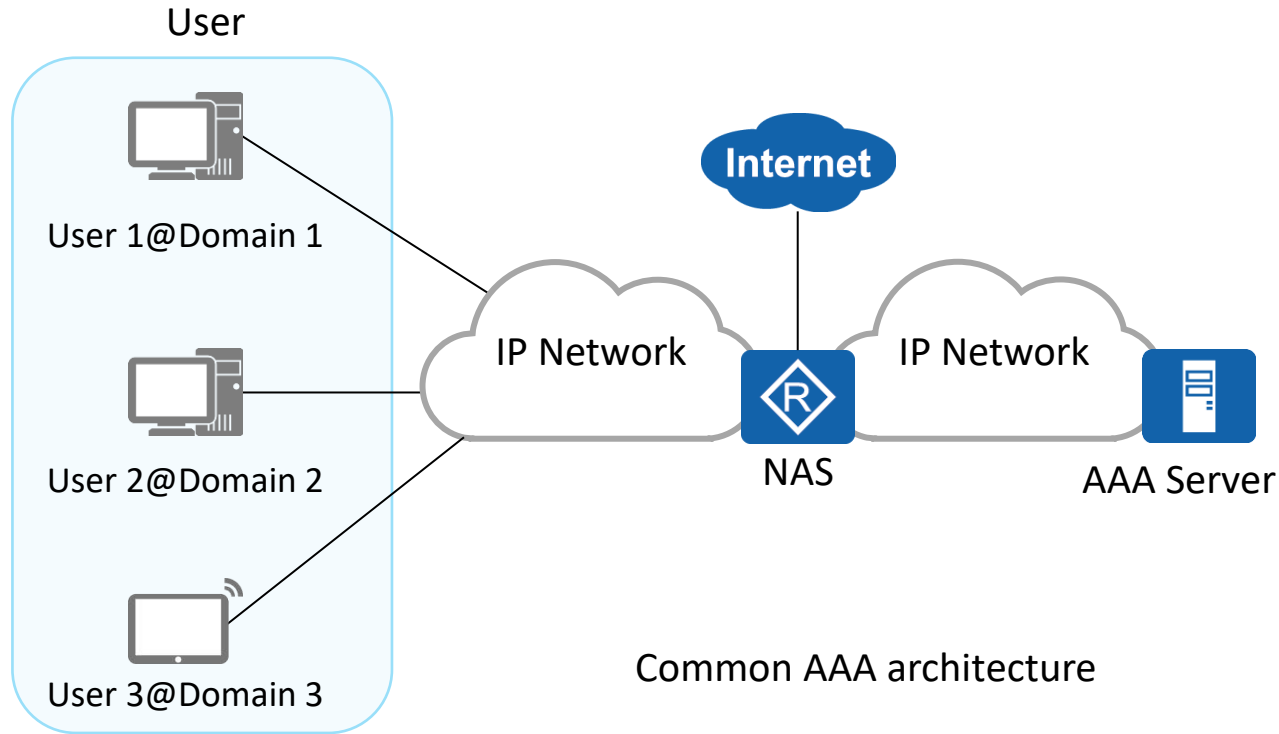
- Authentication, authorization, and accounting (AAA) provides a management mechanism for network security.





Common AAA Architecture

- A common AAA architecture includes the user, network access server (NAS), and AAA server.

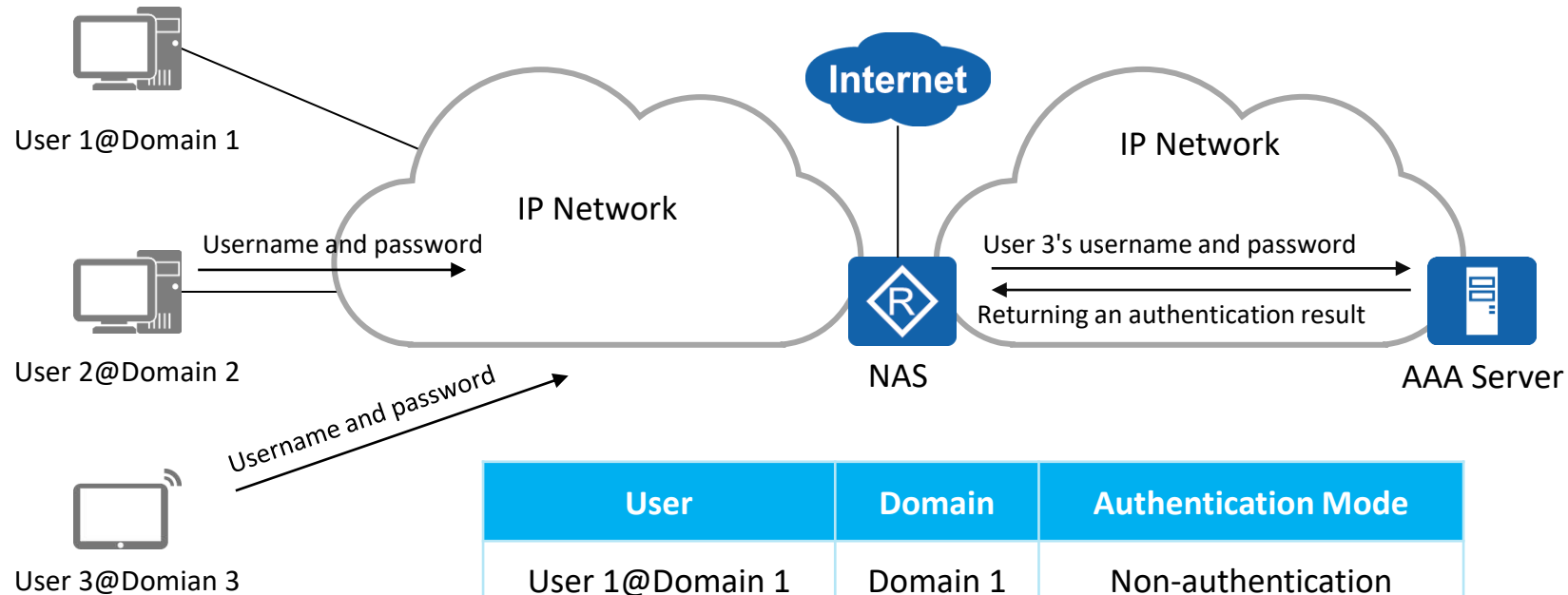


- The NAS collects and manages user access requests in a centralized manner.
- Multiple domains are created on the NAS to manage users. Different domains can be associated with different AAA schemes, which include the authentication scheme, authorization scheme, and accounting scheme.
- When receiving a user access request, the NAS determines the domain to which the user belongs based on the username and performs user management and control based on the AAA schemes configured for the domain.



Authentication

- AAA supports the following authentication modes: non-authentication, local authentication, and remote authentication.

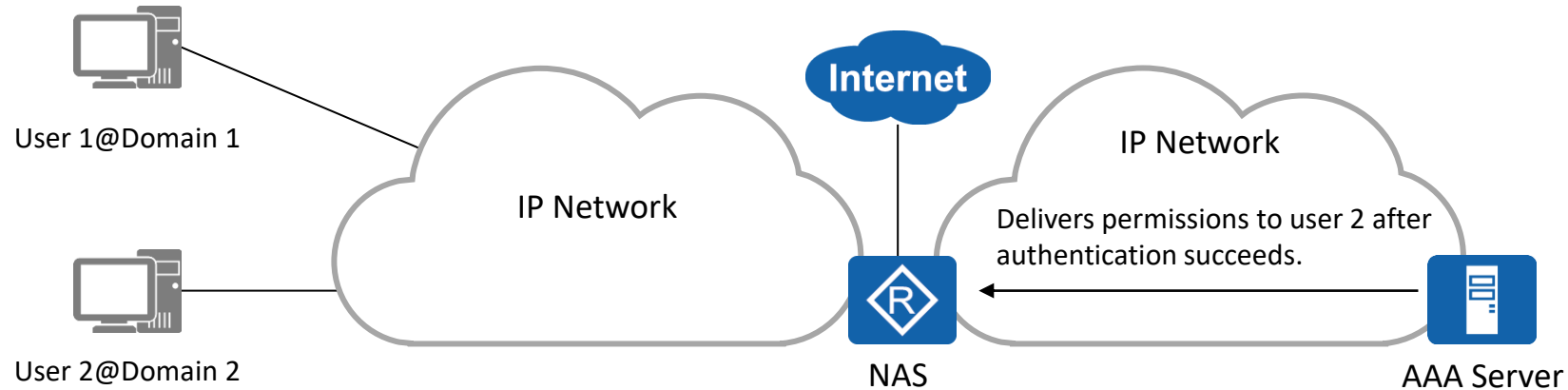


User	Domain	Authentication Mode
User 1@Domain 1	Domain 1	Non-authentication
User 2@Domain 2	Domain 2	Local authentication
User 3@Domain 3	Domain 3	Remote authentication



Authorization

- AAA supports the following authorization modes: non-authorization, local authorization, and remote authorization.
- Authorization information includes the user group, VLAN ID, and ACL number.



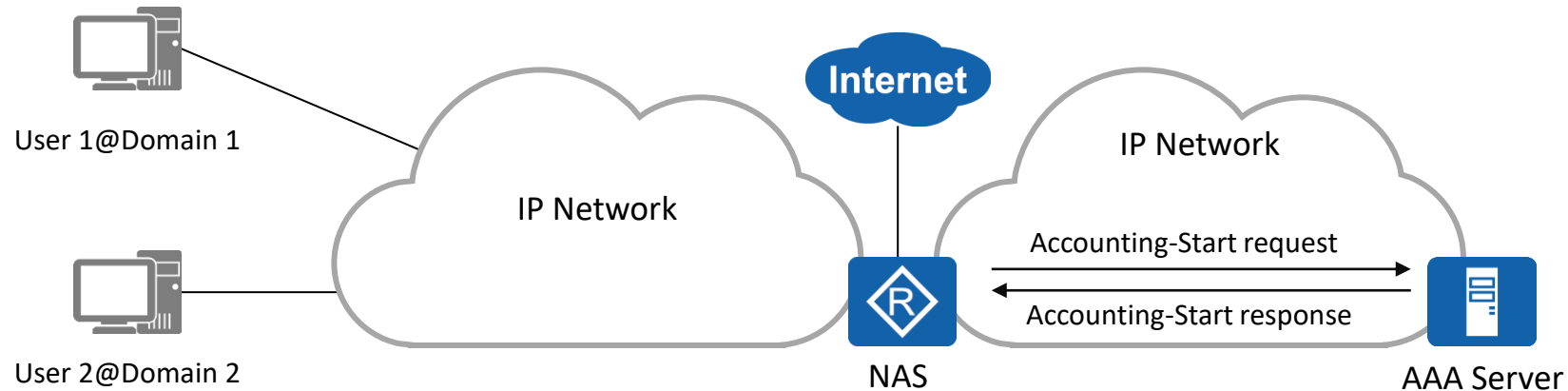
User 3@Domain 3

User	Domain	Authorization Mode	Authorization Content
User 1@Domain 1	Domain 1	Non-authorization	None
User 2@Domain 2	Domain 2	Local authorization	Internet access is allowed.
User 3@Domain 3	Domain 3	Remote authorization	Authorization is granted by a remote server.



Accounting

- The accounting function monitors the network behavior and network resource utilization of authorized users.
- AAA supports two accounting modes: non-accounting and remote accounting.

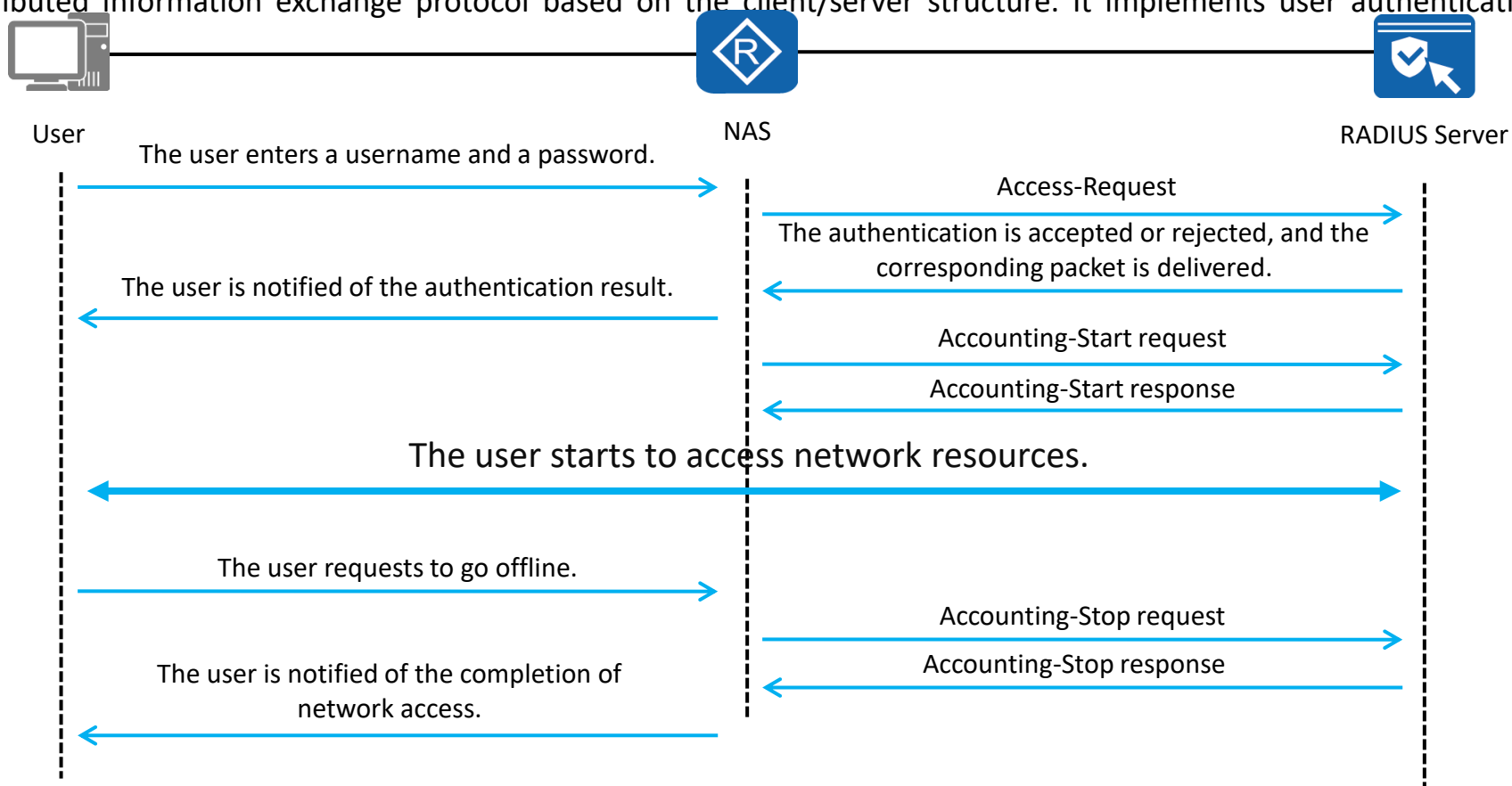


User	Domain	Accounting Mode
User 1@Domain 1	Domain 1	Non-accounting
User 2@Domain 2	Domain 2	Non-accounting
User 3@Domain 3	Domain 3	Remote accounting



AAA Implementation Protocol - RADIUS

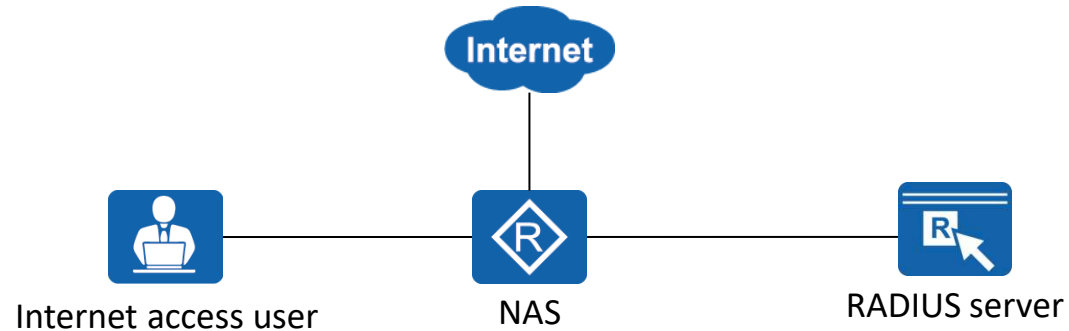
- Of the protocols that are used to implement AAA, RADIUS is the most commonly used.
- RADIUS is a distributed information exchange protocol based on the client/server structure. It implements user authentication, accounting, and authorization.





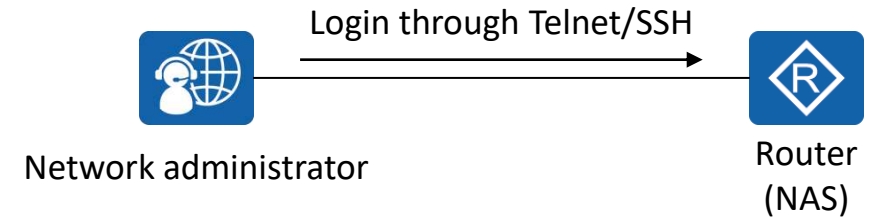
Common AAA Application Scenarios

AAA for Internet Access Users Through RADIUS



- AAA schemes are configured on the NAS to implement interworking between the NAS and RADIUS server.
- After the user enters a username and a password on the client, the NAS sends the username and password to the RADIUS server for authentication.
- If the authentication succeeds, the user is granted the Internet access permission.
- The RADIUS server can record the user's network resource utilization during Internet access.

Local Authentication and Authorization for Administrative Users



- After local AAA schemes are configured on Router, Router compares the username and password of the network administrator with the locally configured username and password when the network administrator logs in to Router.
- After the authentication succeeds, Router grants certain administrator permissions to the network administrator.



Contents

1. AAA Overview
- 2. AAA Configuration**



AAA Configuration (1)

1. Enter the AAA view.

```
[Huawei] aaa
```

Exit the system view and enter the AAA view.

2. Create an authentication scheme.

```
[Huawei-aaa] authentication-scheme authentication-scheme-name
```

Create an authentication scheme and enter the authentication scheme view.

```
[Huawei-aaa-authentication-scheme-name] authentication-mode { hwtacacs | local | radius }
```

Set the authentication mode to local authentication. By default, the authentication mode is local authentication.



AAA Configuration (2)

3. Create a domain and bind an authentication scheme to the domain.

```
[Huawei-aaa] domain domain-name
```

Create a domain and enter the domain view.

```
[Huawei-aaa-domain-name] authentication-scheme authentication-scheme-name
```

Bind the authentication scheme to the domain.

4. Create a user.

```
[Huawei-aaa] local-user user-name password cipher password
```

Create a local user and configure a password for the local user.

- If the username contains a delimiter "@", the character before "@" is the username and the character after "@" is the domain name.
- If the value does not contain "@", the entire character string represents the username and the domain name is the default one.



AAA Configuration (3)

5. Configure a user access type.

```
[Huawei-aaa] local-user user-name service-type { { terminal | telnet | ftp | ssh | snmp | http } | ppp | none }
```

Configure the access type of the local user. By default, all access types are disabled for a local user.

6. Configure a user level.

```
[Huawei-aaa] local-user user-name privilege level level
```

Specify the permission level of the local user.



AAA Configuration Examples

- After a user password and a user level are configured on R1, host A can use the configured username and password to remotely log in to R1.



```
[R1]aaa
[R1-aaa]local-user huawei password cipher huawei123
[R1-aaa]local-user huawei service-type telnet
[R1-aaa]local-user huawei privilege level 0
[R1]user-interface vty 0 4
[R1-ui-vty0-4]authentication-mode aaa
```



Configuration Verification (1)

- In AAA, each domain is associated with an authentication scheme, an authorization scheme, and an accounting scheme. In this example, the default domain is used.

```
[R1]display domain name default_admin
Domain-name:                default_admin
Domain-state:               Active
Authentication-scheme-name:  default
Accounting-scheme-name:     default
Authorization-scheme-name:   -
Service-scheme-name:        -
RADIUS-server-template:     -
HWTACACS-server-template:   -
User-group:                 -
```



Configuration Verification (2)

- After the user properly logs in and logs out, you can view the user record.

```
[R1]display aaa offline-record all
```

```
-----  
User name:          huawei  
Domain name:       default_admin  
User MAC: 00e0-fc12-3456  
User access type:   telnet  
User IP address:    10.1.1.2  
User ID:            1  
User login time:    2019/12/28 17:59:10  
User offline time:  2019/12/28 18:00:04  
User offline reason: user request to offline
```




Quiz

1. What authentication, authorization, and accounting modes are supported by AAA?
 2. When a new common user is configured with local authentication but is not associated with a user-defined domain, which domain does the user belong to?
-
1. AAA supports the following authentication/authorization modes: non-authentication, local authentication, and remote authentication.
AAA supports two accounting modes: non-accounting and remote accounting.
 1. If the domain to which a user belongs is not specified when the user is created, the user is automatically associated with the **default** domain (the administrator is associated with the **default_admin** domain).



Summary

- AAA improves enterprise network security and prevents unauthorized users from logging in to enterprise networks by authenticating the identities of enterprise employees and external users, authorizing accessible resources, and monitoring Internet access behavior.
 - Authentication: determines which users can access the network.
 - Authorization: authorizes users to access specific services.
 - Accounting: records network resource utilization.
- AAA technology can be implemented either locally or through a remote server.
- Of the protocols that are used to implement AAA, RADIUS is the most commonly used.

The image features a blue-tinted background showing silhouettes of several groups of business professionals in a modern office environment. They are standing on a highly reflective floor, and their reflections are clearly visible. The background also shows blurred outlines of skyscrapers, suggesting a high-rise office building. Overlaid on this scene is the text "Thank You" and the website address "www.huawei.com".

Thank You

www.huawei.com



Network Address Translation



Foreword

- With the development of the Internet and the increase of network applications, limited public IPv4 addresses have become the bottleneck of network development. To solve this problem, Network Address Translation (NAT) was introduced.
- NAT enables hosts on an internal network to access an external network. It not only helps alleviate IPv4 address shortage but also improves the security of the internal network as NAT prevents devices on the external network from directly communicating with hosts on the internal network that uses private addresses.
- This course describes the motivation behind NAT, and implementations and application scenarios of different types of NAT.



Objectives

- On completion of this course, you will be able to:
 - Understand the motivation behind NAT.
 - Master NAT classification and implementations.
 - Master NAT selection in different scenarios.



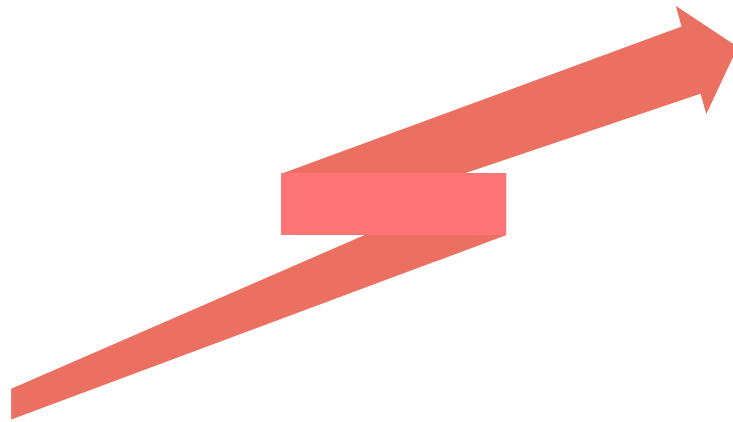
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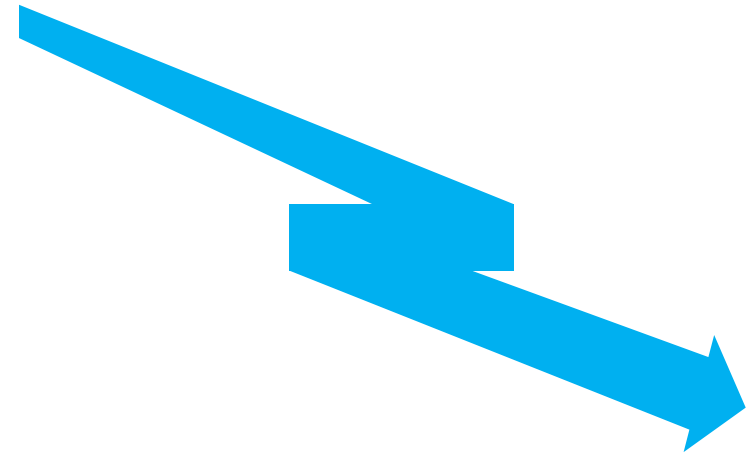


Motivation Behind NAT

- As the number of Internet users increases, public IPv4 addresses become scarcer.
- What's worse, uneven allocation of these addresses has resulted in a severe shortage of available public IPv4 addresses in some areas.
- To overcome public IPv4 address shortage, it is necessary to use transition technologies.



Internet users



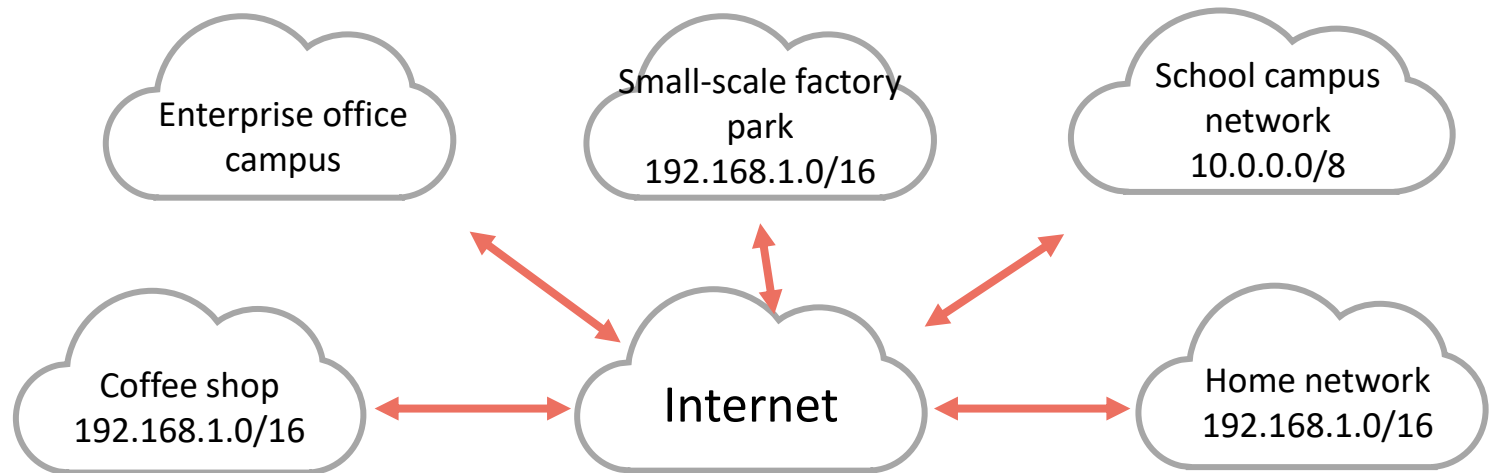
Public IPv4 addresses

0



Private IP Addresses

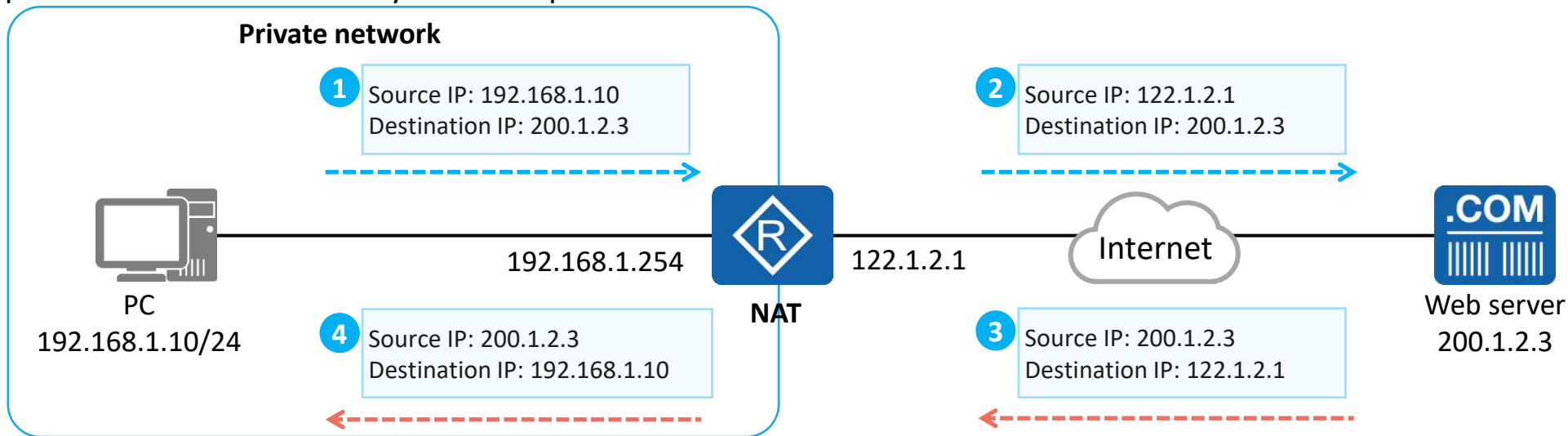
- Public IP addresses: managed and allocated by a dedicated organization and can be used for direct communication on the Internet
- Private IP addresses: can be used by organizations or individuals randomly on internal networks, but cannot be used for direct communication on the Internet
- The following Class A, B, and C addresses are reserved as private IP addresses:
 - Class A: 10.0.0.0–10.255.255.255
 - Class B: 172.16.0.0–172.31.255.255
 - Class C: 192.168.0.0–192.168.255.255





NAT Implementation

- NAT: translates IP addresses in IP data packets. It is widely used on live networks and is usually deployed on network egress devices, such as routers or firewalls.
- Typical NAT application scenario: Private addresses are used on private networks (enterprises or homes), and NAT is deployed on egress devices. For traffic from an internal network to an external network, NAT translates the source addresses of the data packets into specific public addresses. For traffic from an external network to an internal network, NAT translates the destination address of the data packets.
- NAT+private addresses effectively conserve public IPv4 addresses.





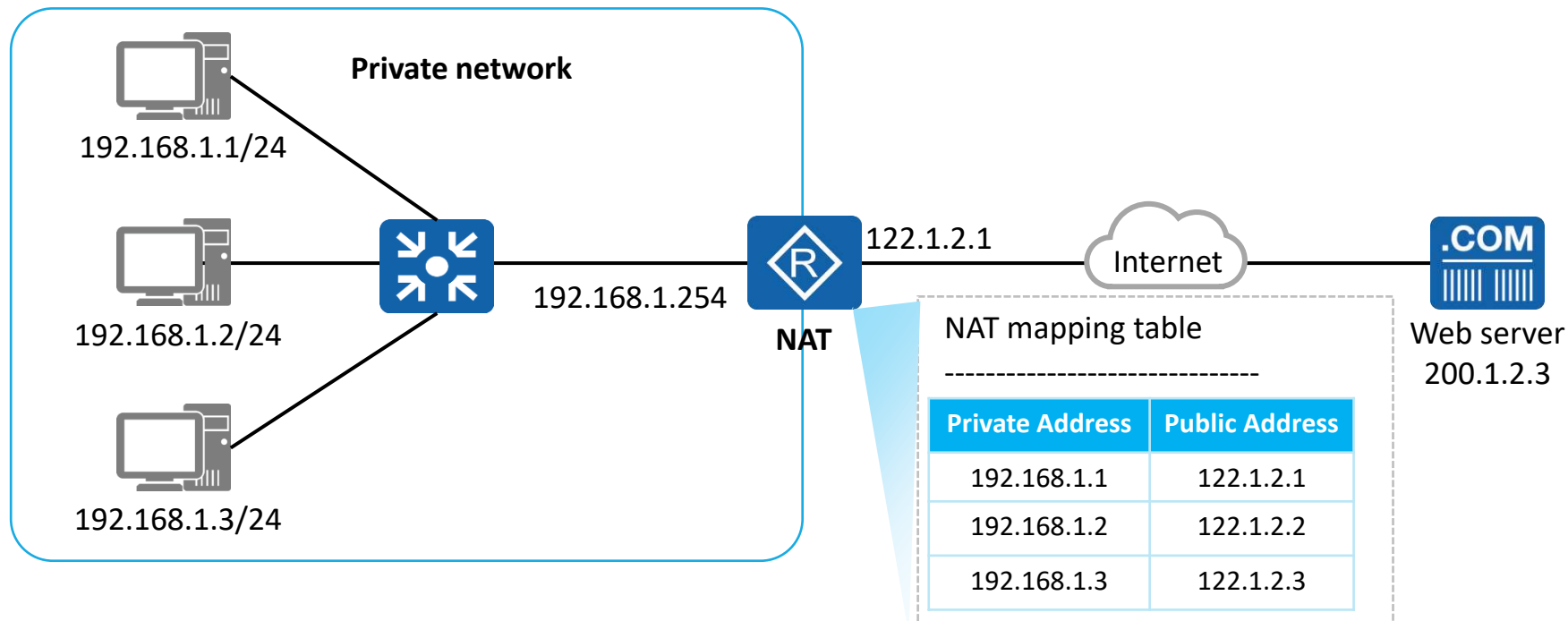
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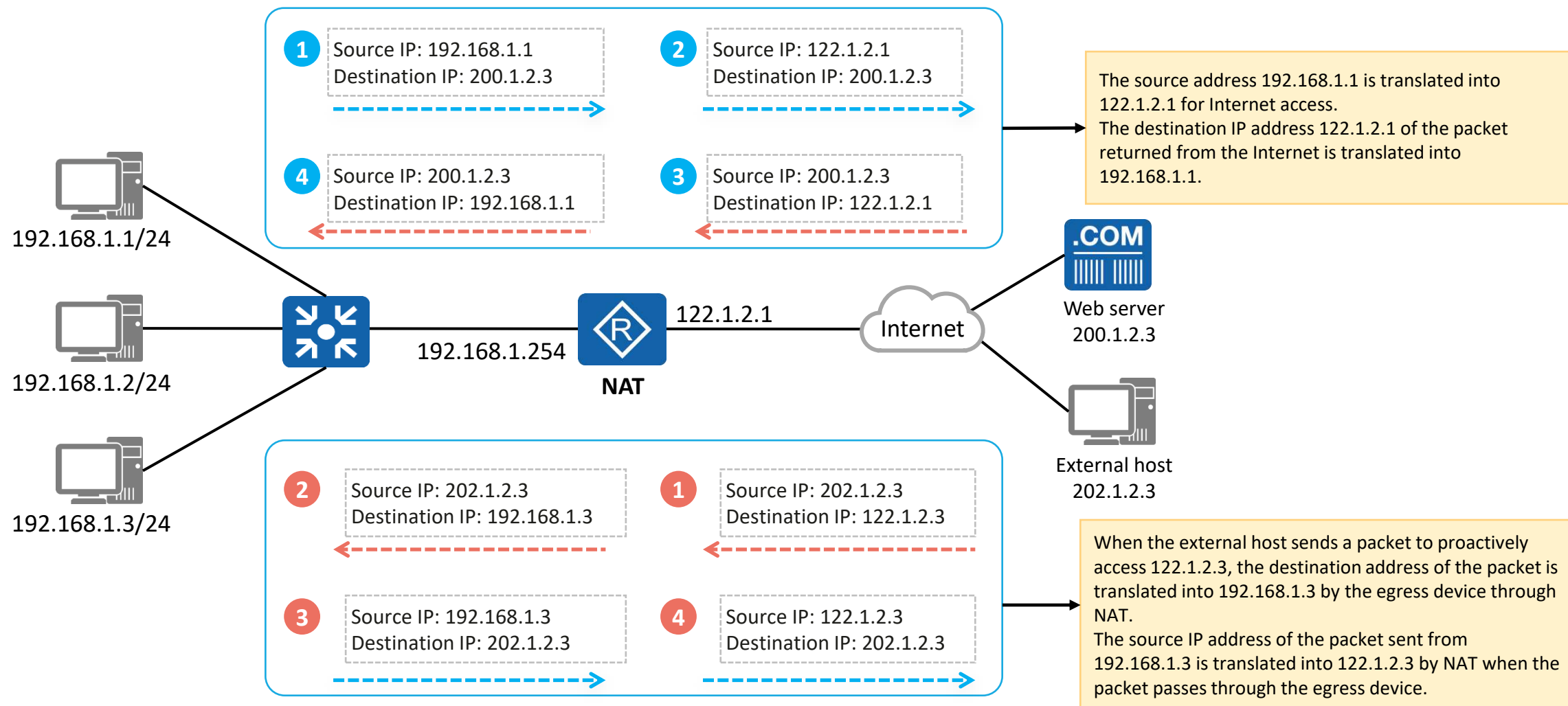
Static NAT Implementation

- Static NAT: A private IP address is mapped to a fixed public IP address.
- Bidirectional access: When an internal host with a private IP address accesses the Internet, the egress NAT device translates the private IP address into a public IP address. Similarly, when an external network device sends packets to access an internal network, the NAT device translates the public address (destination address) carried in the packets into a private address.





Static NAT Example





Configuring Static NAT

1. Method 1: Configure static NAT in the interface view.

```
[Huawei-GigabitEthernet0/0/0] nat static global { global-address } inside { host-address }
```

global { *global-address* } is used to configure an external public IP address, and **inside** { *host-address* } is used to configure an internal private IP address.

2. Method 2: Configure static NAT in the system view.

```
[Huawei] nat static global { global-address } inside { host-address }
```

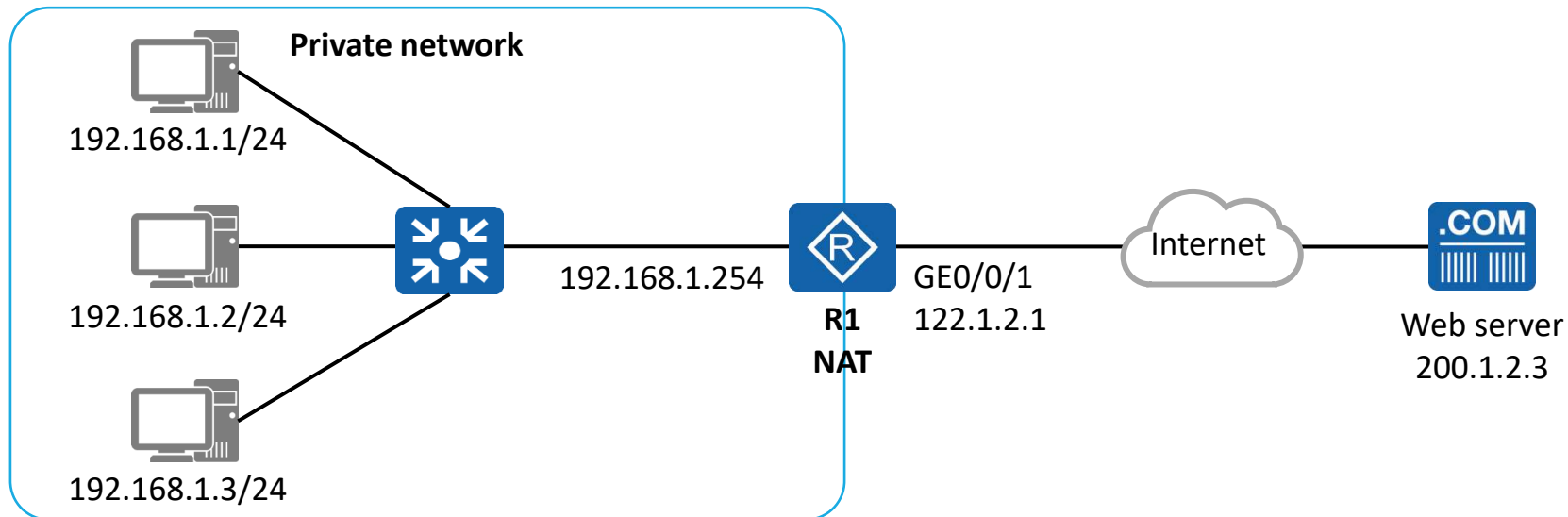
The command format in the system view is the same as that in the interface view. After this configuration, enable static NAT on a specific interface.

```
[Huawei-GigabitEthernet0/0/0] nat static enable
```

This command enables static NAT on the interface.



Example for Configuring Static NAT



- Configure static NAT on R1 to map private addresses of internal hosts to public addresses in one-to-one mode.

```
[R1]interface GigabitEthernet0/0/1
[R1-GigabitEthernet0/0/1]ip address 122.1.2.1 24
[R1-GigabitEthernet0/0/1]nat static global 122.1.2.1 inside 192.168.1.1
[R1-GigabitEthernet0/0/1]nat static global 122.1.2.2 inside 192.168.1.2
[R1-GigabitEthernet0/0/1]nat static global 122.1.2.3 inside 192.168.1.3
```



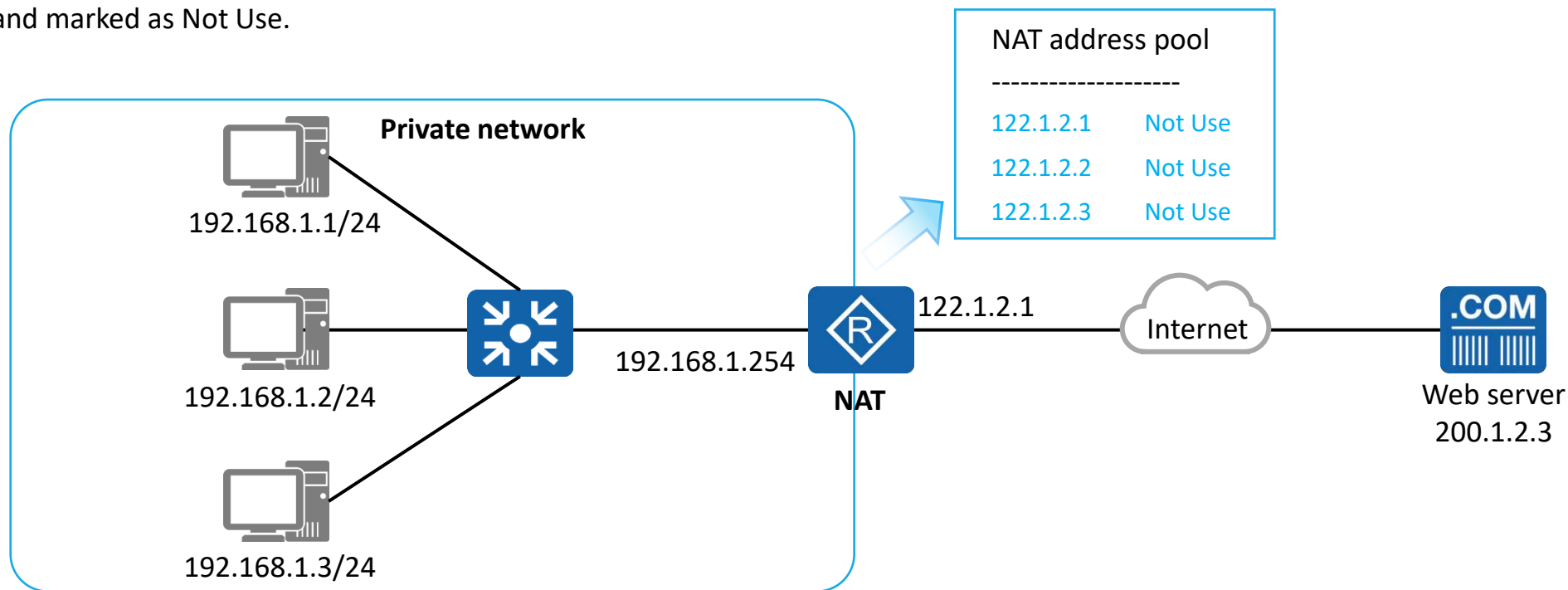
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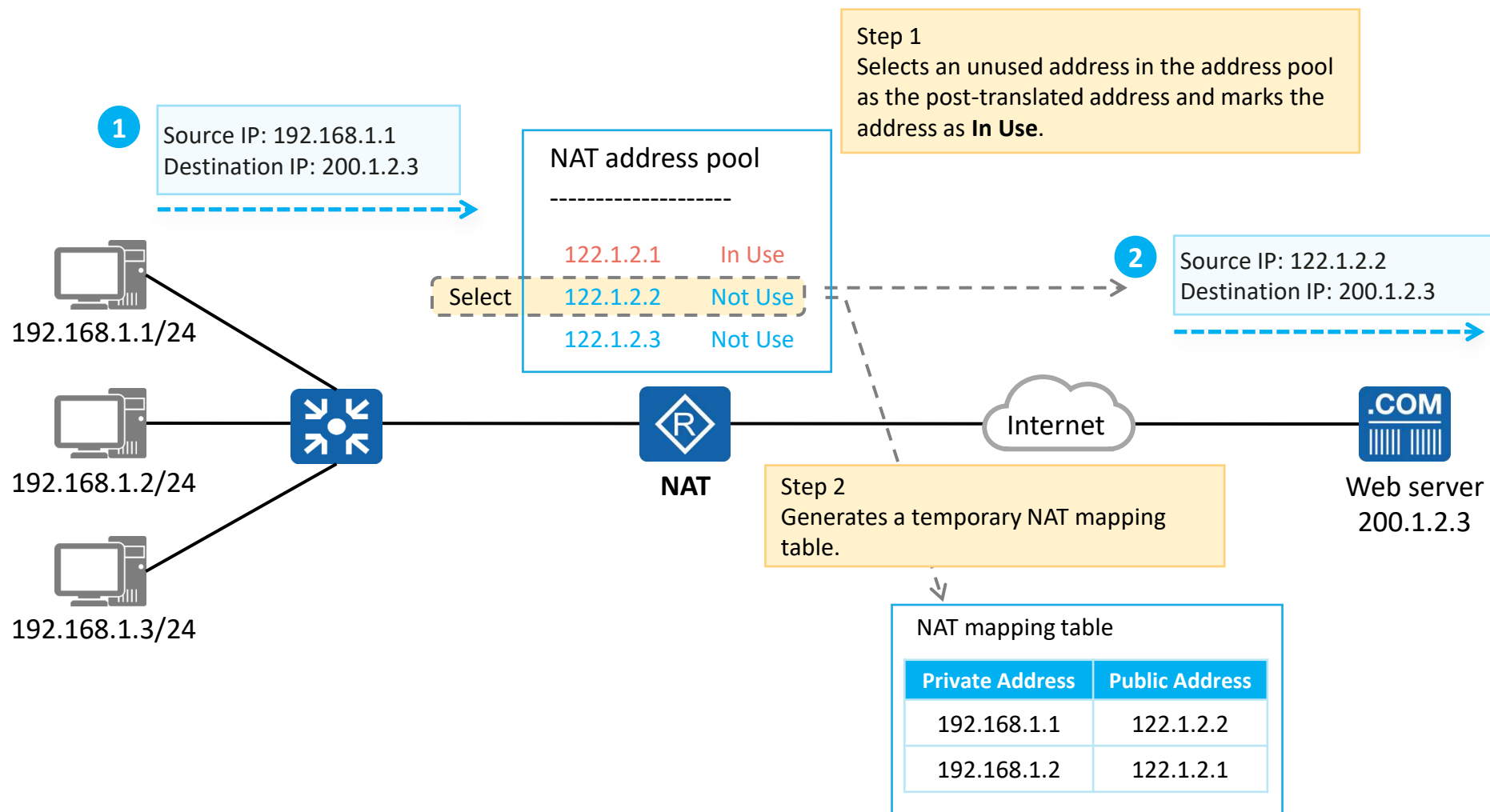
Dynamic NAT Implementation

- Dynamic NAT: A private IP address is mapped to a public IP address from a NAT address pool containing a group of public IP addresses. Static NAT strictly maps addresses in one-to-one mode. As a result, even if an internal host is offline for a long time or does not send data, the public address is still occupied by the host.
- Dynamic NAT prevents such address wastes. When an internal host accesses an external network, an available IP address in a NAT address pool is temporarily assigned to the host and marked as In Use. When the host no longer accesses the external network, the assigned IP address is reclaimed and marked as Not Use.



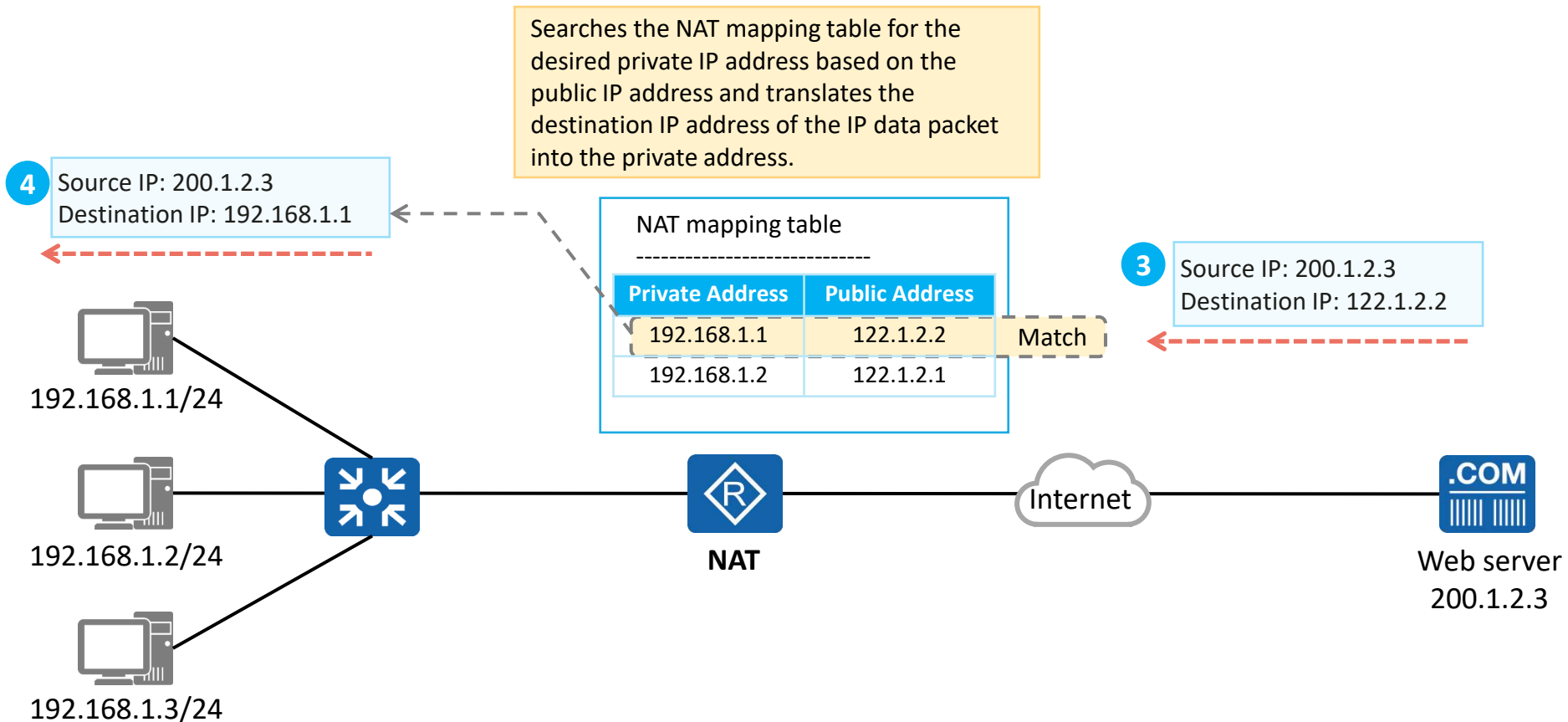


Dynamic NAT Example (1)





Dynamic NAT Example (2)





Configuring Dynamic NAT

1. Create an address pool.

```
[Huawei] nat address-group group-index start-address end-address
```

Configure a public address range. *group-index* specifies the address pool ID, and *start-address* and *end-address* specify the start and end addresses of the address pool, respectively.

2. Configure an ACL rule for NAT.

```
[Huawei] acl number  
[Huawei-acl-basic-number] rule permit source source-address source-wildcard
```

Configure a basic ACL to match the source address range that requires dynamic NAT.

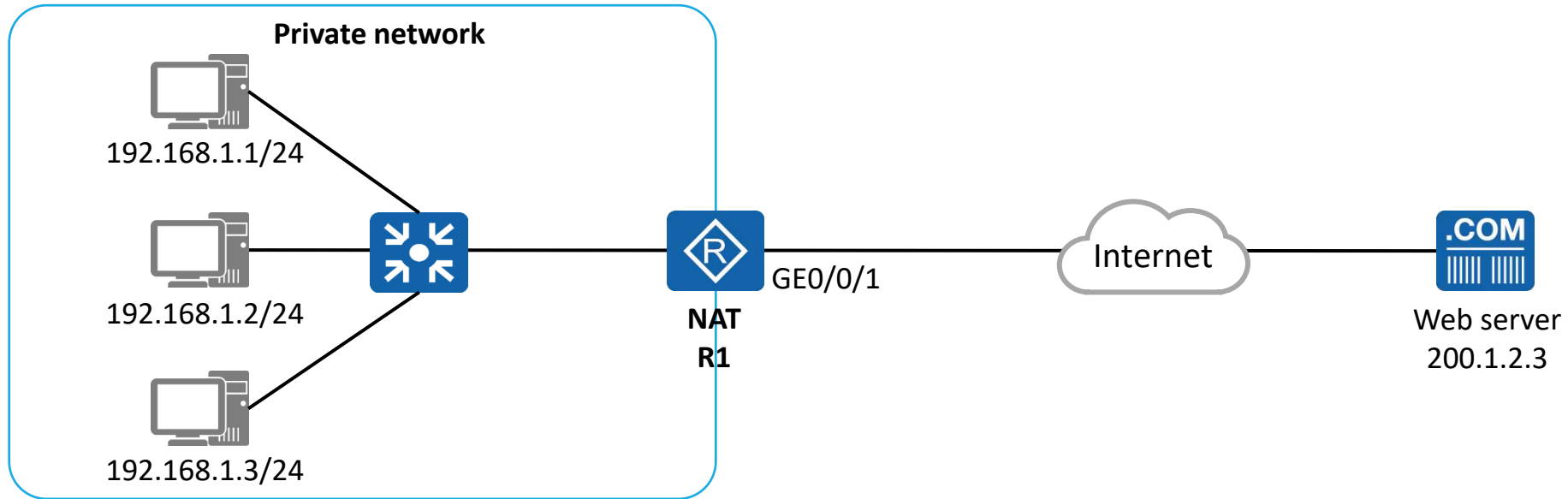
3. Configure outbound NAT with the address pool in the interface view.

```
[Huawei-GigabitEthernet0/0/0] nat outbound acl-number address-group group-index [ no-pat ]
```

Associate the ACL rule with the address pool for dynamic NAT on the interface. The **no-pat** parameter specifies that port translation is not performed.



Example for Configuring Dynamic NAT



- Configure dynamic NAT on R1 to dynamically map private addresses of internal hosts to public addresses.

```
[R1]nat address-group 1 122.1.2.1 122.1.2.3
[R1]acl 2000
[R1-acl-basic-2000]rule 5 permit source 192.168.1.0 0.0.0.255
[R1-acl-basic-2000]quit
[R1]interface GigabitEthernet0/0/1
[R1-GigabitEthernet0/0/1]nat outbound 2000 address-group 1 no-pat
```



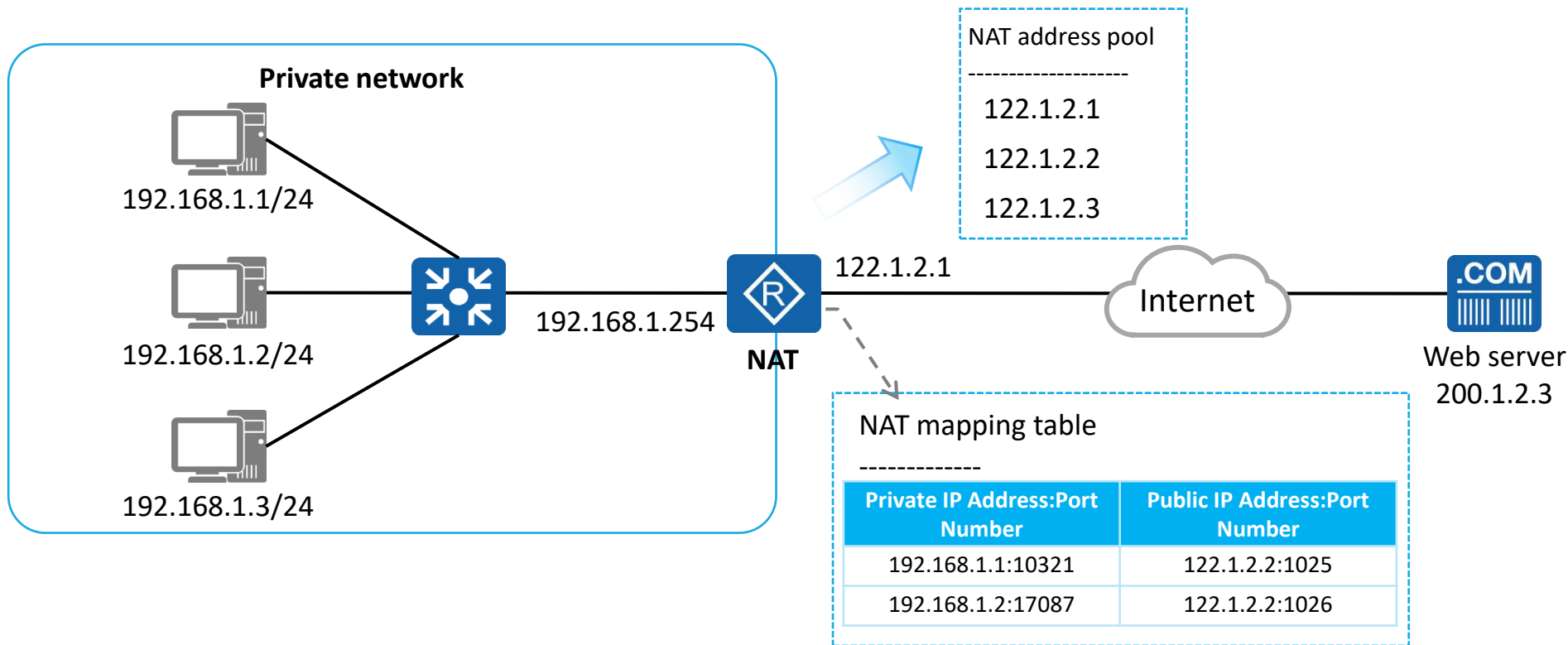
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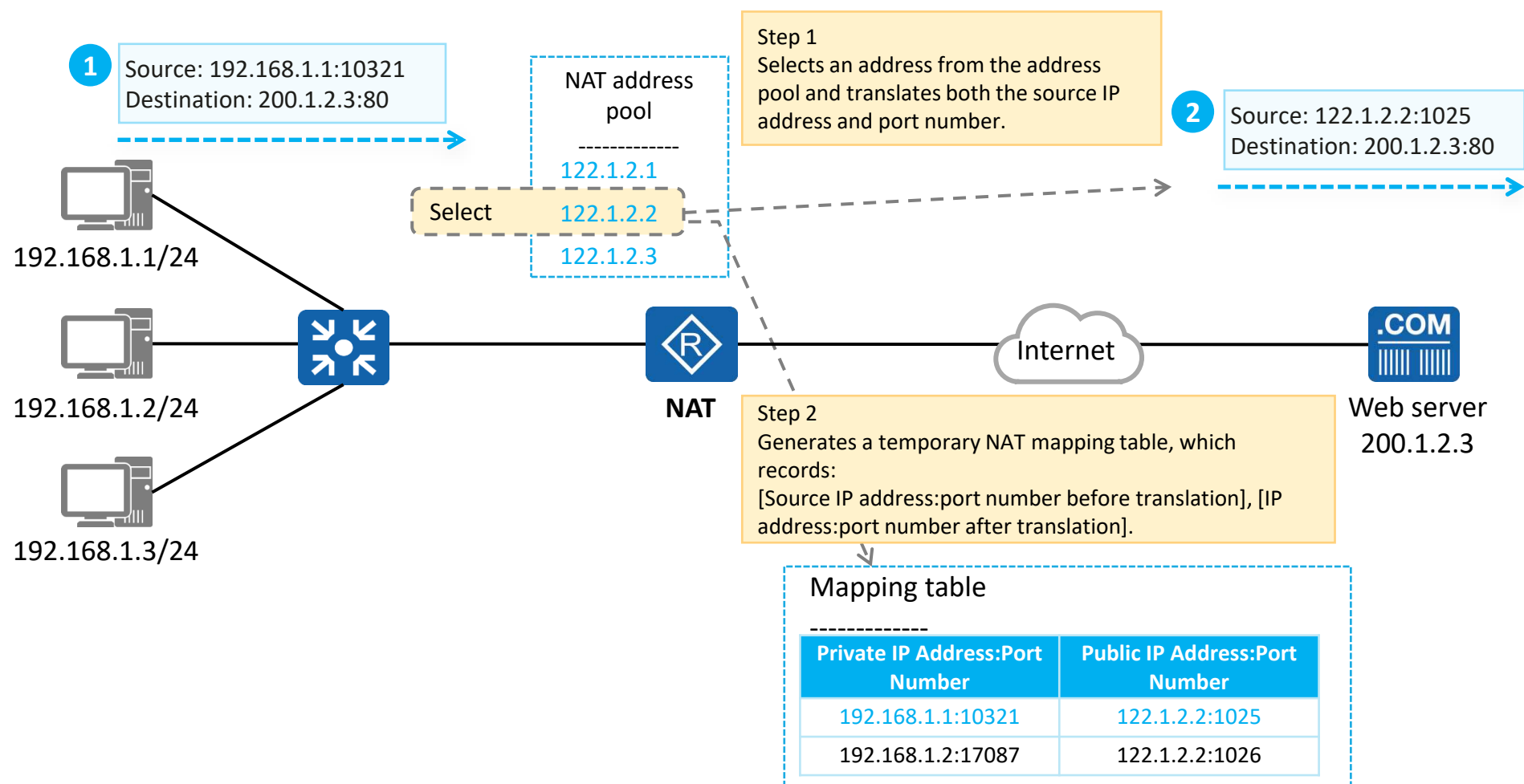
NAPT Implementation

- Dynamic NAT does not translate port numbers. It belongs to No-Port Address Translation (No-PAT). In this mode, the mapping between public and private addresses is still 1:1, which cannot improve public address utilization.
- Network Address and Port Translation (NAPT): translates both IP addresses and port numbers from multiple internal hosts to one public IP address in an address pool. In this way, 1:n mapping between public and private addresses is implemented, which effectively improves public address utilization.



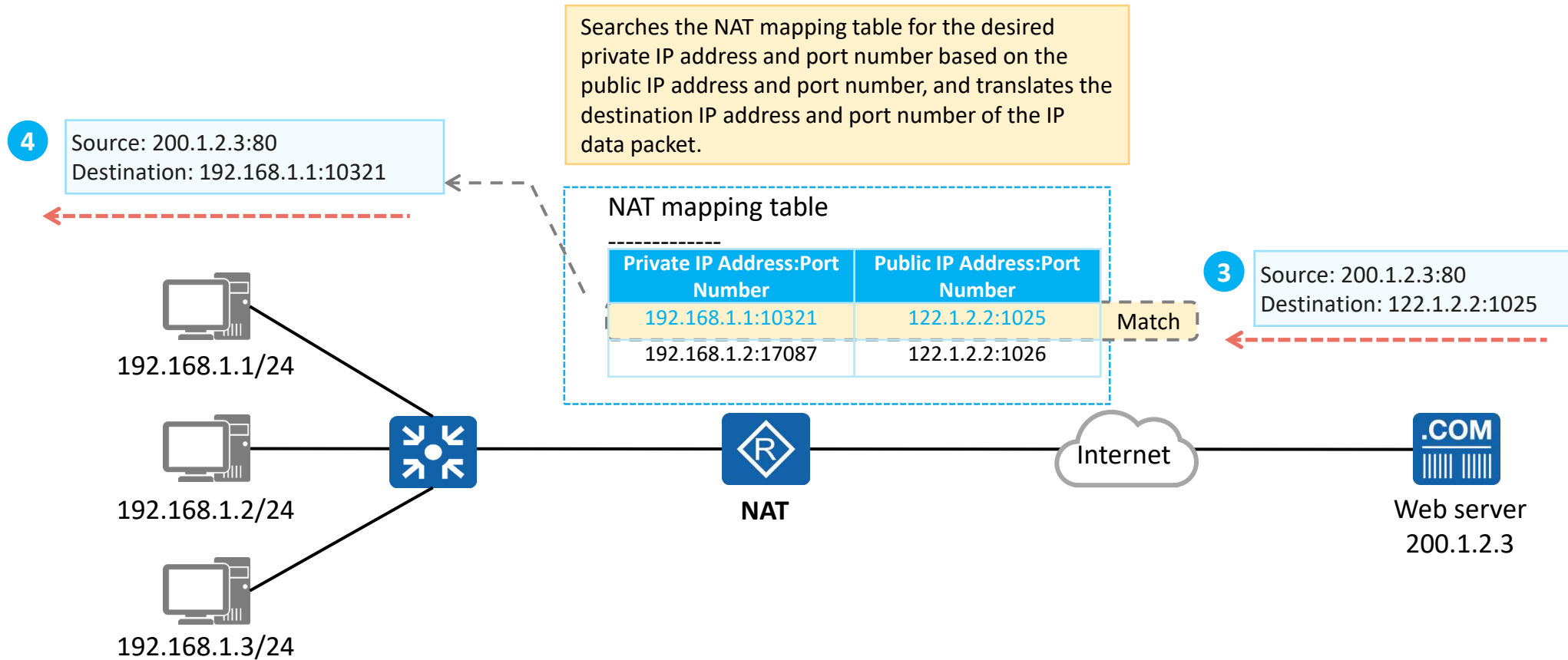


NAPT Example (1)



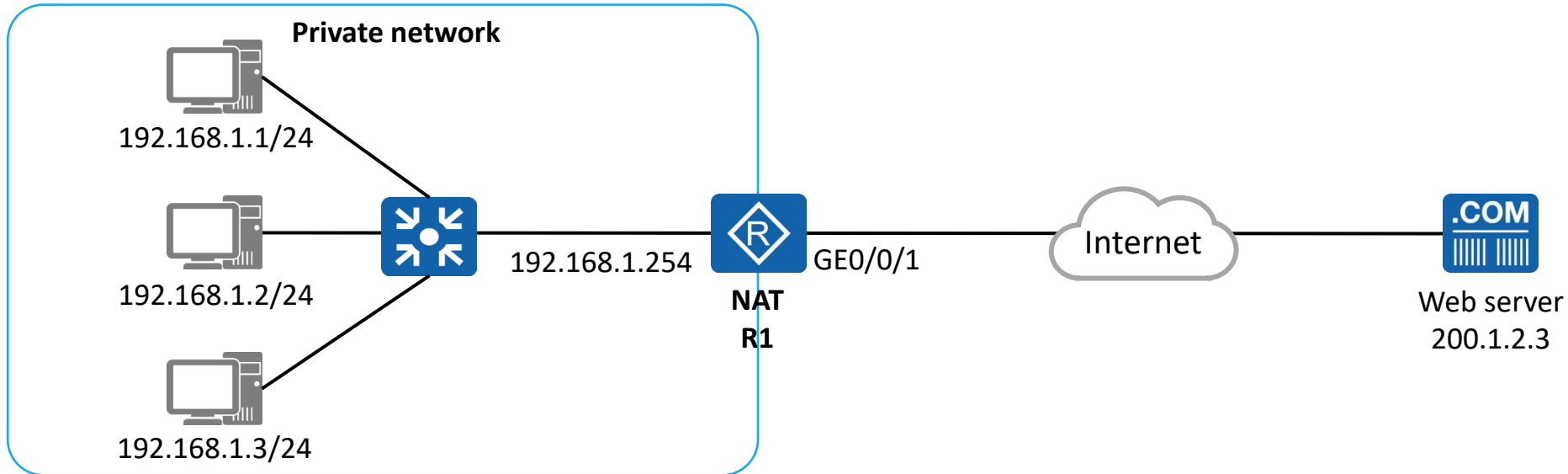


NAPT Example (2)





Example for Configuring NAPT



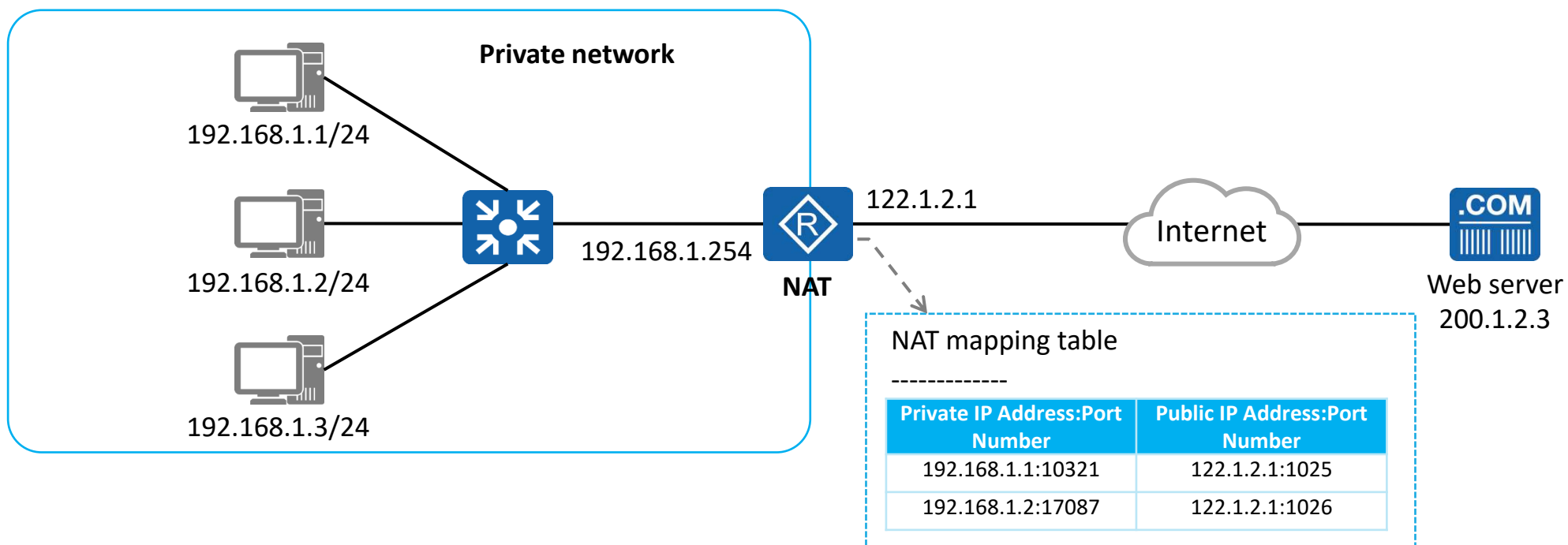
- Configure NAPT on R1 to allow all hosts with private IP addresses on the internal network to access the public network through 122.1.2.1.

```
[R1]nat address-group 1 122.1.2.1 122.1.2.1
[R1]acl 2000
[R1-acl-basic-2000]rule 5 permit source 192.168.1.0 0.0.0.255
[R1-acl-basic-2000]quit
[R1]interface GigabitEthernet0/0/1
[R1-GigabitEthernet0/0/1]nat outbound 2000 address-group 1
```



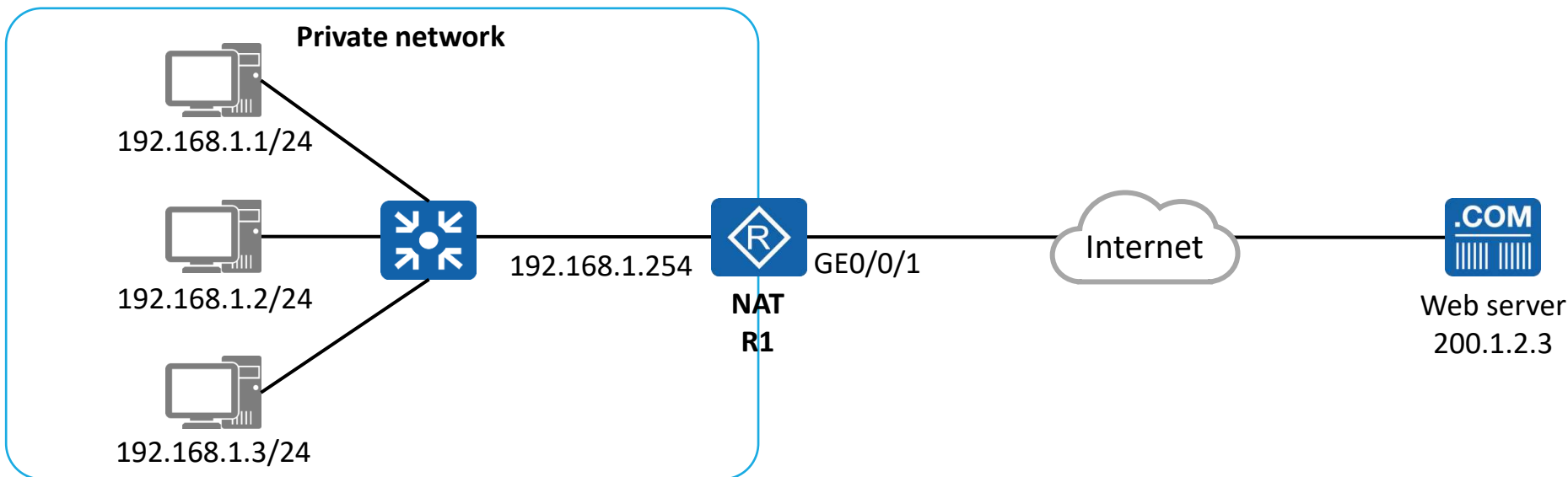
Easy IP

- Easy IP: translates both IP addresses and transport-layer port numbers. The implementation of Easy IP is the same as that of NAPT. The difference is that Easy IP does not involve address pools. It uses an interface address as a public address for NAT.
- Easy IP applies to scenarios where public IP addresses are not fixed, such as scenarios where public IP addresses are dynamically obtained by egress devices on private networks through DHCP or PPPoE dialup.





Example for Configuring Easy IP



- Configure Easy IP on R1 to allow all hosts with private IP addresses on the internal network to access the public network through 122.1.2.1.

```
[R1-acl-basic-2000]rule 5 permit source 192.168.1.0 0.0.0.255
[R1-acl-basic-2000]quit
[R1]interface GigabitEthernet0/0/1
[R1-GigabitEthernet0/0/1]nat outbound 2000
```



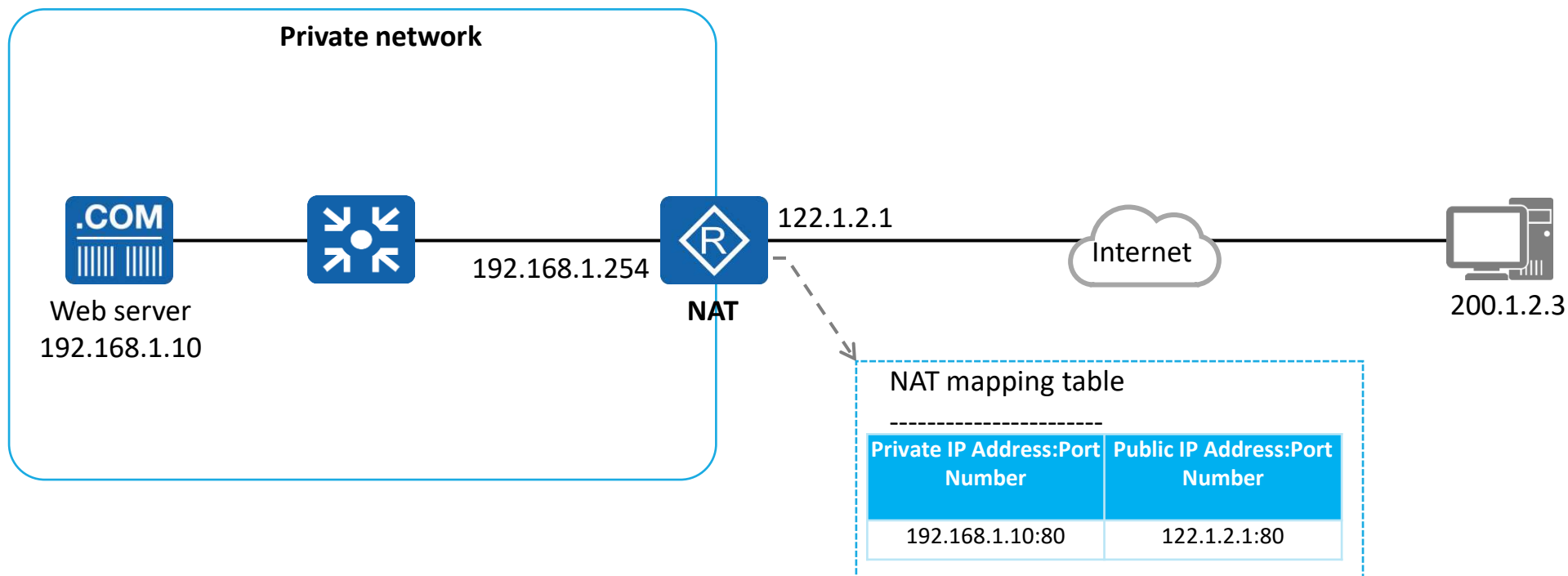
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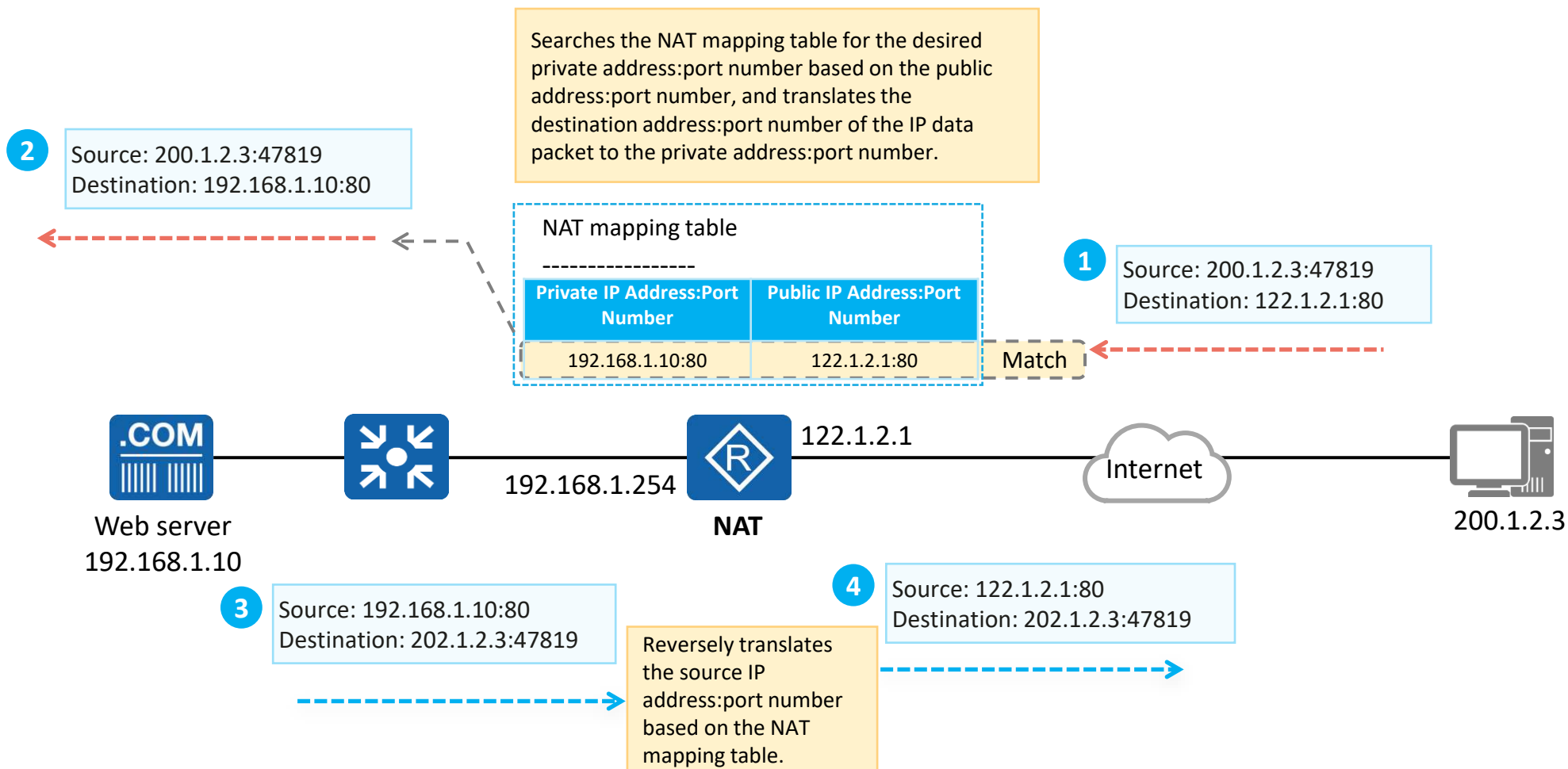
NAT Server

- NAT Server provides a one-to-one mapping between a public IP address:port and a private IP address:port, allowing an internal server to offer services to external users.
- External hosts access the **public IP:port**, and the NAT device translates the traffic to the corresponding **private server IP:port**.
- **Use Case:** Publishing internal servers (such as web, FTP, or mail servers) to the Internet.



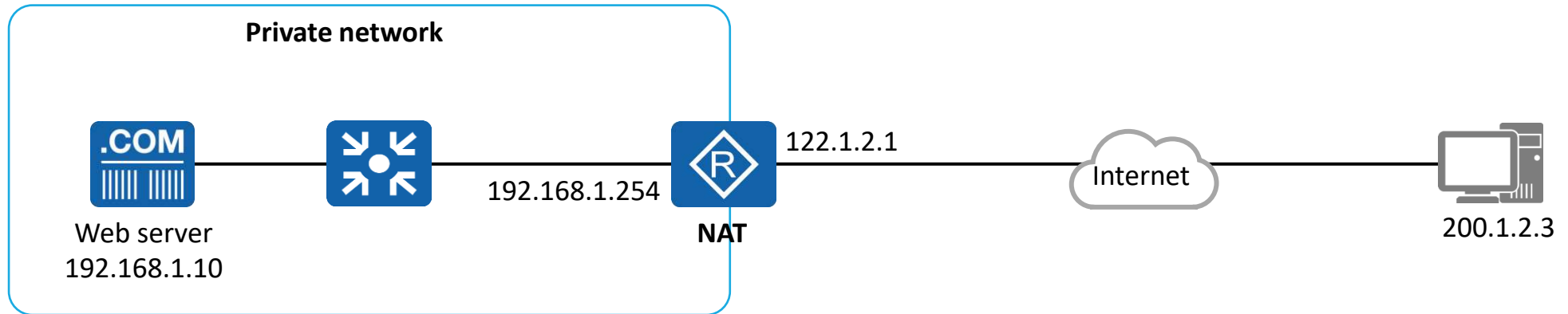


NAT Server Example





Example for Configuring NAT Server



- Configure NAT Server on R1 to map the internal server's IP address 192.168.1.10 and port number 80 to the public IP address 122.1.2.1 and port number 8080.

```
[R1]interface GigabitEthernet0/0/1
[R1-GigabitEthernet0/0/1]ip address 122.1.2.1 24
[R1-GigabitEthernet0/0/1]nat server protocol tcp global 122.1.2.1 www inside 192.168.1.10 8080
```




Quiz

1. What types of NAT can enable external devices to proactively access an internal server?
2. What are the advantages of NAPT over No-PAT?
 1. Static NAT and NAT Server provide a fixed mapping between a public address and an internal server, allowing external hosts to access the server through its public IP address.
 2. NAPT allows multiple private hosts to share a single public IP address by using different port numbers, improving public IP address utilization via conservation and low-cost.



Summary

- Using private addresses on private networks and using NAT at the network egress effectively reduce the number of required public IPv4 addresses. NAT effectively alleviates the shortage of public IPv4 addresses.
- Dynamic NAT, NAPT, and Easy IP provide source address translation for private network hosts to access the public network.
- NAT Server enables internal servers to provide services for public networks.
- Static NAT provides one-to-one mapping and supports bidirectional communication.

The background of the slide features a blue-tinted image of a modern office interior. In the foreground, several groups of business professionals are silhouetted against the bright light coming from large windows. They appear to be in various stages of collaboration, some looking at documents or devices. The floor is highly reflective, mirroring the figures. In the background, through the glass walls, a dense urban skyline with various skyscrapers is visible under a clear sky.

Thank You

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Network Services and Applications



Foreword

- The Internet has become an integral part of our lives, with a wide range of applications such as file transfer, email sending, online video, web browsing, and online gaming. Because of the layered network model, common users can use various services provided by the application layer, without knowing technical details such as communication technology implementations.
- In previous courses, we have learned technologies related to the data link layer, network layer, and transport layer. This chapter will describe common network services and applications such as FTP, DHCP, and HTTP.



Objectives

- On completion of this course, you will be able to:
 - Understand FTP fundamentals.
 - Understand TFTP fundamentals.
 - Understand DHCP fundamentals.
 - Understand Telnet fundamentals.
 - Understand HTTP fundamentals.
 - Understand DNS fundamentals.
 - Understand NTP fundamentals.



Contents

1. File Transfer

- FTP

- TFTP

2. Telnet

3. DHCP

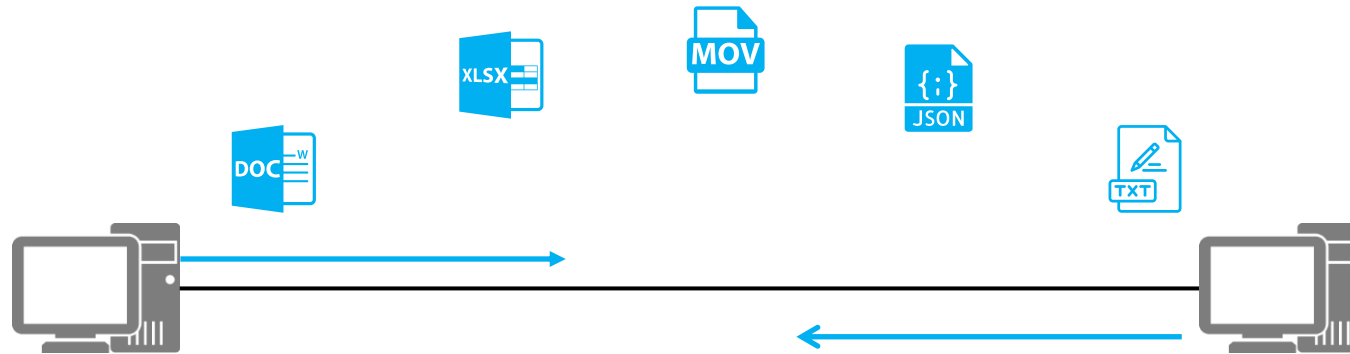
4. HTTP

5. DNS

6. NTP



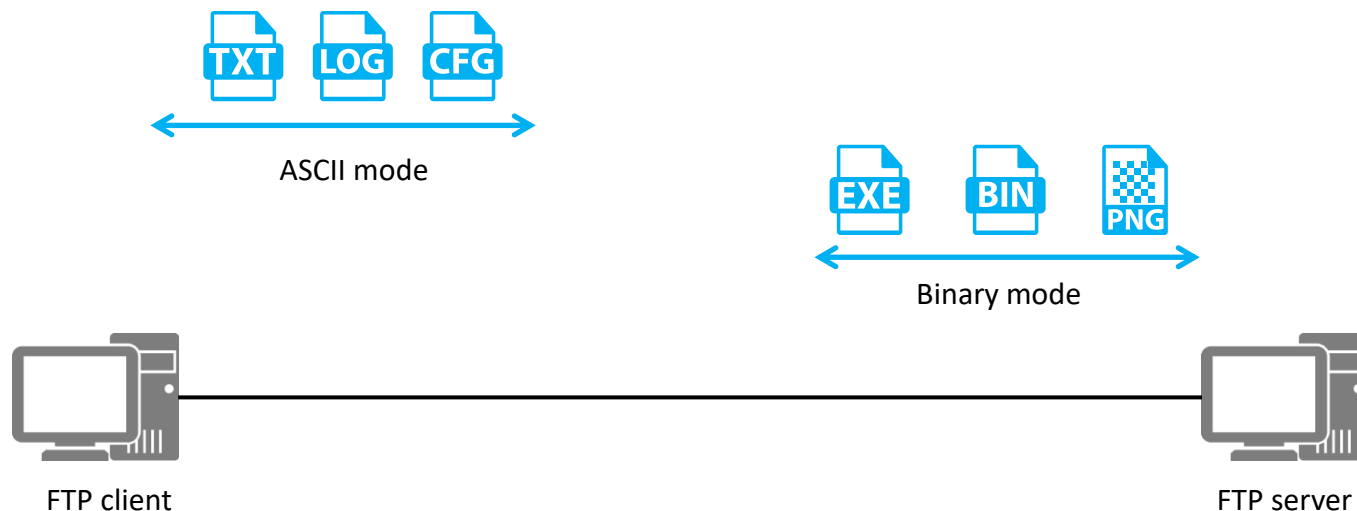
File Transfer Protocols



- File transfer between hosts is an important function of IP networks. Nowadays, people can conveniently transfer files using web pages and mailboxes.
- However, in the early Internet era when the World Wide Web (WWW) did not come into being and operating systems used command-line interfaces, people transferred files via command-line tools. The most commonly used protocols for transferring files at that time are File Transfer Protocol (FTP) and Trivial File Transfer Protocol (TFTP).



Basic Concepts of FTP

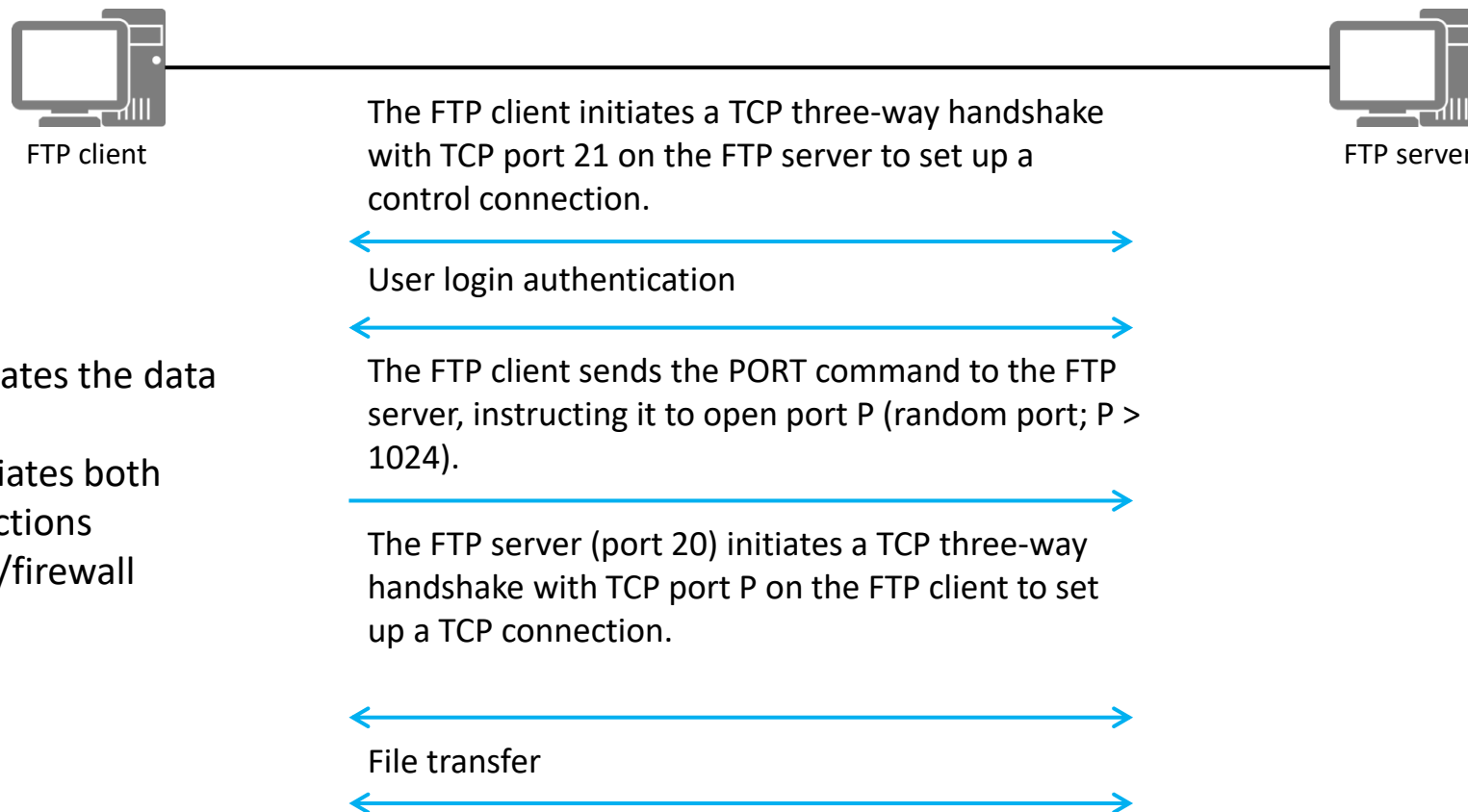


- FTP adopts the typical client/server (C/S) architecture. After an FTP client establishes a TCP connection with an FTP server, files can be uploaded and downloaded.
- FTP uses different transfer modes based on the file type:
 - **ASCII mode:** When a text file (in TXT, LOG, or CFG format) is transferred, the encoding mode of the text content is converted to improve the transfer efficiency. This mode is recommended for transferring configuration files and log files of network devices.
 - **Binary mode:** Non-text files (in CC, BIN, EXE, or PNG format), such as images and executable programs, are transferred in binary mode. This mode is recommended for transferring version files of network devices.



FTP Transfer Process - Active Mode

- FTP works in two modes: active mode (PORT) and passive mode (PASV).

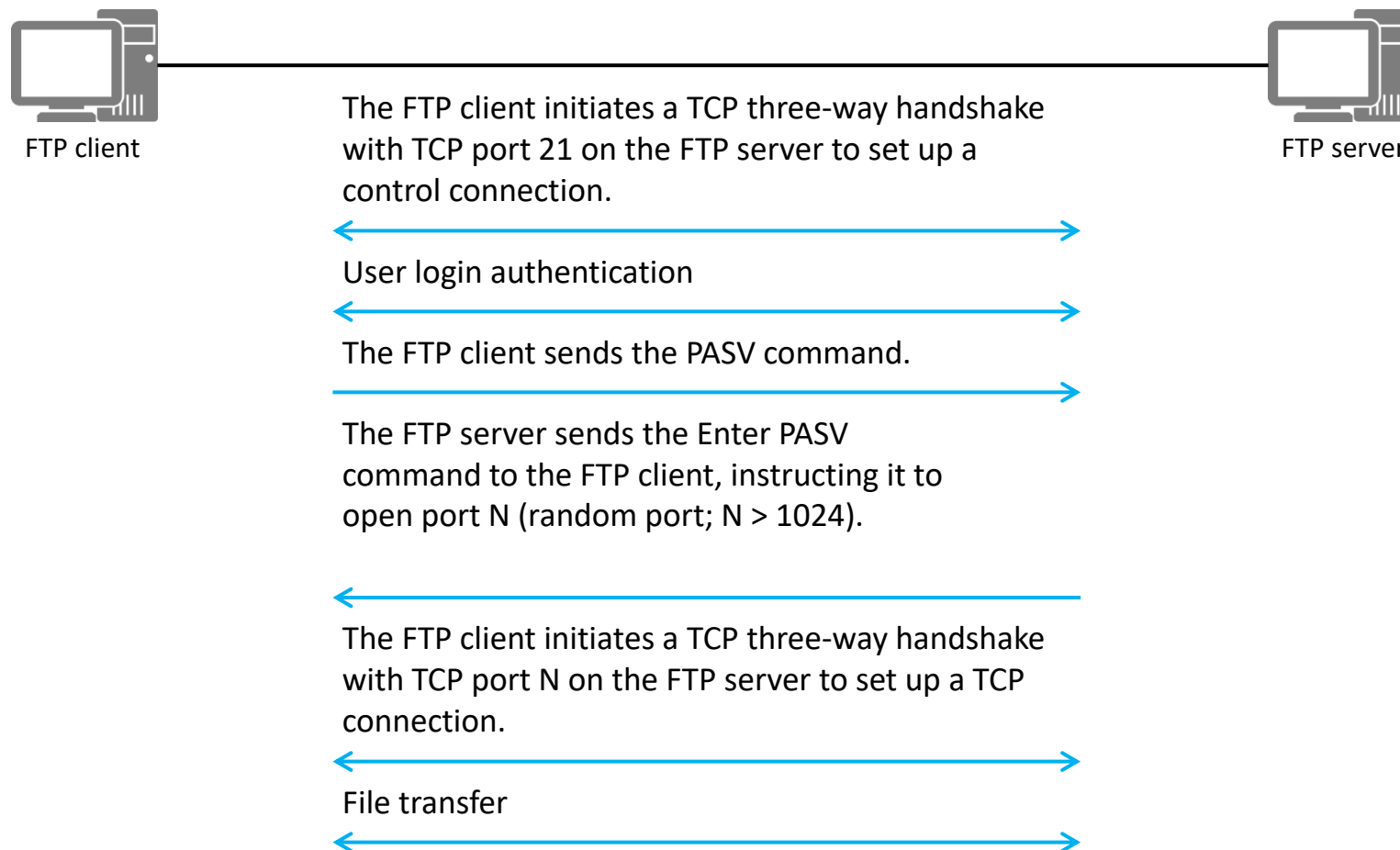


•**Active FTP:** Server initiates the data connection.

•**Passive FTP:** Client initiates both control and data connections (recommended for NAT/firewall environments).



FTP Transfer Process - Passive Mode





Configuration Commands (Device as FTP Server)

A user accesses a device through FTP.

1. Enable the FTP server function.

```
[Huawei]ftp [ ipv6 ] server enable
```

By default, the FTP server function is disabled.

2. Configure a local FTP user.

```
[Huawei]aaa  
[Huawei]local-user user-name password irreversible-cipher password  
[Huawei]local-user user-name privilege level level  
[Huawei]local-user user-name service-type ftp  
[Huawei]local-user user-name ftp-directory directory
```

The privilege level must be set to level 3 or higher. Otherwise, the FTP connection fails.



Configuration Commands (Device as FTP Client)

1. A VRP device that functions as an FTP client accesses an FTP server.

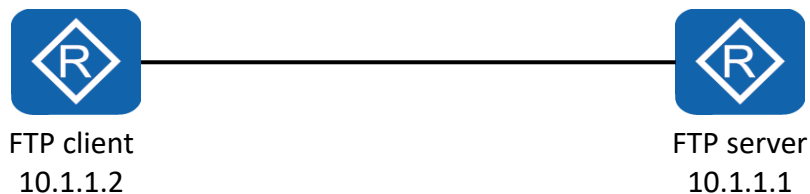
```
<FTP Client>ftp 10.1.1.1
Trying 10.1.1.1 ...
Press CTRL+K to abort
Connected to 10.1.1.1.
220 FTP service ready.
User(10.1.1.1:(none)):ftp
331 Password required for ftp.
Enter password:
230 User logged in.
```

2. Common commands used when the VRP device functions as an FTP client.

ascii	Set the file transfer type to ASCII, and it is the default type
binary	Set the file transfer type to support the binary image
ls	List the contents of the current or remote directory
passive	Set the toggle passive mode, the default is on
get	Download the remote file to the local host
put	Upload a local file to the remote host



Configuration Example



- One router functions as the FTP server, and the other as the FTP client.
- Enable the FTP service on the FTP server and create an FTP login account. Then, the FTP client logs in to the FTP server and runs the **get** command to download a file.

Configurations on the FTP server:

```
<Huawei> system-view
[Huawei] sysname FTP_Server
[FTP_Server] ftp server enable
[FTP_Server] aaa
[FTP_Server-aaa] local-user admin1234 password irreversible-cipher
Helloworld@6789
[FTP_Server-aaa] local-user admin1234 privilege level 15
[FTP_Server-aaa] local-user admin1234 service-type ftp
[FTP_Server-aaa] local-user admin1234 ftp-directory flash:
```

Operations on the FTP client:

```
<FTP Client>ftp 10.1.1.1
[FTP Client-ftp]get sslvpn.zip
200 Port command okay.
FTP: 828482 byte(s) received in 2.990 second(s) 277.08Kbyte(s)/sec.
```



Contents

1. File Transfer

- FTP

- **TFTP**

2. Telnet

3. DHCP

4. HTTP

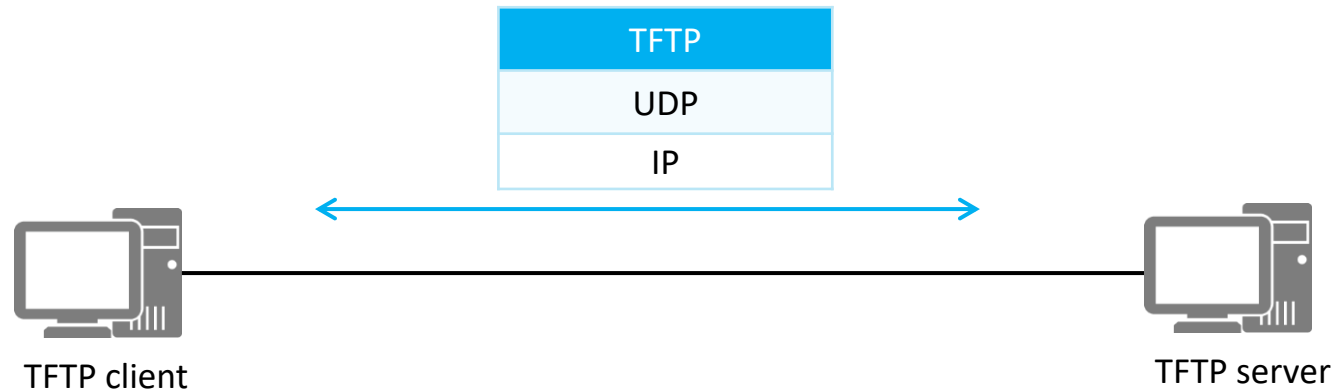
5. DNS

6. NTP



Basic Concepts of TFTP

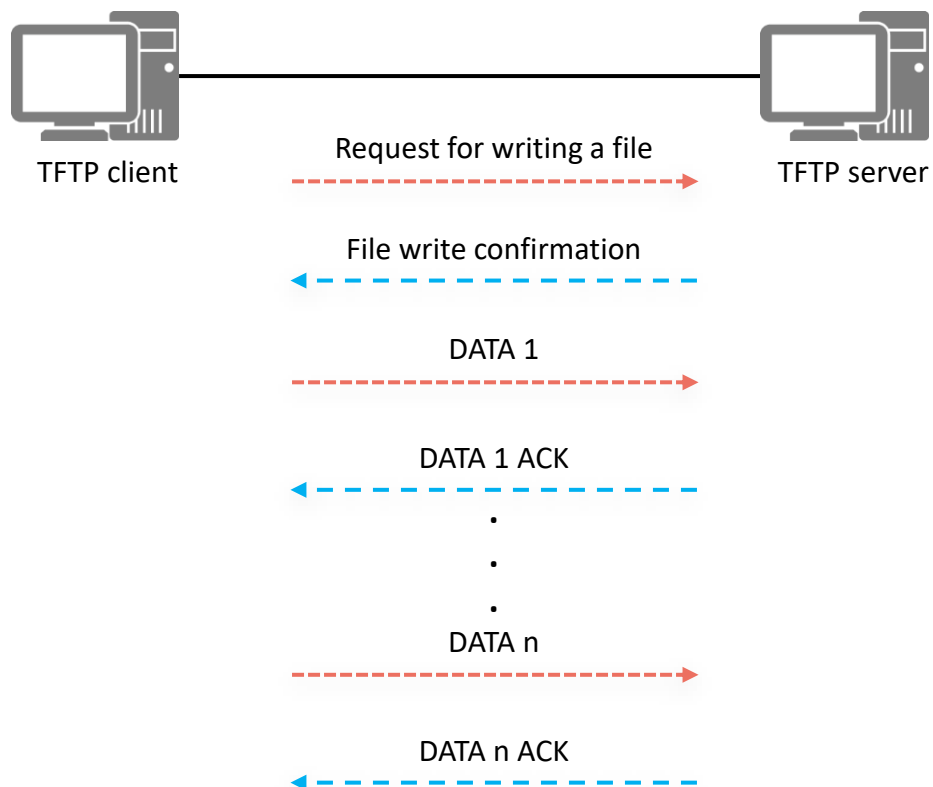
- Compared with FTP, TFTP is designed to transfer small files and is easier to implement.
 - Using UDP (port 69) for transmission
 - Authentication not required
 - You can only request a file from or upload a file to the server, but cannot view the file directory on the server.



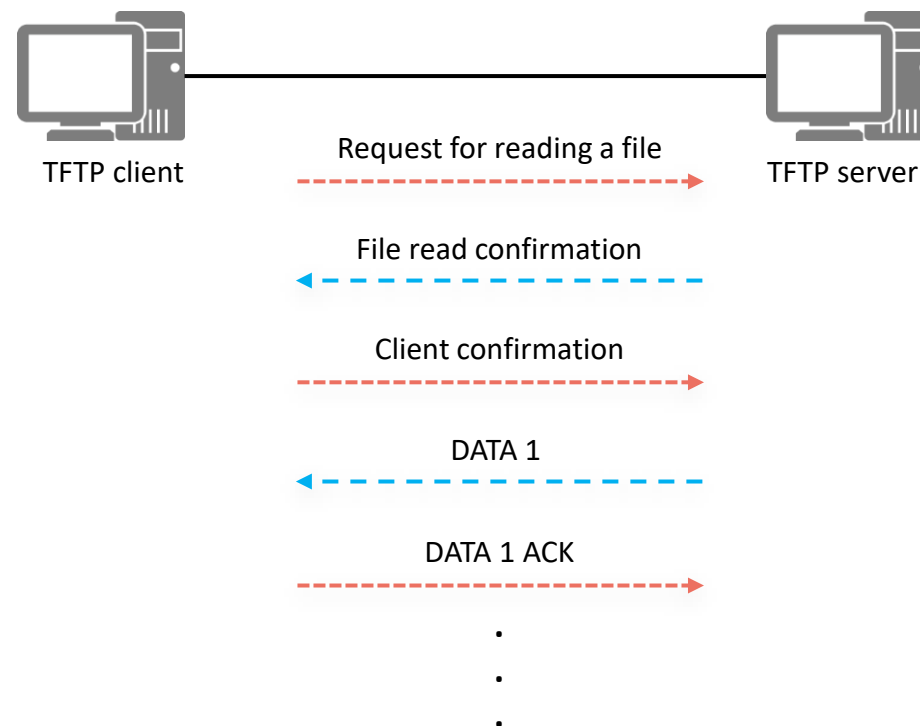


TFTP Transfer Example

Upload a File



Download a File





Configuration Commands (Device as TFTP Client)

1. Download a file (VRP device functioning as a TFTP client).

```
<HUAWEI> tftp tftp_server get filename
```

You do not need to log in to the TFTP server, and only need to enter the IP address of the TFTP server and the corresponding command.

2. Upload a file (VRP device functioning as a TFTP client).

```
<HUAWEI> tftp tftp_server put filename
```

You do not need to log in to the TFTP server, and only need to enter the IP address of the TFTP server and the corresponding command.

Currently, VRP devices can function only as TFTP clients.



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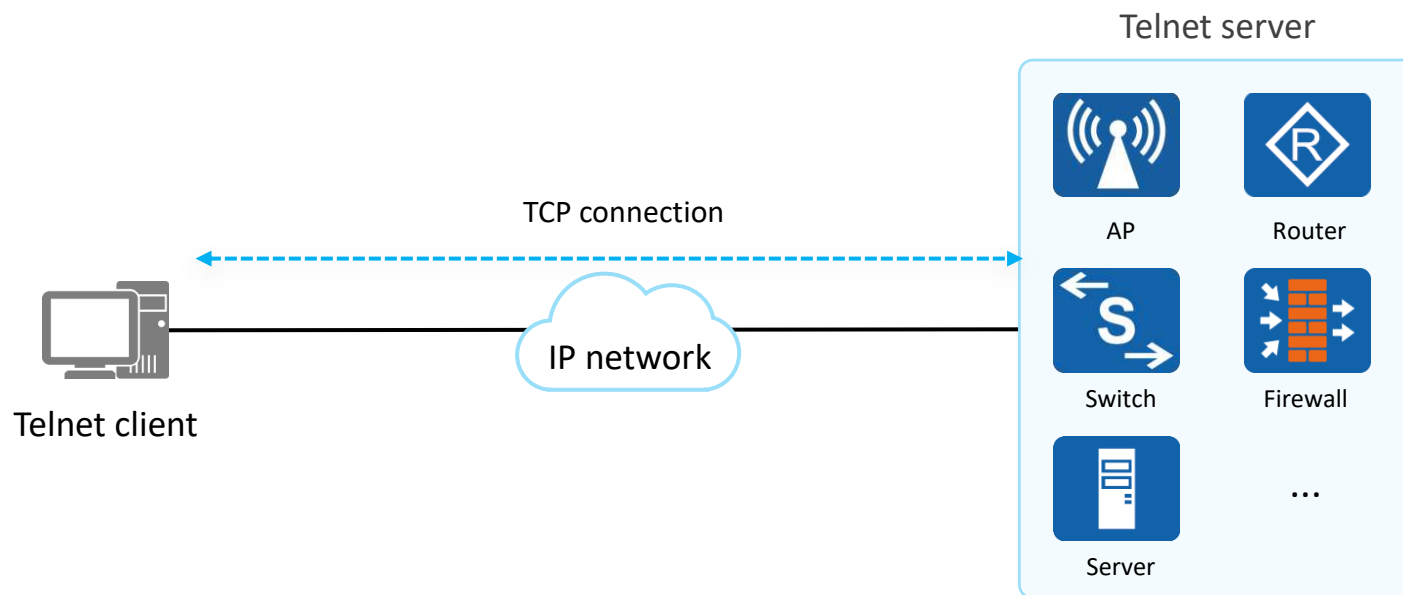
5. DNS

6. NTP



Application Scenario of Telnet

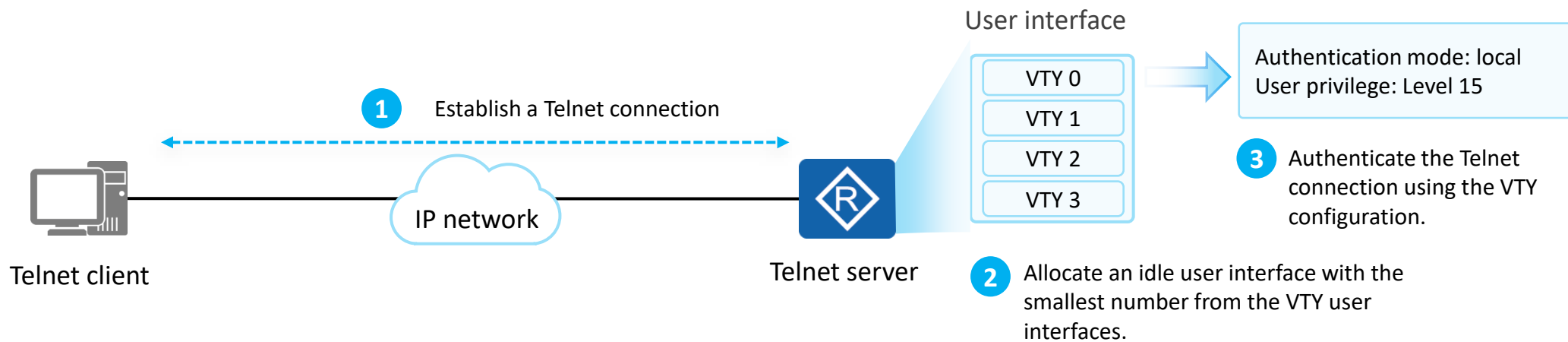
- To facilitate device management using commands, you can use Telnet to manage devices.
- Device management through Telnet is different from that using the console port. In Telnet-based device management mode, no dedicated cable is required to directly connect to the console port of the Telnet server, as long as the Telnet server's IP address is reachable and Telnet clients can communicate with the Telnet server's TCP port 23.
- The device that can be managed through Telnet is called the Telnet server, and the device connecting to the Telnet server is called the Telnet client. Many network devices can act as both the Telnet server and Telnet client.





VTY User Interface

- When a user logs in to a device using the console port or Telnet, the system allocates a user interface to manage and monitor the current session between the device and the user. A series of parameters can be set in each user interface view to specify the authentication mode and user privilege level after login. After a user logs in to a device, user operations that can be performed depend on the configured parameters.
- The user interface type of Telnet is virtual type terminal (VTY) user interface.





Configuration Commands (1)

1. Enable the Telnet server function.

```
[Huawei] telnet server enable
```

The Telnet server function is enabled on the device (disabled by default). To disable this function, run the **undo telnet server enable** command.

2. Enter the user view.

```
[Huawei] user-interface vty first-ui-number [ last-ui-number ]
```

The VTY user interface view is displayed. VTY user interfaces may vary according to device models.

3. Configure protocols supported by the VTY user interface.

```
[Huawei-ui-vty0-4] protocol inbound { all | telnet | ssh }
```

By default, the VTY user interface supports Secure Shell (SSH) and Telnet.



Configuration Commands (2)

4. Configure the authentication mode and the authentication password in password authentication mode.

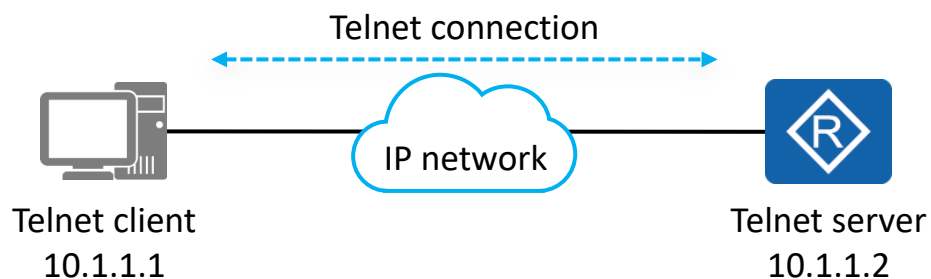
```
[Huawei-ui-vty0-4] authentication-mode {aaa | none | password}  
[Huawei-ui-vty0-4] set authentication password cipher
```

By default, no default authentication mode is available. You need to manually configure an authentication mode.

The **set authentication password cipher** command implementation varies according to VRP versions. In some versions, you need to press **Enter** and then enter the password. In other versions, you can directly enter the password after the command.



Configuration Example (1)



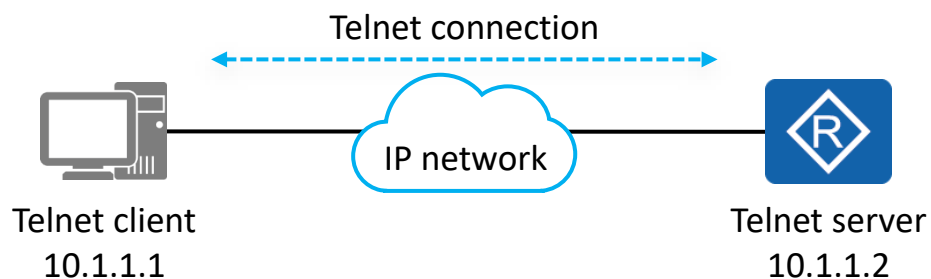
- Configure the router at 10.1.1.2 as the Telnet server and set the authentication mode to AAA local authentication. Create an account named **huawei**, set the password to **Huawei@123**, and set the privilege level to 15.
- Log in to and manage the Telnet server through the Telnet client.

Configurations on the Telnet server:

```
<Huawei> system-view
[Huawei] telnet server enable
[Huawei] aaa
[Huawei-aaa] local-user huawei password irreversible-cipher
Huawei@123
[Huawei-aaa] local-user huawei privilege level 15
[Huawei-aaa] local-user huawei service-type telnet
[Huawei-aaa] quit
[Huawei] user-interface vty 0 4
[Huawei-ui-vty0-4] authentication-mode aaa
```




Configuration Example (2)



- Configure the router at 10.1.1.2 as the Telnet server and set the authentication mode to AAA local authentication. Create an account named **huawei**, set the password to **Huawei@123**, and set the privilege level to 15.
- Log in to and manage the Telnet server through the Telnet client.

Operations on the Telnet client:

```
<Host>telnet 10.1.1.2
```

Login authentication

Username:huawei

Password:

Info: The max number of VTY users is 5, and the number
of current VTY users on line is 1.

The current login time is 2020-01-08 15:37:25.

```
<Huawei>
```



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Issues Faced by Manual Network Parameter Configuration (1)

Too Many Hard-to-Understand Parameters

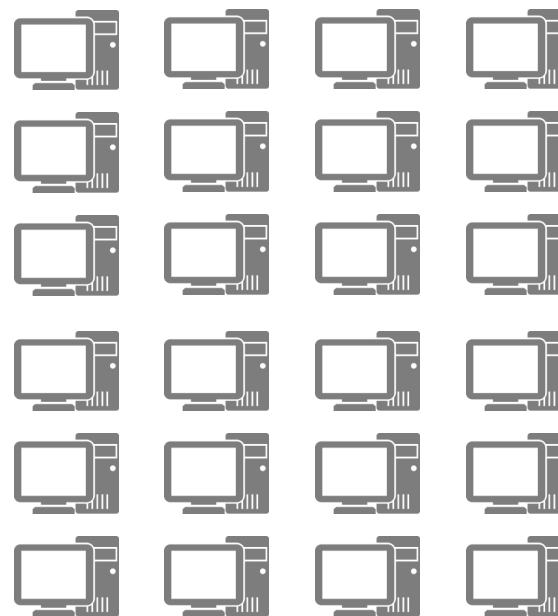
IPv4 address configuration:

IP address	
Mask	
Gateway	



- Common users are not familiar with network parameters and misconfiguration often occurs, resulting in network access failure. Random IP address configuration may cause IP address conflicts.

Huge Workload



Work Plan of This Week	
☐	Address allocation
☐	Address allocation
☐	Address configuration
☐	Address configuration

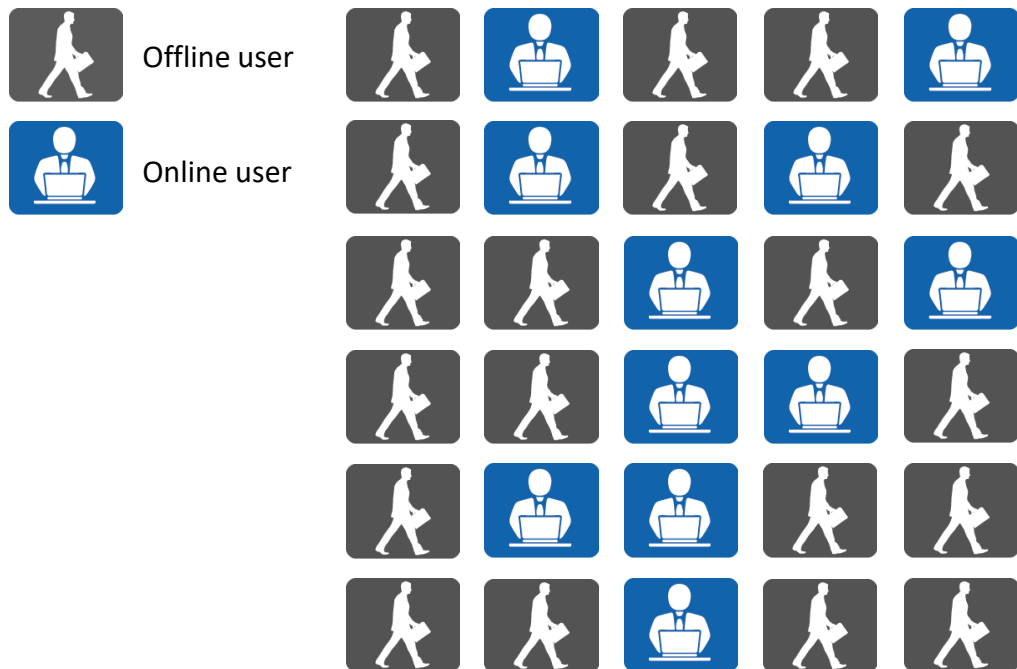


- Network administrators centrally configure network parameters, with heavy workloads and repetitive tasks.
- Network administrators need to plan and allocate IP addresses to users in advance.



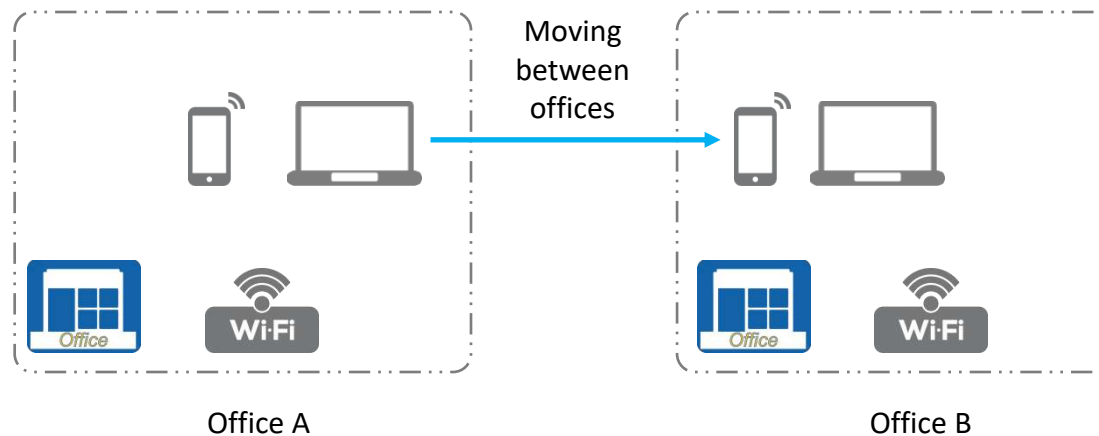
Issues Faced by Manual Network Parameter Configuration (2)

Low Utilization



- On an enterprise network, each user uses a fixed IP address. As a result, the IP address utilization is low, and some IP addresses may remain unused for a long time.

Poor Flexibility

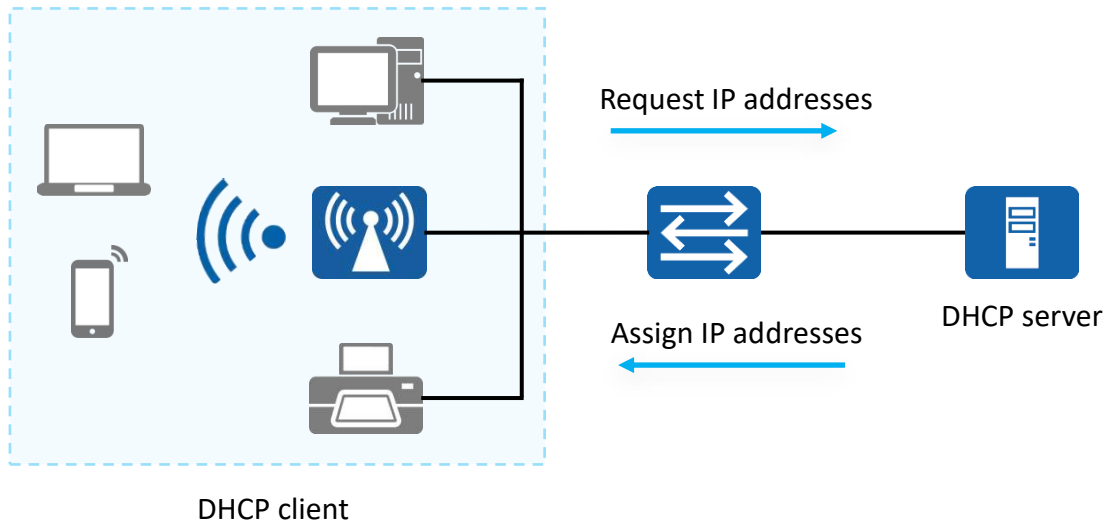


- Wireless local area networks (WLANs) allow for flexible station (STA) access locations. When a STA moves from one wireless coverage area to another, the IP address of the STA may need to be reconfigured.



Basic Concepts of DHCP

DHCP Working Principle



- To overcome the disadvantages of the traditional static IP configuration mode, the Dynamic Host Configuration Protocol (DHCP) is developed to dynamically assign suitable IP addresses to hosts. DHCP follows a client/server architecture in which hosts do not require manual configuration.
- DHCP assigns IP addresses using the **DORA process (Discover, Offer, Request, Acknowledge)**, through which a client automatically obtains an IP address from a DHCP server.
- This enables plug-and-play connectivity allowing devices to join the network automatically.



DHCP (Dynamic Host Configuration Protocol)

Automatically assigns IP addresses and network parameters to clients. Uses UDP (port 67 server, 68 client).

DHCP 4-step Process (DORA):

- 1. Discover: Client broadcasts DHCP Discover to find servers.
- 2. Offer: Server unicasts DHCP Offer with available IP address.
- 3. Request: Client broadcasts DHCP Request to accept offered IP (notifies all servers).
- 4. ACK: Server unicasts DHCP ACK confirming the lease.

LEASE RENEWAL	At 50% of lease (T1): Client unicasts Request to same server. At 87.5% (T2): Client broadcasts Request to any server (rebinding).
------------------	---

[Huawei] dhcp enable

[Huawei-GE0/0/0] dhcp select interface # Use interface subnet as pool

[Huawei-GE0/0/0] dhcp server dns-list 8.8.8.8

[Huawei-GE0/0/0] dhcp server excluded-ip-address 10.1.1.1 # Exclude router IP

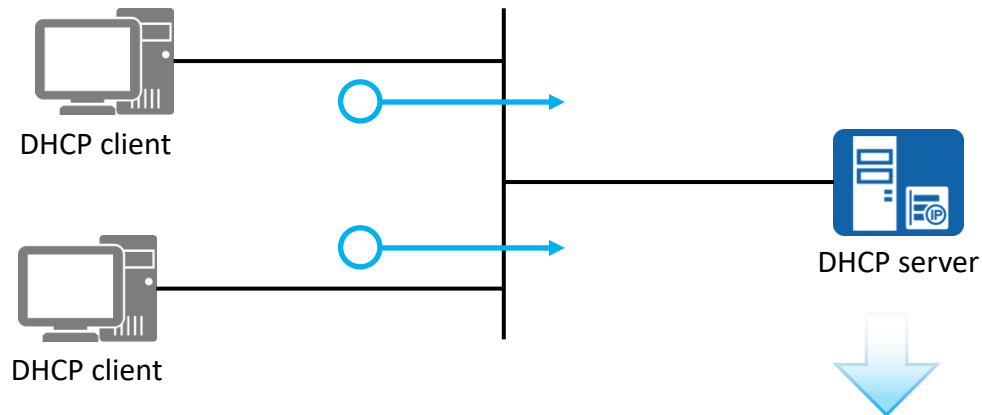
[Huawei-GE0/0/0] dhcp server lease day 3



DHCP Advantages

Unified Management

○ DHCP address request



Pool-No 1

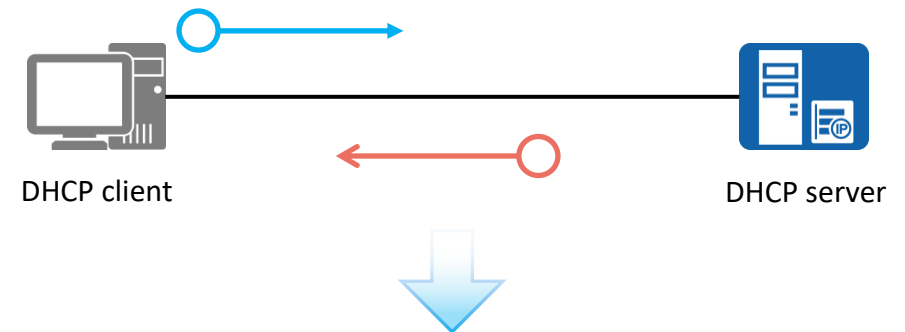
DNS-server 10.1.1.2 | Gateway 10.1.2.1
Network 10.1.2.0 | Mask 255.255.255.0
Total Used
252 2

- IP addresses are obtained from the address pool on the DHCP server. The DHCP server records and maintain the usage status of IP addresses for unified IP address assignment and management.

IP Address Lease

○ DHCP address request

○ DHCP address response

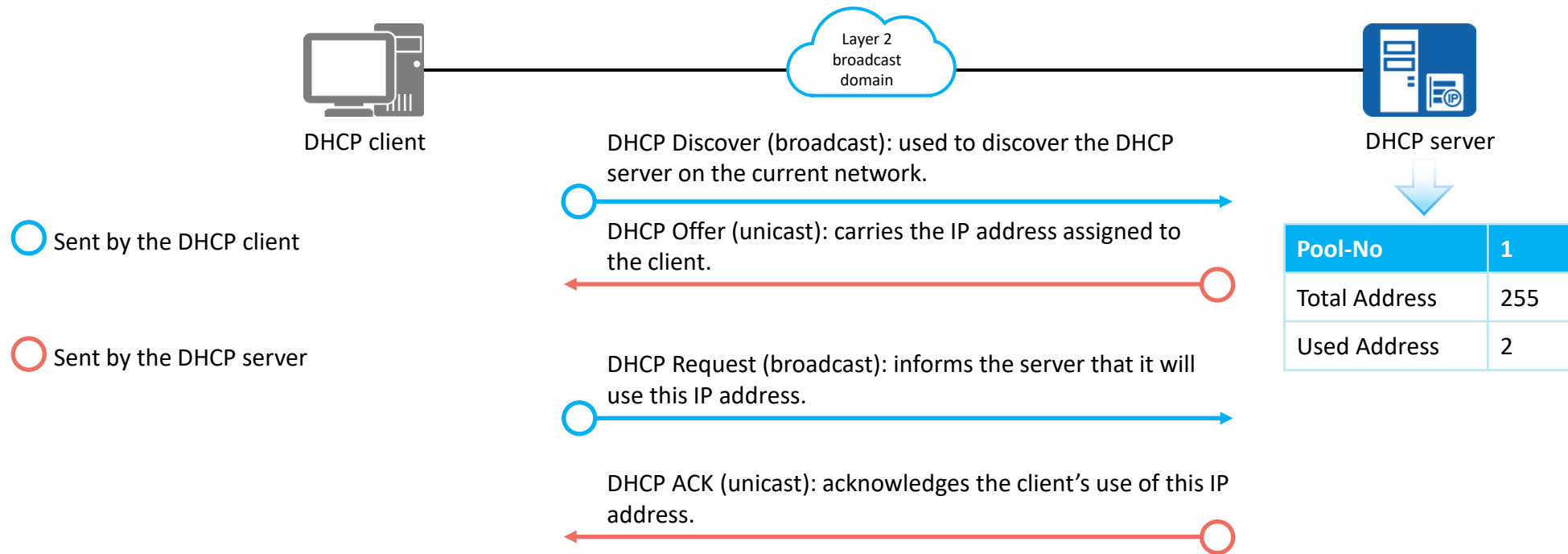


IP:192.168.1.10
Network mask:24
Gateway:192.168.1.1
DNS: 114.114.114.114
Lease: 8 hour

- DHCP defines the lease time to improve IP address utilization.



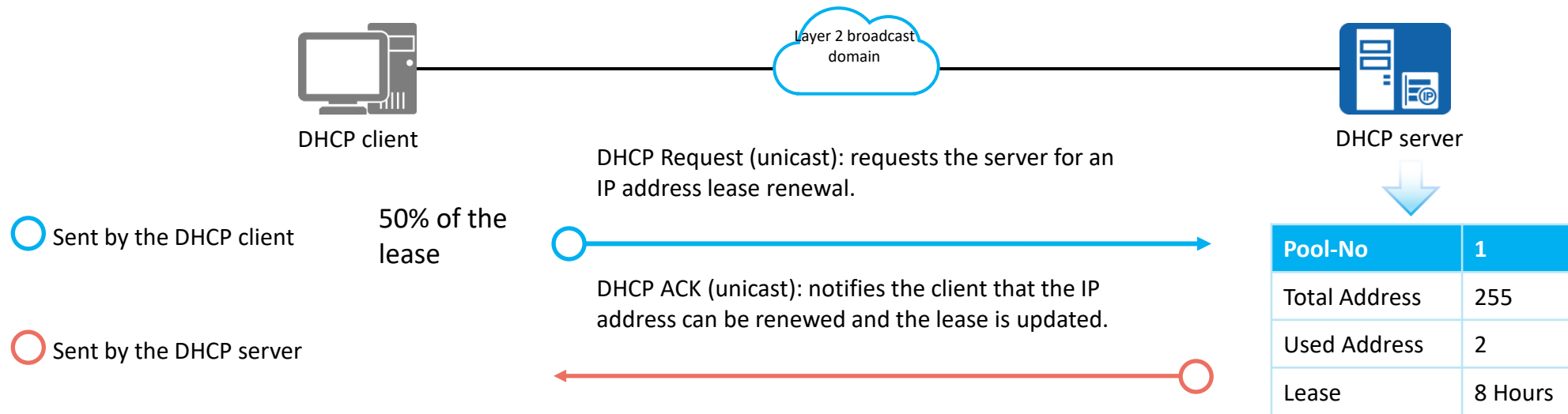
DHCP Working Principle



- Question: Why does a DHCP client need to send a DHCP Request packet to the DHCP server to notify its use of a particular IP address after receiving a DHCP Offer packet?



DHCP Lease Renewal



- If the DHCP client fails to receive a response from the original DHCP server at 50% of the lease (known as T1), the DHCP client waits until 87.5% of the lease (known as T2) has passed. At T2, the client enters the rebinding state, and broadcasts a DHCP Request packet, to which any DHCP server can respond and renew the lease.



Configuration Commands (1)

1. Enable DHCP.

```
[Huawei] dhcp enable
```

2. Enable the interface to use the interface address pool to provide the DHCP server function.

```
[Huawei-GigabitEthernet0/0/0] dhcp select interface
```

3. Specify a DNS server IP address for the interface address pool.

```
[Huawei-GigabitEthernet0/0/0] dhcp server dns-list ip-address
```

4. Configure the range of IP addresses that cannot be automatically assigned to clients from the interface address pool.

```
[Huawei-GigabitEthernet0/0/0] dhcp server excluded-ip-address start-ip-address [ end-ip-address ]
```

5. Configure the lease of IP addresses in the interface address pool of the DHCP server.

```
[Huawei-GigabitEthernet0/0/0] dhcp server lease { day day [ hour hour [ minute minute ] ] | unlimited }
```

By default, the IP address lease is one day.



Configuration Commands (2)

6. Create a global address pool.

```
[Huawei]ip pool ip-pool-name
```

7. Specify the range of IP addresses that can be assigned dynamically in the global address pool.

```
[Huawei-ip-pool-2]network ip-address [ mask { mask | mask-length } ]
```

8. Configure the gateway address for DHCP clients.

```
[Huawei-ip-pool-2]gateway-list ip-address
```

9. Specify the DNS server IP address that the DHCP server delivers to DHCP clients.

```
[Huawei-ip-pool-2]dns-list ip-address
```

10. Set the IP address lease.

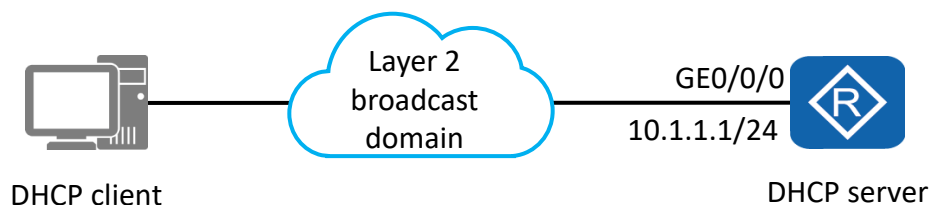
```
[Huawei-ip-pool-2] lease { day day [ hour hour [ minute minute ] ] | unlimited }
```

11. Enable the DHCP server function on the interface.

```
[Huawei-GigabitEthernet0/0/0]dhcp select global
```



DHCP Interface Address Pool Configuration



Requirement:

- Configure a router as the DHCP server, configure the subnet to which GE0/0/0 belongs as the address pool of DHCP clients, set the IP address of GE0/0/0 to that of the DNS server, and set the lease to three days.

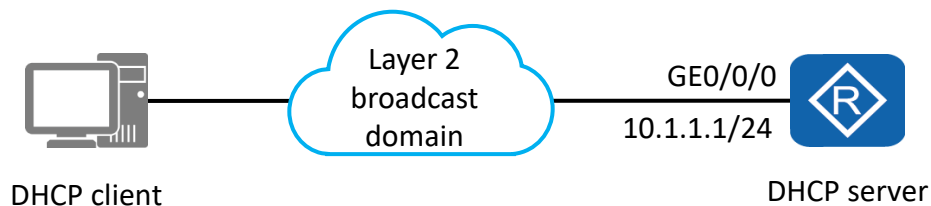
Configuration on the DHCP server:

```
[Huawei]dhcp enable
[Huawei]interface GigabitEthernet0/0/0
[Huawei-GigabitEthernet0/0/0]dhcp select interface
[Huawei-GigabitEthernet0/0/0]dhcp server dns-list 10.1.1.2
[Huawei-GigabitEthernet0/0/0]dhcp server excluded-ip-address 10.1.1.2
[Huawei-GigabitEthernet0/0/0]dhcp server lease day 3
```

Enable the DHCP service globally, enter the interface view, associate the current interface with the DHCP address pool, configure the DNS address and excluded IP address (excluding the interface IP address) in the interface view, and configure the lease of the IP addresses assigned to clients.



DHCP Global Address Pool Configuration



Requirement:

- Configure a router as the DHCP server and configure the global address pool **pool2** to assign IP addresses (on the subnet 1.1.1.0/24) to DHCP clients. Set both the gateway address and DNS address to 1.1.1.1, set the lease to 10 days, and enable GE0/0/0 to use the global address pool.

Configuration on the DHCP server:

```
[Huawei]dhcp enable
[Huawei]ip pool pool2
Info: It's successful to create an IP address pool.
[Huawei-ip-pool-pool2]network 1.1.1.0 mask 24
[Huawei-ip-pool-pool2]gateway-list 1.1.1.1
[Huawei-ip-pool-pool2]dns-list 1.1.1.1
[Huawei-ip-pool-pool2]lease day 10
[Huawei-ip-pool-pool2]quit
[Huawei]interface GigabitEthernet0/0/0
[Huawei-GigabitEthernet0/0/1]dhcp select global
```

- Enable the DHCP service globally and configure the global address pool **pool2**. Configure the address range, gateway address, DNS address, and lease for **pool2**.
- Select the global address pool on a specific interface (GE0/0/0). When GE0/0/0 receives a DHCP request, it assigns an IP address from the global address pool.

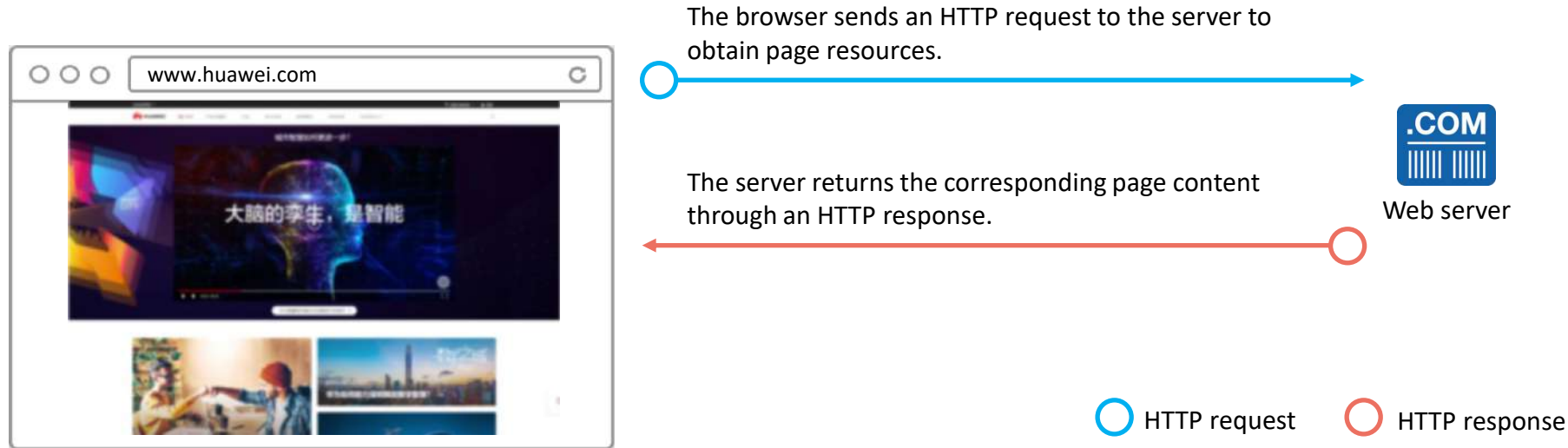


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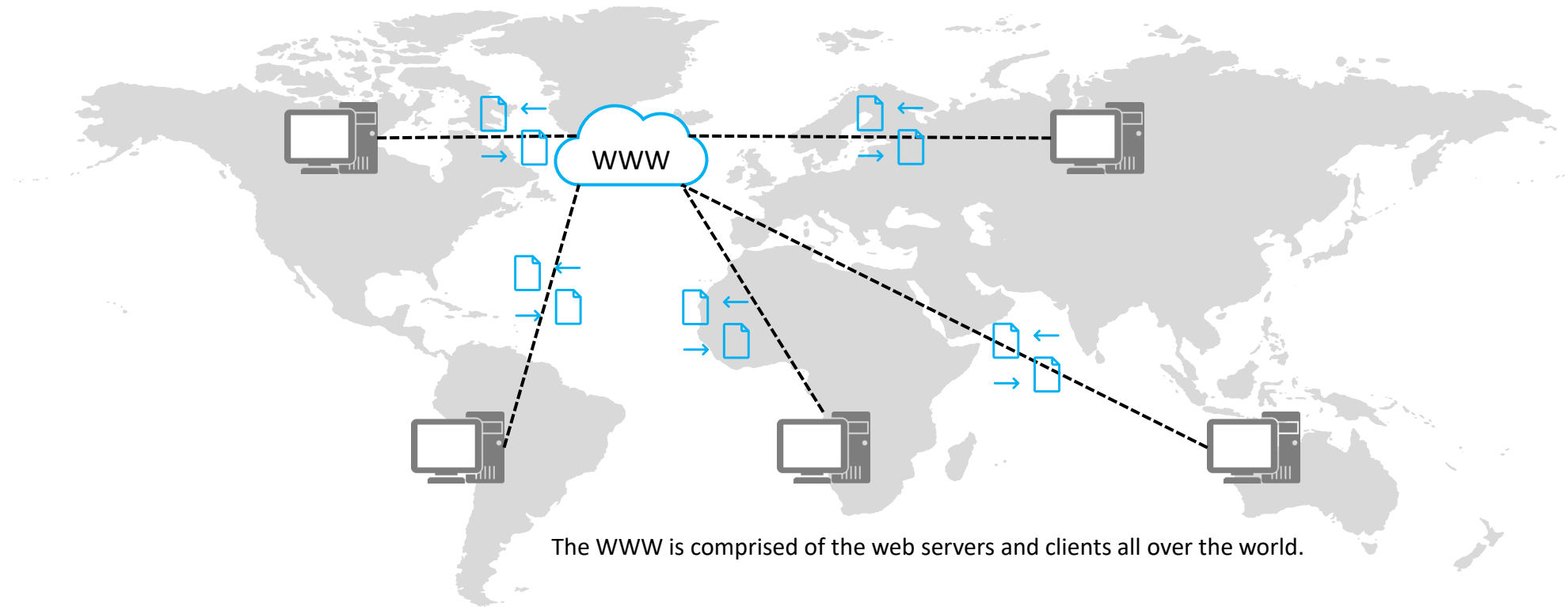
Web Page Access Using a Browser



- When you enter a uniform resource locator (URL) in a browser, the browser can obtain data from a web server and display the content on the page.
- Hypertext Transfer Protocol (HTTP): an application layer protocol for communication between a client browser or another program and a web server
- HTTP adopts the typical C/S architecture, and uses TCP for transmission.



Background



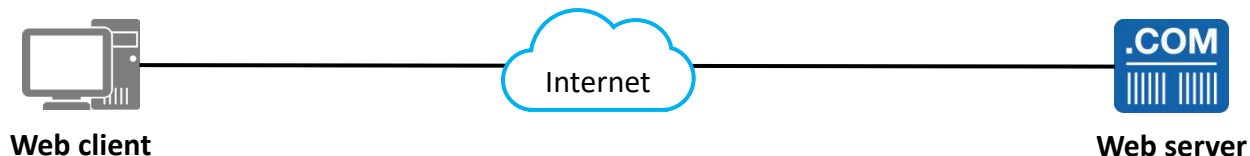
The WWW is comprised of the web servers and clients all over the world.

- In the early days of the Internet, World Wide Web (WWW) was proposed to share documents.
- The WWW consists of three parts: Hypertext Markup Language (HTML) for displaying document content in a browser, HTTP for transmitting documents on the network, and URLs for specifying document locations on the network.
- WWW was actually the name of a client application for browsing HTML documents, and now represents a collection of technologies (HTML + HTTP + URL) and is commonly known as the Web.



Transfer Example (1)

- HTTP request
- HTTP response



The URL www.servs_app.com/web/index.html is entered in the address box of a browser. After obtaining the IP address corresponding to the domain name through DNS resolution, the client sends an HTTP request to the server to request the page.

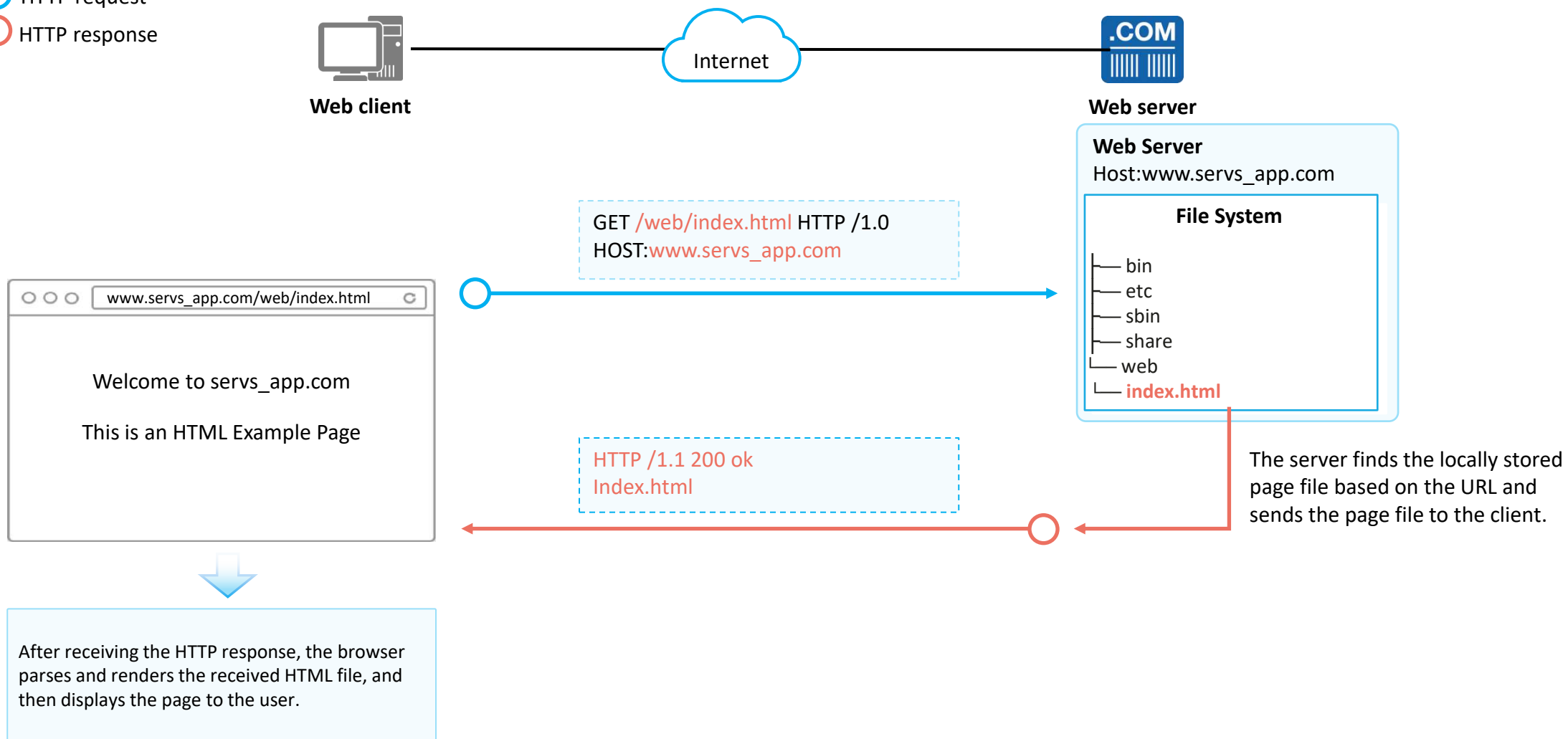
GET </web/index.html> HTTP /1.0
HOST:www.servs_app.com





Transfer Example (2)

- HTTP request
- HTTP response





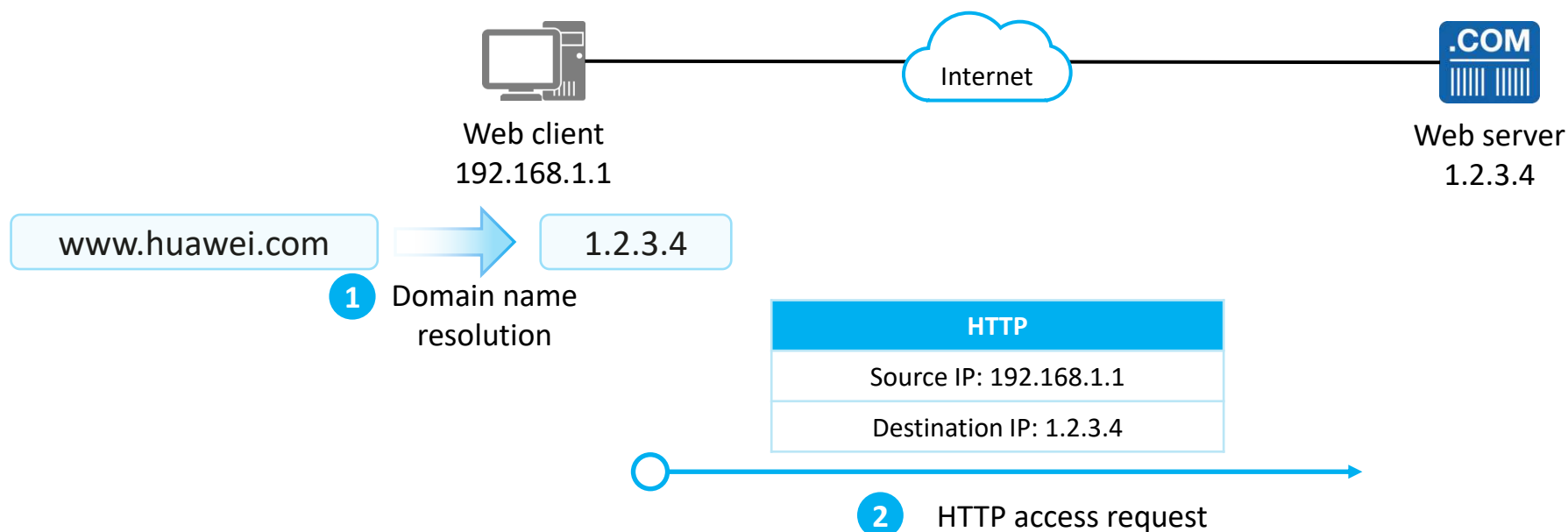
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Birth of DNS

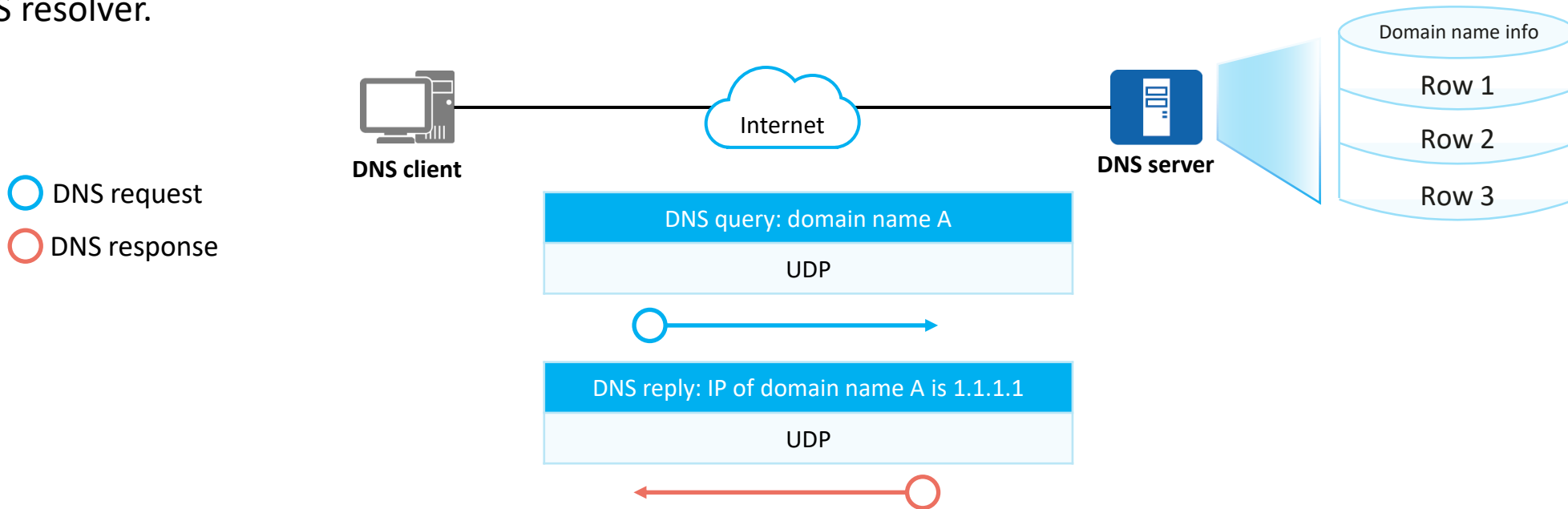
- When you enter a domain name in your browser to access a website, the domain name is resolved to an IP address. The browser actually communicates with this IP address.
- The protocol used for resolving domain names to IP addresses is Domain Name System (DNS).
- Each node on the network has a unique IP address, and nodes can communicate with one another through IP addresses. However, if all nodes communicate through IP addresses, it is difficult to remember so many IP addresses. Therefore, DNS is proposed to map IP addresses to alphanumeric character strings (domain names).





DNS Components

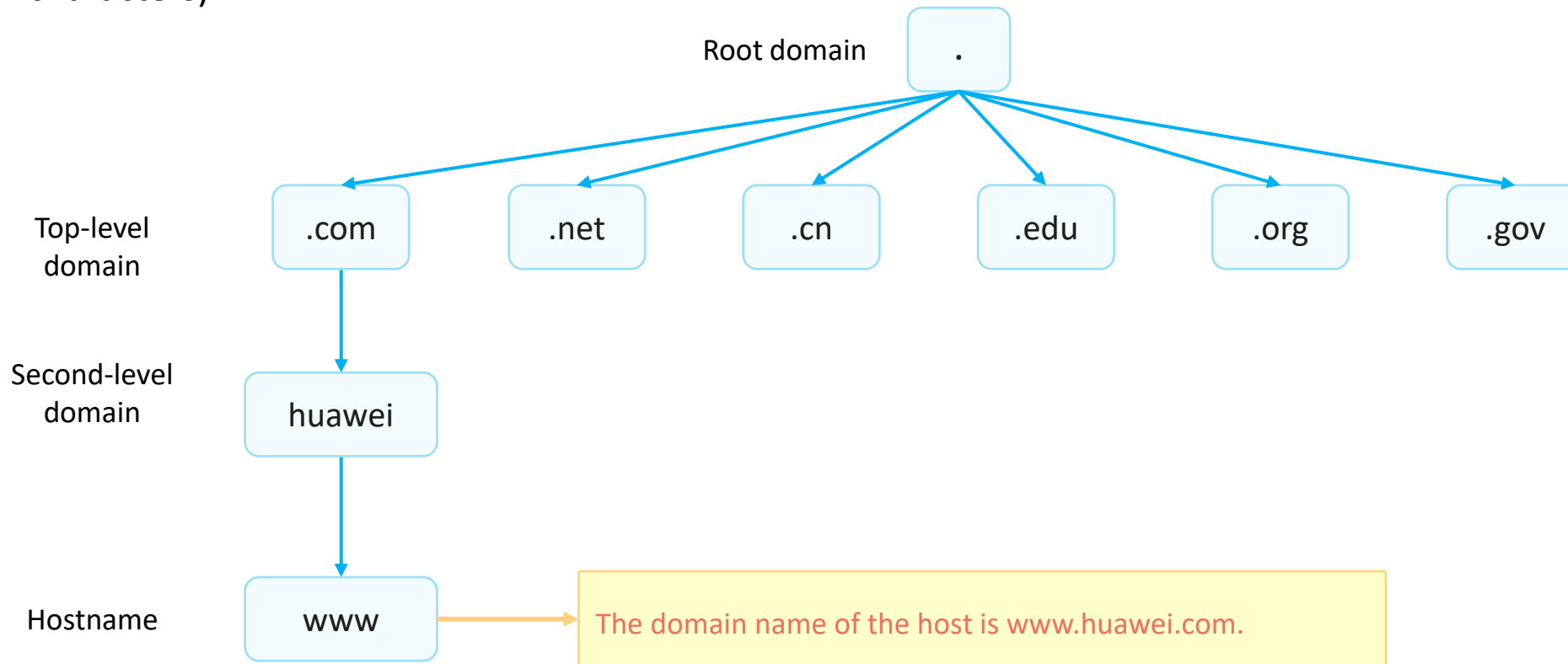
- **Domain name:** a sequence of characters to identify hosts. In most cases, the URL entered in the browser when you visit a website is the domain name of the website.
- **DNS server:** maintains the mappings between domain names and IP addresses and responds to requests from the DNS resolver.





Domain Name Format

- The domain name is in the format of *hostname.second-level domain.top-level domain.root domain*. The root domain is represented by a dot (.). Generally, the root domain is denoted by an empty name (that is, containing no characters).

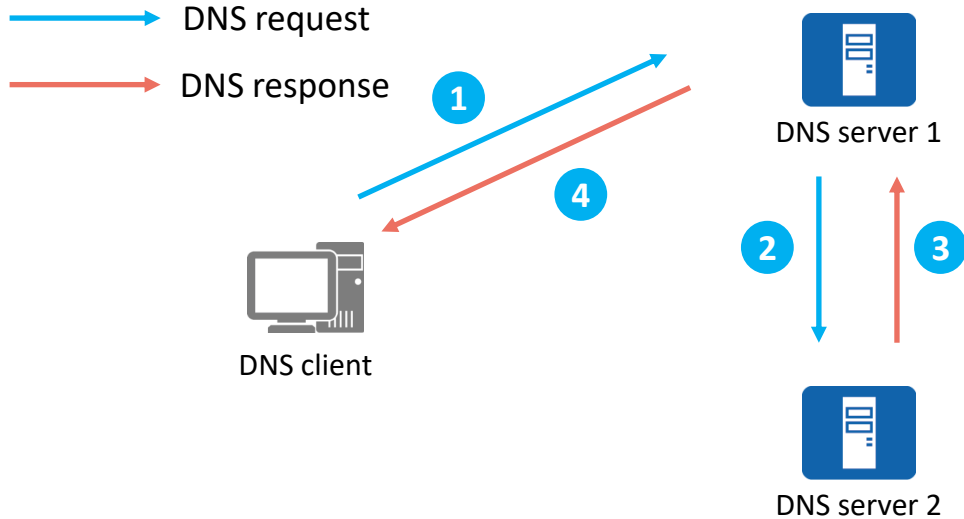




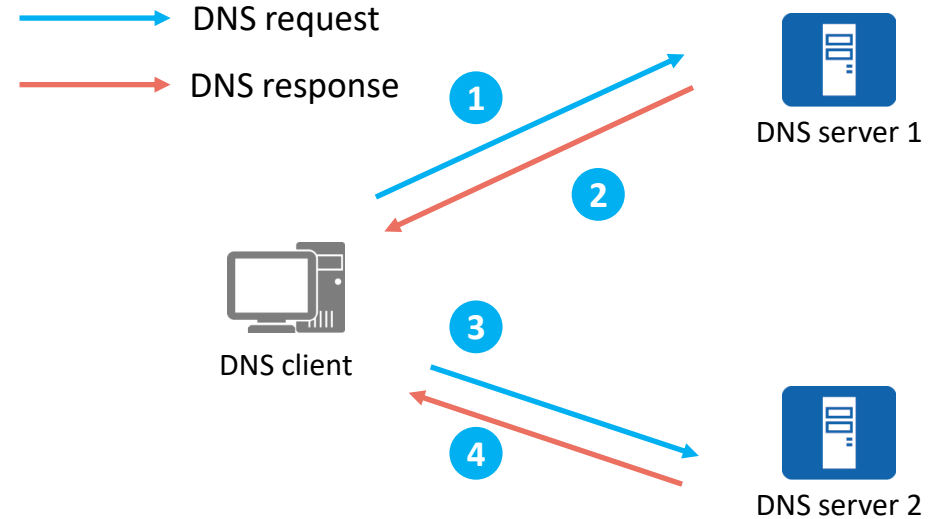
DNS Query Modes

- The DNS is a distributed system. The database of most DNS servers does not have all domain name records. When a client queries a domain name from a DNS server but the DNS server does not have the record of the domain name, the client can continue the query in either of the following ways:
 - **Recursive query:** The DNS server queries other DNS servers and returns the query result to the DNS client.
 - **Iterative query:** The DNS server informs the DNS client of the IP address of another DNS server, from which the DNS client queries the domain name.

Recursive Query



Iterative Query





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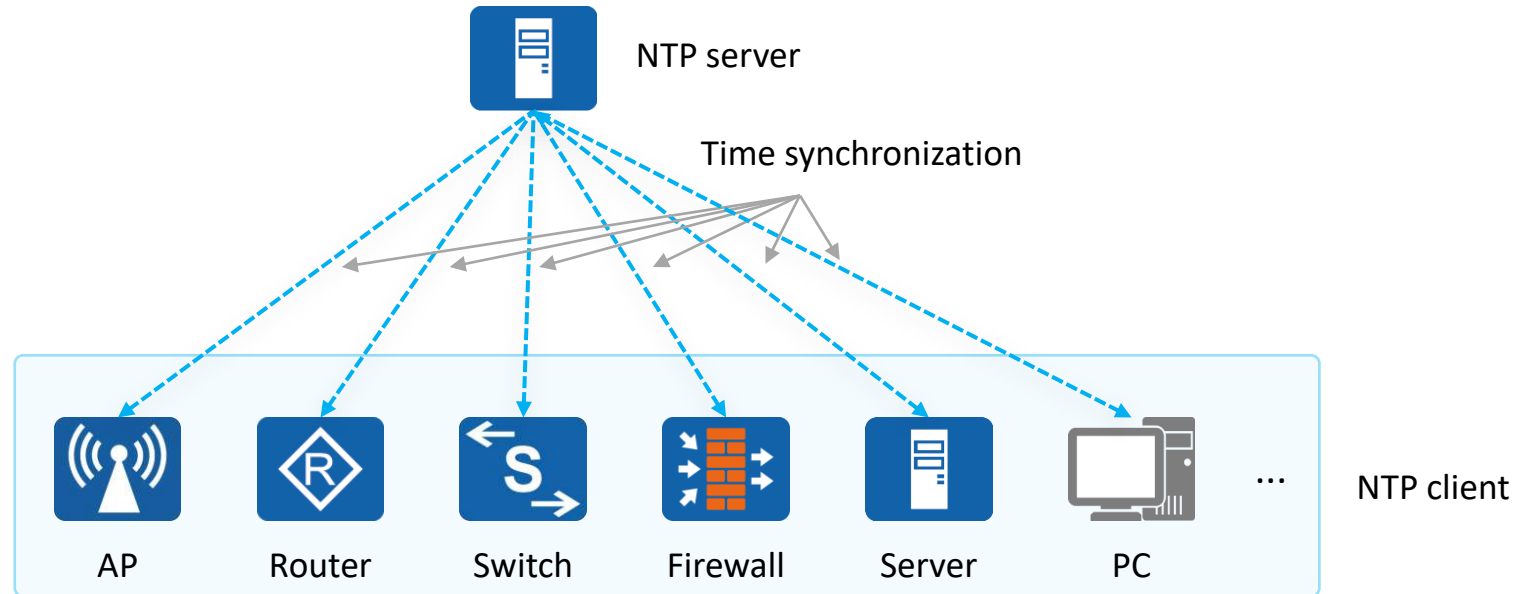
Time Synchronization Requirements

- Consistent clock of all devices is required in many scenarios on enterprise campus networks:
 - Network management: Analysis of logs or debugging messages collected from different routers needs time for reference.
 - Charging system: The clocks of all devices must be consistent.
 - Several systems working together on the same complicate event: Systems have to take the same clock for reference to ensure a proper sequence of implementation.
 - Incremental backup between a backup server and clients: Clocks on the backup server and clients should be synchronized.
 - System time: Some applications need to know the time when users log in to the system and the time when files are modified.



NTP Overview

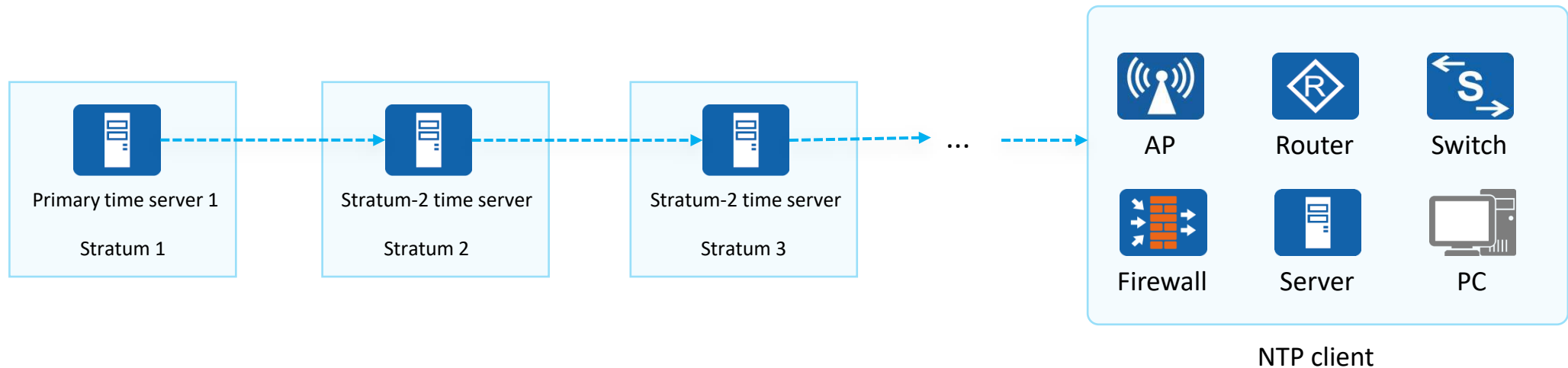
- If the administrator manually enters commands to change the system time for time synchronization, the workload is heavy and the accuracy cannot be ensured. Therefore, the Network Time Protocol (NTP) is designed to synchronize the clocks of devices.
- NTP is an application layer protocol belonging to the TCP/IP suite and synchronizes time between a group of distributed time servers and clients. NTP is based on IP and UDP, and NTP packets are transmitted using UDP on port number 123.





NTP Network Structure

- **Primary time server:** directly synchronizes its clock with a standard reference clock through a cable or radio. Typically, the standard reference clock is either a radio clock or the Global Positioning System (GPS).
- **Stratum-2 time server:** synchronizes its clock with either the primary time server or other stratum-2 time servers within the network. Stratum-2 time servers use NTP to send time information to other hosts in a Local Area Network (LAN).
- **Stratum:** is a hierarchical standard for clock synchronization. It represents the precision of a clock. The value of a stratum ranges from 1 to 15. A smaller value indicates higher precision. The value 1 indicates the highest clock precision, and the value 15 indicates that the clock is not synchronized.





Quiz

1. Which FTP mode is recommended for transferring log and configuration files on network devices?
Why?
2. Why does a DHCP client need to send a DHCP Request packet to the DHCP server to notify its use of a particular IP address after receiving a DHCP Offer packet?
3. What are the functions of HTML, URL, and HTTP?
 1. ASCII mode; The binary mode is more applicable to the transmission of non-text files that cannot be converted, such as EXE, BIN, and CC (VRP version file extension) files.
 2. A client's DHCP Request packet is broadcast, so other DHCP servers on the network know that the client has selected a particular IP address assigned by the DHCP server. This ensures that other DHCP servers can release this IP address assigned to the client through the unicast DHCP Offer packet.
 3. HTML is used to display page content, URL is used to locate the network location of a document or file, and HTTP is used for requesting and transferring files.



Summary

- FTP is used to transfer files. You are advised to use different transfer modes for different files. FTP is based on TCP and therefore can ensure the reliability and efficiency of file transfer.
- Dynamically assigning IP addresses through DHCP reduces the workload of the administrator and avoids IP address conflicts caused by manual configuration of network parameters.
- As the document transfer protocol of WWW, HTTP is widely used in today's network for encoding and transporting information between a client (such as a web browser) and a web server.

The background of the slide features a blue-tinted image of several business professionals in a modern office environment. They are standing on a highly reflective floor, and their silhouettes are clearly visible against the lighter background. The overall aesthetic is professional and corporate.

Thank You
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WLAN Overview



Foreword

- Wired LANs are expensive and lack mobility. The increasing demand for portability and mobility requires wireless local area network (WLAN) technologies.
- WLAN is now the most cost-efficient and convenient network access mode.
- This course introduces the development of WLAN in different phases, concepts related to WLAN technologies, implementation and basic configurations of common WLAN networking architectures, and WLAN development trends.



Objectives

- On completion of this course, you will be able to:
 - Understand basic concepts of WLAN and the history of the 802.11 protocol family.
 - Learn about different WLAN devices.
 - Distinguish between different WLAN networking architectures.
 - Understand the WLAN working process.
 - Complete basic WLAN configurations.



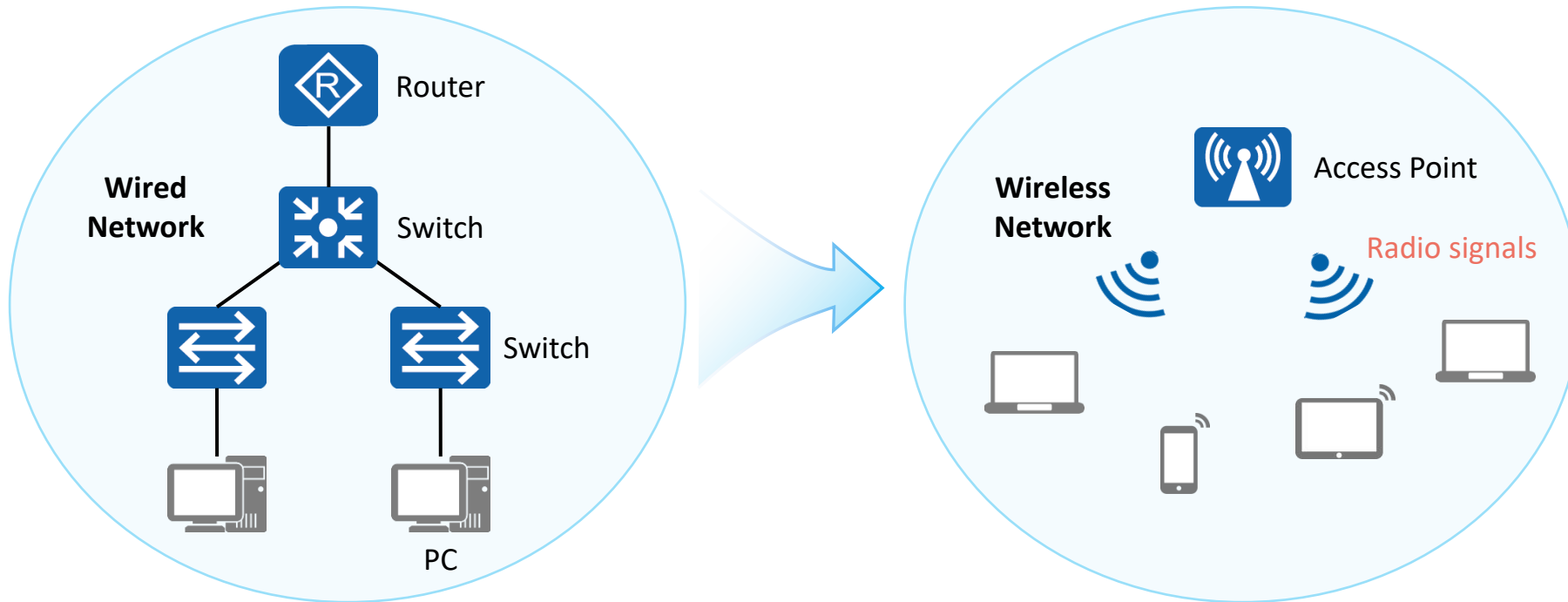
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2. Basic Concepts of WLAN
3. WLAN Fundamentals
4. WLAN Configuration Implementation
5. Next-Generation WLAN Solutions



Introduction to WLAN

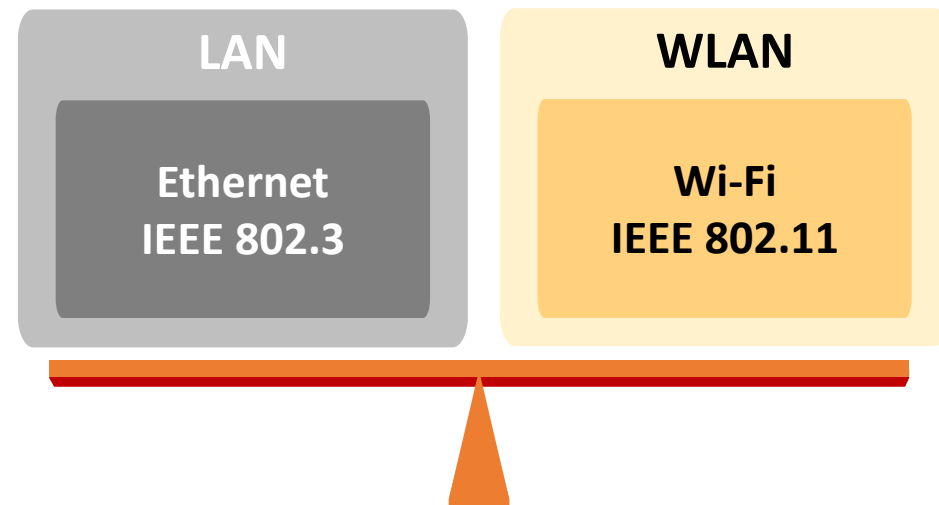
- A wireless local area network (WLAN) is constructed using wireless technologies. It uses high-frequency (2.4 GHz or 5 GHz) signals such as radio waves, lasers, and infrared rays to replace the traditional media used for transmission on a wired LAN.
- WLAN technology allows users to easily access a wireless network and move around within the coverage of the wireless network.





IEEE 802.11、WLAN and Wi-Fi

- IEEE 802.11 suites are standards for WLANs which are defined by the Institute of Electrical and Electronics Engineering (IEEE).
- Wi-Fi Alliance was formed by a group of major manufacturers and the logo "Wi-Fi" was created. The Wi-Fi standards are WLAN technologies based on IEEE 802.11 standards.

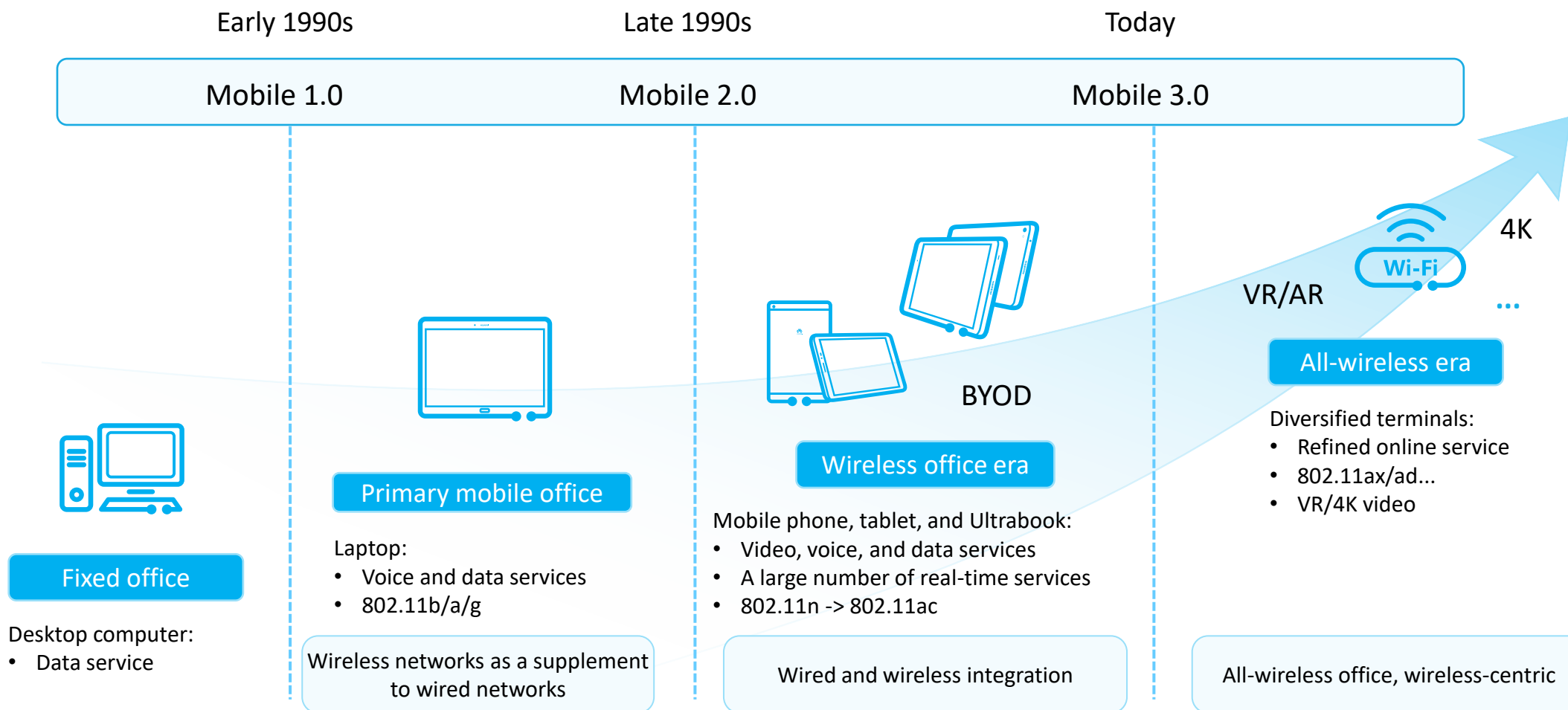


IEEE 802.11 Standards and Wi-Fi Generations

Frequency Band	2.4GHz	2.4GHz	2.4GHz、5GHz	2.4GHz & 5GHz	5GHz	5GHz	2.4GHz & 5GHz
Throughput	2Mbit/s	11Mbit/s	54Mbit/s	300Mbit/s	1300Mbit/s	6.9Gbit/s	9.6Gbit/s
Standard	802.11	802.11b	802.11a、802.11g	802.11n	802.11ac wave1	802.11ac wave2	802.11ax
Wi-Fi	Wi-Fi 1	Wi-Fi 2	Wi-Fi 3	Wi-Fi 4	Wi-Fi 5		Wi-Fi 6
Released In	1997	1999	2003	2009	2013	2015	2018



Wi-Fi Development Trends in Office Scenarios





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WLAN Devices

Home



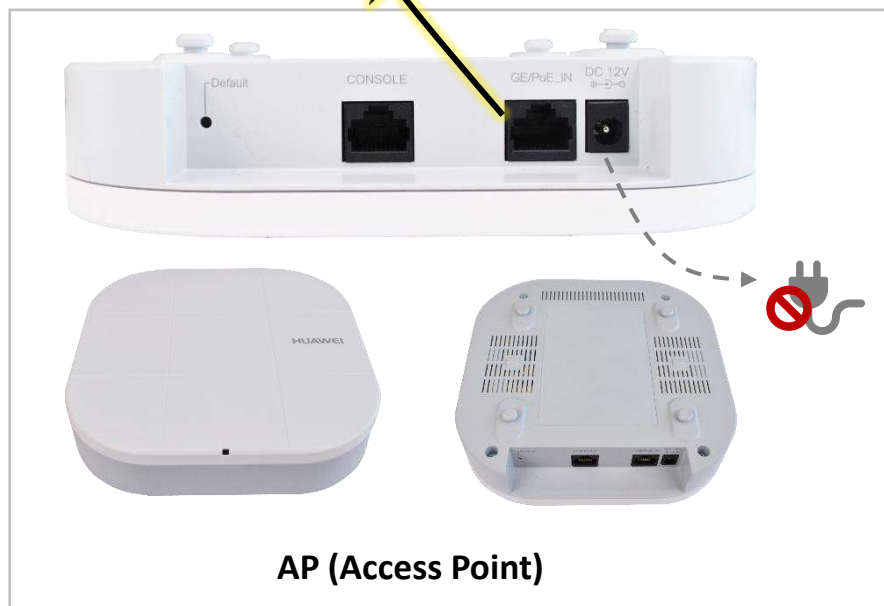
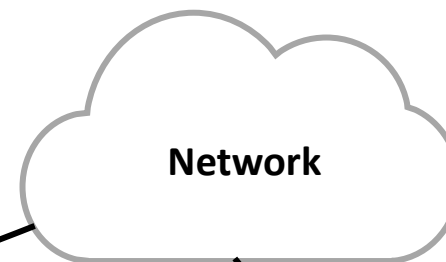
Wireless Router

Enterprise

PoE Switch



Network



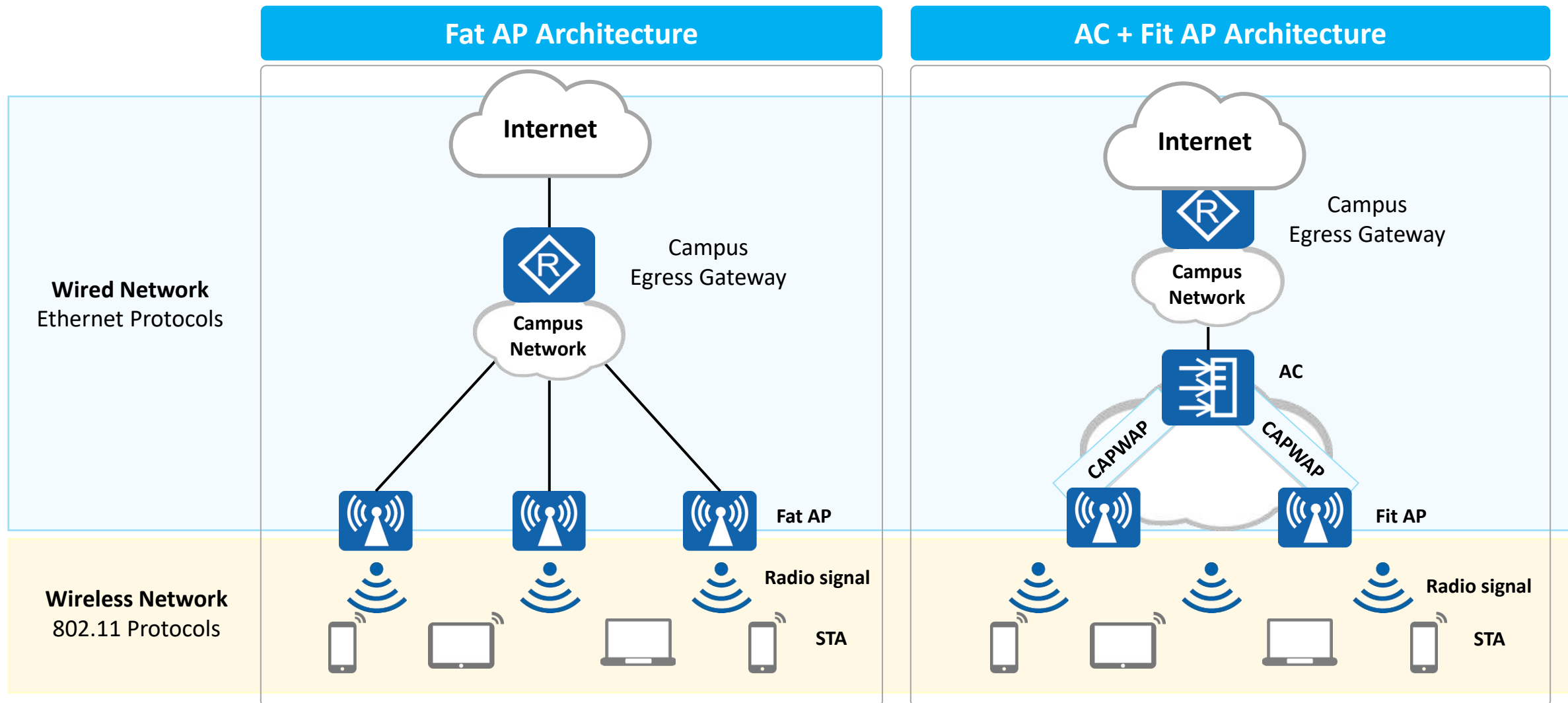
AP (Access Point)



AC (Access Controller)

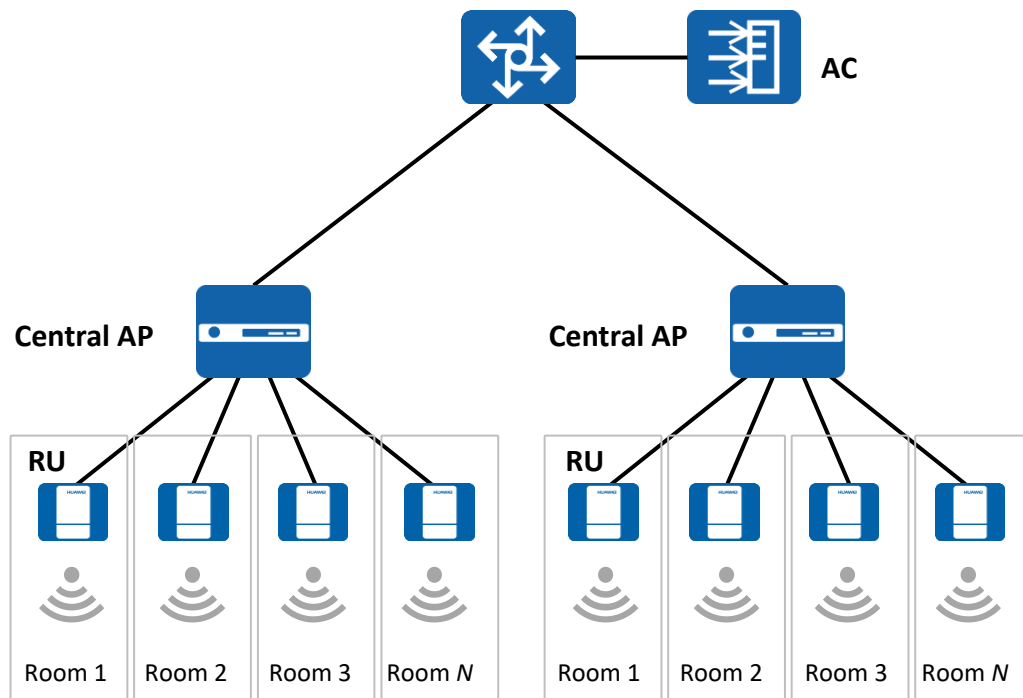


Basic WLAN Networking Architecture





Agile Distributed Architecture

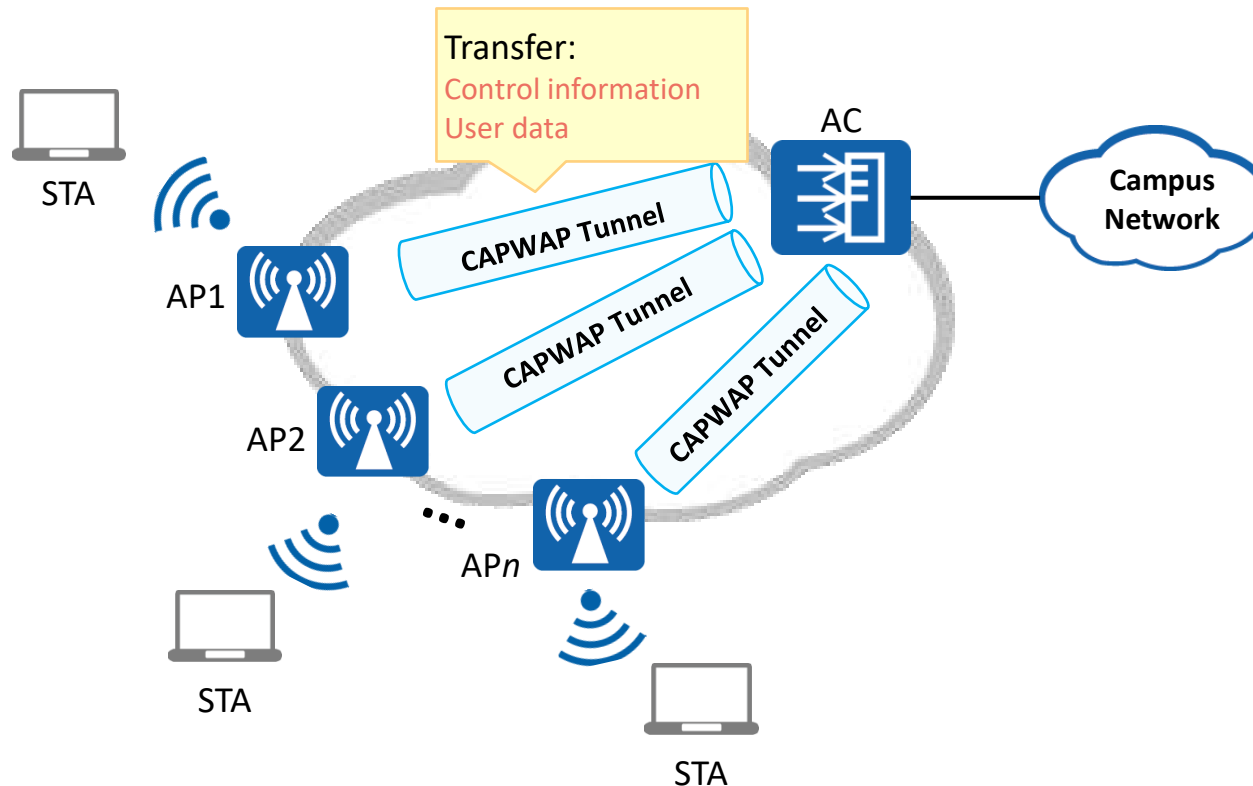


Architecture Characteristics

- The agile distributed architecture divides an AP into a central AP and remote units (RUs). The central AP can manage multiple RUs, which provides good coverage and reduces costs. RUs can be used in the Fat AP, AC + Fit AP, and cloud management architectures.
- Application scope: densely distributed rooms



CAPWAP



What Is a CAPWAP Tunnel?

- **Control And Provisioning of Wireless Access Points (CAPWAP):** defines how to manage and configure APs. That is, an AC manages and controls APs in a centralized manner through CAPWAP tunnels.

CAPWAP Tunnel Functions

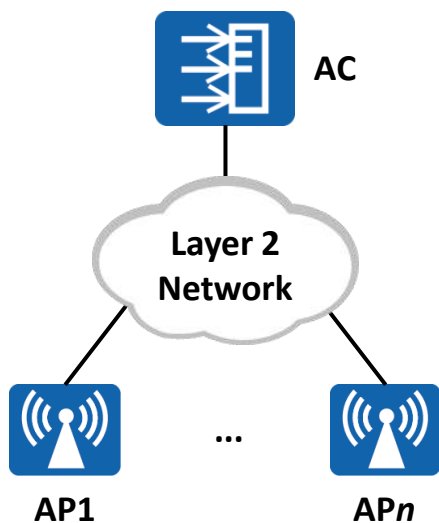
- Maintains the running status of the AC and APs.
- Allows the AC to manage APs and deliver service configurations to the APs.
- Allows APs to exchange data sent by STAs with the AC through CAPWAP tunnels when the tunnel forwarding mode is used.



AP-AC Networking

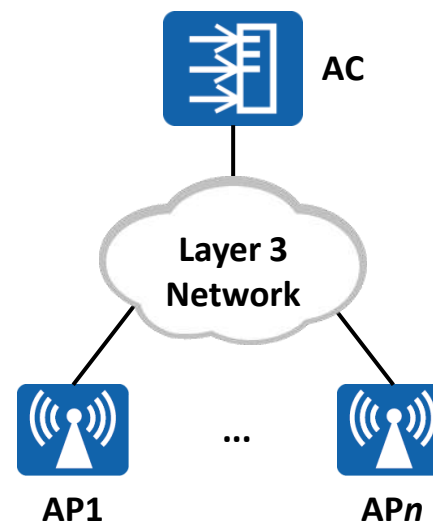
- The AP-AC networking modes are classified into Layer 2 networking and Layer 3 networking.

Layer 2 Networking



- Layer 2 networking: APs are connected to an AC directly or across a Layer 2 network.
- The Layer 2 networking features quick deployment. It is applicable to simple or temporary networking but not to large networking.

Layer 3 Networking



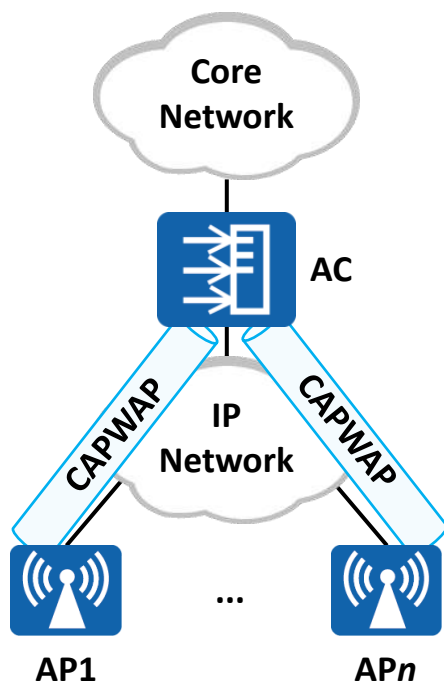
- Layer 3 networking: APs are connected to an AC across a Layer 3 network.
- In the actual networking, an AC can connect to dozens or even hundreds of APs, which is usually complex. In most cases, the Layer 3 networking is used on a large network.



AC Connection Mode

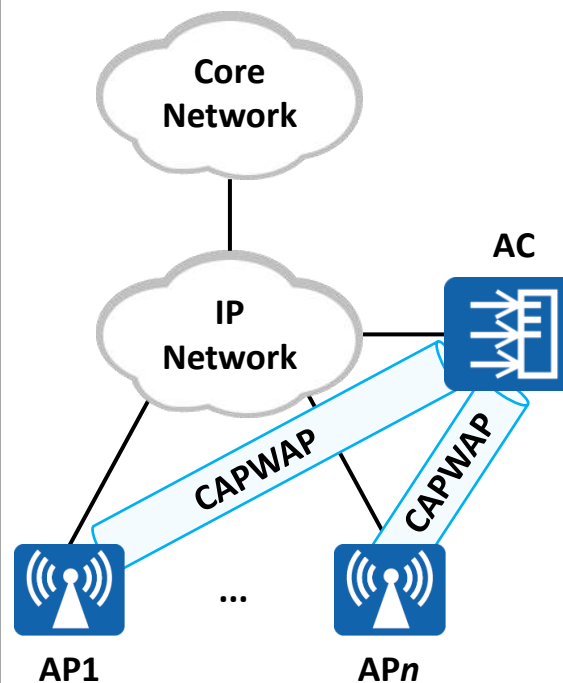
- ACs can be connected in in-path or off-path mode.

In-Path Networking



- In the in-path networking, the APs, AC, and core network are connected in a chain. All data destined for the core layer passes through the AC.
- In this networking, the AC also functions as an aggregation switch to forward and process data traffic and management traffic of APs.

Off-Path Networking

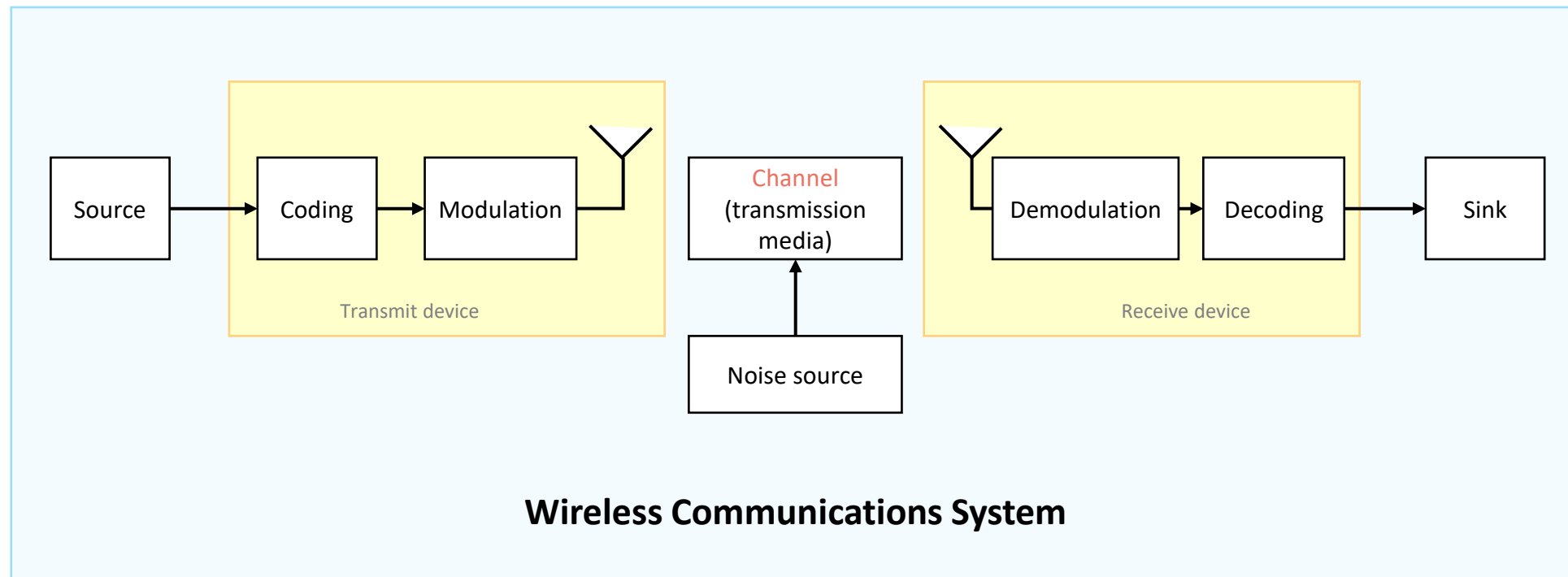


- In the off-path networking, the AC connects to the network between APs and the core network, but does not directly connect to APs.
- In this networking, the AC is connected to APs in off-path mode, the service data of APs reaches the uplink network without passing through the AC.



Wireless Communications System

- In a wireless communications system, information may be an image, a text, a sound, or the like. The transmit device first applies source coding to convert information into digital signals that allow for circuit calculation and processing, and then into **radio waves** by means of channel coding and modulation.

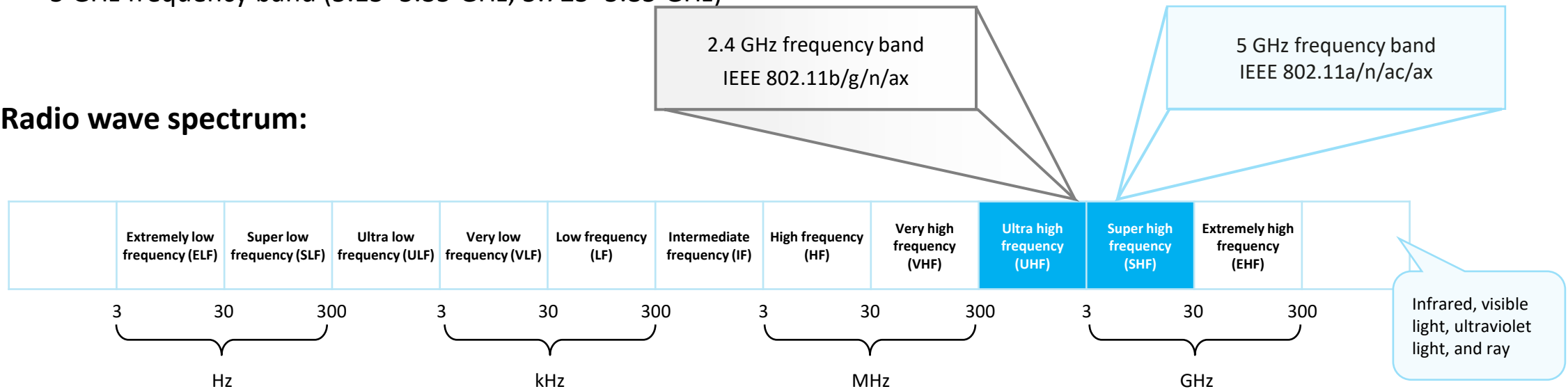




Radio Wave

- A radio wave is an electromagnetic wave whose frequency is between 3 Hz and about 300 GHz. Radio technology converts sound signals or other signals and transmits them by using radio waves.
- WLAN technology enables transmission of information by radio waves over the air. Currently, the WLAN uses the following frequency bands:
 - 2.4 GHz frequency band (2.4–2.4835 GHz)
 - 5 GHz frequency band (5.15–5.35 GHz, 5.725–5.85 GHz)

- **Radio wave spectrum:**





Radio Channel

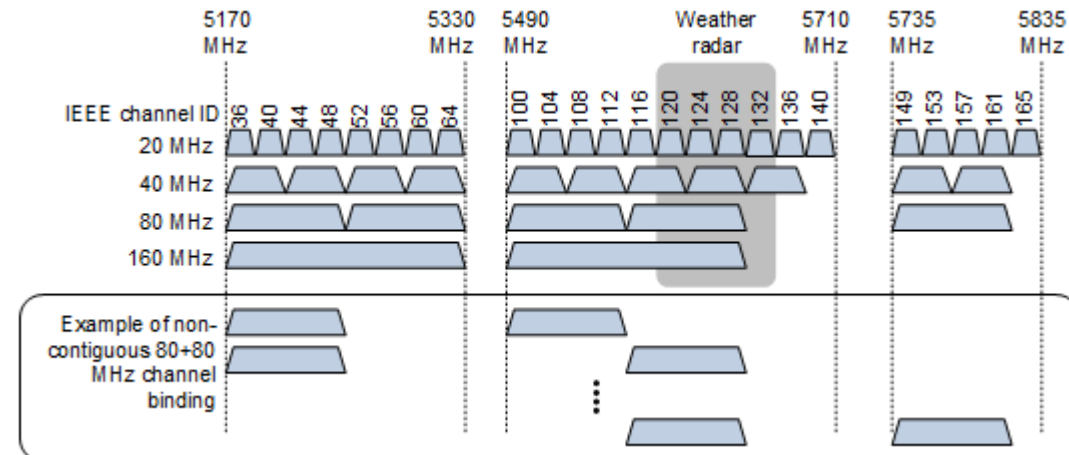
- A channel transmits information, and a radio channel is a radio wave in space. Given that radio waves are ubiquitous, the random use of spectrum resources will cause endless interference issues. Therefore, in addition to defining the usable frequency bands, wireless communication protocols must also accurately divide the frequency ranges. Each frequency range is a channel.

2.4 GHz Frequency Band



- The 2.4 GHz frequency band is divided into 14 channels with overlapping or non-overlapping relationships, each with a bandwidth of 20 MHz.
 - Overlapping channels, such as channels 1 and 2, **interfere with each other**.
 - Non-overlapping channels, such as channels 1 and 6, do not interfere with each other.

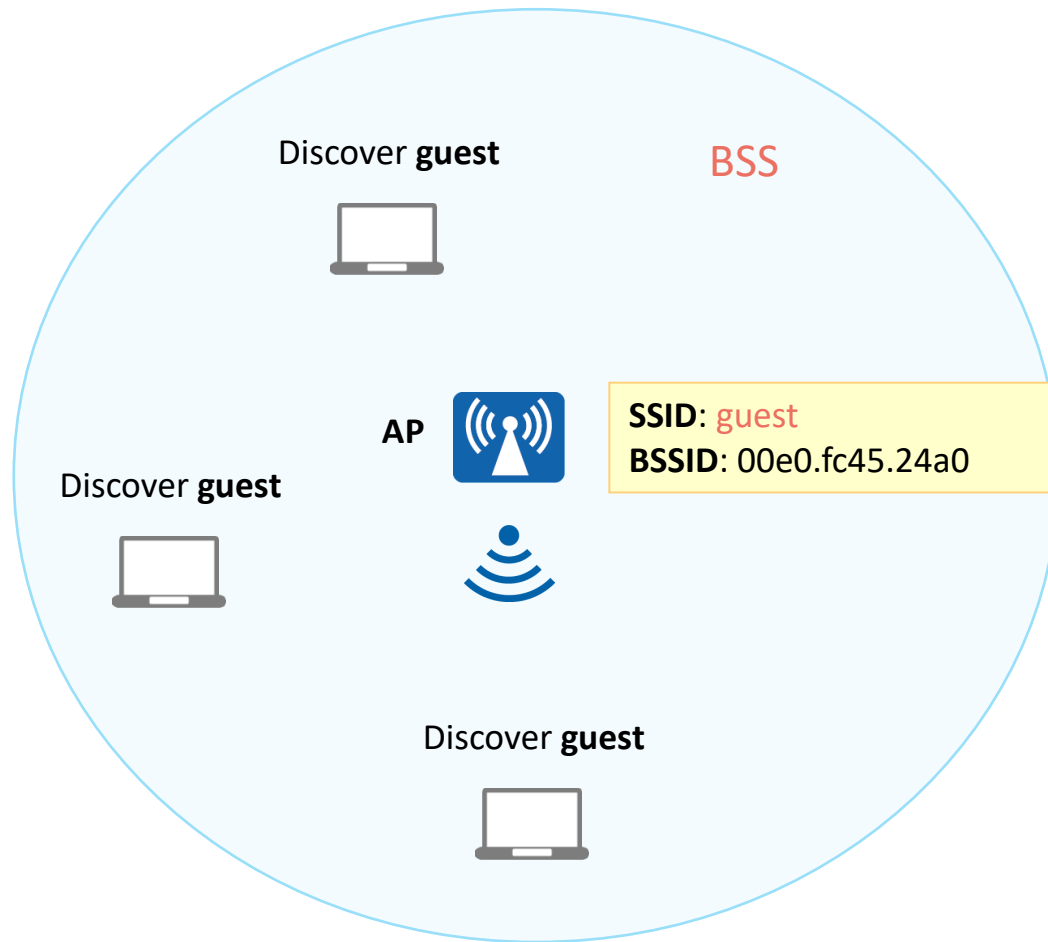
5 GHz Frequency Band



- The 5 GHz frequency band has **richer spectrum resources**. In addition to 20 MHz channels, APs working on the 5 GHz frequency band support 40 MHz, 80 MHz, and higher-bandwidth channels.



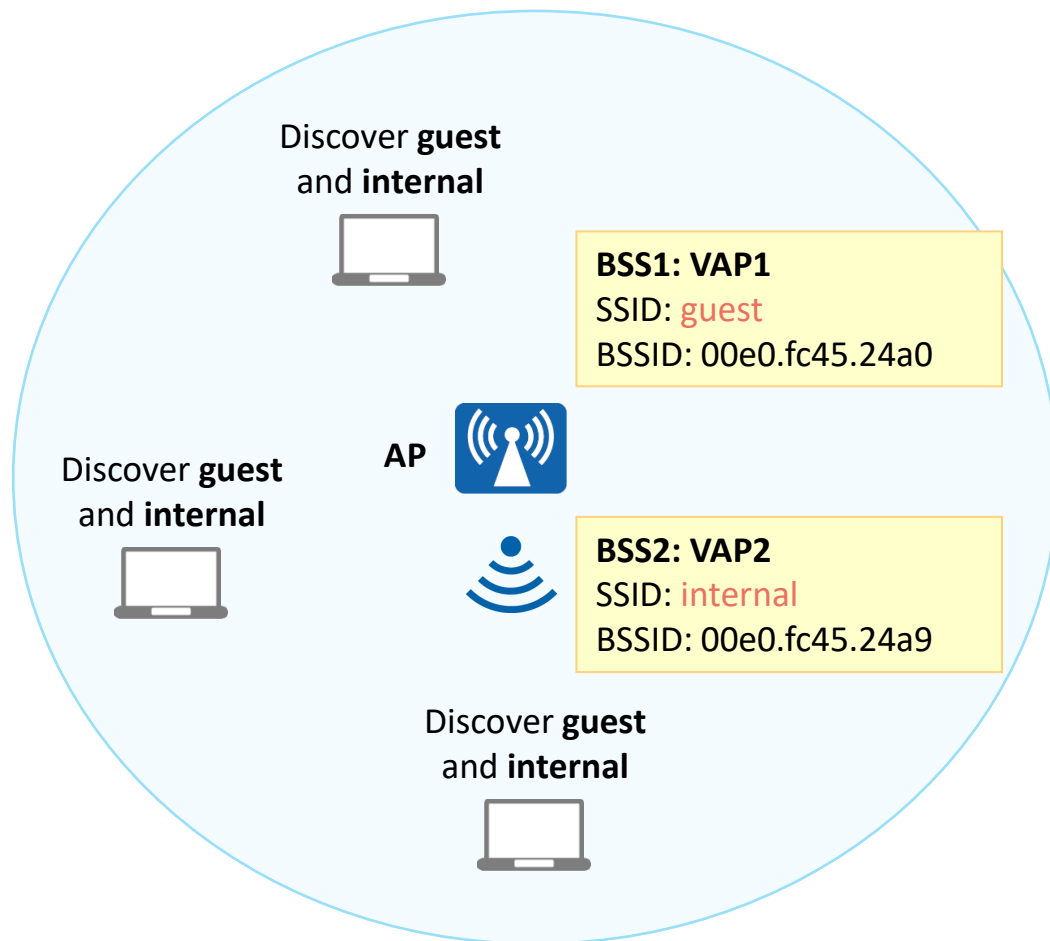
BSS/SSID/BSSID



- Basic service set (BSS):
 - An area covered by an AP.
 - STAs in a BSS can communicate with each other.
- Basic service set identifier (BSSID):
 - An identifier of a WLAN, which is represented by the AP's MAC address.
- Service set identifier (SSID):
 - An identifier of a WLAN, which is represented by a string of characters.
 - SSIDs can replace BSSIDs to help users identify different WLANs.



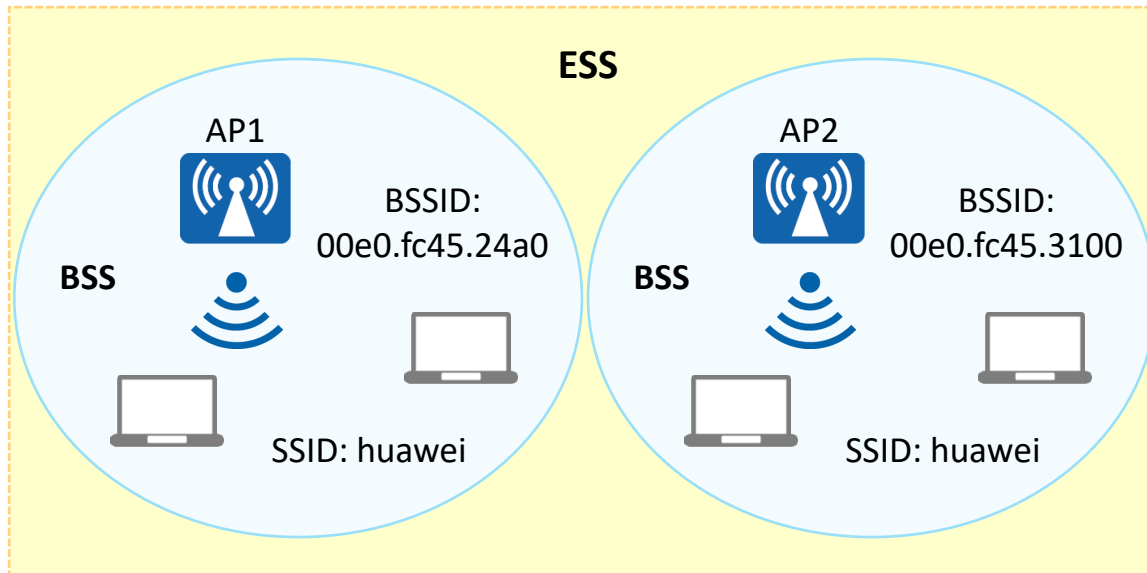
VAP



- In the early stage, APs support only one BSS. If multiple BSSs are deployed in a space, we must also deploy multiple APs, which increases costs and occupies channel resources. To resolve this problem, APs now support creation of multiple virtual access points (VAPs).
- VAP:
 - A physical AP can be virtualized into multiple VAPs, each of which provides the same functions as the physical AP.
 - Each VAP corresponds to one BSS. In this way, one AP may provide multiple BSSs that can have different SSIDs specified.



ESS



- The coverage of a BSS is limited. An extended service set (ESS) can be used to expand the coverage. When a STA moves from one BSS to another BSS, an ESS ensures that the STA does not sense the change of the SSID.
- ESS:
 - A larger-scale virtual BSS that consists of multiple BSSs with the same SSID.

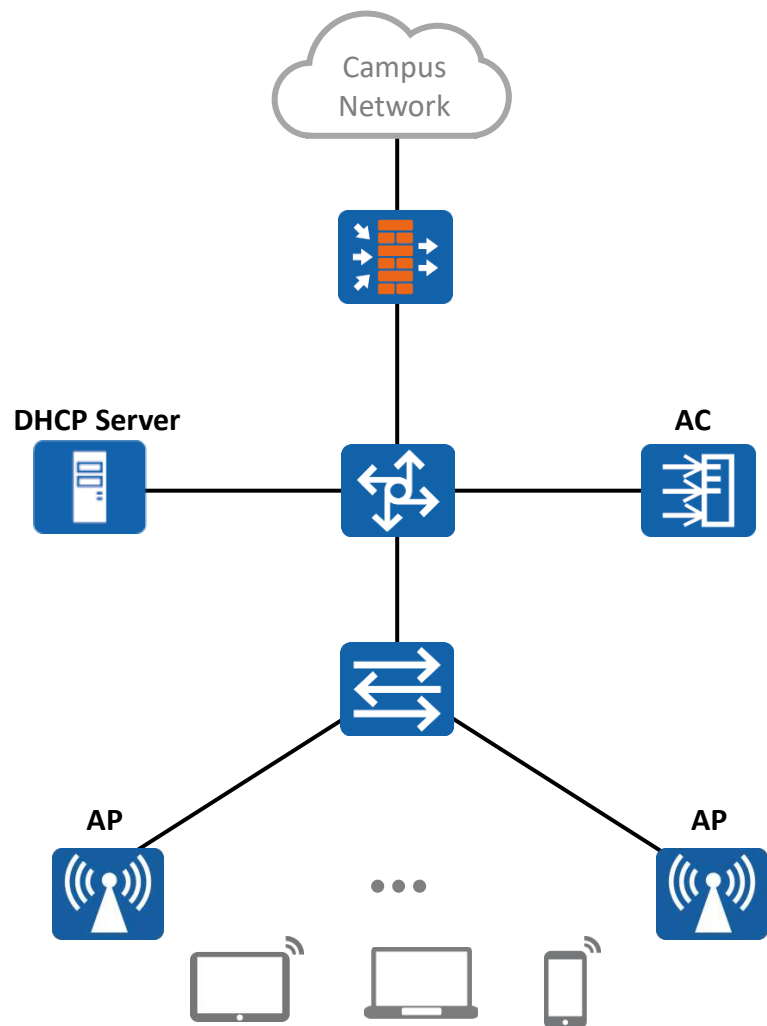


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WLAN Working Process Overview



WLAN Working Process

1 AP onboarding

An AP obtains an IP address, discovers an AC, and sets up a connection with the AC.

2 WLAN service configuration delivery

The AC delivers WLAN service configurations to the AP.

3 STA access

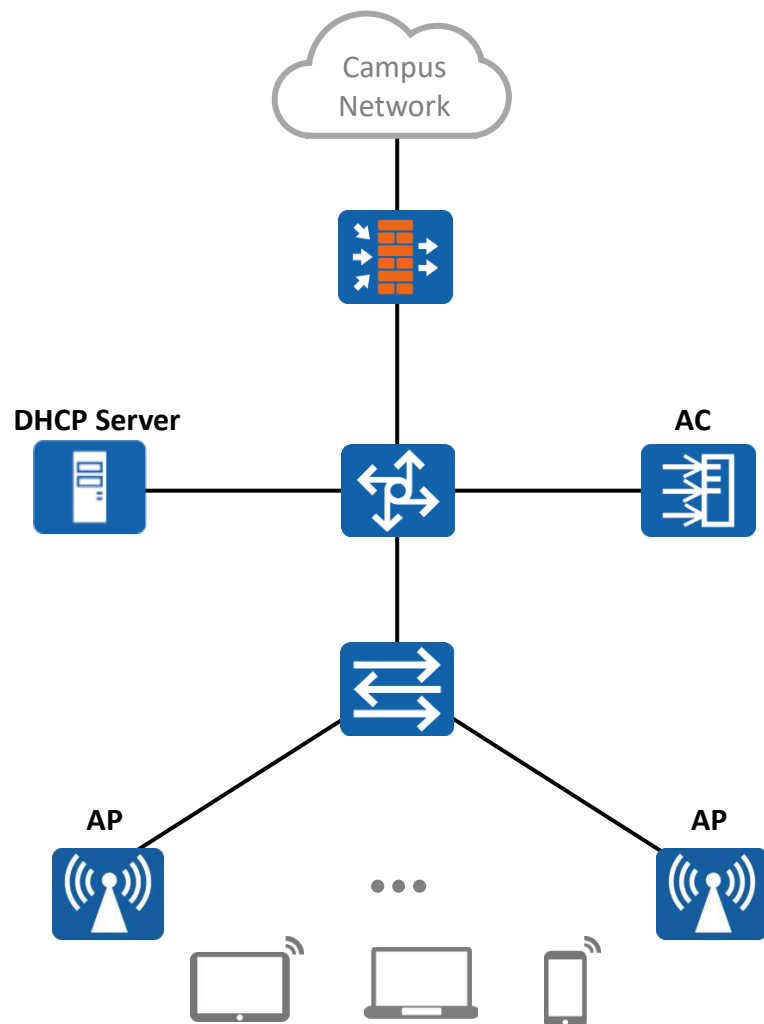
STAs find the SSID transmitted by the AP, connect to the network, and go online.

4 WLAN service data forwarding

The WLAN starts to forward service data.



WLAN Working Process: Step 1



WLAN Working Process

1 AP onboarding

The AC can manage and control Fit APs in a centralized manner and deliver services only after they go online. The procedure is as follows:

1. An AP obtains an IP address.
2. The AP discovers the AC and establishes a CAPWAP tunnel with it.
3. AP access control
4. AP upgrade
5. CAPWAP tunnel maintenance

2 WLAN service configuration delivery

3 STA access

4 WLAN service data forwarding



APs Obtain IP Addresses

- An AP can communicate with an AC only after obtaining an IP address.

IP Address Allocation

- **An AP can obtain an IP address** in either of the following modes:
 - Static mode: A user logs in to the AP and configures its IP address.
 - DHCP mode: The AP serves as a DHCP client and requests an IP address from a DHCP server.
- Typical solutions:
 - Deploy a dedicated DHCP server to assign IP addresses to APs.
 - Configure the AC to assign IP addresses to APs.
 - Use a device on the network, such as a core switch, to assign IP addresses to APs.



IP address allocation



CAPWAP
tunnel establishment



AP access control



AP upgrade
(Optional)

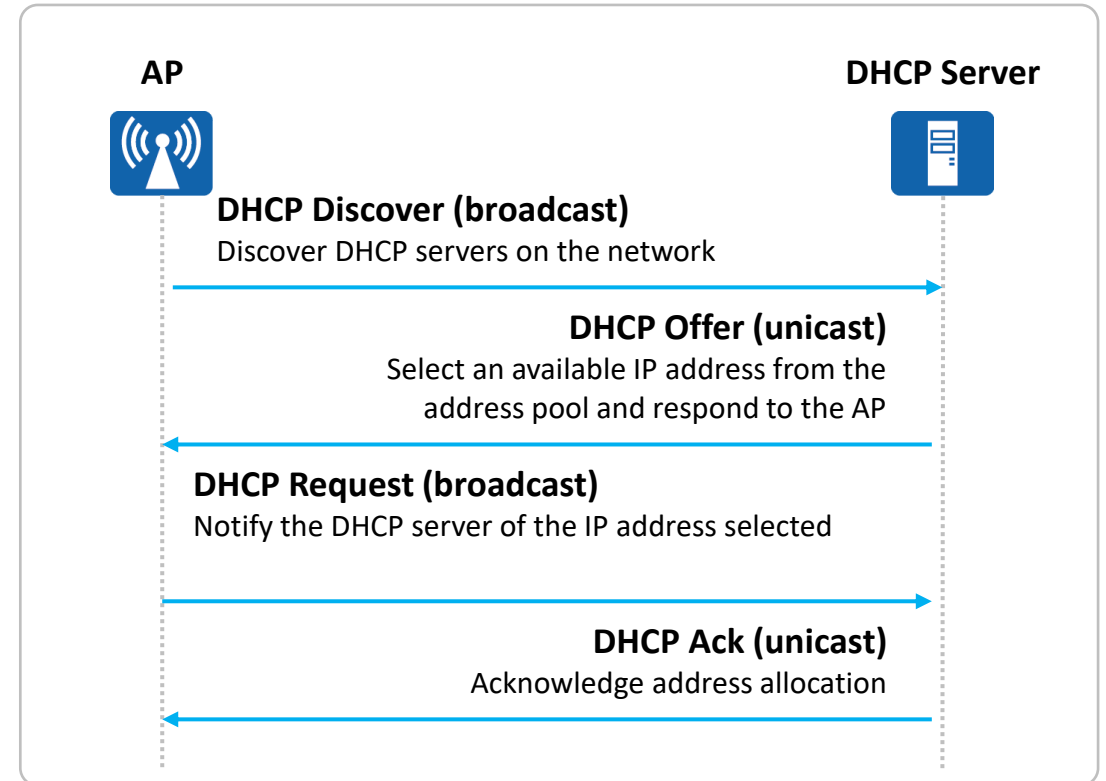
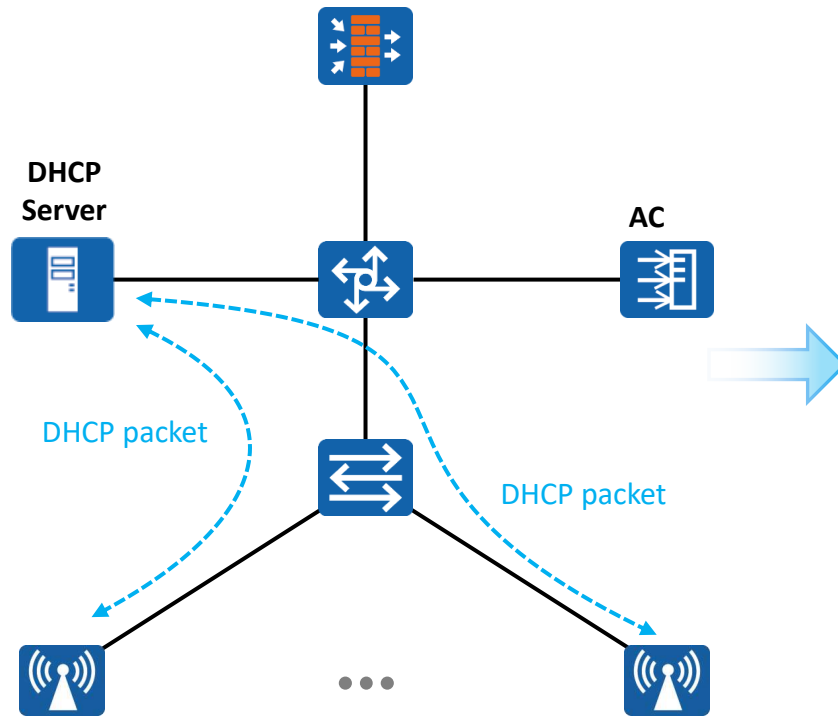


CAPWAP
tunnel maintenance



DHCP IP Address Allocation

- IP address allocation
- CAPWAP tunnel establishment
- AP access control
- AP upgrade (Optional)
- CAPWAP tunnel maintenance





CAPWAP Tunnel Establishment



IP address allocation



**CAPWAP
tunnel establishment**



AP access control



AP upgrade
(Optional)



CAPWAP
tunnel maintenance

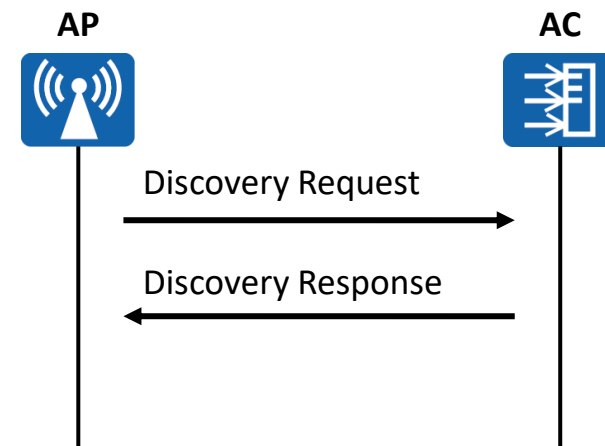
- The AC manages and controls APs in a centralized manner through CAPWAP tunnels.

Step 1: AC Discovery

- An AP sends a Discovery Request packet to **find an available AC**
- APs can discover an AC in either of the following ways:
 - Static: AC IP address list preconfigured on the APs
 - Dynamic: DHCP, DNS, and broadcast

Step 2: CAPWAP Tunnel Establishment

- APs associate with the AC and **establish CAPWAP tunnels** including data tunnels and control
 - Data tunnel: transmits service data packets from APs to the AC for centralized forwarding.
 - Control tunnel: transmits control packets between the AC and APs.

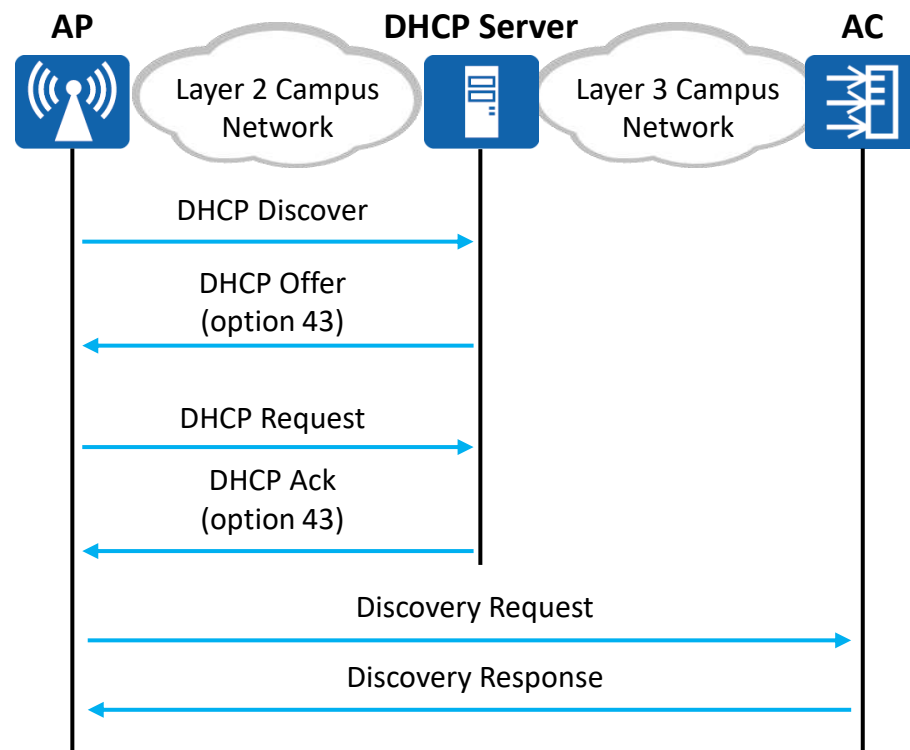




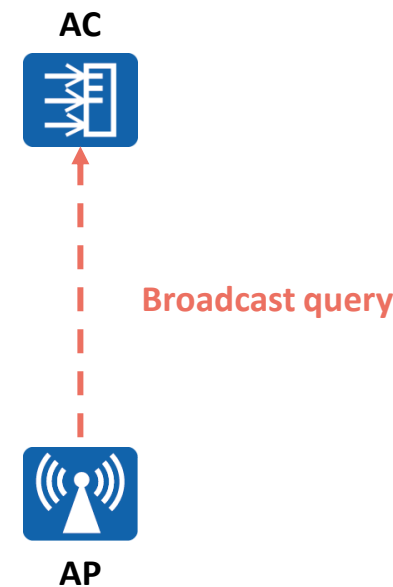
Step 1: APs Dynamically Discover the AC

- IP address allocation
- CAPWAP tunnel establishment
- AP access control
- AP upgrade (Optional)
- CAPWAP tunnel maintenance

DHCP Mode (Layer 3 Networking)



Broadcast Mode (Layer 2 Networking)

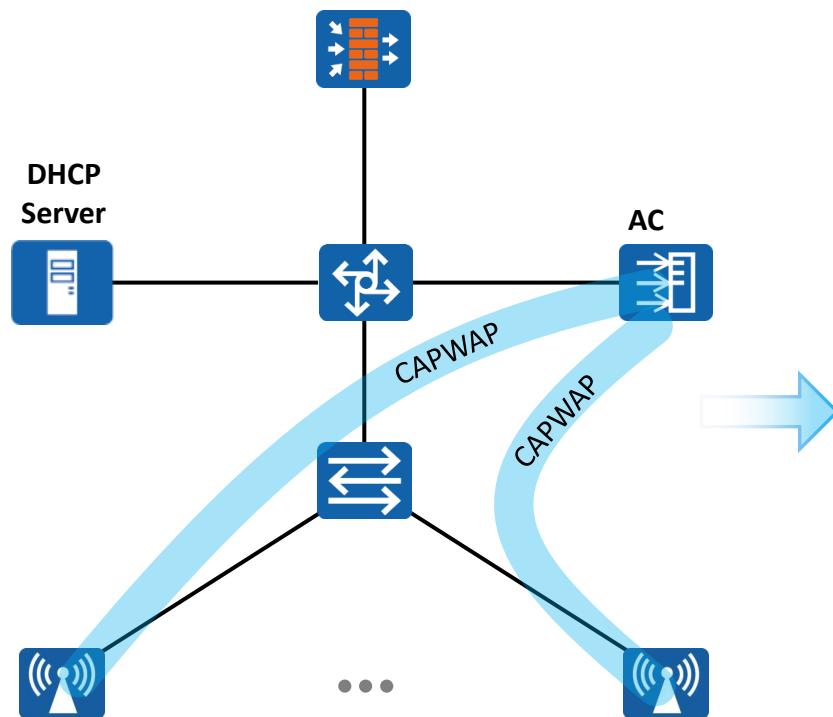




Step 2: CAPWAP Tunnel Establishment

- IP address allocation
- CAPWAP tunnel establishment**
- AP access control
- AP upgrade (Optional)
- CAPWAP tunnel maintenance

CAPWAP tunnel



Step 2: CAPWAP Tunnel Establishment

- APs associate with the AC and **establish CAPWAP tunnels**, including data tunnels and control tunnels.
 - Data tunnel: transmits service data packets from APs to the AC for centralized forwarding. Datagram Transport Layer Security (DTLS) encryption can be enabled over the data tunnel to ensure security of CAPWAP data packets. Subsequently, CAPWAP data packets will be encrypted and decrypted using DTLS.
 - Control tunnel: transmits control packets between the AC and APs. DTLS encryption can be enabled over the control tunnel to ensure security of CAPWAP control packets. Subsequently, CAPWAP control packets will be encrypted and decrypted using DTLS.



AP Access Control

AP Onboarding

Configuration
Delivery

STA Access

Data Forwarding



IP address allocation



CAPWAP
tunnel establishment



AP access control



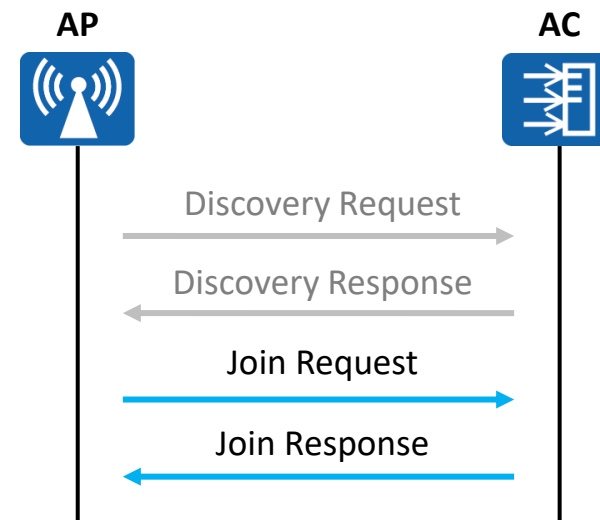
AP upgrade
(Optional)



CAPWAP
tunnel maintenance

AP Access Control

- After discovering and AC, the AP sends a Join Request packet to the AC. The AC then determines **whether to allow the AP access** sends a Join Response packet to the AP.
- The AC supports three AP authentication modes: MAC address authentication, SN authentication, and non-authentication.





AP Upgrade



IP address allocation



CAPWAP
tunnel establishment



AP access control



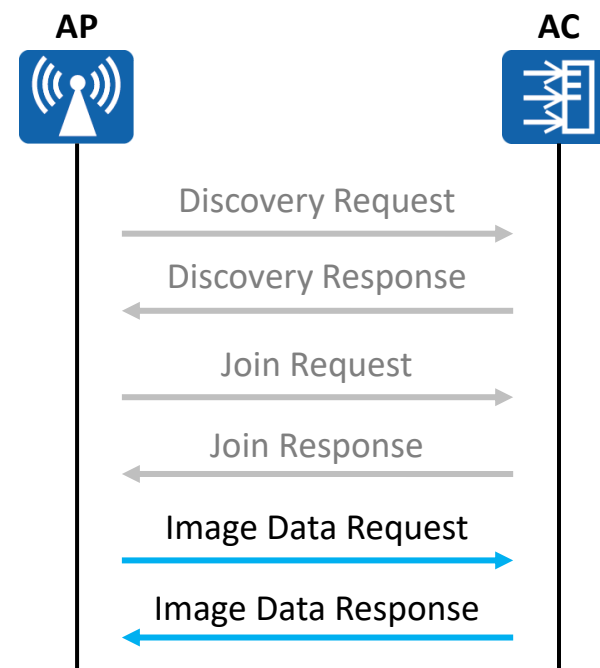
**AP upgrade
(Optional)**



CAPWAP
tunnel maintenance

AP Upgrade

- The AP determines whether its system software version is the same as that specified on the AC according to parameters in the received Join Response packet. If they are different, the AP sends an Image Data Request packet to request the software package and then **upgrades its software version** in AC, FTP, or SFTP mode.
- After the software version is updated, the AP restarts and repeats steps 1 to 3.





CAPWAP Tunnel Maintenance

AP Onboarding

Configuration
Delivery

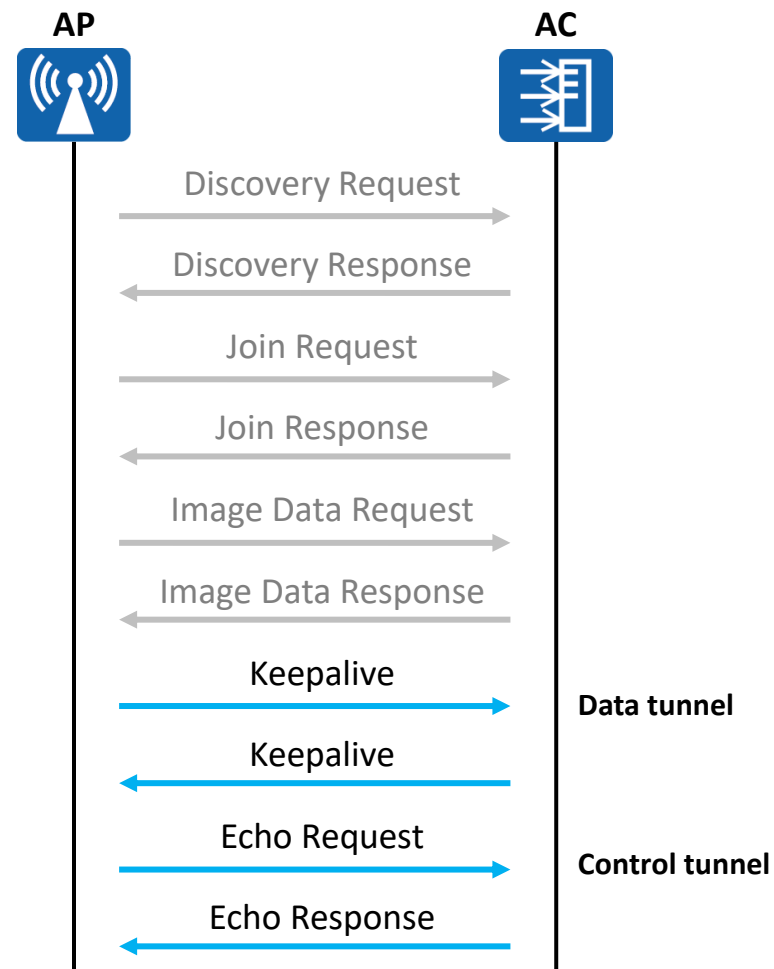
STA Access

Data Forwarding

- IP address allocation
- CAPWAP tunnel establishment
- AP access control
- AP upgrade (Optional)
- CAPWAP tunnel maintenance**

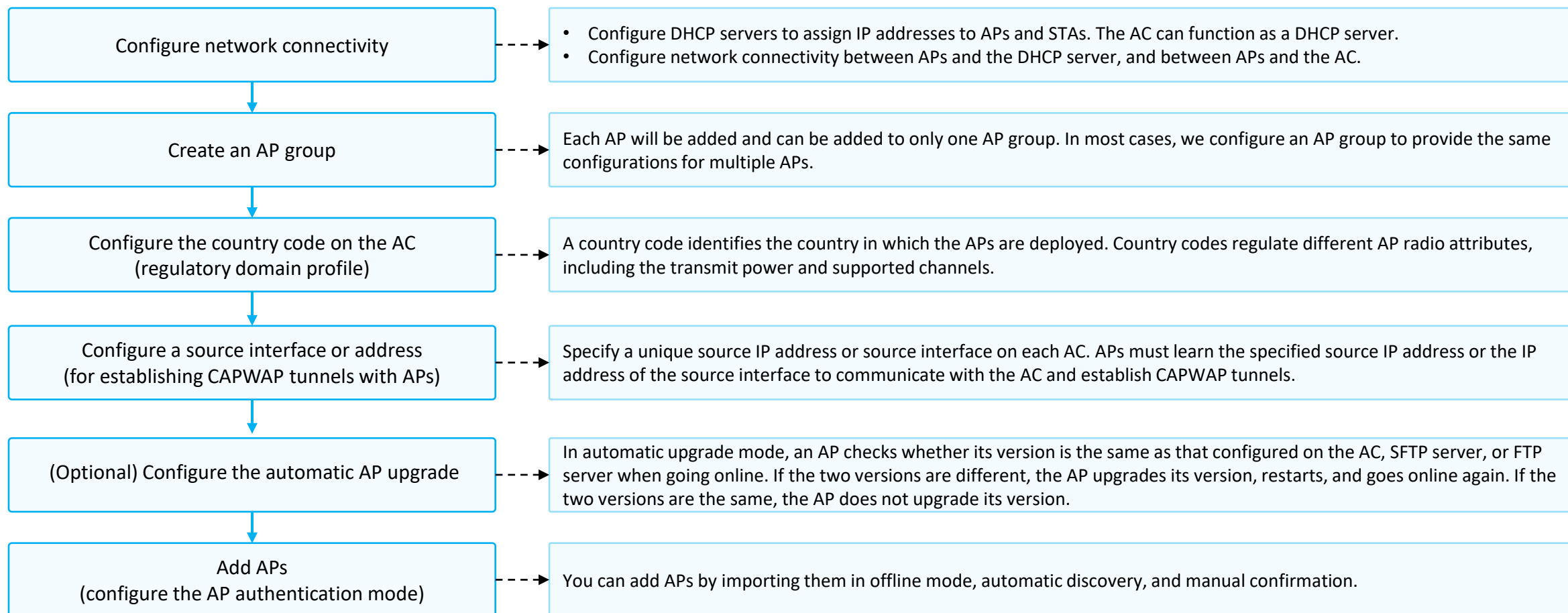
CAPWAP Tunnel Maintenance

- Data tunnel maintenance:
 - The AP and AC exchange Keepalive packets to detect the data tunnel connectivity.
- Control tunnel maintenance:
 - The AP and AC exchange Echo packets to detect the control tunnel connectivity.



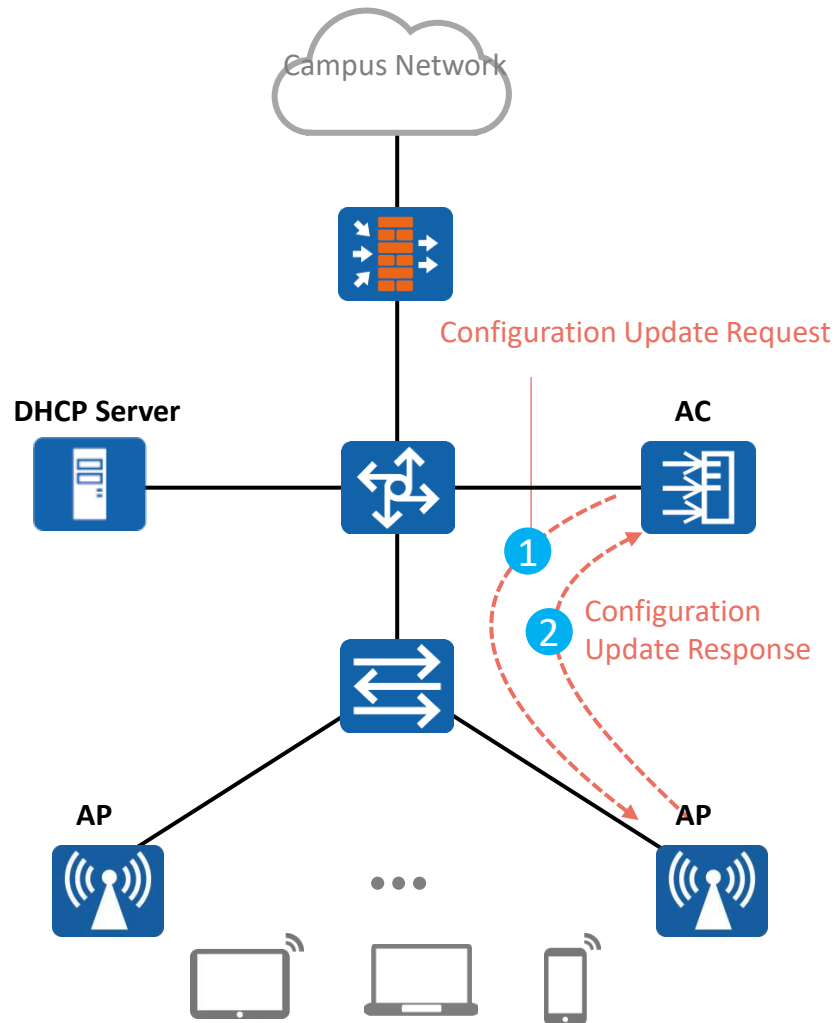


Preconfigurations on the AC for APs to Go Online





WLAN Working Process: Step 2



WLAN Working Process

- ① AP onboarding
- ② WLAN service configuration delivery

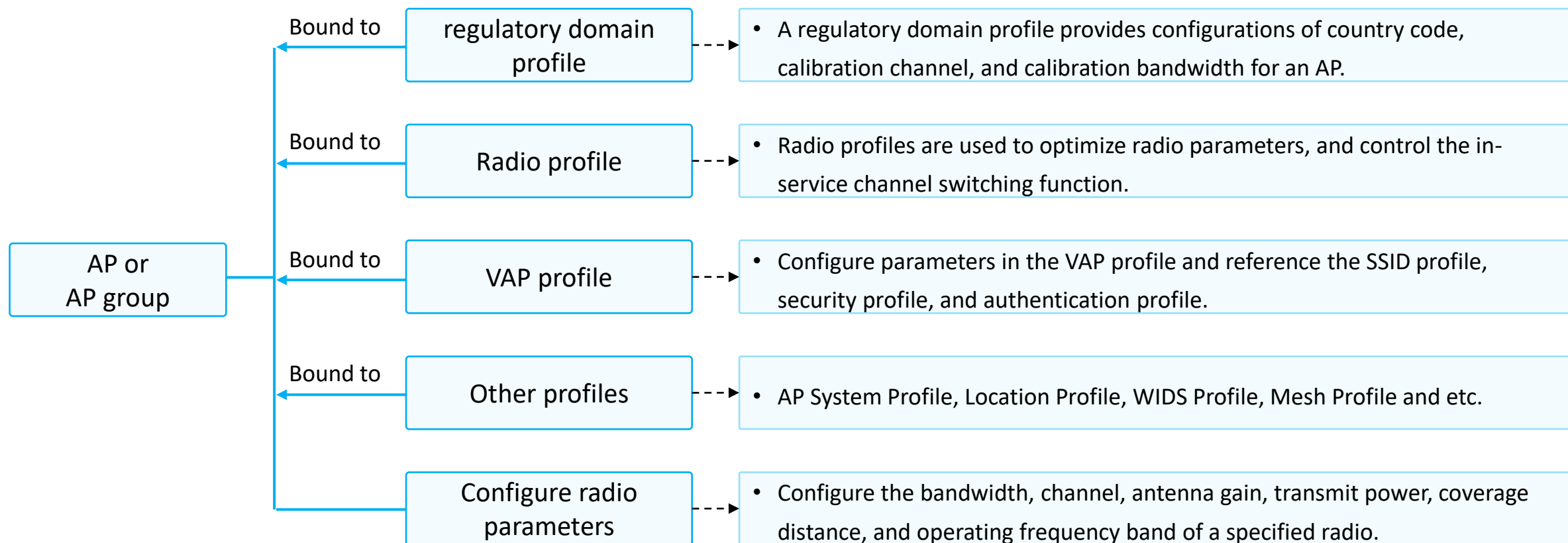
The AC sends a Configuration Update Request to an AP. If the AC receives a Configuration Update Response from the AP, the AC then delivers service configuration to the AP.

- ③ STA access
- ④ WLAN service data forwarding



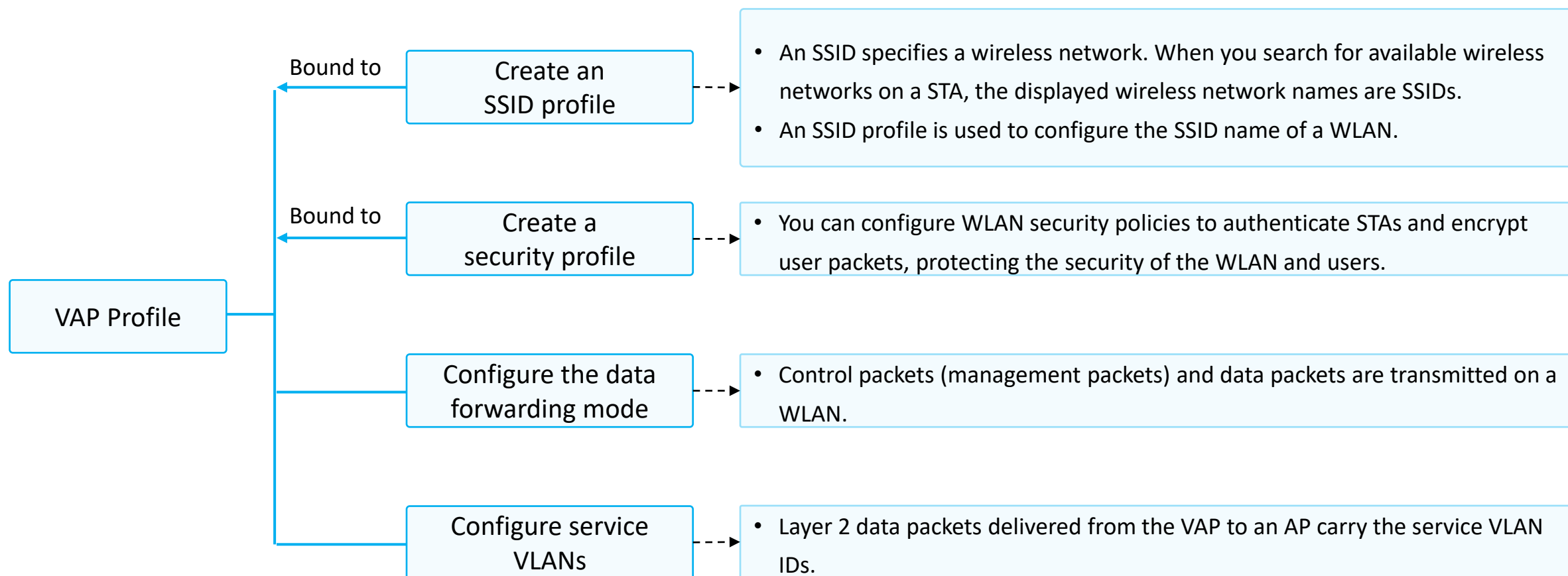
WLAN Profiles

- Various profiles are designed based on different functions and features of WLAN networks to help users configure and maintain functions of WLAN networks. These profiles are called WLAN profiles.



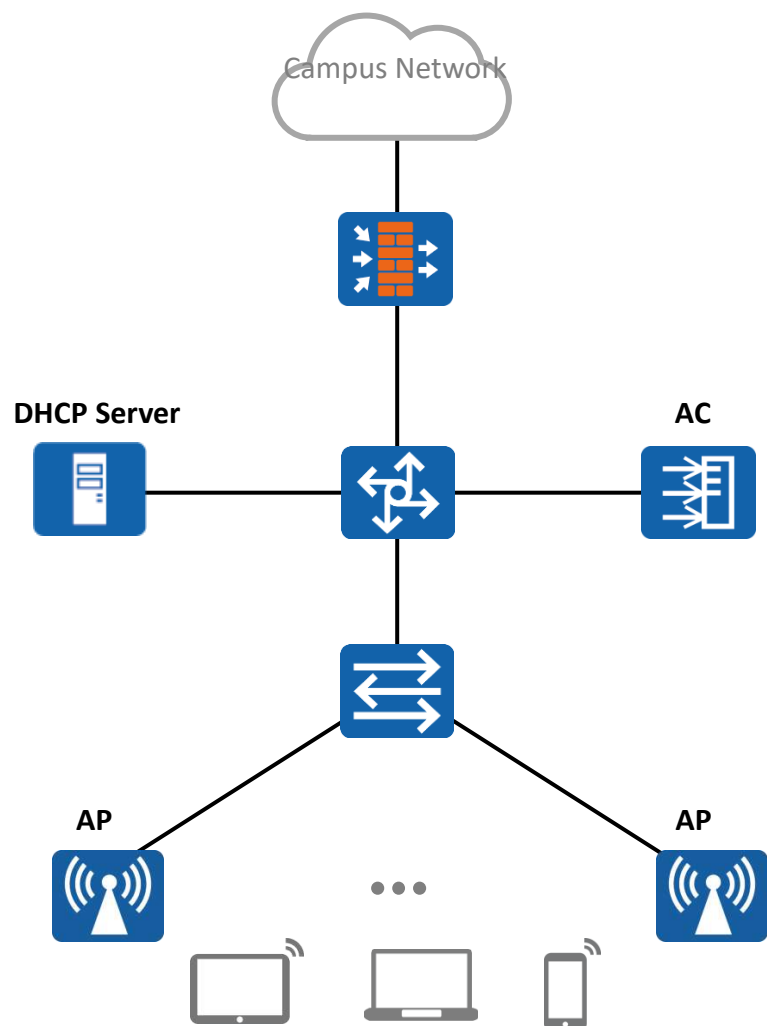


VAP Profile





WLAN Working Process: Step 3



WLAN Working Process

- ① AP onboarding
- ② WLAN service configuration delivery
- ③ STA access

STAs can access a WLAN after CAPWAP tunnels are established. The STA access process consists of six phases: scanning, link authentication, association, access authentication, DHCP, and user authentication.

- ④ WLAN service data forwarding



Scanning



Scanning



Link authentication



Association



Access authentication



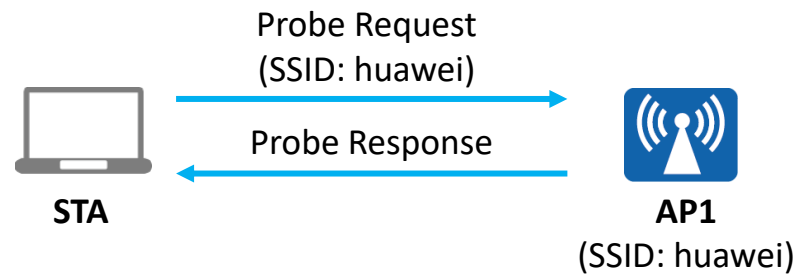
DHCP



User authentication

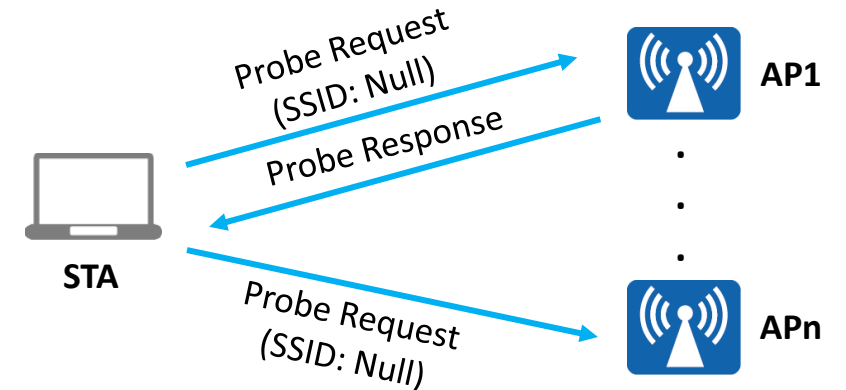
- In active scanning, a STA **periodically searches** for nearby wireless networks.
- The STA can send two types of Probe Request frames: probes containing an SSID and probes that do not contain an SSID.

Active Scanning by Sending a Probe Request Frame Containing an SSID



- The STA sends a Probe Request containing an SSID on each channel to search for the AP with the same SSID. Only the AP with the same SSID will respond to the STA.

Active Scanning by Sending a Probe Request Frame Containing No SSID



- The STA periodically broadcasts a Probe Request frame that does not contain an SSID on the supported channels. The APs return Probe Response frames to notify the STA of the wireless services they can provide.



WLAN Security Protocols



Scanning



Link authentication



Association



Access authentication



DHCP



User authentication

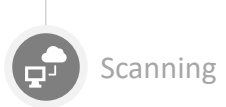
- As WLAN technologies use radio signals to transmit service data, service data can be easily intercepted or tampered with by attackers when being transmitted on open wireless channels. Ensuring WLAN security is crucial to building safe and effective wireless networks.
- Common security policy:

Security Policy	Link Authentication	Access Authentication	Data Encryption	Description
WEP	Open system	N/A	No encryption or WEP	Insecure policy
	Shared-key Authentication	N/A	WEP	Insecure policy
WPA/WPA2-802.1X	Open system	802.1X (EAP)	TKIP or CCMP	A more secure policy, applicable to large enterprises
WPA/WPA2-PSK	Open system	PSK	TKIP or CCMP	More secure policy, applicable to small- and medium-sized enterprises or household users



Link Authentication

- To ensure wireless link security, **an AP needs to authenticate STAs that attempt to access the AP.**
- IEEE 802.11 defines two authentication modes: open system authentication and shared key authentication.



Scanning



Link authentication



Association



Access authentication

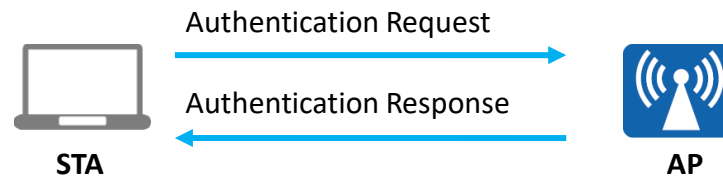


DHCP



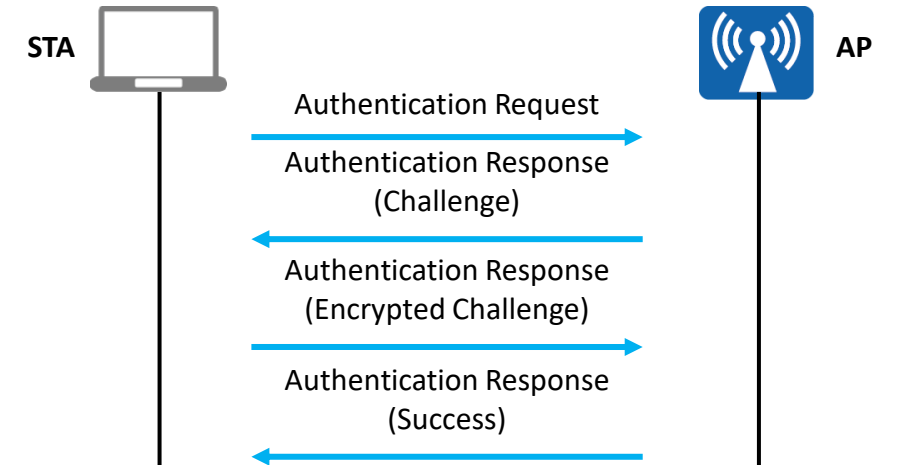
User authentication

Open System Authentication



- Open system authentication requires no authentication, allowing any STA to be successfully authenticated.

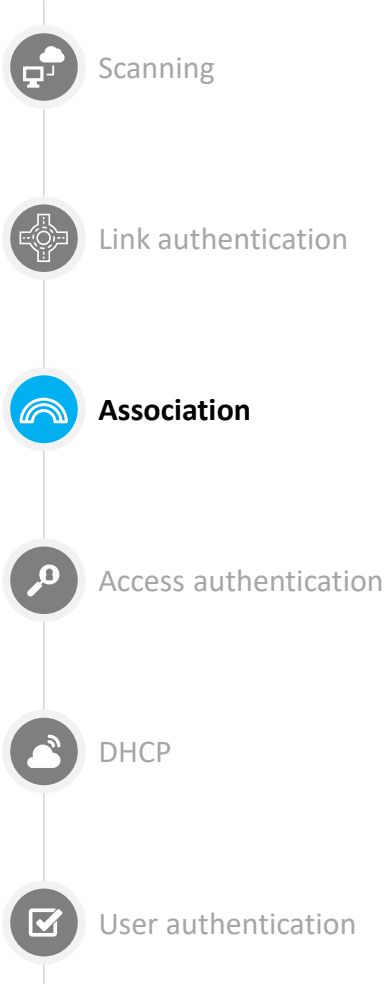
Shared Key Authentication



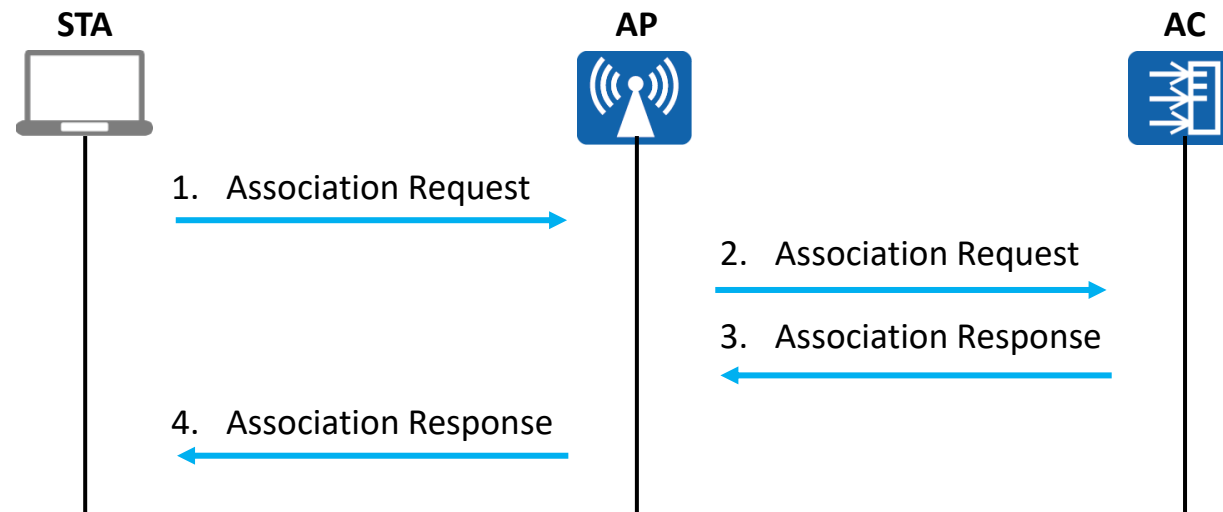
- Shared key authentication requires that the STA and AP have the same shared key preconfigured. The AP checks whether a STA has the same shared key to determine the authentication result. If the STA has the same shared key as the AP, the STA is authenticated. Otherwise, STA authentication fails.



Association



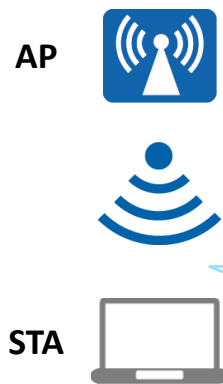
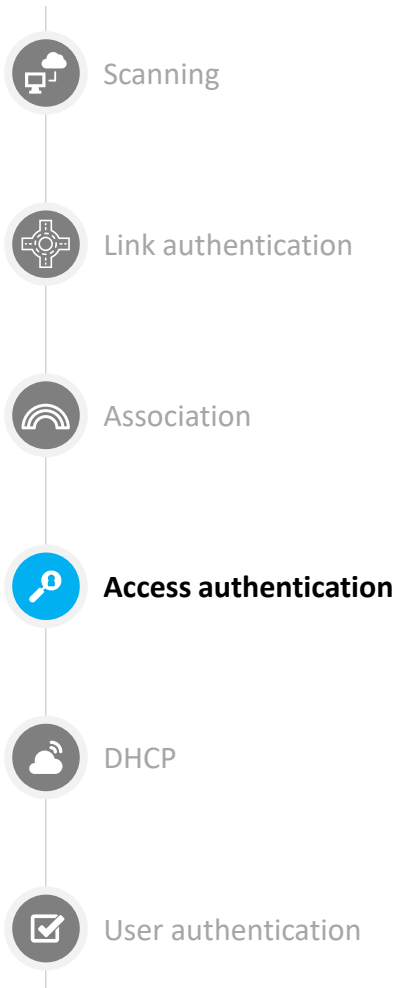
- After link authentication is complete, a STA initiates **link service negotiation** using Association packets.
- The STA association process is actually a link service negotiation process, during which the supported rate, channel, and the like are negotiated.





Access Authentication

- User access authentication **differentiates users** and controls access rights of users. Compared with link authentication, access authentication is more secure.
- Major access authentication modes include PSK authentication and 802.1X authentication.



Access authentication is performed on the wireless-side interface, allowing STAs to send data over wireless links.



STA Address Allocation



Scanning



Link authentication



Association



Access authentication

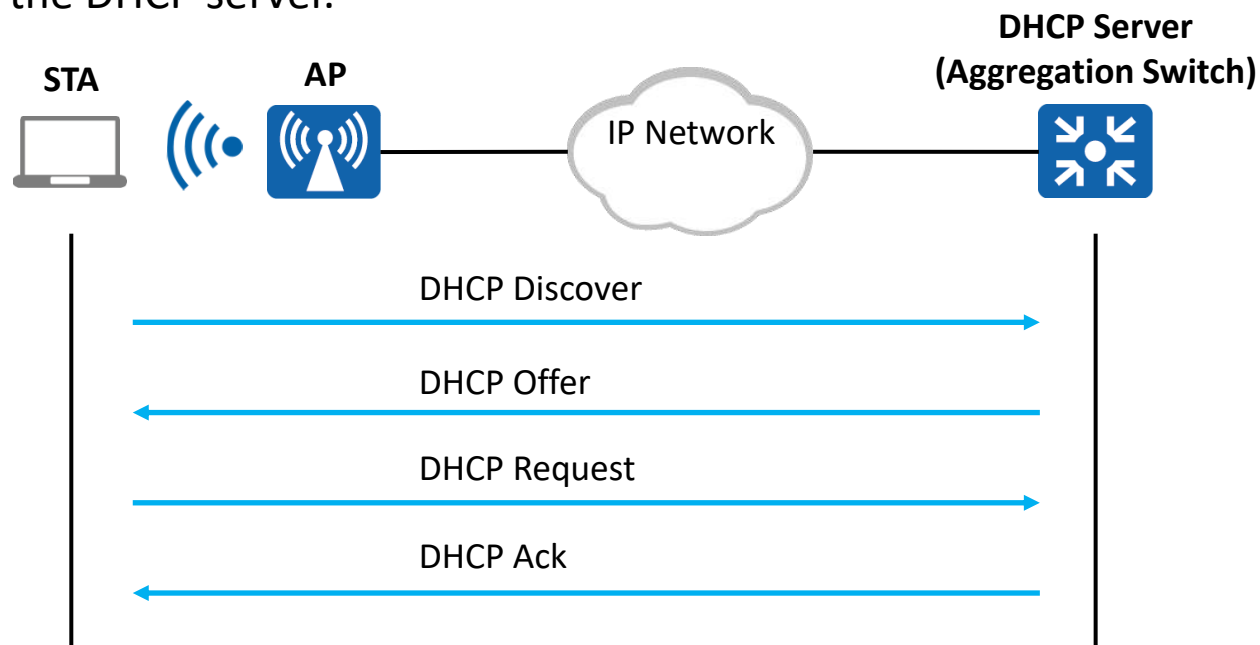


DHCP



User authentication

- The prerequisite for APs and STAs to go online properly is that they have obtained IP addresses.
- If STAs obtain IP addresses through DHCP, the AC or aggregation switch can function as a DHCP server to assign IP addresses to the STAs. In most cases, the aggregation switch is used as the DHCP server.





User Authentication

- User authentication is an end-to-end security architecture, supporting 802.1X, MAC address, and Portal authentication modes.

Portal Authentication

- Portal authentication is also known as web authentication. Portal authentication websites are referred to as web portals.
- To access the Internet, users must be authenticated on web portals. **The users can access network resources only after successful authentication.**



Huawei-Guest

Just for Guest

+86

Phone Number

Password

Get
Password

Login



Scanning



Link authentication



Association



Access authentication



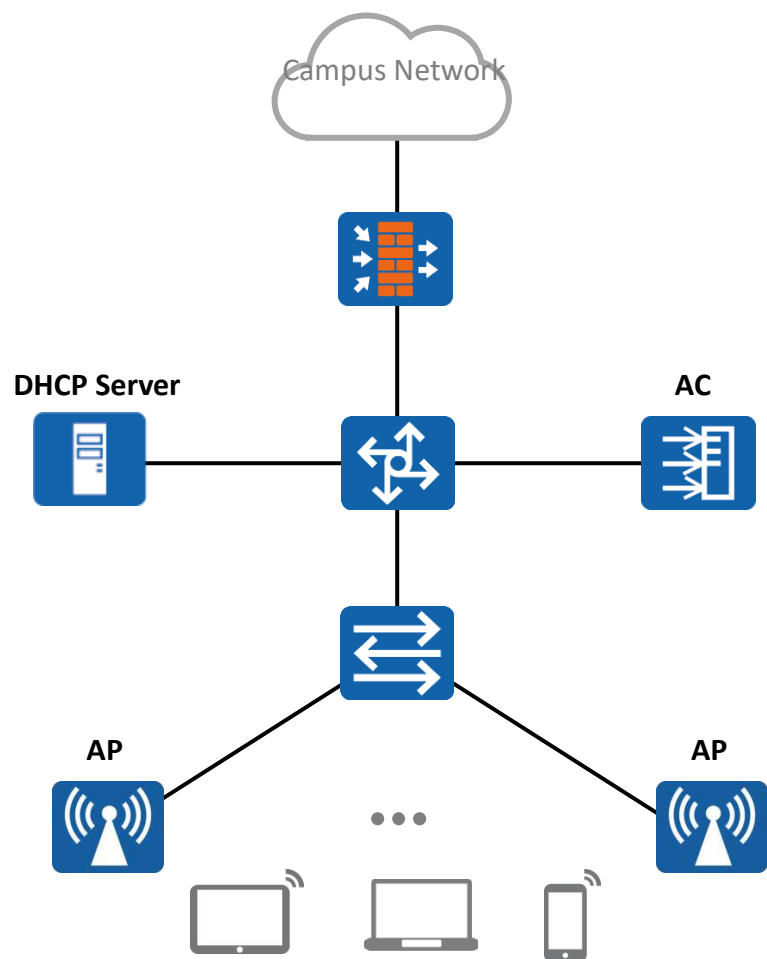
DHCP



User authentication



WLAN Working Process: Step 4



WLAN Working Process

- ① AP onboarding
- ② WLAN service configuration delivery
- ③ STA access
- ④ WLAN service data forwarding

Control packets (management packets) and data packets are transmitted over CAPWAP tunnels.

- Control packets are forwarded through the CAPWAP control tunnel.
- User data packets can be forwarded in tunnel forwarding (centralized forwarding) or direct forwarding (local forwarding) mode.



Data Forwarding Mode

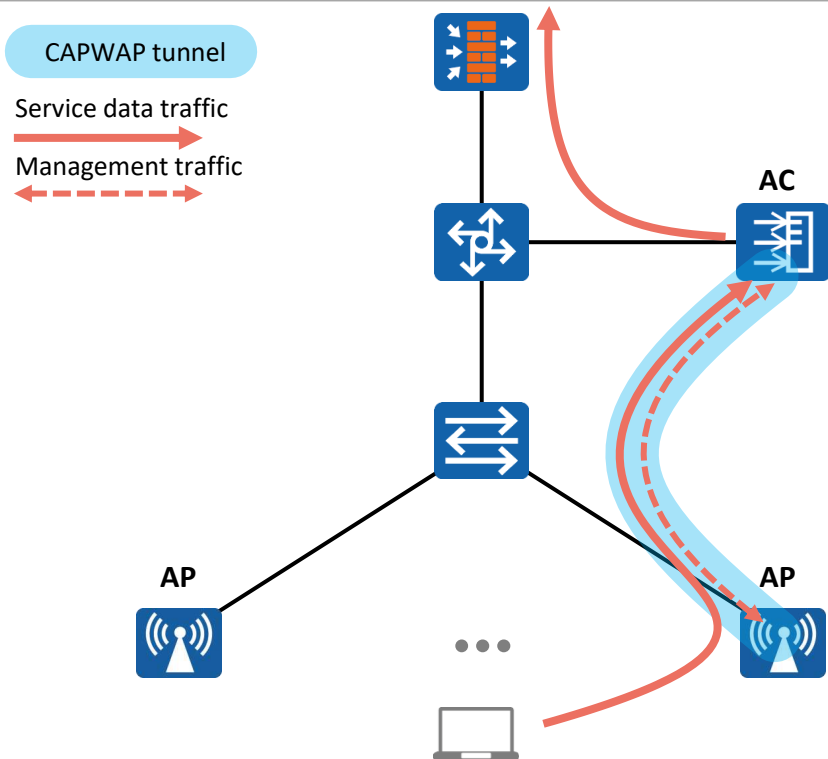
AP Onboarding

Configuration
Delivery

STA Access

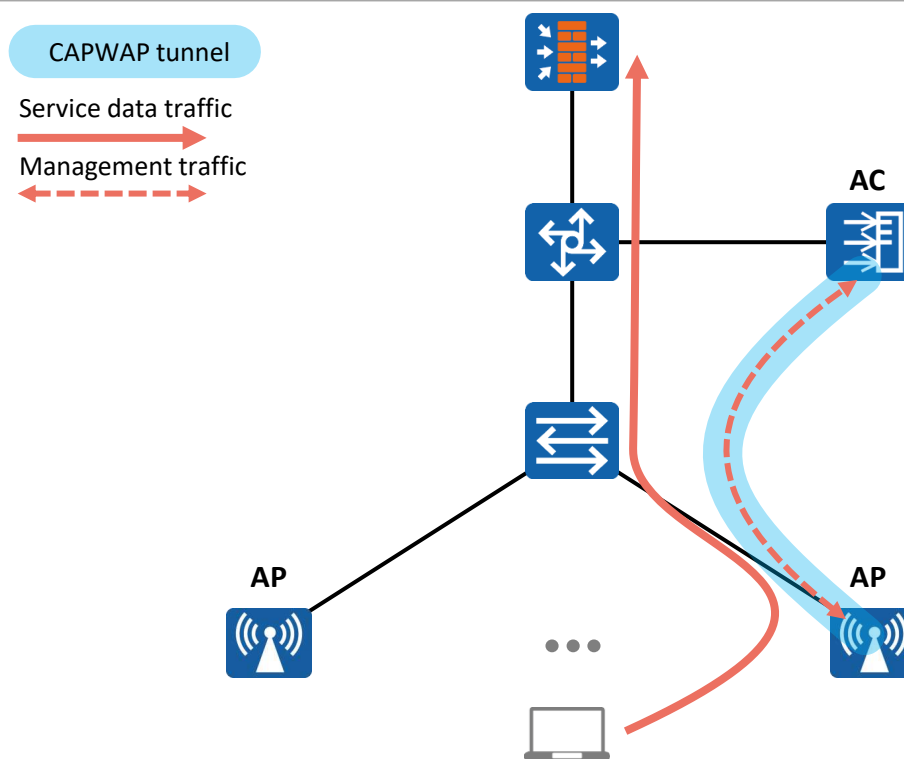
Data Forwarding

Tunnel Forwarding



- In tunnel forwarding mode, APs encapsulate user data packets over a CAPWAP data tunnel and send them to an AC. The AC then forwards these packets to an upper-layer network.

Direct Forwarding



- In direct forwarding mode, an AP directly forwards user data packets to an upper-layer network without encapsulating them over a CAPWAP data tunnel.



Contents

1. WLAN Overview
2. Basic Concepts of WLAN
3. WLAN Fundamentals
- 4. WLAN Configuration Implementation**
5. Next-Generation WLAN Solutions



Basic WLAN Configuration Commands: Configuring an AP to Go Online (1)

1. Configure the AC as a DHCP server and configure the Option 43 field.

```
[AC-ip-pool-pool1] option code [ sub-option sub-code ] { ascii ascii-string | hex hex-string | cipher cipher-string | ip-address ip-address
```

Configure the user-defined option that a DHCP server assigns to a DHCP client.

2. Create a regulatory domain profile and configure the country code.

```
[AC] wlan  
[AC-wlan-view]
```

Enter the WLAN view.

```
[AC-wlan-view] regulatory-domain-profile name profile-name  
[AC-wlan-regulate-domain-profile-name]
```

Create a regulatory domain profile and enter the regulatory domain profile view, or enter the view of an existing regulatory domain profile.

```
[AC-wlan-regulate-domain-profile-name] country-code country-code
```

Configure the country code.



Basic WLAN Configuration Commands: Configuring an AP to Go Online (2)

```
[AC-wlan-view] ap-group name group-name  
[AC-wlan-ap-group-group-name]
```

Create an AP group and enter the AP group view, or enter the view of an existing AP group.

```
[AC-wlan-ap-group-group-name] regulatory-domain-profile profile-name
```

Bind the regulatory domain profile to an AP or AP group.

3. Configure a source interface or address.

```
[AC] capwap source interface { loopback loopback-number | vlanif vlan-id }
```

Specify a source interface on the AC for establishing CAPWAP tunnels with APs.

```
[AC] capwap source ip-address ip-address
```

Configure the source IP address on the AC.



Basic WLAN Configuration Commands: Configuring an AP to Go Online (3)

4. Add APs in offline mode.

```
[AC-wlan-view] ap auth-mode { mac-auth | sn-auth }
```

Set the AP authentication mode to MAC address or SN authentication. By default, MAC address authentication is used.

```
[AC-wlan-view] ap-id ap-id [ [ type-id type-id | ap-type ap-type ] { ap-mac ap-mac | ap-sn ap-sn | ap-mac ap-mac ap-sn ap-sn } ]  
[AC-wlan-ap-ap-id] ap-name ap-name
```

Manually add an AP in offline mode or enter the AP view, and configure the name of a single AP.

```
[AC-wlan-view] ap-id 0  
[AC-wlan-ap-0] ap-group ap-group
```

Add the AP to an AP group.

5. Verify the configuration.

```
[AC] display ap { all | ap-group ap-group }
```

Check AP information.



Basic WLAN Configuration Commands: Configuring Radios (1)

1. Enter the radio view.

```
[AC-wlan-view] ap-id 0  
[AC-wlan-ap-0] radio radio-id  
[AC-wlan-radio-0]
```

2. Configure the working bandwidth and channel for a radio.

```
[AC-wlan-radio-0/0] channel { 20mhz | 40mhz-minus | 40mhz-plus | 80mhz | 160mhz } channel  
Warning: This action may cause service interruption. Continue?[Y/N]y
```

```
[AC-wlan-radio-0/0] channel 80+80mhz channel1 channel2  
Warning: This action may cause service interruption. Continue?[Y/N]y
```

Configure the working bandwidth and channel for all APs in an AP group or for a specified radio of a single AP.

3. Configure the antenna gain.

```
[AC-wlan-radio-0/0] antenna-gain antenna-gain
```

Configure the antenna gain for all APs in an AP group or for a specified radio of a single AP.



Basic WLAN Configuration Commands: Configuring Radios (2)

4. Configure the transmit power for a radio.

```
[AC-wlan-radio-0/0] eirp eirp
```

Configure the transmit power for all APs in an AP group or for a specified radio of a single AP.

5. Configure the radio coverage distance.

```
[AC-wlan-radio-0/0] coverage distance distance
```

Configure the radio coverage distance for all APs in an AP group or for a specified radio of a single AP.

6. Configure the operating frequency for a radio.

```
[AC-wlan-radio-0/0] frequency { 2.4g | 5g }
```



Basic WLAN Configuration Commands: Configuring Radios (3)

7. Create a radio profile.

```
[AC-wlan-view] radio-2g-profile name profile-name
```

Create a 2G radio profile and enter the 2G radio profile view, or enter the view of an existing 2G radio profile.

8. Bind the radio profile.

```
[AC-wlan-view] ap-group name group-name
```

```
[AC-wlan-ap-group-group-name] radio-2g-profile profile-name radio { radio-id | all }
```

Bind the specified 2G radio profile to the 2G radio in the AP group.



Basic WLAN Configuration Commands: Configuring VAPs (1)

1. Create a VAP profile.

```
[AC-wlan-view] vap-profile name profile-name  
[AC-wlan-vap-prof-profile-name]
```

Create a VAP profile and enter the VAP profile view, or enter the view of an existing VAP profile.

2. Configure the data forwarding mode.

```
[AC-wlan-vap-prof-profile-name] forward-mode { direct-forward | tunnel }
```

Set the data forwarding mode in the VAP profile to direct or tunnel.

3. Configure service VLANs.

```
[AC-wlan-vap-prof-profile-name] service-vlan { vlan-id vlan-id | vlan-pool pool-name }
```

Configure service VLANs configured for the VAP.



Basic WLAN Configuration Commands: Configuring VAPs (2)

4. Configure a security profile.

```
[AC-wlan-view] security-profile name profile-name  
[AC-wlan-sec-prof-profile-name]
```

Create a security profile and enter the security profile view.

By default, security profiles **default**, **default-wds**, and **default-mesh** are available in the system.

```
[AC-wlan-view] vap-profile name profile-name  
[AC-wlan-vap-prof-profile-name] security-profile profile-name
```

Bind the security profile to the VAP profile.



Basic WLAN Configuration Commands: Configuring VAPs (3)

5. Configure an SSID profile.

```
[AC-wlan-view] ssid-profile name profile-name  
[AC-wlan-ssid-prof-profile-name]
```

Create an SSID profile and enter the SSID profile view, or enter the view of an existing SSID profile.
By default, the system provides the SSID profile **default**.

```
[AC-wlan-ssid-prof-profile-name] ssid ssid
```

Configure an SSID for the SSID profile.
By default, the SSID **HUAWEI-WLAN** is configured in an SSID profile.

```
[AC-wlan-view] vap-profile name profile-name  
[AC-wlan-vap-prof-profile-name] ssid-profile profile-name
```

Bind the SSID profile to the VAP profile.



Basic WLAN Configuration Commands: Configuring VAPs (4)

6. Bind the VAP profile.

```
[AC-wlan-view] ap-group name group-name  
[AC-wlan-ap-group-group-name] vap-profile profile-name wlan wlan-id radio { radio-id | all } [ service-vlan { vlan-id vlan-id  
| vlan-pool pool-name } ]
```

Bind the specified VAP profile to radios in an AP group.

7. Check VAP information.

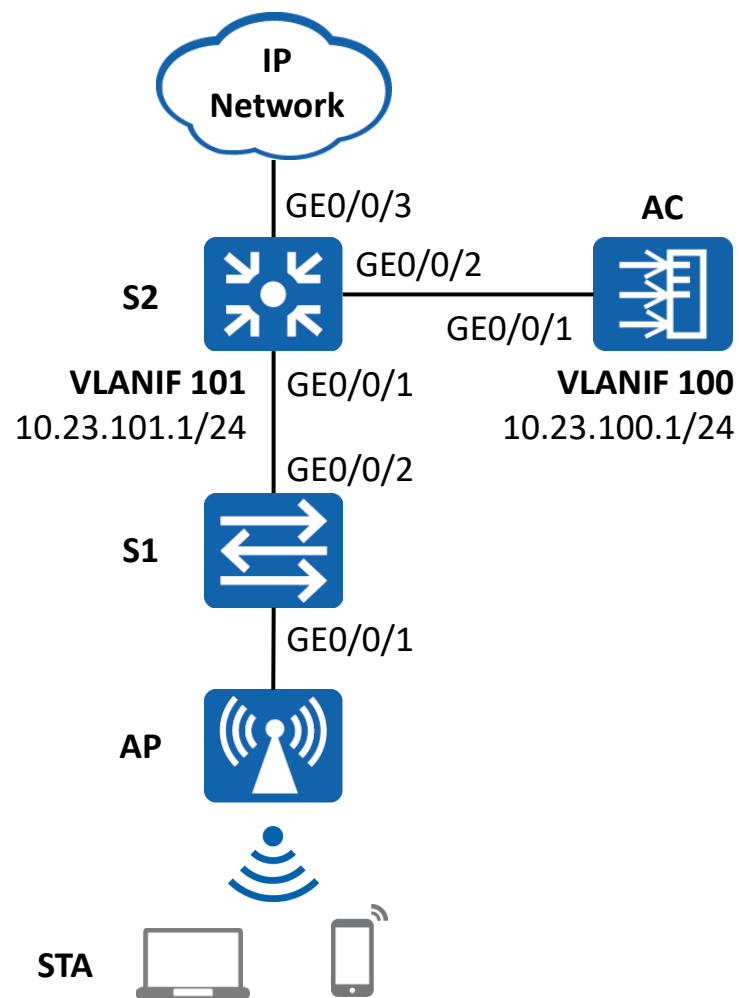
```
[AC] display vap { ap-group ap-group-name | { ap-name ap-name | ap-id ap-id } [ radio radio-id ] } [ ssid ssid ]
```

```
[AC] display vap { all | ssid ssid }
```

Display information about service VAPs.



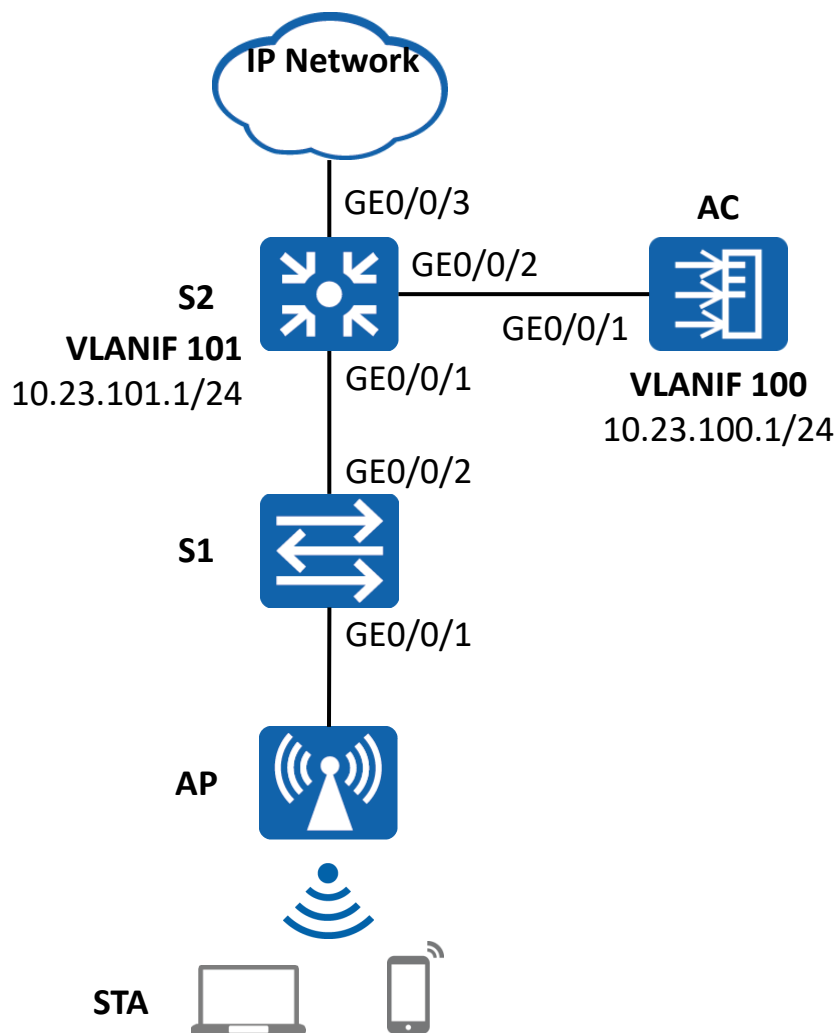
Example for Configuring Layer 2 Tunnel Forwarding in Off-Path Mode



Data	Configuration
Management VLAN for APs	VLAN 100
Service VLAN for STAs	VLAN 101
DHCP server	The AC functions as a DHCP server to assign IP addresses to APs. The aggregation switch S2 functions as a DHCP server to assign IP addresses to STAs. The default gateway address of STAs is 10.23.101.1.
IP address pool for APs	10.23.100.2–10.23.100.254/24
IP address pool for STAs	10.23.101.2–10.23.101.254/24
IP address of the AC's source interface	VLANIF 100: 10.23.100.1/24
AP group	Name: ap-group1 Referenced profiles: VAP profile wlan-net and regulatory domain profile
Regulatory domain profile	Name: default Country code: CN
SSID profile	Name: wlan-net SSID name: wlan-net
Security profile	Name: wlan-net Security policy: WPA-WPA2+PSK+AES Password: a1234567
VAP profile	Name: wlan-net Forwarding mode: tunnel forwarding Service VLAN: VLAN 101 Referenced profiles: SSID profile wlan-net and security profile wlan-net



Configuring Network Connectivity



1. Create VLANs and interfaces on S1, S2, and AC.
2. Configure DHCP servers to assign IP addresses to APs and STAs.

Configure VLANIF 100 on the AC to assign IP address to APs.

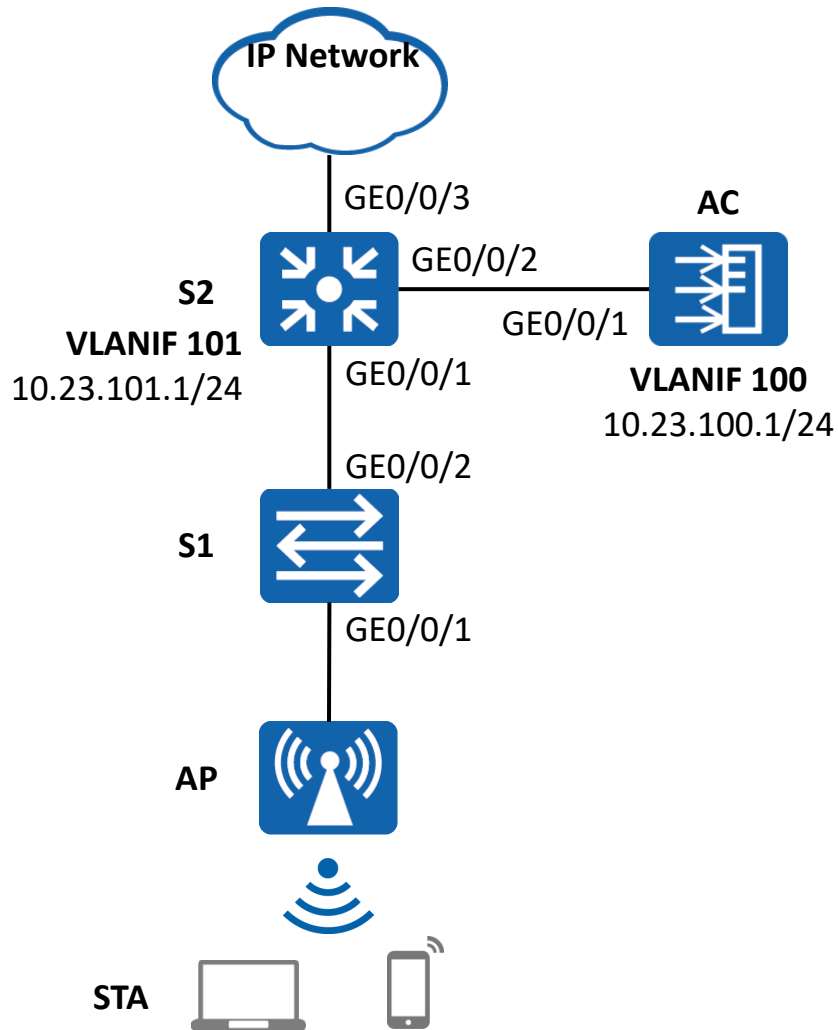
```
[AC] dhcp enable
[AC] interface vlanif 100
[AC-Vlanif100] ip address 10.23.100.1 24
[AC-Vlanif100] dhcp select interface
```

Configure VLANIF 101 on S2 to assign IP addresses to STAs and specify 10.23.101.1 as the default gateway address of the STAs.

```
[S2] dhcp enable
[S2] interface vlanif 101
[S2-Vlanif101] ip address 10.23.101.1 24
[S2-Vlanif101] dhcp select interface
```




Configuring APs to Go Online (1)



1. Create an AP group.

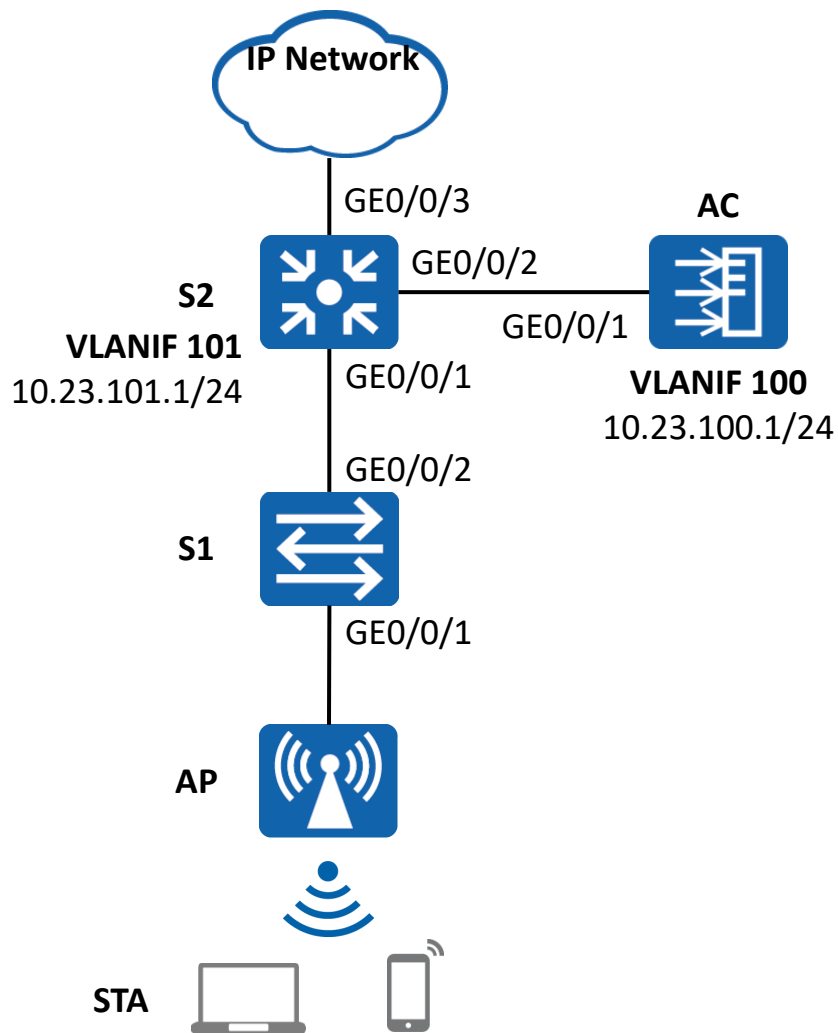
```
[AC] wlan
[AC-wlan-view] ap-group name ap-group1
[AC-wlan-ap-group-ap-group1] quit
```

2. Create a regulatory domain profile and configure the country code.

```
AC-wlan-view] regulatory-domain-profile name default
[AC-wlan-regulate-domain-default] country-code cn
[AC-wlan-regulate-domain-default] quit
[AC-wlan-view] ap-group name ap-group1
[AC-wlan-ap-group-ap-group1] regulatory-domain-profile default
Warning: Modifying the country code will clear channel, power and antenna
gain configurations of the radio and reset the AP. Continue?
e?[Y/N]:y
[AC-wlan-ap-group-ap-group1] quit
[AC-wlan-view] quit
```



Configuring APs to Go Online (2)



3. Configure the AC's source interface.

```
[AC] capwap source interface vlanif 100
```

4. Import an AP in offline mode on the AC.

```
[AC] wlan
```

```
[AC-wlan-view] ap auth-mode mac-auth
```

```
[AC-wlan-view] ap-id 0 ap-mac 60de-4476-e360
```

```
[AC-wlan-ap-0] ap-name area_1
```

Warning: This operation may cause AP reset. Continue? [Y/N]:y

```
[AC-wlan-ap-0] ap-group ap-group1
```

Warning: This operation may cause AP reset. If the country code changes, it will clear channel, power and antenna gain configurations of the radio,

Whether to continue? [Y/N]:y

```
[AC-wlan-ap-0] quit
```



Verifying the AP Onboarding Configuration

- After the AP is powered on, run the **display ap all** command to check the AP state. If the **State** field displays **nor**, the AP has gone online.

```
[AC-wlan-view] display ap all
```

```
Total AP information:
```

```
nor : normal      [1]
```

```
Extra information:
```

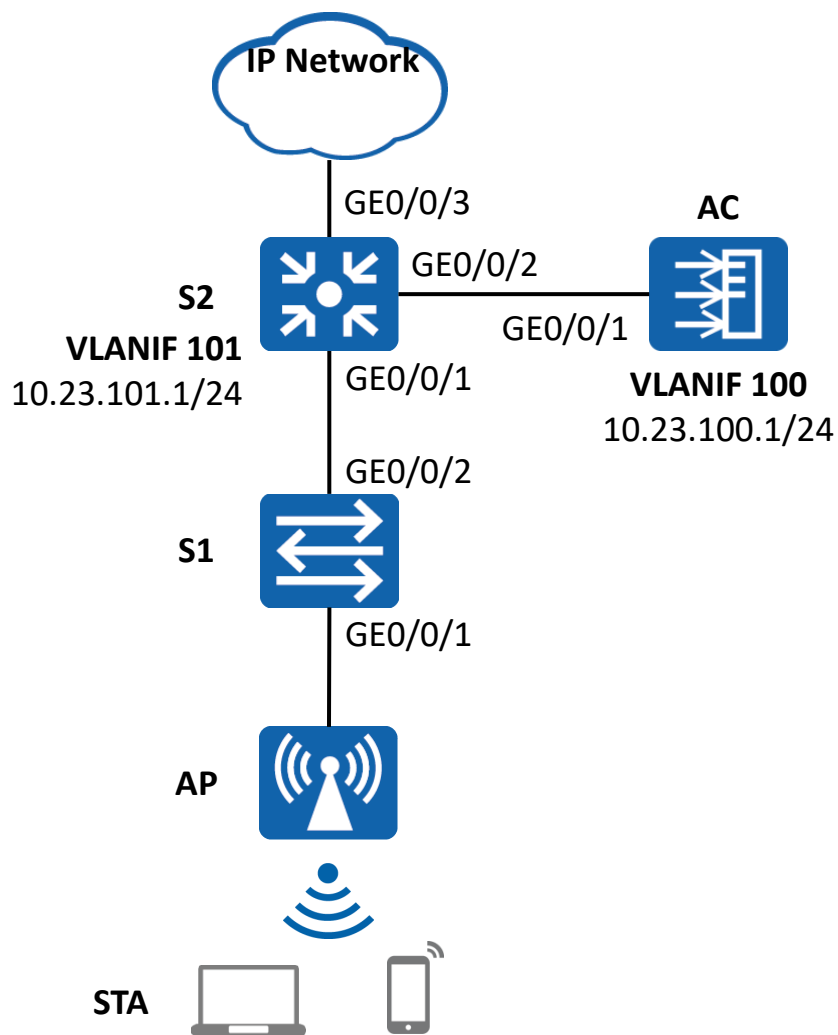
```
P : insufficient power supply
```

ID	MAC	Name	Group	IP	Type	State	STA	Uptime	ExtraInfo
0	60de-4476-e360	area_1	ap-group1	10.23.100.254	AP5030DN	nor	0	10S	-

```
Total: 1
```



Configuring WLAN Service Parameters (1)



1. Create security profile **wlan-net** and configure a security policy.

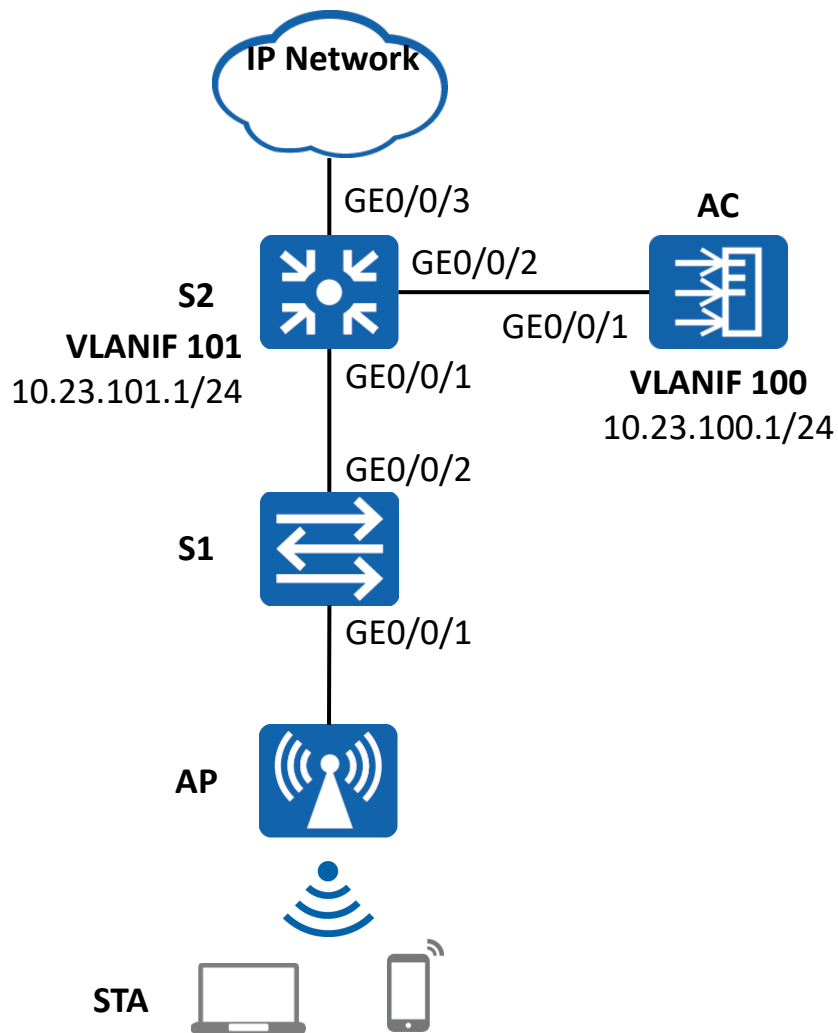
```
[AC-wlan-view] security-profile name wlan-net
[AC-wlan-sec-prof-wlan-net] security wpa-wpa2 psk pass-phrase a1234567
aes
[AC-wlan-sec-prof-wlan-net] quit
```

2. Create SSID profile **wlan-net** and set the SSID name to **wlan-net**.

```
[AC-wlan-view] ssid-profile name wlan-net
[AC-wlan-ssid-prof-wlan-net] ssid wlan-net
[AC-wlan-ssid-prof-wlan-net] quit
```



Configuring WLAN Service Parameters (2)



3. Create VAP profile **wlan-net**, set the data forwarding mode and service VLAN, and bind the security profile and SSID profile to the VAP profile.

```
[AC-wlan-view] vap-profile name wlan-net
[AC-wlan-vap-prof-wlan-net] forward-mode tunnel
[AC-wlan-vap-prof-wlan-net] service-vlan vlan-id 101
[AC-wlan-vap-prof-wlan-net] security-profile wlan-net
[AC-wlan-vap-prof-wlan-net] ssid-profile wlan-net
[AC-wlan-vap-prof-wlan-net] quit
```

4. Bind the VAP profile to the AP group and apply configurations in VAP profile **wlan-net** to radio 0 and radio 1 of the APs in the AP group.

```
[AC-wlan-view] ap-group name ap-group1
[AC-wlan-ap-group-ap-group1] vap-profile wlan-net wlan 1 radio 0
[AC-wlan-ap-group-ap-group1] vap-profile wlan-net wlan 1 radio 1
[AC-wlan-ap-group-ap-group1] quit
```



Checking VAP Profile Information

- The AC automatically delivers WLAN service configuration to the AP. After the service configuration is complete, run the **display vap ssid wlan-net** command. If **Status** in the command output is displayed as **ON**, the VAPs have been successfully created on AP radios.

```
[AC-wlan-view] display vap ssid wlan-net
```

```
WID : WLAN ID
```

AP ID	AP name	RfID	WID	BSSID	Status	Auth type	STA	SSID
0	area_1	0	1	60DE-4476-E360	ON	WPA/WPA2-PSK	0	wlan-net
0	area_1	1	1	60DE-4476-E370	ON	WPA/WPA2-PSK	0	wlan-net

```
Total: 2
```



Contents

1. WLAN Overview
2. Basic Concepts of WLAN
3. WLAN Fundamentals
4. WLAN Configuration Implementation
- 5. Next-Generation WLAN Solutions**



Huawei WLAN Solutions Meet Future Wireless Network Construction Requirements

All-scenario

- Use scenario-based customized solutions for complex and diversified application scenarios
- Complete WLAN deployment and management solutions for campus networks and branch networks

High bandwidth

- 802.11ac Wave 2 protocol, dual-5G radio coverage, and up to 3.46 Gbps wireless access bandwidth
- Huawei is a key contributor to the next-generation 802.11ax standard (Wi-Fi 6) with a single 5 GHz radio rate of up to 9.6 Gbps.
- Roaming and multiple wireless QoS protocols such as Wi-Fi multimedia (WMM) to ensure QoS

High security

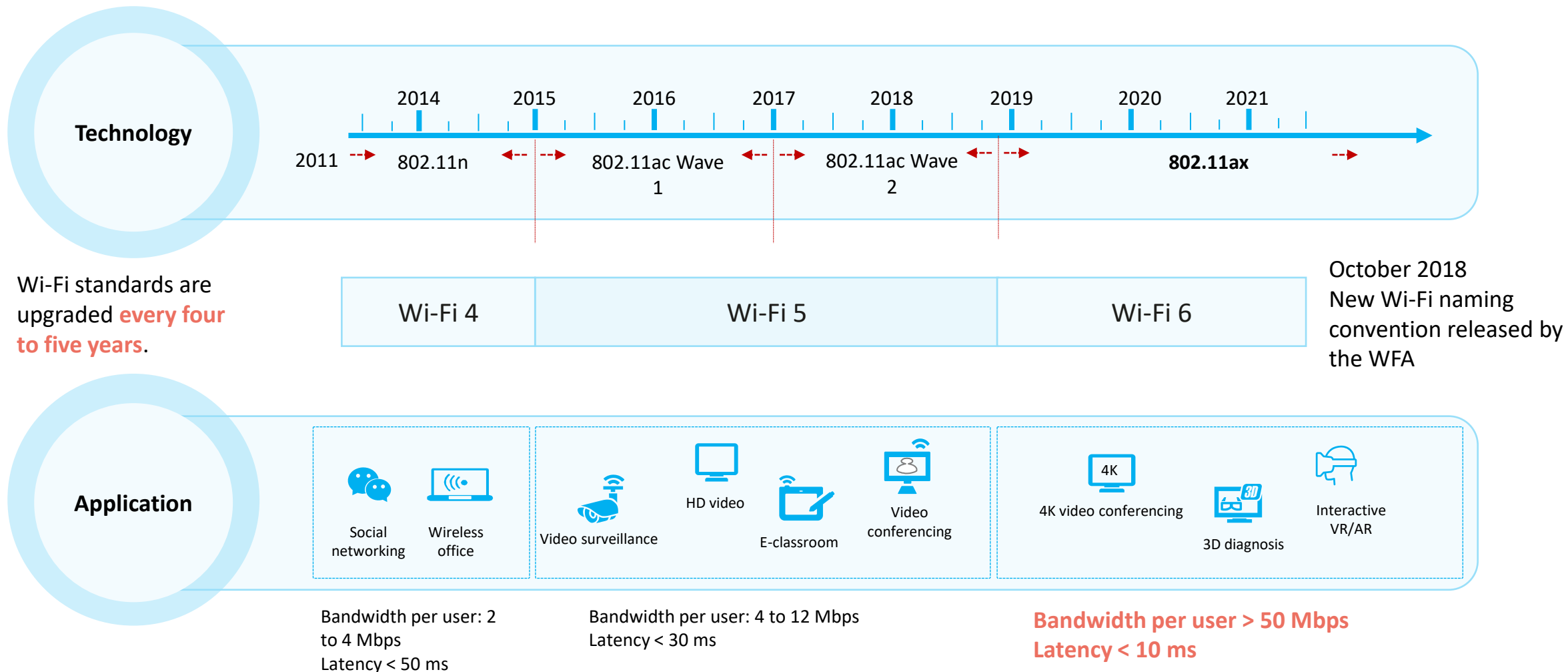
- Mainstream authentication and encryption modes, such as WPA, WPA2, WPA3, and WAPI
- Wireless intrusion detection
- Portal and 802.1X authentication, protecting intranet security

Easy deployment

- APs support plug-and-play, automatic upgrade, automatic channel selection, dynamic rate and power adjustment, and load balancing.
- IoT APs and APs with built-in high-density antennas, simplifying installation and enabling fast deployment
- APs support cloud management and can work in dual-stack mode to smoothly switch between the cloud and local management modes.



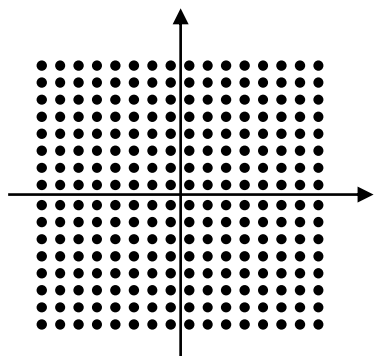
Dual Drivers (Technology Advances + Application Development) Promote the Arrival of the Wi-Fi 6 Era





Wi-Fi 6 Vs. Wi-Fi 5

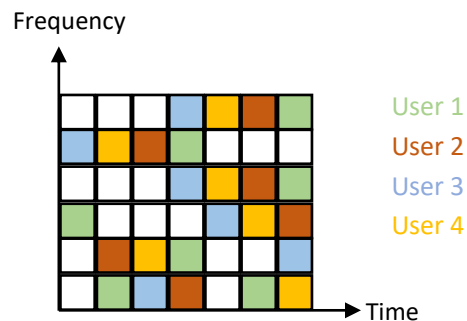
High Bandwidth



1024-QAM
8x8 MU-MIMO

- Rate of up to **9.6 Gbps**
- Bandwidth increased by **4** times

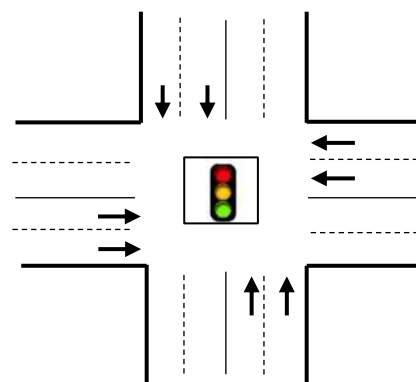
High Concurrency Rate



UL/DL OFDMA
UL/DL MU-MIMO

- Access of **1024** STAs per AP
- Number of concurrent users increased by **4** times

Low Latency



OFDMA
Spatial Reuse

- Service latency reduced to **20 ms**
- Average latency reduced by **30%**

Low Power Consumption

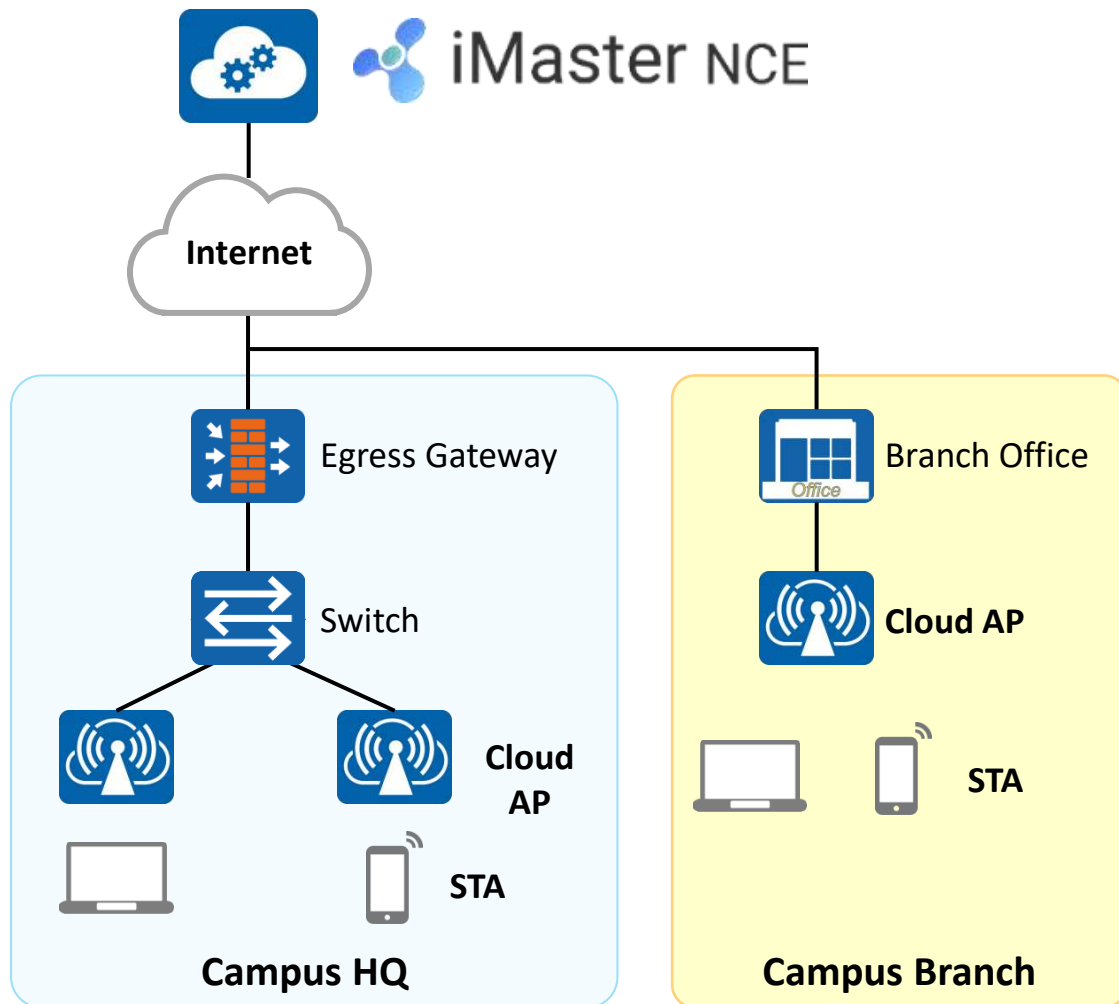


TWT
20 MHz-Only

- Target wakeup time (TWT) mechanism
- STA power consumption reduced by **30%**



Next-Generation Campus Network: Intent-Driven Campus (Small- and Medium-Sized)



Basic Concepts

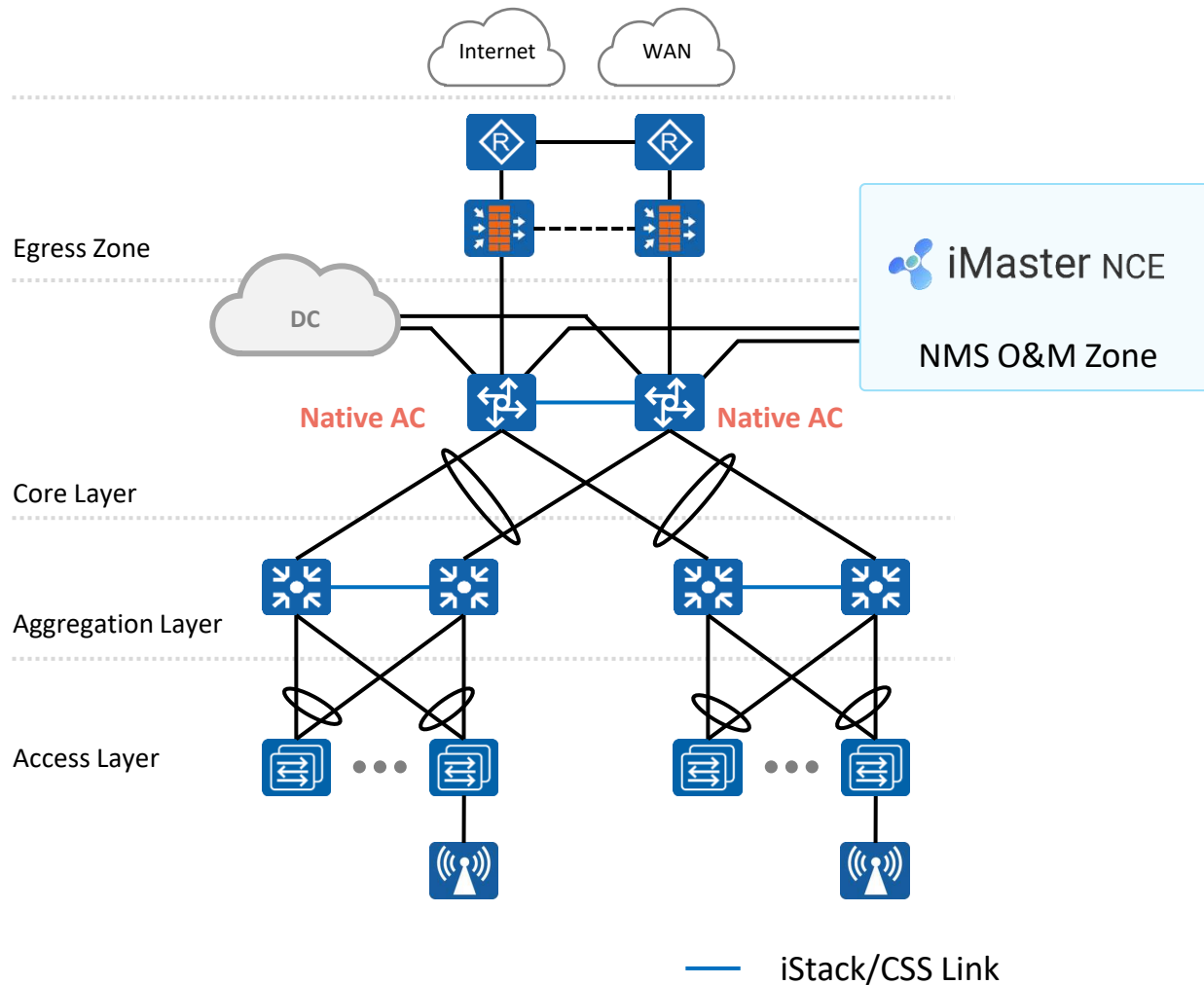
- The cloud management platform allows centralized management and maintenance of devices at any place, greatly reducing network deployment and O&M costs.
- Applicable scope: small- and medium-sized enterprises

Advantages (Compared with the AC + Fit AP Architecture)

- Plug-and-play and automatic deployment reduce network deployment costs.
- All network elements (NEs) are monitored and managed on the cloud management platform in a unified manner.
- Cloud solutions usually provide various tools on the cloud, reducing costs.



Next-Generation Campus Network: Intent-Driven Campus (Medium- and Large-Sized)



Architecture Characteristics

- iMaster NCE manages and configures APs in a unified manner and provides various functions. By further integrating with wired networks and leveraging Big Data and AI technologies, this architecture implements simplified, intelligent, and secure campus networks.
- Applicable scope: medium- and large-sized enterprises



Quiz

1. What are the advantages and disadvantages of in-path and off-path networking modes?
2. (Multiple) Which of the following methods are supported by Fit APs to discover an AC?()
 - A. Static discovery
 - B. Dynamic discovery through DHCP
 - C. Dynamic discovery through FTP
 - D. Dynamic discovery through DNS



Summary

- WLAN technology allows users to easily access a wireless network and freely move around within the coverage of the wireless network, eliminating the constraints of wired networks.
- In this course, we have learned WLAN technologies on enterprise networks, including the basic concepts, fundamentals, network architectures, configuration implementation, and development trend of WLAN technologies.

The background of the slide features a blue-tinted image of several business professionals in a modern office. They are standing on a highly reflective floor, and their silhouettes are clearly visible against the bright background. The overall aesthetic is professional and corporate.

Thank You
www.huawei.com



WAN Technologies



Foreword

- As economic globalization and digital transformation accelerate, enterprises keep expanding their scales. More and more branches locate in different regions, with each branch network being considered as a local area network (LAN). The headquarters and branches need to cross geographical locations to communicate with each other. To better carry out services, an enterprise needs to connect these geographically dispersed branches through a wide area network (WAN).
- The development of the WAN technologies is accompanied by the continuously increased bandwidth. In the early stage, X.25 provided only the bandwidth of 64 kbit/s. Later, the digital data network (DDN) and Frame Relay (FR) increased the bandwidth to 2 Mbit/s. Synchronous digital hierarchy (SDH) and asynchronous transfer mode (ATM) further increased the bandwidth to 10 Gbit/s. Now, the current IP-based WANs provide 10 Gbit/s or even higher bandwidth.
- This course describes the development history of WAN technologies, especially the implementations and configurations of Point-to-Point Protocol (PPP) and Point-to-Point Protocol over Ethernet (PPPoE).



Objectives

- On completion of this course, you will be able to:
 - Understand the basic concepts and development history of WANs.
 - Understand PPP and PPPoE implementations.
 - Master basic PPP and PPPoE configurations.
 - Understand basic MPLS/SR concepts.



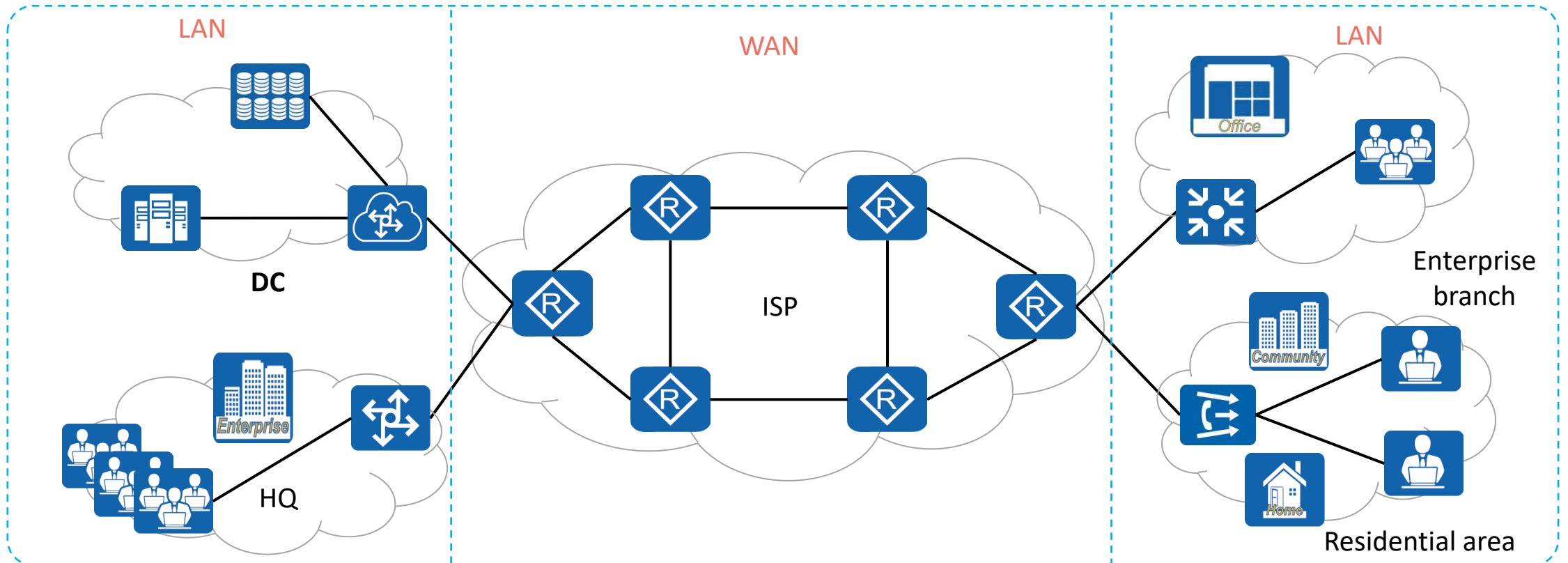
Contents

- 1. Overview of Early WAN Technologies**
2. PPP Implementation and Configuration
3. PPPoE Implementation and Configuration
4. Development of WAN Technologies



What Is a WAN?

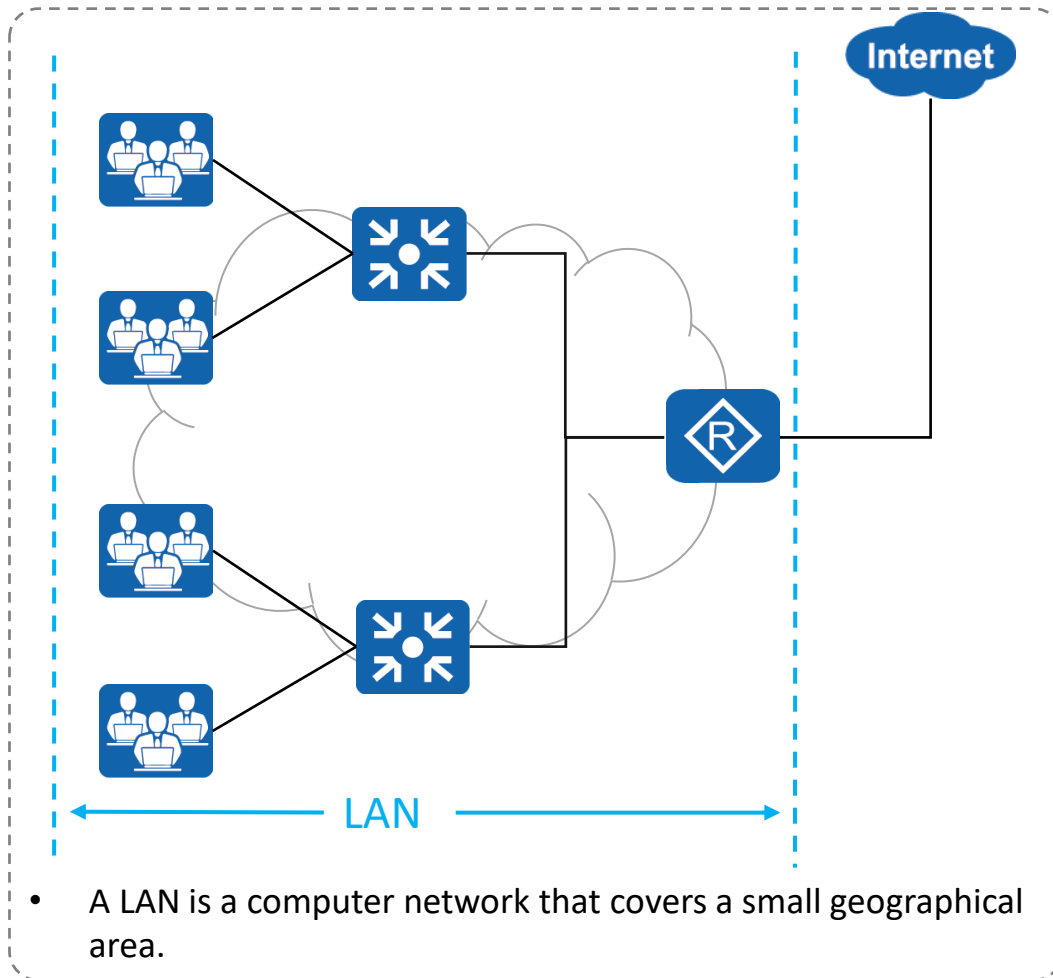
- A **WAN** is a network that connects LANs in different areas. A WAN generally covers tens of kilometers to thousands of kilometers. It can connect multiple regions, cities, and countries, or provide long-distance communication across several continents, forming an international remote network.



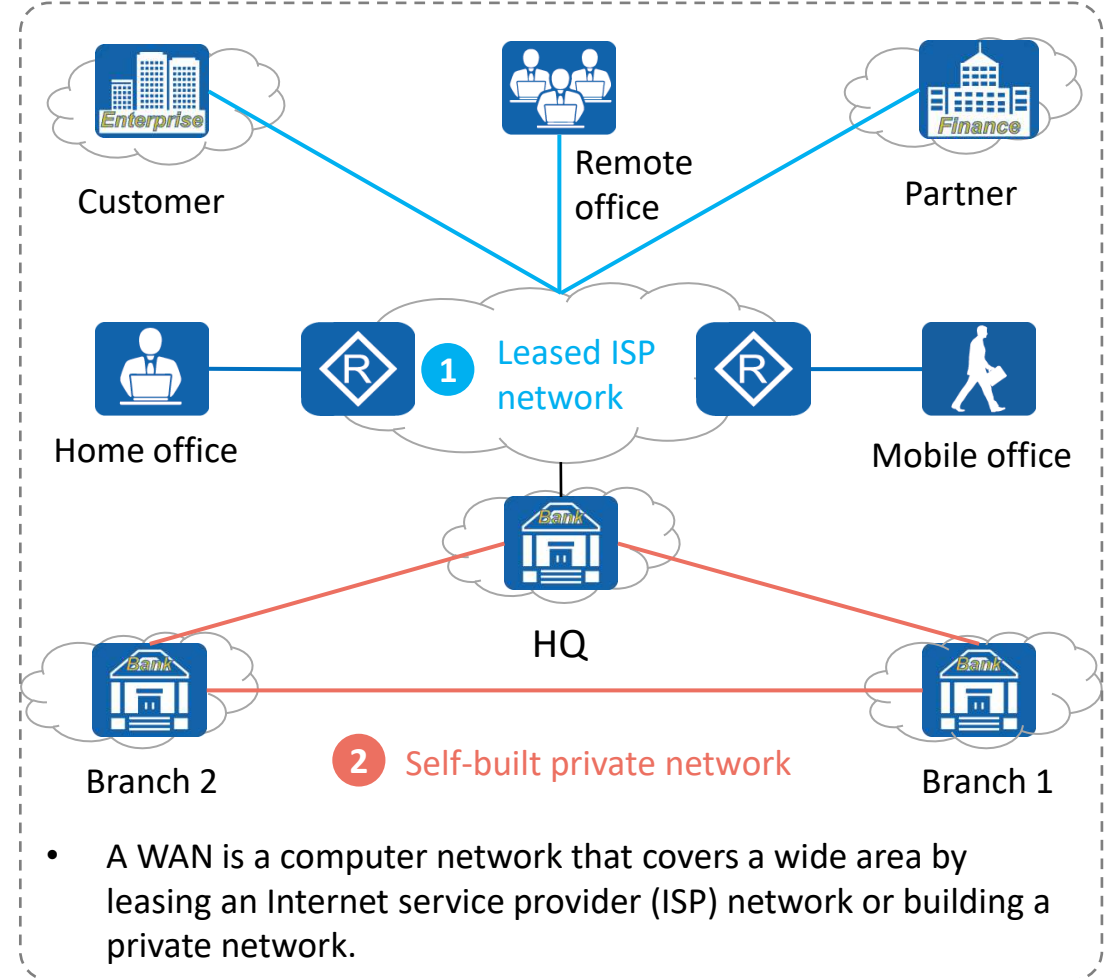


Differences Between a WAN and a LAN

LAN



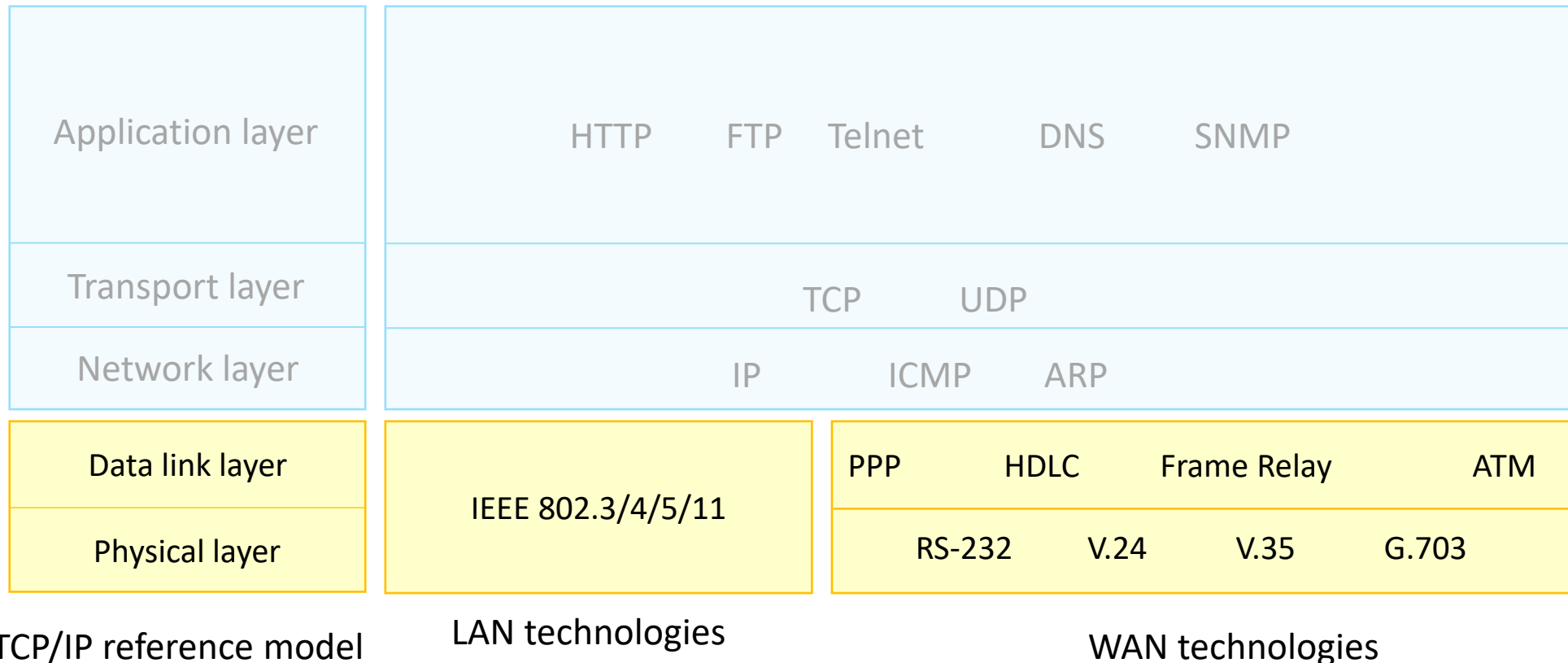
WAN





Overview of Early WAN Technologies

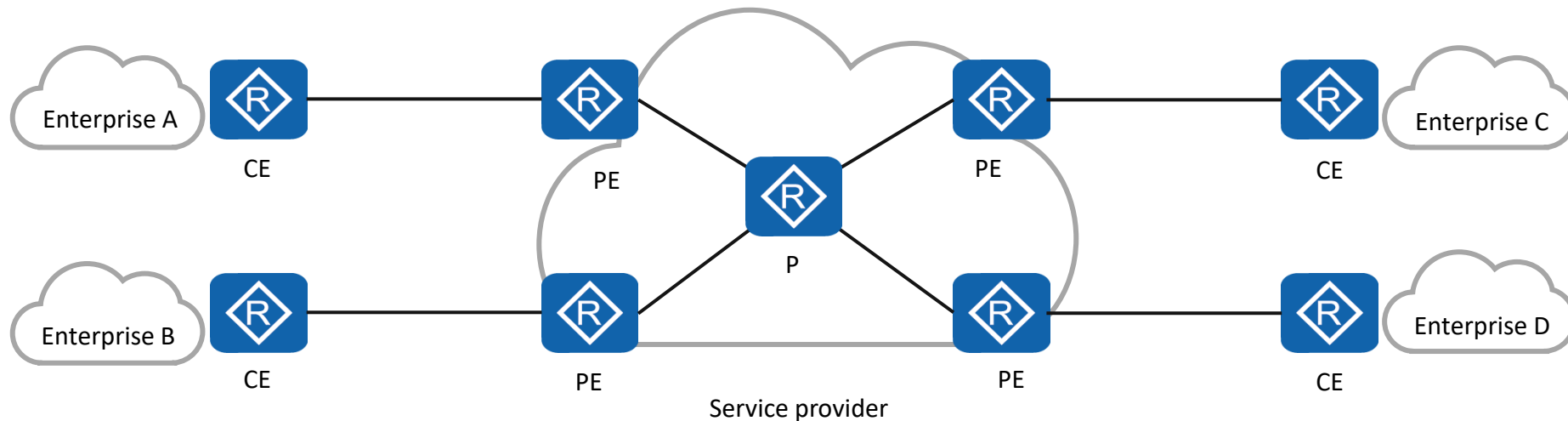
- The early WANs and LANs differ in the data link layer and physical layer and are the same in the other layers in the TCP/IP reference model.





WAN Device Roles

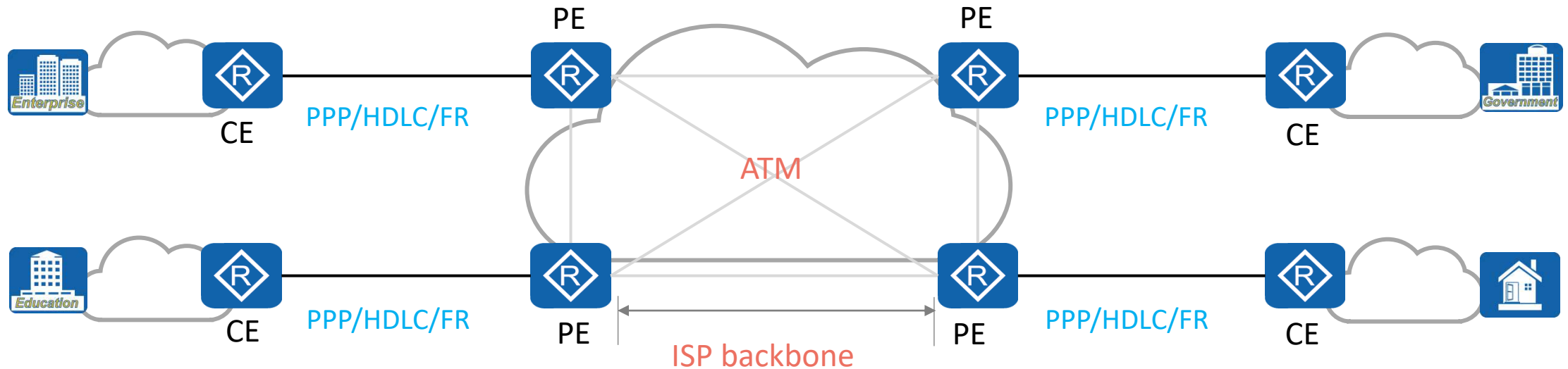
- There are three basic roles of WAN devices: customer edge (CE), provider edge (PE), and provider (P). They are defined as follows:
 - CE: a device located at the customer premises and connected to one or more PEs for user access.
 - PE: a service provider's important edge device that is connected to both a CE and a P.
 - P: a service provider's device that is not connected to any CE.





Application of Early WAN Technologies

- The early WAN technologies perform different Layer 2 encapsulation at the data link layer for different types of physical links. PPP, HDLC, and FR are commonly used between CEs and PEs to implement long-distance transmission of user access packets over a WAN. ATM is commonly used on ISP backbone networks for high-speed forwarding.





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1. Overview of Early WAN Technologies
- 2. PPP Implementation and Configuration**
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PPP Introduction

- PPP is a common WAN data link layer protocol. It is used for P2P data encapsulation and transmission on full-duplex links.
- PPP provides the Password Authentication Protocol (PAP) and Challenge Handshake Authentication Protocol (CHAP).
- PPP features high extensibility. For example, PPP can be extended as Point-to-Point Protocol over Ethernet (PPPoE) when PPP packets need to be transmitted over an Ethernet.
- PPP provides the Link Control Protocol (LCP), which is used to negotiate link layer parameters, such as the maximum receive unit (MRU) and authentication mode.
- PPP provides various Network Control Protocols (NCPs), such as IP Control Protocol (IPCP), for negotiation of network layer parameters and better support for network layer protocols.





PPP Link Setup Process

Feature
Introduction

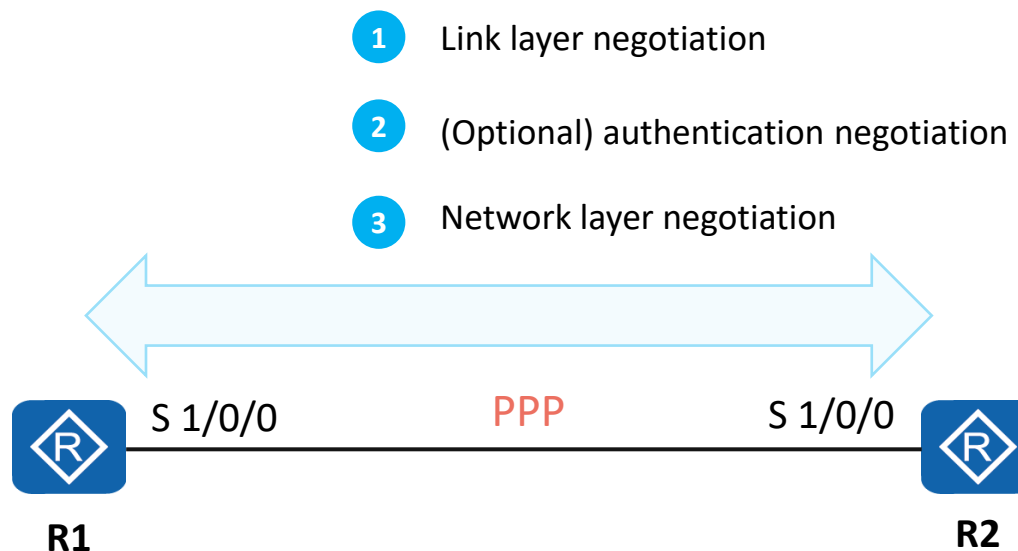
Link Setup

LCP Negotiation

Authentication
Negotiation

NCP Negotiation

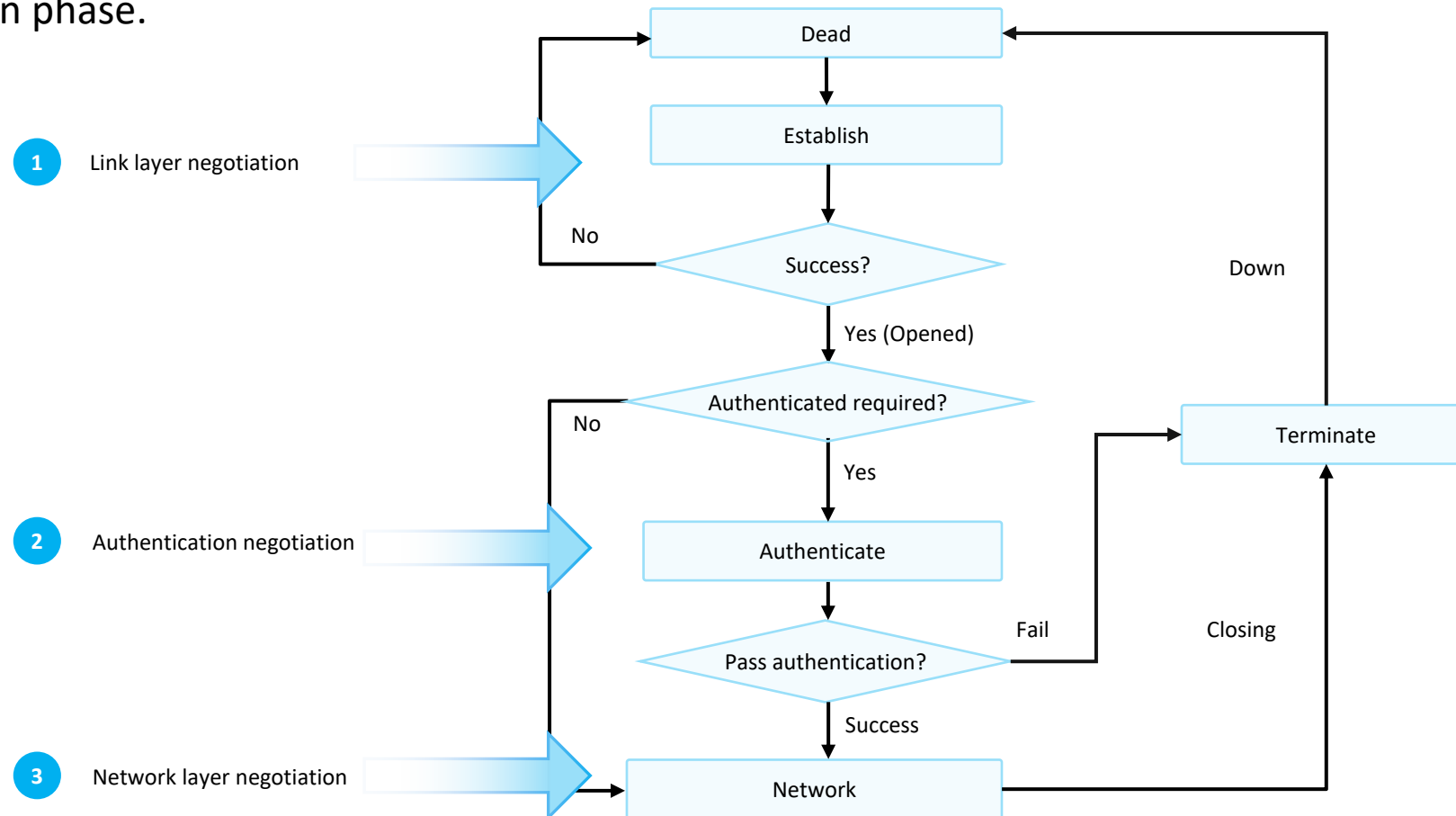
- PPP link setup involves link layer negotiation, optional authentication negotiation, and network layer negotiation.
 - Link layer negotiation: LCP packets are used to negotiate link parameters and establish link layer connections.
 - (Optional) authentication negotiation: The authentication mode negotiated during link layer negotiation is used for link authentication.
 - Network layer negotiation: NCP negotiation is used to select and configure a network layer protocol and negotiate network layer parameters.





State Machine of the PPP Link Interface

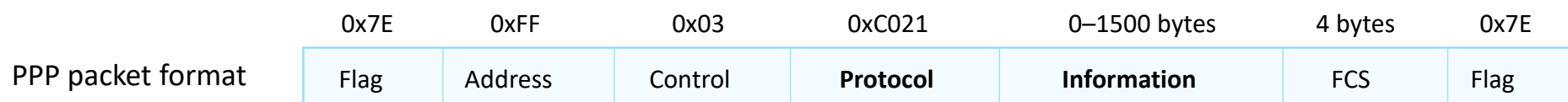
- PPP negotiation is performed by the interfaces at both ends of a link. The interface status indicates the protocol negotiation phase.





LCP Packet Format

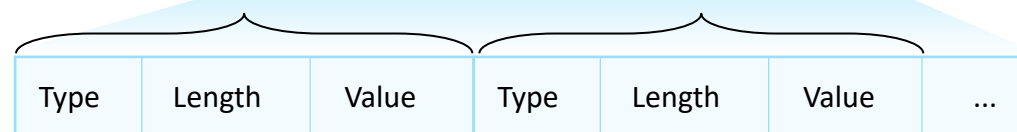
- The Protocol field in a PPP packet identifies the type of the PPP packet. For example, if the Protocol field is 0xC021, the packet is an LCP packet. The Code field is further used to identify different types of LCP packets, as shown in the following table.



0x0021: IP packet
0x8021: IPCP packet
0xC021: LCP packet
0xC023: PAP packet
0xC223: CHAP packet



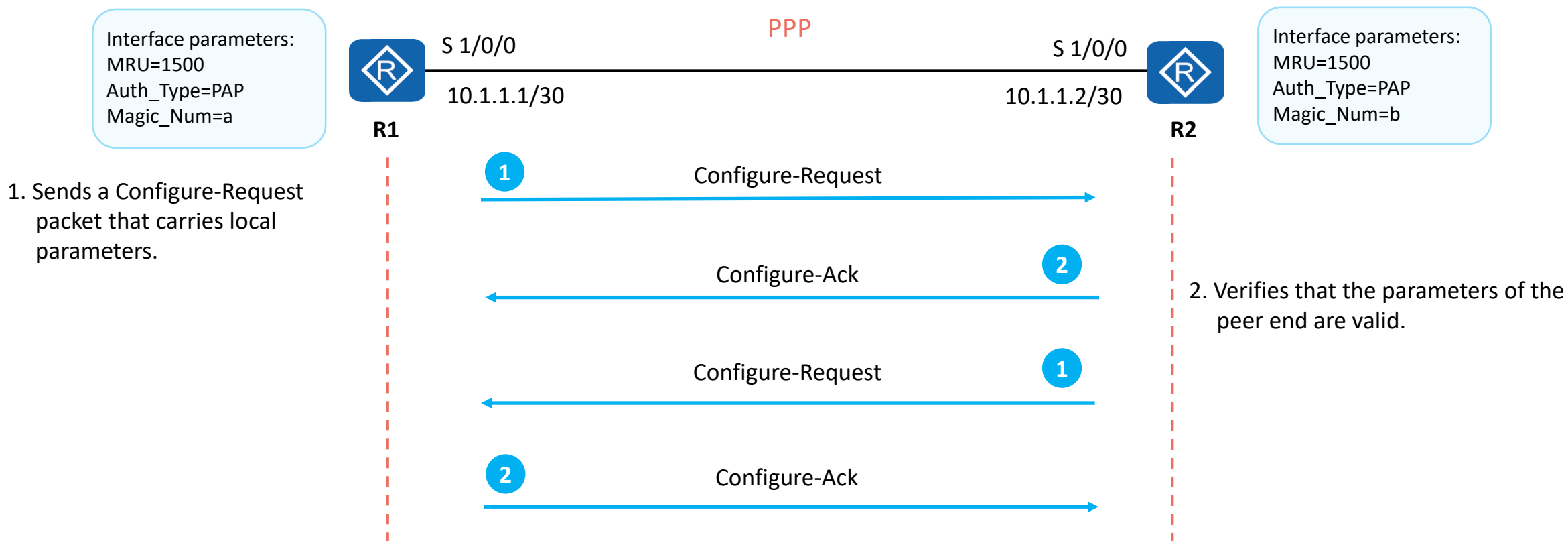
Code	Name	Content
0x01	Configure-Request	Configuration request packet.
0x02	Configure-Ack	Configuration success packet.
0x03	Configure-Nak	Configuration parameters need to be negotiated.
0x04	Configure-Reject	Configuration parameters cannot be identified.



The TLV structure contains common parameters used in LCP negotiation, such as the MRU, authentication protocol, and magic number.

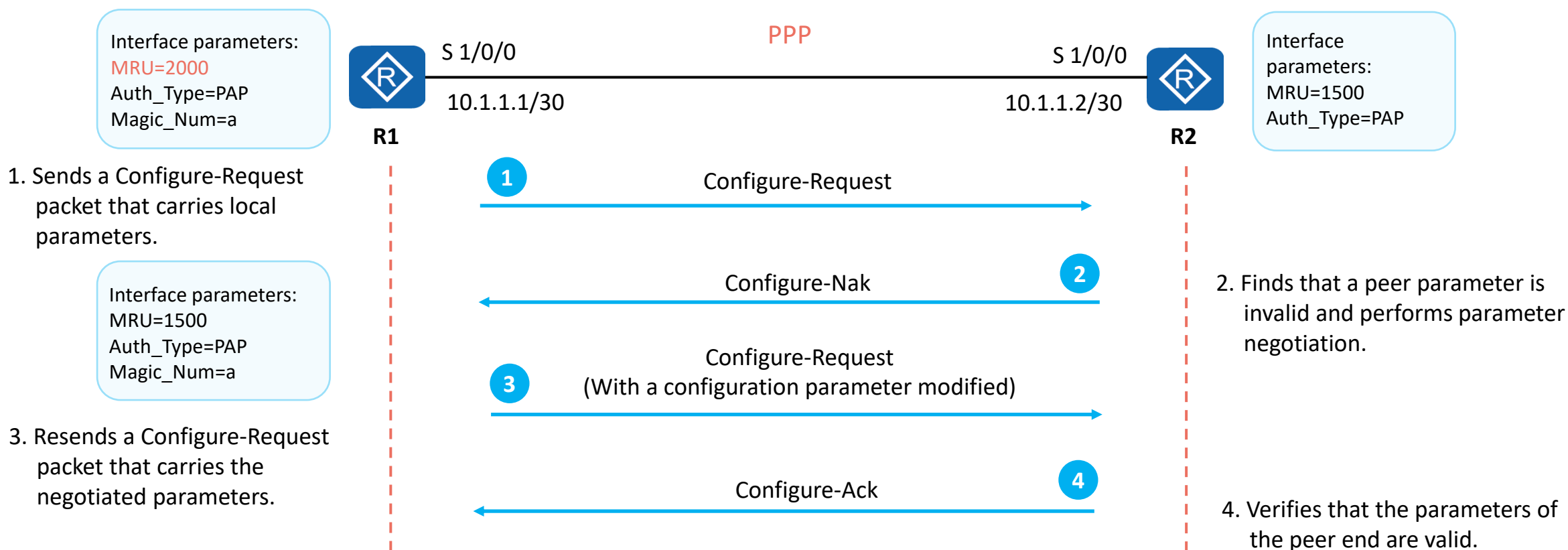
LCP Negotiation Process - Normal Negotiation

- LCP negotiation is implemented by exchanging different LCP packets. The negotiation is initiated by sending a Configure-Request packet from either party. If the peer end identifies and accepts all parameters in the packet, the peer end returns a Configure-Ack packet to the local end, indicating that the negotiation is successful.



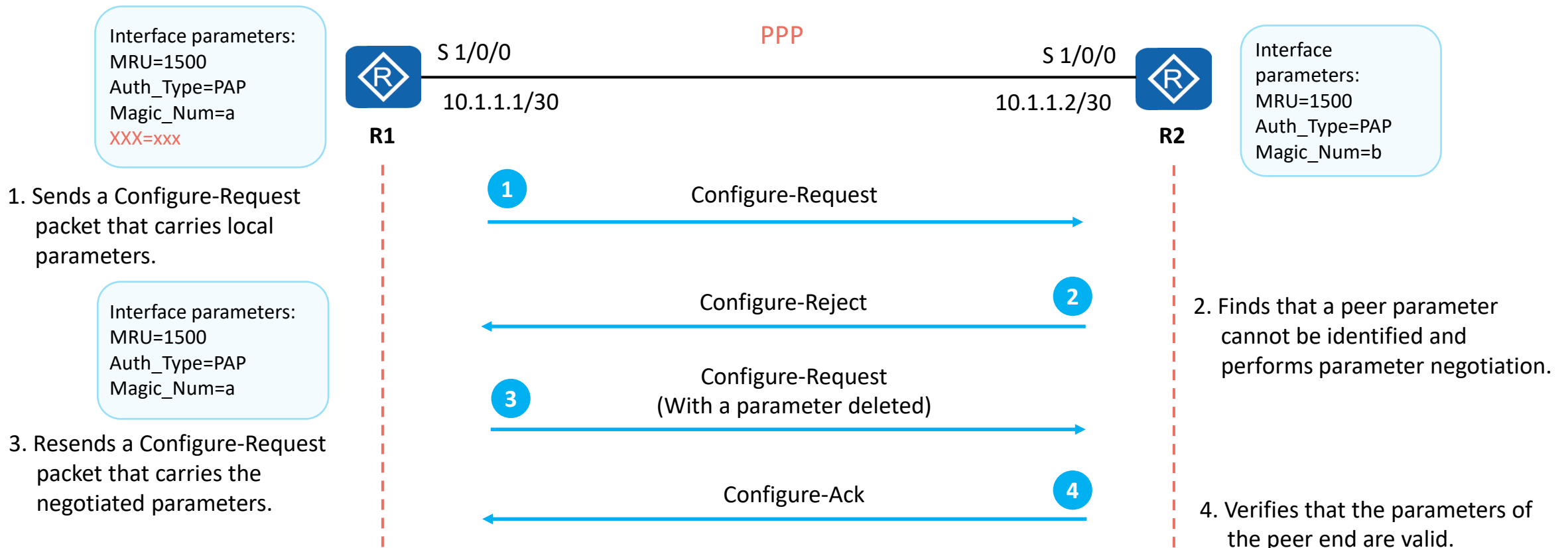
LCP Negotiation Process - Parameter Mismatch

- If LCP parameters do not match during LCP packet exchange, the receiver responds with a Configure-Nak packet to instruct the peer end to modify parameters and perform renegotiation.



LCP Negotiation - Unrecognized Parameters

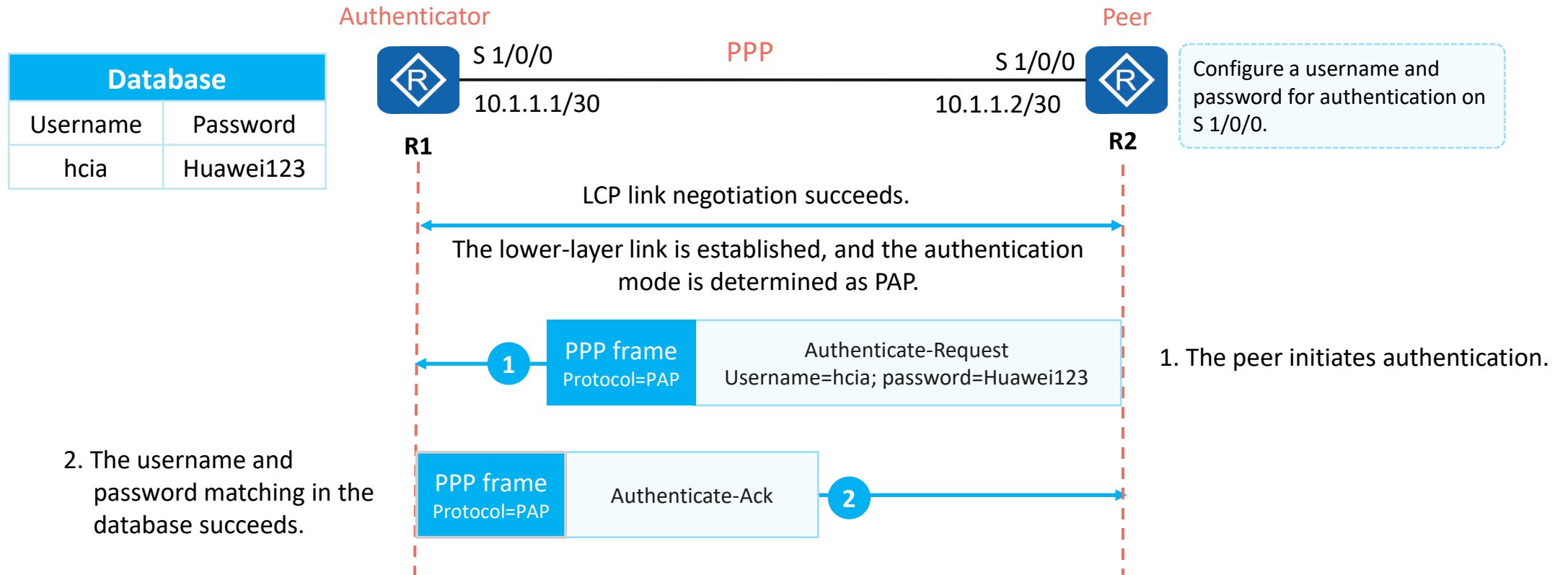
- If LCP parameters cannot be identified during LCP packet exchange, the receiver responds with a Configure-Reject packet to instruct the peer end to delete the unidentifiable parameters and renegotiates with the peer end.





PPP Authentication Mode - PAP

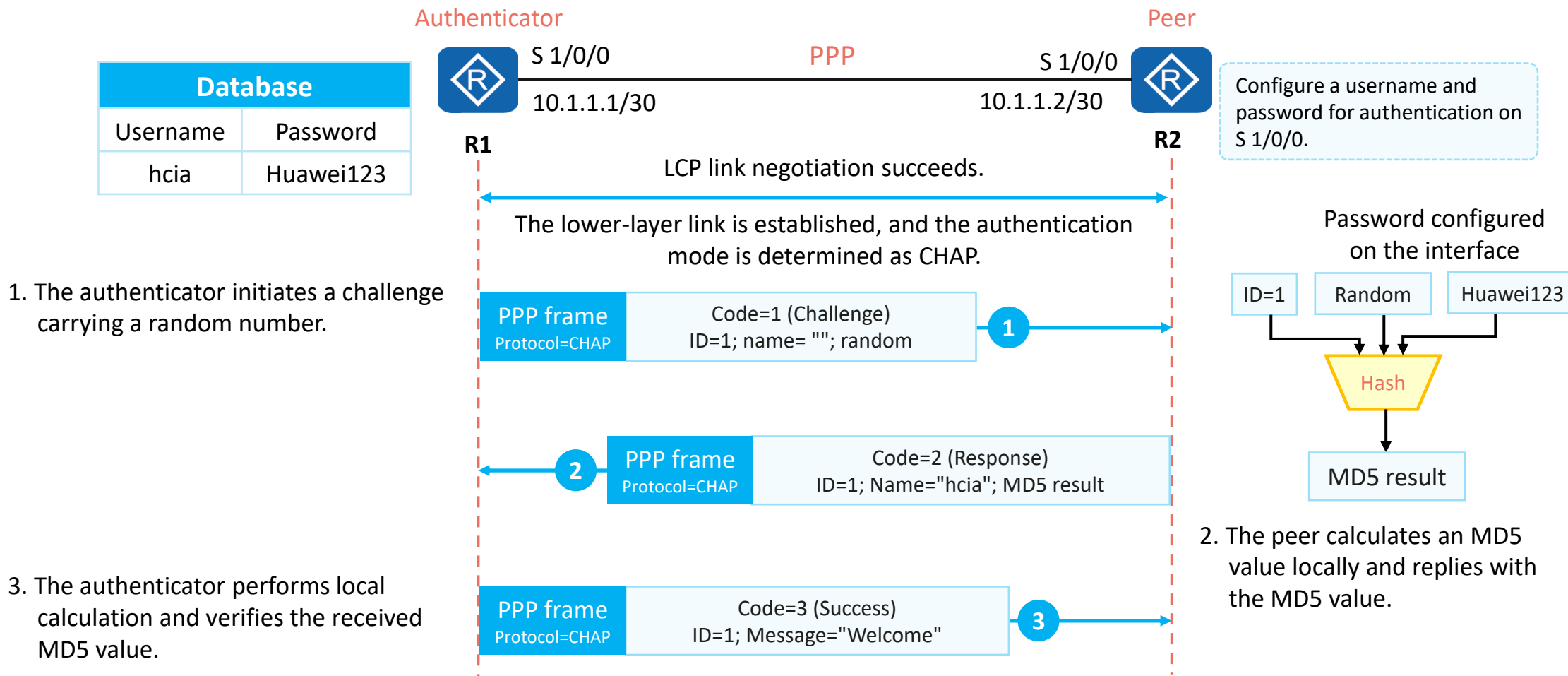
- After the link negotiation is successful, authentication negotiation can be performed. There are two authentication negotiation modes: PAP and CHAP.
- PAP authentication requires a two-way handshake. Negotiation packets are transmitted on the link in clear text.





PPP Authentication Mode - CHAP

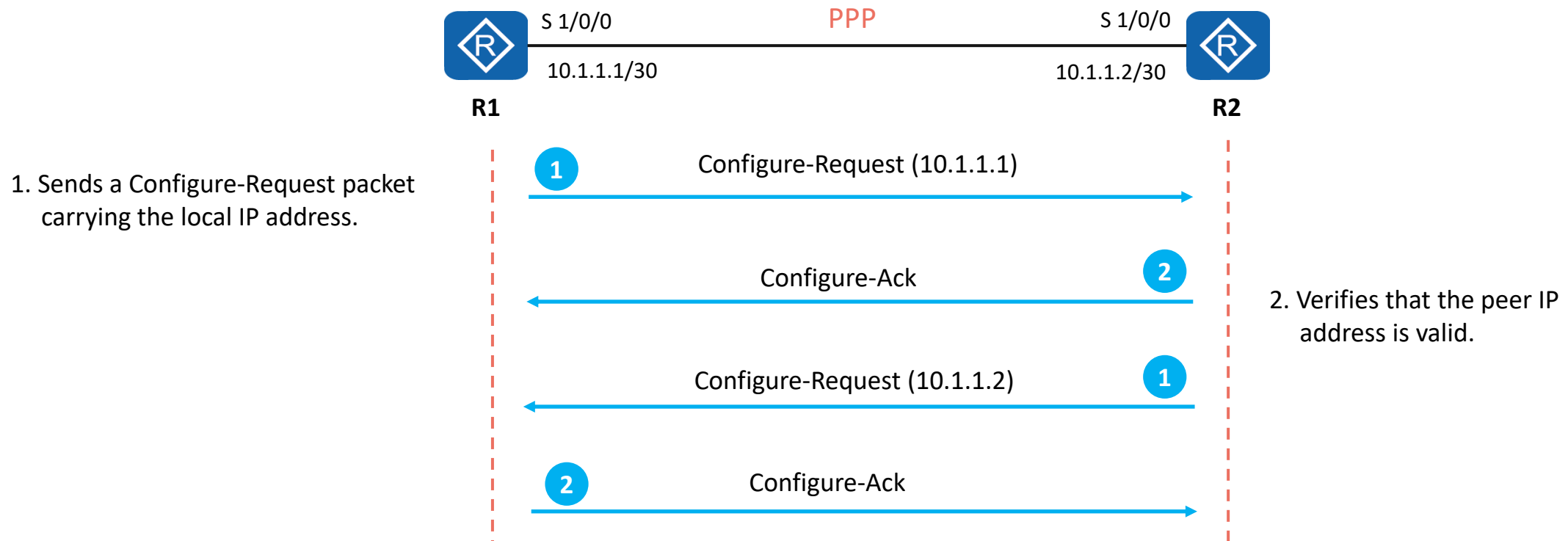
- CHAP authentication requires a three-way handshake. Negotiation packets are encrypted before being transmitted on a link.





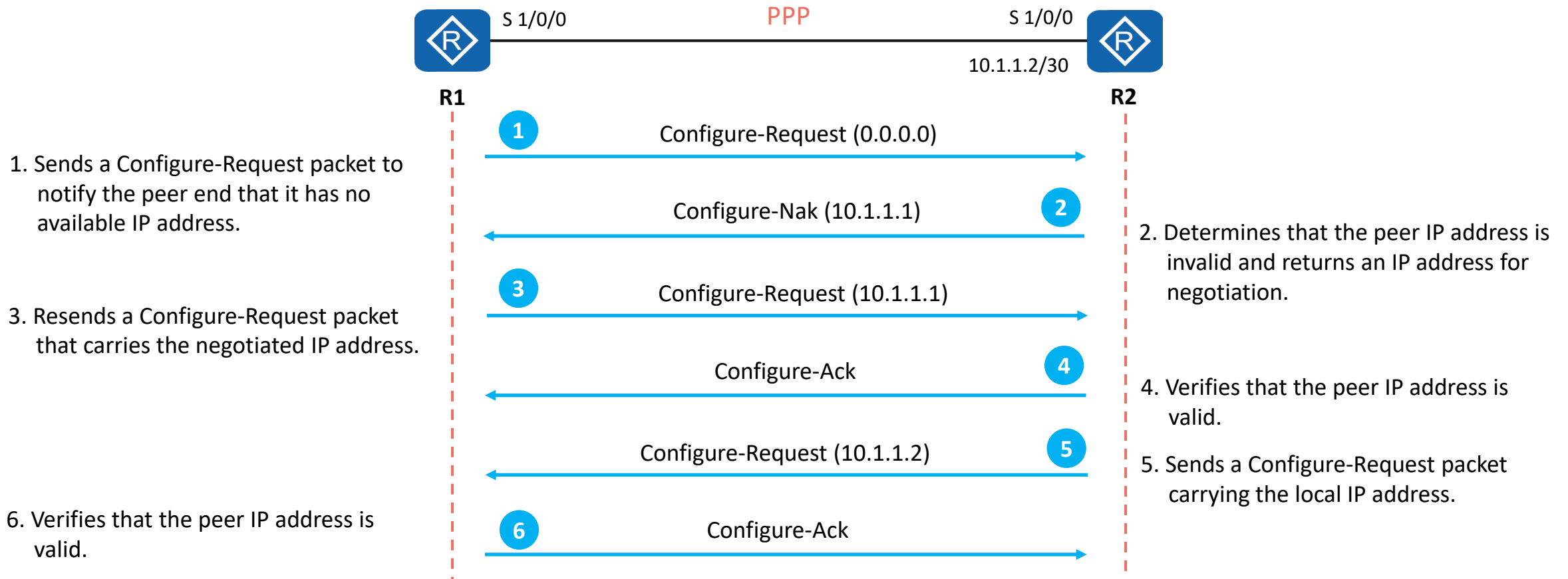
NCP Negotiation - Static IP Address Negotiation

- After PPP authentication negotiation, the two ends enter the NCP negotiation phase to negotiate the format and type of data packets transmitted on the data link. IPCP, for example, is classified into static and dynamic IP address negotiation.
- Static IP address negotiation requires manual configuration of IP addresses at both ends of a link.



NCP Negotiation - Dynamic IP Address Negotiation

- In dynamic IP address negotiation, one end of a PPP link can assign an IP address to the other end.





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Configuring Basic PPP Functions

1. Encapsulate an interface with PPP.

```
[Huawei-Serial0/0/0] link-protocol ppp
```

In the interface view, change the interface encapsulation protocol to PPP. The default encapsulation protocol of Huawei devices' serial interfaces is PPP.

2. Configure a negotiation timeout period.

```
[Huawei-Serial0/0/0] ppp timer negotiate seconds
```

During LCP negotiation, the local end sends an LCP negotiation packet to the peer end. If the local end does not receive a reply packet from the peer end within the specified negotiation timeout period, the local end resends an LCP negotiation packet.



Configuring PAP Authentication

1. Configure an authenticator to authenticate a peer using the PAP mode.

```
[Huawei-aaa] local-user user-name password { cipher | irreversible-cipher } password  
[Huawei-aaa] local-user user-name service-type ppp
```

```
[Huawei-Serial0/0/0] ppp authentication-mode pap
```

Before configuring the authenticator to authenticate a peer using the PAP mode, add the username and password of the peer to the local user list in the AAA view. Then select the PAP authentication mode.

2. Configure the peer to be authenticated by the authenticator in PAP mode.

```
[Huawei-Serial0/0/0] ppp pap local-user user-name password { cipher | simple } password
```

This command configures the peer to send its username and password to the authenticator.



Configuring CHAP Authentication

1. Configure an authenticator to authenticate a peer using CHAP mode.

```
[Huawei-aaa] local-user user-name password { cipher | irreversible-cipher } password  
[Huawei-aaa] local-user user-name service-type ppp
```

```
[Huawei-Serial0/0/0] ppp authentication-mode chap
```

2. Configure the peer to be authenticated by the authenticator in CHAP mode.

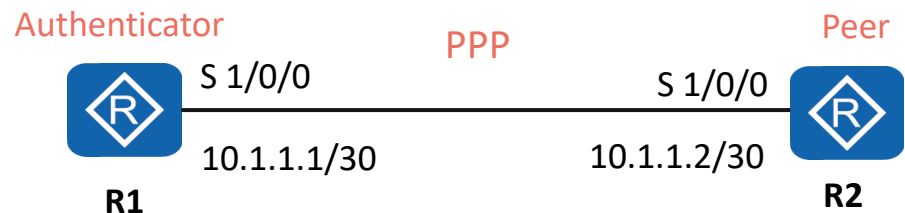
```
[Huawei-Serial0/0/0] ppp chap user user-name
```

```
[Huawei-Serial0/0/0] ppp chap password { cipher | simple } password
```

This command configures a local username and a password for CHAP authentication.



Example for Configuring PAP Authentication



- Experiment requirements:
 1. Enable PAP authentication on the PPP link between R1 and R2.
 2. Configure R1 as the authenticator.
 3. Configure R2 as the peer.

Configurations on R1

```
[R1]aaa # Add information about the user to be authenticated.  
[R1-aaa]local-user huawei password cipher huawei123  
[R1-aaa]local-user huawei service-type ppp  
# Specify the service type of the user to be authenticated.  
[R1]interface Serial 1/0/0  
[R1-Serial1/0/0]link-protocol ppp  
[R1-Serial1/0/0]ppp authentication-mode pap  
# Set the authentication mode to PAP.  
[R1-Serial1/0/0]ip address 10.1.1.1 30
```

Configurations on R2

```
[R2]interface Serial 1/0/0  
[R2-Serial1/0/0]link-protocol ppp  
[R2-Serial1/0/0]ppp pap local-user huawei password cipher huawei123 #  
Add user information for PPP authentication.  
[R2-Serial1/0/0]ip address 10.1.1.2 30
```



Example for Configuring CHAP Authentication



- Experiment requirements:
 1. Enable CHAP authentication on the PPP link between R1 and R2.
 2. Configure R1 as the authenticator.
 3. Configure R2 as the peer.

Configurations on R1

```
[R1]aaa # Add information about the user to be authenticated.  
[R1-aaa]local-user huawei password cipher huawei123  
[R1-aaa]local-user huawei service-type ppp  
# Specify the service type of the user to be authenticated.  
[R1]interface Serial 1/0/0  
[R1-Serial1/0/0]link-protocol ppp  
[R1-Serial1/0/0]ppp authentication-mode chap  
# Set the authentication mode to CHAP.
```

Configurations on R2

```
[R2]interface Serial 1/0/0  
[R2-Serial1/0/0]link-protocol ppp  
[R2-Serial1/0/0]ppp chap user huawei  
[R2-Serial1/0/0]ppp chap password cipher huawei123  
# Add user information for PPP authentication.
```



Contents

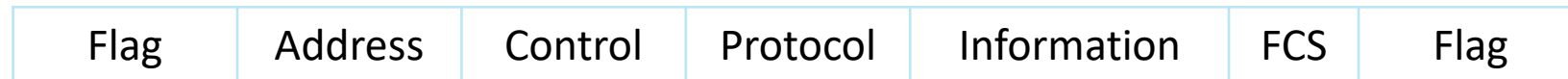
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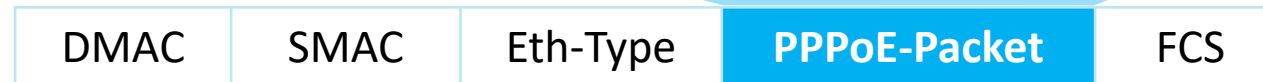
What Is PPPoE?

- PPP over Ethernet (PPPoE) is a link layer protocol that encapsulates PPP frames into Ethernet frames. PPPoE enables multiple hosts on an Ethernet to connect to a broadband remote access server (BRAS).
- PPPoE integrates the advantages of Ethernet and PPP. It has the flexible networking advantage of Ethernet and can use PPP to implement authentication and accounting.

PPP frame structure



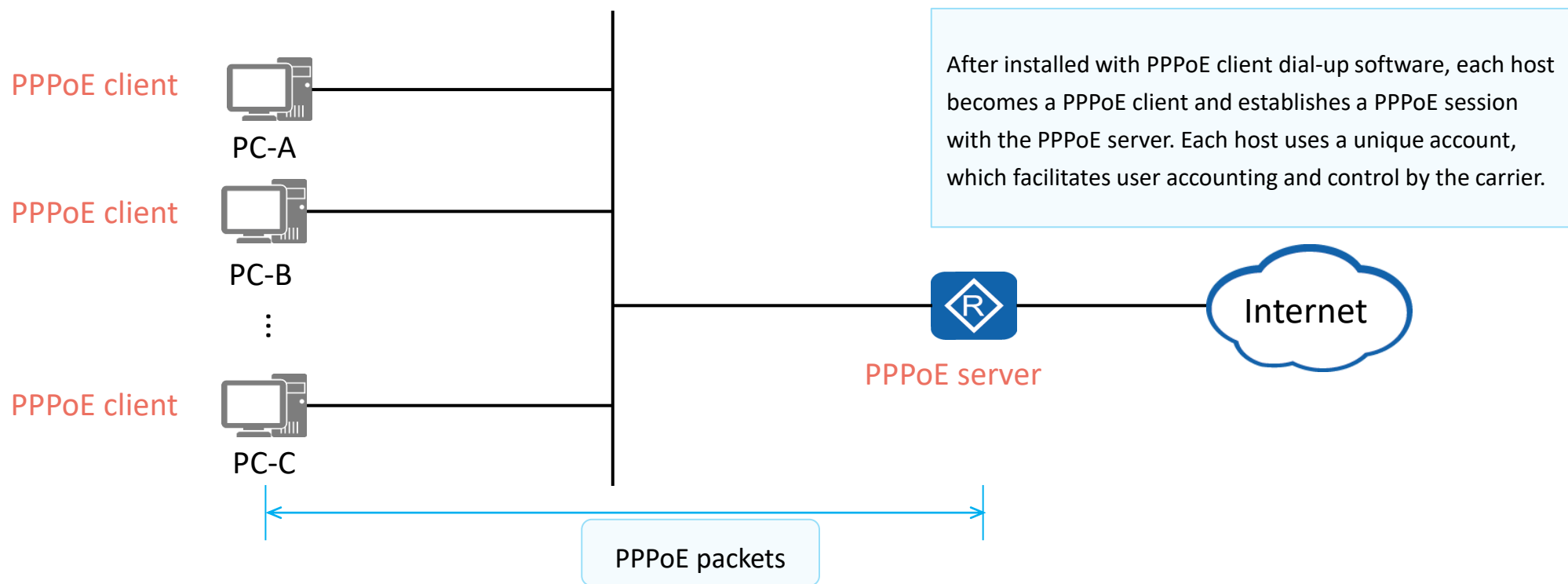
PPPoE frame structure





PPPoE Application Scenarios

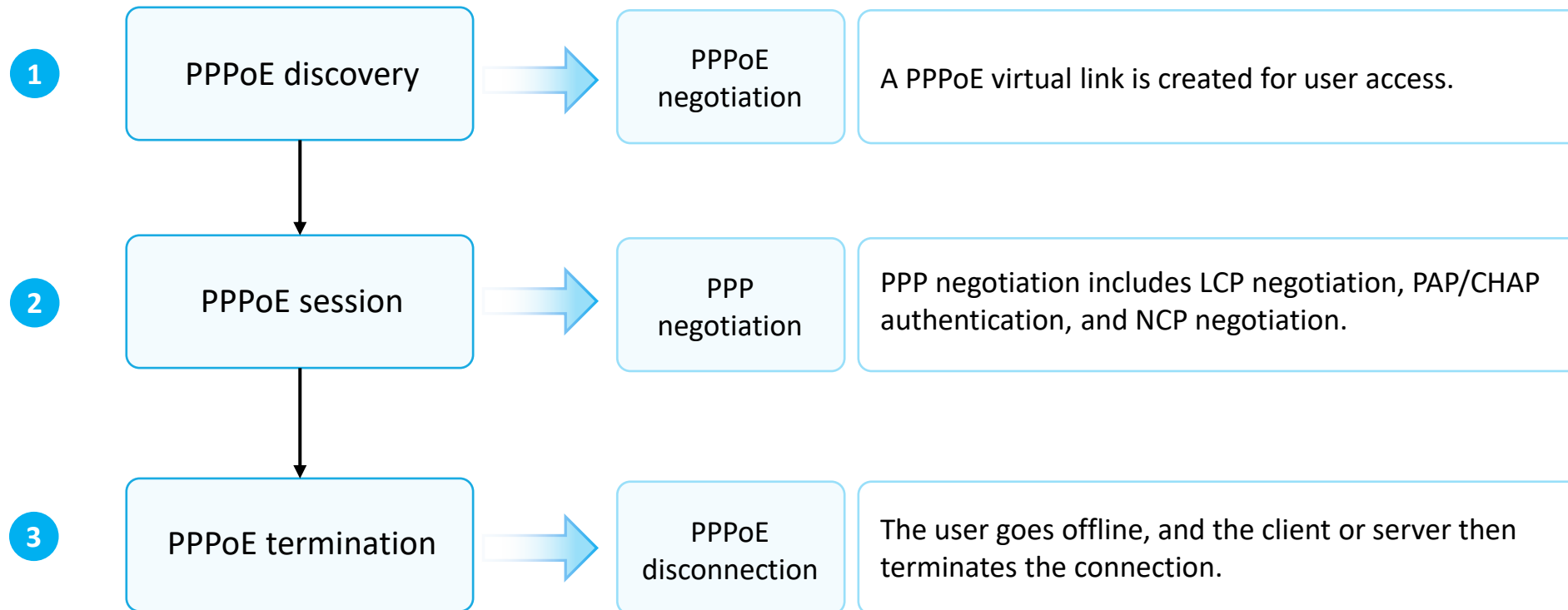
- PPPoE provides P2P connections on an Ethernet. A PPPoE client and a PPPoE server establish a PPP session to encapsulate PPP data packets and provide access services for hosts on the Ethernet, implementing user control and accounting. PPPoE is widely used on enterprise and carrier networks.
- PPPoE is usually used by home users and enterprise users to dial up to access the Internet.





PPPoE Session Establishment

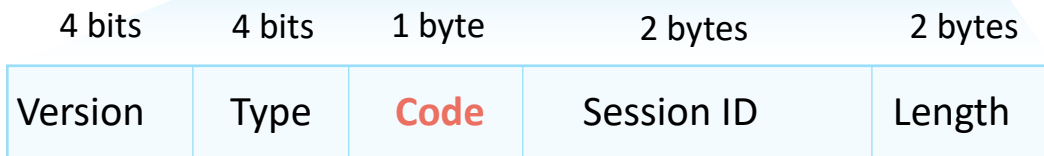
- PPPoE session establishment involves three stages: PPPoE discovery, session, and termination stages.





PPPoE Packets

- A PPPoE session is established by exchanging different PPPoE packets. The PPPoE packet structure and common packet types are as follows.



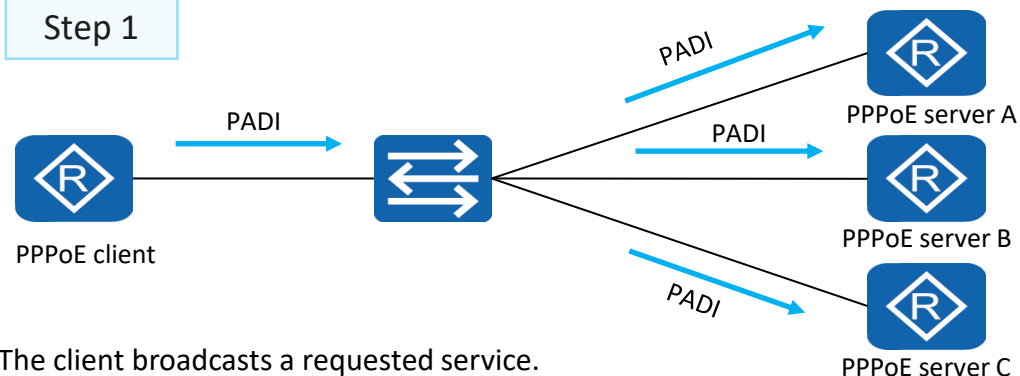
Code	Name	Content
0x09	PADI	PPPoE Active Discovery Initiation packet
0x07	PADO	PPPoE Active Discovery Offer packet
0x19	PADR	PPPoE Active Discovery Request packet
0x65	PADS	PPPoE Active Discovery Session-confirmation packet
0xa7	PADT	PPPoE Active Discovery Terminate packet



PPPoE Discovery Stage

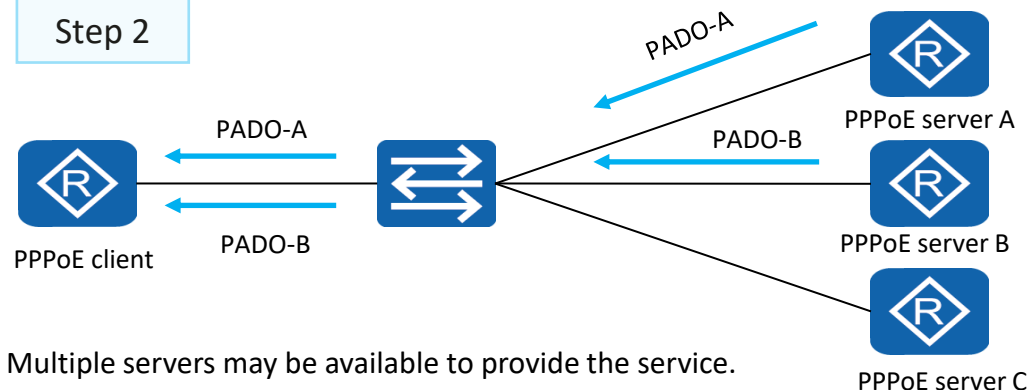
- PPPoE discovery involves four steps: 1) the client sends a request, 2) the servers respond to the request, 3) the client confirms a response and 4) establishes a session.

Step 1



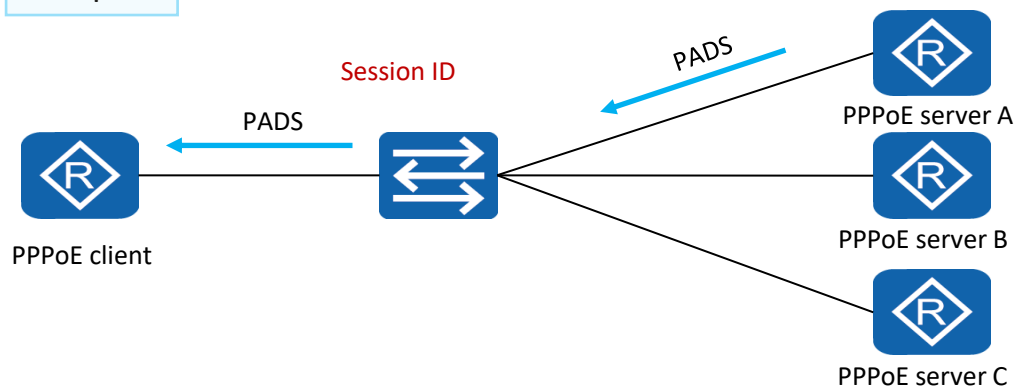
- The client broadcasts a requested service.

Step 2



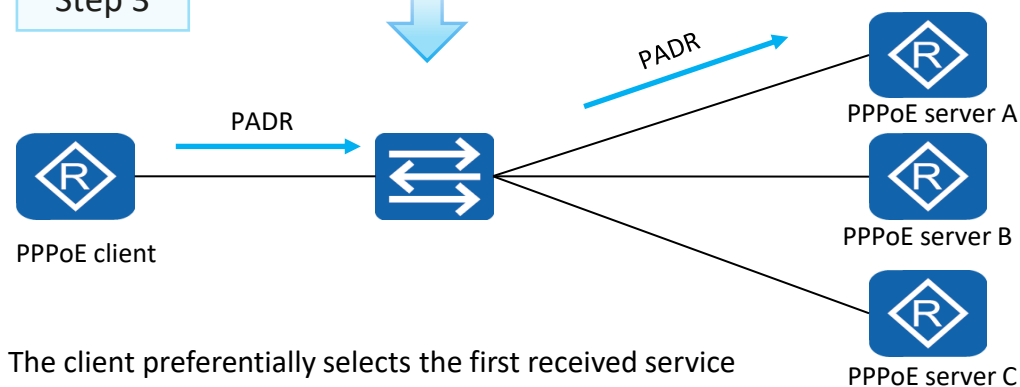
- Multiple servers may be available to provide the service.

Step 4



- The server assigns a session ID to the client to establish a session.

Step 3

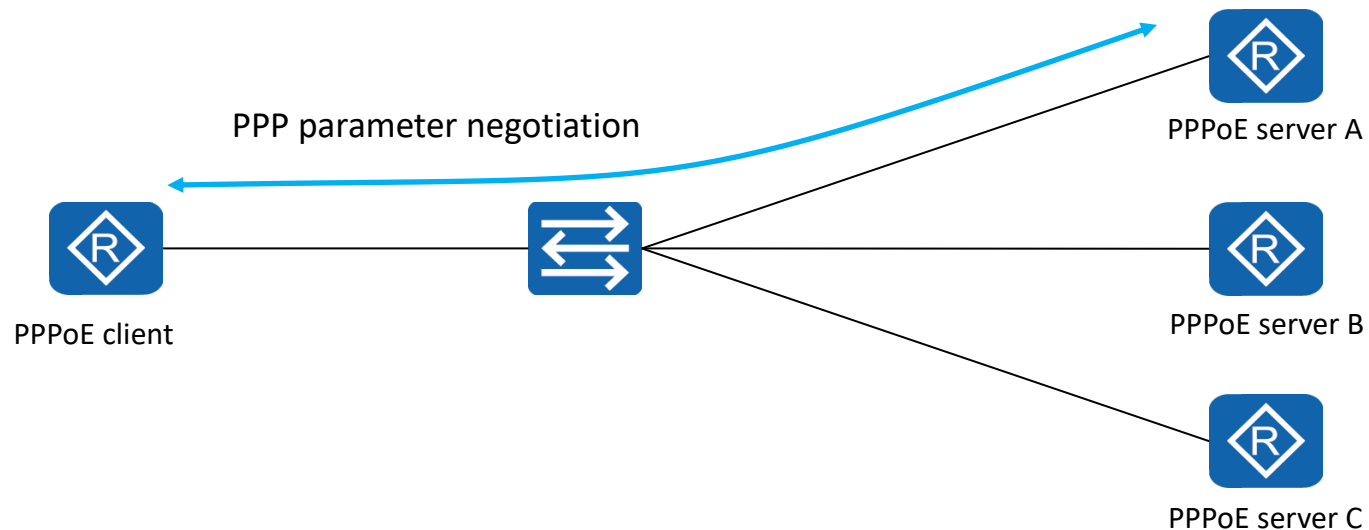


- The client preferentially selects the first received service response and sends a service request.



PPPoE Session Stage

- In the PPPoE session stage, PPP negotiation, including LCP, authentication, and NCP negotiation, is performed.

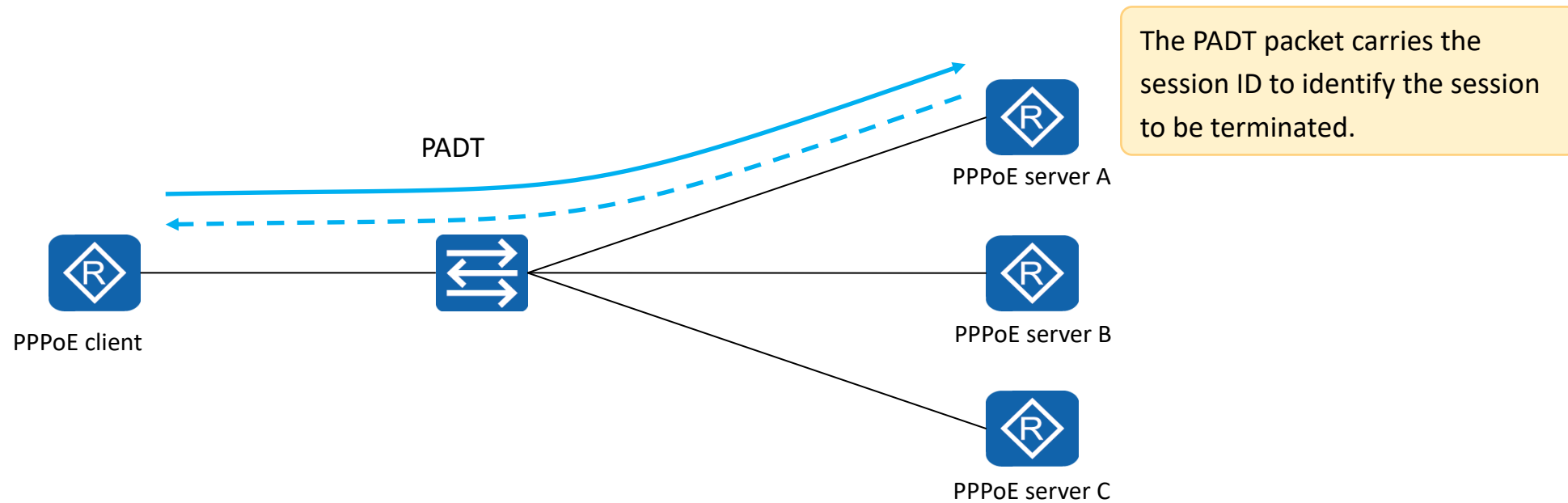


In the entire session stage, the session ID allocated by the PPPoE server remains unchanged.



PPPoE Session Termination Stage

- If the PPPoE client wants to terminate the session, it sends a PADT packet to the PPPoE server.
- Similarly, if the PPPoE server wants to terminate the session, it sends a PADT packet to the PPPoE client.





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Configuring Basic PPPoE Functions

1. Configure a dialer rule and set conditions for initiating a PPPoE session under the rule.

```
[Huawei] dialer-rule
```

Configure a username on the dialer interface. The username must be the same as that of the peer server.

```
[Huawei-Dialer1] dialer user username
```

3. Add the interface to a dialer group.

```
[Huawei-Dialer1] dialer-group group-number
```

4. Specify a dialer bundle for the interface.

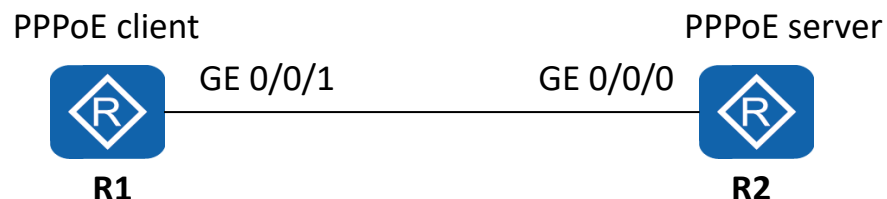
```
[Huawei-Dialer1] dialer-bundle number
```

5. Bind a physical interface to the dialer bundle.

```
[Huawei-Ethernet0/0/0] pppoe-client dial-bundle-number number
```



Example for Configuring a PPPoE Client (1)



- Experiment requirements:

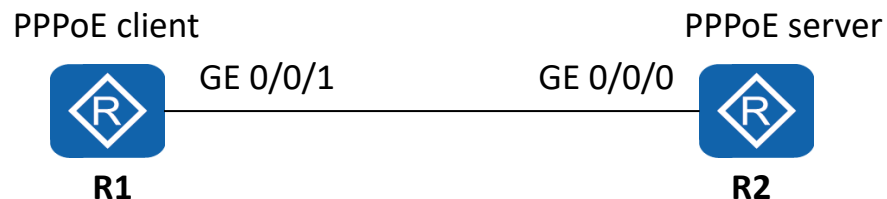
1. Configure R1 as a PPPoE client and R2 as a PPPoE server.
2. Configure a dialer interface for the PPPoE client on R1.
3. Configure the authentication function on the dialer interface on R1.
4. The dialer interface on R1 can obtain the IP address allocated by the PPPoE server.
5. R1 can access the server through the dialer interface.

1. Create a dialer interface and configure a username and password for authentication.

```
[R1]dialer-rule
[R1-dialer-rule]dialer-rule 1 ip permit
[R1-dialer-rule]quit
[R1]interface dialer 1
[R1-Dialer1] dialer user enterprise
[R1-Dialer1] dialer-group 1
[R1-Dialer1] dialer bundle 1
[R1-Dialer1] ppp chap user huawei1
[R1-Dialer1] ppp chap password cipher huawei123
[R1-Dialer1] ip address ppp-negotiate
```



Example for Configuring a PPPoE Client (2)



- Experiment requirements:

1. Configure R1 as a PPPoE client and R2 as a PPPoE server.
2. Configure a dialer interface for the PPPoE client on R1.
3. Configure the authentication function on the dialer interface on R1.
4. The dialer interface on R1 can obtain the IP address allocated by the PPPoE server.
5. R1 can access the server through the dialer interface.

2. Bind the dialer interface to an outbound interface.

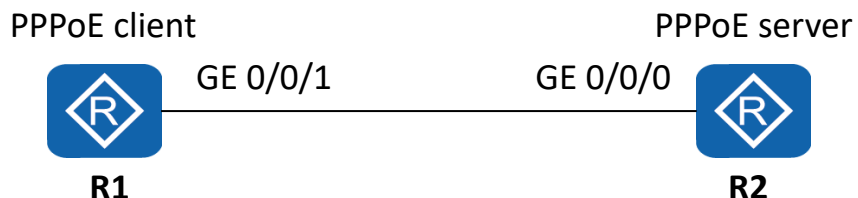
```
[R1]interface GigabitEthernet 0/0/1  
[R1-GigabitEthernet0/0/1]pppoe-client dial-bundle-number 1  
[R1-GigabitEthernet0/0/1]quit
```

3. Configure a default route from the PPPoE client to the server.

```
[R1]ip route-static 0.0.0.0 0.0.0.0 dialer 1
```



Example for Configuring a PPPoE Server



- Experiment requirements:
 1. Create an address pool on the PPPoE server for address allocation to the PPPoE client.
 2. The PPPoE server authenticates the PPPoE client and assigns a valid IP address to the client.

1. Create an address pool and a virtual template.

```
[R2]ip pool pool1 # Create an address pool and specify the range of the IP
addresses to be allocated and a gateway.
[R2-ip-pool-pool1]network 192.168.1.0 mask 255.255.255.0
[R2-ip-pool-pool1]gateway-list 192.168.1.254
[R2]interface Virtual-Template 1 # Create a virtual template interface.
[R2-Virtual-Template1]ppp authentication-mode chap
[R2-Virtual-Template1]ip address 192.168.1.254 255.255.255.0
[R2-Virtual-Template1]remote address pool pool1
```

2. Bind a physical interface to the virtual template.

```
[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]pppoe-server bind virtual-template 1
[R2-GigabitEthernet0/0/0]quit
```

3. Create an access user.

```
[R2]aaa # Add information about the user to be authenticated.
[R2-aaa]local-user huawei1 password cipher huawei123
[R2-aaa]local-user huawei1 service-type ppp
```



Verifying the Configuration

1. Check detailed information about the dialer interface.

```
<R1>display interface Dialer 1
Dialer1 current state: UP
Line protocol current state: UP (spoofing)
Description: HUAWEI, AR Series, Dialer1 Interface
Route Port, The Maximum Transmit Unit is 1500, Hold timer is
10(sec)
Internet Address is negotiated, 192.168.10.254/32
Link layer protocol is PPP
LCP initial
Physical is Dialer
Bound to Dialer1:0:
Dialer1:0 current state : UP
Line protocol current state : UP
Link layer protocol is PPP
LCP opened, IPCP opened
```

2. Check the initial status of the PPPoE session on the client.

```
[R1]display pppoe-client session summary
PPPoE Client Session:
ID  Bundle  Dialer Intf    Client-MAC    Server-MAC    State
0   1        1  GE0/0/1  54899876830c  000000000000  IDLE
```

3. Check the establishment status of the PPPoE session on the client.

```
[R1]display pppoe-client session summary
PPPoE Client Session:
ID  Bundle  Dialer Intf    Client-MAC    Server-MAC    State
1   1        1  GE0/0/1  00e0fc0308f6  00e0fc036781  UP
```




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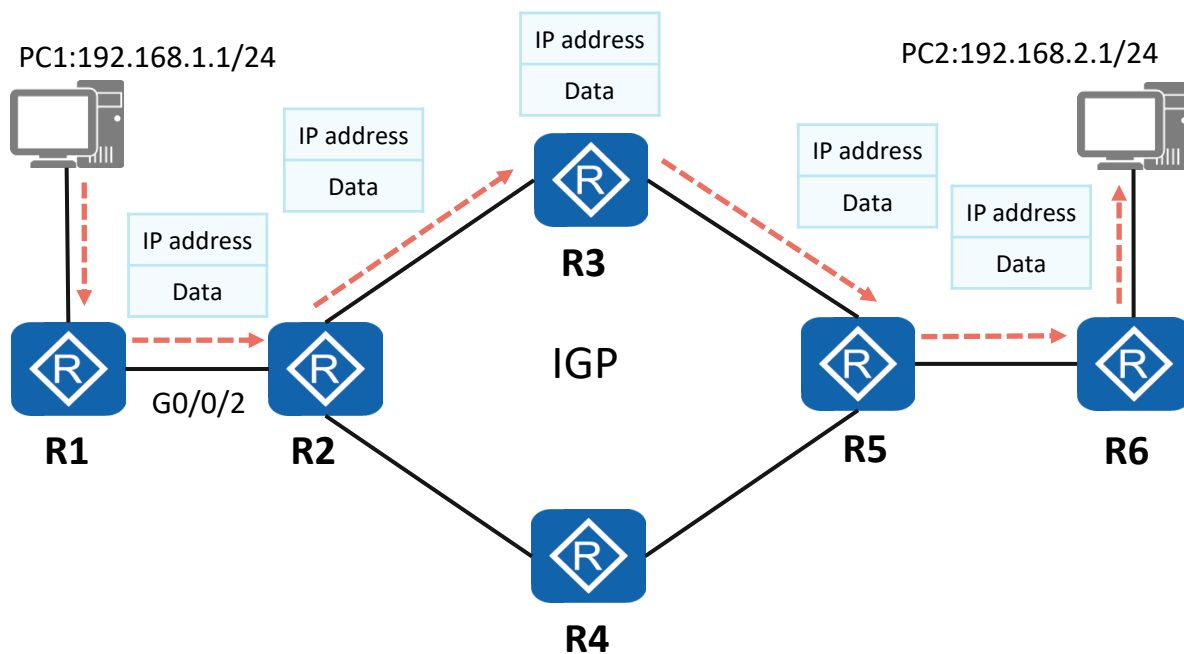
Evolution of WAN Technologies

- The data link layer protocols commonly used on early WANs include PPP, HDLC, and ATM. With the network evolution towards all-IP, the IP-based Internet becomes popular. However, the IP technology based on the longest match rule must use software to search for routes, resulting in low forwarding performance, which has become the bottleneck that restricts the network development.
- Multiprotocol Label Switching (MPLS) was originally proposed to improve the forwarding speeds of routers. Compared with the traditional IP routing mode, MPLS parses IP packet headers only at the network edges during data forwarding. Transit nodes forward packets based on labels, without the need to parse IP packet headers. This speeds up software processing.
- With the improvement of router performance, the route search speed is no longer a bottleneck for network development. Thus, MPLS loses its advantage in fast forwarding speed. However, leveraging support for multi-layer labels and a connection-oriented forwarding plane, MPLS is widely applied in various scenarios, such as virtual private network (VPN), traffic engineering (TE), and quality of service (QoS) scenarios.



Traditional IP Routing and Forwarding

- Traditional IP forwarding uses hop-by-hop forwarding. Each time a data packet passes through a router, the router decapsulates the packet to check the network layer information and searches its routing table based on the longest match rule to guide packet forwarding. The repeat process of decapsulating packets, searching routing tables, and re-encapsulating the packets on routers lead to low forwarding performance.



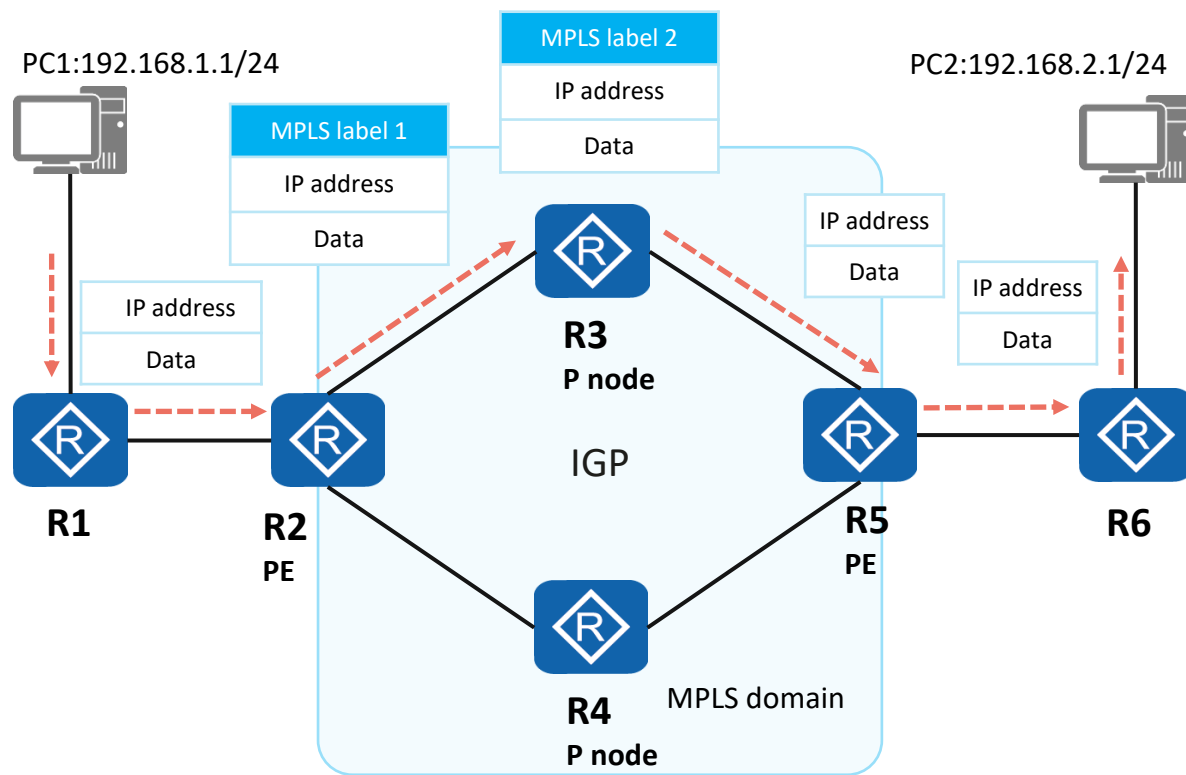
- Characteristics of traditional IP routing and forwarding:
 - All routers need to know the network-wide routes.
 - Traditional IP forwarding is connectionless-oriented and cannot provide good end-to-end QoS guarantee.

R1 routing table

Destination/Mask	Protocol	Preference	Cost	NextHop	Interface
192.168.1.0/24	Direct	0	0	192.168.1.254	GE 0/0/0
192.168.12.0/24	Direct	0	0	192.168.12.1	GE 0/0/2
192.168.2.0/24	OSPF	10	3	192.168.12.2	GE 0/0/2



MPLS Label-based Forwarding

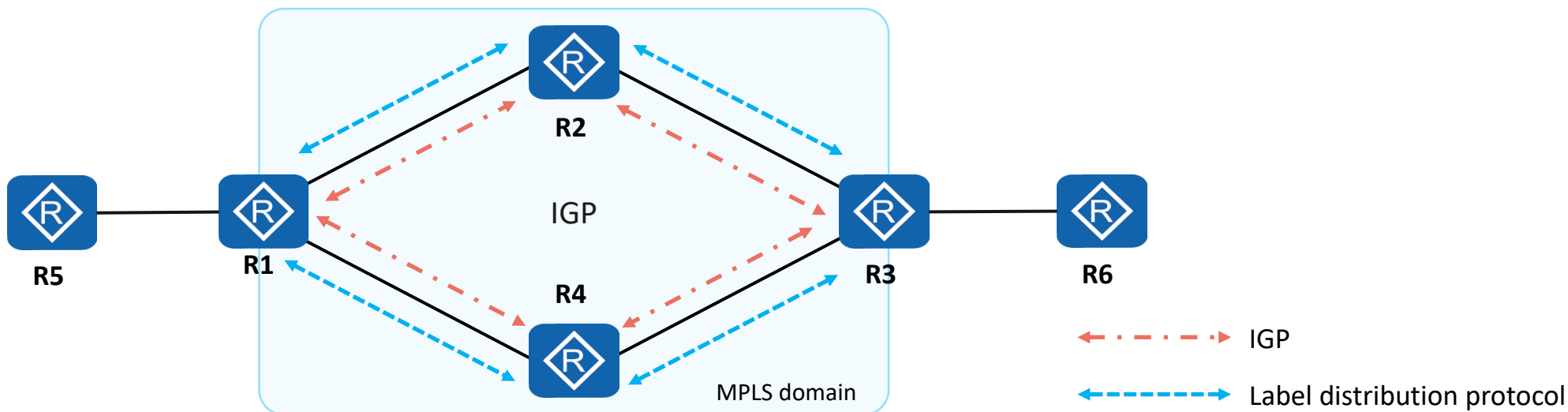


- MPLS is used on IP backbone networks.
- MPLS is a tunneling technology that provides connection-oriented switching for the network layer based on IP routing and control protocols. It provides better QoS guarantee.
- MPLS labels, instead of IP routes, are searched for to forward packets, which greatly improves forwarding efficiency.
- Labels used in MPLS forwarding can be manually configured or dynamically allocated using a label distribution protocol.



MPLS Forwarding Problems

- MPLS labels can be statically or dynamically distributed. The involved problems are as follows:
 - Static label distribution requires manual configuration. As the network scale expands, network topologies are prone to change. Static label configuration cannot meet the requirements of large-scale networks.
 - Some dynamic label distribution protocols do not have the path computation capability and need to use IGPs to compute paths. In addition, the control planes of these protocols are complex, requiring devices to send a large number of messages to maintain peer and path status, wasting link bandwidth and device resources. What is more, despite supporting TE, some label distribution protocols require complex configurations and do not support load balancing. Devices have to send a large number of protocol packets to maintain proper paths. In addition, as devices are independent and know only their own status, they need to exchange signaling packets, which also waste link bandwidth and device resources.





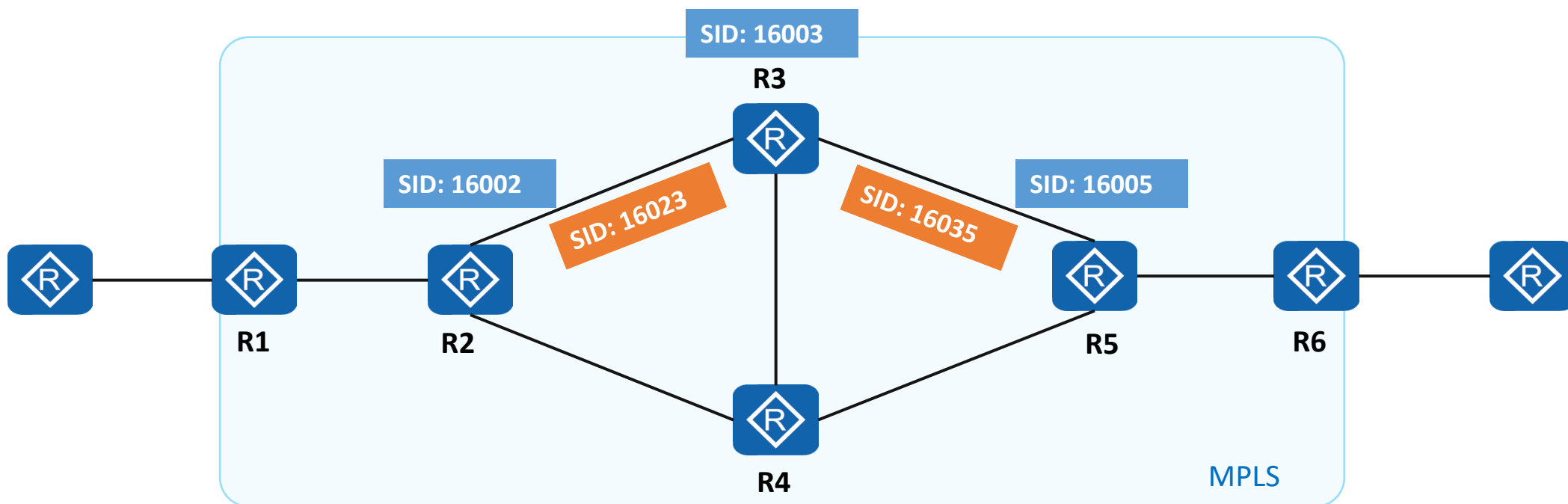
Introduction to Segment Routing

- To solve the problems facing traditional IP forwarding and MPLS forwarding, the industry proposed Segment Routing (SR). SR makes the following improvements:
 1. Extends the existing protocols.
 - The extended IGPs and BGP have the label distribution capability, eliminating the need for other label distribution protocols on networks, and thereby simplifying protocols.
 2. Introduces the source routing mechanism.
 - Using the source routing mechanism, controllers can centrally calculate paths.
 3. Allows networks to be defined by services.
 - Networks are driven by services. After service requirements, such as latency, bandwidth, and packet loss rate requirements, are raised by applications, a controller can collect information such as the network topology, bandwidth usage, and latency, and calculate explicit paths based on these requirements.



SR Forwarding Implementation (1)

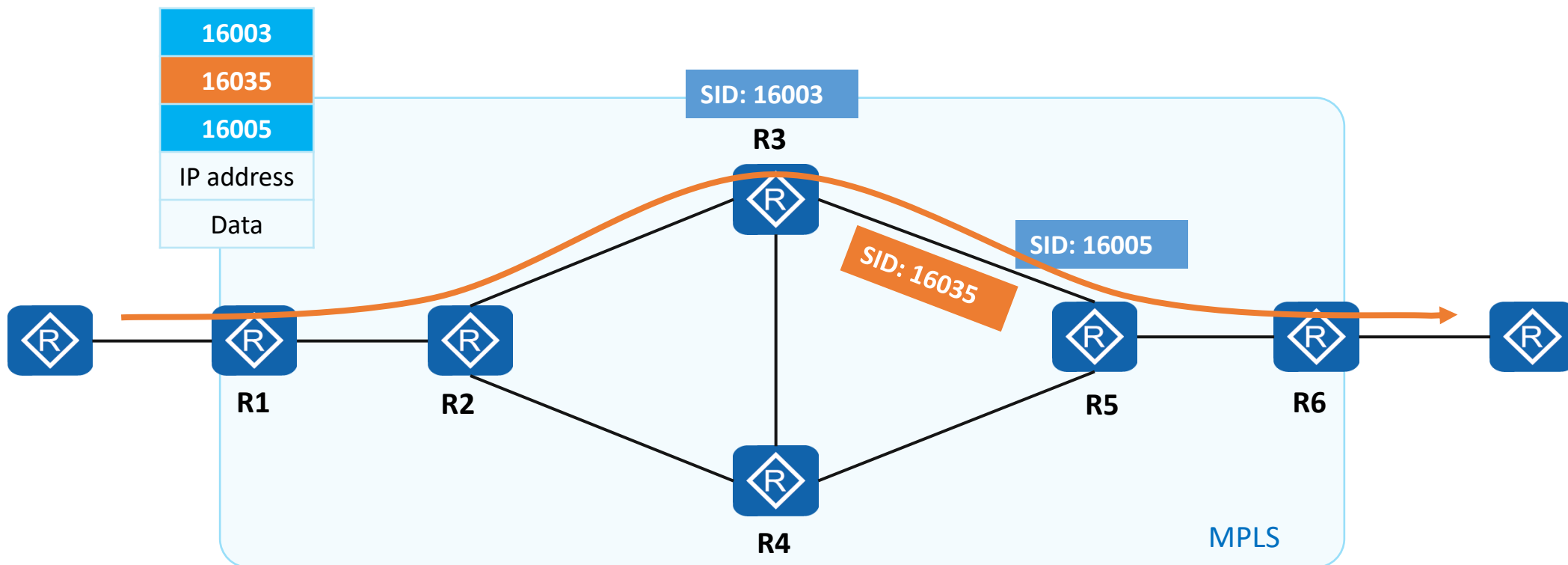
- SR divides a network path into segments and assigns segment IDs (SIDs) to these segments.
- SIDs are allocated to forwarding nodes or adjacency links. In this example, SIDs of the forwarding nodes are expressed in 1600X, where X is a node ID; SIDs of the adjacency links are expressed in 160XX, where XX indicates the node IDs at both ends of a link.





SR Forwarding Implementation (2)

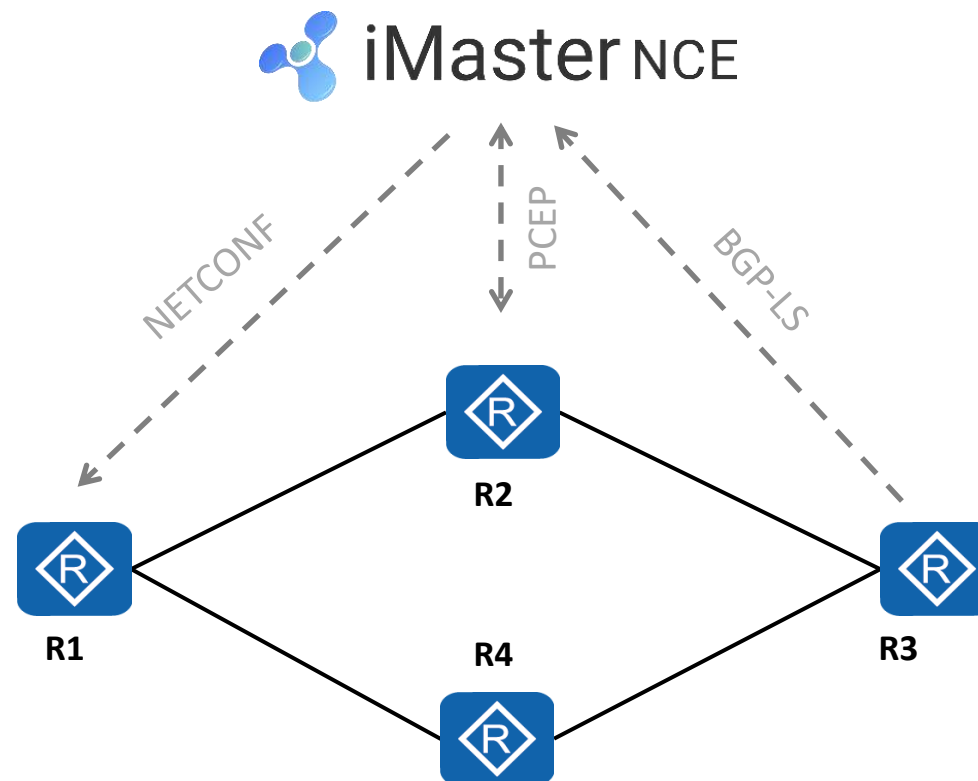
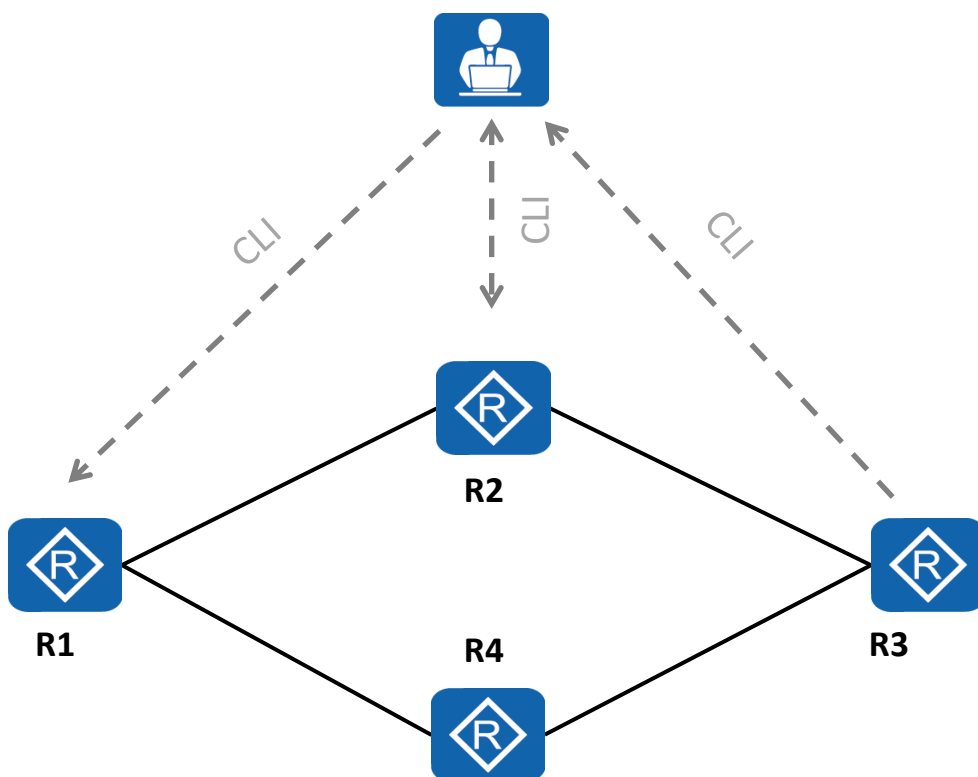
- SIDs of adjacency links and network nodes are arranged in order to form a segment list, which represents a forwarding path. The segment list is encoded by the source node in a header of a data packet, and is transmitted with the data packet. The essence of SR is instructions, which guide where and how packets go.





SR Deployment Modes

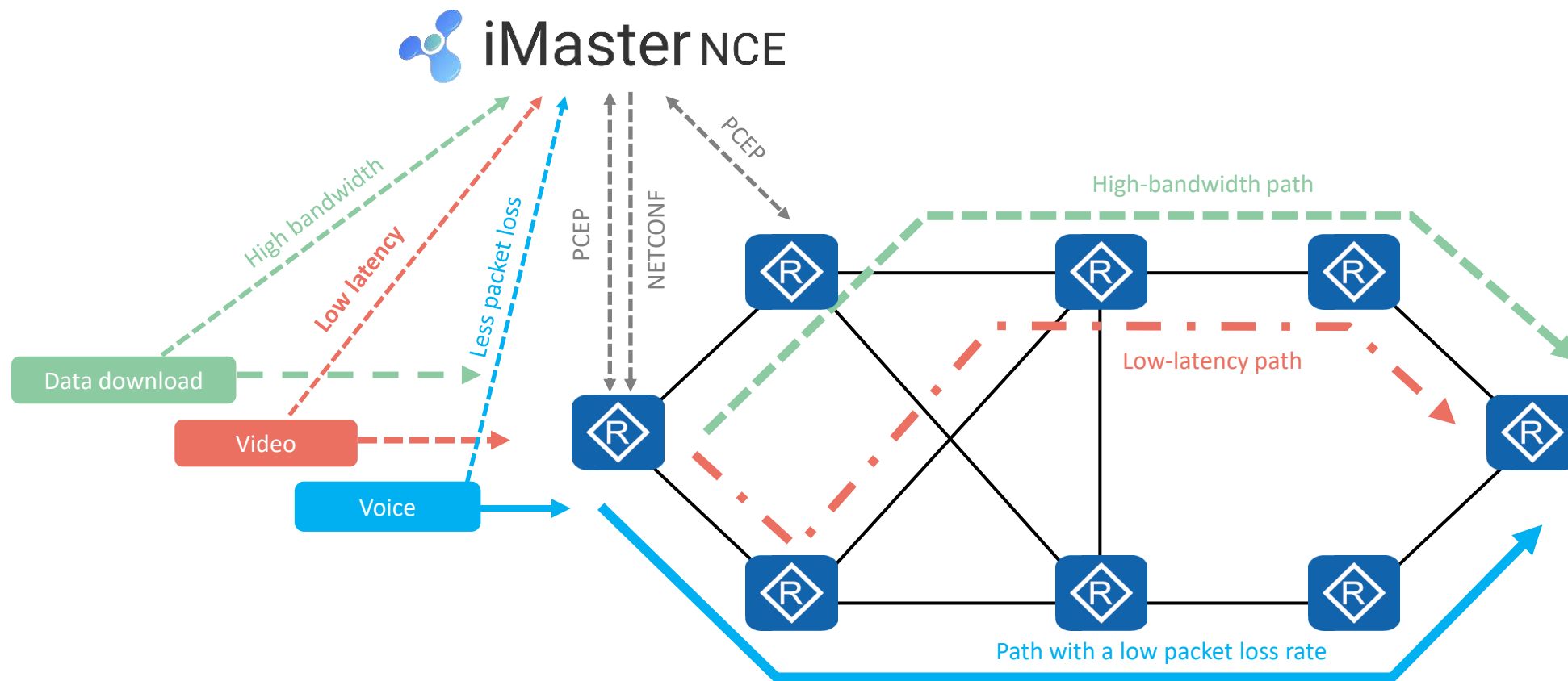
- SR can be deployed with or without a controller. If a controller is used, the controller collects information, reserves path resources, computes paths, and delivers the results to the source node. This mode is preferred.





SR Application

- SR can be used to easily specify packet forwarding paths. On a live network, different paths can be defined for different services. In this example, three explicit paths are defined to implement the service-driven network: one each for data download, video, and voice services. Devices are managed by the controller, which can quickly provision paths in real time.





Quiz

1. (Multiple) Which of the following statements about PPP are true?
 - A. PPP supports the bundling of multiple physical links into a logical link to increase the bandwidth.
 - B. PPP supports cleartext and ciphertext authentication.
 - C. PPP cannot be deployed on Ethernet links because of its poor scalability.
 - D. PPP supports asynchronous and synchronous links for the physical layer.
 - E. PPP supports multiple network layer protocols, such as IPCP.
2. (Single) After a PPPoE client sends a PADI packet to PPPoE servers, the PPPoE servers reply with a PADO packet. Which kind of frame is the PADO packet?
 - A. A. Multicast B. Broadcast C. Unicast D. Anycast
3. (Single) Which of the following values of the Length/Type field in an Ethernet data frame indicates that the Ethernet data frame carries PPPoE discovery packets?
 - A. A. 0x0800 B. 0x8864 C. 0x8863 D. 0x0806



Summary

- This course reviews the types and applications of early WAN technologies and describes the evolution of WANs from the early circuit switching networks to IP networks, MPLS label switching network, and finally to SR networks. With the development of network technologies, networks become more efficient and intelligent.
- The course also describes the implementation of PPP, including parameter negotiation during PPP link establishment, authentication negotiation, and network layer negotiation. It analyzes in detail two PPP authentication protocols – PAP and CHAP, and describes their working processes and differences.
- PPPoE is the most widely used PPP application. By analyzing how a PPPoE session is discovered, negotiated, established, and torn down, this course help you better understand the working mechanism and configuration of PPPoE.



More Information

- (Multimedia) Segment Routing MPLS Advanced Series
 - <https://support.huawei.com/carrier/docview?nid=DOC1100645168&path=PBI1-7275726/PBI1-21782273/PBI1-7275849/PBI1-7276518/PBI1-15837>
- (Multimedia) Segment Routing IPv6 Advanced Series
 - <https://support.huawei.com/enterprise/en/doc/EDOC1100133514?idPath=24030814%7C9856750%7C22715517%7C9858933%7C15837>

The background of the slide features a blue-tinted image of several business professionals in a modern office. They are standing on a highly reflective floor, and their silhouettes are clearly visible. The overall aesthetic is professional and corporate.

Thank You
www.huawei.com



Network Management and O&M



Foreword

- The ever expanding network and increasing network devices present a significant challenge in managing networks effectively and providing high-quality network services.
- There are many network management and O&M methods, of which this course describes some of the most common.



Objectives

- On completion of this course, you will be able to:
 - Understand basic concepts of network management and O&M.
 - Master common network management and O&M methods.
 - Describe basic functions of network management and O&M.
 - Understand the fundamentals of SNMP.
 - Understand Huawei iMaster NCE and related technologies.



Contents

- 1. Basic Concepts of Network Management and O&M**
2. SNMP Fundamentals and Configuration
3. Network Management Based on Huawei iMaster NCE



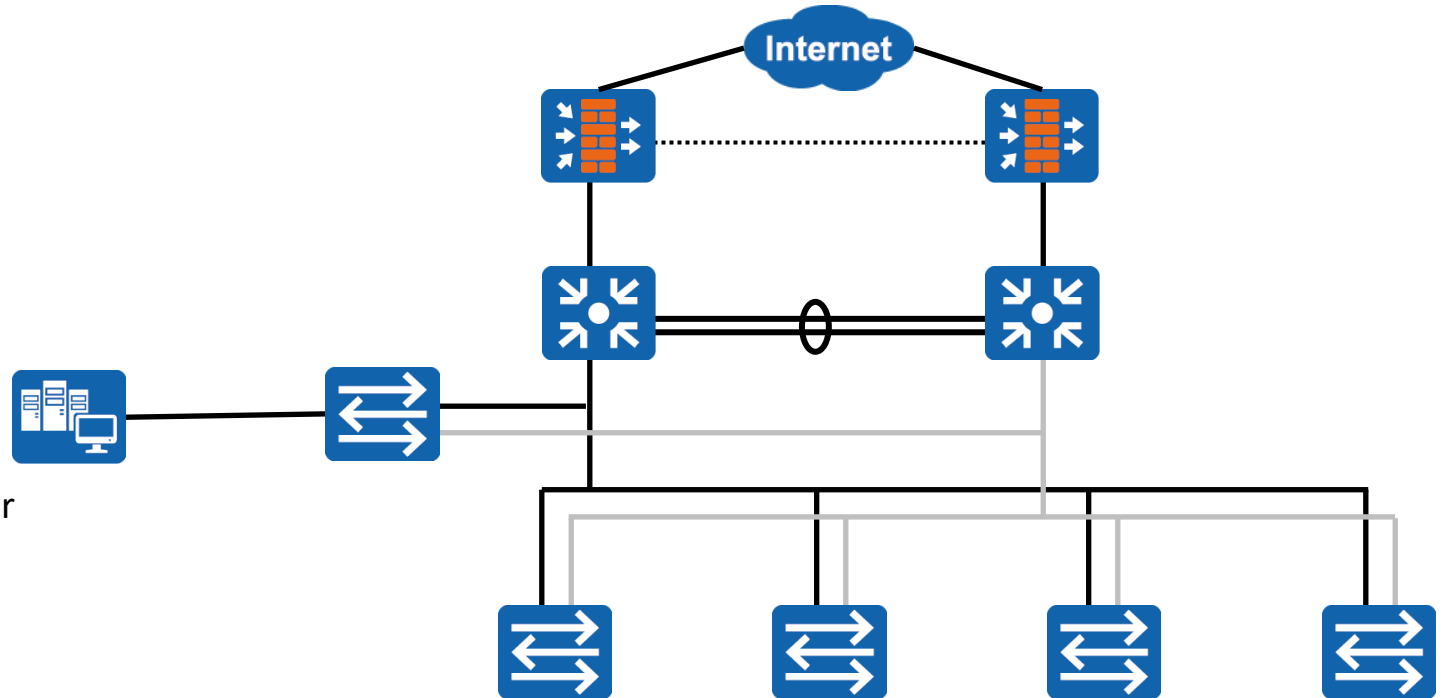
What Is Network Management and O&M?

- Network management and O&M plays an important role on a communications network. It ensures that devices work properly and the communications network runs properly to provide efficient, reliable, and secure communications services.

The network administrator manages and maintains the network for stable operation.



Network administrator



Common enterprise network architecture



Basic Network Management Functions

Configuration
management

Performance
management

Fault
management

Security
management

Accounting
management

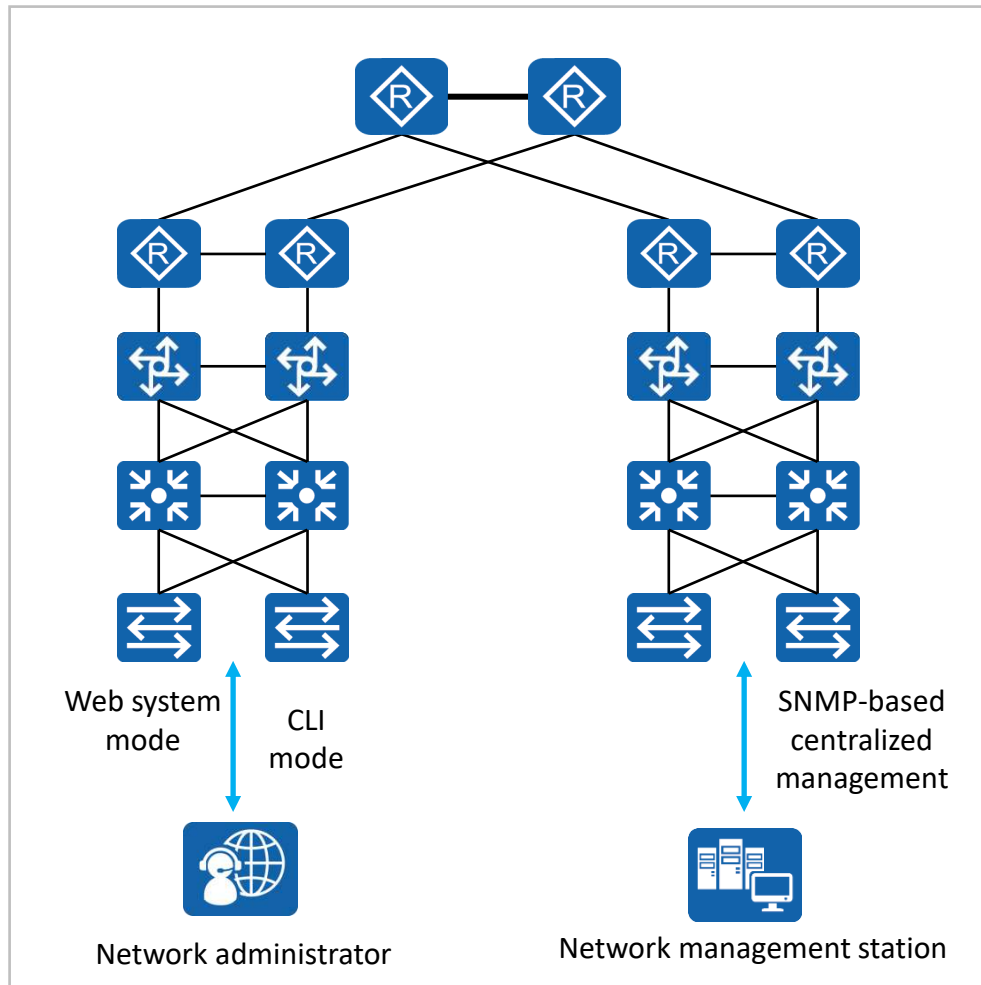
OSI defines five functional models for network management:

- Configuration management: monitors network configuration information so that network administrators can generate, query, and modify hardware and software running parameters and conditions, and configure services.
- Performance management: manages network performance so that the network can provide reliable, continuous, and low-latency communication capabilities with as few network resources as possible.
- Fault management: ensures that the network is always available and rectifies faults as soon as possible.
- Security management: protects networks and systems from unauthorized access and attacks.
- Accounting management: records the network resource usage of users, charges users, and collects statistics on network resource usage.

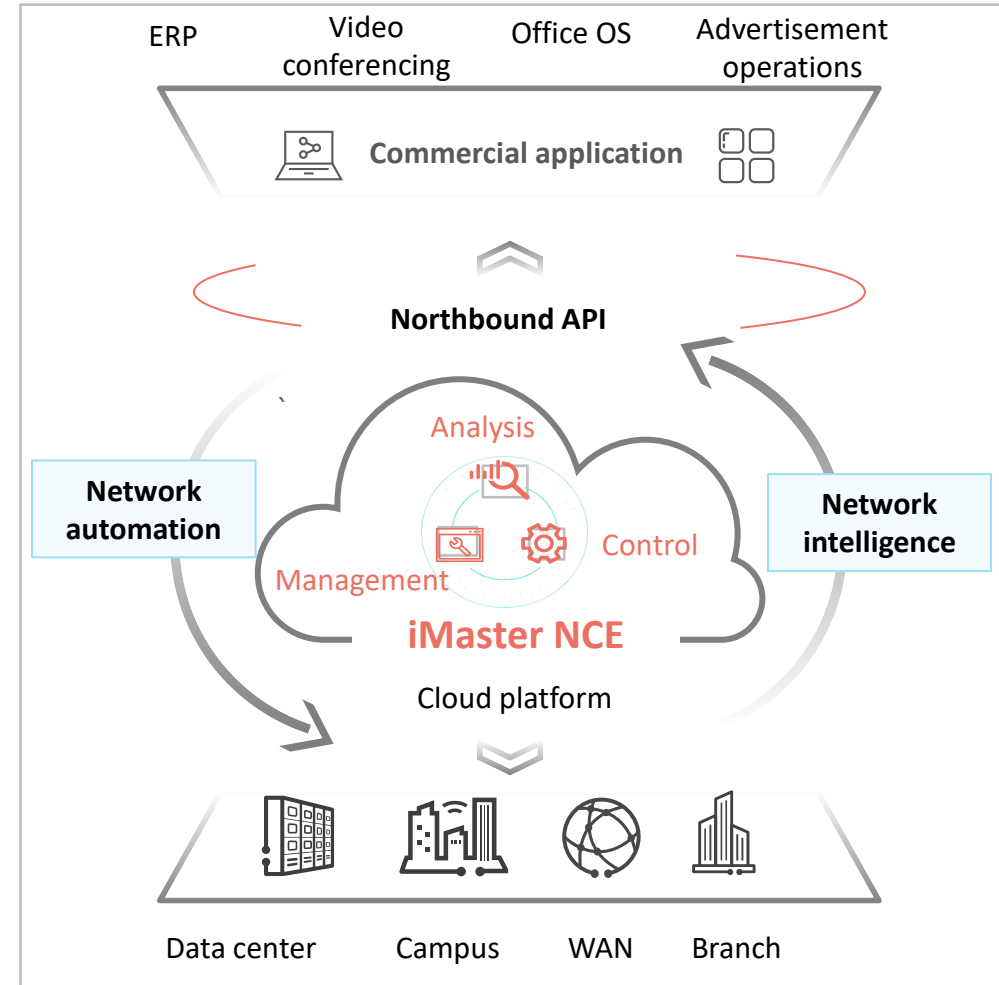


Network Management Modes

Traditional Network Management and O&M



iMaster NCE-based Network Management and O&M





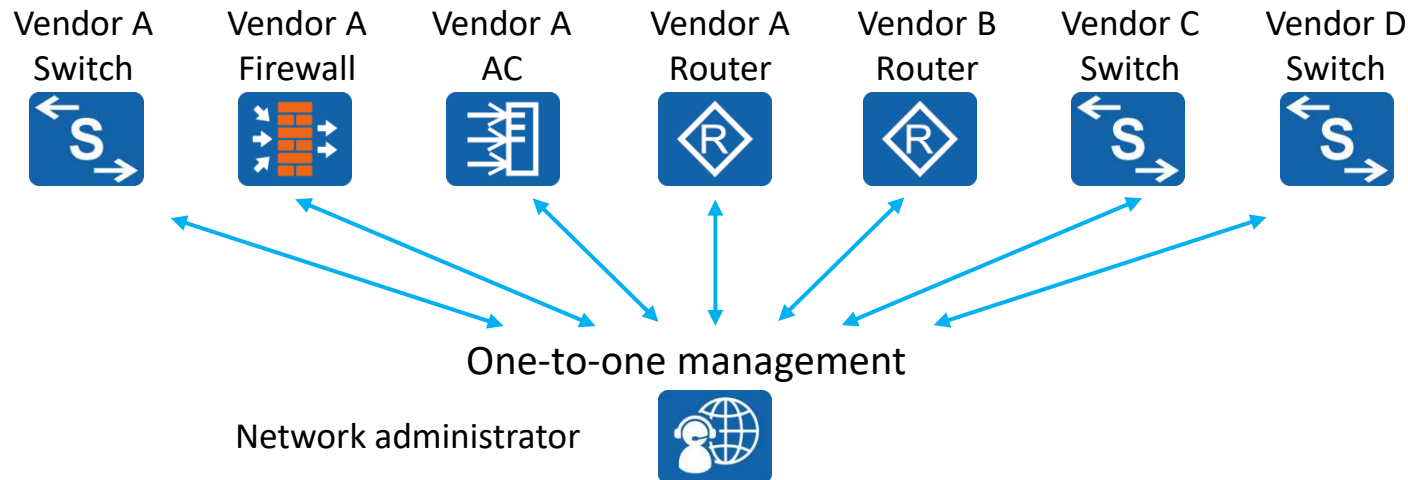
Contents

1. Basic Concepts of Network Management and O&M
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Management Through the CLI or Web System

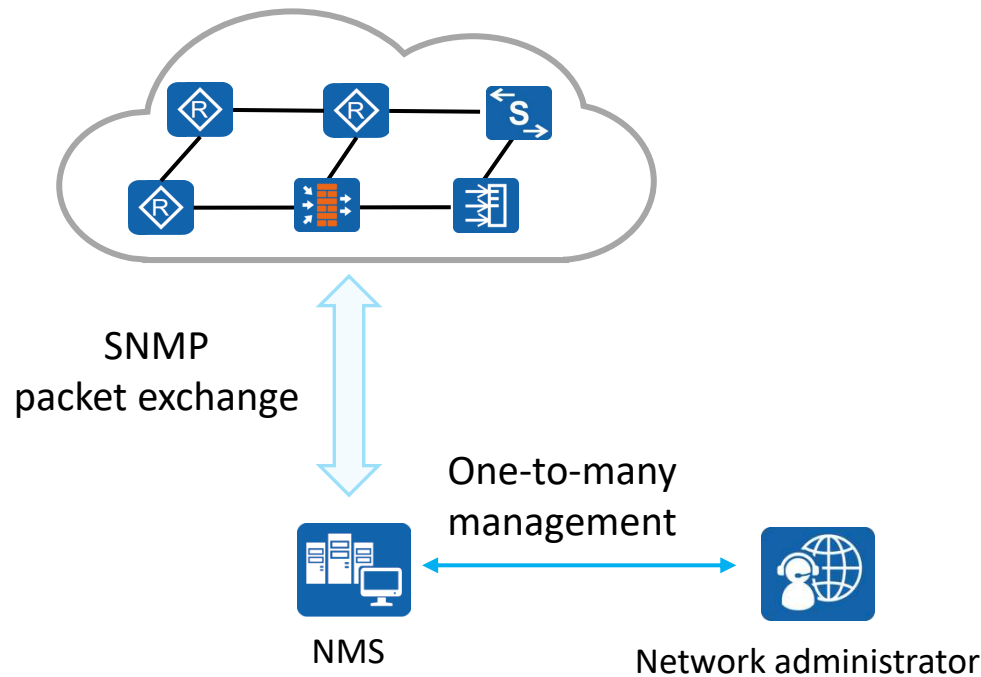
- When the network scale is small, the CLI and web system are generally used for network management.
 - Network administrators can log in to a device through HTTPS, Telnet, or the console port to manage the device.
 - These network management modes do not require any program or server to be installed on the network, and the cost is low.
 - Network administrators must have a good master of network knowledge and vendor-specific network configuration commands.
 - These modes have great limitations when the network scale is large and the network topology is complex.





SNMP-based Centralized Management

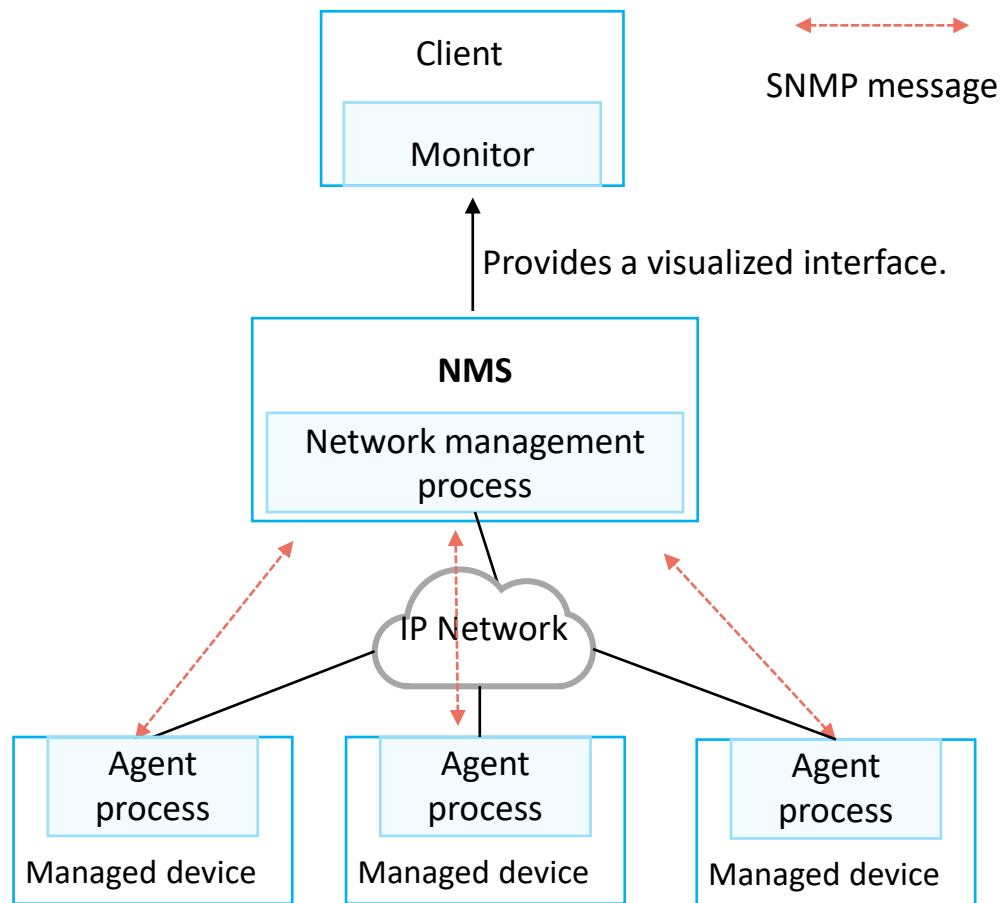
- SNMP is a standard network management protocol widely used on TCP/IP networks. It provides a method for managing NEs through a central computer that runs network management software, that is, a network management station.



- Network administrators can use the NMS to query information, modify information, and troubleshoot faults on any node on the network, improving work efficiency.
- Network devices of different types and vendors are managed in a unified manner.



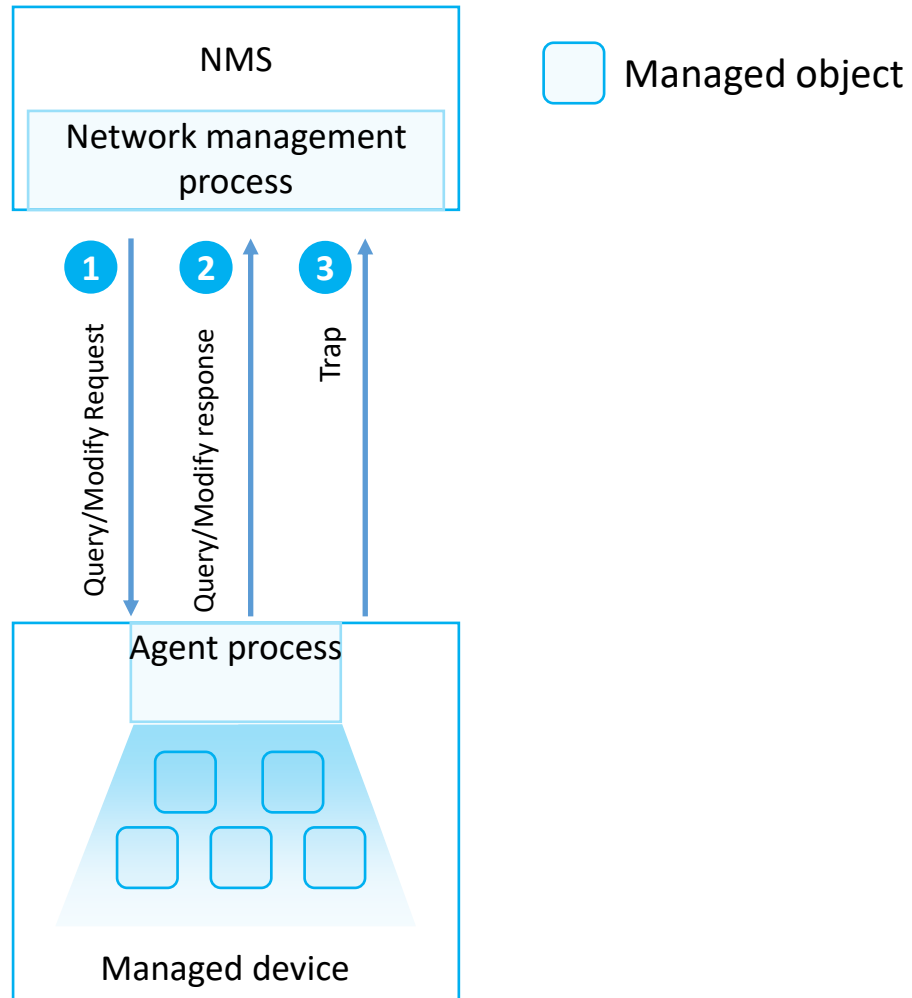
Typical SNMP Architecture



- On a network where SNMP is used for network management, a network management system (NMS) functions as a network management center and runs management processes. Each managed device needs to run an agent process. The management process and agent process communicate with each other through SNMP messages.
- An NMS is a system that uses SNMP to manage and monitor network devices. The NMS software runs on NMS servers.
- Managed devices are devices that are managed by the NMS on the network.
- The agent process runs on managed devices to maintain the information data of the managed devices, respond to the request from the NMS, and report the management data to the NMS that sends the request.



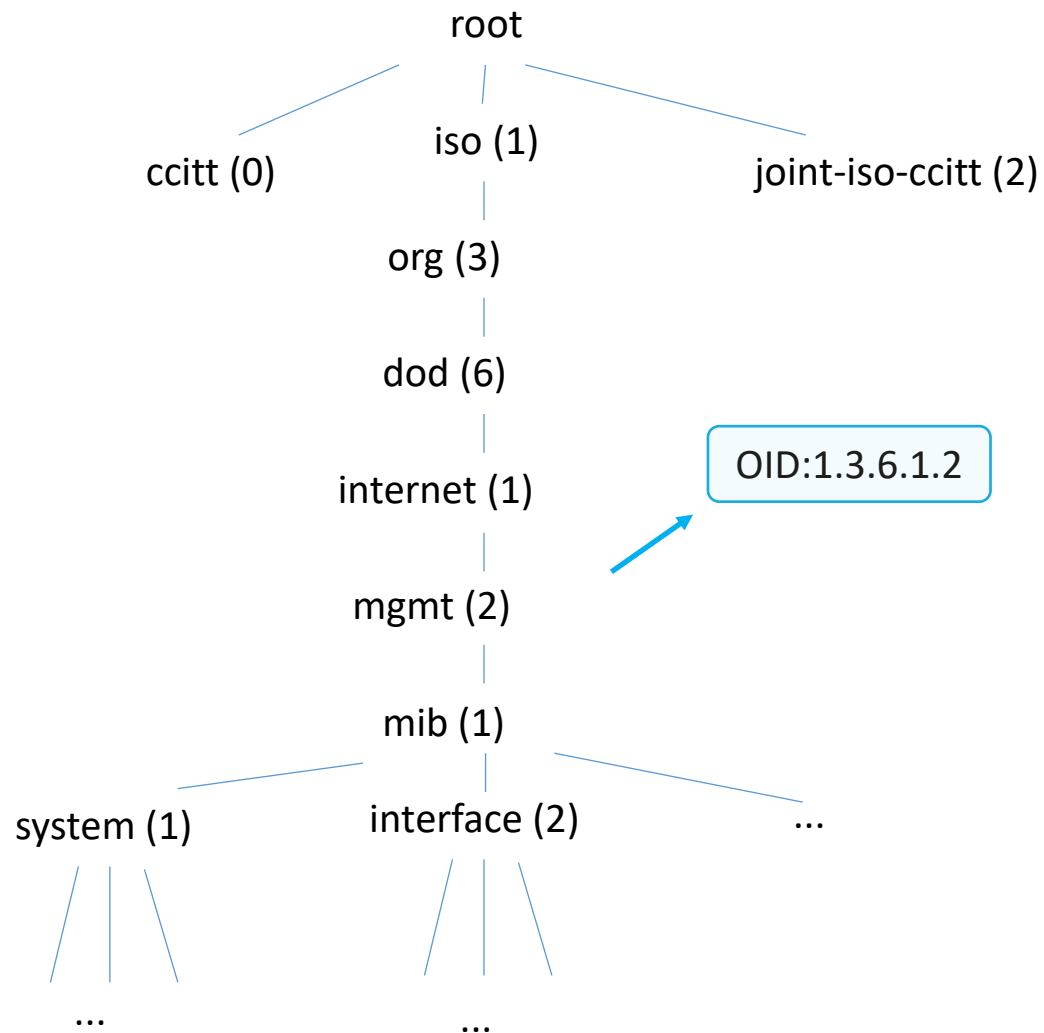
SNMP Message Exchange



- The NMS and managed devices exchange messages in the following modes:
 - The NMS sends a request for modifying or querying configuration information to a managed device through SNMP. The agent process running on the managed device responds to the request from the NMS.
 - The managed device can proactively report traps to the NMS so that the network administrator can detect faults in a timely manner.
- Managed object: Each device may contain multiple managed objects. A managed object can be a hardware component or a set of parameters configured on the hardware or software (such as a routing protocol).
- SNMP uses management information bases (MIBs) to describe a group of objects of a manageable entity.



MIB



- A MIB is a database containing the variables that are maintained by managed devices. (The variables can be queried or set by the agent processes.) The MIB defines the attributes of managed devices in the database.
 - Object identifier (OID) of an object
 - Status of an object
 - Access permission of an object
 - Data types of an object
- A MIB provides a structure that contains data on all NEs that may be managed on the network. Because the data structure is similar to the tree structure, a MIB is also called an object naming tree.



Common MIB Objects

- Objects used for query or modification:

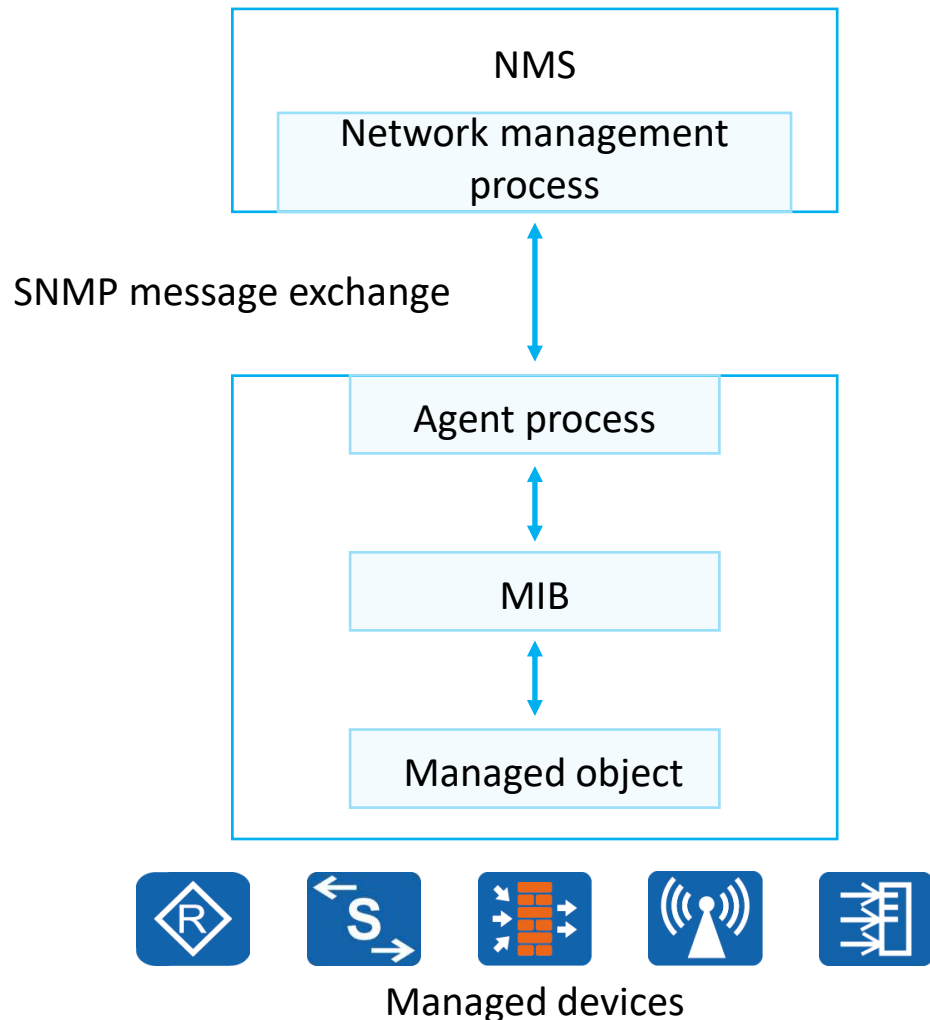
OID	Object Name	Data Type	Maximum Access	Description
1.3.6.1.2.1.2.1	ifNumber	Integer	read-only	Number of network interfaces in the system (regardless of the current interface status)
1.3.6.1.4.1.2011.5.25.41.1.2.1.1.3	hwIpAdEntNetMask	IpAddress	read-create	Subnet mask of an IP address

- Objects used for alarm notification:

OID	Object Name	Bound Variable	Description
3.6.1.6.3.1.1.5.3	linkDown	ifIndex ifAdminStatus ifOperStatus ifDesc	It is detected that one of the communication links in the ifOperStatus object has entered the down state from another state (but not the notPresent state). The original state is indicated by the value of ifOperStatus.



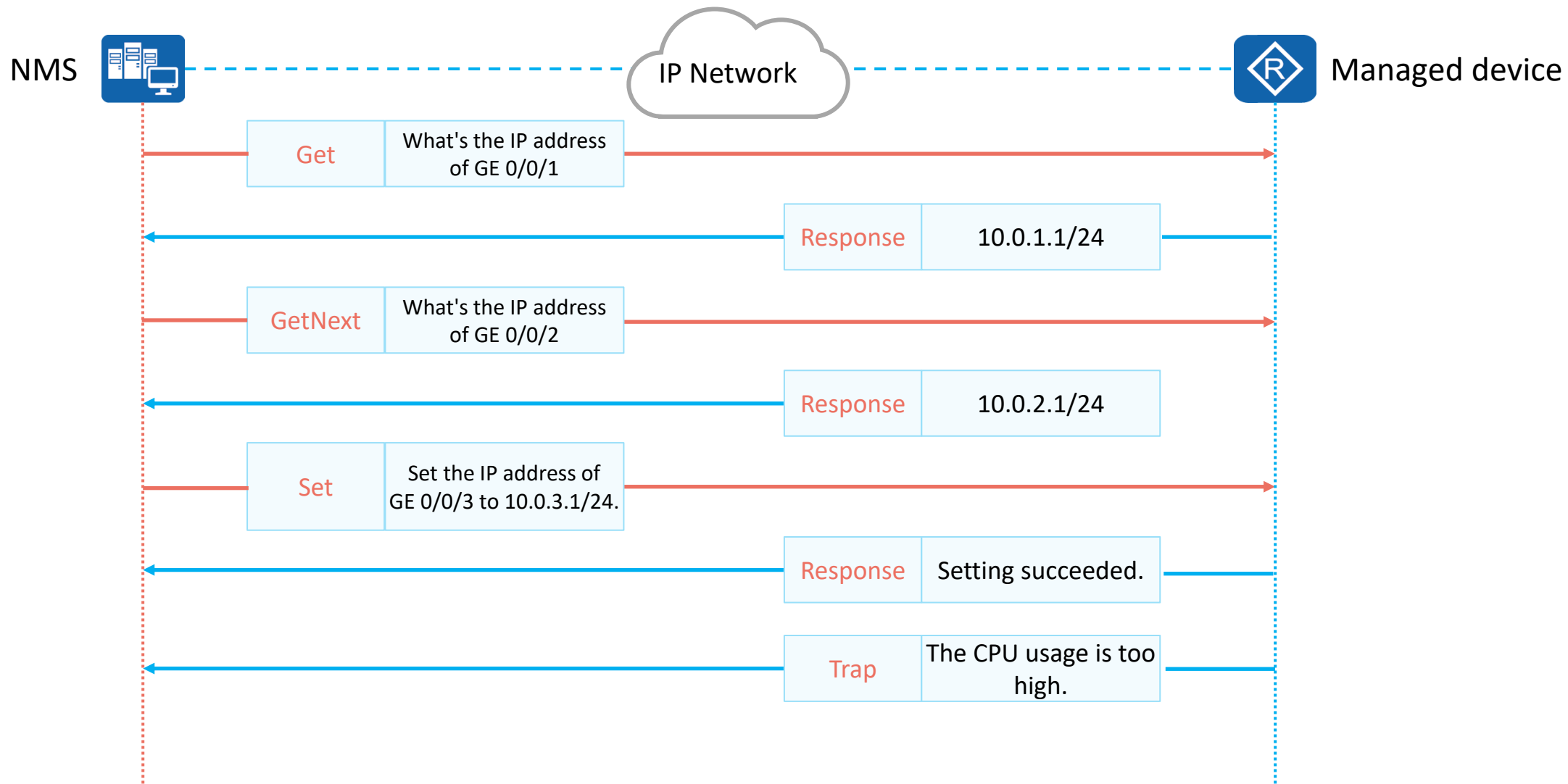
SNMP Management Model



- Query/Modify operation:
 - The NMS sends an SNMP request message to an agent process.
 - The agent process searches the MIB on the device for information to be queried or modified and sends an SNMP response message to the NMS.
- Trap operation:
 - If the trap triggering conditions defined for a module are met, the agent process sends a message to notify the NMS that an event or trap has occurred on a managed object. This helps network administrators promptly process network faults.

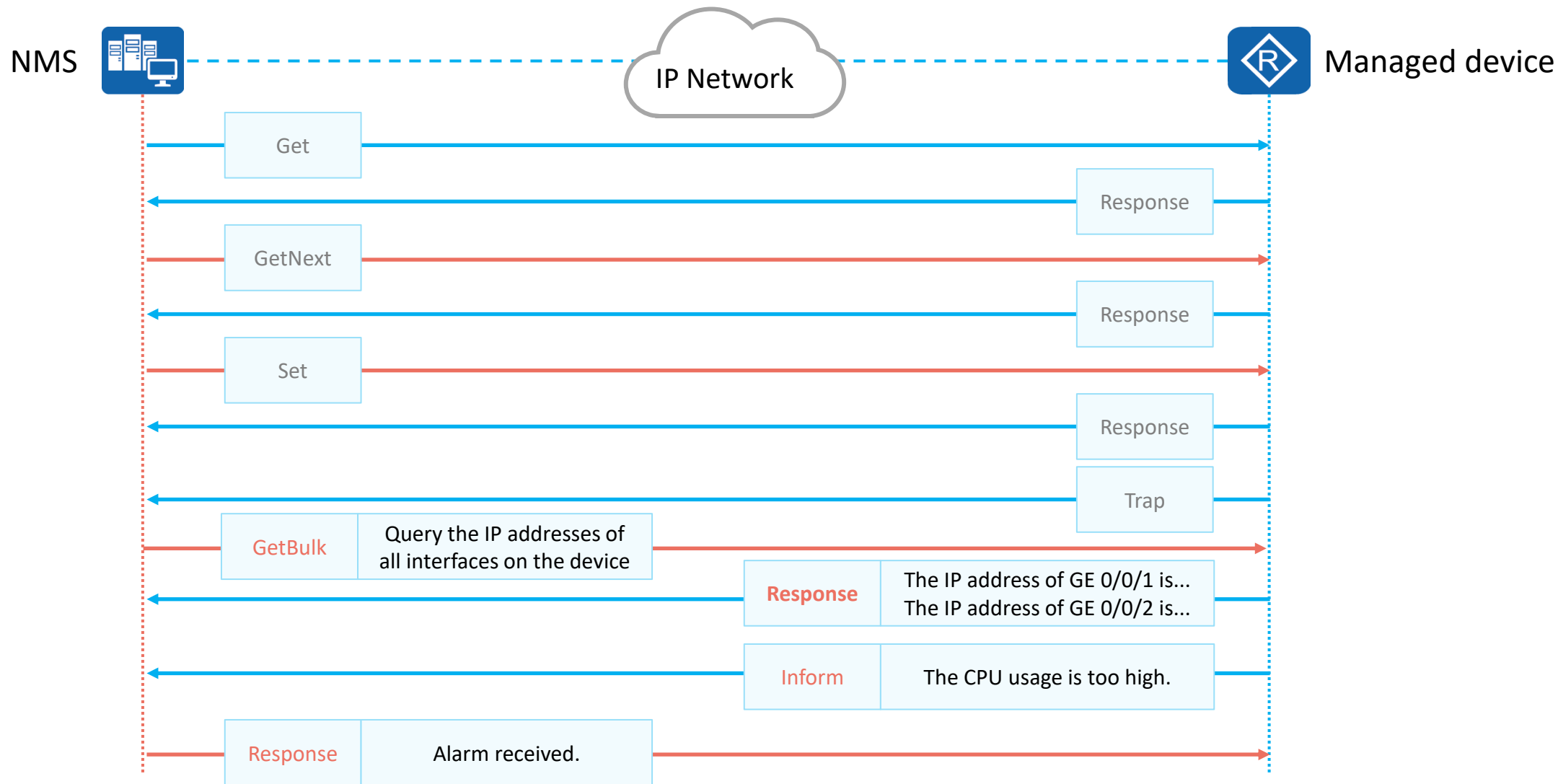


SNMPv1





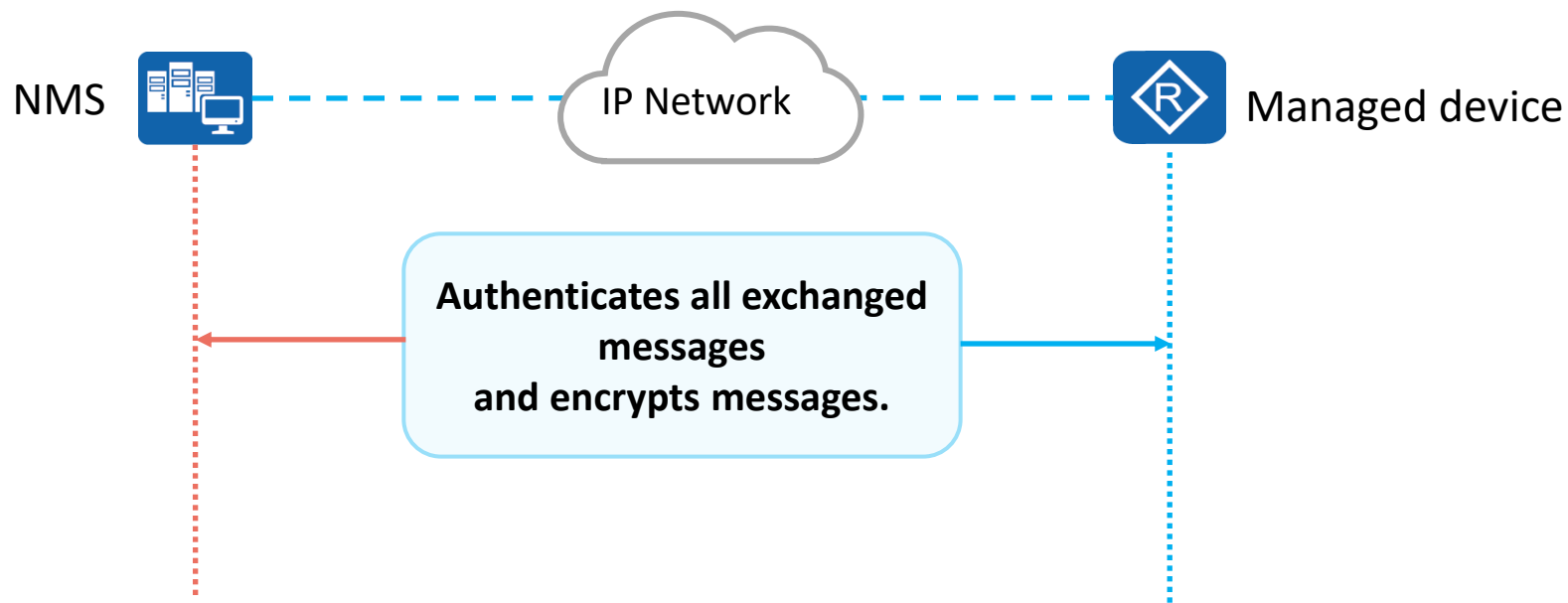
SNMPv2c





SNMPv3

- SNMPv3 has the same working mechanism as SNMPv1 and SNMPv2c, but adds header data and security parameters.
- SNMPv3 messages can be authenticated and encrypted.
- SNMPv3 is applicable to networks of various scales and has high security.





SNMP Summary

- SNMP has the following advantages:
 - Simplicity: SNMP is applicable to networks that require high speed and low cost because it uses a polling mechanism and provides basic network management functions. Moreover, SNMP uses UDP to exchange data and therefore is supported by most products.
 - Convenience: SNMP allows management information exchange between arbitrary devices on a network, so that a network administrator can query information and locate faults on any device.
- SNMPv1 applies to small-scale networks where security requirements are not high or the network environment is safe and stable, such as campus networks and small-sized enterprise networks.
- SNMPv2c applies to medium- and large-sized networks where security requirements are not high or the network environment is safe, but a large volume of traffic exists and traffic congestion may occur.
- SNMPv3 is the recommended version and applies to networks of various scales, especially those networks that have high security requirements and allow only authorized administrators to manage network devices.



Basic SNMP Configuration (1)

1. Enable the SNMP agent function.

```
[Huawei] snmp-agent
```

2. Set the SNMP version.

```
[Huawei] snmp-agent sys-info version [v1 | v2c | v3]
```

You can configure the SNMP version as required. However, the protocol version used on the device must be the same as that used on the NMS.

3. Create or update MIB view information.

```
[Huawei] snmp-agent mib-view view-name { exclude | include } subtree-name [mask mask]
```

4. Add a new SNMP group and map users in this group to the SNMP view.

```
[Huawei] snmp-agent group v3 group-name { authentication | noauth | privacy } [ read-view view-name | write-view view-name | notify-view view-name ]
```

This command is used to create an SNMP group of the SNMPv3 version and specify the authentication and encryption mode and one or more of read-only view, read-write view, and notification view. It is a mandatory command on networks that require high security.



Basic SNMP Configuration (2)

5. Add a user to the SNMP group.

```
[Huawei] snmp-agent usm-user v3 user-name group group-name
```

6. Configure an authentication password for an SNMPv3 user.

```
[Huawei] snmp-agent usm-user v3 user-name authentication-mode { md5 | sha | sha2-256 }
```

7. Configure the SNMPv3 user encryption password.

```
[Huawei] snmp-agent usm-user v3 user-name privacy-mode { aes128 | des56 }
```

8. Set parameters for the device to send traps.

```
[Huawei] snmp-agent target-host trap-paramsname paramsname v3 securityname securityname { authentication | noauthnopriv | privacy }
```



Basic SNMP Configuration (3)

9. Configure the target host of traps.

```
[Huawei] snmp-agent target-host trap-hostname hostname address ipv4-address trap-paramsname paramsname
```

10. Enable all trap functions.

```
[Huawei] snmp-agent trap enable
```

Note that this command is used only to enable the device to send traps. This command must be used together with the **snmp-agent target-host** command. The **snmp-agent target-host** command specifies the device to which traps are sent.

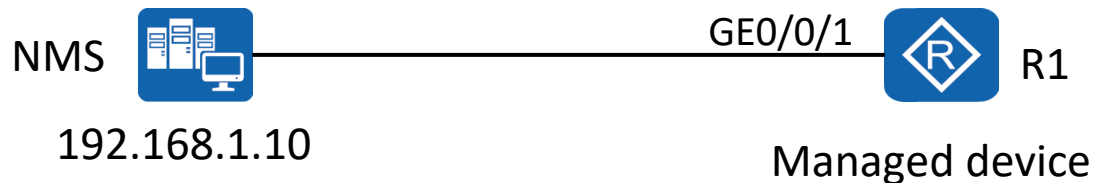
11. Configure the source interface that sends traps.

```
[Huawei] snmp-agent trap source interface-type interface-number
```

Note that a source IP address must have been configured for the interface that sends traps.



SNMP Configuration Example (Network Device Side)



- Enable SNMP on R1 and set the SNMP version to SNMPv3.
- Set the SNMPv3 group name to **test** and encryption authentication mode to **privacy**.
- Create an SNMPv3 user named **R1** and set the authentication and encryption passwords to **HCIA-Datacom123**.
- Create a trap parameter named **param** and set securityname to **sec**.
- Set the IP address of the SNMP target host to 192.168.1.10.
- Enable the trap function and specify GE 0/0/1 as the source interface that sends traps.

R1configuration:

```
[R1]snmp-agent
[R1]snmp-agent sys-info version v3
[R1]snmp-agent group v3 test privacy
[R1]snmp-agent usm-user v3 R1 test authentication-mode md5
HCIA@Datacom123 privacy-mode aes128 HCIA-Datacom123
[R1]snmp-agent target-host trap-paramsname param v3
securityname sec privacy
[R1]snmp-agent target-host trap-hostname nms address
192.168.1.10 trap-paramsname param
[R1]snmp-agent trap source GigabitEthernet 0/0/1
[R1]snmp-agent trap enable
Info: All switches of SNMP trap/notification will be open.
Continue? [Y/N]:y
```



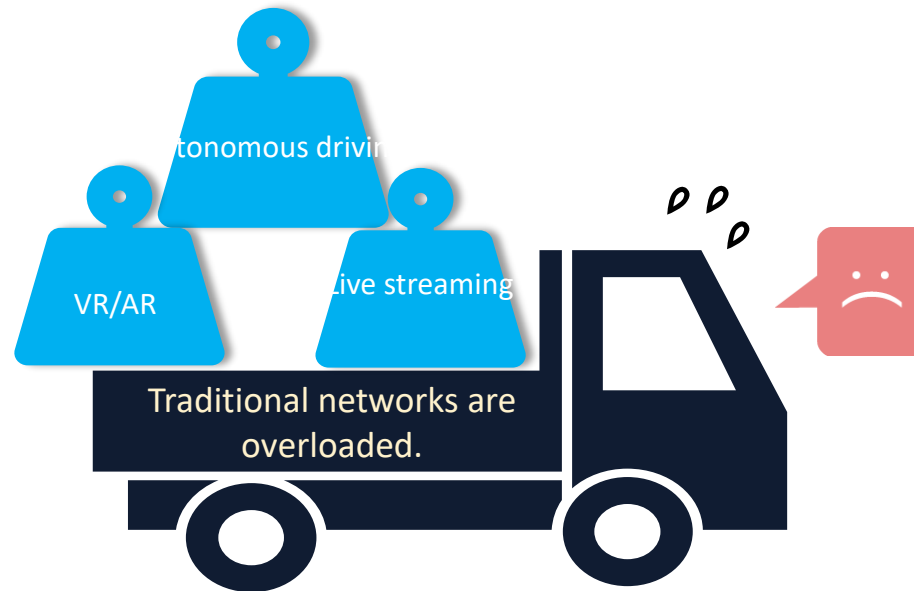
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Transformation and Challenges of the Network Industry

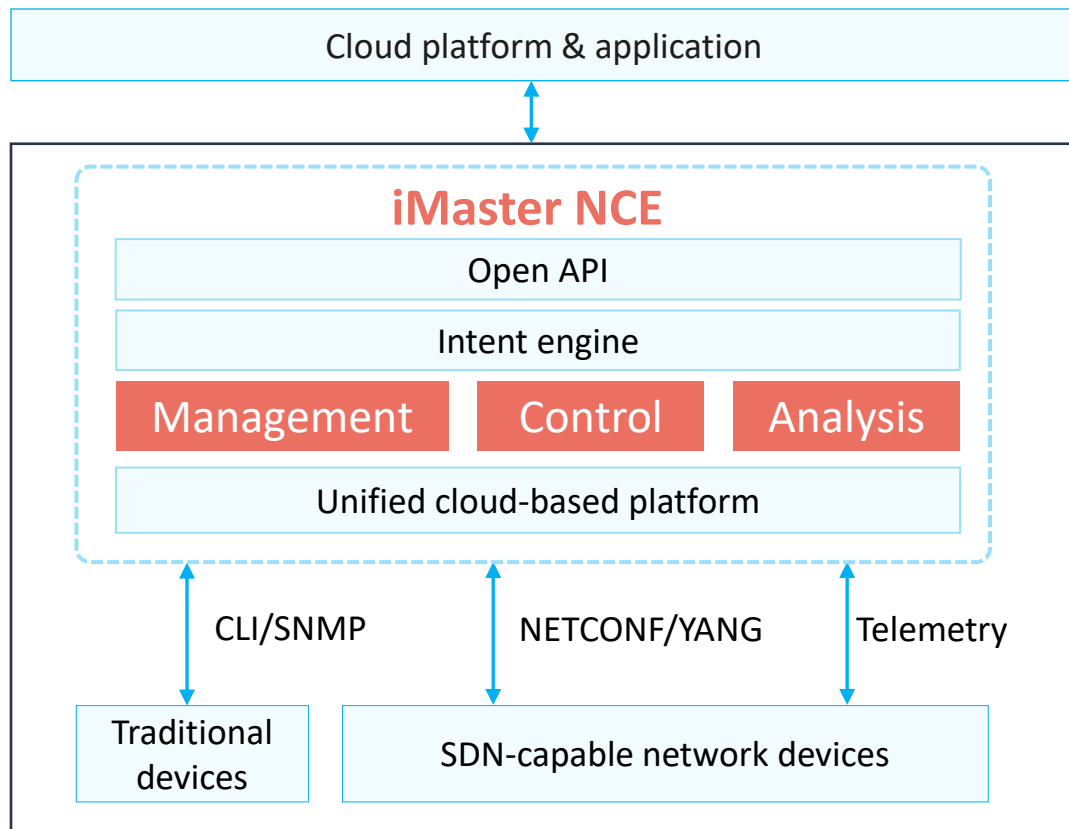
- With the advent of the 5G and cloud era, innovative services such as VR/AR, live streaming, and autonomous driving are emerging, and the entire ICT industry is booming. At the same time, the traffic of the entire network also increases explosively. Huawei Global Industry Vision (GIV) predicts that the amount of new data will reach 180 ZB by 2025. Moreover, the dynamic complexity of services makes the entire network more complex.
- Such challenges can only be overcome by constructing automated and intelligent network systems centered on user experience.





Huawei iMaster NCE

- Huawei iMaster NCE is a network automation and intelligence platform that integrates management, control, analysis, and AI functions.

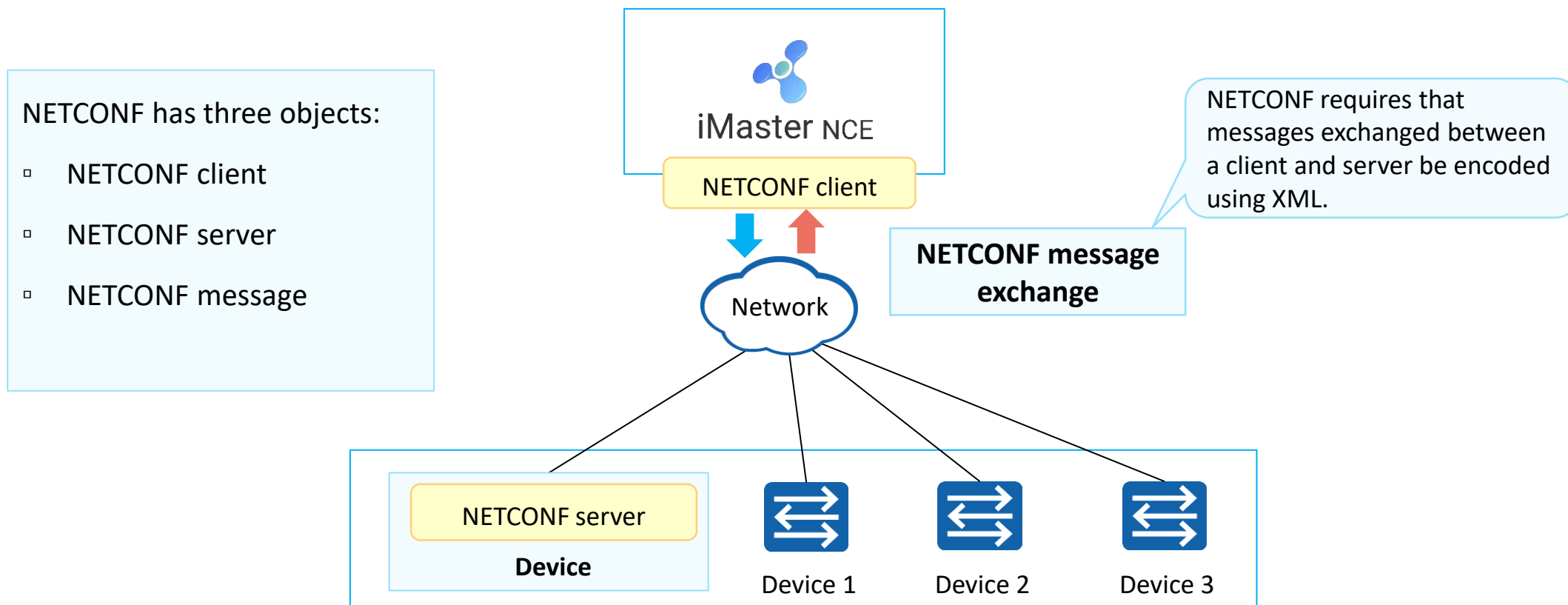


- In terms of management and control, iMaster NCE allows you to:
 - Manage and control traditional devices through traditional technologies such as CLI and SNMP.
 - Manage and control SDN-capable networks through NETCONF (based on the YANG model).
- iMaster NCE collects network data through protocols such as SNMP and telemetry, performs intelligent big data analysis based on AI algorithms, and displays device and network status in multiple dimensions through dashboards and reports, helping O&M personnel quickly detect and handle device and network exceptions and ensuring normal running of devices and networks.



NETCONF Overview

- NETCONF provides a network device management mechanism. You can use NETCONF to add, modify, or delete configurations of network devices, and obtain configurations and status of network devices.



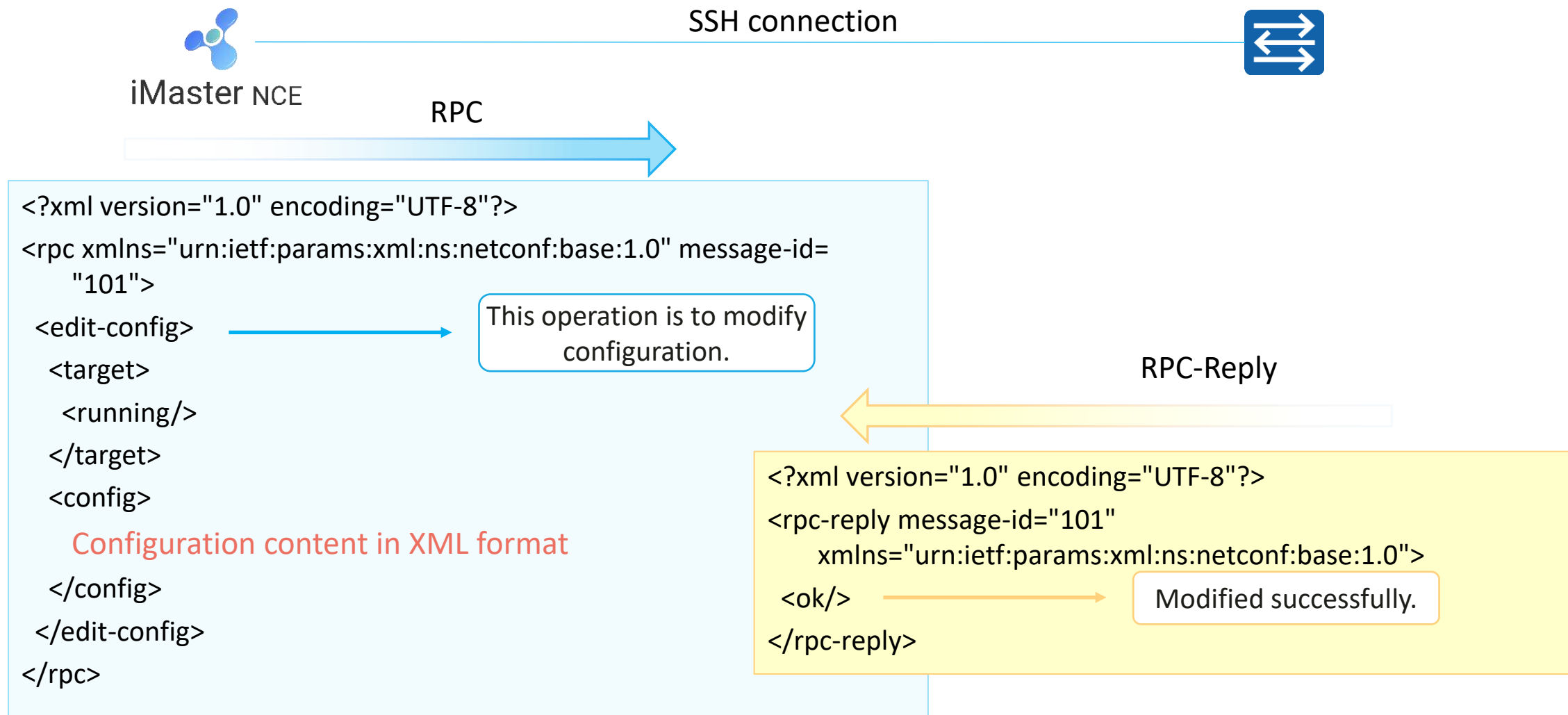


NETCONF Advantages

Function	NETCONF	SNMP	CLI
Interface type	Machine-machine interface: The interface definition is complete and standard, and the interface is easy to control and use.	Machine-to-machine interface	Man-machine interface
Operation efficiency	High: Object-based modeling is supported. Only one interaction is required for object operations. Operations such as filtering and batch processing are supported.	Medium	Low
Scalability	Proprietary protocol capabilities can be extended.	Weak	Moderate
Transaction	Supports transaction processing mechanisms such as trial running, rollback upon errors, and configuration rollback.	Not supported	Partially supported
Secure transmission	Multiple security protocols: SSH, TLS, BEEP/TLS, and SOAP/HTTP/TLS	Only SNMPv3 supports secure transmission.	SSH



Typical NETCONF Interaction





YANG Language Overview

- Yet Another Next Generation (YANG) is a data modeling language that standardizes NETCONF data content.
- The YANG model defines the hierarchical structure of data and can be used for NETCONF-based operations. Modeling objects include configuration, status data, remote procedure calls, and notifications. This allows a complete description of all data exchanged between a NETCONF client and server.

A model is an abstraction and expression of things.

A data model is an abstraction and expression of data features.

Name, gender, height,
weight, age, skin color...



Person



Router

Interface, routing
protocol, IP address,
routing table...



YANG and XML (1)

- A YANG file is loaded on the NETCONF client (such as the NMS or SDN controller).
- The YANG file is used to convert data into XML-format NETCONF messages before they are sent to the device.

```
list server {  
  key "name";  
  unique "ip port";  
  leaf name {  
    type string;  
  }  
  leaf ip {  
    type inet:ip-address;  
  }  
  leaf port {  
    type inet:port-number;  
  }  
}
```

YANG file

+

```
name="smtp"  
ip=192.0.2.1  
port=25
```

```
name="http"  
ip=192.0.2.1  
port=
```

```
name="ftp"  
ip=192.0.2.1  
port=
```

Data

=

```
<server>  
  <name>smtp</name>  
  <ip>192.0.2.1</ip>  
  <port>25</port>  
</server>  
<server>  
  <name>http</name>  
  <ip>192.0.2.1</ip>  
</server>  
<server>  
  <name>ftp</name>  
  <ip>192.0.2.1</ip>  
</server>
```

XML



YANG and XML (2)

- A YANG file is loaded on the NETCONF server (such as a router or switch).
- The YANG file is used to convert received XML-format NETCONF messages into data for subsequent processing.

```
<server>
  <name>smtp</name>
  <ip>192.0.2.1</ip>
  <port>25</port>
</server>
<server>
  <name>http</name>
  <ip>192.0.2.1</ip>
</server>
<server>
  <name>ftp</name>
  <ip>192.0.2.1</ip>
</server>
```

XML

+

```
list server {
  key "name";
  unique "ip port";
  leaf name {
    type string;
  }
  leaf ip {
    type inet:ip-address;
  }
  leaf port {
    type inet:port-number;
  }
}
```

YANG file

=

```
name="smtp"
ip=192.0.2.1
port=25
```

```
name="http"
ip=192.0.2.1
port=
```

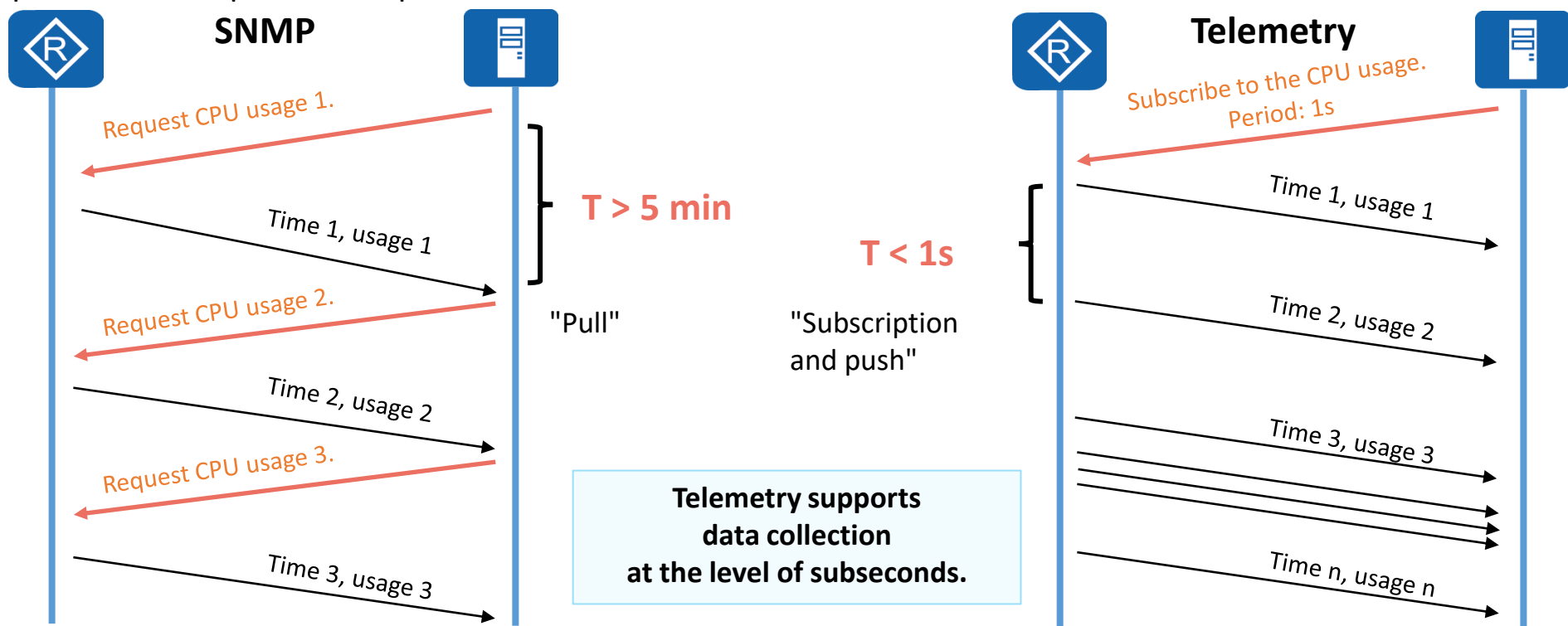
```
name="ftp"
ip=192.0.2.1
port=
```

Data



Telemetry Overview

- Telemetry, also called network telemetry, is a technology that remotely collects data from physical or virtual devices at a high speed.
- Devices periodically send interface traffic statistics, CPU usage, and memory usage to collectors in push mode. Compared with the traditional pull mode, the push mode provides faster and more real-time data collection.





Quiz

1. (Single) On an SNMP-based network, which of the following runs the management process to manage the managed devices? ()
 - A. NMS
 - B. Agent process
 - C. MIB
 - D. SNMP

2. (Single) In SNMPv1, which of the following operations is used by a managed device to report traps? ()
 - A. Get-Request
 - B. Set-Request
 - C. Trap
 - D. Response



Quiz

3. YANG is a data modeling language. ()
 - A. True
 - B. False
4. Telemetry supports data collection at the level of subseconds. ()
 - A. True
 - B. False



Summary

- With the development of network technologies, more and more network management and O&M methods are available. The common methods are as follows:
 - CLI mode or web system
 - SNMP
 - Huawei iMaster NCE's intelligent O&M platform (covering management, control, and analysis)

The image features a blue-tinted background showing silhouettes of several groups of business professionals in a modern office environment. They are standing on a reflective floor, and their reflections are visible below them. The background also shows a blurred view of city buildings through large windows. Overlaid on this scene is the text "Thank You" and the website address "www.huawei.com".

Thank You

www.huawei.com



IPv6 Basics



Foreword

- In the 1980s, the Internet Engineering Task Force (IETF) released RFC 791 – Internet Protocol, which marks the standardization of IPv4. In the following decades, IPv4 has become one of the most popular protocols. Numerous people have developed various applications based on IPv4 and made various supplements and enhancements to IPv4, enabling the Internet to flourish.
- However, with the expansion of the Internet and the development of new technologies such as 5G and Internet of Things (IoT), IPv4 faces more and more challenges. It is imperative to replace IPv4 with IPv6.
- This course describes the reasons for IPv4-to-IPv6 transition and basic IPv6 knowledge.



Objectives

- On completion of this course, you will be able to:
 - Summarize the advantages of IPv6 over IPv4.
 - Describe the basic concepts of IPv6.
 - Describe the formats and functions of IPv6 packet headers.
 - Describe the IPv6 address format and address types.
 - Describe the method and basic procedure for configuring IPv6 addresses.
 - Configure IPv6 addresses and IPv6 static routes.



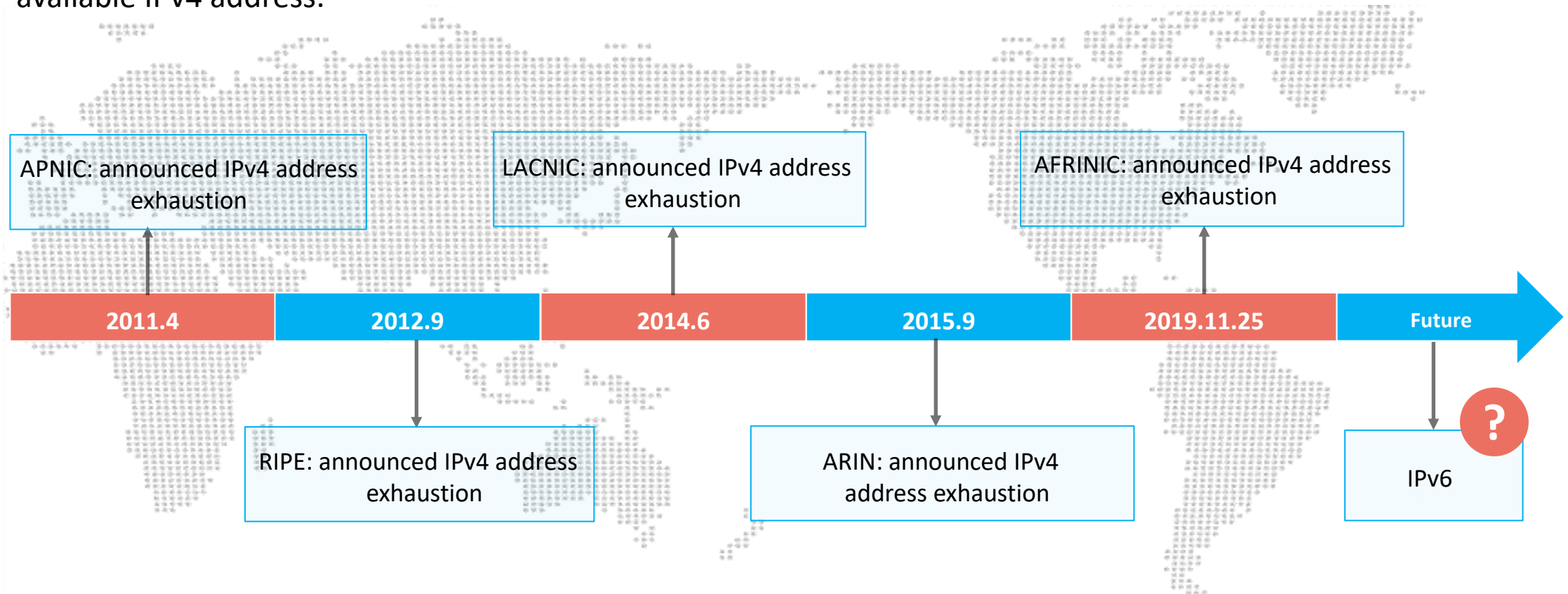
Contents

- 1. IPv6 Overview**
2. IPv6 Address Configuration
3. Typical IPv6 Configuration Examples



IPv4 Status

- On February 3, 2011, the Internet Assigned Numbers Authority (IANA) announced even allocation of its last 4.68 million IPv4 addresses to five Regional Internet Registries (RIRs) around the world. The IANA thereafter had no available IPv4 address.





Why IPv6?

IPv4

Exhausted public IP addresses

Improper packet header design

Large routing table, leading to inefficient
table query

Dependency on ARP causes broadcast
storms

...

vs.

IPv6

Nearly infinite address space

Hierarchical address allocation

Plug-and-play

Simplified packet header

IPv6 security features

Integrity of E2E communication

Support for mobility

Enhanced QoS features

...



IPv6 Advantages

Nearly infinite address space

The 128-bit address length provides numerous addresses, meeting the requirements of emerging services such as the IoT and facilitating service evolution and expansion.

Hierarchical address structure

IPv6 addresses are allocated more properly than IPv4 addresses, facilitating route aggregation (reducing the size of IPv6 routing tables) and fast route query.

Plug-and-play

IPv6 supports stateless address autoconfiguration (SLAAC), simplifying terminal access.

Simplified packet header

The simplified packet header improves forwarding efficiency. New applications can be supported using extension headers, which facilitate the forwarding processing of network devices and reduce investment costs.

Security features

IPsec, source address authentication, and other security features ensure E2E security, preventing NAT from damaging the integrity of E2E communication.

Mobility

Greatly improves real-time communication and performance of mobile networks.

Enhanced QoS features

A Flow Label field is additionally defined and can be used to allocate a specific resource for a special service and data flow.



Basic IPv6 Header

- An IPv6 header consists of a mandatory basic IPv6 header and optional extension headers.
- The basic header provides basic information for packet forwarding and is parsed by all devices on a forwarding path.

IPv4 packet header (20–60 bytes)

Version	HL	ToS	Total Length	
Identification			Flags	Fragment Offset
TTL	Protocol		Head Checksum	
Source Address				
Destination Address				
Options				Padding

Basic IPv6 header (40 bytes)

Version	Traffic Class	Flow Label		
Payload Length		Next Header	Hop Limit	
Source Address				
Destination Address				

~~Deleted~~

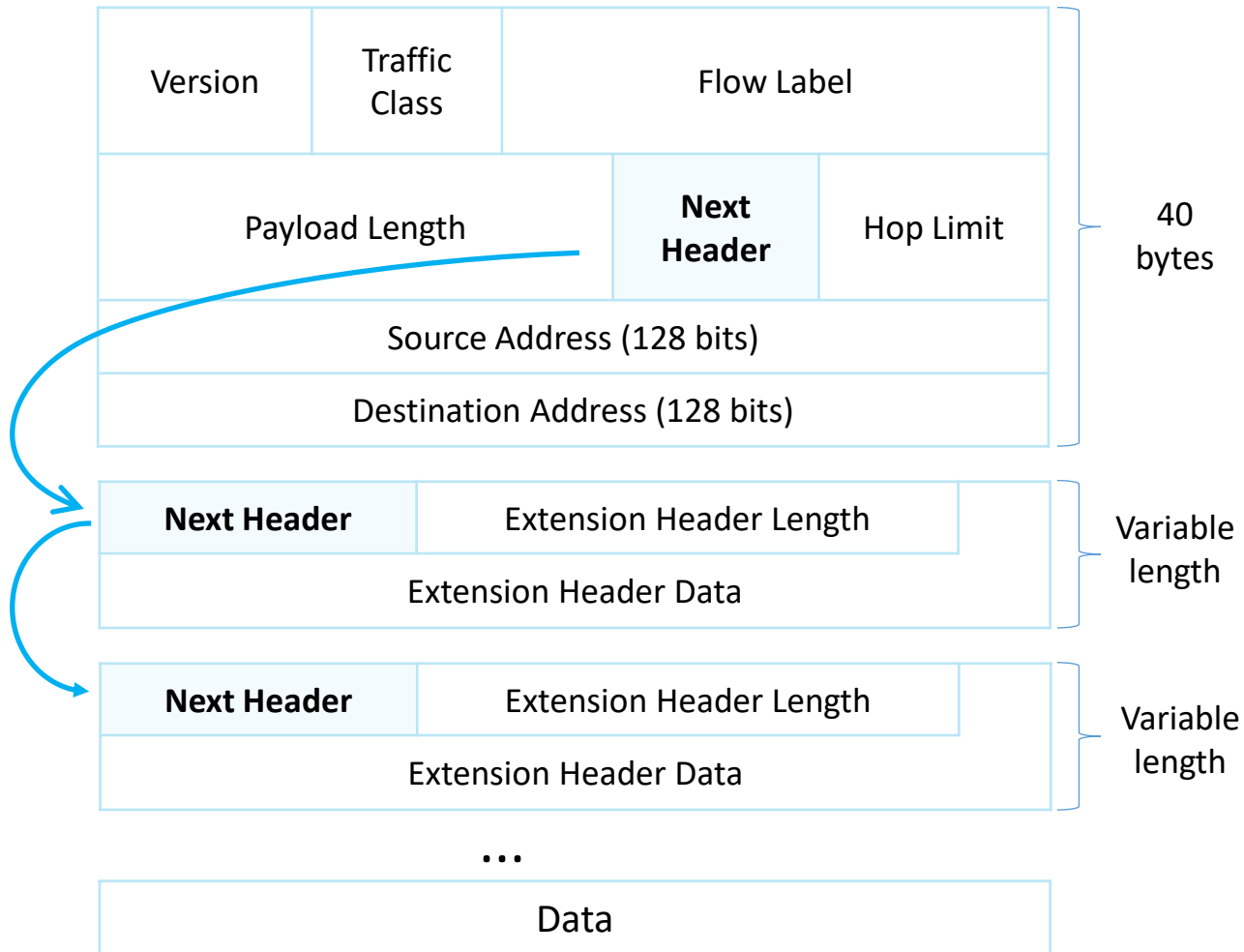
Reserved

Name/Location
changed

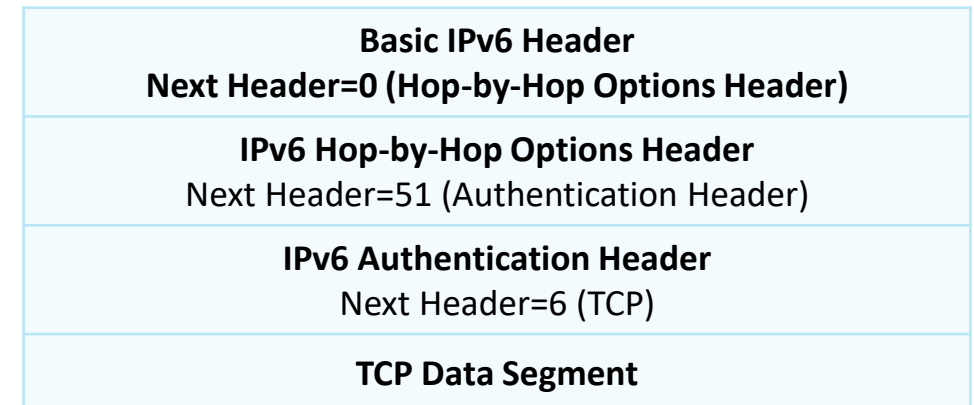
New



IPv6 Extension Header



- **Extension Header Length:** 8 bits long. This field indicates the extension header length excluding the length of the Next Header field.
- **Extension Header Data:** variable length. This field indicates the payload of the extension headers and is a combination of a series of options and padding fields.



IPv6 packet example



IPv6 Packet Processing Mechanism

Basic IPv6 Header Next Header=0 (Hop-by-Hop Options Header)
IPv6 Hop-by-Hop Options Header Next Header=51 (Authentication Header)
IPv6 Authentication Header Next Header=6 (TCP)
TCP Data Segment

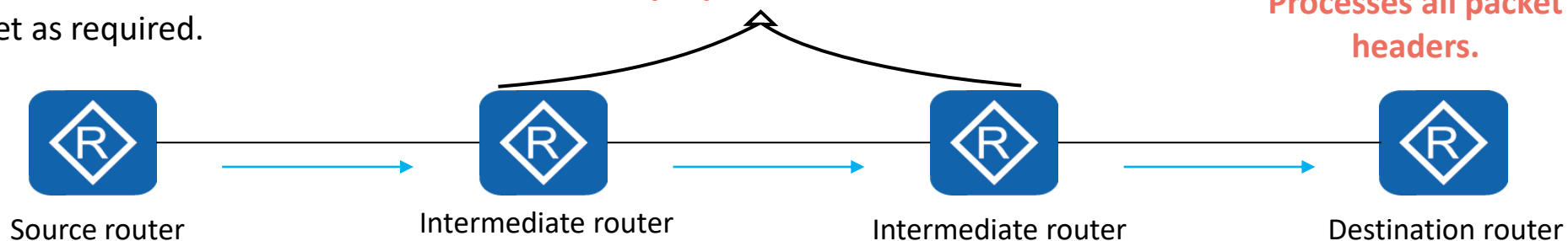
Basic IPv6 Header Next Header=0 (Hop-by-Hop Options Header)
IPv6 Hop-by-Hop Options Header Next Header=51 (Authentication Header)
IPv6 Authentication Header Next Header=6 (TCP)
TCP Data Segment

Basic IPv6 Header Next Header=0 (Hop-by-Hop Options Header)
IPv6 Hop-by-Hop Options Header Next Header=51 (Authentication Header)
IPv6 Authentication Header Next Header=6 (TCP)
TCP Data Segment

Constructs an IPv6 packet as required.

Process the **basic header** and **Hop-by-Hop Options header**.

Processes all packet headers.

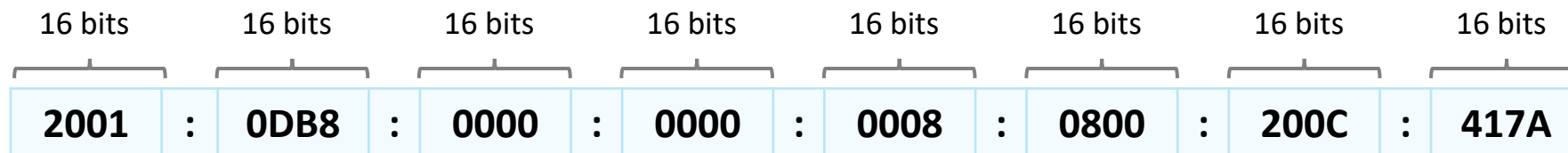


- The length of the basic packet header is fixed, **improving the forwarding efficiency**.
- The extension headers **meet special requirements**.



IPv6 Address

- The length of an IPv6 address is 128 bits. Colons are generally used to divide the IPv6 address into eight segments. Each segment contains 16 bits and is expressed in hexadecimal notation.



The letters in an IPv6 address are case insensitive. For example, **A** is equivalent to **a**.

- Similar to an IPv4 address, an IPv6 address is expressed in the format of IPv6 address/mask length.
 - Example: 2001:0DB8:2345:CD30:1230:4567:89AB:CDEF/64

IPv6 address: 2001:0DB8:2345:CD30:1230:4567:89AB:CDEF

Subnet number: 2001:0DB8:2345:CD30::/64



IPv6 Address Abbreviation Specifications

- For convenience, IPv6 can be abbreviated according to the following rules.

Abbreviation Specifications

2001	:	0DB8	:	0000	:	0000	:	0008	:	0800	:	200C	:	417A
------	---	------	---	------	---	------	---	------	---	------	---	------	---	------

The leading 0s in each 16-bit segment can be omitted. However, if all bits in a 16-bit segment are 0s, at least one 0 must be reserved. The trailing 0s cannot be omitted.



2001	:	DB8	:	0	:	0	:	8	:	800	:	200C	:	417A
------	---	-----	---	---	---	---	---	---	---	-----	---	------	---	------

If one or more consecutive 16-bit segments contain only 0s, a double colon (::) can be used to represent them, but only one :: is allowed in an entire IPv6 address.



2001	:	DB8	:	::	:	8	:	800	:	200C	:	417A
------	---	-----	---	----	---	---	---	-----	---	------	---	------

If an abbreviated IPv6 address contains two double colons (::), the IPv6 address cannot be restored to the original one.

Abbreviation Examples

Before 0000:0000:0000:0000:0000:0000:0000:0001 After
::1

Before 2001:0DB8:0000:0000:FB00:1400:5000:45FF After
2001:DB8::FB00:1400:5000:45FF

Before 2001:0DB8:0000:0000:0000:2A2A:0000:0001 After
2001:DB8::2A2A:0:1

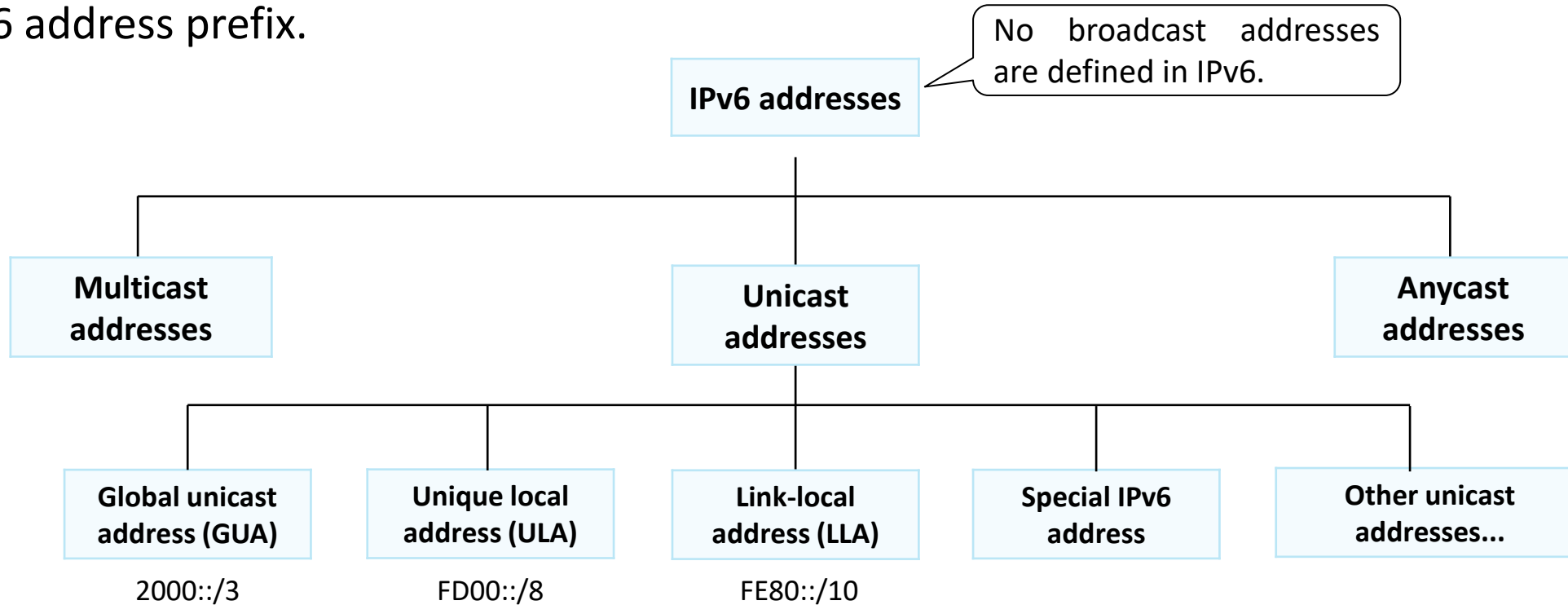
Before 2001:0DB8:0000:1234:FB00:0000:5000:45FF After
2001:DB8::1234:FB00:0:5000:45FF

or 2001:DB8:0:1234:FB00::5000:45FF



IPv6 Address Classification

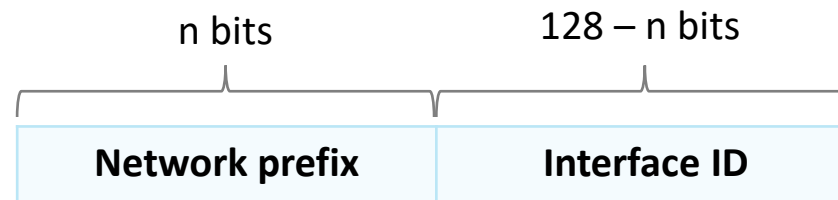
- IPv6 addresses are classified into unicast, multicast, and anycast addresses according to the IPv6 address prefix.





IPv6 Unicast Address Format

- An IPv6 unicast address is composed of two parts:
 - Network prefix: consists of n bits and is parallel to the network ID of an IPv4 address.
 - Interface ID: consists of $(128 - n)$ bits and is parallel to the host ID of an IPv4 address.
- Common IPv6 unicast addresses, such as GUAs and LLAs, require that the network prefix and interface ID be 64 bits.





Interface ID of an IPv6 Unicast Address

- 3 methods to generate an interface ID:
 - Manual configuration
 - Automatic generation by the system
 - Using the IEEE 64-bit extended unique identifier (EUI-64) standard
- EUI-64 is most commonly used. It converts the MAC address of an interface into an IPv6 interface ID.

MAC address (hexadecimal) 3C-52-82-49-7E-9D

MAC address (binary) 00111100-10010010-10000010 - 01001001-01111110-10011101

1 Bit 7 inversion

2 Insert FFFE

EUI-64 ID
(binary)

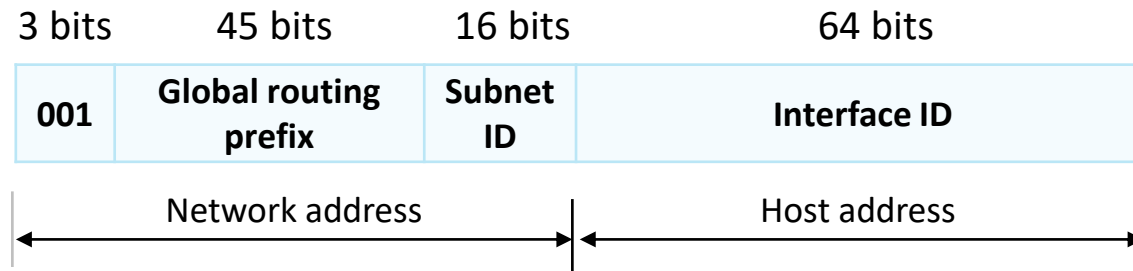
00111110-10010010-10000010-11111111-11111110-01001001-01111110-10011101

EUI-64 ID (hexadecimal) 3E-52-82-FF-FE-49-7E-9D

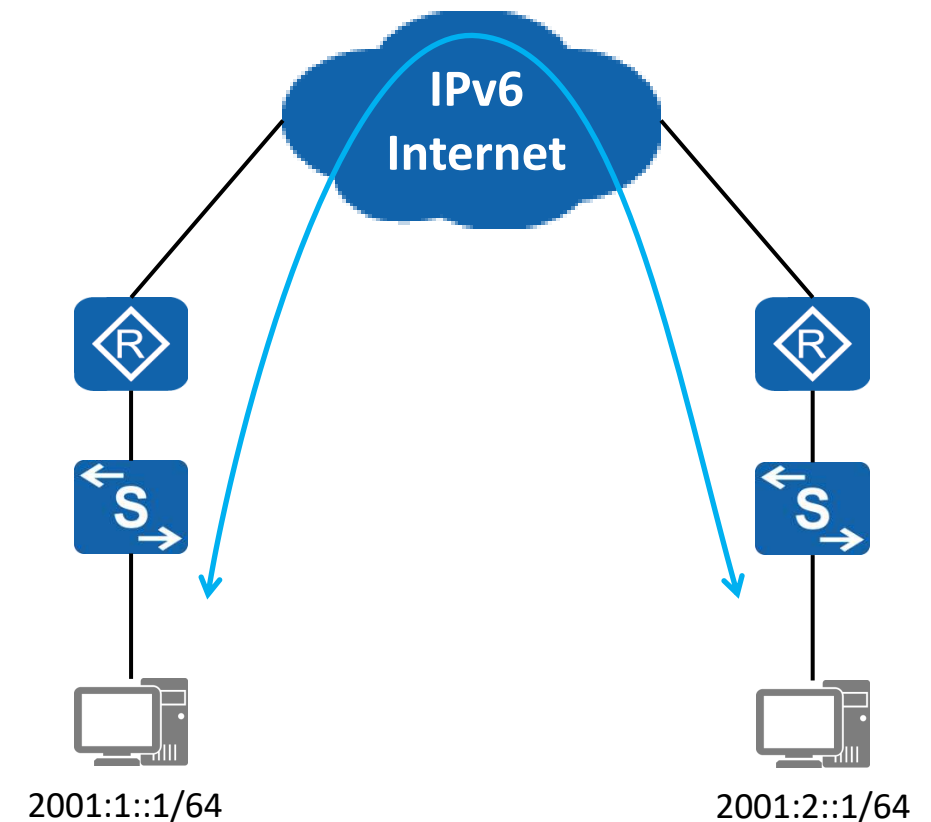


Common IPv6 Unicast Address - GUA

- A GUA is also called an aggregatable GUA. This type of address is globally unique and is used by hosts that need to access the Internet. It is equivalent to a public IPv4 address.



- The network address and interface ID of a GUA are each generally 64 bits long.
- Global routing prefix: is assigned by a provider to an organization and is generally at least 45 bits.
- Subnet ID: An organization can divide subnets based on network requirements.
- Interface ID: identifies a device's interface.





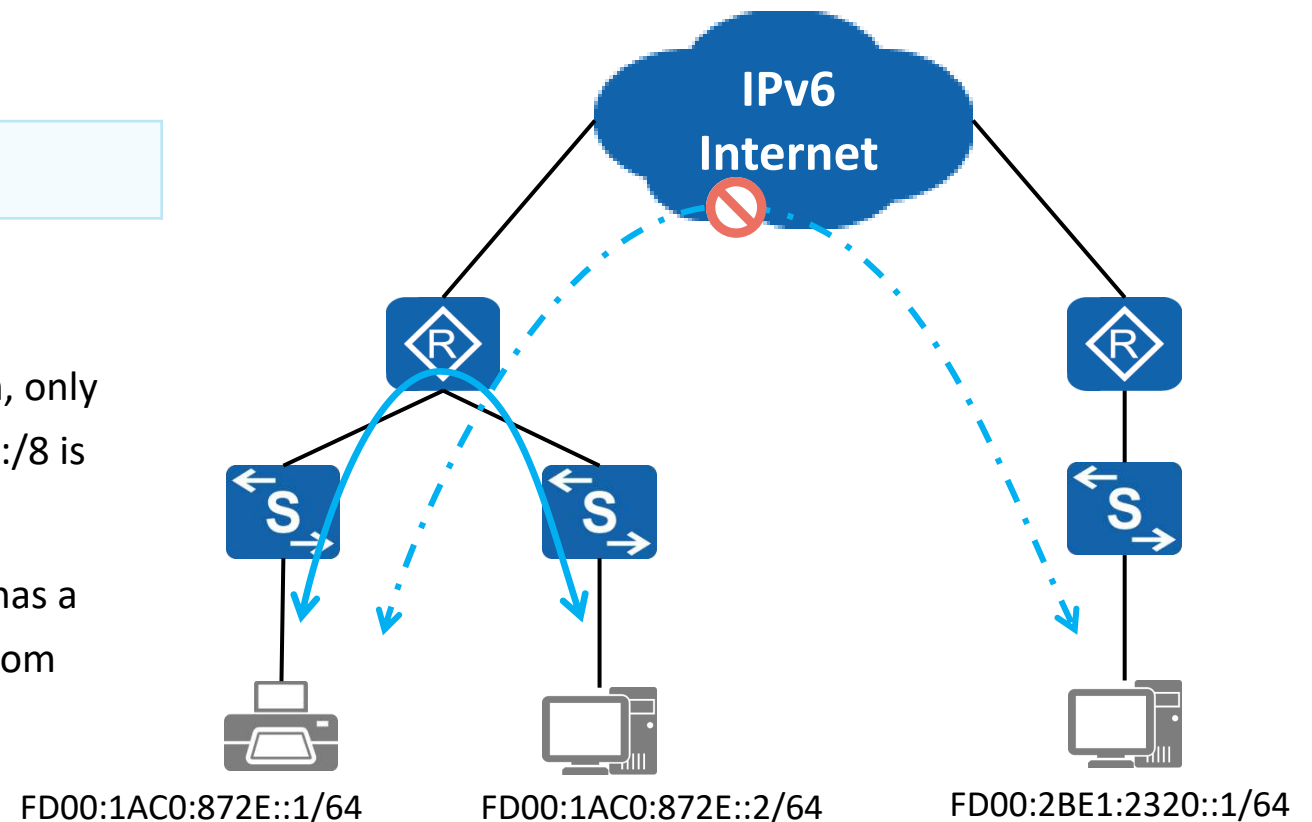
Common IPv6 Unicast Address - ULA

- A ULA is a private IPv6 address that can be used only on an intranet. This type of address cannot be routed on an IPv6 public network and therefore cannot be used to directly access a public network.

8 bits	40 bits	16 bits	64 bits
1111 1101	Global ID	Subnet ID	Interface ID

Generated using a
pseudo-random algorithm

- ULAs use the FC00::/7 address segment, among which, only the FD00::/8 address segment is currently used. FC00::/8 is reserved for future expansion.
- Although a ULA is valid only in a limited range, it also has a globally unique prefix (generated using a pseudo-random algorithm, low conflict probability).



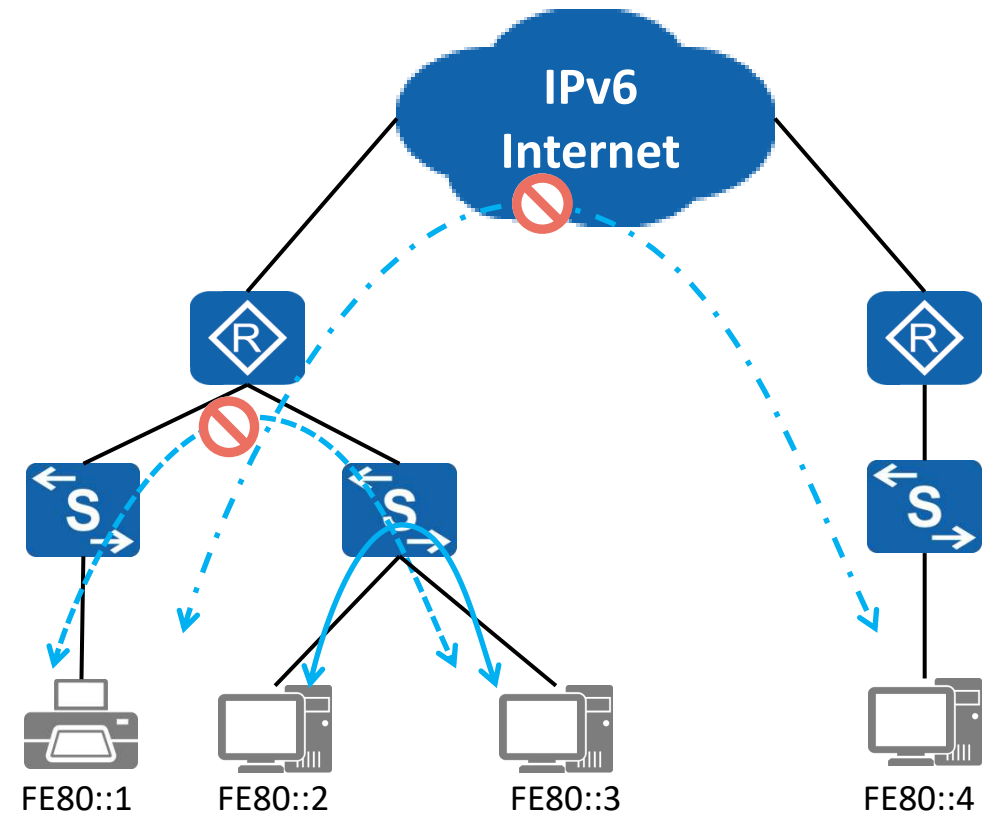


Common IPv6 Unicast Address - LLA

- An LLA is another type of IPv6 address with limited application scope. The valid range of the LLA is the local link, with the prefix of FE80::/10.

10 bit	54 bit	64 bit
1111 1110 10	0	Interface ID
Fixed at 0		

- An LLA is used for communication on a single link, such as during IPv6 SLAAC and IPv6 neighbor discovery.
- Data packets with the source or destination IPv6 address being an LLA are not forwarded out of the originating link. In other words, the valid scope of an LLA is the local link.
- Each IPv6 interface must have an LLA. Huawei devices support automatic generation and manual configuration of LLAs.



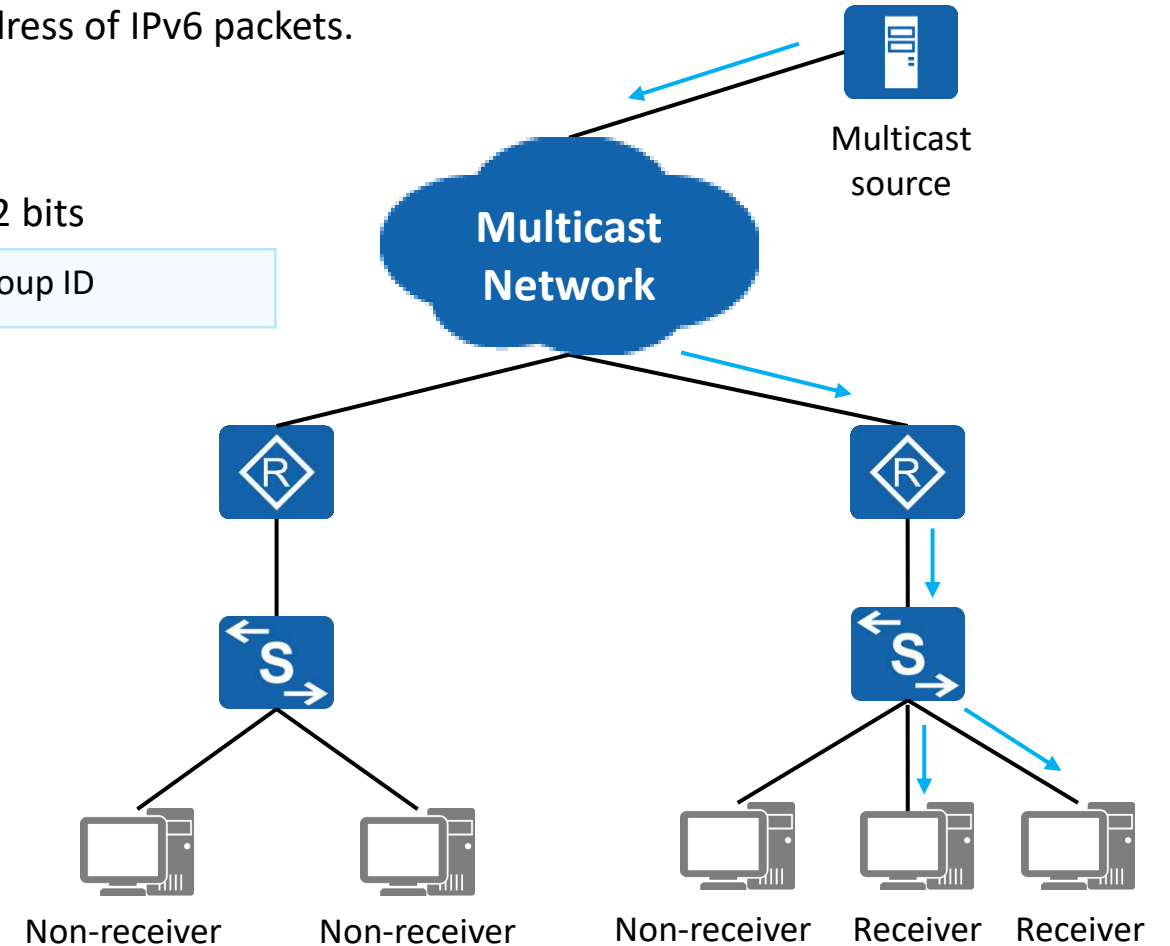


IPv6 Multicast Address

- An IPv6 multicast address identifies multiple interfaces and is generally used in one-to-many communication scenarios.
- An IPv6 multicast address can be used only as the destination address of IPv6 packets.

8 bits	4 bits	4 bits	80 bits	32 bits
11111111	Flags	Scope	Reserved (must be 0)	Group ID

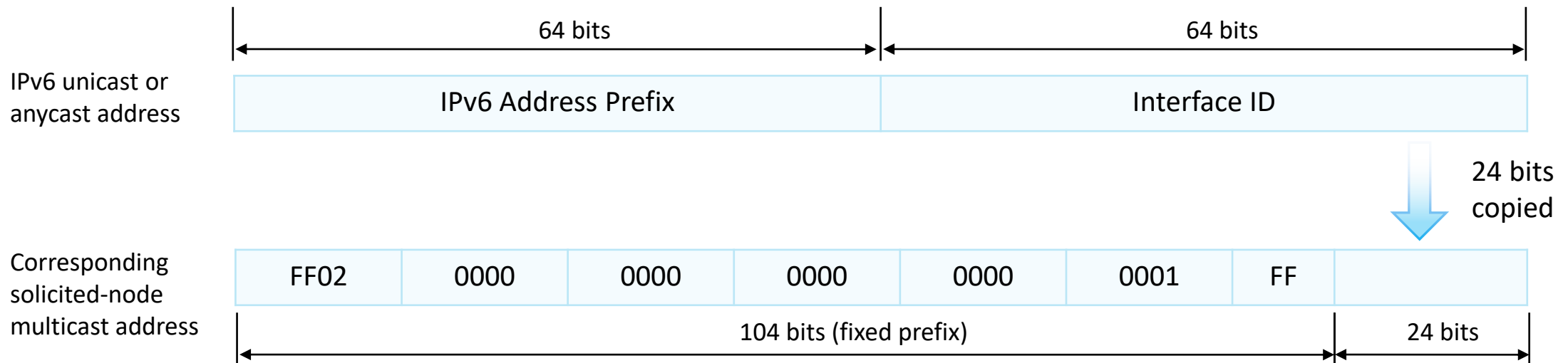
- **Flags:** indicates a permanent or transient multicast group.
- **Scope:** indicates the multicast group scope.
- **Group ID:** indicates a multicast group ID.





Solicited-Node Multicast Address

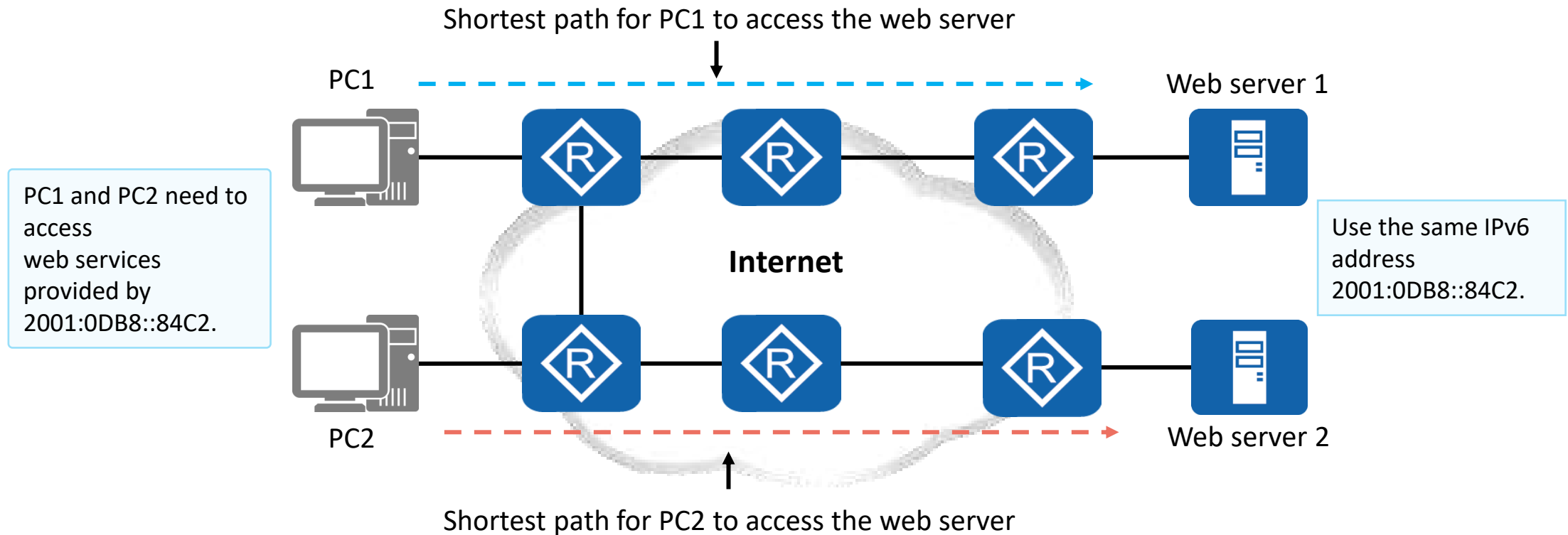
- If a node has an IPv6 unicast or anycast address, a solicited-node multicast address is generated for the address, and the node joins the corresponding multicast group. This address is used for **neighbor discovery** and **duplicate address detection (DAD)**. A solicited-node multicast address is **valid only on the local link**.





IPv6 Anycast Address

- An anycast address identifies a group of network interfaces, which usually belong to different nodes. An anycast address can be used as the source or destination address of IPv6 packets.





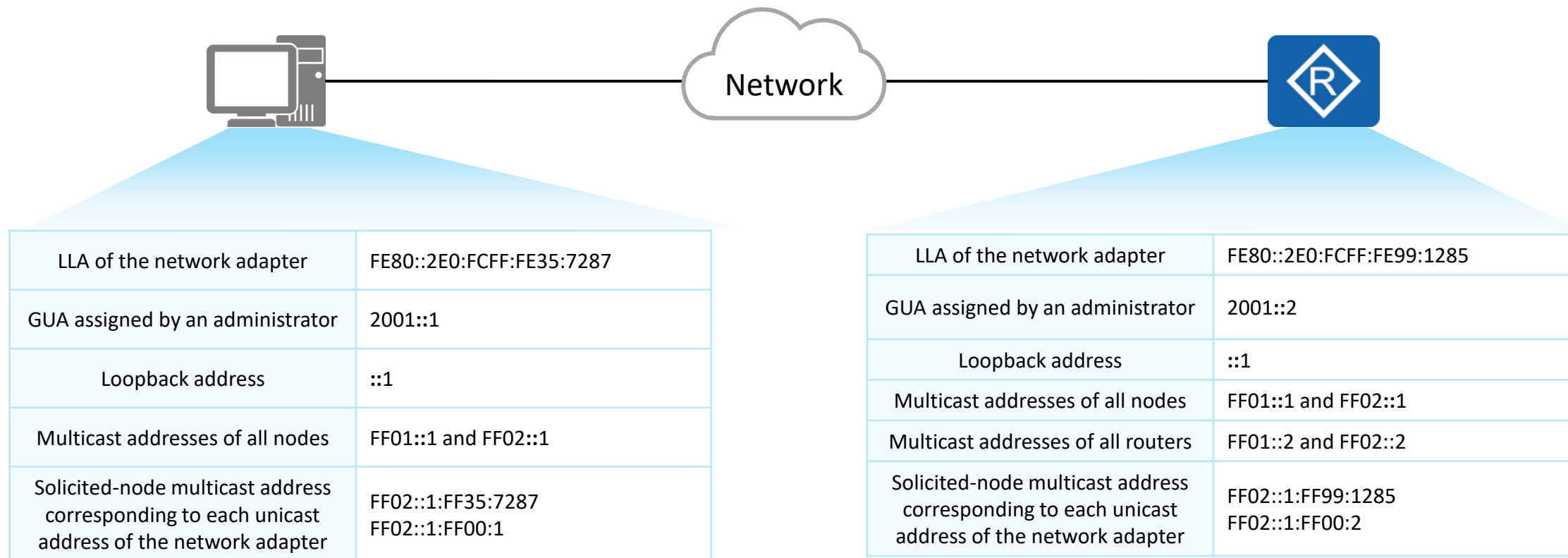
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IPv6 Addresses of Hosts and Routers

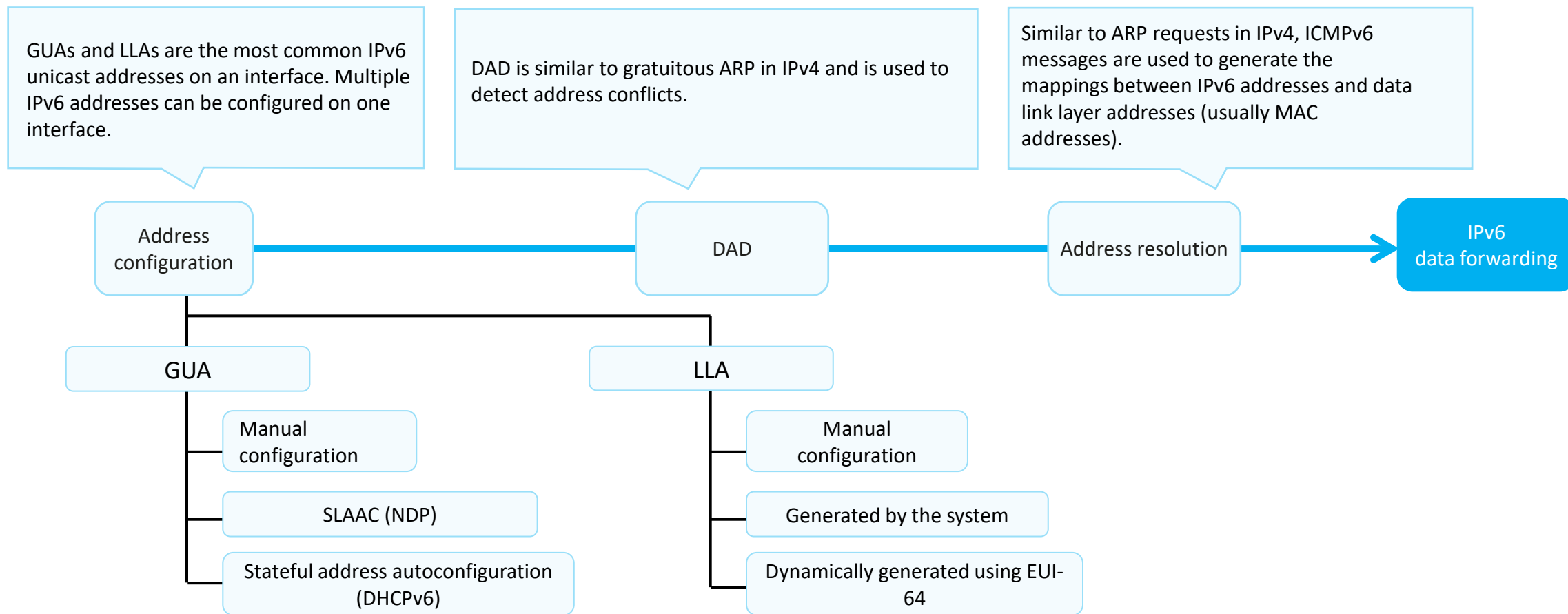
- The unicast IPv6 addresses and multicast addresses of hosts and routers are typically as follows:





Service Process of IPv6 Unicast Addresses

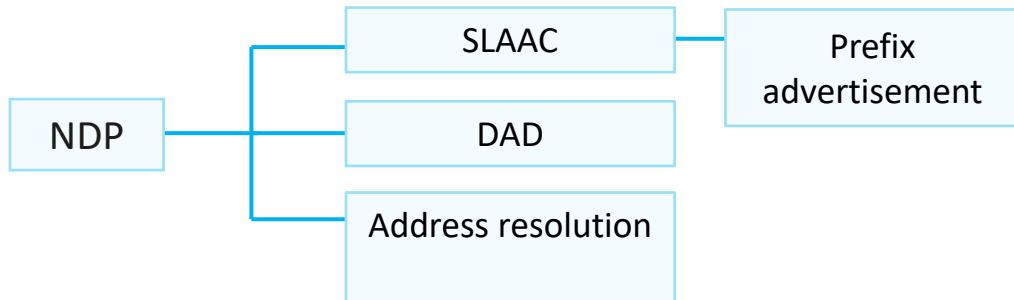
- Before sending IPv6 packets, an interface undergoes address configuration, DAD, and address resolution. During this process, the Neighbor Discovery Protocol (NDP) plays an important role.





NDP

- NDP is defined in RFC 2461, which was replaced by RFC 4861.
- NDP uses ICMPv6 messages to implement its functions.



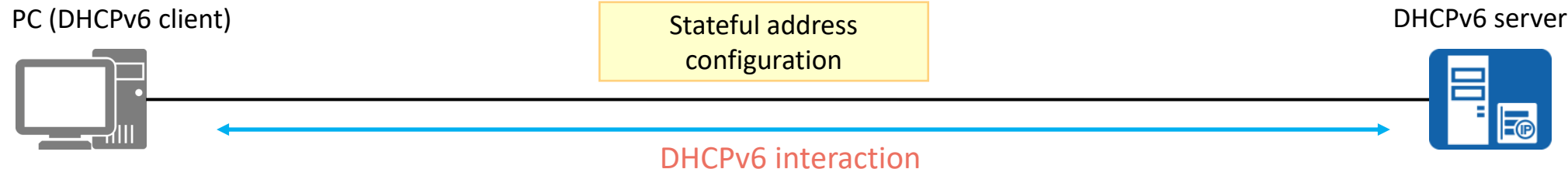
ICMPv6 messages used by NDP

ICMPv6 Type	Message Name
133	Router Solicitation (RS)
134	Router Advertisement (RA)
135	Neighbor Solicitation (NS)
136	Neighbor Advertisement (NA)

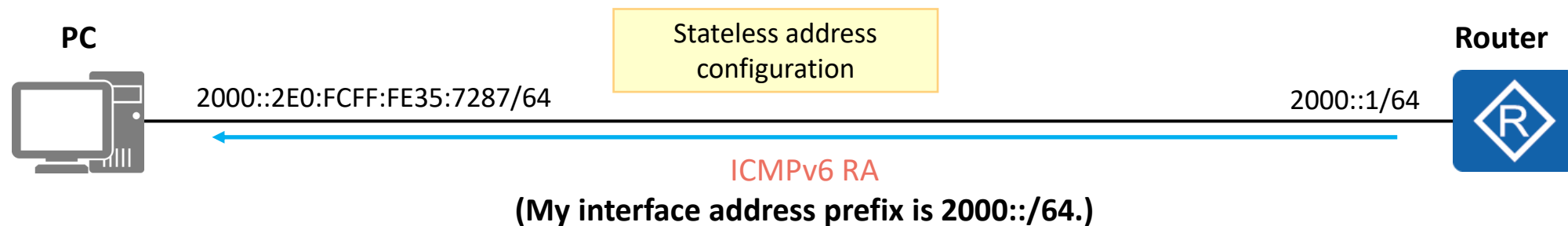
Mechanism	RS 133	RA 134	NS 135	NA 136
Address resolution			✓	✓
Prefix advertisement	✓	✓		
DAD			✓	✓



Dynamic IPv6 Address Configuration



- Through DHCPv6 message exchange, the DHCPv6 server automatically configures IPv6 addresses/prefixes **and other** network configuration parameters (such as DNS, NIS, and SNTP server addresses).

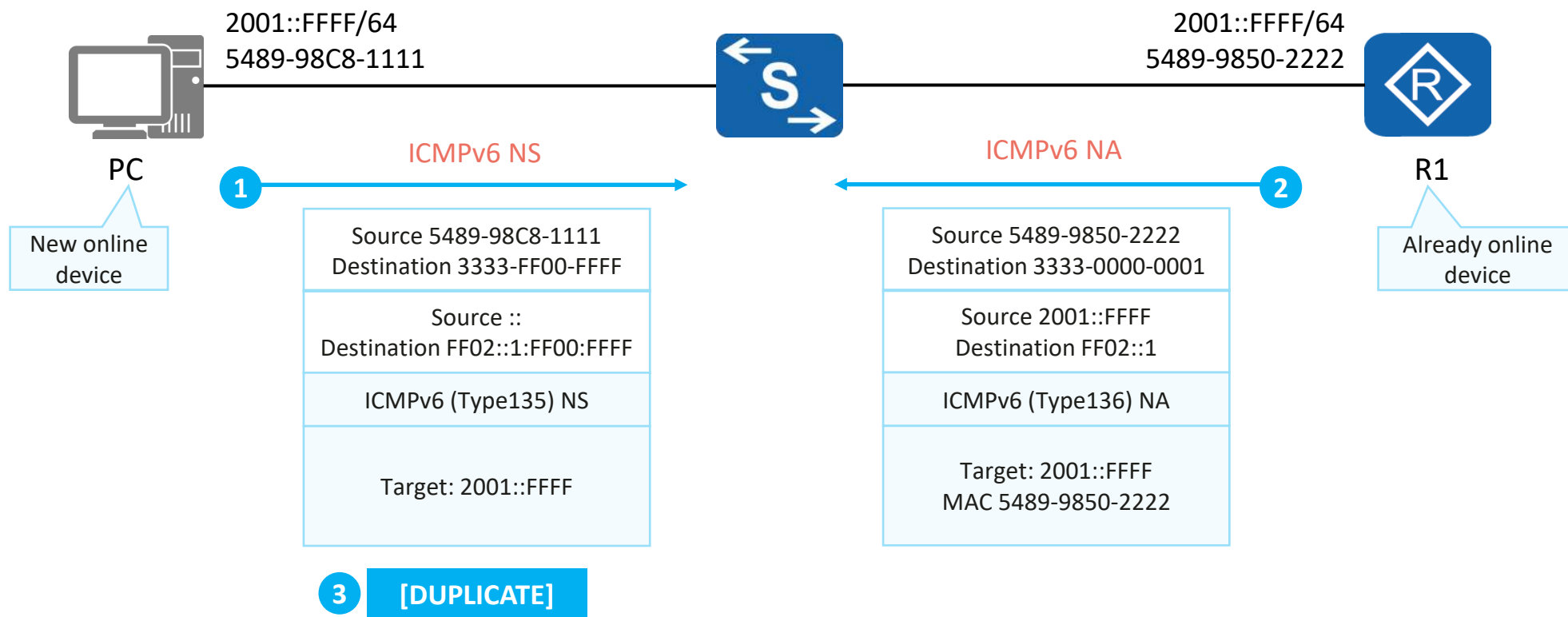


- The PC generates a unicast address based on the address prefix in the RA and the locally generated 64-bit interface ID (for example, using EUI-64).
- Only** IPv6 addresses can be obtained. Parameters such as NIS and SNTP server parameters cannot be obtained. DHCPv6 or manual configuration is required to obtain other configuration information.



DAD

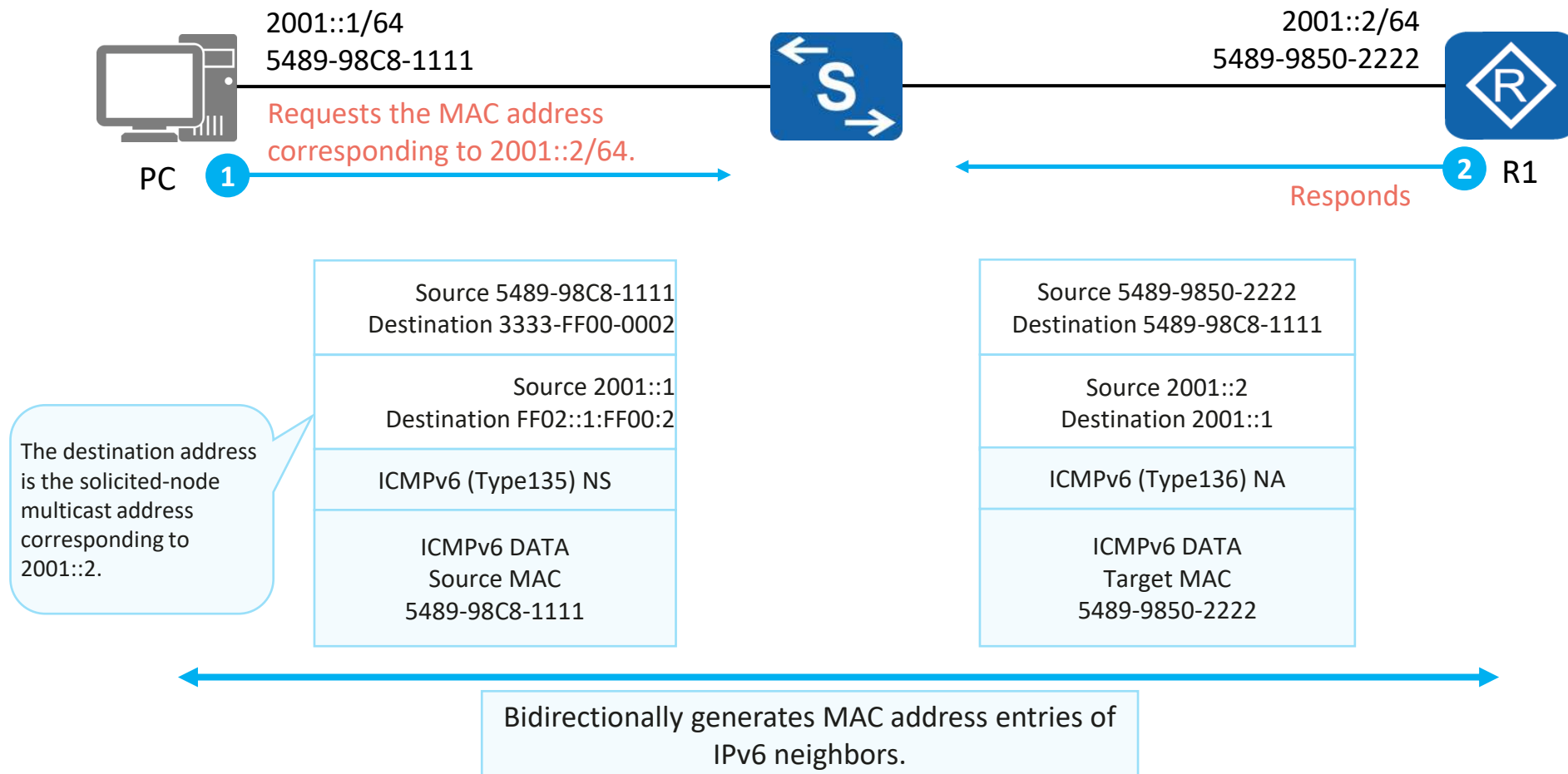
- Regardless of how an IPv6 unicast address is configured, a host or router:
 - Performs DAD through ICMPv6 messages.
 - Uses a unicast address only after passing the DAD procedure.





Address Resolution

- IPv6 uses ICMPv6 NS and NA messages to replace the address resolution function of ARP in IPv4.





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1. IPv6 Overview
2. IPv6 Address Configuration
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Basic IPv6 Configurations (1)

1. Enable IPv6.

```
[Huawei] ipv6
```

Enable the device to send and receive IPv6 unicast packets, including local IPv6 packets.

```
[Huawei-GigabitEthernet0/0/0] ipv6 enable
```

Enable IPv6 on the interface in the interface view.

2. Configure an LLA for the interface.

```
[Huawei-GigabitEthernet0/0/0] ipv6 address ipv6-address link-local
```

```
[Huawei-GigabitEthernet0/0/0] ipv6 address auto link-local
```

Configure an LLA for the interface manually or automatically in the interface view.

3. Configure a GUA for the interface.

```
[Huawei-GigabitEthernet0/0/0] ipv6 address { ipv6-address prefix-length | ipv6-address/prefix-length }
```

```
[Huawei-GigabitEthernet0/0/0] ipv6 address auto { global | dhcp }
```

Configure a GUA for the interface manually or automatically (stateful or stateless) in the interface view.



Basic IPv6 Configurations (2)

4. Configure an IPv6 static route.

```
[Huawei] ipv6 route-static dest-ipv6-address prefix-length { interface-type interface-number [ nexthop-ipv6-address ] | nexthop-ipv6-address } [ preference preference ]
```

5. Display IPv6 information on an interface.

```
[Huawei] display ipv6 interface [ interface-type interface-number | brief ]
```

6. Display neighbor entry information.

```
[Huawei] display ipv6 neighbors
```

7. Enable an interface to send RA messages.

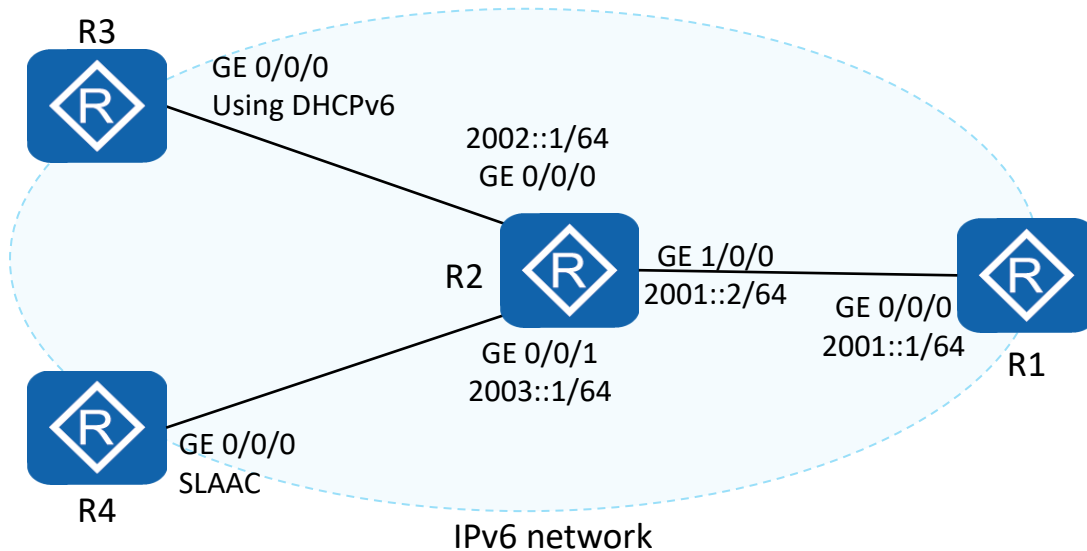
```
[Huawei-GigabitEthernet0/0/0] undo ipv6 nd ra halt
```

By default, a Huawei router's interfaces do not send ICMPv6 RA messages. In this situation, other devices on the links connected to the interfaces cannot perform SLAAC.

To perform SLAAC, you need to manually enable the function of sending RA messages.



Example: Configuring a Small IPv6 Network (1)



• Configuration Requirements

- Connect R1 and R2 through interfaces with static IPv6 addresses.
- Configure R2 as a DHCPv6 server to assign a GUA to GE 0/0/0 of R3.
- Enable R2 to send RA messages, and configure GE 0/0/0 of R4 to automatically perform SLAAC based on the RA messages sent by R2.
- Configure static routes to implement mutual access between the devices.

1. Enable IPv6 globally and on related interfaces of R1, R2, R3, and R4, and enable the interfaces to automatically generate LLAs. The following uses R1 configurations as an example.

```
[R1]ipv6
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ipv6 enable
[R1-GigabitEthernet0/0/0]ipv6 address auto link-local
```

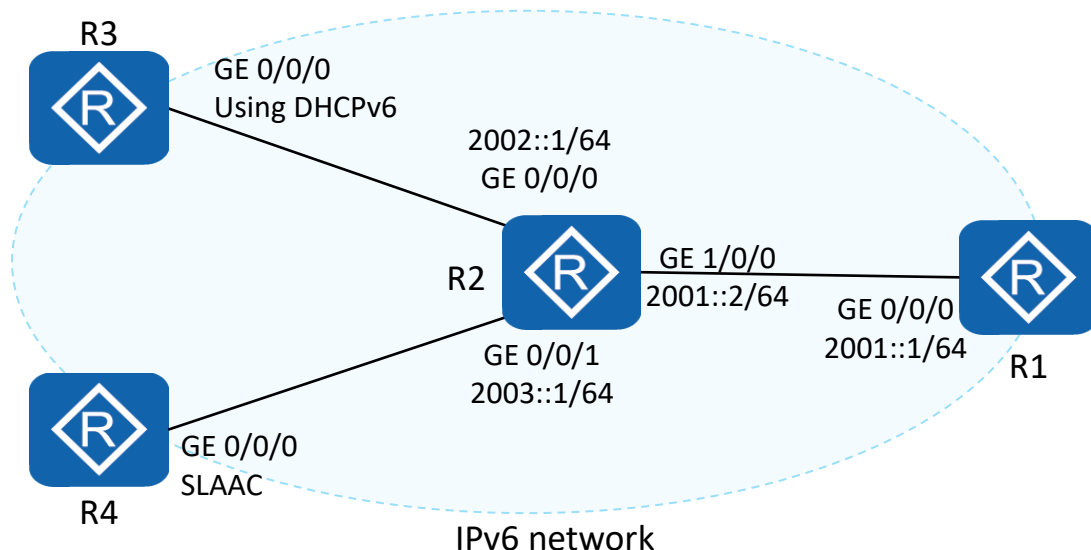
2. Configure static IPv6 GUAs on the related interfaces of R1 and R2.

```
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ipv6 address 2001::1 64
```

```
[R2]interface GigabitEthernet 1/0/0
[R2-GigabitEthernet1/0/0]ipv6 address 2001::2 64
[R2-GigabitEthernet1/0/0]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]ipv6 address 2002::1 64
[R2-GigabitEthernet0/0/0]interface GigabitEthernet 0/0/1
[R2-GigabitEthernet0/0/1]ipv6 address 2003::1 64
```



Example: Configuring a Small IPv6 Network (2)



- Configuration Requirements

- Connect R1 and R2 through interfaces with static IPv6 addresses.
- Configure R2 as a DHCPv6 server to assign a GUA to GE 0/0/0 of R3.
- Enable R2 to send RA messages, and configure GE 0/0/0 of R4 to automatically perform SLAAC based on the RA messages sent by R2.
- Configure static routes to implement mutual access between the devices.

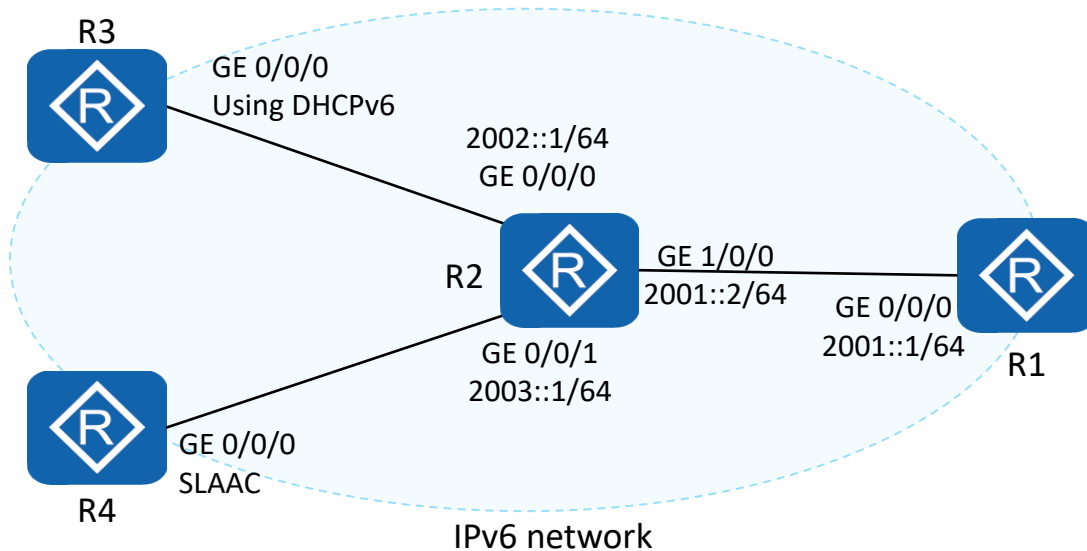
3. Configure R2 as a DHCPv6 server. Configure the related interface of R3 to obtain a GUA using DHCPv6.

```
[R2]dhcp enable
[R2]dhcpv6 pool pool1
[R2-dhcpv6-pool-pool1]address prefix 2002::/64
[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]dhcpv6 server pool1
```

```
[R3]dhcp enable
[R3]interface GigabitEthernet 0/0/0
[R3-GigabitEthernet0/0/0]ipv6 address auto dhcp
```



Example: Configuring a Small IPv6 Network (3)



- Configuration Requirements

- Connect R1 and R2 through interfaces with static IPv6 addresses.
- Configure R2 as a DHCPv6 server to assign a GUA to GE 0/0/0 of R3.
- Enable R2 to send RA messages, and configure GE 0/0/0 of R4 to automatically perform SLAAC based on the RA messages sent by R2.
- Configure static routes to implement mutual access between the devices.

4. Enable R2 to advertise RA messages. Enable R4 to obtain an address through SLAAC based on the RA messages sent by R2.

```
[R2]interface GigabitEthernet 0/0/1
```

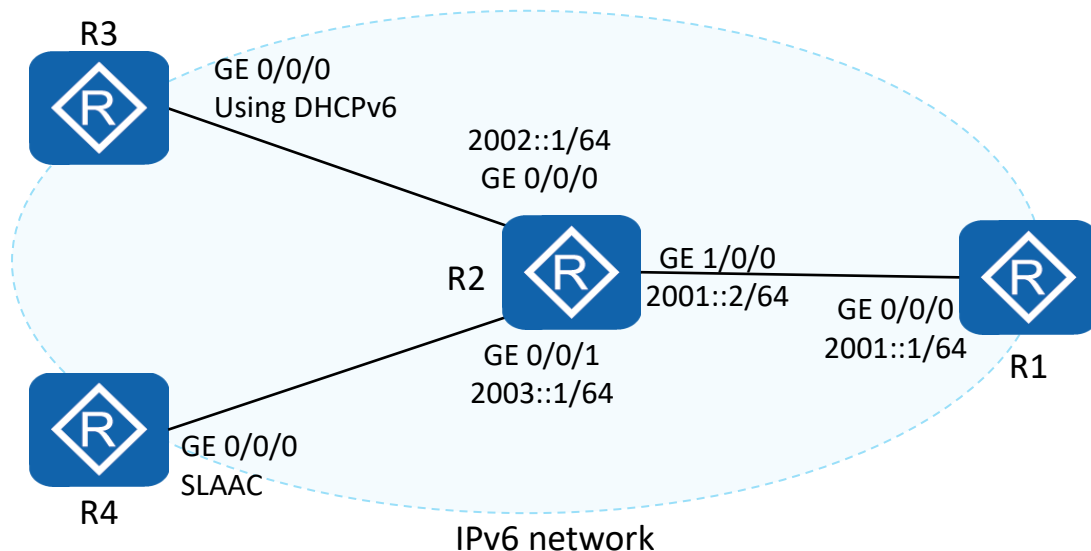
```
[R2-GigabitEthernet0/0/1]undo ipv6 nd ra halt
```

```
[R4]interface GigabitEthernet 0/0/0
```

```
[R4-GigabitEthernet0/0/0]ipv6 address auto global
```



Example: Configuring a Small IPv6 Network (4)



- Configuration Requirements

- Connect R1 and R2 through interfaces with static IPv6 addresses.
- Configure R2 as a DHCPv6 server to assign a GUA to GE 0/0/0 of R3.
- Enable R2 to send RA messages, and configure GE 0/0/0 of R4 to automatically perform SLAAC based on the RA messages sent by R2.
- Configure static routes to implement mutual access between the devices.

5. Configure static routes on R4.

```
[R4]ipv6 route-static 2001:: 64 2003::1  
[R4]ipv6 route-static 2002:: 64 2003::1
```

6. Configure an aggregated static route on R1.

```
[R1]ipv6 route-static 2002:: 15 2001::2
```

7. Configure a default route on R3.

```
[R3]ipv6 route-static :: 0 2002::1
```



Quiz

1. What is the most abbreviated form of the IPv6 address 2001:0DB8:0000:0000:032A:0000:0000:2D70?
2. What is the process of SLAAC for IPv6 hosts?

1. 2001:DB8::32A:0:0:2D70 or 2001:DBB:0:0:32A::2D70

2. SLAAC (Stateless Address Autoconfiguration) is a method that allows IPv6 hosts to automatically configure their own IP address without a DHCP server.

An IPv6 host obtains an address prefix from the Router Advertisement (RA) message sent by the related router interface and then generates an interface ID by inserting a 16-bit FFFE into the existing 48-bit MAC address of the host's interface. After generating an IPv6 address, the IPv6 host checks whether the address is unique through Duplicate Address Detection (DAD).



Summary

Comparison	IPv6	IPv4
Address length	128 bits	32 bits
Packet format	A fixed 40-byte basic packet header+variable-length extension headers	A basic header containing the Options field to support extended features
Address type	Unicast, multicast, and anycast	Unicast, multicast, and broadcast
Address configuration	Static, DHCP, and SLAAC	Static and DHCP
DAD	ICMPv6	Gratuitous ARP
Address resolution	ICMPv6	ARP

The image features a blue-tinted background showing silhouettes of several groups of business professionals in a modern office environment. They are standing on a reflective floor, and their reflections are visible. The background includes architectural elements like glass walls and windows, suggesting a high-rise building. Overlaid on this scene is the text "Thank You" and the website address "www.huawei.com".

Thank You

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SDN and NFV Overview



Foreword

- The open ecosystem of the computing industry brings booming development of multiple fields, such as Commercial Off-the-Shelf (COTS), operating system, virtualization, middleware, cloud computing, and software applications. The network industry is also seeking transformation and development. Software Defined Networking (SDN) and Network Functions Virtualization (NFV) are mainly used.
- This course aims to help engineers understand the development of SDN and NFV and introduce Huawei SDN and NFV solutions.



Objectives

- Upon completion of this course, you will be able to:
 - Describe the development of SDN and NFV.
 - Understand basic principles of OpenFlow.
 - Understand Huawei SDN solution.
 - Understand the standard NFV architecture.
 - Understand Huawei NFV solution.



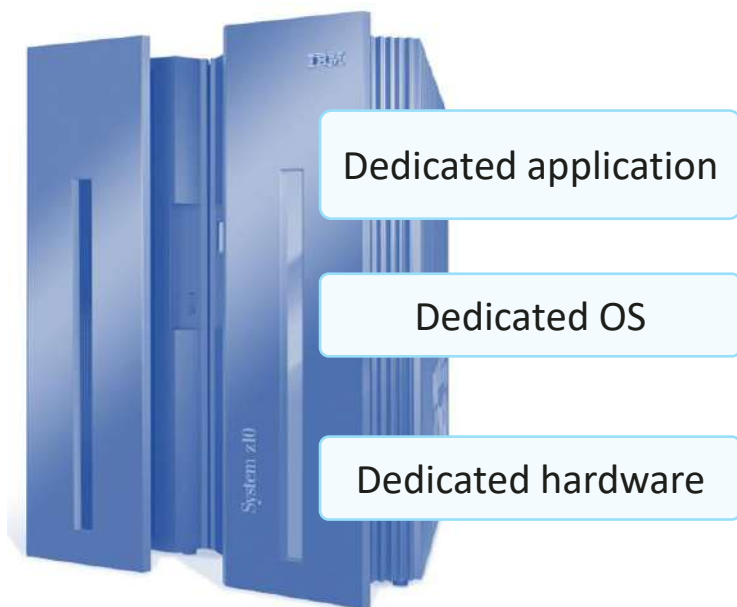
Contents

- 1. SDN Overview**
2. NFV Overview



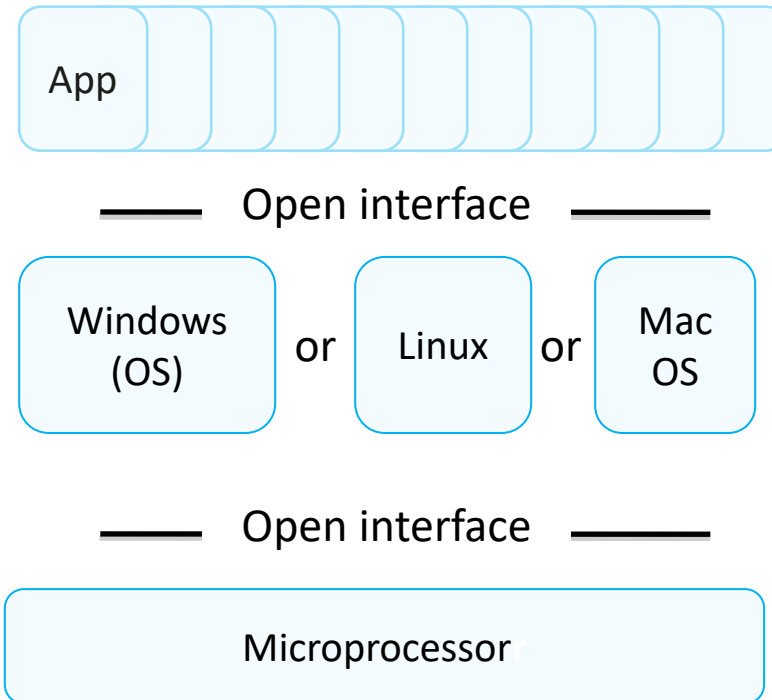
Evolution of the Computer Era

Mainframe



Vertical integration and closed interfaces
Small-scale industry applications

PC (compatible)



Horizontal integration and open interfaces
Large-scale application across industries



Network Industry Development: Implications from the IT Industry

- The transformation of the IT industry has triggered the thinking of the network industry. The industry has proposed the SDN concept and has made attempts to put SDN into commercial use, aiming to make networks more open, flexible, and simple.

Computing Industry Openness Promotes Ecosystem Development

Cloud service



ECS



EVS

Comprehensive cloud services

Database

Middleware

OS

Virtualization

Server, storage device, PC

Universal hardware

DELTA



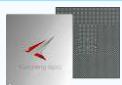
FusionSphere

Various virtualization technologies, operating systems, middleware, database software, etc.



x86/ARM server

Storage array PC ...



x86/ARM chip

Memory Hard disk ...

What About Network Industry Changes



Network application

SDN controller

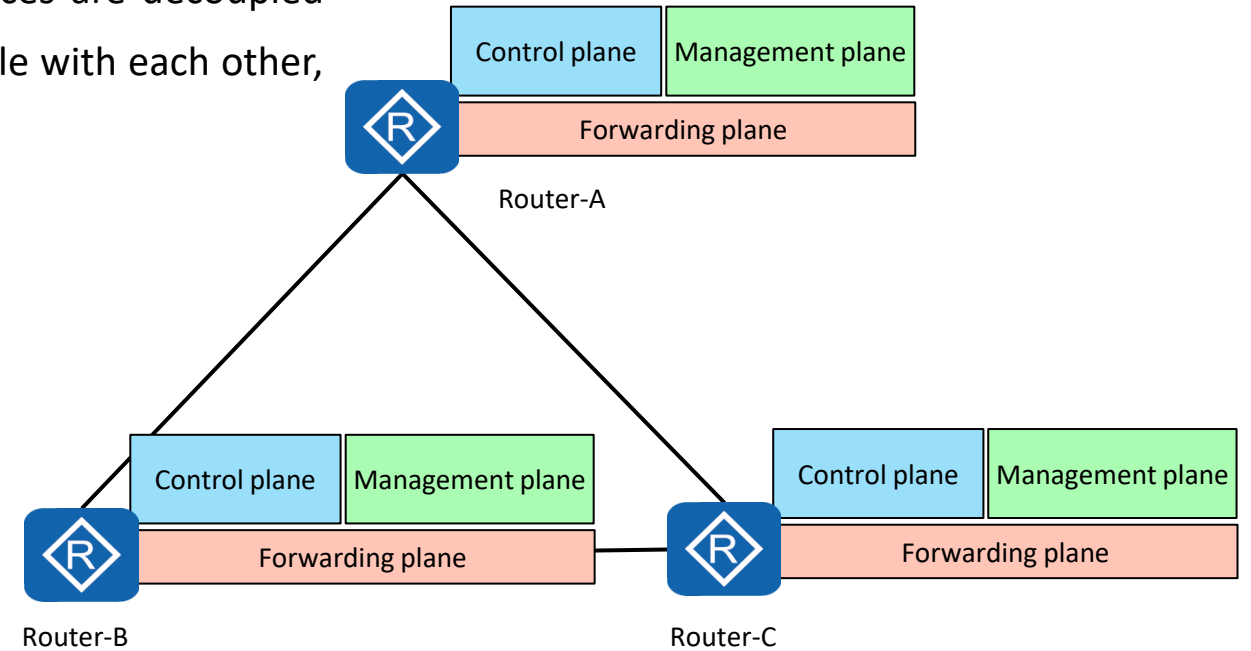
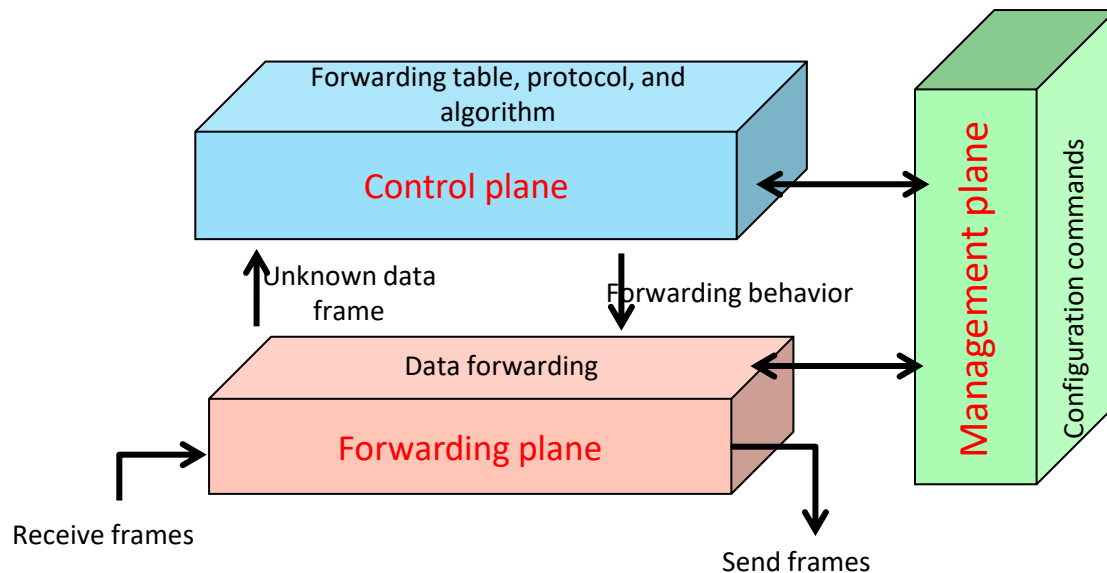
Hardware network device

- Does the network industry build a hierarchical and open ecosystem according to the computing industry?



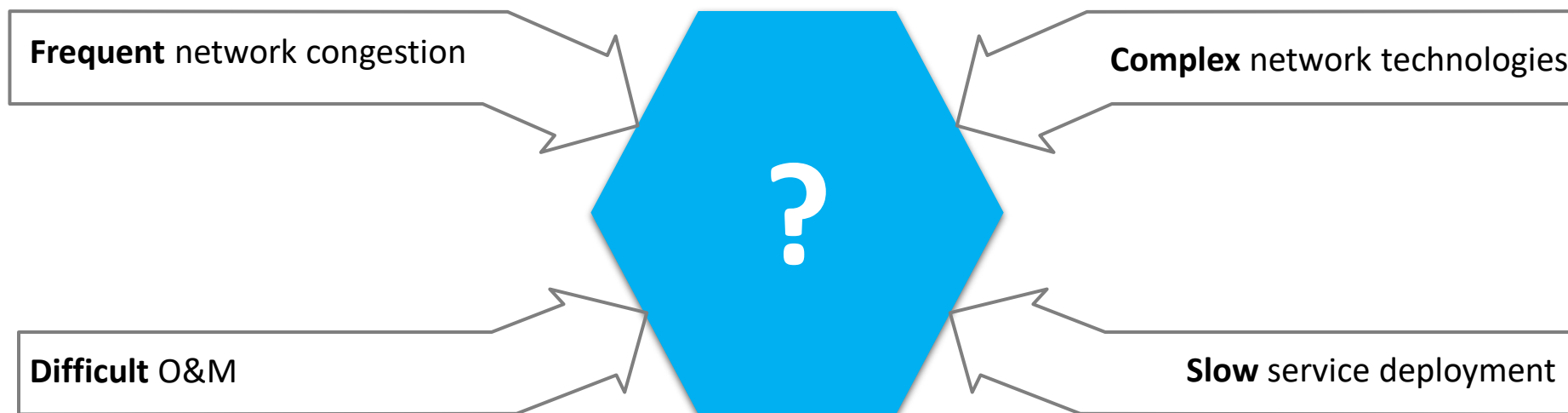
Current Situation of the Network Industry: Typical IP Network - Distributed Network

- The typical IP network is a distributed network with peer-to-peer control. Each network device has independent forwarding, control, and management planes. The control plane of a network device exchanges packets of a routing protocol to generate an independent data plane to guide packet forwarding.
- The advantage of a typical IP network is that network devices are decoupled from protocols, devices from different vendors are compatible with each other, and network convergence is ensured in fault scenarios.





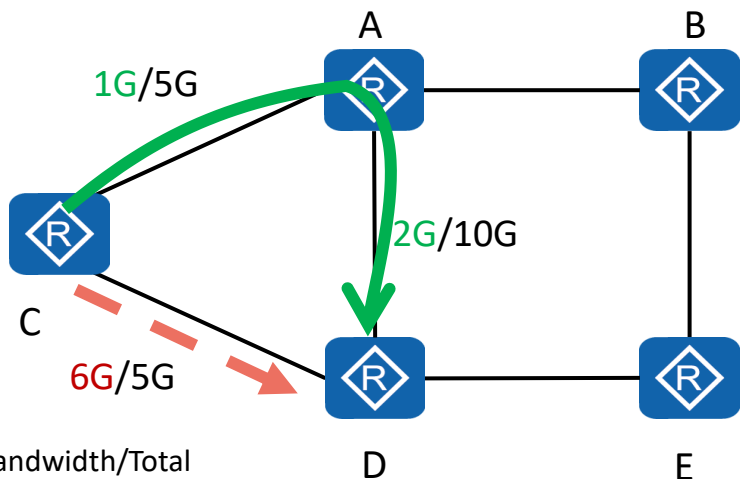
Thinking in the Network Field: Problems Faced by Typical Networks





Frequent Network Congestion

Problem and Solution of Bandwidth-based Route Selection

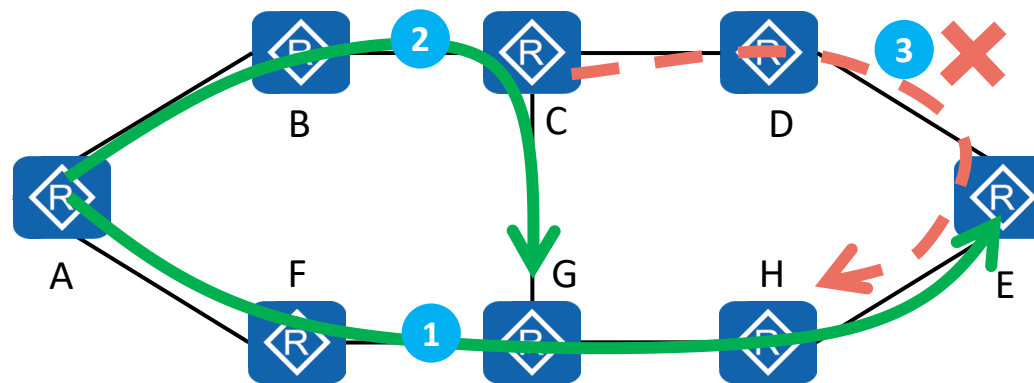


Used bandwidth/Total
bandwidth

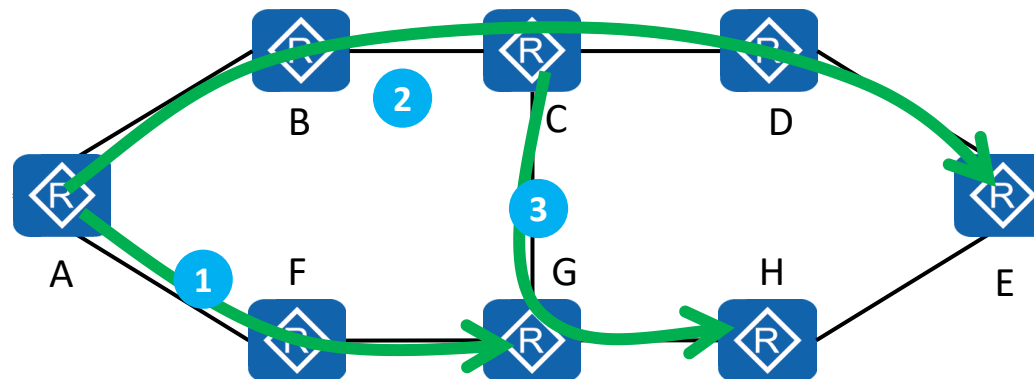
The network computes forwarding paths based on bandwidth. The link from router C to router D is the shortest forwarding path. The volume of service traffic from router C to router D exceeds the bandwidth, causing packet loss. Although other links are idle, the algorithm still selects the shortest path for forwarding. The optimal traffic forwarding path is C-A-D.

Problem and Solution of Tunnel Establishment Based on Fixed Sequence

Tunnels are established in sequence: 1. A-E; 2. A-G; 3. C-H. Tunnel 3 fails to be established due to insufficient bandwidth.



Global path calculation and optimal tunnel path adjustment:



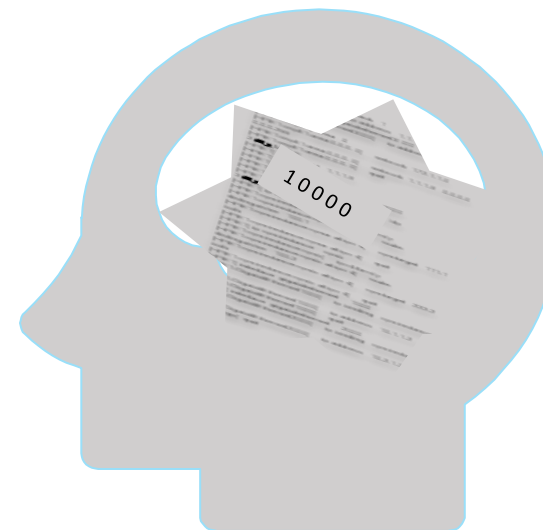
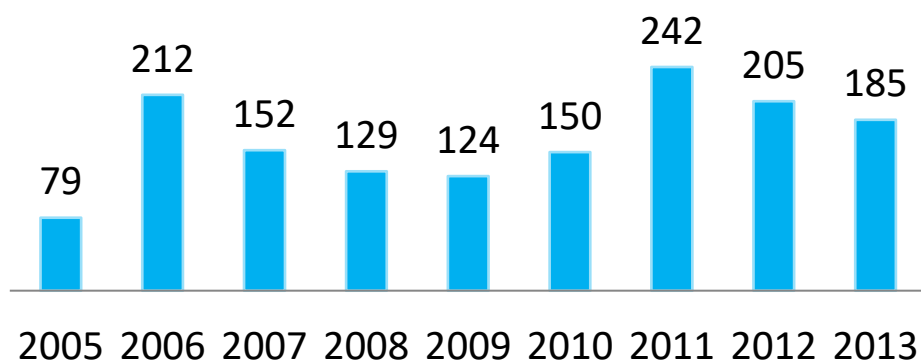


Complex Network Technologies

Many network protocols: Network technology experts need to learn many RFCs related to network devices. Understanding the RFCs takes a long time, and the number of RFCs is still increasing.

Difficult network configuration: To be familiar with devices of a specific vendor, you need to master tens of thousands of commands. Additionally, the number of commands is still increasing.

RFC increase trends





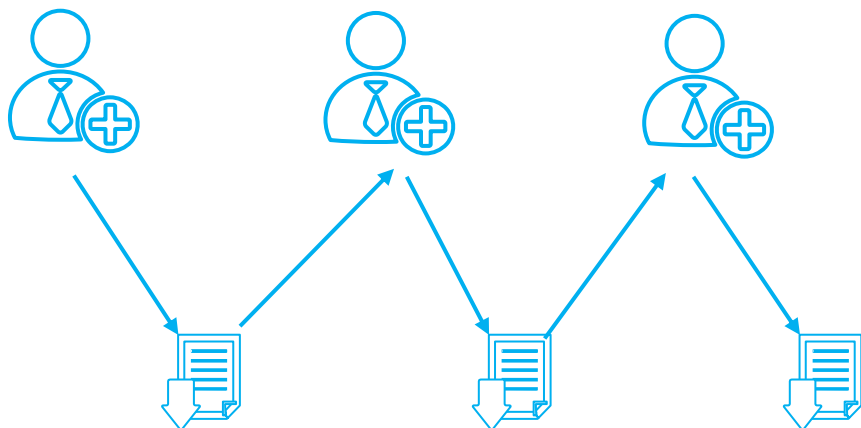
Difficulty in Locating and Analyzing Network Faults

Difficult to Spot Faults

Manual fault
identification

Manual packet obtaining
for locating faults

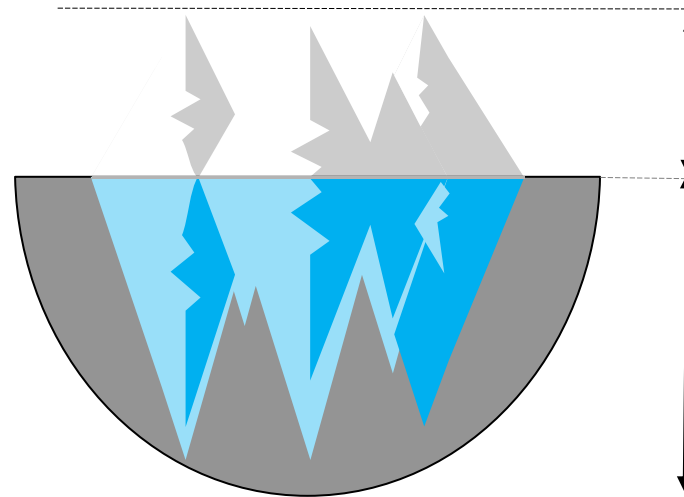
Manual fault diagnosis



- Traditional O&M networks rely on manual fault identification, location, and diagnosis.
- More than 85% of network faults are found only after service complaints. Problems cannot be proactively identified or analyzed.

Difficult to Locate Faults

Abnormal flows account for 3.65% of all flows on the network.



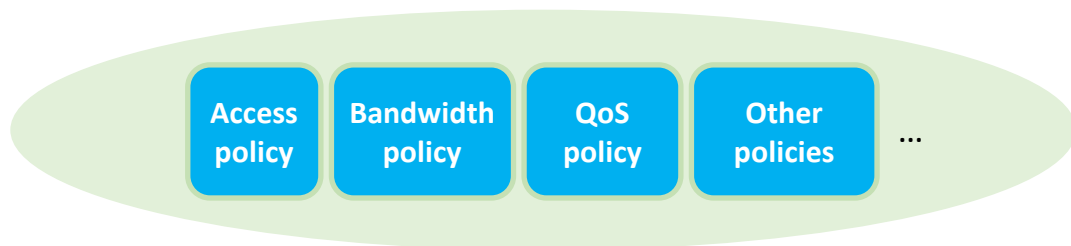
The network faults that are found upon user complaints are just the tip of the iceberg.

- Traditional O&M only monitors device indicators. Some indicators are normal, but user experience is poor. There is no correlated analysis of users and networks.
- According to data center network (DCN) statistics, it takes 76 minutes to locate a fault on average.



Slow Network Service Deployment

Network policy



Complex and inflexible network policy changes:

Network policies cannot be defined by user. Policy changes are complex and cannot be flexibly adjusted.

IP address-based, fixed location, and CLI-based configuration

Service network

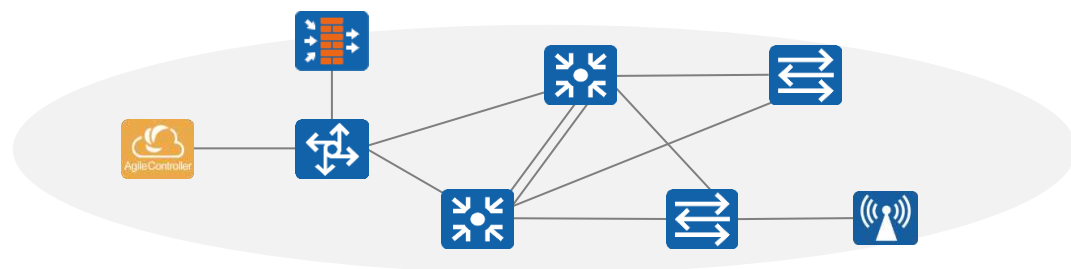


Long service deployment period:

New service deployment involves E2E device configuration modification.

End-to-end configuration using commands

Physical network



Low physical network deployment efficiency:

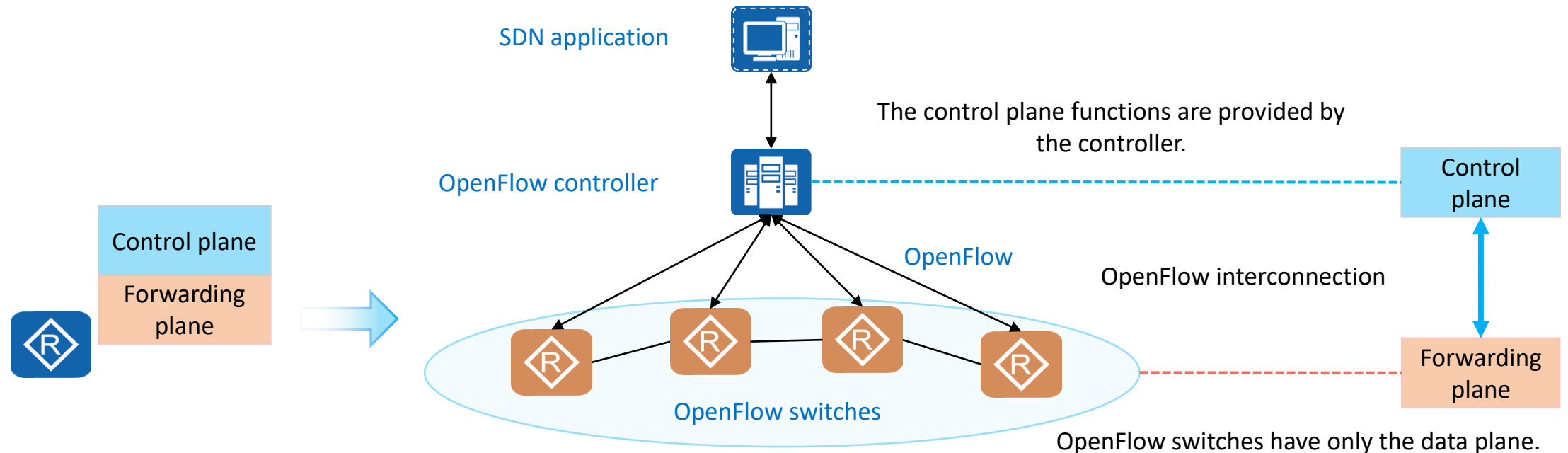
The physical network does not support zero touch provisioning (ZTP).

Command line-based configuration by device



SDN Origin

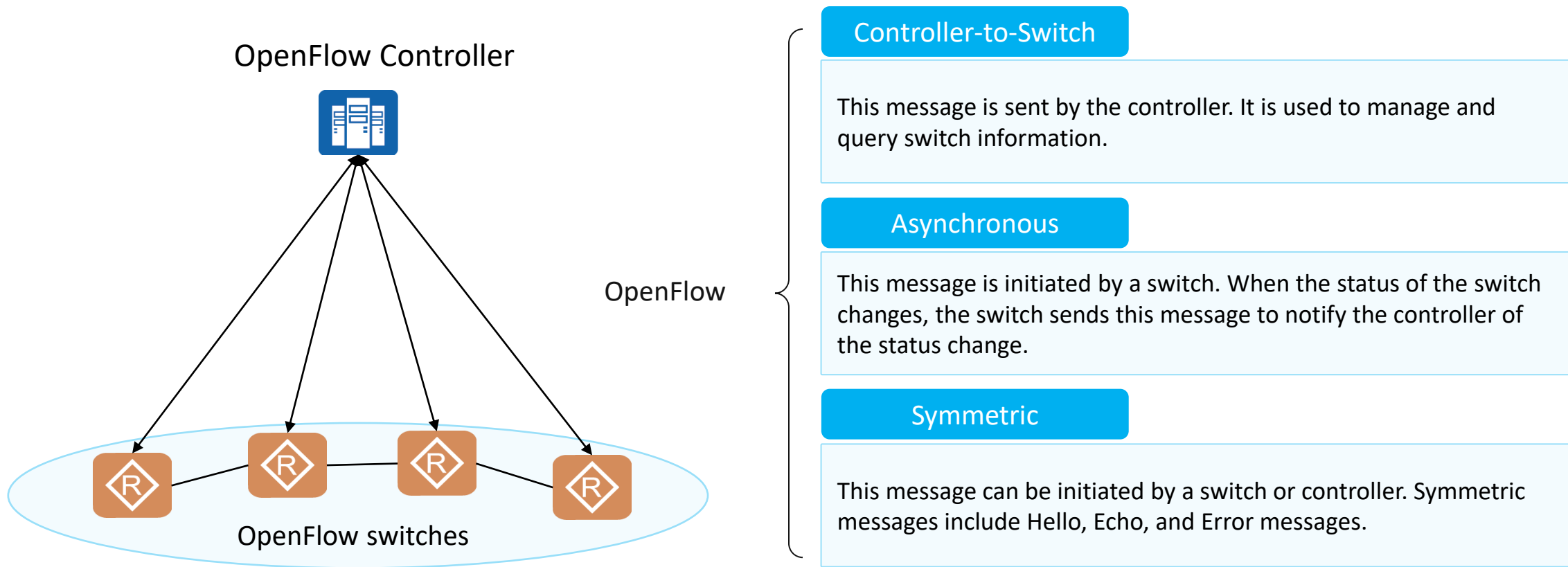
- SDN was developed by the Clean Slate Program at Stanford University as an innovative new network architecture. The core of SDN is to separate the control plane from the data plane of network devices to implement centralized control of the network control plane and provide good support for network application innovation.
- SDN has three characteristics in initial phase: forwarding-control separation, centralized control, and open programmable interfaces.





Basic Concepts of OpenFlow

- OpenFlow is an SBI protocol between a controller and a switch. It defines three types of messages: Controller-to-Switch, Asynchronous, and Symmetric. Each message contains more subtypes.





Flow Table Overview

- OpenFlow switches forward packets based on flow tables.
- Each flow entry includes the Match Fields, Priority, Counters, Instructions, Timeouts, Cookie, and Flags. The Match Fields and Instructions are key fields for packet forwarding.
 - The Match Fields is a field against which a packet is matched and can be customized.
 - The Instructions field indicates OpenFlow processing when a packet matches a flow entry.

Match Fields	Priority	Counters	Instructions	Timeouts	Cookie	Flags
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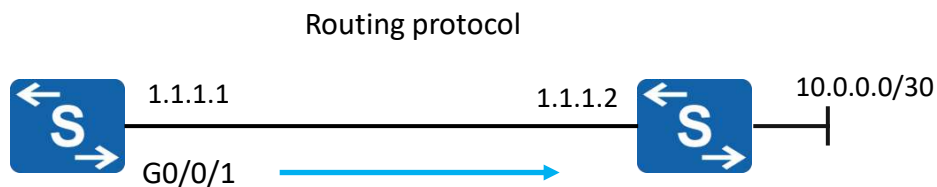
Flow table fields can be customized. The following table is an example.

Ingress Port	Ether Source	Ether Dst	Ether Type	VLAN ID	VLAN Priority	IP Src	IP Dst	TCP Src Port	TCP Dst Port
3	MAC1	MAC2	0x8100	10	7	IP1	IP2	5321	8080



Comparison Between Forwarding Modes

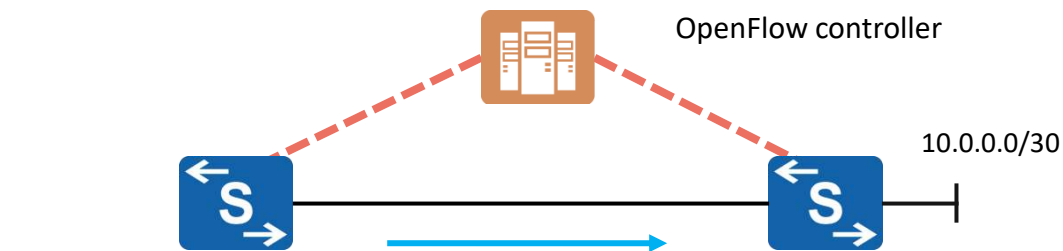
Typical Routing Protocol: Packet Forwarding Based on Routing Tables



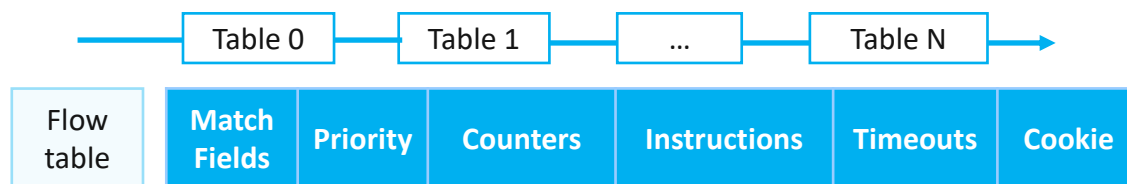
Routing table	Destination Network	Protocol	Next Hop	Outbound Interface
	10.0.0.0/30	OSPF	1.1.1.2	G0/0/1

- In typical cases, network devices query routing tables to guide traffic forwarding.
- Entries in a routing table are calculated by running a routing protocol between network devices.
- The length of the routing table is fixed. Network devices forward packets based on the longest match rule. A network device has only one routing table.

OpenFlow: Packet Forwarding Based on Flow Tables



Flow table matching process:



- OpenFlow is a network protocol. Switches running OpenFlow forward traffic based on flow tables.
- Flow tables are calculated by the OpenFlow controller and then delivered to switches.
- A flow table has variable length and defines various matching and forwarding rules. A network device has multiple flow tables.



Essential Requirements of SDN

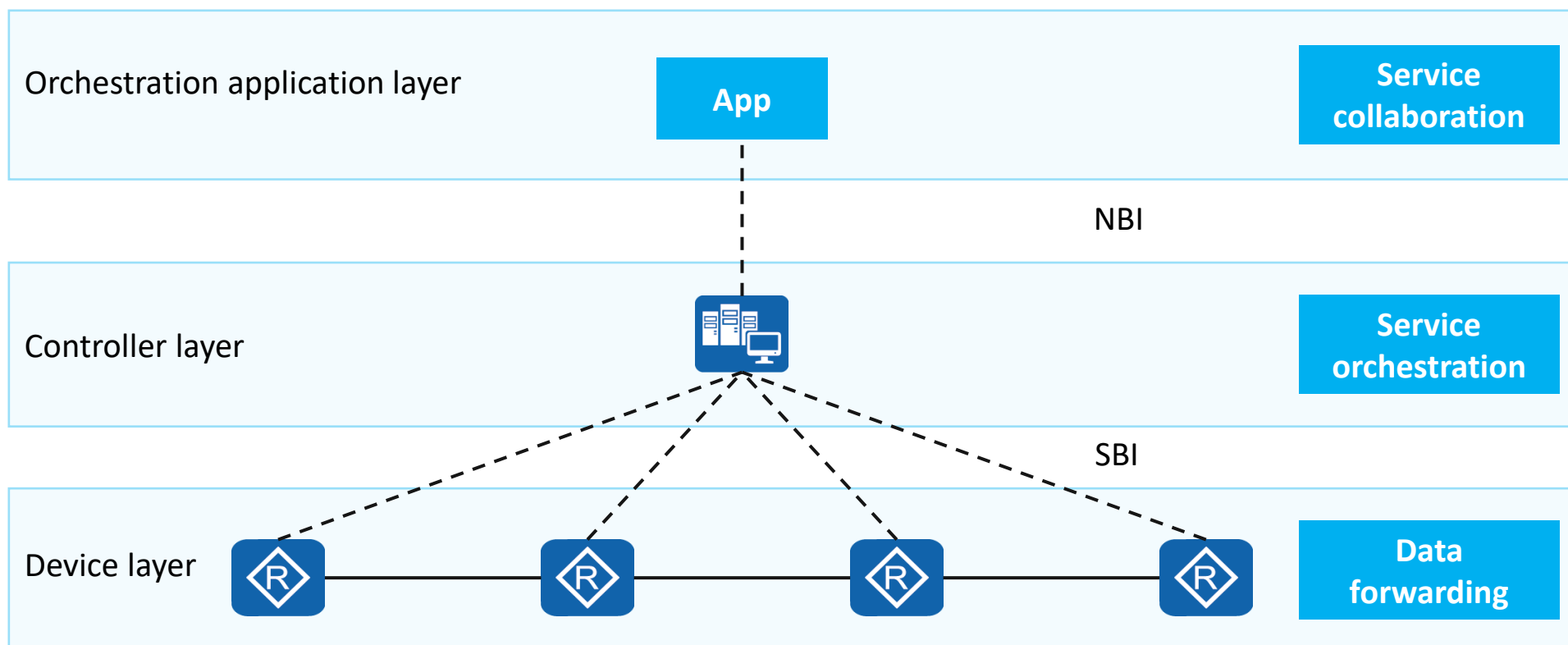
- The essence of SDN is to make networks more open, flexible, and simple. It builds a centralized brain for a network and implements fast service deployment, traffic optimization, or network service openness through centralized control in the global view.
- SDN has the following benefits:
 - Provides centralized management, simplifying network management and O&M.
 - Shields technical differences, simplifies network configuration, and reduces O&M costs.
 - Offers automatic optimization, improving network utilization.
 - Deploys services rapidly, shortening the service rollout time.
 - Builds an open network, supporting open and programmable third-party applications.

SDN transforms network architecture.



SDN Network Architecture

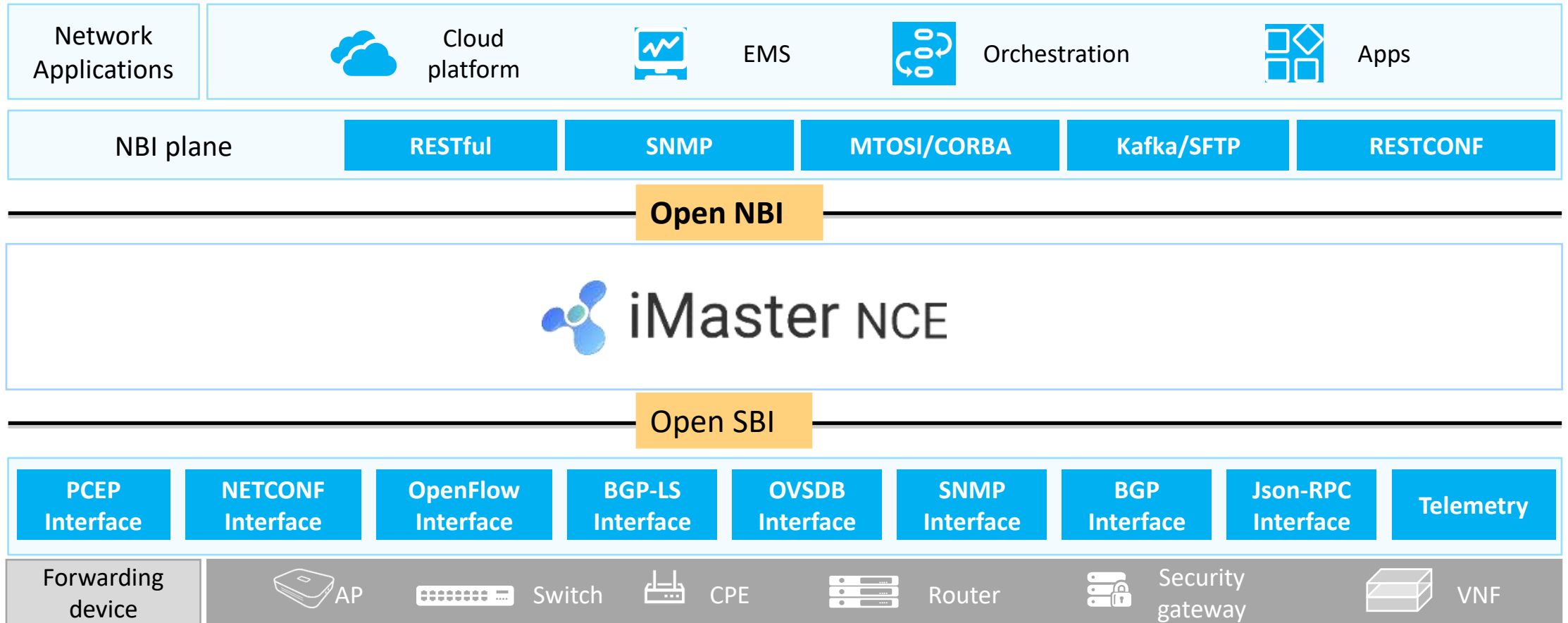
- The SDN network architecture consists of the orchestration application layer, controller layer, and device layer. Different layers are connected through open interfaces. From the perspective of the controller layer, SBIs oriented to the device layer and NBIs oriented to the orchestration application layer are distinguished. OpenFlow is one of SBI protocols.





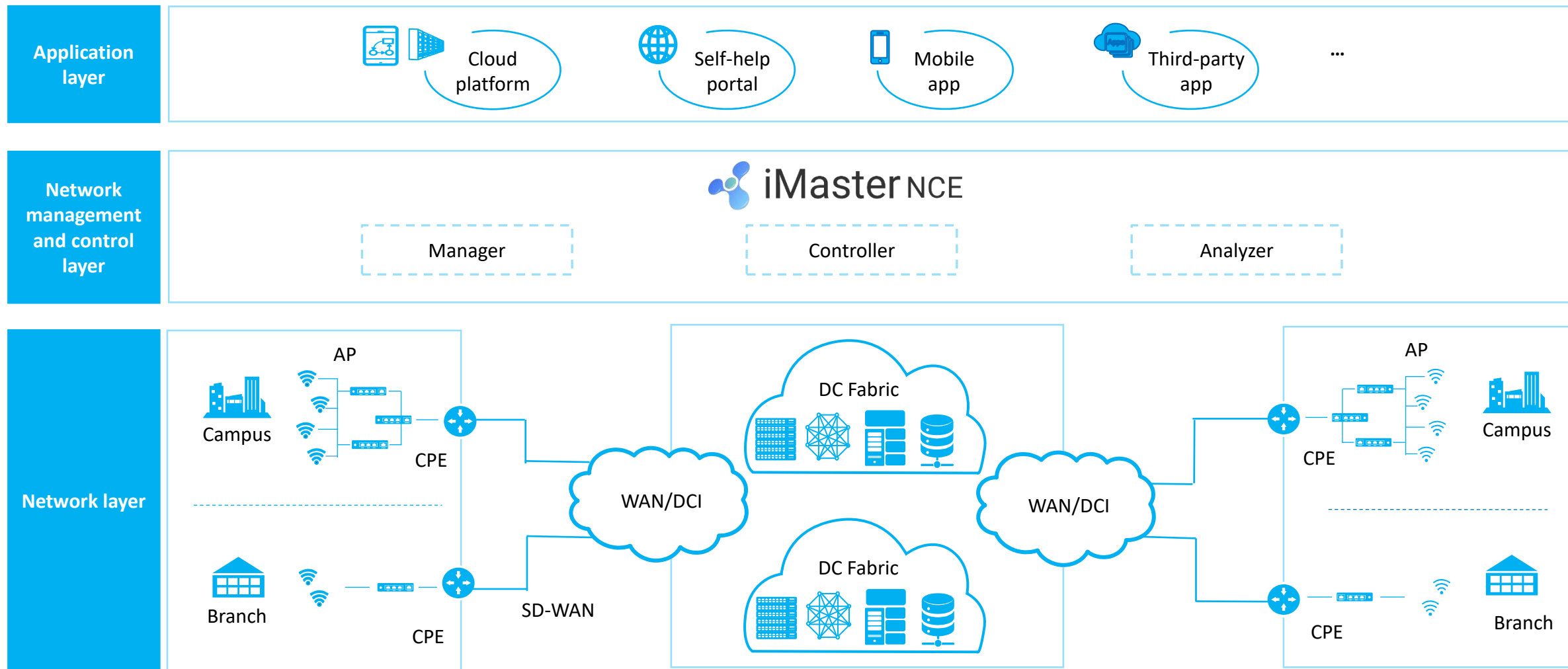
Huawei SDN Network Architecture

- Huawei SDN network architecture supports various SBIs and NBIs, including OpenFlow, OVSDB, NETCONF, PCEP, RESTful, SNMP, BGP, JSON-RPC, and RESTCONF interfaces.





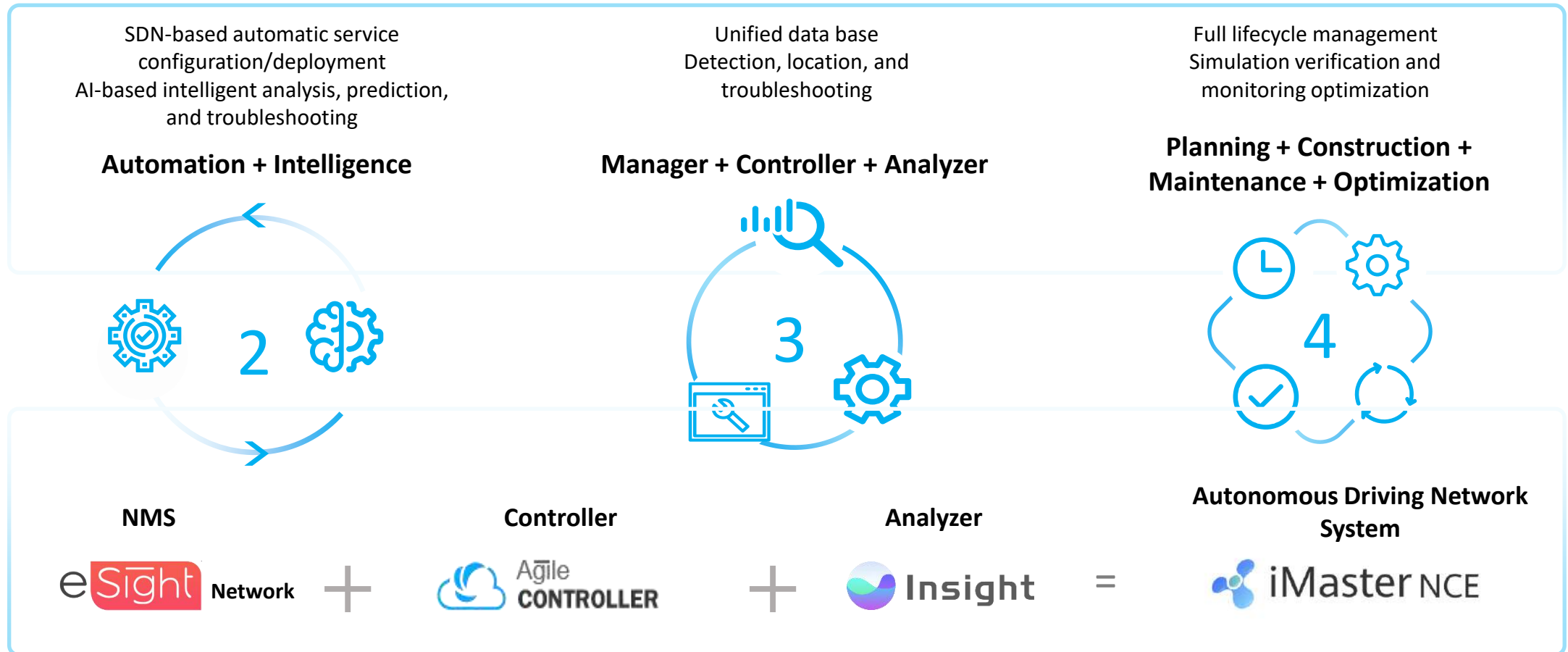
Huawei SDN Solution - Integrating Management, Control, and Analysis to Build an Intent-Driven Network





Introduction to iMaster NCE

- Huawei iMaster NCE is the industry intelligent network automation platform that integrates management, control, analysis, and AI capabilities.





iMaster NCE Application



DC

iMaster NCE-Fabric *



Enterprise
campus

iMaster NCE-Campus *



SD-WAN

iMaster NCE-WAN



IP WAN

iMaster NCE-IP



WAN
transmission

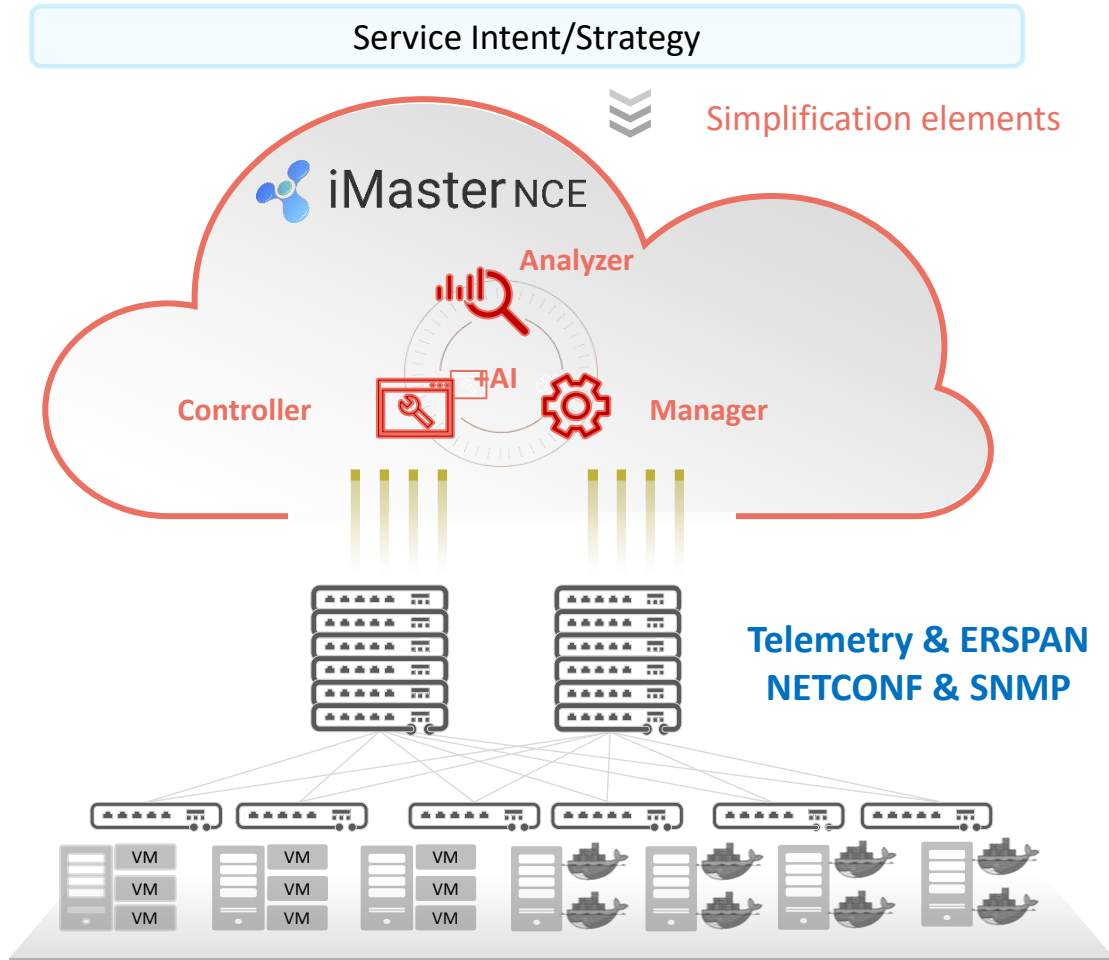
iMaster NCE-T

* iMaster NCE-Fabric and iMaster NCE-Campus are introduced in this document.



Huawei CloudFabric DCN Autonomous Driving Network Solution

- Based on iMaster NCE-Fabric, DCNs provide full-lifecycle services from planning, construction, O&M, to optimization.



Integrated planning and construction:

- The planning tool interconnects with iMaster NCE-Fabric to implement integrated planning and construction.
- Zero Touch Provisioning (ZTP)

Simplified deployment

- Service intent self-understanding and conversion
- Network change simulation and evaluation, eliminating human errors

Intelligent O&M:

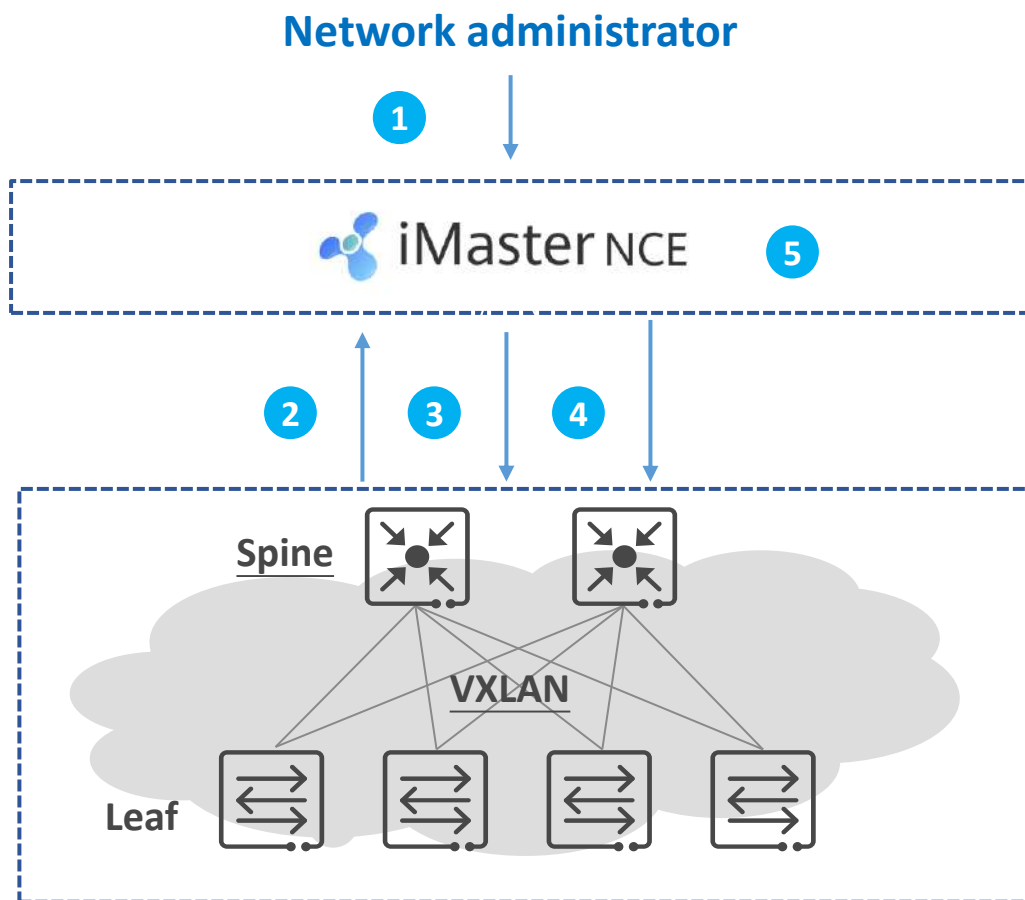
- Rapid fault detection and location based on knowledge graph and expert experience
- Fast fault rectification based on expert experiences and simulation analysis

Real-time optimization:

- AI-Fabric-oriented local traffic inference and online model training and optimization
- User behavior prediction and resource optimization suggestions



Simplified ZTP Deployment



ZTP deployment process:

1. The network administrator clicks the icon on iMaster NCE to start the ZTP task.
2. A device automatically obtains an IP address to access iMaster NCE.
3. iMaster NCE determines the device role (spine or leaf node), delivers configurations such as the management IP address, SNMP configuration, and NETCONF configuration to online devices, and manages the devices through the management IP address.
4. iMaster NCE globally delivers interconnection configurations as well as OSPF or BGP configurations.
5. The device goes online successfully, and the administrator views network-wide information on iMaster NCE.

Note: The DC uses the spine-leaf architecture.



Network Intent Self-understanding and Fast Service Deployment

Network design

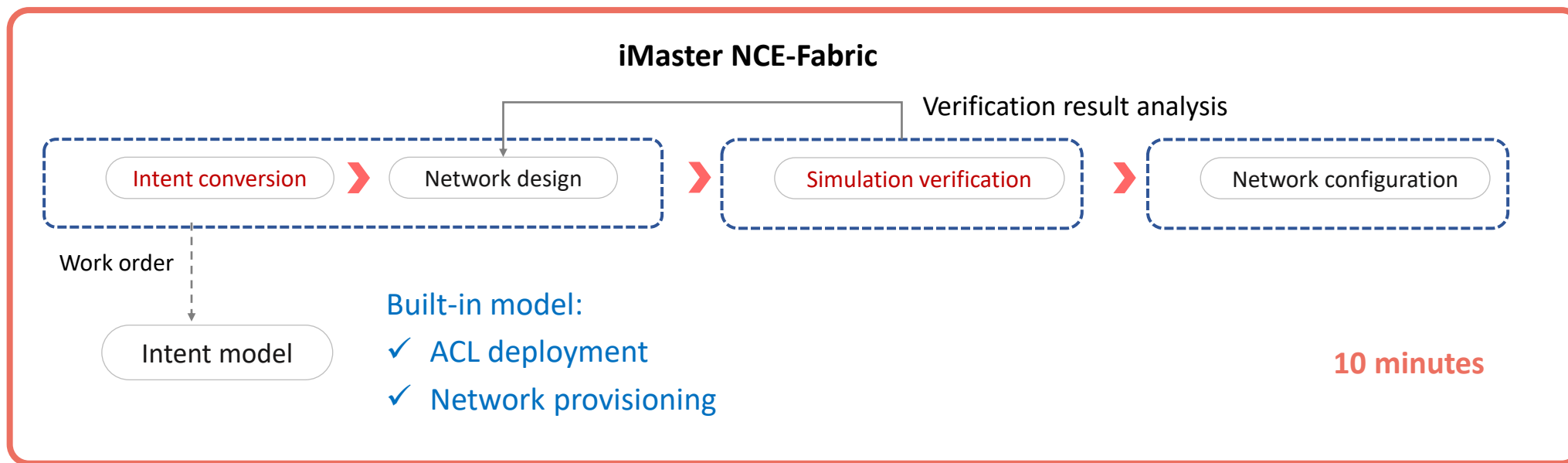
2 to 3 days

Configuration delivery

10 minutes

Service verification

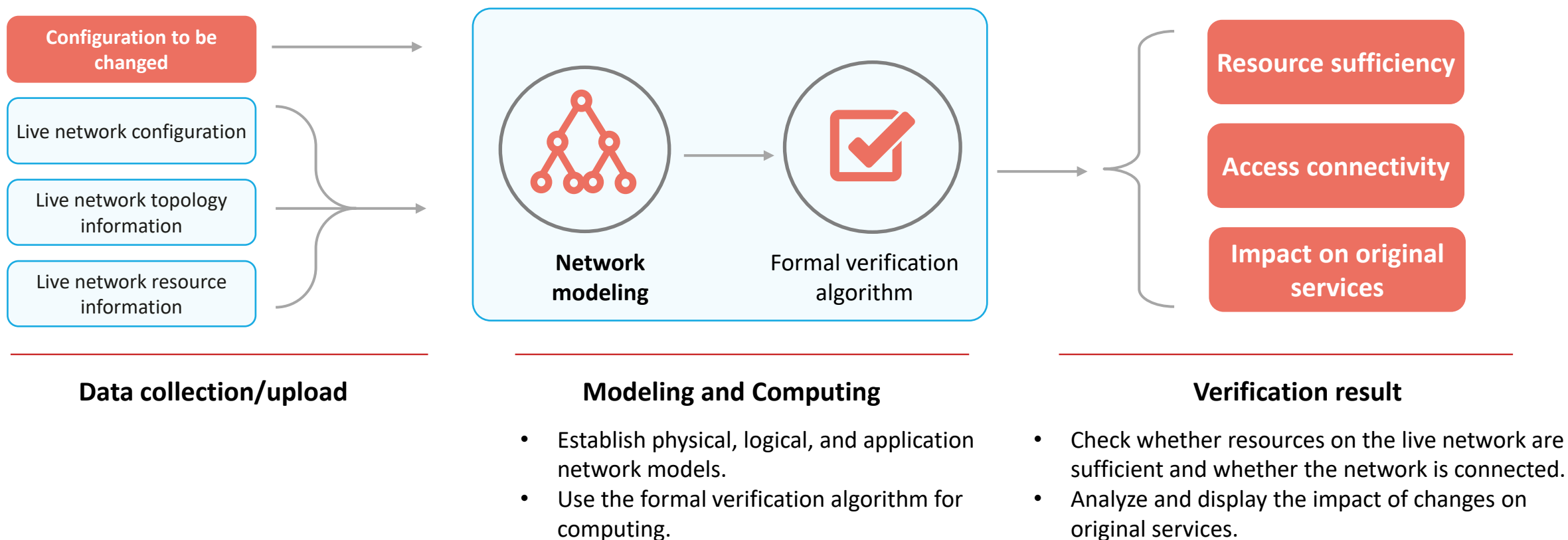
1 to 2 days



Huawei iMaster NCE-Fabric supports automatic and fast deployment of virtualization, cloud computing, and container networks.

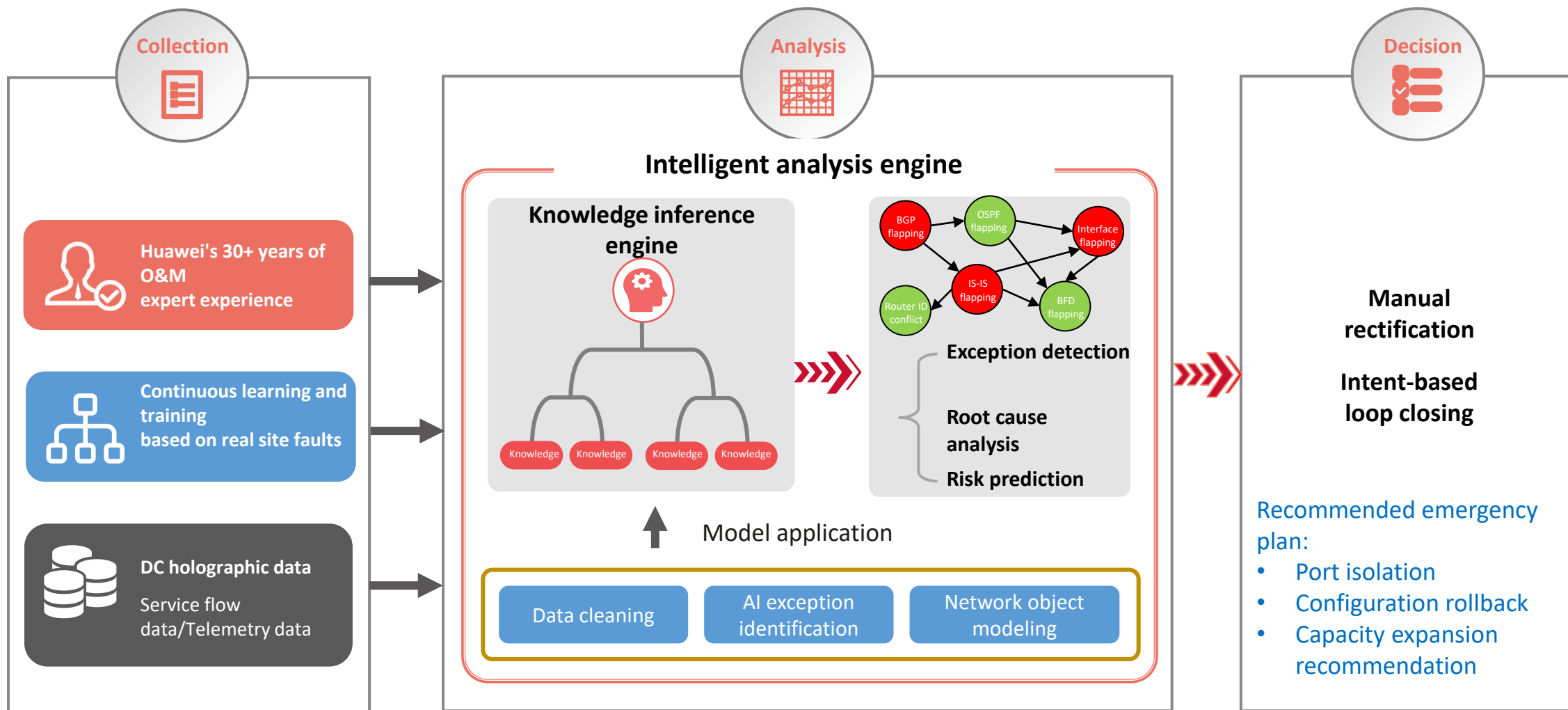


Network Change Simulation and Change Risk Prediction



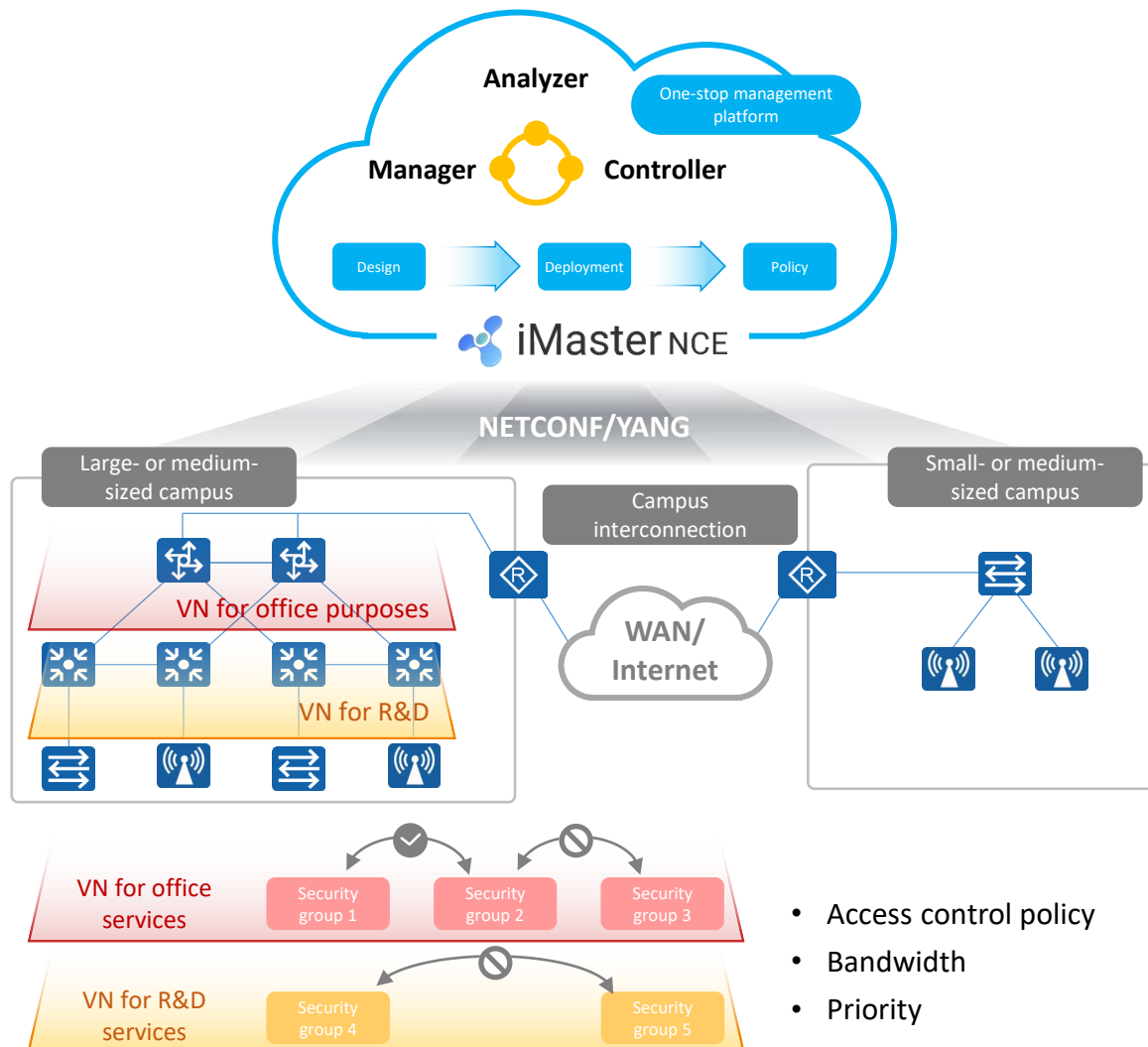


AI-powered Intelligent O&M for DCNs





Huawei CloudCampus Autonomous Driving Network Solution



Fast network deployment, improving deployment efficiency by 600%

- Device plug-and-play: simplified device deployment, scenario navigation, and template-based configuration
- Simplified network deployment: Network resource pooling, multi-purpose network, and automatic service provisioning

Fast service provisioning, improving user experience by 100%

- Free mobility: GUI-based policy configuration, allowing users to access the network anytime and anywhere without changing the roaming permission and user experience
- Intelligent terminal identification: Anti-spoofing for terminal access, with an intelligent terminal identification accuracy of over 95%
- Intelligent HQoS: Application-based scheduling and shaping, and refined bandwidth management, ensuring service experience of key users

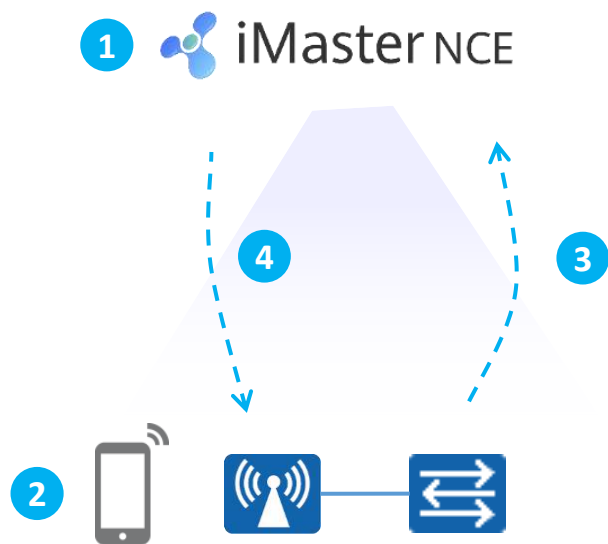
Fast intelligent O&M, improving network performance by over 50%

- Real-time experience visualization: Telemetry-based network experience visualization at each moment, for each user, and in each area
- Precise fault analysis: Proactively identifying 85% of typical network issues and providing suggestions, and comparing and analyzing real-time data to predict faults
- Intelligent network optimization: Predictive optimization of wireless networks based on historical data, improving network-wide performance by over 50% (Source: Tolly Certification)



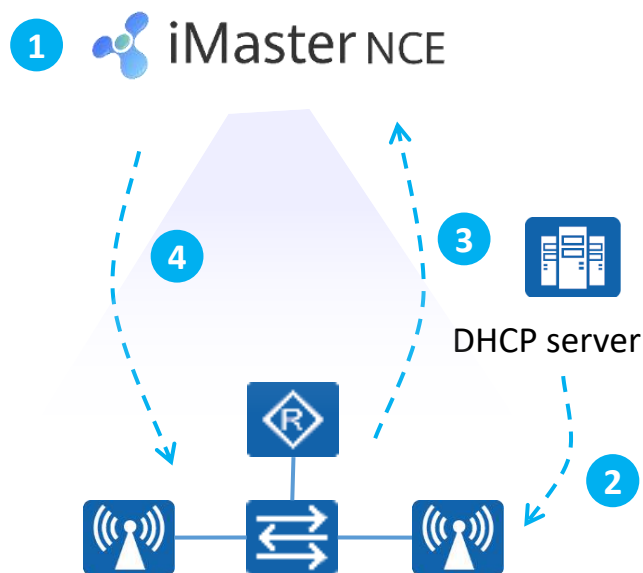
Device Plug-and-Play

Deployment by Scanning Bar Codes



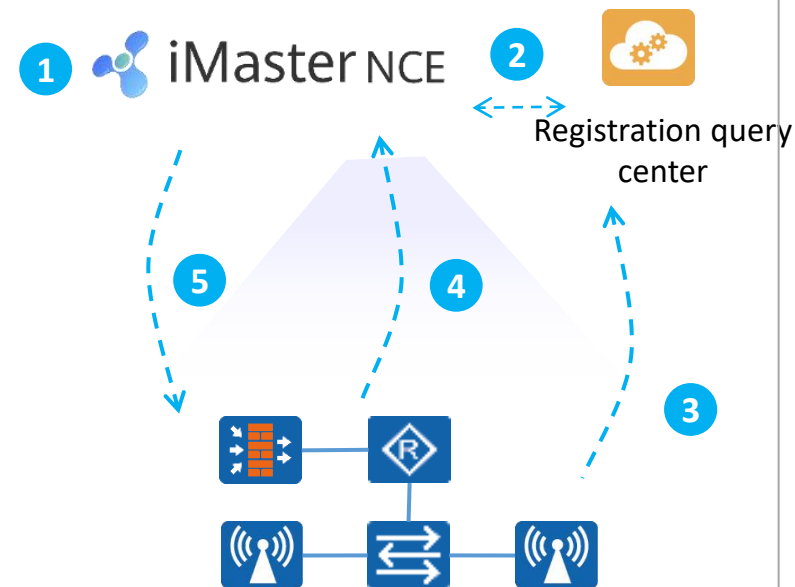
1. Pre-configuration
2. Deployment by scanning bar codes
3. Automatic device registration and login
4. Automatic configuration delivery

DHCP-based Deployment



1. Pre-configuration
2. Obtaining registration information through the DHCP server
3. Automatic device registration and login
4. Automatic configuration delivery

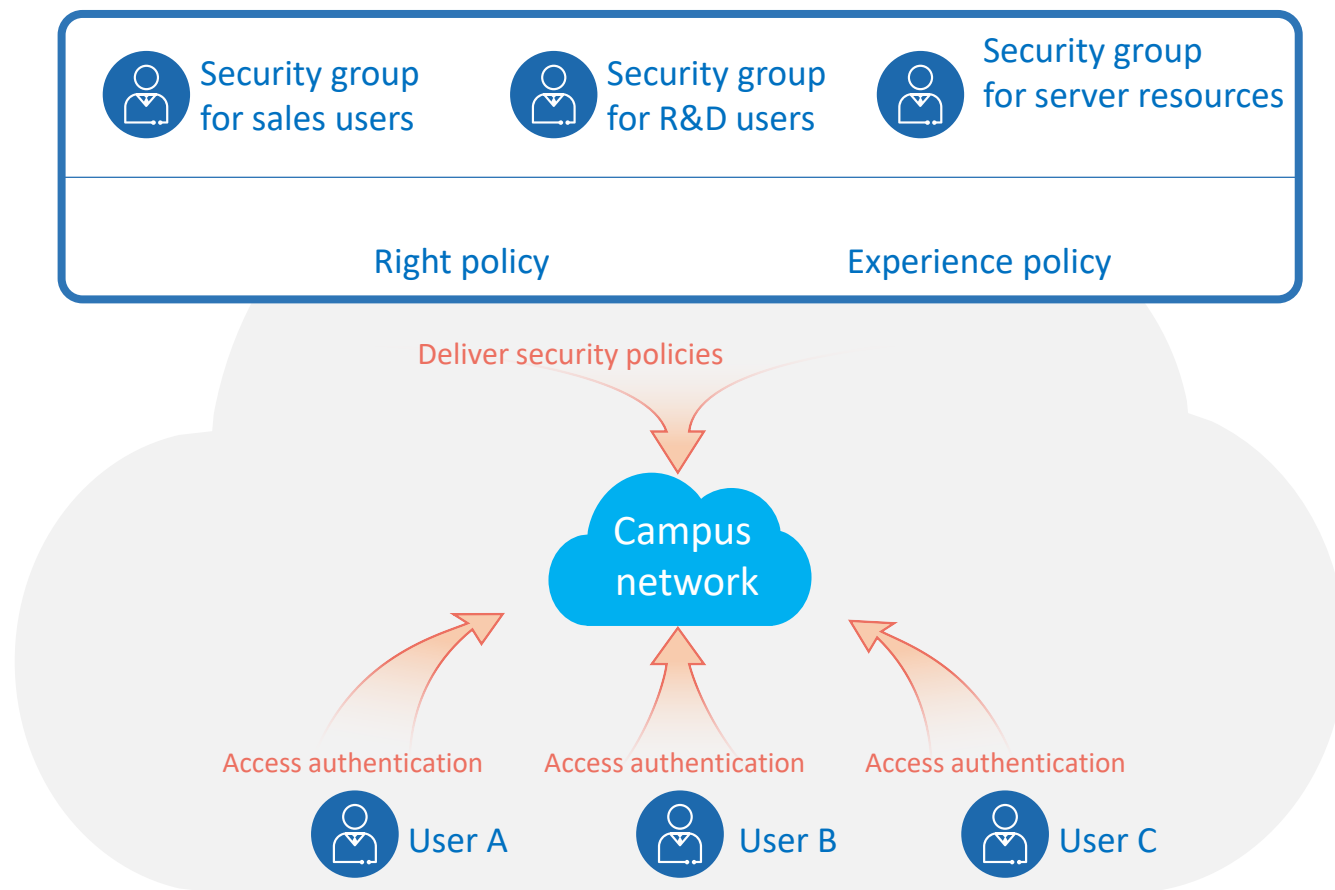
Deployment Through the Registration Query Center



1. Pre-configuration
2. Information synchronization
3. Obtaining registration information through the registration query center
4. Automatic device registration and login
5. Automatic configuration delivery

Free Mobility: Policy Management Based on Security Groups

- Free mobility: Enables users to have consistent network rights and security policies regardless of their locations and IP addresses.

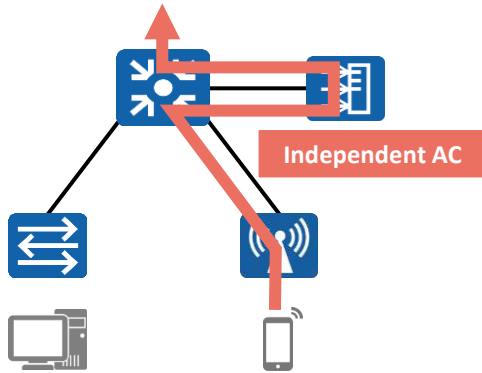


- 1 Use security groups. A security group is a group of users for which the same security policy is used.
- 2 Define security group-based permission control policies and user experience policies and deliver the policies to network devices.
- 3 A security group is authorized to a user after the user passes access authentication.
- 4 After user traffic enters a network, network devices enforce policies based on the source and destination security groups of the traffic.



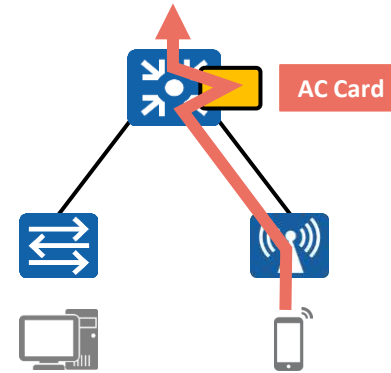
Wired and Wireless Convergence

WLAN Construction Mode 1: Standalone AC



- This mode poses a wireless traffic bottleneck and increases faulty nodes.
- Wired and wireless management is independent.
- Wired and wireless authentication points are separated.

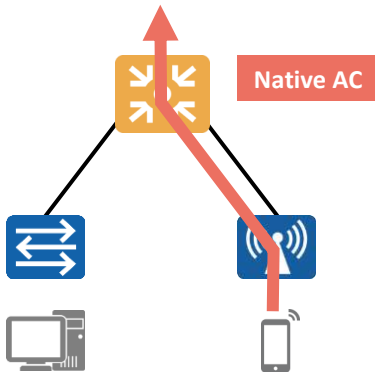
WLAN Construction Mode 2: AC Card



- An AC card is installed on a switch to provide AC functions.
- Hardware-level convergence.

Wired and wireless authentication point separation, distributed policy control, separation of control and data traffic forwarding, and troubleshooting and management difficulties.

Wired and Wireless Convergence (Native AC)



The switch integrates the AC function, eliminating wireless traffic forwarding bottlenecks and reducing fault nodes. Wired and wireless devices are centrally managed.

- Unified management and converged forwarding of wired and wireless services
- Converged management for wired and wireless users and gateway convergence
- Converged authentication points for wired and wireless access
- Unified wired and wireless policy execution



Intelligent Terminal Identification, Ensuring Secure Access

DC

Enterprise
campus

Requirements and Challenges

A university

50+ types of smart terminals

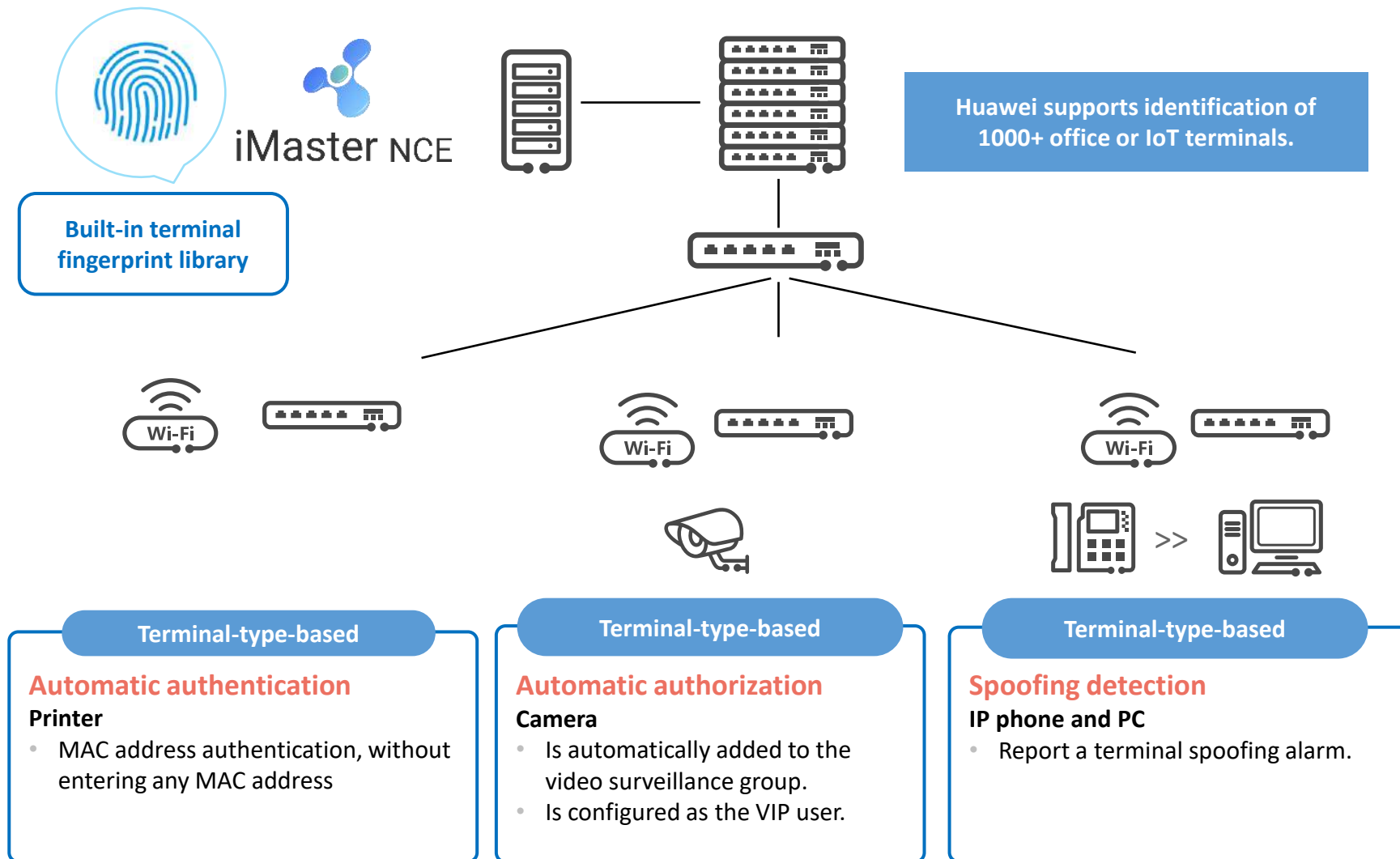
Terminal information is
collected by IT departments
of colleges:
MAC address collection is
difficult and error-prone.

An enterprise

100+ authentication faults
reported per day

It is difficult to locate access
spoofing.

>>





HQoS: User- and Application-based QoS Policy

User- and application-based QoS policies ensure experience of high-priority users and applications

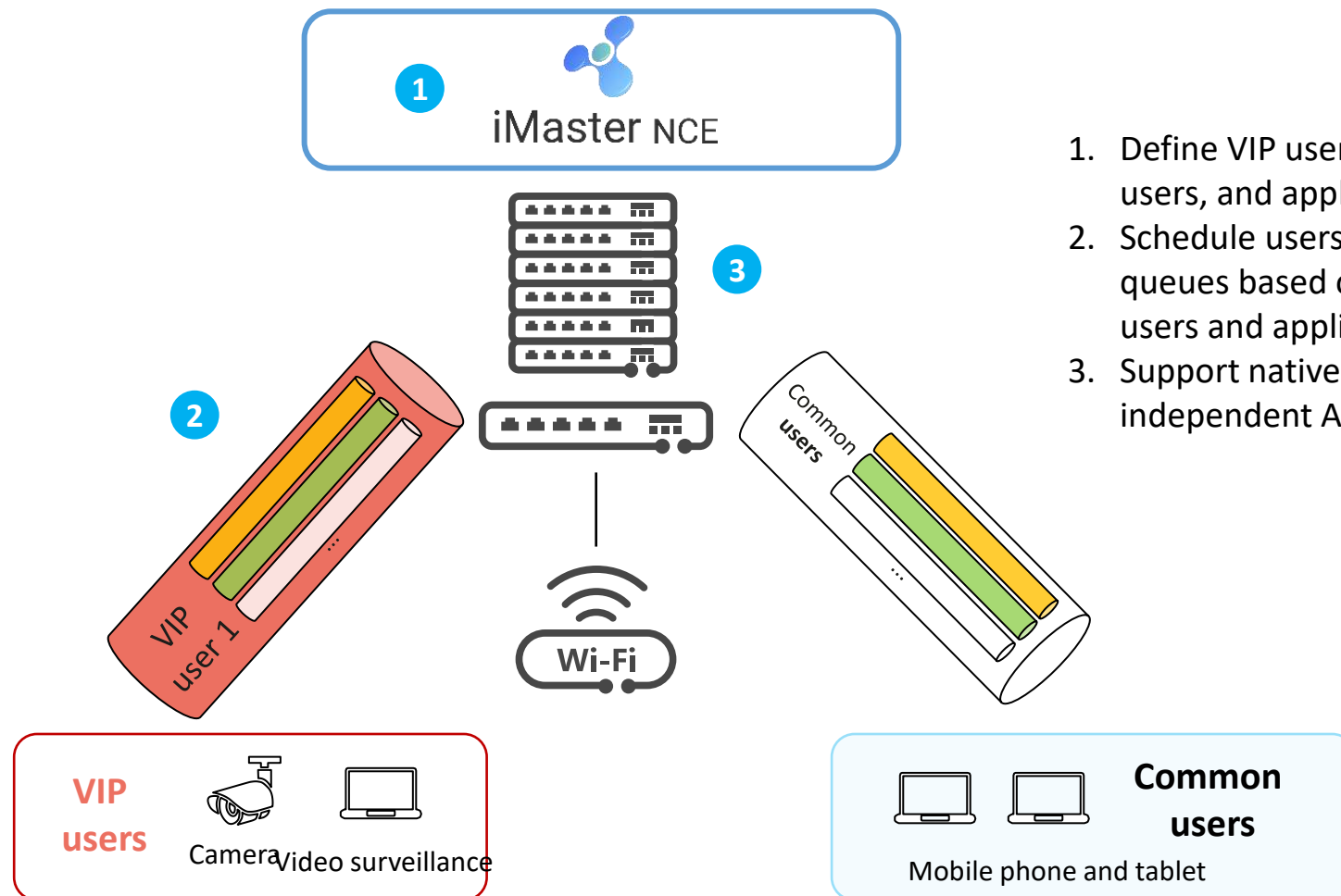
Requirements and Challenges

Traditional QoS policies are invalid for video services.

(Example) Building surveillance scenario:

Wireless video services of common users increase, occupying a large number of network resources and causing network congestion.

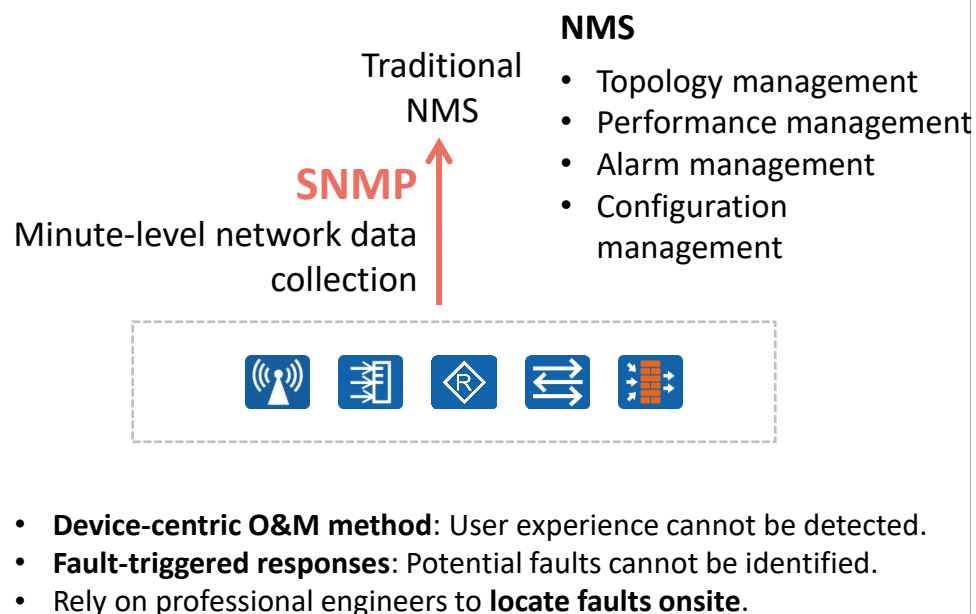
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AI-Powered Intelligent O&M of Campus Networks

As-Is: Device-Centered Network Management



To-Be: User Experience-Centered AI-Powered Intelligent O&M



Telemetry
Second-level network data collection

Intelligent network analyzer

- Visualized user experience management
- User journey playback
- Potential fault identification
- Root cause identification
- Predictive network optimization



Experience visualization: Telemetry-based second-level data collection, visualized experience of each user and each application in real time

Minute-level potential fault identification and root cause location

- Identify potential faults based on dynamic baselines and big data correlation analysis.
- Accurately locate root causes of faults through KPI association analysis and protocol tracing.

Predictive network optimization: AI is used to intelligently analyze the AP load trend and implement predictive optimization on the wireless network.

The efficiency is improved by using algorithms. With scenario-based continuous learning and expert experience, intelligent O&M frees O&M personnel from complex alarms and noises, making O&M more automated and intelligent.



AI-Powered Intelligent Radio Calibration

Traditional radio calibration cannot achieve expected results

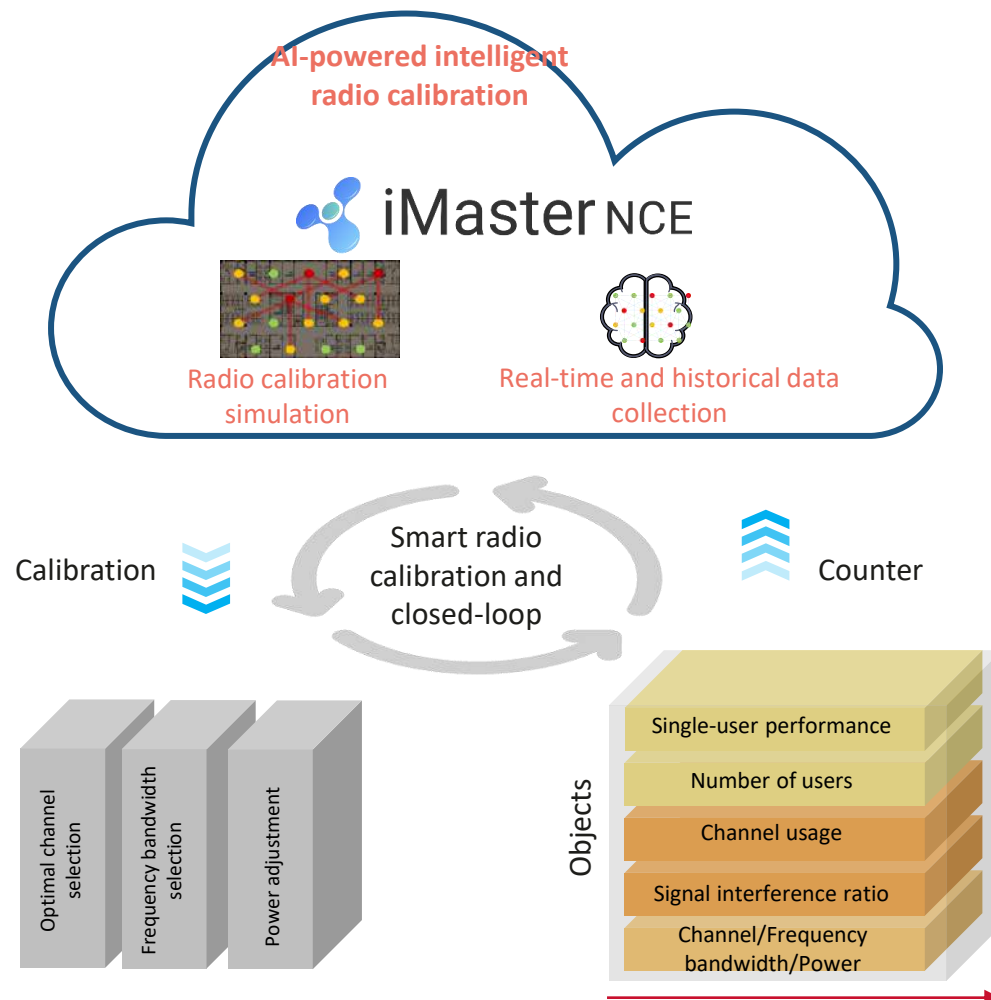
Phase 1: Manual Calibration

Based on engineers' experience, the calibration is time-consuming and error-prone, and the calibration result is unstable.



Phase 2: Automatic Calibration

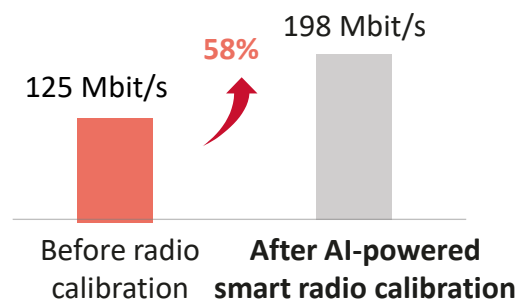
Calibration based on real-time radio interference does not consider device load changes, and the calibration result cannot be ensured.



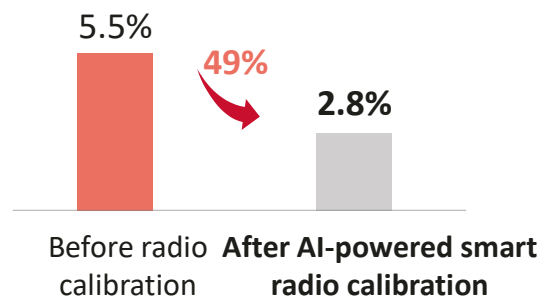
Tested and verified by authoritative organizations



Average downlink rate per terminal:



Average Wi-Fi channel interference





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2. NFV Overview

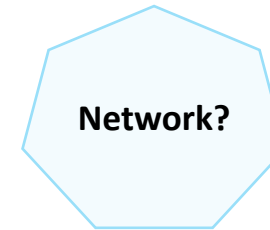
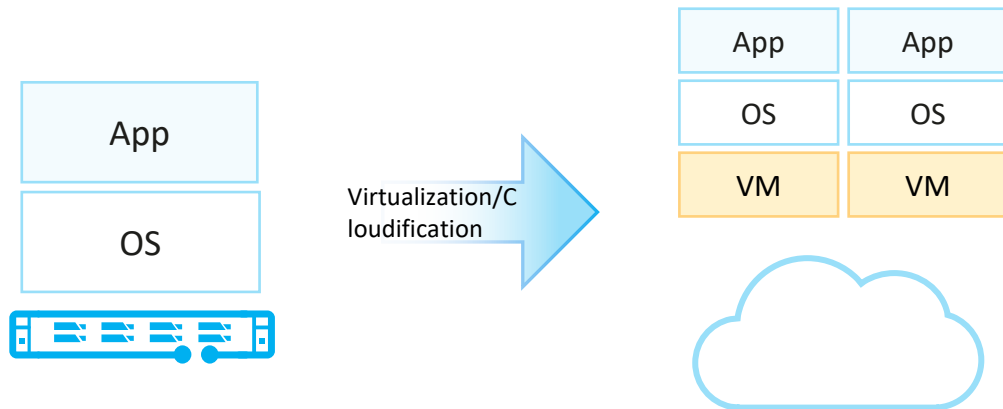


NFV Background: Thinking from IT Industry Transformation

- The IT industry transformation brings thinking on **network architecture** and **device architecture** in the network industry. The network architecture layer involves the SDN controller and the device architecture layer involves the device deployment mode.

IT Industry Transformation

- In recent years, IT technologies such as virtualization and cloud computing have been booming, and applications deployed on hardware have been gradually migrated to the cloud. Applications are deployed on private clouds, public clouds, or hybrid clouds as software.



- Thinking about the network industry: Can network applications be deployed in a software-based manner?
- In the context, Network Functions Virtualization (NFV) is introduced.



Origin of NFV

- In October 2012, 13 top carriers (including AT&T, Verizon, VDF, DT, T-Mobile, BT, and Telefonica) released the first version of NFV White Paper at the SDN and OpenFlow World Congress. In addition, the Industry Specification Group (ISG) was founded to promote the definition of network virtualization requirements and the formulation of the system architecture.
- In 2013, the ETSI NFV ISG conducted the first phase of research and completed the formulation of related standards. The ETSI NFV ISG defined NFV requirements and architecture and sorts out the standardization processes of different interfaces.





NFV Value

- NFV aims to address issues such as complex deployment and O&M and service innovation difficulties due to large numbers of telecom network hardware devices. NFV brings the following benefits to carriers while reconstructing telecom networks:
 - Shortened service rollout time
 - Reduced network construction cost
 - Improved network O&M efficiency
 - Open ecosystem



Key NFV Technologies: Virtualization

- Virtualization is the foundation of NFV, and cloudification is the key.
- On traditional telecom networks, each NE is implemented by dedicated hardware, resulting in high costs and difficult O&M. Virtualization features partition, isolation, encapsulation, and independence from hardware, which can meet NFV requirements. Carriers use virtualization to run software-based NEs on universal infrastructures.

Partition



Multiple VMs can concurrently run on a single physical server.

Isolation



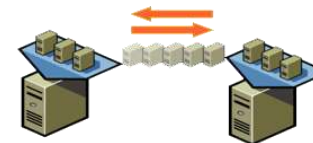
VMs that run on the same server are isolated from each other.

Encapsulation



All data of a VM is saved in files. A VM can be moved and replicated by moving and replicating the files.

Hardware independence



VMs can run on any servers without any modifications.



Key NFV Technology: Cloudification

- As defined by the National Institute of Standards and Technology (NIST), cloud computing is a model that allows users to obtain resources (for example, networks, servers, storage devices, applications, services) in a shared compute resource pool based on their needs anytime, anywhere. This model enables fast resource provisioning and release, and minimizes the resource management workload and interactions with service providers.
- Cloud computing has many advantages. Cloudification of network functions on carriers' networks mainly uses resource pooling and rapid elastic scaling.

Characteristics of Cloud Computing

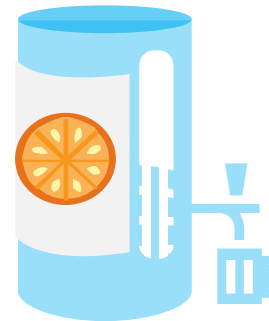
1 On-demand self-service



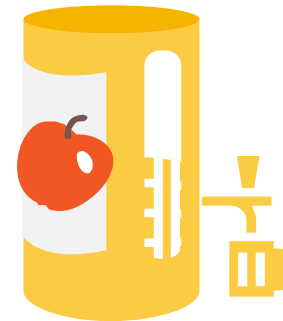
2 Broad network access



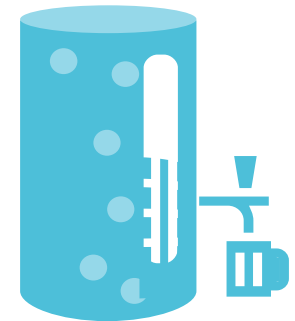
3 Resource pooling



4 Rapid elasticity



5 Measured service





Introduction to the NFV Architecture

- The NFV architecture includes the network functions virtualization infrastructure (NFVI), a virtualized network function (VNF), and management and orchestration (MANO). In addition, the NFV architecture needs to support the existing business support system (BSS) or operations support system (OSS).

OSS/BSS: is an existing operation/O&M support system.

VNF: uses cloud resources to construct software NEs.

NFVI: provides cloud-based resource pools.

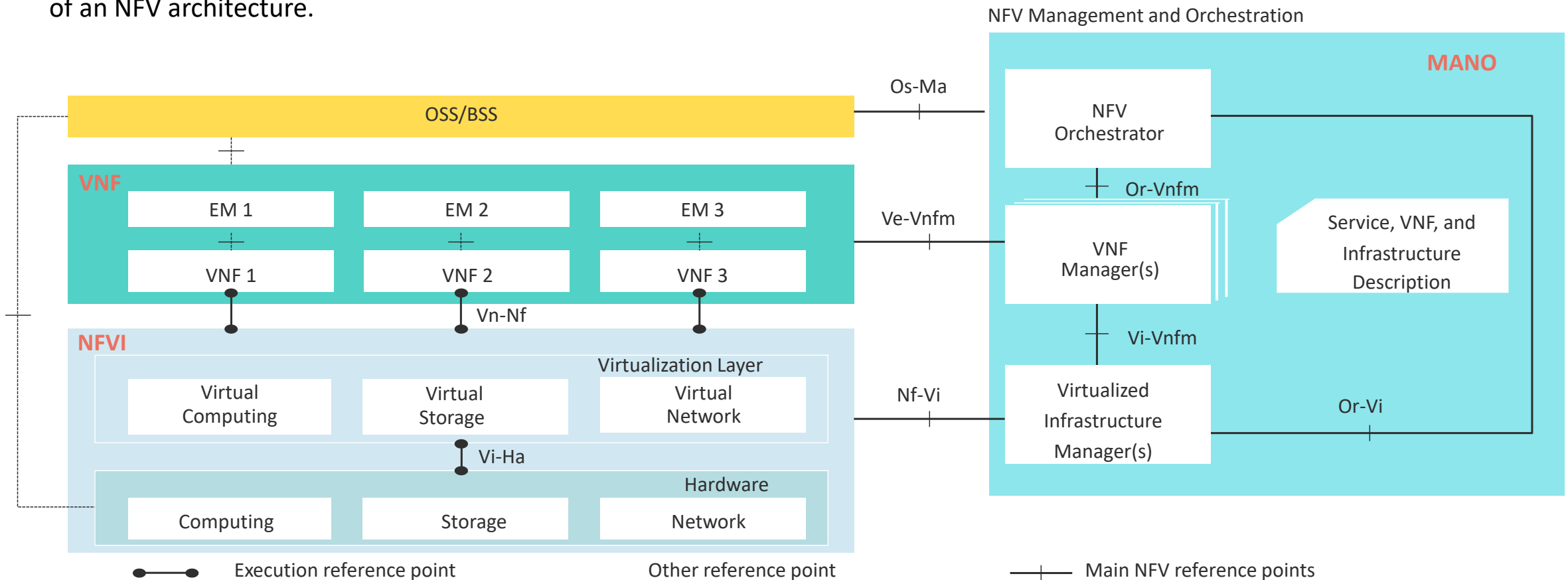
MANO:

Provides functions such as service orchestration, service management, and resource management.



Standard NFV Architecture

- ETSI defines the standard NFV architecture, which consists of the NFVI, VNF, and MANO. The NFVI includes the universal hardware layer and virtualization layer. The VNF is implemented using software, and the MANO implements management and orchestration of an NFV architecture.





Functional Modules of the NFV Architecture

- Main functional modules defined in the standard NFV architecture:

OSS or BSS

Management system for a service provider. It is not a functional component in the NFV architecture, but the MANO must provide an interface for interoperation with the OSS or BSS.

MANO

NFV management and orchestration. The MANO includes the VIM, VNFM, and NFVO, and provides unified management and orchestration for VNFs and the NFVI.

- **VIM:** NFVI management module that runs on an infrastructure site. The VIM provides functions such as resource discovery, virtual resource management and allocation, and fault handling.
- **VNFM:** It controls the VNF lifecycle (including instantiation, configuration, and shutdown).
- **NFVO:** It orchestrates and manages all the software resources and network services on an NFV network.

VNF

VNFs refer to VMs as well as service NEs and network function software deployed on the VMs.

NFVI

NFV infrastructure, including required hardware and software. The NFVI provides a running environment for VNFs.

- **Hardware layer:** includes hardware devices that provide compute, network, and storage resources.
- **Virtualization layer:** abstracts hardware resources to form virtual resources, such as virtual compute, storage, and network resources. The virtualization function is implemented by Hypervisor^[1].



NFV Architecture Interfaces

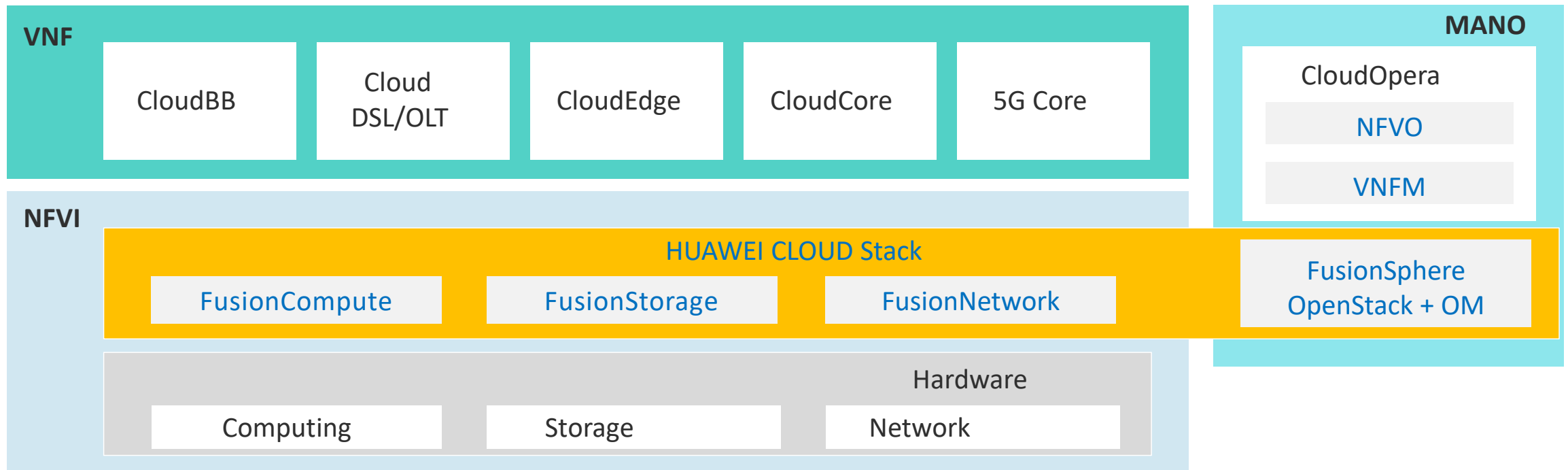
- Main interfaces of the standard NFV architecture:

Interface	Description
Vi-Ha	Is used between the virtualization layer and hardware layer. The virtualization layer meets basic hardware compatibility requirements.
Vn-Nf	Is used between a VM and the NFVI. It ensures that VMs can be deployed on the NFVI to meet performance, reliability, and scalability requirements. The NFVI meets VMs' OS compatibility requirements.
Nf-Vi	Is used between the virtualization layer management software and NFVI. It provides management of virtual computing, storage, and network systems of NFVI, virtual infrastructure configuration and connections, as well as system usage, performance monitoring, and fault management.
Ve-Vnfm	Is used between the VNFM and a VNF, implementing VNF lifecycle management, VNF configuration, VNF performance, and fault management.
OS-Ma	Manages lifecycles of network services and VNFs.
Vi-Vnfm	Is used for interaction between the service application management system or service orchestration system and virtualization layer management software.
Or-Vnfm	Sends configuration information to the VNFM, configures the VNFM, and connects the orchestrator and VNFM. It exchanges information with the NFVI resources allocated to VNFs and information between VNFs.
Or-Vi	Is used to send resource reservation and resource allocation requests required by the orchestrator and exchange virtual hardware resource configurations and status information.



Huawei's NFV Solution

- In the Huawei NFV architecture, functions of the virtualization layer and VIM are implemented by the HUAWEI CLOUD Stack NFVI platform. HUAWEI CLOUD Stack can virtualize compute, storage, and network resources and centrally manage, monitor, and optimize physical virtualization resources.
- Huawei provides cloud-based solutions for carriers' wireless networks, bearer networks, transport networks, access networks, and core networks.





FAQ

- Q1: What is the relationship between SDN and NFV in the industry?
- A: Both SDN and NFV involve network transformation and the NFV concept was proposed at the SDN and OpenFlow World Congress. However, they are independent of each other. SDN mainly affects the network architecture, and NFV mainly affects the NE deployment mode.
- Q2: What is the relationship between SDN and NFV in Huawei solutions?
- A: Huawei provides different solutions for SDN and NFV, but they are associated. Huawei NFVI solution is provided by HUAWEI CLOUD Stack.



Quiz

1. (Multiple) Which of the following statements about Huawei SDN solution are true? ()
 - A. The solution supports various SBI protocols, such as RESTful, NETCONF, and OVSDB.
 - B. OpenFlow can be used as the SBI protocol.
 - C. The solution integrates management, control, and analysis to build a simplified network.
 - D. The solution provides open and programmable network interfaces to support third-party application development and system interconnection.
2. Please briefly describe the benefits of NFV.



Summary

- With the transformation and development of the network industry, SDN and NFV are proposed.
- SDN is an innovation of network architecture. It uses a controller to make networks more open, flexible, and simple.
- NFV is an innovation in the deployment of telecom network devices. Based on virtualization and cloud computing, NFV helps reconstruct telecom networks.



More Information

- For more information about OpenFlow, visit <https://www.opennetworking.org/> .
- For more information about Huawei SDN solution, see the HCIP course.

The background of the slide features a blue-tinted image of several business professionals in a modern office environment. They are standing on a highly reflective floor, and their silhouettes are clearly visible against the lighter background. The overall aesthetic is professional and corporate.

Thank You
www.huawei.com



Network Programmability and Automation



Foreword

- New protocols, technologies, and delivery and O&M modes are emerging in the network engineering field. Conventional networks face challenges from new connection requirements, such as requirements for cloud computing and artificial intelligence (AI). Enterprises are also pursuing service agility, flexibility, and elasticity. Against this backdrop, network automation becomes increasingly important.
- Network programmability and automation is to simplify network configuration, management, monitoring, and operations for engineers and improve deployment and O&M efficiency. This course is to help network engineers understand Python programming and implement network automation.



Objectives

- On completion of this course, you will be able to:
 - Describe the difficulties of conventional network O&M.
 - Understand the implementation of network automation.
 - Understand the classification of programming languages.
 - Describe the Python code style.
 - Describe the basic usage of Python telnetlib.



Contents

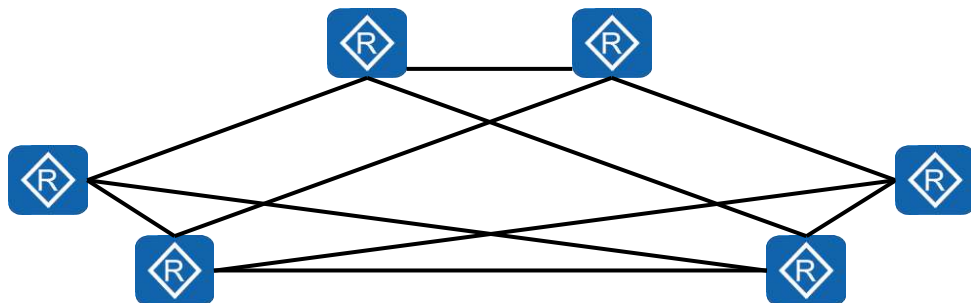
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2. Overview of Programming Language and Python
3. Cases



Background: Difficulties in Conventional Network O&M

- Conventional network O&M requires network engineers to manually log in to network devices, query and execute configuration commands, and filter command output. This highly human-dependent working mode is time-consuming, inefficient, and difficult to audit.

Numerous devices
Complex operations
Low efficiency



Network device

Typical O&M Scenarios

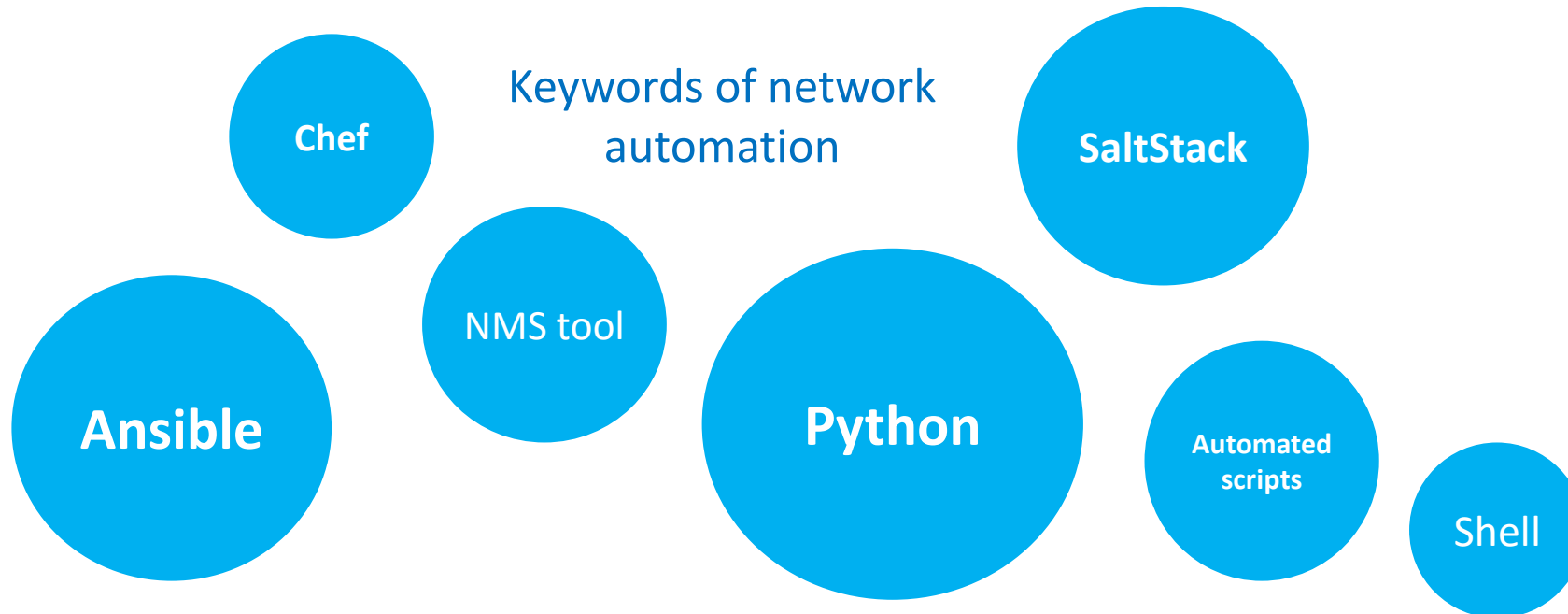
Are the following working scenes familiar to you?

1. Device upgrade: Thousands of network devices reside on a live network. You have to periodically upgrade the devices in batches.
2. Configuration audit: An enterprise needs to audit the configuration of devices every year. For example, the enterprise requires that STelnet be enabled on all devices and spanning tree security be configured on Ethernet switches. In this case, you have to quickly find out the devices that do not meet the requirements.
3. Configuration change: Due to network security requirements, device accounts and passwords need to be changed every three months. You have to delete the original account and create an account on thousands of network devices.



Network Automation

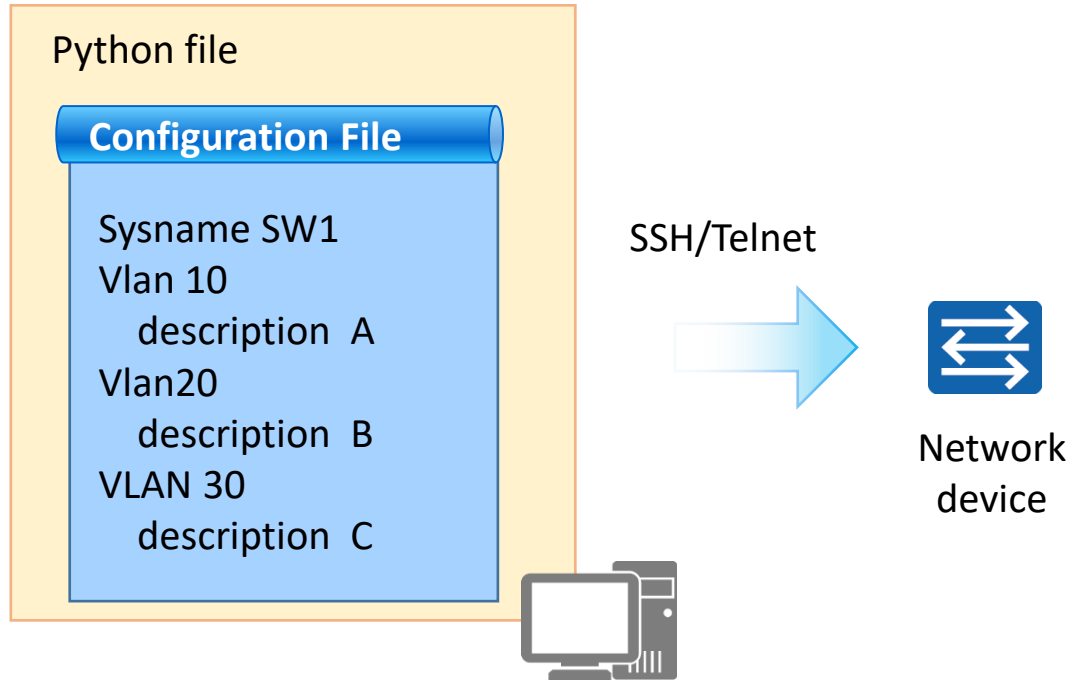
- Network automation: Tools are used to implement automated network deployment, operations, and O&M, gradually reducing dependency on human. This solves the conventional network O&M problems.
- Many open-source tools, such as Ansible, SaltStack, Puppet, and Chef, are available for network automation in the industry. From the perspective of network engineering capability construction, it is recommended that engineers acquire the code programming capability.





Programming-based Network Automation

- In recent years, with the emergence of network automation technologies, Python-based programming capabilities have become a new skill requirement for network engineers.
- Automation script written in Python can execute repeated, time-consuming, and rule-based operations.



Example: Implementing automated device configuration using Python

- What can network automation do? The most intuitive example of network automation is automated device configuration. This process can be divided into two steps: writing a configuration file, and writing Python code to push the configuration file to a device.
- Write the configuration script in command line interface (CLI) mode, and then upload the script to the device using Telnet/SSH. This method is easy to understand for network engineers who are beginning to learn network programmability and automation. This presentation describes how to implement network automation.



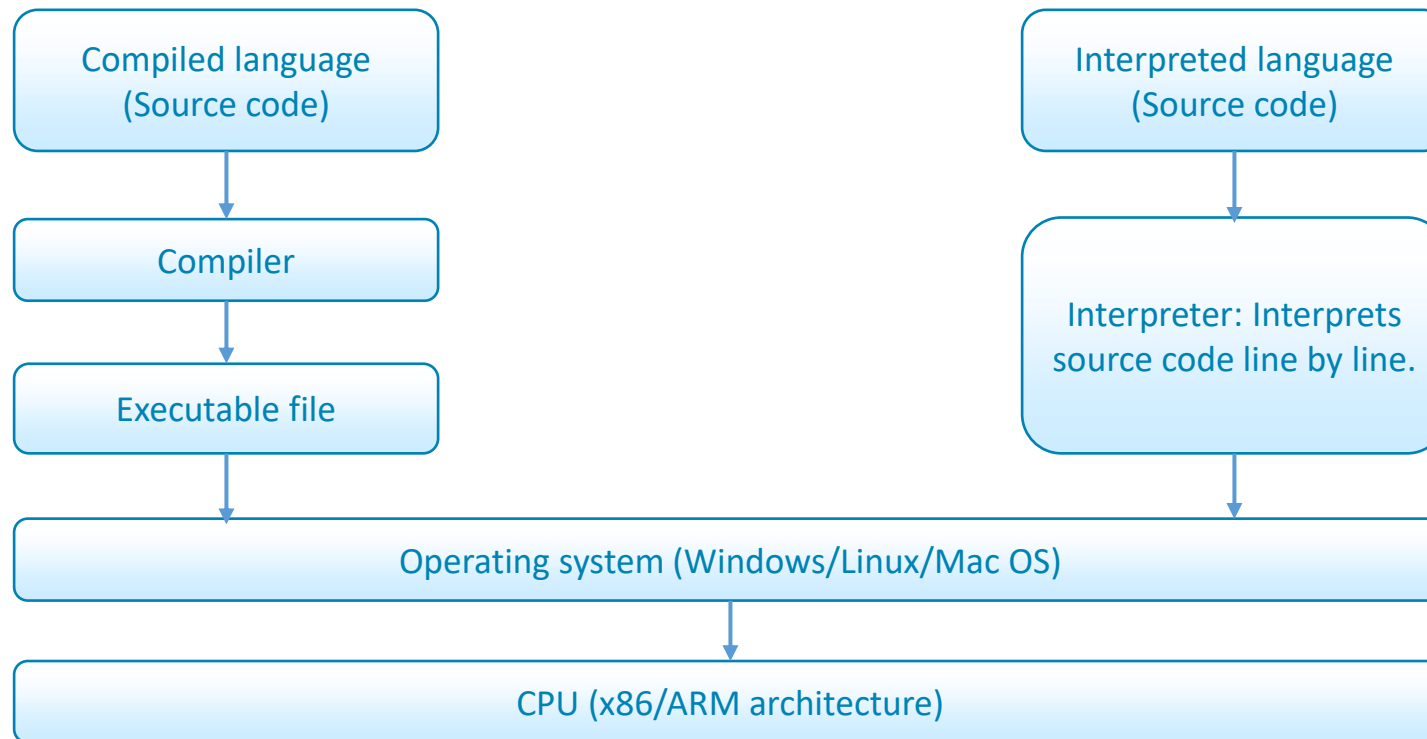
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Programming Languages

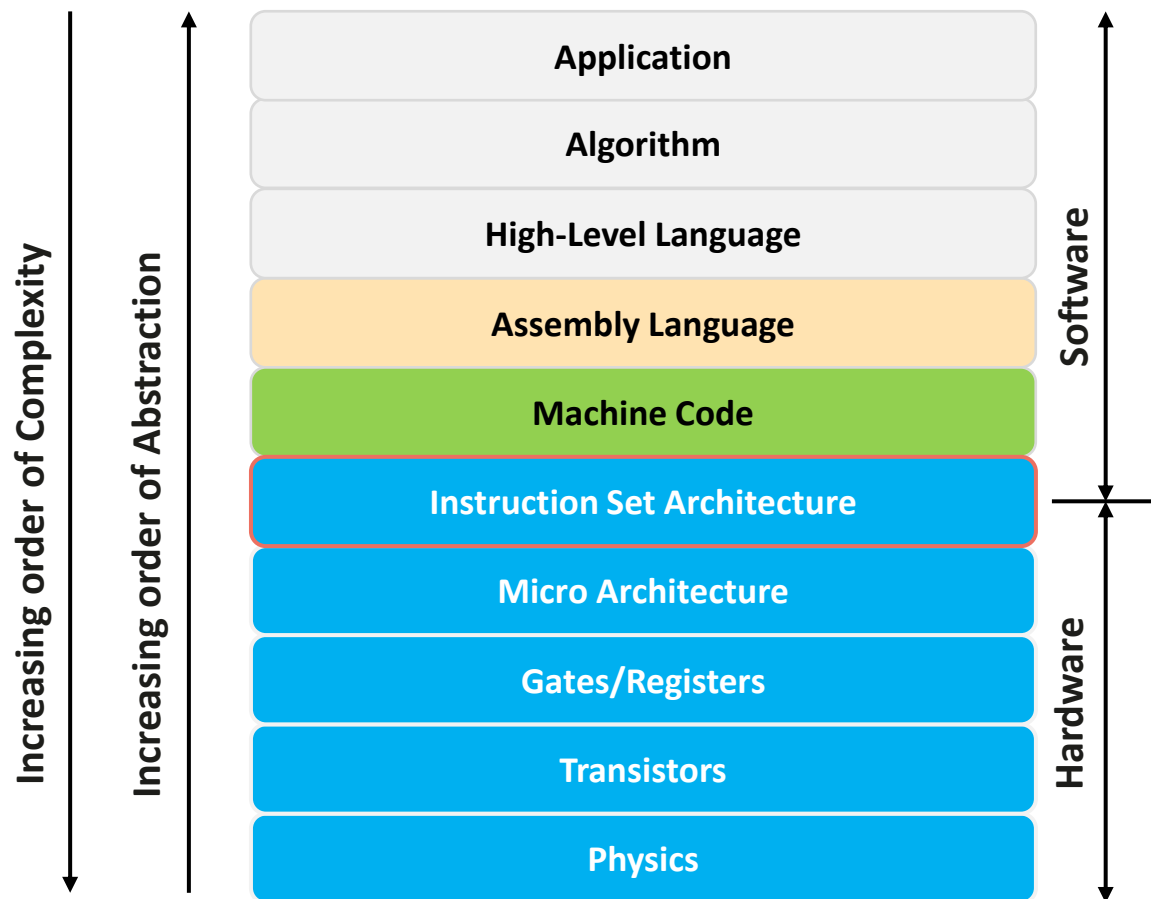
- A programming language is used to write a computer program and control behavior of a computer.
- According to whether compilation is required before execution of a language, the programming language may be classified into the compiled language, and interpreted language that does not need to be compiled.



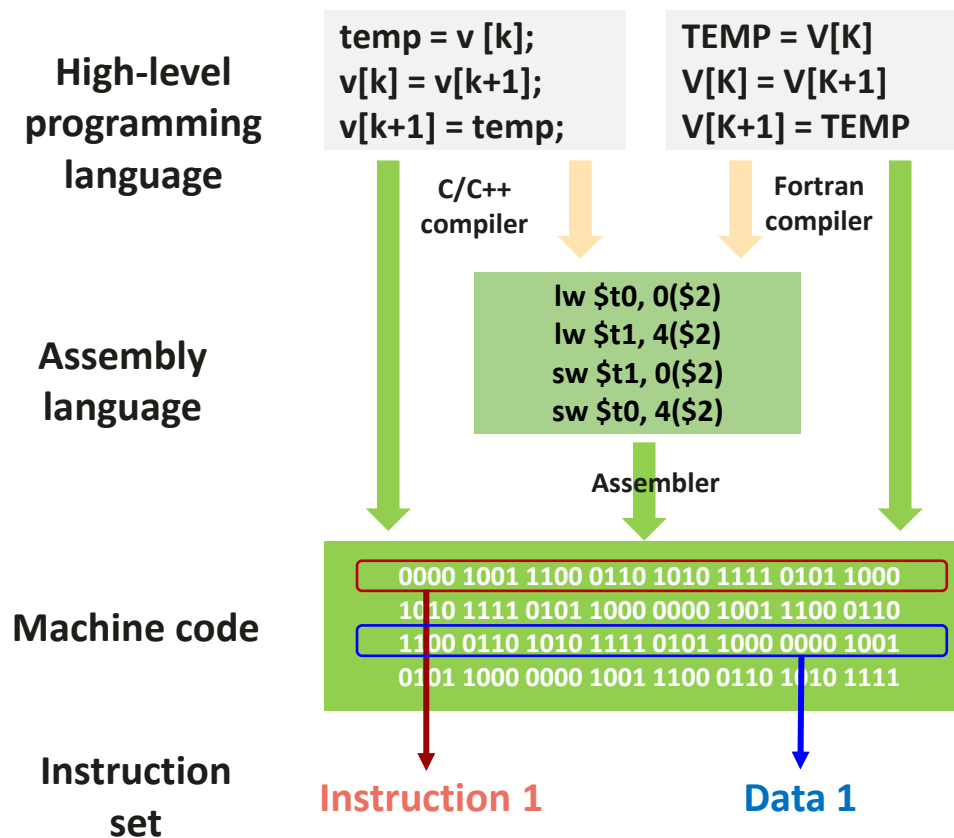


Computing Technology Stack and Program Execution Process

Computing Technology Stack



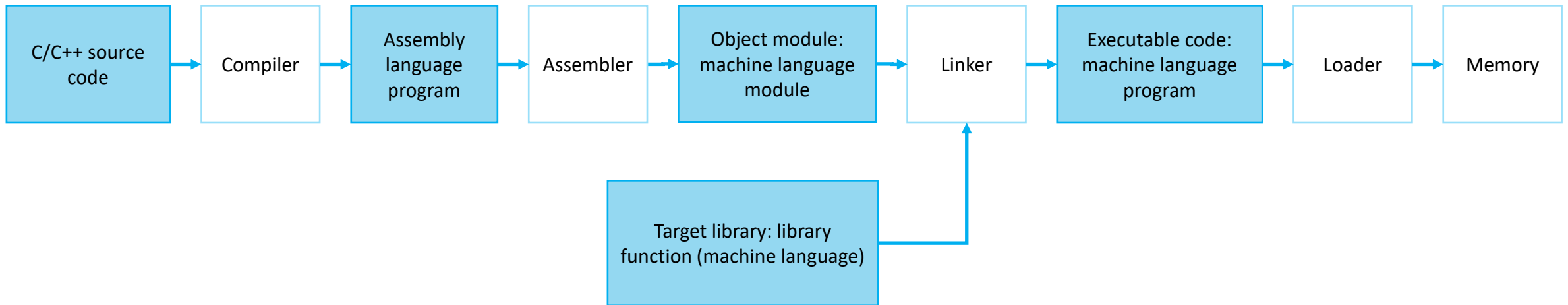
Program Execution Process





High-level Programming Language - Compiled Language

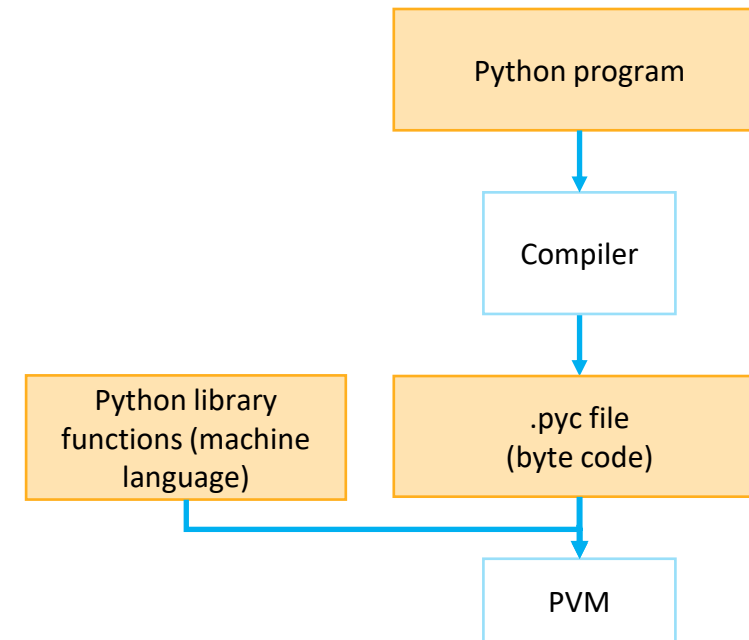
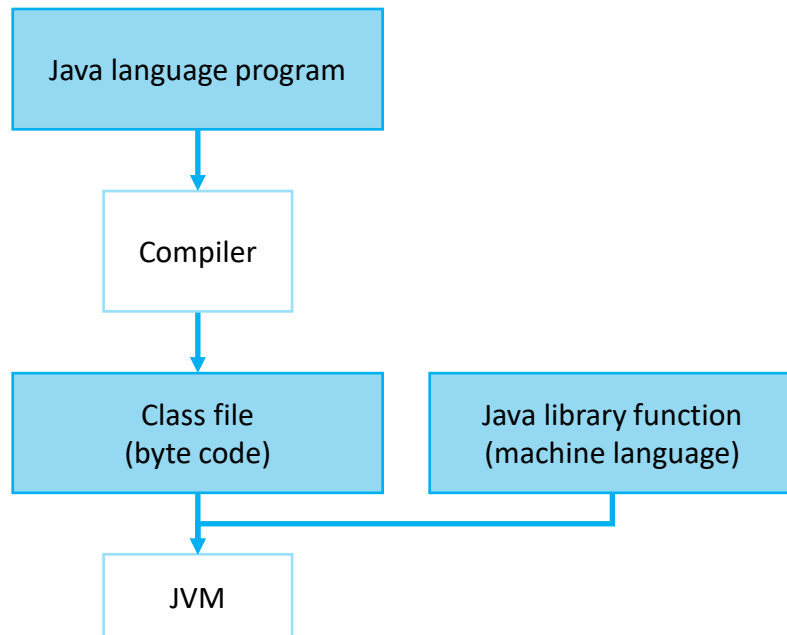
- Compiled language: Before a program in a compiled language is executed, a compilation process is performed to compile the program into a machine language file. The compilation result can be directly used without re-translation during running. Typical compiled languages include C/C++ and Go.
- From source code to program: The source code needs to be translated into machine instructions by the compiler and assembler, and then the linker uses the link library function to generate the machine language program. The machine language must match the instruction set of the CPU, which is loaded to the memory by the loader during running and executed by the CPU.





High-level Programming Language - Interpreted Language

- Interpreted language: Interpreted language programs do not need to be compiled before running. They are translated line by line when running. Typically, Java and Python are interpreted languages.
- Process from source code to programs: Source code of an interpreted language is generated by the compiler and then interpreted and executed by a virtual machine (VM) (for example, JVM/PVM). The VM shields the differences between CPU instruction sets. Therefore, portability of the interpreted language is relatively good.





What Is Python?

- Python is a fully-open-source high-level programming language. Its author is Guido Van Rossum.

Advantages of Python:

- Is a dynamically typed interpreted language with elegant syntax. It allows learners to focus on program logic rather than syntax detail learning.
- Supports both process- and object-oriented programming.
- Provides abundant third-party libraries.
- Is nicknamed the glue language because it can call code written in other languages.

Disadvantages of Python:

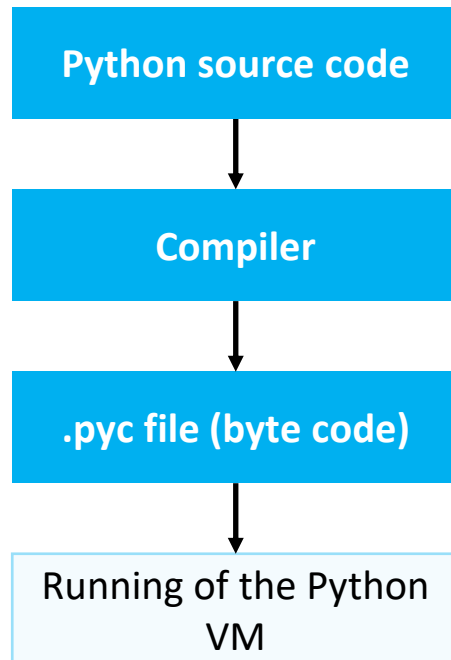
- Runs slow. Is an interpreted language that runs without being compiled. Code is translated line by line at run time into machine code that the CPU can understand, which is time-consuming.

With support for abundant third-party libraries and advantages of the Python language, Python can be used in many fields, such as AI, data science, apps, and scripts for automated O&M.



Python Code Execution Process

Process of compiling and running a Python program



Operation

1. Install Python and the running environment in an operating system.
2. Compile Python source code.
3. The compiler runs the Python source code and generates a .pyc file (byte code).
4. A Python VM converts the byte code into the machine language.
5. Hardware executes the machine language.



Getting Started with Python Code - Interactive Running

- Python runs in either interactive or script mode.
- Interactive programming does not require script files to be created, and code is written in the interactive mode of the Python interpreter.

1. Input --
2. Output --
3. Input --
4. Input --
5. Input --
6. Output --

```
C:\Users\Richard>python
Python 3.7.4 (default, Aug 9 2019, 18:34:13) [MSC v.1915 64 bit (AMD64)] :: Anaconda, Inc.
on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print ("hello world")
hello world
>>> a = 1
>>> b = 2
>>> print ( a + b )
3
>>>
```



Getting Started with Python Code - Script-based Running

- Code in script mode can run on various Python compilers or in integrated development environments. For example, IDLE, Atom, Visual Studio, Pycharm, and Anaconda provided by Python can be used.

demo.py

```
print ("hello world")  
a = 1  
b = 2  
print ( a + b )
```

1 Write a Python script file (.py).

1. Input --
2. Output --
3. Output --

```
C:\Users\Richard>python demo.py  
hello world  
3
```

2 Execute the script file.



Code Style Guide for Python

- Code style rules refer to naming rules, code indentation, and code and statement segmentation modes that must be complied with when Python is used to write code. Good style rules help improve code readability and facilitate code maintenance and modification.
- For example, the following rules for using semicolons, parentheses, blank lines, and spaces are recommended:

Semicolon

- A semicolon can be added at the end of a line in Python, but is not recommended to separate statements.
- It is recommended that each sentence be in a separate line.

Blank line

- Different functions or statement blocks can be separated by spaces. A blank line helps differentiate two segments of code, improving code readability.

Parentheses

- Parentheses can be used for the continuation of long statements. Avoid unnecessary parentheses.

Space

- Spaces are not recommended in parentheses.
- You can determine whether to add spaces on both ends of an operator.



Code Style Guide for Python - Identifier Naming

- A Python identifier represents the name of a constant, variable, function, or another object.
- An identifier is usually composed of letters, digits, and underscores, but cannot start with a digit. Identifiers are case sensitive and must be unique. If an identifier does not comply with the rules, the compiler will output a `SyntaxError` message when running the code.

1. Assign a value

--

```
User_ID = 10
```

2. Assign a value

--

```
user_id = 20
```

3. Assign a string

--

```
User_Name = 'Richard'
```

4. Assign a value

--

```
Count = 1 + 1
```

5. Incorrect identifier

--

```
4_passwd = "Huawei"
```

```
print ( User_ID )
```

```
print ( user_id )
```

```
print ( User_Name )
```

```
print ( Count )
```

```
print ( 4_passwd )
```

`print()` is a built-in function of Python and is used to output content in parentheses.

Question: What is the output of the print command on the right?



Code Style Guide for Python - Code Indentation

- In Python programs, code indentation represents the scope of a code block. If a code block contains two or more statements, the statements must have the same indentation. For Python, code indentation is a syntax rule that uses code indentation and colons to distinguish between layers of code.
- When writing code, you are advised to use four spaces for indentation. If incorrect indentation is used in the program code, an IndentationError error message is displayed during code running.

Correct indentation --

Correct indentation --

Incorrect indentation --

```
if True:
    print ("Hello")
else:
    print (0)

a = "Python"
    print (a)
```



Code Style Guide for Python - Using Comments

- Comments are explanations added to programs to improve program readability. In the Python program, comments are classified into single-line comments and multi-line comments.
- A single-line comment starts with a pound sign (#).
- A multi-line comment can contain multiple lines, which are enclosed in a pair of three quotation marks ('''...''' or '''' ... '''').

Single-line comment --

```
# Assign a string to a.  
a = "Python"  
print (a)
```

Multi-line comment --

```
''''  
The output is Python.  
''''
```




Code Style Guide for Python - Source Code File Structure

- A complete Python source code file generally consists of interpreter and encoding format declaration, document string, module import, and running code.
- If you need to call a class of a standard library or a third-party library in a program, use "import" or "from... import" statement to import related modules. The import statement is always after the module comment or document string (docstring) at the top of the file.

Interpreter declaration	--	<code>#!/usr/bin/env python</code>
Encoding format declaration	--	<code>#-*- coding:utf-8 -*-</code>
Module comment or document string	--	Description of a document (docstring) This document is intended for... <code>"""</code>
Time when a module is imported	--	<code>import time</code>
Code is running	--	<code>...</code>



Python Functions and Modules

- A function is a block of organized, reusable code that is used to perform a single, related action. It can improve the modularity of the program and code utilization. The function is defined using the `def` keyword.
- A module is a saved Python file. Modules can contain definitions of functions, classes, and variables that can then be utilized in other Python programs. The only difference between a module and a regular Python program is that the module is used for importing by other programs. Therefore, a module usually does not have a `main` function.

`demo.py`

```
def sit(): #Define a function.  
    print ('A dog is now sitting')  
  
sit() #Call a function.
```

Execution result:

A dog is now sitting.

1 Write a Python file (.py).

`test.py`

```
import demo #Import a module.  
  
demo.sit() #Call a function.
```

Execution result:

A dog is now sitting.
A dog is now sitting.

2 Import a module.



Python Classes and Methods

- A class is a collection of properties and methods that are the same. The class keyword is used to define a class.
- The function of an instantiated class is called a method. When you define a method, a class must carry the **self** keyword, which indicates the instance of the class.

demo.py

```
class Dog(): #Define a class.  
    def sit(self): #Define a method.  
        print("A dog is now sitting.")  
  
Richard = Dog() #The class is instantiated.  
print (type(Richard.sit)) #The function of an instantiated type is called a  
method.  
print (type(Dog.sit)) #The type is function.
```

Execution result:

```
<class 'method'>  
<class 'function'>
```

1 Write a Python file (.py).

test.py

```
import demo  
  
demo.Dog.sit
```

Execution result:

```
A dog is now sitting.  
<class 'method'>  
<class 'function'>
```

2 Import a module.



Introduction to telnetlib

- telnetlib is a module in the standard Python library. It provides the telnetlib.Telnet class for implementing the Telnet function.
- Different methods in the telnetlib.Telnet class are called to implement different functions.

Import the Telnet class of the telnetlib module. --
Create a Telnet connection to a specified server. --
Invoke the read_all() method. --

```
from telnetlib import Telnet
tn = Telnet(host=None, port=0[, timeout])
tn.read_all()
...
```

Method	Function
Telnet.read_until (expected, timeout=None)	Read data until a given byte string, <i>expected</i> , is encountered or until <i>timeout</i> seconds have passed.
Telnet.read_all ()	Read all data until EOF as bytes; block until connection closed.
Telnet.read_very_eager()	Read everything that can be without blocking in I/O (eager). Raise EOFError if connection closed and no cooked data available. Return b" if no cooked data available otherwise. Do not block unless in the midst of an IAC sequence.
Telnet.write(buffer)	Write a byte string to the socket, doubling any IAC characters.
Telnet.close()	Close the connection.



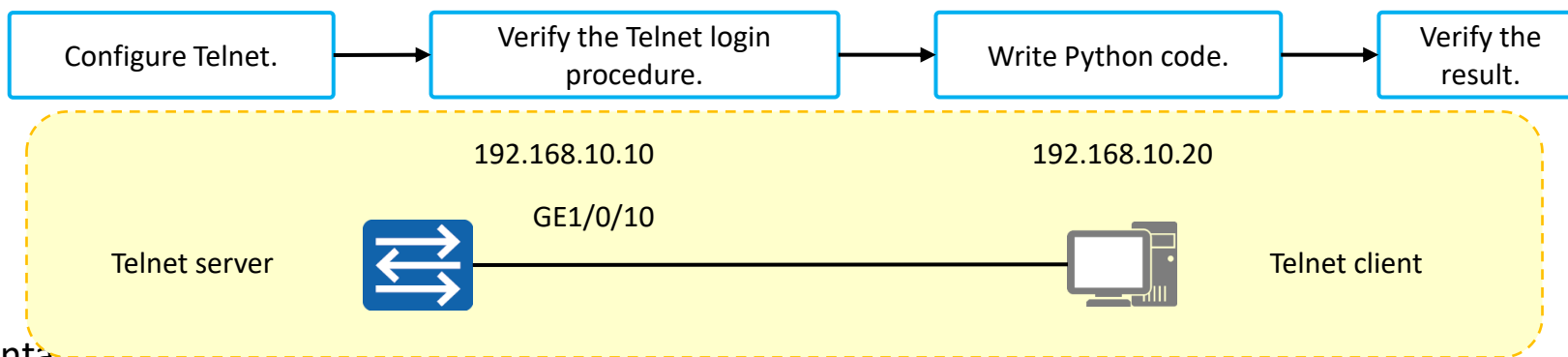
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Case: Logging In to a Device Using telnetlib

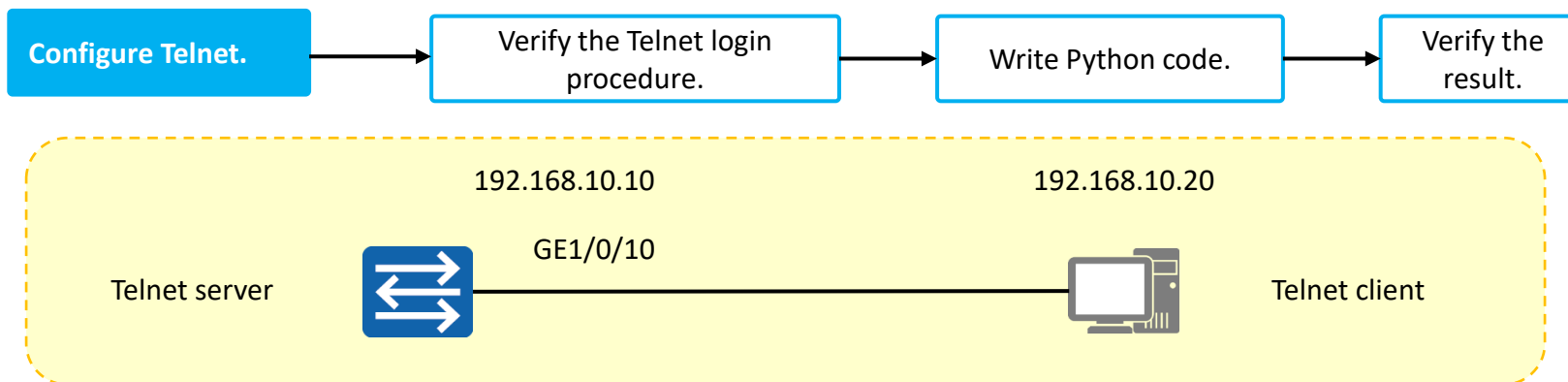
- Case description :
- A network device functions as a Telnet server, and the Python telnetlib needs to be used as a Telnet client to log in to the device.



- The implementation process is as follows.
- Configure the Telnet service.
- Manually verify and view the Telnet login procedure as a reference for code implementation.
- Compile and run Python code.
- Verify the result.



Case: Logging In to a Device Using telnetlib



Configure the IP address of interface on the device:

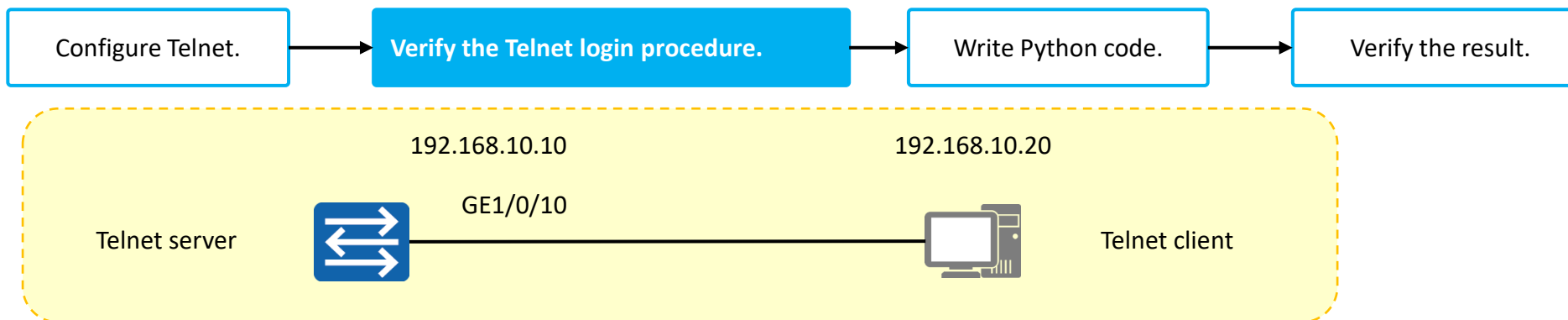
```
[Huawei] interface GE 1/0/0
[Huawei -GE1/0/0] ip add 192.168.10.10 24
[Huawei -GE1/0/0] quit
```

Configure the Telnet service:

```
[Huawei] user-interface vty 0 4
[Huawei-ui-vty0-4] authentication-mode password
[Huawei-ui-vty0-4] set authentication password simple Huawei@123
[Huawei-ui-vty0-4] protocol inbound telnet
[Huawei-ui-vty0-4] user privilege level 15
[Huawei-ui-vty0-4] quit
[Huawei] telnet server enable
```



Case: Logging In to a Device Using telnetlib



Telnet login:

- 1 Run a login command.
Command output

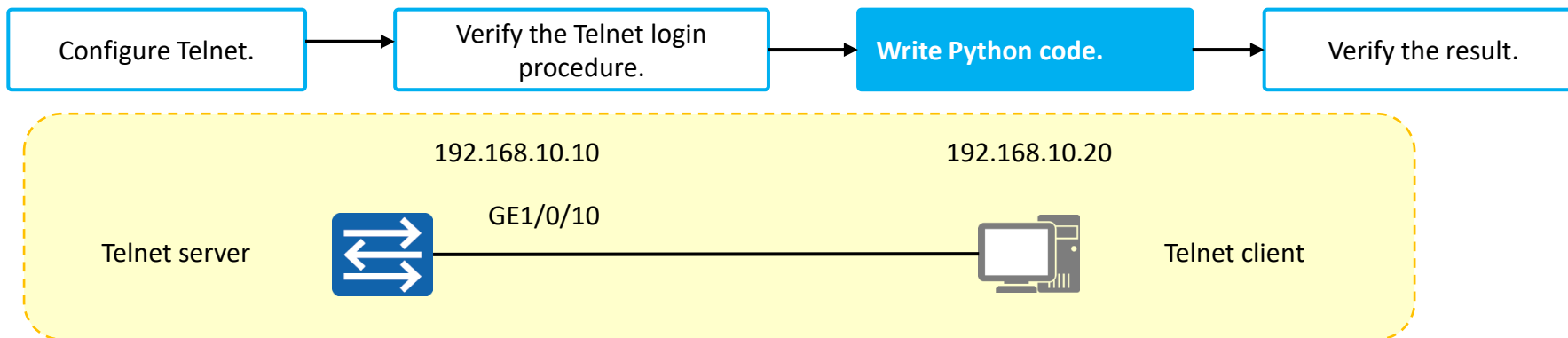
```
C:\Users\Richard>telnet 192.168.10.10
Login authentication
```

- 2 Enter a password.
Command output

```
Password:
Info: The max number of VTY users is 5, and the number of current VTY users on line is 1.
The current login time is 2020-01-15 21:12:57.
<Huawei>
```




Case: Logging In to a Device Using telnetlib



Imports the module.

Sets the IP address for a host.

Sets the password for logging in to the device.

Logs in to the host through Telnet.

Prints data until **Password:** is displayed.

Sets an ASCII password and starts a new line.

Prints data until **<Huawei>** is displayed.

Closes the Telnet connection.

```
-- import telnetlib
-- host = '192.168.10.10'
-- password = 'Huawei@123'

-- tn = telnetlib.Telnet(host)
-- tn.read_until(b'Password:')
-- tn.write(password.encode('ascii') + b"\n")
-- print (tn.read_until(b'<Huawei>').decode('ascii'))
-- tn.close()
```



Case: Running Result Comparison

Configure Telnet.

Verify the Telnet login
procedure.

Write Python code.

Verify the result.

Manual Telnet login result:

```
C:\Users\Richard>telnet 192.168.10.10
```

```
Login authentication
```

```
Password:
```

```
Info: The max number of VTY users is 5, and the number of current VTY users on line is 1. The current login time is 2020-01-15 21:12:57.
```

```
<Huawei>
```

Python code execution
result:

```
#Run Python code in the compiler.
```

```
Info: The max number of VTY users is 5, and the number  
of current VTY users on line is 1.
```

```
The current login time is 2020-01-15 22:12:57.
```

```
<Huawei>
```



Quiz

1. Python is a compiled language. ()
 - A. True
 - B. False
2. How to create VLAN 10 using telnetlib?



Summary

- Network automation uses tools to implement automated network deployment, operation, and O&M, gradually reducing dependency on people. You can use a programming language or tool to implement the network automation.
- Python is a fully-open-source high-level programming language that is simple syntax and is easy to learn. It has rich standard libraries and third-party libraries, which are applicable to the network engineering field.
- The telnetlib module of Python provides the telnetlib.Telnet class for implementing the Telnet function. It helps you enter the network programmability and automation world!



More Information

- For more information about Python, visit <https://www.python.org/>.

The image features a blue-tinted background showing silhouettes of several groups of business professionals in a modern office environment. They are standing on a highly reflective floor, and their reflections are clearly visible. The background also shows blurred outlines of skyscrapers, suggesting a high-rise office building. Overlaid on this scene is the text "Thank You" and the website address "www.huawei.com".

Thank You

www.huawei.com

Revision Record

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Course Code	Product	Product Version	Course Version

Author/ID	Date	Reviewer/ID	New/ Update
Zhang Linrui /zwx570554			



Typical Campus Network Architectures and Practices



Foreword

- A broad range of places, such as campuses, office spaces, and shopping malls, are covered by networks. You can access internal resources of your school, access internal printers of your company to print documents, or access the Internet to browse news through the networks.
- These networks belong to campus networks and are generally constructed by enterprises or organizations. Campus networks not only improve the operational efficiency of enterprises, but also provide network access services for external users.
- This chapter describes the basic architecture of a campus network and details how to build a campus network.



Objectives

- Upon completion of this course, you will be able to:
 - Understand the definition of campus networks.
 - Understand the typical networking architectures of campus networks.
 - Master the planning and design methods of small campus networks.
 - Master the deployment and implementation methods of small campus networks.
 - Understand the small campus network O&M concepts.
 - Understand the small campus network optimization concepts.
 - Independently complete a campus network project.

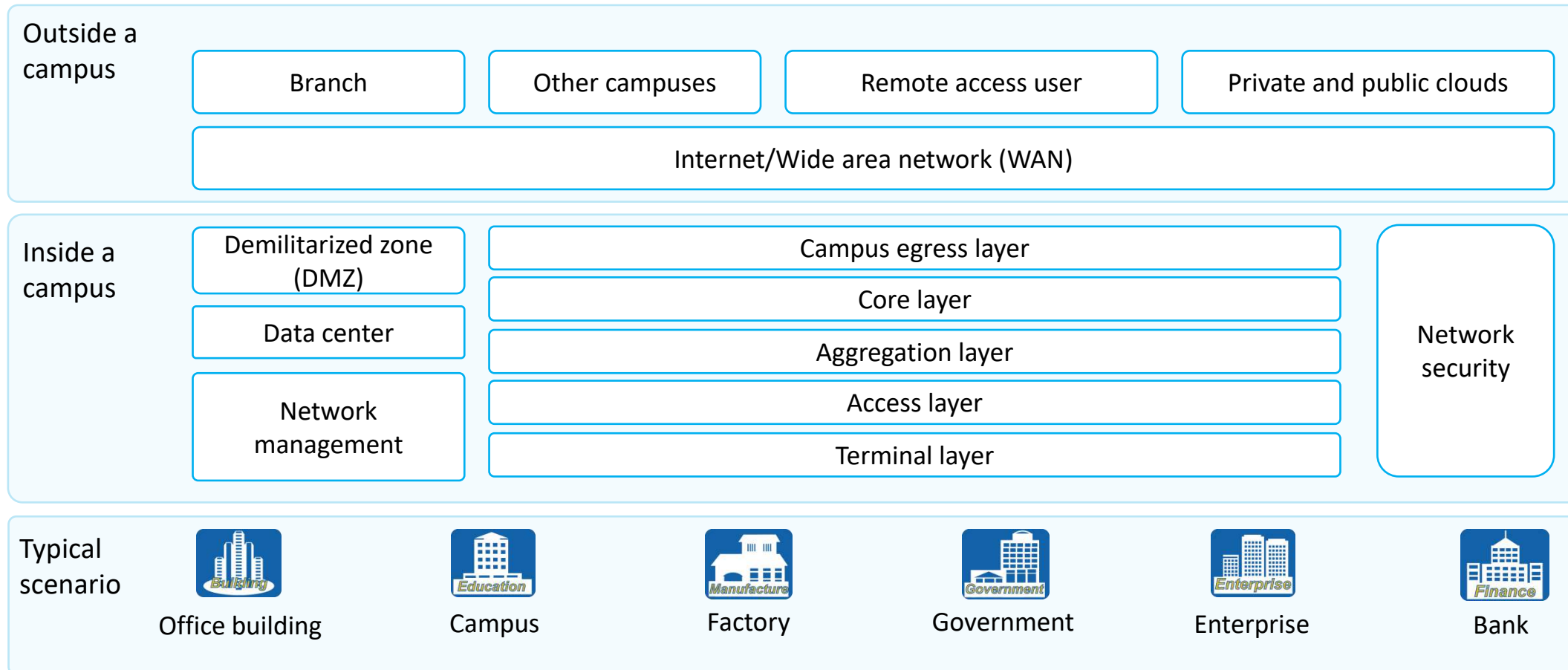


Contents

- 1. Basic Concepts of Campus Networks**
2. Typical Campus Network Construction Process



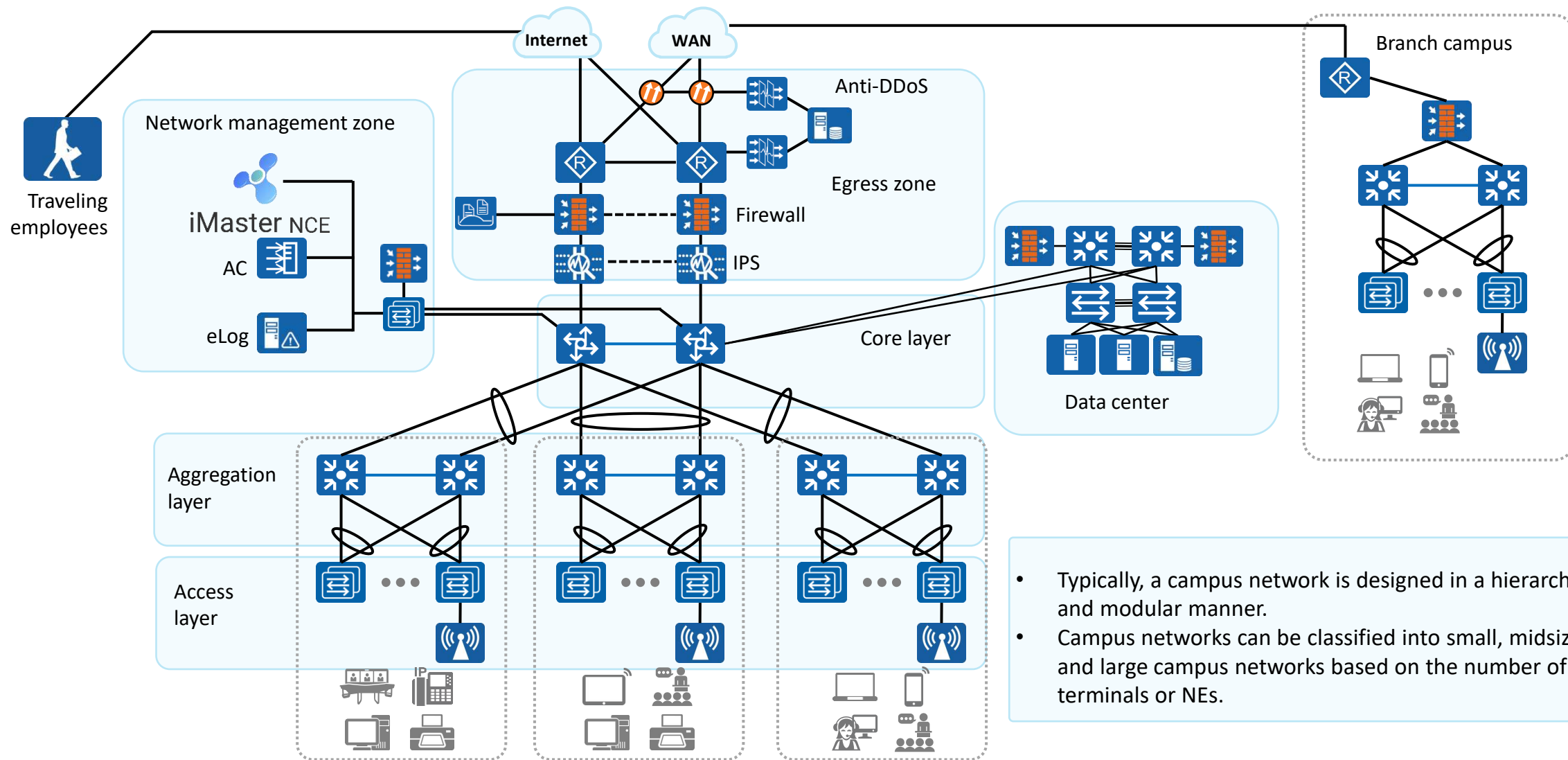
What Is a Campus Network?



A campus network is a local area network (LAN) that connects people and things in a specified area. Typically, a campus network has only one management entity. If there are multiple management entries in an area, the area is considered to have multiple campus networks.



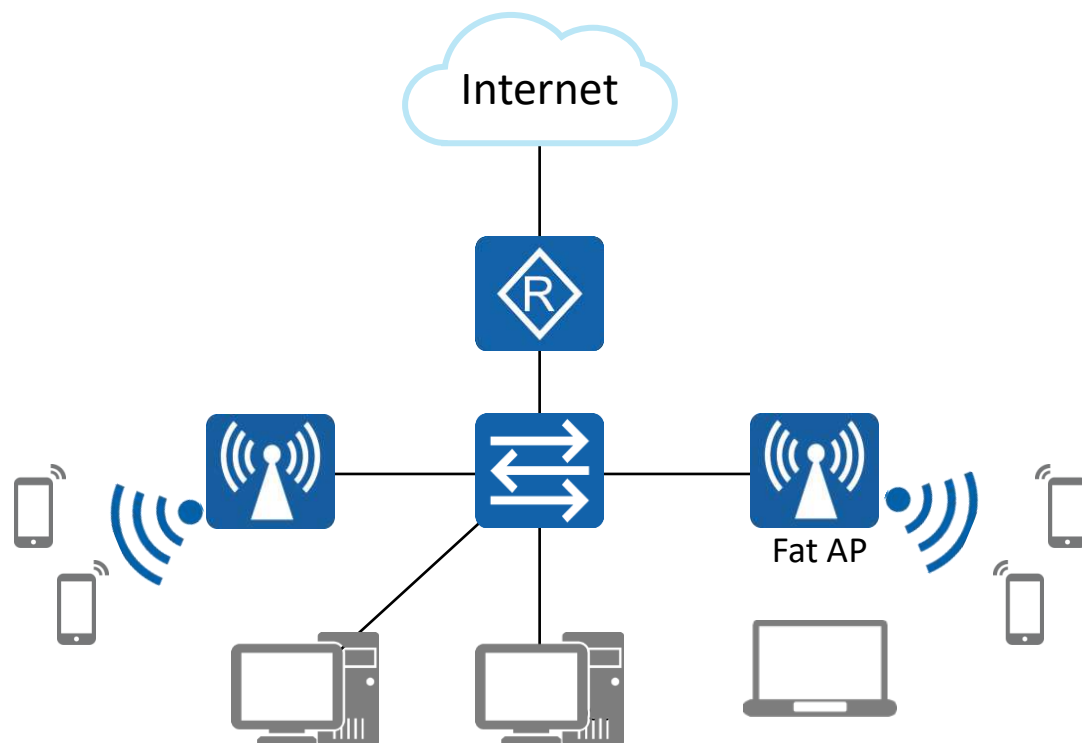
Typical Campus Network Architecture



- Typically, a campus network is designed in a hierarchical and modular manner.
- Campus networks can be classified into small, midsize, and large campus networks based on the number of terminals or NEs.



Typical Architecture of Small Campus Networks



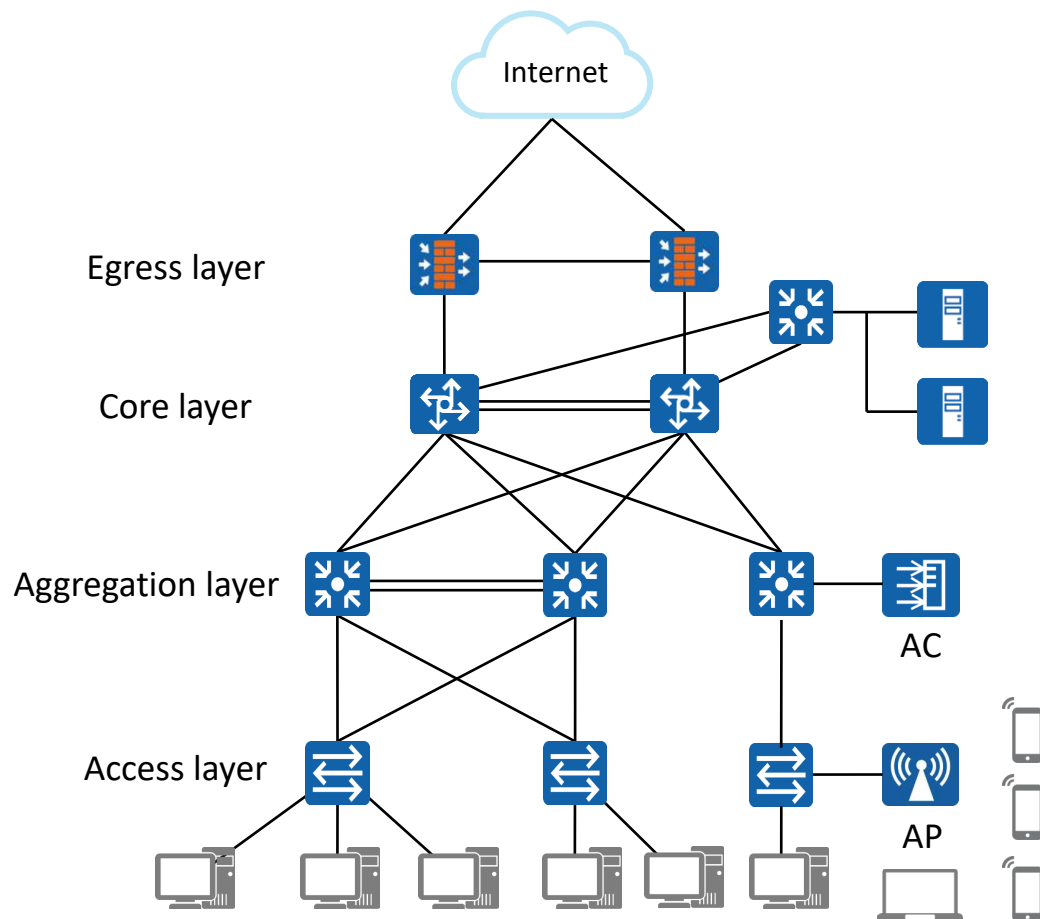
Network topology of a chain cafe

- Small campus networks are typically deployed in scenarios where the number of access users is small (several or dozens of users). A small campus network can cover only one location, has a simple architecture, and is constructed to enable mutual access between internal resources.
- Characteristics of small campus networks:
 - Small number of users
 - Only one location
 - Simple network architecture
 - Simple network requirements

Number of terminals	< 200
Number of NEs	< 25



Typical Architecture of Midsize Campus Networks



Network topology of a foreign trade company

- A midsize campus network supports access of hundreds to thousands of users.
- The modular design is introduced to midsize campus networks, that is, the networks can be partitioned by function. However, the number of function modules is small. In most cases, a midsize campus network is flexibly partitioned based on service requirements.
- Characteristics of midsize campus networks:

- Midsize network scale

- Most commonly used

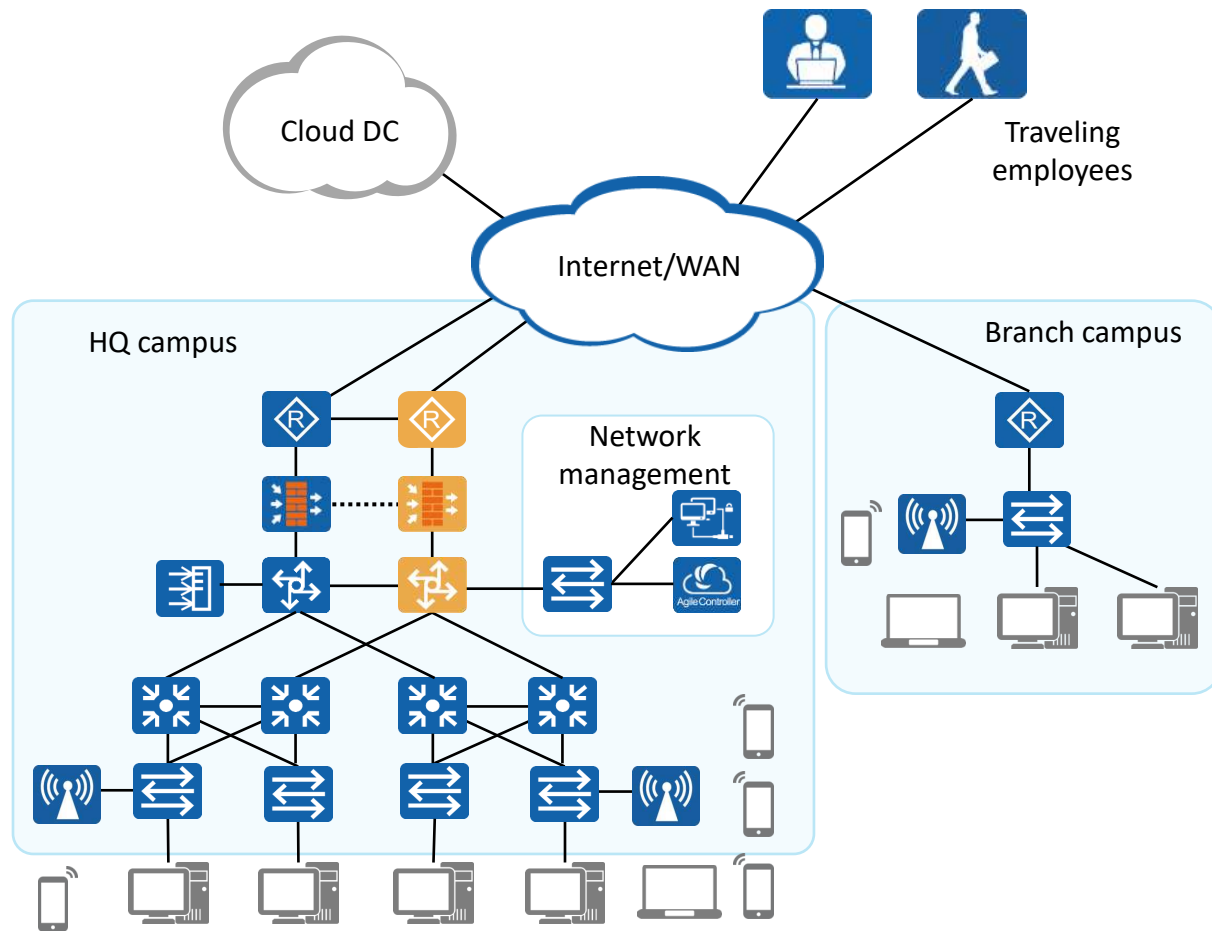
- Function partition

- Typical three-layer network architecture: core, aggregation, and access

Number of terminals	200 to 2000
Number of NEs	25 to 100



Typical Architecture of Large Campus Networks



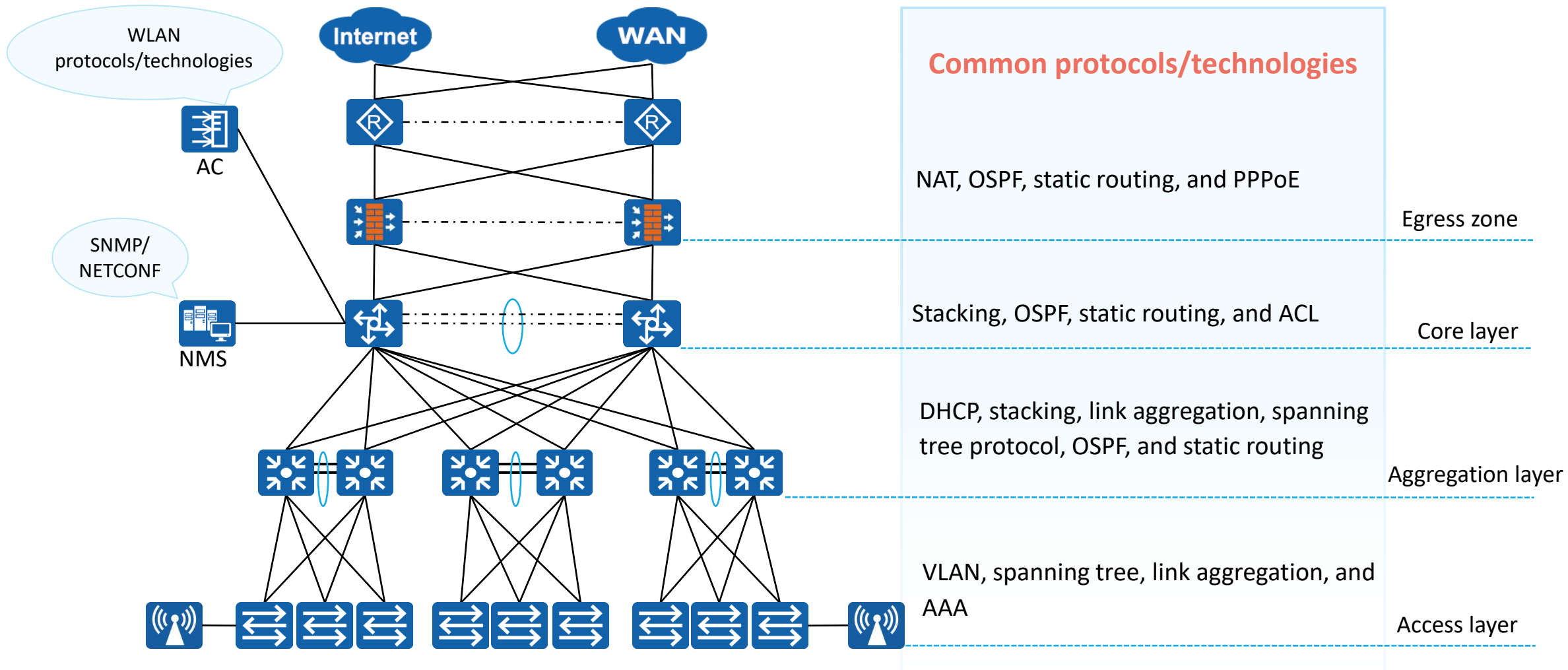
Network topology of a large enterprise

- A large campus network can cover multiple buildings and connect to multiple campuses in a city through WANs. Typically, a large campus network provides access services and allows traveling employees to access their company's internal network through technologies such as Virtual Private Network (VPN).
- Characteristics of large campus networks:
 - Wide coverage
 - Large number of users
 - Complex network requirements
 - Comprehensive function modules
 - Complex network architecture

Number of terminals	> 2000
Number of NEs	> 100



Main Protocols and Technologies of Campus Networks





Contents

1. Basic Concepts of Campus Networks
- 2. Campus Network Project Practice**



Networking Requirements

- A company (with about 200 employees) plans to build a brand-new campus network to meet service development requirements. The network requirements are as follows:
 - Meet the current services requirements of the company.
 - Use a simple network topology for easy O&M.
 - Provide wired access for employees and wireless access for guests.
 - Implement simple network traffic management.
 - Ensure network security.



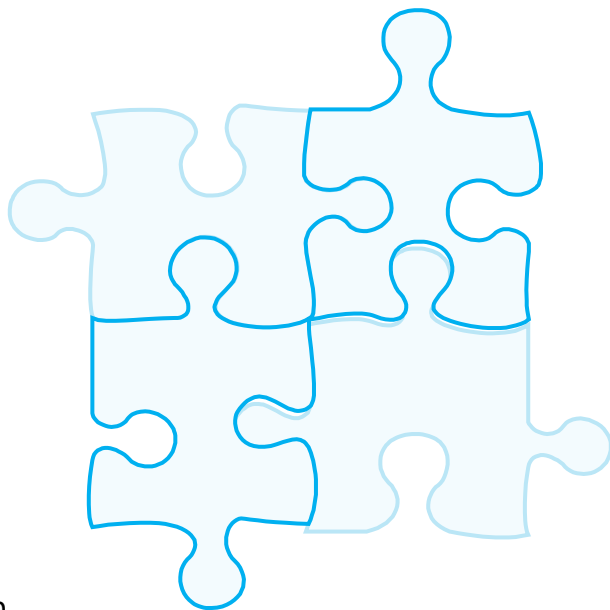
Campus Network Project Lifecycle

1 Planning and design

- Device model selection
- Physical topology
- Logical topology
- Technologies and protocols

3 Network O&M

- Routine maintenance
- Software and configuration backup
- Centralized monitoring via the network management system (NMS)
- Software upgrade



2 Deployment and implementation

- Device installation
- Single UPS commissioning
- Joint commissioning test
- Network migration and integration

4 Network optimization

- Network security improvement
- Software and configuration backup
- User experience improvement



Small Campus Network Design

Planning and
Design

Deployment and
Implementation

Network O&M

Network
Optimization

1. Networking solution design

Device model selection

Physical topology

2. Network design

Basic service

WLAN

Layer 2 loop prevention

Network reliability

3. Security design

Egress security

Intranet wired security

Intranet wireless security

4. Network O&M and management design

Basic network
management

Intelligent O&M

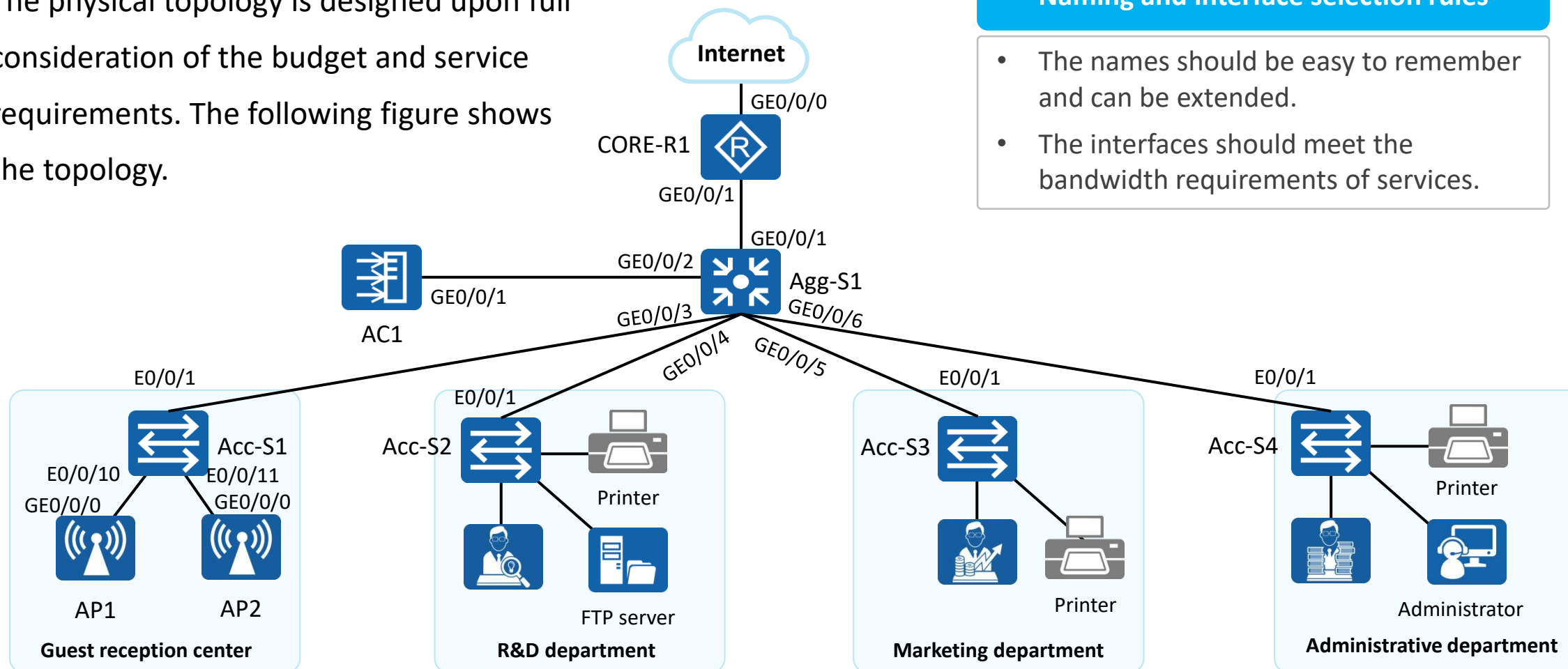


Networking Solution Design

The physical topology is designed upon full consideration of the budget and service requirements. The following figure shows the topology.

Naming and interface selection rules

- The names should be easy to remember and can be extended.
- The interfaces should meet the bandwidth requirements of services.





Basic Service Design: VLAN Design

- You are advised to assign consecutive VLAN IDs to ensure proper use of VLAN resources.
- VLANs can be classified into service VLANs, management VLANs, and interconnection VLANs as required.
- Typically, VLANs are assigned based on interfaces.

Service VLAN design

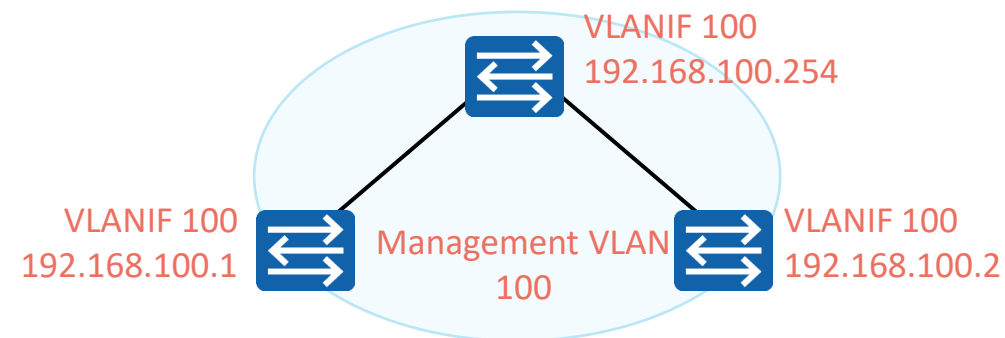
VLAN assignment by
geographic area

VLAN assignment by logical
area

VLAN assignment by
personnel structure

VLAN assignment by service
type

Management VLAN design



In most cases, Layer 2 switches use VLANIF interface addresses as management addresses. It is recommended that all switches on the same Layer 2 network use the same management VLAN and their management IP addresses be on the same network segment.



VLAN Planning

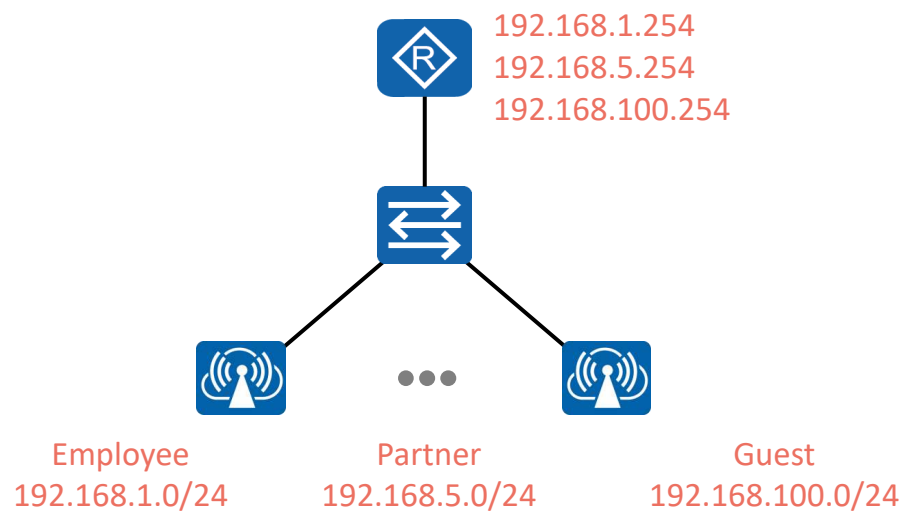
- A management VLAN is reserved for Layer 2 devices.
- VLANs are classified into the guest VLAN, R&D department VLAN, marketing department VLAN, and administrative department VLAN.
- Layer 3 switches need to be connected to routers through VLANIF interfaces. Therefore, interconnection VLANs need to be reserved.
- A VLAN is established for CAPWAP tunnels between APs and ACs.

VLAN ID	VLAN Description
1	Guest VLAN or WLAN service VLAN
2	R&D department VLAN
3	Marketing department VLAN
4	Administrative department VLAN
100	Management VLAN of Layer 2 devices
101	Management VLAN of WLAN services
102	Interconnection VLAN between Agg-S1 and CORE-R1



Basic Service Design: IP Address Design

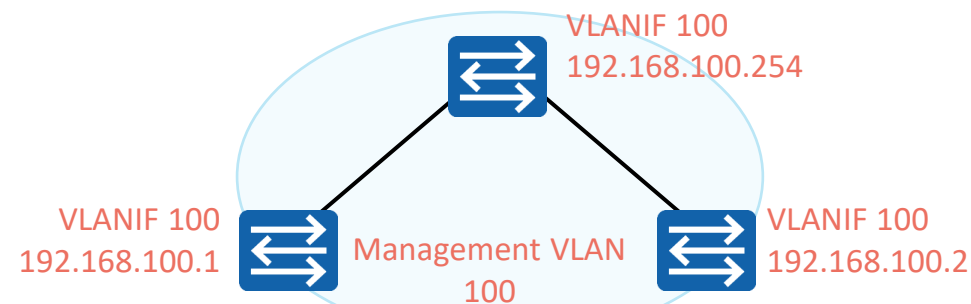
Service IP address



The service IP addresses are the IP addresses of servers, hosts, or gateways.

- It is recommended that the gateway IP addresses use the same rightmost digits, such as .254.
- The IP address ranges of different services must be clearly distinguished. The IP addresses of each type of service terminals must be continuous and can be aggregated.
- An IP address segment with a 24-bit mask is recommended.

Management IP address



Layer 2 devices use VLANIF interface IP addresses as the management IP addresses. It is recommended that all Layer 2 switches connected to a gateway use on the same network segment.

IP address for network device interconnection

It is recommended that the interconnection IP addresses use a 30-bit mask, and core devices use smaller host IP addresses.



IP Address Planning

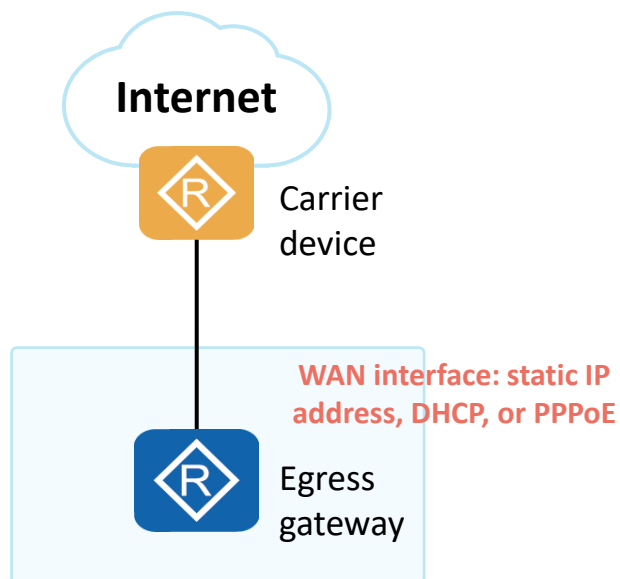
- Reserve sufficient IP addresses based on the number of clients to be accessed and plan network segments and gateway addresses for each type of service.
- Plan network segments for management IP addresses.
- Divide network segments for interconnection IP addresses.

IP Network Segment/Mask	Gateway Address	Network Segment Description
192.168.1.0/24	192.168.1.254	Network segment to which wireless access guests belong, with the gateway located on Agg-S1
192.168.2.0/24	192.168.2.254	Network segment to which the R&D department belongs, with the gateway located on Agg-S1
192.168.3.0/24	192.168.3.254	Network segment to which the marketing department belongs, with the gateway located on Agg-S1
192.168.4.0/24	192.168.4.254	Network segment to which the administrative department belongs, with the gateway located on Agg-S1
192.168.100.0/24	192.168.100.254	Management network segment of Layer 2 devices, with the gateway located on Agg-S1
192.168.101.0/24	N/A	Management network segment of WLAN services
192.168.102.0/30	N/A	Network segment between Agg-S1 and CORE-R1
1.1.1.1/32	N/A	Loopback interface address on CORE-R1, which is used as the management IP address



Basic Service Design: IP Address Allocation Mode Design

Egress gateway

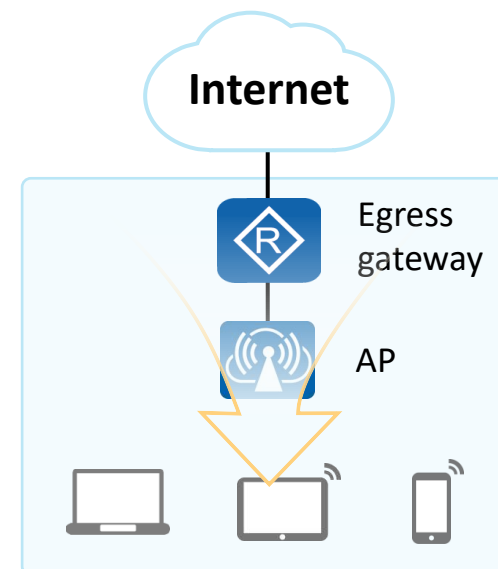


IP addresses of WAN interfaces are assigned by the carrier in static, DHCP, or PPPoE mode. The IP addresses of the egress gateways need to be obtained from the carrier in advance.

Devices such as servers and printers

It is recommended that servers and special terminals (such as punch-card machines, printing servers, and IP video surveillance devices) use statically bound IP addresses.

End users



It is recommended that IP addresses of end users are allocated by gateways through DHCP.



IP Address Allocation Mode Planning

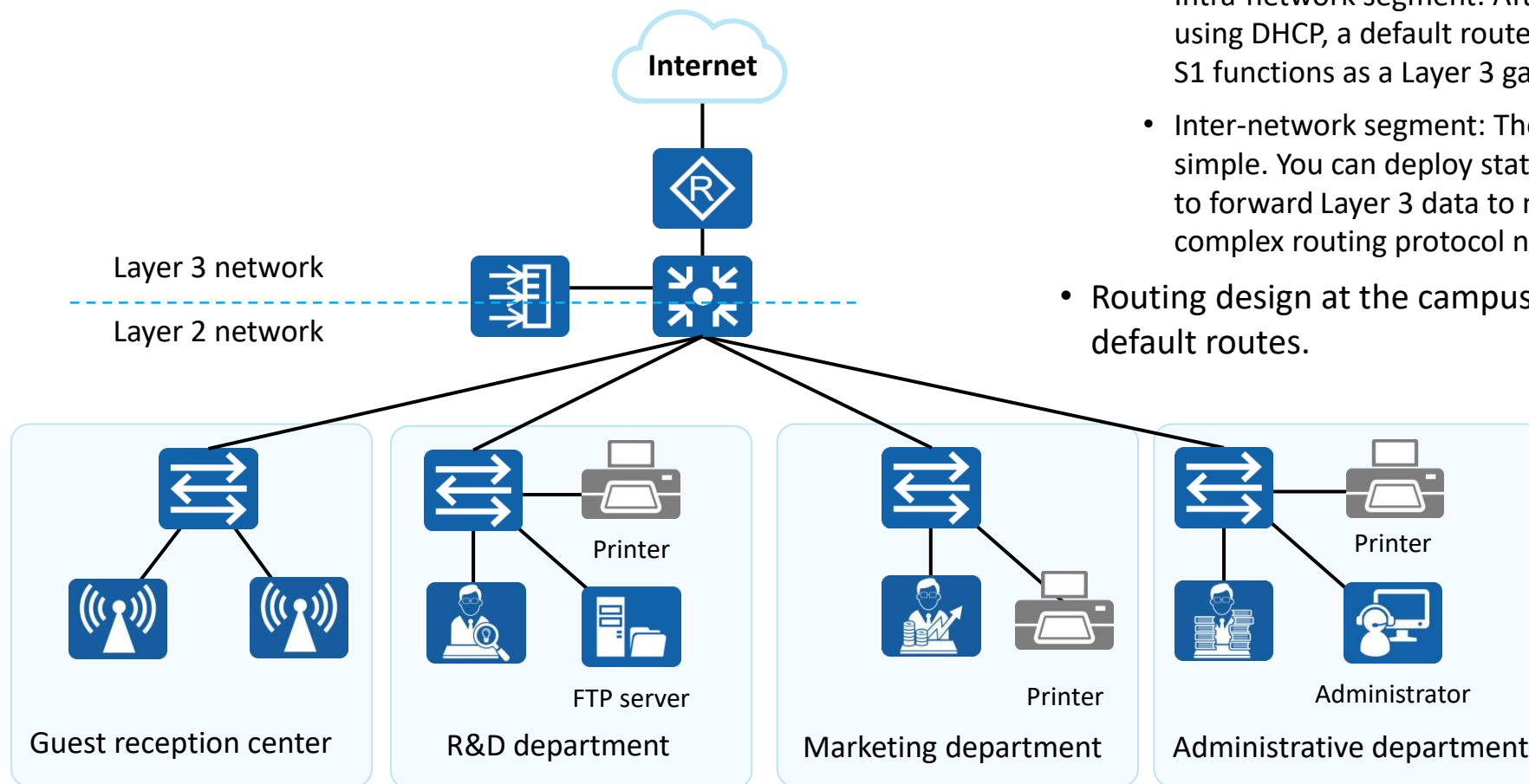
- The egress gateway obtains an IP address through PPPoE.
- All terminals obtain IP addresses through DHCP. The servers and printers are assigned fixed IP addresses.
- IP addresses of all network devices (except APs) are statically configured.

IP Network Segment/Interface	Allocation Mode	Allocation Mode Description
192.168.1.0/24 192.168.2.0/24 192.168.3.0/24 192.168.4.0/24	DHCP	Allocated by Agg-S1. Agg-S1 allocates fixed IP addresses to fixed devices such as servers and printers.
192.168.100.0/24	Static	Device management IP addresses, which are statically configured
192.168.101.0/24	DHCP	IP addresses of ACs are statically configured, and IP addresses of APs are allocated by Agg-S1.
192.168.102.0/30	Static	Interconnection IP address, which is statically configured
GE0/0/0 on CORE-R1	PPPoE	IP address assigned by the carrier



Basic Service Design: Routing Design

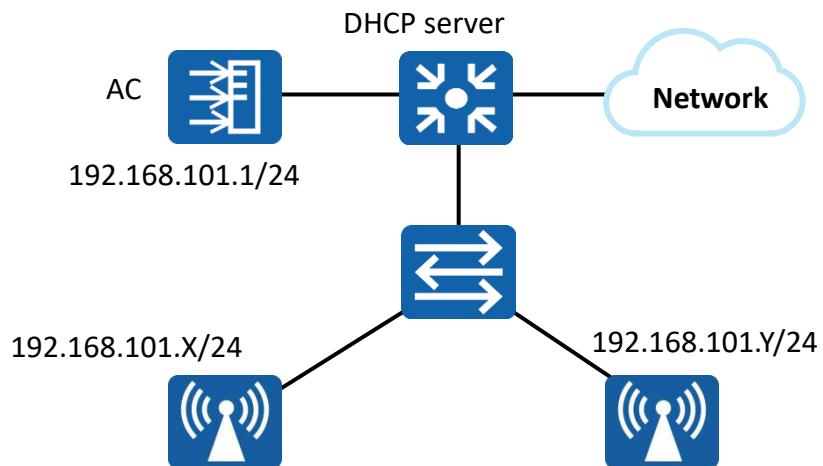
- Routing design inside a campus network:
 - Intra-network segment: After an IP address is allocated using DHCP, a default route is generated by default and Agg-S1 functions as a Layer 3 gateway.
 - Inter-network segment: The current network topology is simple. You can deploy static routes on all devices that need to forward Layer 3 data to meet the requirements. No complex routing protocol needs to be deployed.
- Routing design at the campus egress: Configure static default routes.





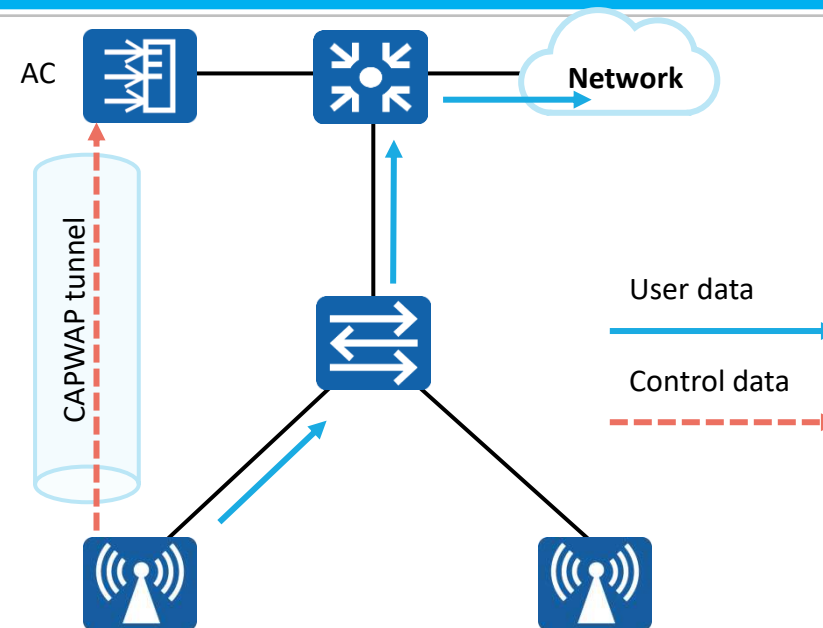
WLAN Design

WLAN networking design



- Based on the IP addresses of the AC and APs and whether data traffic passes through the AC, the networking can be divided into:
 - Inline Layer 2 networking
 - Bypass Layer 2 networking
 - Inline Layer 3 networking
 - Bypass Layer 3 networking
- This example uses the bypass Layer 2 networking.

WLAN data forwarding design



- Control packets and data packets are transmitted on a WLAN.
 - Control packets are forwarded through CAPWAP tunnels.
 - User data packets are forwarded in tunnel or direct mode.
- This example uses the direct forwarding mode.

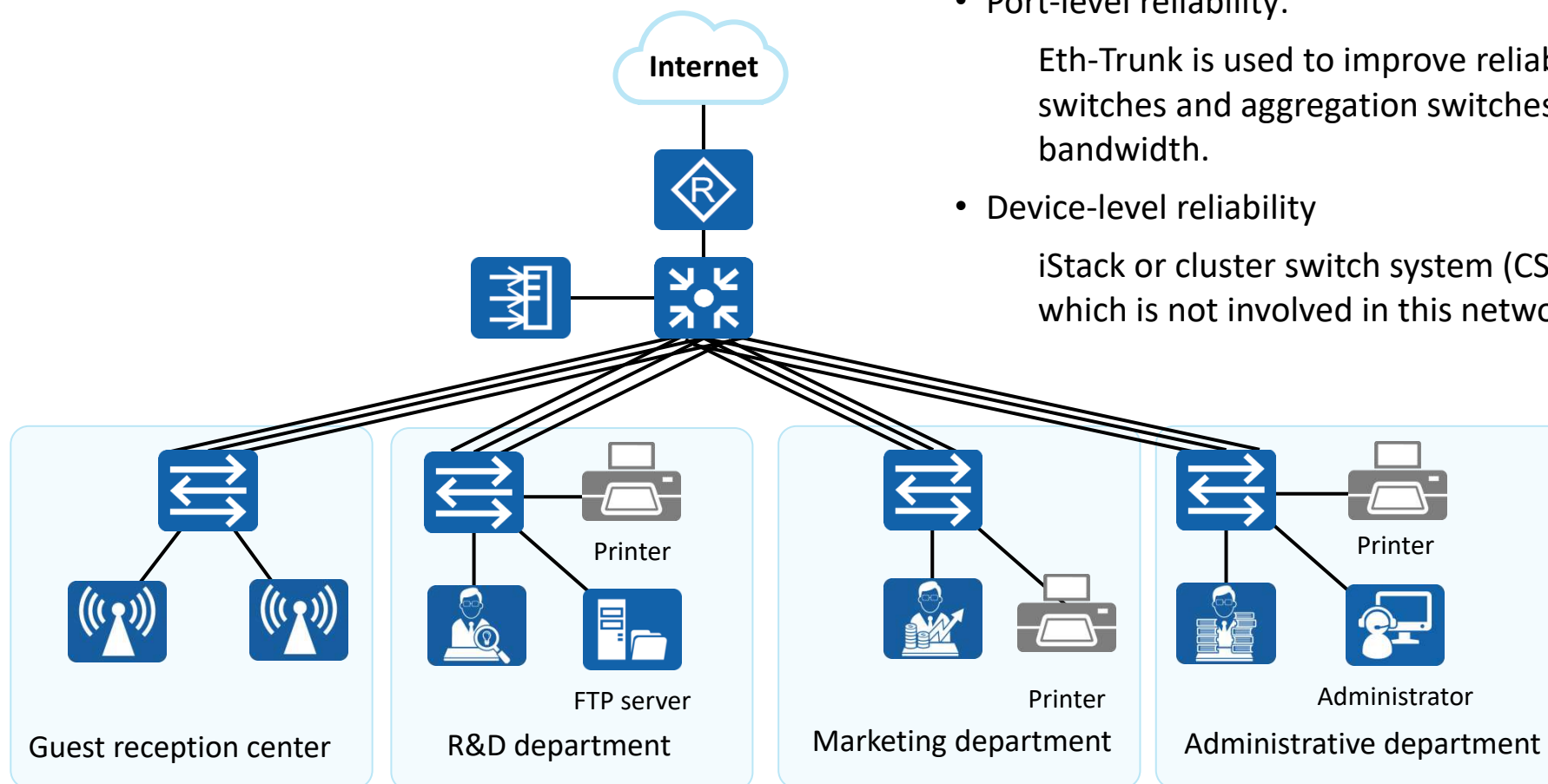


WLAN Data Plan

Item	Value
Management VLAN for APs	VLAN 101
Service VLAN for STAs	VLAN 1
DHCP server	Agg-S1 functions as a DHCP server to allocate IP addresses to APs and STAs. The default gateway address of STAs is 192.168.1.254.
IP address pool for APs	192.168.101.2 to 192.168.101.253/24
IP address pool for STAs	192.168.1.1 to 192.168.1.253/24
Source interface address of the AC	VLANIF 101: 192.168.101.1/24
AP group	Name: ap-group1 Referenced profiles: VAP profile WLAN-Guest and regulatory domain profile default
Regulatory domain profile	Name: default Country code: CN
SSID profile	Name: WLAN-Guest SSID name: WLAN-Guest
Security profile	Name: WLAN-Guest Security policy: WPA-WPA2+PSK+AES Password: WLAN@Guest123
VAP profile	Name: WLAN-Guest Forwarding mode: direct forwarding Service VLAN: VLAN 1 Referenced profiles: SSID profile WLAN-Guest and security profile WLAN-Guest



Reliability Design



- Port-level reliability:

Eth-Trunk is used to improve reliability between access switches and aggregation switches and increase link bandwidth.

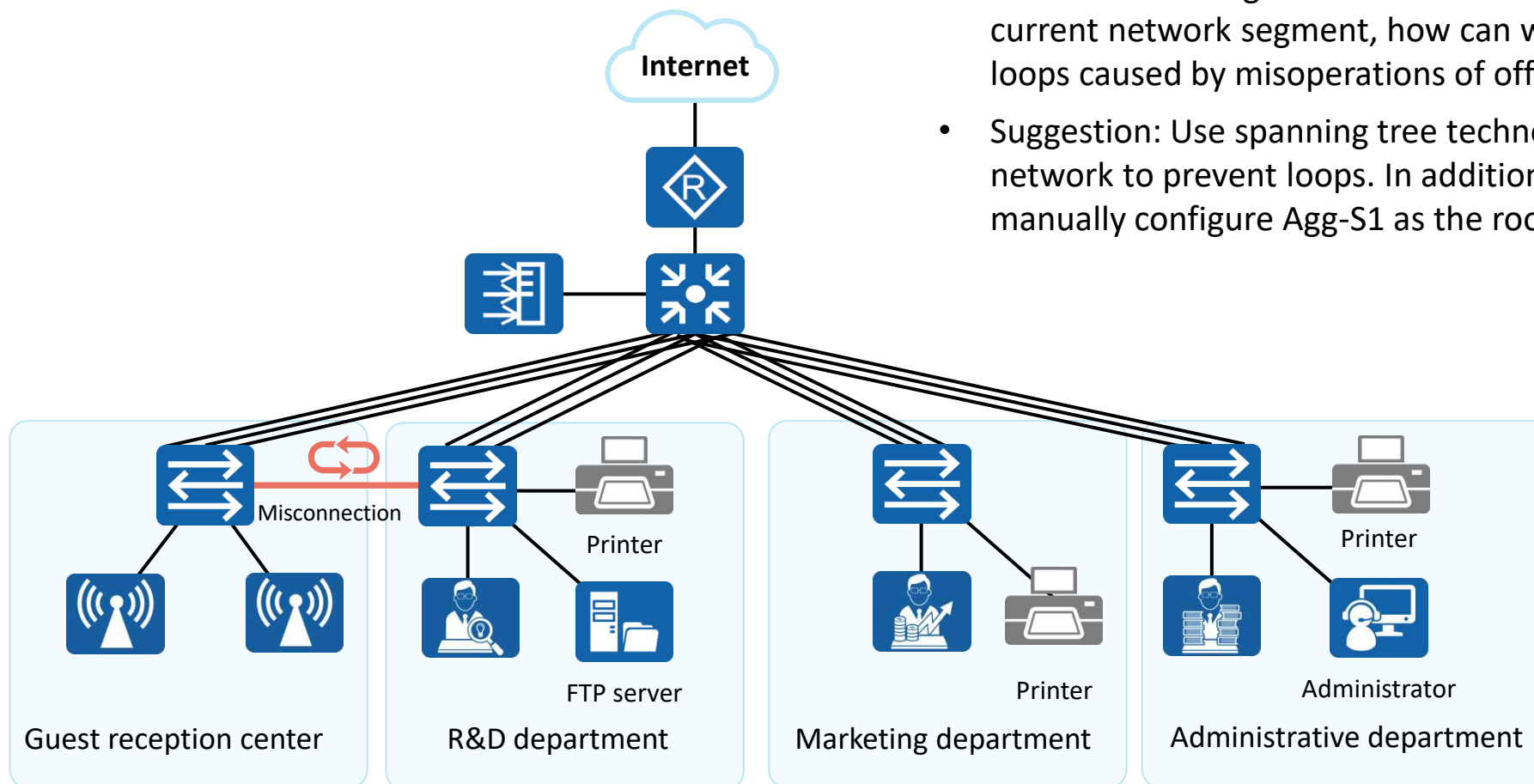
- Device-level reliability

iStack or cluster switch system (CSS) technology can be used, which is not involved in this networking.



Layer 2 Loop Prevention

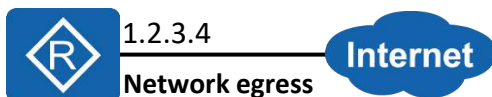
- Question: Although no redundant link is introduced to the current network segment, how can we prevent Layer 2 network loops caused by misoperations of office personnel?
- Suggestion: Use spanning tree technology on the Layer 2 network to prevent loops. In addition, you are advised to manually configure Agg-S1 as the root bridge.





Egress NAT Design

Static NAT

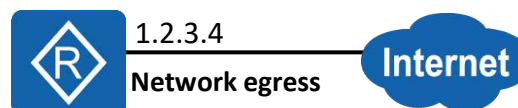


NAT mapping table

Private IP Address	Public IP Address
192.168.1.1	1.2.3.1
192.168.1.2	1.2.3.2

- Static NAT applies to scenarios where a large number of static IP addresses are configured and clients need to use fixed IP addresses.

Dynamic NAT



NAT address pool

1.2.3.1	Not in use
1.2.3.2	Not in use
1.3.3.3	Not in use

- Dynamic NAT introduces the address pool concept. Available IP addresses in the address pool are allocated to clients for Internet access.

NAPT and Easy IP



NAT mapping table

Private IP Address:Port Number	Public IP Address:Port Number
192.168.1.10:80	1.2.3.4:10335

- NAPT translates port numbers based on dynamic NAT to improve public address usage.
- Easy IP applies to scenarios where IP addresses of outbound network interfaces are dynamically allocated.

NAT Server



NAT mapping table

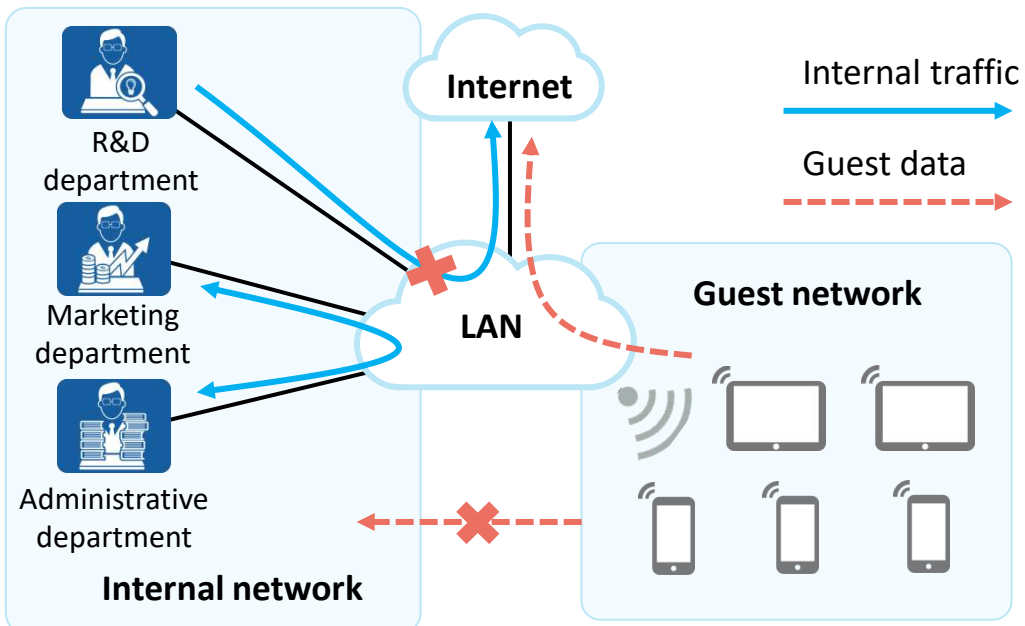
Private IP Address:Port Number	Public IP Address:Port Number
192.168.1.1:10321	1.2.3.4:1025
192.168.1.2:17087	1.2.3.4:1026

The NAT server applies to scenarios where a server on the intranet needs to externally provide services.



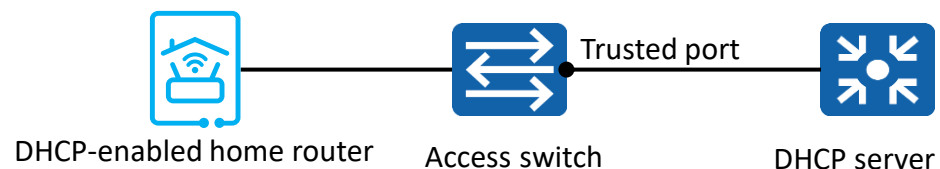
Security Design

Traffic Control



- Different departments can access each other but cannot access the Internet.
- Guests can access the Internet but cannot access the internal network.
- You can use technologies such as traffic policing and traffic filtering to isolate the internal network from the external network and use NAT to control the internal network's access to the Internet.

DHCP Security



- On a campus network, employees often connect unauthorized DHCP-enabled wireless routers to the network, causing private address disorders, address conflicts, and Internet access failures.
- In most cases, DHCP snooping is enabled on access switches to prevent this issue.

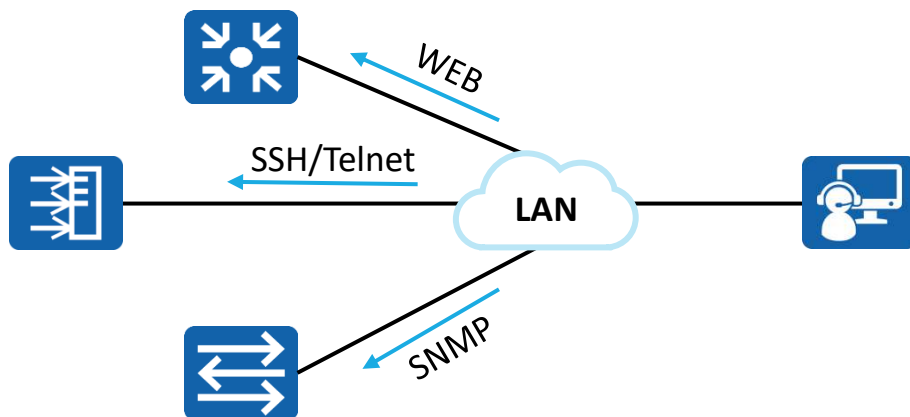
Network Management Security

- When network devices are managed through Telnet or the web system, you can use access control list (ACL) technology to allow only users with fixed IP addresses to log in to the devices.
- For the centralized NMS, SNMPv3 supports identity authentication and encryption, significantly enhancing the NMS security.



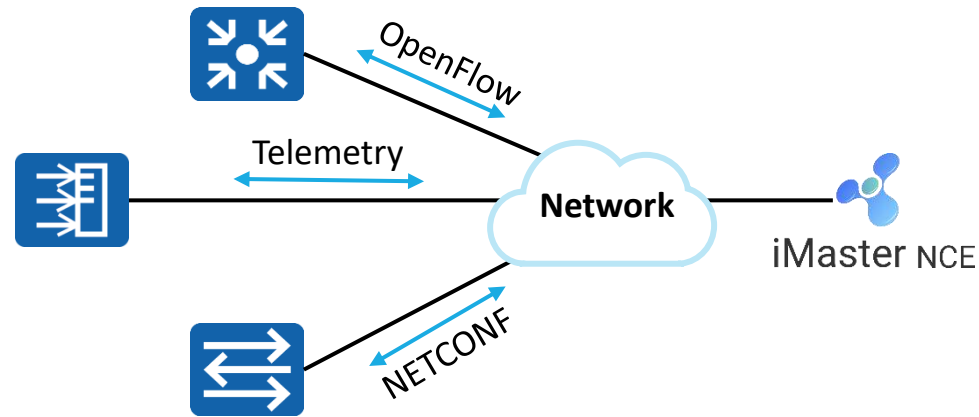
Network O&M and Management Design

Traditional Device Management



- When the network administrator and devices' IP addresses are routable to each other, you can manage the devices through Telnet, the web system, or SSH.
- When there are a large number of devices on a network, you can deploy an SNMP-based unified NMS for network O&M and management.

Management Based on iMaster NCE



- In addition to the SNMP-based traditional NMS, Huawei iMaster NCE can also be used for network management and O&M to implement autonomous network driving.



Small Campus Network Deployment and Implementation

- The project deployment and implementation process must include:
 - Solution formulation
 - Device installation
 - Network commissioning
 - Network migration and integration
 - Transfer-to-maintenance (ETM) training
 - Project acceptance
- The specific process is determined based on the actual situation.



Configuration Scheme (1)

1. Connect network devices using physical cables, configure link aggregation, and add interface description. For details, see the following two tables.

Device	Interface	Configuration
Acc-S1	Eth-trunk 1	Mode: LACP-static Trunkport: GE0/0/1, GE0/0/2, GE0/0/3 Description: to Agg-S1's eth-trunk 1
	E0/0/10	Description: to AP1
	E0/0/11	Description: to AP2
Acc-S2	Eth-trunk 1	Mode: LACP-static Trunkport: GE0/0/1, GE0/0/2, GE0/0/3 Description: to Agg-S1's eth-trunk 2
Acc-S3	Eth-trunk 1	Mode: LACP-static Trunkport: GE0/0/1, GE0/0/2, GE0/0/3 Description: to Agg-S1's eth-trunk 3
Acc-S4	Eth-trunk 1	Mode: LACP-static Trunkport: GE0/0/1, GE0/0/2, GE0/0/3 Description: to Agg-S1's eth-trunk 4
AC1	GE0/0/1	Description: to Agg-S1's GE0/0/2
CORE-R1	GE0/0/1	Description: to Agg-S1's GE0/0/1

Device	Interface	Configuration
Agg-S1	Eth-trunk 1	Mode: LACP-static Trunkport: GE0/0/3, GE0/0/7, GE0/0/8 Description: to Acc-S1's eth-trunk 1
	Eth-trunk 2	Mode: LACP-static Trunkport: GE0/0/4, GE0/0/9, GE0/0/10 Description: to Acc-S2's eth-trunk 1
	Eth-trunk 3	Mode: LACP-static Trunkport: GE0/0/5, GE0/0/11, GE0/0/12 Description: to Acc-S3's eth-trunk 1
	Eth-trunk 4	Mode: LACP-static Trunkport: GE0/0/6, GE0/0/13, GE0/0/14 Description: to Acc-S4's eth-trunk 1
	GE0/0/1	Description: to CORE-R1's GE0/0/1
	GE0/0/2	Description: to AC1's GE0/0/1



Configuration Scheme (2)

2. Assign VLANs based on interfaces. For details, see the following two tables.

Device	Interface	Type	Configuration
Acc-S1	Eth-trunk 1	Trunk	PVID:100 Allow-pass VLAN 1, 100, 101
	E0/0/10		PVID:101 Allow-pass VLAN 1, 101
	E0/0/11		
Acc-S2	Eth-trunk 1	Trunk	PVID:100 Allow pass VLAN 2, 100
	Other ports	Access	Default VLAN 2
Acc-S3	Eth-trunk 1	Trunk	PVID:100 Allow pass VLAN 3, 100
	Other ports	Access	Default VLAN 3
Acc-S4	Eth-trunk 1	Trunk	PVID:100 Allow pass VLAN 4, 100
	Other ports	Access	Default VLAN 4

Device	Interface	Type	Configuration
Agg-S1	Eth-trunk 1	Trunk	PVID:100 Allow-pass VLAN 1, 100, 101
	Eth-trunk 2	Trunk	PVID:100 Allow pass VLAN 2, 100
	Eth-trunk 3	Trunk	PVID:100 Allow pass VLAN 3, 100
	Eth-trunk 4	Trunk	PVID:100 Allow pass VLAN 4, 100
	GE0/0/2	Access	Default VLAN 101
	GE0/0/1	Access	Default VLAN 102
AC1	GE0/0/1	Access	Default VLAN 101



Configuration Scheme (3)

3. Allocate IP addresses to STAs and APs using DHCP and statically configure IP addresses for network devices. For details, see the following two tables.

Device	Interface	Address/Mask
Agg-S1	VLANIF 1	192.168.1.254/24
	VLANIF 2	192.168.2.254/24
	VLANIF 3	192.168.3.254/24
	VLANIF 4	192.168.4.254/24
	VLANIF 100	192.168.100.254/24
	VLANIF 101	192.168.101.254/24
	VLANIF 102	192.168.102.2/30
CORE-R1	GE0/0/1	192.168.102.1/30
	GE0/0/0	Automatic obtaining via PPPoE
	Loopback0	1.1.1.1/32

Device	Interface	Address/Mask
Acc-S1	VLANIF 100	192.168.100.1/24
Acc-S2	VLANIF 100	192.168.100.2/24
Acc-S3	VLANIF 100	192.168.100.3/24
Acc-S4	VLANIF 100	192.168.100.4/24
AC1	VLANIF 101	192.168.1.101/24



Configuration Scheme (4)

4. Configure the IP address allocation mode. For details about DHCP, see the following table.

Network Segment	Other Parameters	Remarks
192.168.1.0/24	Gateway:192.168.1.254 DNS:192.168.1.254	Agg-S1 functions as a DHCP server.
192.168.2.0/24	Gateway:192.168.2.254 DNS:192.168.2.254	Agg-S1 functions as a DHCP server. Fixed IP addresses are allocated to printer (1) and the FTP server.
192.168.3.0/24	Gateway:192.168.3.254 DNS:192.168.3.254	Agg-S1 functions as a DHCP server. A fixed IP address is allocated to printer (2).
192.168.3.0/24	Gateway:192.168.4.254 DNS:192.168.4.254	Agg-S1 functions as a DHCP server. Fixed IP addresses are allocated to printer (3) and the network administrator.
192.168.101.0/24	N/A	Agg-S1 functions as a DHCP server. The IP address (192.168.101.1) occupied by the AC is not allocated.



Configuration Scheme (5)

5. Configure routes. Static routes are used because the network scale is small and the number of NEs is also small. For details, see the following table.

Device	Route Configuration	Remarks
Acc-S1	0.0.0.0 0 192.168.100.254	Route that enables the network administrator to access Layer 2 switches across network segments.
Acc-S2		
Acc-S3		
Acc-S4		
AC1	0.0.0.0 0 192.168.101.254	Route that enables the administrator to access AC1 across network segments.
Agg-S1	0.0.0.0 0 192.168.102.1	Route that matches the traffic destined for the Internet
CORE-R1	192.168.0.0 20 192.168.102.2	Aggregated route for the core router to access the intranet
	Default route	Route pointing to an interface on the external network



Configuration Scheme (6)

6. Configure network management. Set the network management mode to Telnet-based remote management and authentication mode to Authentication, Authorization, and Accounting (AAA). For details, see the following table.

Device	Management Mode	Authentication Mode	Remarks
Acc-S1	Telnet	AAA	The user name and password must be complex and different. In addition, record them.
Acc-S2			
Acc-S3			
Acc-S4			
Agg-S1			
CORE-R1			
AC1			
AP1&AP2	Centralized control and management by the AC	N/A	N/A

7. Network egress configuration

Device	Interface	Access Mode	NAT Mode	Remarks
CORE-R1	GE0/0/0	PPPoE	Easy IP	User name: PPPoEUser123 Password: Huawei@123



Configuration Scheme (7)

8. Configure the WLAN as planned.
9. Perform security-related configurations. For details, see the following table.

Module	Related Technology	Configuration
Traffic monitoring	Traffic policy, NAT, and ACL	1. Configure an advanced ACL to block the traffic from 192.168.1.0/24 to the service network segment on the intranet and allow other traffic to pass through. Configure a traffic filtering policy to reference this ACL and apply the policy to an interface. 2. Configure a basic ACL to permit only the traffic from 192.168.1.0/24 and apply this ACL to the NAT configuration on an outbound network interface.
Network management security	AAA and ACL	Configure a basic ACL to permit only the packets whose source IP address is the administrator's IP address and wildcard mask is 0, and apply the ACL to the VTY interfaces of all managed devices.
DHCP security	DHCP snooping	Enable DHCP snooping on all access switches and configure the uplink interfaces as trusted interfaces.



Small Campus Network Commissioning

1. Connectivity Test

Basic link interconnection test

Layer 2 interoperability test

Layer 3 interoperability test

2. High Reliability Commissioning

Loop prevention function test

Path switchover test

Hot Standby (HSB) test

3. Service Performance Test

Service traffic test

Access control test



Small Campus Network O&M

- After a small campus network is provisioned, it enters the O&M phase. Common O&M methods include:
 - Device environment check
 - Basic device information check
 - Device running status check
 - Service check
 - Alarm handling
- When the network scale reaches a certain level, the network management software can be used for network management and O&M to improve efficiency.



Small Campus Network Optimization

- Network optimization can comprehensively improve the reliability and robustness of networks and better support the development of enterprise services. Common network optimization solutions include but are not limited to:
 - Device performance optimization, such as hardware upgrade and software version update
 - Basic network optimization, such as network architecture optimization and routing protocol adjustment
 - Service quality optimization, such as preferential forwarding of voice and video services
- Formulate an appropriate network optimization solution based on network requirements and actual conditions.



Quiz

1. What is the complete lifecycle of a campus network?
2. What is the function of a management IP address?



Summary

- This chapter describes the concepts, types, and common technologies of campus networks.
- Understand the lifecycle of campus networks:
 - Planning and design
 - Deployment and implementation
 - Network O&M
 - Network optimization
- Based on the previous courses, this course focuses on the planning, design, deployment, and implementation of campus networks and details how to establish a small campus network.

The image features a blue-tinted background showing silhouettes of several groups of business professionals in a modern office environment. They are standing on a highly reflective floor, and their reflections are clearly visible. The background also shows architectural elements like glass walls and windows. Overlaid on this scene is the text "Thank You" and the website address "www.huawei.com".

Thank You

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